Tables for Evaluating Local Structural-After-Measurement (LSAM) and Traditional Approaches for the Estimation of Complex Nonlinear Effects Among Latent Variables

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Performance Measures

Table S1

Complete performance measures for evaluating parameter estimates in the simulation studies

Performance Measure	Empirical Calculation	MCSE
Mean-Based Metrics		
Absolute Bias	$\bar{B} - \beta$	$\sqrt{S_B^2/K}$
Relative Bias	$(\frac{\bar{B}}{\beta}) - 1$	$\sqrt{S_B^2/(K\beta^2)}$
Percent Relative Bias	$\left[(\frac{\bar{B}}{\beta}) - 1\right] \times 100\%$	—
MSE	$\frac{1}{K}\sum_{k=1}^{K}(B_k - \beta)^2$	$\sqrt{\frac{1}{K}[S_B^4(k_B - 1) + 4S_B^3 g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
RMSE	$\sqrt{\frac{1}{K}\sum_{k=1}^{K}(B_k - \beta)^2}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\text{RMSE}_{(j)} - \text{RMSE})^2}$
Relative MSE	$\frac{(\bar{B}-\beta)^2 + S_B^2}{\beta^2}$	$\sqrt{\frac{1}{K\beta^4}[S_B^4(k_B - 1) + 4S_B^3 g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
Relative RMSE	$\sqrt{\frac{(\bar{B}-\beta)^2 + S_B^2}{\beta^2}}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\text{rRMSE}_{(j)} - \text{rRMSE})^2}$
SE/SD Ratio	$\overline{SE}/S_B$	—
Rejection Rate	$r_\alpha = \frac{1}{K}\sum_{k=1}^{K} I(P_k < \alpha)$	$\sqrt{r_\alpha(1 - r_\alpha)/K}$
Coverage	$c_\beta = \frac{1}{K}\sum_{k=1}^{K} I(L_k \leq \beta \leq U_k)$	$\sqrt{c_\beta(1 - c_\beta)/K}$
Average Width	$\bar{W} = \bar{U} - \bar{L}$	$\sqrt{S_W^2/K}$
Median-Based Metrics		
Median Bias	$Mdn(B) - \beta$	—
Relative Median Bias	$\frac{Mdn(B)-\beta}{\beta}$	—
Percent Relative Median Bias	$\frac{Mdn(B)-\beta}{\beta} \times 100\%$	—
MAD	$Mdn(\lVert B_k - Mdn(B) \rVert)$	—
Median RMSE	$\sqrt{\text{Bias}_{Mdn}^2 + \text{MAD}^2}$	—

Note. B = estimator for parameter β; K = number of replications producing estimates $B_1, ..., B_K$; $\bar{B} = (1/K)\sum_{k=1}^{K} B_k$ = sample mean; $S_B^2 = (1/(K-1))\sum_{k=1}^{K}(B_k - \bar{B})^2$ = sample variance; $S_B = \sqrt{S_B^2}$ = sample standard deviation; $\overline{SE} = (1/K)\sum_{k=1}^{K}\widehat{SE}_k$ = mean standard error; k_B = standardized kurtosis; g_B = standardized skewness; $I(\cdot)$ = indicator function; P_k = p-value from replication k; α = significance level; L_k, U_k = lower and upper CI endpoints; S_W^2 = variance of CI widths; r_α and c_β denote rejection rate and coverage, respectively; $Mdn(B)$ = sample median of estimates; MAD = median absolute deviation with scaling factor 1.4826. MCSE = Monte Carlo Standard Error, dashes indicate MCSEs not computed. MSE = Mean Squared Error. RMSE = Root Mean Squared Error.

Complete Results Tables — Simulation 1

Table S1

Simulation 1 - Condition 1: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.297	-0.019 ± 0.003	-0.060 ± 0.010	0.010 ± 0.000	0.098 ± 0.003	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.009	0.290	-0.026	-8.12	0.0914	0.0949	1.007	93.929 ± 0.793	0.387 ± 0.002	88.742 ± 1.050
LMS	beta32	0.316	0.301	-0.015 ± 0.003	-0.046 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.005	0.307 ± 0.009	0.301	-0.015	-4.90	0.0964	0.0977	1.025	94.923 ± 0.729	0.386 ± 0.002	90.177 ± 0.989
LMS	beta33	0.139	0.110	-0.029 ± 0.005	-0.212 ± 0.033	0.019 ± 0.001	0.137 ± 0.004	0.020 ± 0.001	0.140 ± 0.004	1.015 ± 0.050	1.007 ± 0.030	0.102	-0.037	-26.73	0.1307	0.1358	1.012	94.371 ± 0.766	0.543 ± 0.006	14.680 ± 1.176
LMS	beta34	0.101	0.109	0.008 ± 0.003	0.076 ± 0.031	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.876 ± 0.054	0.936 ± 0.033	0.104	0.003	2.59	0.0894	0.0894	0.997	95.585 ± 0.682	0.369 ± 0.005	22.406 ± 1.385
LMS	beta35	0.101	0.103	0.002 ± 0.003	0.021 ± 0.030	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.803 ± 0.043	0.896 ± 0.028	0.101	0.000	-0.06	0.0899	0.0899	1.009	95.033 ± 0.722	0.358 ± 0.005	22.075 ± 1.378
LSAM	beta31	0.316	0.321	0.005 ± 0.004	0.015 ± 0.012	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	0.140 ± 0.008	0.374 ± 0.012	0.316	0.000	-0.01	0.1212	0.1212	1.073	96.799 ± 0.585	0.497 ± 0.011	78.918 ± 1.355
LSAM	beta32	0.316	0.326	0.010 ± 0.004	0.033 ± 0.012	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.127 ± 0.006	0.356 ± 0.011	0.326	0.010	3.11	0.1157	0.1161	1.125	98.124 ± 0.451	0.494 ± 0.009	80.684 ± 1.312
LSAM	beta33	0.139	0.149	0.010 ± 0.006	0.075 ± 0.041	0.030 ± 0.002	0.172 ± 0.005	0.030 ± 0.002	0.172 ± 0.006	1.535 ± 0.094	1.239 ± 0.043	0.142	0.003	2.30	0.1624	0.1625	1.125	98.013 ± 0.464	0.759 ± 0.024	13.466 ± 1.134
LSAM	beta34	0.101	0.107	0.006 ± 0.004	0.057 ± 0.040	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.004	1.469 ± 0.088	1.212 ± 0.042	0.102	0.001	1.10	0.1168	0.1168	1.087	97.792 ± 0.488	0.522 ± 0.013	10.265 ± 1.008
LSAM	beta35	0.101	0.104	0.003 ± 0.004	0.031 ± 0.042	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.127 ± 0.004	1.587 ± 0.098	1.260 ± 0.044	0.097	-0.004	-3.80	0.1169	0.1170	1.072	97.792 ± 0.488	0.535 ± 0.019	11.921 ± 1.077
QML	beta31	0.316	0.298	-0.018 ± 0.003	-0.058 ± 0.010	0.010 ± 0.000	0.098 ± 0.003	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.009	0.290	-0.026	-8.33	0.0920	0.0957	1.009	93.929 ± 0.793	0.388 ± 0.002	88.631 ± 1.055
QML	beta32	0.316	0.302	-0.014 ± 0.003	-0.044 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.095 ± 0.005	0.308 ± 0.009	0.299	-0.017	-5.26	0.0963	0.0977	1.025	94.812 ± 0.737	0.387 ± 0.002	90.618 ± 0.969
QML	beta33	0.139	0.111	-0.028 ± 0.005	-0.199 ± 0.033	0.019 ± 0.001	0.138 ± 0.004	0.020 ± 0.001	0.140 ± 0.004	1.022 ± 0.052	1.011 ± 0.031	0.105	-0.034	-24.52	0.1348	0.1391	1.018	94.481 ± 0.759	0.550 ± 0.007	14.901 ± 1.183
QML	beta34	0.101	0.108	0.007 ± 0.003	0.066 ± 0.031	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.878 ± 0.052	0.937 ± 0.032	0.104	0.003	2.82	0.0889	0.0889	1.006	95.585 ± 0.682	0.373 ± 0.005	21.854 ± 1.373
QML	beta35	0.101	0.103	0.002 ± 0.003	0.016 ± 0.030	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	0.823 ± 0.045	0.907 ± 0.029	0.100	-0.001	-1.11	0.0913	0.0913	1.009	94.481 ± 0.759	0.363 ± 0.005	21.854 ± 1.373
UPI	beta31	0.316	0.320	0.004 ± 0.004	0.012 ± 0.013	0.015 ± 0.001	0.122 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	0.150 ± 0.008	0.387 ± 0.012	0.313	-0.003	-1.06	0.1208	0.1209	1.131	96.689 ± 0.594	0.543 ± 0.009	71.523 ± 1.499
UPI	beta32	0.316	0.327	0.011 ± 0.004	0.036 ± 0.012	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.003	0.135 ± 0.006	0.367 ± 0.011	0.325	0.009	2.77	0.1201	0.1204	1.188	98.234 ± 0.438	0.538 ± 0.009	74.503 ± 1.448
UPI	beta33	0.139	0.165	0.026 ± 0.007	0.187 ± 0.047	0.039 ± 0.002	0.198 ± 0.006	0.040 ± 0.003	0.199 ± 0.007	2.059 ± 0.132	1.435 ± 0.052	0.145	0.006	4.58	0.1682	0.1684	1.298	98.896 ± 0.347	1.006 ± 0.054	7.837 ± 0.893
UPI	beta34	0.101	0.110	0.009 ± 0.005	0.084 ± 0.048	0.021 ± 0.001	0.146 ± 0.005	0.021 ± 0.001	0.146 ± 0.005	2.091 ± 0.130	1.446 ± 0.051	0.100	-0.001	-0.99	0.1268	0.1268	1.259	98.124 ± 0.451	0.720 ± 0.022	4.415 ± 0.682
UPI	beta35	0.101	0.098	-0.003 ± 0.005	-0.029 ± 0.049	0.022 ± 0.001	0.149 ± 0.005	0.022 ± 0.001	0.149 ± 0.005	2.170 ± 0.131	1.473 ± 0.051	0.091	-0.010	-9.73	0.1309	0.1313	1.259	97.903 ± 0.476	0.735 ± 0.028	5.298 ± 0.744

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S2

Simulation 1 - Condition 2: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.297	-0.019 ± 0.002	-0.061 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.199 ± 0.005	0.297	-0.019	-6.11	0.0581	0.0612	1.035	93.461 ± 0.784	0.243 ± 0.003	99.598 ± 0.201
LMS	beta32	0.316	0.295	-0.021 ± 0.002	-0.067 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.195 ± 0.005	0.292	-0.024	-7.63	0.0612	0.0658	1.066	94.165 ± 0.743	0.242 ± 0.003	99.698 ± 0.174
LMS	beta33	0.139	0.106	-0.033 ± 0.003	-0.236 ± 0.020	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.436 ± 0.021	0.660 ± 0.018	0.110	-0.029	-20.65	0.0813	0.0862	1.084	93.260 ± 0.795	0.364 ± 0.028	25.050 ± 1.374
LMS	beta34	0.101	0.109	0.008 ± 0.002	0.084 ± 0.018	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.323 ± 0.017	0.568 ± 0.017	0.107	0.006	6.28	0.0532	0.0536	1.094	94.769 ± 0.706	0.244 ± 0.020	49.799 ± 1.586
LMS	beta35	0.101	0.108	0.007 ± 0.002	0.071 ± 0.018	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.324 ± 0.015	0.569 ± 0.016	0.107	0.006	6.06	0.0542	0.0545	1.045	94.769 ± 0.706	0.234 ± 0.011	49.296 ± 1.586
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.317	0.001	0.36	0.0663	0.0663	1.017	95.674 ± 0.645	0.273 ± 0.001	99.396 ± 0.246
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.209 ± 0.006	0.314	-0.002	-0.52	0.0697	0.0697	1.051	96.781 ± 0.560	0.272 ± 0.001	99.698 ± 0.174
LSAM	beta33	0.139	0.138	-0.001 ± 0.003	-0.004 ± 0.021	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.459 ± 0.024	0.677 ± 0.020	0.141	0.002	1.54	0.0896	0.0896	1.008	96.579 ± 0.576	0.372 ± 0.004	36.016 ± 1.523
LSAM	beta34	0.101	0.107	0.006 ± 0.002	0.055 ± 0.021	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.447 ± 0.025	0.668 ± 0.021	0.104	0.003	3.39	0.0620	0.0621	1.000	95.875 ± 0.631	0.264 ± 0.002	36.720 ± 1.529
LSAM	beta35	0.101	0.107	0.006 ± 0.002	0.060 ± 0.022	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.464 ± 0.024	0.681 ± 0.021	0.106	0.005	4.55	0.0673	0.0674	0.995	95.775 ± 0.638	0.267 ± 0.003	36.922 ± 1.531
QML	beta31	0.316	0.297	-0.019 ± 0.002	-0.061 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.199 ± 0.005	0.297	-0.019	-6.11	0.0584	0.0615	1.019	93.461 ± 0.784	0.240 ± 0.001	99.799 ± 0.142
QML	beta32	0.316	0.295	-0.021 ± 0.002	-0.067 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.195 ± 0.005	0.292	-0.024	-7.46	0.0615	0.0659	1.056	94.266 ± 0.737	0.240 ± 0.001	99.799 ± 0.142
QML	beta33	0.139	0.108	-0.031 ± 0.003	-0.222 ± 0.020	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.091 ± 0.002	0.431 ± 0.020	0.657 ± 0.018	0.112	-0.027	-19.23	0.0819	0.0861	1.000	93.561 ± 0.778	0.337 ± 0.002	26.258 ± 1.396
QML	beta34	0.101	0.108	0.007 ± 0.002	0.072 ± 0.018	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.320 ± 0.017	0.566 ± 0.017	0.106	0.005	4.85	0.0532	0.0534	1.004	94.668 ± 0.713	0.223 ± 0.002	49.296 ± 1.586
QML	beta35	0.101	0.107	0.006 ± 0.002	0.059 ± 0.018	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.324 ± 0.015	0.569 ± 0.016	0.107	0.006	5.51	0.0552	0.0555	0.996	94.869 ± 0.700	0.223 ± 0.002	48.592 ± 1.585
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.008 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.002	0.220 ± 0.006	0.318	0.002	0.77	0.0667	0.0667	1.025	95.875 ± 0.631	0.279 ± 0.001	99.296 ± 0.265
UPI	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.208 ± 0.006	0.315	-0.001	-0.26	0.0682	0.0682	1.071	97.183 ± 0.525	0.276 ± 0.001	99.799 ± 0.142
UPI	beta33	0.139	0.142	0.003 ± 0.003	0.021 ± 0.023	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.515 ± 0.026	0.718 ± 0.022	0.140	0.001	0.86	0.0935	0.0935	1.029	97.082 ± 0.534	0.402 ± 0.004	31.489 ± 1.473
UPI	beta34	0.101	0.108	0.007 ± 0.002	0.070 ± 0.024	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.554 ± 0.031	0.744 ± 0.024	0.103	0.002	2.19	0.0703	0.0704	1.012	97.183 ± 0.525	0.297 ± 0.003	29.175 ± 1.442
UPI	beta35	0.101	0.106	0.005 ± 0.002	0.050 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.573 ± 0.031	0.757 ± 0.024	0.102	0.001	0.89	0.0696	0.0696	0.996	95.372 ± 0.666	0.298 ± 0.004	28.873 ± 1.437

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S3

Simulation 1 - Condition 3: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.305	-0.011	-3.56	0.0675	0.0684	1.050	95.190 ± 0.677	0.280 ± 0.002	99.499 ± 0.223
LMS	beta32	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.226 ± 0.006	0.303	-0.013	-4.17	0.0708	0.0720	1.008	95.190 ± 0.677	0.279 ± 0.001	99.098 ± 0.299
LMS	beta33	0.139	0.122	-0.017 ± 0.003	-0.123 ± 0.022	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.488 ± 0.025	0.698 ± 0.021	0.121	-0.018	-12.75	0.0875	0.0893	0.916	91.884 ± 0.864	0.343 ± 0.003	32.565 ± 1.483
LMS	beta34	0.101	0.104	0.003 ± 0.002	0.034 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.380 ± 0.017	0.617 ± 0.017	0.104	0.003	3.09	0.0643	0.0644	0.948	93.888 ± 0.758	0.231 ± 0.002	44.689 ± 1.574
LMS	beta35	0.101	0.103	0.002 ± 0.002	0.022 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.382 ± 0.018	0.618 ± 0.018	0.101	0.000	0.28	0.0583	0.0583	0.952	93.888 ± 0.758	0.233 ± 0.002	44.689 ± 1.574
LSAM	beta31	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.231 ± 0.007	0.318	0.002	0.72	0.0712	0.0712	1.043	95.892 ± 0.628	0.299 ± 0.001	98.798 ± 0.345
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.240 ± 0.007	0.315	-0.001	-0.40	0.0752	0.0752	1.004	95.691 ± 0.643	0.299 ± 0.001	98.798 ± 0.345
LSAM	beta33	0.139	0.141	0.002 ± 0.003	0.016 ± 0.023	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.533 ± 0.029	0.730 ± 0.023	0.140	0.001	0.77	0.0915	0.0915	0.907	92.786 ± 0.819	0.361 ± 0.003	37.776 ± 1.535
LSAM	beta34	0.101	0.104	0.003 ± 0.002	0.028 ± 0.022	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.465 ± 0.023	0.682 ± 0.020	0.103	0.002	1.86	0.0675	0.0675	0.943	95.090 ± 0.684	0.255 ± 0.002	39.178 ± 1.545
LSAM	beta35	0.101	0.103	0.002 ± 0.002	0.023 ± 0.021	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.451 ± 0.021	0.671 ± 0.019	0.100	-0.001	-1.07	0.0621	0.0621	0.957	94.389 ± 0.728	0.254 ± 0.002	38.377 ± 1.539
QML	beta31	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.305	-0.011	-3.50	0.0672	0.0681	1.044	95.291 ± 0.671	0.279 ± 0.001	99.599 ± 0.200
QML	beta32	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.225 ± 0.006	0.303	-0.013	-4.23	0.0704	0.0717	1.005	95.190 ± 0.677	0.278 ± 0.001	99.198 ± 0.282
QML	beta33	0.139	0.123	-0.016 ± 0.003	-0.118 ± 0.022	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.485 ± 0.025	0.696 ± 0.021	0.122	-0.017	-12.42	0.0869	0.0886	0.919	92.285 ± 0.845	0.344 ± 0.003	32.966 ± 1.488
QML	beta34	0.101	0.104	0.003 ± 0.002	0.027 ± 0.019	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.378 ± 0.017	0.615 ± 0.017	0.103	0.002	2.39	0.0643	0.0644	0.952	94.289 ± 0.735	0.232 ± 0.002	44.790 ± 1.574
QML	beta35	0.101	0.102	0.001 ± 0.002	0.015 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.382 ± 0.018	0.618 ± 0.017	0.101	0.000	-0.00	0.0590	0.0590	0.953	93.888 ± 0.758	0.233 ± 0.002	44.088 ± 1.572
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.318	0.002	0.61	0.0714	0.0714	1.022	95.992 ± 0.621	0.295 ± 0.001	98.798 ± 0.345
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.011 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.240 ± 0.007	0.316	0.000	0.10	0.0749	0.0749	0.995	95.190 ± 0.677	0.295 ± 0.001	98.898 ± 0.330
UPI	beta33	0.139	0.143	0.004 ± 0.003	0.032 ± 0.024	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	0.568 ± 0.031	0.754 ± 0.024	0.141	0.002	1.57	0.0966	0.0967	0.926	93.687 ± 0.770	0.380 ± 0.003	34.870 ± 1.509
UPI	beta34	0.101	0.104	0.003 ± 0.002	0.031 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.523 ± 0.026	0.723 ± 0.021	0.103	0.002	2.14	0.0702	0.0703	0.956	96.293 ± 0.598	0.274 ± 0.003	34.168 ± 1.501
UPI	beta35	0.101	0.103	0.002 ± 0.002	0.016 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.487 ± 0.023	0.698 ± 0.020	0.100	-0.001	-1.46	0.0643	0.0643	0.998	95.090 ± 0.684	0.276 ± 0.003	31.563 ± 1.471

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S4

Simulation 1 - Condition 4: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.304	-0.012 ± 0.001	-0.038 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.304	-0.012	-3.79	0.0413	0.0430	1.027	93.800 ± 0.763	0.175 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.305	-0.011 ± 0.001	-0.034 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.147 ± 0.004	0.304	-0.012	-3.90	0.0460	0.0476	0.986	93.700 ± 0.768	0.174 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.120	-0.019 ± 0.002	-0.135 ± 0.012	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.166 ± 0.007	0.407 ± 0.011	0.118	-0.021	-14.88	0.0538	0.0577	1.012	94.100 ± 0.745	0.212 ± 0.001	59.800 ± 1.550
LMS	beta34	0.101	0.106	0.005 ± 0.001	0.045 ± 0.011	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.133 ± 0.006	0.364 ± 0.010	0.105	0.004	3.71	0.0353	0.0355	1.004	94.300 ± 0.733	0.144 ± 0.001	84.000 ± 1.159
LMS	beta35	0.101	0.105	0.004 ± 0.001	0.037 ± 0.011	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.131 ± 0.006	0.363 ± 0.010	0.105	0.004	4.22	0.0367	0.0369	1.009	95.300 ± 0.669	0.144 ± 0.001	81.200 ± 1.236
LSAM	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.315	-0.001	-0.24	0.0432	0.0432	1.032	95.000 ± 0.689	0.185 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.315	-0.001	-0.21	0.0504	0.0504	0.991	95.400 ± 0.662	0.185 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.137	-0.002 ± 0.002	-0.012 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.150 ± 0.007	0.387 ± 0.011	0.137	-0.002	-1.41	0.0543	0.0544	1.017	95.600 ± 0.649	0.214 ± 0.001	71.900 ± 1.421
LSAM	beta34	0.101	0.103	0.002 ± 0.001	0.019 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.148 ± 0.007	0.385 ± 0.011	0.101	0.000	0.38	0.0385	0.0385	0.999	94.800 ± 0.702	0.152 ± 0.001	77.600 ± 1.318
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.012 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.148 ± 0.007	0.385 ± 0.011	0.102	0.001	0.73	0.0398	0.0398	1.003	94.900 ± 0.696	0.153 ± 0.001	75.000 ± 1.369
QML	beta31	0.316	0.304	-0.012 ± 0.001	-0.038 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.304	-0.012	-3.86	0.0409	0.0427	1.026	93.700 ± 0.768	0.175 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.305	-0.011 ± 0.001	-0.033 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.304	-0.012	-3.83	0.0459	0.0475	0.985	93.600 ± 0.774	0.174 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.121	-0.018 ± 0.002	-0.132 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.165 ± 0.007	0.407 ± 0.011	0.119	-0.020	-14.61	0.0541	0.0578	1.011	94.400 ± 0.727	0.212 ± 0.001	60.400 ± 1.547
QML	beta34	0.101	0.105	0.004 ± 0.001	0.040 ± 0.011	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.132 ± 0.006	0.364 ± 0.010	0.104	0.003	3.19	0.0352	0.0354	1.003	94.400 ± 0.727	0.144 ± 0.001	83.500 ± 1.174
QML	beta35	0.101	0.104	0.003 ± 0.001	0.032 ± 0.011	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.131 ± 0.006	0.362 ± 0.010	0.105	0.004	3.87	0.0372	0.0374	1.008	95.300 ± 0.669	0.144 ± 0.001	80.800 ± 1.246
UPI	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.315	-0.001	-0.18	0.0429	0.0429	1.011	94.700 ± 0.708	0.182 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.316	0.000	0.06	0.0494	0.0494	0.971	95.100 ± 0.683	0.182 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.138	-0.001 ± 0.002	-0.005 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.138	-0.001	-0.43	0.0528	0.0528	1.007	95.200 ± 0.676	0.219 ± 0.001	70.900 ± 1.436
UPI	beta34	0.101	0.102	0.001 ± 0.001	0.012 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.155 ± 0.008	0.394 ± 0.011	0.101	0.000	-0.09	0.0384	0.0384	1.006	95.600 ± 0.649	0.157 ± 0.001	74.300 ± 1.382
UPI	beta35	0.101	0.102	0.001 ± 0.001	0.009 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.153 ± 0.007	0.391 ± 0.011	0.102	0.001	0.74	0.0389	0.0389	1.019	95.400 ± 0.662	0.158 ± 0.001	73.100 ± 1.402

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S5

Simulation 1 - Condition 5: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.313	-0.003	-1.11	0.0584	0.0585	0.970	94.200 ± 0.739	0.227 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.310	-0.006 ± 0.002	-0.018 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.310	-0.006	-1.92	0.0587	0.0590	0.980	93.500 ± 0.780	0.227 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.134	-0.005 ± 0.002	-0.033 ± 0.016	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.241 ± 0.011	0.491 ± 0.013	0.134	-0.005	-3.57	0.0674	0.0676	0.935	93.100 ± 0.801	0.250 ± 0.001	55.300 ± 1.572
LMS	beta34	0.101	0.102	0.001 ± 0.001	0.014 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.206 ± 0.009	0.454 ± 0.012	0.100	-0.001	-0.56	0.0454	0.0454	0.962	93.600 ± 0.774	0.173 ± 0.001	63.600 ± 1.522
LMS	beta35	0.101	0.101	0.000 ± 0.001	0.001 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.209 ± 0.009	0.457 ± 0.012	0.102	0.001	0.73	0.0454	0.0454	0.956	94.000 ± 0.751	0.173 ± 0.001	63.000 ± 1.527
LSAM	beta31	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.194 ± 0.005	0.317	0.001	0.35	0.0605	0.0605	0.971	94.400 ± 0.727	0.233 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.316	0.000	-0.06	0.0595	0.0595	0.985	94.000 ± 0.751	0.233 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.142	0.003 ± 0.002	0.020 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.248 ± 0.011	0.498 ± 0.014	0.142	0.003	1.97	0.0694	0.0695	0.928	92.800 ± 0.817	0.252 ± 0.001	58.800 ± 1.556
LSAM	beta34	0.101	0.102	0.001 ± 0.002	0.012 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.221 ± 0.010	0.470 ± 0.013	0.100	-0.001	-1.34	0.0479	0.0479	0.953	93.300 ± 0.791	0.178 ± 0.001	60.700 ± 1.545
LSAM	beta35	0.101	0.101	-0.000 ± 0.002	-0.003 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.225 ± 0.010	0.474 ± 0.013	0.101	0.000	-0.45	0.0482	0.0482	0.945	93.800 ± 0.763	0.177 ± 0.001	60.700 ± 1.545
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.312	-0.004	-1.14	0.0581	0.0582	0.970	94.200 ± 0.739	0.227 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.310	-0.006 ± 0.002	-0.019 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.310	-0.006	-1.93	0.0586	0.0589	0.981	93.300 ± 0.791	0.227 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.134	-0.005 ± 0.002	-0.034 ± 0.016	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.241 ± 0.011	0.491 ± 0.013	0.134	-0.005	-3.87	0.0681	0.0683	0.936	93.100 ± 0.801	0.250 ± 0.001	55.100 ± 1.573
QML	beta34	0.101	0.102	0.001 ± 0.001	0.013 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.206 ± 0.009	0.454 ± 0.012	0.100	-0.001	-0.61	0.0454	0.0454	0.961	93.700 ± 0.768	0.173 ± 0.001	63.400 ± 1.523
QML	beta35	0.101	0.101	-0.000 ± 0.001	-0.000 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.209 ± 0.009	0.457 ± 0.012	0.102	0.001	0.58	0.0457	0.0457	0.955	93.900 ± 0.757	0.173 ± 0.001	63.100 ± 1.526
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.317	0.001	0.32	0.0598	0.0598	0.939	93.300 ± 0.791	0.225 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.316	0.000	0.06	0.0594	0.0594	0.949	93.300 ± 0.791	0.225 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.143	0.004 ± 0.002	0.025 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.255 ± 0.012	0.505 ± 0.014	0.142	0.003	1.83	0.0701	0.0702	0.927	92.900 ± 0.812	0.255 ± 0.001	58.600 ± 1.558
UPI	beta34	0.101	0.102	0.001 ± 0.002	0.005 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.224 ± 0.010	0.473 ± 0.013	0.099	-0.002	-1.93	0.0461	0.0462	0.963	93.700 ± 0.768	0.180 ± 0.001	59.000 ± 1.555
UPI	beta35	0.101	0.101	-0.000 ± 0.002	-0.003 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.240 ± 0.011	0.489 ± 0.014	0.100	-0.001	-0.50	0.0484	0.0484	0.934	94.200 ± 0.739	0.181 ± 0.001	59.600 ± 1.552

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S6

Simulation 1 - Condition 6: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.015 ± 0.001	0.123 ± 0.003	0.312	-0.004	-1.25	0.0390	0.0392	0.950	93.600 ± 0.774	0.144 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.120 ± 0.003	0.310	-0.006	-1.97	0.0372	0.0377	0.976	94.500 ± 0.721	0.144 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.134	-0.005 ± 0.001	-0.036 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.132	-0.007	-5.31	0.0390	0.0397	1.014	95.400 ± 0.662	0.159 ± 0.001	92.100 ± 0.853
LMS	beta34	0.101	0.102	0.001 ± 0.001	0.008 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.083 ± 0.004	0.287 ± 0.008	0.101	0.000	-0.36	0.0277	0.0277	0.967	93.300 ± 0.791	0.110 ± 0.000	94.100 ± 0.745
LMS	beta35	0.101	0.101	0.000 ± 0.001	0.003 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.078 ± 0.004	0.279 ± 0.008	0.102	0.001	0.92	0.0282	0.0282	0.991	95.300 ± 0.669	0.109 ± 0.000	94.300 ± 0.733
LSAM	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.004	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.016 ± 0.001	0.125 ± 0.003	0.317	0.001	0.22	0.0398	0.0398	0.953	93.700 ± 0.768	0.147 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.315	-0.001	-0.17	0.0382	0.0382	0.980	94.800 ± 0.702	0.147 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.141	0.002 ± 0.001	0.016 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.139	0.000	-0.02	0.0382	0.0382	1.016	95.600 ± 0.649	0.158 ± 0.001	94.600 ± 0.715
LSAM	beta34	0.101	0.101	-0.000 ± 0.001	-0.003 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.101	0.000	-0.27	0.0286	0.0286	0.955	93.100 ± 0.801	0.112 ± 0.000	92.600 ± 0.828
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.007 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.082 ± 0.004	0.286 ± 0.008	0.100	-0.001	-0.63	0.0292	0.0292	0.983	95.200 ± 0.676	0.111 ± 0.000	93.600 ± 0.774
QML	beta31	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.015 ± 0.001	0.123 ± 0.003	0.312	-0.004	-1.23	0.0393	0.0395	0.955	93.400 ± 0.785	0.145 ± 0.001	99.900 ± 0.100
QML	beta32	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.120 ± 0.003	0.310	-0.006	-1.98	0.0370	0.0375	0.984	94.400 ± 0.727	0.145 ± 0.001	99.900 ± 0.100
QML	beta33	0.139	0.134	-0.005 ± 0.001	-0.038 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.131	-0.008	-5.46	0.0388	0.0395	1.014	95.500 ± 0.656	0.159 ± 0.001	92.100 ± 0.853
QML	beta34	0.101	0.102	0.001 ± 0.001	0.008 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.101	0.000	-0.21	0.0276	0.0276	0.966	93.200 ± 0.796	0.110 ± 0.000	94.100 ± 0.745
QML	beta35	0.101	0.101	0.000 ± 0.001	0.002 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.078 ± 0.004	0.279 ± 0.008	0.102	0.001	0.87	0.0283	0.0283	0.991	95.200 ± 0.676	0.109 ± 0.000	94.200 ± 0.739
UPI	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.016 ± 0.001	0.125 ± 0.003	0.317	0.001	0.20	0.0401	0.0401	0.921	92.400 ± 0.838	0.143 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.315	-0.001	-0.27	0.0386	0.0386	0.951	93.600 ± 0.774	0.143 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.141	0.002 ± 0.001	0.017 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.138	-0.001	-0.39	0.0387	0.0387	1.015	94.900 ± 0.696	0.159 ± 0.001	94.500 ± 0.721
UPI	beta34	0.101	0.101	-0.000 ± 0.001	-0.002 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.090 ± 0.004	0.300 ± 0.009	0.100	-0.001	-0.69	0.0291	0.0291	0.953	93.600 ± 0.774	0.113 ± 0.000	92.600 ± 0.828
UPI	beta35	0.101	0.100	-0.001 ± 0.001	-0.007 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.101	0.000	-0.45	0.0296	0.0296	0.988	95.400 ± 0.662	0.113 ± 0.000	93.400 ± 0.785

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S7

Simulation 1 - Condition 7: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.344	0.028 ± 0.004	0.087 ± 0.012	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.124 ± 0.007	0.352 ± 0.011	0.342	0.026	8.13	0.1067	0.1098	1.033	95.972 ± 0.677	0.437 ± 0.003	89.692 ± 1.047
LMS	beta32	0.316	0.340	0.024 ± 0.004	0.075 ± 0.012	0.011 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.109 ± 0.003	0.120 ± 0.006	0.346 ± 0.010	0.334	0.018	5.82	0.1115	0.1130	1.046	96.564 ± 0.627	0.438 ± 0.003	88.389 ± 1.103
LMS	beta33	0.139	0.013	-0.126 ± 0.005	-0.904 ± 0.033	0.018 ± 0.001	0.135 ± 0.004	0.034 ± 0.002	0.184 ± 0.004	1.754 ± 0.079	1.324 ± 0.032	0.012	-0.127	-91.38	0.1417	0.1903	1.060	83.412 ± 1.280	0.559 ± 0.007	4.502 ± 0.714
LMS	beta34	0.101	0.220	0.119 ± 0.003	1.181 ± 0.030	0.008 ± 0.000	0.089 ± 0.002	0.022 ± 0.001	0.149 ± 0.003	2.167 ± 0.091	1.472 ± 0.032	0.214	0.113	111.84	0.0846	0.1412	1.044	79.265 ± 1.395	0.363 ± 0.005	75.000 ± 1.490
LMS	beta35	0.101	0.218	0.117 ± 0.003	1.162 ± 0.030	0.008 ± 0.000	0.089 ± 0.003	0.022 ± 0.001	0.147 ± 0.003	2.121 ± 0.096	1.457 ± 0.034	0.213	0.112	110.84	0.0855	0.1409	1.039	80.450 ± 1.365	0.362 ± 0.005	75.474 ± 1.481
LSAM	beta31	0.316	0.320	0.004 ± 0.008	0.014 ± 0.025	0.054 ± 0.003	0.233 ± 0.007	0.054 ± 0.003	0.233 ± 0.008	0.543 ± 0.030	0.737 ± 0.024	0.322	0.006	1.77	0.2303	0.2304	1.139	98.578 ± 0.408	1.040 ± 0.023	29.858 ± 1.575
LSAM	beta32	0.316	0.318	0.002 ± 0.008	0.007 ± 0.026	0.057 ± 0.003	0.239 ± 0.007	0.057 ± 0.003	0.239 ± 0.008	0.571 ± 0.032	0.755 ± 0.025	0.307	-0.009	-2.92	0.2360	0.2362	1.158	98.223 ± 0.455	1.084 ± 0.028	29.028 ± 1.562
LSAM	beta33	0.139	0.168	0.029 ± 0.006	0.209 ± 0.045	0.033 ± 0.002	0.181 ± 0.005	0.034 ± 0.002	0.183 ± 0.006	1.740 ± 0.102	1.319 ± 0.045	0.165	0.026	18.89	0.1715	0.1735	1.116	96.209 ± 0.657	0.792 ± 0.020	11.137 ± 1.083
LSAM	beta34	0.101	0.104	0.003 ± 0.004	0.025 ± 0.035	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	1.062 ± 0.062	1.031 ± 0.035	0.096	-0.005	-5.17	0.0952	0.0954	1.061	97.275 ± 0.560	0.433 ± 0.011	23.223 ± 1.453
LSAM	beta35	0.101	0.102	0.001 ± 0.004	0.009 ± 0.036	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	1.107 ± 0.071	1.052 ± 0.038	0.103	0.002	1.73	0.0946	0.0947	1.112	97.275 ± 0.560	0.463 ± 0.018	24.289 ± 1.476
QML	beta31	0.316	0.344	0.028 ± 0.004	0.088 ± 0.012	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.112 ± 0.003	0.125 ± 0.006	0.353 ± 0.011	0.341	0.025	7.84	0.1093	0.1121	1.032	95.972 ± 0.677	0.438 ± 0.003	88.863 ± 1.083
QML	beta32	0.316	0.340	0.024 ± 0.004	0.075 ± 0.012	0.012 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.121 ± 0.006	0.348 ± 0.010	0.334	0.018	5.83	0.1107	0.1122	1.044	96.327 ± 0.647	0.440 ± 0.003	87.796 ± 1.127
QML	beta33	0.139	0.015	-0.124 ± 0.005	-0.892 ± 0.034	0.019 ± 0.001	0.137 ± 0.004	0.034 ± 0.002	0.185 ± 0.005	1.763 ± 0.085	1.328 ± 0.034	0.016	-0.123	-88.21	0.1389	0.1853	1.053	84.834 ± 1.235	0.565 ± 0.007	4.621 ± 0.723
QML	beta34	0.101	0.220	0.119 ± 0.003	1.182 ± 0.031	0.008 ± 0.000	0.091 ± 0.003	0.023 ± 0.001	0.150 ± 0.003	2.213 ± 0.099	1.488 ± 0.035	0.214	0.113	112.27	0.0854	0.1420	1.016	79.265 ± 1.395	0.363 ± 0.006	74.052 ± 1.509
QML	beta35	0.101	0.217	0.116 ± 0.003	1.151 ± 0.030	0.008 ± 0.000	0.089 ± 0.003	0.021 ± 0.001	0.146 ± 0.003	2.100 ± 0.096	1.449 ± 0.035	0.212	0.111	109.95	0.0866	0.1408	1.025	80.569 ± 1.362	0.358 ± 0.005	75.355 ± 1.483
UPI	beta31	0.316	0.336	0.020 ± 0.008	0.062 ± 0.026	0.055 ± 0.003	0.234 ± 0.007	0.055 ± 0.003	0.235 ± 0.008	0.552 ± 0.033	0.743 ± 0.026	0.329	0.013	4.15	0.2397	0.2401	1.234	98.697 ± 0.390	1.133 ± 0.036	31.991 ± 1.606
UPI	beta32	0.316	0.323	0.007 ± 0.008	0.021 ± 0.026	0.058 ± 0.003	0.242 ± 0.007	0.058 ± 0.003	0.242 ± 0.008	0.584 ± 0.035	0.765 ± 0.026	0.322	0.006	1.99	0.2263	0.2264	1.270	99.171 ± 0.312	1.203 ± 0.053	29.976 ± 1.577
UPI	beta33	0.139	0.143	0.004 ± 0.007	0.026 ± 0.050	0.041 ± 0.003	0.201 ± 0.007	0.041 ± 0.003	0.201 ± 0.008	2.097 ± 0.153	1.448 ± 0.059	0.133	-0.006	-4.04	0.1738	0.1739	1.468	96.090 ± 0.667	1.159 ± 0.068	5.924 ± 0.813
UPI	beta34	0.101	0.101	0.000 ± 0.004	0.004 ± 0.036	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	1.112 ± 0.072	1.055 ± 0.039	0.095	-0.006	-6.31	0.0989	0.0991	1.183	96.564 ± 0.627	0.494 ± 0.019	18.839 ± 1.346
UPI	beta35	0.101	0.104	0.003 ± 0.004	0.030 ± 0.038	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	1.202 ± 0.080	1.096 ± 0.041	0.095	-0.006	-6.32	0.0916	0.0918	1.235	97.393 ± 0.548	0.536 ± 0.028	21.327 ± 1.410

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S8

Simulation 1 - Condition 8: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.340	0.024 ± 0.002	0.075 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.227 ± 0.007	0.338	0.022	7.08	0.0640	0.0678	1.057	94.796 ± 0.710	0.281 ± 0.007	99.490 ± 0.228
LMS	beta32	0.316	0.339	0.023 ± 0.002	0.072 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.048 ± 0.002	0.220 ± 0.006	0.338	0.022	7.09	0.0686	0.0722	1.116	95.918 ± 0.632	0.288 ± 0.018	99.592 ± 0.204
LMS	beta33	0.139	0.010	-0.129 ± 0.003	-0.927 ± 0.020	0.008 ± 0.000	0.087 ± 0.002	0.024 ± 0.001	0.156 ± 0.003	1.253 ± 0.043	1.119 ± 0.020	0.010	-0.129	-92.53	0.0843	0.1538	1.079	64.898 ± 1.525	0.369 ± 0.037	6.939 ± 0.812
LMS	beta34	0.101	0.223	0.122 ± 0.002	1.211 ± 0.018	0.003 ± 0.000	0.057 ± 0.001	0.018 ± 0.001	0.135 ± 0.002	1.780 ± 0.050	1.334 ± 0.019	0.218	0.117	116.01	0.0545	0.1292	1.046	35.102 ± 1.525	0.232 ± 0.018	99.286 ± 0.269
LMS	beta35	0.101	0.225	0.124 ± 0.002	1.228 ± 0.017	0.003 ± 0.000	0.052 ± 0.001	0.018 ± 0.000	0.135 ± 0.002	1.776 ± 0.045	1.333 ± 0.017	0.223	0.122	120.85	0.0541	0.1335	1.094	33.367 ± 1.506	0.224 ± 0.011	99.592 ± 0.204
LSAM	beta31	0.316	0.317	0.001 ± 0.004	0.003 ± 0.013	0.018 ± 0.001	0.132 ± 0.003	0.018 ± 0.001	0.132 ± 0.004	0.175 ± 0.009	0.419 ± 0.013	0.314	-0.002	-0.64	0.1236	0.1236	1.006	95.714 ± 0.647	0.522 ± 0.004	68.878 ± 1.479
LSAM	beta32	0.316	0.314	-0.002 ± 0.004	-0.006 ± 0.013	0.017 ± 0.001	0.132 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	0.175 ± 0.009	0.418 ± 0.012	0.309	-0.007	-2.33	0.1250	0.1252	1.006	96.837 ± 0.559	0.521 ± 0.004	70.102 ± 1.462
LSAM	beta33	0.139	0.150	0.011 ± 0.003	0.080 ± 0.021	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.456 ± 0.025	0.675 ± 0.021	0.141	0.002	1.51	0.0841	0.0841	0.976	95.204 ± 0.683	0.357 ± 0.004	36.939 ± 1.542
LSAM	beta34	0.101	0.102	0.001 ± 0.002	0.015 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.258 ± 0.013	0.508 ± 0.015	0.100	-0.001	-0.66	0.0484	0.0484	0.960	94.490 ± 0.729	0.193 ± 0.002	60.204 ± 1.564
LSAM	beta35	0.101	0.104	0.003 ± 0.002	0.030 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.260 ± 0.014	0.510 ± 0.016	0.102	0.001	0.96	0.0483	0.0483	0.941	94.898 ± 0.703	0.190 ± 0.002	62.959 ± 1.543
QML	beta31	0.316	0.340	0.024 ± 0.002	0.076 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.228 ± 0.007	0.339	0.023	7.41	0.0646	0.0687	1.026	94.388 ± 0.735	0.273 ± 0.005	99.796 ± 0.144
QML	beta32	0.316	0.339	0.023 ± 0.002	0.073 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.221 ± 0.006	0.339	0.023	7.25	0.0677	0.0715	1.062	95.918 ± 0.632	0.275 ± 0.006	99.796 ± 0.144
QML	beta33	0.139	0.013	-0.126 ± 0.003	-0.904 ± 0.020	0.008 ± 0.000	0.088 ± 0.002	0.023 ± 0.001	0.153 ± 0.003	1.214 ± 0.042	1.102 ± 0.020	0.014	-0.125	-89.76	0.0848	0.1508	0.971	66.429 ± 1.509	0.334 ± 0.003	7.653 ± 0.849
QML	beta34	0.101	0.222	0.121 ± 0.002	1.196 ± 0.018	0.003 ± 0.000	0.057 ± 0.001	0.018 ± 0.001	0.134 ± 0.002	1.747 ± 0.050	1.322 ± 0.019	0.216	0.115	114.28	0.0563	0.1284	0.947	36.224 ± 1.535	0.211 ± 0.002	99.592 ± 0.204
QML	beta35	0.101	0.223	0.122 ± 0.002	1.213 ± 0.017	0.003 ± 0.000	0.052 ± 0.001	0.018 ± 0.000	0.133 ± 0.002	1.737 ± 0.044	1.318 ± 0.017	0.222	0.121	119.54	0.0533	0.1320	1.029	33.265 ± 1.505	0.211 ± 0.002	99.592 ± 0.204
UPI	beta31	0.316	0.321	0.005 ± 0.004	0.015 ± 0.014	0.018 ± 0.001	0.135 ± 0.003	0.018 ± 0.001	0.135 ± 0.004	0.183 ± 0.009	0.428 ± 0.012	0.321	0.005	1.52	0.1267	0.1268	1.015	96.327 ± 0.601	0.537 ± 0.005	67.551 ± 1.496
UPI	beta32	0.316	0.319	0.003 ± 0.004	0.010 ± 0.014	0.019 ± 0.001	0.136 ± 0.003	0.019 ± 0.001	0.136 ± 0.004	0.186 ± 0.009	0.431 ± 0.013	0.315	-0.001	-0.37	0.1338	0.1338	1.006	96.735 ± 0.568	0.538 ± 0.005	68.265 ± 1.487
UPI	beta33	0.139	0.146	0.007 ± 0.003	0.052 ± 0.025	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.611 ± 0.039	0.782 ± 0.028	0.133	-0.006	-3.96	0.0919	0.0921	1.063	95.408 ± 0.669	0.452 ± 0.012	25.306 ± 1.389
UPI	beta34	0.101	0.102	0.001 ± 0.002	0.010 ± 0.017	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.286 ± 0.014	0.535 ± 0.016	0.100	-0.001	-1.28	0.0523	0.0523	0.960	94.694 ± 0.716	0.203 ± 0.003	57.143 ± 1.581
UPI	beta35	0.101	0.103	0.002 ± 0.002	0.021 ± 0.017	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.291 ± 0.016	0.540 ± 0.017	0.098	-0.003	-2.89	0.0480	0.0481	0.942	94.898 ± 0.703	0.201 ± 0.003	60.408 ± 1.562

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S9

Simulation 1 - Condition 9: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.324	0.008 ± 0.003	0.026 ± 0.009	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.078 ± 0.004	0.279 ± 0.008	0.323	0.007	2.08	0.0870	0.0872	1.004	94.929 ± 0.699	0.345 ± 0.001	95.436 ± 0.665
LMS	beta32	0.316	0.328	0.012 ± 0.003	0.038 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.077 ± 0.004	0.277 ± 0.008	0.324	0.008	2.53	0.0858	0.0862	1.014	95.436 ± 0.665	0.345 ± 0.001	96.247 ± 0.605
LMS	beta33	0.139	0.083	-0.056 ± 0.003	-0.406 ± 0.021	0.008 ± 0.000	0.091 ± 0.002	0.011 ± 0.000	0.107 ± 0.003	0.595 ± 0.025	0.771 ± 0.019	0.079	-0.060	-42.97	0.0907	0.1086	0.955	87.120 ± 1.067	0.341 ± 0.003	18.966 ± 1.248
LMS	beta34	0.101	0.155	0.054 ± 0.002	0.535 ± 0.017	0.003 ± 0.000	0.052 ± 0.001	0.006 ± 0.000	0.075 ± 0.002	0.555 ± 0.023	0.745 ± 0.017	0.153	0.052	51.19	0.0515	0.0730	0.958	81.744 ± 1.230	0.197 ± 0.002	88.134 ± 1.030
LMS	beta35	0.101	0.154	0.053 ± 0.002	0.522 ± 0.017	0.003 ± 0.000	0.053 ± 0.001	0.006 ± 0.000	0.075 ± 0.002	0.550 ± 0.023	0.741 ± 0.017	0.154	0.053	52.55	0.0505	0.0733	0.947	81.643 ± 1.233	0.198 ± 0.002	87.424 ± 1.056
LSAM	beta31	0.316	0.320	0.004 ± 0.004	0.012 ± 0.014	0.019 ± 0.001	0.139 ± 0.003	0.019 ± 0.001	0.139 ± 0.004	0.194 ± 0.009	0.440 ± 0.013	0.317	0.001	0.21	0.1334	0.1334	1.001	95.943 ± 0.628	0.545 ± 0.004	65.416 ± 1.515
LSAM	beta32	0.316	0.325	0.009 ± 0.004	0.029 ± 0.014	0.020 ± 0.001	0.140 ± 0.003	0.020 ± 0.001	0.140 ± 0.004	0.197 ± 0.009	0.444 ± 0.013	0.318	0.002	0.64	0.1296	0.1296	0.996	95.233 ± 0.679	0.546 ± 0.004	67.647 ± 1.490
LSAM	beta33	0.139	0.152	0.013 ± 0.003	0.095 ± 0.023	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.001	0.099 ± 0.003	0.512 ± 0.027	0.716 ± 0.022	0.149	0.010	6.95	0.0956	0.0961	0.943	94.726 ± 0.712	0.365 ± 0.004	40.061 ± 1.561
LSAM	beta34	0.101	0.100	-0.001 ± 0.002	-0.012 ± 0.018	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.317 ± 0.017	0.563 ± 0.018	0.100	-0.001	-0.82	0.0488	0.0488	0.917	93.306 ± 0.796	0.204 ± 0.003	56.288 ± 1.580
LSAM	beta35	0.101	0.097	-0.004 ± 0.002	-0.036 ± 0.018	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.336 ± 0.018	0.580 ± 0.018	0.098	-0.003	-2.76	0.0507	0.0508	0.901	91.176 ± 0.903	0.206 ± 0.003	53.651 ± 1.588
QML	beta31	0.316	0.325	0.009 ± 0.003	0.027 ± 0.009	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.323	0.007	2.23	0.0863	0.0866	1.002	95.030 ± 0.692	0.346 ± 0.001	95.436 ± 0.665
QML	beta32	0.316	0.328	0.012 ± 0.003	0.039 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.077 ± 0.004	0.278 ± 0.008	0.325	0.009	2.73	0.0854	0.0858	1.013	95.436 ± 0.665	0.345 ± 0.001	96.247 ± 0.605
QML	beta33	0.139	0.084	-0.055 ± 0.003	-0.399 ± 0.021	0.008 ± 0.000	0.091 ± 0.002	0.011 ± 0.000	0.107 ± 0.003	0.591 ± 0.025	0.769 ± 0.019	0.080	-0.059	-42.79	0.0897	0.1076	0.952	87.221 ± 1.063	0.341 ± 0.003	19.371 ± 1.259
QML	beta34	0.101	0.154	0.053 ± 0.002	0.526 ± 0.017	0.003 ± 0.000	0.053 ± 0.001	0.006 ± 0.000	0.075 ± 0.002	0.546 ± 0.023	0.739 ± 0.017	0.152	0.051	50.58	0.0513	0.0724	0.951	81.136 ± 1.246	0.196 ± 0.002	87.525 ± 1.052
QML	beta35	0.101	0.153	0.052 ± 0.002	0.513 ± 0.017	0.003 ± 0.000	0.053 ± 0.001	0.005 ± 0.000	0.074 ± 0.002	0.539 ± 0.023	0.734 ± 0.017	0.153	0.052	51.97	0.0497	0.0723	0.943	81.643 ± 1.233	0.196 ± 0.002	87.120 ± 1.067
UPI	beta31	0.316	0.327	0.011 ± 0.004	0.034 ± 0.014	0.019 ± 0.001	0.139 ± 0.003	0.019 ± 0.001	0.139 ± 0.004	0.193 ± 0.009	0.440 ± 0.012	0.323	0.007	2.06	0.1324	0.1326	1.001	95.842 ± 0.636	0.544 ± 0.004	67.748 ± 1.489
UPI	beta32	0.316	0.330	0.014 ± 0.004	0.044 ± 0.014	0.019 ± 0.001	0.139 ± 0.003	0.020 ± 0.001	0.140 ± 0.004	0.195 ± 0.010	0.442 ± 0.013	0.324	0.008	2.44	0.1260	0.1262	0.999	95.132 ± 0.685	0.545 ± 0.004	69.473 ± 1.467
UPI	beta33	0.139	0.149	0.010 ± 0.003	0.071 ± 0.024	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.553 ± 0.030	0.743 ± 0.023	0.142	0.003	1.86	0.0992	0.0992	1.033	95.233 ± 0.679	0.417 ± 0.007	30.933 ± 1.472
UPI	beta34	0.101	0.098	-0.003 ± 0.002	-0.032 ± 0.018	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.328 ± 0.018	0.573 ± 0.018	0.098	-0.003	-2.48	0.0517	0.0518	0.919	92.799 ± 0.823	0.208 ± 0.003	54.361 ± 1.586
UPI	beta35	0.101	0.096	-0.005 ± 0.002	-0.047 ± 0.019	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.359 ± 0.021	0.599 ± 0.020	0.095	-0.006	-5.69	0.0507	0.0511	0.897	91.785 ± 0.874	0.212 ± 0.003	52.840 ± 1.590

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S10

Simulation 1 - Condition 10: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.029 ± 0.001	0.170 ± 0.005	0.319	0.003	1.03	0.0516	0.0517	1.029	94.985 ± 0.691	0.215 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.322	0.006	1.75	0.0571	0.0574	0.978	95.186 ± 0.678	0.215 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.084	-0.055 ± 0.002	-0.395 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.006 ± 0.000	0.079 ± 0.002	0.321 ± 0.014	0.567 ± 0.013	0.086	-0.053	-38.29	0.0540	0.0759	0.942	80.642 ± 1.251	0.209 ± 0.001	37.513 ± 1.533
LMS	beta34	0.101	0.158	0.057 ± 0.001	0.564 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.004 ± 0.000	0.066 ± 0.001	0.423 ± 0.014	0.650 ± 0.011	0.156	0.055	54.74	0.0325	0.0641	0.951	55.366 ± 1.574	0.122 ± 0.001	100.000 ± 0.000
LMS	beta35	0.101	0.157	0.056 ± 0.001	0.555 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.004 ± 0.000	0.065 ± 0.001	0.413 ± 0.013	0.643 ± 0.011	0.155	0.054	53.67	0.0311	0.0625	0.941	53.661 ± 1.579	0.121 ± 0.001	99.799 ± 0.142
LSAM	beta31	0.316	0.316	0.000 ± 0.003	0.001 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.068 ± 0.003	0.260 ± 0.008	0.314	-0.002	-0.61	0.0771	0.0771	1.002	94.985 ± 0.691	0.323 ± 0.001	95.988 ± 0.622
LSAM	beta32	0.316	0.314	-0.002 ± 0.003	-0.005 ± 0.008	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.069 ± 0.003	0.262 ± 0.007	0.314	-0.002	-0.51	0.0819	0.0819	0.993	94.383 ± 0.729	0.322 ± 0.001	95.888 ± 0.629
LSAM	beta33	0.139	0.144	0.005 ± 0.002	0.037 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.145	0.006	4.18	0.0511	0.0514	0.954	93.882 ± 0.759	0.203 ± 0.001	81.143 ± 1.239
LSAM	beta34	0.101	0.102	0.001 ± 0.001	0.007 ± 0.010	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.099 ± 0.005	0.315 ± 0.010	0.101	0.000	0.30	0.0315	0.0315	0.922	92.578 ± 0.830	0.115 ± 0.001	90.772 ± 0.917
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.013 ± 0.010	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.100 ± 0.005	0.317 ± 0.010	0.103	0.002	1.70	0.0303	0.0304	0.914	91.976 ± 0.860	0.115 ± 0.001	90.672 ± 0.921
QML	beta31	0.316	0.322	0.006 ± 0.002	0.017 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.029 ± 0.001	0.170 ± 0.005	0.319	0.003	1.10	0.0516	0.0517	1.039	94.985 ± 0.691	0.218 ± 0.002	99.799 ± 0.142
QML	beta32	0.316	0.321	0.005 ± 0.002	0.017 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.322	0.006	1.75	0.0574	0.0577	0.985	94.985 ± 0.691	0.217 ± 0.001	99.799 ± 0.142
QML	beta33	0.139	0.085	-0.054 ± 0.002	-0.392 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.006 ± 0.000	0.079 ± 0.002	0.321 ± 0.014	0.566 ± 0.013	0.086	-0.053	-37.86	0.0542	0.0755	0.936	80.542 ± 1.254	0.208 ± 0.001	38.415 ± 1.540
QML	beta34	0.101	0.157	0.056 ± 0.001	0.559 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.004 ± 0.000	0.065 ± 0.001	0.418 ± 0.014	0.646 ± 0.011	0.156	0.055	53.99	0.0326	0.0635	0.943	54.965 ± 1.576	0.121 ± 0.001	100.000 ± 0.000
QML	beta35	0.101	0.157	0.056 ± 0.001	0.550 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.004 ± 0.000	0.064 ± 0.001	0.407 ± 0.013	0.638 ± 0.011	0.155	0.054	53.24	0.0314	0.0623	0.935	54.062 ± 1.578	0.120 ± 0.001	99.799 ± 0.142
UPI	beta31	0.316	0.319	0.003 ± 0.003	0.010 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.068 ± 0.003	0.260 ± 0.008	0.319	0.003	0.80	0.0766	0.0766	0.997	94.985 ± 0.691	0.322 ± 0.001	96.590 ± 0.575
UPI	beta32	0.316	0.317	0.001 ± 0.003	0.003 ± 0.008	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.068 ± 0.003	0.261 ± 0.007	0.318	0.002	0.57	0.0816	0.0816	0.989	94.283 ± 0.735	0.320 ± 0.001	95.587 ± 0.650
UPI	beta33	0.139	0.145	0.006 ± 0.002	0.041 ± 0.013	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.174 ± 0.009	0.417 ± 0.012	0.144	0.005	3.52	0.0560	0.0562	0.961	94.183 ± 0.741	0.217 ± 0.002	76.530 ± 1.342
UPI	beta34	0.101	0.101	0.000 ± 0.001	0.002 ± 0.010	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.100	-0.001	-0.58	0.0321	0.0321	0.924	92.076 ± 0.855	0.117 ± 0.001	89.970 ± 0.951
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.004 ± 0.010	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.102 ± 0.005	0.320 ± 0.010	0.101	0.000	0.09	0.0308	0.0308	0.925	92.277 ± 0.845	0.117 ± 0.001	90.070 ± 0.947

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S11

Simulation 1 - Condition 11: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.003	0.007 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.322	0.006	1.89	0.0765	0.0768	0.990	94.784 ± 0.704	0.308 ± 0.001	96.991 ± 0.541
LMS	beta32	0.316	0.321	0.005 ± 0.003	0.015 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.254 ± 0.007	0.321	0.005	1.43	0.0746	0.0747	0.982	94.684 ± 0.711	0.309 ± 0.001	97.292 ± 0.514
LMS	beta33	0.139	0.116	-0.023 ± 0.002	-0.168 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.241 ± 0.011	0.491 ± 0.014	0.116	-0.023	-16.84	0.0628	0.0670	0.949	91.174 ± 0.898	0.239 ± 0.002	49.248 ± 1.583
LMS	beta34	0.101	0.121	0.020 ± 0.001	0.202 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.177 ± 0.009	0.421 ± 0.012	0.121	0.020	19.52	0.0341	0.0394	0.917	88.365 ± 1.015	0.134 ± 0.001	90.471 ± 0.930
LMS	beta35	0.101	0.120	0.019 ± 0.001	0.188 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.177 ± 0.009	0.421 ± 0.012	0.119	0.018	17.37	0.0362	0.0402	0.891	87.462 ± 1.049	0.133 ± 0.001	90.271 ± 0.939
LSAM	beta31	0.316	0.317	0.001 ± 0.003	0.003 ± 0.010	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.315 ± 0.009	0.322	0.006	1.75	0.0973	0.0974	0.975	94.283 ± 0.735	0.381 ± 0.002	87.964 ± 1.030
LSAM	beta32	0.316	0.321	0.005 ± 0.003	0.016 ± 0.010	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.102 ± 0.005	0.319 ± 0.010	0.321	0.005	1.49	0.0948	0.0949	0.965	93.480 ± 0.782	0.381 ± 0.002	89.769 ± 0.960
LSAM	beta33	0.139	0.139	0.000 ± 0.002	0.001 ± 0.015	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.219 ± 0.011	0.468 ± 0.014	0.137	-0.002	-1.64	0.0618	0.0618	0.930	92.076 ± 0.855	0.237 ± 0.002	63.892 ± 1.521
LSAM	beta34	0.101	0.101	0.000 ± 0.001	0.005 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.156 ± 0.008	0.395 ± 0.012	0.101	0.000	-0.27	0.0375	0.0375	0.881	91.474 ± 0.884	0.138 ± 0.002	79.137 ± 1.287
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.012 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.161 ± 0.008	0.401 ± 0.012	0.098	-0.003	-2.69	0.0374	0.0375	0.865	91.274 ± 0.894	0.137 ± 0.002	76.329 ± 1.346
QML	beta31	0.316	0.318	0.002 ± 0.003	0.007 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.063 ± 0.003	0.252 ± 0.007	0.322	0.006	1.87	0.0772	0.0774	0.988	94.584 ± 0.717	0.308 ± 0.001	96.991 ± 0.541
QML	beta32	0.316	0.320	0.004 ± 0.003	0.014 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.065 ± 0.003	0.254 ± 0.007	0.320	0.004	1.21	0.0746	0.0747	0.981	94.584 ± 0.717	0.308 ± 0.001	97.392 ± 0.505
QML	beta33	0.139	0.116	-0.023 ± 0.002	-0.167 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.242 ± 0.011	0.492 ± 0.014	0.116	-0.023	-16.64	0.0621	0.0663	0.946	91.174 ± 0.898	0.239 ± 0.002	49.749 ± 1.583
QML	beta34	0.101	0.121	0.020 ± 0.001	0.199 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.175 ± 0.009	0.419 ± 0.012	0.120	0.019	19.17	0.0343	0.0394	0.908	87.964 ± 1.030	0.133 ± 0.001	90.572 ± 0.925
QML	beta35	0.101	0.120	0.019 ± 0.001	0.188 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.177 ± 0.009	0.421 ± 0.012	0.119	0.018	17.39	0.0362	0.0403	0.886	87.061 ± 1.063	0.132 ± 0.001	90.471 ± 0.930
UPI	beta31	0.316	0.320	0.004 ± 0.003	0.012 ± 0.010	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.009	0.324	0.008	2.39	0.0965	0.0968	0.959	93.882 ± 0.759	0.375 ± 0.002	89.870 ± 0.956
UPI	beta32	0.316	0.323	0.007 ± 0.003	0.022 ± 0.010	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.102 ± 0.005	0.319 ± 0.009	0.320	0.004	1.42	0.0948	0.0949	0.952	93.480 ± 0.782	0.375 ± 0.002	90.070 ± 0.947
UPI	beta33	0.139	0.140	0.001 ± 0.002	0.004 ± 0.015	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.233 ± 0.012	0.483 ± 0.014	0.140	0.001	0.63	0.0650	0.0650	0.937	92.879 ± 0.815	0.246 ± 0.002	62.086 ± 1.537
UPI	beta34	0.101	0.100	-0.001 ± 0.001	-0.007 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.160 ± 0.009	0.400 ± 0.012	0.100	-0.001	-0.62	0.0374	0.0374	0.874	91.374 ± 0.889	0.139 ± 0.002	77.332 ± 1.326
UPI	beta35	0.101	0.099	-0.002 ± 0.001	-0.016 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.164 ± 0.009	0.405 ± 0.012	0.097	-0.004	-4.02	0.0367	0.0369	0.868	89.669 ± 0.964	0.139 ± 0.002	76.329 ± 1.346

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S12

Simulation 1 - Condition 12: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.316	0.000	0.12	0.0469	0.0469	1.034	96.396 ± 0.590	0.194 ± 0.001	99.900 ± 0.100
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.159 ± 0.004	0.314	-0.002	-0.60	0.0481	0.0481	0.980	93.794 ± 0.763	0.193 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.120	-0.019 ± 0.001	-0.140 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.119	-0.020	-14.16	0.0404	0.0450	0.933	88.689 ± 1.002	0.149 ± 0.001	86.486 ± 1.082
LMS	beta34	0.101	0.122	0.021 ± 0.001	0.210 ± 0.006	0.000 ± 0.000	0.021 ± 0.000	0.001 ± 0.000	0.030 ± 0.001	0.086 ± 0.004	0.294 ± 0.006	0.122	0.021	20.64	0.0202	0.0290	1.011	83.584 ± 1.172	0.082 ± 0.001	100.000 ± 0.000
LMS	beta35	0.101	0.122	0.021 ± 0.001	0.208 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.097 ± 0.004	0.312 ± 0.007	0.122	0.021	20.33	0.0229	0.0308	0.884	79.580 ± 1.275	0.081 ± 0.001	99.800 ± 0.141
LSAM	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.315	-0.001	-0.32	0.0568	0.0568	1.026	96.096 ± 0.613	0.234 ± 0.001	99.800 ± 0.141
LSAM	beta32	0.316	0.313	-0.003 ± 0.002	-0.011 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.195 ± 0.005	0.314	-0.002	-0.55	0.0581	0.0581	0.969	93.594 ± 0.775	0.234 ± 0.001	99.900 ± 0.100
LSAM	beta33	0.139	0.140	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.139	0.000	0.33	0.0395	0.0395	0.919	91.692 ± 0.873	0.144 ± 0.001	94.895 ± 0.696
LSAM	beta34	0.101	0.102	0.001 ± 0.001	0.008 ± 0.007	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.101	0.000	-0.24	0.0204	0.0204	0.979	93.994 ± 0.752	0.081 ± 0.001	98.799 ± 0.345
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.102	0.001	0.76	0.0226	0.0226	0.870	90.290 ± 0.937	0.081 ± 0.001	98.599 ± 0.372
QML	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.317	0.001	0.26	0.0470	0.0470	1.027	96.096 ± 0.613	0.193 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.160 ± 0.004	0.314	-0.002	-0.60	0.0483	0.0483	0.976	93.694 ± 0.769	0.193 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.120	-0.019 ± 0.001	-0.138 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.105 ± 0.005	0.325 ± 0.009	0.120	-0.019	-13.83	0.0404	0.0448	0.928	88.689 ± 1.002	0.149 ± 0.001	86.587 ± 1.078
QML	beta34	0.101	0.122	0.021 ± 0.001	0.205 ± 0.006	0.000 ± 0.000	0.021 ± 0.000	0.001 ± 0.000	0.029 ± 0.001	0.084 ± 0.003	0.289 ± 0.006	0.121	0.020	19.77	0.0208	0.0289	0.996	83.483 ± 1.175	0.081 ± 0.001	100.000 ± 0.000
QML	beta35	0.101	0.122	0.021 ± 0.001	0.208 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.097 ± 0.004	0.312 ± 0.007	0.122	0.021	20.30	0.0228	0.0307	0.880	79.580 ± 1.275	0.081 ± 0.001	99.800 ± 0.141
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.005 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.316	0.000	0.09	0.0566	0.0566	1.008	95.295 ± 0.670	0.231 ± 0.001	99.800 ± 0.141
UPI	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.195 ± 0.006	0.316	0.000	-0.14	0.0577	0.0577	0.956	93.493 ± 0.780	0.231 ± 0.001	99.800 ± 0.141
UPI	beta33	0.139	0.140	0.001 ± 0.001	0.005 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.086 ± 0.004	0.294 ± 0.008	0.139	0.000	0.26	0.0406	0.0406	0.922	91.592 ± 0.878	0.148 ± 0.001	94.294 ± 0.734
UPI	beta34	0.101	0.101	0.000 ± 0.001	0.002 ± 0.007	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.045 ± 0.002	0.211 ± 0.006	0.101	0.000	-0.29	0.0206	0.0206	0.972	93.293 ± 0.791	0.081 ± 0.001	98.699 ± 0.359
UPI	beta35	0.101	0.102	0.001 ± 0.001	0.012 ± 0.008	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.101	0.000	-0.14	0.0228	0.0228	0.872	90.490 ± 0.928	0.082 ± 0.001	98.699 ± 0.359

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S13

Simulation 1 - Condition 13: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.305	-0.011 ± 0.004	-0.036 ± 0.013	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.093 ± 0.006	0.305 ± 0.012	0.308	-0.008	-2.41	0.0987	0.0990	1.009	94.203 ± 0.995	0.379 ± 0.003	90.036 ± 1.275
LMS	beta32	0.316	0.307	-0.009 ± 0.004	-0.027 ± 0.012	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.076 ± 0.005	0.276 ± 0.010	0.302	-0.014	-4.43	0.0965	0.0975	1.114	96.920 ± 0.735	0.380 ± 0.003	94.022 ± 1.009
LMS	beta33	0.139	0.131	-0.008 ± 0.006	-0.060 ± 0.041	0.018 ± 0.001	0.133 ± 0.005	0.018 ± 0.001	0.133 ± 0.006	0.914 ± 0.067	0.956 ± 0.041	0.123	-0.016	-11.24	0.1264	0.1274	1.005	94.928 ± 0.934	0.523 ± 0.008	16.667 ± 1.586
LMS	beta34	0.101	0.046	-0.055 ± 0.004	-0.545 ± 0.040	0.009 ± 0.001	0.095 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	1.172 ± 0.066	1.082 ± 0.035	0.049	-0.052	-51.16	0.0931	0.1065	1.011	87.319 ± 1.416	0.375 ± 0.005	8.333 ± 1.176
LMS	beta35	0.101	0.045	-0.056 ± 0.004	-0.552 ± 0.041	0.009 ± 0.001	0.096 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	1.212 ± 0.078	1.101 ± 0.040	0.041	-0.060	-59.11	0.0974	0.1142	0.996	87.681 ± 1.399	0.376 ± 0.005	7.428 ± 1.116
LSAM	beta31	0.316	0.325	0.009 ± 0.005	0.029 ± 0.016	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.146 ± 0.009	0.382 ± 0.015	0.328	0.012	3.75	0.1258	0.1263	1.254	96.014 ± 0.833	0.592 ± 0.020	71.196 ± 1.927
LSAM	beta32	0.316	0.325	0.009 ± 0.005	0.028 ± 0.015	0.013 ± 0.001	0.114 ± 0.004	0.013 ± 0.001	0.114 ± 0.005	0.130 ± 0.009	0.360 ± 0.015	0.323	0.007	2.32	0.1148	0.1151	1.364	98.913 ± 0.441	0.607 ± 0.024	74.638 ± 1.852
LSAM	beta33	0.139	0.142	0.003 ± 0.009	0.024 ± 0.062	0.041 ± 0.003	0.201 ± 0.008	0.040 ± 0.003	0.201 ± 0.009	2.095 ± 0.168	1.448 ± 0.066	0.131	-0.008	-5.78	0.1780	0.1781	1.311	98.732 ± 0.476	1.034 ± 0.057	7.971 ± 1.153
LSAM	beta34	0.101	0.108	0.007 ± 0.010	0.066 ± 0.099	0.055 ± 0.004	0.234 ± 0.009	0.055 ± 0.004	0.234 ± 0.010	5.357 ± 0.419	2.315 ± 0.103	0.108	0.007	7.02	0.2195	0.2196	1.377	99.275 ± 0.361	1.263 ± 0.077	1.449 ± 0.509
LSAM	beta35	0.101	0.110	0.009 ± 0.011	0.087 ± 0.107	0.064 ± 0.005	0.253 ± 0.011	0.064 ± 0.005	0.253 ± 0.012	6.278 ± 0.538	2.506 ± 0.120	0.101	0.000	-0.49	0.2287	0.2287	1.318	99.094 ± 0.403	1.308 ± 0.078	0.725 ± 0.361
QML	beta31	0.316	0.305	-0.011 ± 0.004	-0.034 ± 0.013	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.094 ± 0.006	0.306 ± 0.012	0.306	-0.010	-3.16	0.0975	0.0980	1.006	93.659 ± 1.037	0.379 ± 0.003	90.217 ± 1.264
QML	beta32	0.316	0.308	-0.008 ± 0.004	-0.025 ± 0.012	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.077 ± 0.005	0.277 ± 0.010	0.302	-0.014	-4.48	0.0974	0.0984	1.111	96.739 ± 0.756	0.380 ± 0.003	94.203 ± 0.995
QML	beta33	0.139	0.133	-0.006 ± 0.006	-0.045 ± 0.042	0.018 ± 0.001	0.136 ± 0.005	0.018 ± 0.001	0.136 ± 0.006	0.952 ± 0.071	0.976 ± 0.042	0.129	-0.010	-7.15	0.1249	0.1253	0.997	94.928 ± 0.934	0.530 ± 0.008	15.942 ± 1.558
QML	beta34	0.101	0.045	-0.056 ± 0.004	-0.553 ± 0.041	0.009 ± 0.001	0.097 ± 0.004	0.013 ± 0.001	0.112 ± 0.004	1.228 ± 0.074	1.108 ± 0.038	0.047	-0.054	-53.26	0.0925	0.1070	0.998	87.681 ± 1.399	0.380 ± 0.006	7.065 ± 1.091
QML	beta35	0.101	0.045	-0.056 ± 0.004	-0.558 ± 0.042	0.010 ± 0.001	0.099 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	1.265 ± 0.084	1.125 ± 0.042	0.042	-0.059	-58.39	0.0954	0.1122	0.984	86.957 ± 1.433	0.380 ± 0.006	7.246 ± 1.103
UPI	beta31	0.316	0.326	0.010 ± 0.005	0.031 ± 0.017	0.016 ± 0.001	0.128 ± 0.004	0.016 ± 0.001	0.128 ± 0.005	0.164 ± 0.011	0.405 ± 0.016	0.319	0.003	0.92	0.1340	0.1340	1.579	96.196 ± 0.814	0.790 ± 0.043	60.870 ± 2.077
UPI	beta32	0.316	0.325	0.009 ± 0.005	0.028 ± 0.016	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.117 ± 0.005	0.138 ± 0.010	0.372 ± 0.016	0.323	0.007	2.07	0.1176	0.1177	1.698	98.913 ± 0.441	0.780 ± 0.037	60.688 ± 2.079
UPI	beta33	0.139	0.163	0.024 ± 0.009	0.176 ± 0.062	0.041 ± 0.003	0.202 ± 0.008	0.041 ± 0.003	0.203 ± 0.009	2.139 ± 0.174	1.462 ± 0.067	0.149	0.010	7.30	0.1864	0.1867	1.715	99.275 ± 0.361	1.358 ± 0.069	3.986 ± 0.833
UPI	beta34	0.101	0.101	-0.000 ± 0.012	-0.003 ± 0.121	0.082 ± 0.007	0.287 ± 0.012	0.082 ± 0.007	0.287 ± 0.014	8.072 ± 0.690	2.841 ± 0.136	0.069	-0.032	-31.57	0.2441	0.2462	2.066	97.464 ± 0.669	2.326 ± 0.162	0.181 ± 0.181
UPI	beta35	0.101	0.094	-0.007 ± 0.012	-0.071 ± 0.122	0.084 ± 0.008	0.289 ± 0.013	0.084 ± 0.008	0.289 ± 0.015	8.194 ± 0.746	2.862 ± 0.145	0.064	-0.037	-36.15	0.2322	0.2351	1.961	96.739 ± 0.756	2.224 ± 0.144	0.362 ± 0.256

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S14

Simulation 1 - Condition 14: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.006	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.006	0.307	-0.009	-2.91	0.0593	0.0600	0.983	93.770 ± 0.799	0.236 ± 0.001	99.891 ± 0.109
LMS	beta32	0.316	0.302	-0.014 ± 0.002	-0.045 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.005	0.299	-0.017	-5.52	0.0647	0.0670	0.966	93.005 ± 0.843	0.235 ± 0.001	99.891 ± 0.109
LMS	beta33	0.139	0.137	-0.002 ± 0.003	-0.012 ± 0.020	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.358 ± 0.018	0.598 ± 0.018	0.138	-0.001	-0.75	0.0809	0.0809	1.006	93.880 ± 0.792	0.328 ± 0.003	40.656 ± 1.624
LMS	beta34	0.101	0.042	-0.059 ± 0.002	-0.586 ± 0.019	0.003 ± 0.000	0.059 ± 0.001	0.007 ± 0.000	0.084 ± 0.002	0.684 ± 0.026	0.827 ± 0.017	0.039	-0.062	-61.02	0.0626	0.0878	1.023	80.328 ± 1.314	0.236 ± 0.002	10.601 ± 1.018
LMS	beta35	0.101	0.040	-0.061 ± 0.002	-0.608 ± 0.019	0.004 ± 0.000	0.059 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.714 ± 0.029	0.845 ± 0.018	0.042	-0.059	-58.81	0.0560	0.0816	0.998	80.000 ± 1.322	0.232 ± 0.002	10.273 ± 1.004
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.221 ± 0.007	0.322	0.006	1.99	0.0672	0.0675	1.035	95.738 ± 0.668	0.283 ± 0.002	99.563 ± 0.218
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.316	0.000	-0.07	0.0705	0.0705	1.027	96.393 ± 0.616	0.283 ± 0.002	98.907 ± 0.344
LSAM	beta33	0.139	0.138	-0.001 ± 0.003	-0.010 ± 0.024	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.520 ± 0.029	0.721 ± 0.024	0.140	0.001	0.54	0.0913	0.0913	1.035	96.612 ± 0.598	0.407 ± 0.005	34.645 ± 1.573
LSAM	beta34	0.101	0.111	0.010 ± 0.004	0.101 ± 0.042	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.004	1.655 ± 0.100	1.286 ± 0.044	0.097	-0.004	-3.88	0.1230	0.1231	1.062	97.268 ± 0.539	0.539 ± 0.008	7.978 ± 0.896
LSAM	beta35	0.101	0.102	0.001 ± 0.004	0.006 ± 0.042	0.016 ± 0.001	0.128 ± 0.004	0.016 ± 0.001	0.128 ± 0.004	1.600 ± 0.096	1.265 ± 0.043	0.094	-0.007	-7.10	0.1165	0.1167	1.040	97.268 ± 0.539	0.521 ± 0.009	7.760 ± 0.884
QML	beta31	0.316	0.307	-0.009 ± 0.002	-0.027 ± 0.006	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.006	0.308	-0.008	-2.67	0.0588	0.0594	0.982	93.770 ± 0.799	0.236 ± 0.001	99.891 ± 0.109
QML	beta32	0.316	0.302	-0.014 ± 0.002	-0.044 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.299	-0.017	-5.39	0.0650	0.0672	0.966	93.005 ± 0.843	0.235 ± 0.001	99.891 ± 0.109
QML	beta33	0.139	0.138	-0.001 ± 0.003	-0.005 ± 0.020	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.359 ± 0.018	0.599 ± 0.018	0.139	0.000	0.35	0.0817	0.0817	1.005	94.426 ± 0.758	0.328 ± 0.002	41.093 ± 1.627
QML	beta34	0.101	0.041	-0.060 ± 0.002	-0.593 ± 0.019	0.003 ± 0.000	0.059 ± 0.001	0.007 ± 0.000	0.084 ± 0.002	0.692 ± 0.026	0.832 ± 0.017	0.039	-0.062	-61.58	0.0618	0.0877	1.022	79.891 ± 1.325	0.236 ± 0.002	9.945 ± 0.989
QML	beta35	0.101	0.039	-0.062 ± 0.002	-0.615 ± 0.019	0.004 ± 0.000	0.060 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.725 ± 0.029	0.852 ± 0.018	0.042	-0.059	-58.91	0.0558	0.0816	0.991	79.235 ± 1.341	0.231 ± 0.002	10.273 ± 1.004
UPI	beta31	0.316	0.322	0.006 ± 0.002	0.019 ± 0.008	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.228 ± 0.007	0.321	0.005	1.43	0.0731	0.0732	1.106	96.284 ± 0.625	0.312 ± 0.003	97.268 ± 0.539
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.008	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.007	0.315	-0.001	-0.42	0.0724	0.0724	1.109	96.721 ± 0.589	0.314 ± 0.004	96.831 ± 0.579
UPI	beta33	0.139	0.146	0.007 ± 0.003	0.050 ± 0.025	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.558 ± 0.031	0.747 ± 0.024	0.142	0.003	1.86	0.0951	0.0951	1.184	97.596 ± 0.506	0.481 ± 0.010	25.246 ± 1.436
UPI	beta34	0.101	0.117	0.016 ± 0.005	0.160 ± 0.051	0.025 ± 0.002	0.157 ± 0.005	0.025 ± 0.002	0.158 ± 0.006	2.445 ± 0.170	1.564 ± 0.060	0.097	-0.004	-3.66	0.1389	0.1390	1.294	97.268 ± 0.539	0.798 ± 0.022	1.639 ± 0.420
UPI	beta35	0.101	0.108	0.007 ± 0.005	0.067 ± 0.051	0.025 ± 0.002	0.157 ± 0.005	0.025 ± 0.002	0.157 ± 0.006	2.426 ± 0.157	1.558 ± 0.057	0.090	-0.011	-10.74	0.1369	0.1373	1.280	96.503 ± 0.607	0.789 ± 0.023	1.202 ± 0.360

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S15

Simulation 1 - Condition 15: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.308	-0.008 ± 0.002	-0.025 ± 0.007	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.214 ± 0.006	0.307	-0.009	-2.84	0.0652	0.0658	1.041	96.061 ± 0.618	0.274 ± 0.001	99.293 ± 0.266
LMS	beta32	0.316	0.307	-0.009 ± 0.002	-0.029 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.305	-0.011	-3.47	0.0696	0.0704	0.992	94.343 ± 0.734	0.273 ± 0.001	99.495 ± 0.225
LMS	beta33	0.139	0.143	0.004 ± 0.003	0.031 ± 0.020	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.389 ± 0.018	0.624 ± 0.017	0.140	0.001	0.44	0.0871	0.0871	1.002	95.253 ± 0.676	0.340 ± 0.003	38.889 ± 1.549
LMS	beta34	0.101	0.058	-0.043 ± 0.002	-0.429 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.677 ± 0.029	0.823 ± 0.020	0.059	-0.042	-41.50	0.0677	0.0796	0.994	87.475 ± 1.052	0.276 ± 0.004	13.131 ± 1.073
LMS	beta35	0.101	0.057	-0.044 ± 0.002	-0.437 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.679 ± 0.029	0.824 ± 0.020	0.056	-0.045	-44.84	0.0667	0.0806	0.995	89.091 ± 0.991	0.276 ± 0.002	13.030 ± 1.070
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.002	0.233 ± 0.006	0.315	-0.001	-0.43	0.0734	0.0735	1.044	95.960 ± 0.626	0.301 ± 0.001	98.788 ± 0.348
LSAM	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.057 ± 0.003	0.239 ± 0.007	0.316	0.000	0.05	0.0771	0.0771	1.018	95.657 ± 0.648	0.301 ± 0.001	98.485 ± 0.388
LSAM	beta33	0.139	0.141	0.002 ± 0.003	0.016 ± 0.022	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.462 ± 0.024	0.680 ± 0.021	0.138	-0.001	-0.36	0.0936	0.0936	0.984	95.960 ± 0.626	0.365 ± 0.003	35.051 ± 1.516
LSAM	beta34	0.101	0.103	0.002 ± 0.004	0.016 ± 0.036	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.004	1.317 ± 0.069	1.147 ± 0.035	0.101	0.000	0.21	0.1097	0.1097	0.992	95.354 ± 0.669	0.451 ± 0.004	13.232 ± 1.077
LSAM	beta35	0.101	0.102	0.001 ± 0.004	0.010 ± 0.037	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	1.359 ± 0.073	1.166 ± 0.037	0.096	-0.005	-4.74	0.1121	0.1122	0.971	95.960 ± 0.626	0.448 ± 0.004	12.222 ± 1.041
QML	beta31	0.316	0.308	-0.008 ± 0.002	-0.025 ± 0.007	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.214 ± 0.006	0.307	-0.009	-2.79	0.0649	0.0655	1.038	96.263 ± 0.603	0.273 ± 0.001	99.394 ± 0.247
QML	beta32	0.316	0.307	-0.009 ± 0.002	-0.029 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.305	-0.011	-3.58	0.0696	0.0705	0.990	94.343 ± 0.734	0.273 ± 0.001	99.495 ± 0.225
QML	beta33	0.139	0.144	0.005 ± 0.003	0.035 ± 0.020	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.391 ± 0.018	0.625 ± 0.018	0.140	0.001	0.95	0.0871	0.0871	1.000	95.253 ± 0.676	0.340 ± 0.003	39.394 ± 1.553
QML	beta34	0.101	0.057	-0.044 ± 0.002	-0.438 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.684 ± 0.030	0.827 ± 0.020	0.057	-0.044	-43.10	0.0672	0.0801	0.988	87.879 ± 1.037	0.275 ± 0.003	12.525 ± 1.052
QML	beta35	0.101	0.056	-0.045 ± 0.002	-0.445 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.689 ± 0.029	0.830 ± 0.020	0.055	-0.046	-45.92	0.0674	0.0818	1.003	89.293 ± 0.983	0.278 ± 0.003	12.525 ± 1.052
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.007	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.236 ± 0.007	0.315	-0.001	-0.17	0.0730	0.0730	1.037	96.263 ± 0.603	0.303 ± 0.002	98.081 ± 0.436
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.241 ± 0.007	0.317	0.001	0.34	0.0776	0.0776	1.011	94.545 ± 0.722	0.302 ± 0.002	98.283 ± 0.413
UPI	beta33	0.139	0.148	0.009 ± 0.003	0.063 ± 0.023	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.528 ± 0.030	0.727 ± 0.024	0.146	0.007	4.75	0.0972	0.0974	0.997	96.768 ± 0.562	0.394 ± 0.004	34.848 ± 1.514
UPI	beta34	0.101	0.104	0.003 ± 0.004	0.032 ± 0.041	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.004	1.684 ± 0.095	1.298 ± 0.042	0.102	0.001	0.51	0.1219	0.1219	1.086	96.566 ± 0.579	0.558 ± 0.009	6.970 ± 0.809
UPI	beta35	0.101	0.105	0.004 ± 0.004	0.043 ± 0.041	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.004	1.687 ± 0.095	1.299 ± 0.042	0.098	-0.003	-2.84	0.1158	0.1159	1.078	96.364 ± 0.595	0.555 ± 0.008	6.566 ± 0.787

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S16

Simulation 1 - Condition 16: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.001	-0.021 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.310	-0.006	-1.87	0.0422	0.0426	1.008	94.383 ± 0.729	0.171 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.308	-0.008 ± 0.001	-0.024 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.309	-0.007	-2.32	0.0420	0.0426	1.026	94.885 ± 0.698	0.171 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.142	0.003 ± 0.002	0.020 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.150 ± 0.007	0.387 ± 0.011	0.140	0.001	0.70	0.0534	0.0535	1.002	94.483 ± 0.723	0.211 ± 0.001	75.727 ± 1.358
LMS	beta34	0.101	0.059	-0.042 ± 0.001	-0.417 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.358 ± 0.014	0.598 ± 0.013	0.058	-0.043	-42.63	0.0412	0.0596	1.003	83.651 ± 1.171	0.170 ± 0.001	25.777 ± 1.385
LMS	beta35	0.101	0.060	-0.041 ± 0.001	-0.407 ± 0.013	0.002 ± 0.000	0.043 ± 0.001	0.004 ± 0.000	0.059 ± 0.001	0.344 ± 0.013	0.587 ± 0.012	0.060	-0.041	-40.68	0.0429	0.0594	1.014	83.651 ± 1.171	0.170 ± 0.001	27.884 ± 1.420
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.318	0.002	0.67	0.0455	0.0456	1.024	95.687 ± 0.643	0.183 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.317	0.001	0.20	0.0450	0.0450	1.040	96.189 ± 0.606	0.183 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.140	0.001 ± 0.002	0.009 ± 0.012	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.140	0.001	0.58	0.0533	0.0533	1.009	95.085 ± 0.685	0.216 ± 0.001	72.919 ± 1.407
LSAM	beta34	0.101	0.103	0.002 ± 0.002	0.015 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.431 ± 0.021	0.656 ± 0.019	0.098	-0.003	-2.48	0.0635	0.0636	1.023	95.286 ± 0.671	0.266 ± 0.001	31.194 ± 1.467
LSAM	beta35	0.101	0.103	0.002 ± 0.002	0.022 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.432 ± 0.020	0.657 ± 0.018	0.101	0.000	0.39	0.0667	0.0667	1.022	94.985 ± 0.691	0.266 ± 0.001	32.397 ± 1.482
QML	beta31	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.310	-0.006	-1.75	0.0419	0.0423	1.009	94.584 ± 0.717	0.171 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.308	-0.008 ± 0.001	-0.024 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.309	-0.007	-2.29	0.0422	0.0428	1.027	94.985 ± 0.691	0.171 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.142	0.003 ± 0.002	0.024 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.150 ± 0.007	0.387 ± 0.011	0.141	0.002	1.13	0.0531	0.0531	1.005	94.483 ± 0.723	0.212 ± 0.001	75.426 ± 1.363
QML	beta34	0.101	0.058	-0.043 ± 0.001	-0.424 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.363 ± 0.014	0.603 ± 0.012	0.057	-0.044	-43.46	0.0410	0.0600	1.005	83.450 ± 1.177	0.170 ± 0.001	25.476 ± 1.380
QML	beta35	0.101	0.059	-0.042 ± 0.001	-0.414 ± 0.013	0.002 ± 0.000	0.043 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.350 ± 0.014	0.591 ± 0.012	0.059	-0.042	-41.68	0.0425	0.0598	1.019	83.049 ± 1.188	0.170 ± 0.001	27.683 ± 1.417
UPI	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.319	0.003	0.95	0.0454	0.0455	1.005	95.587 ± 0.650	0.180 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.317	0.001	0.20	0.0442	0.0442	1.021	95.988 ± 0.622	0.180 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.142	0.003 ± 0.002	0.019 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.158 ± 0.007	0.398 ± 0.011	0.142	0.003	2.02	0.0518	0.0519	1.027	95.687 ± 0.643	0.222 ± 0.001	71.916 ± 1.423
UPI	beta34	0.101	0.103	0.002 ± 0.002	0.024 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.478 ± 0.023	0.691 ± 0.020	0.098	-0.003	-3.07	0.0658	0.0658	1.044	96.189 ± 0.606	0.286 ± 0.002	26.680 ± 1.401
UPI	beta35	0.101	0.105	0.004 ± 0.002	0.037 ± 0.023	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.512 ± 0.026	0.715 ± 0.022	0.101	0.000	-0.09	0.0686	0.0686	1.017	95.386 ± 0.664	0.288 ± 0.002	28.385 ± 1.428

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S17

Simulation 1 - Condition 17: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.002	-0.016 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.310	-0.006	-1.92	0.0580	0.0583	1.056	95.700 ± 0.641	0.234 ± 0.011	99.900 ± 0.100
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.314	-0.002	-0.73	0.0593	0.0593	1.029	95.000 ± 0.689	0.231 ± 0.008	99.900 ± 0.100
LMS	beta33	0.139	0.145	0.006 ± 0.002	0.046 ± 0.015	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.215 ± 0.009	0.464 ± 0.012	0.147	0.008	5.87	0.0644	0.0649	0.987	94.500 ± 0.721	0.248 ± 0.001	62.900 ± 1.528
LMS	beta34	0.101	0.076	-0.025 ± 0.002	-0.249 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.392 ± 0.017	0.626 ± 0.016	0.077	-0.024	-23.65	0.0572	0.0620	1.004	92.400 ± 0.838	0.228 ± 0.001	25.400 ± 1.377
LMS	beta35	0.101	0.077	-0.024 ± 0.002	-0.236 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.380 ± 0.016	0.617 ± 0.015	0.075	-0.026	-25.43	0.0568	0.0623	1.012	93.200 ± 0.796	0.228 ± 0.001	24.800 ± 1.366
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.314	-0.002	-0.62	0.0579	0.0579	1.011	96.000 ± 0.620	0.231 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.317	0.001	0.33	0.0596	0.0596	0.997	94.600 ± 0.715	0.231 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.143	0.004 ± 0.002	0.031 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.216 ± 0.009	0.465 ± 0.012	0.143	0.004	3.19	0.0632	0.0634	0.983	94.200 ± 0.739	0.249 ± 0.001	61.900 ± 1.536
LSAM	beta34	0.101	0.098	-0.003 ± 0.002	-0.027 ± 0.023	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.515 ± 0.022	0.718 ± 0.019	0.100	-0.001	-1.00	0.0733	0.0733	0.999	94.700 ± 0.708	0.284 ± 0.001	27.200 ± 1.407
LSAM	beta35	0.101	0.099	-0.002 ± 0.002	-0.017 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.500 ± 0.023	0.707 ± 0.020	0.097	-0.004	-3.82	0.0712	0.0713	1.015	95.800 ± 0.634	0.284 ± 0.001	27.000 ± 1.404
QML	beta31	0.316	0.311	-0.005 ± 0.002	-0.017 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.310	-0.006	-1.92	0.0579	0.0582	1.006	95.500 ± 0.656	0.223 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.313	-0.003	-0.83	0.0591	0.0592	0.992	95.000 ± 0.689	0.223 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.145	0.006 ± 0.002	0.045 ± 0.015	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.215 ± 0.009	0.463 ± 0.012	0.147	0.008	5.76	0.0646	0.0651	0.987	94.200 ± 0.739	0.248 ± 0.001	62.900 ± 1.528
QML	beta34	0.101	0.075	-0.026 ± 0.002	-0.252 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.395 ± 0.018	0.628 ± 0.016	0.076	-0.025	-24.33	0.0573	0.0623	1.002	92.400 ± 0.838	0.228 ± 0.001	25.400 ± 1.377
QML	beta35	0.101	0.077	-0.024 ± 0.002	-0.240 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.382 ± 0.016	0.618 ± 0.015	0.075	-0.026	-25.64	0.0569	0.0625	1.013	93.600 ± 0.774	0.229 ± 0.001	24.500 ± 1.360
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.314	-0.002	-0.68	0.0585	0.0585	0.973	94.600 ± 0.715	0.223 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.316	0.000	-0.06	0.0599	0.0599	0.962	93.900 ± 0.757	0.223 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.144	0.005 ± 0.002	0.039 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.223 ± 0.010	0.472 ± 0.013	0.145	0.006	4.27	0.0640	0.0643	0.984	94.500 ± 0.721	0.253 ± 0.001	62.300 ± 1.533
UPI	beta34	0.101	0.099	-0.002 ± 0.002	-0.019 ± 0.023	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.544 ± 0.024	0.738 ± 0.020	0.100	-0.001	-1.29	0.0743	0.0743	1.010	95.100 ± 0.683	0.295 ± 0.001	25.200 ± 1.373
UPI	beta35	0.101	0.100	-0.001 ± 0.002	-0.011 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.527 ± 0.025	0.726 ± 0.021	0.098	-0.003	-3.07	0.0677	0.0678	1.029	95.700 ± 0.641	0.296 ± 0.001	24.000 ± 1.351

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S18

Simulation 1 - Condition 18: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.314	-0.002	-0.65	0.0358	0.0358	1.007	95.100 ± 0.683	0.141 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.312	-0.004 ± 0.001	-0.014 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.311	-0.005	-1.56	0.0361	0.0364	0.979	95.200 ± 0.676	0.141 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.142	0.003 ± 0.001	0.019 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.081 ± 0.004	0.284 ± 0.008	0.140	0.001	0.70	0.0390	0.0390	1.021	94.900 ± 0.696	0.158 ± 0.000	94.400 ± 0.727
LMS	beta34	0.101	0.076	-0.025 ± 0.001	-0.243 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.191 ± 0.008	0.437 ± 0.011	0.076	-0.025	-25.18	0.0367	0.0447	1.001	90.500 ± 0.927	0.144 ± 0.000	56.100 ± 1.569
LMS	beta35	0.101	0.078	-0.023 ± 0.001	-0.227 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.189 ± 0.008	0.435 ± 0.010	0.078	-0.023	-22.72	0.0372	0.0438	0.979	89.900 ± 0.953	0.144 ± 0.000	55.800 ± 1.570
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.318	0.002	0.78	0.0367	0.0367	1.010	95.500 ± 0.656	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.315	-0.001	-0.44	0.0378	0.0378	0.980	95.100 ± 0.683	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.140	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.079 ± 0.004	0.281 ± 0.008	0.138	-0.001	-0.50	0.0394	0.0394	1.024	95.600 ± 0.649	0.157 ± 0.000	94.200 ± 0.739
LSAM	beta34	0.101	0.099	-0.002 ± 0.001	-0.024 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.207 ± 0.010	0.454 ± 0.013	0.099	-0.002	-2.41	0.0453	0.0454	0.993	94.600 ± 0.715	0.179 ± 0.000	58.900 ± 1.556
LSAM	beta35	0.101	0.101	-0.000 ± 0.001	-0.004 ± 0.015	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.212 ± 0.009	0.460 ± 0.013	0.100	-0.001	-0.52	0.0463	0.0463	0.975	94.000 ± 0.751	0.178 ± 0.000	60.000 ± 1.549
QML	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.314	-0.002	-0.67	0.0359	0.0359	1.011	94.800 ± 0.702	0.141 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.312	-0.004 ± 0.001	-0.014 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.311	-0.005	-1.58	0.0362	0.0365	0.988	95.400 ± 0.662	0.142 ± 0.001	99.900 ± 0.100
QML	beta33	0.139	0.141	0.002 ± 0.001	0.017 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.080 ± 0.004	0.284 ± 0.008	0.140	0.001	0.40	0.0392	0.0392	1.021	95.000 ± 0.689	0.158 ± 0.000	94.400 ± 0.727
QML	beta34	0.101	0.076	-0.025 ± 0.001	-0.245 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.193 ± 0.008	0.439 ± 0.011	0.076	-0.025	-25.11	0.0370	0.0448	1.000	90.300 ± 0.936	0.144 ± 0.000	55.600 ± 1.571
QML	beta35	0.101	0.078	-0.023 ± 0.001	-0.229 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.191 ± 0.008	0.437 ± 0.010	0.078	-0.023	-23.06	0.0370	0.0437	0.977	89.600 ± 0.965	0.144 ± 0.000	55.600 ± 1.571
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.318	0.002	0.50	0.0366	0.0366	0.974	94.400 ± 0.727	0.140 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	-0.50	0.0381	0.0381	0.947	94.200 ± 0.739	0.140 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.140	0.001 ± 0.001	0.005 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.080 ± 0.004	0.283 ± 0.008	0.138	-0.001	-0.50	0.0397	0.0397	1.025	95.600 ± 0.649	0.158 ± 0.000	94.200 ± 0.739
UPI	beta34	0.101	0.099	-0.002 ± 0.001	-0.016 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.217 ± 0.010	0.466 ± 0.013	0.099	-0.002	-1.96	0.0468	0.0468	0.987	94.500 ± 0.721	0.182 ± 0.001	58.500 ± 1.558
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.001 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.220 ± 0.010	0.469 ± 0.013	0.100	-0.001	-0.61	0.0471	0.0471	0.974	94.100 ± 0.745	0.181 ± 0.001	58.300 ± 1.559

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S19

Simulation 1 - Condition 19: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.302	-0.014 ± 0.007	-0.046 ± 0.023	0.011 ± 0.002	0.105 ± 0.008	0.011 ± 0.002	0.106 ± 0.008	0.112 ± 0.016	0.335 ± 0.027	0.297	-0.019	-6.09	0.1101	0.1117	1.046	93.532 ± 1.735	0.432 ± 0.022	84.080 ± 2.581
LMS	beta32	0.316	0.304	-0.012 ± 0.008	-0.038 ± 0.026	0.013 ± 0.002	0.115 ± 0.008	0.013 ± 0.002	0.115 ± 0.009	0.133 ± 0.018	0.365 ± 0.028	0.298	-0.018	-5.74	0.1332	0.1344	0.963	95.522 ± 1.459	0.434 ± 0.020	81.592 ± 2.734
LMS	beta33	0.139	0.189	0.050 ± 0.010	0.356 ± 0.073	0.021 ± 0.003	0.144 ± 0.010	0.023 ± 0.003	0.152 ± 0.011	1.198 ± 0.151	1.094 ± 0.078	0.203	0.064	46.09	0.1489	0.1621	1.454	96.517 ± 1.293	0.822 ± 0.173	32.338 ± 3.299
LMS	beta34	0.101	0.061	-0.040 ± 0.007	-0.398 ± 0.072	0.011 ± 0.002	0.103 ± 0.008	0.012 ± 0.002	0.110 ± 0.008	1.185 ± 0.151	1.089 ± 0.078	0.035	-0.066	-65.18	0.1083	0.1268	1.334	83.582 ± 2.613	0.536 ± 0.099	3.980 ± 1.379
LMS	beta35	0.101	0.066	-0.035 ± 0.007	-0.349 ± 0.073	0.011 ± 0.001	0.104 ± 0.007	0.012 ± 0.001	0.110 ± 0.007	1.178 ± 0.131	1.086 ± 0.070	0.045	-0.056	-55.38	0.1060	0.1198	1.287	87.065 ± 2.367	0.525 ± 0.099	8.955 ± 2.014
LSAM	beta31	0.316	0.327	0.011 ± 0.011	0.035 ± 0.036	0.025 ± 0.003	0.160 ± 0.011	0.025 ± 0.003	0.160 ± 0.012	0.255 ± 0.034	0.505 ± 0.039	0.326	0.010	3.30	0.1529	0.1532	1.140	96.517 ± 1.293	0.713 ± 0.038	53.731 ± 3.517
LSAM	beta32	0.316	0.334	0.018 ± 0.012	0.056 ± 0.037	0.027 ± 0.003	0.165 ± 0.009	0.027 ± 0.003	0.165 ± 0.011	0.274 ± 0.030	0.523 ± 0.034	0.329	0.013	4.20	0.1636	0.1641	1.118	97.512 ± 1.099	0.722 ± 0.033	54.229 ± 3.514
LSAM	beta33	0.139	0.290	0.151 ± 0.015	1.085 ± 0.105	0.042 ± 0.006	0.206 ± 0.016	0.065 ± 0.009	0.255 ± 0.020	3.365 ± 0.490	1.834 ± 0.142	0.279	0.140	100.42	0.1847	0.2315	1.042	96.517 ± 1.293	0.842 ± 0.052	27.861 ± 3.162
LSAM	beta34	0.101	0.004	-0.097 ± 0.009	-0.965 ± 0.089	0.016 ± 0.002	0.128 ± 0.009	0.026 ± 0.003	0.160 ± 0.011	2.524 ± 0.315	1.589 ± 0.106	-0.003	-0.104	-102.60	0.1189	0.1577	1.085	85.075 ± 2.513	0.544 ± 0.035	1.493 ± 0.855
LSAM	beta35	0.101	0.010	-0.091 ± 0.010	-0.899 ± 0.096	0.019 ± 0.003	0.138 ± 0.011	0.027 ± 0.003	0.165 ± 0.011	2.671 ± 0.313	1.634 ± 0.105	-0.008	-0.109	-107.92	0.1202	0.1623	1.045	86.070 ± 2.442	0.566 ± 0.036	4.478 ± 1.459
QML	beta31	0.316	0.290	-0.026 ± 0.006	-0.081 ± 0.020	0.008 ± 0.001	0.092 ± 0.005	0.009 ± 0.001	0.095 ± 0.006	0.090 ± 0.009	0.300 ± 0.018	0.295	-0.021	-6.52	0.0976	0.0997	1.061	92.537 ± 1.854	0.381 ± 0.005	85.572 ± 2.478
QML	beta32	0.316	0.291	-0.025 ± 0.007	-0.078 ± 0.022	0.010 ± 0.001	0.098 ± 0.005	0.010 ± 0.001	0.100 ± 0.006	0.101 ± 0.009	0.317 ± 0.018	0.294	-0.022	-6.85	0.1159	0.1179	1.017	96.020 ± 1.379	0.389 ± 0.005	84.080 ± 2.581
QML	beta33	0.139	0.184	0.045 ± 0.009	0.325 ± 0.067	0.017 ± 0.002	0.132 ± 0.008	0.019 ± 0.002	0.139 ± 0.009	1.003 ± 0.110	1.002 ± 0.064	0.201	0.062	44.52	0.1462	0.1587	1.051	94.527 ± 1.604	0.544 ± 0.014	33.333 ± 3.325
QML	beta34	0.101	0.057	-0.044 ± 0.006	-0.436 ± 0.063	0.008 ± 0.001	0.091 ± 0.005	0.010 ± 0.001	0.101 ± 0.005	0.993 ± 0.091	0.997 ± 0.054	0.039	-0.062	-61.15	0.0905	0.1096	0.948	83.582 ± 2.613	0.337 ± 0.009	8.458 ± 1.963
QML	beta35	0.101	0.062	-0.039 ± 0.007	-0.382 ± 0.065	0.009 ± 0.001	0.093 ± 0.006	0.010 ± 0.001	0.101 ± 0.006	0.996 ± 0.103	0.998 ± 0.060	0.043	-0.058	-57.55	0.0913	0.1083	0.990	86.567 ± 2.405	0.362 ± 0.010	12.438 ± 2.328
UPI	beta31	0.316	0.308	-0.008 ± 0.016	-0.024 ± 0.050	0.049 ± 0.007	0.222 ± 0.017	0.049 ± 0.007	0.221 ± 0.019	0.491 ± 0.075	0.701 ± 0.060	0.335	0.019	5.95	0.2499	0.2506	1.987	92.040 ± 1.909	1.728 ± 0.233	26.368 ± 3.108
UPI	beta32	0.316	0.329	0.013 ± 0.017	0.041 ± 0.054	0.059 ± 0.007	0.243 ± 0.015	0.059 ± 0.007	0.243 ± 0.018	0.591 ± 0.074	0.769 ± 0.056	0.323	0.007	2.06	0.2651	0.2651	1.770	89.055 ± 2.202	1.687 ± 0.198	25.373 ± 3.069
UPI	beta33	0.139	0.260	0.121 ± 0.019	0.869 ± 0.134	0.070 ± 0.011	0.264 ± 0.021	0.084 ± 0.014	0.290 ± 0.025	4.354 ± 0.709	2.087 ± 0.183	0.222	0.083	59.82	0.2374	0.2516	1.832	99.005 ± 0.700	1.898 ± 0.222	7.960 ± 1.909
UPI	beta34	0.101	-0.037	-0.138 ± 0.019	-1.369 ± 0.189	0.073 ± 0.011	0.271 ± 0.021	0.092 ± 0.013	0.303 ± 0.024	9.020 ± 1.307	3.003 ± 0.236	-0.019	-0.120	-118.89	0.2473	0.2749	1.790	89.055 ± 2.202	1.899 ± 0.236	2.985 ± 1.200
UPI	beta35	0.101	-0.040	-0.141 ± 0.018	-1.394 ± 0.181	0.067 ± 0.010	0.259 ± 0.019	0.087 ± 0.012	0.294 ± 0.022	8.492 ± 1.141	2.914 ± 0.213	-0.019	-0.120	-118.54	0.2386	0.2669	1.667	86.567 ± 2.405	1.693 ± 0.176	2.985 ± 1.200

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S20

Simulation 1 - Condition 20: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.296	-0.020 ± 0.004	-0.062 ± 0.012	0.004 ± 0.000	0.063 ± 0.003	0.004 ± 0.000	0.066 ± 0.003	0.043 ± 0.004	0.208 ± 0.011	0.291	-0.025	-7.80	0.0736	0.0776	1.006	95.276 ± 1.331	0.248 ± 0.005	99.606 ± 0.393
LMS	beta32	0.316	0.295	-0.021 ± 0.004	-0.066 ± 0.012	0.004 ± 0.000	0.060 ± 0.003	0.004 ± 0.000	0.064 ± 0.004	0.041 ± 0.004	0.202 ± 0.011	0.301	-0.015	-4.89	0.0692	0.0709	1.038	93.307 ± 1.568	0.246 ± 0.005	98.819 ± 0.678
LMS	beta33	0.139	0.208	0.069 ± 0.005	0.494 ± 0.038	0.007 ± 0.001	0.084 ± 0.005	0.012 ± 0.001	0.108 ± 0.004	0.609 ± 0.044	0.780 ± 0.032	0.214	0.075	53.86	0.1001	0.1250	1.081	87.795 ± 2.054	0.357 ± 0.012	69.291 ± 2.894
LMS	beta34	0.101	0.046	-0.055 ± 0.004	-0.546 ± 0.038	0.004 ± 0.000	0.060 ± 0.004	0.007 ± 0.001	0.082 ± 0.004	0.654 ± 0.052	0.809 ± 0.035	0.046	-0.055	-54.26	0.0684	0.0876	1.010	75.197 ± 2.710	0.239 ± 0.015	9.449 ± 1.835
LMS	beta35	0.101	0.050	-0.051 ± 0.004	-0.508 ± 0.036	0.003 ± 0.000	0.058 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	0.584 ± 0.037	0.764 ± 0.028	0.041	-0.060	-58.97	0.0639	0.0874	0.990	74.803 ± 2.724	0.224 ± 0.005	10.236 ± 1.902
LSAM	beta31	0.316	0.330	0.014 ± 0.006	0.045 ± 0.020	0.010 ± 0.001	0.099 ± 0.005	0.010 ± 0.001	0.100 ± 0.006	0.100 ± 0.010	0.317 ± 0.019	0.318	0.002	0.61	0.0932	0.0932	0.963	97.638 ± 0.953	0.375 ± 0.005	93.701 ± 1.524
LSAM	beta32	0.316	0.323	0.007 ± 0.006	0.021 ± 0.018	0.008 ± 0.001	0.089 ± 0.005	0.008 ± 0.001	0.089 ± 0.006	0.080 ± 0.008	0.282 ± 0.017	0.318	0.002	0.74	0.0821	0.0821	1.063	96.063 ± 1.220	0.371 ± 0.005	94.094 ± 1.479
LSAM	beta33	0.139	0.279	0.140 ± 0.007	1.008 ± 0.048	0.011 ± 0.001	0.106 ± 0.005	0.031 ± 0.002	0.176 ± 0.007	1.598 ± 0.123	1.264 ± 0.051	0.263	0.124	89.07	0.0958	0.1565	0.987	80.709 ± 2.476	0.411 ± 0.008	82.283 ± 2.396
LSAM	beta34	0.101	-0.002	-0.103 ± 0.005	-1.019 ± 0.046	0.006 ± 0.001	0.075 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	1.581 ± 0.115	1.258 ± 0.048	-0.001	-0.102	-101.41	0.0651	0.1214	0.921	61.024 ± 3.060	0.269 ± 0.005	3.937 ± 1.220
LSAM	beta35	0.101	0.003	-0.098 ± 0.004	-0.971 ± 0.043	0.005 ± 0.001	0.069 ± 0.004	0.014 ± 0.001	0.120 ± 0.004	1.412 ± 0.094	1.188 ± 0.042	0.003	-0.098	-96.54	0.0624	0.1158	0.976	66.929 ± 2.952	0.265 ± 0.005	3.937 ± 1.220
QML	beta31	0.316	0.293	-0.023 ± 0.004	-0.074 ± 0.012	0.004 ± 0.000	0.061 ± 0.003	0.004 ± 0.000	0.065 ± 0.003	0.042 ± 0.003	0.206 ± 0.010	0.287	-0.029	-9.09	0.0652	0.0713	1.006	94.882 ± 1.383	0.240 ± 0.002	99.606 ± 0.393
QML	beta32	0.316	0.291	-0.025 ± 0.004	-0.078 ± 0.012	0.003 ± 0.000	0.058 ± 0.003	0.004 ± 0.000	0.063 ± 0.003	0.040 ± 0.004	0.199 ± 0.010	0.295	-0.021	-6.73	0.0605	0.0642	1.044	92.126 ± 1.690	0.238 ± 0.001	99.606 ± 0.393
QML	beta33	0.139	0.205	0.066 ± 0.005	0.473 ± 0.037	0.007 ± 0.001	0.082 ± 0.004	0.011 ± 0.001	0.105 ± 0.004	0.574 ± 0.043	0.758 ± 0.032	0.208	0.069	49.89	0.0905	0.1140	1.109	88.189 ± 2.025	0.358 ± 0.017	69.291 ± 2.894
QML	beta34	0.101	0.042	-0.059 ± 0.004	-0.588 ± 0.037	0.004 ± 0.000	0.059 ± 0.003	0.007 ± 0.001	0.084 ± 0.004	0.691 ± 0.055	0.831 ± 0.036	0.040	-0.061	-60.07	0.0618	0.0866	1.005	74.803 ± 2.724	0.234 ± 0.015	10.236 ± 1.902
QML	beta35	0.101	0.045	-0.056 ± 0.004	-0.551 ± 0.035	0.003 ± 0.000	0.056 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	0.612 ± 0.039	0.783 ± 0.028	0.036	-0.065	-64.19	0.0560	0.0857	0.999	75.197 ± 2.710	0.220 ± 0.005	12.598 ± 2.082
UPI	beta31	0.316	0.312	-0.004 ± 0.014	-0.012 ± 0.045	0.051 ± 0.006	0.225 ± 0.013	0.051 ± 0.006	0.225 ± 0.015	0.507 ± 0.058	0.712 ± 0.047	0.333	0.017	5.52	0.2598	0.2604	1.646	83.465 ± 2.331	1.455 ± 0.178	38.583 ± 3.054
UPI	beta32	0.316	0.309	-0.007 ± 0.014	-0.023 ± 0.044	0.050 ± 0.007	0.223 ± 0.015	0.049 ± 0.007	0.222 ± 0.017	0.494 ± 0.067	0.703 ± 0.053	0.334	0.018	5.76	0.2202	0.2209	1.771	84.252 ± 2.286	1.545 ± 0.207	47.244 ± 3.133
UPI	beta33	0.139	0.273	0.134 ± 0.012	0.966 ± 0.085	0.036 ± 0.005	0.189 ± 0.012	0.054 ± 0.007	0.232 ± 0.015	2.777 ± 0.347	1.666 ± 0.111	0.231	0.092	66.49	0.1591	0.1840	1.663	98.425 ± 0.781	1.233 ± 0.116	24.803 ± 2.710
UPI	beta34	0.101	-0.093	-0.194 ± 0.015	-1.917 ± 0.146	0.055 ± 0.007	0.235 ± 0.016	0.092 ± 0.011	0.304 ± 0.020	9.058 ± 1.114	3.010 ± 0.196	-0.083	-0.184	-182.50	0.2450	0.3066	1.766	81.890 ± 2.416	1.625 ± 0.201	1.575 ± 0.781
UPI	beta35	0.101	-0.052	-0.153 ± 0.013	-1.513 ± 0.132	0.045 ± 0.005	0.212 ± 0.013	0.068 ± 0.007	0.261 ± 0.014	6.687 ± 0.668	2.586 ± 0.141	-0.049	-0.150	-148.18	0.2186	0.2650	1.719	82.283 ± 2.396	1.430 ± 0.157	3.543 ± 1.160

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S21

Simulation 1 - Condition 21: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.302	-0.014 ± 0.002	-0.044 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.301	-0.015	-4.90	0.0723	0.0740	1.044	93.023 ± 0.810	0.296 ± 0.013	98.382 ± 0.401
LMS	beta32	0.316	0.303	-0.013 ± 0.002	-0.040 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.243 ± 0.007	0.301	-0.015	-4.69	0.0734	0.0749	0.979	92.214 ± 0.852	0.291 ± 0.008	98.787 ± 0.348
LMS	beta33	0.139	0.153	0.014 ± 0.003	0.103 ± 0.020	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.403 ± 0.017	0.635 ± 0.017	0.152	0.013	9.51	0.0862	0.0872	1.030	95.147 ± 0.683	0.352 ± 0.005	44.995 ± 1.582
LMS	beta34	0.101	0.083	-0.018 ± 0.002	-0.174 ± 0.019	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.388 ± 0.017	0.623 ± 0.016	0.082	-0.019	-18.82	0.0601	0.0631	0.985	92.315 ± 0.847	0.233 ± 0.002	30.536 ± 1.464
LMS	beta35	0.101	0.084	-0.017 ± 0.002	-0.171 ± 0.019	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.375 ± 0.018	0.613 ± 0.017	0.084	-0.017	-16.60	0.0573	0.0597	1.026	92.720 ± 0.826	0.239 ± 0.002	29.929 ± 1.456
LSAM	beta31	0.316	0.312	-0.004 ± 0.003	-0.012 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.312	-0.004	-1.38	0.0793	0.0794	1.007	94.944 ± 0.697	0.313 ± 0.001	97.270 ± 0.518
LSAM	beta32	0.316	0.314	-0.002 ± 0.003	-0.007 ± 0.009	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.071 ± 0.003	0.267 ± 0.008	0.312	-0.004	-1.14	0.0795	0.0795	0.945	92.518 ± 0.837	0.313 ± 0.001	97.776 ± 0.469
LSAM	beta33	0.139	0.178	0.039 ± 0.003	0.280 ± 0.022	0.009 ± 0.000	0.095 ± 0.002	0.011 ± 0.000	0.103 ± 0.003	0.545 ± 0.026	0.738 ± 0.020	0.176	0.037	26.93	0.0924	0.0997	0.988	93.630 ± 0.777	0.368 ± 0.003	50.961 ± 1.590
LSAM	beta34	0.101	0.072	-0.029 ± 0.002	-0.284 ± 0.021	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.515 ± 0.023	0.717 ± 0.019	0.071	-0.030	-29.45	0.0665	0.0729	0.968	90.394 ± 0.937	0.253 ± 0.002	21.436 ± 1.305
LSAM	beta35	0.101	0.073	-0.028 ± 0.002	-0.275 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.502 ± 0.024	0.709 ± 0.020	0.075	-0.026	-25.51	0.0616	0.0668	0.998	90.597 ± 0.928	0.258 ± 0.002	21.031 ± 1.296
QML	beta31	0.316	0.302	-0.014 ± 0.002	-0.044 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.300	-0.016	-5.02	0.0718	0.0736	0.991	92.720 ± 0.826	0.281 ± 0.001	98.686 ± 0.362
QML	beta32	0.316	0.303	-0.013 ± 0.002	-0.041 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.243 ± 0.007	0.301	-0.015	-4.71	0.0737	0.0752	0.949	92.012 ± 0.862	0.282 ± 0.001	98.989 ± 0.318
QML	beta33	0.139	0.154	0.015 ± 0.003	0.107 ± 0.020	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.402 ± 0.017	0.634 ± 0.017	0.153	0.014	9.80	0.0854	0.0865	1.019	95.248 ± 0.677	0.347 ± 0.003	45.298 ± 1.583
QML	beta34	0.101	0.082	-0.019 ± 0.002	-0.186 ± 0.019	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.390 ± 0.017	0.624 ± 0.016	0.081	-0.020	-19.88	0.0605	0.0637	0.981	92.518 ± 0.837	0.232 ± 0.002	30.435 ± 1.463
QML	beta35	0.101	0.082	-0.019 ± 0.002	-0.184 ± 0.019	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.376 ± 0.018	0.613 ± 0.017	0.084	-0.017	-17.17	0.0579	0.0605	1.021	92.619 ± 0.831	0.236 ± 0.002	29.929 ± 1.456
UPI	beta31	0.316	0.321	0.005 ± 0.003	0.015 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.068 ± 0.003	0.260 ± 0.008	0.321	0.005	1.52	0.0773	0.0774	1.013	94.742 ± 0.710	0.326 ± 0.003	96.057 ± 0.619
UPI	beta32	0.316	0.323	0.007 ± 0.003	0.022 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.076 ± 0.004	0.276 ± 0.008	0.322	0.006	1.86	0.0830	0.0832	0.956	93.428 ± 0.788	0.326 ± 0.004	96.663 ± 0.571
UPI	beta33	0.139	0.166	0.027 ± 0.003	0.191 ± 0.023	0.010 ± 0.000	0.100 ± 0.002	0.011 ± 0.001	0.104 ± 0.003	0.558 ± 0.028	0.747 ± 0.022	0.162	0.023	16.63	0.0972	0.0999	1.017	96.360 ± 0.596	0.400 ± 0.005	40.849 ± 1.563
UPI	beta34	0.101	0.088	-0.013 ± 0.002	-0.124 ± 0.025	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.613 ± 0.030	0.783 ± 0.023	0.089	-0.012	-12.15	0.0742	0.0752	0.981	93.428 ± 0.788	0.300 ± 0.004	24.570 ± 1.369
UPI	beta35	0.101	0.088	-0.013 ± 0.002	-0.132 ± 0.023	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.554 ± 0.028	0.744 ± 0.022	0.086	-0.015	-14.65	0.0676	0.0692	1.056	95.147 ± 0.683	0.306 ± 0.005	23.660 ± 1.351

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S22

Simulation 1 - Condition 22: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.303	-0.013 ± 0.002	-0.041 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.161 ± 0.004	0.301	-0.015	-4.68	0.0502	0.0524	0.928	91.600 ± 0.877	0.179 ± 0.002	99.900 ± 0.100
LMS	beta32	0.316	0.304	-0.012 ± 0.001	-0.039 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.304	-0.012	-3.85	0.0442	0.0459	1.011	94.300 ± 0.733	0.178 ± 0.001	99.900 ± 0.100
LMS	beta33	0.139	0.158	0.019 ± 0.002	0.135 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.189 ± 0.009	0.434 ± 0.012	0.159	0.020	14.54	0.0556	0.0591	0.973	93.000 ± 0.807	0.219 ± 0.001	79.700 ± 1.272
LMS	beta34	0.101	0.080	-0.021 ± 0.001	-0.212 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.191 ± 0.008	0.437 ± 0.011	0.080	-0.021	-20.74	0.0376	0.0431	0.980	89.100 ± 0.985	0.148 ± 0.001	56.000 ± 1.570
LMS	beta35	0.101	0.080	-0.021 ± 0.001	-0.206 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.187 ± 0.008	0.432 ± 0.011	0.080	-0.021	-21.21	0.0384	0.0440	0.977	89.100 ± 0.985	0.147 ± 0.001	56.700 ± 1.567
LSAM	beta31	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.029 ± 0.001	0.169 ± 0.005	0.311	-0.005	-1.61	0.0527	0.0529	0.927	92.800 ± 0.817	0.194 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.313	-0.003	-0.99	0.0472	0.0473	1.002	95.600 ± 0.649	0.193 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.179	0.040 ± 0.002	0.288 ± 0.013	0.004 ± 0.000	0.059 ± 0.001	0.005 ± 0.000	0.072 ± 0.002	0.265 ± 0.012	0.515 ± 0.013	0.179	0.040	28.51	0.0584	0.0706	0.955	88.000 ± 1.028	0.222 ± 0.001	88.600 ± 1.005
LSAM	beta34	0.101	0.069	-0.032 ± 0.001	-0.319 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.274 ± 0.011	0.523 ± 0.012	0.068	-0.033	-32.44	0.0422	0.0535	0.952	84.900 ± 1.132	0.156 ± 0.001	41.700 ± 1.559
LSAM	beta35	0.101	0.070	-0.031 ± 0.001	-0.310 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.267 ± 0.011	0.517 ± 0.012	0.068	-0.033	-32.28	0.0414	0.0527	0.948	83.900 ± 1.162	0.155 ± 0.001	42.200 ± 1.562
QML	beta31	0.316	0.303	-0.013 ± 0.002	-0.041 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.161 ± 0.004	0.301	-0.015	-4.81	0.0507	0.0529	0.921	91.700 ± 0.872	0.178 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.304	-0.012 ± 0.001	-0.040 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.303	-0.013	-4.07	0.0441	0.0459	1.005	94.200 ± 0.739	0.177 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.158	0.019 ± 0.002	0.136 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.159	0.020	14.33	0.0545	0.0581	0.974	93.200 ± 0.796	0.219 ± 0.001	80.100 ± 1.263
QML	beta34	0.101	0.079	-0.022 ± 0.001	-0.221 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.194 ± 0.008	0.440 ± 0.011	0.079	-0.022	-21.36	0.0375	0.0432	0.979	88.800 ± 0.997	0.148 ± 0.001	55.900 ± 1.570
QML	beta35	0.101	0.079	-0.022 ± 0.001	-0.215 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.190 ± 0.008	0.436 ± 0.011	0.078	-0.023	-22.43	0.0382	0.0444	0.977	88.800 ± 0.997	0.147 ± 0.001	55.900 ± 1.570
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.029 ± 0.001	0.171 ± 0.005	0.316	0.000	0.10	0.0532	0.0532	0.913	92.000 ± 0.858	0.194 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.318	0.002	0.63	0.0475	0.0476	0.998	95.000 ± 0.689	0.193 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.159	0.020 ± 0.002	0.145 ± 0.014	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.217 ± 0.011	0.466 ± 0.014	0.157	0.018	12.63	0.0589	0.0614	0.962	94.000 ± 0.751	0.232 ± 0.001	77.500 ± 1.321
UPI	beta34	0.101	0.084	-0.017 ± 0.001	-0.166 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.246 ± 0.012	0.496 ± 0.014	0.083	-0.018	-18.27	0.0472	0.0507	0.951	91.700 ± 0.872	0.176 ± 0.001	48.400 ± 1.580
UPI	beta35	0.101	0.085	-0.016 ± 0.001	-0.159 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.230 ± 0.010	0.480 ± 0.013	0.083	-0.018	-17.44	0.0449	0.0482	0.968	91.100 ± 0.900	0.174 ± 0.001	48.700 ± 1.581

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S23

Simulation 1 - Condition 23: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.002	-0.021 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.307	-0.009	-3.00	0.0603	0.0610	0.994	94.600 ± 0.715	0.229 ± 0.001	99.900 ± 0.100
LMS	beta32	0.316	0.311	-0.005 ± 0.002	-0.017 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.312	-0.004	-1.39	0.0585	0.0586	1.018	96.000 ± 0.620	0.229 ± 0.001	99.900 ± 0.100
LMS	beta33	0.139	0.140	0.001 ± 0.002	0.010 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.217 ± 0.010	0.466 ± 0.013	0.139	0.000	-0.14	0.0597	0.0597	1.003	94.600 ± 0.715	0.255 ± 0.001	59.000 ± 1.555
LMS	beta34	0.101	0.096	-0.005 ± 0.001	-0.052 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.210 ± 0.009	0.459 ± 0.012	0.094	-0.007	-6.65	0.0464	0.0469	0.971	94.100 ± 0.745	0.175 ± 0.001	57.300 ± 1.564
LMS	beta35	0.101	0.095	-0.006 ± 0.001	-0.061 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.199 ± 0.009	0.446 ± 0.012	0.093	-0.008	-7.71	0.0436	0.0442	1.010	94.400 ± 0.727	0.177 ± 0.001	55.500 ± 1.572
LSAM	beta31	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.310	-0.006	-1.88	0.0610	0.0613	1.000	95.600 ± 0.649	0.235 ± 0.001	99.900 ± 0.100
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.001	0.187 ± 0.005	0.316	0.000	0.08	0.0583	0.0583	1.018	95.700 ± 0.641	0.235 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.148	0.009 ± 0.002	0.063 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.222 ± 0.010	0.472 ± 0.013	0.148	0.009	6.15	0.0620	0.0626	1.007	94.200 ± 0.739	0.256 ± 0.001	63.000 ± 1.527
LSAM	beta34	0.101	0.095	-0.006 ± 0.002	-0.064 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.229 ± 0.010	0.478 ± 0.013	0.091	-0.010	-9.51	0.0475	0.0485	0.962	93.700 ± 0.768	0.181 ± 0.001	54.500 ± 1.575
LSAM	beta35	0.101	0.093	-0.008 ± 0.001	-0.077 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.215 ± 0.010	0.463 ± 0.013	0.092	-0.009	-9.14	0.0454	0.0463	1.004	93.900 ± 0.757	0.182 ± 0.001	52.400 ± 1.579
QML	beta31	0.316	0.309	-0.007 ± 0.002	-0.022 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.306	-0.010	-3.02	0.0599	0.0607	0.993	94.600 ± 0.715	0.228 ± 0.001	99.900 ± 0.100
QML	beta32	0.316	0.310	-0.006 ± 0.002	-0.018 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.183 ± 0.005	0.311	-0.005	-1.51	0.0585	0.0587	1.015	95.800 ± 0.634	0.229 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.140	0.001 ± 0.002	0.009 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.217 ± 0.010	0.466 ± 0.013	0.139	0.000	0.01	0.0602	0.0602	1.003	94.500 ± 0.721	0.255 ± 0.001	58.900 ± 1.556
QML	beta34	0.101	0.096	-0.005 ± 0.001	-0.053 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.211 ± 0.009	0.459 ± 0.012	0.094	-0.007	-7.13	0.0464	0.0469	0.970	94.100 ± 0.745	0.175 ± 0.001	57.400 ± 1.564
QML	beta35	0.101	0.095	-0.006 ± 0.001	-0.062 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.199 ± 0.009	0.446 ± 0.012	0.093	-0.008	-7.90	0.0433	0.0441	1.009	94.300 ± 0.733	0.177 ± 0.001	55.900 ± 1.570
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.315	-0.001	-0.39	0.0608	0.0608	0.968	94.600 ± 0.715	0.226 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.001	0.186 ± 0.005	0.321	0.005	1.49	0.0575	0.0577	0.986	94.800 ± 0.702	0.227 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.144	0.005 ± 0.002	0.035 ± 0.015	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.227 ± 0.010	0.476 ± 0.013	0.145	0.006	4.51	0.0643	0.0646	0.990	94.100 ± 0.745	0.256 ± 0.001	61.500 ± 1.539
UPI	beta34	0.101	0.099	-0.002 ± 0.002	-0.018 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.238 ± 0.011	0.488 ± 0.014	0.096	-0.005	-4.79	0.0487	0.0489	0.949	94.000 ± 0.751	0.183 ± 0.001	57.000 ± 1.566
UPI	beta35	0.101	0.097	-0.004 ± 0.001	-0.039 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.219 ± 0.010	0.468 ± 0.013	0.095	-0.006	-5.74	0.0452	0.0456	0.993	94.600 ± 0.715	0.184 ± 0.001	55.500 ± 1.572

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S24

Simulation 1 - Condition 24: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.001	-0.023 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.308	-0.008	-2.47	0.0365	0.0374	1.016	95.095 ± 0.683	0.144 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.310	-0.006	-1.83	0.0356	0.0360	0.998	94.094 ± 0.746	0.144 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.138	-0.001 ± 0.001	-0.008 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.005	0.310 ± 0.009	0.138	-0.001	-0.56	0.0418	0.0418	0.947	93.093 ± 0.802	0.160 ± 0.001	89.690 ± 0.962
LMS	beta34	0.101	0.099	-0.002 ± 0.001	-0.024 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.078 ± 0.003	0.280 ± 0.007	0.099	-0.002	-2.35	0.0298	0.0299	1.003	95.495 ± 0.656	0.111 ± 0.000	93.994 ± 0.752
LMS	beta35	0.101	0.096	-0.005 ± 0.001	-0.050 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.082 ± 0.004	0.287 ± 0.008	0.096	-0.005	-5.39	0.0282	0.0287	0.992	93.994 ± 0.752	0.111 ± 0.000	91.992 ± 0.859
LSAM	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.313	-0.003	-1.07	0.0381	0.0383	1.019	95.596 ± 0.649	0.148 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	-0.48	0.0368	0.0368	1.009	94.595 ± 0.715	0.148 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.145	0.006 ± 0.001	0.043 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.145	0.006	4.61	0.0426	0.0431	0.955	93.594 ± 0.775	0.160 ± 0.001	93.393 ± 0.786
LSAM	beta34	0.101	0.097	-0.004 ± 0.001	-0.043 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.096	-0.005	-4.55	0.0310	0.0313	0.995	94.595 ± 0.715	0.113 ± 0.000	91.792 ± 0.868
LSAM	beta35	0.101	0.094	-0.007 ± 0.001	-0.071 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.093	-0.008	-8.15	0.0299	0.0310	0.991	93.493 ± 0.780	0.113 ± 0.000	89.590 ± 0.966
QML	beta31	0.316	0.309	-0.007 ± 0.001	-0.023 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.308	-0.008	-2.55	0.0367	0.0376	1.033	95.195 ± 0.677	0.146 ± 0.002	99.900 ± 0.100
QML	beta32	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.310	-0.006	-1.87	0.0356	0.0361	1.019	94.094 ± 0.746	0.147 ± 0.003	99.900 ± 0.100
QML	beta33	0.139	0.138	-0.001 ± 0.001	-0.009 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.005	0.310 ± 0.009	0.138	-0.001	-0.80	0.0413	0.0413	0.947	93.193 ± 0.797	0.160 ± 0.001	89.790 ± 0.958
QML	beta34	0.101	0.099	-0.002 ± 0.001	-0.025 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.078 ± 0.003	0.280 ± 0.007	0.098	-0.003	-2.49	0.0298	0.0299	1.002	95.495 ± 0.656	0.111 ± 0.000	94.094 ± 0.746
QML	beta35	0.101	0.096	-0.005 ± 0.001	-0.051 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.082 ± 0.004	0.287 ± 0.008	0.095	-0.006	-5.47	0.0284	0.0290	0.989	93.994 ± 0.752	0.111 ± 0.000	91.892 ± 0.864
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.314	-0.002	-0.55	0.0373	0.0374	0.980	94.795 ± 0.703	0.143 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.317	0.001	0.31	0.0377	0.0377	0.971	93.694 ± 0.769	0.143 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.141	0.002 ± 0.001	0.012 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.140	0.001	0.52	0.0430	0.0430	0.949	93.193 ± 0.797	0.160 ± 0.001	91.792 ± 0.868
UPI	beta34	0.101	0.101	0.000 ± 0.001	0.001 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.101	0.000	-0.16	0.0310	0.0310	0.996	95.495 ± 0.656	0.114 ± 0.000	93.193 ± 0.797
UPI	beta35	0.101	0.098	-0.003 ± 0.001	-0.031 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.097	-0.004	-3.76	0.0303	0.0306	0.994	94.394 ± 0.728	0.114 ± 0.000	91.391 ± 0.887

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S25

Simulation 1 - Condition 25: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.330	0.014 ± 0.005	0.045 ± 0.016	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.128 ± 0.005	0.163 ± 0.011	0.404 ± 0.016	0.335	0.019	6.16	0.1330	0.1344	1.042	95.638 ± 0.837	0.519 ± 0.018	77.349 ± 1.715
LMS	beta32	0.316	0.336	0.020 ± 0.005	0.062 ± 0.017	0.017 ± 0.001	0.130 ± 0.005	0.017 ± 0.001	0.131 ± 0.005	0.172 ± 0.012	0.415 ± 0.017	0.342	0.026	8.15	0.1348	0.1373	1.031	95.805 ± 0.821	0.524 ± 0.019	77.852 ± 1.701
LMS	beta33	0.139	0.088	-0.051 ± 0.008	-0.369 ± 0.060	0.042 ± 0.003	0.204 ± 0.007	0.044 ± 0.003	0.210 ± 0.009	2.290 ± 0.171	1.513 ± 0.064	0.112	-0.027	-19.32	0.2035	0.2052	1.048	94.295 ± 0.950	0.839 ± 0.038	15.940 ± 1.499
LMS	beta34	0.101	0.199	0.098 ± 0.005	0.967 ± 0.050	0.015 ± 0.001	0.123 ± 0.005	0.025 ± 0.002	0.157 ± 0.006	2.426 ± 0.184	1.558 ± 0.063	0.178	0.077	76.44	0.1097	0.1342	1.064	91.779 ± 1.125	0.515 ± 0.039	46.309 ± 2.042
LMS	beta35	0.101	0.192	0.091 ± 0.005	0.900 ± 0.045	0.012 ± 0.001	0.111 ± 0.005	0.021 ± 0.002	0.143 ± 0.006	2.016 ± 0.155	1.420 ± 0.058	0.180	0.079	78.71	0.1127	0.1379	1.106	92.282 ± 1.093	0.482 ± 0.025	44.799 ± 2.037
LSAM	beta31	0.316	0.350	0.034 ± 0.053	0.107 ± 0.169	1.692 ± 0.107	1.301 ± 0.042	1.690 ± 0.108	1.300 ± 0.049	16.925 ± 1.079	4.114 ± 0.156	0.360	0.044	13.96	1.4482	1.4488	2.576	99.832 ± 0.168	13.136 ± 0.460	2.013 ± 0.575
LSAM	beta32	0.316	0.455	0.139 ± 0.055	0.440 ± 0.174	1.810 ± 0.115	1.345 ± 0.043	1.826 ± 0.118	1.351 ± 0.052	18.287 ± 1.182	4.276 ± 0.164	0.374	0.058	18.41	1.3955	1.3967	2.546	100.000 ± 0.000	13.427 ± 0.482	2.517 ± 0.642
LSAM	beta33	0.139	0.295	0.156 ± 0.016	1.126 ± 0.113	0.146 ± 0.013	0.382 ± 0.017	0.170 ± 0.016	0.413 ± 0.021	8.814 ± 0.849	2.969 ± 0.153	0.252	0.113	81.14	0.3062	0.3263	2.162	99.161 ± 0.374	3.240 ± 0.153	4.530 ± 0.852
LSAM	beta34	0.101	0.041	-0.060 ± 0.020	-0.598 ± 0.200	0.244 ± 0.020	0.494 ± 0.020	0.247 ± 0.020	0.497 ± 0.023	24.248 ± 1.986	4.924 ± 0.226	0.057	-0.044	-43.21	0.4500	0.4521	2.477	99.329 ± 0.334	4.798 ± 0.198	2.852 ± 0.682
LSAM	beta35	0.101	0.000	-0.101 ± 0.022	-1.005 ± 0.216	0.284 ± 0.022	0.533 ± 0.021	0.294 ± 0.023	0.542 ± 0.024	28.776 ± 2.293	5.364 ± 0.239	0.047	-0.054	-53.25	0.4153	0.4188	2.390	99.664 ± 0.237	4.990 ± 0.211	3.020 ± 0.701
QML	beta31	0.316	0.333	0.017 ± 0.005	0.055 ± 0.015	0.014 ± 0.001	0.119 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.146 ± 0.010	0.382 ± 0.015	0.333	0.017	5.33	0.1251	0.1262	1.003	95.470 ± 0.852	0.470 ± 0.004	80.537 ± 1.622
QML	beta32	0.316	0.337	0.021 ± 0.005	0.068 ± 0.016	0.015 ± 0.001	0.123 ± 0.004	0.016 ± 0.001	0.125 ± 0.005	0.157 ± 0.011	0.396 ± 0.016	0.340	0.024	7.45	0.1260	0.1282	0.978	95.805 ± 0.821	0.473 ± 0.004	81.879 ± 1.578
QML	beta33	0.139	0.101	-0.038 ± 0.007	-0.273 ± 0.051	0.030 ± 0.002	0.173 ± 0.005	0.031 ± 0.002	0.177 ± 0.007	1.626 ± 0.103	1.275 ± 0.048	0.111	-0.028	-20.14	0.1852	0.1873	0.989	94.463 ± 0.937	0.672 ± 0.013	16.611 ± 1.524
QML	beta34	0.101	0.185	0.084 ± 0.004	0.827 ± 0.043	0.011 ± 0.001	0.105 ± 0.004	0.018 ± 0.001	0.134 ± 0.005	1.765 ± 0.118	1.329 ± 0.048	0.177	0.076	75.58	0.0981	0.1243	0.955	89.094 ± 1.277	0.393 ± 0.008	52.517 ± 2.045
QML	beta35	0.101	0.179	0.078 ± 0.004	0.776 ± 0.038	0.009 ± 0.001	0.095 ± 0.003	0.015 ± 0.001	0.123 ± 0.004	1.480 ± 0.085	1.217 ± 0.038	0.178	0.077	76.59	0.1010	0.1273	1.039	89.094 ± 1.277	0.386 ± 0.008	51.342 ± 2.047
UPI	beta31	0.316	0.330	0.014 ± 0.022	0.045 ± 0.069	0.281 ± 0.023	0.530 ± 0.021	0.281 ± 0.023	0.530 ± 0.024	2.809 ± 0.226	1.676 ± 0.076	0.337	0.021	6.74	0.4934	0.4939	1.993	99.329 ± 0.334	4.140 ± 0.193	5.201 ± 0.910
UPI	beta32	0.316	0.356	0.040 ± 0.022	0.126 ± 0.068	0.278 ± 0.022	0.527 ± 0.021	0.279 ± 0.022	0.528 ± 0.024	2.791 ± 0.225	1.671 ± 0.076	0.336	0.020	6.47	0.4902	0.4907	2.033	99.161 ± 0.374	4.199 ± 0.192	5.369 ± 0.923
UPI	beta33	0.139	0.177	0.038 ± 0.012	0.273 ± 0.084	0.082 ± 0.007	0.287 ± 0.013	0.084 ± 0.007	0.289 ± 0.014	4.323 ± 0.388	2.079 ± 0.103	0.158	0.019	13.40	0.2442	0.2449	1.924	97.651 ± 0.620	2.163 ± 0.114	3.020 ± 0.701
UPI	beta34	0.101	0.071	-0.030 ± 0.009	-0.300 ± 0.085	0.044 ± 0.004	0.209 ± 0.009	0.045 ± 0.004	0.211 ± 0.010	4.363 ± 0.370	2.089 ± 0.099	0.066	-0.035	-34.64	0.1893	0.1925	2.013	98.154 ± 0.551	1.649 ± 0.082	3.020 ± 0.701
UPI	beta35	0.101	0.063	-0.038 ± 0.008	-0.376 ± 0.082	0.041 ± 0.004	0.202 ± 0.009	0.042 ± 0.004	0.205 ± 0.010	4.118 ± 0.371	2.029 ± 0.100	0.061	-0.040	-39.27	0.1859	0.1901	2.053	98.154 ± 0.551	1.622 ± 0.078	1.678 ± 0.526

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S26

Simulation 1 - Condition 26: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.338	0.022 ± 0.003	0.071 ± 0.009	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.055 ± 0.003	0.234 ± 0.008	0.339	0.023	7.12	0.0767	0.0800	1.068	95.305 ± 0.837	0.295 ± 0.002	99.061 ± 0.382
LMS	beta32	0.316	0.335	0.019 ± 0.003	0.061 ± 0.009	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	0.054 ± 0.003	0.232 ± 0.008	0.337	0.021	6.76	0.0765	0.0794	1.063	95.618 ± 0.810	0.296 ± 0.002	98.592 ± 0.466
LMS	beta33	0.139	0.117	-0.022 ± 0.005	-0.155 ± 0.033	0.013 ± 0.001	0.114 ± 0.004	0.014 ± 0.001	0.116 ± 0.004	0.700 ± 0.045	0.837 ± 0.031	0.121	-0.018	-12.90	0.1238	0.1250	1.009	94.679 ± 0.888	0.452 ± 0.007	24.570 ± 1.703
LMS	beta34	0.101	0.184	0.083 ± 0.002	0.817 ± 0.024	0.004 ± 0.000	0.060 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	1.024 ± 0.051	1.012 ± 0.026	0.178	0.077	76.65	0.0658	0.1016	1.068	78.404 ± 1.628	0.253 ± 0.004	85.759 ± 1.382
LMS	beta35	0.101	0.183	0.082 ± 0.002	0.809 ± 0.024	0.004 ± 0.000	0.062 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	1.029 ± 0.053	1.014 ± 0.027	0.178	0.077	76.46	0.0649	0.1009	1.030	79.499 ± 1.597	0.250 ± 0.004	85.915 ± 1.376
LSAM	beta31	0.316	0.395	0.079 ± 0.023	0.249 ± 0.073	0.344 ± 0.029	0.587 ± 0.025	0.350 ± 0.030	0.592 ± 0.028	3.506 ± 0.301	1.872 ± 0.089	0.368	0.052	16.47	0.6449	0.6470	1.927	99.687 ± 0.221	4.434 ± 0.261	8.451 ± 1.100
LSAM	beta32	0.316	0.340	0.024 ± 0.023	0.075 ± 0.073	0.342 ± 0.028	0.585 ± 0.024	0.342 ± 0.028	0.585 ± 0.027	3.425 ± 0.283	1.851 ± 0.085	0.351	0.035	11.11	0.6723	0.6732	1.904	99.061 ± 0.382	4.364 ± 0.254	11.268 ± 1.251
LSAM	beta33	0.139	0.231	0.092 ± 0.005	0.661 ± 0.039	0.019 ± 0.002	0.138 ± 0.006	0.027 ± 0.002	0.166 ± 0.007	1.419 ± 0.112	1.191 ± 0.050	0.239	0.100	72.27	0.1529	0.1829	1.485	96.714 ± 0.705	0.803 ± 0.037	33.646 ± 1.869
LSAM	beta34	0.101	0.043	-0.058 ± 0.008	-0.575 ± 0.075	0.036 ± 0.003	0.190 ± 0.009	0.040 ± 0.004	0.199 ± 0.010	3.880 ± 0.346	1.970 ± 0.095	0.054	-0.047	-46.46	0.2023	0.2077	1.936	98.435 ± 0.491	1.445 ± 0.093	6.260 ± 0.958
LSAM	beta35	0.101	0.057	-0.044 ± 0.008	-0.435 ± 0.076	0.038 ± 0.003	0.195 ± 0.009	0.040 ± 0.004	0.200 ± 0.010	3.919 ± 0.361	1.980 ± 0.099	0.067	-0.034	-33.86	0.2060	0.2089	1.848	97.966 ± 0.558	1.414 ± 0.091	6.729 ± 0.991
QML	beta31	0.316	0.338	0.022 ± 0.003	0.071 ± 0.009	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.053 ± 0.003	0.229 ± 0.008	0.338	0.022	7.11	0.0740	0.0773	1.059	94.836 ± 0.875	0.286 ± 0.001	99.374 ± 0.312
QML	beta32	0.316	0.335	0.019 ± 0.003	0.061 ± 0.009	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.003	0.225 ± 0.008	0.335	0.019	6.14	0.0740	0.0765	1.060	95.149 ± 0.850	0.285 ± 0.001	99.531 ± 0.270
QML	beta33	0.139	0.112	-0.027 ± 0.004	-0.192 ± 0.029	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	0.584 ± 0.035	0.764 ± 0.027	0.113	-0.026	-18.72	0.1083	0.1114	1.001	94.053 ± 0.936	0.403 ± 0.004	23.318 ± 1.673
QML	beta34	0.101	0.181	0.080 ± 0.002	0.789 ± 0.022	0.003 ± 0.000	0.055 ± 0.002	0.009 ± 0.000	0.097 ± 0.002	0.922 ± 0.041	0.960 ± 0.022	0.177	0.076	75.64	0.0623	0.0986	1.054	73.709 ± 1.741	0.229 ± 0.003	88.263 ± 1.273
QML	beta35	0.101	0.180	0.079 ± 0.002	0.785 ± 0.022	0.003 ± 0.000	0.057 ± 0.002	0.010 ± 0.000	0.098 ± 0.002	0.934 ± 0.044	0.966 ± 0.024	0.178	0.077	75.90	0.0601	0.0974	1.013	73.396 ± 1.748	0.226 ± 0.003	89.515 ± 1.212
UPI	beta31	0.316	0.313	-0.003 ± 0.015	-0.010 ± 0.047	0.143 ± 0.011	0.378 ± 0.014	0.143 ± 0.011	0.378 ± 0.016	1.430 ± 0.106	1.196 ± 0.050	0.318	0.002	0.68	0.3355	0.3355	1.843	100.000 ± 0.000	2.732 ± 0.135	9.859 ± 1.179
UPI	beta32	0.316	0.318	0.002 ± 0.016	0.006 ± 0.050	0.162 ± 0.012	0.402 ± 0.016	0.161 ± 0.012	0.402 ± 0.017	1.616 ± 0.125	1.271 ± 0.055	0.310	-0.006	-1.82	0.3420	0.3421	1.813	99.531 ± 0.270	2.858 ± 0.154	11.268 ± 1.251
UPI	beta33	0.139	0.171	0.032 ± 0.007	0.231 ± 0.048	0.028 ± 0.002	0.168 ± 0.007	0.029 ± 0.002	0.171 ± 0.008	1.509 ± 0.127	1.228 ± 0.057	0.161	0.022	16.19	0.1516	0.1532	1.757	97.966 ± 0.558	1.156 ± 0.069	14.397 ± 1.389
UPI	beta34	0.101	0.072	-0.029 ± 0.005	-0.290 ± 0.052	0.017 ± 0.001	0.132 ± 0.005	0.018 ± 0.001	0.135 ± 0.006	1.782 ± 0.141	1.335 ± 0.059	0.067	-0.034	-33.49	0.1230	0.1276	1.903	98.748 ± 0.440	0.983 ± 0.053	3.599 ± 0.737
UPI	beta35	0.101	0.070	-0.031 ± 0.005	-0.303 ± 0.053	0.019 ± 0.002	0.136 ± 0.006	0.020 ± 0.002	0.140 ± 0.006	1.915 ± 0.154	1.384 ± 0.062	0.069	-0.032	-31.89	0.1202	0.1245	1.875	98.122 ± 0.537	1.003 ± 0.057	8.451 ± 1.100

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S27

Simulation 1 - Condition 27: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.323	0.007 ± 0.003	0.022 ± 0.009	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.079 ± 0.004	0.281 ± 0.008	0.324	0.008	2.45	0.0949	0.0952	1.047	95.810 ± 0.674	0.364 ± 0.002	93.998 ± 0.799
LMS	beta32	0.316	0.322	0.006 ± 0.003	0.018 ± 0.010	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	0.084 ± 0.005	0.290 ± 0.009	0.321	0.005	1.60	0.0987	0.0989	1.016	95.357 ± 0.708	0.364 ± 0.003	92.752 ± 0.873
LMS	beta33	0.139	0.109	-0.030 ± 0.003	-0.219 ± 0.023	0.009 ± 0.000	0.094 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.510 ± 0.026	0.714 ± 0.021	0.108	-0.031	-22.23	0.0961	0.1010	0.997	93.771 ± 0.813	0.369 ± 0.004	23.783 ± 1.433
LMS	beta34	0.101	0.145	0.044 ± 0.002	0.439 ± 0.019	0.003 ± 0.000	0.056 ± 0.001	0.005 ± 0.000	0.071 ± 0.002	0.501 ± 0.024	0.708 ± 0.018	0.142	0.041	40.77	0.0564	0.0698	0.943	86.183 ± 1.161	0.207 ± 0.002	78.256 ± 1.388
LMS	beta35	0.101	0.144	0.043 ± 0.002	0.424 ± 0.019	0.003 ± 0.000	0.056 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.485 ± 0.025	0.696 ± 0.019	0.142	0.041	40.41	0.0573	0.0703	0.945	86.523 ± 1.149	0.207 ± 0.003	78.482 ± 1.383
LSAM	beta31	0.316	0.326	0.010 ± 0.006	0.030 ± 0.018	0.030 ± 0.002	0.174 ± 0.004	0.030 ± 0.002	0.174 ± 0.005	0.302 ± 0.016	0.550 ± 0.017	0.321	0.005	1.72	0.1806	0.1806	1.065	95.470 ± 0.700	0.724 ± 0.012	46.999 ± 1.680
LSAM	beta32	0.316	0.322	0.006 ± 0.006	0.019 ± 0.019	0.032 ± 0.002	0.177 ± 0.005	0.032 ± 0.002	0.177 ± 0.006	0.315 ± 0.016	0.562 ± 0.017	0.317	0.001	0.25	0.1761	0.1761	1.050	96.263 ± 0.638	0.730 ± 0.013	48.018 ± 1.681
LSAM	beta33	0.139	0.169	0.030 ± 0.003	0.218 ± 0.024	0.010 ± 0.001	0.100 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	0.566 ± 0.031	0.753 ± 0.024	0.161	0.022	16.14	0.1013	0.1037	0.993	93.658 ± 0.820	0.390 ± 0.005	40.544 ± 1.652
LSAM	beta34	0.101	0.090	-0.011 ± 0.002	-0.106 ± 0.023	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.493 ± 0.028	0.702 ± 0.023	0.088	-0.013	-12.92	0.0672	0.0685	0.958	92.639 ± 0.879	0.263 ± 0.005	38.165 ± 1.635
LSAM	beta35	0.101	0.089	-0.012 ± 0.002	-0.117 ± 0.024	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.520 ± 0.030	0.721 ± 0.024	0.088	-0.013	-12.47	0.0684	0.0695	0.965	93.431 ± 0.834	0.272 ± 0.006	36.353 ± 1.619
QML	beta31	0.316	0.324	0.008 ± 0.003	0.024 ± 0.009	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.079 ± 0.004	0.282 ± 0.008	0.324	0.008	2.51	0.0909	0.0913	1.047	95.696 ± 0.683	0.364 ± 0.003	94.451 ± 0.770
QML	beta32	0.316	0.322	0.006 ± 0.003	0.018 ± 0.010	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.082 ± 0.004	0.286 ± 0.009	0.322	0.006	1.76	0.0974	0.0975	1.023	95.017 ± 0.732	0.362 ± 0.002	92.865 ± 0.866
QML	beta33	0.139	0.109	-0.030 ± 0.003	-0.217 ± 0.023	0.009 ± 0.000	0.094 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.506 ± 0.025	0.712 ± 0.021	0.106	-0.033	-23.49	0.0978	0.1031	0.989	92.978 ± 0.860	0.365 ± 0.003	23.783 ± 1.433
QML	beta34	0.101	0.145	0.044 ± 0.002	0.432 ± 0.019	0.003 ± 0.000	0.056 ± 0.001	0.005 ± 0.000	0.071 ± 0.002	0.493 ± 0.024	0.702 ± 0.018	0.142	0.041	40.21	0.0548	0.0682	0.936	85.504 ± 1.185	0.205 ± 0.002	78.256 ± 1.388
QML	beta35	0.101	0.143	0.042 ± 0.002	0.417 ± 0.019	0.003 ± 0.000	0.056 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.476 ± 0.024	0.690 ± 0.019	0.141	0.040	39.92	0.0560	0.0690	0.935	86.183 ± 1.161	0.204 ± 0.002	78.709 ± 1.378
UPI	beta31	0.316	0.326	0.010 ± 0.007	0.033 ± 0.023	0.046 ± 0.003	0.215 ± 0.007	0.046 ± 0.003	0.215 ± 0.007	0.463 ± 0.028	0.680 ± 0.024	0.334	0.018	5.58	0.2148	0.2155	1.151	97.395 ± 0.536	0.969 ± 0.036	40.317 ± 1.651
UPI	beta32	0.316	0.327	0.011 ± 0.007	0.035 ± 0.023	0.047 ± 0.003	0.216 ± 0.006	0.047 ± 0.003	0.217 ± 0.007	0.470 ± 0.027	0.685 ± 0.023	0.324	0.008	2.57	0.1991	0.1993	1.143	97.395 ± 0.536	0.969 ± 0.035	41.450 ± 1.658
UPI	beta33	0.139	0.154	0.015 ± 0.004	0.111 ± 0.027	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	0.676 ± 0.044	0.822 ± 0.030	0.144	0.005	3.93	0.1116	0.1117	1.063	94.904 ± 0.740	0.472 ± 0.011	28.313 ± 1.516
UPI	beta34	0.101	0.092	-0.009 ± 0.003	-0.092 ± 0.028	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	0.718 ± 0.048	0.847 ± 0.032	0.091	-0.010	-10.10	0.0765	0.0772	1.062	93.658 ± 0.820	0.354 ± 0.014	33.069 ± 1.583
UPI	beta35	0.101	0.089	-0.012 ± 0.003	-0.119 ± 0.029	0.007 ± 0.000	0.086 ± 0.003	0.007 ± 0.000	0.087 ± 0.003	0.735 ± 0.048	0.857 ± 0.032	0.090	-0.011	-11.36	0.0735	0.0744	1.075	93.998 ± 0.799	0.362 ± 0.016	33.749 ± 1.591

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S28

Simulation 1 - Condition 28: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.324	0.008 ± 0.002	0.027 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.180 ± 0.005	0.324	0.008	2.66	0.0624	0.0629	1.011	95.349 ± 0.685	0.224 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.322	0.006 ± 0.002	0.019 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.320	0.004	1.32	0.0564	0.0565	1.028	95.455 ± 0.677	0.224 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.116	-0.023 ± 0.002	-0.169 ± 0.014	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.211 ± 0.010	0.459 ± 0.013	0.116	-0.023	-16.55	0.0601	0.0644	0.966	90.486 ± 0.954	0.225 ± 0.001	53.066 ± 1.623
LMS	beta34	0.101	0.143	0.042 ± 0.001	0.418 ± 0.011	0.001 ± 0.000	0.033 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.279 ± 0.010	0.528 ± 0.010	0.142	0.041	40.85	0.0318	0.0521	0.983	74.101 ± 1.424	0.126 ± 0.001	97.992 ± 0.456
LMS	beta35	0.101	0.144	0.043 ± 0.001	0.426 ± 0.011	0.001 ± 0.000	0.033 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.286 ± 0.011	0.535 ± 0.010	0.145	0.044	43.08	0.0318	0.0539	0.988	71.353 ± 1.470	0.127 ± 0.001	97.886 ± 0.468
LSAM	beta31	0.316	0.327	0.011 ± 0.003	0.034 ± 0.010	0.010 ± 0.000	0.101 ± 0.002	0.010 ± 0.000	0.101 ± 0.003	0.102 ± 0.005	0.320 ± 0.009	0.328	0.012	3.93	0.1068	0.1075	1.018	95.666 ± 0.662	0.401 ± 0.002	88.372 ± 1.042
LSAM	beta32	0.316	0.320	0.004 ± 0.003	0.011 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.004	0.306 ± 0.008	0.317	0.001	0.45	0.1009	0.1009	1.060	95.772 ± 0.654	0.402 ± 0.003	88.055 ± 1.054
LSAM	beta33	0.139	0.162	0.023 ± 0.002	0.164 ± 0.014	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.200 ± 0.010	0.447 ± 0.012	0.161	0.022	15.66	0.0569	0.0609	0.960	92.072 ± 0.878	0.217 ± 0.001	81.607 ± 1.260
LSAM	beta34	0.101	0.089	-0.012 ± 0.001	-0.115 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.145 ± 0.007	0.381 ± 0.011	0.089	-0.012	-11.52	0.0355	0.0374	0.975	92.283 ± 0.868	0.140 ± 0.002	70.085 ± 1.489
LSAM	beta35	0.101	0.092	-0.009 ± 0.001	-0.094 ± 0.012	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.138 ± 0.007	0.372 ± 0.011	0.092	-0.009	-8.95	0.0332	0.0344	0.997	92.706 ± 0.845	0.142 ± 0.002	70.296 ± 1.486
QML	beta31	0.316	0.325	0.009 ± 0.002	0.028 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.325	0.009	2.87	0.0624	0.0631	1.010	95.243 ± 0.692	0.224 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.322	0.006 ± 0.002	0.019 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.320	0.004	1.22	0.0561	0.0562	1.026	95.349 ± 0.685	0.224 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.116	-0.023 ± 0.002	-0.168 ± 0.014	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.208 ± 0.010	0.456 ± 0.013	0.117	-0.022	-16.13	0.0590	0.0631	0.966	90.169 ± 0.968	0.223 ± 0.001	53.594 ± 1.621
QML	beta34	0.101	0.143	0.042 ± 0.001	0.414 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.275 ± 0.010	0.524 ± 0.010	0.142	0.041	40.51	0.0311	0.0514	0.977	73.679 ± 1.432	0.125 ± 0.001	98.203 ± 0.432
QML	beta35	0.101	0.144	0.043 ± 0.001	0.423 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.283 ± 0.010	0.532 ± 0.010	0.144	0.043	42.52	0.0307	0.0528	0.988	71.459 ± 1.468	0.126 ± 0.001	98.203 ± 0.432
UPI	beta31	0.316	0.323	0.007 ± 0.004	0.022 ± 0.012	0.014 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.116 ± 0.003	0.136 ± 0.007	0.368 ± 0.011	0.323	0.007	2.23	0.1199	0.1201	1.030	96.195 ± 0.622	0.470 ± 0.006	78.541 ± 1.335
UPI	beta32	0.316	0.314	-0.002 ± 0.004	-0.007 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.127 ± 0.006	0.357 ± 0.010	0.310	-0.006	-1.92	0.1131	0.1132	1.077	96.512 ± 0.597	0.476 ± 0.008	77.061 ± 1.367
UPI	beta33	0.139	0.150	0.011 ± 0.002	0.076 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.195 ± 0.010	0.441 ± 0.013	0.149	0.010	6.89	0.0587	0.0595	0.974	93.869 ± 0.780	0.231 ± 0.002	74.736 ± 1.413
UPI	beta34	0.101	0.092	-0.009 ± 0.001	-0.091 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.178 ± 0.011	0.422 ± 0.014	0.092	-0.009	-8.74	0.0374	0.0384	0.986	92.918 ± 0.834	0.161 ± 0.002	65.433 ± 1.546
UPI	beta35	0.101	0.094	-0.007 ± 0.001	-0.068 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.174 ± 0.010	0.417 ± 0.014	0.093	-0.008	-7.90	0.0388	0.0396	1.029	94.503 ± 0.741	0.168 ± 0.004	65.856 ± 1.542

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S29

Simulation 1 - Condition 29: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.003	0.002 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.253 ± 0.007	0.316	0.000	0.05	0.0755	0.0755	1.010	95.000 ± 0.689	0.316 ± 0.001	97.000 ± 0.539
LMS	beta32	0.316	0.318	0.002 ± 0.003	0.005 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.065 ± 0.003	0.255 ± 0.007	0.318	0.002	0.58	0.0798	0.0798	1.000	94.900 ± 0.696	0.315 ± 0.001	96.900 ± 0.548
LMS	beta33	0.139	0.122	-0.017 ± 0.002	-0.126 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.254 ± 0.013	0.504 ± 0.015	0.123	-0.016	-11.79	0.0654	0.0674	0.924	92.100 ± 0.853	0.246 ± 0.002	52.900 ± 1.578
LMS	beta34	0.101	0.121	0.020 ± 0.001	0.198 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.195 ± 0.009	0.442 ± 0.012	0.121	0.020	20.07	0.0366	0.0418	0.888	86.300 ± 1.087	0.139 ± 0.001	88.000 ± 1.028
LMS	beta35	0.101	0.120	0.019 ± 0.001	0.189 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.184 ± 0.009	0.429 ± 0.012	0.120	0.019	18.65	0.0376	0.0420	0.883	88.100 ± 1.024	0.135 ± 0.001	88.400 ± 1.013
LSAM	beta31	0.316	0.315	-0.001 ± 0.003	-0.002 ± 0.010	0.011 ± 0.000	0.104 ± 0.002	0.011 ± 0.000	0.104 ± 0.003	0.108 ± 0.005	0.329 ± 0.009	0.313	-0.003	-0.95	0.0990	0.0991	0.983	94.400 ± 0.727	0.401 ± 0.002	85.700 ± 1.107
LSAM	beta32	0.316	0.317	0.001 ± 0.003	0.004 ± 0.010	0.011 ± 0.000	0.104 ± 0.002	0.011 ± 0.000	0.104 ± 0.003	0.108 ± 0.005	0.329 ± 0.009	0.314	-0.002	-0.52	0.1041	0.1041	0.977	94.200 ± 0.739	0.399 ± 0.002	86.100 ± 1.094
LSAM	beta33	0.139	0.145	0.006 ± 0.002	0.040 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.251 ± 0.013	0.501 ± 0.015	0.142	0.003	2.11	0.0668	0.0669	0.900	92.100 ± 0.853	0.245 ± 0.002	64.300 ± 1.515
LSAM	beta34	0.101	0.101	0.000 ± 0.001	0.003 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.185 ± 0.010	0.430 ± 0.014	0.103	0.002	1.62	0.0384	0.0385	0.863	90.800 ± 0.914	0.147 ± 0.002	74.900 ± 1.371
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.006 ± 0.013	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.181 ± 0.010	0.425 ± 0.013	0.100	-0.001	-0.58	0.0395	0.0395	0.845	90.000 ± 0.949	0.142 ± 0.002	76.500 ± 1.341
QML	beta31	0.316	0.317	0.001 ± 0.003	0.002 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.253 ± 0.007	0.317	0.001	0.23	0.0763	0.0763	1.008	94.900 ± 0.696	0.316 ± 0.002	96.900 ± 0.548
QML	beta32	0.316	0.317	0.001 ± 0.003	0.005 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.065 ± 0.003	0.255 ± 0.007	0.318	0.002	0.67	0.0802	0.0802	0.994	94.700 ± 0.708	0.314 ± 0.001	97.000 ± 0.539
QML	beta33	0.139	0.121	-0.018 ± 0.002	-0.126 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.255 ± 0.013	0.505 ± 0.015	0.124	-0.015	-11.15	0.0662	0.0679	0.920	91.700 ± 0.872	0.245 ± 0.002	52.900 ± 1.578
QML	beta34	0.101	0.121	0.020 ± 0.001	0.195 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.194 ± 0.009	0.441 ± 0.012	0.121	0.020	19.60	0.0364	0.0414	0.877	85.800 ± 1.104	0.137 ± 0.001	88.200 ± 1.020
QML	beta35	0.101	0.120	0.019 ± 0.001	0.188 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.184 ± 0.009	0.429 ± 0.012	0.120	0.019	18.34	0.0375	0.0419	0.878	87.700 ± 1.039	0.134 ± 0.001	88.400 ± 1.013
UPI	beta31	0.316	0.319	0.003 ± 0.003	0.010 ± 0.010	0.011 ± 0.000	0.103 ± 0.002	0.011 ± 0.000	0.103 ± 0.003	0.105 ± 0.005	0.325 ± 0.009	0.317	0.001	0.40	0.0973	0.0974	0.976	93.600 ± 0.774	0.392 ± 0.002	87.800 ± 1.035
UPI	beta32	0.316	0.322	0.006 ± 0.003	0.020 ± 0.010	0.011 ± 0.000	0.103 ± 0.002	0.011 ± 0.000	0.103 ± 0.003	0.106 ± 0.005	0.326 ± 0.009	0.321	0.005	1.60	0.1019	0.1021	0.968	93.600 ± 0.774	0.390 ± 0.002	87.900 ± 1.031
UPI	beta33	0.139	0.140	0.001 ± 0.002	0.007 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.255 ± 0.013	0.505 ± 0.015	0.138	-0.001	-0.79	0.0663	0.0663	0.903	92.400 ± 0.838	0.248 ± 0.002	62.100 ± 1.534
UPI	beta34	0.101	0.102	0.001 ± 0.001	0.006 ± 0.013	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.182 ± 0.010	0.426 ± 0.013	0.102	0.001	1.14	0.0381	0.0381	0.859	90.400 ± 0.932	0.145 ± 0.002	75.700 ± 1.356
UPI	beta35	0.101	0.100	-0.001 ± 0.001	-0.011 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.175 ± 0.010	0.418 ± 0.013	0.101	0.000	-0.23	0.0377	0.0377	0.852	90.300 ± 0.936	0.141 ± 0.002	75.500 ± 1.360

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S30

Simulation 1 - Condition 30: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.005	0.315	-0.001	-0.29	0.0486	0.0486	1.011	95.100 ± 0.683	0.196 ± 0.000	99.900 ± 0.100
LMS	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.159 ± 0.004	0.316	0.000	-0.03	0.0510	0.0510	0.994	95.700 ± 0.641	0.196 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.125	-0.014 ± 0.001	-0.102 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.005	0.310 ± 0.009	0.123	-0.016	-11.32	0.0402	0.0432	0.951	92.900 ± 0.812	0.152 ± 0.001	87.600 ± 1.042
LMS	beta34	0.101	0.120	0.019 ± 0.001	0.187 ± 0.007	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.084 ± 0.004	0.289 ± 0.007	0.119	0.018	17.95	0.0215	0.0281	0.950	84.100 ± 1.156	0.083 ± 0.001	99.700 ± 0.173
LMS	beta35	0.101	0.119	0.018 ± 0.001	0.182 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.029 ± 0.001	0.080 ± 0.003	0.283 ± 0.006	0.119	0.018	18.26	0.0214	0.0282	0.952	82.900 ± 1.191	0.082 ± 0.001	99.800 ± 0.141
LSAM	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.006	0.315	-0.001	-0.29	0.0590	0.0590	1.006	94.800 ± 0.702	0.242 ± 0.001	99.700 ± 0.173
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.005	0.316	0.000	-0.08	0.0626	0.0626	0.997	95.600 ± 0.649	0.242 ± 0.001	99.900 ± 0.100
LSAM	beta33	0.139	0.143	0.004 ± 0.001	0.032 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.086 ± 0.004	0.294 ± 0.009	0.142	0.003	2.32	0.0395	0.0396	0.927	93.400 ± 0.785	0.148 ± 0.001	95.300 ± 0.669
LSAM	beta34	0.101	0.100	-0.001 ± 0.001	-0.010 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.052 ± 0.002	0.227 ± 0.007	0.100	-0.001	-1.38	0.0223	0.0223	0.926	92.900 ± 0.812	0.083 ± 0.001	97.700 ± 0.474
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.012 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.100	-0.001	-0.84	0.0211	0.0211	0.928	92.200 ± 0.848	0.083 ± 0.001	98.200 ± 0.420
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.005	0.316	0.000	-0.10	0.0487	0.0487	1.011	95.000 ± 0.689	0.196 ± 0.000	99.900 ± 0.100
QML	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.316	0.000	-0.02	0.0512	0.0512	0.996	95.700 ± 0.641	0.196 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.125	-0.014 ± 0.001	-0.102 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.005	0.310 ± 0.009	0.124	-0.015	-11.15	0.0405	0.0433	0.947	92.600 ± 0.828	0.151 ± 0.001	87.500 ± 1.046
QML	beta34	0.101	0.119	0.018 ± 0.001	0.183 ± 0.007	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.082 ± 0.004	0.285 ± 0.007	0.119	0.018	17.41	0.0211	0.0274	0.943	84.200 ± 1.153	0.082 ± 0.001	99.700 ± 0.173
QML	beta35	0.101	0.119	0.018 ± 0.001	0.182 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.029 ± 0.001	0.080 ± 0.003	0.283 ± 0.006	0.119	0.018	18.11	0.0212	0.0280	0.947	82.500 ± 1.202	0.081 ± 0.001	99.800 ± 0.141
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.006	0.317	0.001	0.40	0.0596	0.0596	0.995	94.500 ± 0.721	0.238 ± 0.001	99.700 ± 0.173
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.317	0.001	0.40	0.0620	0.0620	0.990	95.600 ± 0.649	0.237 ± 0.001	99.900 ± 0.100
UPI	beta33	0.139	0.140	0.001 ± 0.001	0.009 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.083 ± 0.004	0.287 ± 0.008	0.139	0.000	0.28	0.0396	0.0396	0.936	93.400 ± 0.785	0.147 ± 0.001	94.900 ± 0.696
UPI	beta34	0.101	0.101	-0.000 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.100	-0.001	-0.73	0.0224	0.0225	0.923	92.900 ± 0.812	0.083 ± 0.001	98.100 ± 0.432
UPI	beta35	0.101	0.100	-0.001 ± 0.001	-0.006 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.101	0.000	-0.29	0.0217	0.0217	0.928	92.700 ± 0.823	0.082 ± 0.001	98.200 ± 0.420

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S31

Simulation 1 - Condition 31: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.012	-0.004 ± 0.038	0.009 ± 0.002	0.093 ± 0.011	0.009 ± 0.002	0.093 ± 0.012	0.086 ± 0.019	0.293 ± 0.038	0.320	0.004	1.29	0.1012	0.1013	1.140	91.667 ± 3.568	0.417 ± 0.043	93.333 ± 3.220
LMS	beta32	0.316	0.314	-0.002 ± 0.015	-0.006 ± 0.048	0.014 ± 0.004	0.117 ± 0.019	0.013 ± 0.004	0.116 ± 0.020	0.135 ± 0.039	0.367 ± 0.062	0.312	-0.004	-1.11	0.1201	0.1202	0.850	90.000 ± 3.873	0.390 ± 0.018	93.333 ± 3.220
LMS	beta33	0.139	0.221	0.082 ± 0.017	0.592 ± 0.119	0.016 ± 0.003	0.128 ± 0.013	0.023 ± 0.004	0.151 ± 0.016	1.181 ± 0.221	1.087 ± 0.116	0.209	0.070	50.18	0.1595	0.1741	1.245	88.333 ± 4.144	0.624 ± 0.132	56.667 ± 6.397
LMS	beta34	0.101	-0.035	-0.136 ± 0.010	-1.351 ± 0.100	0.006 ± 0.001	0.078 ± 0.009	0.025 ± 0.003	0.157 ± 0.009	2.414 ± 0.254	1.554 ± 0.086	-0.043	-0.144	-142.29	0.1221	0.1886	1.994	46.667 ± 6.441	0.611 ± 0.222	5.000 ± 2.814
LMS	beta35	0.101	-0.034	-0.135 ± 0.012	-1.335 ± 0.119	0.009 ± 0.003	0.093 ± 0.018	0.027 ± 0.005	0.164 ± 0.018	2.622 ± 0.537	1.619 ± 0.177	-0.032	-0.133	-132.18	0.0942	0.1634	0.913	51.667 ± 6.451	0.334 ± 0.030	8.333 ± 3.568
LSAM	beta31	0.316	0.349	0.033 ± 0.025	0.104 ± 0.078	0.037 ± 0.007	0.191 ± 0.020	0.037 ± 0.008	0.192 ± 0.024	0.371 ± 0.077	0.609 ± 0.077	0.322	0.006	1.76	0.2036	0.2037	1.173	96.667 ± 2.317	0.879 ± 0.095	51.667 ± 6.451
LSAM	beta32	0.316	0.341	0.025 ± 0.022	0.080 ± 0.069	0.029 ± 0.006	0.169 ± 0.019	0.029 ± 0.006	0.169 ± 0.022	0.287 ± 0.064	0.536 ± 0.071	0.329	0.013	4.10	0.1621	0.1626	1.302	98.333 ± 1.653	0.862 ± 0.083	45.000 ± 6.423
LSAM	beta33	0.139	0.317	0.178 ± 0.026	1.280 ± 0.184	0.039 ± 0.010	0.198 ± 0.026	0.070 ± 0.016	0.265 ± 0.033	3.630 ± 0.827	1.905 ± 0.236	0.254	0.115	82.82	0.1543	0.1925	1.061	96.667 ± 2.317	0.822 ± 0.096	41.667 ± 6.365
LSAM	beta34	0.101	-0.089	-0.190 ± 0.020	-1.877 ± 0.199	0.024 ± 0.005	0.156 ± 0.018	0.060 ± 0.009	0.245 ± 0.020	5.867 ± 0.896	2.422 ± 0.199	-0.065	-0.166	-164.18	0.1752	0.2412	1.294	70.000 ± 5.916	0.791 ± 0.092	0.000 ± 0.000
LSAM	beta35	0.101	-0.065	-0.166 ± 0.022	-1.641 ± 0.216	0.028 ± 0.007	0.169 ± 0.024	0.055 ± 0.009	0.235 ± 0.021	5.436 ± 0.874	2.331 ± 0.206	-0.063	-0.164	-162.37	0.1449	0.2189	1.281	80.000 ± 5.164	0.847 ± 0.143	0.000 ± 0.000
QML	beta31	0.316	0.304	-0.012 ± 0.012	-0.037 ± 0.037	0.008 ± 0.002	0.091 ± 0.011	0.008 ± 0.002	0.091 ± 0.012	0.084 ± 0.019	0.289 ± 0.039	0.313	-0.003	-1.00	0.0873	0.0874	1.005	91.667 ± 3.568	0.360 ± 0.007	93.333 ± 3.220
QML	beta32	0.316	0.299	-0.017 ± 0.012	-0.053 ± 0.036	0.008 ± 0.001	0.089 ± 0.009	0.008 ± 0.001	0.090 ± 0.010	0.081 ± 0.014	0.285 ± 0.032	0.304	-0.012	-3.78	0.0884	0.0892	1.043	90.000 ± 3.873	0.364 ± 0.008	93.333 ± 3.220
QML	beta33	0.139	0.223	0.084 ± 0.017	0.604 ± 0.122	0.017 ± 0.003	0.131 ± 0.013	0.024 ± 0.004	0.155 ± 0.016	1.237 ± 0.221	1.112 ± 0.114	0.208	0.069	49.38	0.1264	0.1439	0.885	90.000 ± 3.873	0.454 ± 0.016	58.333 ± 6.365
QML	beta34	0.101	-0.039	-0.140 ± 0.010	-1.388 ± 0.100	0.006 ± 0.001	0.078 ± 0.008	0.026 ± 0.003	0.160 ± 0.009	2.513 ± 0.272	1.585 ± 0.090	-0.036	-0.137	-136.07	0.0891	0.1638	1.021	50.000 ± 6.455	0.313 ± 0.014	6.667 ± 3.220
QML	beta35	0.101	-0.031	-0.132 ± 0.009	-1.306 ± 0.088	0.005 ± 0.001	0.069 ± 0.005	0.022 ± 0.002	0.149 ± 0.009	2.164 ± 0.244	1.471 ± 0.086	-0.039	-0.140	-138.49	0.0732	0.1579	1.151	50.000 ± 6.455	0.311 ± 0.014	10.000 ± 3.873
UPI	beta31	0.316	0.268	-0.048 ± 0.030	-0.153 ± 0.094	0.053 ± 0.011	0.230 ± 0.025	0.054 ± 0.011	0.233 ± 0.028	0.543 ± 0.106	0.737 ± 0.088	0.336	0.020	6.46	0.3278	0.3284	1.677	86.667 ± 4.389	1.510 ± 0.231	20.000 ± 5.164
UPI	beta32	0.316	0.387	0.071 ± 0.044	0.225 ± 0.141	0.119 ± 0.024	0.344 ± 0.037	0.122 ± 0.027	0.349 ± 0.046	1.218 ± 0.272	1.104 ± 0.146	0.288	-0.028	-8.84	0.4373	0.4382	1.145	88.333 ± 4.144	1.546 ± 0.199	26.667 ± 5.709
UPI	beta33	0.139	0.232	0.093 ± 0.030	0.668 ± 0.214	0.053 ± 0.014	0.231 ± 0.035	0.061 ± 0.018	0.247 ± 0.042	3.151 ± 0.954	1.775 ± 0.303	0.201	0.062	44.28	0.2634	0.2705	1.322	96.667 ± 2.317	1.194 ± 0.165	1.667 ± 1.653
UPI	beta34	0.101	-0.172	-0.273 ± 0.073	-2.699 ± 0.719	0.316 ± 0.145	0.562 ± 0.153	0.385 ± 0.175	0.621 ± 0.164	37.780 ± 17.141	6.147 ± 1.619	-0.145	-0.246	-243.37	0.5718	0.6224	1.254	81.667 ± 4.995	2.765 ± 0.574	0.000 ± 0.000
UPI	beta35	0.101	-0.265	-0.366 ± 0.066	-3.624 ± 0.651	0.260 ± 0.076	0.509 ± 0.082	0.389 ± 0.117	0.624 ± 0.102	38.152 ± 11.430	6.177 ± 1.009	-0.123	-0.224	-221.81	0.5567	0.6001	1.392	83.333 ± 4.811	2.780 ± 0.702	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S32

Simulation 1 - Condition 32: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.298	-0.018 ± 0.007	-0.056 ± 0.023	0.003 ± 0.000	0.051 ± 0.005	0.003 ± 0.000	0.053 ± 0.006	0.028 ± 0.005	0.168 ± 0.019	0.303	-0.013	-4.22	0.0611	0.0626	1.315	96.000 ± 2.771	0.261 ± 0.033	98.000 ± 1.980
LMS	beta32	0.316	0.316	-0.000 ± 0.009	-0.002 ± 0.028	0.004 ± 0.001	0.062 ± 0.008	0.004 ± 0.001	0.061 ± 0.009	0.038 ± 0.009	0.195 ± 0.029	0.297	-0.019	-5.88	0.0722	0.0746	0.974	98.000 ± 1.980	0.237 ± 0.007	100.000 ± 0.000
LMS	beta33	0.139	0.211	0.072 ± 0.010	0.518 ± 0.073	0.005 ± 0.001	0.072 ± 0.007	0.010 ± 0.002	0.101 ± 0.010	0.530 ± 0.094	0.728 ± 0.071	0.207	0.068	49.20	0.0859	0.1098	1.184	90.000 ± 4.243	0.334 ± 0.041	80.000 ± 5.657
LMS	beta34	0.101	-0.018	-0.119 ± 0.009	-1.179 ± 0.089	0.004 ± 0.001	0.064 ± 0.009	0.018 ± 0.003	0.135 ± 0.010	1.781 ± 0.254	1.335 ± 0.100	-0.025	-0.126	-125.18	0.0735	0.1463	1.472	44.000 ± 7.020	0.368 ± 0.156	4.000 ± 2.771
LMS	beta35	0.101	-0.011	-0.112 ± 0.007	-1.106 ± 0.071	0.003 ± 0.001	0.050 ± 0.006	0.015 ± 0.001	0.122 ± 0.006	1.468 ± 0.146	1.212 ± 0.062	-0.016	-0.117	-116.21	0.0554	0.1298	1.083	40.000 ± 6.928	0.214 ± 0.008	4.000 ± 2.771
LSAM	beta31	0.316	0.330	0.014 ± 0.017	0.044 ± 0.054	0.015 ± 0.004	0.121 ± 0.016	0.014 ± 0.004	0.120 ± 0.019	0.145 ± 0.038	0.381 ± 0.060	0.338	0.022	6.87	0.1237	0.1256	0.962	100.000 ± 0.000	0.456 ± 0.026	86.000 ± 4.907
LSAM	beta32	0.316	0.329	0.013 ± 0.018	0.041 ± 0.058	0.017 ± 0.005	0.130 ± 0.019	0.017 ± 0.004	0.129 ± 0.020	0.167 ± 0.044	0.409 ± 0.064	0.312	-0.004	-1.20	0.1041	0.1042	0.923	96.000 ± 2.771	0.470 ± 0.026	86.000 ± 4.907
LSAM	beta33	0.139	0.257	0.118 ± 0.013	0.846 ± 0.096	0.009 ± 0.003	0.094 ± 0.015	0.023 ± 0.005	0.150 ± 0.017	1.166 ± 0.250	1.080 ± 0.124	0.265	0.126	90.74	0.1019	0.1622	1.025	90.000 ± 4.243	0.378 ± 0.015	84.000 ± 5.185
LSAM	beta34	0.101	-0.044	-0.145 ± 0.017	-1.437 ± 0.167	0.014 ± 0.004	0.119 ± 0.018	0.035 ± 0.007	0.187 ± 0.020	3.429 ± 0.692	1.852 ± 0.201	-0.050	-0.151	-149.75	0.0954	0.1788	0.978	64.000 ± 6.788	0.457 ± 0.031	2.000 ± 1.980
LSAM	beta35	0.101	-0.034	-0.135 ± 0.018	-1.336 ± 0.182	0.017 ± 0.004	0.130 ± 0.019	0.035 ± 0.005	0.186 ± 0.015	3.408 ± 0.484	1.846 ± 0.148	-0.052	-0.153	-151.77	0.1072	0.1871	0.957	70.000 ± 6.481	0.488 ± 0.042	4.000 ± 2.771
QML	beta31	0.316	0.300	-0.016 ± 0.007	-0.052 ± 0.023	0.003 ± 0.000	0.051 ± 0.005	0.003 ± 0.001	0.053 ± 0.006	0.028 ± 0.005	0.168 ± 0.019	0.301	-0.015	-4.83	0.0579	0.0599	1.195	96.000 ± 2.771	0.239 ± 0.010	100.000 ± 0.000
QML	beta32	0.316	0.309	-0.007 ± 0.008	-0.022 ± 0.024	0.003 ± 0.001	0.054 ± 0.007	0.003 ± 0.001	0.054 ± 0.008	0.029 ± 0.007	0.172 ± 0.024	0.294	-0.022	-7.01	0.0686	0.0721	1.319	98.000 ± 1.980	0.281 ± 0.049	98.000 ± 1.980
QML	beta33	0.139	0.213	0.074 ± 0.011	0.531 ± 0.077	0.006 ± 0.001	0.075 ± 0.008	0.011 ± 0.002	0.105 ± 0.011	0.570 ± 0.104	0.755 ± 0.076	0.217	0.078	56.44	0.0838	0.1148	1.891	92.000 ± 3.837	0.559 ± 0.254	76.000 ± 6.040
QML	beta34	0.101	-0.020	-0.121 ± 0.008	-1.203 ± 0.081	0.003 ± 0.001	0.058 ± 0.005	0.018 ± 0.002	0.134 ± 0.008	1.771 ± 0.209	1.331 ± 0.081	-0.029	-0.130	-129.11	0.0705	0.1482	2.115	44.000 ± 7.020	0.482 ± 0.266	10.000 ± 4.243
QML	beta35	0.101	-0.019	-0.120 ± 0.007	-1.188 ± 0.069	0.002 ± 0.001	0.049 ± 0.006	0.017 ± 0.002	0.129 ± 0.007	1.642 ± 0.161	1.281 ± 0.065	-0.029	-0.130	-128.83	0.0559	0.1416	2.641	40.000 ± 6.928	0.507 ± 0.293	6.000 ± 3.359
UPI	beta31	0.316	0.355	0.039 ± 0.085	0.125 ± 0.269	0.362 ± 0.126	0.602 ± 0.119	0.357 ± 0.131	0.597 ± 0.128	3.572 ± 1.312	1.890 ± 0.404	0.449	0.133	41.95	0.7560	0.7676	1.267	86.000 ± 4.907	2.990 ± 0.597	14.000 ± 4.907
UPI	beta32	0.316	0.415	0.099 ± 0.090	0.314 ± 0.285	0.405 ± 0.122	0.637 ± 0.105	0.407 ± 0.135	0.638 ± 0.121	4.076 ± 1.349	2.019 ± 0.383	0.307	-0.009	-2.72	0.6962	0.6963	1.185	94.000 ± 3.359	2.957 ± 0.721	22.000 ± 5.858
UPI	beta33	0.139	0.159	0.020 ± 0.054	0.144 ± 0.392	0.148 ± 0.054	0.385 ± 0.078	0.146 ± 0.054	0.382 ± 0.081	7.536 ± 2.802	2.745 ± 0.582	0.170	0.031	22.52	0.3736	0.3750	1.386	98.000 ± 1.980	2.091 ± 0.379	6.000 ± 3.359
UPI	beta34	0.101	-0.678	-0.779 ± 0.261	-7.711 ± 2.585	3.407 ± 1.138	1.846 ± 0.342	3.946 ± 1.484	1.986 ± 0.417	386.817 ± 145.514	19.668 ± 4.132	-0.290	-0.391	-387.30	1.1281	1.1940	0.930	90.000 ± 4.243	6.727 ± 1.415	2.000 ± 1.980
UPI	beta35	0.101	-0.583	-0.684 ± 0.236	-6.768 ± 2.336	2.784 ± 1.035	1.668 ± 0.352	3.195 ± 1.273	1.787 ± 0.404	313.217 ± 124.768	17.698 ± 4.005	-0.390	-0.491	-486.42	1.3315	1.4193	0.938	94.000 ± 3.359	6.133 ± 1.147	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S33

Simulation 1 - Condition 33: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.306	-0.010 ± 0.003	-0.032 ± 0.008	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.003	0.223 ± 0.008	0.303	-0.013	-4.03	0.0671	0.0683	1.018	94.286 ± 0.877	0.279 ± 0.003	99.143 ± 0.348
LMS	beta32	0.316	0.310	-0.006 ± 0.003	-0.019 ± 0.008	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.044 ± 0.002	0.211 ± 0.007	0.308	-0.008	-2.57	0.0668	0.0673	1.066	96.714 ± 0.674	0.278 ± 0.003	99.429 ± 0.285
LMS	beta33	0.139	0.172	0.033 ± 0.003	0.238 ± 0.023	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.420 ± 0.023	0.648 ± 0.021	0.172	0.033	23.50	0.0780	0.0845	1.046	93.857 ± 0.908	0.344 ± 0.006	54.571 ± 1.882
LMS	beta34	0.101	0.030	-0.071 ± 0.003	-0.701 ± 0.026	0.005 ± 0.000	0.071 ± 0.002	0.010 ± 0.000	0.100 ± 0.002	0.980 ± 0.044	0.990 ± 0.024	0.028	-0.073	-72.01	0.0726	0.1028	1.007	76.714 ± 1.597	0.279 ± 0.003	7.000 ± 0.964
LMS	beta35	0.101	0.029	-0.072 ± 0.003	-0.712 ± 0.026	0.005 ± 0.000	0.068 ± 0.002	0.010 ± 0.000	0.099 ± 0.002	0.966 ± 0.042	0.983 ± 0.023	0.027	-0.074	-73.10	0.0708	0.1023	1.042	77.286 ± 1.584	0.280 ± 0.008	6.714 ± 0.946
LSAM	beta31	0.316	0.312	-0.004 ± 0.003	-0.011 ± 0.010	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.064 ± 0.004	0.253 ± 0.009	0.308	-0.008	-2.37	0.0771	0.0774	1.014	95.571 ± 0.778	0.317 ± 0.002	97.143 ± 0.630
LSAM	beta32	0.316	0.316	-0.000 ± 0.003	-0.001 ± 0.009	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.059 ± 0.003	0.243 ± 0.008	0.315	-0.001	-0.36	0.0767	0.0767	1.050	97.857 ± 0.547	0.316 ± 0.002	97.857 ± 0.547
LSAM	beta33	0.139	0.180	0.041 ± 0.003	0.298 ± 0.025	0.008 ± 0.000	0.090 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	0.512 ± 0.029	0.715 ± 0.023	0.181	0.042	30.51	0.0817	0.0920	1.027	93.571 ± 0.927	0.364 ± 0.003	52.571 ± 1.887
LSAM	beta34	0.101	0.043	-0.058 ± 0.004	-0.573 ± 0.042	0.012 ± 0.001	0.112 ± 0.004	0.016 ± 0.001	0.126 ± 0.004	1.545 ± 0.089	1.243 ± 0.040	0.042	-0.059	-58.90	0.1056	0.1212	0.997	86.429 ± 1.294	0.436 ± 0.005	5.714 ± 0.877
LSAM	beta35	0.101	0.044	-0.057 ± 0.004	-0.565 ± 0.040	0.012 ± 0.001	0.108 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	1.453 ± 0.072	1.205 ± 0.035	0.037	-0.064	-62.96	0.1012	0.1195	1.016	89.714 ± 1.148	0.428 ± 0.005	5.000 ± 0.824
QML	beta31	0.316	0.306	-0.010 ± 0.003	-0.033 ± 0.008	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.003	0.224 ± 0.008	0.303	-0.013	-4.12	0.0667	0.0680	1.002	94.286 ± 0.877	0.275 ± 0.001	99.429 ± 0.285
QML	beta32	0.316	0.310	-0.006 ± 0.003	-0.019 ± 0.008	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.211 ± 0.007	0.308	-0.008	-2.60	0.0662	0.0667	1.050	96.714 ± 0.674	0.274 ± 0.001	99.714 ± 0.202
QML	beta33	0.139	0.173	0.034 ± 0.003	0.245 ± 0.023	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.425 ± 0.024	0.652 ± 0.021	0.172	0.033	24.05	0.0782	0.0851	1.028	94.429 ± 0.867	0.339 ± 0.003	55.143 ± 1.880
QML	beta34	0.101	0.028	-0.073 ± 0.003	-0.719 ± 0.026	0.005 ± 0.000	0.070 ± 0.002	0.010 ± 0.000	0.101 ± 0.002	1.000 ± 0.045	1.000 ± 0.024	0.027	-0.074	-73.19	0.0699	0.1017	1.006	75.857 ± 1.617	0.277 ± 0.002	6.857 ± 0.955
QML	beta35	0.101	0.027	-0.074 ± 0.003	-0.730 ± 0.026	0.005 ± 0.000	0.068 ± 0.002	0.010 ± 0.000	0.100 ± 0.002	0.990 ± 0.043	0.995 ± 0.023	0.025	-0.076	-74.93	0.0697	0.1029	1.014	77.571 ± 1.577	0.272 ± 0.002	6.857 ± 0.955
UPI	beta31	0.316	0.324	0.008 ± 0.004	0.025 ± 0.011	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.093 ± 0.007	0.305 ± 0.012	0.322	0.006	1.87	0.0897	0.0899	1.581	97.429 ± 0.598	0.595 ± 0.033	79.286 ± 1.532
UPI	beta32	0.316	0.326	0.010 ± 0.004	0.031 ± 0.012	0.010 ± 0.001	0.100 ± 0.004	0.010 ± 0.001	0.100 ± 0.004	0.100 ± 0.007	0.317 ± 0.013	0.324	0.008	2.63	0.0940	0.0944	1.616	99.143 ± 0.348	0.631 ± 0.049	79.714 ± 1.520
UPI	beta33	0.139	0.190	0.051 ± 0.004	0.369 ± 0.027	0.010 ± 0.001	0.100 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	0.657 ± 0.042	0.810 ± 0.029	0.183	0.044	31.66	0.0921	0.1020	1.300	97.857 ± 0.547	0.511 ± 0.017	38.286 ± 1.837
UPI	beta34	0.101	0.056	-0.045 ± 0.007	-0.442 ± 0.072	0.037 ± 0.003	0.191 ± 0.007	0.039 ± 0.003	0.196 ± 0.008	3.778 ± 0.266	1.944 ± 0.077	0.054	-0.047	-46.57	0.1667	0.1732	1.321	94.286 ± 0.877	0.991 ± 0.049	4.714 ± 0.801
UPI	beta35	0.101	0.057	-0.044 ± 0.007	-0.432 ± 0.068	0.033 ± 0.002	0.181 ± 0.006	0.034 ± 0.002	0.186 ± 0.007	3.377 ± 0.240	1.838 ± 0.073	0.047	-0.054	-53.42	0.1569	0.1659	1.383	93.857 ± 0.908	0.978 ± 0.063	4.286 ± 0.766

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S34

Simulation 1 - Condition 34: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.305	-0.011 ± 0.002	-0.036 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.304	-0.012	-3.79	0.0446	0.0461	0.977	92.931 ± 0.919	0.172 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.308	-0.008 ± 0.002	-0.024 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	0.020 ± 0.001	0.142 ± 0.005	0.310	-0.006	-2.05	0.0413	0.0418	0.996	93.059 ± 0.911	0.172 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.172	0.033 ± 0.002	0.235 ± 0.013	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.195 ± 0.009	0.442 ± 0.012	0.172	0.033	23.82	0.0516	0.0613	1.029	91.131 ± 1.019	0.210 ± 0.001	88.689 ± 1.136
LMS	beta34	0.101	0.032	-0.069 ± 0.002	-0.683 ± 0.015	0.002 ± 0.000	0.043 ± 0.001	0.007 ± 0.000	0.081 ± 0.001	0.647 ± 0.023	0.804 ± 0.015	0.032	-0.069	-68.80	0.0435	0.0820	1.023	64.524 ± 1.715	0.172 ± 0.001	8.740 ± 1.013
LMS	beta35	0.101	0.029	-0.072 ± 0.002	-0.711 ± 0.015	0.002 ± 0.000	0.043 ± 0.001	0.007 ± 0.000	0.084 ± 0.001	0.689 ± 0.023	0.830 ± 0.014	0.029	-0.072	-71.13	0.0428	0.0836	1.005	60.540 ± 1.752	0.170 ± 0.001	9.383 ± 1.045
LSAM	beta31	0.316	0.309	-0.007 ± 0.002	-0.023 ± 0.006	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.026 ± 0.001	0.160 ± 0.005	0.307	-0.009	-2.74	0.0501	0.0509	0.987	94.859 ± 0.792	0.194 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.025 ± 0.001	0.159 ± 0.005	0.314	-0.002	-0.58	0.0462	0.0463	0.986	94.344 ± 0.828	0.194 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.179	0.040 ± 0.002	0.287 ± 0.014	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.227 ± 0.010	0.477 ± 0.012	0.179	0.040	29.06	0.0533	0.0668	1.035	89.846 ± 1.083	0.215 ± 0.001	90.488 ± 1.052
LSAM	beta34	0.101	0.048	-0.053 ± 0.002	-0.529 ± 0.024	0.005 ± 0.000	0.067 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.721 ± 0.033	0.849 ± 0.022	0.046	-0.055	-54.07	0.0681	0.0873	0.981	83.805 ± 1.321	0.258 ± 0.002	8.997 ± 1.026
LSAM	beta35	0.101	0.042	-0.059 ± 0.002	-0.585 ± 0.024	0.004 ± 0.000	0.067 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.779 ± 0.034	0.883 ± 0.021	0.040	-0.061	-60.35	0.0618	0.0868	0.985	82.391 ± 1.366	0.258 ± 0.002	8.869 ± 1.019
QML	beta31	0.316	0.305	-0.011 ± 0.002	-0.036 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.304	-0.012	-3.85	0.0444	0.0460	0.982	92.802 ± 0.927	0.173 ± 0.001	99.871 ± 0.128
QML	beta32	0.316	0.308	-0.008 ± 0.002	-0.025 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	0.020 ± 0.001	0.142 ± 0.005	0.310	-0.006	-1.98	0.0412	0.0416	1.000	93.316 ± 0.895	0.173 ± 0.001	99.871 ± 0.128
QML	beta33	0.139	0.172	0.033 ± 0.002	0.238 ± 0.013	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.198 ± 0.009	0.444 ± 0.012	0.172	0.033	23.72	0.0511	0.0608	1.034	90.617 ± 1.045	0.212 ± 0.002	88.303 ± 1.152
QML	beta34	0.101	0.031	-0.070 ± 0.002	-0.697 ± 0.015	0.002 ± 0.000	0.043 ± 0.001	0.007 ± 0.000	0.082 ± 0.001	0.664 ± 0.023	0.815 ± 0.015	0.031	-0.070	-69.80	0.0436	0.0829	1.039	63.239 ± 1.729	0.174 ± 0.003	8.226 ± 0.985
QML	beta35	0.101	0.028	-0.073 ± 0.002	-0.725 ± 0.015	0.002 ± 0.000	0.043 ± 0.001	0.007 ± 0.000	0.085 ± 0.001	0.707 ± 0.023	0.841 ± 0.014	0.028	-0.073	-72.27	0.0428	0.0846	1.016	59.897 ± 1.757	0.172 ± 0.002	9.254 ± 1.039
UPI	beta31	0.316	0.313	-0.003 ± 0.002	-0.008 ± 0.008	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.046 ± 0.003	0.213 ± 0.007	0.316	0.000	-0.05	0.0665	0.0665	1.396	96.787 ± 0.632	0.369 ± 0.033	90.874 ± 1.032
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.003	0.208 ± 0.007	0.318	0.002	0.62	0.0618	0.0619	1.353	96.915 ± 0.620	0.349 ± 0.020	92.416 ± 0.949
UPI	beta33	0.139	0.180	0.041 ± 0.002	0.292 ± 0.016	0.004 ± 0.000	0.060 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.273 ± 0.013	0.522 ± 0.014	0.180	0.041	29.70	0.0608	0.0735	1.225	94.344 ± 0.828	0.289 ± 0.008	74.550 ± 1.562
UPI	beta34	0.101	0.054	-0.047 ± 0.004	-0.464 ± 0.042	0.014 ± 0.001	0.118 ± 0.004	0.016 ± 0.001	0.127 ± 0.004	1.587 ± 0.089	1.260 ± 0.040	0.052	-0.049	-48.10	0.1070	0.1175	1.270	92.545 ± 0.942	0.589 ± 0.043	8.740 ± 1.013
UPI	beta35	0.101	0.045	-0.056 ± 0.004	-0.556 ± 0.040	0.013 ± 0.001	0.113 ± 0.004	0.016 ± 0.001	0.126 ± 0.004	1.558 ± 0.088	1.248 ± 0.040	0.042	-0.059	-58.55	0.1006	0.1167	1.243	92.288 ± 0.956	0.550 ± 0.021	7.712 ± 0.956

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S35

Simulation 1 - Condition 35: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.002	-0.011 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.313	-0.003	-0.85	0.0547	0.0548	1.022	95.100 ± 0.683	0.223 ± 0.001	99.900 ± 0.100
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.001	0.183 ± 0.005	0.314	-0.002	-0.69	0.0580	0.0581	0.984	94.700 ± 0.708	0.224 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.150	0.011 ± 0.002	0.083 ± 0.015	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.217 ± 0.009	0.466 ± 0.012	0.151	0.012	8.75	0.0652	0.0663	1.006	94.300 ± 0.733	0.252 ± 0.001	65.100 ± 1.507
LMS	beta34	0.101	0.069	-0.032 ± 0.002	-0.322 ± 0.019	0.004 ± 0.000	0.060 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.454 ± 0.020	0.674 ± 0.017	0.069	-0.032	-31.29	0.0605	0.0682	0.988	89.200 ± 0.982	0.232 ± 0.001	22.500 ± 1.321
LMS	beta35	0.101	0.070	-0.031 ± 0.002	-0.310 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.429 ± 0.018	0.655 ± 0.016	0.069	-0.032	-31.24	0.0564	0.0646	1.013	91.200 ± 0.896	0.232 ± 0.001	22.000 ± 1.310
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.315	-0.001	-0.25	0.0575	0.0575	1.030	95.700 ± 0.641	0.232 ± 0.001	99.900 ± 0.100
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.318	0.002	0.49	0.0615	0.0615	0.989	95.000 ± 0.689	0.232 ± 0.001	99.900 ± 0.100
LSAM	beta33	0.139	0.149	0.010 ± 0.002	0.072 ± 0.015	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.218 ± 0.009	0.467 ± 0.012	0.149	0.010	7.13	0.0647	0.0655	1.004	95.100 ± 0.683	0.253 ± 0.001	65.000 ± 1.508
LSAM	beta34	0.101	0.087	-0.014 ± 0.002	-0.139 ± 0.023	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.563 ± 0.026	0.750 ± 0.021	0.086	-0.015	-14.37	0.0739	0.0753	0.986	92.800 ± 0.817	0.288 ± 0.001	22.600 ± 1.323
LSAM	beta35	0.101	0.088	-0.013 ± 0.002	-0.131 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.532 ± 0.023	0.730 ± 0.020	0.087	-0.014	-13.90	0.0694	0.0708	1.010	93.700 ± 0.768	0.287 ± 0.001	22.500 ± 1.321
QML	beta31	0.316	0.312	-0.004 ± 0.002	-0.011 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.313	-0.003	-0.88	0.0552	0.0553	1.023	95.100 ± 0.683	0.224 ± 0.001	99.900 ± 0.100
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.001	0.184 ± 0.005	0.314	-0.002	-0.76	0.0582	0.0582	0.986	94.600 ± 0.715	0.224 ± 0.001	99.900 ± 0.100
QML	beta33	0.139	0.151	0.012 ± 0.002	0.083 ± 0.015	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.217 ± 0.009	0.466 ± 0.012	0.151	0.012	8.72	0.0655	0.0666	1.006	94.400 ± 0.727	0.252 ± 0.001	65.200 ± 1.506
QML	beta34	0.101	0.068	-0.033 ± 0.002	-0.326 ± 0.019	0.004 ± 0.000	0.060 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.458 ± 0.020	0.677 ± 0.017	0.069	-0.032	-31.82	0.0603	0.0683	0.988	89.500 ± 0.969	0.232 ± 0.001	22.000 ± 1.310
QML	beta35	0.101	0.069	-0.032 ± 0.002	-0.314 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.432 ± 0.018	0.657 ± 0.016	0.069	-0.032	-31.33	0.0563	0.0646	1.013	91.000 ± 0.905	0.232 ± 0.001	21.800 ± 1.306
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.013 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.321	0.005	1.61	0.0567	0.0570	0.992	94.300 ± 0.733	0.223 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.321	0.005	1.68	0.0611	0.0613	0.952	94.400 ± 0.727	0.224 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.145	0.006 ± 0.002	0.046 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.220 ± 0.010	0.469 ± 0.013	0.145	0.006	4.33	0.0653	0.0656	0.998	95.200 ± 0.676	0.254 ± 0.001	62.800 ± 1.528
UPI	beta34	0.101	0.096	-0.005 ± 0.002	-0.046 ± 0.024	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.597 ± 0.029	0.772 ± 0.022	0.095	-0.006	-5.88	0.0754	0.0756	0.987	94.200 ± 0.739	0.302 ± 0.002	24.800 ± 1.366
UPI	beta35	0.101	0.096	-0.005 ± 0.002	-0.045 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.567 ± 0.026	0.753 ± 0.021	0.098	-0.003	-3.40	0.0738	0.0738	1.014	94.800 ± 0.702	0.302 ± 0.002	24.400 ± 1.358

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S36

Simulation 1 - Condition 36: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.310	-0.006	-2.02	0.0356	0.0361	1.015	95.000 ± 0.689	0.141 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.312	-0.004 ± 0.001	-0.012 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.313	-0.003	-0.93	0.0375	0.0376	0.992	94.700 ± 0.708	0.141 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.149	0.010 ± 0.001	0.075 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.090 ± 0.004	0.299 ± 0.008	0.149	0.010	7.06	0.0403	0.0414	1.006	94.700 ± 0.708	0.159 ± 0.000	95.400 ± 0.662
LMS	beta34	0.101	0.071	-0.030 ± 0.001	-0.293 ± 0.011	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.214 ± 0.008	0.463 ± 0.010	0.072	-0.029	-29.06	0.0371	0.0473	1.028	87.300 ± 1.053	0.146 ± 0.000	48.100 ± 1.580
LMS	beta35	0.101	0.069	-0.032 ± 0.001	-0.315 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.242 ± 0.010	0.492 ± 0.011	0.069	-0.032	-32.03	0.0377	0.0497	0.980	85.800 ± 1.104	0.146 ± 0.000	45.200 ± 1.574
LSAM	beta31	0.316	0.312	-0.004 ± 0.001	-0.011 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.116 ± 0.003	0.312	-0.004	-1.19	0.0367	0.0369	1.019	95.100 ± 0.683	0.146 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.316	0.000	-0.08	0.0392	0.0392	1.000	95.100 ± 0.683	0.146 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.149	0.010 ± 0.001	0.070 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.148	0.009	6.55	0.0401	0.0411	1.003	94.600 ± 0.715	0.158 ± 0.000	95.200 ± 0.676
LSAM	beta34	0.101	0.090	-0.011 ± 0.001	-0.108 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.207 ± 0.009	0.455 ± 0.012	0.089	-0.012	-11.62	0.0457	0.0472	1.027	94.700 ± 0.708	0.180 ± 0.000	49.100 ± 1.581
LSAM	beta35	0.101	0.087	-0.014 ± 0.001	-0.140 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.236 ± 0.011	0.486 ± 0.013	0.086	-0.015	-14.86	0.0465	0.0489	0.980	93.200 ± 0.796	0.181 ± 0.001	47.600 ± 1.579
QML	beta31	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.310	-0.006	-2.03	0.0357	0.0363	1.033	95.100 ± 0.683	0.144 ± 0.002	99.900 ± 0.100
QML	beta32	0.316	0.312	-0.004 ± 0.001	-0.013 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.313	-0.003	-1.04	0.0376	0.0377	1.005	94.800 ± 0.702	0.143 ± 0.001	99.900 ± 0.100
QML	beta33	0.139	0.149	0.010 ± 0.001	0.074 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.090 ± 0.004	0.299 ± 0.008	0.149	0.010	7.00	0.0403	0.0415	1.005	94.600 ± 0.715	0.159 ± 0.000	95.400 ± 0.662
QML	beta34	0.101	0.071	-0.030 ± 0.001	-0.296 ± 0.011	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.216 ± 0.008	0.465 ± 0.010	0.071	-0.030	-29.55	0.0369	0.0474	1.026	86.800 ± 1.070	0.146 ± 0.000	47.800 ± 1.580
QML	beta35	0.101	0.069	-0.032 ± 0.001	-0.318 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.244 ± 0.010	0.494 ± 0.011	0.068	-0.033	-32.31	0.0377	0.0499	0.978	85.500 ± 1.113	0.146 ± 0.000	45.100 ± 1.574
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.315	-0.001	-0.45	0.0360	0.0360	0.982	93.800 ± 0.763	0.141 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.318	0.002	0.68	0.0393	0.0393	0.966	94.200 ± 0.739	0.141 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.143	0.004 ± 0.001	0.030 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.143	0.004	2.73	0.0407	0.0409	0.989	94.900 ± 0.696	0.159 ± 0.000	93.400 ± 0.785
UPI	beta34	0.101	0.099	-0.002 ± 0.001	-0.018 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.209 ± 0.010	0.457 ± 0.013	0.100	-0.001	-1.40	0.0473	0.0473	1.034	96.000 ± 0.620	0.187 ± 0.001	55.000 ± 1.573
UPI	beta35	0.101	0.096	-0.005 ± 0.002	-0.047 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.231 ± 0.011	0.481 ± 0.014	0.097	-0.004	-4.27	0.0454	0.0456	0.988	93.900 ± 0.757	0.187 ± 0.001	52.900 ± 1.578

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S37

Simulation 1 - Condition 37: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.004	-0.000 ± 0.012	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.114 ± 0.006	0.338 ± 0.011	0.306	-0.010	-3.16	0.1057	0.1061	1.105	97.265 ± 0.562	0.463 ± 0.007	83.115 ± 1.292
LMS	beta32	0.316	0.310	-0.006 ± 0.004	-0.019 ± 0.012	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	0.117 ± 0.006	0.342 ± 0.011	0.308	-0.008	-2.52	0.1131	0.1134	1.077	95.957 ± 0.679	0.455 ± 0.006	82.521 ± 1.310
LMS	beta33	0.139	0.047	-0.092 ± 0.009	-0.660 ± 0.066	0.072 ± 0.005	0.268 ± 0.010	0.080 ± 0.006	0.283 ± 0.012	4.149 ± 0.332	2.037 ± 0.088	0.092	-0.047	-33.65	0.2191	0.2241	0.974	95.363 ± 0.725	1.023 ± 0.040	17.717 ± 1.317
LMS	beta34	0.101	0.175	0.074 ± 0.006	0.729 ± 0.059	0.030 ± 0.002	0.174 ± 0.007	0.036 ± 0.003	0.188 ± 0.008	3.481 ± 0.293	1.866 ± 0.083	0.137	0.036	35.92	0.1330	0.1378	0.992	94.768 ± 0.768	0.675 ± 0.025	8.799 ± 0.977
LMS	beta35	0.101	0.172	0.071 ± 0.006	0.701 ± 0.058	0.029 ± 0.002	0.171 ± 0.007	0.034 ± 0.003	0.185 ± 0.009	3.360 ± 0.293	1.833 ± 0.085	0.136	0.035	34.71	0.1361	0.1405	1.006	95.957 ± 0.679	0.675 ± 0.025	9.156 ± 0.994
LSAM	beta31	0.316	0.326	0.010 ± 0.004	0.033 ± 0.012	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.132 ± 0.008	0.363 ± 0.013	0.322	0.006	1.75	0.1114	0.1115	1.065	97.146 ± 0.574	0.478 ± 0.005	81.688 ± 1.334
LSAM	beta32	0.316	0.323	0.007 ± 0.004	0.021 ± 0.013	0.013 ± 0.001	0.115 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.132 ± 0.007	0.364 ± 0.012	0.320	0.004	1.37	0.1132	0.1133	1.068	96.790 ± 0.608	0.481 ± 0.007	78.835 ± 1.409
LSAM	beta33	0.139	0.155	0.016 ± 0.006	0.115 ± 0.044	0.031 ± 0.002	0.175 ± 0.006	0.031 ± 0.002	0.176 ± 0.006	1.603 ± 0.102	1.266 ± 0.046	0.148	0.009	6.53	0.1597	0.1600	1.080	98.335 ± 0.441	0.742 ± 0.025	14.863 ± 1.227
LSAM	beta34	0.101	0.105	0.004 ± 0.004	0.039 ± 0.041	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	1.446 ± 0.095	1.202 ± 0.045	0.098	-0.003	-3.07	0.1112	0.1112	1.074	97.860 ± 0.499	0.511 ± 0.011	8.680 ± 0.971
LSAM	beta35	0.101	0.108	0.007 ± 0.004	0.069 ± 0.042	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.004	1.462 ± 0.090	1.209 ± 0.043	0.103	0.002	2.18	0.1074	0.1074	1.093	97.265 ± 0.562	0.522 ± 0.015	9.750 ± 1.023
QML	beta31	0.316	0.303	-0.013 ± 0.003	-0.041 ± 0.011	0.010 ± 0.000	0.098 ± 0.003	0.010 ± 0.000	0.099 ± 0.003	0.097 ± 0.005	0.312 ± 0.009	0.297	-0.019	-5.98	0.0993	0.1011	1.050	96.076 ± 0.670	0.402 ± 0.003	87.872 ± 1.126
QML	beta32	0.316	0.299	-0.017 ± 0.003	-0.053 ± 0.011	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.009	0.295	-0.021	-6.63	0.1007	0.1028	1.038	95.125 ± 0.743	0.400 ± 0.003	87.872 ± 1.126
QML	beta33	0.139	0.110	-0.029 ± 0.005	-0.207 ± 0.035	0.020 ± 0.001	0.143 ± 0.004	0.021 ± 0.001	0.146 ± 0.005	1.098 ± 0.062	1.048 ± 0.034	0.112	-0.027	-19.12	0.1406	0.1431	1.043	96.195 ± 0.660	0.584 ± 0.008	16.647 ± 1.284
QML	beta34	0.101	0.113	0.012 ± 0.003	0.120 ± 0.033	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.911 ± 0.053	0.954 ± 0.032	0.105	0.004	4.39	0.0969	0.0970	1.049	95.125 ± 0.743	0.393 ± 0.006	13.793 ± 1.189
QML	beta35	0.101	0.113	0.012 ± 0.003	0.118 ± 0.032	0.009 ± 0.000	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.889 ± 0.051	0.943 ± 0.032	0.106	0.005	4.90	0.0943	0.0944	1.067	95.957 ± 0.679	0.395 ± 0.006	14.031 ± 1.198
UPI	beta31	0.316	0.329	0.013 ± 0.004	0.040 ± 0.013	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	0.138 ± 0.008	0.372 ± 0.013	0.322	0.006	1.97	0.1179	0.1181	1.150	98.335 ± 0.441	0.527 ± 0.007	77.527 ± 1.439
UPI	beta32	0.316	0.324	0.008 ± 0.004	0.025 ± 0.013	0.014 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.136 ± 0.007	0.369 ± 0.012	0.321	0.005	1.55	0.1167	0.1168	1.179	98.454 ± 0.425	0.538 ± 0.010	74.554 ± 1.502
UPI	beta33	0.139	0.174	0.035 ± 0.007	0.249 ± 0.051	0.043 ± 0.003	0.207 ± 0.007	0.044 ± 0.003	0.209 ± 0.008	2.270 ± 0.152	1.507 ± 0.057	0.158	0.019	13.32	0.1873	0.1883	1.189	99.287 ± 0.290	0.963 ± 0.036	8.680 ± 0.971
UPI	beta34	0.101	0.102	0.001 ± 0.005	0.014 ± 0.050	0.022 ± 0.001	0.148 ± 0.005	0.022 ± 0.001	0.148 ± 0.005	2.134 ± 0.140	1.461 ± 0.054	0.094	-0.007	-7.17	0.1283	0.1285	1.236	97.384 ± 0.550	0.715 ± 0.021	4.875 ± 0.743
UPI	beta35	0.101	0.108	0.007 ± 0.005	0.071 ± 0.048	0.020 ± 0.001	0.141 ± 0.005	0.020 ± 0.001	0.141 ± 0.005	1.960 ± 0.136	1.400 ± 0.054	0.099	-0.002	-1.64	0.1233	0.1233	1.289	97.146 ± 0.574	0.714 ± 0.025	4.994 ± 0.751

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S38

Simulation 1 - Condition 38: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.002	-0.019 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.208 ± 0.006	0.308	-0.008	-2.57	0.0671	0.0676	1.062	96.004 ± 0.652	0.272 ± 0.004	98.779 ± 0.366
LMS	beta32	0.316	0.312	-0.004 ± 0.002	-0.012 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.206 ± 0.006	0.307	-0.009	-2.79	0.0637	0.0643	1.068	96.670 ± 0.598	0.273 ± 0.004	98.890 ± 0.349
LMS	beta33	0.139	0.118	-0.021 ± 0.004	-0.151 ± 0.027	0.013 ± 0.001	0.115 ± 0.004	0.014 ± 0.001	0.116 ± 0.005	0.702 ± 0.052	0.838 ± 0.034	0.118	-0.021	-15.20	0.1110	0.1130	1.221	96.892 ± 0.578	0.548 ± 0.021	29.301 ± 1.516
LMS	beta34	0.101	0.130	0.029 ± 0.003	0.291 ± 0.025	0.006 ± 0.000	0.076 ± 0.003	0.007 ± 0.001	0.081 ± 0.003	0.651 ± 0.052	0.807 ± 0.034	0.131	0.030	29.30	0.0768	0.0823	1.233	97.336 ± 0.536	0.368 ± 0.014	26.970 ± 1.479
LMS	beta35	0.101	0.131	0.030 ± 0.003	0.296 ± 0.028	0.007 ± 0.000	0.083 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	0.769 ± 0.059	0.877 ± 0.036	0.128	0.027	26.70	0.0794	0.0839	1.119	96.448 ± 0.617	0.366 ± 0.012	27.636 ± 1.490
LSAM	beta31	0.316	0.320	0.004 ± 0.002	0.013 ± 0.007	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.213 ± 0.006	0.315	-0.001	-0.25	0.0671	0.0671	1.028	96.337 ± 0.626	0.271 ± 0.001	99.778 ± 0.157
LSAM	beta32	0.316	0.323	0.007 ± 0.002	0.022 ± 0.007	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.214 ± 0.006	0.321	0.005	1.51	0.0679	0.0681	1.029	96.337 ± 0.626	0.271 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.157	0.018 ± 0.003	0.129 ± 0.022	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.447 ± 0.024	0.669 ± 0.021	0.146	0.007	5.32	0.0972	0.0974	1.042	97.558 ± 0.514	0.373 ± 0.004	41.287 ± 1.640
LSAM	beta34	0.101	0.100	-0.001 ± 0.002	-0.010 ± 0.021	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.386 ± 0.021	0.621 ± 0.020	0.104	0.003	3.32	0.0624	0.0625	1.068	97.003 ± 0.568	0.263 ± 0.003	31.521 ± 1.548
LSAM	beta35	0.101	0.098	-0.003 ± 0.002	-0.031 ± 0.022	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.452 ± 0.024	0.672 ± 0.021	0.099	-0.002	-1.57	0.0662	0.0662	0.990	95.006 ± 0.726	0.263 ± 0.003	34.184 ± 1.580
QML	beta31	0.316	0.297	-0.019 ± 0.002	-0.059 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.202 ± 0.006	0.296	-0.020	-6.35	0.0631	0.0663	1.022	93.896 ± 0.798	0.245 ± 0.001	99.667 ± 0.192
QML	beta32	0.316	0.300	-0.016 ± 0.002	-0.050 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.005	0.295	-0.021	-6.65	0.0597	0.0633	1.022	95.006 ± 0.726	0.245 ± 0.001	99.889 ± 0.111
QML	beta33	0.139	0.119	-0.020 ± 0.003	-0.145 ± 0.020	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.380 ± 0.019	0.617 ± 0.018	0.113	-0.026	-18.79	0.0871	0.0909	1.070	95.560 ± 0.686	0.350 ± 0.003	29.634 ± 1.521
QML	beta34	0.101	0.105	0.004 ± 0.002	0.038 ± 0.018	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.280 ± 0.015	0.529 ± 0.016	0.109	0.008	8.36	0.0566	0.0572	1.113	96.115 ± 0.644	0.233 ± 0.002	42.064 ± 1.645
QML	beta35	0.101	0.105	0.004 ± 0.002	0.038 ± 0.019	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.335 ± 0.017	0.579 ± 0.017	0.108	0.007	7.03	0.0566	0.0570	1.021	94.562 ± 0.755	0.234 ± 0.002	42.397 ± 1.646
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.014 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.214 ± 0.006	0.317	0.001	0.42	0.0686	0.0686	1.039	96.781 ± 0.588	0.275 ± 0.001	99.778 ± 0.157
UPI	beta32	0.316	0.323	0.007 ± 0.002	0.021 ± 0.007	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.214 ± 0.006	0.321	0.005	1.68	0.0657	0.0659	1.043	96.892 ± 0.578	0.275 ± 0.001	99.889 ± 0.111
UPI	beta33	0.139	0.163	0.024 ± 0.003	0.170 ± 0.023	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.526 ± 0.030	0.725 ± 0.024	0.151	0.012	8.68	0.0997	0.1004	1.069	98.446 ± 0.412	0.411 ± 0.005	38.291 ± 1.619
UPI	beta34	0.101	0.100	-0.001 ± 0.002	-0.010 ± 0.023	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.459 ± 0.025	0.677 ± 0.022	0.100	-0.001	-0.88	0.0678	0.0678	1.094	97.114 ± 0.558	0.294 ± 0.004	24.972 ± 1.442
UPI	beta35	0.101	0.099	-0.002 ± 0.002	-0.024 ± 0.024	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.536 ± 0.032	0.732 ± 0.025	0.098	-0.003	-3.03	0.0695	0.0696	1.026	95.117 ± 0.718	0.297 ± 0.004	25.971 ± 1.461

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S39

Simulation 1 - Condition 39: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.002	-0.019 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.312	-0.004	-1.39	0.0771	0.0772	1.025	95.301 ± 0.676	0.300 ± 0.002	97.140 ± 0.533
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.008	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.235 ± 0.007	0.314	-0.002	-0.78	0.0717	0.0717	1.033	95.199 ± 0.683	0.301 ± 0.003	98.264 ± 0.417
LMS	beta33	0.139	0.118	-0.021 ± 0.003	-0.150 ± 0.023	0.010 ± 0.001	0.101 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	0.551 ± 0.034	0.742 ± 0.025	0.123	-0.016	-11.21	0.0924	0.0937	1.092	96.425 ± 0.593	0.433 ± 0.010	32.074 ± 1.492
LMS	beta34	0.101	0.119	0.018 ± 0.002	0.178 ± 0.023	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.562 ± 0.035	0.750 ± 0.026	0.108	0.007	6.59	0.0660	0.0663	1.026	95.301 ± 0.676	0.296 ± 0.007	30.950 ± 1.477
LMS	beta35	0.101	0.119	0.018 ± 0.002	0.183 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.551 ± 0.036	0.743 ± 0.027	0.111	0.010	9.45	0.0661	0.0668	1.050	96.527 ± 0.585	0.299 ± 0.006	30.644 ± 1.473
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.008	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.002	0.235 ± 0.006	0.318	0.002	0.78	0.0715	0.0716	1.019	94.995 ± 0.697	0.297 ± 0.001	98.774 ± 0.352
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.010 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.057 ± 0.003	0.238 ± 0.007	0.316	0.000	0.13	0.0740	0.0740	1.011	95.812 ± 0.640	0.298 ± 0.001	99.081 ± 0.305
LSAM	beta33	0.139	0.142	0.003 ± 0.003	0.024 ± 0.020	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.385 ± 0.018	0.620 ± 0.018	0.141	0.002	1.44	0.0826	0.0827	1.040	95.914 ± 0.633	0.351 ± 0.003	38.509 ± 1.555
LSAM	beta34	0.101	0.103	0.002 ± 0.002	0.024 ± 0.020	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.408 ± 0.020	0.639 ± 0.019	0.098	-0.003	-2.93	0.0612	0.0613	0.984	94.382 ± 0.736	0.249 ± 0.002	35.853 ± 1.533
LSAM	beta35	0.101	0.102	0.001 ± 0.002	0.009 ± 0.020	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.386 ± 0.019	0.621 ± 0.018	0.098	-0.003	-2.48	0.0599	0.0600	1.018	96.016 ± 0.625	0.251 ± 0.002	35.444 ± 1.529
QML	beta31	0.316	0.305	-0.011 ± 0.002	-0.034 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.002	0.233 ± 0.006	0.307	-0.009	-2.89	0.0735	0.0741	1.010	94.076 ± 0.755	0.288 ± 0.001	98.570 ± 0.379
QML	beta32	0.316	0.307	-0.009 ± 0.002	-0.027 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.003	0.231 ± 0.007	0.308	-0.008	-2.54	0.0698	0.0703	1.014	94.791 ± 0.710	0.289 ± 0.001	98.979 ± 0.321
QML	beta33	0.139	0.119	-0.020 ± 0.003	-0.145 ± 0.020	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.419 ± 0.021	0.647 ± 0.019	0.120	-0.019	-13.58	0.0851	0.0872	1.044	95.608 ± 0.655	0.359 ± 0.003	31.971 ± 1.491
QML	beta34	0.101	0.108	0.007 ± 0.002	0.070 ± 0.020	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.395 ± 0.020	0.629 ± 0.019	0.102	0.001	0.94	0.0583	0.0583	0.995	94.178 ± 0.748	0.246 ± 0.003	37.589 ± 1.548
QML	beta35	0.101	0.108	0.007 ± 0.002	0.071 ± 0.020	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.381 ± 0.020	0.617 ± 0.019	0.103	0.002	2.24	0.0592	0.0592	1.021	95.812 ± 0.640	0.248 ± 0.003	37.385 ± 1.546
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.002	0.237 ± 0.006	0.317	0.001	0.46	0.0730	0.0730	1.002	94.586 ± 0.723	0.295 ± 0.001	98.672 ± 0.366
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.057 ± 0.003	0.240 ± 0.007	0.318	0.002	0.53	0.0765	0.0765	0.994	95.608 ± 0.655	0.295 ± 0.001	98.876 ± 0.337
UPI	beta33	0.139	0.146	0.007 ± 0.003	0.050 ± 0.021	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.430 ± 0.021	0.656 ± 0.019	0.141	0.002	1.34	0.0869	0.0869	1.040	97.140 ± 0.533	0.371 ± 0.003	36.159 ± 1.536
UPI	beta34	0.101	0.102	0.001 ± 0.002	0.013 ± 0.022	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.465 ± 0.023	0.682 ± 0.020	0.097	-0.004	-4.28	0.0643	0.0645	0.982	94.791 ± 0.710	0.265 ± 0.002	31.767 ± 1.488
UPI	beta35	0.101	0.102	0.001 ± 0.002	0.014 ± 0.021	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.443 ± 0.024	0.665 ± 0.021	0.101	0.000	-0.38	0.0614	0.0614	1.027	96.221 ± 0.609	0.271 ± 0.003	29.111 ± 1.452

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S40

Simulation 1 - Condition 40: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.002	-0.015 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.310	-0.006	-2.01	0.0471	0.0476	0.982	94.478 ± 0.724	0.184 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.310	-0.006 ± 0.001	-0.018 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.309	-0.007	-2.30	0.0476	0.0482	0.996	94.779 ± 0.705	0.184 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.125	-0.014 ± 0.002	-0.103 ± 0.014	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.214 ± 0.011	0.463 ± 0.013	0.127	-0.012	-8.88	0.0604	0.0616	0.998	94.378 ± 0.730	0.245 ± 0.002	58.233 ± 1.563
LMS	beta34	0.101	0.114	0.013 ± 0.001	0.125 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.218 ± 0.013	0.467 ± 0.015	0.110	0.009	9.26	0.0425	0.0436	0.967	96.185 ± 0.607	0.172 ± 0.002	80.422 ± 1.257
LMS	beta35	0.101	0.111	0.010 ± 0.001	0.102 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.195 ± 0.011	0.441 ± 0.014	0.109	0.008	7.45	0.0412	0.0419	0.996	95.683 ± 0.644	0.169 ± 0.002	81.627 ± 1.227
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.315	-0.001	-0.25	0.0462	0.0462	0.989	94.980 ± 0.692	0.184 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.315	-0.001	-0.19	0.0482	0.0482	1.001	95.080 ± 0.685	0.184 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.139	-0.000 ± 0.002	-0.001 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.165 ± 0.008	0.406 ± 0.011	0.138	-0.001	-0.75	0.0554	0.0554	0.960	93.876 ± 0.760	0.213 ± 0.001	72.289 ± 1.418
LSAM	beta34	0.101	0.103	0.002 ± 0.001	0.016 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.165 ± 0.009	0.406 ± 0.012	0.102	0.001	1.17	0.0400	0.0400	0.945	94.478 ± 0.724	0.152 ± 0.001	75.904 ± 1.355
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.008 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.152 ± 0.007	0.390 ± 0.011	0.100	-0.001	-0.90	0.0399	0.0400	0.983	94.980 ± 0.692	0.152 ± 0.001	75.502 ± 1.363
QML	beta31	0.316	0.306	-0.010 ± 0.001	-0.032 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.304	-0.012	-3.82	0.0460	0.0475	0.981	93.273 ± 0.794	0.180 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.305	-0.011 ± 0.001	-0.034 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.304	-0.012	-3.88	0.0473	0.0489	0.993	94.177 ± 0.742	0.180 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.120	-0.019 ± 0.002	-0.138 ± 0.013	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.192 ± 0.009	0.438 ± 0.012	0.120	-0.019	-13.34	0.0563	0.0593	0.982	93.976 ± 0.754	0.223 ± 0.001	58.534 ± 1.561
QML	beta34	0.101	0.106	0.005 ± 0.001	0.051 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.163 ± 0.008	0.404 ± 0.012	0.105	0.004	3.99	0.0394	0.0396	0.961	95.382 ± 0.665	0.153 ± 0.001	80.622 ± 1.252
QML	beta35	0.101	0.104	0.003 ± 0.001	0.034 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.154 ± 0.007	0.392 ± 0.011	0.103	0.002	1.56	0.0386	0.0386	0.986	94.378 ± 0.730	0.153 ± 0.001	81.928 ± 1.219
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.315	-0.001	-0.29	0.0474	0.0474	0.973	94.578 ± 0.718	0.181 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.316	0.000	-0.12	0.0498	0.0498	0.980	94.578 ± 0.718	0.182 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.140	0.001 ± 0.002	0.009 ± 0.013	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.174 ± 0.008	0.418 ± 0.012	0.138	-0.001	-0.55	0.0532	0.0532	0.961	94.076 ± 0.748	0.219 ± 0.001	71.386 ± 1.432
UPI	beta34	0.101	0.102	0.001 ± 0.001	0.012 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.176 ± 0.009	0.420 ± 0.013	0.101	0.000	0.39	0.0412	0.0412	0.949	94.779 ± 0.705	0.158 ± 0.001	73.996 ± 1.390
UPI	beta35	0.101	0.100	-0.001 ± 0.001	-0.010 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.164 ± 0.008	0.405 ± 0.011	0.100	-0.001	-1.04	0.0401	0.0401	0.983	94.578 ± 0.718	0.158 ± 0.001	73.996 ± 1.390

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S41

Simulation 1 - Condition 41: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.002	-0.017 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.308	-0.008	-2.55	0.0546	0.0552	0.997	94.494 ± 0.722	0.231 ± 0.001	99.800 ± 0.141
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.313	-0.003	-1.03	0.0593	0.0594	0.994	94.695 ± 0.709	0.232 ± 0.001	99.800 ± 0.141
LMS	beta33	0.139	0.132	-0.007 ± 0.002	-0.048 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.247 ± 0.012	0.497 ± 0.014	0.134	-0.005	-3.91	0.0673	0.0675	0.975	94.595 ± 0.715	0.263 ± 0.002	54.755 ± 1.575
LMS	beta34	0.101	0.106	0.005 ± 0.002	0.046 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.224 ± 0.011	0.474 ± 0.014	0.103	0.002	1.70	0.0464	0.0464	0.980	92.893 ± 0.813	0.183 ± 0.002	63.263 ± 1.525
LMS	beta35	0.101	0.107	0.006 ± 0.002	0.055 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.230 ± 0.011	0.479 ± 0.014	0.105	0.004	3.72	0.0460	0.0461	0.975	94.094 ± 0.746	0.184 ± 0.002	65.065 ± 1.508
LSAM	beta31	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.006	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.006	0.312	-0.004	-1.12	0.0565	0.0566	0.996	94.795 ± 0.703	0.232 ± 0.001	99.900 ± 0.100
LSAM	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.316	0.000	0.00	0.0587	0.0587	0.997	94.795 ± 0.703	0.233 ± 0.001	99.900 ± 0.100
LSAM	beta33	0.139	0.140	0.001 ± 0.002	0.010 ± 0.015	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.225 ± 0.011	0.475 ± 0.014	0.141	0.002	1.78	0.0640	0.0641	0.965	94.494 ± 0.722	0.250 ± 0.002	59.660 ± 1.552
LSAM	beta34	0.101	0.102	0.001 ± 0.001	0.011 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.214 ± 0.010	0.462 ± 0.013	0.100	-0.001	-0.79	0.0465	0.0465	0.966	92.092 ± 0.854	0.177 ± 0.001	63.664 ± 1.522
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.013 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.212 ± 0.010	0.460 ± 0.013	0.101	0.000	0.38	0.0446	0.0446	0.969	93.794 ± 0.763	0.177 ± 0.001	63.964 ± 1.519
QML	beta31	0.316	0.309	-0.007 ± 0.002	-0.022 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.306	-0.010	-3.07	0.0544	0.0553	0.996	94.394 ± 0.728	0.231 ± 0.001	99.900 ± 0.100
QML	beta32	0.316	0.311	-0.005 ± 0.002	-0.015 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.312	-0.004	-1.38	0.0597	0.0598	0.995	94.595 ± 0.715	0.231 ± 0.001	99.900 ± 0.100
QML	beta33	0.139	0.131	-0.008 ± 0.002	-0.058 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.240 ± 0.011	0.490 ± 0.014	0.132	-0.007	-5.08	0.0664	0.0668	0.972	94.695 ± 0.709	0.258 ± 0.002	54.555 ± 1.575
QML	beta34	0.101	0.104	0.003 ± 0.001	0.031 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.215 ± 0.010	0.464 ± 0.013	0.102	0.001	0.65	0.0456	0.0456	0.978	92.793 ± 0.818	0.179 ± 0.001	63.564 ± 1.523
QML	beta35	0.101	0.105	0.004 ± 0.001	0.038 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.218 ± 0.010	0.467 ± 0.013	0.104	0.003	2.60	0.0452	0.0452	0.976	93.994 ± 0.752	0.180 ± 0.001	65.465 ± 1.504
UPI	beta31	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.006	0.313	-0.003	-1.06	0.0568	0.0569	0.965	94.194 ± 0.740	0.225 ± 0.001	99.900 ± 0.100
UPI	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.317	0.001	0.38	0.0587	0.0587	0.965	93.694 ± 0.769	0.225 ± 0.001	99.900 ± 0.100
UPI	beta33	0.139	0.141	0.002 ± 0.002	0.016 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.236 ± 0.011	0.486 ± 0.014	0.142	0.003	2.43	0.0651	0.0652	0.958	94.194 ± 0.740	0.253 ± 0.002	58.659 ± 1.558
UPI	beta34	0.101	0.102	0.001 ± 0.002	0.010 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.223 ± 0.011	0.472 ± 0.013	0.102	0.001	0.99	0.0449	0.0449	0.965	91.892 ± 0.864	0.180 ± 0.001	61.662 ± 1.538
UPI	beta35	0.101	0.102	0.001 ± 0.002	0.014 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.221 ± 0.011	0.470 ± 0.014	0.101	0.000	-0.37	0.0457	0.0457	0.970	93.994 ± 0.752	0.181 ± 0.001	62.062 ± 1.535

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S42

Simulation 1 - Condition 42: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.001	-0.015 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.311	-0.005	-1.69	0.0389	0.0393	0.980	95.100 ± 0.683	0.146 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.313	-0.003 ± 0.001	-0.009 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.313	-0.003	-0.96	0.0388	0.0389	1.003	95.300 ± 0.669	0.145 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.133	-0.006 ± 0.001	-0.040 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.092 ± 0.004	0.304 ± 0.008	0.135	-0.004	-2.95	0.0391	0.0393	0.988	95.800 ± 0.634	0.162 ± 0.001	85.400 ± 1.117
LMS	beta34	0.101	0.104	0.003 ± 0.001	0.028 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.103	0.002	2.02	0.0288	0.0289	0.990	95.100 ± 0.683	0.114 ± 0.001	97.600 ± 0.484
LMS	beta35	0.101	0.103	0.002 ± 0.001	0.019 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.102	0.001	1.01	0.0294	0.0294	1.009	94.500 ± 0.721	0.115 ± 0.001	97.100 ± 0.531
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	-0.52	0.0399	0.0399	0.989	95.600 ± 0.649	0.146 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.316	0.000	0.13	0.0387	0.0387	1.004	95.500 ± 0.656	0.146 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.140	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.141	0.002	1.23	0.0380	0.0381	0.985	95.200 ± 0.676	0.155 ± 0.001	91.000 ± 0.905
LSAM	beta34	0.101	0.101	-0.000 ± 0.001	-0.001 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.079 ± 0.004	0.281 ± 0.008	0.101	0.000	0.08	0.0275	0.0275	0.989	94.500 ± 0.721	0.110 ± 0.001	96.800 ± 0.557
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.008 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.077 ± 0.004	0.278 ± 0.008	0.100	-0.001	-0.96	0.0274	0.0274	1.006	93.600 ± 0.774	0.111 ± 0.001	95.700 ± 0.641
QML	beta31	0.316	0.310	-0.006 ± 0.001	-0.019 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.309	-0.007	-2.15	0.0387	0.0393	0.980	95.100 ± 0.683	0.145 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.312	-0.004 ± 0.001	-0.013 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.312	-0.004	-1.38	0.0386	0.0388	1.002	95.700 ± 0.641	0.145 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.132	-0.007 ± 0.001	-0.053 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.133	-0.006	-4.27	0.0387	0.0392	0.987	95.400 ± 0.662	0.160 ± 0.001	85.100 ± 1.126
QML	beta34	0.101	0.103	0.002 ± 0.001	0.016 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.102	0.001	0.83	0.0286	0.0287	0.989	94.800 ± 0.702	0.113 ± 0.001	97.500 ± 0.494
QML	beta35	0.101	0.102	0.001 ± 0.001	0.008 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.080 ± 0.004	0.283 ± 0.008	0.101	0.000	-0.07	0.0289	0.0289	1.007	94.100 ± 0.745	0.113 ± 0.001	97.000 ± 0.539
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	-0.68	0.0396	0.0397	0.959	94.900 ± 0.696	0.142 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.316	0.000	0.12	0.0390	0.0390	0.971	95.000 ± 0.689	0.141 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.140	0.001 ± 0.001	0.007 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.141	0.002	1.26	0.0406	0.0406	0.978	94.500 ± 0.721	0.156 ± 0.001	90.600 ± 0.923
UPI	beta34	0.101	0.101	-0.000 ± 0.001	-0.001 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.101	0.000	0.14	0.0286	0.0286	0.991	94.100 ± 0.745	0.112 ± 0.001	96.500 ± 0.581
UPI	beta35	0.101	0.100	-0.001 ± 0.001	-0.009 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.078 ± 0.004	0.280 ± 0.008	0.100	-0.001	-1.18	0.0280	0.0280	1.012	94.200 ± 0.739	0.112 ± 0.001	95.900 ± 0.627

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S43

Simulation 1 - Condition 43: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.338	0.022 ± 0.005	0.070 ± 0.015	0.019 ± 0.001	0.138 ± 0.004	0.019 ± 0.001	0.139 ± 0.005	0.194 ± 0.011	0.441 ± 0.015	0.345	0.029	9.28	0.1387	0.1418	1.110	95.309 ± 0.743	0.599 ± 0.018	70.247 ± 1.606
LMS	beta32	0.316	0.333	0.017 ± 0.005	0.054 ± 0.015	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.138 ± 0.005	0.191 ± 0.012	0.438 ± 0.015	0.340	0.024	7.49	0.1327	0.1348	1.129	96.790 ± 0.619	0.608 ± 0.021	69.012 ± 1.625
LMS	beta33	0.139	-0.143	-0.282 ± 0.014	-2.031 ± 0.101	0.159 ± 0.010	0.399 ± 0.013	0.238 ± 0.017	0.488 ± 0.018	12.339 ± 0.879	3.513 ± 0.132	-0.027	-0.166	-119.76	0.3203	0.3610	0.997	90.494 ± 1.031	1.558 ± 0.083	8.642 ± 0.987
LMS	beta34	0.101	0.326	0.225 ± 0.008	2.229 ± 0.078	0.050 ± 0.004	0.224 ± 0.008	0.101 ± 0.007	0.317 ± 0.011	9.882 ± 0.653	3.144 ± 0.108	0.257	0.156	154.81	0.1512	0.2175	1.019	93.457 ± 0.869	0.895 ± 0.044	46.173 ± 1.752
LMS	beta35	0.101	0.323	0.222 ± 0.008	2.198 ± 0.077	0.048 ± 0.004	0.220 ± 0.008	0.098 ± 0.007	0.313 ± 0.011	9.574 ± 0.649	3.094 ± 0.109	0.267	0.166	164.36	0.1625	0.2323	1.072	93.580 ± 0.861	0.925 ± 0.045	45.309 ± 1.749
LSAM	beta31	0.316	0.323	0.007 ± 0.008	0.023 ± 0.026	0.055 ± 0.004	0.235 ± 0.008	0.055 ± 0.004	0.235 ± 0.009	0.554 ± 0.036	0.744 ± 0.028	0.326	0.010	3.01	0.2292	0.2294	1.121	97.778 ± 0.518	1.033 ± 0.023	29.136 ± 1.597
LSAM	beta32	0.316	0.318	0.002 ± 0.008	0.007 ± 0.026	0.056 ± 0.003	0.237 ± 0.007	0.056 ± 0.003	0.237 ± 0.008	0.562 ± 0.035	0.749 ± 0.027	0.308	-0.008	-2.38	0.2259	0.2260	1.107	98.889 ± 0.368	1.028 ± 0.032	31.605 ± 1.634
LSAM	beta33	0.139	0.168	0.029 ± 0.006	0.209 ± 0.046	0.032 ± 0.002	0.180 ± 0.006	0.033 ± 0.002	0.182 ± 0.007	1.719 ± 0.112	1.311 ± 0.048	0.164	0.025	17.93	0.1738	0.1756	1.126	96.296 ± 0.664	0.794 ± 0.039	10.988 ± 1.099
LSAM	beta34	0.101	0.098	-0.003 ± 0.003	-0.026 ± 0.033	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.888 ± 0.055	0.942 ± 0.034	0.098	-0.003	-3.37	0.0901	0.0902	1.131	97.531 ± 0.545	0.422 ± 0.014	23.580 ± 1.492
LSAM	beta35	0.101	0.103	0.002 ± 0.004	0.021 ± 0.035	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.989 ± 0.068	0.994 ± 0.039	0.100	-0.001	-1.46	0.0932	0.0932	1.063	96.914 ± 0.608	0.419 ± 0.016	25.679 ± 1.535
QML	beta31	0.316	0.342	0.026 ± 0.004	0.081 ± 0.013	0.013 ± 0.001	0.115 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	0.139 ± 0.008	0.373 ± 0.012	0.342	0.026	8.33	0.1190	0.1219	1.024	95.556 ± 0.724	0.462 ± 0.003	83.210 ± 1.313
QML	beta32	0.316	0.339	0.023 ± 0.004	0.074 ± 0.013	0.013 ± 0.001	0.114 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.136 ± 0.007	0.369 ± 0.011	0.339	0.023	7.31	0.1171	0.1194	1.032	95.926 ± 0.695	0.463 ± 0.004	82.840 ± 1.325
QML	beta33	0.139	0.009	-0.130 ± 0.005	-0.938 ± 0.038	0.023 ± 0.001	0.152 ± 0.004	0.040 ± 0.002	0.200 ± 0.005	2.073 ± 0.101	1.440 ± 0.038	0.012	-0.127	-91.58	0.1497	0.1965	1.033	88.642 ± 1.115	0.615 ± 0.009	6.667 ± 0.876
QML	beta34	0.101	0.223	0.122 ± 0.003	1.211 ± 0.032	0.008 ± 0.001	0.092 ± 0.003	0.023 ± 0.001	0.153 ± 0.004	2.289 ± 0.108	1.513 ± 0.037	0.216	0.115	114.28	0.0897	0.1462	1.075	85.802 ± 1.226	0.386 ± 0.007	71.111 ± 1.593
QML	beta35	0.101	0.222	0.121 ± 0.003	1.196 ± 0.032	0.009 ± 0.000	0.092 ± 0.003	0.023 ± 0.001	0.152 ± 0.004	2.265 ± 0.104	1.505 ± 0.036	0.216	0.115	113.78	0.0931	0.1479	1.045	83.827 ± 1.294	0.378 ± 0.006	73.086 ± 1.558
UPI	beta31	0.316	0.329	0.013 ± 0.008	0.041 ± 0.027	0.057 ± 0.004	0.238 ± 0.008	0.057 ± 0.004	0.239 ± 0.009	0.570 ± 0.036	0.755 ± 0.028	0.343	0.027	8.65	0.2335	0.2351	1.185	97.901 ± 0.504	1.108 ± 0.031	31.111 ± 1.627
UPI	beta32	0.316	0.334	0.018 ± 0.008	0.056 ± 0.026	0.056 ± 0.004	0.237 ± 0.008	0.056 ± 0.004	0.237 ± 0.009	0.564 ± 0.037	0.751 ± 0.028	0.318	0.002	0.60	0.2268	0.2268	1.136	98.025 ± 0.489	1.055 ± 0.023	32.222 ± 1.642
UPI	beta33	0.139	0.153	0.014 ± 0.007	0.098 ± 0.051	0.041 ± 0.003	0.203 ± 0.007	0.041 ± 0.003	0.203 ± 0.008	2.140 ± 0.155	1.463 ± 0.059	0.134	-0.005	-3.37	0.1750	0.1750	1.378	95.802 ± 0.705	1.096 ± 0.049	5.926 ± 0.830
UPI	beta34	0.101	0.100	-0.001 ± 0.003	-0.010 ± 0.035	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.966 ± 0.061	0.983 ± 0.036	0.097	-0.004	-4.34	0.0916	0.0917	1.208	97.037 ± 0.596	0.470 ± 0.017	20.864 ± 1.428
UPI	beta35	0.101	0.099	-0.002 ± 0.003	-0.019 ± 0.034	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.960 ± 0.064	0.980 ± 0.037	0.098	-0.003	-2.58	0.0907	0.0908	1.125	96.049 ± 0.684	0.437 ± 0.013	21.605 ± 1.446

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S44

Simulation 1 - Condition 44: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.341	0.025 ± 0.003	0.078 ± 0.010	0.008 ± 0.000	0.092 ± 0.003	0.009 ± 0.000	0.095 ± 0.003	0.090 ± 0.005	0.300 ± 0.009	0.341	0.025	7.87	0.0887	0.0921	0.980	92.043 ± 0.887	0.352 ± 0.006	92.043 ± 0.887
LMS	beta32	0.316	0.332	0.016 ± 0.003	0.050 ± 0.009	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.083 ± 0.004	0.289 ± 0.009	0.334	0.018	5.61	0.0902	0.0919	1.011	93.548 ± 0.806	0.356 ± 0.007	89.570 ± 1.002
LMS	beta33	0.139	-0.111	-0.250 ± 0.011	-1.795 ± 0.078	0.108 ± 0.007	0.329 ± 0.010	0.171 ± 0.011	0.413 ± 0.015	8.828 ± 0.594	2.971 ± 0.105	-0.016	-0.155	-111.82	0.2143	0.2648	0.795	87.527 ± 1.083	1.026 ± 0.044	12.903 ± 1.099
LMS	beta34	0.101	0.302	0.201 ± 0.005	1.992 ± 0.052	0.026 ± 0.002	0.161 ± 0.006	0.066 ± 0.004	0.258 ± 0.007	6.519 ± 0.363	2.553 ± 0.073	0.254	0.153	151.97	0.1044	0.1856	0.901	71.290 ± 1.484	0.570 ± 0.027	81.720 ± 1.267
LMS	beta35	0.101	0.312	0.211 ± 0.006	2.092 ± 0.056	0.029 ± 0.002	0.172 ± 0.006	0.074 ± 0.004	0.272 ± 0.008	7.256 ± 0.404	2.694 ± 0.077	0.265	0.164	162.53	0.1090	0.1971	0.863	69.140 ± 1.515	0.580 ± 0.023	79.892 ± 1.314
LSAM	beta31	0.316	0.325	0.009 ± 0.004	0.030 ± 0.014	0.017 ± 0.001	0.131 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	0.173 ± 0.009	0.416 ± 0.013	0.316	0.000	-0.05	0.1331	0.1331	1.001	95.591 ± 0.673	0.515 ± 0.004	71.075 ± 1.487
LSAM	beta32	0.316	0.313	-0.003 ± 0.004	-0.010 ± 0.014	0.017 ± 0.001	0.132 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	0.175 ± 0.009	0.418 ± 0.013	0.308	-0.008	-2.41	0.1301	0.1303	1.000	96.882 ± 0.570	0.518 ± 0.004	69.140 ± 1.515
LSAM	beta33	0.139	0.156	0.017 ± 0.003	0.124 ± 0.023	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.490 ± 0.028	0.700 ± 0.023	0.147	0.008	5.48	0.0870	0.0873	0.947	95.161 ± 0.704	0.356 ± 0.004	41.720 ± 1.617
LSAM	beta34	0.101	0.099	-0.002 ± 0.002	-0.019 ± 0.016	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.234 ± 0.013	0.484 ± 0.015	0.099	-0.002	-1.64	0.0470	0.0470	0.991	95.376 ± 0.689	0.190 ± 0.002	59.677 ± 1.609
LSAM	beta35	0.101	0.104	0.003 ± 0.002	0.026 ± 0.017	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.281 ± 0.017	0.530 ± 0.018	0.104	0.003	3.22	0.0504	0.0505	0.929	96.129 ± 0.633	0.195 ± 0.002	61.183 ± 1.598
QML	beta31	0.316	0.344	0.028 ± 0.002	0.089 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.252 ± 0.007	0.342	0.026	8.11	0.0747	0.0789	0.970	91.935 ± 0.893	0.284 ± 0.001	99.032 ± 0.321
QML	beta32	0.316	0.338	0.022 ± 0.002	0.070 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.057 ± 0.003	0.239 ± 0.007	0.336	0.020	6.39	0.0752	0.0779	1.010	93.118 ± 0.830	0.286 ± 0.002	98.817 ± 0.355
QML	beta33	0.139	0.016	-0.123 ± 0.003	-0.882 ± 0.024	0.010 ± 0.001	0.101 ± 0.003	0.025 ± 0.001	0.159 ± 0.003	1.306 ± 0.052	1.143 ± 0.024	0.014	-0.125	-90.14	0.0992	0.1598	0.932	74.624 ± 1.427	0.369 ± 0.003	8.172 ± 0.898
QML	beta34	0.101	0.221	0.120 ± 0.002	1.184 ± 0.018	0.003 ± 0.000	0.056 ± 0.001	0.017 ± 0.001	0.132 ± 0.002	1.713 ± 0.050	1.309 ± 0.020	0.218	0.117	115.88	0.0546	0.1291	1.022	43.656 ± 1.626	0.226 ± 0.002	99.140 ± 0.303
QML	beta35	0.101	0.225	0.124 ± 0.002	1.226 ± 0.019	0.004 ± 0.000	0.059 ± 0.002	0.019 ± 0.001	0.137 ± 0.002	1.846 ± 0.057	1.359 ± 0.021	0.221	0.120	118.34	0.0565	0.1322	0.990	41.505 ± 1.616	0.230 ± 0.002	98.925 ± 0.338
UPI	beta31	0.316	0.331	0.015 ± 0.004	0.048 ± 0.014	0.018 ± 0.001	0.133 ± 0.003	0.018 ± 0.001	0.134 ± 0.004	0.179 ± 0.009	0.423 ± 0.013	0.327	0.011	3.47	0.1307	0.1312	1.013	95.591 ± 0.673	0.528 ± 0.005	72.796 ± 1.459
UPI	beta32	0.316	0.317	0.001 ± 0.004	0.002 ± 0.014	0.018 ± 0.001	0.133 ± 0.003	0.018 ± 0.001	0.133 ± 0.004	0.177 ± 0.009	0.420 ± 0.013	0.310	-0.006	-2.00	0.1332	0.1334	1.018	96.882 ± 0.570	0.531 ± 0.005	68.817 ± 1.519
UPI	beta33	0.139	0.156	0.017 ± 0.004	0.121 ± 0.027	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	0.668 ± 0.043	0.817 ± 0.029	0.140	0.001	0.74	0.0975	0.0976	1.034	94.624 ± 0.740	0.455 ± 0.016	27.849 ± 1.470
UPI	beta34	0.101	0.097	-0.004 ± 0.002	-0.041 ± 0.016	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.239 ± 0.013	0.489 ± 0.015	0.096	-0.005	-4.78	0.0459	0.0462	1.027	95.591 ± 0.673	0.198 ± 0.003	54.946 ± 1.632
UPI	beta35	0.101	0.104	0.003 ± 0.002	0.027 ± 0.018	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.298 ± 0.018	0.546 ± 0.019	0.103	0.002	2.39	0.0507	0.0508	0.950	96.022 ± 0.641	0.205 ± 0.003	57.419 ± 1.621

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S45

Simulation 1 - Condition 45: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.320	0.004 ± 0.003	0.012 ± 0.010	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.090 ± 0.004	0.300 ± 0.009	0.322	0.006	1.89	0.0959	0.0961	1.041	95.899 ± 0.643	0.386 ± 0.006	86.435 ± 1.110
LMS	beta32	0.316	0.327	0.011 ± 0.003	0.035 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.093 ± 0.005	0.305 ± 0.009	0.327	0.011	3.51	0.0916	0.0923	1.006	94.217 ± 0.757	0.378 ± 0.004	89.169 ± 1.008
LMS	beta33	0.139	0.087	-0.052 ± 0.004	-0.377 ± 0.026	0.012 ± 0.001	0.111 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	0.775 ± 0.047	0.880 ± 0.029	0.094	-0.045	-32.44	0.1001	0.1098	1.048	93.270 ± 0.812	0.454 ± 0.012	21.030 ± 1.321
LMS	beta34	0.101	0.169	0.068 ± 0.002	0.670 ± 0.021	0.004 ± 0.000	0.067 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.884 ± 0.048	0.940 ± 0.026	0.159	0.058	57.73	0.0591	0.0830	1.010	89.590 ± 0.990	0.264 ± 0.008	82.650 ± 1.228
LMS	beta35	0.101	0.163	0.062 ± 0.002	0.611 ± 0.019	0.004 ± 0.000	0.059 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.718 ± 0.037	0.847 ± 0.023	0.159	0.058	56.94	0.0547	0.0793	1.073	89.485 ± 0.995	0.249 ± 0.006	83.491 ± 1.204
LSAM	beta31	0.316	0.314	-0.002 ± 0.005	-0.007 ± 0.015	0.021 ± 0.001	0.144 ± 0.004	0.021 ± 0.001	0.144 ± 0.004	0.208 ± 0.011	0.456 ± 0.014	0.316	0.000	0.03	0.1407	0.1407	0.969	95.584 ± 0.666	0.548 ± 0.004	64.143 ± 1.555
LSAM	beta32	0.316	0.327	0.011 ± 0.005	0.035 ± 0.015	0.022 ± 0.001	0.147 ± 0.004	0.022 ± 0.001	0.147 ± 0.005	0.218 ± 0.011	0.466 ± 0.014	0.323	0.007	2.35	0.1402	0.1404	0.943	93.481 ± 0.801	0.544 ± 0.004	65.825 ± 1.538
LSAM	beta33	0.139	0.154	0.015 ± 0.003	0.111 ± 0.022	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.475 ± 0.024	0.689 ± 0.020	0.151	0.012	8.88	0.0925	0.0934	0.968	93.796 ± 0.782	0.359 ± 0.004	43.638 ± 1.608
LSAM	beta34	0.101	0.105	0.004 ± 0.002	0.037 ± 0.019	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.346 ± 0.019	0.588 ± 0.019	0.103	0.002	2.24	0.0545	0.0546	0.897	92.429 ± 0.858	0.209 ± 0.003	58.885 ± 1.596
LSAM	beta35	0.101	0.099	-0.002 ± 0.002	-0.021 ± 0.018	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.304 ± 0.017	0.551 ± 0.017	0.102	0.001	1.00	0.0516	0.0516	0.940	92.324 ± 0.863	0.205 ± 0.003	53.523 ± 1.617
QML	beta31	0.316	0.321	0.005 ± 0.003	0.017 ± 0.009	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.082 ± 0.004	0.287 ± 0.008	0.322	0.006	2.04	0.0911	0.0914	1.023	95.478 ± 0.674	0.363 ± 0.006	91.062 ± 0.925
QML	beta32	0.316	0.328	0.012 ± 0.003	0.038 ± 0.010	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.087 ± 0.004	0.295 ± 0.009	0.326	0.010	3.27	0.0906	0.0912	0.987	94.006 ± 0.770	0.359 ± 0.006	93.165 ± 0.818
QML	beta33	0.139	0.085	-0.054 ± 0.003	-0.392 ± 0.022	0.009 ± 0.000	0.095 ± 0.002	0.012 ± 0.001	0.110 ± 0.003	0.624 ± 0.032	0.790 ± 0.023	0.088	-0.051	-36.57	0.0882	0.1018	0.972	90.011 ± 0.972	0.363 ± 0.004	21.136 ± 1.324
QML	beta34	0.101	0.161	0.060 ± 0.002	0.596 ± 0.019	0.003 ± 0.000	0.058 ± 0.001	0.007 ± 0.000	0.083 ± 0.002	0.680 ± 0.032	0.825 ± 0.020	0.155	0.054	53.47	0.0562	0.0780	0.945	85.699 ± 1.135	0.214 ± 0.003	87.802 ± 1.061
QML	beta35	0.101	0.156	0.055 ± 0.002	0.546 ± 0.017	0.003 ± 0.000	0.052 ± 0.001	0.006 ± 0.000	0.076 ± 0.002	0.561 ± 0.025	0.749 ± 0.018	0.154	0.053	52.56	0.0497	0.0727	1.017	85.174 ± 1.152	0.206 ± 0.003	87.592 ± 1.069
UPI	beta31	0.316	0.318	0.002 ± 0.005	0.007 ± 0.015	0.021 ± 0.001	0.145 ± 0.004	0.021 ± 0.001	0.144 ± 0.004	0.209 ± 0.011	0.457 ± 0.014	0.322	0.006	1.94	0.1467	0.1468	0.964	95.478 ± 0.674	0.546 ± 0.004	66.141 ± 1.535
UPI	beta32	0.316	0.332	0.016 ± 0.005	0.050 ± 0.015	0.021 ± 0.001	0.146 ± 0.004	0.022 ± 0.001	0.147 ± 0.005	0.217 ± 0.012	0.466 ± 0.014	0.328	0.012	3.77	0.1424	0.1429	0.942	93.481 ± 0.801	0.541 ± 0.004	67.298 ± 1.521
UPI	beta33	0.139	0.152	0.013 ± 0.003	0.092 ± 0.024	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.560 ± 0.033	0.749 ± 0.025	0.143	0.004	3.22	0.0953	0.0954	1.032	95.163 ± 0.696	0.418 ± 0.008	33.964 ± 1.536
UPI	beta34	0.101	0.104	0.003 ± 0.002	0.028 ± 0.020	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.378 ± 0.022	0.615 ± 0.021	0.102	0.001	0.66	0.0562	0.0562	0.880	90.641 ± 0.944	0.214 ± 0.003	54.154 ± 1.616
UPI	beta35	0.101	0.098	-0.003 ± 0.002	-0.032 ± 0.018	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.311 ± 0.018	0.557 ± 0.018	0.100	-0.001	-1.36	0.0508	0.0508	0.949	92.955 ± 0.830	0.209 ± 0.003	50.578 ± 1.621

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S46

Simulation 1 - Condition 46: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.322	0.006	1.79	0.0565	0.0568	0.996	94.769 ± 0.706	0.227 ± 0.001	99.598 ± 0.201
LMS	beta32	0.316	0.323	0.007 ± 0.002	0.022 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.325	0.009	2.84	0.0570	0.0577	0.992	94.064 ± 0.749	0.228 ± 0.001	99.497 ± 0.224
LMS	beta33	0.139	0.087	-0.052 ± 0.002	-0.374 ± 0.016	0.005 ± 0.000	0.068 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.379 ± 0.019	0.616 ± 0.017	0.091	-0.048	-34.77	0.0633	0.0796	0.955	89.034 ± 0.991	0.255 ± 0.003	36.419 ± 1.526
LMS	beta34	0.101	0.161	0.060 ± 0.001	0.594 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.005 ± 0.000	0.071 ± 0.001	0.493 ± 0.019	0.702 ± 0.014	0.156	0.055	54.13	0.0345	0.0646	0.940	63.883 ± 1.524	0.140 ± 0.002	99.698 ± 0.174
LMS	beta35	0.101	0.162	0.061 ± 0.001	0.601 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.005 ± 0.000	0.072 ± 0.001	0.501 ± 0.019	0.708 ± 0.014	0.158	0.057	56.50	0.0348	0.0669	0.958	64.085 ± 1.522	0.142 ± 0.002	99.598 ± 0.201
LSAM	beta31	0.316	0.316	-0.000 ± 0.003	-0.001 ± 0.008	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.071 ± 0.003	0.266 ± 0.008	0.317	0.001	0.47	0.0818	0.0818	0.973	93.964 ± 0.755	0.321 ± 0.001	95.372 ± 0.666
LSAM	beta32	0.316	0.318	0.002 ± 0.003	0.006 ± 0.008	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.071 ± 0.003	0.266 ± 0.008	0.318	0.002	0.56	0.0813	0.0813	0.973	93.863 ± 0.761	0.321 ± 0.001	95.372 ± 0.666
LSAM	beta33	0.139	0.141	0.002 ± 0.002	0.014 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.159 ± 0.008	0.399 ± 0.012	0.138	-0.001	-0.97	0.0519	0.0519	0.936	93.360 ± 0.790	0.203 ± 0.001	76.459 ± 1.346
LSAM	beta34	0.101	0.102	0.001 ± 0.001	0.006 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.105 ± 0.006	0.324 ± 0.010	0.100	-0.001	-1.36	0.0296	0.0296	0.889	93.461 ± 0.784	0.114 ± 0.001	91.247 ± 0.896
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.006 ± 0.010	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.097 ± 0.005	0.312 ± 0.009	0.100	-0.001	-0.94	0.0303	0.0303	0.923	92.656 ± 0.827	0.114 ± 0.001	92.052 ± 0.858
QML	beta31	0.316	0.321	0.005 ± 0.002	0.015 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.321	0.005	1.51	0.0552	0.0554	0.992	94.366 ± 0.731	0.220 ± 0.001	99.899 ± 0.101
QML	beta32	0.316	0.323	0.007 ± 0.002	0.021 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.323	0.007	2.16	0.0557	0.0562	0.985	93.863 ± 0.761	0.220 ± 0.001	99.899 ± 0.101
QML	beta33	0.139	0.079	-0.060 ± 0.002	-0.430 ± 0.014	0.004 ± 0.000	0.061 ± 0.001	0.007 ± 0.000	0.085 ± 0.002	0.376 ± 0.016	0.613 ± 0.014	0.081	-0.058	-42.07	0.0577	0.0822	0.939	81.891 ± 1.221	0.224 ± 0.002	34.507 ± 1.508
QML	beta34	0.101	0.158	0.057 ± 0.001	0.563 ± 0.011	0.001 ± 0.000	0.035 ± 0.001	0.004 ± 0.000	0.067 ± 0.001	0.437 ± 0.016	0.661 ± 0.012	0.154	0.053	52.53	0.0329	0.0624	0.929	60.563 ± 1.550	0.128 ± 0.001	99.899 ± 0.101
QML	beta35	0.101	0.158	0.057 ± 0.001	0.567 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.004 ± 0.000	0.067 ± 0.001	0.436 ± 0.015	0.660 ± 0.012	0.156	0.055	54.31	0.0329	0.0640	0.949	60.362 ± 1.551	0.127 ± 0.001	100.000 ± 0.000
UPI	beta31	0.316	0.318	0.002 ± 0.003	0.006 ± 0.008	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.071 ± 0.003	0.266 ± 0.008	0.319	0.003	1.04	0.0828	0.0829	0.972	94.668 ± 0.713	0.320 ± 0.001	95.674 ± 0.645
UPI	beta32	0.316	0.321	0.005 ± 0.003	0.016 ± 0.008	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.071 ± 0.003	0.266 ± 0.007	0.322	0.006	1.98	0.0797	0.0800	0.972	93.461 ± 0.784	0.320 ± 0.001	95.875 ± 0.631
UPI	beta33	0.139	0.141	0.002 ± 0.002	0.013 ± 0.013	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.174 ± 0.009	0.418 ± 0.013	0.137	-0.002	-1.65	0.0541	0.0541	0.967	94.165 ± 0.743	0.220 ± 0.002	72.435 ± 1.417
UPI	beta34	0.101	0.101	0.000 ± 0.001	0.000 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.107 ± 0.006	0.327 ± 0.010	0.099	-0.002	-2.13	0.0306	0.0307	0.904	92.656 ± 0.827	0.117 ± 0.001	90.241 ± 0.941
UPI	beta35	0.101	0.101	-0.000 ± 0.001	-0.004 ± 0.010	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.099 ± 0.005	0.315 ± 0.009	0.099	-0.002	-1.72	0.0307	0.0308	0.928	92.455 ± 0.838	0.116 ± 0.001	91.247 ± 0.896

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S47

Simulation 1 - Condition 47: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.003	-0.004 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.314	-0.002	-0.63	0.0782	0.0782	0.999	95.477 ± 0.659	0.310 ± 0.001	95.477 ± 0.659
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.314	-0.002	-0.79	0.0766	0.0766	1.015	94.874 ± 0.699	0.311 ± 0.001	96.080 ± 0.615
LMS	beta33	0.139	0.115	-0.024 ± 0.002	-0.171 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.268 ± 0.013	0.518 ± 0.015	0.117	-0.022	-15.52	0.0630	0.0666	0.941	92.060 ± 0.857	0.251 ± 0.002	49.246 ± 1.585
LMS	beta34	0.101	0.125	0.024 ± 0.001	0.237 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.215 ± 0.012	0.463 ± 0.014	0.121	0.020	19.96	0.0367	0.0419	0.878	89.347 ± 0.978	0.139 ± 0.002	93.367 ± 0.789
LMS	beta35	0.101	0.124	0.023 ± 0.001	0.225 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.196 ± 0.010	0.443 ± 0.013	0.120	0.019	18.75	0.0364	0.0411	0.919	90.955 ± 0.909	0.139 ± 0.002	92.161 ± 0.852
LSAM	beta31	0.316	0.315	-0.001 ± 0.003	-0.004 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.004	0.312 ± 0.009	0.315	-0.001	-0.38	0.0982	0.0982	0.985	94.975 ± 0.693	0.381 ± 0.002	87.035 ± 1.065
LSAM	beta32	0.316	0.314	-0.002 ± 0.003	-0.007 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.095 ± 0.004	0.309 ± 0.009	0.311	-0.005	-1.44	0.0950	0.0951	0.999	94.573 ± 0.718	0.382 ± 0.002	88.543 ± 1.010
LSAM	beta33	0.139	0.138	-0.001 ± 0.002	-0.005 ± 0.015	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.226 ± 0.011	0.476 ± 0.014	0.137	-0.002	-1.16	0.0630	0.0630	0.909	93.266 ± 0.794	0.236 ± 0.002	64.020 ± 1.522
LSAM	beta34	0.101	0.103	0.002 ± 0.001	0.023 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.160 ± 0.008	0.400 ± 0.012	0.101	0.000	0.14	0.0360	0.0360	0.866	91.558 ± 0.881	0.137 ± 0.002	81.307 ± 1.236
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.013 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.151 ± 0.008	0.389 ± 0.012	0.100	-0.001	-0.78	0.0354	0.0355	0.895	92.462 ± 0.837	0.138 ± 0.002	81.206 ± 1.238
QML	beta31	0.316	0.315	-0.001 ± 0.003	-0.004 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.314	-0.002	-0.62	0.0790	0.0791	0.993	95.377 ± 0.666	0.309 ± 0.001	96.080 ± 0.615
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.312	-0.004	-1.22	0.0775	0.0776	1.011	94.673 ± 0.712	0.310 ± 0.001	96.382 ± 0.592
QML	beta33	0.139	0.113	-0.026 ± 0.002	-0.187 ± 0.015	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.268 ± 0.013	0.517 ± 0.015	0.115	-0.024	-17.21	0.0633	0.0676	0.935	91.457 ± 0.886	0.246 ± 0.002	49.447 ± 1.585
QML	beta34	0.101	0.124	0.023 ± 0.001	0.230 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.209 ± 0.011	0.457 ± 0.013	0.121	0.020	19.55	0.0369	0.0418	0.872	88.241 ± 1.021	0.136 ± 0.002	93.367 ± 0.789
QML	beta35	0.101	0.123	0.022 ± 0.001	0.220 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.192 ± 0.010	0.438 ± 0.012	0.119	0.018	18.29	0.0367	0.0410	0.913	90.754 ± 0.918	0.137 ± 0.002	92.161 ± 0.852
UPI	beta31	0.316	0.318	0.002 ± 0.003	0.006 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.097 ± 0.004	0.311 ± 0.009	0.318	0.002	0.74	0.0962	0.0962	0.973	93.869 ± 0.761	0.376 ± 0.002	88.744 ± 1.002
UPI	beta32	0.316	0.316	0.000 ± 0.003	0.000 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.004	0.306 ± 0.008	0.313	-0.003	-1.09	0.0939	0.0940	0.994	94.372 ± 0.731	0.377 ± 0.002	88.442 ± 1.014
UPI	beta33	0.139	0.139	0.000 ± 0.002	0.003 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.248 ± 0.013	0.498 ± 0.015	0.137	-0.002	-1.14	0.0650	0.0650	0.911	92.764 ± 0.821	0.247 ± 0.003	62.613 ± 1.534
UPI	beta34	0.101	0.102	0.001 ± 0.001	0.010 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.164 ± 0.009	0.405 ± 0.012	0.100	-0.001	-0.69	0.0378	0.0378	0.863	91.256 ± 0.896	0.139 ± 0.002	78.392 ± 1.305
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.002 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.146 ± 0.007	0.382 ± 0.012	0.100	-0.001	-0.61	0.0365	0.0365	0.916	92.261 ± 0.847	0.139 ± 0.002	79.899 ± 1.270

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S48

Simulation 1 - Condition 48: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.159 ± 0.004	0.317	0.001	0.45	0.0524	0.0524	0.988	95.100 ± 0.683	0.194 ± 0.001	99.900 ± 0.100
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.316	0.000	0.00	0.0484	0.0484	1.009	95.700 ± 0.641	0.193 ± 0.001	99.900 ± 0.100
LMS	beta33	0.139	0.124	-0.015 ± 0.001	-0.104 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.099 ± 0.004	0.315 ± 0.008	0.125	-0.014	-10.21	0.0424	0.0447	0.965	93.200 ± 0.796	0.156 ± 0.001	85.600 ± 1.110
LMS	beta34	0.101	0.122	0.021 ± 0.001	0.211 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.031 ± 0.001	0.093 ± 0.004	0.305 ± 0.007	0.121	0.020	19.39	0.0219	0.0294	0.973	85.200 ± 1.123	0.085 ± 0.001	99.900 ± 0.100
LMS	beta35	0.101	0.122	0.021 ± 0.001	0.206 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.120	0.019	19.28	0.0206	0.0283	0.945	87.200 ± 1.056	0.084 ± 0.001	100.000 ± 0.000
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.317	0.001	0.29	0.0615	0.0615	0.977	94.700 ± 0.708	0.234 ± 0.001	99.900 ± 0.100
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.316	0.000	-0.02	0.0608	0.0608	0.997	95.600 ± 0.649	0.233 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.143	0.004 ± 0.001	0.028 ± 0.009	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.078 ± 0.004	0.279 ± 0.008	0.142	0.003	2.22	0.0386	0.0387	0.955	95.000 ± 0.689	0.144 ± 0.001	96.200 ± 0.605
LSAM	beta34	0.101	0.101	0.000 ± 0.001	0.001 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.047 ± 0.002	0.216 ± 0.006	0.100	-0.001	-1.41	0.0211	0.0212	0.945	93.100 ± 0.801	0.081 ± 0.001	99.400 ± 0.244
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.007 ± 0.007	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.049 ± 0.002	0.221 ± 0.007	0.101	0.000	-0.10	0.0195	0.0195	0.918	92.300 ± 0.843	0.080 ± 0.001	99.100 ± 0.299
QML	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.159 ± 0.004	0.317	0.001	0.38	0.0515	0.0515	0.980	94.800 ± 0.702	0.193 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.315	-0.001	-0.35	0.0488	0.0488	1.003	95.600 ± 0.649	0.193 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.122	-0.017 ± 0.001	-0.121 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.101 ± 0.004	0.318 ± 0.008	0.123	-0.016	-11.56	0.0424	0.0454	0.960	92.300 ± 0.843	0.154 ± 0.001	85.300 ± 1.120
QML	beta34	0.101	0.122	0.021 ± 0.001	0.205 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.030 ± 0.001	0.089 ± 0.004	0.299 ± 0.007	0.120	0.019	18.92	0.0217	0.0289	0.968	85.400 ± 1.117	0.084 ± 0.001	99.900 ± 0.100
QML	beta35	0.101	0.122	0.021 ± 0.001	0.204 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.120	0.019	19.08	0.0206	0.0282	0.938	86.200 ± 1.091	0.083 ± 0.001	100.000 ± 0.000
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.318	0.002	0.67	0.0626	0.0626	0.968	94.200 ± 0.739	0.231 ± 0.001	99.900 ± 0.100
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.188 ± 0.005	0.318	0.002	0.53	0.0592	0.0592	0.986	95.300 ± 0.669	0.230 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.143	0.004 ± 0.001	0.030 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.080 ± 0.004	0.283 ± 0.008	0.142	0.003	2.27	0.0387	0.0388	0.966	93.900 ± 0.757	0.148 ± 0.001	95.800 ± 0.634
UPI	beta34	0.101	0.101	-0.000 ± 0.001	-0.005 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.048 ± 0.002	0.218 ± 0.006	0.100	-0.001	-1.26	0.0208	0.0208	0.945	92.500 ± 0.833	0.082 ± 0.001	99.500 ± 0.223
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.005 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.050 ± 0.003	0.224 ± 0.007	0.100	-0.001	-0.84	0.0196	0.0196	0.919	91.900 ± 0.863	0.082 ± 0.001	99.000 ± 0.315

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S49

Simulation 1 - Condition 49: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.005	-0.001 ± 0.015	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.005	0.124 ± 0.010	0.352 ± 0.016	0.315	-0.001	-0.23	0.1048	0.1048	1.026	94.206 ± 1.010	0.447 ± 0.010	84.486 ± 1.565
LMS	beta32	0.316	0.315	-0.001 ± 0.005	-0.005 ± 0.015	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	0.114 ± 0.007	0.337 ± 0.013	0.314	-0.002	-0.75	0.1174	0.1174	1.054	95.514 ± 0.895	0.440 ± 0.009	86.542 ± 1.475
LMS	beta33	0.139	0.100	-0.039 ± 0.008	-0.282 ± 0.061	0.038 ± 0.004	0.195 ± 0.010	0.039 ± 0.004	0.199 ± 0.012	2.042 ± 0.230	1.429 ± 0.086	0.117	-0.022	-15.99	0.1833	0.1846	1.077	96.449 ± 0.800	0.823 ± 0.037	21.682 ± 1.782
LMS	beta34	0.101	0.097	-0.004 ± 0.007	-0.036 ± 0.069	0.026 ± 0.003	0.162 ± 0.009	0.026 ± 0.003	0.162 ± 0.009	2.579 ± 0.266	1.606 ± 0.090	0.074	-0.027	-26.75	0.1301	0.1329	1.001	86.729 ± 1.467	0.637 ± 0.035	3.925 ± 0.840
LMS	beta35	0.101	0.095	-0.006 ± 0.007	-0.057 ± 0.065	0.023 ± 0.002	0.152 ± 0.008	0.023 ± 0.002	0.152 ± 0.008	2.256 ± 0.228	1.502 ± 0.083	0.073	-0.028	-27.85	0.1219	0.1251	1.037	88.785 ± 1.364	0.617 ± 0.031	4.486 ± 0.895
LSAM	beta31	0.316	0.322	0.006 ± 0.005	0.019 ± 0.017	0.015 ± 0.001	0.122 ± 0.005	0.015 ± 0.001	0.122 ± 0.005	0.149 ± 0.011	0.386 ± 0.017	0.316	0.000	0.02	0.1223	0.1223	1.319	97.196 ± 0.714	0.631 ± 0.029	73.645 ± 1.905
LSAM	beta32	0.316	0.321	0.005 ± 0.005	0.016 ± 0.016	0.014 ± 0.001	0.120 ± 0.004	0.014 ± 0.001	0.120 ± 0.005	0.145 ± 0.010	0.380 ± 0.015	0.328	0.012	3.74	0.1331	0.1337	1.372	97.570 ± 0.666	0.646 ± 0.033	71.028 ± 1.961
LSAM	beta33	0.139	0.163	0.024 ± 0.008	0.176 ± 0.059	0.037 ± 0.003	0.191 ± 0.009	0.037 ± 0.004	0.192 ± 0.010	1.917 ± 0.182	1.384 ± 0.073	0.142	0.003	2.10	0.1836	0.1836	1.493	98.131 ± 0.586	1.118 ± 0.077	12.336 ± 1.422
LSAM	beta34	0.101	0.086	-0.015 ± 0.010	-0.145 ± 0.101	0.056 ± 0.005	0.237 ± 0.011	0.056 ± 0.005	0.237 ± 0.012	5.498 ± 0.497	2.345 ± 0.118	0.097	-0.004	-4.32	0.2090	0.2091	1.533	98.318 ± 0.556	1.421 ± 0.111	0.935 ± 0.416
LSAM	beta35	0.101	0.100	-0.001 ± 0.011	-0.006 ± 0.107	0.063 ± 0.005	0.251 ± 0.011	0.063 ± 0.005	0.250 ± 0.012	6.150 ± 0.533	2.480 ± 0.121	0.108	0.007	6.63	0.2306	0.2307	1.463	98.879 ± 0.455	1.438 ± 0.095	1.495 ± 0.525
QML	beta31	0.316	0.306	-0.010 ± 0.004	-0.032 ± 0.014	0.010 ± 0.001	0.101 ± 0.004	0.010 ± 0.001	0.101 ± 0.004	0.103 ± 0.007	0.321 ± 0.013	0.303	-0.013	-4.00	0.1001	0.1009	0.992	93.645 ± 1.055	0.392 ± 0.007	90.841 ± 1.247
QML	beta32	0.316	0.304	-0.012 ± 0.004	-0.037 ± 0.013	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	0.097 ± 0.006	0.311 ± 0.011	0.308	-0.008	-2.61	0.1051	0.1055	1.091	94.019 ± 1.025	0.418 ± 0.031	89.346 ± 1.334
QML	beta33	0.139	0.133	-0.006 ± 0.006	-0.043 ± 0.040	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	0.841 ± 0.051	0.917 ± 0.034	0.132	-0.007	-4.85	0.1382	0.1384	1.106	96.075 ± 0.840	0.553 ± 0.011	22.243 ± 1.798
QML	beta34	0.101	0.047	-0.054 ± 0.004	-0.531 ± 0.042	0.010 ± 0.001	0.098 ± 0.004	0.012 ± 0.001	0.112 ± 0.004	1.220 ± 0.071	1.105 ± 0.037	0.048	-0.053	-52.40	0.0977	0.1111	1.035	84.299 ± 1.573	0.397 ± 0.008	3.925 ± 0.840
QML	beta35	0.101	0.049	-0.052 ± 0.004	-0.510 ± 0.041	0.009 ± 0.001	0.096 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	1.162 ± 0.070	1.078 ± 0.037	0.050	-0.051	-50.97	0.0963	0.1092	1.076	86.729 ± 1.467	0.405 ± 0.008	5.047 ± 0.946
UPI	beta31	0.316	0.322	0.006 ± 0.005	0.018 ± 0.017	0.015 ± 0.001	0.124 ± 0.005	0.015 ± 0.001	0.124 ± 0.006	0.155 ± 0.013	0.394 ± 0.018	0.315	-0.001	-0.28	0.1262	0.1262	1.599	97.383 ± 0.690	0.780 ± 0.043	62.617 ± 2.092
UPI	beta32	0.316	0.322	0.006 ± 0.005	0.018 ± 0.017	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.005	0.148 ± 0.011	0.385 ± 0.016	0.321	0.005	1.64	0.1301	0.1302	1.685	97.944 ± 0.614	0.804 ± 0.044	61.308 ± 2.106
UPI	beta33	0.139	0.180	0.041 ± 0.009	0.298 ± 0.062	0.040 ± 0.003	0.199 ± 0.009	0.041 ± 0.004	0.204 ± 0.010	2.143 ± 0.186	1.464 ± 0.071	0.146	0.007	4.89	0.1917	0.1919	1.803	98.879 ± 0.455	1.410 ± 0.079	5.047 ± 0.946
UPI	beta34	0.101	0.095	-0.006 ± 0.013	-0.056 ± 0.128	0.089 ± 0.008	0.298 ± 0.014	0.089 ± 0.008	0.298 ± 0.015	8.704 ± 0.814	2.950 ± 0.153	0.079	-0.022	-22.17	0.2381	0.2391	1.990	96.262 ± 0.820	2.327 ± 0.159	0.374 ± 0.264
UPI	beta35	0.101	0.083	-0.018 ± 0.012	-0.180 ± 0.123	0.082 ± 0.008	0.287 ± 0.014	0.082 ± 0.008	0.287 ± 0.015	8.082 ± 0.770	2.843 ± 0.149	0.067	-0.034	-33.25	0.2477	0.2500	1.975	95.514 ± 0.895	2.220 ± 0.154	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S50

Simulation 1 - Condition 50: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.196 ± 0.006	0.309	-0.007	-2.15	0.0629	0.0632	1.058	97.213 ± 0.561	0.257 ± 0.002	99.419 ± 0.259
LMS	beta32	0.316	0.309	-0.007 ± 0.002	-0.021 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.206 ± 0.006	0.310	-0.006	-1.83	0.0660	0.0663	1.016	94.425 ± 0.782	0.258 ± 0.003	99.535 ± 0.232
LMS	beta33	0.139	0.141	0.002 ± 0.003	0.014 ± 0.023	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	0.437 ± 0.026	0.661 ± 0.023	0.138	-0.001	-0.68	0.0946	0.0946	1.208	96.167 ± 0.654	0.435 ± 0.017	44.135 ± 1.692
LMS	beta34	0.101	0.064	-0.037 ± 0.003	-0.366 ± 0.025	0.005 ± 0.000	0.074 ± 0.003	0.007 ± 0.000	0.083 ± 0.002	0.668 ± 0.032	0.818 ± 0.023	0.060	-0.041	-40.12	0.0689	0.0799	1.200	83.740 ± 1.258	0.348 ± 0.015	1.974 ± 0.474
LMS	beta35	0.101	0.066	-0.035 ± 0.003	-0.349 ± 0.026	0.006 ± 0.000	0.076 ± 0.003	0.007 ± 0.000	0.084 ± 0.002	0.687 ± 0.034	0.829 ± 0.023	0.061	-0.040	-39.54	0.0702	0.0808	1.160	83.391 ± 1.268	0.345 ± 0.016	2.787 ± 0.561
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.210 ± 0.006	0.316	0.000	-0.04	0.0654	0.0654	1.081	98.026 ± 0.474	0.281 ± 0.002	98.722 ± 0.383
LSAM	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.008	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.050 ± 0.002	0.223 ± 0.007	0.313	-0.003	-0.87	0.0708	0.0708	1.021	95.935 ± 0.673	0.282 ± 0.002	99.071 ± 0.327
LSAM	beta33	0.139	0.149	0.010 ± 0.003	0.070 ± 0.024	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	0.506 ± 0.031	0.712 ± 0.025	0.141	0.002	1.23	0.0950	0.0950	1.054	96.980 ± 0.583	0.407 ± 0.007	35.772 ± 1.634
LSAM	beta34	0.101	0.097	-0.004 ± 0.004	-0.036 ± 0.041	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	1.446 ± 0.097	1.202 ± 0.045	0.102	0.001	0.82	0.1189	0.1189	1.104	97.561 ± 0.526	0.526 ± 0.010	5.226 ± 0.758
LSAM	beta35	0.101	0.099	-0.002 ± 0.004	-0.017 ± 0.042	0.016 ± 0.001	0.125 ± 0.004	0.016 ± 0.001	0.125 ± 0.005	1.529 ± 0.098	1.236 ± 0.045	0.101	0.000	0.36	0.1206	0.1206	1.074	98.026 ± 0.474	0.526 ± 0.009	6.852 ± 0.861
QML	beta31	0.316	0.304	-0.012 ± 0.002	-0.038 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.188 ± 0.005	0.301	-0.015	-4.69	0.0581	0.0600	1.040	96.051 ± 0.664	0.238 ± 0.001	99.884 ± 0.116
QML	beta32	0.316	0.301	-0.015 ± 0.002	-0.048 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.202 ± 0.006	0.299	-0.017	-5.26	0.0618	0.0640	0.979	92.799 ± 0.881	0.238 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.140	0.001 ± 0.003	0.009 ± 0.020	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.334 ± 0.016	0.578 ± 0.017	0.134	-0.005	-3.43	0.0820	0.0821	1.047	95.819 ± 0.682	0.330 ± 0.003	44.367 ± 1.693
QML	beta34	0.101	0.039	-0.062 ± 0.002	-0.610 ± 0.019	0.003 ± 0.000	0.055 ± 0.001	0.007 ± 0.000	0.083 ± 0.002	0.672 ± 0.026	0.820 ± 0.017	0.040	-0.061	-60.19	0.0576	0.0838	1.108	76.887 ± 1.437	0.241 ± 0.002	4.762 ± 0.726
QML	beta35	0.101	0.041	-0.060 ± 0.002	-0.594 ± 0.020	0.003 ± 0.000	0.058 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.684 ± 0.027	0.827 ± 0.018	0.041	-0.060	-59.17	0.0592	0.0841	1.059	78.746 ± 1.394	0.242 ± 0.002	6.620 ± 0.847
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.003	0.220 ± 0.007	0.315	-0.001	-0.21	0.0723	0.0723	1.151	98.026 ± 0.474	0.313 ± 0.005	97.096 ± 0.572
UPI	beta32	0.316	0.315	-0.001 ± 0.003	-0.003 ± 0.008	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.315	-0.001	-0.32	0.0744	0.0744	1.095	96.167 ± 0.654	0.316 ± 0.005	96.748 ± 0.605
UPI	beta33	0.139	0.159	0.020 ± 0.004	0.145 ± 0.027	0.012 ± 0.001	0.111 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	0.656 ± 0.045	0.810 ± 0.031	0.147	0.008	5.42	0.1017	0.1020	1.160	98.258 ± 0.446	0.504 ± 0.015	27.526 ± 1.522
UPI	beta34	0.101	0.097	-0.004 ± 0.005	-0.043 ± 0.051	0.023 ± 0.002	0.152 ± 0.005	0.023 ± 0.002	0.152 ± 0.006	2.256 ± 0.153	1.502 ± 0.057	0.094	-0.007	-7.11	0.1350	0.1352	1.324	97.445 ± 0.538	0.788 ± 0.027	0.813 ± 0.306
UPI	beta35	0.101	0.102	0.001 ± 0.005	0.012 ± 0.054	0.025 ± 0.002	0.159 ± 0.005	0.025 ± 0.002	0.159 ± 0.006	2.480 ± 0.162	1.575 ± 0.058	0.096	-0.005	-4.96	0.1363	0.1364	1.288	96.980 ± 0.583	0.803 ± 0.033	2.207 ± 0.501

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S51

Simulation 1 - Condition 51: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.002	0.230 ± 0.007	0.312	-0.004	-1.13	0.0720	0.0721	1.022	95.258 ± 0.682	0.291 ± 0.002	98.660 ± 0.369
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.228 ± 0.007	0.310	-0.006	-1.76	0.0692	0.0694	1.032	95.464 ± 0.668	0.291 ± 0.003	98.454 ± 0.396
LMS	beta33	0.139	0.139	0.000 ± 0.003	0.002 ± 0.022	0.009 ± 0.000	0.096 ± 0.003	0.009 ± 0.000	0.096 ± 0.003	0.473 ± 0.025	0.688 ± 0.022	0.147	0.008	5.96	0.0883	0.0887	1.058	95.567 ± 0.661	0.397 ± 0.010	42.680 ± 1.588
LMS	beta34	0.101	0.076	-0.025 ± 0.003	-0.243 ± 0.027	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.785 ± 0.038	0.886 ± 0.025	0.070	-0.031	-30.86	0.0798	0.0856	1.048	86.392 ± 1.101	0.354 ± 0.010	7.113 ± 0.825
LMS	beta35	0.101	0.077	-0.024 ± 0.003	-0.237 ± 0.027	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.749 ± 0.037	0.865 ± 0.025	0.069	-0.032	-32.08	0.0733	0.0801	1.069	88.969 ± 1.006	0.352 ± 0.008	5.361 ± 0.723
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.241 ± 0.007	0.319	0.003	0.86	0.0774	0.0775	1.004	95.979 ± 0.631	0.300 ± 0.001	98.144 ± 0.433
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.005 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.232 ± 0.007	0.317	0.001	0.42	0.0728	0.0729	1.037	95.670 ± 0.653	0.298 ± 0.001	98.144 ± 0.433
LSAM	beta33	0.139	0.146	0.007 ± 0.003	0.052 ± 0.021	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.434 ± 0.021	0.659 ± 0.019	0.146	0.007	4.93	0.0857	0.0860	0.996	95.567 ± 0.661	0.357 ± 0.003	41.443 ± 1.582
LSAM	beta34	0.101	0.099	-0.002 ± 0.004	-0.017 ± 0.036	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	1.276 ± 0.069	1.129 ± 0.035	0.099	-0.002	-1.49	0.1052	0.1052	0.989	94.948 ± 0.703	0.442 ± 0.004	12.990 ± 1.079
LSAM	beta35	0.101	0.100	-0.001 ± 0.004	-0.011 ± 0.035	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	1.192 ± 0.064	1.092 ± 0.034	0.100	-0.001	-0.50	0.1067	0.1067	1.025	95.876 ± 0.638	0.443 ± 0.004	9.897 ± 0.959
QML	beta31	0.316	0.310	-0.006 ± 0.002	-0.019 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.226 ± 0.006	0.309	-0.007	-2.24	0.0713	0.0717	1.003	94.639 ± 0.723	0.280 ± 0.001	99.175 ± 0.290
QML	beta32	0.316	0.308	-0.008 ± 0.002	-0.024 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.307	-0.009	-2.91	0.0685	0.0691	1.018	94.845 ± 0.710	0.279 ± 0.001	99.072 ± 0.308
QML	beta33	0.139	0.141	0.002 ± 0.003	0.017 ± 0.020	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.400 ± 0.019	0.633 ± 0.018	0.146	0.007	5.00	0.0841	0.0844	1.009	95.361 ± 0.675	0.348 ± 0.004	44.124 ± 1.594
QML	beta34	0.101	0.061	-0.040 ± 0.002	-0.395 ± 0.024	0.006 ± 0.000	0.075 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.701 ± 0.029	0.837 ± 0.020	0.059	-0.042	-41.69	0.0751	0.0861	0.992	84.948 ± 1.148	0.290 ± 0.003	8.969 ± 0.917
QML	beta35	0.101	0.061	-0.040 ± 0.002	-0.395 ± 0.023	0.005 ± 0.000	0.071 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.649 ± 0.028	0.806 ± 0.020	0.060	-0.041	-40.18	0.0650	0.0767	1.046	86.907 ± 1.083	0.291 ± 0.003	7.423 ± 0.842
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.244 ± 0.007	0.319	0.003	0.90	0.0741	0.0741	0.997	95.464 ± 0.668	0.301 ± 0.002	98.144 ± 0.433
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.315	-0.001	-0.18	0.0732	0.0732	1.025	95.052 ± 0.696	0.300 ± 0.002	98.144 ± 0.433
UPI	beta33	0.139	0.151	0.012 ± 0.003	0.083 ± 0.022	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.496 ± 0.028	0.705 ± 0.023	0.148	0.009	6.63	0.0913	0.0918	1.010	95.876 ± 0.638	0.385 ± 0.003	37.835 ± 1.557
UPI	beta34	0.101	0.103	0.002 ± 0.004	0.023 ± 0.042	0.018 ± 0.001	0.133 ± 0.004	0.018 ± 0.001	0.133 ± 0.004	1.733 ± 0.099	1.317 ± 0.043	0.098	-0.003	-3.00	0.1173	0.1173	1.065	96.289 ± 0.607	0.555 ± 0.009	7.423 ± 0.842
UPI	beta35	0.101	0.099	-0.002 ± 0.004	-0.019 ± 0.040	0.016 ± 0.001	0.126 ± 0.004	0.016 ± 0.001	0.126 ± 0.004	1.562 ± 0.087	1.250 ± 0.040	0.097	-0.004	-3.61	0.1113	0.1113	1.099	96.392 ± 0.599	0.544 ± 0.008	6.082 ± 0.767

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S52

Simulation 1 - Condition 52: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.312	-0.004 ± 0.001	-0.014 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.312	-0.004	-1.33	0.0440	0.0442	1.008	95.745 ± 0.642	0.178 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.313	-0.003 ± 0.001	-0.009 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.313	-0.003	-1.10	0.0459	0.0460	1.026	95.643 ± 0.650	0.178 ± 0.001	99.899 ± 0.101
LMS	beta33	0.139	0.140	0.001 ± 0.002	0.008 ± 0.013	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.178 ± 0.009	0.422 ± 0.013	0.142	0.003	1.82	0.0558	0.0558	1.003	95.441 ± 0.664	0.231 ± 0.003	70.618 ± 1.450
LMS	beta34	0.101	0.070	-0.031 ± 0.002	-0.305 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.059 ± 0.001	0.344 ± 0.013	0.587 ± 0.013	0.066	-0.035	-35.08	0.0469	0.0588	1.023	84.802 ± 1.143	0.203 ± 0.002	19.048 ± 1.250
LMS	beta35	0.101	0.073	-0.028 ± 0.002	-0.277 ± 0.017	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.059 ± 0.001	0.347 ± 0.013	0.589 ± 0.013	0.067	-0.034	-33.55	0.0484	0.0591	1.002	85.106 ± 1.133	0.206 ± 0.003	20.162 ± 1.277
LSAM	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.315	-0.001	-0.16	0.0446	0.0446	1.016	95.643 ± 0.650	0.183 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.318	0.002	0.74	0.0465	0.0466	1.044	96.960 ± 0.546	0.183 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.139	-0.000 ± 0.002	-0.001 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.155 ± 0.007	0.394 ± 0.011	0.139	0.000	-0.12	0.0543	0.0543	0.998	95.238 ± 0.678	0.214 ± 0.001	70.922 ± 1.445
LSAM	beta34	0.101	0.099	-0.002 ± 0.002	-0.024 ± 0.021	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.448 ± 0.020	0.669 ± 0.018	0.098	-0.003	-3.45	0.0649	0.0650	1.000	95.035 ± 0.691	0.265 ± 0.001	29.179 ± 1.447
LSAM	beta35	0.101	0.102	0.001 ± 0.002	0.010 ± 0.022	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.457 ± 0.022	0.676 ± 0.020	0.098	-0.003	-2.69	0.0673	0.0673	0.991	95.339 ± 0.671	0.265 ± 0.001	30.091 ± 1.460
QML	beta31	0.316	0.308	-0.008 ± 0.001	-0.026 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.307	-0.009	-2.78	0.0425	0.0434	1.089	94.934 ± 0.698	0.188 ± 0.013	99.797 ± 0.143
QML	beta32	0.316	0.309	-0.007 ± 0.001	-0.023 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.309	-0.007	-2.20	0.0437	0.0442	1.049	95.137 ± 0.685	0.179 ± 0.003	99.797 ± 0.143
QML	beta33	0.139	0.138	-0.001 ± 0.002	-0.010 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.157 ± 0.008	0.396 ± 0.011	0.138	-0.001	-0.40	0.0533	0.0533	0.995	95.137 ± 0.685	0.215 ± 0.001	71.834 ± 1.432
QML	beta34	0.101	0.059	-0.042 ± 0.001	-0.415 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.368 ± 0.013	0.607 ± 0.012	0.057	-0.044	-43.58	0.0423	0.0611	1.018	80.243 ± 1.267	0.178 ± 0.001	20.770 ± 1.291
QML	beta35	0.101	0.061	-0.040 ± 0.001	-0.394 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.359 ± 0.013	0.599 ± 0.012	0.058	-0.043	-42.20	0.0441	0.0613	0.996	80.547 ± 1.260	0.178 ± 0.001	21.986 ± 1.318
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.316	0.000	0.00	0.0452	0.0452	0.999	95.238 ± 0.678	0.180 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.318	0.002	0.51	0.0468	0.0468	1.020	95.846 ± 0.635	0.180 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.141	0.002 ± 0.002	0.012 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.165 ± 0.007	0.406 ± 0.011	0.138	-0.001	-0.44	0.0576	0.0576	0.999	95.441 ± 0.664	0.221 ± 0.001	70.517 ± 1.451
UPI	beta34	0.101	0.099	-0.002 ± 0.002	-0.019 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.498 ± 0.025	0.705 ± 0.021	0.098	-0.003	-2.58	0.0684	0.0685	1.032	95.441 ± 0.664	0.288 ± 0.002	23.810 ± 1.356
UPI	beta35	0.101	0.102	0.001 ± 0.002	0.014 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.494 ± 0.025	0.703 ± 0.021	0.098	-0.003	-2.48	0.0694	0.0695	1.029	96.049 ± 0.620	0.287 ± 0.002	25.532 ± 1.388

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S53

Simulation 1 - Condition 53: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.314	-0.002	-0.75	0.0593	0.0594	0.961	94.494 ± 0.722	0.226 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.006	0.313	-0.003	-0.93	0.0607	0.0608	0.974	95.295 ± 0.670	0.227 ± 0.001	99.700 ± 0.173
LMS	beta33	0.139	0.141	0.002 ± 0.002	0.016 ± 0.016	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.241 ± 0.012	0.491 ± 0.014	0.143	0.004	2.74	0.0607	0.0608	0.969	94.595 ± 0.715	0.259 ± 0.002	61.562 ± 1.539
LMS	beta34	0.101	0.083	-0.018 ± 0.002	-0.174 ± 0.020	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.438 ± 0.019	0.662 ± 0.017	0.080	-0.021	-20.79	0.0660	0.0692	0.976	90.891 ± 0.910	0.247 ± 0.002	21.321 ± 1.296
LMS	beta35	0.101	0.087	-0.014 ± 0.002	-0.139 ± 0.020	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.421 ± 0.019	0.648 ± 0.018	0.083	-0.018	-17.82	0.0622	0.0648	0.988	92.092 ± 0.854	0.248 ± 0.002	21.221 ± 1.294
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.315	-0.001	-0.17	0.0602	0.0602	0.958	94.394 ± 0.728	0.229 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.006	0.318	0.002	0.62	0.0612	0.0612	0.976	95.696 ± 0.642	0.230 ± 0.001	99.700 ± 0.173
LSAM	beta33	0.139	0.142	0.003 ± 0.002	0.019 ± 0.015	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.222 ± 0.010	0.471 ± 0.013	0.143	0.004	2.96	0.0621	0.0622	0.967	93.894 ± 0.758	0.248 ± 0.001	62.062 ± 1.535
LSAM	beta34	0.101	0.098	-0.003 ± 0.002	-0.025 ± 0.023	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.539 ± 0.024	0.734 ± 0.020	0.096	-0.005	-4.81	0.0755	0.0756	0.975	94.194 ± 0.740	0.283 ± 0.001	27.127 ± 1.407
LSAM	beta35	0.101	0.103	0.002 ± 0.002	0.015 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.521 ± 0.024	0.722 ± 0.020	0.102	0.001	0.79	0.0713	0.0713	0.986	95.596 ± 0.649	0.282 ± 0.001	27.928 ± 1.419
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.008 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.312	-0.004	-1.37	0.0593	0.0595	0.959	94.394 ± 0.728	0.225 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.006	0.312	-0.004	-1.37	0.0606	0.0607	0.972	95.295 ± 0.670	0.226 ± 0.001	99.700 ± 0.173
QML	beta33	0.139	0.141	0.002 ± 0.002	0.012 ± 0.015	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.233 ± 0.011	0.483 ± 0.014	0.142	0.003	1.81	0.0604	0.0605	0.967	94.494 ± 0.722	0.255 ± 0.002	61.862 ± 1.537
QML	beta34	0.101	0.080	-0.021 ± 0.002	-0.212 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.424 ± 0.017	0.651 ± 0.016	0.077	-0.024	-23.41	0.0645	0.0687	0.975	90.090 ± 0.945	0.238 ± 0.002	22.122 ± 1.313
QML	beta35	0.101	0.083	-0.018 ± 0.002	-0.178 ± 0.019	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.402 ± 0.017	0.634 ± 0.016	0.080	-0.021	-20.77	0.0606	0.0642	0.988	90.991 ± 0.906	0.238 ± 0.002	23.123 ± 1.334
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.315	-0.001	-0.29	0.0603	0.0603	0.919	93.293 ± 0.791	0.221 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.006	0.318	0.002	0.50	0.0616	0.0616	0.938	94.795 ± 0.703	0.222 ± 0.001	99.600 ± 0.200
UPI	beta33	0.139	0.142	0.003 ± 0.002	0.024 ± 0.015	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.230 ± 0.010	0.480 ± 0.013	0.143	0.004	2.89	0.0619	0.0620	0.962	93.493 ± 0.780	0.251 ± 0.001	63.463 ± 1.524
UPI	beta34	0.101	0.099	-0.002 ± 0.002	-0.016 ± 0.024	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.574 ± 0.026	0.758 ± 0.021	0.099	-0.002	-1.89	0.0785	0.0785	0.981	95.095 ± 0.683	0.294 ± 0.002	25.626 ± 1.381
UPI	beta35	0.101	0.103	0.002 ± 0.002	0.024 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.562 ± 0.027	0.750 ± 0.021	0.102	0.001	1.15	0.0746	0.0746	0.985	94.995 ± 0.690	0.292 ± 0.002	26.026 ± 1.388

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S54

Simulation 1 - Condition 54: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.314	-0.002	-0.62	0.0357	0.0358	0.997	95.095 ± 0.683	0.143 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.312	-0.004 ± 0.001	-0.012 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.311	-0.005	-1.68	0.0369	0.0373	0.972	93.493 ± 0.780	0.143 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.142	0.003 ± 0.001	0.023 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.092 ± 0.004	0.303 ± 0.009	0.142	0.003	2.48	0.0412	0.0414	0.986	94.494 ± 0.722	0.162 ± 0.001	89.890 ± 0.954
LMS	beta34	0.101	0.084	-0.017 ± 0.001	-0.171 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.174 ± 0.007	0.417 ± 0.011	0.082	-0.019	-18.98	0.0378	0.0424	1.030	90.591 ± 0.924	0.155 ± 0.001	56.957 ± 1.567
LMS	beta35	0.101	0.083	-0.018 ± 0.001	-0.177 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.184 ± 0.007	0.428 ± 0.010	0.083	-0.018	-17.91	0.0407	0.0446	1.001	90.991 ± 0.906	0.155 ± 0.001	55.455 ± 1.572
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.317	0.001	0.40	0.0372	0.0372	0.999	94.995 ± 0.690	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.120 ± 0.003	0.314	-0.002	-0.66	0.0380	0.0380	0.970	93.794 ± 0.763	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.140	0.001 ± 0.001	0.007 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.139	0.000	-0.07	0.0397	0.0397	0.990	94.895 ± 0.696	0.157 ± 0.001	92.292 ± 0.844
LSAM	beta34	0.101	0.102	0.001 ± 0.001	0.009 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.196 ± 0.009	0.442 ± 0.012	0.103	0.002	1.78	0.0442	0.0442	1.018	95.596 ± 0.649	0.178 ± 0.001	61.862 ± 1.537
LSAM	beta35	0.101	0.101	0.000 ± 0.001	0.003 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.205 ± 0.009	0.453 ± 0.012	0.102	0.001	0.51	0.0477	0.0477	0.995	95.195 ± 0.677	0.179 ± 0.001	60.360 ± 1.548
QML	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.313	-0.003	-1.03	0.0356	0.0358	0.996	95.195 ± 0.677	0.142 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.310	-0.006	-2.00	0.0371	0.0376	0.972	93.594 ± 0.775	0.143 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.141	0.002 ± 0.001	0.013 ± 0.009	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.141	0.002	1.20	0.0409	0.0409	0.986	94.394 ± 0.728	0.161 ± 0.001	90.290 ± 0.937
QML	beta34	0.101	0.080	-0.021 ± 0.001	-0.203 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.177 ± 0.007	0.421 ± 0.010	0.079	-0.022	-21.30	0.0368	0.0426	1.030	89.890 ± 0.954	0.150 ± 0.001	55.856 ± 1.571
QML	beta35	0.101	0.080	-0.021 ± 0.001	-0.208 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.187 ± 0.007	0.433 ± 0.010	0.080	-0.021	-21.00	0.0400	0.0452	0.998	88.889 ± 0.994	0.150 ± 0.001	54.755 ± 1.575
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.317	0.001	0.33	0.0370	0.0370	0.965	94.494 ± 0.722	0.140 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.314	-0.002	-0.57	0.0377	0.0378	0.936	92.593 ± 0.829	0.140 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.140	0.001 ± 0.001	0.009 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.140	0.001	0.89	0.0399	0.0399	0.990	94.494 ± 0.722	0.158 ± 0.001	92.092 ± 0.854
UPI	beta34	0.101	0.102	0.001 ± 0.001	0.011 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.203 ± 0.010	0.450 ± 0.013	0.103	0.002	1.61	0.0461	0.0461	1.021	95.495 ± 0.656	0.182 ± 0.001	61.261 ± 1.541
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.004 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.213 ± 0.009	0.461 ± 0.013	0.101	0.000	0.36	0.0477	0.0477	0.996	95.195 ± 0.677	0.182 ± 0.001	59.359 ± 1.554

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S55

Simulation 1 - Condition 55: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.312	-0.004 ± 0.010	-0.014 ± 0.032	0.023 ± 0.003	0.151 ± 0.011	0.023 ± 0.003	0.150 ± 0.012	0.226 ± 0.032	0.476 ± 0.038	0.302	-0.014	-4.46	0.1337	0.1344	1.172	94.907 ± 1.496	0.692 ± 0.055	69.907 ± 3.121
LMS	beta32	0.316	0.322	0.006 ± 0.010	0.018 ± 0.033	0.023 ± 0.003	0.153 ± 0.009	0.023 ± 0.003	0.153 ± 0.010	0.235 ± 0.027	0.484 ± 0.033	0.315	-0.001	-0.30	0.1568	0.1568	1.167	93.981 ± 1.618	0.701 ± 0.056	67.593 ± 3.185
LMS	beta33	0.139	0.045	-0.094 ± 0.034	-0.677 ± 0.248	0.256 ± 0.039	0.506 ± 0.039	0.263 ± 0.044	0.513 ± 0.047	13.637 ± 2.296	3.693 ± 0.338	0.179	0.040	29.04	0.2586	0.2617	1.016	90.741 ± 1.972	2.015 ± 0.216	31.944 ± 3.173
LMS	beta34	0.101	0.146	0.045 ± 0.017	0.450 ± 0.171	0.064 ± 0.009	0.253 ± 0.019	0.066 ± 0.010	0.257 ± 0.022	6.458 ± 1.024	2.541 ± 0.221	0.090	-0.011	-11.32	0.1748	0.1751	1.461	85.648 ± 2.386	1.450 ± 0.234	5.556 ± 1.559
LMS	beta35	0.101	0.186	0.085 ± 0.024	0.843 ± 0.233	0.120 ± 0.018	0.346 ± 0.027	0.127 ± 0.022	0.356 ± 0.033	12.414 ± 2.114	3.523 ± 0.324	0.086	-0.015	-14.75	0.1485	0.1493	1.152	87.963 ± 2.214	1.564 ± 0.267	3.241 ± 1.205
LSAM	beta31	0.316	0.338	0.022 ± 0.010	0.068 ± 0.032	0.022 ± 0.002	0.148 ± 0.008	0.022 ± 0.003	0.149 ± 0.010	0.223 ± 0.026	0.472 ± 0.032	0.325	0.009	2.74	0.1468	0.1470	1.101	97.222 ± 1.118	0.638 ± 0.018	62.037 ± 3.302
LSAM	beta32	0.316	0.317	0.001 ± 0.010	0.003 ± 0.031	0.021 ± 0.002	0.145 ± 0.007	0.021 ± 0.002	0.145 ± 0.008	0.210 ± 0.019	0.458 ± 0.026	0.324	0.008	2.54	0.1529	0.1531	1.120	97.222 ± 1.118	0.637 ± 0.015	56.019 ± 3.377
LSAM	beta33	0.139	0.282	0.143 ± 0.013	1.029 ± 0.092	0.035 ± 0.004	0.187 ± 0.010	0.055 ± 0.006	0.235 ± 0.013	2.862 ± 0.299	1.692 ± 0.096	0.288	0.149	107.47	0.1887	0.2407	1.096	96.759 ± 1.205	0.804 ± 0.034	33.333 ± 3.208
LSAM	beta34	0.101	-0.002	-0.103 ± 0.008	-1.020 ± 0.080	0.014 ± 0.001	0.119 ± 0.006	0.025 ± 0.002	0.157 ± 0.008	2.426 ± 0.220	1.558 ± 0.077	-0.003	-0.104	-102.72	0.1102	0.1514	1.074	83.333 ± 2.536	0.502 ± 0.019	1.852 ± 0.917
LSAM	beta35	0.101	0.011	-0.090 ± 0.008	-0.887 ± 0.083	0.015 ± 0.002	0.123 ± 0.008	0.023 ± 0.002	0.152 ± 0.008	2.267 ± 0.207	1.506 ± 0.077	0.002	-0.099	-97.88	0.1168	0.1530	1.096	85.648 ± 2.386	0.529 ± 0.026	4.167 ± 1.360
QML	beta31	0.316	0.295	-0.021 ± 0.006	-0.067 ± 0.020	0.009 ± 0.001	0.094 ± 0.005	0.009 ± 0.001	0.096 ± 0.006	0.093 ± 0.010	0.305 ± 0.019	0.295	-0.021	-6.55	0.0958	0.0980	1.072	95.833 ± 1.360	0.396 ± 0.005	88.889 ± 2.138
QML	beta32	0.316	0.291	-0.025 ± 0.007	-0.079 ± 0.022	0.011 ± 0.001	0.103 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	0.111 ± 0.010	0.334 ± 0.019	0.295	-0.021	-6.72	0.1018	0.1040	0.984	93.981 ± 1.618	0.396 ± 0.005	82.870 ± 2.564
QML	beta33	0.139	0.183	0.044 ± 0.010	0.313 ± 0.072	0.021 ± 0.003	0.147 ± 0.009	0.023 ± 0.003	0.153 ± 0.010	1.205 ± 0.131	1.098 ± 0.070	0.191	0.052	37.34	0.1458	0.1548	0.988	93.056 ± 1.730	0.568 ± 0.016	35.185 ± 3.249
QML	beta34	0.101	0.057	-0.044 ± 0.006	-0.436 ± 0.064	0.009 ± 0.001	0.095 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	1.076 ± 0.096	1.037 ± 0.055	0.053	-0.048	-47.09	0.1019	0.1124	0.983	81.019 ± 2.668	0.367 ± 0.011	6.019 ± 1.618
QML	beta35	0.101	0.063	-0.038 ± 0.006	-0.378 ± 0.064	0.009 ± 0.001	0.094 ± 0.006	0.010 ± 0.001	0.102 ± 0.006	1.014 ± 0.102	1.007 ± 0.059	0.052	-0.049	-48.81	0.0947	0.1067	0.997	86.111 ± 2.353	0.369 ± 0.012	7.407 ± 1.782
UPI	beta31	0.316	0.335	0.019 ± 0.016	0.059 ± 0.050	0.054 ± 0.007	0.233 ± 0.015	0.054 ± 0.007	0.233 ± 0.017	0.543 ± 0.071	0.737 ± 0.055	0.340	0.024	7.71	0.2287	0.2300	1.842	94.907 ± 1.496	1.679 ± 0.194	36.111 ± 3.268
UPI	beta32	0.316	0.321	0.005 ± 0.016	0.016 ± 0.049	0.052 ± 0.008	0.228 ± 0.018	0.052 ± 0.008	0.228 ± 0.020	0.518 ± 0.081	0.720 ± 0.062	0.318	0.002	0.69	0.2201	0.2201	1.641	92.593 ± 1.782	1.466 ± 0.174	33.333 ± 3.208
UPI	beta33	0.139	0.266	0.127 ± 0.016	0.916 ± 0.116	0.056 ± 0.007	0.237 ± 0.015	0.072 ± 0.009	0.269 ± 0.019	3.744 ± 0.489	1.935 ± 0.138	0.259	0.120	86.57	0.2270	0.2570	1.655	98.611 ± 0.796	1.541 ± 0.124	12.500 ± 2.250
UPI	beta34	0.101	-0.050	-0.151 ± 0.020	-1.490 ± 0.194	0.083 ± 0.012	0.288 ± 0.021	0.105 ± 0.014	0.325 ± 0.024	10.325 ± 1.376	3.213 ± 0.233	-0.019	-0.120	-118.99	0.2215	0.2520	1.686	91.204 ± 1.927	1.905 ± 0.249	0.926 ± 0.652
UPI	beta35	0.101	-0.019	-0.120 ± 0.019	-1.192 ± 0.191	0.081 ± 0.012	0.284 ± 0.022	0.095 ± 0.013	0.308 ± 0.024	9.285 ± 1.311	3.047 ± 0.235	-0.007	-0.108	-106.85	0.2562	0.2780	1.768	83.333 ± 2.536	1.967 ± 0.275	0.926 ± 0.652

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S56

Simulation 1 - Condition 56: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.005	-0.021 ± 0.016	0.005 ± 0.001	0.071 ± 0.004	0.005 ± 0.001	0.072 ± 0.005	0.051 ± 0.006	0.226 ± 0.016	0.307	-0.009	-2.88	0.0827	0.0832	1.065	96.135 ± 1.340	0.298 ± 0.033	98.551 ± 0.831
LMS	beta32	0.316	0.311	-0.005 ± 0.005	-0.016 ± 0.015	0.005 ± 0.001	0.070 ± 0.005	0.005 ± 0.001	0.070 ± 0.005	0.050 ± 0.006	0.223 ± 0.016	0.298	-0.018	-5.62	0.0720	0.0741	1.127	96.618 ± 1.256	0.311 ± 0.044	99.034 ± 0.680
LMS	beta33	0.139	0.226	0.087 ± 0.008	0.629 ± 0.055	0.012 ± 0.002	0.110 ± 0.011	0.020 ± 0.003	0.140 ± 0.011	1.015 ± 0.151	1.007 ± 0.080	0.217	0.078	56.39	0.1295	0.1514	1.302	80.676 ± 2.744	0.560 ± 0.096	63.285 ± 3.350
LMS	beta34	0.101	0.071	-0.030 ± 0.005	-0.300 ± 0.051	0.006 ± 0.001	0.075 ± 0.006	0.006 ± 0.001	0.080 ± 0.006	0.632 ± 0.083	0.795 ± 0.058	0.078	-0.023	-22.54	0.0952	0.0978	1.398	83.575 ± 2.575	0.408 ± 0.085	0.483 ± 0.482
LMS	beta35	0.101	0.063	-0.038 ± 0.004	-0.376 ± 0.039	0.003 ± 0.000	0.057 ± 0.003	0.005 ± 0.000	0.068 ± 0.003	0.457 ± 0.037	0.676 ± 0.032	0.057	-0.044	-43.07	0.0732	0.0852	1.493	85.990 ± 2.412	0.333 ± 0.038	3.865 ± 1.340
LSAM	beta31	0.316	0.318	0.002 ± 0.007	0.008 ± 0.021	0.009 ± 0.001	0.095 ± 0.005	0.009 ± 0.001	0.095 ± 0.006	0.091 ± 0.009	0.301 ± 0.019	0.324	0.008	2.46	0.0929	0.0932	0.983	93.237 ± 1.745	0.367 ± 0.005	93.237 ± 1.745
LSAM	beta32	0.316	0.331	0.015 ± 0.007	0.048 ± 0.022	0.010 ± 0.001	0.099 ± 0.007	0.010 ± 0.001	0.100 ± 0.008	0.101 ± 0.014	0.318 ± 0.025	0.324	0.008	2.56	0.0952	0.0956	0.965	95.169 ± 1.490	0.376 ± 0.006	95.169 ± 1.490
LSAM	beta33	0.139	0.272	0.133 ± 0.007	0.960 ± 0.052	0.011 ± 0.001	0.104 ± 0.005	0.029 ± 0.002	0.169 ± 0.007	1.483 ± 0.124	1.218 ± 0.054	0.270	0.131	93.92	0.1049	0.1674	1.008	79.710 ± 2.795	0.412 ± 0.008	79.227 ± 2.820
LSAM	beta34	0.101	0.009	-0.092 ± 0.005	-0.908 ± 0.045	0.004 ± 0.000	0.066 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	1.247 ± 0.093	1.117 ± 0.044	0.009	-0.092	-90.62	0.0618	0.1104	1.022	68.599 ± 3.226	0.263 ± 0.006	3.865 ± 1.340
LSAM	beta35	0.101	-0.001	-0.102 ± 0.004	-1.010 ± 0.042	0.004 ± 0.000	0.061 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	1.382 ± 0.101	1.176 ± 0.045	-0.002	-0.103	-102.10	0.0576	0.1181	1.103	64.251 ± 3.331	0.264 ± 0.005	3.382 ± 1.256
QML	beta31	0.316	0.292	-0.024 ± 0.005	-0.077 ± 0.014	0.004 ± 0.000	0.065 ± 0.004	0.005 ± 0.000	0.069 ± 0.004	0.048 ± 0.005	0.219 ± 0.013	0.298	-0.018	-5.85	0.0649	0.0675	0.939	91.787 ± 1.908	0.239 ± 0.002	100.000 ± 0.000
QML	beta32	0.316	0.294	-0.022 ± 0.005	-0.068 ± 0.015	0.004 ± 0.001	0.066 ± 0.004	0.005 ± 0.001	0.069 ± 0.004	0.048 ± 0.005	0.219 ± 0.014	0.292	-0.024	-7.67	0.0660	0.0703	0.932	94.686 ± 1.559	0.241 ± 0.003	100.000 ± 0.000
QML	beta33	0.139	0.199	0.060 ± 0.006	0.429 ± 0.044	0.008 ± 0.001	0.088 ± 0.004	0.011 ± 0.001	0.106 ± 0.006	0.587 ± 0.057	0.766 ± 0.042	0.196	0.057	41.26	0.0986	0.1141	1.021	86.473 ± 2.377	0.354 ± 0.009	63.285 ± 3.350
QML	beta34	0.101	0.051	-0.050 ± 0.004	-0.497 ± 0.039	0.003 ± 0.000	0.057 ± 0.003	0.006 ± 0.001	0.076 ± 0.004	0.563 ± 0.054	0.750 ± 0.039	0.049	-0.052	-51.88	0.0576	0.0779	1.016	79.710 ± 2.795	0.227 ± 0.006	8.213 ± 1.908
QML	beta35	0.101	0.045	-0.056 ± 0.003	-0.553 ± 0.032	0.002 ± 0.000	0.047 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	0.518 ± 0.039	0.720 ± 0.029	0.036	-0.065	-63.98	0.0463	0.0795	1.228	77.295 ± 2.912	0.225 ± 0.005	8.213 ± 1.908
UPI	beta31	0.316	0.321	0.005 ± 0.015	0.015 ± 0.048	0.049 ± 0.006	0.220 ± 0.014	0.048 ± 0.006	0.220 ± 0.016	0.484 ± 0.061	0.696 ± 0.051	0.338	0.022	7.01	0.2233	0.2244	1.877	91.787 ± 1.908	1.622 ± 0.221	46.860 ± 3.468
UPI	beta32	0.316	0.312	-0.004 ± 0.019	-0.012 ± 0.061	0.076 ± 0.012	0.276 ± 0.023	0.076 ± 0.012	0.276 ± 0.025	0.760 ± 0.123	0.872 ± 0.078	0.327	0.011	3.61	0.2739	0.2741	1.530	83.575 ± 2.575	1.657 ± 0.254	40.097 ± 3.406
UPI	beta33	0.139	0.259	0.120 ± 0.015	0.862 ± 0.108	0.047 ± 0.007	0.217 ± 0.017	0.061 ± 0.010	0.247 ± 0.021	3.160 ± 0.512	1.778 ± 0.154	0.227	0.088	63.19	0.1738	0.1947	1.725	98.068 ± 0.957	1.465 ± 0.198	19.807 ± 2.770
UPI	beta34	0.101	-0.098	-0.199 ± 0.017	-1.967 ± 0.169	0.060 ± 0.009	0.246 ± 0.020	0.100 ± 0.014	0.316 ± 0.024	9.758 ± 1.378	3.124 ± 0.234	-0.064	-0.165	-163.70	0.2395	0.2911	1.667	85.507 ± 2.447	1.605 ± 0.173	2.415 ± 1.067
UPI	beta35	0.101	-0.082	-0.183 ± 0.019	-1.807 ± 0.185	0.073 ± 0.011	0.269 ± 0.020	0.106 ± 0.013	0.325 ± 0.021	10.347 ± 1.239	3.217 ± 0.209	-0.081	-0.182	-180.21	0.2725	0.3277	1.563	84.541 ± 2.513	1.651 ± 0.204	2.899 ± 1.166

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S57

Simulation 1 - Condition 57: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.002	-0.023 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.307	-0.009	-2.78	0.0729	0.0734	1.026	95.670 ± 0.653	0.295 ± 0.002	97.526 ± 0.499
LMS	beta32	0.316	0.307	-0.009 ± 0.002	-0.030 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.243 ± 0.007	0.306	-0.010	-3.17	0.0724	0.0731	0.991	95.155 ± 0.689	0.296 ± 0.002	96.598 ± 0.582
LMS	beta33	0.139	0.156	0.017 ± 0.003	0.119 ± 0.024	0.010 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.552 ± 0.029	0.743 ± 0.023	0.163	0.024	17.12	0.0955	0.0984	1.026	93.918 ± 0.767	0.410 ± 0.008	46.804 ± 1.602
LMS	beta34	0.101	0.090	-0.011 ± 0.002	-0.108 ± 0.023	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.515 ± 0.028	0.718 ± 0.022	0.081	-0.020	-19.58	0.0669	0.0698	1.004	89.381 ± 0.989	0.282 ± 0.005	14.948 ± 1.145
LMS	beta35	0.101	0.091	-0.010 ± 0.002	-0.095 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.498 ± 0.026	0.705 ± 0.022	0.083	-0.018	-18.11	0.0642	0.0668	1.022	91.134 ± 0.913	0.283 ± 0.005	16.495 ± 1.192
LSAM	beta31	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.060 ± 0.003	0.246 ± 0.007	0.312	-0.004	-1.41	0.0767	0.0769	1.020	96.289 ± 0.607	0.311 ± 0.001	97.526 ± 0.499
LSAM	beta32	0.316	0.313	-0.003 ± 0.003	-0.011 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.065 ± 0.003	0.256 ± 0.007	0.308	-0.008	-2.49	0.0752	0.0756	0.986	94.742 ± 0.717	0.312 ± 0.001	97.010 ± 0.547
LSAM	beta33	0.139	0.183	0.044 ± 0.003	0.320 ± 0.022	0.009 ± 0.000	0.094 ± 0.002	0.011 ± 0.001	0.104 ± 0.003	0.564 ± 0.026	0.751 ± 0.020	0.181	0.042	30.16	0.0931	0.1021	0.994	92.990 ± 0.820	0.368 ± 0.003	54.433 ± 1.599
LSAM	beta34	0.101	0.066	-0.035 ± 0.002	-0.344 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.540 ± 0.023	0.735 ± 0.018	0.065	-0.036	-35.88	0.0670	0.0762	0.993	87.113 ± 1.076	0.255 ± 0.002	15.567 ± 1.164
LSAM	beta35	0.101	0.068	-0.033 ± 0.002	-0.323 ± 0.021	0.005 ± 0.000	0.067 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.546 ± 0.025	0.739 ± 0.019	0.066	-0.035	-35.14	0.0632	0.0725	0.972	88.763 ± 1.014	0.256 ± 0.002	18.041 ± 1.235
QML	beta31	0.316	0.304	-0.012 ± 0.002	-0.039 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.003	0.231 ± 0.007	0.302	-0.014	-4.43	0.0721	0.0734	1.017	95.155 ± 0.689	0.287 ± 0.001	98.041 ± 0.445
QML	beta32	0.316	0.302	-0.014 ± 0.002	-0.045 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.240 ± 0.007	0.302	-0.014	-4.58	0.0713	0.0728	0.986	94.742 ± 0.717	0.288 ± 0.001	97.423 ± 0.509
QML	beta33	0.139	0.153	0.014 ± 0.003	0.104 ± 0.021	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.458 ± 0.022	0.677 ± 0.019	0.158	0.019	13.63	0.0912	0.0932	0.993	94.433 ± 0.736	0.362 ± 0.004	46.495 ± 1.601
QML	beta34	0.101	0.081	-0.020 ± 0.002	-0.200 ± 0.020	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.437 ± 0.019	0.661 ± 0.017	0.076	-0.025	-25.21	0.0632	0.0681	0.987	88.351 ± 1.030	0.246 ± 0.003	18.763 ± 1.254
QML	beta35	0.101	0.082	-0.019 ± 0.002	-0.189 ± 0.020	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.423 ± 0.018	0.650 ± 0.017	0.076	-0.025	-24.60	0.0596	0.0646	0.993	89.897 ± 0.968	0.245 ± 0.003	20.515 ± 1.297
UPI	beta31	0.316	0.324	0.008 ± 0.003	0.026 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.065 ± 0.003	0.255 ± 0.008	0.322	0.006	2.00	0.0792	0.0794	1.025	96.598 ± 0.582	0.323 ± 0.003	96.598 ± 0.582
UPI	beta32	0.316	0.321	0.005 ± 0.003	0.015 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.068 ± 0.003	0.260 ± 0.008	0.318	0.002	0.68	0.0779	0.0779	0.999	95.155 ± 0.689	0.322 ± 0.002	97.113 ± 0.538
UPI	beta33	0.139	0.175	0.036 ± 0.003	0.261 ± 0.024	0.011 ± 0.001	0.104 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.631 ± 0.033	0.795 ± 0.024	0.166	0.027	19.77	0.0950	0.0989	0.972	94.845 ± 0.710	0.398 ± 0.004	44.536 ± 1.596
UPI	beta34	0.101	0.081	-0.020 ± 0.003	-0.196 ± 0.026	0.007 ± 0.000	0.081 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	0.683 ± 0.040	0.826 ± 0.027	0.080	-0.021	-21.16	0.0772	0.0801	0.990	92.990 ± 0.820	0.315 ± 0.005	18.351 ± 1.243
UPI	beta35	0.101	0.083	-0.018 ± 0.002	-0.177 ± 0.025	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.615 ± 0.031	0.784 ± 0.023	0.079	-0.022	-21.89	0.0688	0.0723	0.995	94.124 ± 0.755	0.301 ± 0.004	20.928 ± 1.306

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S58

Simulation 1 - Condition 58: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.001	-0.021 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.310	-0.006	-1.97	0.0436	0.0440	1.018	94.869 ± 0.700	0.184 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.309	-0.007 ± 0.001	-0.022 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.308	-0.008	-2.57	0.0449	0.0456	1.019	95.674 ± 0.645	0.184 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.164	0.025 ± 0.002	0.182 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.217 ± 0.010	0.466 ± 0.012	0.166	0.027	19.73	0.0581	0.0642	1.024	91.046 ± 0.906	0.240 ± 0.002	75.050 ± 1.373
LMS	beta34	0.101	0.088	-0.013 ± 0.001	-0.129 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.198 ± 0.009	0.445 ± 0.012	0.085	-0.016	-15.78	0.0415	0.0445	0.996	90.744 ± 0.919	0.168 ± 0.002	55.332 ± 1.577
LMS	beta35	0.101	0.087	-0.014 ± 0.001	-0.137 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.193 ± 0.008	0.439 ± 0.011	0.084	-0.017	-17.03	0.0401	0.0436	1.020	90.342 ± 0.937	0.169 ± 0.002	53.320 ± 1.582
LSAM	beta31	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.313	-0.003	-1.08	0.0480	0.0481	1.037	95.775 ± 0.638	0.194 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.313	-0.003	-1.05	0.0469	0.0470	1.030	95.976 ± 0.623	0.195 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.179	0.040 ± 0.002	0.290 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.005 ± 0.000	0.069 ± 0.002	0.247 ± 0.011	0.497 ± 0.012	0.177	0.038	27.63	0.0561	0.0680	1.010	88.129 ± 1.026	0.222 ± 0.001	88.028 ± 1.030
LSAM	beta34	0.101	0.070	-0.031 ± 0.001	-0.312 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.253 ± 0.010	0.503 ± 0.011	0.069	-0.032	-31.87	0.0394	0.0509	1.001	86.217 ± 1.093	0.156 ± 0.001	41.247 ± 1.561
LSAM	beta35	0.101	0.069	-0.032 ± 0.001	-0.319 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.258 ± 0.011	0.508 ± 0.012	0.068	-0.033	-32.24	0.0395	0.0512	1.007	85.312 ± 1.123	0.158 ± 0.001	39.235 ± 1.549
QML	beta31	0.316	0.304	-0.012 ± 0.001	-0.038 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.304	-0.012	-3.75	0.0436	0.0452	1.020	94.266 ± 0.737	0.181 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.303	-0.013 ± 0.001	-0.040 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.302	-0.014	-4.30	0.0452	0.0472	1.014	95.473 ± 0.659	0.181 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.156	0.017 ± 0.002	0.126 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.182 ± 0.008	0.426 ± 0.011	0.157	0.018	13.24	0.0555	0.0584	1.017	93.260 ± 0.795	0.226 ± 0.001	74.849 ± 1.376
QML	beta34	0.101	0.082	-0.019 ± 0.001	-0.191 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.188 ± 0.008	0.434 ± 0.011	0.080	-0.021	-20.76	0.0393	0.0445	1.002	89.235 ± 0.983	0.155 ± 0.001	55.131 ± 1.578
QML	beta35	0.101	0.081	-0.020 ± 0.001	-0.197 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.187 ± 0.007	0.432 ± 0.010	0.079	-0.022	-22.09	0.0386	0.0446	1.019	87.726 ± 1.041	0.155 ± 0.001	53.521 ± 1.582
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.317	0.001	0.17	0.0466	0.0466	1.023	95.473 ± 0.659	0.193 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.316	0.000	-0.07	0.0463	0.0463	1.015	95.775 ± 0.638	0.194 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.160	0.021 ± 0.002	0.151 ± 0.013	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.198 ± 0.010	0.445 ± 0.013	0.159	0.020	14.49	0.0573	0.0608	1.011	94.567 ± 0.719	0.230 ± 0.001	78.571 ± 1.301
UPI	beta34	0.101	0.084	-0.017 ± 0.001	-0.166 ± 0.014	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.220 ± 0.010	0.469 ± 0.013	0.084	-0.017	-16.54	0.0422	0.0454	1.014	92.555 ± 0.833	0.176 ± 0.001	46.982 ± 1.583
UPI	beta35	0.101	0.083	-0.018 ± 0.001	-0.174 ± 0.014	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.223 ± 0.010	0.473 ± 0.013	0.083	-0.018	-18.10	0.0438	0.0475	1.007	93.159 ± 0.801	0.175 ± 0.001	46.680 ± 1.582

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S59

Simulation 1 - Condition 59: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.308	-0.008 ± 0.002	-0.025 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.310	-0.006	-1.86	0.0602	0.0605	1.022	94.800 ± 0.702	0.232 ± 0.001	99.500 ± 0.223
LMS	beta32	0.316	0.310	-0.006 ± 0.002	-0.018 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.311	-0.005	-1.58	0.0611	0.0613	0.976	95.100 ± 0.683	0.232 ± 0.001	99.400 ± 0.244
LMS	beta33	0.139	0.139	-0.000 ± 0.002	-0.003 ± 0.016	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.270 ± 0.013	0.520 ± 0.015	0.144	0.005	3.94	0.0688	0.0690	0.934	93.300 ± 0.791	0.265 ± 0.002	58.700 ± 1.557
LMS	beta34	0.101	0.098	-0.003 ± 0.002	-0.029 ± 0.016	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.245 ± 0.011	0.495 ± 0.014	0.095	-0.006	-5.95	0.0490	0.0494	0.939	92.400 ± 0.838	0.184 ± 0.002	56.400 ± 1.568
LMS	beta35	0.101	0.099	-0.002 ± 0.002	-0.024 ± 0.016	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.248 ± 0.012	0.498 ± 0.015	0.095	-0.006	-6.31	0.0474	0.0479	0.947	92.500 ± 0.833	0.187 ± 0.002	55.700 ± 1.571
LSAM	beta31	0.316	0.311	-0.005 ± 0.002	-0.014 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.313	-0.003	-1.03	0.0604	0.0605	1.015	95.100 ± 0.683	0.234 ± 0.001	99.600 ± 0.200
LSAM	beta32	0.316	0.313	-0.003 ± 0.002	-0.008 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.313	-0.003	-1.02	0.0605	0.0606	0.982	95.300 ± 0.669	0.235 ± 0.001	99.700 ± 0.173
LSAM	beta33	0.139	0.148	0.009 ± 0.002	0.063 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.260 ± 0.012	0.510 ± 0.014	0.151	0.012	8.96	0.0685	0.0697	0.915	92.500 ± 0.833	0.252 ± 0.002	64.200 ± 1.516
LSAM	beta34	0.101	0.093	-0.008 ± 0.002	-0.079 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.241 ± 0.010	0.491 ± 0.013	0.091	-0.010	-10.25	0.0509	0.0520	0.923	92.000 ± 0.858	0.177 ± 0.001	53.900 ± 1.576
LSAM	beta35	0.101	0.093	-0.008 ± 0.002	-0.076 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.239 ± 0.011	0.488 ± 0.014	0.092	-0.009	-8.76	0.0471	0.0479	0.940	92.600 ± 0.828	0.180 ± 0.001	53.600 ± 1.577
QML	beta31	0.316	0.307	-0.009 ± 0.002	-0.030 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.309	-0.007	-2.28	0.0605	0.0609	1.022	94.600 ± 0.715	0.231 ± 0.001	99.500 ± 0.223
QML	beta32	0.316	0.309	-0.007 ± 0.002	-0.023 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.309	-0.007	-2.19	0.0607	0.0611	0.978	94.900 ± 0.696	0.232 ± 0.001	99.300 ± 0.264
QML	beta33	0.139	0.137	-0.002 ± 0.002	-0.011 ± 0.016	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.262 ± 0.013	0.512 ± 0.015	0.142	0.003	2.27	0.0692	0.0692	0.929	93.000 ± 0.807	0.259 ± 0.002	58.900 ± 1.556
QML	beta34	0.101	0.097	-0.004 ± 0.002	-0.045 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.237 ± 0.011	0.487 ± 0.013	0.094	-0.007	-7.13	0.0484	0.0489	0.931	91.700 ± 0.872	0.179 ± 0.001	56.900 ± 1.566
QML	beta35	0.101	0.097	-0.004 ± 0.002	-0.040 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.240 ± 0.012	0.490 ± 0.014	0.094	-0.007	-7.34	0.0465	0.0471	0.937	92.100 ± 0.853	0.181 ± 0.002	56.000 ± 1.570
UPI	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.319	0.003	0.93	0.0601	0.0601	0.981	94.700 ± 0.708	0.225 ± 0.001	99.800 ± 0.141
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.317	0.001	0.20	0.0620	0.0620	0.947	93.900 ± 0.757	0.226 ± 0.001	99.700 ± 0.173
UPI	beta33	0.139	0.146	0.007 ± 0.002	0.050 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.257 ± 0.012	0.507 ± 0.014	0.148	0.009	6.82	0.0678	0.0684	0.923	92.700 ± 0.823	0.254 ± 0.002	62.400 ± 1.532
UPI	beta34	0.101	0.097	-0.004 ± 0.002	-0.041 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.253 ± 0.011	0.503 ± 0.014	0.094	-0.007	-7.04	0.0501	0.0506	0.910	92.400 ± 0.838	0.181 ± 0.001	56.900 ± 1.566
UPI	beta35	0.101	0.097	-0.004 ± 0.002	-0.035 ± 0.016	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.245 ± 0.012	0.495 ± 0.015	0.095	-0.006	-5.61	0.0470	0.0474	0.941	93.100 ± 0.801	0.184 ± 0.001	55.400 ± 1.572

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S60

Simulation 1 - Condition 60: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.312	-0.004 ± 0.001	-0.014 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.312	-0.004	-1.32	0.0377	0.0379	1.003	95.300 ± 0.669	0.146 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.312	-0.004 ± 0.001	-0.013 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.311	-0.005	-1.46	0.0374	0.0377	0.982	94.600 ± 0.715	0.147 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.139	-0.000 ± 0.001	-0.002 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.139	0.000	0.13	0.0404	0.0404	1.013	94.200 ± 0.739	0.165 ± 0.001	86.900 ± 1.067
LMS	beta34	0.101	0.099	-0.002 ± 0.001	-0.021 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.097	-0.004	-3.99	0.0294	0.0297	1.005	94.600 ± 0.715	0.116 ± 0.001	95.500 ± 0.656
LMS	beta35	0.101	0.100	-0.001 ± 0.001	-0.010 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.097	-0.004	-4.06	0.0294	0.0297	0.989	95.100 ± 0.683	0.116 ± 0.001	95.700 ± 0.641
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.316	0.000	-0.13	0.0389	0.0389	0.998	95.800 ± 0.634	0.148 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.315	-0.001	-0.33	0.0363	0.0363	0.987	94.800 ± 0.702	0.148 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.145	0.006 ± 0.001	0.044 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.145	0.006	4.27	0.0383	0.0388	1.007	94.100 ± 0.745	0.157 ± 0.001	92.600 ± 0.828
LSAM	beta34	0.101	0.095	-0.006 ± 0.001	-0.059 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.095	-0.006	-6.22	0.0297	0.0303	0.998	94.300 ± 0.733	0.112 ± 0.001	93.700 ± 0.768
LSAM	beta35	0.101	0.096	-0.005 ± 0.001	-0.050 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.094	-0.007	-6.75	0.0289	0.0297	0.978	94.300 ± 0.733	0.112 ± 0.001	94.000 ± 0.751
QML	beta31	0.316	0.310	-0.006 ± 0.001	-0.018 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.311	-0.005	-1.72	0.0377	0.0380	1.001	95.000 ± 0.689	0.146 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.311	-0.005 ± 0.001	-0.017 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.122 ± 0.003	0.310	-0.006	-1.84	0.0374	0.0378	0.980	94.500 ± 0.721	0.146 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.137	-0.002 ± 0.001	-0.016 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.087 ± 0.004	0.296 ± 0.008	0.137	-0.002	-1.19	0.0405	0.0405	1.010	94.600 ± 0.715	0.163 ± 0.001	86.700 ± 1.074
QML	beta34	0.101	0.098	-0.003 ± 0.001	-0.033 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.097	-0.004	-4.33	0.0293	0.0297	1.003	94.600 ± 0.715	0.114 ± 0.001	95.300 ± 0.669
QML	beta35	0.101	0.099	-0.002 ± 0.001	-0.022 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.096	-0.005	-4.97	0.0291	0.0296	0.987	94.500 ± 0.721	0.114 ± 0.001	95.600 ± 0.649
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.318	0.002	0.54	0.0386	0.0386	0.966	95.000 ± 0.689	0.142 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.318	0.002 ± 0.001	0.006 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.120 ± 0.003	0.317	0.001	0.44	0.0363	0.0364	0.960	94.000 ± 0.751	0.143 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.141	0.002 ± 0.001	0.012 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.140	0.001	0.94	0.0389	0.0389	0.993	94.700 ± 0.708	0.157 ± 0.001	91.300 ± 0.891
UPI	beta34	0.101	0.099	-0.002 ± 0.001	-0.016 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.099	-0.002	-2.38	0.0293	0.0294	0.989	94.000 ± 0.751	0.114 ± 0.001	94.900 ± 0.696
UPI	beta35	0.101	0.100	-0.001 ± 0.001	-0.009 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.097	-0.004	-3.95	0.0294	0.0296	0.971	94.400 ± 0.727	0.113 ± 0.001	95.200 ± 0.676

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S61

Simulation 1 - Condition 61: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.325	0.009 ± 0.007	0.027 ± 0.023	0.032 ± 0.003	0.179 ± 0.007	0.032 ± 0.003	0.179 ± 0.008	0.322 ± 0.026	0.567 ± 0.026	0.340	0.024	7.66	0.1631	0.1649	1.173	95.798 ± 0.822	0.824 ± 0.045	56.134 ± 2.034
LMS	beta32	0.316	0.307	-0.009 ± 0.008	-0.027 ± 0.024	0.035 ± 0.003	0.187 ± 0.007	0.035 ± 0.003	0.187 ± 0.008	0.352 ± 0.026	0.593 ± 0.025	0.343	0.027	8.67	0.1717	0.1739	1.130	94.790 ± 0.911	0.830 ± 0.036	53.782 ± 2.044
LMS	beta33	0.139	-0.320	-0.459 ± 0.030	-3.301 ± 0.219	0.553 ± 0.032	0.744 ± 0.021	0.763 ± 0.055	0.873 ± 0.034	39.479 ± 2.839	6.283 ± 0.245	0.019	-0.120	-86.31	0.5633	0.5760	0.805	85.210 ± 1.455	2.348 ± 0.138	23.193 ± 1.730
LMS	beta34	0.101	0.436	0.335 ± 0.018	3.322 ± 0.183	0.203 ± 0.017	0.451 ± 0.019	0.316 ± 0.028	0.562 ± 0.026	30.939 ± 2.761	5.562 ± 0.260	0.249	0.148	146.93	0.2328	0.2760	0.919	95.126 ± 0.883	1.624 ± 0.115	22.857 ± 1.721
LMS	beta35	0.101	0.471	0.370 ± 0.020	3.662 ± 0.194	0.229 ± 0.017	0.479 ± 0.018	0.365 ± 0.030	0.604 ± 0.026	35.819 ± 2.922	5.985 ± 0.257	0.268	0.167	165.48	0.2649	0.3132	0.974	93.950 ± 0.977	1.828 ± 0.149	25.042 ± 1.776
LSAM	beta31	0.316	0.392	0.076 ± 0.050	0.241 ± 0.158	1.485 ± 0.100	1.219 ± 0.041	1.489 ± 0.100	1.220 ± 0.048	14.909 ± 1.002	3.861 ± 0.152	0.366	0.050	15.84	1.2678	1.2688	2.564	100.000 ± 0.000	12.248 ± 0.457	2.689 ± 0.663
LSAM	beta32	0.316	0.392	0.076 ± 0.051	0.241 ± 0.160	1.524 ± 0.103	1.235 ± 0.042	1.527 ± 0.104	1.236 ± 0.049	15.297 ± 1.044	3.911 ± 0.156	0.410	0.094	29.70	1.3126	1.3159	2.524	99.664 ± 0.237	12.215 ± 0.463	2.689 ± 0.663
LSAM	beta33	0.139	0.259	0.120 ± 0.014	0.865 ± 0.104	0.124 ± 0.011	0.352 ± 0.016	0.138 ± 0.013	0.372 ± 0.018	7.158 ± 0.655	2.675 ± 0.133	0.251	0.112	80.63	0.3131	0.3326	2.177	98.319 ± 0.527	3.006 ± 0.155	6.218 ± 0.990
LSAM	beta34	0.101	0.033	-0.068 ± 0.020	-0.674 ± 0.196	0.233 ± 0.019	0.483 ± 0.020	0.238 ± 0.020	0.488 ± 0.023	23.302 ± 1.931	4.827 ± 0.223	0.055	-0.046	-45.91	0.3934	0.3961	2.391	99.496 ± 0.290	4.529 ± 0.197	2.689 ± 0.663
LSAM	beta35	0.101	0.037	-0.064 ± 0.020	-0.636 ± 0.197	0.235 ± 0.019	0.484 ± 0.020	0.238 ± 0.019	0.488 ± 0.022	23.365 ± 1.908	4.834 ± 0.221	0.043	-0.058	-57.24	0.4029	0.4070	2.390	98.824 ± 0.442	4.538 ± 0.203	2.689 ± 0.663
QML	beta31	0.316	0.340	0.024 ± 0.005	0.075 ± 0.015	0.013 ± 0.001	0.114 ± 0.004	0.014 ± 0.001	0.116 ± 0.004	0.136 ± 0.008	0.368 ± 0.014	0.341	0.025	7.93	0.1210	0.1235	1.091	96.471 ± 0.756	0.488 ± 0.005	80.168 ± 1.635
QML	beta32	0.316	0.337	0.021 ± 0.005	0.066 ± 0.016	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.124 ± 0.005	0.154 ± 0.010	0.393 ± 0.015	0.341	0.025	7.99	0.1208	0.1234	1.013	95.630 ± 0.838	0.486 ± 0.005	77.647 ± 1.708
QML	beta33	0.139	0.087	-0.052 ± 0.008	-0.375 ± 0.056	0.035 ± 0.002	0.188 ± 0.006	0.038 ± 0.002	0.195 ± 0.007	1.974 ± 0.128	1.405 ± 0.053	0.106	-0.033	-23.87	0.1965	0.1993	0.996	93.782 ± 0.990	0.735 ± 0.018	14.958 ± 1.462
QML	beta34	0.101	0.183	0.082 ± 0.004	0.816 ± 0.042	0.011 ± 0.001	0.103 ± 0.004	0.017 ± 0.001	0.132 ± 0.005	1.710 ± 0.123	1.308 ± 0.050	0.178	0.077	76.57	0.0951	0.1225	1.038	91.933 ± 1.116	0.420 ± 0.012	50.588 ± 2.050
QML	beta35	0.101	0.190	0.089 ± 0.005	0.880 ± 0.045	0.012 ± 0.001	0.111 ± 0.004	0.020 ± 0.001	0.142 ± 0.005	1.987 ± 0.142	1.410 ± 0.054	0.175	0.074	73.44	0.1044	0.1281	0.973	91.261 ± 1.158	0.424 ± 0.011	49.748 ± 2.050
UPI	beta31	0.316	0.379	0.063 ± 0.021	0.200 ± 0.068	0.274 ± 0.022	0.523 ± 0.021	0.277 ± 0.023	0.526 ± 0.024	2.775 ± 0.227	1.666 ± 0.076	0.308	-0.008	-2.60	0.4894	0.4895	2.165	99.664 ± 0.237	4.439 ± 0.219	6.555 ± 1.015
UPI	beta32	0.316	0.307	-0.009 ± 0.022	-0.029 ± 0.069	0.286 ± 0.023	0.535 ± 0.022	0.285 ± 0.023	0.534 ± 0.025	2.858 ± 0.234	1.691 ± 0.078	0.296	-0.020	-6.24	0.4650	0.4655	2.184	99.832 ± 0.168	4.577 ± 0.223	5.378 ± 0.925
UPI	beta33	0.139	0.161	0.022 ± 0.012	0.159 ± 0.085	0.082 ± 0.007	0.287 ± 0.012	0.083 ± 0.007	0.287 ± 0.013	4.276 ± 0.353	2.068 ± 0.096	0.151	0.012	8.84	0.2731	0.2733	2.009	98.151 ± 0.552	2.259 ± 0.128	3.025 ± 0.702
UPI	beta34	0.101	0.051	-0.050 ± 0.008	-0.493 ± 0.080	0.039 ± 0.003	0.197 ± 0.009	0.041 ± 0.004	0.203 ± 0.010	4.026 ± 0.354	2.007 ± 0.097	0.065	-0.036	-35.18	0.1864	0.1897	2.250	99.160 ± 0.374	1.734 ± 0.096	2.185 ± 0.599
UPI	beta35	0.101	0.076	-0.025 ± 0.009	-0.243 ± 0.085	0.044 ± 0.004	0.210 ± 0.009	0.045 ± 0.004	0.211 ± 0.010	4.366 ± 0.379	2.089 ± 0.100	0.083	-0.018	-17.78	0.1760	0.1769	2.197	99.496 ± 0.290	1.806 ± 0.096	3.025 ± 0.702

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S62

Simulation 1 - Condition 62: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.327	0.011 ± 0.004	0.035 ± 0.014	0.011 ± 0.001	0.106 ± 0.004	0.011 ± 0.001	0.107 ± 0.005	0.114 ± 0.009	0.337 ± 0.015	0.340	0.024	7.68	0.0950	0.0980	1.048	94.894 ± 0.924	0.436 ± 0.014	81.690 ± 1.623
LMS	beta32	0.316	0.332	0.016 ± 0.005	0.049 ± 0.015	0.013 ± 0.001	0.113 ± 0.005	0.013 ± 0.001	0.114 ± 0.005	0.129 ± 0.011	0.360 ± 0.017	0.348	0.032	10.22	0.1020	0.1070	1.083	92.430 ± 1.110	0.478 ± 0.028	81.162 ± 1.641
LMS	beta33	0.139	-0.045	-0.184 ± 0.022	-1.321 ± 0.158	0.274 ± 0.023	0.523 ± 0.022	0.307 ± 0.030	0.554 ± 0.029	15.900 ± 1.547	3.988 ± 0.209	0.134	-0.005	-3.46	0.2468	0.2468	0.646	81.162 ± 1.641	1.325 ± 0.089	37.500 ± 2.031
LMS	beta34	0.101	0.293	0.192 ± 0.011	1.904 ± 0.109	0.069 ± 0.007	0.262 ± 0.013	0.106 ± 0.011	0.325 ± 0.017	10.344 ± 1.043	3.216 ± 0.169	0.210	0.109	107.84	0.1205	0.1625	0.704	90.141 ± 1.251	0.723 ± 0.047	60.387 ± 2.052
LMS	beta35	0.101	0.307	0.206 ± 0.012	2.041 ± 0.122	0.086 ± 0.008	0.294 ± 0.014	0.129 ± 0.013	0.359 ± 0.019	12.631 ± 1.273	3.554 ± 0.187	0.208	0.107	106.03	0.1239	0.1638	0.758	86.620 ± 1.428	0.874 ± 0.080	63.028 ± 2.025
LSAM	beta31	0.316	0.331	0.015 ± 0.024	0.047 ± 0.074	0.314 ± 0.029	0.560 ± 0.026	0.314 ± 0.029	0.560 ± 0.028	3.142 ± 0.287	1.772 ± 0.090	0.331	0.015	4.78	0.5906	0.5908	1.888	99.120 ± 0.392	4.148 ± 0.267	13.732 ± 1.444
LSAM	beta32	0.316	0.405	0.089 ± 0.024	0.282 ± 0.076	0.328 ± 0.029	0.573 ± 0.025	0.336 ± 0.030	0.579 ± 0.029	3.361 ± 0.302	1.833 ± 0.091	0.370	0.054	17.04	0.6203	0.6226	1.949	99.472 ± 0.304	4.377 ± 0.298	12.324 ± 1.379
LSAM	beta33	0.139	0.228	0.089 ± 0.006	0.637 ± 0.042	0.019 ± 0.002	0.138 ± 0.006	0.027 ± 0.002	0.164 ± 0.007	1.386 ± 0.113	1.177 ± 0.051	0.221	0.082	58.95	0.1513	0.1721	1.461	95.423 ± 0.877	0.789 ± 0.043	39.789 ± 2.054
LSAM	beta34	0.101	0.060	-0.041 ± 0.008	-0.405 ± 0.075	0.033 ± 0.003	0.181 ± 0.009	0.034 ± 0.003	0.185 ± 0.010	3.360 ± 0.322	1.833 ± 0.096	0.069	-0.032	-32.01	0.1853	0.1881	1.794	97.711 ± 0.627	1.271 ± 0.083	8.451 ± 1.167
LSAM	beta35	0.101	0.039	-0.062 ± 0.008	-0.611 ± 0.079	0.036 ± 0.003	0.191 ± 0.008	0.040 ± 0.004	0.200 ± 0.010	3.925 ± 0.355	1.981 ± 0.098	0.050	-0.051	-50.50	0.1971	0.2036	1.964	97.535 ± 0.651	1.467 ± 0.112	7.570 ± 1.110
QML	beta31	0.316	0.334	0.018 ± 0.003	0.057 ± 0.009	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	0.052 ± 0.003	0.228 ± 0.008	0.335	0.019	6.15	0.0731	0.0757	1.107	95.599 ± 0.861	0.303 ± 0.011	99.120 ± 0.392
QML	beta32	0.316	0.342	0.026 ± 0.003	0.083 ± 0.009	0.005 ± 0.000	0.070 ± 0.002	0.006 ± 0.000	0.074 ± 0.003	0.055 ± 0.003	0.235 ± 0.008	0.341	0.025	7.97	0.0737	0.0779	1.087	95.070 ± 0.908	0.297 ± 0.005	98.768 ± 0.463
QML	beta33	0.139	0.114	-0.025 ± 0.005	-0.179 ± 0.034	0.012 ± 0.001	0.112 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	0.675 ± 0.044	0.822 ± 0.031	0.111	-0.028	-19.85	0.1071	0.1106	0.951	92.958 ± 1.074	0.416 ± 0.005	23.944 ± 1.791
QML	beta34	0.101	0.178	0.077 ± 0.002	0.764 ± 0.022	0.003 ± 0.000	0.052 ± 0.002	0.009 ± 0.000	0.093 ± 0.002	0.853 ± 0.038	0.924 ± 0.022	0.181	0.080	79.30	0.0611	0.1007	1.125	77.993 ± 1.738	0.232 ± 0.003	89.789 ± 1.271
QML	beta35	0.101	0.179	0.078 ± 0.003	0.771 ± 0.026	0.004 ± 0.000	0.062 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.966 ± 0.048	0.983 ± 0.026	0.178	0.077	75.93	0.0634	0.0995	0.961	75.528 ± 1.804	0.232 ± 0.003	88.204 ± 1.353
UPI	beta31	0.316	0.352	0.036 ± 0.017	0.113 ± 0.053	0.158 ± 0.013	0.397 ± 0.016	0.159 ± 0.013	0.398 ± 0.018	1.589 ± 0.128	1.260 ± 0.058	0.321	0.005	1.62	0.3517	0.3517	1.801	99.472 ± 0.304	2.803 ± 0.156	12.676 ± 1.396
UPI	beta32	0.316	0.307	-0.009 ± 0.016	-0.027 ± 0.051	0.145 ± 0.012	0.381 ± 0.016	0.145 ± 0.012	0.380 ± 0.018	1.449 ± 0.118	1.204 ± 0.056	0.291	-0.025	-7.77	0.3301	0.3310	1.855	99.472 ± 0.304	2.768 ± 0.148	10.387 ± 1.280
UPI	beta33	0.139	0.176	0.037 ± 0.007	0.263 ± 0.050	0.027 ± 0.002	0.165 ± 0.007	0.029 ± 0.002	0.169 ± 0.008	1.476 ± 0.124	1.215 ± 0.057	0.158	0.019	13.51	0.1582	0.1593	1.664	98.063 ± 0.578	1.077 ± 0.060	13.732 ± 1.444
UPI	beta34	0.101	0.059	-0.042 ± 0.006	-0.416 ± 0.056	0.018 ± 0.001	0.134 ± 0.005	0.020 ± 0.002	0.141 ± 0.006	1.938 ± 0.150	1.392 ± 0.060	0.066	-0.035	-34.83	0.1193	0.1244	1.782	98.415 ± 0.524	0.938 ± 0.052	4.930 ± 0.908
UPI	beta35	0.101	0.075	-0.026 ± 0.005	-0.262 ± 0.054	0.017 ± 0.001	0.130 ± 0.005	0.018 ± 0.001	0.133 ± 0.006	1.722 ± 0.132	1.312 ± 0.057	0.077	-0.024	-23.67	0.1169	0.1193	1.832	99.296 ± 0.351	0.934 ± 0.051	5.986 ± 0.995

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S63

Simulation 1 - Condition 63: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.003	0.005 ± 0.011	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.101 ± 0.005	0.317 ± 0.010	0.321	0.005	1.49	0.1069	0.1070	0.989	94.273 ± 0.786	0.389 ± 0.004	85.452 ± 1.193
LMS	beta32	0.316	0.326	0.010 ± 0.003	0.031 ± 0.010	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.083 ± 0.004	0.287 ± 0.008	0.326	0.010	3.08	0.0956	0.0961	1.078	96.564 ± 0.617	0.381 ± 0.004	89.118 ± 1.054
LMS	beta33	0.139	0.117	-0.022 ± 0.004	-0.161 ± 0.028	0.014 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	0.724 ± 0.040	0.851 ± 0.027	0.117	-0.022	-15.51	0.1174	0.1194	1.037	94.731 ± 0.756	0.473 ± 0.013	27.033 ± 1.503
LMS	beta34	0.101	0.152	0.051 ± 0.002	0.502 ± 0.022	0.004 ± 0.000	0.065 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.671 ± 0.041	0.819 ± 0.027	0.148	0.047	46.26	0.0631	0.0785	1.028	93.013 ± 0.863	0.264 ± 0.008	71.707 ± 1.524
LMS	beta35	0.101	0.148	0.047 ± 0.002	0.461 ± 0.020	0.003 ± 0.000	0.058 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.546 ± 0.031	0.739 ± 0.022	0.144	0.043	42.71	0.0548	0.0697	1.080	93.929 ± 0.808	0.247 ± 0.006	74.227 ± 1.480
LSAM	beta31	0.316	0.323	0.007 ± 0.006	0.022 ± 0.019	0.033 ± 0.002	0.181 ± 0.005	0.033 ± 0.002	0.181 ± 0.005	0.330 ± 0.017	0.574 ± 0.017	0.329	0.013	4.09	0.1823	0.1828	0.994	94.845 ± 0.748	0.707 ± 0.008	49.943 ± 1.692
LSAM	beta32	0.316	0.329	0.013 ± 0.006	0.042 ± 0.018	0.028 ± 0.001	0.169 ± 0.004	0.029 ± 0.001	0.169 ± 0.005	0.286 ± 0.015	0.535 ± 0.017	0.329	0.013	4.24	0.1708	0.1713	1.043	96.793 ± 0.596	0.689 ± 0.007	51.546 ± 1.691
LSAM	beta33	0.139	0.172	0.033 ± 0.004	0.238 ± 0.026	0.011 ± 0.001	0.105 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.628 ± 0.037	0.793 ± 0.027	0.163	0.024	17.56	0.1045	0.1074	0.934	92.784 ± 0.876	0.385 ± 0.005	44.903 ± 1.683
LSAM	beta34	0.101	0.088	-0.013 ± 0.002	-0.127 ± 0.024	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.515 ± 0.027	0.718 ± 0.023	0.090	-0.011	-10.45	0.0707	0.0715	0.947	90.951 ± 0.971	0.265 ± 0.005	37.686 ± 1.640
LSAM	beta35	0.101	0.087	-0.014 ± 0.002	-0.136 ± 0.022	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.440 ± 0.026	0.663 ± 0.022	0.089	-0.012	-11.97	0.0622	0.0634	0.985	93.700 ± 0.822	0.253 ± 0.004	38.832 ± 1.649
QML	beta31	0.316	0.320	0.004 ± 0.003	0.012 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.092 ± 0.004	0.303 ± 0.009	0.323	0.007	2.09	0.1027	0.1029	0.974	94.044 ± 0.801	0.365 ± 0.002	89.462 ± 1.039
QML	beta32	0.316	0.326	0.010 ± 0.003	0.032 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.077 ± 0.004	0.278 ± 0.008	0.325	0.009	2.69	0.0921	0.0925	1.050	96.220 ± 0.645	0.359 ± 0.002	91.982 ± 0.919
QML	beta33	0.139	0.110	-0.029 ± 0.003	-0.212 ± 0.025	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.589 ± 0.029	0.767 ± 0.022	0.110	-0.029	-20.77	0.1084	0.1122	0.965	93.242 ± 0.850	0.388 ± 0.005	26.231 ± 1.489
QML	beta34	0.101	0.146	0.045 ± 0.002	0.441 ± 0.019	0.003 ± 0.000	0.057 ± 0.001	0.005 ± 0.000	0.072 ± 0.002	0.513 ± 0.026	0.716 ± 0.020	0.143	0.042	41.87	0.0587	0.0723	0.975	89.347 ± 1.044	0.218 ± 0.003	77.549 ± 1.412
QML	beta35	0.101	0.143	0.042 ± 0.002	0.416 ± 0.018	0.003 ± 0.000	0.053 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.450 ± 0.023	0.671 ± 0.019	0.141	0.040	39.89	0.0502	0.0644	1.014	89.691 ± 1.029	0.211 ± 0.003	79.611 ± 1.364
UPI	beta31	0.316	0.328	0.012 ± 0.007	0.036 ± 0.023	0.046 ± 0.003	0.215 ± 0.006	0.046 ± 0.003	0.215 ± 0.007	0.462 ± 0.026	0.680 ± 0.022	0.331	0.015	4.65	0.2124	0.2129	1.109	95.647 ± 0.691	0.933 ± 0.029	42.153 ± 1.671
UPI	beta32	0.316	0.335	0.019 ± 0.007	0.061 ± 0.021	0.040 ± 0.002	0.199 ± 0.006	0.040 ± 0.002	0.200 ± 0.007	0.399 ± 0.022	0.632 ± 0.021	0.332	0.016	5.19	0.1973	0.1979	1.164	97.136 ± 0.564	0.907 ± 0.024	44.903 ± 1.683
UPI	beta33	0.139	0.163	0.024 ± 0.004	0.172 ± 0.028	0.013 ± 0.001	0.116 ± 0.004	0.014 ± 0.001	0.118 ± 0.005	0.726 ± 0.050	0.852 ± 0.033	0.154	0.015	10.58	0.1037	0.1047	1.045	95.189 ± 0.724	0.475 ± 0.013	32.875 ± 1.590
UPI	beta34	0.101	0.087	-0.014 ± 0.003	-0.141 ± 0.028	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	0.721 ± 0.047	0.849 ± 0.031	0.087	-0.014	-14.12	0.0777	0.0790	1.054	92.211 ± 0.907	0.349 ± 0.013	34.708 ± 1.611
UPI	beta35	0.101	0.087	-0.014 ± 0.003	-0.138 ± 0.026	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.608 ± 0.038	0.780 ± 0.028	0.088	-0.013	-12.56	0.0689	0.0701	1.106	95.418 ± 0.708	0.336 ± 0.011	34.364 ± 1.607

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S64

Simulation 1 - Condition 64: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.322	0.006 ± 0.002	0.018 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.183 ± 0.005	0.324	0.008	2.58	0.0581	0.0587	1.034	95.478 ± 0.674	0.234 ± 0.002	99.264 ± 0.277
LMS	beta32	0.316	0.320	0.004 ± 0.002	0.013 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.323	0.007	2.16	0.0615	0.0619	1.009	94.953 ± 0.710	0.233 ± 0.001	98.843 ± 0.347
LMS	beta33	0.139	0.124	-0.015 ± 0.002	-0.111 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.257 ± 0.014	0.507 ± 0.016	0.125	-0.014	-10.11	0.0653	0.0668	0.973	94.742 ± 0.724	0.262 ± 0.003	50.368 ± 1.621
LMS	beta34	0.101	0.146	0.045 ± 0.001	0.450 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.003 ± 0.000	0.059 ± 0.001	0.336 ± 0.014	0.580 ± 0.013	0.144	0.043	42.15	0.0361	0.0558	0.973	78.128 ± 1.340	0.141 ± 0.002	98.633 ± 0.377
LMS	beta35	0.101	0.147	0.046 ± 0.001	0.454 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.353 ± 0.017	0.594 ± 0.015	0.142	0.041	41.03	0.0365	0.0552	0.924	78.970 ± 1.321	0.140 ± 0.002	99.369 ± 0.257
LSAM	beta31	0.316	0.322	0.006 ± 0.003	0.018 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.005	0.312 ± 0.009	0.318	0.002	0.55	0.0993	0.0993	1.023	95.584 ± 0.666	0.395 ± 0.002	88.854 ± 1.020
LSAM	beta32	0.316	0.319	0.003 ± 0.003	0.011 ± 0.010	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.101 ± 0.005	0.317 ± 0.009	0.319	0.003	1.06	0.1015	0.1015	1.003	95.268 ± 0.688	0.394 ± 0.002	87.277 ± 1.081
LSAM	beta33	0.139	0.161	0.022 ± 0.002	0.159 ± 0.014	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.211 ± 0.011	0.459 ± 0.013	0.157	0.018	13.25	0.0553	0.0583	0.928	91.377 ± 0.910	0.218 ± 0.001	82.860 ± 1.222
LSAM	beta34	0.101	0.090	-0.011 ± 0.001	-0.105 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.158 ± 0.009	0.397 ± 0.013	0.091	-0.010	-9.42	0.0361	0.0374	0.914	89.695 ± 0.986	0.139 ± 0.002	70.978 ± 1.472
LSAM	beta35	0.101	0.091	-0.010 ± 0.001	-0.098 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.153 ± 0.007	0.391 ± 0.011	0.091	-0.010	-10.22	0.0365	0.0380	0.909	89.485 ± 0.995	0.136 ± 0.002	71.714 ± 1.460
QML	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.322	0.006	1.91	0.0565	0.0568	1.018	95.268 ± 0.688	0.226 ± 0.001	99.895 ± 0.105
QML	beta32	0.316	0.320	0.004 ± 0.002	0.011 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.183 ± 0.005	0.321	0.005	1.49	0.0595	0.0597	0.996	94.637 ± 0.731	0.225 ± 0.001	99.790 ± 0.149
QML	beta33	0.139	0.112	-0.027 ± 0.002	-0.194 ± 0.015	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.243 ± 0.012	0.493 ± 0.014	0.112	-0.027	-19.52	0.0598	0.0657	0.949	91.167 ± 0.920	0.234 ± 0.002	48.265 ± 1.620
QML	beta34	0.101	0.144	0.043 ± 0.001	0.428 ± 0.011	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.301 ± 0.012	0.549 ± 0.011	0.143	0.042	41.67	0.0340	0.0541	0.952	74.763 ± 1.409	0.130 ± 0.001	98.843 ± 0.347
QML	beta35	0.101	0.145	0.044 ± 0.001	0.435 ± 0.012	0.001 ± 0.000	0.036 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.317 ± 0.014	0.563 ± 0.013	0.142	0.041	40.61	0.0353	0.0541	0.908	75.184 ± 1.401	0.129 ± 0.001	99.369 ± 0.257
UPI	beta31	0.316	0.319	0.003 ± 0.004	0.009 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.128 ± 0.007	0.357 ± 0.011	0.315	-0.001	-0.44	0.1131	0.1131	1.032	95.584 ± 0.666	0.457 ± 0.005	79.811 ± 1.302
UPI	beta32	0.316	0.315	-0.001 ± 0.004	-0.004 ± 0.012	0.013 ± 0.001	0.115 ± 0.003	0.013 ± 0.001	0.115 ± 0.003	0.132 ± 0.007	0.363 ± 0.011	0.313	-0.003	-1.00	0.1167	0.1167	1.013	96.109 ± 0.627	0.455 ± 0.005	77.603 ± 1.352
UPI	beta33	0.139	0.151	0.012 ± 0.002	0.087 ± 0.014	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.203 ± 0.011	0.450 ± 0.014	0.147	0.008	5.78	0.0576	0.0582	0.976	94.427 ± 0.744	0.235 ± 0.003	76.341 ± 1.378
UPI	beta34	0.101	0.091	-0.010 ± 0.001	-0.095 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.190 ± 0.011	0.436 ± 0.014	0.092	-0.009	-8.83	0.0377	0.0388	0.939	91.693 ± 0.895	0.158 ± 0.002	64.984 ± 1.547
UPI	beta35	0.101	0.093	-0.008 ± 0.001	-0.076 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.179 ± 0.009	0.423 ± 0.013	0.092	-0.009	-8.76	0.0392	0.0402	0.943	91.272 ± 0.915	0.155 ± 0.002	67.298 ± 1.521

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S65

Simulation 1 - Condition 65: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.002	-0.015 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.248 ± 0.007	0.310	-0.006	-1.92	0.0775	0.0778	1.023	95.565 ± 0.654	0.313 ± 0.001	96.069 ± 0.617
LMS	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.320	0.004	1.35	0.0795	0.0796	1.023	95.766 ± 0.639	0.313 ± 0.001	97.177 ± 0.526
LMS	beta33	0.139	0.124	-0.015 ± 0.002	-0.109 ± 0.017	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.287 ± 0.015	0.535 ± 0.016	0.126	-0.013	-9.58	0.0657	0.0671	0.897	92.036 ± 0.860	0.256 ± 0.003	54.032 ± 1.582
LMS	beta34	0.101	0.121	0.020 ± 0.001	0.197 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.202 ± 0.011	0.449 ± 0.013	0.120	0.019	18.89	0.0378	0.0423	0.884	90.827 ± 0.916	0.141 ± 0.002	88.810 ± 1.001
LMS	beta35	0.101	0.120	0.019 ± 0.001	0.188 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.200 ± 0.012	0.448 ± 0.014	0.116	0.015	15.28	0.0359	0.0391	0.885	90.625 ± 0.925	0.142 ± 0.002	90.323 ± 0.939
LSAM	beta31	0.316	0.311	-0.005 ± 0.003	-0.016 ± 0.010	0.010 ± 0.000	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.101 ± 0.005	0.317 ± 0.009	0.310	-0.006	-1.82	0.0981	0.0983	1.007	95.565 ± 0.654	0.395 ± 0.002	85.786 ± 1.109
LSAM	beta32	0.316	0.321	0.005 ± 0.003	0.015 ± 0.010	0.010 ± 0.000	0.101 ± 0.002	0.010 ± 0.000	0.101 ± 0.003	0.102 ± 0.005	0.319 ± 0.009	0.321	0.005	1.50	0.1017	0.1018	1.002	95.665 ± 0.647	0.395 ± 0.002	86.593 ± 1.082
LSAM	beta33	0.139	0.145	0.006 ± 0.002	0.041 ± 0.016	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.261 ± 0.013	0.511 ± 0.016	0.145	0.006	3.96	0.0651	0.0653	0.866	90.726 ± 0.921	0.240 ± 0.002	67.641 ± 1.485
LSAM	beta34	0.101	0.100	-0.001 ± 0.001	-0.010 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.176 ± 0.009	0.420 ± 0.013	0.101	0.000	-0.47	0.0391	0.0391	0.857	90.927 ± 0.912	0.143 ± 0.002	76.210 ± 1.352
LSAM	beta35	0.101	0.098	-0.003 ± 0.001	-0.025 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.175 ± 0.010	0.419 ± 0.013	0.096	-0.005	-4.58	0.0380	0.0383	0.874	92.137 ± 0.855	0.145 ± 0.002	74.294 ± 1.388
QML	beta31	0.316	0.311	-0.005 ± 0.002	-0.015 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.062 ± 0.003	0.248 ± 0.007	0.310	-0.006	-1.90	0.0784	0.0787	1.040	95.262 ± 0.675	0.319 ± 0.007	96.169 ± 0.609
QML	beta32	0.316	0.319	0.003 ± 0.002	0.010 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.248 ± 0.007	0.320	0.004	1.39	0.0782	0.0783	1.032	95.766 ± 0.639	0.317 ± 0.004	97.177 ± 0.526
QML	beta33	0.139	0.122	-0.017 ± 0.002	-0.124 ± 0.016	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.284 ± 0.015	0.533 ± 0.016	0.123	-0.016	-11.54	0.0651	0.0670	0.887	91.028 ± 0.907	0.251 ± 0.002	54.032 ± 1.582
QML	beta34	0.101	0.120	0.019 ± 0.001	0.191 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.197 ± 0.010	0.444 ± 0.013	0.119	0.018	18.12	0.0374	0.0416	0.871	90.020 ± 0.952	0.138 ± 0.002	88.710 ± 1.005
QML	beta35	0.101	0.120	0.019 ± 0.001	0.184 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.198 ± 0.011	0.445 ± 0.014	0.116	0.015	15.03	0.0356	0.0387	0.876	90.121 ± 0.947	0.140 ± 0.002	90.625 ± 0.925
UPI	beta31	0.316	0.316	0.000 ± 0.003	0.000 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.005	0.312 ± 0.009	0.312	-0.004	-1.18	0.0959	0.0960	1.001	95.161 ± 0.681	0.388 ± 0.002	89.214 ± 0.985
UPI	beta32	0.316	0.325	0.009 ± 0.003	0.029 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.004	0.314 ± 0.009	0.326	0.010	3.20	0.1019	0.1024	1.000	95.565 ± 0.654	0.387 ± 0.002	88.609 ± 1.009
UPI	beta33	0.139	0.143	0.004 ± 0.002	0.025 ± 0.016	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.260 ± 0.013	0.509 ± 0.015	0.142	0.003	2.26	0.0677	0.0678	0.878	92.440 ± 0.839	0.244 ± 0.003	65.827 ± 1.506
UPI	beta34	0.101	0.100	-0.001 ± 0.001	-0.012 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.176 ± 0.009	0.420 ± 0.013	0.100	-0.001	-0.99	0.0393	0.0393	0.855	90.827 ± 0.916	0.142 ± 0.002	76.613 ± 1.344
UPI	beta35	0.101	0.098	-0.003 ± 0.001	-0.028 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.170 ± 0.009	0.412 ± 0.013	0.098	-0.003	-3.41	0.0371	0.0372	0.877	91.230 ± 0.898	0.143 ± 0.002	73.891 ± 1.395

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S66

Simulation 1 - Condition 66: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.162 ± 0.005	0.316	0.000	0.02	0.0502	0.0502	0.983	93.800 ± 0.763	0.197 ± 0.001	99.900 ± 0.100
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.319	0.003	0.80	0.0487	0.0488	1.009	95.200 ± 0.676	0.197 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.128	-0.011 ± 0.001	-0.082 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.102 ± 0.005	0.319 ± 0.009	0.129	-0.010	-7.04	0.0426	0.0438	0.971	92.000 ± 0.858	0.163 ± 0.005	85.300 ± 1.120
LMS	beta34	0.101	0.120	0.019 ± 0.001	0.192 ± 0.007	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.119	0.018	18.18	0.0223	0.0289	0.933	87.300 ± 1.053	0.086 ± 0.001	99.600 ± 0.200
LMS	beta35	0.101	0.120	0.019 ± 0.001	0.187 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.119	0.018	17.54	0.0230	0.0290	0.941	88.100 ± 1.024	0.086 ± 0.001	99.900 ± 0.100
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.313	-0.003	-0.79	0.0627	0.0627	0.976	94.300 ± 0.733	0.243 ± 0.001	99.700 ± 0.173
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.005	0.316	0.000	0.11	0.0601	0.0601	1.006	95.400 ± 0.662	0.244 ± 0.001	99.800 ± 0.141
LSAM	beta33	0.139	0.144	0.005 ± 0.001	0.034 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.143	0.004	2.91	0.0406	0.0408	0.926	91.200 ± 0.896	0.146 ± 0.001	95.200 ± 0.676
LSAM	beta34	0.101	0.100	-0.001 ± 0.001	-0.009 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.054 ± 0.003	0.232 ± 0.007	0.099	-0.002	-1.52	0.0226	0.0227	0.911	93.600 ± 0.774	0.084 ± 0.001	98.100 ± 0.432
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.010 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.053 ± 0.003	0.231 ± 0.007	0.100	-0.001	-1.42	0.0219	0.0220	0.926	92.500 ± 0.833	0.085 ± 0.001	98.400 ± 0.397
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.162 ± 0.005	0.316	0.000	-0.09	0.0504	0.0504	0.980	93.700 ± 0.768	0.197 ± 0.001	99.900 ± 0.100
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.318	0.002	0.58	0.0488	0.0489	1.013	95.400 ± 0.662	0.198 ± 0.001	99.900 ± 0.100
QML	beta33	0.139	0.125	-0.014 ± 0.001	-0.100 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.126	-0.013	-9.15	0.0422	0.0441	0.937	91.300 ± 0.891	0.156 ± 0.001	85.000 ± 1.129
QML	beta34	0.101	0.120	0.019 ± 0.001	0.187 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.119	0.018	17.72	0.0220	0.0283	0.919	86.600 ± 1.077	0.084 ± 0.001	99.700 ± 0.173
QML	beta35	0.101	0.120	0.019 ± 0.001	0.186 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.087 ± 0.004	0.296 ± 0.008	0.119	0.018	17.53	0.0228	0.0289	0.934	87.700 ± 1.039	0.085 ± 0.001	100.000 ± 0.000
UPI	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.316	0.000	-0.04	0.0634	0.0634	0.960	93.900 ± 0.757	0.239 ± 0.001	99.800 ± 0.141
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.005	0.318	0.002	0.51	0.0603	0.0603	0.988	95.000 ± 0.689	0.240 ± 0.001	99.800 ± 0.141
UPI	beta33	0.139	0.141	0.002 ± 0.001	0.017 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.141	0.002	1.49	0.0397	0.0397	0.919	92.300 ± 0.843	0.146 ± 0.001	95.400 ± 0.662
UPI	beta34	0.101	0.101	-0.000 ± 0.001	-0.003 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.053 ± 0.003	0.231 ± 0.007	0.101	0.000	-0.01	0.0218	0.0218	0.908	93.300 ± 0.791	0.083 ± 0.001	98.500 ± 0.384
UPI	beta35	0.101	0.101	-0.000 ± 0.001	-0.002 ± 0.007	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.100	-0.001	-0.89	0.0218	0.0219	0.913	93.100 ± 0.801	0.084 ± 0.001	98.400 ± 0.397

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S67

Simulation 1 - Condition 67: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.303	-0.013 ± 0.014	-0.041 ± 0.045	0.008 ± 0.002	0.089 ± 0.013	0.008 ± 0.002	0.088 ± 0.014	0.078 ± 0.020	0.279 ± 0.044	0.313	-0.003	-0.92	0.1160	0.1161	1.141	97.368 ± 2.597	0.396 ± 0.016	86.842 ± 5.484
LMS	beta32	0.316	0.308	-0.008 ± 0.015	-0.026 ± 0.048	0.009 ± 0.002	0.093 ± 0.010	0.008 ± 0.002	0.092 ± 0.012	0.085 ± 0.017	0.291 ± 0.040	0.309	-0.007	-2.07	0.1441	0.1443	1.080	94.737 ± 3.622	0.394 ± 0.017	86.842 ± 5.484
LMS	beta33	0.139	0.190	0.051 ± 0.026	0.366 ± 0.190	0.027 ± 0.010	0.163 ± 0.035	0.028 ± 0.009	0.169 ± 0.031	1.470 ± 0.447	1.212 ± 0.219	0.162	0.023	16.42	0.1872	0.1886	0.894	100.000 ± 0.000	0.571 ± 0.044	47.368 ± 8.100
LMS	beta34	0.101	-0.002	-0.103 ± 0.017	-1.020 ± 0.164	0.010 ± 0.004	0.102 ± 0.026	0.021 ± 0.003	0.144 ± 0.013	2.037 ± 0.330	1.427 ± 0.133	-0.009	-0.110	-108.71	0.1008	0.1490	0.977	76.316 ± 6.897	0.391 ± 0.027	10.526 ± 4.978
LMS	beta35	0.101	0.006	-0.095 ± 0.024	-0.943 ± 0.241	0.022 ± 0.015	0.150 ± 0.068	0.031 ± 0.011	0.176 ± 0.038	3.034 ± 1.101	1.742 ± 0.376	0.018	-0.083	-81.86	0.1478	0.1694	0.791	60.526 ± 7.929	0.464 ± 0.061	10.526 ± 4.978
LSAM	beta31	0.316	0.312	-0.004 ± 0.040	-0.012 ± 0.126	0.060 ± 0.019	0.245 ± 0.045	0.059 ± 0.019	0.242 ± 0.048	0.586 ± 0.194	0.766 ± 0.151	0.346	0.030	9.36	0.1899	0.1922	1.459	100.000 ± 0.000	1.402 ± 0.373	44.737 ± 8.066
LSAM	beta32	0.316	0.385	0.069 ± 0.036	0.219 ± 0.115	0.050 ± 0.012	0.225 ± 0.030	0.054 ± 0.013	0.232 ± 0.034	0.539 ± 0.128	0.734 ± 0.107	0.332	0.016	5.05	0.2423	0.2428	1.649	100.000 ± 0.000	1.451 ± 0.369	36.842 ± 7.825
LSAM	beta33	0.139	0.310	0.171 ± 0.033	1.228 ± 0.237	0.041 ± 0.009	0.203 ± 0.025	0.069 ± 0.018	0.263 ± 0.038	3.592 ± 0.926	1.895 ± 0.270	0.281	0.142	102.50	0.2033	0.2483	1.406	100.000 ± 0.000	1.121 ± 0.235	21.053 ± 6.613
LSAM	beta34	0.101	-0.071	-0.172 ± 0.027	-1.699 ± 0.271	0.028 ± 0.007	0.168 ± 0.023	0.057 ± 0.013	0.239 ± 0.029	5.594 ± 1.233	2.365 ± 0.286	-0.059	-0.160	-158.59	0.1892	0.2479	1.466	84.211 ± 5.915	0.968 ± 0.166	2.632 ± 2.597
LSAM	beta35	0.101	-0.117	-0.218 ± 0.048	-2.159 ± 0.476	0.088 ± 0.031	0.296 ± 0.060	0.133 ± 0.029	0.365 ± 0.045	13.038 ± 2.813	3.611 ± 0.449	-0.047	-0.148	-146.97	0.2713	0.3092	1.434	89.474 ± 4.978	1.665 ± 0.455	0.000 ± 0.000
QML	beta31	0.316	0.300	-0.016 ± 0.014	-0.051 ± 0.045	0.008 ± 0.002	0.087 ± 0.012	0.008 ± 0.002	0.087 ± 0.014	0.076 ± 0.019	0.276 ± 0.043	0.317	0.001	0.25	0.0824	0.0824	1.098	97.368 ± 2.597	0.374 ± 0.012	89.474 ± 4.978
QML	beta32	0.316	0.308	-0.008 ± 0.014	-0.026 ± 0.045	0.008 ± 0.002	0.088 ± 0.009	0.008 ± 0.002	0.088 ± 0.012	0.077 ± 0.015	0.277 ± 0.037	0.293	-0.023	-7.39	0.0959	0.0987	1.085	94.737 ± 3.622	0.376 ± 0.011	92.105 ± 4.374
QML	beta33	0.139	0.213	0.074 ± 0.019	0.530 ± 0.136	0.014 ± 0.003	0.116 ± 0.012	0.019 ± 0.004	0.136 ± 0.017	0.963 ± 0.210	0.981 ± 0.125	0.195	0.056	40.18	0.1516	0.1616	1.223	100.000 ± 0.000	0.558 ± 0.033	42.105 ± 8.009
QML	beta34	0.101	-0.033	-0.134 ± 0.013	-1.326 ± 0.125	0.006 ± 0.001	0.078 ± 0.009	0.024 ± 0.003	0.154 ± 0.011	2.337 ± 0.326	1.529 ± 0.112	-0.037	-0.138	-136.83	0.0937	0.1669	1.195	65.789 ± 7.696	0.365 ± 0.025	7.895 ± 4.374
QML	beta35	0.101	-0.034	-0.135 ± 0.014	-1.339 ± 0.138	0.007 ± 0.002	0.086 ± 0.017	0.025 ± 0.003	0.160 ± 0.011	2.494 ± 0.333	1.579 ± 0.112	-0.006	-0.107	-105.87	0.1135	0.1559	1.173	63.158 ± 7.825	0.394 ± 0.030	7.895 ± 4.374
UPI	beta31	0.316	0.227	-0.089 ± 0.059	-0.283 ± 0.188	0.134 ± 0.063	0.366 ± 0.106	0.138 ± 0.069	0.372 ± 0.114	1.385 ± 0.695	1.177 ± 0.361	0.367	0.051	16.13	0.5065	0.5090	1.743	92.105 ± 4.374	2.500 ± 0.698	21.053 ± 6.613
UPI	beta32	0.316	0.366	0.050 ± 0.056	0.159 ± 0.176	0.117 ± 0.029	0.343 ± 0.047	0.117 ± 0.030	0.342 ± 0.054	1.170 ± 0.299	1.082 ± 0.171	0.257	-0.059	-18.72	0.4635	0.4672	1.502	92.105 ± 4.374	2.017 ± 0.400	15.789 ± 5.915
UPI	beta33	0.139	0.180	0.041 ± 0.036	0.293 ± 0.256	0.048 ± 0.012	0.219 ± 0.031	0.048 ± 0.014	0.220 ± 0.039	2.507 ± 0.739	1.583 ± 0.279	0.124	-0.015	-10.70	0.3027	0.3030	2.766	100.000 ± 0.000	2.377 ± 0.715	2.632 ± 2.597
UPI	beta34	0.101	-0.104	-0.205 ± 0.091	-2.034 ± 0.902	0.315 ± 0.100	0.561 ± 0.100	0.349 ± 0.113	0.591 ± 0.111	34.221 ± 11.118	5.850 ± 1.100	-0.177	-0.278	-275.08	0.6789	0.7336	1.792	97.368 ± 2.597	3.944 ± 1.064	0.000 ± 0.000
UPI	beta35	0.101	-0.294	-0.395 ± 0.093	-3.910 ± 0.916	0.326 ± 0.132	0.571 ± 0.138	0.473 ± 0.190	0.688 ± 0.158	46.361 ± 18.618	6.809 ± 1.568	-0.086	-0.187	-185.05	0.6912	0.7160	1.791	86.842 ± 5.484	4.005 ± 0.932	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S68

Simulation 1 - Condition 68: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.012	0.001 ± 0.038	0.006 ± 0.001	0.076 ± 0.008	0.006 ± 0.001	0.075 ± 0.010	0.056 ± 0.012	0.237 ± 0.032	0.317	0.001	0.20	0.0798	0.0798	0.809	92.500 ± 4.165	0.240 ± 0.006	100.000 ± 0.000
LMS	beta32	0.316	0.320	0.004 ± 0.010	0.014 ± 0.031	0.004 ± 0.001	0.063 ± 0.006	0.004 ± 0.001	0.062 ± 0.008	0.039 ± 0.007	0.197 ± 0.024	0.320	0.004	1.38	0.0802	0.0804	1.073	97.500 ± 2.469	0.265 ± 0.016	100.000 ± 0.000
LMS	beta33	0.139	0.229	0.090 ± 0.016	0.649 ± 0.113	0.010 ± 0.004	0.100 ± 0.023	0.018 ± 0.006	0.134 ± 0.025	0.923 ± 0.301	0.960 ± 0.177	0.211	0.072	51.50	0.1104	0.1316	0.919	82.500 ± 6.008	0.359 ± 0.025	80.000 ± 6.325
LMS	beta34	0.101	0.004	-0.097 ± 0.009	-0.958 ± 0.086	0.003 ± 0.001	0.055 ± 0.008	0.012 ± 0.001	0.111 ± 0.007	1.209 ± 0.142	1.100 ± 0.068	-0.013	-0.114	-112.42	0.0687	0.1327	1.215	50.000 ± 7.906	0.263 ± 0.030	7.500 ± 4.165
LMS	beta35	0.101	0.017	-0.084 ± 0.012	-0.830 ± 0.123	0.006 ± 0.002	0.078 ± 0.018	0.013 ± 0.004	0.114 ± 0.018	1.277 ± 0.359	1.130 ± 0.177	0.011	-0.090	-89.51	0.0702	0.1145	1.033	72.500 ± 7.060	0.318 ± 0.036	5.000 ± 3.446
LSAM	beta31	0.316	0.347	0.031 ± 0.019	0.099 ± 0.061	0.015 ± 0.002	0.123 ± 0.011	0.016 ± 0.003	0.125 ± 0.015	0.157 ± 0.028	0.396 ± 0.048	0.343	0.027	8.57	0.1438	0.1463	0.980	100.000 ± 0.000	0.471 ± 0.045	85.000 ± 5.646
LSAM	beta32	0.316	0.314	-0.002 ± 0.018	-0.008 ± 0.058	0.014 ± 0.003	0.117 ± 0.013	0.013 ± 0.003	0.115 ± 0.016	0.133 ± 0.028	0.365 ± 0.050	0.314	-0.002	-0.61	0.1178	0.1179	0.985	97.500 ± 2.469	0.451 ± 0.037	82.500 ± 6.008
LSAM	beta33	0.139	0.262	0.123 ± 0.015	0.884 ± 0.109	0.009 ± 0.002	0.096 ± 0.013	0.024 ± 0.004	0.155 ± 0.015	1.244 ± 0.222	1.115 ± 0.108	0.253	0.114	82.01	0.0957	0.1488	1.076	82.500 ± 6.008	0.403 ± 0.023	85.000 ± 5.646
LSAM	beta34	0.101	-0.072	-0.173 ± 0.019	-1.710 ± 0.185	0.014 ± 0.005	0.118 ± 0.027	0.043 ± 0.011	0.208 ± 0.028	4.257 ± 1.047	2.063 ± 0.274	-0.077	-0.178	-176.00	0.1202	0.2146	1.020	60.000 ± 7.746	0.472 ± 0.067	10.000 ± 4.743
LSAM	beta35	0.101	-0.020	-0.121 ± 0.017	-1.199 ± 0.165	0.011 ± 0.002	0.105 ± 0.011	0.025 ± 0.004	0.160 ± 0.014	2.496 ± 0.403	1.580 ± 0.140	-0.056	-0.157	-155.00	0.0905	0.1808	1.053	67.500 ± 7.406	0.435 ± 0.038	5.000 ± 3.446
QML	beta31	0.316	0.306	-0.010 ± 0.011	-0.032 ± 0.035	0.005 ± 0.001	0.070 ± 0.008	0.005 ± 0.001	0.070 ± 0.009	0.050 ± 0.010	0.223 ± 0.030	0.305	-0.011	-3.64	0.0638	0.0649	0.821	92.500 ± 4.165	0.227 ± 0.004	100.000 ± 0.000
QML	beta32	0.316	0.307	-0.009 ± 0.009	-0.030 ± 0.029	0.003 ± 0.001	0.058 ± 0.006	0.003 ± 0.001	0.058 ± 0.007	0.034 ± 0.006	0.183 ± 0.023	0.291	-0.025	-7.85	0.0586	0.0636	1.026	95.000 ± 3.446	0.233 ± 0.003	100.000 ± 0.000
QML	beta33	0.139	0.210	0.071 ± 0.012	0.511 ± 0.087	0.006 ± 0.001	0.076 ± 0.010	0.011 ± 0.002	0.104 ± 0.010	0.556 ± 0.096	0.745 ± 0.073	0.213	0.074	52.99	0.0703	0.1018	1.089	82.500 ± 6.008	0.326 ± 0.011	77.500 ± 6.603
QML	beta34	0.101	-0.021	-0.122 ± 0.008	-1.205 ± 0.078	0.002 ± 0.001	0.050 ± 0.006	0.017 ± 0.002	0.131 ± 0.007	1.686 ± 0.177	1.299 ± 0.070	-0.037	-0.138	-136.38	0.0502	0.1466	1.179	42.500 ± 7.816	0.229 ± 0.011	7.500 ± 4.165
QML	beta35	0.101	-0.004	-0.105 ± 0.009	-1.041 ± 0.084	0.003 ± 0.001	0.054 ± 0.006	0.014 ± 0.002	0.118 ± 0.008	1.362 ± 0.177	1.167 ± 0.079	-0.017	-0.118	-116.73	0.0498	0.1280	1.093	50.000 ± 7.906	0.231 ± 0.011	12.500 ± 5.229
UPI	beta31	0.316	0.336	0.020 ± 0.079	0.063 ± 0.250	0.250 ± 0.094	0.500 ± 0.108	0.244 ± 0.095	0.494 ± 0.113	2.444 ± 0.954	1.563 ± 0.358	0.392	0.076	24.10	0.5135	0.5192	1.288	77.500 ± 6.603	2.525 ± 0.609	22.500 ± 6.603
UPI	beta32	0.316	0.265	-0.051 ± 0.111	-0.162 ± 0.351	0.493 ± 0.222	0.702 ± 0.189	0.484 ± 0.226	0.695 ± 0.194	4.844 ± 2.265	2.201 ± 0.615	0.290	-0.026	-8.16	0.5685	0.5691	1.120	85.000 ± 5.646	3.085 ± 0.999	20.000 ± 6.325
UPI	beta33	0.139	0.142	0.003 ± 0.052	0.019 ± 0.372	0.107 ± 0.045	0.327 ± 0.080	0.104 ± 0.045	0.323 ± 0.082	5.389 ± 2.306	2.321 ± 0.587	0.174	0.035	25.05	0.3351	0.3369	1.578	100.000 ± 0.000	2.021 ± 0.457	25.000 ± 6.847
UPI	beta34	0.101	-0.515	-0.616 ± 0.224	-6.104 ± 2.214	2.000 ± 0.654	1.414 ± 0.258	2.330 ± 0.873	1.526 ± 0.322	228.410 ± 85.546	15.113 ± 3.184	-0.507	-0.608	-601.81	1.3286	1.4611	0.969	90.000 ± 4.743	5.369 ± 1.305	2.500 ± 2.469
UPI	beta35	0.101	-0.164	-0.265 ± 0.168	-2.624 ± 1.661	1.126 ± 0.445	1.061 ± 0.239	1.168 ± 0.501	1.081 ± 0.267	114.498 ± 49.115	10.700 ± 2.641	-0.144	-0.245	-242.77	1.1353	1.1614	0.930	87.500 ± 5.229	3.868 ± 1.029	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S69

Simulation 1 - Condition 69: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.308	-0.008 ± 0.003	-0.024 ± 0.009	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.003	0.226 ± 0.007	0.309	-0.007	-2.32	0.0738	0.0741	1.029	95.065 ± 0.825	0.286 ± 0.003	98.549 ± 0.456
LMS	beta32	0.316	0.314	-0.002 ± 0.003	-0.008 ± 0.009	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.235 ± 0.008	0.311	-0.005	-1.61	0.0761	0.0763	0.992	95.646 ± 0.777	0.289 ± 0.006	97.678 ± 0.574
LMS	beta33	0.139	0.170	0.031 ± 0.003	0.221 ± 0.024	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	0.459 ± 0.027	0.678 ± 0.023	0.172	0.033	23.39	0.0844	0.0905	1.097	91.147 ± 1.082	0.383 ± 0.014	53.701 ± 1.900
LMS	beta34	0.101	0.049	-0.052 ± 0.003	-0.515 ± 0.031	0.007 ± 0.000	0.083 ± 0.003	0.010 ± 0.000	0.098 ± 0.003	0.943 ± 0.041	0.971 ± 0.025	0.036	-0.065	-64.24	0.0763	0.1002	1.025	77.213 ± 1.598	0.334 ± 0.009	5.225 ± 0.848
LMS	beta35	0.101	0.045	-0.056 ± 0.003	-0.556 ± 0.029	0.006 ± 0.000	0.078 ± 0.003	0.009 ± 0.000	0.096 ± 0.002	0.907 ± 0.039	0.953 ± 0.024	0.038	-0.063	-62.32	0.0725	0.0960	1.120	76.343 ± 1.619	0.343 ± 0.017	2.032 ± 0.538
LSAM	beta31	0.316	0.310	-0.006 ± 0.003	-0.018 ± 0.010	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.063 ± 0.003	0.251 ± 0.008	0.312	-0.004	-1.17	0.0796	0.0797	1.015	94.775 ± 0.848	0.316 ± 0.002	96.517 ± 0.699
LSAM	beta32	0.316	0.317	0.001 ± 0.003	0.003 ± 0.010	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	0.066 ± 0.004	0.257 ± 0.008	0.318	0.002	0.62	0.0816	0.0816	0.993	95.646 ± 0.777	0.316 ± 0.002	96.517 ± 0.699
LSAM	beta33	0.139	0.187	0.048 ± 0.003	0.349 ± 0.025	0.008 ± 0.000	0.091 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.551 ± 0.033	0.742 ± 0.025	0.187	0.048	34.74	0.0829	0.0959	1.015	91.292 ± 1.074	0.363 ± 0.003	55.152 ± 1.895
LSAM	beta34	0.101	0.046	-0.055 ± 0.004	-0.547 ± 0.043	0.013 ± 0.001	0.113 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	1.542 ± 0.080	1.242 ± 0.038	0.034	-0.067	-66.23	0.1059	0.1253	0.975	86.357 ± 1.308	0.431 ± 0.006	5.951 ± 0.901
LSAM	beta35	0.101	0.037	-0.064 ± 0.004	-0.638 ± 0.039	0.011 ± 0.001	0.103 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	1.453 ± 0.074	1.205 ± 0.035	0.036	-0.065	-64.78	0.1027	0.1217	1.047	86.067 ± 1.319	0.424 ± 0.005	3.048 ± 0.655
QML	beta31	0.316	0.305	-0.011 ± 0.003	-0.036 ± 0.008	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.003	0.223 ± 0.007	0.305	-0.011	-3.62	0.0724	0.0733	1.013	94.775 ± 0.848	0.277 ± 0.002	98.984 ± 0.382
QML	beta32	0.316	0.310	-0.006 ± 0.003	-0.019 ± 0.009	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.003	0.230 ± 0.008	0.309	-0.007	-2.06	0.0738	0.0741	0.973	95.065 ± 0.825	0.276 ± 0.001	98.403 ± 0.478
QML	beta33	0.139	0.170	0.031 ± 0.003	0.226 ± 0.023	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.421 ± 0.023	0.649 ± 0.021	0.171	0.032	22.89	0.0822	0.0882	1.037	91.582 ± 1.058	0.344 ± 0.004	53.701 ± 1.900
QML	beta34	0.101	0.035	-0.066 ± 0.003	-0.655 ± 0.028	0.005 ± 0.000	0.073 ± 0.002	0.010 ± 0.000	0.098 ± 0.002	0.951 ± 0.039	0.975 ± 0.023	0.027	-0.074	-73.20	0.0709	0.1024	0.990	74.020 ± 1.671	0.283 ± 0.003	6.821 ± 0.960
QML	beta35	0.101	0.031	-0.070 ± 0.003	-0.695 ± 0.026	0.005 ± 0.000	0.068 ± 0.002	0.010 ± 0.000	0.098 ± 0.002	0.933 ± 0.038	0.966 ± 0.022	0.029	-0.072	-70.85	0.0702	0.1002	1.074	74.601 ± 1.658	0.286 ± 0.004	3.338 ± 0.684
UPI	beta31	0.316	0.320	0.004 ± 0.004	0.012 ± 0.012	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.100 ± 0.006	0.316 ± 0.011	0.325	0.009	2.81	0.0954	0.0959	1.313	96.952 ± 0.655	0.513 ± 0.019	79.245 ± 1.545
UPI	beta32	0.316	0.325	0.009 ± 0.004	0.030 ± 0.012	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	0.102 ± 0.006	0.320 ± 0.011	0.327	0.011	3.63	0.0980	0.0987	1.290	97.242 ± 0.624	0.509 ± 0.020	79.826 ± 1.529
UPI	beta33	0.139	0.196	0.057 ± 0.004	0.409 ± 0.026	0.009 ± 0.001	0.096 ± 0.003	0.013 ± 0.001	0.112 ± 0.004	0.647 ± 0.038	0.804 ± 0.026	0.193	0.054	38.71	0.0921	0.1067	1.263	95.501 ± 0.790	0.477 ± 0.009	44.412 ± 1.893
UPI	beta34	0.101	0.054	-0.047 ± 0.007	-0.467 ± 0.072	0.036 ± 0.003	0.191 ± 0.007	0.039 ± 0.003	0.196 ± 0.008	3.777 ± 0.274	1.944 ± 0.079	0.046	-0.055	-54.10	0.1746	0.1830	1.239	95.065 ± 0.825	0.926 ± 0.041	5.225 ± 0.848
UPI	beta35	0.101	0.053	-0.048 ± 0.007	-0.475 ± 0.066	0.030 ± 0.002	0.174 ± 0.006	0.032 ± 0.002	0.180 ± 0.007	3.186 ± 0.214	1.785 ± 0.068	0.055	-0.046	-45.74	0.1544	0.1611	1.275	94.049 ± 0.901	0.869 ± 0.036	3.628 ± 0.712

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S70

Simulation 1 - Condition 70: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.002	-0.019 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.021 ± 0.001	0.144 ± 0.005	0.310	-0.006	-1.82	0.0451	0.0455	0.987	95.059 ± 0.782	0.174 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.313	-0.003	-0.90	0.0449	0.0450	1.002	95.449 ± 0.752	0.175 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.175	0.036 ± 0.002	0.261 ± 0.014	0.003 ± 0.000	0.054 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.220 ± 0.010	0.469 ± 0.012	0.178	0.039	28.10	0.0523	0.0653	1.015	87.906 ± 1.176	0.215 ± 0.002	86.736 ± 1.223
LMS	beta34	0.101	0.040	-0.061 ± 0.002	-0.599 ± 0.018	0.003 ± 0.000	0.050 ± 0.002	0.006 ± 0.000	0.079 ± 0.001	0.608 ± 0.021	0.780 ± 0.014	0.035	-0.066	-65.37	0.0464	0.0807	0.985	64.109 ± 1.730	0.195 ± 0.003	6.112 ± 0.864
LMS	beta35	0.101	0.042	-0.059 ± 0.002	-0.587 ± 0.018	0.003 ± 0.000	0.050 ± 0.001	0.006 ± 0.000	0.078 ± 0.001	0.591 ± 0.020	0.769 ± 0.014	0.035	-0.066	-65.73	0.0447	0.0800	0.991	66.190 ± 1.706	0.195 ± 0.003	7.282 ± 0.937
LSAM	beta31	0.316	0.311	-0.005 ± 0.002	-0.014 ± 0.006	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.026 ± 0.001	0.161 ± 0.005	0.312	-0.004	-1.38	0.0493	0.0495	0.971	94.148 ± 0.846	0.193 ± 0.001	99.870 ± 0.130
LSAM	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.005	0.310	-0.006	-1.80	0.0498	0.0501	1.007	95.059 ± 0.782	0.193 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.181	0.042 ± 0.002	0.301 ± 0.014	0.003 ± 0.000	0.055 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.249 ± 0.012	0.499 ± 0.014	0.181	0.042	30.18	0.0537	0.0682	0.982	88.687 ± 1.142	0.213 ± 0.001	89.857 ± 1.089
LSAM	beta34	0.101	0.041	-0.060 ± 0.002	-0.597 ± 0.024	0.005 ± 0.000	0.067 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	0.799 ± 0.035	0.894 ± 0.021	0.037	-0.064	-63.14	0.0621	0.0890	0.967	79.974 ± 1.443	0.255 ± 0.002	9.753 ± 1.070
LSAM	beta35	0.101	0.043	-0.058 ± 0.002	-0.573 ± 0.025	0.005 ± 0.000	0.069 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	0.791 ± 0.032	0.889 ± 0.020	0.038	-0.063	-62.69	0.0649	0.0907	0.955	80.104 ± 1.440	0.257 ± 0.002	9.623 ± 1.063
QML	beta31	0.316	0.306	-0.010 ± 0.002	-0.030 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.021 ± 0.001	0.144 ± 0.005	0.307	-0.009	-2.79	0.0447	0.0455	0.986	93.888 ± 0.864	0.172 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.309	-0.007 ± 0.002	-0.022 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.309	-0.007	-2.25	0.0451	0.0457	0.996	95.059 ± 0.782	0.172 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.172	0.033 ± 0.002	0.237 ± 0.014	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.203 ± 0.010	0.450 ± 0.012	0.174	0.035	24.93	0.0528	0.0631	1.017	89.077 ± 1.125	0.212 ± 0.003	86.736 ± 1.223
QML	beta34	0.101	0.030	-0.071 ± 0.002	-0.703 ± 0.016	0.002 ± 0.000	0.046 ± 0.001	0.007 ± 0.000	0.084 ± 0.001	0.699 ± 0.023	0.836 ± 0.014	0.026	-0.075	-74.60	0.0429	0.0867	1.013	58.388 ± 1.777	0.182 ± 0.004	8.453 ± 1.003
QML	beta35	0.101	0.031	-0.070 ± 0.002	-0.694 ± 0.016	0.002 ± 0.000	0.046 ± 0.001	0.007 ± 0.000	0.084 ± 0.001	0.688 ± 0.022	0.830 ± 0.014	0.026	-0.075	-73.88	0.0430	0.0861	0.973	58.778 ± 1.775	0.175 ± 0.001	8.062 ± 0.982
UPI	beta31	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.008	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.047 ± 0.003	0.217 ± 0.008	0.316	0.000	0.04	0.0636	0.0636	1.225	97.399 ± 0.574	0.329 ± 0.013	92.848 ± 0.929
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.003	0.200 ± 0.007	0.311	-0.005	-1.44	0.0633	0.0634	1.290	98.049 ± 0.499	0.320 ± 0.010	93.238 ± 0.905
UPI	beta33	0.139	0.183	0.044 ± 0.002	0.317 ± 0.016	0.004 ± 0.000	0.063 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.306 ± 0.016	0.553 ± 0.016	0.186	0.047	33.59	0.0607	0.0766	1.130	93.758 ± 0.872	0.279 ± 0.004	76.723 ± 1.524
UPI	beta34	0.101	0.047	-0.054 ± 0.004	-0.534 ± 0.042	0.014 ± 0.001	0.117 ± 0.004	0.017 ± 0.001	0.129 ± 0.004	1.623 ± 0.097	1.274 ± 0.043	0.040	-0.061	-60.12	0.1063	0.1225	1.187	92.198 ± 0.967	0.544 ± 0.031	8.453 ± 1.003
UPI	beta35	0.101	0.051	-0.050 ± 0.004	-0.496 ± 0.041	0.013 ± 0.001	0.116 ± 0.004	0.016 ± 0.001	0.126 ± 0.004	1.565 ± 0.091	1.251 ± 0.041	0.045	-0.056	-55.57	0.1008	0.1153	1.171	94.148 ± 0.846	0.533 ± 0.015	7.022 ± 0.921

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S71

Simulation 1 - Condition 71: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.314	-0.002	-0.48	0.0557	0.0557	0.985	94.494 ± 0.722	0.227 ± 0.001	99.700 ± 0.173
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.316	0.000	0.00	0.0608	0.0608	0.965	94.695 ± 0.709	0.228 ± 0.001	99.800 ± 0.141
LMS	beta33	0.139	0.148	0.009 ± 0.002	0.066 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.243 ± 0.011	0.493 ± 0.014	0.152	0.013	9.34	0.0691	0.0703	0.978	94.595 ± 0.715	0.260 ± 0.002	64.164 ± 1.517
LMS	beta34	0.101	0.077	-0.024 ± 0.002	-0.240 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.483 ± 0.021	0.695 ± 0.018	0.071	-0.030	-29.34	0.0641	0.0706	0.969	88.689 ± 1.002	0.250 ± 0.002	16.717 ± 1.181
LMS	beta35	0.101	0.080	-0.021 ± 0.002	-0.211 ± 0.022	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.506 ± 0.023	0.711 ± 0.019	0.074	-0.027	-26.98	0.0684	0.0736	0.923	88.689 ± 1.002	0.248 ± 0.002	20.521 ± 1.278
LSAM	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.317	0.001	0.36	0.0583	0.0583	0.984	94.695 ± 0.709	0.232 ± 0.001	99.600 ± 0.200
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.317	0.001	0.35	0.0609	0.0609	0.971	95.095 ± 0.683	0.233 ± 0.001	99.800 ± 0.141
LSAM	beta33	0.139	0.150	0.011 ± 0.002	0.076 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.225 ± 0.010	0.474 ± 0.013	0.152	0.013	9.56	0.0673	0.0686	0.978	95.095 ± 0.683	0.250 ± 0.001	65.966 ± 1.499
LSAM	beta34	0.101	0.088	-0.013 ± 0.002	-0.127 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.585 ± 0.026	0.765 ± 0.021	0.083	-0.018	-17.40	0.0754	0.0774	0.956	93.193 ± 0.797	0.286 ± 0.001	20.320 ± 1.273
LSAM	beta35	0.101	0.091	-0.010 ± 0.002	-0.100 ± 0.024	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.601 ± 0.028	0.775 ± 0.022	0.088	-0.013	-12.38	0.0762	0.0772	0.935	92.993 ± 0.808	0.285 ± 0.001	23.624 ± 1.344
QML	beta31	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.313	-0.003	-0.81	0.0552	0.0552	0.982	94.394 ± 0.728	0.226 ± 0.001	99.800 ± 0.141
QML	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.315	-0.001	-0.45	0.0605	0.0605	0.964	94.595 ± 0.715	0.227 ± 0.001	99.800 ± 0.141
QML	beta33	0.139	0.148	0.009 ± 0.002	0.065 ± 0.015	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.235 ± 0.011	0.484 ± 0.013	0.152	0.013	9.10	0.0683	0.0695	0.977	94.595 ± 0.715	0.256 ± 0.002	64.364 ± 1.515
QML	beta34	0.101	0.073	-0.028 ± 0.002	-0.280 ± 0.020	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.473 ± 0.020	0.688 ± 0.017	0.069	-0.032	-32.15	0.0620	0.0700	0.966	87.688 ± 1.040	0.241 ± 0.002	17.718 ± 1.208
QML	beta35	0.101	0.075	-0.026 ± 0.002	-0.253 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.487 ± 0.021	0.698 ± 0.018	0.071	-0.030	-29.44	0.0665	0.0729	0.932	88.188 ± 1.021	0.240 ± 0.002	21.421 ± 1.298
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.013 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.321	0.005	1.61	0.0594	0.0596	0.938	93.694 ± 0.769	0.223 ± 0.001	99.800 ± 0.141
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.195 ± 0.006	0.322	0.006	1.94	0.0604	0.0607	0.928	93.594 ± 0.775	0.224 ± 0.001	99.800 ± 0.141
UPI	beta33	0.139	0.147	0.008 ± 0.002	0.054 ± 0.015	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.225 ± 0.010	0.474 ± 0.013	0.147	0.008	5.66	0.0663	0.0667	0.980	94.294 ± 0.734	0.252 ± 0.001	65.365 ± 1.505
UPI	beta34	0.101	0.097	-0.004 ± 0.003	-0.036 ± 0.025	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.644 ± 0.030	0.803 ± 0.023	0.093	-0.008	-7.53	0.0796	0.0800	0.952	93.794 ± 0.763	0.302 ± 0.002	23.023 ± 1.332
UPI	beta35	0.101	0.100	-0.001 ± 0.003	-0.005 ± 0.025	0.006 ± 0.000	0.081 ± 0.002	0.006 ± 0.000	0.081 ± 0.002	0.636 ± 0.030	0.797 ± 0.023	0.098	-0.003	-2.72	0.0830	0.0830	0.952	94.895 ± 0.696	0.301 ± 0.002	26.126 ± 1.390

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S72

Simulation 1 - Condition 72: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.001	-0.011 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.313	-0.003	-0.98	0.0379	0.0380	0.989	94.900 ± 0.696	0.143 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.311	-0.005 ± 0.001	-0.017 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.311	-0.005	-1.65	0.0364	0.0367	0.990	94.500 ± 0.721	0.143 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.150	0.011 ± 0.001	0.083 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.152	0.013	9.18	0.0381	0.0402	1.000	93.500 ± 0.780	0.161 ± 0.001	92.800 ± 0.817
LMS	beta34	0.101	0.075	-0.026 ± 0.001	-0.262 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.221 ± 0.009	0.470 ± 0.011	0.073	-0.028	-27.92	0.0385	0.0478	1.002	86.700 ± 1.074	0.155 ± 0.001	45.600 ± 1.575
LMS	beta35	0.101	0.074	-0.027 ± 0.001	-0.265 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.224 ± 0.009	0.473 ± 0.011	0.073	-0.028	-28.14	0.0382	0.0476	0.989	87.000 ± 1.063	0.154 ± 0.001	45.500 ± 1.575
LSAM	beta31	0.316	0.314	-0.002 ± 0.001	-0.005 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	-0.63	0.0378	0.0378	0.985	95.000 ± 0.689	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.312	-0.004 ± 0.001	-0.012 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.120 ± 0.003	0.312	-0.004	-1.30	0.0383	0.0385	0.985	94.600 ± 0.715	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.150	0.011 ± 0.001	0.076 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.150	0.011	7.80	0.0388	0.0403	0.993	94.000 ± 0.751	0.157 ± 0.001	94.200 ± 0.739
LSAM	beta34	0.101	0.088	-0.013 ± 0.001	-0.125 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.222 ± 0.010	0.472 ± 0.012	0.088	-0.013	-12.71	0.0450	0.0468	0.997	93.400 ± 0.785	0.180 ± 0.001	48.200 ± 1.580
LSAM	beta35	0.101	0.088	-0.013 ± 0.001	-0.130 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.224 ± 0.010	0.473 ± 0.012	0.086	-0.015	-14.47	0.0451	0.0474	0.994	94.100 ± 0.745	0.179 ± 0.001	48.100 ± 1.580
QML	beta31	0.316	0.312	-0.004 ± 0.001	-0.014 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.312	-0.004	-1.38	0.0378	0.0381	0.986	94.600 ± 0.715	0.142 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.309	-0.007 ± 0.001	-0.021 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.309	-0.007	-2.06	0.0365	0.0370	0.986	94.400 ± 0.727	0.142 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.149	0.010 ± 0.001	0.075 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.150	0.011	8.20	0.0382	0.0398	1.000	93.900 ± 0.757	0.159 ± 0.001	93.000 ± 0.807
QML	beta34	0.101	0.071	-0.030 ± 0.001	-0.294 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.231 ± 0.009	0.480 ± 0.011	0.070	-0.031	-30.64	0.0381	0.0491	0.998	84.400 ± 1.147	0.150 ± 0.001	44.700 ± 1.572
QML	beta35	0.101	0.071	-0.030 ± 0.001	-0.296 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.234 ± 0.009	0.483 ± 0.011	0.070	-0.031	-30.99	0.0372	0.0486	0.987	84.500 ± 1.144	0.150 ± 0.001	44.500 ± 1.572
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.317	0.001	0.38	0.0383	0.0383	0.951	94.600 ± 0.715	0.140 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.005 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.315	-0.001	-0.42	0.0388	0.0388	0.951	94.000 ± 0.751	0.140 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.144	0.005 ± 0.001	0.036 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.087 ± 0.004	0.296 ± 0.008	0.144	0.005	3.60	0.0385	0.0388	0.984	94.300 ± 0.733	0.157 ± 0.001	93.000 ± 0.807
UPI	beta34	0.101	0.098	-0.003 ± 0.001	-0.034 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.220 ± 0.010	0.469 ± 0.013	0.096	-0.005	-4.78	0.0476	0.0479	1.005	95.300 ± 0.669	0.186 ± 0.001	53.900 ± 1.576
UPI	beta35	0.101	0.097	-0.004 ± 0.002	-0.043 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.225 ± 0.010	0.474 ± 0.013	0.095	-0.006	-6.23	0.0462	0.0467	0.990	94.700 ± 0.708	0.185 ± 0.001	53.400 ± 1.577

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S73

Simulation 1 - Condition 73: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.312	-0.004 ± 0.003	-0.013 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.095 ± 0.005	0.308 ± 0.009	0.308	-0.008	-2.51	0.0957	0.0960	1.008	94.562 ± 0.755	0.385 ± 0.002	—
LMS	beta32	0.316	0.318	0.002 ± 0.003	0.007 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.101 ± 0.005	0.317 ± 0.009	0.316	0.000	0.07	0.1019	0.1019	0.985	94.673 ± 0.748	0.387 ± 0.002	—
LMS	beta33	0.000	-0.039	-0.039 ± 0.004	—	0.017 ± 0.001	0.131 ± 0.003	0.019 ± 0.001	0.137 ± 0.004	—	—	-0.036	-0.036	—	0.1320	0.1367	1.038	93.119 ± 0.843	0.534 ± 0.007	6.881 ± 0.843
LMS	beta34	0.000	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.004	0.004	—	0.0855	0.0856	1.007	94.562 ± 0.755	0.349 ± 0.005	5.438 ± 0.755
LMS	beta35	0.000	0.004	0.004 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.002	0.002	—	0.0857	0.0857	1.053	95.782 ± 0.670	0.344 ± 0.004	4.218 ± 0.670
LSAM	beta31	0.316	0.320	0.004 ± 0.004	0.013 ± 0.012	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	0.131 ± 0.007	0.362 ± 0.011	0.315	-0.001	-0.18	0.1108	0.1108	1.036	96.115 ± 0.644	0.464 ± 0.005	—
LSAM	beta32	0.316	0.325	0.009 ± 0.004	0.028 ± 0.012	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.003	0.135 ± 0.007	0.367 ± 0.011	0.326	0.010	3.27	0.1210	0.1214	1.022	95.339 ± 0.702	0.463 ± 0.005	—
LSAM	beta33	0.000	0.001	0.001 ± 0.005	—	0.027 ± 0.002	0.165 ± 0.005	0.027 ± 0.002	0.165 ± 0.006	—	—	0.005	0.005	—	0.1517	0.1518	1.105	98.335 ± 0.426	0.714 ± 0.020	1.665 ± 0.426
LSAM	beta34	0.000	-0.002	-0.002 ± 0.004	—	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.119 ± 0.004	—	—	-0.002	-0.002	—	0.1081	0.1081	1.054	97.780 ± 0.491	0.493 ± 0.012	2.220 ± 0.491
LSAM	beta35	0.000	-0.002	-0.002 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	—	—	-0.002	-0.002	—	0.1005	0.1006	1.111	98.668 ± 0.382	0.491 ± 0.011	1.332 ± 0.382
QML	beta31	0.316	0.312	-0.004 ± 0.003	-0.012 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.095 ± 0.005	0.308 ± 0.009	0.308	-0.008	-2.56	0.0948	0.0951	1.006	94.562 ± 0.755	0.384 ± 0.002	—
QML	beta32	0.316	0.318	0.002 ± 0.003	0.007 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.317 ± 0.009	0.316	0.000	-0.05	0.1017	0.1017	0.985	94.895 ± 0.733	0.387 ± 0.002	—
QML	beta33	0.000	-0.039	-0.039 ± 0.004	—	0.017 ± 0.001	0.130 ± 0.004	0.018 ± 0.001	0.136 ± 0.004	—	—	-0.036	-0.036	—	0.1315	0.1363	1.058	93.785 ± 0.804	0.540 ± 0.010	6.215 ± 0.804
QML	beta34	0.000	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.004	0.004	—	0.0856	0.0857	1.021	94.673 ± 0.748	0.352 ± 0.006	5.327 ± 0.748
QML	beta35	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	—	—	0.002	0.002	—	0.0852	0.0852	1.079	95.671 ± 0.678	0.347 ± 0.005	4.329 ± 0.678
UPI	beta31	0.316	0.321	0.005 ± 0.004	0.015 ± 0.012	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.137 ± 0.007	0.370 ± 0.012	0.316	0.000	-0.11	0.1147	0.1147	1.094	96.892 ± 0.578	0.502 ± 0.008	—
UPI	beta32	0.316	0.325	0.009 ± 0.004	0.030 ± 0.013	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	0.143 ± 0.008	0.378 ± 0.012	0.324	0.008	2.65	0.1221	0.1224	1.084	96.892 ± 0.578	0.506 ± 0.009	—
UPI	beta33	0.000	0.004	0.004 ± 0.006	—	0.033 ± 0.002	0.181 ± 0.006	0.033 ± 0.002	0.181 ± 0.006	—	—	0.005	0.005	—	0.1670	0.1671	1.198	99.445 ± 0.247	0.851 ± 0.024	0.555 ± 0.247
UPI	beta34	0.000	-0.004	-0.004 ± 0.004	—	0.017 ± 0.001	0.132 ± 0.004	0.017 ± 0.001	0.132 ± 0.005	—	—	-0.002	-0.002	—	0.1162	0.1162	1.252	98.890 ± 0.349	0.647 ± 0.021	1.110 ± 0.349
UPI	beta35	0.000	0.001	0.001 ± 0.005	—	0.018 ± 0.001	0.136 ± 0.005	0.018 ± 0.001	0.135 ± 0.005	—	—	0.002	0.002	—	0.1097	0.1097	1.243	99.334 ± 0.271	0.661 ± 0.021	0.666 ± 0.271

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S74

Simulation 1 - Condition 74: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.008 ± 0.006	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.006	0.313	-0.003	-1.01	0.0571	0.0572	1.029	95.787 ± 0.636	0.237 ± 0.001	—
LMS	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.315	-0.001	-0.18	0.0610	0.0610	1.016	94.885 ± 0.698	0.238 ± 0.001	—
LMS	beta33	0.000	-0.044	-0.044 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	—	—	-0.043	-0.043	—	0.0838	0.0942	0.974	90.371 ± 0.934	0.332 ± 0.003	9.629 ± 0.934
LMS	beta34	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.006	0.006	—	0.0547	0.0551	0.964	93.480 ± 0.782	0.212 ± 0.002	6.520 ± 0.782
LMS	beta35	0.000	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.006	0.006	—	0.0510	0.0514	1.047	95.186 ± 0.678	0.216 ± 0.003	4.814 ± 0.678
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.206 ± 0.006	0.318	0.002	0.55	0.0641	0.0641	1.022	95.988 ± 0.622	0.260 ± 0.001	—
LSAM	beta32	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.208 ± 0.006	0.320	0.004	1.28	0.0674	0.0675	1.015	95.486 ± 0.657	0.261 ± 0.001	—
LSAM	beta33	0.000	-0.004	-0.004 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	-0.004	-0.004	—	0.0836	0.0837	0.977	94.885 ± 0.698	0.351 ± 0.003	5.115 ± 0.698
LSAM	beta34	0.000	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.001	0.001	—	0.0606	0.0606	0.973	95.587 ± 0.650	0.245 ± 0.002	4.413 ± 0.650
LSAM	beta35	0.000	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	—	0.0564	0.0564	1.032	96.891 ± 0.550	0.243 ± 0.002	3.109 ± 0.550
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.008 ± 0.006	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.006	0.313	-0.003	-1.05	0.0571	0.0572	1.043	95.787 ± 0.636	0.240 ± 0.004	—
QML	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.315	-0.001	-0.27	0.0611	0.0611	1.025	95.085 ± 0.685	0.240 ± 0.002	—
QML	beta33	0.000	-0.043	-0.043 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	—	—	-0.042	-0.042	—	0.0817	0.0920	0.989	90.772 ± 0.917	0.334 ± 0.005	9.228 ± 0.917
QML	beta34	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.006	0.006	—	0.0540	0.0543	0.976	93.781 ± 0.765	0.213 ± 0.002	6.219 ± 0.765
QML	beta35	0.000	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.006	0.006	—	0.0504	0.0507	1.070	95.687 ± 0.643	0.218 ± 0.005	4.313 ± 0.643
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.205 ± 0.006	0.317	0.001	0.35	0.0633	0.0634	1.037	96.289 ± 0.599	0.264 ± 0.001	—
UPI	beta32	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.209 ± 0.006	0.320	0.004	1.24	0.0666	0.0667	1.027	95.787 ± 0.636	0.265 ± 0.001	—
UPI	beta33	0.000	-0.004	-0.004 ± 0.003	—	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	—	—	-0.004	-0.004	—	0.0844	0.0845	0.996	95.787 ± 0.636	0.370 ± 0.004	4.213 ± 0.636
UPI	beta34	0.000	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.002	0.002	—	0.0618	0.0618	1.013	96.991 ± 0.541	0.268 ± 0.003	3.009 ± 0.541
UPI	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.002	0.002	—	0.0592	0.0592	1.053	97.894 ± 0.455	0.265 ± 0.003	2.106 ± 0.455

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S75

Simulation 1 - Condition 75: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.213 ± 0.006	0.314	-0.002	-0.64	0.0710	0.0710	1.030	95.888 ± 0.629	0.272 ± 0.001	—
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.218 ± 0.006	0.313	-0.003	-1.05	0.0669	0.0670	1.010	94.784 ± 0.704	0.272 ± 0.001	—
LMS	beta33	0.000	-0.022	-0.022 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	—	—	-0.021	-0.021	—	0.0857	0.0883	0.973	93.681 ± 0.771	0.330 ± 0.002	6.319 ± 0.771
LMS	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0570	0.0570	0.980	93.480 ± 0.782	0.221 ± 0.002	6.520 ± 0.782
LMS	beta35	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.005	0.005	—	0.0544	0.0547	1.011	94.784 ± 0.704	0.224 ± 0.002	5.216 ± 0.704
LSAM	beta31	0.316	0.322	0.006 ± 0.002	0.017 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.229 ± 0.006	0.319	0.003	0.82	0.0745	0.0745	1.017	95.286 ± 0.671	0.287 ± 0.001	—
LSAM	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.002	0.232 ± 0.007	0.314	-0.002	-0.48	0.0714	0.0714	0.999	95.186 ± 0.678	0.287 ± 0.001	—
LSAM	beta33	0.000	0.000	-0.000 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	—	—	-0.002	-0.002	—	0.0879	0.0879	0.957	94.383 ± 0.729	0.340 ± 0.003	5.617 ± 0.729
LSAM	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.002	-0.002	—	0.0615	0.0615	0.956	93.982 ± 0.753	0.240 ± 0.002	6.018 ± 0.753
LSAM	beta35	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.004	0.004	—	0.0607	0.0608	1.001	96.088 ± 0.614	0.244 ± 0.002	3.912 ± 0.614
QML	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.213 ± 0.006	0.314	-0.002	-0.64	0.0712	0.0713	1.029	95.888 ± 0.629	0.272 ± 0.001	—
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.218 ± 0.006	0.313	-0.003	-1.02	0.0670	0.0671	1.009	94.885 ± 0.698	0.272 ± 0.001	—
QML	beta33	0.000	-0.021	-0.021 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	—	—	-0.021	-0.021	—	0.0859	0.0885	0.983	93.681 ± 0.771	0.334 ± 0.004	6.319 ± 0.771
QML	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0570	0.0570	0.994	93.581 ± 0.776	0.224 ± 0.003	6.419 ± 0.776
QML	beta35	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.005	0.005	—	0.0543	0.0545	1.027	95.186 ± 0.678	0.227 ± 0.003	4.814 ± 0.678
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.002	0.230 ± 0.006	0.318	0.002	0.50	0.0748	0.0748	1.020	95.687 ± 0.643	0.289 ± 0.001	—
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.316	0.000	-0.15	0.0718	0.0718	1.002	95.085 ± 0.685	0.289 ± 0.001	—
UPI	beta33	0.000	0.002	0.002 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	-0.002	-0.002	—	0.0916	0.0917	0.959	95.587 ± 0.650	0.353 ± 0.003	4.413 ± 0.650
UPI	beta34	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.004	-0.004	—	0.0635	0.0637	0.983	95.186 ± 0.678	0.254 ± 0.002	4.814 ± 0.678
UPI	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.003	0.003	—	0.0636	0.0636	1.025	96.389 ± 0.591	0.260 ± 0.003	3.611 ± 0.591

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S76

Simulation 1 - Condition 76: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.008 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.313	-0.003	-1.07	0.0454	0.0455	0.985	94.300 ± 0.733	0.170 ± 0.000	—
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.313	-0.003	-0.86	0.0427	0.0428	0.990	94.400 ± 0.727	0.171 ± 0.000	—
LMS	beta33	0.000	-0.021	-0.021 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	—	—	-0.022	-0.022	—	0.0530	0.0574	0.989	93.400 ± 0.785	0.207 ± 0.001	6.600 ± 0.785
LMS	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.005	0.005	—	0.0344	0.0348	0.990	94.100 ± 0.745	0.139 ± 0.001	5.900 ± 0.745
LMS	beta35	0.000	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.003	0.003	—	0.0373	0.0375	0.973	94.700 ± 0.708	0.139 ± 0.001	5.300 ± 0.708
LSAM	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.315	-0.001	-0.36	0.0478	0.0479	0.988	94.800 ± 0.702	0.178 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.316	0.000	-0.01	0.0438	0.0438	0.985	94.600 ± 0.715	0.178 ± 0.000	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.002	-0.002	—	0.0515	0.0515	0.983	94.700 ± 0.708	0.205 ± 0.001	5.300 ± 0.708
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.002	0.002	—	0.0358	0.0359	0.991	94.800 ± 0.702	0.145 ± 0.001	5.200 ± 0.702
LSAM	beta35	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.002	0.002	—	0.0392	0.0393	0.960	94.700 ± 0.708	0.145 ± 0.001	5.300 ± 0.708
QML	beta31	0.316	0.314	-0.002 ± 0.001	-0.008 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.312	-0.004	-1.12	0.0451	0.0452	0.986	94.300 ± 0.733	0.170 ± 0.000	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.313	-0.003	-0.87	0.0428	0.0429	0.991	94.300 ± 0.733	0.171 ± 0.000	—
QML	beta33	0.000	-0.021	-0.021 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	—	—	-0.022	-0.022	—	0.0525	0.0569	1.029	93.300 ± 0.791	0.215 ± 0.008	6.700 ± 0.791
QML	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.005	0.005	—	0.0344	0.0348	1.016	94.300 ± 0.733	0.142 ± 0.003	5.700 ± 0.733
QML	beta35	0.000	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.003	0.003	—	0.0374	0.0375	1.001	94.800 ± 0.702	0.142 ± 0.004	5.200 ± 0.702
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.315	-0.001	-0.42	0.0482	0.0482	0.987	94.900 ± 0.696	0.178 ± 0.000	—
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.315	-0.001	-0.16	0.0441	0.0441	0.985	94.600 ± 0.715	0.178 ± 0.000	—
UPI	beta33	0.000	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0523	0.0524	0.979	94.700 ± 0.708	0.208 ± 0.001	5.300 ± 0.708
UPI	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.002	0.002	—	0.0357	0.0357	1.011	95.500 ± 0.656	0.148 ± 0.001	4.500 ± 0.656
UPI	beta35	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0401	0.0401	0.972	95.300 ± 0.669	0.148 ± 0.001	4.700 ± 0.669

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S77

Simulation 1 - Condition 77: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.317	0.001	0.31	0.0538	0.0538	0.989	94.895 ± 0.696	0.220 ± 0.001	—
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.312	-0.004	-1.35	0.0573	0.0575	0.989	94.394 ± 0.728	0.220 ± 0.001	—
LMS	beta33	0.000	-0.009	-0.009 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.009	-0.009	—	0.0643	0.0649	0.949	93.193 ± 0.797	0.246 ± 0.001	6.807 ± 0.797
LMS	beta34	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	—	0.0444	0.0444	0.966	93.594 ± 0.775	0.170 ± 0.001	6.406 ± 0.775
LMS	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.003	0.003	—	0.0481	0.0482	0.980	95.195 ± 0.677	0.170 ± 0.001	4.805 ± 0.677
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.319	0.003	0.81	0.0563	0.0564	0.984	94.895 ± 0.696	0.225 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.313	-0.003	-0.96	0.0578	0.0579	0.981	94.194 ± 0.740	0.224 ± 0.001	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.001	-0.001	—	0.0649	0.0649	0.944	93.293 ± 0.791	0.247 ± 0.001	6.707 ± 0.791
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.001	-0.001	—	0.0463	0.0463	0.957	93.794 ± 0.763	0.175 ± 0.001	6.206 ± 0.763
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.002	0.002	—	0.0487	0.0487	0.968	95.295 ± 0.670	0.174 ± 0.001	4.705 ± 0.670
QML	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.317	0.001	0.31	0.0537	0.0537	0.988	94.795 ± 0.703	0.220 ± 0.001	—
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.312	-0.004	-1.29	0.0571	0.0573	0.988	94.394 ± 0.728	0.219 ± 0.001	—
QML	beta33	0.000	-0.009	-0.009 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.010	-0.010	—	0.0642	0.0649	0.952	93.193 ± 0.797	0.247 ± 0.001	6.807 ± 0.797
QML	beta34	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0444	0.0444	0.968	93.694 ± 0.769	0.171 ± 0.001	6.306 ± 0.769
QML	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.003	0.003	—	0.0482	0.0483	0.978	95.195 ± 0.677	0.170 ± 0.001	4.805 ± 0.677
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.318	0.002	0.76	0.0564	0.0565	0.981	94.094 ± 0.746	0.225 ± 0.001	—
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.313	-0.003	-0.86	0.0576	0.0576	0.980	93.794 ± 0.763	0.224 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.002	-0.002	—	0.0651	0.0652	0.946	93.193 ± 0.797	0.248 ± 0.001	6.807 ± 0.797
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.001	-0.001	—	0.0464	0.0464	0.959	93.594 ± 0.775	0.177 ± 0.001	6.406 ± 0.775
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0478	0.0478	0.977	95.696 ± 0.642	0.177 ± 0.001	4.304 ± 0.642

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S78

Simulation 1 - Condition 78: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.313	-0.003	-0.88	0.0359	0.0360	1.001	94.600 ± 0.715	0.138 ± 0.000	—
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.314	-0.002	-0.48	0.0340	0.0340	0.987	93.200 ± 0.796	0.138 ± 0.000	—
LMS	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.008	-0.008	—	0.0400	0.0407	0.998	94.100 ± 0.745	0.156 ± 0.001	5.900 ± 0.745
LMS	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.002	0.002	—	0.0267	0.0268	0.994	94.000 ± 0.751	0.107 ± 0.000	6.000 ± 0.751
LMS	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.002	0.002	—	0.0268	0.0268	0.991	94.200 ± 0.739	0.108 ± 0.000	5.800 ± 0.739
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.314	-0.002	-0.52	0.0362	0.0363	0.999	94.300 ± 0.733	0.141 ± 0.000	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.315	-0.001	-0.23	0.0351	0.0351	0.986	93.700 ± 0.768	0.141 ± 0.000	—
LSAM	beta33	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0388	0.0388	0.993	94.500 ± 0.721	0.154 ± 0.001	5.500 ± 0.721
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0271	0.0271	0.989	93.900 ± 0.757	0.108 ± 0.000	6.100 ± 0.757
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0281	0.0281	0.981	93.500 ± 0.780	0.109 ± 0.000	6.500 ± 0.780
QML	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.112 ± 0.003	0.313	-0.003	-0.89	0.0358	0.0359	1.000	94.500 ± 0.721	0.138 ± 0.000	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.314	-0.002	-0.51	0.0339	0.0340	0.985	93.200 ± 0.796	0.138 ± 0.000	—
QML	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.008	-0.008	—	0.0397	0.0405	0.992	94.000 ± 0.751	0.155 ± 0.001	6.000 ± 0.751
QML	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.002	0.002	—	0.0267	0.0268	0.992	94.000 ± 0.751	0.107 ± 0.000	6.000 ± 0.751
QML	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.002	0.002	—	0.0270	0.0270	0.985	94.300 ± 0.733	0.107 ± 0.000	5.700 ± 0.733
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.314	-0.002	-0.60	0.0367	0.0368	0.997	94.300 ± 0.733	0.141 ± 0.000	—
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.315	-0.001	-0.32	0.0348	0.0348	0.985	93.600 ± 0.774	0.140 ± 0.000	—
UPI	beta33	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.000	0.000	—	0.0381	0.0381	0.989	94.700 ± 0.708	0.154 ± 0.001	5.300 ± 0.708
UPI	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0277	0.0277	0.987	93.900 ± 0.757	0.109 ± 0.000	6.100 ± 0.757
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0275	0.0275	0.978	93.700 ± 0.768	0.109 ± 0.000	6.300 ± 0.768

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S79

Simulation 1 - Condition 79: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.276	-0.040 ± 0.004	-0.127 ± 0.012	0.011 ± 0.001	0.106 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.129 ± 0.006	0.359 ± 0.010	0.271	-0.045	-14.19	0.1092	0.1181	1.029	92.399 ± 0.913	0.428 ± 0.003	—
LMS	beta32	0.316	0.279	-0.037 ± 0.003	-0.118 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.011 ± 0.001	0.107 ± 0.003	0.114 ± 0.005	0.338 ± 0.009	0.276	-0.040	-12.66	0.1082	0.1153	1.091	93.468 ± 0.852	0.429 ± 0.003	—
LMS	beta33	0.000	-0.042	-0.042 ± 0.004	—	0.014 ± 0.001	0.120 ± 0.004	0.016 ± 0.001	0.127 ± 0.004	—	—	-0.042	-0.042	—	0.1202	0.1274	1.045	95.012 ± 0.750	0.493 ± 0.008	4.988 ± 0.750
LMS	beta34	0.000	0.072	0.072 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	—	—	0.065	0.065	—	0.0718	0.0968	1.033	87.411 ± 1.143	0.295 ± 0.005	12.589 ± 1.143
LMS	beta35	0.000	0.066	0.066 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	—	—	0.062	0.062	—	0.0706	0.0939	1.018	86.105 ± 1.192	0.287 ± 0.006	13.895 ± 1.192
LSAM	beta31	0.316	0.319	0.003 ± 0.008	0.009 ± 0.024	0.048 ± 0.003	0.220 ± 0.007	0.048 ± 0.003	0.220 ± 0.008	0.483 ± 0.031	0.695 ± 0.025	0.304	-0.012	-3.75	0.2222	0.2226	1.207	97.743 ± 0.512	1.040 ± 0.046	—
LSAM	beta32	0.316	0.328	0.012 ± 0.007	0.039 ± 0.023	0.046 ± 0.003	0.214 ± 0.006	0.046 ± 0.003	0.214 ± 0.007	0.460 ± 0.027	0.678 ± 0.023	0.322	0.006	1.96	0.2165	0.2166	1.246	98.100 ± 0.471	1.045 ± 0.040	—
LSAM	beta33	0.000	0.010	0.010 ± 0.005	—	0.019 ± 0.001	0.138 ± 0.004	0.019 ± 0.001	0.138 ± 0.005	—	—	0.009	0.009	—	0.1340	0.1343	1.247	97.268 ± 0.562	0.673 ± 0.032	2.732 ± 0.562
LSAM	beta34	0.000	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.087 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	—	—	0.000	0.000	—	0.0741	0.0741	1.230	98.100 ± 0.471	0.417 ± 0.025	1.900 ± 0.471
LSAM	beta35	0.000	-0.005	-0.005 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.001	0.084 ± 0.003	—	—	-0.002	-0.002	—	0.0740	0.0740	1.229	97.981 ± 0.485	0.406 ± 0.019	2.019 ± 0.485
QML	beta31	0.316	0.276	-0.040 ± 0.004	-0.127 ± 0.012	0.011 ± 0.001	0.106 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.129 ± 0.006	0.360 ± 0.010	0.271	-0.045	-14.20	0.1083	0.1172	1.024	92.280 ± 0.920	0.428 ± 0.003	—
QML	beta32	0.316	0.279	-0.037 ± 0.003	-0.117 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.011 ± 0.001	0.107 ± 0.003	0.114 ± 0.005	0.338 ± 0.009	0.275	-0.041	-13.00	0.1068	0.1144	1.087	93.468 ± 0.852	0.427 ± 0.003	—
QML	beta33	0.000	-0.041	-0.041 ± 0.004	—	0.015 ± 0.001	0.121 ± 0.004	0.016 ± 0.001	0.127 ± 0.004	—	—	-0.042	-0.042	—	0.1195	0.1265	1.040	95.131 ± 0.742	0.492 ± 0.007	4.869 ± 0.742
QML	beta34	0.000	0.071	0.071 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	—	—	0.064	0.064	—	0.0714	0.0958	1.034	87.530 ± 1.139	0.295 ± 0.005	12.470 ± 1.139
QML	beta35	0.000	0.066	0.066 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.010 ± 0.001	0.098 ± 0.003	—	—	0.062	0.062	—	0.0707	0.0943	1.005	85.867 ± 1.201	0.284 ± 0.005	14.133 ± 1.201
UPI	beta31	0.316	0.319	0.003 ± 0.007	0.010 ± 0.024	0.047 ± 0.003	0.217 ± 0.007	0.047 ± 0.003	0.217 ± 0.008	0.470 ± 0.029	0.686 ± 0.024	0.307	-0.009	-3.00	0.2173	0.2175	1.192	98.219 ± 0.456	1.013 ± 0.027	—
UPI	beta32	0.316	0.331	0.015 ± 0.007	0.047 ± 0.023	0.043 ± 0.002	0.208 ± 0.005	0.043 ± 0.002	0.208 ± 0.007	0.433 ± 0.023	0.658 ± 0.021	0.321	0.005	1.64	0.2166	0.2167	1.248	98.456 ± 0.425	1.016 ± 0.028	—
UPI	beta33	0.000	0.009	0.009 ± 0.005	—	0.025 ± 0.002	0.157 ± 0.005	0.025 ± 0.002	0.157 ± 0.006	—	—	0.009	0.009	—	0.1314	0.1317	1.438	99.169 ± 0.313	0.883 ± 0.038	0.831 ± 0.313
UPI	beta34	0.000	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.087 ± 0.003	0.007 ± 0.001	0.087 ± 0.003	—	—	-0.002	-0.002	—	0.0755	0.0755	1.233	98.219 ± 0.456	0.419 ± 0.013	1.781 ± 0.456
UPI	beta35	0.000	-0.006	-0.006 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	—	—	-0.002	-0.002	—	0.0781	0.0782	1.217	98.456 ± 0.425	0.403 ± 0.014	1.544 ± 0.425

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S80

Simulation 1 - Condition 80: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.276	-0.040 ± 0.002	-0.126 ± 0.007	0.004 ± 0.000	0.065 ± 0.001	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.002	0.240 ± 0.006	0.277	-0.039	-12.49	0.0670	0.0777	1.035	90.442 ± 0.943	0.262 ± 0.001	—
LMS	beta32	0.316	0.272	-0.044 ± 0.002	-0.139 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.006 ± 0.000	0.080 ± 0.002	0.063 ± 0.003	0.252 ± 0.006	0.270	-0.046	-14.44	0.0682	0.0821	1.012	88.695 ± 1.015	0.263 ± 0.001	—
LMS	beta33	0.000	-0.046	-0.046 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	—	—	-0.043	-0.043	—	0.0727	0.0846	1.027	92.395 ± 0.850	0.303 ± 0.003	7.605 ± 0.850
LMS	beta34	0.000	0.070	0.070 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.007 ± 0.000	0.084 ± 0.002	—	—	0.067	0.067	—	0.0439	0.0801	0.971	64.029 ± 1.539	0.175 ± 0.002	35.971 ± 1.539
LMS	beta35	0.000	0.069	0.069 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.007 ± 0.000	0.083 ± 0.002	—	—	0.068	0.068	—	0.0424	0.0802	0.989	64.748 ± 1.532	0.174 ± 0.002	35.252 ± 1.532
LSAM	beta31	0.316	0.320	0.004 ± 0.004	0.013 ± 0.013	0.016 ± 0.001	0.125 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	0.157 ± 0.008	0.396 ± 0.012	0.318	0.002	0.71	0.1245	0.1246	1.037	96.917 ± 0.554	0.509 ± 0.004	—
LSAM	beta32	0.316	0.313	-0.003 ± 0.004	-0.008 ± 0.013	0.016 ± 0.001	0.128 ± 0.003	0.016 ± 0.001	0.128 ± 0.004	0.165 ± 0.008	0.406 ± 0.011	0.313	-0.003	-0.92	0.1299	0.1299	1.018	96.403 ± 0.597	0.512 ± 0.004	—
LSAM	beta33	0.000	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.000	0.000	—	0.0706	0.0706	1.034	96.300 ± 0.605	0.290 ± 0.003	3.700 ± 0.605
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.001	0.001	—	0.0392	0.0392	0.985	96.506 ± 0.589	0.173 ± 0.002	3.494 ± 0.589
LSAM	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	—	0.0404	0.0404	0.997	97.122 ± 0.536	0.173 ± 0.002	2.878 ± 0.536
QML	beta31	0.316	0.276	-0.040 ± 0.002	-0.125 ± 0.007	0.004 ± 0.000	0.065 ± 0.001	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.002	0.240 ± 0.006	0.276	-0.040	-12.53	0.0675	0.0782	1.030	90.339 ± 0.947	0.261 ± 0.001	—
QML	beta32	0.316	0.272	-0.044 ± 0.002	-0.138 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.006 ± 0.000	0.080 ± 0.002	0.063 ± 0.003	0.252 ± 0.006	0.271	-0.045	-14.18	0.0682	0.0816	1.011	88.695 ± 1.015	0.263 ± 0.001	—
QML	beta33	0.000	-0.044	-0.044 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	—	—	-0.042	-0.042	—	0.0721	0.0834	1.051	92.600 ± 0.839	0.308 ± 0.006	7.400 ± 0.839
QML	beta34	0.000	0.070	0.070 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.007 ± 0.000	0.083 ± 0.002	—	—	0.066	0.066	—	0.0428	0.0785	0.980	64.440 ± 1.535	0.175 ± 0.003	35.560 ± 1.535
QML	beta35	0.000	0.068	0.068 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.007 ± 0.000	0.082 ± 0.002	—	—	0.068	0.068	—	0.0416	0.0795	1.012	65.159 ± 1.527	0.176 ± 0.004	34.841 ± 1.527
UPI	beta31	0.316	0.320	0.004 ± 0.004	0.013 ± 0.013	0.015 ± 0.001	0.124 ± 0.003	0.015 ± 0.001	0.124 ± 0.004	0.153 ± 0.007	0.392 ± 0.011	0.319	0.003	0.95	0.1249	0.1249	1.052	97.122 ± 0.536	0.510 ± 0.005	—
UPI	beta32	0.316	0.313	-0.003 ± 0.004	-0.009 ± 0.013	0.016 ± 0.001	0.127 ± 0.003	0.016 ± 0.001	0.127 ± 0.004	0.162 ± 0.008	0.402 ± 0.011	0.314	-0.002	-0.72	0.1267	0.1267	1.031	97.020 ± 0.545	0.513 ± 0.004	—
UPI	beta33	0.000	0.001	0.001 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.001	-0.001	—	0.0689	0.0689	1.139	97.842 ± 0.466	0.335 ± 0.006	2.158 ± 0.466
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.001	-0.001	—	0.0388	0.0388	1.014	96.814 ± 0.563	0.175 ± 0.003	3.186 ± 0.563
UPI	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.001	-0.001	—	0.0390	0.0390	1.019	97.225 ± 0.527	0.175 ± 0.002	2.775 ± 0.527

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S81

Simulation 1 - Condition 81: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.283	-0.033 ± 0.003	-0.104 ± 0.009	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.092 ± 0.002	0.085 ± 0.004	0.292 ± 0.007	0.280	-0.036	-11.45	0.0847	0.0921	1.011	94.010 ± 0.756	0.341 ± 0.001	—
LMS	beta32	0.316	0.293	-0.023 ± 0.003	-0.072 ± 0.009	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.080 ± 0.003	0.283 ± 0.007	0.292	-0.024	-7.50	0.0895	0.0925	1.004	93.807 ± 0.768	0.340 ± 0.001	—
LMS	beta33	0.000	-0.018	-0.018 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.016	-0.016	—	0.0794	0.0811	0.971	93.198 ± 0.802	0.308 ± 0.003	6.802 ± 0.802
LMS	beta34	0.000	0.036	0.036 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.059 ± 0.001	—	—	0.035	0.035	—	0.0462	0.0579	0.951	84.873 ± 1.142	0.174 ± 0.002	15.127 ± 1.142
LMS	beta35	0.000	0.032	0.032 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.028	0.028	—	0.0489	0.0563	0.878	85.482 ± 1.122	0.170 ± 0.002	14.518 ± 1.122
LSAM	beta31	0.316	0.310	-0.006 ± 0.004	-0.018 ± 0.014	0.019 ± 0.001	0.139 ± 0.003	0.019 ± 0.001	0.139 ± 0.004	0.195 ± 0.009	0.441 ± 0.013	0.312	-0.004	-1.19	0.1339	0.1340	0.985	95.635 ± 0.651	0.538 ± 0.004	—
LSAM	beta32	0.316	0.329	0.013 ± 0.004	0.041 ± 0.014	0.020 ± 0.001	0.140 ± 0.003	0.020 ± 0.001	0.141 ± 0.004	0.198 ± 0.009	0.445 ± 0.013	0.330	0.014	4.28	0.1400	0.1406	0.975	95.533 ± 0.658	0.535 ± 0.004	—
LSAM	beta33	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	—	—	0.003	0.003	—	0.0813	0.0813	0.954	94.619 ± 0.719	0.316 ± 0.003	5.381 ± 0.719
LSAM	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.002	0.002	—	0.0477	0.0478	0.947	94.010 ± 0.756	0.195 ± 0.003	5.990 ± 0.756
LSAM	beta35	0.000	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.002	-0.002	—	0.0509	0.0509	0.892	94.518 ± 0.725	0.192 ± 0.003	5.482 ± 0.725
QML	beta31	0.316	0.283	-0.033 ± 0.003	-0.104 ± 0.009	0.007 ± 0.000	0.087 ± 0.002	0.009 ± 0.000	0.092 ± 0.002	0.086 ± 0.004	0.293 ± 0.008	0.280	-0.036	-11.43	0.0854	0.0927	1.008	93.807 ± 0.768	0.342 ± 0.002	—
QML	beta32	0.316	0.293	-0.023 ± 0.003	-0.071 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	0.080 ± 0.004	0.284 ± 0.008	0.293	-0.023	-7.39	0.0901	0.0931	1.000	93.807 ± 0.768	0.340 ± 0.002	—
QML	beta33	0.000	-0.018	-0.018 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.015	-0.015	—	0.0799	0.0814	0.969	93.198 ± 0.802	0.307 ± 0.003	6.802 ± 0.802
QML	beta34	0.000	0.036	0.036 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.059 ± 0.001	—	—	0.035	0.035	—	0.0464	0.0579	0.953	85.076 ± 1.135	0.174 ± 0.002	14.924 ± 1.135
QML	beta35	0.000	0.032	0.032 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.028	0.028	—	0.0488	0.0561	0.877	85.584 ± 1.119	0.170 ± 0.002	14.416 ± 1.119
UPI	beta31	0.316	0.310	-0.006 ± 0.004	-0.018 ± 0.014	0.019 ± 0.001	0.138 ± 0.003	0.019 ± 0.001	0.138 ± 0.004	0.191 ± 0.009	0.437 ± 0.013	0.312	-0.004	-1.24	0.1303	0.1304	0.989	95.635 ± 0.651	0.535 ± 0.004	—
UPI	beta32	0.316	0.330	0.014 ± 0.004	0.045 ± 0.014	0.019 ± 0.001	0.139 ± 0.003	0.020 ± 0.001	0.140 ± 0.004	0.195 ± 0.009	0.442 ± 0.013	0.331	0.015	4.63	0.1372	0.1380	0.977	95.330 ± 0.672	0.532 ± 0.004	—
UPI	beta33	0.000	0.002	0.002 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	—	—	0.003	0.003	—	0.0794	0.0795	1.029	95.635 ± 0.651	0.352 ± 0.006	4.365 ± 0.651
UPI	beta34	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.002	0.002	—	0.0467	0.0467	0.955	94.619 ± 0.719	0.196 ± 0.003	5.381 ± 0.719
UPI	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.003	-0.003	—	0.0504	0.0505	0.907	94.010 ± 0.756	0.193 ± 0.003	5.990 ± 0.756

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S82

Simulation 1 - Condition 82: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.286	-0.030 ± 0.002	-0.095 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.285	-0.031	-9.82	0.0549	0.0631	1.014	91.291 ± 0.892	0.210 ± 0.001	—
LMS	beta32	0.316	0.287	-0.029 ± 0.002	-0.091 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.285	-0.031	-9.84	0.0532	0.0616	1.011	91.692 ± 0.873	0.212 ± 0.001	—
LMS	beta33	0.000	-0.015	-0.015 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	-0.013	-0.013	—	0.0484	0.0502	1.009	94.094 ± 0.746	0.190 ± 0.002	5.906 ± 0.746
LMS	beta34	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.030	0.030	—	0.0261	0.0398	0.946	75.976 ± 1.352	0.101 ± 0.001	24.024 ± 1.352
LMS	beta35	0.000	0.032	0.032 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.030	0.030	—	0.0261	0.0396	0.943	77.878 ± 1.313	0.104 ± 0.001	22.122 ± 1.313
LSAM	beta31	0.316	0.316	0.000 ± 0.003	0.000 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.065 ± 0.003	0.254 ± 0.007	0.316	0.000	-0.10	0.0835	0.0835	0.999	95.095 ± 0.683	0.315 ± 0.001	—
LSAM	beta32	0.316	0.318	0.002 ± 0.003	0.006 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.066 ± 0.003	0.256 ± 0.008	0.315	-0.001	-0.17	0.0767	0.0767	1.002	94.595 ± 0.715	0.318 ± 0.001	—
LSAM	beta33	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.001	0.001	—	0.0453	0.0454	1.004	95.395 ± 0.663	0.178 ± 0.001	4.605 ± 0.663
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0264	0.0264	0.953	95.596 ± 0.649	0.103 ± 0.001	4.404 ± 0.649
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0240	0.0240	0.960	93.994 ± 0.752	0.106 ± 0.001	6.006 ± 0.752
QML	beta31	0.316	0.286	-0.030 ± 0.002	-0.095 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.285	-0.031	-9.87	0.0553	0.0635	1.008	91.291 ± 0.892	0.210 ± 0.001	—
QML	beta32	0.316	0.287	-0.029 ± 0.002	-0.091 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.285	-0.031	-9.86	0.0535	0.0619	1.007	91.592 ± 0.878	0.212 ± 0.001	—
QML	beta33	0.000	-0.015	-0.015 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	-0.013	-0.013	—	0.0482	0.0499	1.009	93.994 ± 0.752	0.190 ± 0.001	6.006 ± 0.752
QML	beta34	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.030	0.030	—	0.0264	0.0400	0.949	76.076 ± 1.350	0.102 ± 0.001	23.924 ± 1.350
QML	beta35	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.030	0.030	—	0.0260	0.0395	0.945	77.978 ± 1.311	0.104 ± 0.001	22.022 ± 1.311
UPI	beta31	0.316	0.316	0.000 ± 0.003	0.001 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.253 ± 0.007	0.317	0.001	0.39	0.0859	0.0859	1.001	95.395 ± 0.663	0.314 ± 0.001	—
UPI	beta32	0.316	0.318	0.002 ± 0.003	0.007 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.065 ± 0.003	0.256 ± 0.008	0.315	-0.001	-0.30	0.0762	0.0762	1.001	94.795 ± 0.703	0.317 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0455	0.0455	1.007	95.295 ± 0.670	0.185 ± 0.002	4.705 ± 0.670
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0263	0.0263	0.951	96.096 ± 0.613	0.104 ± 0.001	3.904 ± 0.613
UPI	beta35	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0247	0.0247	0.954	94.494 ± 0.722	0.106 ± 0.001	5.506 ± 0.722

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S83

Simulation 1 - Condition 83: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.301	-0.015 ± 0.002	-0.047 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.060 ± 0.003	0.246 ± 0.007	0.301	-0.015	-4.67	0.0764	0.0778	1.010	94.795 ± 0.703	0.302 ± 0.001	—
LMS	beta32	0.316	0.303	-0.013 ± 0.002	-0.040 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.302	-0.014	-4.51	0.0763	0.0776	0.991	93.493 ± 0.780	0.302 ± 0.001	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0617	0.0617	0.915	92.392 ± 0.839	0.225 ± 0.002	7.608 ± 0.839
LMS	beta34	0.000	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.012	0.012	—	0.0325	0.0346	0.884	89.289 ± 0.978	0.123 ± 0.001	10.711 ± 0.978
LMS	beta35	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.011	0.011	—	0.0342	0.0360	0.887	89.590 ± 0.966	0.123 ± 0.001	10.410 ± 0.966
LSAM	beta31	0.316	0.317	0.001 ± 0.003	0.002 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.091 ± 0.004	0.302 ± 0.008	0.317	0.001	0.45	0.0967	0.0967	0.999	95.596 ± 0.649	0.374 ± 0.002	—
LSAM	beta32	0.316	0.319	0.003 ± 0.003	0.010 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.004	0.307 ± 0.009	0.317	0.001	0.19	0.0939	0.0939	0.986	93.794 ± 0.763	0.375 ± 0.002	—
LSAM	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.004	0.004	—	0.0601	0.0603	0.912	92.793 ± 0.818	0.226 ± 0.002	7.207 ± 0.818
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0323	0.0323	0.892	91.692 ± 0.873	0.132 ± 0.002	8.308 ± 0.873
LSAM	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0350	0.0350	0.899	91.692 ± 0.873	0.132 ± 0.002	8.308 ± 0.873
QML	beta31	0.316	0.301	-0.015 ± 0.002	-0.047 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.301	-0.015	-4.82	0.0772	0.0787	1.004	94.795 ± 0.703	0.301 ± 0.001	—
QML	beta32	0.316	0.303	-0.013 ± 0.002	-0.040 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.250 ± 0.007	0.302	-0.014	-4.51	0.0769	0.0782	0.986	93.493 ± 0.780	0.302 ± 0.001	—
QML	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0617	0.0617	0.916	92.492 ± 0.834	0.225 ± 0.002	7.508 ± 0.834
QML	beta34	0.000	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.012	0.012	—	0.0326	0.0346	0.881	89.289 ± 0.978	0.123 ± 0.001	10.711 ± 0.978
QML	beta35	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.011	0.011	—	0.0344	0.0362	0.885	89.790 ± 0.958	0.122 ± 0.001	10.210 ± 0.958
UPI	beta31	0.316	0.317	0.001 ± 0.003	0.003 ± 0.010	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.091 ± 0.004	0.301 ± 0.008	0.317	0.001	0.44	0.0942	0.0943	0.999	95.696 ± 0.642	0.373 ± 0.002	—
UPI	beta32	0.316	0.319	0.003 ± 0.003	0.011 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.004	0.306 ± 0.009	0.317	0.001	0.33	0.0918	0.0918	0.986	93.594 ± 0.775	0.374 ± 0.002	—
UPI	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.005	0.005	—	0.0607	0.0609	0.919	93.393 ± 0.786	0.230 ± 0.002	6.607 ± 0.786
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0327	0.0327	0.897	91.692 ± 0.873	0.132 ± 0.002	8.308 ± 0.873
UPI	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0353	0.0353	0.899	91.391 ± 0.887	0.132 ± 0.002	8.609 ± 0.887

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S84

Simulation 1 - Condition 84: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.303	-0.013 ± 0.002	-0.043 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.303	-0.013	-4.06	0.0474	0.0492	0.992	93.900 ± 0.757	0.187 ± 0.000	—
LMS	beta32	0.316	0.301	-0.015 ± 0.002	-0.048 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.027 ± 0.001	0.164 ± 0.004	0.302	-0.014	-4.41	0.0480	0.0500	0.966	93.100 ± 0.801	0.188 ± 0.000	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.003	-0.003	—	0.0368	0.0369	0.915	92.300 ± 0.843	0.139 ± 0.001	7.700 ± 0.843
LMS	beta34	0.000	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.011	0.011	—	0.0199	0.0226	0.913	90.200 ± 0.940	0.073 ± 0.001	9.800 ± 0.940
LMS	beta35	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	—	0.0200	0.0231	0.907	88.000 ± 1.028	0.074 ± 0.001	12.000 ± 1.028
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.318	0.002	0.72	0.0585	0.0585	0.983	94.200 ± 0.739	0.228 ± 0.001	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.006	0.317	0.001	0.19	0.0581	0.0581	0.958	93.300 ± 0.791	0.229 ± 0.001	—
LSAM	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0363	0.0363	0.912	92.600 ± 0.828	0.134 ± 0.001	7.400 ± 0.828
LSAM	beta34	0.000	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0202	0.0202	0.923	93.400 ± 0.785	0.075 ± 0.001	6.600 ± 0.785
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0205	0.0205	0.913	93.200 ± 0.796	0.076 ± 0.001	6.800 ± 0.796
QML	beta31	0.316	0.303	-0.013 ± 0.002	-0.043 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.159 ± 0.004	0.303	-0.013	-4.14	0.0479	0.0496	0.987	93.800 ± 0.763	0.187 ± 0.000	—
QML	beta32	0.316	0.301	-0.015 ± 0.002	-0.048 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.027 ± 0.001	0.165 ± 0.004	0.302	-0.014	-4.42	0.0483	0.0503	0.961	92.800 ± 0.817	0.188 ± 0.000	—
QML	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.003	-0.003	—	0.0367	0.0368	0.916	92.200 ± 0.848	0.138 ± 0.001	7.800 ± 0.848
QML	beta34	0.000	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.011	0.011	—	0.0199	0.0226	0.912	89.900 ± 0.953	0.073 ± 0.001	10.100 ± 0.953
QML	beta35	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	—	0.0201	0.0231	0.905	87.800 ± 1.035	0.074 ± 0.001	12.200 ± 1.035
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.318	0.002	0.63	0.0581	0.0581	0.983	94.400 ± 0.727	0.228 ± 0.001	—
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.006	0.316	0.000	0.09	0.0582	0.0582	0.961	94.000 ± 0.751	0.229 ± 0.001	—
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0359	0.0359	0.923	93.100 ± 0.801	0.135 ± 0.001	6.900 ± 0.801
UPI	beta34	0.000	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.001	-0.001	—	0.0202	0.0202	0.922	93.200 ± 0.796	0.075 ± 0.001	6.800 ± 0.796
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0203	0.0203	0.920	93.000 ± 0.807	0.076 ± 0.001	7.000 ± 0.807

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S85

Simulation 1 - Condition 85: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.004	-0.011 ± 0.013	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.089 ± 0.006	0.298 ± 0.011	0.316	0.000	-0.02	0.0923	0.0923	1.019	95.358 ± 0.925	0.376 ± 0.003	—
LMS	beta32	0.316	0.311	-0.005 ± 0.004	-0.014 ± 0.014	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.097 ± 0.006	0.312 ± 0.012	0.309	-0.007	-2.09	0.0972	0.0975	0.985	94.971 ± 0.961	0.381 ± 0.003	—
LMS	beta33	0.000	-0.013	-0.013 ± 0.006	—	0.016 ± 0.001	0.127 ± 0.005	0.016 ± 0.001	0.128 ± 0.006	—	—	-0.015	-0.015	—	0.1229	0.1237	1.032	95.358 ± 0.925	0.514 ± 0.008	4.642 ± 0.925
LMS	beta34	0.000	0.004	0.004 ± 0.004	—	0.009 ± 0.001	0.095 ± 0.004	0.009 ± 0.001	0.095 ± 0.004	—	—	0.005	0.005	—	0.1013	0.1014	0.993	94.584 ± 0.995	0.368 ± 0.006	5.416 ± 0.995
LMS	beta35	0.000	-0.001	-0.001 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	—	—	-0.001	-0.001	—	0.0921	0.0921	1.074	96.905 ± 0.762	0.374 ± 0.006	3.095 ± 0.762
LSAM	beta31	0.316	0.316	0.000 ± 0.005	0.001 ± 0.017	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.005	0.149 ± 0.011	0.386 ± 0.016	0.321	0.005	1.64	0.1254	0.1255	1.413	96.905 ± 0.762	0.676 ± 0.034	—
LSAM	beta32	0.316	0.322	0.006 ± 0.005	0.020 ± 0.017	0.015 ± 0.001	0.124 ± 0.004	0.015 ± 0.001	0.124 ± 0.005	0.153 ± 0.011	0.391 ± 0.016	0.316	0.000	-0.08	0.1304	0.1304	1.318	97.292 ± 0.714	0.639 ± 0.027	—
LSAM	beta33	0.000	0.006	0.006 ± 0.008	—	0.037 ± 0.003	0.192 ± 0.009	0.037 ± 0.003	0.192 ± 0.010	—	—	-0.002	-0.002	—	0.1797	0.1798	1.527	98.839 ± 0.471	1.147 ± 0.083	1.161 ± 0.471
LSAM	beta34	0.000	0.017	0.017 ± 0.012	—	0.070 ± 0.006	0.265 ± 0.012	0.070 ± 0.007	0.265 ± 0.014	—	—	0.008	0.008	—	0.2300	0.2301	1.469	100.000 ± 0.000	1.525 ± 0.115	0.000 ± 0.000
LSAM	beta35	0.000	-0.008	-0.008 ± 0.011	—	0.057 ± 0.005	0.239 ± 0.011	0.057 ± 0.005	0.239 ± 0.012	—	—	0.001	0.001	—	0.2041	0.2041	1.572	100.000 ± 0.000	1.474 ± 0.110	0.000 ± 0.000
QML	beta31	0.316	0.313	-0.003 ± 0.004	-0.010 ± 0.013	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.089 ± 0.006	0.298 ± 0.011	0.316	0.000	-0.08	0.0902	0.0902	1.025	95.358 ± 0.925	0.378 ± 0.003	—
QML	beta32	0.316	0.312	-0.004 ± 0.004	-0.013 ± 0.014	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.097 ± 0.006	0.312 ± 0.012	0.308	-0.008	-2.39	0.0964	0.0967	0.994	94.971 ± 0.961	0.384 ± 0.004	—
QML	beta33	0.000	-0.013	-0.013 ± 0.006	—	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.006	—	—	-0.017	-0.017	—	0.1239	0.1251	1.059	95.745 ± 0.888	0.534 ± 0.012	4.255 ± 0.888
QML	beta34	0.000	0.004	0.004 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	—	—	0.007	0.007	—	0.0959	0.0961	1.017	94.584 ± 0.995	0.377 ± 0.006	5.416 ± 0.995
QML	beta35	0.000	-0.001	-0.001 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	—	—	-0.003	-0.003	—	0.0887	0.0888	1.103	96.905 ± 0.762	0.387 ± 0.009	3.095 ± 0.762
UPI	beta31	0.316	0.324	0.008 ± 0.005	0.026 ± 0.016	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.117 ± 0.005	0.137 ± 0.009	0.371 ± 0.015	0.325	0.009	2.72	0.1282	0.1285	1.704	97.099 ± 0.738	0.781 ± 0.057	—
UPI	beta32	0.316	0.314	-0.002 ± 0.005	-0.006 ± 0.017	0.015 ± 0.001	0.123 ± 0.004	0.015 ± 0.001	0.122 ± 0.005	0.150 ± 0.010	0.388 ± 0.016	0.311	-0.005	-1.47	0.1280	0.1280	1.510	97.679 ± 0.662	0.726 ± 0.040	—
UPI	beta33	0.000	0.009	0.009 ± 0.008	—	0.033 ± 0.003	0.182 ± 0.008	0.033 ± 0.003	0.183 ± 0.009	—	—	0.005	0.005	—	0.1643	0.1644	1.730	99.807 ± 0.193	1.238 ± 0.070	0.193 ± 0.193
UPI	beta34	0.000	0.005	0.005 ± 0.012	—	0.069 ± 0.006	0.263 ± 0.012	0.069 ± 0.006	0.262 ± 0.014	—	—	0.005	0.005	—	0.2403	0.2404	2.125	99.807 ± 0.193	2.187 ± 0.150	0.193 ± 0.193
UPI	beta35	0.000	0.017	0.017 ± 0.012	—	0.070 ± 0.007	0.264 ± 0.013	0.070 ± 0.007	0.265 ± 0.014	—	—	0.005	0.005	—	0.2389	0.2389	2.177	100.000 ± 0.000	2.256 ± 0.179	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S86

Simulation 1 - Condition 86: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.314	-0.002	-0.69	0.0613	0.0614	1.021	96.107 ± 0.645	0.236 ± 0.001	—
LMS	beta32	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.006	0.316	0.000	0.15	0.0629	0.0629	0.982	93.771 ± 0.806	0.236 ± 0.001	—
LMS	beta33	0.000	-0.021	-0.021 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.019	-0.019	—	0.0798	0.0821	1.054	93.660 ± 0.813	0.330 ± 0.004	6.340 ± 0.813
LMS	beta34	0.000	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.005	-0.005	—	0.0556	0.0558	1.045	95.884 ± 0.663	0.236 ± 0.003	4.116 ± 0.663
LMS	beta35	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.002	0.002	—	0.0563	0.0564	1.046	95.106 ± 0.720	0.239 ± 0.005	4.894 ± 0.720
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.014 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.215 ± 0.007	0.317	0.001	0.43	0.0669	0.0669	1.049	97.219 ± 0.548	0.278 ± 0.002	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.002	0.218 ± 0.007	0.317	0.001	0.36	0.0663	0.0663	1.025	95.662 ± 0.679	0.277 ± 0.002	—
LSAM	beta33	0.000	0.001	0.001 ± 0.003	—	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	—	—	0.002	0.002	—	0.0951	0.0952	1.076	96.997 ± 0.569	0.399 ± 0.006	3.003 ± 0.569
LSAM	beta34	0.000	-0.011	-0.011 ± 0.004	—	0.015 ± 0.001	0.123 ± 0.004	0.015 ± 0.001	0.124 ± 0.004	—	—	-0.009	-0.009	—	0.1126	0.1130	1.069	98.331 ± 0.427	0.517 ± 0.010	1.669 ± 0.427
LSAM	beta35	0.000	0.005	0.005 ± 0.004	—	0.016 ± 0.001	0.126 ± 0.004	0.016 ± 0.001	0.126 ± 0.004	—	—	0.007	0.007	—	0.1129	0.1131	1.043	98.220 ± 0.441	0.513 ± 0.008	1.780 ± 0.441
QML	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.314	-0.002	-0.68	0.0620	0.0620	1.020	96.107 ± 0.645	0.236 ± 0.001	—
QML	beta32	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.006	0.316	0.000	-0.06	0.0625	0.0625	0.982	93.660 ± 0.813	0.236 ± 0.001	—
QML	beta33	0.000	-0.021	-0.021 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.019	-0.019	—	0.0792	0.0813	1.047	93.993 ± 0.792	0.328 ± 0.003	6.007 ± 0.792
QML	beta34	0.000	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.005	-0.005	—	0.0539	0.0541	1.041	95.884 ± 0.663	0.235 ± 0.002	4.116 ± 0.663
QML	beta35	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.001	0.001	—	0.0562	0.0563	1.031	95.884 ± 0.663	0.234 ± 0.002	4.116 ± 0.663
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.003	0.219 ± 0.007	0.320	0.004	1.30	0.0676	0.0678	1.169	98.109 ± 0.454	0.316 ± 0.006	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.003	0.225 ± 0.007	0.318	0.002	0.55	0.0700	0.0701	1.128	96.107 ± 0.645	0.314 ± 0.006	—
UPI	beta33	0.000	0.002	0.002 ± 0.003	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	—	—	0.004	0.004	—	0.0909	0.0910	1.223	98.443 ± 0.413	0.474 ± 0.011	1.557 ± 0.413
UPI	beta34	0.000	-0.011	-0.011 ± 0.005	—	0.025 ± 0.002	0.159 ± 0.005	0.025 ± 0.002	0.159 ± 0.006	—	—	-0.009	-0.009	—	0.1288	0.1291	1.296	99.333 ± 0.272	0.808 ± 0.029	0.667 ± 0.272
UPI	beta35	0.000	0.010	0.010 ± 0.005	—	0.023 ± 0.001	0.152 ± 0.005	0.023 ± 0.001	0.152 ± 0.005	—	—	0.003	0.003	—	0.1289	0.1290	1.344	99.555 ± 0.222	0.802 ± 0.032	0.445 ± 0.222

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S87

Simulation 1 - Condition 87: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.314	-0.002	-0.78	0.0743	0.0743	0.979	94.652 ± 0.715	0.269 ± 0.001	—
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.316	0.000	-0.00	0.0686	0.0686	0.980	95.055 ± 0.689	0.270 ± 0.001	—
LMS	beta33	0.000	-0.012	-0.012 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	—	—	-0.015	-0.015	—	0.0859	0.0873	0.964	93.946 ± 0.758	0.332 ± 0.002	6.054 ± 0.758
LMS	beta34	0.000	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.000	0.000	—	0.0714	0.0714	0.940	93.946 ± 0.758	0.269 ± 0.002	6.054 ± 0.758
LMS	beta35	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.000	0.000	—	0.0670	0.0670	1.029	95.459 ± 0.661	0.270 ± 0.002	4.541 ± 0.661
LSAM	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.316	0.000	-0.10	0.0779	0.0779	0.989	94.854 ± 0.702	0.291 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.008 ± 0.007	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.002	0.235 ± 0.006	0.318	0.002	0.65	0.0730	0.0730	0.996	95.560 ± 0.654	0.290 ± 0.001	—
LSAM	beta33	0.000	-0.003	-0.003 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.005	-0.005	—	0.0874	0.0875	0.971	94.551 ± 0.721	0.355 ± 0.003	5.449 ± 0.721
LSAM	beta34	0.000	0.005	0.005 ± 0.004	—	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.003	—	—	0.001	0.001	—	0.1084	0.1084	0.941	95.762 ± 0.640	0.439 ± 0.004	4.238 ± 0.640
LSAM	beta35	0.000	0.000	-0.000 ± 0.003	—	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	—	—	-0.002	-0.002	—	0.1039	0.1039	1.030	96.367 ± 0.594	0.428 ± 0.003	3.633 ± 0.594
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.313	-0.003	-0.86	0.0736	0.0737	0.978	94.753 ± 0.708	0.269 ± 0.001	—
QML	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.316	0.000	0.03	0.0685	0.0685	0.980	95.055 ± 0.689	0.270 ± 0.001	—
QML	beta33	0.000	-0.012	-0.012 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	—	—	-0.015	-0.015	—	0.0849	0.0863	0.967	93.744 ± 0.769	0.333 ± 0.002	6.256 ± 0.769
QML	beta34	0.000	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.001	-0.001	—	0.0711	0.0711	0.948	94.147 ± 0.746	0.270 ± 0.002	5.853 ± 0.746
QML	beta35	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.000	0.000	—	0.0661	0.0661	1.033	95.762 ± 0.640	0.270 ± 0.002	4.238 ± 0.640
UPI	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.242 ± 0.007	0.315	-0.001	-0.17	0.0768	0.0768	0.986	94.955 ± 0.695	0.295 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.318	0.002	0.69	0.0746	0.0746	1.005	95.560 ± 0.654	0.295 ± 0.001	—
UPI	beta33	0.000	-0.003	-0.003 ± 0.003	—	0.009 ± 0.000	0.097 ± 0.003	0.009 ± 0.000	0.097 ± 0.003	—	—	-0.005	-0.005	—	0.0884	0.0885	0.989	96.065 ± 0.618	0.376 ± 0.003	3.935 ± 0.618
UPI	beta34	0.000	0.004	0.004 ± 0.004	—	0.018 ± 0.001	0.135 ± 0.004	0.018 ± 0.001	0.135 ± 0.004	—	—	0.003	0.003	—	0.1203	0.1204	1.007	97.175 ± 0.526	0.533 ± 0.008	2.825 ± 0.526
UPI	beta35	0.000	0.001	0.001 ± 0.004	—	0.015 ± 0.001	0.122 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	—	—	0.000	0.000	—	0.1107	0.1107	1.104	97.477 ± 0.498	0.526 ± 0.009	2.523 ± 0.498

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S88

Simulation 1 - Condition 88: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.316	0.000	0.12	0.0447	0.0447	0.988	94.595 ± 0.715	0.169 ± 0.000	—
LMS	beta32	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.130 ± 0.004	0.317	0.001	0.19	0.0410	0.0410	1.049	96.396 ± 0.590	0.169 ± 0.000	—
LMS	beta33	0.000	-0.008	-0.008 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	-0.005	-0.005	—	0.0479	0.0482	1.071	95.095 ± 0.683	0.209 ± 0.002	4.905 ± 0.683
LMS	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.004	-0.004	—	0.0435	0.0437	1.009	94.795 ± 0.703	0.169 ± 0.002	5.205 ± 0.703
LMS	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0435	0.0435	1.008	94.595 ± 0.715	0.170 ± 0.001	5.405 ± 0.715
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.318	0.002	0.49	0.0462	0.0462	0.989	94.795 ± 0.703	0.176 ± 0.000	—
LSAM	beta32	0.316	0.318	0.002 ± 0.001	0.008 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.136 ± 0.004	0.318	0.002	0.73	0.0418	0.0419	1.053	96.697 ± 0.565	0.177 ± 0.000	—
LSAM	beta33	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.004	0.004	—	0.0498	0.0500	1.039	95.195 ± 0.677	0.209 ± 0.001	4.805 ± 0.677
LSAM	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.002	-0.002	—	0.0661	0.0661	0.984	94.995 ± 0.690	0.257 ± 0.001	5.005 ± 0.690
LSAM	beta35	0.000	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.001	0.001	—	0.0649	0.0649	0.994	95.596 ± 0.649	0.257 ± 0.001	4.404 ± 0.649
QML	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.316	0.000	0.11	0.0447	0.0447	0.987	94.494 ± 0.722	0.169 ± 0.000	—
QML	beta32	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.130 ± 0.004	0.317	0.001	0.20	0.0410	0.0410	1.047	96.396 ± 0.590	0.169 ± 0.000	—
QML	beta33	0.000	-0.008	-0.008 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	-0.005	-0.005	—	0.0481	0.0484	1.064	94.995 ± 0.690	0.207 ± 0.001	5.005 ± 0.690
QML	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.004	-0.004	—	0.0430	0.0432	1.007	94.895 ± 0.696	0.168 ± 0.001	5.105 ± 0.696
QML	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0430	0.0430	1.009	94.795 ± 0.703	0.169 ± 0.001	5.205 ± 0.703
UPI	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.318	0.002	0.53	0.0451	0.0452	0.991	94.695 ± 0.709	0.177 ± 0.000	—
UPI	beta32	0.316	0.319	0.003 ± 0.001	0.008 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.318	0.002	0.59	0.0420	0.0420	1.048	96.496 ± 0.582	0.178 ± 0.000	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.005	0.005	—	0.0502	0.0505	1.042	95.495 ± 0.656	0.212 ± 0.001	4.505 ± 0.656
UPI	beta34	0.000	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.004	-0.004	—	0.0685	0.0686	1.005	96.296 ± 0.598	0.274 ± 0.002	3.704 ± 0.598
UPI	beta35	0.000	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.001	0.001	—	0.0685	0.0685	1.012	96.897 ± 0.549	0.278 ± 0.002	3.103 ± 0.549

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S89

Simulation 1 - Condition 89: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.029 ± 0.001	0.169 ± 0.005	0.315	-0.001	-0.18	0.0537	0.0537	1.038	95.900 ± 0.627	0.217 ± 0.001	—
LMS	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.029 ± 0.001	0.171 ± 0.005	0.317	0.001	0.20	0.0535	0.0535	1.033	95.500 ± 0.656	0.218 ± 0.001	—
LMS	beta33	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.001	-0.001	—	0.0611	0.0611	0.981	93.400 ± 0.785	0.249 ± 0.001	6.600 ± 0.785
LMS	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.000	0.000	—	0.0580	0.0580	1.010	95.500 ± 0.656	0.226 ± 0.001	4.500 ± 0.656
LMS	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0572	0.0573	0.992	94.200 ± 0.739	0.228 ± 0.001	5.800 ± 0.739
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.030 ± 0.001	0.172 ± 0.005	0.315	-0.001	-0.20	0.0555	0.0555	1.041	95.900 ± 0.627	0.222 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.174 ± 0.005	0.317	0.001	0.45	0.0549	0.0549	1.031	95.600 ± 0.649	0.223 ± 0.001	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.002	0.002	—	0.0614	0.0614	0.962	93.200 ± 0.796	0.247 ± 0.001	6.800 ± 0.796
LSAM	beta34	0.000	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.002	0.002	—	0.0713	0.0713	1.007	95.800 ± 0.634	0.279 ± 0.001	4.200 ± 0.634
LSAM	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.004	-0.004	—	0.0723	0.0723	0.982	94.100 ± 0.745	0.281 ± 0.001	5.900 ± 0.745
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.029 ± 0.001	0.169 ± 0.005	0.315	-0.001	-0.16	0.0541	0.0541	1.037	95.800 ± 0.634	0.217 ± 0.001	—
QML	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.029 ± 0.001	0.171 ± 0.005	0.317	0.001	0.17	0.0535	0.0535	1.029	95.400 ± 0.662	0.218 ± 0.000	—
QML	beta33	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.001	-0.001	—	0.0610	0.0610	0.977	93.400 ± 0.785	0.248 ± 0.001	6.600 ± 0.785
QML	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.000	0.000	—	0.0581	0.0581	1.014	95.500 ± 0.656	0.226 ± 0.001	4.500 ± 0.656
QML	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0564	0.0565	0.992	94.200 ± 0.739	0.227 ± 0.001	5.800 ± 0.739
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.030 ± 0.001	0.172 ± 0.005	0.316	0.000	0.10	0.0557	0.0557	1.035	95.900 ± 0.627	0.221 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.031 ± 0.001	0.175 ± 0.005	0.318	0.002	0.49	0.0555	0.0555	1.024	95.600 ± 0.649	0.221 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.001	0.001	—	0.0600	0.0601	0.956	93.700 ± 0.768	0.249 ± 0.001	6.300 ± 0.768
UPI	beta34	0.000	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.004	0.004	—	0.0703	0.0704	1.032	95.900 ± 0.627	0.289 ± 0.001	4.100 ± 0.627
UPI	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	-0.002	-0.002	—	0.0719	0.0719	0.998	94.200 ± 0.739	0.290 ± 0.001	5.800 ± 0.739

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S90

Simulation 1 - Condition 90: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.000	0.108 ± 0.003	0.316	0.000	0.02	0.0344	0.0344	1.023	95.800 ± 0.634	0.137 ± 0.000	—
LMS	beta32	0.316	0.316	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.315	-0.001	-0.26	0.0355	0.0355	0.996	95.200 ± 0.676	0.137 ± 0.000	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.004	-0.004	—	0.0427	0.0429	0.970	94.500 ± 0.721	0.157 ± 0.001	5.500 ± 0.721
LMS	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0376	0.0376	0.980	95.300 ± 0.669	0.144 ± 0.001	4.700 ± 0.669
LMS	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0358	0.0359	0.956	93.400 ± 0.785	0.143 ± 0.000	6.600 ± 0.785
LSAM	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.316	0.000	0.10	0.0347	0.0347	1.022	95.300 ± 0.669	0.140 ± 0.000	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.315	-0.001	-0.28	0.0365	0.0365	0.995	95.400 ± 0.662	0.140 ± 0.000	—
LSAM	beta33	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0422	0.0422	0.966	94.300 ± 0.733	0.156 ± 0.000	5.700 ± 0.733
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0460	0.0460	0.975	95.100 ± 0.683	0.177 ± 0.000	4.900 ± 0.683
LSAM	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.003	-0.003	—	0.0442	0.0444	0.950	93.400 ± 0.785	0.176 ± 0.000	6.600 ± 0.785
QML	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.000	0.108 ± 0.003	0.316	0.000	0.01	0.0344	0.0344	1.022	95.900 ± 0.627	0.137 ± 0.000	—
QML	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.315	-0.001	-0.29	0.0356	0.0356	0.995	95.200 ± 0.676	0.137 ± 0.000	—
QML	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.004	-0.004	—	0.0429	0.0431	0.969	94.400 ± 0.727	0.157 ± 0.000	5.600 ± 0.727
QML	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0376	0.0376	0.983	95.300 ± 0.669	0.144 ± 0.001	4.700 ± 0.669
QML	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	—	0.0359	0.0360	0.959	93.400 ± 0.785	0.143 ± 0.000	6.600 ± 0.785
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.316	0.000	0.13	0.0346	0.0346	1.019	95.300 ± 0.669	0.139 ± 0.000	—
UPI	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.315	-0.001	-0.21	0.0364	0.0364	0.992	95.300 ± 0.669	0.139 ± 0.000	—
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0420	0.0420	0.967	93.700 ± 0.768	0.156 ± 0.000	6.300 ± 0.768
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0473	0.0473	0.973	94.600 ± 0.715	0.179 ± 0.001	5.400 ± 0.715
UPI	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.003	-0.003	—	0.0448	0.0449	0.955	93.500 ± 0.780	0.179 ± 0.001	6.500 ± 0.780

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S91

Simulation 1 - Condition 91: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.328	0.012 ± 0.007	0.040 ± 0.023	0.012 ± 0.002	0.110 ± 0.007	0.012 ± 0.002	0.111 ± 0.008	0.123 ± 0.017	0.351 ± 0.027	0.324	0.008	2.44	0.1174	0.1177	1.022	95.671 ± 1.339	0.442 ± 0.035	—
LMS	beta32	0.316	0.320	0.004 ± 0.007	0.013 ± 0.021	0.011 ± 0.001	0.103 ± 0.007	0.011 ± 0.001	0.103 ± 0.008	0.106 ± 0.014	0.326 ± 0.025	0.332	0.016	5.18	0.1255	0.1265	0.985	93.939 ± 1.570	0.399 ± 0.016	—
LMS	beta33	0.000	0.016	0.016 ± 0.009	—	0.019 ± 0.002	0.137 ± 0.007	0.019 ± 0.002	0.138 ± 0.009	—	—	0.023	0.023	—	0.1442	0.1461	1.154	95.238 ± 1.401	0.620 ± 0.050	4.762 ± 1.401
LMS	beta34	0.000	-0.043	-0.043 ± 0.006	—	0.009 ± 0.001	0.093 ± 0.007	0.010 ± 0.001	0.102 ± 0.007	—	—	-0.046	-0.046	—	0.0970	0.1075	1.241	92.208 ± 1.764	0.454 ± 0.088	7.792 ± 1.764
LMS	beta35	0.000	-0.039	-0.039 ± 0.005	—	0.006 ± 0.001	0.078 ± 0.005	0.008 ± 0.001	0.087 ± 0.005	—	—	-0.045	-0.045	—	0.0802	0.0922	1.455	90.909 ± 1.891	0.445 ± 0.093	9.091 ± 1.891
LSAM	beta31	0.316	0.344	0.028 ± 0.010	0.087 ± 0.032	0.024 ± 0.003	0.155 ± 0.009	0.025 ± 0.003	0.157 ± 0.011	0.247 ± 0.029	0.497 ± 0.034	0.332	0.016	5.20	0.1484	0.1493	1.111	97.835 ± 0.957	0.675 ± 0.026	—
LSAM	beta32	0.316	0.347	0.031 ± 0.010	0.099 ± 0.032	0.023 ± 0.003	0.153 ± 0.010	0.024 ± 0.003	0.155 ± 0.011	0.242 ± 0.029	0.492 ± 0.034	0.348	0.032	10.01	0.1575	0.1607	1.154	96.970 ± 1.128	0.690 ± 0.042	—
LSAM	beta33	0.000	0.113	0.113 ± 0.011	—	0.026 ± 0.003	0.162 ± 0.008	0.039 ± 0.004	0.197 ± 0.011	—	—	0.100	0.100	—	0.1582	0.1871	1.172	95.238 ± 1.401	0.745 ± 0.042	4.762 ± 1.401
LSAM	beta34	0.000	-0.077	-0.077 ± 0.008	—	0.014 ± 0.001	0.116 ± 0.006	0.019 ± 0.002	0.140 ± 0.008	—	—	-0.063	-0.063	—	0.1005	0.1187	1.156	95.238 ± 1.401	0.527 ± 0.032	4.762 ± 1.401
LSAM	beta35	0.000	-0.071	-0.071 ± 0.007	—	0.011 ± 0.001	0.104 ± 0.006	0.016 ± 0.002	0.126 ± 0.008	—	—	-0.069	-0.069	—	0.0876	0.1114	1.230	94.372 ± 1.516	0.501 ± 0.043	5.628 ± 1.516
QML	beta31	0.316	0.319	0.003 ± 0.006	0.009 ± 0.020	0.009 ± 0.001	0.097 ± 0.005	0.009 ± 0.001	0.097 ± 0.006	0.093 ± 0.011	0.306 ± 0.020	0.313	-0.003	-1.05	0.0988	0.0989	1.008	95.238 ± 1.401	0.382 ± 0.005	—
QML	beta32	0.316	0.313	-0.003 ± 0.006	-0.009 ± 0.019	0.008 ± 0.001	0.092 ± 0.004	0.008 ± 0.001	0.092 ± 0.005	0.084 ± 0.007	0.290 ± 0.016	0.326	0.010	3.14	0.1088	0.1092	1.031	94.372 ± 1.516	0.371 ± 0.005	—
QML	beta33	0.000	0.025	0.025 ± 0.008	—	0.016 ± 0.002	0.128 ± 0.006	0.017 ± 0.002	0.130 ± 0.007	—	—	0.034	0.034	—	0.1267	0.1312	1.018	95.238 ± 1.401	0.511 ± 0.013	4.762 ± 1.401
QML	beta34	0.000	-0.044	-0.044 ± 0.005	—	0.007 ± 0.001	0.081 ± 0.004	0.008 ± 0.001	0.092 ± 0.005	—	—	-0.046	-0.046	—	0.0803	0.0926	1.048	92.208 ± 1.764	0.333 ± 0.009	7.792 ± 1.764
QML	beta35	0.000	-0.041	-0.041 ± 0.005	—	0.006 ± 0.001	0.076 ± 0.004	0.007 ± 0.001	0.086 ± 0.005	—	—	-0.047	-0.047	—	0.0760	0.0894	1.038	93.506 ± 1.621	0.310 ± 0.009	6.494 ± 1.621
UPI	beta31	0.316	0.362	0.046 ± 0.015	0.145 ± 0.048	0.054 ± 0.008	0.232 ± 0.017	0.056 ± 0.008	0.236 ± 0.018	0.558 ± 0.077	0.747 ± 0.057	0.366	0.050	15.97	0.2529	0.2579	2.038	94.372 ± 1.516	1.853 ± 0.187	—
UPI	beta32	0.316	0.337	0.021 ± 0.017	0.067 ± 0.052	0.063 ± 0.008	0.252 ± 0.016	0.064 ± 0.008	0.252 ± 0.019	0.636 ± 0.083	0.798 ± 0.059	0.319	0.003	0.81	0.2654	0.2654	1.678	89.610 ± 2.008	1.655 ± 0.188	—
UPI	beta33	0.000	0.061	0.061 ± 0.016	—	0.060 ± 0.008	0.244 ± 0.016	0.063 ± 0.009	0.251 ± 0.019	—	—	0.021	0.021	—	0.2239	0.2249	1.627	98.701 ± 0.745	1.556 ± 0.143	1.299 ± 0.745
UPI	beta34	0.000	-0.134	-0.134 ± 0.017	—	0.063 ± 0.009	0.251 ± 0.018	0.081 ± 0.011	0.284 ± 0.021	—	—	-0.101	-0.101	—	0.2133	0.2361	1.926	100.000 ± 0.000	1.897 ± 0.177	0.000 ± 0.000
UPI	beta35	0.000	-0.103	-0.103 ± 0.016	—	0.060 ± 0.008	0.244 ± 0.017	0.070 ± 0.010	0.265 ± 0.021	—	—	-0.078	-0.078	—	0.2222	0.2354	1.633	99.567 ± 0.432	1.563 ± 0.182	0.433 ± 0.432

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S92

Simulation 1 - Condition 92: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.325	0.009 ± 0.004	0.029 ± 0.012	0.004 ± 0.000	0.062 ± 0.003	0.004 ± 0.000	0.062 ± 0.004	0.039 ± 0.004	0.197 ± 0.011	0.318	0.002	0.57	0.0694	0.0694	0.950	93.443 ± 1.585	0.230 ± 0.002	—
LMS	beta32	0.316	0.312	-0.004 ± 0.004	-0.013 ± 0.012	0.003 ± 0.000	0.058 ± 0.003	0.003 ± 0.000	0.058 ± 0.003	0.034 ± 0.003	0.184 ± 0.010	0.315	-0.001	-0.33	0.0663	0.0663	1.002	95.902 ± 1.269	0.228 ± 0.002	—
LMS	beta33	0.000	0.050	0.050 ± 0.005	—	0.006 ± 0.001	0.080 ± 0.004	0.009 ± 0.001	0.094 ± 0.005	—	—	0.051	0.051	—	0.0893	0.1028	0.962	87.295 ± 2.132	0.302 ± 0.005	12.705 ± 2.132
LMS	beta34	0.000	-0.056	-0.056 ± 0.003	—	0.002 ± 0.000	0.046 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.057	-0.057	—	0.0582	0.0816	1.049	76.639 ± 2.709	0.187 ± 0.003	23.361 ± 2.709
LMS	beta35	0.000	-0.051	-0.051 ± 0.003	—	0.003 ± 0.000	0.052 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	—	—	-0.054	-0.054	—	0.0549	0.0767	0.930	75.820 ± 2.741	0.190 ± 0.003	24.180 ± 2.741
LSAM	beta31	0.316	0.359	0.043 ± 0.006	0.136 ± 0.018	0.008 ± 0.001	0.090 ± 0.004	0.010 ± 0.001	0.100 ± 0.005	0.099 ± 0.008	0.315 ± 0.016	0.350	0.034	10.74	0.0903	0.0964	1.028	95.082 ± 1.384	0.363 ± 0.005	—
LSAM	beta32	0.316	0.338	0.022 ± 0.006	0.070 ± 0.019	0.009 ± 0.001	0.093 ± 0.004	0.009 ± 0.001	0.095 ± 0.006	0.091 ± 0.009	0.301 ± 0.018	0.336	0.020	6.43	0.0975	0.0996	0.995	95.902 ± 1.269	0.362 ± 0.005	—
LSAM	beta33	0.000	0.124	0.124 ± 0.006	—	0.010 ± 0.001	0.099 ± 0.006	0.025 ± 0.002	0.158 ± 0.007	—	—	0.119	0.119	—	0.0966	0.1531	0.937	79.918 ± 2.565	0.364 ± 0.007	20.082 ± 2.565
LSAM	beta34	0.000	-0.086	-0.086 ± 0.004	—	0.004 ± 0.000	0.063 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	—	—	-0.085	-0.085	—	0.0660	0.1077	1.028	71.721 ± 2.883	0.252 ± 0.006	28.279 ± 2.883
LSAM	beta35	0.000	-0.077	-0.077 ± 0.004	—	0.005 ± 0.001	0.068 ± 0.004	0.011 ± 0.001	0.103 ± 0.005	—	—	-0.075	-0.075	—	0.0624	0.0974	0.939	77.869 ± 2.658	0.248 ± 0.006	22.131 ± 2.658
QML	beta31	0.316	0.327	0.011 ± 0.004	0.034 ± 0.013	0.004 ± 0.000	0.063 ± 0.003	0.004 ± 0.000	0.063 ± 0.004	0.040 ± 0.004	0.200 ± 0.011	0.321	0.005	1.66	0.0647	0.0649	0.948	93.852 ± 1.538	0.233 ± 0.002	—
QML	beta32	0.316	0.313	-0.003 ± 0.004	-0.010 ± 0.012	0.004 ± 0.000	0.059 ± 0.003	0.004 ± 0.000	0.059 ± 0.003	0.035 ± 0.003	0.188 ± 0.010	0.316	0.000	-0.06	0.0603	0.0603	0.992	95.492 ± 1.328	0.231 ± 0.002	—
QML	beta33	0.000	0.052	0.052 ± 0.005	—	0.007 ± 0.001	0.084 ± 0.004	0.010 ± 0.001	0.099 ± 0.005	—	—	0.055	0.055	—	0.0840	0.1006	0.946	88.934 ± 2.008	0.310 ± 0.004	11.066 ± 2.008
QML	beta34	0.000	-0.060	-0.060 ± 0.003	—	0.002 ± 0.000	0.048 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	-0.062	-0.062	—	0.0527	0.0816	1.031	77.459 ± 2.675	0.193 ± 0.003	22.541 ± 2.675
QML	beta35	0.000	-0.056	-0.056 ± 0.003	—	0.003 ± 0.000	0.054 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	—	—	-0.059	-0.059	—	0.0519	0.0783	0.925	75.820 ± 2.741	0.197 ± 0.003	24.180 ± 2.741
UPI	beta31	0.316	0.387	0.071 ± 0.020	0.224 ± 0.062	0.094 ± 0.011	0.306 ± 0.018	0.098 ± 0.011	0.314 ± 0.021	0.986 ± 0.114	0.993 ± 0.066	0.381	0.065	20.46	0.3195	0.3260	1.357	86.475 ± 2.189	1.629 ± 0.165	—
UPI	beta32	0.316	0.313	-0.003 ± 0.019	-0.008 ± 0.061	0.091 ± 0.014	0.302 ± 0.023	0.091 ± 0.013	0.301 ± 0.025	0.909 ± 0.135	0.953 ± 0.079	0.363	0.047	14.83	0.3433	0.3465	1.290	83.607 ± 2.370	1.527 ± 0.190	—
UPI	beta33	0.000	0.058	0.058 ± 0.016	—	0.062 ± 0.008	0.250 ± 0.017	0.065 ± 0.009	0.256 ± 0.020	—	—	-0.017	-0.017	—	0.1935	0.1942	1.453	97.131 ± 1.069	1.421 ± 0.141	2.869 ± 1.069
UPI	beta34	0.000	-0.240	-0.240 ± 0.022	—	0.122 ± 0.017	0.349 ± 0.024	0.179 ± 0.024	0.423 ± 0.030	—	—	-0.200	-0.200	—	0.3196	0.3773	1.368	97.541 ± 0.991	1.871 ± 0.156	2.459 ± 0.991
UPI	beta35	0.000	-0.200	-0.200 ± 0.025	—	0.149 ± 0.024	0.385 ± 0.031	0.188 ± 0.029	0.434 ± 0.036	—	—	-0.178	-0.178	—	0.3405	0.3841	1.215	98.770 ± 0.705	1.836 ± 0.198	1.230 ± 0.705

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S93

Simulation 1 - Condition 93: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.215 ± 0.006	0.308	-0.008	-2.59	0.0660	0.0665	1.021	95.147 ± 0.683	0.272 ± 0.001	—
LMS	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.316	0.000	0.12	0.0690	0.0690	0.993	94.540 ± 0.722	0.272 ± 0.001	—
LMS	beta33	0.000	0.007	0.007 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.008	0.008	—	0.0846	0.0850	0.969	94.843 ± 0.703	0.333 ± 0.003	5.157 ± 0.703
LMS	beta34	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.018	-0.018	—	0.0544	0.0573	0.991	93.428 ± 0.788	0.220 ± 0.002	6.572 ± 0.788
LMS	beta35	0.000	-0.016	-0.016 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.020	-0.020	—	0.0565	0.0598	0.977	91.608 ± 0.882	0.221 ± 0.002	8.392 ± 0.882
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.057 ± 0.003	0.238 ± 0.007	0.315	-0.001	-0.41	0.0724	0.0724	1.017	94.641 ± 0.716	0.300 ± 0.001	—
LSAM	beta32	0.316	0.326	0.010 ± 0.002	0.031 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.324	0.008	2.41	0.0731	0.0735	0.996	94.742 ± 0.710	0.301 ± 0.001	—
LSAM	beta33	0.000	0.032	0.032 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	0.034	0.034	—	0.0856	0.0921	0.956	93.832 ± 0.765	0.345 ± 0.003	6.168 ± 0.765
LSAM	beta34	0.000	-0.024	-0.024 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.024	-0.024	—	0.0591	0.0639	1.001	94.439 ± 0.729	0.240 ± 0.002	5.561 ± 0.729
LSAM	beta35	0.000	-0.024	-0.024 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.025	-0.025	—	0.0615	0.0664	0.973	92.417 ± 0.842	0.241 ± 0.002	7.583 ± 0.842
QML	beta31	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.047 ± 0.002	0.216 ± 0.006	0.308	-0.008	-2.51	0.0658	0.0662	1.020	95.046 ± 0.690	0.273 ± 0.001	—
QML	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.316	0.000	0.12	0.0694	0.0694	0.992	94.439 ± 0.729	0.273 ± 0.001	—
QML	beta33	0.000	0.007	0.007 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.008	0.008	—	0.0841	0.0845	0.972	94.843 ± 0.703	0.333 ± 0.002	5.157 ± 0.703
QML	beta34	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.018	-0.018	—	0.0550	0.0578	0.995	93.529 ± 0.782	0.220 ± 0.002	6.471 ± 0.782
QML	beta35	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.020	-0.020	—	0.0563	0.0596	0.979	91.810 ± 0.872	0.221 ± 0.002	8.190 ± 0.872
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.242 ± 0.007	0.316	0.000	-0.04	0.0750	0.0750	1.076	95.854 ± 0.634	0.323 ± 0.007	—
UPI	beta32	0.316	0.327	0.011 ± 0.002	0.035 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.250 ± 0.007	0.325	0.009	2.82	0.0743	0.0748	1.070	95.753 ± 0.641	0.328 ± 0.010	—
UPI	beta33	0.000	0.024	0.024 ± 0.003	—	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	—	—	0.027	0.027	—	0.0885	0.0925	0.984	94.237 ± 0.741	0.375 ± 0.005	5.763 ± 0.741
UPI	beta34	0.000	-0.016	-0.016 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.017	-0.017	—	0.0655	0.0676	1.063	95.854 ± 0.634	0.289 ± 0.005	4.146 ± 0.634
UPI	beta35	0.000	-0.014	-0.014 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	-0.015	-0.015	—	0.0709	0.0725	1.064	95.046 ± 0.690	0.303 ± 0.012	4.954 ± 0.690

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S94

Simulation 1 - Condition 94: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.311	-0.005	-1.50	0.0407	0.0410	1.028	94.895 ± 0.696	0.171 ± 0.000	—
LMS	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.316	0.000	-0.10	0.0467	0.0467	0.984	94.494 ± 0.722	0.171 ± 0.000	—
LMS	beta33	0.000	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	—	—	0.010	0.010	—	0.0538	0.0547	1.008	93.994 ± 0.752	0.210 ± 0.002	6.006 ± 0.752
LMS	beta34	0.000	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.019	-0.019	—	0.0371	0.0418	0.992	90.691 ± 0.919	0.139 ± 0.002	9.309 ± 0.919
LMS	beta35	0.000	-0.021	-0.021 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.021	-0.021	—	0.0324	0.0384	1.036	91.491 ± 0.883	0.139 ± 0.001	8.509 ± 0.883
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.317	0.001	0.19	0.0457	0.0457	1.019	95.596 ± 0.649	0.186 ± 0.000	—
LSAM	beta32	0.316	0.321	0.005 ± 0.002	0.017 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.321	0.005	1.51	0.0507	0.0509	0.980	94.695 ± 0.709	0.187 ± 0.000	—
LSAM	beta33	0.000	0.033	0.033 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.034	0.034	—	0.0521	0.0621	1.004	91.391 ± 0.887	0.207 ± 0.001	8.609 ± 0.887
LSAM	beta34	0.000	-0.025	-0.025 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.025	-0.025	—	0.0378	0.0452	0.970	88.689 ± 1.002	0.145 ± 0.001	11.311 ± 1.002
LSAM	beta35	0.000	-0.027	-0.027 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.027	-0.027	—	0.0341	0.0434	1.011	89.489 ± 0.970	0.145 ± 0.001	10.511 ± 0.970
QML	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.312	-0.004	-1.42	0.0410	0.0412	1.024	94.895 ± 0.696	0.171 ± 0.000	—
QML	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.316	0.000	-0.07	0.0467	0.0467	0.981	94.494 ± 0.722	0.171 ± 0.000	—
QML	beta33	0.000	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	—	—	0.010	0.010	—	0.0541	0.0550	1.001	94.094 ± 0.746	0.209 ± 0.001	5.906 ± 0.746
QML	beta34	0.000	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.019	-0.019	—	0.0372	0.0420	0.980	90.891 ± 0.910	0.138 ± 0.001	9.109 ± 0.910
QML	beta35	0.000	-0.021	-0.021 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.021	-0.021	—	0.0325	0.0386	1.030	91.491 ± 0.883	0.138 ± 0.001	8.509 ± 0.883
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.316	0.000	-0.08	0.0463	0.0463	1.019	95.696 ± 0.642	0.192 ± 0.001	—
UPI	beta32	0.316	0.321	0.005 ± 0.002	0.015 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.319	0.003	0.93	0.0495	0.0495	0.992	94.695 ± 0.709	0.192 ± 0.001	—
UPI	beta33	0.000	0.024	0.024 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.023	0.023	—	0.0543	0.0591	1.011	93.794 ± 0.763	0.220 ± 0.001	6.206 ± 0.763
UPI	beta34	0.000	-0.018	-0.018 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.016	-0.016	—	0.0412	0.0442	0.987	93.293 ± 0.791	0.166 ± 0.001	6.707 ± 0.791
UPI	beta35	0.000	-0.020	-0.020 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.020	-0.020	—	0.0393	0.0439	1.018	94.394 ± 0.728	0.168 ± 0.002	5.606 ± 0.728

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S95

Simulation 1 - Condition 95: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.316	0.000	0.07	0.0555	0.0555	0.993	95.295 ± 0.670	0.220 ± 0.001	—
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.311	-0.005	-1.57	0.0577	0.0579	0.990	94.995 ± 0.690	0.220 ± 0.001	—
LMS	beta33	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	0.001	0.001	—	0.0665	0.0665	0.951	94.094 ± 0.746	0.246 ± 0.001	5.906 ± 0.746
LMS	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.008	-0.008	—	0.0446	0.0453	0.979	94.294 ± 0.734	0.171 ± 0.001	5.706 ± 0.734
LMS	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.005	-0.005	—	0.0453	0.0456	0.959	92.793 ± 0.818	0.169 ± 0.001	7.207 ± 0.818
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.317	0.001	0.34	0.0575	0.0575	0.992	95.195 ± 0.677	0.225 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.312	-0.004	-1.12	0.0612	0.0613	0.986	94.895 ± 0.696	0.226 ± 0.001	—
LSAM	beta33	0.000	0.011	0.011 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.011	0.011	—	0.0651	0.0660	0.950	93.293 ± 0.791	0.247 ± 0.001	6.707 ± 0.791
LSAM	beta34	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.009	-0.009	—	0.0464	0.0474	0.971	93.894 ± 0.758	0.175 ± 0.001	6.106 ± 0.758
LSAM	beta35	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.006	-0.006	—	0.0461	0.0465	0.956	92.993 ± 0.808	0.173 ± 0.001	7.007 ± 0.808
QML	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.316	0.000	0.06	0.0556	0.0556	0.993	95.295 ± 0.670	0.220 ± 0.001	—
QML	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.311	-0.005	-1.52	0.0581	0.0583	0.991	94.995 ± 0.690	0.220 ± 0.001	—
QML	beta33	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	0.002	0.002	—	0.0661	0.0661	0.951	94.094 ± 0.746	0.246 ± 0.001	5.906 ± 0.746
QML	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.008	-0.008	—	0.0445	0.0452	0.975	94.194 ± 0.740	0.170 ± 0.001	5.806 ± 0.740
QML	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.005	-0.005	—	0.0453	0.0455	0.962	92.693 ± 0.823	0.169 ± 0.001	7.307 ± 0.823
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.318	0.002	0.55	0.0559	0.0560	0.991	94.995 ± 0.690	0.224 ± 0.001	—
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.312	-0.004	-1.12	0.0601	0.0602	0.985	94.795 ± 0.703	0.224 ± 0.001	—
UPI	beta33	0.000	0.007	0.007 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.007	0.007	—	0.0674	0.0677	0.949	93.894 ± 0.758	0.248 ± 0.001	6.106 ± 0.758
UPI	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.006	-0.006	—	0.0472	0.0476	0.965	94.494 ± 0.722	0.177 ± 0.001	5.506 ± 0.722
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.003	-0.003	—	0.0458	0.0459	0.962	93.093 ± 0.802	0.176 ± 0.001	6.907 ± 0.802

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S96

Simulation 1 - Condition 96: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.316	0.000	-0.14	0.0341	0.0341	0.988	93.794 ± 0.763	0.139 ± 0.001	—
LMS	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.316	0.000	-0.10	0.0367	0.0367	0.997	95.395 ± 0.663	0.139 ± 0.000	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.003	-0.003	—	0.0370	0.0371	1.044	95.295 ± 0.670	0.158 ± 0.003	4.705 ± 0.670
LMS	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0269	0.0275	1.038	93.894 ± 0.758	0.111 ± 0.004	6.106 ± 0.758
LMS	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.004	-0.004	—	0.0275	0.0278	1.030	94.995 ± 0.690	0.107 ± 0.001	5.005 ± 0.690
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.317	0.001	0.27	0.0353	0.0354	0.982	93.694 ± 0.769	0.142 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.317	0.001	0.19	0.0379	0.0379	0.997	95.896 ± 0.628	0.142 ± 0.000	—
LSAM	beta33	0.000	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.006	0.006	—	0.0366	0.0370	1.024	95.295 ± 0.670	0.153 ± 0.001	4.705 ± 0.670
LSAM	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.007	-0.007	—	0.0276	0.0284	0.995	94.094 ± 0.746	0.109 ± 0.000	5.906 ± 0.746
LSAM	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.005	-0.005	—	0.0281	0.0285	1.022	94.895 ± 0.696	0.109 ± 0.000	5.105 ± 0.696
QML	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.316	0.000	-0.15	0.0342	0.0342	0.993	93.794 ± 0.763	0.140 ± 0.001	—
QML	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.316	0.000	-0.11	0.0366	0.0366	0.999	95.395 ± 0.663	0.139 ± 0.001	—
QML	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.003	-0.003	—	0.0370	0.0371	1.056	95.395 ± 0.663	0.159 ± 0.004	4.605 ± 0.663
QML	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0268	0.0274	1.058	94.194 ± 0.740	0.113 ± 0.006	5.806 ± 0.740
QML	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.004	-0.004	—	0.0275	0.0278	1.028	94.895 ± 0.696	0.107 ± 0.000	5.105 ± 0.696
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.317	0.001	0.24	0.0347	0.0347	0.982	93.594 ± 0.775	0.141 ± 0.000	—
UPI	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.317	0.001	0.33	0.0382	0.0382	0.995	95.596 ± 0.649	0.141 ± 0.000	—
UPI	beta33	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0369	0.0369	1.023	95.796 ± 0.635	0.154 ± 0.001	4.204 ± 0.635
UPI	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.003	-0.003	—	0.0277	0.0278	0.997	94.695 ± 0.709	0.110 ± 0.000	5.305 ± 0.709
UPI	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.002	-0.002	—	0.0289	0.0290	1.027	96.296 ± 0.598	0.110 ± 0.000	3.704 ± 0.598

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S97

Simulation 1 - Condition 97: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.280	-0.036 ± 0.005	-0.114 ± 0.015	0.012 ± 0.001	0.108 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	0.129 ± 0.008	0.359 ± 0.013	0.280	-0.036	-11.37	0.1171	0.1225	1.083	95.792 ± 0.899	0.457 ± 0.007	—
LMS	beta32	0.316	0.282	-0.034 ± 0.005	-0.107 ± 0.016	0.013 ± 0.001	0.112 ± 0.004	0.014 ± 0.001	0.117 ± 0.005	0.137 ± 0.009	0.371 ± 0.015	0.290	-0.026	-8.29	0.1280	0.1307	1.014	93.186 ± 1.128	0.446 ± 0.005	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.006	—	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.005	—	—	0.011	0.011	—	0.1523	0.1528	1.133	96.393 ± 0.835	0.574 ± 0.029	3.607 ± 0.835
LMS	beta34	0.000	0.040	0.040 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.003	0.008 ± 0.001	0.087 ± 0.004	—	—	0.033	0.033	—	0.0799	0.0866	1.091	96.192 ± 0.857	0.329 ± 0.012	3.808 ± 0.857
LMS	beta35	0.000	0.040	0.040 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.003	0.007 ± 0.001	0.085 ± 0.004	—	—	0.033	0.033	—	0.0765	0.0832	1.128	96.192 ± 0.857	0.330 ± 0.027	3.808 ± 0.857
LSAM	beta31	0.316	0.468	0.152 ± 0.052	0.481 ± 0.164	1.333 ± 0.106	1.155 ± 0.046	1.353 ± 0.109	1.163 ± 0.054	13.554 ± 1.091	3.682 ± 0.170	0.419	0.103	32.50	1.2043	1.2087	2.659	99.800 ± 0.200	12.033 ± 0.493	—
LSAM	beta32	0.316	0.468	0.152 ± 0.051	0.482 ± 0.162	1.301 ± 0.101	1.141 ± 0.045	1.322 ± 0.105	1.150 ± 0.052	13.236 ± 1.048	3.638 ± 0.165	0.454	0.138	43.66	1.1810	1.1890	2.615	99.399 ± 0.346	11.693 ± 0.482	—
LSAM	beta33	0.000	0.115	0.115 ± 0.013	—	0.087 ± 0.008	0.295 ± 0.014	0.100 ± 0.009	0.316 ± 0.016	—	—	0.084	0.084	—	0.2454	0.2593	2.432	99.399 ± 0.346	2.812 ± 0.150	0.601 ± 0.346
LSAM	beta34	0.000	-0.105	-0.105 ± 0.020	—	0.195 ± 0.019	0.442 ± 0.021	0.206 ± 0.020	0.454 ± 0.024	—	—	-0.054	-0.054	—	0.3650	0.3690	2.528	99.599 ± 0.283	4.381 ± 0.211	0.401 ± 0.283
LSAM	beta35	0.000	-0.105	-0.105 ± 0.020	—	0.191 ± 0.019	0.437 ± 0.022	0.202 ± 0.021	0.449 ± 0.025	—	—	-0.065	-0.065	—	0.3631	0.3689	2.437	99.599 ± 0.283	4.173 ± 0.205	0.401 ± 0.283
QML	beta31	0.316	0.283	-0.033 ± 0.005	-0.106 ± 0.015	0.012 ± 0.001	0.108 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	0.128 ± 0.008	0.357 ± 0.013	0.279	-0.037	-11.69	0.1111	0.1171	1.059	94.990 ± 0.977	0.448 ± 0.004	—
QML	beta32	0.316	0.285	-0.031 ± 0.005	-0.099 ± 0.016	0.013 ± 0.001	0.112 ± 0.004	0.014 ± 0.001	0.116 ± 0.004	0.135 ± 0.008	0.368 ± 0.014	0.293	-0.023	-7.28	0.1210	0.1232	1.006	93.186 ± 1.128	0.442 ± 0.004	—
QML	beta33	0.000	0.000	0.000 ± 0.005	—	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	—	—	0.007	0.007	—	0.1374	0.1375	1.065	96.192 ± 0.857	0.506 ± 0.010	3.808 ± 0.857
QML	beta34	0.000	0.035	0.035 ± 0.003	—	0.005 ± 0.000	0.074 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	0.032	0.032	—	0.0737	0.0805	1.034	95.190 ± 0.958	0.299 ± 0.007	4.810 ± 0.958
QML	beta35	0.000	0.035	0.035 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	—	—	0.031	0.031	—	0.0698	0.0764	1.036	95.391 ± 0.939	0.286 ± 0.007	4.609 ± 0.939
UPI	beta31	0.316	0.427	0.111 ± 0.028	0.351 ± 0.090	0.400 ± 0.037	0.633 ± 0.029	0.412 ± 0.037	0.642 ± 0.032	4.122 ± 0.370	2.030 ± 0.102	0.467	0.151	47.85	0.6376	0.6553	1.756	97.796 ± 0.657	4.354 ± 0.240	—
UPI	beta32	0.316	0.440	0.124 ± 0.028	0.391 ± 0.087	0.381 ± 0.031	0.617 ± 0.025	0.395 ± 0.032	0.629 ± 0.029	3.960 ± 0.317	1.990 ± 0.091	0.427	0.111	35.18	0.6697	0.6789	1.838	98.196 ± 0.596	4.447 ± 0.229	—
UPI	beta33	0.000	0.042	0.042 ± 0.014	—	0.098 ± 0.010	0.313 ± 0.016	0.099 ± 0.010	0.315 ± 0.017	—	—	0.050	0.050	—	0.2814	0.2857	1.836	99.599 ± 0.283	2.251 ± 0.145	0.401 ± 0.283
UPI	beta34	0.000	-0.063	-0.063 ± 0.012	—	0.069 ± 0.007	0.263 ± 0.014	0.073 ± 0.008	0.270 ± 0.015	—	—	-0.078	-0.078	—	0.2458	0.2578	1.719	99.800 ± 0.200	1.773 ± 0.104	0.200 ± 0.200
UPI	beta35	0.000	-0.059	-0.059 ± 0.011	—	0.063 ± 0.006	0.250 ± 0.012	0.066 ± 0.006	0.257 ± 0.012	—	—	-0.065	-0.065	—	0.2421	0.2506	1.815	99.599 ± 0.283	1.779 ± 0.103	0.401 ± 0.283

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S98

Simulation 1 - Condition 98: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.284	-0.032 ± 0.003	-0.100 ± 0.009	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.232 ± 0.008	0.286	-0.030	-9.37	0.0780	0.0834	1.037	93.043 ± 1.061	0.269 ± 0.001	—
LMS	beta32	0.316	0.287	-0.029 ± 0.003	-0.091 ± 0.008	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.003	0.219 ± 0.007	0.288	-0.028	-8.77	0.0715	0.0766	1.089	95.478 ± 0.867	0.269 ± 0.002	—
LMS	beta33	0.000	0.014	0.014 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	—	—	0.024	0.024	—	0.0850	0.0882	0.985	94.957 ± 0.913	0.322 ± 0.005	5.043 ± 0.913
LMS	beta34	0.000	0.032	0.032 ± 0.002	—	0.002 ± 0.000	0.044 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.026	0.026	—	0.0468	0.0538	1.000	90.957 ± 1.196	0.173 ± 0.003	9.043 ± 1.196
LMS	beta35	0.000	0.033	0.033 ± 0.002	—	0.002 ± 0.000	0.044 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	0.031	0.031	—	0.0453	0.0547	1.014	94.087 ± 0.984	0.173 ± 0.003	5.913 ± 0.984
LSAM	beta31	0.316	0.434	0.118 ± 0.024	0.374 ± 0.075	0.322 ± 0.029	0.567 ± 0.025	0.335 ± 0.030	0.579 ± 0.029	3.358 ± 0.302	1.832 ± 0.091	0.396	0.080	25.18	0.6190	0.6241	1.824	99.304 ± 0.347	4.057 ± 0.262	—
LSAM	beta32	0.316	0.417	0.101 ± 0.024	0.320 ± 0.075	0.324 ± 0.030	0.569 ± 0.026	0.334 ± 0.031	0.578 ± 0.029	3.344 ± 0.306	1.829 ± 0.092	0.396	0.080	25.19	0.5870	0.5924	1.779	99.826 ± 0.174	3.970 ± 0.251	—
LSAM	beta33	0.000	0.071	0.071 ± 0.005	—	0.013 ± 0.001	0.116 ± 0.005	0.018 ± 0.002	0.136 ± 0.006	—	—	0.068	0.068	—	0.1177	0.1357	1.493	95.304 ± 0.882	0.678 ± 0.037	4.696 ± 0.882
LSAM	beta34	0.000	-0.068	-0.068 ± 0.008	—	0.034 ± 0.003	0.183 ± 0.009	0.038 ± 0.004	0.195 ± 0.010	—	—	-0.043	-0.043	—	0.1784	0.1836	1.830	98.957 ± 0.424	1.313 ± 0.094	1.043 ± 0.424
LSAM	beta35	0.000	-0.063	-0.063 ± 0.008	—	0.034 ± 0.003	0.184 ± 0.009	0.038 ± 0.004	0.195 ± 0.011	—	—	-0.048	-0.048	—	0.1737	0.1802	1.726	99.652 ± 0.246	1.247 ± 0.083	0.348 ± 0.246
QML	beta31	0.316	0.286	-0.030 ± 0.003	-0.096 ± 0.009	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.233 ± 0.008	0.287	-0.029	-9.07	0.0741	0.0795	1.034	93.391 ± 1.036	0.273 ± 0.001	—
QML	beta32	0.316	0.289	-0.027 ± 0.003	-0.086 ± 0.008	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.048 ± 0.003	0.220 ± 0.007	0.289	-0.027	-8.69	0.0678	0.0732	1.084	95.304 ± 0.882	0.272 ± 0.001	—
QML	beta33	0.000	0.011	0.011 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.007 ± 0.000	0.081 ± 0.003	—	—	0.017	0.017	—	0.0785	0.0804	0.974	94.609 ± 0.942	0.306 ± 0.003	5.391 ± 0.942
QML	beta34	0.000	0.031	0.031 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.026	0.026	—	0.0448	0.0519	0.995	89.391 ± 1.284	0.169 ± 0.002	10.609 ± 1.284
QML	beta35	0.000	0.032	0.032 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	0.028	0.028	—	0.0432	0.0518	1.007	92.348 ± 1.109	0.167 ± 0.002	7.652 ± 1.109
UPI	beta31	0.316	0.531	0.215 ± 0.024	0.679 ± 0.074	0.318 ± 0.025	0.564 ± 0.022	0.363 ± 0.028	0.603 ± 0.026	3.637 ± 0.278	1.907 ± 0.081	0.588	0.272	86.22	0.6037	0.6623	1.587	93.565 ± 1.023	3.505 ± 0.183	—
UPI	beta32	0.316	0.529	0.213 ± 0.025	0.674 ± 0.080	0.370 ± 0.028	0.608 ± 0.023	0.415 ± 0.028	0.644 ± 0.025	4.152 ± 0.284	2.038 ± 0.080	0.573	0.257	81.30	0.6334	0.6835	1.527	95.478 ± 0.867	3.639 ± 0.202	—
UPI	beta33	0.000	0.107	0.107 ± 0.010	—	0.054 ± 0.005	0.232 ± 0.011	0.065 ± 0.006	0.255 ± 0.012	—	—	0.097	0.097	—	0.1993	0.2215	1.523	99.652 ± 0.246	1.384 ± 0.073	0.348 ± 0.246
UPI	beta34	0.000	-0.100	-0.100 ± 0.009	—	0.043 ± 0.004	0.207 ± 0.009	0.053 ± 0.004	0.230 ± 0.010	—	—	-0.110	-0.110	—	0.1860	0.2161	1.545	97.043 ± 0.706	1.251 ± 0.064	2.957 ± 0.706
UPI	beta35	0.000	-0.102	-0.102 ± 0.009	—	0.050 ± 0.004	0.225 ± 0.010	0.061 ± 0.005	0.247 ± 0.011	—	—	-0.113	-0.113	—	0.2050	0.2340	1.503	97.565 ± 0.643	1.323 ± 0.074	2.435 ± 0.643

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S99

Simulation 1 - Condition 99: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.293	-0.023 ± 0.003	-0.074 ± 0.009	0.008 ± 0.000	0.089 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	0.085 ± 0.004	0.292 ± 0.008	0.294	-0.022	-6.86	0.0941	0.0965	0.995	93.393 ± 0.831	0.348 ± 0.002	—
LMS	beta32	0.316	0.296	-0.020 ± 0.003	-0.065 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.079 ± 0.004	0.282 ± 0.008	0.293	-0.023	-7.42	0.0949	0.0977	1.022	95.073 ± 0.724	0.347 ± 0.002	—
LMS	beta33	0.000	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	0.004	0.004	—	0.0854	0.0855	1.010	94.961 ± 0.732	0.320 ± 0.015	5.039 ± 0.732
LMS	beta34	0.000	0.022	0.022 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.018	0.018	—	0.0461	0.0496	0.941	90.034 ± 1.002	0.171 ± 0.005	9.966 ± 1.002
LMS	beta35	0.000	0.020	0.020 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.018	0.018	—	0.0434	0.0470	0.951	92.049 ± 0.905	0.171 ± 0.003	7.951 ± 0.905
LSAM	beta31	0.316	0.335	0.019 ± 0.006	0.061 ± 0.018	0.029 ± 0.001	0.171 ± 0.004	0.030 ± 0.001	0.172 ± 0.005	0.297 ± 0.014	0.545 ± 0.016	0.339	0.023	7.26	0.1707	0.1722	1.023	96.193 ± 0.640	0.686 ± 0.009	—
LSAM	beta32	0.316	0.342	0.026 ± 0.006	0.083 ± 0.018	0.028 ± 0.001	0.166 ± 0.004	0.028 ± 0.002	0.168 ± 0.005	0.284 ± 0.015	0.533 ± 0.017	0.332	0.016	5.19	0.1711	0.1719	1.050	97.200 ± 0.552	0.685 ± 0.009	—
LSAM	beta33	0.000	0.023	0.023 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.021	0.021	—	0.0807	0.0833	0.988	94.961 ± 0.732	0.329 ± 0.004	5.039 ± 0.732
LSAM	beta34	0.000	-0.015	-0.015 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.014	-0.014	—	0.0591	0.0608	0.982	94.065 ± 0.791	0.240 ± 0.005	5.935 ± 0.791
LSAM	beta35	0.000	-0.018	-0.018 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.014	-0.014	—	0.0552	0.0568	0.982	95.185 ± 0.716	0.242 ± 0.004	4.815 ± 0.716
QML	beta31	0.316	0.293	-0.023 ± 0.003	-0.073 ± 0.010	0.008 ± 0.000	0.090 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.086 ± 0.004	0.293 ± 0.008	0.294	-0.022	-6.89	0.0930	0.0955	0.984	93.617 ± 0.818	0.346 ± 0.002	—
QML	beta32	0.316	0.296	-0.020 ± 0.003	-0.063 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	0.080 ± 0.004	0.283 ± 0.008	0.293	-0.023	-7.39	0.0933	0.0961	1.012	95.073 ± 0.724	0.347 ± 0.002	—
QML	beta33	0.000	0.001	0.001 ± 0.003	—	0.006 ± 0.000	0.081 ± 0.002	0.006 ± 0.000	0.081 ± 0.003	—	—	0.003	0.003	—	0.0829	0.0830	0.961	95.073 ± 0.724	0.304 ± 0.003	4.927 ± 0.724
QML	beta34	0.000	0.022	0.022 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.018	0.018	—	0.0451	0.0486	0.914	90.034 ± 1.002	0.166 ± 0.002	9.966 ± 1.002
QML	beta35	0.000	0.020	0.020 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	0.019	0.019	—	0.0428	0.0467	0.938	91.825 ± 0.917	0.169 ± 0.002	8.175 ± 0.917
UPI	beta31	0.316	0.339	0.023 ± 0.007	0.074 ± 0.022	0.044 ± 0.002	0.209 ± 0.006	0.044 ± 0.002	0.210 ± 0.007	0.442 ± 0.025	0.665 ± 0.022	0.336	0.020	6.36	0.2031	0.2041	1.180	96.865 ± 0.583	0.966 ± 0.049	—
UPI	beta32	0.316	0.356	0.040 ± 0.007	0.127 ± 0.022	0.044 ± 0.003	0.210 ± 0.006	0.046 ± 0.003	0.213 ± 0.007	0.456 ± 0.028	0.675 ± 0.023	0.343	0.027	8.57	0.1938	0.1956	1.189	97.872 ± 0.483	0.977 ± 0.047	—
UPI	beta33	0.000	0.026	0.026 ± 0.003	—	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	—	—	0.023	0.023	—	0.0876	0.0907	1.158	96.865 ± 0.583	0.440 ± 0.020	3.135 ± 0.583
UPI	beta34	0.000	-0.018	-0.018 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	—	—	-0.014	-0.014	—	0.0675	0.0690	1.141	95.745 ± 0.675	0.342 ± 0.018	4.255 ± 0.675
UPI	beta35	0.000	-0.026	-0.026 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.001	0.086 ± 0.003	—	—	-0.017	-0.017	—	0.0630	0.0653	1.111	96.976 ± 0.573	0.359 ± 0.020	3.024 ± 0.573

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S100

Simulation 1 - Condition 100: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.294	-0.022 ± 0.002	-0.069 ± 0.006	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.296	-0.020	-6.40	0.0573	0.0608	1.017	93.270 ± 0.812	0.215 ± 0.001	—
LMS	beta32	0.316	0.293	-0.023 ± 0.002	-0.073 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.292	-0.024	-7.48	0.0564	0.0611	1.002	92.744 ± 0.841	0.216 ± 0.001	—
LMS	beta33	0.000	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.003	0.003	—	0.0485	0.0486	1.005	94.953 ± 0.710	0.192 ± 0.002	5.047 ± 0.710
LMS	beta34	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.018	0.018	—	0.0264	0.0321	0.962	87.592 ± 1.069	0.102 ± 0.001	12.408 ± 1.069
LMS	beta35	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.017	0.017	—	0.0271	0.0320	0.944	87.907 ± 1.057	0.104 ± 0.002	12.093 ± 1.057
LSAM	beta31	0.316	0.337	0.021 ± 0.003	0.065 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.096 ± 0.005	0.309 ± 0.009	0.334	0.018	5.79	0.0958	0.0975	1.018	94.848 ± 0.717	0.382 ± 0.002	—
LSAM	beta32	0.316	0.335	0.019 ± 0.003	0.060 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.097 ± 0.004	0.311 ± 0.009	0.334	0.018	5.64	0.0971	0.0987	1.013	96.004 ± 0.635	0.383 ± 0.002	—
LSAM	beta33	0.000	0.018	0.018 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.014	0.014	—	0.0442	0.0464	0.973	93.375 ± 0.807	0.181 ± 0.001	6.625 ± 0.807
LSAM	beta34	0.000	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.012	-0.012	—	0.0307	0.0331	0.979	93.060 ± 0.824	0.124 ± 0.001	6.940 ± 0.824
LSAM	beta35	0.000	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.014	-0.014	—	0.0310	0.0342	0.966	94.217 ± 0.757	0.126 ± 0.001	5.783 ± 0.757
QML	beta31	0.316	0.294	-0.022 ± 0.002	-0.068 ± 0.006	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.296	-0.020	-6.36	0.0552	0.0588	1.012	93.481 ± 0.801	0.215 ± 0.001	—
QML	beta32	0.316	0.293	-0.023 ± 0.002	-0.072 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.293	-0.023	-7.41	0.0553	0.0601	0.993	92.534 ± 0.852	0.216 ± 0.001	—
QML	beta33	0.000	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.002	0.002	—	0.0477	0.0478	1.002	94.848 ± 0.717	0.191 ± 0.001	5.152 ± 0.717
QML	beta34	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.018	0.018	—	0.0260	0.0317	0.962	87.487 ± 1.073	0.102 ± 0.001	12.513 ± 1.073
QML	beta35	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.017	0.017	—	0.0265	0.0318	0.934	87.907 ± 1.057	0.103 ± 0.001	12.093 ± 1.057
UPI	beta31	0.316	0.344	0.028 ± 0.004	0.087 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.136 ± 0.007	0.369 ± 0.011	0.341	0.025	7.78	0.1132	0.1158	1.025	95.689 ± 0.659	0.456 ± 0.005	—
UPI	beta32	0.316	0.341	0.025 ± 0.004	0.080 ± 0.012	0.013 ± 0.001	0.114 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	0.137 ± 0.007	0.370 ± 0.011	0.337	0.021	6.80	0.1124	0.1145	1.018	96.530 ± 0.593	0.455 ± 0.005	—
UPI	beta33	0.000	0.020	0.020 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.017	0.017	—	0.0492	0.0519	0.987	94.532 ± 0.737	0.198 ± 0.002	5.468 ± 0.737
UPI	beta34	0.000	-0.017	-0.017 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.014	-0.014	—	0.0343	0.0372	0.991	94.637 ± 0.731	0.149 ± 0.002	5.363 ± 0.731
UPI	beta35	0.000	-0.018	-0.018 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.017	-0.017	—	0.0347	0.0385	0.958	95.058 ± 0.703	0.149 ± 0.002	4.942 ± 0.703

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S101

Simulation 1 - Condition 101: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.298	-0.018 ± 0.002	-0.058 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.254 ± 0.007	0.296	-0.020	-6.28	0.0754	0.0779	0.997	94.076 ± 0.748	0.305 ± 0.001	—
LMS	beta32	0.316	0.308	-0.008 ± 0.003	-0.025 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.252 ± 0.007	0.302	-0.014	-4.48	0.0797	0.0810	0.985	94.980 ± 0.692	0.306 ± 0.001	—
LMS	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.003	0.003	—	0.0578	0.0579	0.932	92.570 ± 0.831	0.224 ± 0.002	7.430 ± 0.831
LMS	beta34	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.010	0.010	—	0.0315	0.0329	0.900	89.659 ± 0.965	0.124 ± 0.001	10.341 ± 0.965
LMS	beta35	0.000	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.009	0.009	—	0.0324	0.0336	0.961	92.470 ± 0.836	0.124 ± 0.001	7.530 ± 0.836
LSAM	beta31	0.316	0.313	-0.003 ± 0.003	-0.010 ± 0.010	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.101 ± 0.005	0.317 ± 0.009	0.311	-0.005	-1.55	0.0928	0.0930	0.992	95.482 ± 0.658	0.390 ± 0.002	—
LSAM	beta32	0.316	0.326	0.010 ± 0.003	0.032 ± 0.010	0.010 ± 0.000	0.102 ± 0.002	0.010 ± 0.000	0.102 ± 0.003	0.104 ± 0.005	0.323 ± 0.009	0.319	0.003	0.84	0.1003	0.1003	0.984	95.080 ± 0.685	0.392 ± 0.002	—
LSAM	beta33	0.000	0.008	0.008 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.007	0.007	—	0.0575	0.0579	0.931	92.771 ± 0.821	0.224 ± 0.002	7.229 ± 0.821
LSAM	beta34	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0369	0.0370	0.913	92.369 ± 0.841	0.137 ± 0.002	7.631 ± 0.841
LSAM	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0343	0.0345	0.958	93.976 ± 0.754	0.138 ± 0.002	6.024 ± 0.754
QML	beta31	0.316	0.298	-0.018 ± 0.002	-0.058 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.065 ± 0.003	0.255 ± 0.007	0.296	-0.020	-6.33	0.0753	0.0779	0.990	94.076 ± 0.748	0.305 ± 0.001	—
QML	beta32	0.316	0.308	-0.008 ± 0.003	-0.025 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.254 ± 0.007	0.302	-0.014	-4.57	0.0806	0.0819	0.979	94.880 ± 0.698	0.306 ± 0.001	—
QML	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.004	0.004	—	0.0574	0.0575	0.929	92.671 ± 0.826	0.223 ± 0.002	7.329 ± 0.826
QML	beta34	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.009	0.009	—	0.0315	0.0329	0.896	89.458 ± 0.973	0.124 ± 0.001	10.542 ± 0.973
QML	beta35	0.000	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.009	0.009	—	0.0322	0.0333	0.960	92.169 ± 0.851	0.124 ± 0.001	7.831 ± 0.851
UPI	beta31	0.316	0.313	-0.003 ± 0.003	-0.011 ± 0.010	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.010	0.309	-0.007	-2.10	0.0933	0.0935	0.985	94.277 ± 0.736	0.386 ± 0.002	—
UPI	beta32	0.316	0.326	0.010 ± 0.003	0.031 ± 0.010	0.010 ± 0.000	0.101 ± 0.002	0.010 ± 0.000	0.102 ± 0.003	0.104 ± 0.005	0.322 ± 0.009	0.318	0.002	0.72	0.1002	0.1003	0.977	94.980 ± 0.692	0.388 ± 0.002	—
UPI	beta33	0.000	0.007	0.007 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.007	0.007	—	0.0558	0.0562	0.950	94.277 ± 0.736	0.226 ± 0.002	5.723 ± 0.736
UPI	beta34	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0357	0.0358	0.904	92.269 ± 0.846	0.136 ± 0.002	7.731 ± 0.846
UPI	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0344	0.0346	0.948	94.378 ± 0.730	0.137 ± 0.002	5.622 ± 0.730

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S102

Simulation 1 - Condition 102: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.305	-0.011 ± 0.002	-0.034 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.305	-0.011	-3.52	0.0468	0.0481	1.000	94.400 ± 0.727	0.190 ± 0.000	—
LMS	beta32	0.316	0.305	-0.011 ± 0.002	-0.036 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.305	-0.011	-3.52	0.0500	0.0512	1.007	93.800 ± 0.763	0.190 ± 0.000	—
LMS	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0363	0.0363	0.979	94.400 ± 0.727	0.138 ± 0.001	5.600 ± 0.727
LMS	beta34	0.000	0.008	0.008 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.008	0.008	—	0.0192	0.0206	0.896	91.300 ± 0.891	0.074 ± 0.001	8.700 ± 0.891
LMS	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	0.009	0.009	—	0.0185	0.0204	0.920	89.500 ± 0.969	0.073 ± 0.001	10.500 ± 0.969
LSAM	beta31	0.316	0.322	0.006 ± 0.002	0.018 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.195 ± 0.005	0.321	0.005	1.60	0.0601	0.0603	0.983	94.100 ± 0.745	0.236 ± 0.001	—
LSAM	beta32	0.316	0.321	0.005 ± 0.002	0.014 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.320	0.004	1.41	0.0620	0.0621	0.992	94.800 ± 0.702	0.235 ± 0.001	—
LSAM	beta33	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.004	0.004	—	0.0348	0.0350	0.972	94.200 ± 0.739	0.133 ± 0.001	5.800 ± 0.739
LSAM	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0204	0.0207	0.898	91.600 ± 0.877	0.078 ± 0.001	8.400 ± 0.877
LSAM	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0192	0.0194	0.922	93.100 ± 0.801	0.077 ± 0.001	6.900 ± 0.801
QML	beta31	0.316	0.305	-0.011 ± 0.002	-0.035 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.305	-0.011	-3.44	0.0470	0.0483	0.997	94.300 ± 0.733	0.190 ± 0.000	—
QML	beta32	0.316	0.305	-0.011 ± 0.002	-0.036 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.305	-0.011	-3.61	0.0504	0.0517	1.004	93.700 ± 0.768	0.190 ± 0.001	—
QML	beta33	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0359	0.0359	1.039	94.300 ± 0.733	0.146 ± 0.009	5.700 ± 0.733
QML	beta34	0.000	0.008	0.008 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.008	0.008	—	0.0192	0.0207	0.915	90.800 ± 0.914	0.075 ± 0.002	9.200 ± 0.914
QML	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	0.009	0.009	—	0.0187	0.0205	0.959	89.600 ± 0.965	0.077 ± 0.004	10.400 ± 0.965
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.321	0.005	1.44	0.0592	0.0594	0.986	94.100 ± 0.745	0.235 ± 0.001	—
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.320	0.004	1.39	0.0611	0.0613	0.994	94.500 ± 0.721	0.234 ± 0.001	—
UPI	beta33	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.002	0.002	—	0.0338	0.0339	0.974	94.400 ± 0.727	0.134 ± 0.001	5.600 ± 0.727
UPI	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0198	0.0201	0.903	92.100 ± 0.853	0.077 ± 0.001	7.900 ± 0.853
UPI	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.002	-0.002	—	0.0189	0.0190	0.924	93.300 ± 0.791	0.077 ± 0.001	6.700 ± 0.791

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S103

Simulation 1 - Condition 103: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.301	-0.015 ± 0.011	-0.048 ± 0.035	0.008 ± 0.002	0.090 ± 0.009	0.008 ± 0.002	0.090 ± 0.011	0.081 ± 0.016	0.285 ± 0.034	0.303	-0.013	-4.13	0.1005	0.1013	1.070	93.939 ± 2.937	0.376 ± 0.016	—
LMS	beta32	0.316	0.323	0.007 ± 0.012	0.022 ± 0.039	0.010 ± 0.002	0.099 ± 0.009	0.010 ± 0.002	0.099 ± 0.010	0.098 ± 0.016	0.313 ± 0.033	0.341	0.025	7.75	0.1186	0.1211	1.047	96.970 ± 2.110	0.408 ± 0.028	—
LMS	beta33	0.000	0.025	0.025 ± 0.017	—	0.019 ± 0.004	0.139 ± 0.015	0.020 ± 0.004	0.140 ± 0.016	—	—	0.049	0.049	—	0.1414	0.1495	1.112	93.939 ± 2.937	0.604 ± 0.086	6.061 ± 2.937
LMS	beta34	0.000	-0.058	-0.058 ± 0.011	—	0.009 ± 0.002	0.092 ± 0.011	0.012 ± 0.003	0.108 ± 0.014	—	—	-0.055	-0.055	—	0.1190	0.1312	0.923	80.303 ± 4.895	0.334 ± 0.018	19.697 ± 4.895
LMS	beta35	0.000	-0.072	-0.072 ± 0.014	—	0.012 ± 0.003	0.111 ± 0.016	0.017 ± 0.005	0.132 ± 0.019	—	—	-0.070	-0.070	—	0.1166	0.1359	1.209	80.303 ± 4.895	0.527 ± 0.139	19.697 ± 4.895
LSAM	beta31	0.316	0.304	-0.012 ± 0.019	-0.037 ± 0.061	0.025 ± 0.006	0.158 ± 0.020	0.025 ± 0.006	0.157 ± 0.022	0.246 ± 0.060	0.496 ± 0.070	0.306	-0.010	-3.11	0.1474	0.1477	1.463	100.000 ± 0.000	0.903 ± 0.122	—
LSAM	beta32	0.316	0.384	0.068 ± 0.022	0.214 ± 0.070	0.033 ± 0.007	0.181 ± 0.021	0.037 ± 0.008	0.192 ± 0.025	0.367 ± 0.082	0.606 ± 0.078	0.373	0.057	18.04	0.1522	0.1626	1.407	100.000 ± 0.000	0.996 ± 0.164	—
LSAM	beta33	0.000	0.103	0.103 ± 0.019	—	0.025 ± 0.006	0.158 ± 0.021	0.035 ± 0.008	0.188 ± 0.025	—	—	0.098	0.098	—	0.1600	0.1878	1.297	98.485 ± 1.504	0.805 ± 0.076	1.515 ± 1.504
LSAM	beta34	0.000	-0.085	-0.085 ± 0.020	—	0.028 ± 0.008	0.166 ± 0.025	0.035 ± 0.008	0.186 ± 0.023	—	—	-0.079	-0.079	—	0.1452	0.1655	1.691	96.970 ± 2.110	1.102 ± 0.224	3.030 ± 2.110
LSAM	beta35	0.000	-0.150	-0.150 ± 0.023	—	0.036 ± 0.009	0.189 ± 0.027	0.058 ± 0.014	0.240 ± 0.032	—	—	-0.128	-0.128	—	0.1576	0.2029	1.240	96.970 ± 2.110	0.921 ± 0.137	3.030 ± 2.110
QML	beta31	0.316	0.297	-0.019 ± 0.010	-0.059 ± 0.033	0.007 ± 0.001	0.084 ± 0.009	0.007 ± 0.001	0.086 ± 0.010	0.074 ± 0.014	0.272 ± 0.032	0.303	-0.013	-4.21	0.0913	0.0923	1.076	93.939 ± 2.937	0.356 ± 0.007	—
QML	beta32	0.316	0.318	0.002 ± 0.012	0.007 ± 0.037	0.009 ± 0.001	0.096 ± 0.008	0.009 ± 0.001	0.095 ± 0.010	0.091 ± 0.014	0.302 ± 0.030	0.329	0.013	4.21	0.1070	0.1078	0.961	96.970 ± 2.110	0.362 ± 0.007	—
QML	beta33	0.000	0.048	0.048 ± 0.016	—	0.016 ± 0.003	0.127 ± 0.013	0.018 ± 0.004	0.135 ± 0.016	—	—	0.059	0.059	—	0.1267	0.1396	0.965	93.939 ± 2.937	0.480 ± 0.021	6.061 ± 2.937
QML	beta34	0.000	-0.062	-0.062 ± 0.011	—	0.008 ± 0.002	0.087 ± 0.011	0.011 ± 0.003	0.106 ± 0.013	—	—	-0.051	-0.051	—	0.1041	0.1161	0.970	83.333 ± 4.587	0.330 ± 0.014	16.667 ± 4.587
QML	beta35	0.000	-0.076	-0.076 ± 0.012	—	0.009 ± 0.002	0.097 ± 0.008	0.015 ± 0.003	0.123 ± 0.012	—	—	-0.074	-0.074	—	0.1094	0.1322	0.889	80.303 ± 4.895	0.338 ± 0.017	19.697 ± 4.895
UPI	beta31	0.316	0.267	-0.049 ± 0.039	-0.155 ± 0.124	0.101 ± 0.032	0.318 ± 0.056	0.102 ± 0.030	0.320 ± 0.054	1.022 ± 0.301	1.011 ± 0.172	0.295	-0.021	-6.62	0.3905	0.3911	1.521	80.303 ± 4.895	1.897 ± 0.347	—
UPI	beta32	0.316	0.331	0.015 ± 0.044	0.047 ± 0.139	0.128 ± 0.023	0.358 ± 0.033	0.126 ± 0.023	0.355 ± 0.040	1.263 ± 0.228	1.124 ± 0.126	0.280	-0.036	-11.26	0.4033	0.4048	1.491	83.333 ± 4.587	2.089 ± 0.358	—
UPI	beta33	0.000	0.004	0.004 ± 0.026	—	0.045 ± 0.013	0.213 ± 0.032	0.045 ± 0.013	0.211 ± 0.034	—	—	-0.024	-0.024	—	0.2163	0.2176	1.987	100.000 ± 0.000	1.658 ± 0.245	0.000 ± 0.000
UPI	beta34	0.000	-0.151	-0.151 ± 0.052	—	0.175 ± 0.044	0.419 ± 0.057	0.195 ± 0.052	0.442 ± 0.067	—	—	-0.087	-0.087	—	0.3544	0.3650	1.809	98.485 ± 1.504	2.969 ± 0.671	1.515 ± 1.504
UPI	beta35	0.000	-0.210	-0.210 ± 0.072	—	0.343 ± 0.094	0.585 ± 0.086	0.382 ± 0.106	0.618 ± 0.096	—	—	-0.119	-0.119	—	0.4729	0.4876	1.758	100.000 ± 0.000	4.035 ± 0.751	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S104

Simulation 1 - Condition 104: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.007	0.003 ± 0.023	0.003 ± 0.001	0.053 ± 0.005	0.003 ± 0.001	0.052 ± 0.007	0.027 ± 0.005	0.166 ± 0.021	0.317	0.001	0.24	0.0581	0.0581	1.074	96.154 ± 2.667	0.222 ± 0.003	—
LMS	beta32	0.316	0.314	-0.002 ± 0.008	-0.007 ± 0.025	0.003 ± 0.001	0.057 ± 0.006	0.003 ± 0.001	0.057 ± 0.007	0.032 ± 0.006	0.179 ± 0.022	0.314	-0.002	-0.65	0.0505	0.0505	0.993	96.154 ± 2.667	0.223 ± 0.003	—
LMS	beta33	0.000	0.052	0.052 ± 0.009	—	0.004 ± 0.001	0.062 ± 0.007	0.006 ± 0.001	0.080 ± 0.009	—	—	0.046	0.046	—	0.0775	0.0903	1.181	94.231 ± 3.233	0.286 ± 0.008	5.769 ± 3.233
LMS	beta34	0.000	-0.063	-0.063 ± 0.007	—	0.003 ± 0.000	0.051 ± 0.004	0.007 ± 0.001	0.081 ± 0.006	—	—	-0.068	-0.068	—	0.0576	0.0889	0.985	75.000 ± 6.005	0.199 ± 0.006	25.000 ± 6.005
LMS	beta35	0.000	-0.058	-0.058 ± 0.005	—	0.001 ± 0.000	0.038 ± 0.005	0.005 ± 0.001	0.069 ± 0.005	—	—	-0.053	-0.053	—	0.0473	0.0713	1.337	82.692 ± 5.246	0.201 ± 0.008	17.308 ± 5.246
LSAM	beta31	0.316	0.353	0.037 ± 0.013	0.117 ± 0.042	0.009 ± 0.002	0.095 ± 0.010	0.010 ± 0.002	0.101 ± 0.013	0.102 ± 0.022	0.320 ± 0.040	0.352	0.036	11.41	0.0948	0.1014	1.101	96.154 ± 2.667	0.410 ± 0.015	—
LSAM	beta32	0.316	0.342	0.026 ± 0.014	0.082 ± 0.045	0.010 ± 0.003	0.102 ± 0.014	0.011 ± 0.003	0.104 ± 0.017	0.109 ± 0.030	0.330 ± 0.053	0.337	0.021	6.77	0.1079	0.1100	1.001	96.154 ± 2.667	0.401 ± 0.015	—
LSAM	beta33	0.000	0.091	0.091 ± 0.011	—	0.006 ± 0.002	0.080 ± 0.011	0.015 ± 0.003	0.121 ± 0.014	—	—	0.091	0.091	—	0.0927	0.1296	1.089	90.385 ± 4.088	0.342 ± 0.012	9.615 ± 4.088
LSAM	beta34	0.000	-0.113	-0.113 ± 0.014	—	0.010 ± 0.002	0.099 ± 0.009	0.023 ± 0.004	0.150 ± 0.013	—	—	-0.122	-0.122	—	0.1010	0.1586	0.983	78.846 ± 5.663	0.383 ± 0.019	21.154 ± 5.663
LSAM	beta35	0.000	-0.119	-0.119 ± 0.015	—	0.011 ± 0.003	0.106 ± 0.013	0.025 ± 0.004	0.158 ± 0.014	—	—	-0.110	-0.110	—	0.0963	0.1465	1.032	82.692 ± 5.246	0.427 ± 0.030	17.308 ± 5.246
QML	beta31	0.316	0.319	0.003 ± 0.007	0.009 ± 0.024	0.003 ± 0.001	0.054 ± 0.006	0.003 ± 0.001	0.053 ± 0.007	0.028 ± 0.006	0.168 ± 0.021	0.321	0.005	1.68	0.0612	0.0615	1.071	96.154 ± 2.667	0.225 ± 0.003	—
QML	beta32	0.316	0.315	-0.001 ± 0.008	-0.004 ± 0.025	0.003 ± 0.001	0.057 ± 0.006	0.003 ± 0.001	0.057 ± 0.007	0.032 ± 0.006	0.180 ± 0.022	0.313	-0.003	-1.02	0.0498	0.0499	0.996	94.231 ± 3.233	0.224 ± 0.003	—
QML	beta33	0.000	0.053	0.053 ± 0.009	—	0.004 ± 0.001	0.065 ± 0.008	0.007 ± 0.002	0.083 ± 0.010	—	—	0.045	0.045	—	0.0794	0.0914	1.152	94.231 ± 3.233	0.293 ± 0.008	5.769 ± 3.233
QML	beta34	0.000	-0.068	-0.068 ± 0.007	—	0.003 ± 0.000	0.053 ± 0.004	0.007 ± 0.001	0.086 ± 0.007	—	—	-0.068	-0.068	—	0.0576	0.0888	0.956	71.154 ± 6.283	0.200 ± 0.007	28.846 ± 6.283
QML	beta35	0.000	-0.063	-0.063 ± 0.005	—	0.002 ± 0.000	0.039 ± 0.004	0.005 ± 0.001	0.074 ± 0.005	—	—	-0.056	-0.056	—	0.0473	0.0735	1.323	82.692 ± 5.246	0.202 ± 0.007	17.308 ± 5.246
UPI	beta31	0.316	0.372	0.056 ± 0.097	0.178 ± 0.308	0.493 ± 0.136	0.702 ± 0.105	0.487 ± 0.144	0.698 ± 0.119	4.874 ± 1.439	2.208 ± 0.376	0.425	0.109	34.35	0.7991	0.8064	1.162	76.923 ± 5.843	3.198 ± 0.742	—
UPI	beta32	0.316	0.407	0.091 ± 0.110	0.287 ± 0.348	0.629 ± 0.281	0.793 ± 0.207	0.626 ± 0.291	0.791 ± 0.216	6.265 ± 2.911	2.503 ± 0.682	0.371	0.055	17.44	0.7502	0.7522	1.156	73.077 ± 6.151	3.597 ± 0.776	—
UPI	beta33	0.000	-0.050	-0.050 ± 0.078	—	0.320 ± 0.107	0.566 ± 0.104	0.316 ± 0.107	0.562 ± 0.109	—	—	-0.042	-0.042	—	0.3126	0.3154	1.239	96.154 ± 2.667	2.748 ± 0.667	3.846 ± 2.667
UPI	beta34	0.000	-0.451	-0.451 ± 0.248	—	3.188 ± 1.262	1.785 ± 0.397	3.330 ± 1.309	1.825 ± 0.404	—	—	-0.421	-0.421	—	1.2217	1.2921	0.817	96.154 ± 2.667	5.715 ± 1.109	3.846 ± 2.667
UPI	beta35	0.000	-0.844	-0.844 ± 0.205	—	2.188 ± 0.434	1.479 ± 0.158	2.858 ± 0.740	1.690 ± 0.243	—	—	-0.367	-0.367	—	1.6473	1.6878	1.340	98.077 ± 1.904	7.771 ± 1.492	1.923 ± 1.904

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S105

Simulation 1 - Condition 105: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.003	0.007 ± 0.008	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.003	0.218 ± 0.007	0.315	-0.001	-0.30	0.0674	0.0674	1.005	94.775 ± 0.848	0.272 ± 0.001	—
LMS	beta32	0.316	0.318	0.002 ± 0.003	0.007 ± 0.009	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.003	0.224 ± 0.008	0.316	0.000	0.06	0.0732	0.0732	0.970	94.775 ± 0.848	0.270 ± 0.001	—
LMS	beta33	0.000	0.016	0.016 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.017	0.017	—	0.0781	0.0799	1.050	95.791 ± 0.765	0.334 ± 0.004	4.209 ± 0.765
LMS	beta34	0.000	-0.031	-0.031 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.034	-0.034	—	0.0599	0.0687	1.041	92.308 ± 1.015	0.268 ± 0.003	7.692 ± 1.015
LMS	beta35	0.000	-0.030	-0.030 ± 0.003	—	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.036	-0.036	—	0.0701	0.0787	0.982	89.985 ± 1.144	0.264 ± 0.003	10.015 ± 1.144
LSAM	beta31	0.316	0.321	0.005 ± 0.003	0.015 ± 0.010	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.063 ± 0.004	0.250 ± 0.009	0.320	0.004	1.33	0.0791	0.0793	1.008	95.646 ± 0.777	0.312 ± 0.002	—
LSAM	beta32	0.316	0.323	0.007 ± 0.003	0.021 ± 0.010	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	0.066 ± 0.004	0.256 ± 0.009	0.319	0.003	0.99	0.0814	0.0814	0.982	95.791 ± 0.765	0.311 ± 0.002	—
LSAM	beta33	0.000	0.033	0.033 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	—	—	0.034	0.034	—	0.0749	0.0821	1.061	95.646 ± 0.777	0.348 ± 0.003	4.354 ± 0.777
LSAM	beta34	0.000	-0.048	-0.048 ± 0.004	—	0.011 ± 0.001	0.105 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	—	—	-0.050	-0.050	—	0.0908	0.1038	1.013	93.614 ± 0.931	0.417 ± 0.005	6.386 ± 0.931
LSAM	beta35	0.000	-0.049	-0.049 ± 0.004	—	0.011 ± 0.001	0.106 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	—	—	-0.051	-0.051	—	0.1013	0.1135	0.977	91.727 ± 1.049	0.407 ± 0.005	8.273 ± 1.049
QML	beta31	0.316	0.319	0.003 ± 0.003	0.008 ± 0.008	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.003	0.219 ± 0.007	0.315	-0.001	-0.19	0.0674	0.0674	1.004	94.775 ± 0.848	0.272 ± 0.001	—
QML	beta32	0.316	0.318	0.002 ± 0.003	0.008 ± 0.009	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.003	0.225 ± 0.008	0.316	0.000	0.12	0.0726	0.0726	0.969	94.920 ± 0.837	0.270 ± 0.001	—
QML	beta33	0.000	0.016	0.016 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.017	0.017	—	0.0773	0.0791	1.042	95.791 ± 0.765	0.332 ± 0.003	4.209 ± 0.765
QML	beta34	0.000	-0.032	-0.032 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.036	-0.036	—	0.0598	0.0696	1.031	92.308 ± 1.015	0.266 ± 0.002	7.692 ± 1.015
QML	beta35	0.000	-0.030	-0.030 ± 0.003	—	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.036	-0.036	—	0.0688	0.0778	0.980	89.985 ± 1.144	0.263 ± 0.002	10.015 ± 1.144
UPI	beta31	0.316	0.330	0.014 ± 0.004	0.045 ± 0.012	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.109 ± 0.007	0.330 ± 0.013	0.324	0.008	2.63	0.1005	0.1008	1.547	97.388 ± 0.608	0.626 ± 0.047	—
UPI	beta32	0.316	0.331	0.015 ± 0.004	0.048 ± 0.013	0.011 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	0.116 ± 0.007	0.340 ± 0.013	0.328	0.012	3.86	0.0987	0.0994	1.438	97.823 ± 0.556	0.601 ± 0.044	—
UPI	beta33	0.000	0.036	0.036 ± 0.004	—	0.009 ± 0.001	0.095 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	—	—	0.034	0.034	—	0.0922	0.0982	1.332	97.678 ± 0.574	0.498 ± 0.023	2.322 ± 0.574
UPI	beta34	0.000	-0.055	-0.055 ± 0.007	—	0.035 ± 0.003	0.187 ± 0.007	0.038 ± 0.003	0.195 ± 0.008	—	—	-0.060	-0.060	—	0.1657	0.1764	1.372	97.968 ± 0.538	1.006 ± 0.062	2.032 ± 0.538
UPI	beta35	0.000	-0.042	-0.042 ± 0.007	—	0.031 ± 0.002	0.175 ± 0.006	0.032 ± 0.002	0.180 ± 0.007	—	—	-0.043	-0.043	—	0.1541	0.1601	1.374	98.113 ± 0.518	0.944 ± 0.050	1.887 ± 0.518

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S106

Simulation 1 - Condition 106: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.316	0.000	-0.07	0.0457	0.0457	0.981	94.974 ± 0.795	0.170 ± 0.001	—
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.136 ± 0.004	0.316	0.000	-0.09	0.0420	0.0420	1.007	94.444 ± 0.833	0.169 ± 0.001	—
LMS	beta33	0.000	0.020	0.020 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.024	0.024	—	0.0511	0.0566	1.043	93.519 ± 0.895	0.213 ± 0.007	6.481 ± 0.895
LMS	beta34	0.000	-0.031	-0.031 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.033	-0.033	—	0.0406	0.0526	1.051	87.302 ± 1.211	0.174 ± 0.012	12.698 ± 1.211
LMS	beta35	0.000	-0.032	-0.032 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.032	-0.032	—	0.0405	0.0517	1.008	87.037 ± 1.222	0.163 ± 0.001	12.963 ± 1.222
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.008 ± 0.006	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.024 ± 0.001	0.156 ± 0.005	0.318	0.002	0.75	0.0509	0.0509	0.984	94.841 ± 0.804	0.189 ± 0.001	—
LSAM	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.024 ± 0.001	0.155 ± 0.005	0.319	0.003	0.96	0.0493	0.0494	0.990	95.106 ± 0.785	0.189 ± 0.001	—
LSAM	beta33	0.000	0.035	0.035 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.038	0.038	—	0.0494	0.0623	1.013	89.683 ± 1.106	0.207 ± 0.001	10.317 ± 1.106
LSAM	beta34	0.000	-0.048	-0.048 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.052	-0.052	—	0.0612	0.0801	0.990	86.905 ± 1.227	0.243 ± 0.002	13.095 ± 1.227
LSAM	beta35	0.000	-0.050	-0.050 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	—	—	-0.050	-0.050	—	0.0576	0.0765	0.987	86.508 ± 1.243	0.243 ± 0.002	13.492 ± 1.243
QML	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.316	0.000	0.00	0.0458	0.0458	0.976	94.841 ± 0.804	0.169 ± 0.000	—
QML	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.005	0.316	0.000	0.07	0.0422	0.0422	1.002	94.444 ± 0.833	0.169 ± 0.000	—
QML	beta33	0.000	0.020	0.020 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.025	0.025	—	0.0513	0.0568	1.007	93.519 ± 0.895	0.206 ± 0.001	6.481 ± 0.895
QML	beta34	0.000	-0.032	-0.032 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.034	-0.034	—	0.0405	0.0527	0.984	87.169 ± 1.216	0.163 ± 0.001	12.831 ± 1.216
QML	beta35	0.000	-0.033	-0.033 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.033	-0.033	—	0.0407	0.0522	1.004	87.037 ± 1.222	0.163 ± 0.001	12.963 ± 1.222
UPI	beta31	0.316	0.326	0.010 ± 0.003	0.032 ± 0.009	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.058 ± 0.004	0.240 ± 0.009	0.321	0.005	1.57	0.0735	0.0737	1.415	98.545 ± 0.436	0.418 ± 0.025	—
UPI	beta32	0.316	0.328	0.012 ± 0.003	0.039 ± 0.009	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	0.063 ± 0.004	0.252 ± 0.010	0.325	0.009	2.93	0.0723	0.0729	1.349	98.016 ± 0.507	0.416 ± 0.024	—
UPI	beta33	0.000	0.049	0.049 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	0.051	0.051	—	0.0654	0.0828	1.227	94.180 ± 0.851	0.300 ± 0.009	5.820 ± 0.851
UPI	beta34	0.000	-0.065	-0.065 ± 0.004	—	0.015 ± 0.001	0.121 ± 0.004	0.019 ± 0.001	0.137 ± 0.005	—	—	-0.062	-0.062	—	0.1084	0.1251	1.279	96.825 ± 0.638	0.606 ± 0.035	3.175 ± 0.638
UPI	beta35	0.000	-0.067	-0.067 ± 0.004	—	0.015 ± 0.001	0.121 ± 0.004	0.019 ± 0.001	0.138 ± 0.005	—	—	-0.059	-0.059	—	0.1030	0.1188	1.329	95.503 ± 0.754	0.629 ± 0.039	4.497 ± 0.754

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S107

Simulation 1 - Condition 107: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.314	-0.002	-0.55	0.0591	0.0592	0.947	92.893 ± 0.813	0.218 ± 0.000	—
LMS	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.175 ± 0.005	0.316	0.000	-0.04	0.0563	0.0563	1.006	95.295 ± 0.670	0.217 ± 0.000	—
LMS	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	—	—	0.003	0.003	—	0.0675	0.0675	0.944	93.493 ± 0.780	0.246 ± 0.001	6.507 ± 0.780
LMS	beta34	0.000	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.008	-0.008	—	0.0577	0.0582	1.000	94.494 ± 0.722	0.228 ± 0.001	5.506 ± 0.722
LMS	beta35	0.000	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.007	-0.007	—	0.0545	0.0549	1.025	94.995 ± 0.690	0.225 ± 0.001	5.005 ± 0.690
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.315	-0.001	-0.31	0.0601	0.0601	0.952	92.593 ± 0.829	0.225 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.318	0.002	0.62	0.0582	0.0582	1.009	95.596 ± 0.649	0.224 ± 0.001	—
LSAM	beta33	0.000	0.008	0.008 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	—	—	0.008	0.008	—	0.0686	0.0691	0.945	93.493 ± 0.780	0.246 ± 0.001	6.507 ± 0.780
LSAM	beta34	0.000	-0.011	-0.011 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.009	-0.009	—	0.0719	0.0725	0.991	94.394 ± 0.728	0.280 ± 0.001	5.606 ± 0.728
LSAM	beta35	0.000	-0.008	-0.008 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.008	-0.008	—	0.0672	0.0676	1.020	95.495 ± 0.656	0.278 ± 0.001	4.505 ± 0.656
QML	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.314	-0.002	-0.57	0.0591	0.0591	0.947	92.793 ± 0.818	0.218 ± 0.000	—
QML	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.175 ± 0.005	0.316	0.000	-0.02	0.0562	0.0562	1.005	95.195 ± 0.677	0.217 ± 0.000	—
QML	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	—	—	0.003	0.003	—	0.0678	0.0678	0.944	93.493 ± 0.780	0.246 ± 0.001	6.507 ± 0.780
QML	beta34	0.000	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.008	-0.008	—	0.0577	0.0583	1.001	94.494 ± 0.722	0.227 ± 0.001	5.506 ± 0.722
QML	beta35	0.000	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.007	-0.007	—	0.0546	0.0550	1.029	94.895 ± 0.696	0.225 ± 0.001	5.105 ± 0.696
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.315	-0.001	-0.20	0.0610	0.0610	0.940	92.793 ± 0.818	0.223 ± 0.001	—
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.317	0.001	0.38	0.0585	0.0585	1.004	95.596 ± 0.649	0.222 ± 0.001	—
UPI	beta33	0.000	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	—	—	0.003	0.003	—	0.0688	0.0689	0.936	93.093 ± 0.802	0.248 ± 0.001	6.907 ± 0.802
UPI	beta34	0.000	-0.006	-0.006 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.006	-0.006	—	0.0756	0.0758	1.004	94.995 ± 0.690	0.296 ± 0.002	5.005 ± 0.690
UPI	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.002	-0.002	—	0.0713	0.0713	1.016	95.295 ± 0.670	0.291 ± 0.002	4.705 ± 0.670

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S108

Simulation 1 - Condition 108: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.315	-0.001	-0.34	0.0354	0.0355	0.994	94.500 ± 0.721	0.137 ± 0.000	—
LMS	beta32	0.316	0.318	0.002 ± 0.001	0.006 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.011 ± 0.001	0.107 ± 0.003	0.318	0.002	0.67	0.0329	0.0330	1.034	95.200 ± 0.676	0.137 ± 0.000	—
LMS	beta33	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.004	0.004	—	0.0412	0.0414	0.986	93.500 ± 0.780	0.156 ± 0.001	6.500 ± 0.780
LMS	beta34	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.010	-0.010	—	0.0343	0.0358	1.019	94.800 ± 0.702	0.143 ± 0.001	5.200 ± 0.702
LMS	beta35	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.010	-0.010	—	0.0395	0.0407	0.971	93.300 ± 0.791	0.143 ± 0.001	6.700 ± 0.791
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.316	0.000	-0.01	0.0370	0.0370	0.995	95.100 ± 0.683	0.140 ± 0.000	—
LSAM	beta32	0.316	0.318	0.002 ± 0.001	0.008 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.319	0.003	0.87	0.0344	0.0346	1.036	95.200 ± 0.676	0.140 ± 0.000	—
LSAM	beta33	0.000	0.009	0.009 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.009	0.009	—	0.0405	0.0415	0.977	93.600 ± 0.774	0.154 ± 0.000	6.400 ± 0.774
LSAM	beta34	0.000	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.011	-0.011	—	0.0417	0.0431	1.009	94.500 ± 0.721	0.175 ± 0.000	5.500 ± 0.721
LSAM	beta35	0.000	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.012	-0.012	—	0.0481	0.0497	0.961	93.400 ± 0.785	0.174 ± 0.000	6.600 ± 0.785
QML	beta31	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.315	-0.001	-0.35	0.0354	0.0354	0.994	94.500 ± 0.721	0.137 ± 0.000	—
QML	beta32	0.316	0.318	0.002 ± 0.001	0.006 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.011 ± 0.001	0.107 ± 0.003	0.318	0.002	0.64	0.0329	0.0330	1.035	95.200 ± 0.676	0.137 ± 0.000	—
QML	beta33	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.005	0.005	—	0.0412	0.0414	0.985	93.600 ± 0.774	0.156 ± 0.001	6.400 ± 0.774
QML	beta34	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.010	-0.010	—	0.0341	0.0356	1.027	94.900 ± 0.696	0.144 ± 0.001	5.100 ± 0.696
QML	beta35	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.010	-0.010	—	0.0395	0.0408	0.967	93.200 ± 0.796	0.142 ± 0.000	6.800 ± 0.796
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.316	0.000	0.13	0.0366	0.0366	0.994	94.800 ± 0.702	0.140 ± 0.000	—
UPI	beta32	0.316	0.319	0.003 ± 0.001	0.008 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.318	0.002	0.78	0.0342	0.0343	1.030	95.000 ± 0.689	0.140 ± 0.000	—
UPI	beta33	0.000	0.005	0.005 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.004	0.004	—	0.0411	0.0413	0.974	93.400 ± 0.785	0.155 ± 0.000	6.600 ± 0.785
UPI	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.006	-0.006	—	0.0449	0.0453	1.008	94.700 ± 0.708	0.181 ± 0.001	5.300 ± 0.708
UPI	beta35	0.000	-0.005	-0.005 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.006	-0.006	—	0.0492	0.0496	0.968	94.500 ± 0.721	0.180 ± 0.001	5.500 ± 0.721

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S109

Simulation 1 - Condition 109: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.004	-0.007 ± 0.011	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.102 ± 0.005	0.319 ± 0.010	0.310	-0.006	-1.98	0.1067	0.1069	1.116	95.068 ± 0.760	0.442 ± 0.014	—
LMS	beta32	0.316	0.314	-0.002 ± 0.003	-0.007 ± 0.010	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	0.088 ± 0.005	0.296 ± 0.009	0.312	-0.004	-1.11	0.0972	0.0973	1.164	97.164 ± 0.583	0.427 ± 0.012	—
LMS	beta33	0.000	-0.081	-0.081 ± 0.007	—	0.042 ± 0.004	0.204 ± 0.009	0.048 ± 0.004	0.220 ± 0.011	—	—	-0.056	-0.056	—	0.1660	0.1751	1.113	97.041 ± 0.595	0.892 ± 0.052	2.959 ± 0.595
LMS	beta34	0.000	0.033	0.033 ± 0.004	—	0.016 ± 0.001	0.128 ± 0.006	0.017 ± 0.002	0.132 ± 0.007	—	—	0.021	0.021	—	0.1079	0.1100	1.145	96.424 ± 0.652	0.574 ± 0.052	3.576 ± 0.652
LMS	beta35	0.000	0.041	0.041 ± 0.004	—	0.016 ± 0.001	0.125 ± 0.005	0.017 ± 0.002	0.132 ± 0.006	—	—	0.029	0.029	—	0.1071	0.1108	1.071	96.424 ± 0.652	0.525 ± 0.019	3.576 ± 0.652
LSAM	beta31	0.316	0.321	0.005 ± 0.004	0.015 ± 0.012	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.123 ± 0.006	0.351 ± 0.011	0.313	-0.003	-0.91	0.1169	0.1169	1.041	96.547 ± 0.641	0.452 ± 0.004	—
LSAM	beta32	0.316	0.320	0.004 ± 0.004	0.014 ± 0.012	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	0.114 ± 0.006	0.337 ± 0.011	0.320	0.004	1.13	0.1031	0.1032	1.095	97.041 ± 0.595	0.457 ± 0.006	—
LSAM	beta33	0.000	0.013	0.013 ± 0.005	—	0.024 ± 0.002	0.154 ± 0.005	0.024 ± 0.002	0.155 ± 0.006	—	—	0.000	0.000	—	0.1440	0.1440	1.124	98.150 ± 0.473	0.680 ± 0.029	1.850 ± 0.473
LSAM	beta34	0.000	-0.010	-0.010 ± 0.004	—	0.012 ± 0.001	0.109 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	—	—	-0.006	-0.006	—	0.1028	0.1030	1.107	98.150 ± 0.473	0.473 ± 0.017	1.850 ± 0.473
LSAM	beta35	0.000	-0.001	-0.001 ± 0.004	—	0.013 ± 0.001	0.115 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	—	—	0.006	0.006	—	0.1084	0.1086	1.062	98.027 ± 0.488	0.477 ± 0.015	1.973 ± 0.488
QML	beta31	0.316	0.314	-0.002 ± 0.003	-0.006 ± 0.011	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.005	0.306 ± 0.009	0.311	-0.005	-1.72	0.0990	0.0992	1.024	95.314 ± 0.742	0.388 ± 0.003	—
QML	beta32	0.316	0.312	-0.004 ± 0.003	-0.012 ± 0.010	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.080 ± 0.004	0.283 ± 0.009	0.311	-0.005	-1.67	0.0917	0.0918	1.101	96.671 ± 0.630	0.386 ± 0.003	—
QML	beta33	0.000	-0.041	-0.041 ± 0.004	—	0.016 ± 0.001	0.128 ± 0.004	0.018 ± 0.001	0.134 ± 0.005	—	—	-0.048	-0.048	—	0.1310	0.1396	1.101	96.424 ± 0.652	0.552 ± 0.009	3.576 ± 0.652
QML	beta34	0.000	0.004	0.004 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.003	0.007 ± 0.001	0.086 ± 0.003	—	—	0.006	0.006	—	0.0882	0.0884	1.077	96.671 ± 0.630	0.365 ± 0.008	3.329 ± 0.630
QML	beta35	0.000	0.012	0.012 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	—	—	0.013	0.013	—	0.0872	0.0882	1.072	96.054 ± 0.684	0.361 ± 0.007	3.946 ± 0.684
UPI	beta31	0.316	0.320	0.004 ± 0.004	0.014 ± 0.012	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.122 ± 0.006	0.350 ± 0.011	0.315	-0.001	-0.34	0.1195	0.1195	1.112	97.534 ± 0.545	0.481 ± 0.005	—
UPI	beta32	0.316	0.322	0.006 ± 0.004	0.020 ± 0.012	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	0.113 ± 0.006	0.337 ± 0.011	0.319	0.003	0.97	0.1031	0.1032	1.162	97.411 ± 0.558	0.484 ± 0.005	—
UPI	beta33	0.000	0.017	0.017 ± 0.006	—	0.027 ± 0.002	0.164 ± 0.006	0.027 ± 0.002	0.165 ± 0.006	—	—	0.002	0.002	—	0.1507	0.1507	1.230	99.507 ± 0.246	0.793 ± 0.020	0.493 ± 0.246
UPI	beta34	0.000	-0.009	-0.009 ± 0.004	—	0.014 ± 0.001	0.118 ± 0.004	0.014 ± 0.001	0.119 ± 0.004	—	—	-0.003	-0.003	—	0.1022	0.1022	1.296	99.137 ± 0.325	0.601 ± 0.017	0.863 ± 0.325
UPI	beta35	0.000	-0.005	-0.005 ± 0.004	—	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	—	—	0.003	0.003	—	0.1107	0.1107	1.160	99.507 ± 0.246	0.577 ± 0.014	0.493 ± 0.246

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S110

Simulation 1 - Condition 110: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.006	0.315	-0.001	-0.32	0.0609	0.0609	1.041	96.237 ± 0.624	0.252 ± 0.002	—
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.313	-0.003	-0.84	0.0672	0.0673	1.011	96.022 ± 0.641	0.252 ± 0.002	—
LMS	beta33	0.000	-0.046	-0.046 ± 0.003	—	0.011 ± 0.001	0.105 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	—	—	-0.041	-0.041	—	0.1015	0.1096	1.180	97.527 ± 0.509	0.485 ± 0.025	2.473 ± 0.509
LMS	beta34	0.000	0.018	0.018 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	—	—	0.013	0.013	—	0.0638	0.0652	1.134	97.419 ± 0.520	0.293 ± 0.011	2.581 ± 0.520
LMS	beta35	0.000	0.017	0.017 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.014	0.014	—	0.0620	0.0636	1.204	97.527 ± 0.509	0.293 ± 0.012	2.473 ± 0.509
LSAM	beta31	0.316	0.322	0.006 ± 0.002	0.020 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.205 ± 0.006	0.319	0.003	1.04	0.0644	0.0645	1.020	95.376 ± 0.689	0.258 ± 0.001	—
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.210 ± 0.006	0.318	0.002	0.71	0.0673	0.0673	0.992	95.376 ± 0.689	0.258 ± 0.001	—
LSAM	beta33	0.000	0.006	0.006 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	-0.005	-0.005	—	0.0869	0.0870	1.036	95.806 ± 0.657	0.343 ± 0.003	4.194 ± 0.657
LSAM	beta34	0.000	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0593	0.0593	1.022	96.237 ± 0.624	0.241 ± 0.002	3.763 ± 0.624
LSAM	beta35	0.000	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	—	0.0600	0.0600	1.075	97.527 ± 0.509	0.241 ± 0.002	2.473 ± 0.509
QML	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.315	-0.001	-0.34	0.0592	0.0592	1.015	95.269 ± 0.696	0.237 ± 0.001	—
QML	beta32	0.316	0.313	-0.003 ± 0.002	-0.011 ± 0.006	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.006	0.314	-0.002	-0.70	0.0622	0.0623	0.983	94.839 ± 0.725	0.236 ± 0.001	—
QML	beta33	0.000	-0.036	-0.036 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	—	—	-0.041	-0.041	—	0.0864	0.0954	1.013	94.194 ± 0.767	0.327 ± 0.003	5.806 ± 0.767
QML	beta34	0.000	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.007	0.007	—	0.0533	0.0537	1.030	96.022 ± 0.641	0.213 ± 0.002	3.978 ± 0.641
QML	beta35	0.000	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.005	0.005	—	0.0551	0.0554	1.072	96.559 ± 0.598	0.214 ± 0.002	3.441 ± 0.598
UPI	beta31	0.316	0.323	0.007 ± 0.002	0.021 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.318	0.002	0.70	0.0641	0.0641	1.031	95.591 ± 0.673	0.263 ± 0.001	—
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.007	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.044 ± 0.002	0.211 ± 0.006	0.318	0.002	0.65	0.0670	0.0671	1.004	95.484 ± 0.681	0.262 ± 0.001	—
UPI	beta33	0.000	0.008	0.008 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	-0.002	-0.002	—	0.0854	0.0854	1.053	96.882 ± 0.570	0.363 ± 0.004	3.118 ± 0.570
UPI	beta34	0.000	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.004	-0.004	—	0.0618	0.0619	1.011	97.419 ± 0.520	0.264 ± 0.003	2.581 ± 0.520
UPI	beta35	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.001	-0.001	—	0.0607	0.0607	1.111	98.602 ± 0.385	0.265 ± 0.003	1.398 ± 0.385

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S111

Simulation 1 - Condition 111: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.318	0.002	0.75	0.0692	0.0692	1.013	95.243 ± 0.677	0.278 ± 0.002	—
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.232 ± 0.007	0.312	-0.004	-1.39	0.0713	0.0714	0.967	94.636 ± 0.717	0.278 ± 0.002	—
LMS	beta33	0.000	-0.026	-0.026 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	-0.019	-0.019	—	0.0894	0.0913	1.040	95.749 ± 0.642	0.380 ± 0.009	4.251 ± 0.642
LMS	beta34	0.000	0.008	0.008 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.004	0.004	—	0.0580	0.0581	1.026	95.040 ± 0.691	0.246 ± 0.004	4.960 ± 0.691
LMS	beta35	0.000	0.012	0.012 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.005	0.005	—	0.0547	0.0550	1.043	95.243 ± 0.677	0.253 ± 0.006	4.757 ± 0.677
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.323	0.007	2.31	0.0726	0.0730	0.993	94.838 ± 0.704	0.286 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.242 ± 0.007	0.317	0.001	0.24	0.0748	0.0748	0.949	94.433 ± 0.729	0.284 ± 0.001	—
LSAM	beta33	0.000	0.003	0.003 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.002	0.002	—	0.0860	0.0860	0.968	94.838 ± 0.704	0.339 ± 0.003	5.162 ± 0.704
LSAM	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	—	0.0601	0.0602	0.979	95.344 ± 0.670	0.239 ± 0.002	4.656 ± 0.670
LSAM	beta35	0.000	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.001	-0.001	—	0.0579	0.0579	0.997	94.838 ± 0.704	0.237 ± 0.002	5.162 ± 0.704
QML	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.002	0.220 ± 0.006	0.318	0.002	0.78	0.0682	0.0683	1.006	95.040 ± 0.691	0.274 ± 0.001	—
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.229 ± 0.007	0.312	-0.004	-1.32	0.0717	0.0719	0.964	94.636 ± 0.717	0.274 ± 0.001	—
QML	beta33	0.000	-0.023	-0.023 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.020	-0.020	—	0.0860	0.0882	1.006	95.040 ± 0.691	0.338 ± 0.004	4.960 ± 0.691
QML	beta34	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	—	0.0553	0.0553	1.007	94.433 ± 0.729	0.226 ± 0.002	5.567 ± 0.729
QML	beta35	0.000	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.002	0.002	—	0.0540	0.0540	1.030	95.243 ± 0.677	0.229 ± 0.003	4.757 ± 0.677
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.234 ± 0.007	0.320	0.004	1.40	0.0748	0.0749	0.993	95.142 ± 0.684	0.287 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.242 ± 0.007	0.315	-0.001	-0.19	0.0748	0.0748	0.953	94.737 ± 0.710	0.286 ± 0.001	—
UPI	beta33	0.000	0.003	0.003 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	0.001	0.001	—	0.0853	0.0853	0.980	95.749 ± 0.642	0.352 ± 0.003	4.251 ± 0.642
UPI	beta34	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.003	-0.003	—	0.0590	0.0591	0.980	95.850 ± 0.634	0.254 ± 0.003	4.150 ± 0.634
UPI	beta35	0.000	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.001	-0.001	—	0.0588	0.0588	1.026	95.749 ± 0.642	0.251 ± 0.003	4.251 ± 0.642

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S112

Simulation 1 - Condition 112: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.314	-0.002	-0.56	0.0452	0.0452	0.996	94.578 ± 0.718	0.172 ± 0.001	—
LMS	beta32	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.314	-0.002	-0.74	0.0421	0.0422	1.016	95.281 ± 0.672	0.173 ± 0.001	—
LMS	beta33	0.000	-0.023	-0.023 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.019	-0.019	—	0.0518	0.0553	1.013	94.880 ± 0.698	0.229 ± 0.003	5.120 ± 0.698
LMS	beta34	0.000	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.004	0.004	—	0.0359	0.0361	1.035	96.185 ± 0.607	0.151 ± 0.003	3.815 ± 0.607
LMS	beta35	0.000	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.004	0.004	—	0.0362	0.0363	0.975	96.084 ± 0.615	0.150 ± 0.002	3.916 ± 0.615
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.316	0.000	0.12	0.0462	0.0462	0.992	94.779 ± 0.705	0.178 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.317	0.001	0.27	0.0446	0.0446	1.010	95.281 ± 0.672	0.178 ± 0.000	—
LSAM	beta33	0.000	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.000	0.000	—	0.0485	0.0485	0.998	93.976 ± 0.754	0.205 ± 0.001	6.024 ± 0.754
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0363	0.0363	0.989	95.281 ± 0.672	0.145 ± 0.001	4.719 ± 0.672
LSAM	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0369	0.0369	0.965	94.378 ± 0.730	0.146 ± 0.001	5.622 ± 0.730
QML	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.314	-0.002	-0.72	0.0448	0.0448	0.995	94.679 ± 0.711	0.171 ± 0.000	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.313	-0.003	-0.87	0.0427	0.0428	1.014	95.181 ± 0.679	0.171 ± 0.000	—
QML	beta33	0.000	-0.023	-0.023 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.021	-0.021	—	0.0503	0.0546	0.998	92.871 ± 0.815	0.213 ± 0.001	7.129 ± 0.815
QML	beta34	0.000	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0347	0.0347	1.012	95.582 ± 0.651	0.142 ± 0.001	4.418 ± 0.651
QML	beta35	0.000	0.005	0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0355	0.0355	0.979	95.281 ± 0.672	0.143 ± 0.001	4.719 ± 0.672
UPI	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.317	0.001	0.24	0.0462	0.0462	0.994	95.181 ± 0.679	0.178 ± 0.000	—
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.317	0.001	0.35	0.0448	0.0449	1.014	95.181 ± 0.679	0.178 ± 0.000	—
UPI	beta33	0.000	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.001	0.001	—	0.0507	0.0507	0.998	94.980 ± 0.692	0.207 ± 0.001	5.020 ± 0.692
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0365	0.0366	0.996	95.582 ± 0.651	0.148 ± 0.001	4.418 ± 0.651
UPI	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0371	0.0372	0.979	95.181 ± 0.679	0.148 ± 0.001	4.819 ± 0.679

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S113

Simulation 1 - Condition 113: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.314	-0.002	-0.59	0.0570	0.0571	0.991	95.700 ± 0.641	0.220 ± 0.001	—
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.180 ± 0.005	0.316	0.000	-0.12	0.0533	0.0533	0.990	95.000 ± 0.689	0.221 ± 0.001	—
LMS	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.009	-0.009	—	0.0685	0.0691	0.923	94.000 ± 0.751	0.252 ± 0.002	6.000 ± 0.751
LMS	beta34	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.002	0.002	—	0.0439	0.0439	0.969	94.100 ± 0.745	0.173 ± 0.002	5.900 ± 0.745
LMS	beta35	0.000	0.005	0.005 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.003	0.003	—	0.0460	0.0460	0.968	94.100 ± 0.745	0.174 ± 0.002	5.900 ± 0.745
LSAM	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.183 ± 0.005	0.316	0.000	-0.14	0.0575	0.0575	0.990	95.300 ± 0.669	0.224 ± 0.001	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.316	0.000	0.10	0.0544	0.0544	0.985	94.800 ± 0.702	0.224 ± 0.001	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	—	0.0683	0.0683	0.919	93.200 ± 0.796	0.245 ± 0.002	6.800 ± 0.796
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0459	0.0459	0.962	93.800 ± 0.763	0.173 ± 0.001	6.200 ± 0.763
LSAM	beta35	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0470	0.0470	0.959	94.000 ± 0.751	0.174 ± 0.001	6.000 ± 0.751
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.314	-0.002	-0.71	0.0576	0.0576	0.991	95.600 ± 0.649	0.220 ± 0.001	—
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.180 ± 0.005	0.316	0.000	-0.08	0.0532	0.0532	0.990	95.000 ± 0.689	0.221 ± 0.001	—
QML	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.010	-0.010	—	0.0681	0.0688	0.923	93.600 ± 0.774	0.249 ± 0.002	6.400 ± 0.774
QML	beta34	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.002	0.002	—	0.0442	0.0442	0.969	93.900 ± 0.757	0.172 ± 0.001	6.100 ± 0.757
QML	beta35	0.000	0.005	0.005 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.003	0.003	—	0.0458	0.0459	0.970	94.000 ± 0.751	0.173 ± 0.001	6.000 ± 0.751
UPI	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.316	0.000	0.03	0.0575	0.0575	0.990	95.000 ± 0.689	0.224 ± 0.001	—
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.316	0.000	-0.03	0.0538	0.0538	0.984	94.900 ± 0.696	0.224 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	—	0.0686	0.0686	0.922	92.900 ± 0.812	0.247 ± 0.002	7.100 ± 0.812
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0453	0.0453	0.962	93.800 ± 0.763	0.175 ± 0.001	6.200 ± 0.763
UPI	beta35	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	—	0.0460	0.0460	0.958	93.700 ± 0.768	0.176 ± 0.001	6.300 ± 0.768

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S114

Simulation 1 - Condition 114: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.316	0.000	0.15	0.0354	0.0354	0.969	93.393 ± 0.786	0.138 ± 0.000	—
LMS	beta32	0.316	0.316	0.000 ± 0.001	0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.318	0.002	0.56	0.0361	0.0361	0.976	94.795 ± 0.703	0.138 ± 0.000	—
LMS	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.005	-0.005	—	0.0409	0.0412	0.970	93.894 ± 0.758	0.158 ± 0.001	6.106 ± 0.758
LMS	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0260	0.0260	0.973	94.695 ± 0.709	0.108 ± 0.001	5.305 ± 0.709
LMS	beta35	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	—	0.0274	0.0274	1.016	96.096 ± 0.613	0.109 ± 0.001	3.904 ± 0.613
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.318	0.002	0.58	0.0367	0.0368	0.965	93.594 ± 0.775	0.141 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.319	0.003	1.04	0.0359	0.0361	0.973	94.595 ± 0.715	0.141 ± 0.000	—
LSAM	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	—	0.0401	0.0402	0.965	94.094 ± 0.746	0.153 ± 0.001	5.906 ± 0.746
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.001	-0.001	—	0.0270	0.0271	0.971	94.394 ± 0.728	0.108 ± 0.000	5.606 ± 0.728
LSAM	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0281	0.0281	1.003	95.696 ± 0.642	0.108 ± 0.001	4.304 ± 0.642
QML	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.317	0.001	0.17	0.0356	0.0356	0.969	93.393 ± 0.786	0.138 ± 0.000	—
QML	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.318	0.002	0.49	0.0361	0.0362	0.976	94.695 ± 0.709	0.138 ± 0.000	—
QML	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.006	-0.006	—	0.0407	0.0411	0.979	93.594 ± 0.775	0.158 ± 0.003	6.406 ± 0.775
QML	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0262	0.0262	0.997	94.595 ± 0.715	0.110 ± 0.003	5.405 ± 0.715
QML	beta35	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	—	0.0274	0.0274	1.007	96.096 ± 0.613	0.108 ± 0.001	3.904 ± 0.613
UPI	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.318	0.002	0.53	0.0362	0.0362	0.964	93.594 ± 0.775	0.140 ± 0.000	—
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.319	0.003	1.02	0.0363	0.0364	0.973	94.695 ± 0.709	0.141 ± 0.000	—
UPI	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.003	0.003	—	0.0397	0.0398	0.962	93.694 ± 0.769	0.154 ± 0.001	6.306 ± 0.769
UPI	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.001	-0.001	—	0.0265	0.0265	0.971	94.094 ± 0.746	0.108 ± 0.001	5.906 ± 0.746
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0275	0.0275	1.006	95.095 ± 0.683	0.109 ± 0.001	4.905 ± 0.683

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S115

Simulation 1 - Condition 115: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.274	-0.042 ± 0.004	-0.134 ± 0.013	0.014 ± 0.001	0.117 ± 0.004	0.015 ± 0.001	0.124 ± 0.004	0.154 ± 0.010	0.393 ± 0.014	0.274	-0.042	-13.24	0.1215	0.1285	1.048	94.993 ± 0.792	0.480 ± 0.007	—
LMS	beta32	0.316	0.271	-0.045 ± 0.004	-0.143 ± 0.014	0.015 ± 0.001	0.121 ± 0.004	0.017 ± 0.001	0.129 ± 0.004	0.166 ± 0.010	0.408 ± 0.014	0.274	-0.042	-13.24	0.1203	0.1273	1.031	94.071 ± 0.857	0.488 ± 0.008	—
LMS	beta33	0.000	-0.106	-0.106 ± 0.009	—	0.065 ± 0.006	0.255 ± 0.012	0.076 ± 0.008	0.276 ± 0.014	—	—	-0.067	-0.067	—	0.1824	0.1942	0.951	96.311 ± 0.684	0.950 ± 0.044	3.689 ± 0.684
LMS	beta34	0.000	0.108	0.108 ± 0.005	—	0.017 ± 0.002	0.129 ± 0.006	0.028 ± 0.002	0.168 ± 0.007	—	—	0.088	0.088	—	0.1054	0.1372	1.038	95.125 ± 0.782	0.524 ± 0.025	4.875 ± 0.782
LMS	beta35	0.000	0.110	0.110 ± 0.005	—	0.021 ± 0.002	0.144 ± 0.007	0.033 ± 0.003	0.181 ± 0.008	—	—	0.087	0.087	—	0.1138	0.1434	0.953	94.598 ± 0.821	0.538 ± 0.026	5.402 ± 0.821
LSAM	beta31	0.316	0.329	0.013 ± 0.008	0.041 ± 0.027	0.054 ± 0.003	0.233 ± 0.007	0.055 ± 0.003	0.233 ± 0.008	0.546 ± 0.034	0.739 ± 0.027	0.322	0.006	1.87	0.2408	0.2408	1.092	97.892 ± 0.521	0.999 ± 0.020	—
LSAM	beta32	0.316	0.327	0.011 ± 0.009	0.034 ± 0.027	0.057 ± 0.003	0.239 ± 0.007	0.057 ± 0.003	0.239 ± 0.008	0.570 ± 0.033	0.755 ± 0.026	0.314	-0.002	-0.48	0.2366	0.2366	1.084	97.365 ± 0.581	1.014 ± 0.021	—
LSAM	beta33	0.000	0.017	0.017 ± 0.006	—	0.024 ± 0.002	0.156 ± 0.006	0.025 ± 0.002	0.157 ± 0.006	—	—	0.005	0.005	—	0.1450	0.1451	1.102	98.287 ± 0.471	0.676 ± 0.021	1.713 ± 0.471
LSAM	beta34	0.000	-0.006	-0.006 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	—	—	0.000	0.000	—	0.0745	0.0745	1.143	98.287 ± 0.471	0.382 ± 0.011	1.713 ± 0.471
LSAM	beta35	0.000	-0.006	-0.006 ± 0.003	—	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	—	—	0.000	0.000	—	0.0849	0.0849	1.068	98.419 ± 0.453	0.396 ± 0.012	1.581 ± 0.453
QML	beta31	0.316	0.279	-0.037 ± 0.004	-0.116 ± 0.012	0.011 ± 0.001	0.107 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.128 ± 0.007	0.357 ± 0.011	0.277	-0.039	-12.31	0.1163	0.1226	1.034	94.203 ± 0.848	0.433 ± 0.003	—
QML	beta32	0.316	0.276	-0.040 ± 0.004	-0.127 ± 0.013	0.012 ± 0.001	0.111 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	0.140 ± 0.007	0.374 ± 0.012	0.277	-0.039	-12.40	0.1084	0.1152	1.006	93.281 ± 0.909	0.438 ± 0.003	—
QML	beta33	0.000	-0.045	-0.045 ± 0.004	—	0.015 ± 0.001	0.123 ± 0.004	0.017 ± 0.001	0.131 ± 0.004	—	—	-0.050	-0.050	—	0.1300	0.1393	1.079	96.443 ± 0.672	0.521 ± 0.009	3.557 ± 0.672
QML	beta34	0.000	0.071	0.071 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	—	—	0.070	0.070	—	0.0747	0.1021	1.105	92.885 ± 0.933	0.298 ± 0.006	7.115 ± 0.933
QML	beta35	0.000	0.071	0.071 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.011 ± 0.001	0.105 ± 0.003	—	—	0.070	0.070	—	0.0820	0.1080	1.017	90.909 ± 1.043	0.307 ± 0.006	9.091 ± 1.043
UPI	beta31	0.316	0.329	0.013 ± 0.008	0.040 ± 0.027	0.054 ± 0.003	0.232 ± 0.007	0.054 ± 0.003	0.232 ± 0.008	0.541 ± 0.034	0.736 ± 0.027	0.321	0.005	1.71	0.2296	0.2297	1.164	98.024 ± 0.505	1.060 ± 0.034	—
UPI	beta32	0.316	0.326	0.010 ± 0.009	0.032 ± 0.027	0.057 ± 0.004	0.239 ± 0.007	0.057 ± 0.004	0.239 ± 0.009	0.573 ± 0.036	0.757 ± 0.027	0.318	0.002	0.61	0.2283	0.2283	1.167	97.892 ± 0.521	1.094 ± 0.039	—
UPI	beta33	0.000	0.012	0.012 ± 0.006	—	0.028 ± 0.002	0.166 ± 0.006	0.028 ± 0.002	0.167 ± 0.007	—	—	-0.006	-0.006	—	0.1378	0.1379	1.487	99.078 ± 0.347	0.969 ± 0.061	0.922 ± 0.347
UPI	beta34	0.000	-0.005	-0.005 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	—	—	0.000	0.000	—	0.0766	0.0766	1.257	98.551 ± 0.434	0.424 ± 0.019	1.449 ± 0.434
UPI	beta35	0.000	-0.004	-0.004 ± 0.003	—	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	—	—	0.002	0.002	—	0.0850	0.0850	1.158	98.551 ± 0.434	0.431 ± 0.017	1.449 ± 0.434

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S116

Simulation 1 - Condition 116: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.272	-0.044 ± 0.002	-0.141 ± 0.007	0.004 ± 0.000	0.063 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.006	0.269	-0.047	-14.95	0.0666	0.0817	1.173	91.694 ± 0.918	0.292 ± 0.011	—
LMS	beta32	0.316	0.273	-0.043 ± 0.002	-0.135 ± 0.007	0.004 ± 0.000	0.067 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.250 ± 0.007	0.272	-0.044	-13.78	0.0673	0.0802	1.077	90.587 ± 0.972	0.282 ± 0.005	—
LMS	beta33	0.000	-0.039	-0.039 ± 0.003	—	0.009 ± 0.001	0.096 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	—	—	-0.040	-0.040	—	0.0933	0.1013	1.529	97.785 ± 0.490	0.575 ± 0.058	2.215 ± 0.490
LMS	beta34	0.000	0.073	0.073 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	—	—	0.069	0.069	—	0.0515	0.0864	1.507	86.157 ± 1.149	0.314 ± 0.036	13.843 ± 1.149
LMS	beta35	0.000	0.074	0.074 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.009 ± 0.000	0.093 ± 0.002	—	—	0.072	0.072	—	0.0557	0.0913	1.296	85.382 ± 1.176	0.283 ± 0.022	14.618 ± 1.176
LSAM	beta31	0.316	0.318	0.002 ± 0.004	0.005 ± 0.013	0.014 ± 0.001	0.120 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	0.143 ± 0.007	0.379 ± 0.012	0.317	0.001	0.26	0.1202	0.1202	1.073	97.342 ± 0.535	0.503 ± 0.004	—
LSAM	beta32	0.316	0.323	0.007 ± 0.004	0.023 ± 0.013	0.016 ± 0.001	0.126 ± 0.003	0.016 ± 0.001	0.126 ± 0.004	0.158 ± 0.008	0.398 ± 0.012	0.312	-0.004	-1.14	0.1244	0.1244	1.024	96.567 ± 0.606	0.504 ± 0.004	—
LSAM	beta33	0.000	0.013	0.013 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.005	0.005	—	0.0667	0.0669	1.075	96.678 ± 0.596	0.285 ± 0.003	3.322 ± 0.596
LSAM	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	—	0.0376	0.0377	1.087	96.567 ± 0.606	0.168 ± 0.002	3.433 ± 0.606
LSAM	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.002	-0.002	—	0.0394	0.0395	0.995	96.124 ± 0.642	0.170 ± 0.002	3.876 ± 0.642
QML	beta31	0.316	0.274	-0.042 ± 0.002	-0.134 ± 0.007	0.004 ± 0.000	0.063 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.002	0.240 ± 0.006	0.272	-0.044	-13.99	0.0643	0.0780	1.063	90.587 ± 0.972	0.262 ± 0.001	—
QML	beta32	0.316	0.276	-0.040 ± 0.002	-0.128 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.244 ± 0.006	0.275	-0.041	-12.95	0.0648	0.0767	1.013	89.147 ± 1.035	0.261 ± 0.001	—
QML	beta33	0.000	-0.032	-0.032 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	-0.040	-0.040	—	0.0740	0.0839	1.126	96.456 ± 0.615	0.309 ± 0.003	3.544 ± 0.615
QML	beta34	0.000	0.063	0.063 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.006 ± 0.000	0.075 ± 0.001	—	—	0.062	0.062	—	0.0428	0.0751	1.081	75.858 ± 1.424	0.175 ± 0.002	24.142 ± 1.424
QML	beta35	0.000	0.064	0.064 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.006 ± 0.000	0.078 ± 0.002	—	—	0.064	0.064	—	0.0462	0.0790	1.021	75.083 ± 1.439	0.176 ± 0.002	24.917 ± 1.439
UPI	beta31	0.316	0.319	0.003 ± 0.004	0.008 ± 0.013	0.014 ± 0.001	0.120 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	0.144 ± 0.007	0.380 ± 0.012	0.317	0.001	0.17	0.1177	0.1177	1.075	97.231 ± 0.546	0.506 ± 0.004	—
UPI	beta32	0.316	0.323	0.007 ± 0.004	0.023 ± 0.013	0.016 ± 0.001	0.125 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	0.157 ± 0.008	0.396 ± 0.012	0.317	0.001	0.21	0.1260	0.1260	1.033	96.567 ± 0.606	0.506 ± 0.005	—
UPI	beta33	0.000	0.013	0.013 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.074 ± 0.003	—	—	0.002	0.002	—	0.0667	0.0667	1.168	98.228 ± 0.439	0.336 ± 0.009	1.772 ± 0.439
UPI	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0378	0.0378	1.108	97.121 ± 0.556	0.172 ± 0.002	2.879 ± 0.556
UPI	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.004	-0.004	—	0.0409	0.0411	1.008	96.678 ± 0.596	0.173 ± 0.002	3.322 ± 0.596

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S117

Simulation 1 - Condition 117: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.285	-0.031 ± 0.003	-0.098 ± 0.008	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.080 ± 0.004	0.282 ± 0.008	0.284	-0.032	-10.24	0.0805	0.0867	1.061	95.062 ± 0.695	0.348 ± 0.005	—
LMS	beta32	0.316	0.286	-0.030 ± 0.003	-0.094 ± 0.009	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.080 ± 0.004	0.282 ± 0.008	0.290	-0.026	-8.08	0.0787	0.0828	1.067	95.062 ± 0.695	0.352 ± 0.014	—
LMS	beta33	0.000	-0.024	-0.024 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.003	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.018	-0.018	—	0.0842	0.0861	1.038	95.782 ± 0.645	0.366 ± 0.009	4.218 ± 0.645
LMS	beta34	0.000	0.040	0.040 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.032	0.032	—	0.0458	0.0558	0.989	93.210 ± 0.807	0.199 ± 0.006	6.790 ± 0.807
LMS	beta35	0.000	0.038	0.038 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.032	0.032	—	0.0423	0.0530	1.029	92.078 ± 0.866	0.194 ± 0.006	7.922 ± 0.866
LSAM	beta31	0.316	0.320	0.004 ± 0.004	0.014 ± 0.014	0.018 ± 0.001	0.136 ± 0.004	0.018 ± 0.001	0.136 ± 0.004	0.184 ± 0.010	0.429 ± 0.013	0.314	-0.002	-0.64	0.1258	0.1259	1.014	95.885 ± 0.637	0.539 ± 0.004	—
LSAM	beta32	0.316	0.322	0.006 ± 0.004	0.018 ± 0.014	0.018 ± 0.001	0.135 ± 0.003	0.018 ± 0.001	0.135 ± 0.004	0.183 ± 0.009	0.427 ± 0.012	0.324	0.008	2.58	0.1289	0.1291	1.018	95.576 ± 0.660	0.539 ± 0.004	—
LSAM	beta33	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.002	0.002	—	0.0760	0.0760	0.974	94.547 ± 0.728	0.319 ± 0.004	5.453 ± 0.728
LSAM	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	—	0.0467	0.0467	0.949	94.650 ± 0.722	0.195 ± 0.003	5.350 ± 0.722
LSAM	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.002	-0.002	—	0.0431	0.0432	0.966	95.165 ± 0.688	0.194 ± 0.003	4.835 ± 0.688
QML	beta31	0.316	0.287	-0.029 ± 0.003	-0.092 ± 0.009	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.079 ± 0.004	0.281 ± 0.008	0.285	-0.031	-9.85	0.0801	0.0859	1.027	94.547 ± 0.728	0.338 ± 0.002	—
QML	beta32	0.316	0.288	-0.028 ± 0.003	-0.088 ± 0.009	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.079 ± 0.004	0.282 ± 0.008	0.291	-0.025	-7.79	0.0793	0.0831	1.015	94.650 ± 0.722	0.336 ± 0.002	—
QML	beta33	0.000	-0.023	-0.023 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	—	—	-0.020	-0.020	—	0.0780	0.0805	1.000	94.959 ± 0.702	0.322 ± 0.004	5.041 ± 0.702
QML	beta34	0.000	0.037	0.037 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.031	0.031	—	0.0459	0.0552	0.940	91.358 ± 0.901	0.177 ± 0.003	8.642 ± 0.901
QML	beta35	0.000	0.035	0.035 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.031	0.031	—	0.0420	0.0520	0.977	91.461 ± 0.896	0.175 ± 0.003	8.539 ± 0.896
UPI	beta31	0.316	0.321	0.005 ± 0.004	0.015 ± 0.014	0.018 ± 0.001	0.135 ± 0.004	0.018 ± 0.001	0.135 ± 0.004	0.184 ± 0.010	0.429 ± 0.013	0.315	-0.001	-0.43	0.1263	0.1264	1.011	95.473 ± 0.667	0.537 ± 0.004	—
UPI	beta32	0.316	0.323	0.007 ± 0.004	0.021 ± 0.014	0.018 ± 0.001	0.134 ± 0.003	0.018 ± 0.001	0.134 ± 0.004	0.181 ± 0.009	0.425 ± 0.012	0.325	0.009	2.71	0.1268	0.1271	1.018	95.782 ± 0.645	0.535 ± 0.004	—
UPI	beta33	0.000	0.002	0.002 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	0.000	0.000	—	0.0725	0.0725	1.061	95.679 ± 0.652	0.354 ± 0.006	4.321 ± 0.652
UPI	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0458	0.0458	0.957	95.679 ± 0.652	0.197 ± 0.003	4.321 ± 0.652
UPI	beta35	0.000	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.003	-0.003	—	0.0437	0.0438	0.957	95.473 ± 0.667	0.195 ± 0.003	4.527 ± 0.667

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S118

Simulation 1 - Condition 118: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.284	-0.032 ± 0.002	-0.103 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.284	-0.032	-10.18	0.0522	0.0613	1.021	91.405 ± 0.891	0.205 ± 0.002	—
LMS	beta32	0.316	0.288	-0.028 ± 0.002	-0.090 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.289	-0.027	-8.56	0.0488	0.0558	1.034	92.012 ± 0.862	0.203 ± 0.001	—
LMS	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.011	-0.011	—	0.0541	0.0551	0.956	95.956 ± 0.626	0.205 ± 0.003	4.044 ± 0.626
LMS	beta34	0.000	0.033	0.033 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.029	0.029	—	0.0265	0.0393	0.961	82.912 ± 1.197	0.108 ± 0.002	17.088 ± 1.197
LMS	beta35	0.000	0.030	0.030 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.029	0.029	—	0.0258	0.0388	1.004	85.238 ± 1.128	0.106 ± 0.001	14.762 ± 1.128
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.248 ± 0.007	0.315	-0.001	-0.27	0.0772	0.0772	1.020	96.158 ± 0.611	0.313 ± 0.001	—
LSAM	beta32	0.316	0.322	0.006 ± 0.002	0.020 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.323	0.007	2.23	0.0771	0.0774	1.028	96.057 ± 0.619	0.313 ± 0.001	—
LSAM	beta33	0.000	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0465	0.0466	0.943	93.529 ± 0.782	0.175 ± 0.001	6.471 ± 0.782
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.001	-0.001	—	0.0241	0.0241	0.983	95.652 ± 0.648	0.103 ± 0.001	4.348 ± 0.648
LSAM	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.002	-0.002	—	0.0226	0.0226	0.995	95.450 ± 0.663	0.102 ± 0.001	4.550 ± 0.663
QML	beta31	0.316	0.285	-0.031 ± 0.002	-0.099 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.285	-0.031	-9.86	0.0533	0.0618	1.009	91.001 ± 0.910	0.206 ± 0.001	—
QML	beta32	0.316	0.289	-0.027 ± 0.002	-0.086 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.183 ± 0.005	0.290	-0.026	-8.32	0.0503	0.0567	1.027	92.315 ± 0.847	0.205 ± 0.001	—
QML	beta33	0.000	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.014	-0.014	—	0.0525	0.0544	0.948	94.843 ± 0.703	0.193 ± 0.002	5.157 ± 0.703
QML	beta34	0.000	0.032	0.032 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.028	0.028	—	0.0259	0.0384	0.955	82.103 ± 1.219	0.104 ± 0.001	17.897 ± 1.219
QML	beta35	0.000	0.029	0.029 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.028	0.028	—	0.0254	0.0380	0.995	84.833 ± 1.141	0.102 ± 0.001	15.167 ± 1.141
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.314	-0.002	-0.52	0.0768	0.0768	1.020	96.158 ± 0.611	0.312 ± 0.001	—
UPI	beta32	0.316	0.323	0.007 ± 0.002	0.021 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.324	0.008	2.69	0.0770	0.0775	1.032	95.854 ± 0.634	0.313 ± 0.001	—
UPI	beta33	0.000	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	—	0.0486	0.0486	0.950	95.248 ± 0.677	0.183 ± 0.002	4.752 ± 0.677
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.001	-0.001	—	0.0243	0.0243	0.983	95.956 ± 0.626	0.103 ± 0.001	4.044 ± 0.626
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.001	-0.001	—	0.0232	0.0232	1.000	94.742 ± 0.710	0.103 ± 0.001	5.258 ± 0.710

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S119

Simulation 1 - Condition 119: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.299	-0.017 ± 0.002	-0.053 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.299	-0.017	-5.37	0.0789	0.0807	0.975	94.854 ± 0.702	0.294 ± 0.001	—
LMS	beta32	0.316	0.301	-0.015 ± 0.003	-0.048 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.067 ± 0.003	0.260 ± 0.007	0.299	-0.017	-5.36	0.0798	0.0815	0.931	92.735 ± 0.825	0.295 ± 0.001	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.002	-0.002	—	0.0564	0.0564	0.906	93.138 ± 0.803	0.225 ± 0.002	6.862 ± 0.803
LMS	beta34	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.009	0.009	—	0.0315	0.0327	0.894	92.735 ± 0.825	0.122 ± 0.002	7.265 ± 0.825
LMS	beta35	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.010	0.010	—	0.0310	0.0325	0.859	91.120 ± 0.904	0.122 ± 0.002	8.880 ± 0.904
LSAM	beta31	0.316	0.316	0.000 ± 0.003	0.001 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.004	0.314 ± 0.009	0.316	0.000	-0.02	0.0991	0.0991	0.964	94.349 ± 0.733	0.375 ± 0.002	—
LSAM	beta32	0.316	0.317	0.001 ± 0.003	0.004 ± 0.010	0.011 ± 0.000	0.103 ± 0.002	0.011 ± 0.000	0.103 ± 0.003	0.107 ± 0.005	0.327 ± 0.009	0.316	0.000	-0.12	0.1019	0.1019	0.922	93.239 ± 0.798	0.374 ± 0.002	—
LSAM	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.004	0.004	—	0.0579	0.0581	0.911	92.634 ± 0.830	0.219 ± 0.002	7.366 ± 0.830
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0342	0.0344	0.907	93.643 ± 0.775	0.132 ± 0.002	6.357 ± 0.775
LSAM	beta35	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0330	0.0331	0.863	91.423 ± 0.890	0.130 ± 0.002	8.577 ± 0.890
QML	beta31	0.316	0.300	-0.016 ± 0.002	-0.052 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.063 ± 0.003	0.252 ± 0.007	0.299	-0.017	-5.25	0.0804	0.0821	0.970	94.753 ± 0.708	0.296 ± 0.001	—
QML	beta32	0.316	0.301	-0.015 ± 0.003	-0.047 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.069 ± 0.003	0.263 ± 0.007	0.300	-0.016	-5.19	0.0813	0.0830	0.925	92.634 ± 0.830	0.296 ± 0.001	—
QML	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.002	-0.002	—	0.0565	0.0566	0.905	93.037 ± 0.808	0.223 ± 0.002	6.963 ± 0.808
QML	beta34	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.009	0.009	—	0.0320	0.0331	0.893	92.533 ± 0.835	0.122 ± 0.001	7.467 ± 0.835
QML	beta35	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.009	0.009	—	0.0311	0.0325	0.857	90.817 ± 0.917	0.122 ± 0.002	9.183 ± 0.917
UPI	beta31	0.316	0.316	0.000 ± 0.003	0.002 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.004	0.313 ± 0.009	0.314	-0.002	-0.54	0.0972	0.0972	0.964	94.652 ± 0.715	0.373 ± 0.002	—
UPI	beta32	0.316	0.318	0.002 ± 0.003	0.005 ± 0.010	0.011 ± 0.000	0.103 ± 0.002	0.011 ± 0.000	0.103 ± 0.003	0.106 ± 0.005	0.325 ± 0.009	0.316	0.000	0.01	0.1010	0.1010	0.922	93.340 ± 0.792	0.372 ± 0.002	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.002	0.002	—	0.0580	0.0581	0.937	93.138 ± 0.803	0.227 ± 0.003	6.862 ± 0.803
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0330	0.0332	0.910	93.845 ± 0.763	0.132 ± 0.002	6.155 ± 0.763
UPI	beta35	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0334	0.0335	0.861	91.726 ± 0.875	0.131 ± 0.002	8.274 ± 0.875

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S120

Simulation 1 - Condition 120: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.299	-0.017 ± 0.001	-0.052 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.301	-0.015	-4.62	0.0457	0.0479	1.028	94.300 ± 0.733	0.184 ± 0.001	—
LMS	beta32	0.316	0.299	-0.017 ± 0.001	-0.055 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.298	-0.018	-5.70	0.0454	0.0489	1.018	93.300 ± 0.791	0.183 ± 0.001	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0384	0.0385	0.956	95.000 ± 0.689	0.141 ± 0.001	5.000 ± 0.689
LMS	beta34	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.010	0.010	—	0.0183	0.0208	0.928	92.200 ± 0.848	0.074 ± 0.001	7.800 ± 0.848
LMS	beta35	0.000	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.012	0.012	—	0.0189	0.0222	0.946	92.500 ± 0.833	0.073 ± 0.001	7.500 ± 0.833
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.317	0.001	0.38	0.0570	0.0570	1.017	95.400 ± 0.662	0.228 ± 0.001	—
LSAM	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.313	-0.003	-0.91	0.0552	0.0553	1.014	94.600 ± 0.715	0.228 ± 0.001	—
LSAM	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.000	0.000	—	0.0356	0.0356	0.961	94.400 ± 0.727	0.133 ± 0.001	5.600 ± 0.727
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0180	0.0180	0.942	94.400 ± 0.727	0.075 ± 0.001	5.600 ± 0.727
LSAM	beta35	0.000	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0181	0.0181	0.957	93.400 ± 0.785	0.075 ± 0.001	6.600 ± 0.785
QML	beta31	0.316	0.300	-0.016 ± 0.001	-0.051 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.302	-0.014	-4.51	0.0460	0.0482	1.023	94.300 ± 0.733	0.186 ± 0.001	—
QML	beta32	0.316	0.299	-0.017 ± 0.001	-0.054 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.298	-0.018	-5.66	0.0459	0.0493	1.012	93.200 ± 0.796	0.185 ± 0.001	—
QML	beta33	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0383	0.0385	0.961	95.100 ± 0.683	0.141 ± 0.001	4.900 ± 0.683
QML	beta34	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.010	0.010	—	0.0183	0.0208	0.932	92.000 ± 0.858	0.075 ± 0.001	8.000 ± 0.858
QML	beta35	0.000	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.012	0.012	—	0.0191	0.0223	0.945	92.100 ± 0.853	0.073 ± 0.001	7.900 ± 0.853
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.317	0.001	0.45	0.0564	0.0564	1.014	95.500 ± 0.656	0.228 ± 0.001	—
UPI	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.181 ± 0.005	0.313	-0.003	-1.00	0.0545	0.0546	1.012	95.000 ± 0.689	0.227 ± 0.001	—
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0365	0.0366	0.962	94.400 ± 0.727	0.135 ± 0.001	5.600 ± 0.727
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0181	0.0181	0.938	94.400 ± 0.727	0.075 ± 0.001	5.600 ± 0.727
UPI	beta35	0.000	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0178	0.0178	0.954	93.900 ± 0.757	0.075 ± 0.001	6.100 ± 0.757

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S121

Simulation 1 - Condition 121: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.321	0.005 ± 0.005	0.017 ± 0.015	0.011 ± 0.001	0.104 ± 0.004	0.011 ± 0.001	0.104 ± 0.004	0.108 ± 0.008	0.329 ± 0.014	0.325	0.009	2.99	0.1044	0.1049	1.042	94.737 ± 0.986	0.425 ± 0.008	—
LMS	beta32	0.316	0.318	0.002 ± 0.005	0.006 ± 0.015	0.012 ± 0.001	0.108 ± 0.004	0.012 ± 0.001	0.108 ± 0.004	0.116 ± 0.008	0.341 ± 0.014	0.317	0.001	0.37	0.1092	0.1092	0.986	94.152 ± 1.036	0.417 ± 0.008	—
LMS	beta33	0.000	-0.068	-0.068 ± 0.010	—	0.047 ± 0.005	0.217 ± 0.012	0.051 ± 0.006	0.227 ± 0.015	—	—	-0.038	-0.038	—	0.1670	0.1712	1.010	95.517 ± 0.914	0.858 ± 0.077	4.483 ± 0.914
LMS	beta34	0.000	0.052	0.052 ± 0.007	—	0.026 ± 0.003	0.161 ± 0.008	0.028 ± 0.003	0.169 ± 0.010	—	—	0.026	0.026	—	0.1279	0.1305	0.967	94.737 ± 0.986	0.609 ± 0.049	5.263 ± 0.986
LMS	beta35	0.000	0.035	0.035 ± 0.006	—	0.017 ± 0.002	0.132 ± 0.007	0.019 ± 0.002	0.137 ± 0.009	—	—	0.018	0.018	—	0.1172	0.1185	1.082	95.906 ± 0.875	0.561 ± 0.038	4.094 ± 0.875
LSAM	beta31	0.316	0.321	0.005 ± 0.005	0.016 ± 0.016	0.013 ± 0.001	0.114 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	0.130 ± 0.008	0.361 ± 0.014	0.326	0.010	3.12	0.1268	0.1272	1.222	97.271 ± 0.719	0.546 ± 0.017	—
LSAM	beta32	0.316	0.327	0.011 ± 0.005	0.036 ± 0.017	0.014 ± 0.001	0.118 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	0.141 ± 0.010	0.375 ± 0.016	0.324	0.008	2.42	0.1232	0.1234	1.176	96.686 ± 0.790	0.545 ± 0.016	—
LSAM	beta33	0.000	0.017	0.017 ± 0.007	—	0.027 ± 0.002	0.165 ± 0.007	0.028 ± 0.002	0.166 ± 0.008	—	—	0.003	0.003	—	0.1664	0.1665	1.327	99.220 ± 0.388	0.860 ± 0.038	0.780 ± 0.388
LSAM	beta34	0.000	-0.001	-0.001 ± 0.009	—	0.046 ± 0.004	0.215 ± 0.010	0.046 ± 0.004	0.215 ± 0.011	—	—	-0.002	-0.002	—	0.2131	0.2131	1.254	99.220 ± 0.388	1.057 ± 0.055	0.780 ± 0.388
LSAM	beta35	0.000	-0.022	-0.022 ± 0.009	—	0.041 ± 0.003	0.202 ± 0.008	0.041 ± 0.003	0.203 ± 0.010	—	—	-0.004	-0.004	—	0.1934	0.1935	1.449	99.415 ± 0.337	1.147 ± 0.067	0.585 ± 0.337
QML	beta31	0.316	0.316	0.000 ± 0.004	0.001 ± 0.013	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.092 ± 0.006	0.304 ± 0.012	0.325	0.009	2.90	0.0996	0.1000	1.019	94.932 ± 0.968	0.384 ± 0.003	—
QML	beta32	0.316	0.319	0.003 ± 0.005	0.010 ± 0.014	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	0.105 ± 0.007	0.325 ± 0.013	0.316	0.000	0.12	0.1047	0.1047	0.952	94.542 ± 1.003	0.383 ± 0.003	—
QML	beta33	0.000	-0.020	-0.020 ± 0.006	—	0.017 ± 0.001	0.131 ± 0.005	0.017 ± 0.001	0.132 ± 0.006	—	—	-0.026	-0.026	—	0.1371	0.1395	1.051	95.517 ± 0.914	0.539 ± 0.011	4.483 ± 0.914
QML	beta34	0.000	0.007	0.007 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.004	0.010 ± 0.001	0.101 ± 0.005	—	—	0.007	0.007	—	0.0985	0.0987	1.006	95.906 ± 0.875	0.398 ± 0.008	4.094 ± 0.875
QML	beta35	0.000	-0.001	-0.001 ± 0.004	—	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.088 ± 0.004	—	—	0.001	0.001	—	0.0978	0.0978	1.130	96.491 ± 0.812	0.389 ± 0.008	3.509 ± 0.812
UPI	beta31	0.316	0.315	-0.001 ± 0.005	-0.002 ± 0.016	0.013 ± 0.001	0.116 ± 0.004	0.013 ± 0.001	0.116 ± 0.005	0.134 ± 0.009	0.366 ± 0.015	0.317	0.001	0.16	0.1260	0.1260	1.550	98.246 ± 0.580	0.704 ± 0.036	—
UPI	beta32	0.316	0.333	0.017 ± 0.006	0.052 ± 0.018	0.016 ± 0.001	0.128 ± 0.005	0.017 ± 0.001	0.129 ± 0.006	0.167 ± 0.014	0.409 ± 0.019	0.326	0.010	3.14	0.1239	0.1243	1.495	97.661 ± 0.667	0.751 ± 0.043	—
UPI	beta33	0.000	0.021	0.021 ± 0.008	—	0.029 ± 0.003	0.170 ± 0.008	0.029 ± 0.003	0.171 ± 0.009	—	—	0.007	0.007	—	0.1597	0.1599	1.923	99.610 ± 0.275	1.283 ± 0.091	0.390 ± 0.275
UPI	beta34	0.000	0.014	0.014 ± 0.011	—	0.066 ± 0.006	0.258 ± 0.012	0.066 ± 0.006	0.258 ± 0.013	—	—	0.009	0.009	—	0.2107	0.2109	2.006	100.000 ± 0.000	2.026 ± 0.140	0.000 ± 0.000
UPI	beta35	0.000	-0.025	-0.025 ± 0.011	—	0.058 ± 0.005	0.241 ± 0.011	0.059 ± 0.005	0.242 ± 0.012	—	—	-0.001	-0.001	—	0.2202	0.2202	2.106	99.805 ± 0.195	1.991 ± 0.151	0.195 ± 0.195

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S122

Simulation 1 - Condition 122: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.203 ± 0.007	0.318	0.002	0.73	0.0643	0.0643	1.001	95.136 ± 0.741	0.252 ± 0.003	—
LMS	beta32	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.007	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.039 ± 0.002	0.198 ± 0.006	0.313	-0.003	-0.84	0.0645	0.0646	1.018	95.848 ± 0.687	0.250 ± 0.002	—
LMS	beta33	0.000	-0.022	-0.022 ± 0.003	—	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	—	—	-0.025	-0.025	—	0.0888	0.0923	1.288	97.272 ± 0.561	0.446 ± 0.020	2.728 ± 0.561
LMS	beta34	0.000	0.015	0.015 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	—	—	0.012	0.012	—	0.0647	0.0657	1.231	97.272 ± 0.561	0.321 ± 0.015	2.728 ± 0.561
LMS	beta35	0.000	0.012	0.012 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	—	—	0.012	0.012	—	0.0662	0.0674	1.189	95.848 ± 0.687	0.303 ± 0.009	4.152 ± 0.687
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.047 ± 0.002	0.216 ± 0.007	0.318	0.002	0.76	0.0664	0.0664	1.005	94.899 ± 0.758	0.269 ± 0.002	—
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.044 ± 0.002	0.211 ± 0.006	0.315	-0.001	-0.33	0.0664	0.0664	1.033	96.085 ± 0.668	0.270 ± 0.002	—
LSAM	beta33	0.000	0.009	0.009 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	-0.002	-0.002	—	0.0864	0.0865	1.107	97.865 ± 0.498	0.370 ± 0.005	2.135 ± 0.498
LSAM	beta34	0.000	-0.012	-0.012 ± 0.004	—	0.013 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.116 ± 0.004	—	—	-0.002	-0.002	—	0.1105	0.1105	1.093	98.814 ± 0.373	0.496 ± 0.008	1.186 ± 0.373
LSAM	beta35	0.000	-0.010	-0.010 ± 0.004	—	0.015 ± 0.001	0.123 ± 0.004	0.015 ± 0.001	0.123 ± 0.004	—	—	-0.001	-0.001	—	0.1175	0.1175	1.050	98.458 ± 0.424	0.505 ± 0.009	1.542 ± 0.424
QML	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.195 ± 0.006	0.316	0.000	0.07	0.0610	0.0610	0.969	94.306 ± 0.798	0.234 ± 0.001	—
QML	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.007	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.006	0.313	-0.003	-0.85	0.0615	0.0615	0.989	95.611 ± 0.706	0.234 ± 0.001	—
QML	beta33	0.000	-0.015	-0.015 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.022	-0.022	—	0.0774	0.0804	1.104	96.323 ± 0.648	0.318 ± 0.003	3.677 ± 0.648
QML	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0581	0.0581	1.085	96.323 ± 0.648	0.232 ± 0.002	3.677 ± 0.648
QML	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.001	0.001	—	0.0565	0.0565	1.083	95.018 ± 0.749	0.232 ± 0.002	4.982 ± 0.749
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.003	0.221 ± 0.007	0.316	0.000	0.02	0.0689	0.0689	1.086	95.611 ± 0.706	0.297 ± 0.004	—
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.215 ± 0.007	0.315	-0.001	-0.26	0.0692	0.0692	1.124	96.797 ± 0.606	0.299 ± 0.004	—
UPI	beta33	0.000	0.009	0.009 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	—	—	0.001	0.001	—	0.0873	0.0873	1.262	98.458 ± 0.424	0.438 ± 0.008	1.542 ± 0.424
UPI	beta34	0.000	-0.011	-0.011 ± 0.005	—	0.021 ± 0.001	0.147 ± 0.005	0.022 ± 0.001	0.147 ± 0.006	—	—	0.000	0.000	—	0.1261	0.1261	1.362	99.526 ± 0.237	0.783 ± 0.035	0.474 ± 0.237
UPI	beta35	0.000	-0.007	-0.007 ± 0.005	—	0.022 ± 0.001	0.148 ± 0.005	0.022 ± 0.001	0.148 ± 0.005	—	—	0.002	0.002	—	0.1296	0.1296	1.334	99.644 ± 0.205	0.776 ± 0.026	0.356 ± 0.205

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S123

Simulation 1 - Condition 123: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.317	0.001	0.17	0.0667	0.0667	1.008	95.697 ± 0.650	0.280 ± 0.002	—
LMS	beta32	0.316	0.320	0.004 ± 0.002	0.011 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.228 ± 0.007	0.323	0.007	2.08	0.0715	0.0718	1.000	94.570 ± 0.725	0.282 ± 0.003	—
LMS	beta33	0.000	-0.016	-0.016 ± 0.003	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	-0.006	-0.006	—	0.0919	0.0921	1.070	95.492 ± 0.664	0.402 ± 0.014	4.508 ± 0.664
LMS	beta34	0.000	0.017	0.017 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	0.008	0.008	—	0.0693	0.0698	1.111	95.594 ± 0.657	0.327 ± 0.012	4.406 ± 0.657
LMS	beta35	0.000	0.014	0.014 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.005	0.005	—	0.0735	0.0737	1.059	95.492 ± 0.664	0.326 ± 0.007	4.508 ± 0.664
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.316	0.000	0.00	0.0688	0.0688	0.996	94.467 ± 0.732	0.287 ± 0.001	—
LSAM	beta32	0.316	0.321	0.005 ± 0.002	0.014 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.057 ± 0.003	0.240 ± 0.007	0.323	0.007	2.34	0.0759	0.0762	0.973	94.365 ± 0.738	0.289 ± 0.001	—
LSAM	beta33	0.000	0.006	0.006 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	—	—	0.009	0.009	—	0.0893	0.0898	0.976	94.980 ± 0.699	0.346 ± 0.003	5.020 ± 0.699
LSAM	beta34	0.000	0.003	0.003 ± 0.003	—	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	—	—	0.003	0.003	—	0.0999	0.0999	1.033	96.516 ± 0.587	0.421 ± 0.004	3.484 ± 0.587
LSAM	beta35	0.000	-0.005	-0.005 ± 0.004	—	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	—	—	-0.006	-0.006	—	0.1091	0.1093	0.995	96.721 ± 0.570	0.430 ± 0.004	3.279 ± 0.570
QML	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.048 ± 0.002	0.220 ± 0.006	0.316	0.000	-0.10	0.0652	0.0652	0.994	95.184 ± 0.685	0.271 ± 0.001	—
QML	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.225 ± 0.007	0.322	0.006	2.04	0.0721	0.0724	0.976	94.262 ± 0.744	0.272 ± 0.001	—
QML	beta33	0.000	-0.011	-0.011 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.006	-0.006	—	0.0896	0.0897	0.977	95.492 ± 0.664	0.337 ± 0.003	4.508 ± 0.664
QML	beta34	0.000	0.008	0.008 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.003	0.003	—	0.0661	0.0662	1.040	95.082 ± 0.692	0.276 ± 0.002	4.918 ± 0.692
QML	beta35	0.000	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.000	0.000	—	0.0700	0.0700	1.007	94.672 ± 0.719	0.279 ± 0.003	5.328 ± 0.719
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.319	0.003	0.85	0.0725	0.0725	1.002	94.775 ± 0.712	0.293 ± 0.002	—
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.014 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.240 ± 0.007	0.323	0.007	2.34	0.0753	0.0756	0.986	94.980 ± 0.699	0.293 ± 0.001	—
UPI	beta33	0.000	0.005	0.005 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	0.008	0.008	—	0.0899	0.0903	0.998	95.799 ± 0.642	0.368 ± 0.004	4.201 ± 0.642
UPI	beta34	0.000	0.003	0.003 ± 0.004	—	0.015 ± 0.001	0.121 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	—	—	0.005	0.005	—	0.1074	0.1076	1.094	97.951 ± 0.453	0.520 ± 0.008	2.049 ± 0.453
UPI	beta35	0.000	-0.003	-0.003 ± 0.004	—	0.015 ± 0.001	0.122 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	—	—	-0.004	-0.004	—	0.1133	0.1134	1.100	97.643 ± 0.486	0.528 ± 0.008	2.357 ± 0.486

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S124

Simulation 1 - Condition 124: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.001	0.007 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.317	0.001	0.25	0.0441	0.0441	1.001	95.678 ± 0.645	0.172 ± 0.001	—
LMS	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.131 ± 0.004	0.315	-0.001	-0.16	0.0405	0.0405	1.057	96.583 ± 0.576	0.172 ± 0.001	—
LMS	beta33	0.000	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.009	-0.009	—	0.0529	0.0536	0.993	96.382 ± 0.592	0.226 ± 0.003	3.618 ± 0.592
LMS	beta34	0.000	0.006	0.006 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.003	0.003	—	0.0444	0.0445	1.013	95.678 ± 0.645	0.184 ± 0.002	4.322 ± 0.645
LMS	beta35	0.000	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.004	0.004	—	0.0435	0.0437	1.016	95.779 ± 0.637	0.185 ± 0.003	4.221 ± 0.637
LSAM	beta31	0.316	0.319	0.003 ± 0.001	0.009 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.318	0.002	0.77	0.0438	0.0439	1.001	94.975 ± 0.693	0.177 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.315	-0.001	-0.16	0.0435	0.0435	1.045	95.779 ± 0.637	0.177 ± 0.000	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0517	0.0517	0.961	94.070 ± 0.749	0.208 ± 0.001	5.930 ± 0.749
LSAM	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.002	-0.002	—	0.0670	0.0670	0.985	94.673 ± 0.712	0.258 ± 0.001	5.327 ± 0.712
LSAM	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0653	0.0653	0.993	95.075 ± 0.686	0.258 ± 0.001	4.925 ± 0.686
QML	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.317	0.001	0.47	0.0433	0.0433	0.997	95.176 ± 0.679	0.170 ± 0.000	—
QML	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.130 ± 0.004	0.315	-0.001	-0.39	0.0406	0.0406	1.053	96.583 ± 0.576	0.170 ± 0.000	—
QML	beta33	0.000	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.010	-0.010	—	0.0518	0.0527	0.974	94.372 ± 0.731	0.209 ± 0.001	5.628 ± 0.731
QML	beta34	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.001	-0.001	—	0.0432	0.0432	1.000	94.874 ± 0.699	0.172 ± 0.001	5.126 ± 0.699
QML	beta35	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	—	0.0421	0.0421	1.009	94.975 ± 0.693	0.173 ± 0.001	5.025 ± 0.693
UPI	beta31	0.316	0.319	0.003 ± 0.001	0.009 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.319	0.003	0.92	0.0442	0.0443	1.001	94.171 ± 0.743	0.177 ± 0.000	—
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.316	0.000	-0.12	0.0445	0.0445	1.044	95.779 ± 0.637	0.177 ± 0.000	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0527	0.0527	0.964	94.070 ± 0.749	0.211 ± 0.001	5.930 ± 0.749
UPI	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.001	-0.001	—	0.0647	0.0647	1.016	95.578 ± 0.652	0.277 ± 0.002	4.422 ± 0.652
UPI	beta35	0.000	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.000	0.000	—	0.0677	0.0677	1.019	96.080 ± 0.615	0.277 ± 0.002	3.920 ± 0.615

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S125

Simulation 1 - Condition 125: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.319	0.003 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.031 ± 0.001	0.176 ± 0.005	0.318	0.002	0.75	0.0558	0.0559	0.996	95.386 ± 0.664	0.217 ± 0.001	—
LMS	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.313	-0.003	-0.94	0.0549	0.0550	1.021	95.286 ± 0.671	0.217 ± 0.001	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.000	0.000	—	0.0583	0.0583	1.006	93.781 ± 0.765	0.251 ± 0.003	6.219 ± 0.765
LMS	beta34	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.001	-0.001	—	0.0621	0.0621	0.963	93.581 ± 0.776	0.234 ± 0.002	6.419 ± 0.776
LMS	beta35	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.000	0.000	—	0.0632	0.0632	0.976	95.486 ± 0.657	0.235 ± 0.002	4.514 ± 0.657
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.318	0.002	0.71	0.0579	0.0580	0.996	94.985 ± 0.691	0.221 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.030 ± 0.001	0.175 ± 0.005	0.314	-0.002	-0.76	0.0562	0.0562	1.023	95.386 ± 0.664	0.221 ± 0.001	—
LSAM	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.003	0.003	—	0.0605	0.0606	0.987	93.180 ± 0.798	0.242 ± 0.001	6.820 ± 0.798
LSAM	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.004	-0.004	—	0.0751	0.0752	0.943	93.180 ± 0.798	0.279 ± 0.001	6.820 ± 0.798
LSAM	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	-0.003	-0.003	—	0.0765	0.0766	0.962	95.085 ± 0.685	0.281 ± 0.001	4.915 ± 0.685
QML	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.031 ± 0.001	0.175 ± 0.005	0.318	0.002	0.78	0.0558	0.0558	0.996	95.186 ± 0.678	0.216 ± 0.001	—
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.313	-0.003	-1.01	0.0547	0.0548	1.021	95.186 ± 0.678	0.217 ± 0.001	—
QML	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0584	0.0584	0.997	93.681 ± 0.771	0.246 ± 0.002	6.319 ± 0.771
QML	beta34	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.002	-0.002	—	0.0608	0.0608	0.958	93.180 ± 0.798	0.229 ± 0.002	6.820 ± 0.798
QML	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	—	0.0630	0.0630	0.974	95.386 ± 0.664	0.231 ± 0.002	4.614 ± 0.664
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.319	0.003	0.80	0.0576	0.0576	0.989	94.885 ± 0.698	0.220 ± 0.001	—
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.031 ± 0.001	0.175 ± 0.005	0.312	-0.004	-1.16	0.0562	0.0564	1.016	95.085 ± 0.685	0.220 ± 0.001	—
UPI	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.002	0.002	—	0.0606	0.0606	0.989	93.280 ± 0.793	0.244 ± 0.001	6.720 ± 0.793
UPI	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	-0.004	-0.004	—	0.0747	0.0748	0.958	93.882 ± 0.759	0.288 ± 0.002	6.118 ± 0.759
UPI	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.002	-0.002	—	0.0758	0.0758	0.979	95.888 ± 0.629	0.290 ± 0.002	4.112 ± 0.629

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S126

Simulation 1 - Condition 126: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.316	0.000	-0.04	0.0349	0.0349	0.998	94.900 ± 0.696	0.136 ± 0.000	—
LMS	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.001	0.108 ± 0.003	0.316	0.000	-0.13	0.0326	0.0326	1.019	95.800 ± 0.634	0.137 ± 0.000	—
LMS	beta33	0.000	-0.007	-0.007 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.005	-0.005	—	0.0389	0.0393	1.003	95.000 ± 0.689	0.158 ± 0.001	5.000 ± 0.689
LMS	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0373	0.0373	0.987	94.800 ± 0.702	0.145 ± 0.001	5.200 ± 0.702
LMS	beta35	0.000	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0344	0.0345	1.048	96.300 ± 0.597	0.146 ± 0.001	3.700 ± 0.597
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.317	0.001	0.19	0.0354	0.0354	0.998	94.900 ± 0.696	0.139 ± 0.000	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.317	0.001	0.27	0.0335	0.0335	1.014	95.300 ± 0.669	0.139 ± 0.000	—
LSAM	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0383	0.0383	0.998	94.600 ± 0.715	0.154 ± 0.001	5.400 ± 0.715
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0454	0.0454	0.986	94.500 ± 0.721	0.175 ± 0.001	5.500 ± 0.721
LSAM	beta35	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.002	0.002	—	0.0438	0.0439	1.035	95.600 ± 0.649	0.176 ± 0.001	4.400 ± 0.649
QML	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.316	0.000	-0.12	0.0349	0.0349	1.000	94.800 ± 0.702	0.137 ± 0.000	—
QML	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.001	0.108 ± 0.003	0.316	0.000	-0.15	0.0325	0.0325	1.019	95.700 ± 0.641	0.137 ± 0.000	—
QML	beta33	0.000	-0.007	-0.007 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.006	-0.006	—	0.0389	0.0393	1.009	95.100 ± 0.683	0.158 ± 0.001	4.900 ± 0.683
QML	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0371	0.0371	0.988	94.500 ± 0.721	0.144 ± 0.001	5.500 ± 0.721
QML	beta35	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	—	0.0342	0.0343	1.070	96.100 ± 0.612	0.148 ± 0.003	3.900 ± 0.612
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.317	0.001	0.21	0.0355	0.0355	0.995	94.700 ± 0.708	0.139 ± 0.000	—
UPI	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.317	0.001	0.24	0.0333	0.0333	1.012	95.400 ± 0.662	0.139 ± 0.000	—
UPI	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	—	0.0380	0.0380	0.998	94.800 ± 0.702	0.155 ± 0.001	5.200 ± 0.702
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0445	0.0445	0.987	94.000 ± 0.751	0.177 ± 0.001	6.000 ± 0.751
UPI	beta35	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.002	0.002	—	0.0433	0.0433	1.037	96.100 ± 0.612	0.179 ± 0.001	3.900 ± 0.612

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S127

Simulation 1 - Condition 127: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.345	0.029 ± 0.010	0.092 ± 0.030	0.015 ± 0.002	0.122 ± 0.010	0.016 ± 0.003	0.125 ± 0.012	0.158 ± 0.026	0.397 ± 0.037	0.339	0.023	7.24	0.1345	0.1364	1.089	94.444 ± 1.800	0.522 ± 0.041	—
LMS	beta32	0.316	0.345	0.029 ± 0.010	0.093 ± 0.032	0.017 ± 0.003	0.130 ± 0.012	0.018 ± 0.003	0.133 ± 0.014	0.178 ± 0.035	0.422 ± 0.045	0.331	0.015	4.87	0.1348	0.1357	0.926	95.679 ± 1.598	0.474 ± 0.033	—
LMS	beta33	0.000	0.026	0.026 ± 0.010	—	0.015 ± 0.002	0.124 ± 0.009	0.016 ± 0.002	0.126 ± 0.009	—	—	0.016	0.016	—	0.1777	0.1784	1.476	95.062 ± 1.702	0.716 ± 0.058	4.938 ± 1.702
LMS	beta34	0.000	-0.030	-0.030 ± 0.007	—	0.008 ± 0.001	0.087 ± 0.008	0.008 ± 0.001	0.092 ± 0.008	—	—	-0.030	-0.030	—	0.1043	0.1086	1.658	88.889 ± 2.469	0.565 ± 0.085	11.111 ± 2.469
LMS	beta35	0.000	-0.047	-0.047 ± 0.008	—	0.010 ± 0.002	0.099 ± 0.010	0.012 ± 0.002	0.109 ± 0.011	—	—	-0.044	-0.044	—	0.1006	0.1098	1.238	86.420 ± 2.692	0.480 ± 0.056	13.580 ± 2.692
LSAM	beta31	0.316	0.359	0.043 ± 0.012	0.136 ± 0.039	0.024 ± 0.003	0.155 ± 0.010	0.026 ± 0.003	0.160 ± 0.012	0.257 ± 0.035	0.507 ± 0.040	0.364	0.048	15.34	0.1550	0.1624	1.006	95.679 ± 1.598	0.611 ± 0.019	—
LSAM	beta32	0.316	0.356	0.040 ± 0.012	0.125 ± 0.037	0.022 ± 0.002	0.147 ± 0.008	0.023 ± 0.003	0.152 ± 0.011	0.231 ± 0.028	0.480 ± 0.034	0.352	0.036	11.26	0.1503	0.1545	1.063	98.148 ± 1.059	0.613 ± 0.019	—
LSAM	beta33	0.000	0.122	0.122 ± 0.011	—	0.021 ± 0.003	0.144 ± 0.011	0.035 ± 0.005	0.188 ± 0.014	—	—	0.122	0.122	—	0.1745	0.2132	1.173	96.914 ± 1.359	0.661 ± 0.026	3.086 ± 1.359
LSAM	beta34	0.000	-0.079	-0.079 ± 0.008	—	0.009 ± 0.001	0.097 ± 0.008	0.016 ± 0.002	0.125 ± 0.009	—	—	-0.073	-0.073	—	0.1060	0.1284	1.133	98.148 ± 1.059	0.430 ± 0.017	1.852 ± 1.059
LSAM	beta35	0.000	-0.079	-0.079 ± 0.007	—	0.009 ± 0.001	0.094 ± 0.006	0.015 ± 0.002	0.123 ± 0.007	—	—	-0.080	-0.080	—	0.1031	0.1307	1.176	91.358 ± 2.208	0.434 ± 0.017	8.642 ± 2.208
QML	beta31	0.316	0.325	0.009 ± 0.008	0.030 ± 0.025	0.010 ± 0.001	0.100 ± 0.006	0.010 ± 0.001	0.100 ± 0.007	0.100 ± 0.012	0.316 ± 0.023	0.328	0.012	3.85	0.1016	0.1024	0.992	96.296 ± 1.484	0.387 ± 0.007	—
QML	beta32	0.316	0.326	0.010 ± 0.007	0.032 ± 0.024	0.009 ± 0.001	0.095 ± 0.005	0.009 ± 0.001	0.095 ± 0.007	0.090 ± 0.010	0.300 ± 0.021	0.325	0.009	2.90	0.1011	0.1015	1.018	94.444 ± 1.800	0.377 ± 0.007	—
QML	beta33	0.000	0.038	0.038 ± 0.009	—	0.013 ± 0.001	0.112 ± 0.007	0.014 ± 0.002	0.118 ± 0.008	—	—	0.036	0.036	—	0.1478	0.1521	1.183	95.062 ± 1.702	0.521 ± 0.020	4.938 ± 1.702
QML	beta34	0.000	-0.046	-0.046 ± 0.005	—	0.005 ± 0.001	0.068 ± 0.004	0.007 ± 0.001	0.082 ± 0.005	—	—	-0.050	-0.050	—	0.0796	0.0942	1.233	91.358 ± 2.208	0.329 ± 0.011	8.642 ± 2.208
QML	beta35	0.000	-0.047	-0.047 ± 0.006	—	0.005 ± 0.001	0.073 ± 0.005	0.008 ± 0.001	0.087 ± 0.005	—	—	-0.054	-0.054	—	0.0813	0.0975	1.148	85.185 ± 2.791	0.328 ± 0.014	14.815 ± 2.791
UPI	beta31	0.316	0.351	0.035 ± 0.019	0.111 ± 0.059	0.056 ± 0.007	0.236 ± 0.014	0.056 ± 0.007	0.238 ± 0.017	0.565 ± 0.069	0.752 ± 0.055	0.360	0.044	13.86	0.2865	0.2898	1.585	91.975 ± 2.134	1.465 ± 0.134	—
UPI	beta32	0.316	0.365	0.049 ± 0.018	0.156 ± 0.057	0.053 ± 0.007	0.230 ± 0.016	0.055 ± 0.007	0.235 ± 0.018	0.552 ± 0.074	0.743 ± 0.058	0.360	0.044	14.07	0.2671	0.2708	1.796	95.679 ± 1.598	1.622 ± 0.177	—
UPI	beta33	0.000	0.058	0.058 ± 0.019	—	0.061 ± 0.011	0.248 ± 0.022	0.064 ± 0.011	0.254 ± 0.025	—	—	0.012	0.012	—	0.2525	0.2527	1.382	98.148 ± 1.059	1.343 ± 0.127	1.852 ± 1.059
UPI	beta34	0.000	-0.120	-0.120 ± 0.020	—	0.064 ± 0.013	0.253 ± 0.027	0.078 ± 0.015	0.279 ± 0.029	—	—	-0.095	-0.095	—	0.2612	0.2781	1.553	100.000 ± 0.000	1.537 ± 0.148	0.000 ± 0.000
UPI	beta35	0.000	-0.146	-0.146 ± 0.021	—	0.075 ± 0.013	0.273 ± 0.024	0.096 ± 0.017	0.309 ± 0.030	—	—	-0.088	-0.088	—	0.2542	0.2689	1.590	98.148 ± 1.059	1.702 ± 0.198	1.852 ± 1.059

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S128

Simulation 1 - Condition 128: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.321	0.005 ± 0.004	0.015 ± 0.014	0.004 ± 0.000	0.060 ± 0.004	0.004 ± 0.001	0.060 ± 0.005	0.036 ± 0.005	0.191 ± 0.015	0.316	0.000	0.06	0.0764	0.0764	0.984	95.876 ± 1.428	0.232 ± 0.007	—
LMS	beta32	0.316	0.324	0.008 ± 0.004	0.027 ± 0.014	0.004 ± 0.000	0.060 ± 0.003	0.004 ± 0.000	0.060 ± 0.004	0.036 ± 0.004	0.190 ± 0.012	0.322	0.006	1.87	0.0856	0.0858	0.968	95.361 ± 1.510	0.226 ± 0.003	—
LMS	beta33	0.000	0.050	0.050 ± 0.006	—	0.006 ± 0.002	0.077 ± 0.011	0.008 ± 0.001	0.092 ± 0.008	—	—	0.041	0.041	—	0.0894	0.0986	1.042	85.567 ± 2.523	0.315 ± 0.022	14.433 ± 2.523
LMS	beta34	0.000	-0.048	-0.048 ± 0.004	—	0.003 ± 0.001	0.055 ± 0.006	0.005 ± 0.001	0.073 ± 0.006	—	—	-0.034	-0.034	—	0.0718	0.0796	0.920	74.227 ± 3.140	0.197 ± 0.013	25.773 ± 3.140
LMS	beta35	0.000	-0.046	-0.046 ± 0.004	—	0.003 ± 0.000	0.051 ± 0.004	0.005 ± 0.000	0.069 ± 0.003	—	—	-0.045	-0.045	—	0.0699	0.0831	1.001	65.464 ± 3.414	0.201 ± 0.014	34.536 ± 3.414
LSAM	beta31	0.316	0.353	0.037 ± 0.006	0.118 ± 0.020	0.008 ± 0.001	0.088 ± 0.004	0.009 ± 0.001	0.096 ± 0.006	0.092 ± 0.010	0.303 ± 0.019	0.345	0.029	9.28	0.0883	0.0930	1.046	95.876 ± 1.428	0.363 ± 0.006	—
LSAM	beta32	0.316	0.354	0.038 ± 0.006	0.121 ± 0.020	0.008 ± 0.001	0.088 ± 0.005	0.009 ± 0.001	0.096 ± 0.006	0.091 ± 0.010	0.302 ± 0.019	0.350	0.034	10.63	0.0884	0.0946	1.031	94.845 ± 1.587	0.355 ± 0.006	—
LSAM	beta33	0.000	0.126	0.126 ± 0.006	—	0.007 ± 0.001	0.083 ± 0.006	0.023 ± 0.002	0.151 ± 0.007	—	—	0.119	0.119	—	0.0774	0.1422	1.089	78.351 ± 2.957	0.354 ± 0.008	21.649 ± 2.957
LSAM	beta34	0.000	-0.085	-0.085 ± 0.004	—	0.004 ± 0.000	0.061 ± 0.004	0.011 ± 0.001	0.105 ± 0.005	—	—	-0.082	-0.082	—	0.0640	0.1039	1.051	77.320 ± 3.007	0.250 ± 0.006	22.680 ± 3.007
LSAM	beta35	0.000	-0.080	-0.080 ± 0.004	—	0.004 ± 0.000	0.059 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	—	—	-0.084	-0.084	—	0.0591	0.1026	1.021	70.619 ± 3.270	0.237 ± 0.005	29.381 ± 3.270
QML	beta31	0.316	0.321	0.005 ± 0.004	0.015 ± 0.013	0.003 ± 0.000	0.056 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	0.031 ± 0.004	0.177 ± 0.012	0.323	0.007	2.26	0.0578	0.0583	1.033	95.876 ± 1.428	0.226 ± 0.002	—
QML	beta32	0.316	0.326	0.010 ± 0.004	0.031 ± 0.013	0.004 ± 0.000	0.059 ± 0.003	0.004 ± 0.000	0.060 ± 0.004	0.036 ± 0.004	0.190 ± 0.012	0.323	0.007	2.16	0.0638	0.0641	0.975	95.361 ± 1.510	0.226 ± 0.002	—
QML	beta33	0.000	0.054	0.054 ± 0.005	—	0.005 ± 0.001	0.069 ± 0.005	0.008 ± 0.001	0.088 ± 0.005	—	—	0.054	0.054	—	0.0674	0.0863	1.101	86.598 ± 2.446	0.298 ± 0.006	13.402 ± 2.446
QML	beta34	0.000	-0.058	-0.058 ± 0.003	—	0.002 ± 0.000	0.047 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.055	-0.055	—	0.0507	0.0748	1.014	71.134 ± 3.253	0.187 ± 0.004	28.866 ± 3.253
QML	beta35	0.000	-0.059	-0.059 ± 0.004	—	0.002 ± 0.000	0.050 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	—	—	-0.063	-0.063	—	0.0476	0.0787	0.972	63.402 ± 3.458	0.190 ± 0.004	36.598 ± 3.458
UPI	beta31	0.316	0.353	0.037 ± 0.025	0.116 ± 0.078	0.116 ± 0.016	0.341 ± 0.024	0.117 ± 0.016	0.342 ± 0.027	1.174 ± 0.165	1.083 ± 0.087	0.378	0.062	19.59	0.3810	0.3860	1.393	86.082 ± 2.485	1.864 ± 0.248	—
UPI	beta32	0.316	0.376	0.060 ± 0.024	0.189 ± 0.077	0.114 ± 0.017	0.338 ± 0.025	0.117 ± 0.018	0.343 ± 0.030	1.175 ± 0.183	1.084 ± 0.094	0.378	0.062	19.63	0.3222	0.3281	1.349	91.237 ± 2.030	1.788 ± 0.232	—
UPI	beta33	0.000	0.090	0.090 ± 0.021	—	0.085 ± 0.015	0.292 ± 0.027	0.093 ± 0.018	0.305 ± 0.032	—	—	0.012	0.012	—	0.2203	0.2206	1.298	95.876 ± 1.428	1.487 ± 0.195	4.124 ± 1.428
UPI	beta34	0.000	-0.205	-0.205 ± 0.024	—	0.110 ± 0.016	0.332 ± 0.025	0.151 ± 0.020	0.389 ± 0.028	—	—	-0.204	-0.204	—	0.3031	0.3655	1.691	97.423 ± 1.138	2.199 ± 0.289	2.577 ± 1.138
UPI	beta35	0.000	-0.186	-0.186 ± 0.023	—	0.100 ± 0.015	0.317 ± 0.025	0.134 ± 0.021	0.367 ± 0.030	—	—	-0.200	-0.200	—	0.3137	0.3720	1.609	96.907 ± 1.243	2.000 ± 0.287	3.093 ± 1.243

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S129

Simulation 1 - Condition 129: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.318	0.002	0.61	0.0729	0.0729	0.988	95.135 ± 0.692	0.274 ± 0.002	—
LMS	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.319	0.003	0.87	0.0703	0.0704	0.996	95.549 ± 0.664	0.276 ± 0.002	—
LMS	beta33	0.000	0.003	0.003 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.009	0.009	—	0.0866	0.0870	1.064	94.099 ± 0.758	0.368 ± 0.007	5.901 ± 0.758
LMS	beta34	0.000	-0.012	-0.012 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.014	-0.014	—	0.0583	0.0600	1.004	90.787 ± 0.931	0.240 ± 0.005	9.213 ± 0.931
LMS	beta35	0.000	-0.015	-0.015 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	-0.017	-0.017	—	0.0517	0.0545	1.049	91.201 ± 0.911	0.237 ± 0.004	8.799 ± 0.911
LSAM	beta31	0.316	0.323	0.007 ± 0.003	0.021 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.062 ± 0.003	0.248 ± 0.007	0.323	0.007	2.34	0.0804	0.0808	0.979	94.928 ± 0.706	0.300 ± 0.001	—
LSAM	beta32	0.316	0.323	0.007 ± 0.003	0.023 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.323	0.007	2.22	0.0795	0.0798	0.972	94.410 ± 0.739	0.302 ± 0.001	—
LSAM	beta33	0.000	0.038	0.038 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	—	—	0.034	0.034	—	0.0896	0.0960	1.007	91.822 ± 0.882	0.345 ± 0.003	8.178 ± 0.882
LSAM	beta34	0.000	-0.029	-0.029 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.027	-0.027	—	0.0613	0.0670	0.958	90.373 ± 0.949	0.239 ± 0.002	9.627 ± 0.949
LSAM	beta35	0.000	-0.031	-0.031 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.032	-0.032	—	0.0567	0.0649	1.009	90.166 ± 0.958	0.242 ± 0.002	9.834 ± 0.958
QML	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.318	0.002	0.65	0.0735	0.0735	0.982	94.928 ± 0.706	0.272 ± 0.001	—
QML	beta32	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.318	0.002	0.76	0.0695	0.0695	0.988	95.445 ± 0.671	0.274 ± 0.001	—
QML	beta33	0.000	0.006	0.006 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	—	—	0.007	0.007	—	0.0842	0.0844	1.035	93.892 ± 0.770	0.341 ± 0.004	6.108 ± 0.770
QML	beta34	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.017	-0.017	—	0.0573	0.0597	0.985	90.166 ± 0.958	0.224 ± 0.003	9.834 ± 0.958
QML	beta35	0.000	-0.019	-0.019 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.021	-0.021	—	0.0514	0.0553	1.036	90.994 ± 0.921	0.227 ± 0.003	9.006 ± 0.921
UPI	beta31	0.316	0.324	0.008 ± 0.003	0.025 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.067 ± 0.003	0.259 ± 0.007	0.323	0.007	2.32	0.0805	0.0809	1.007	95.652 ± 0.656	0.321 ± 0.003	—
UPI	beta32	0.316	0.324	0.008 ± 0.003	0.026 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.069 ± 0.004	0.262 ± 0.008	0.325	0.009	2.84	0.0809	0.0814	0.994	95.238 ± 0.685	0.321 ± 0.003	—
UPI	beta33	0.000	0.032	0.032 ± 0.003	—	0.009 ± 0.000	0.095 ± 0.002	0.010 ± 0.001	0.101 ± 0.003	—	—	0.029	0.029	—	0.0961	0.1005	1.009	94.928 ± 0.706	0.377 ± 0.004	5.072 ± 0.706
UPI	beta34	0.000	-0.022	-0.022 ± 0.002	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	-0.019	-0.019	—	0.0682	0.0708	1.007	94.410 ± 0.739	0.293 ± 0.004	5.590 ± 0.739
UPI	beta35	0.000	-0.024	-0.024 ± 0.002	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.022	-0.022	—	0.0695	0.0728	1.016	94.306 ± 0.746	0.296 ± 0.004	5.694 ± 0.746

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S130

Simulation 1 - Condition 130: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.317	0.001	0.23	0.0438	0.0439	0.992	95.075 ± 0.686	0.168 ± 0.000	—
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.005 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.315	-0.001	-0.44	0.0419	0.0419	1.006	95.276 ± 0.673	0.168 ± 0.001	—
LMS	beta33	0.000	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.011	0.011	—	0.0521	0.0533	1.005	93.467 ± 0.783	0.211 ± 0.002	6.533 ± 0.783
LMS	beta34	0.000	-0.018	-0.018 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.020	-0.020	—	0.0332	0.0386	0.980	86.432 ± 1.086	0.136 ± 0.001	13.568 ± 1.086
LMS	beta35	0.000	-0.015	-0.015 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.017	-0.017	—	0.0322	0.0365	1.023	89.447 ± 0.974	0.142 ± 0.002	10.553 ± 0.974
LSAM	beta31	0.316	0.322	0.006 ± 0.002	0.020 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.322	0.006	1.91	0.0478	0.0481	0.987	94.472 ± 0.724	0.185 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.001	0.009 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.321	0.005	1.47	0.0458	0.0460	1.007	94.472 ± 0.724	0.185 ± 0.001	—
LSAM	beta33	0.000	0.033	0.033 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.033	0.033	—	0.0520	0.0619	0.991	89.246 ± 0.982	0.205 ± 0.001	10.754 ± 0.982
LSAM	beta34	0.000	-0.029	-0.029 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.029	-0.029	—	0.0352	0.0456	0.972	84.523 ± 1.147	0.143 ± 0.001	15.477 ± 1.147
LSAM	beta35	0.000	-0.025	-0.025 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.026	-0.026	—	0.0362	0.0447	0.980	87.538 ± 1.047	0.144 ± 0.001	12.462 ± 1.047
QML	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.317	0.001	0.23	0.0433	0.0433	0.991	95.176 ± 0.679	0.168 ± 0.000	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.005 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.315	-0.001	-0.46	0.0419	0.0419	1.006	94.874 ± 0.699	0.169 ± 0.001	—
QML	beta33	0.000	0.008	0.008 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.010	0.010	—	0.0522	0.0531	0.995	93.869 ± 0.761	0.206 ± 0.001	6.131 ± 0.761
QML	beta34	0.000	-0.021	-0.021 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.022	-0.022	—	0.0336	0.0402	0.982	86.030 ± 1.099	0.136 ± 0.001	13.970 ± 1.099
QML	beta35	0.000	-0.018	-0.018 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.020	-0.020	—	0.0329	0.0383	0.998	89.246 ± 0.982	0.138 ± 0.001	10.754 ± 0.982
UPI	beta31	0.316	0.322	0.006 ± 0.002	0.020 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.322	0.006	1.82	0.0503	0.0506	0.991	95.377 ± 0.666	0.190 ± 0.001	—
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.007 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.320	0.004	1.14	0.0480	0.0482	1.015	94.573 ± 0.718	0.190 ± 0.001	—
UPI	beta33	0.000	0.025	0.025 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.023	0.023	—	0.0556	0.0603	0.964	91.960 ± 0.862	0.220 ± 0.001	8.040 ± 0.862
UPI	beta34	0.000	-0.022	-0.022 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.022	-0.022	—	0.0409	0.0466	0.977	90.754 ± 0.918	0.164 ± 0.001	9.246 ± 0.918
UPI	beta35	0.000	-0.018	-0.018 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.018	-0.018	—	0.0410	0.0446	0.988	92.462 ± 0.837	0.168 ± 0.001	7.538 ± 0.837

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S131

Simulation 1 - Condition 131: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.317	0.001	0.37	0.0547	0.0547	0.969	94.400 ± 0.727	0.218 ± 0.001	—
LMS	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.031 ± 0.002	0.176 ± 0.005	0.315	-0.001	-0.23	0.0537	0.0537	1.013	95.100 ± 0.683	0.220 ± 0.001	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.003	0.003	—	0.0672	0.0673	0.924	94.500 ± 0.721	0.249 ± 0.002	5.500 ± 0.721
LMS	beta34	0.000	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.007	-0.007	—	0.0441	0.0446	0.914	92.100 ± 0.853	0.170 ± 0.002	7.900 ± 0.853
LMS	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.004	-0.004	—	0.0439	0.0441	0.949	92.800 ± 0.817	0.171 ± 0.002	7.200 ± 0.817
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.319	0.003	0.93	0.0564	0.0565	0.967	93.500 ± 0.780	0.224 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.181 ± 0.005	0.316	0.000	0.02	0.0550	0.0550	1.004	94.200 ± 0.739	0.226 ± 0.001	—
LSAM	beta33	0.000	0.010	0.010 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.012	0.012	—	0.0655	0.0666	0.922	93.400 ± 0.785	0.243 ± 0.002	6.600 ± 0.785
LSAM	beta34	0.000	-0.008	-0.008 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.010	-0.010	—	0.0442	0.0453	0.921	92.100 ± 0.853	0.172 ± 0.001	7.900 ± 0.853
LSAM	beta35	0.000	-0.007	-0.007 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.007	-0.007	—	0.0444	0.0449	0.936	92.500 ± 0.833	0.172 ± 0.001	7.500 ± 0.833
QML	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.317	0.001	0.46	0.0546	0.0546	0.968	94.400 ± 0.727	0.218 ± 0.001	—
QML	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.002	0.176 ± 0.005	0.315	-0.001	-0.21	0.0534	0.0534	1.012	95.200 ± 0.676	0.220 ± 0.001	—
QML	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.003	0.003	—	0.0675	0.0675	0.924	94.600 ± 0.715	0.247 ± 0.002	5.400 ± 0.715
QML	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.007	-0.007	—	0.0441	0.0447	0.917	92.000 ± 0.858	0.169 ± 0.001	8.000 ± 0.858
QML	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.005	-0.005	—	0.0438	0.0440	0.949	92.700 ± 0.823	0.170 ± 0.001	7.300 ± 0.823
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.318	0.002	0.63	0.0556	0.0556	0.964	93.400 ± 0.785	0.222 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.181 ± 0.005	0.317	0.001	0.22	0.0557	0.0557	1.000	94.300 ± 0.733	0.224 ± 0.001	—
UPI	beta33	0.000	0.006	0.006 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.009	0.009	—	0.0687	0.0693	0.908	93.800 ± 0.763	0.246 ± 0.002	6.200 ± 0.763
UPI	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.006	-0.006	—	0.0439	0.0443	0.925	92.800 ± 0.817	0.174 ± 0.001	7.200 ± 0.817
UPI	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.004	-0.004	—	0.0454	0.0456	0.939	93.000 ± 0.807	0.175 ± 0.001	7.000 ± 0.807

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S132

Simulation 1 - Condition 132: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.314	-0.002	-0.64	0.0358	0.0358	0.956	93.800 ± 0.763	0.138 ± 0.000	—
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.001	0.108 ± 0.003	0.313	-0.003	-0.90	0.0339	0.0340	1.030	96.600 ± 0.573	0.138 ± 0.000	—
LMS	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0390	0.0390	0.973	94.100 ± 0.745	0.157 ± 0.001	5.900 ± 0.745
LMS	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.006	-0.006	—	0.0289	0.0295	0.966	93.700 ± 0.768	0.107 ± 0.001	6.300 ± 0.768
LMS	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0281	0.0289	0.980	93.500 ± 0.780	0.108 ± 0.001	6.500 ± 0.780
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.120 ± 0.003	0.315	-0.001	-0.19	0.0375	0.0375	0.953	93.700 ± 0.768	0.142 ± 0.000	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.315	-0.001	-0.43	0.0337	0.0337	1.028	96.300 ± 0.597	0.142 ± 0.000	—
LSAM	beta33	0.000	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.009	0.009	—	0.0382	0.0393	0.967	92.400 ± 0.838	0.153 ± 0.001	7.600 ± 0.838
LSAM	beta34	0.000	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.007	-0.007	—	0.0293	0.0302	0.964	93.300 ± 0.791	0.109 ± 0.001	6.700 ± 0.791
LSAM	beta35	0.000	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.008	-0.008	—	0.0288	0.0299	0.965	92.800 ± 0.817	0.108 ± 0.001	7.200 ± 0.817
QML	beta31	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.314	-0.002	-0.66	0.0357	0.0357	0.955	93.700 ± 0.768	0.138 ± 0.000	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.008 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.313	-0.003	-0.90	0.0337	0.0338	1.029	96.500 ± 0.581	0.138 ± 0.000	—
QML	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0390	0.0390	0.967	93.900 ± 0.757	0.155 ± 0.001	6.100 ± 0.757
QML	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.006	-0.006	—	0.0291	0.0297	0.967	93.500 ± 0.780	0.107 ± 0.001	6.500 ± 0.780
QML	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0281	0.0288	0.968	93.600 ± 0.774	0.107 ± 0.001	6.400 ± 0.774
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.120 ± 0.003	0.315	-0.001	-0.21	0.0371	0.0371	0.952	93.800 ± 0.763	0.141 ± 0.000	—
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.315	-0.001	-0.42	0.0340	0.0340	1.025	96.400 ± 0.589	0.141 ± 0.000	—
UPI	beta33	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.005	0.005	—	0.0397	0.0399	0.966	93.800 ± 0.763	0.153 ± 0.001	6.200 ± 0.763
UPI	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.004	-0.004	—	0.0292	0.0296	0.957	93.100 ± 0.801	0.110 ± 0.001	6.900 ± 0.801
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.003	-0.003	—	0.0295	0.0297	0.960	93.700 ± 0.768	0.109 ± 0.001	6.300 ± 0.768

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S133

Simulation 1 - Condition 133: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.271	-0.045 ± 0.006	-0.142 ± 0.018	0.015 ± 0.001	0.123 ± 0.006	0.017 ± 0.002	0.131 ± 0.007	0.172 ± 0.016	0.414 ± 0.022	0.270	-0.046	-14.41	0.1301	0.1378	1.105	96.008 ± 0.897	0.534 ± 0.021	—
LMS	beta32	0.316	0.251	-0.065 ± 0.006	-0.206 ± 0.019	0.018 ± 0.002	0.134 ± 0.007	0.022 ± 0.002	0.149 ± 0.008	0.222 ± 0.023	0.471 ± 0.026	0.266	-0.050	-15.76	0.1334	0.1424	0.995	94.748 ± 1.022	0.523 ± 0.016	—
LMS	beta33	0.000	-0.100	-0.100 ± 0.019	—	0.169 ± 0.022	0.411 ± 0.027	0.179 ± 0.025	0.423 ± 0.031	—	—	0.001	0.001	—	0.2179	0.2179	0.649	93.277 ± 1.148	1.047 ± 0.068	6.723 ± 1.148
LMS	beta34	0.000	0.106	0.106 ± 0.010	—	0.049 ± 0.007	0.220 ± 0.016	0.060 ± 0.009	0.244 ± 0.019	—	—	0.050	0.050	—	0.1116	0.1222	0.722	97.479 ± 0.719	0.624 ± 0.054	2.521 ± 0.719
LMS	beta35	0.000	0.104	0.104 ± 0.010	—	0.050 ± 0.007	0.223 ± 0.015	0.060 ± 0.009	0.246 ± 0.018	—	—	0.051	0.051	—	0.1133	0.1243	0.718	96.218 ± 0.874	0.627 ± 0.050	3.782 ± 0.874
LSAM	beta31	0.316	0.443	0.127 ± 0.053	0.401 ± 0.168	1.348 ± 0.102	1.161 ± 0.044	1.361 ± 0.106	1.167 ± 0.053	13.633 ± 1.063	3.692 ± 0.167	0.409	0.093	29.43	1.1948	1.1984	2.613	99.790 ± 0.210	11.895 ± 0.506	—
LSAM	beta32	0.316	0.482	0.166 ± 0.054	0.526 ± 0.170	1.380 ± 0.106	1.175 ± 0.045	1.405 ± 0.110	1.185 ± 0.054	14.066 ± 1.098	3.750 ± 0.170	0.422	0.106	33.56	1.2522	1.2567	2.644	99.790 ± 0.210	12.174 ± 0.520	—
LSAM	beta33	0.000	0.136	0.136 ± 0.014	—	0.091 ± 0.009	0.302 ± 0.016	0.109 ± 0.011	0.331 ± 0.019	—	—	0.115	0.115	—	0.2622	0.2863	2.401	99.160 ± 0.418	2.843 ± 0.159	0.840 ± 0.418
LSAM	beta34	0.000	-0.100	-0.100 ± 0.020	—	0.196 ± 0.019	0.443 ± 0.022	0.205 ± 0.021	0.453 ± 0.025	—	—	-0.049	-0.049	—	0.3733	0.3765	2.446	99.790 ± 0.210	4.242 ± 0.210	0.210 ± 0.210
LSAM	beta35	0.000	-0.109	-0.109 ± 0.021	—	0.203 ± 0.018	0.451 ± 0.020	0.214 ± 0.019	0.463 ± 0.023	—	—	-0.061	-0.061	—	0.3888	0.3936	2.531	100.000 ± 0.000	4.470 ± 0.222	0.000 ± 0.000
QML	beta31	0.316	0.282	-0.034 ± 0.005	-0.109 ± 0.015	0.011 ± 0.001	0.105 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	0.122 ± 0.008	0.349 ± 0.014	0.282	-0.034	-10.76	0.1111	0.1161	1.075	96.008 ± 0.897	0.442 ± 0.005	—
QML	beta32	0.316	0.270	-0.046 ± 0.005	-0.146 ± 0.015	0.011 ± 0.001	0.106 ± 0.004	0.013 ± 0.001	0.116 ± 0.004	0.134 ± 0.009	0.367 ± 0.014	0.277	-0.039	-12.29	0.1144	0.1208	1.058	93.908 ± 1.096	0.441 ± 0.005	—
QML	beta33	0.000	0.014	0.014 ± 0.006	—	0.017 ± 0.001	0.130 ± 0.006	0.017 ± 0.001	0.131 ± 0.006	—	—	0.012	0.012	—	0.1287	0.1292	1.015	96.218 ± 0.874	0.517 ± 0.012	3.782 ± 0.874
QML	beta34	0.000	0.037	0.037 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.007 ± 0.001	0.082 ± 0.004	—	—	0.033	0.033	—	0.0706	0.0780	1.020	96.008 ± 0.897	0.291 ± 0.008	3.992 ± 0.897
QML	beta35	0.000	0.034	0.034 ± 0.004	—	0.006 ± 0.001	0.077 ± 0.004	0.007 ± 0.001	0.084 ± 0.004	—	—	0.031	0.031	—	0.0770	0.0831	0.973	96.008 ± 0.897	0.294 ± 0.008	3.992 ± 0.897
UPI	beta31	0.316	0.430	0.114 ± 0.029	0.361 ± 0.093	0.413 ± 0.035	0.643 ± 0.028	0.425 ± 0.037	0.652 ± 0.032	4.256 ± 0.367	2.063 ± 0.101	0.435	0.119	37.62	0.6292	0.6403	1.852	97.899 ± 0.657	4.664 ± 0.263	—
UPI	beta32	0.316	0.413	0.097 ± 0.032	0.306 ± 0.103	0.501 ± 0.046	0.708 ± 0.033	0.509 ± 0.047	0.714 ± 0.037	5.098 ± 0.474	2.258 ± 0.117	0.377	0.061	19.23	0.6608	0.6636	1.801	98.739 ± 0.511	4.995 ± 0.289	—
UPI	beta33	0.000	0.047	0.047 ± 0.014	—	0.100 ± 0.010	0.316 ± 0.016	0.102 ± 0.010	0.319 ± 0.017	—	—	0.040	0.040	—	0.2795	0.2824	1.764	99.790 ± 0.210	2.184 ± 0.133	0.210 ± 0.210
UPI	beta34	0.000	-0.057	-0.057 ± 0.012	—	0.065 ± 0.006	0.256 ± 0.012	0.069 ± 0.007	0.262 ± 0.014	—	—	-0.058	-0.058	—	0.2269	0.2342	1.811	99.370 ± 0.363	1.816 ± 0.109	0.630 ± 0.363
UPI	beta35	0.000	-0.057	-0.057 ± 0.013	—	0.082 ± 0.008	0.287 ± 0.014	0.085 ± 0.008	0.292 ± 0.015	—	—	-0.053	-0.053	—	0.2539	0.2593	1.745	99.580 ± 0.296	1.962 ± 0.118	0.420 ± 0.296

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S134

Simulation 1 - Condition 134: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.278	-0.038 ± 0.003	-0.119 ± 0.009	0.004 ± 0.000	0.067 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.058 ± 0.004	0.242 ± 0.009	0.283	-0.033	-10.44	0.0810	0.0874	1.079	93.629 ± 1.073	0.282 ± 0.007	—
LMS	beta32	0.316	0.284	-0.032 ± 0.003	-0.103 ± 0.008	0.004 ± 0.000	0.060 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.047 ± 0.003	0.216 ± 0.008	0.282	-0.034	-10.73	0.0796	0.0865	1.174	96.718 ± 0.783	0.277 ± 0.009	—
LMS	beta33	0.000	0.019	0.019 ± 0.004	—	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	0.012	0.012	—	0.1102	0.1108	1.422	94.015 ± 1.042	0.489 ± 0.056	5.985 ± 1.042
LMS	beta34	0.000	0.039	0.039 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.002	0.004 ± 0.000	0.063 ± 0.003	—	—	0.037	0.037	—	0.0625	0.0728	1.288	95.946 ± 0.867	0.251 ± 0.025	4.054 ± 0.867
LMS	beta35	0.000	0.034	0.034 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	0.039	0.039	—	0.0574	0.0695	1.454	96.718 ± 0.783	0.244 ± 0.030	3.282 ± 0.783
LSAM	beta31	0.316	0.402	0.086 ± 0.027	0.272 ± 0.084	0.369 ± 0.037	0.607 ± 0.031	0.375 ± 0.039	0.613 ± 0.034	3.760 ± 0.388	1.939 ± 0.109	0.410	0.094	29.87	0.6558	0.6625	1.813	99.035 ± 0.430	4.315 ± 0.293	—
LSAM	beta32	0.316	0.429	0.113 ± 0.026	0.358 ± 0.081	0.342 ± 0.036	0.585 ± 0.031	0.355 ± 0.036	0.596 ± 0.033	3.552 ± 0.362	1.885 ± 0.105	0.390	0.074	23.41	0.6109	0.6154	1.823	99.614 ± 0.272	4.181 ± 0.278	—
LSAM	beta33	0.000	0.068	0.068 ± 0.005	—	0.013 ± 0.001	0.116 ± 0.005	0.018 ± 0.001	0.134 ± 0.006	—	—	0.066	0.066	—	0.1238	0.1402	1.598	95.946 ± 0.867	0.727 ± 0.043	4.054 ± 0.867
LSAM	beta34	0.000	-0.056	-0.056 ± 0.009	—	0.040 ± 0.004	0.199 ± 0.010	0.043 ± 0.004	0.207 ± 0.011	—	—	-0.049	-0.049	—	0.2009	0.2068	1.783	98.649 ± 0.507	1.392 ± 0.099	1.351 ± 0.507
LSAM	beta35	0.000	-0.065	-0.065 ± 0.008	—	0.036 ± 0.004	0.188 ± 0.010	0.040 ± 0.004	0.199 ± 0.011	—	—	-0.044	-0.044	—	0.1812	0.1866	1.770	99.035 ± 0.430	1.307 ± 0.091	0.965 ± 0.430
QML	beta31	0.316	0.281	-0.035 ± 0.003	-0.109 ± 0.010	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.059 ± 0.004	0.243 ± 0.009	0.289	-0.027	-8.47	0.0722	0.0770	1.003	91.313 ± 1.237	0.269 ± 0.002	—
QML	beta32	0.316	0.286	-0.030 ± 0.003	-0.094 ± 0.009	0.004 ± 0.000	0.062 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.003	0.219 ± 0.008	0.286	-0.030	-9.43	0.0682	0.0744	1.093	95.946 ± 0.867	0.267 ± 0.001	—
QML	beta33	0.000	0.011	0.011 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	—	—	0.006	0.006	—	0.0854	0.0857	1.038	95.753 ± 0.886	0.307 ± 0.004	4.247 ± 0.886
QML	beta34	0.000	0.033	0.033 ± 0.002	—	0.002 ± 0.000	0.045 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	0.029	0.029	—	0.0479	0.0559	0.967	92.085 ± 1.186	0.171 ± 0.003	7.915 ± 1.186
QML	beta35	0.000	0.029	0.029 ± 0.002	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	—	—	0.032	0.032	—	0.0461	0.0561	1.085	93.243 ± 1.103	0.165 ± 0.003	6.757 ± 1.103
UPI	beta31	0.316	0.571	0.255 ± 0.026	0.806 ± 0.082	0.351 ± 0.029	0.592 ± 0.025	0.415 ± 0.035	0.644 ± 0.030	4.155 ± 0.349	2.038 ± 0.094	0.581	0.265	83.80	0.6153	0.6699	1.452	94.015 ± 1.042	3.371 ± 0.177	—
UPI	beta32	0.316	0.542	0.226 ± 0.025	0.715 ± 0.079	0.322 ± 0.027	0.567 ± 0.024	0.372 ± 0.028	0.610 ± 0.026	3.729 ± 0.283	1.931 ± 0.082	0.583	0.267	84.61	0.6063	0.6627	1.444	93.822 ± 1.058	3.211 ± 0.168	—
UPI	beta33	0.000	0.112	0.112 ± 0.010	—	0.057 ± 0.005	0.238 ± 0.011	0.069 ± 0.006	0.263 ± 0.013	—	—	0.086	0.086	—	0.1885	0.2073	1.438	99.807 ± 0.193	1.341 ± 0.077	0.193 ± 0.193
UPI	beta34	0.000	-0.119	-0.119 ± 0.010	—	0.048 ± 0.004	0.220 ± 0.010	0.062 ± 0.006	0.249 ± 0.012	—	—	-0.105	-0.105	—	0.1996	0.2257	1.468	97.683 ± 0.661	1.264 ± 0.069	2.317 ± 0.661
UPI	beta35	0.000	-0.110	-0.110 ± 0.009	—	0.042 ± 0.004	0.206 ± 0.010	0.054 ± 0.004	0.233 ± 0.010	—	—	-0.112	-0.112	—	0.1916	0.2220	1.445	97.104 ± 0.737	1.166 ± 0.061	2.896 ± 0.737

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S135

Simulation 1 - Condition 135: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.290	-0.026 ± 0.003	-0.083 ± 0.009	0.007 ± 0.000	0.083 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.076 ± 0.004	0.275 ± 0.009	0.290	-0.026	-8.26	0.0845	0.0885	1.067	95.986 ± 0.665	0.347 ± 0.006	—
LMS	beta32	0.316	0.291	-0.025 ± 0.003	-0.078 ± 0.009	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.072 ± 0.004	0.269 ± 0.008	0.296	-0.020	-6.39	0.0852	0.0876	1.067	96.101 ± 0.656	0.341 ± 0.003	—
LMS	beta33	0.000	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	—	—	0.000	0.000	—	0.0821	0.0821	1.011	94.266 ± 0.787	0.347 ± 0.007	5.734 ± 0.787
LMS	beta34	0.000	0.024	0.024 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	0.023	0.023	—	0.0460	0.0515	0.999	94.381 ± 0.780	0.188 ± 0.006	5.619 ± 0.780
LMS	beta35	0.000	0.022	0.022 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	—	—	0.016	0.016	—	0.0434	0.0462	0.967	95.183 ± 0.725	0.179 ± 0.004	4.817 ± 0.725
LSAM	beta31	0.316	0.340	0.024 ± 0.005	0.074 ± 0.017	0.026 ± 0.001	0.161 ± 0.004	0.026 ± 0.001	0.162 ± 0.005	0.264 ± 0.014	0.514 ± 0.016	0.337	0.021	6.74	0.1550	0.1565	1.077	96.445 ± 0.627	0.679 ± 0.008	—
LSAM	beta32	0.316	0.340	0.024 ± 0.005	0.076 ± 0.017	0.025 ± 0.001	0.158 ± 0.004	0.025 ± 0.001	0.159 ± 0.005	0.254 ± 0.014	0.504 ± 0.016	0.339	0.023	7.32	0.1611	0.1627	1.089	96.789 ± 0.597	0.673 ± 0.008	—
LSAM	beta33	0.000	0.023	0.023 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	—	—	0.021	0.021	—	0.0771	0.0800	0.959	92.775 ± 0.877	0.327 ± 0.004	7.225 ± 0.877
LSAM	beta34	0.000	-0.018	-0.018 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.017	-0.017	—	0.0548	0.0574	1.010	94.839 ± 0.749	0.238 ± 0.005	5.161 ± 0.749
LSAM	beta35	0.000	-0.018	-0.018 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.018	-0.018	—	0.0534	0.0564	1.027	94.954 ± 0.741	0.235 ± 0.005	5.046 ± 0.741
QML	beta31	0.316	0.292	-0.024 ± 0.003	-0.076 ± 0.009	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.076 ± 0.004	0.276 ± 0.008	0.291	-0.025	-7.86	0.0845	0.0881	1.040	95.642 ± 0.691	0.342 ± 0.002	—
QML	beta32	0.316	0.294	-0.022 ± 0.003	-0.071 ± 0.009	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.073 ± 0.004	0.270 ± 0.008	0.297	-0.019	-5.95	0.0848	0.0869	1.054	95.413 ± 0.708	0.341 ± 0.002	—
QML	beta33	0.000	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.001	-0.001	—	0.0780	0.0780	0.981	93.463 ± 0.837	0.315 ± 0.004	6.537 ± 0.837
QML	beta34	0.000	0.021	0.021 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	0.020	0.020	—	0.0438	0.0482	0.963	93.119 ± 0.857	0.172 ± 0.003	6.881 ± 0.857
QML	beta35	0.000	0.020	0.020 ± 0.002	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.015	0.015	—	0.0415	0.0442	0.958	94.266 ± 0.787	0.170 ± 0.003	5.734 ± 0.787
UPI	beta31	0.316	0.347	0.031 ± 0.007	0.098 ± 0.021	0.038 ± 0.002	0.194 ± 0.006	0.039 ± 0.002	0.196 ± 0.007	0.386 ± 0.022	0.621 ± 0.021	0.337	0.021	6.76	0.1804	0.1817	1.176	97.592 ± 0.519	0.894 ± 0.030	—
UPI	beta32	0.316	0.351	0.035 ± 0.007	0.112 ± 0.021	0.038 ± 0.002	0.195 ± 0.006	0.039 ± 0.002	0.198 ± 0.007	0.394 ± 0.024	0.627 ± 0.022	0.349	0.033	10.60	0.1874	0.1904	1.178	97.706 ± 0.507	0.901 ± 0.032	—
UPI	beta33	0.000	0.027	0.027 ± 0.003	—	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	—	—	0.023	0.023	—	0.0875	0.0905	1.070	95.183 ± 0.725	0.411 ± 0.010	4.817 ± 0.725
UPI	beta34	0.000	-0.023	-0.023 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	-0.017	-0.017	—	0.0654	0.0677	1.100	96.904 ± 0.587	0.318 ± 0.012	3.096 ± 0.587
UPI	beta35	0.000	-0.025	-0.025 ± 0.003	—	0.006 ± 0.000	0.074 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	—	—	-0.020	-0.020	—	0.0608	0.0639	1.128	96.560 ± 0.617	0.330 ± 0.017	3.440 ± 0.617

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S136

Simulation 1 - Condition 136: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.294	-0.022 ± 0.002	-0.071 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.179 ± 0.005	0.294	-0.022	-7.05	0.0550	0.0593	1.020	92.733 ± 0.855	0.208 ± 0.001	—
LMS	beta32	0.316	0.291	-0.025 ± 0.002	-0.079 ± 0.006	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.291	-0.025	-7.85	0.0538	0.0593	0.978	91.866 ± 0.900	0.207 ± 0.001	—
LMS	beta33	0.000	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.008	0.008	—	0.0530	0.0536	0.972	94.143 ± 0.773	0.203 ± 0.002	5.857 ± 0.773
LMS	beta34	0.000	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.019	0.019	—	0.0286	0.0341	0.939	91.432 ± 0.922	0.105 ± 0.002	8.568 ± 0.922
LMS	beta35	0.000	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.018	0.018	—	0.0279	0.0334	0.974	91.432 ± 0.922	0.105 ± 0.001	8.568 ± 0.922
LSAM	beta31	0.316	0.340	0.024 ± 0.003	0.076 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.001	0.101 ± 0.003	0.103 ± 0.005	0.320 ± 0.009	0.342	0.026	8.23	0.1000	0.1033	1.005	95.987 ± 0.646	0.387 ± 0.003	—
LSAM	beta32	0.316	0.334	0.018 ± 0.003	0.057 ± 0.010	0.010 ± 0.000	0.100 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.103 ± 0.005	0.320 ± 0.010	0.333	0.017	5.50	0.0974	0.0990	0.985	93.926 ± 0.787	0.385 ± 0.003	—
LSAM	beta33	0.000	0.018	0.018 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.017	0.017	—	0.0480	0.0509	0.939	92.842 ± 0.849	0.179 ± 0.001	7.158 ± 0.849
LSAM	beta34	0.000	-0.015	-0.015 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.014	-0.014	—	0.0322	0.0351	0.946	91.866 ± 0.900	0.126 ± 0.001	8.134 ± 0.900
LSAM	beta35	0.000	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.012	-0.012	—	0.0303	0.0326	0.962	92.408 ± 0.872	0.125 ± 0.001	7.592 ± 0.872
QML	beta31	0.316	0.295	-0.021 ± 0.002	-0.067 ± 0.006	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.296	-0.020	-6.43	0.0562	0.0598	1.007	92.408 ± 0.872	0.212 ± 0.001	—
QML	beta32	0.316	0.292	-0.024 ± 0.002	-0.075 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.293	-0.023	-7.37	0.0555	0.0602	0.970	91.649 ± 0.911	0.211 ± 0.001	—
QML	beta33	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.005	0.005	—	0.0494	0.0496	0.966	94.035 ± 0.780	0.195 ± 0.002	5.965 ± 0.780
QML	beta34	0.000	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.017	0.017	—	0.0290	0.0338	0.932	90.889 ± 0.948	0.103 ± 0.001	9.111 ± 0.948
QML	beta35	0.000	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.018	0.018	—	0.0268	0.0321	0.973	90.456 ± 0.968	0.105 ± 0.001	9.544 ± 0.968
UPI	beta31	0.316	0.348	0.032 ± 0.004	0.100 ± 0.012	0.014 ± 0.001	0.120 ± 0.003	0.015 ± 0.001	0.124 ± 0.004	0.153 ± 0.008	0.392 ± 0.012	0.349	0.033	10.60	0.1177	0.1223	1.003	96.095 ± 0.638	0.471 ± 0.006	—
UPI	beta32	0.316	0.340	0.024 ± 0.004	0.077 ± 0.013	0.014 ± 0.001	0.120 ± 0.003	0.015 ± 0.001	0.123 ± 0.004	0.151 ± 0.008	0.388 ± 0.012	0.335	0.019	5.86	0.1137	0.1152	0.996	95.011 ± 0.717	0.469 ± 0.006	—
UPI	beta33	0.000	0.019	0.019 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.017	0.017	—	0.0490	0.0518	0.966	94.794 ± 0.732	0.197 ± 0.002	5.206 ± 0.732
UPI	beta34	0.000	-0.019	-0.019 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	-0.016	-0.016	—	0.0364	0.0398	0.929	93.926 ± 0.787	0.154 ± 0.003	6.074 ± 0.787
UPI	beta35	0.000	-0.017	-0.017 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.002	—	—	-0.012	-0.012	—	0.0356	0.0377	0.955	94.794 ± 0.732	0.154 ± 0.003	5.206 ± 0.732

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S137

Simulation 1 - Condition 137: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.300	-0.016 ± 0.002	-0.049 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.242 ± 0.006	0.299	-0.017	-5.29	0.0770	0.0788	1.016	95.951 ± 0.627	0.299 ± 0.002	—
LMS	beta32	0.316	0.299	-0.017 ± 0.002	-0.053 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.302	-0.014	-4.42	0.0721	0.0735	1.040	95.344 ± 0.670	0.298 ± 0.001	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.002	-0.002	—	0.0578	0.0578	0.947	94.332 ± 0.736	0.226 ± 0.003	5.668 ± 0.736
LMS	beta34	0.000	0.010	0.010 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.006	0.006	—	0.0300	0.0306	0.920	92.611 ± 0.832	0.122 ± 0.002	7.389 ± 0.832
LMS	beta35	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.009	0.009	—	0.0305	0.0318	0.935	92.308 ± 0.848	0.125 ± 0.002	7.692 ± 0.848
LSAM	beta31	0.316	0.319	0.003 ± 0.003	0.010 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.097 ± 0.004	0.311 ± 0.008	0.318	0.002	0.75	0.1015	0.1016	1.006	94.939 ± 0.697	0.388 ± 0.002	—
LSAM	beta32	0.316	0.316	0.000 ± 0.003	0.001 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.092 ± 0.004	0.303 ± 0.009	0.318	0.002	0.60	0.0909	0.0910	1.036	95.749 ± 0.642	0.388 ± 0.002	—
LSAM	beta33	0.000	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	0.003	0.003	—	0.0569	0.0570	0.958	94.332 ± 0.736	0.221 ± 0.002	5.668 ± 0.736
LSAM	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.006	-0.006	—	0.0320	0.0325	0.921	92.915 ± 0.816	0.134 ± 0.002	7.085 ± 0.816
LSAM	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.003	-0.003	—	0.0325	0.0326	0.967	92.915 ± 0.816	0.137 ± 0.002	7.085 ± 0.816
QML	beta31	0.316	0.301	-0.015 ± 0.002	-0.048 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.006	0.300	-0.016	-5.02	0.0783	0.0799	1.004	95.749 ± 0.642	0.299 ± 0.001	—
QML	beta32	0.316	0.300	-0.016 ± 0.002	-0.051 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.240 ± 0.007	0.302	-0.014	-4.42	0.0725	0.0738	1.031	95.040 ± 0.691	0.300 ± 0.001	—
QML	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0575	0.0575	0.948	94.534 ± 0.723	0.225 ± 0.003	5.466 ± 0.723
QML	beta34	0.000	0.010	0.010 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.006	0.006	—	0.0301	0.0307	0.914	92.611 ± 0.832	0.122 ± 0.002	7.389 ± 0.832
QML	beta35	0.000	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.009	0.009	—	0.0308	0.0322	0.936	92.004 ± 0.863	0.125 ± 0.002	7.996 ± 0.863
UPI	beta31	0.316	0.319	0.003 ± 0.003	0.009 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.004	0.306 ± 0.008	0.320	0.004	1.12	0.0966	0.0967	1.010	95.142 ± 0.684	0.383 ± 0.002	—
UPI	beta32	0.316	0.316	0.000 ± 0.003	0.001 ± 0.010	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.090 ± 0.004	0.300 ± 0.008	0.319	0.003	0.80	0.0903	0.0903	1.037	95.749 ± 0.642	0.385 ± 0.002	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	0.002	0.002	—	0.0572	0.0573	0.966	94.028 ± 0.754	0.225 ± 0.002	5.972 ± 0.754
UPI	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0311	0.0314	0.918	92.713 ± 0.827	0.133 ± 0.002	7.287 ± 0.827
UPI	beta35	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0328	0.0328	0.968	93.016 ± 0.811	0.137 ± 0.002	6.984 ± 0.811

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S138

Simulation 1 - Condition 138: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.302	-0.014 ± 0.001	-0.046 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.302	-0.014	-4.44	0.0436	0.0458	1.045	94.289 ± 0.735	0.188 ± 0.003	—
LMS	beta32	0.316	0.306	-0.010 ± 0.002	-0.032 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.305	-0.011	-3.47	0.0468	0.0480	1.028	94.289 ± 0.735	0.193 ± 0.007	—
LMS	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0366	0.0366	0.975	95.992 ± 0.621	0.142 ± 0.002	4.008 ± 0.621
LMS	beta34	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.007	0.007	—	0.0191	0.0203	0.933	93.487 ± 0.781	0.075 ± 0.001	6.513 ± 0.781
LMS	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.007	0.007	—	0.0206	0.0218	0.896	92.184 ± 0.850	0.075 ± 0.001	7.816 ± 0.850
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.317	0.001	0.46	0.0546	0.0546	1.019	95.591 ± 0.650	0.234 ± 0.001	—
LSAM	beta32	0.316	0.323	0.007 ± 0.002	0.024 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.005	0.321	0.005	1.73	0.0612	0.0614	0.979	93.487 ± 0.781	0.236 ± 0.001	—
LSAM	beta33	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.003	0.003	—	0.0349	0.0350	0.959	94.589 ± 0.716	0.132 ± 0.001	5.411 ± 0.716
LSAM	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.004	-0.004	—	0.0194	0.0198	0.919	92.585 ± 0.829	0.077 ± 0.001	7.415 ± 0.829
LSAM	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.004	-0.004	—	0.0209	0.0212	0.905	92.184 ± 0.850	0.079 ± 0.001	7.816 ± 0.850
QML	beta31	0.316	0.302	-0.014 ± 0.001	-0.045 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.302	-0.014	-4.35	0.0443	0.0464	1.021	94.188 ± 0.741	0.186 ± 0.001	—
QML	beta32	0.316	0.306	-0.010 ± 0.002	-0.031 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.157 ± 0.004	0.305	-0.011	-3.37	0.0476	0.0488	0.981	94.188 ± 0.741	0.186 ± 0.001	—
QML	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0363	0.0363	0.958	95.691 ± 0.643	0.139 ± 0.001	4.309 ± 0.643
QML	beta34	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.007	0.007	—	0.0192	0.0203	0.913	93.287 ± 0.792	0.073 ± 0.001	6.713 ± 0.792
QML	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.007	0.007	—	0.0207	0.0219	0.887	92.084 ± 0.855	0.075 ± 0.001	7.916 ± 0.855
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.318	0.002	0.72	0.0555	0.0555	1.021	95.691 ± 0.643	0.233 ± 0.001	—
UPI	beta32	0.316	0.322	0.006 ± 0.002	0.020 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.320	0.004	1.29	0.0632	0.0634	0.982	94.389 ± 0.728	0.235 ± 0.001	—
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.002	0.002	—	0.0353	0.0353	0.959	94.890 ± 0.697	0.133 ± 0.001	5.110 ± 0.697
UPI	beta34	0.000	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0192	0.0195	0.924	93.387 ± 0.787	0.076 ± 0.001	6.613 ± 0.787
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0210	0.0213	0.901	92.685 ± 0.824	0.079 ± 0.001	7.315 ± 0.824

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S139

Simulation 1 - Condition 139: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.012	-0.004 ± 0.038	0.007 ± 0.002	0.086 ± 0.012	0.007 ± 0.002	0.085 ± 0.013	0.073 ± 0.018	0.270 ± 0.041	0.310	-0.006	-1.86	0.1157	0.1159	1.393	96.154 ± 2.667	0.471 ± 0.065	—
LMS	beta32	0.316	0.316	-0.000 ± 0.016	-0.001 ± 0.050	0.013 ± 0.003	0.114 ± 0.014	0.013 ± 0.003	0.113 ± 0.016	0.129 ± 0.029	0.359 ± 0.050	0.340	0.024	7.53	0.1585	0.1603	0.900	94.231 ± 3.233	0.404 ± 0.032	—
LMS	beta33	0.000	-0.013	-0.013 ± 0.031	—	0.049 ± 0.020	0.222 ± 0.052	0.048 ± 0.021	0.220 ± 0.055	—	—	0.028	0.028	—	0.2120	0.2138	1.031	86.538 ± 4.733	0.897 ± 0.350	13.462 ± 4.733
LMS	beta34	0.000	0.014	0.014 ± 0.035	—	0.063 ± 0.032	0.252 ± 0.073	0.062 ± 0.033	0.250 ± 0.075	—	—	-0.024	-0.024	—	0.1370	0.1391	0.904	80.769 ± 5.465	0.891 ± 0.307	19.231 ± 5.465
LMS	beta35	0.000	-0.040	-0.040 ± 0.015	—	0.012 ± 0.006	0.110 ± 0.032	0.013 ± 0.007	0.116 ± 0.035	—	—	-0.035	-0.035	—	0.1499	0.1539	1.154	88.462 ± 4.430	0.498 ± 0.147	11.538 ± 4.430
LSAM	beta31	0.316	0.343	0.027 ± 0.023	0.086 ± 0.074	0.028 ± 0.007	0.168 ± 0.023	0.028 ± 0.007	0.168 ± 0.025	0.283 ± 0.072	0.532 ± 0.080	0.357	0.041	12.96	0.1856	0.1901	1.189	100.000 ± 0.000	0.781 ± 0.061	—
LSAM	beta32	0.316	0.383	0.067 ± 0.023	0.214 ± 0.074	0.029 ± 0.005	0.169 ± 0.016	0.033 ± 0.007	0.181 ± 0.022	0.327 ± 0.067	0.572 ± 0.070	0.356	0.040	12.79	0.1994	0.2034	1.220	98.077 ± 1.904	0.809 ± 0.069	—
LSAM	beta33	0.000	0.099	0.099 ± 0.020	—	0.022 ± 0.005	0.147 ± 0.019	0.031 ± 0.008	0.176 ± 0.025	—	—	0.079	0.079	—	0.1456	0.1658	1.184	96.154 ± 2.667	0.682 ± 0.066	3.846 ± 2.667
LSAM	beta34	0.000	-0.115	-0.115 ± 0.022	—	0.024 ± 0.008	0.156 ± 0.027	0.037 ± 0.010	0.193 ± 0.029	—	—	-0.108	-0.108	—	0.1321	0.1708	1.212	94.231 ± 3.233	0.740 ± 0.112	5.769 ± 3.233
LSAM	beta35	0.000	-0.129	-0.129 ± 0.021	—	0.023 ± 0.005	0.151 ± 0.019	0.039 ± 0.010	0.197 ± 0.027	—	—	-0.095	-0.095	—	0.1780	0.2017	1.275	94.231 ± 3.233	0.756 ± 0.101	5.769 ± 3.233
QML	beta31	0.316	0.313	-0.003 ± 0.011	-0.011 ± 0.036	0.007 ± 0.002	0.082 ± 0.012	0.007 ± 0.002	0.081 ± 0.013	0.066 ± 0.018	0.256 ± 0.041	0.313	-0.003	-0.87	0.0756	0.0757	1.105	96.154 ± 2.667	0.354 ± 0.006	—
QML	beta32	0.316	0.328	0.012 ± 0.014	0.037 ± 0.043	0.010 ± 0.002	0.098 ± 0.012	0.010 ± 0.002	0.098 ± 0.014	0.096 ± 0.023	0.311 ± 0.045	0.318	0.002	0.78	0.1038	0.1038	0.912	94.231 ± 3.233	0.352 ± 0.011	—
QML	beta33	0.000	0.046	0.046 ± 0.015	—	0.012 ± 0.002	0.110 ± 0.012	0.014 ± 0.003	0.118 ± 0.015	—	—	0.051	0.051	—	0.1118	0.1230	1.047	90.385 ± 4.088	0.450 ± 0.025	9.615 ± 4.088
QML	beta34	0.000	-0.059	-0.059 ± 0.010	—	0.005 ± 0.001	0.073 ± 0.008	0.009 ± 0.002	0.094 ± 0.010	—	—	-0.055	-0.055	—	0.0735	0.0916	1.024	80.769 ± 5.465	0.295 ± 0.018	19.231 ± 5.465
QML	beta35	0.000	-0.059	-0.059 ± 0.009	—	0.004 ± 0.001	0.067 ± 0.007	0.008 ± 0.001	0.089 ± 0.009	—	—	-0.060	-0.060	—	0.0971	0.1140	1.180	84.615 ± 5.003	0.310 ± 0.025	15.385 ± 5.003
UPI	beta31	0.316	0.320	0.004 ± 0.049	0.013 ± 0.156	0.126 ± 0.030	0.355 ± 0.045	0.124 ± 0.030	0.352 ± 0.051	1.238 ± 0.298	1.113 ± 0.161	0.290	-0.026	-8.20	0.4128	0.4136	1.563	84.615 ± 5.003	2.175 ± 0.522	—
UPI	beta32	0.316	0.409	0.093 ± 0.075	0.293 ± 0.236	0.290 ± 0.101	0.539 ± 0.104	0.293 ± 0.108	0.542 ± 0.113	2.937 ± 1.077	1.714 ± 0.357	0.344	0.028	8.94	0.4899	0.4907	1.801	88.462 ± 4.430	3.804 ± 0.982	—
UPI	beta33	0.000	-0.028	-0.028 ± 0.027	—	0.037 ± 0.010	0.191 ± 0.029	0.037 ± 0.010	0.192 ± 0.029	—	—	-0.031	-0.031	—	0.2296	0.2317	2.129	100.000 ± 0.000	1.597 ± 0.324	0.000 ± 0.000
UPI	beta34	0.000	-0.201	-0.201 ± 0.067	—	0.237 ± 0.073	0.487 ± 0.082	0.273 ± 0.092	0.522 ± 0.099	—	—	-0.099	-0.099	—	0.4772	0.4873	1.648	100.000 ± 0.000	3.143 ± 0.757	0.000 ± 0.000
UPI	beta35	0.000	-0.342	-0.342 ± 0.132	—	0.908 ± 0.437	0.953 ± 0.273	1.007 ± 0.506	1.003 ± 0.297	—	—	-0.219	-0.219	—	0.5491	0.5910	1.656	100.000 ± 0.000	6.184 ± 1.325	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S140

Simulation 1 - Condition 140: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.009	-0.016 ± 0.027	0.003 ± 0.001	0.055 ± 0.007	0.003 ± 0.001	0.055 ± 0.008	0.030 ± 0.008	0.173 ± 0.027	0.316	0.000	-0.02	0.0853	0.0853	0.986	95.122 ± 3.364	0.213 ± 0.004	—
LMS	beta32	0.316	0.312	-0.004 ± 0.009	-0.011 ± 0.030	0.004 ± 0.001	0.060 ± 0.010	0.004 ± 0.001	0.059 ± 0.011	0.035 ± 0.011	0.188 ± 0.034	0.308	-0.008	-2.58	0.0716	0.0721	0.935	92.683 ± 4.067	0.220 ± 0.005	—
LMS	beta33	0.000	0.058	0.058 ± 0.011	—	0.005 ± 0.001	0.072 ± 0.010	0.008 ± 0.002	0.092 ± 0.012	—	—	0.052	0.052	—	0.1055	0.1176	0.956	80.488 ± 6.189	0.271 ± 0.018	19.512 ± 6.189
LMS	beta34	0.000	-0.050	-0.050 ± 0.007	—	0.002 ± 0.001	0.046 ± 0.008	0.005 ± 0.001	0.067 ± 0.006	—	—	-0.046	-0.046	—	0.0776	0.0901	0.998	65.854 ± 7.406	0.180 ± 0.012	34.146 ± 7.406
LMS	beta35	0.000	-0.056	-0.056 ± 0.007	—	0.002 ± 0.000	0.047 ± 0.005	0.005 ± 0.001	0.072 ± 0.007	—	—	-0.046	-0.046	—	0.0793	0.0914	1.040	68.293 ± 7.267	0.191 ± 0.014	31.707 ± 7.267
LSAM	beta31	0.316	0.344	0.028 ± 0.017	0.089 ± 0.055	0.013 ± 0.003	0.112 ± 0.013	0.013 ± 0.003	0.114 ± 0.016	0.130 ± 0.029	0.361 ± 0.050	0.348	0.032	10.01	0.0834	0.0892	0.918	97.561 ± 2.409	0.402 ± 0.020	—
LSAM	beta32	0.316	0.355	0.039 ± 0.018	0.124 ± 0.056	0.013 ± 0.003	0.114 ± 0.016	0.014 ± 0.004	0.119 ± 0.018	0.143 ± 0.037	0.378 ± 0.058	0.348	0.032	10.00	0.0778	0.0840	0.956	90.244 ± 4.634	0.428 ± 0.022	—
LSAM	beta33	0.000	0.110	0.110 ± 0.013	—	0.007 ± 0.001	0.085 ± 0.009	0.019 ± 0.004	0.139 ± 0.014	—	—	0.111	0.111	—	0.0945	0.1454	1.045	82.927 ± 5.876	0.348 ± 0.022	17.073 ± 5.876
LSAM	beta34	0.000	-0.113	-0.113 ± 0.015	—	0.009 ± 0.002	0.094 ± 0.010	0.021 ± 0.004	0.146 ± 0.015	—	—	-0.117	-0.117	—	0.0859	0.1453	0.966	80.488 ± 6.189	0.355 ± 0.029	19.512 ± 6.189
LSAM	beta35	0.000	-0.129	-0.129 ± 0.013	—	0.007 ± 0.001	0.086 ± 0.006	0.024 ± 0.004	0.155 ± 0.012	—	—	-0.117	-0.117	—	0.0945	0.1508	1.144	73.171 ± 6.920	0.387 ± 0.024	26.829 ± 6.920
QML	beta31	0.316	0.314	-0.002 ± 0.009	-0.008 ± 0.028	0.003 ± 0.001	0.056 ± 0.007	0.003 ± 0.001	0.055 ± 0.008	0.031 ± 0.007	0.175 ± 0.027	0.322	0.006	1.82	0.0571	0.0574	0.975	95.122 ± 3.364	0.214 ± 0.003	—
QML	beta32	0.316	0.316	-0.000 ± 0.010	-0.000 ± 0.030	0.004 ± 0.001	0.062 ± 0.010	0.004 ± 0.001	0.061 ± 0.011	0.037 ± 0.012	0.192 ± 0.036	0.311	-0.005	-1.52	0.0569	0.0571	0.916	92.683 ± 4.067	0.221 ± 0.004	—
QML	beta33	0.000	0.061	0.061 ± 0.012	—	0.006 ± 0.001	0.078 ± 0.010	0.010 ± 0.002	0.098 ± 0.013	—	—	0.081	0.081	—	0.0788	0.1128	0.905	80.488 ± 6.189	0.276 ± 0.012	19.512 ± 6.189
QML	beta34	0.000	-0.064	-0.064 ± 0.007	—	0.002 ± 0.001	0.048 ± 0.007	0.006 ± 0.001	0.080 ± 0.006	—	—	-0.072	-0.072	—	0.0476	0.0862	0.953	65.854 ± 7.406	0.178 ± 0.008	34.146 ± 7.406
QML	beta35	0.000	-0.070	-0.070 ± 0.008	—	0.003 ± 0.001	0.053 ± 0.006	0.008 ± 0.001	0.088 ± 0.009	—	—	-0.072	-0.072	—	0.0556	0.0910	0.939	63.415 ± 7.522	0.195 ± 0.009	36.585 ± 7.522
UPI	beta31	0.316	0.370	0.054 ± 0.052	0.171 ± 0.165	0.112 ± 0.048	0.334 ± 0.086	0.112 ± 0.052	0.335 ± 0.093	1.121 ± 0.518	1.059 ± 0.296	0.398	0.082	25.96	0.5803	0.5861	1.387	87.805 ± 5.110	1.818 ± 0.430	—
UPI	beta32	0.316	0.278	-0.038 ± 0.067	-0.120 ± 0.212	0.185 ± 0.061	0.430 ± 0.080	0.182 ± 0.060	0.426 ± 0.083	1.818 ± 0.600	1.348 ± 0.263	0.384	0.068	21.39	0.5551	0.5592	1.070	73.171 ± 6.920	1.802 ± 0.494	—
UPI	beta33	0.000	-0.099	-0.099 ± 0.040	—	0.065 ± 0.025	0.255 ± 0.055	0.073 ± 0.031	0.270 ± 0.064	—	—	-0.037	-0.037	—	0.2910	0.2934	1.057	97.561 ± 2.409	1.056 ± 0.292	2.439 ± 2.409
UPI	beta34	0.000	-0.368	-0.368 ± 0.203	—	1.689 ± 0.825	1.299 ± 0.372	1.783 ± 0.871	1.335 ± 0.379	—	—	-0.399	-0.399	—	1.0997	1.1698	0.927	97.561 ± 2.409	4.723 ± 1.293	2.439 ± 2.409
UPI	beta35	0.000	-0.482	-0.482 ± 0.256	—	2.695 ± 1.367	1.642 ± 0.503	2.862 ± 1.565	1.692 ± 0.551	—	—	-0.557	-0.557	—	1.3769	1.4853	0.701	95.122 ± 3.364	4.510 ± 1.088	4.878 ± 3.364

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S141

Simulation 1 - Condition 141: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.003	-0.000 ± 0.009	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.049 ± 0.003	0.222 ± 0.008	0.309	-0.007	-2.27	0.0707	0.0710	0.989	94.969 ± 0.867	0.273 ± 0.002	—
LMS	beta32	0.316	0.318	0.002 ± 0.003	0.006 ± 0.009	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.047 ± 0.003	0.217 ± 0.008	0.321	0.005	1.51	0.0672	0.0673	1.007	95.283 ± 0.841	0.271 ± 0.002	—
LMS	beta33	0.000	0.011	0.011 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	0.018	0.018	—	0.0831	0.0851	1.069	92.138 ± 1.067	0.374 ± 0.010	7.862 ± 1.067
LMS	beta34	0.000	-0.012	-0.012 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	—	—	-0.015	-0.015	—	0.0704	0.0719	1.088	91.509 ± 1.105	0.304 ± 0.008	8.491 ± 1.105
LMS	beta35	0.000	-0.013	-0.013 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.074 ± 0.003	—	—	-0.020	-0.020	—	0.0696	0.0725	1.046	87.421 ± 1.315	0.298 ± 0.009	12.579 ± 1.315
LSAM	beta31	0.316	0.320	0.004 ± 0.003	0.013 ± 0.010	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.064 ± 0.004	0.252 ± 0.009	0.316	0.000	-0.12	0.0807	0.0807	0.994	95.912 ± 0.785	0.310 ± 0.002	—
LSAM	beta32	0.316	0.321	0.005 ± 0.003	0.016 ± 0.010	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.059 ± 0.003	0.243 ± 0.008	0.324	0.008	2.58	0.0720	0.0725	1.020	94.969 ± 0.867	0.307 ± 0.002	—
LSAM	beta33	0.000	0.039	0.039 ± 0.004	—	0.008 ± 0.000	0.089 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	0.037	0.037	—	0.0842	0.0921	0.989	91.981 ± 1.077	0.344 ± 0.004	8.019 ± 1.077
LSAM	beta34	0.000	-0.042	-0.042 ± 0.004	—	0.012 ± 0.001	0.108 ± 0.004	0.013 ± 0.001	0.116 ± 0.004	—	—	-0.039	-0.039	—	0.1039	0.1110	0.987	93.868 ± 0.951	0.418 ± 0.006	6.132 ± 0.951
LSAM	beta35	0.000	-0.042	-0.042 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	—	—	-0.049	-0.049	—	0.0990	0.1102	1.001	91.038 ± 1.133	0.395 ± 0.005	8.962 ± 1.133
QML	beta31	0.316	0.316	-0.000 ± 0.003	-0.000 ± 0.009	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.003	0.221 ± 0.008	0.311	-0.005	-1.62	0.0698	0.0700	0.984	95.283 ± 0.841	0.270 ± 0.002	—
QML	beta32	0.316	0.318	0.002 ± 0.003	0.006 ± 0.009	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	0.047 ± 0.003	0.217 ± 0.008	0.320	0.004	1.19	0.0655	0.0656	1.008	94.811 ± 0.879	0.271 ± 0.003	—
QML	beta33	0.000	0.015	0.015 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	—	—	0.019	0.019	—	0.0793	0.0815	1.045	91.824 ± 1.086	0.347 ± 0.016	8.176 ± 1.086
QML	beta34	0.000	-0.020	-0.020 ± 0.003	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.020	-0.020	—	0.0674	0.0703	1.043	91.195 ± 1.124	0.276 ± 0.005	8.805 ± 1.124
QML	beta35	0.000	-0.021	-0.021 ± 0.003	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.026	-0.026	—	0.0676	0.0724	1.077	87.264 ± 1.322	0.285 ± 0.021	12.736 ± 1.322
UPI	beta31	0.316	0.326	0.010 ± 0.004	0.031 ± 0.013	0.011 ± 0.001	0.106 ± 0.004	0.011 ± 0.001	0.106 ± 0.004	0.113 ± 0.008	0.336 ± 0.014	0.318	0.002	0.61	0.1041	0.1041	1.408	97.642 ± 0.602	0.585 ± 0.031	—
UPI	beta32	0.316	0.333	0.017 ± 0.004	0.054 ± 0.013	0.011 ± 0.001	0.107 ± 0.004	0.012 ± 0.001	0.108 ± 0.005	0.117 ± 0.009	0.342 ± 0.014	0.332	0.016	5.17	0.1028	0.1041	1.394	97.170 ± 0.658	0.584 ± 0.033	—
UPI	beta33	0.000	0.044	0.044 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	—	—	0.039	0.039	—	0.0919	0.0997	1.206	96.541 ± 0.725	0.476 ± 0.019	3.459 ± 0.725
UPI	beta34	0.000	-0.043	-0.043 ± 0.007	—	0.034 ± 0.003	0.184 ± 0.007	0.036 ± 0.003	0.189 ± 0.008	—	—	-0.037	-0.037	—	0.1644	0.1686	1.312	96.698 ± 0.709	0.947 ± 0.044	3.302 ± 0.709
UPI	beta35	0.000	-0.059	-0.059 ± 0.007	—	0.034 ± 0.003	0.186 ± 0.008	0.038 ± 0.003	0.195 ± 0.009	—	—	-0.052	-0.052	—	0.1676	0.1755	1.365	97.642 ± 0.602	0.993 ± 0.071	2.358 ± 0.602

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S142

Simulation 1 - Condition 142: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.005	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.127 ± 0.004	0.318	0.002	0.59	0.0388	0.0389	1.063	96.505 ± 0.673	0.167 ± 0.001	—
LMS	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.315	-0.001	-0.40	0.0429	0.0429	1.000	95.027 ± 0.797	0.167 ± 0.001	—
LMS	beta33	0.000	0.019	0.019 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.021	0.021	—	0.0506	0.0547	1.046	90.860 ± 1.056	0.212 ± 0.004	9.140 ± 1.056
LMS	beta34	0.000	-0.026	-0.026 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.029	-0.029	—	0.0409	0.0503	1.061	83.737 ± 1.353	0.176 ± 0.008	16.263 ± 1.353
LMS	beta35	0.000	-0.023	-0.023 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.026	-0.026	—	0.0397	0.0474	0.999	86.290 ± 1.261	0.169 ± 0.002	13.710 ± 1.261
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.005	0.322	0.006	1.74	0.0435	0.0438	1.064	95.699 ± 0.744	0.190 ± 0.001	—
LSAM	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.024 ± 0.001	0.154 ± 0.005	0.318	0.002	0.74	0.0481	0.0482	0.998	94.624 ± 0.827	0.191 ± 0.001	—
LSAM	beta33	0.000	0.037	0.037 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.037	0.037	—	0.0503	0.0626	1.021	88.172 ± 1.184	0.207 ± 0.001	11.828 ± 1.184
LSAM	beta34	0.000	-0.052	-0.052 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	—	—	-0.051	-0.051	—	0.0567	0.0762	0.992	85.349 ± 1.296	0.245 ± 0.002	14.651 ± 1.296
LSAM	beta35	0.000	-0.047	-0.047 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	—	—	-0.049	-0.049	—	0.0620	0.0788	0.967	86.156 ± 1.266	0.246 ± 0.002	13.844 ± 1.266
QML	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.005	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.127 ± 0.004	0.318	0.002	0.62	0.0394	0.0395	1.059	96.102 ± 0.710	0.166 ± 0.000	—
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.316	0.000	-0.16	0.0430	0.0430	0.997	94.892 ± 0.807	0.167 ± 0.001	—
QML	beta33	0.000	0.019	0.019 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.020	0.020	—	0.0511	0.0548	1.021	91.667 ± 1.013	0.206 ± 0.002	8.333 ± 1.013
QML	beta34	0.000	-0.032	-0.032 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.034	-0.034	—	0.0416	0.0536	1.004	83.333 ± 1.366	0.165 ± 0.001	16.667 ± 1.366
QML	beta35	0.000	-0.029	-0.029 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.030	-0.030	—	0.0393	0.0497	0.993	85.618 ± 1.286	0.166 ± 0.001	14.382 ± 1.286
UPI	beta31	0.316	0.327	0.011 ± 0.003	0.034 ± 0.008	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.049 ± 0.003	0.221 ± 0.008	0.327	0.011	3.36	0.0648	0.0657	1.394	98.387 ± 0.462	0.377 ± 0.015	—
UPI	beta32	0.316	0.328	0.012 ± 0.003	0.039 ± 0.008	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.003	0.221 ± 0.007	0.327	0.011	3.43	0.0664	0.0673	1.437	98.253 ± 0.480	0.387 ± 0.018	—
UPI	beta33	0.000	0.049	0.049 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.046	0.046	—	0.0622	0.0777	1.201	92.876 ± 0.943	0.290 ± 0.007	7.124 ± 0.943
UPI	beta34	0.000	-0.064	-0.064 ± 0.004	—	0.014 ± 0.001	0.117 ± 0.004	0.018 ± 0.001	0.133 ± 0.005	—	—	-0.061	-0.061	—	0.0954	0.1132	1.289	95.430 ± 0.766	0.590 ± 0.028	4.570 ± 0.766
UPI	beta35	0.000	-0.064	-0.064 ± 0.004	—	0.014 ± 0.001	0.119 ± 0.004	0.018 ± 0.001	0.135 ± 0.005	—	—	-0.065	-0.065	—	0.1046	0.1231	1.225	94.892 ± 0.807	0.572 ± 0.025	5.108 ± 0.807

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S143

Simulation 1 - Condition 143: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.318	0.002	0.74	0.0555	0.0556	0.999	94.689 ± 0.710	0.219 ± 0.001	—
LMS	beta32	0.316	0.319	0.003 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.314	-0.002	-0.49	0.0554	0.0555	0.984	94.088 ± 0.747	0.218 ± 0.001	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.001	0.001	—	0.0642	0.0642	0.985	94.890 ± 0.697	0.257 ± 0.003	5.110 ± 0.697
LMS	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.008	-0.008	—	0.0597	0.0601	0.964	93.487 ± 0.781	0.234 ± 0.002	6.513 ± 0.781
LMS	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.008	-0.008	—	0.0590	0.0595	0.984	93.086 ± 0.803	0.236 ± 0.003	6.914 ± 0.803
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.320	0.004	1.14	0.0576	0.0577	0.998	94.689 ± 0.710	0.225 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.316	0.000	-0.16	0.0588	0.0588	0.984	93.888 ± 0.758	0.225 ± 0.001	—
LSAM	beta33	0.000	0.006	0.006 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.006	0.006	—	0.0639	0.0641	0.980	93.988 ± 0.752	0.247 ± 0.002	6.012 ± 0.752
LSAM	beta34	0.000	-0.009	-0.009 ± 0.002	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.012	-0.012	—	0.0727	0.0738	0.955	93.287 ± 0.792	0.278 ± 0.001	6.713 ± 0.792
LSAM	beta35	0.000	-0.011	-0.011 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	-0.011	-0.011	—	0.0726	0.0734	0.967	93.086 ± 0.803	0.279 ± 0.001	6.914 ± 0.803
QML	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.319	0.003	0.80	0.0559	0.0560	0.999	94.689 ± 0.710	0.219 ± 0.001	—
QML	beta32	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.179 ± 0.005	0.314	-0.002	-0.58	0.0557	0.0557	0.983	94.188 ± 0.741	0.218 ± 0.001	—
QML	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.001	0.001	—	0.0644	0.0644	0.981	94.890 ± 0.697	0.252 ± 0.002	5.110 ± 0.697
QML	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.008	-0.008	—	0.0598	0.0603	0.964	93.387 ± 0.787	0.230 ± 0.002	6.613 ± 0.787
QML	beta35	0.000	-0.006	-0.006 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.009	-0.009	—	0.0583	0.0590	0.974	93.086 ± 0.803	0.230 ± 0.002	6.914 ± 0.803
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.183 ± 0.005	0.319	0.003	1.10	0.0576	0.0577	0.986	94.289 ± 0.735	0.223 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.316	0.000	0.04	0.0570	0.0570	0.980	93.788 ± 0.764	0.223 ± 0.001	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	0.005	0.005	—	0.0648	0.0650	0.967	93.988 ± 0.752	0.249 ± 0.002	6.012 ± 0.752
UPI	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.004	-0.004	—	0.0766	0.0767	0.966	94.489 ± 0.722	0.296 ± 0.002	5.511 ± 0.722
UPI	beta35	0.000	-0.007	-0.007 ± 0.002	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.007	-0.007	—	0.0769	0.0773	0.957	94.088 ± 0.747	0.294 ± 0.002	5.912 ± 0.747

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S144

Simulation 1 - Condition 144: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.315	-0.001	-0.31	0.0328	0.0328	0.995	95.200 ± 0.676	0.136 ± 0.000	—
LMS	beta32	0.316	0.316	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.112 ± 0.003	0.316	0.000	0.03	0.0351	0.0351	0.987	95.200 ± 0.676	0.137 ± 0.000	—
LMS	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0397	0.0398	0.995	94.500 ± 0.721	0.157 ± 0.001	5.500 ± 0.721
LMS	beta34	0.000	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.009	-0.009	—	0.0364	0.0375	1.017	94.200 ± 0.739	0.144 ± 0.001	5.800 ± 0.739
LMS	beta35	0.000	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.009	-0.009	—	0.0353	0.0365	0.988	92.400 ± 0.838	0.144 ± 0.001	7.600 ± 0.838
LSAM	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.315	-0.001	-0.21	0.0337	0.0337	0.997	94.900 ± 0.696	0.140 ± 0.000	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.317	0.001	0.21	0.0353	0.0353	0.993	95.400 ± 0.662	0.141 ± 0.000	—
LSAM	beta33	0.000	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.008	0.008	—	0.0389	0.0396	0.997	94.000 ± 0.751	0.155 ± 0.001	6.000 ± 0.751
LSAM	beta34	0.000	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.012	-0.012	—	0.0448	0.0463	1.009	94.800 ± 0.702	0.176 ± 0.001	5.200 ± 0.702
LSAM	beta35	0.000	-0.011	-0.011 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.011	-0.011	—	0.0447	0.0461	0.987	92.900 ± 0.812	0.176 ± 0.001	7.100 ± 0.812
QML	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.315	-0.001	-0.26	0.0330	0.0330	0.995	95.200 ± 0.676	0.136 ± 0.000	—
QML	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.112 ± 0.003	0.316	0.000	0.03	0.0352	0.0352	0.988	95.200 ± 0.676	0.137 ± 0.000	—
QML	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0392	0.0393	0.997	94.600 ± 0.715	0.157 ± 0.001	5.400 ± 0.715
QML	beta34	0.000	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.010	-0.010	—	0.0364	0.0377	1.017	94.200 ± 0.739	0.143 ± 0.001	5.800 ± 0.739
QML	beta35	0.000	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.010	-0.010	—	0.0351	0.0365	0.994	92.100 ± 0.853	0.144 ± 0.001	7.900 ± 0.853
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.316	0.000	-0.12	0.0335	0.0335	0.991	94.600 ± 0.715	0.140 ± 0.000	—
UPI	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.316	0.000	0.04	0.0352	0.0352	0.986	95.000 ± 0.689	0.140 ± 0.000	—
UPI	beta33	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0391	0.0392	0.993	94.600 ± 0.715	0.156 ± 0.001	5.400 ± 0.715
UPI	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.006	-0.006	—	0.0466	0.0469	1.001	95.000 ± 0.689	0.181 ± 0.001	5.000 ± 0.689
UPI	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.004	-0.004	—	0.0455	0.0457	0.986	94.100 ± 0.745	0.182 ± 0.001	5.900 ± 0.745

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Complete Results Tables — Simulation 2

Table S145

Simulation 2 - Condition 1: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.004	0.009 ± 0.018	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.240 ± 0.014	0.490 ± 0.017	0.201	0.001	0.49	0.0946	0.0946	1.003	96.399 ± 0.693	0.385 ± 0.004	56.233 ± 1.846
LSAM	beta42	0.20	0.200	0.000 ± 0.004	0.000 ± 0.018	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.238 ± 0.014	0.488 ± 0.017	0.197	-0.003	-1.38	0.0964	0.0964	1.033	96.953 ± 0.640	0.396 ± 0.004	55.125 ± 1.851
LSAM	beta43	0.20	0.208	0.008 ± 0.003	0.041 ± 0.016	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.187 ± 0.010	0.432 ± 0.014	0.207	0.007	3.38	0.0887	0.0889	1.047	96.676 ± 0.667	0.354 ± 0.003	67.590 ± 1.742
LSAM	beta44	0.11	0.110	0.000 ± 0.005	0.004 ± 0.046	0.018 ± 0.001	0.136 ± 0.005	0.018 ± 0.001	0.136 ± 0.005	1.523 ± 0.104	1.234 ± 0.048	0.105	-0.005	-4.63	0.1239	0.1240	1.092	97.922 ± 0.531	0.581 ± 0.010	10.803 ± 1.155
LSAM	beta45	0.11	0.112	0.002 ± 0.004	0.017 ± 0.038	0.012 ± 0.001	0.112 ± 0.004	0.012 ± 0.001	0.112 ± 0.004	1.031 ± 0.068	1.016 ± 0.039	0.109	-0.001	-1.23	0.0987	0.0987	1.053	97.507 ± 0.580	0.461 ± 0.006	12.742 ± 1.241
LSAM	beta46	0.08	0.095	0.015 ± 0.004	0.184 ± 0.047	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	1.606 ± 0.103	1.267 ± 0.047	0.087	0.007	9.30	0.0986	0.0989	1.027	97.230 ± 0.611	0.404 ± 0.007	12.327 ± 1.223
LSAM	beta47	0.08	0.085	0.005 ± 0.004	0.066 ± 0.044	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	1.398 ± 0.092	1.182 ± 0.045	0.077	-0.003	-3.17	0.0919	0.0920	1.064	96.814 ± 0.654	0.394 ± 0.008	11.634 ± 1.193
LSAM	beta51	0.16	0.157	-0.003 ± 0.005	-0.020 ± 0.032	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.137 ± 0.005	0.732 ± 0.043	0.856 ± 0.030	0.154	-0.006	-3.95	0.1248	0.1250	0.877	93.629 ± 0.909	0.471 ± 0.004	28.809 ± 1.685
LSAM	beta52	0.16	0.164	0.004 ± 0.004	0.027 ± 0.022	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.361 ± 0.020	0.601 ± 0.020	0.167	0.007	4.64	0.0931	0.0934	1.013	95.429 ± 0.777	0.382 ± 0.003	40.720 ± 1.828
LSAM	beta53	0.16	0.168	0.008 ± 0.004	0.050 ± 0.023	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.387 ± 0.020	0.622 ± 0.020	0.166	0.006	3.44	0.1004	0.1005	1.024	97.507 ± 0.580	0.399 ± 0.003	38.366 ± 1.810
LSAM	beta54	0.16	0.164	0.004 ± 0.004	0.027 ± 0.023	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.370 ± 0.022	0.608 ± 0.021	0.160	0.000	-0.25	0.0920	0.0920	0.972	95.568 ± 0.766	0.371 ± 0.003	43.075 ± 1.843
LSAM	beta55	0.08	0.084	0.004 ± 0.006	0.052 ± 0.070	0.023 ± 0.001	0.151 ± 0.005	0.023 ± 0.001	0.151 ± 0.006	3.560 ± 0.229	1.887 ± 0.070	0.076	-0.004	-4.63	0.1436	0.1436	1.044	97.922 ± 0.531	0.618 ± 0.011	6.648 ± 0.927
LSAM	beta56	0.08	0.086	0.006 ± 0.005	0.073 ± 0.065	0.020 ± 0.001	0.140 ± 0.004	0.020 ± 0.001	0.140 ± 0.005	3.066 ± 0.193	1.751 ± 0.064	0.091	0.011	14.10	0.1319	0.1324	0.988	96.122 ± 0.719	0.543 ± 0.008	8.587 ± 1.043
LSAM	beta57	0.06	0.071	0.011 ± 0.004	0.191 ± 0.064	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	3.012 ± 0.183	1.736 ± 0.062	0.072	0.012	19.42	0.0961	0.0968	0.985	96.537 ± 0.680	0.400 ± 0.007	8.587 ± 1.043
LSAM	beta58	0.06	0.059	-0.001 ± 0.003	-0.023 ± 0.048	0.006 ± 0.000	0.077 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	1.648 ± 0.109	1.284 ± 0.049	0.059	-0.001	-1.50	0.0688	0.0688	1.029	97.645 ± 0.564	0.311 ± 0.004	7.479 ± 0.979
QML	beta41	0.20	0.183	-0.017 ± 0.003	-0.083 ± 0.016	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.186 ± 0.010	0.431 ± 0.014	0.183	-0.017	-8.44	0.0823	0.0840	0.962	92.936 ± 0.954	0.319 ± 0.002	63.850 ± 1.788
QML	beta42	0.20	0.187	-0.013 ± 0.003	-0.064 ± 0.016	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.177 ± 0.009	0.421 ± 0.014	0.184	-0.016	-7.79	0.0790	0.0805	1.017	95.152 ± 0.799	0.332 ± 0.003	62.188 ± 1.805
QML	beta43	0.20	0.187	-0.013 ± 0.003	-0.064 ± 0.014	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.153 ± 0.008	0.391 ± 0.012	0.187	-0.013	-6.29	0.0779	0.0789	1.035	94.598 ± 0.841	0.313 ± 0.002	68.144 ± 1.734
QML	beta44	0.11	0.096	-0.014 ± 0.004	-0.125 ± 0.036	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.974 ± 0.053	0.987 ± 0.032	0.094	-0.016	-14.17	0.1116	0.1127	1.089	95.429 ± 0.777	0.460 ± 0.006	13.019 ± 1.252
QML	beta45	0.11	0.102	-0.008 ± 0.003	-0.071 ± 0.030	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.655 ± 0.035	0.809 ± 0.026	0.101	-0.009	-8.58	0.0899	0.0904	1.070	96.537 ± 0.680	0.372 ± 0.004	18.144 ± 1.434
QML	beta46	0.08	0.087	0.007 ± 0.003	0.082 ± 0.036	0.006 ± 0.000	0.078 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	0.957 ± 0.065	0.978 ± 0.038	0.082	0.002	2.30	0.0731	0.0731	1.017	95.706 ± 0.754	0.311 ± 0.005	17.729 ± 1.421
QML	beta47	0.08	0.077	-0.003 ± 0.003	-0.040 ± 0.034	0.005 ± 0.000	0.074 ± 0.003	0.005 ± 0.000	0.074 ± 0.003	0.855 ± 0.058	0.924 ± 0.036	0.076	-0.004	-5.24	0.0674	0.0675	1.030	95.014 ± 0.810	0.299 ± 0.004	16.898 ± 1.395
QML	beta51	0.16	0.180	0.020 ± 0.005	0.126 ± 0.031	0.017 ± 0.001	0.132 ± 0.004	0.018 ± 0.001	0.133 ± 0.005	0.691 ± 0.040	0.831 ± 0.029	0.176	0.016	9.92	0.1256	0.1266	0.962	95.429 ± 0.777	0.496 ± 0.005	31.440 ± 1.728
QML	beta52	0.16	0.151	-0.009 ± 0.004	-0.059 ± 0.024	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.410 ± 0.023	0.640 ± 0.022	0.153	-0.007	-4.51	0.0954	0.0957	1.033	94.875 ± 0.821	0.413 ± 0.005	33.518 ± 1.757
QML	beta53	0.16	0.137	-0.023 ± 0.004	-0.144 ± 0.022	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.366 ± 0.019	0.605 ± 0.019	0.135	-0.025	-15.63	0.0975	0.1006	1.060	95.706 ± 0.754	0.391 ± 0.003	28.116 ± 1.673
QML	beta54	0.16	0.161	0.001 ± 0.003	0.009 ± 0.020	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.287 ± 0.016	0.536 ± 0.018	0.159	-0.001	-0.76	0.0859	0.0859	1.008	96.122 ± 0.719	0.339 ± 0.002	47.230 ± 1.858
QML	beta55	0.08	0.091	0.011 ± 0.005	0.132 ± 0.062	0.018 ± 0.001	0.134 ± 0.004	0.018 ± 0.001	0.135 ± 0.004	2.832 ± 0.157	1.683 ± 0.056	0.092	0.012	15.25	0.1378	0.1383	1.069	96.399 ± 0.693	0.563 ± 0.008	9.834 ± 1.108
QML	beta56	0.08	0.083	0.003 ± 0.004	0.034 ± 0.056	0.014 ± 0.001	0.120 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	2.230 ± 0.127	1.493 ± 0.051	0.089	0.009	11.65	0.1207	0.1210	1.050	95.152 ± 0.799	0.492 ± 0.006	9.280 ± 1.080
QML	beta57	0.06	0.051	-0.009 ± 0.003	-0.155 ± 0.051	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	1.868 ± 0.104	1.367 ± 0.046	0.049	-0.011	-18.49	0.0800	0.0808	1.003	93.075 ± 0.945	0.320 ± 0.005	9.972 ± 1.115
QML	beta58	0.06	0.047	-0.013 ± 0.002	-0.219 ± 0.039	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	1.134 ± 0.069	1.065 ± 0.038	0.051	-0.009	-15.61	0.0567	0.0574	1.013	94.460 ± 0.851	0.248 ± 0.003	10.803 ± 1.155
UPI	beta41	0.20	0.206	0.006 ± 0.004	0.029 ± 0.020	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	0.278 ± 0.017	0.527 ± 0.019	0.205	0.005	2.28	0.1044	0.1045	1.221	97.368 ± 0.596	0.504 ± 0.010	42.936 ± 1.842
UPI	beta42	0.20	0.200	-0.000 ± 0.004	-0.000 ± 0.020	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	0.278 ± 0.016	0.527 ± 0.018	0.203	0.003	1.55	0.1013	0.1013	1.252	98.061 ± 0.513	0.518 ± 0.012	41.967 ± 1.837
UPI	beta43	0.20	0.208	0.008 ± 0.003	0.040 ± 0.017	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	0.212 ± 0.012	0.460 ± 0.016	0.206	0.006	2.79	0.0961	0.0962	1.218	98.338 ± 0.476	0.438 ± 0.007	52.216 ± 1.859
UPI	beta44	0.11	0.129	0.019 ± 0.007	0.174 ± 0.063	0.034 ± 0.002	0.186 ± 0.007	0.035 ± 0.003	0.187 ± 0.008	2.876 ± 0.210	1.696 ± 0.070	0.109	-0.001	-0.61	0.1683	0.1683	1.484	99.446 ± 0.276	1.080 ± 0.039	3.878 ± 0.719
UPI	beta45	0.11	0.119	0.009 ± 0.005	0.077 ± 0.049	0.021 ± 0.002	0.144 ± 0.005	0.021 ± 0.002	0.144 ± 0.006	1.706 ± 0.127	1.306 ± 0.055	0.111	0.001	0.97	0.1228	0.1228	1.481	98.892 ± 0.390	0.833 ± 0.029	3.324 ± 0.667
UPI	beta46	0.08	0.089	0.009 ± 0.005	0.113 ± 0.067	0.021 ± 0.002	0.144 ± 0.006	0.021 ± 0.002	0.144 ± 0.006	3.236 ± 0.256	1.799 ± 0.079	0.086	0.006	7.14	0.1223	0.1224	1.313	98.753 ± 0.413	0.740 ± 0.025	2.770 ± 0.611

Table S145

Simulation 2 - Condition 1: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.086	0.006 ± 0.005	0.080 ± 0.058	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.125 ± 0.005	2.448 ± 0.191	1.565 ± 0.068	0.073	-0.007	-8.30	0.1068	0.1070	1.416	98.338 ± 0.476	0.694 ± 0.025	3.601 ± 0.693
UPI	beta51	0.16	0.166	0.006 ± 0.005	0.040 ± 0.030	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	0.664 ± 0.039	0.815 ± 0.029	0.162	0.002	1.25	0.1196	0.1196	1.032	96.399 ± 0.693	0.527 ± 0.007	23.961 ± 1.589
UPI	beta52	0.16	0.166	0.006 ± 0.004	0.037 ± 0.022	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.363 ± 0.021	0.602 ± 0.021	0.173	0.013	8.08	0.0944	0.0953	1.175	96.953 ± 0.640	0.443 ± 0.008	33.380 ± 1.755
UPI	beta53	0.16	0.164	0.004 ± 0.004	0.028 ± 0.023	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.376 ± 0.019	0.613 ± 0.019	0.166	0.006	3.86	0.1004	0.1006	1.214	98.061 ± 0.513	0.467 ± 0.007	30.194 ± 1.709
UPI	beta54	0.16	0.162	0.002 ± 0.004	0.013 ± 0.022	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.364 ± 0.020	0.603 ± 0.020	0.158	-0.002	-0.99	0.0924	0.0924	1.125	96.814 ± 0.654	0.426 ± 0.006	33.795 ± 1.760
UPI	beta55	0.08	0.089	0.009 ± 0.006	0.118 ± 0.080	0.030 ± 0.002	0.173 ± 0.006	0.030 ± 0.002	0.173 ± 0.007	4.674 ± 0.321	2.162 ± 0.085	0.083	0.003	3.65	0.1525	0.1526	1.395	99.030 ± 0.365	0.945 ± 0.038	2.493 ± 0.580
UPI	beta56	0.08	0.080	-0.000 ± 0.006	-0.004 ± 0.074	0.025 ± 0.002	0.159 ± 0.006	0.025 ± 0.002	0.159 ± 0.007	3.936 ± 0.298	1.984 ± 0.084	0.078	-0.002	-2.77	0.1390	0.1391	1.367	98.615 ± 0.435	0.851 ± 0.033	1.801 ± 0.495
UPI	beta57	0.06	0.069	0.009 ± 0.004	0.153 ± 0.072	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.117 ± 0.005	3.809 ± 0.283	1.952 ± 0.081	0.068	0.008	13.77	0.1074	0.1077	1.330	97.645 ± 0.564	0.609 ± 0.027	2.632 ± 0.596
UPI	beta58	0.06	0.062	0.002 ± 0.003	0.038 ± 0.055	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	2.213 ± 0.157	1.488 ± 0.060	0.058	-0.002	-3.00	0.0826	0.0826	1.423	97.784 ± 0.548	0.498 ± 0.017	2.355 ± 0.564

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S146

Simulation 2 - Condition 2: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.007 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.003	0.271 ± 0.008	0.200	0.000	0.20	0.0574	0.0574	1.023	96.443 ± 0.590	0.218 ± 0.001	96.748 ± 0.565
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.075 ± 0.004	0.274 ± 0.008	0.201	0.001	0.42	0.0545	0.0545	1.040	95.020 ± 0.693	0.223 ± 0.001	96.748 ± 0.565
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.066 ± 0.003	0.257 ± 0.008	0.198	-0.002	-1.02	0.0481	0.0481	1.012	95.528 ± 0.659	0.204 ± 0.001	98.272 ± 0.415
LSAM	beta44	0.11	0.116	0.006 ± 0.003	0.055 ± 0.023	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.511 ± 0.027	0.715 ± 0.022	0.116	0.006	5.61	0.0745	0.0747	0.990	96.443 ± 0.590	0.304 ± 0.002	35.772 ± 1.528
LSAM	beta45	0.11	0.112	0.002 ± 0.002	0.021 ± 0.018	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.330 ± 0.016	0.575 ± 0.016	0.110	0.000	0.01	0.0575	0.0575	1.014	96.138 ± 0.614	0.251 ± 0.002	42.073 ± 1.574
LSAM	beta46	0.08	0.083	0.003 ± 0.002	0.041 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.451 ± 0.023	0.672 ± 0.021	0.082	0.002	2.55	0.0527	0.0527	1.020	95.528 ± 0.659	0.215 ± 0.002	32.825 ± 1.497
LSAM	beta47	0.08	0.084	0.004 ± 0.002	0.045 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.417 ± 0.020	0.646 ± 0.019	0.082	0.002	2.74	0.0478	0.0479	1.023	96.138 ± 0.614	0.207 ± 0.002	36.585 ± 1.536
LSAM	beta51	0.16	0.156	-0.004 ± 0.002	-0.028 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.010	0.449 ± 0.013	0.153	-0.007	-4.08	0.0682	0.0685	0.983	94.919 ± 0.700	0.277 ± 0.001	59.858 ± 1.563
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.006	0.345 ± 0.010	0.163	0.003	1.59	0.0526	0.0527	1.003	95.122 ± 0.687	0.217 ± 0.001	84.146 ± 1.164
LSAM	beta53	0.16	0.163	0.003 ± 0.002	0.020 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.006	0.363 ± 0.010	0.162	0.002	1.00	0.0571	0.0571	1.015	95.122 ± 0.687	0.231 ± 0.001	81.402 ± 1.240
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.012 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.006	0.342 ± 0.010	0.161	0.001	0.35	0.0541	0.0541	1.002	95.325 ± 0.673	0.215 ± 0.001	85.467 ± 1.124
LSAM	beta55	0.08	0.083	0.003 ± 0.003	0.036 ± 0.034	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	1.130 ± 0.053	1.063 ± 0.030	0.080	0.000	-0.60	0.0826	0.0826	0.961	94.919 ± 0.700	0.321 ± 0.002	19.207 ± 1.256
LSAM	beta56	0.08	0.081	0.001 ± 0.002	0.008 ± 0.031	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.949 ± 0.045	0.974 ± 0.028	0.080	0.000	-0.42	0.0756	0.0756	0.972	95.224 ± 0.680	0.297 ± 0.002	19.004 ± 1.251
LSAM	beta57	0.06	0.067	0.007 ± 0.002	0.110 ± 0.029	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.811 ± 0.046	0.900 ± 0.029	0.064	0.004	6.55	0.0517	0.0519	0.996	96.748 ± 0.565	0.209 ± 0.002	21.443 ± 1.308
LSAM	beta58	0.06	0.063	0.003 ± 0.001	0.042 ± 0.023	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.534 ± 0.027	0.730 ± 0.022	0.062	0.002	2.67	0.0418	0.0418	1.015	95.833 ± 0.637	0.174 ± 0.001	28.760 ± 1.443
QML	beta41	0.20	0.184	-0.016 ± 0.002	-0.082 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.256 ± 0.007	0.183	-0.017	-8.25	0.0509	0.0535	1.016	93.598 ± 0.780	0.194 ± 0.001	97.459 ± 0.502
QML	beta42	0.20	0.190	-0.010 ± 0.002	-0.049 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.189	-0.011	-5.61	0.0488	0.0501	1.040	95.732 ± 0.644	0.200 ± 0.001	98.069 ± 0.439
QML	beta43	0.20	0.182	-0.018 ± 0.002	-0.090 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.181	-0.019	-9.44	0.0472	0.0508	1.018	93.089 ± 0.809	0.189 ± 0.001	97.967 ± 0.450
QML	beta44	0.11	0.103	-0.007 ± 0.002	-0.062 ± 0.019	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.375 ± 0.018	0.612 ± 0.018	0.103	-0.007	-6.16	0.0680	0.0683	1.017	95.224 ± 0.680	0.267 ± 0.002	34.146 ± 1.512
QML	beta45	0.11	0.107	-0.003 ± 0.002	-0.032 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.261 ± 0.012	0.511 ± 0.014	0.108	-0.002	-2.05	0.0568	0.0568	1.008	94.512 ± 0.726	0.222 ± 0.001	47.459 ± 1.592
QML	beta46	0.08	0.078	-0.002 ± 0.001	-0.020 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.287 ± 0.014	0.536 ± 0.015	0.079	-0.001	-1.07	0.0412	0.0412	1.067	95.224 ± 0.680	0.179 ± 0.001	41.362 ± 1.570
QML	beta47	0.08	0.077	-0.003 ± 0.001	-0.041 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.288 ± 0.013	0.537 ± 0.015	0.075	-0.005	-5.66	0.0407	0.0409	1.031	95.427 ± 0.666	0.173 ± 0.001	41.362 ± 1.570
QML	beta51	0.16	0.176	0.016 ± 0.002	0.102 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.213 ± 0.010	0.462 ± 0.013	0.175	0.015	9.12	0.0707	0.0722	1.019	95.122 ± 0.687	0.288 ± 0.001	68.496 ± 1.481
QML	beta52	0.16	0.148	-0.012 ± 0.002	-0.077 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.007	0.386 ± 0.011	0.151	-0.009	-5.78	0.0564	0.0572	1.003	94.106 ± 0.751	0.238 ± 0.001	70.122 ± 1.459
QML	beta53	0.16	0.137	-0.023 ± 0.002	-0.146 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.143 ± 0.006	0.379 ± 0.010	0.136	-0.024	-14.79	0.0552	0.0601	1.040	94.004 ± 0.757	0.228 ± 0.001	66.565 ± 1.504
QML	beta54	0.16	0.160	0.000 ± 0.002	0.000 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.159	-0.001	-0.63	0.0519	0.0519	1.002	94.411 ± 0.732	0.204 ± 0.001	88.415 ± 1.020
QML	beta55	0.08	0.094	0.014 ± 0.003	0.171 ± 0.033	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	1.094 ± 0.048	1.046 ± 0.028	0.094	0.014	17.13	0.0837	0.0848	1.003	94.817 ± 0.707	0.325 ± 0.002	22.866 ± 1.339
QML	beta56	0.08	0.075	-0.005 ± 0.002	-0.068 ± 0.029	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.858 ± 0.039	0.926 ± 0.026	0.073	-0.007	-8.96	0.0713	0.0717	0.981	94.411 ± 0.732	0.284 ± 0.002	18.089 ± 1.227
QML	beta57	0.06	0.050	-0.010 ± 0.001	-0.170 ± 0.025	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.635 ± 0.030	0.797 ± 0.022	0.048	-0.012	-19.67	0.0454	0.0469	1.017	93.394 ± 0.792	0.186 ± 0.002	17.683 ± 1.216
QML	beta58	0.06	0.051	-0.009 ± 0.001	-0.149 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.425 ± 0.020	0.652 ± 0.018	0.051	-0.009	-15.74	0.0373	0.0384	1.012	93.902 ± 0.763	0.151 ± 0.001	26.626 ± 1.409
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.075 ± 0.004	0.275 ± 0.008	0.201	0.001	0.36	0.0553	0.0553	1.045	95.528 ± 0.659	0.225 ± 0.001	95.833 ± 0.637
UPI	beta42	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.077 ± 0.004	0.277 ± 0.008	0.200	0.000	0.21	0.0552	0.0552	1.067	95.630 ± 0.652	0.231 ± 0.001	95.528 ± 0.659
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.003	0.258 ± 0.008	0.199	-0.001	-0.45	0.0488	0.0488	1.045	95.935 ± 0.630	0.212 ± 0.001	97.764 ± 0.471
UPI	beta44	0.11	0.120	0.010 ± 0.003	0.091 ± 0.025	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.635 ± 0.036	0.797 ± 0.026	0.119	0.009	7.78	0.0770	0.0775	1.075	97.764 ± 0.471	0.367 ± 0.005	26.931 ± 1.414
UPI	beta45	0.11	0.114	0.004 ± 0.002	0.038 ± 0.020	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.397 ± 0.019	0.630 ± 0.018	0.110	0.000	-0.05	0.0662	0.0662	1.101	96.748 ± 0.565	0.299 ± 0.003	33.130 ± 1.500
UPI	beta46	0.08	0.082	0.002 ± 0.002	0.019 ± 0.024	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.581 ± 0.033	0.762 ± 0.025	0.079	-0.001	-1.72	0.0561	0.0561	1.092	97.358 ± 0.511	0.261 ± 0.004	21.545 ± 1.311

Table S146

Simulation 2 - Condition 2: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.084	0.004 ± 0.002	0.053 ± 0.022	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.496 ± 0.027	0.704 ± 0.022	0.081	0.001	1.09	0.0537	0.0537	1.132	97.764 ± 0.471	0.249 ± 0.003	23.577 ± 1.353
UPI	beta51	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.009	0.437 ± 0.013	0.155	-0.005	-3.00	0.0676	0.0678	1.020	95.732 ± 0.644	0.279 ± 0.001	61.585 ± 1.551
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.014 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.006	0.345 ± 0.010	0.164	0.004	2.38	0.0542	0.0543	1.022	95.935 ± 0.630	0.221 ± 0.001	82.927 ± 1.200
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.014 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.127 ± 0.006	0.356 ± 0.010	0.161	0.001	0.75	0.0570	0.0570	1.056	95.833 ± 0.637	0.236 ± 0.001	79.472 ± 1.288
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.010 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.006	0.339 ± 0.010	0.160	0.000	0.23	0.0539	0.0539	1.032	96.341 ± 0.598	0.219 ± 0.001	84.451 ± 1.155
UPI	beta55	0.08	0.085	0.005 ± 0.003	0.061 ± 0.035	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	1.183 ± 0.062	1.087 ± 0.033	0.080	0.000	-0.06	0.0847	0.0847	1.044	97.561 ± 0.492	0.356 ± 0.004	13.516 ± 1.090
UPI	beta56	0.08	0.080	0.000 ± 0.003	0.001 ± 0.032	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.989 ± 0.051	0.994 ± 0.030	0.077	-0.003	-3.20	0.0738	0.0739	1.074	96.951 ± 0.548	0.335 ± 0.004	12.805 ± 1.065
UPI	beta57	0.06	0.066	0.006 ± 0.002	0.098 ± 0.030	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.900 ± 0.051	0.949 ± 0.031	0.062	0.002	3.40	0.0537	0.0537	1.055	97.154 ± 0.530	0.234 ± 0.003	16.159 ± 1.173
UPI	beta58	0.06	0.063	0.003 ± 0.002	0.052 ± 0.026	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.660 ± 0.039	0.812 ± 0.027	0.060	0.000	0.74	0.0466	0.0466	1.081	96.850 ± 0.557	0.206 ± 0.003	19.207 ± 1.256

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S147

Simulation 2 - Condition 3: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.005 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.004	0.313 ± 0.009	0.198	-0.002	-1.10	0.0630	0.0630	0.987	94.254 ± 0.739	0.243 ± 0.001	90.726 ± 0.921
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.315 ± 0.009	0.201	0.001	0.46	0.0658	0.0658	1.005	95.464 ± 0.661	0.248 ± 0.001	89.919 ± 0.956
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.290 ± 0.008	0.202	0.002	1.05	0.0592	0.0592	1.007	95.363 ± 0.668	0.229 ± 0.001	94.657 ± 0.714
LSAM	beta44	0.11	0.111	0.001 ± 0.002	0.009 ± 0.022	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.475 ± 0.023	0.689 ± 0.020	0.106	-0.004	-3.82	0.0703	0.0704	0.978	93.448 ± 0.786	0.291 ± 0.002	34.577 ± 1.510
LSAM	beta45	0.11	0.113	0.003 ± 0.002	0.023 ± 0.019	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.344 ± 0.016	0.587 ± 0.016	0.115	0.005	4.09	0.0634	0.0636	0.989	94.657 ± 0.714	0.250 ± 0.002	44.960 ± 1.579
LSAM	beta46	0.08	0.082	0.002 ± 0.002	0.024 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.467 ± 0.023	0.683 ± 0.020	0.080	0.000	-0.02	0.0554	0.0554	0.955	94.052 ± 0.751	0.205 ± 0.002	36.391 ± 1.528
LSAM	beta47	0.08	0.083	0.003 ± 0.002	0.034 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.451 ± 0.023	0.672 ± 0.020	0.079	-0.001	-1.32	0.0518	0.0518	0.961	93.750 ± 0.769	0.202 ± 0.002	36.895 ± 1.532
LSAM	beta51	0.16	0.164	0.004 ± 0.003	0.022 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.250 ± 0.011	0.500 ± 0.013	0.160	0.000	-0.04	0.0801	0.0801	0.989	94.456 ± 0.727	0.310 ± 0.001	53.931 ± 1.583
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.012 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.158	-0.002	-1.42	0.0596	0.0596	1.018	95.464 ± 0.661	0.251 ± 0.001	73.488 ± 1.401
LSAM	beta53	0.16	0.160	0.000 ± 0.002	0.000 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.180 ± 0.008	0.425 ± 0.012	0.163	0.003	1.94	0.0667	0.0667	0.992	94.758 ± 0.708	0.264 ± 0.001	68.044 ± 1.481
LSAM	beta54	0.16	0.158	-0.002 ± 0.002	-0.015 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.157 ± 0.007	0.397 ± 0.011	0.157	-0.003	-1.84	0.0661	0.0661	0.990	95.363 ± 0.668	0.246 ± 0.001	71.774 ± 1.429
LSAM	beta55	0.08	0.087	0.007 ± 0.003	0.082 ± 0.033	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	1.095 ± 0.055	1.047 ± 0.031	0.085	0.005	5.99	0.0791	0.0793	0.992	94.859 ± 0.701	0.325 ± 0.002	17.944 ± 1.218
LSAM	beta56	0.08	0.082	0.002 ± 0.003	0.023 ± 0.033	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	1.051 ± 0.054	1.025 ± 0.031	0.080	0.000	-0.50	0.0785	0.0785	0.931	94.456 ± 0.727	0.300 ± 0.002	20.262 ± 1.276
LSAM	beta57	0.06	0.057	-0.003 ± 0.002	-0.042 ± 0.030	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.874 ± 0.040	0.935 ± 0.026	0.056	-0.004	-6.97	0.0543	0.0545	0.965	93.649 ± 0.774	0.212 ± 0.002	20.363 ± 1.279
LSAM	beta58	0.06	0.062	0.002 ± 0.002	0.034 ± 0.025	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.638 ± 0.032	0.799 ± 0.024	0.060	0.000	0.14	0.0451	0.0451	0.965	95.464 ± 0.661	0.181 ± 0.001	28.327 ± 1.431
QML	beta41	0.20	0.191	-0.009 ± 0.002	-0.044 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.004	0.295 ± 0.008	0.188	-0.012	-6.12	0.0585	0.0598	0.998	94.456 ± 0.727	0.228 ± 0.001	91.431 ± 0.889
QML	beta42	0.20	0.196	-0.004 ± 0.002	-0.022 ± 0.009	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.004	0.300 ± 0.008	0.195	-0.005	-2.57	0.0607	0.0609	1.003	95.363 ± 0.668	0.235 ± 0.001	91.230 ± 0.898
QML	beta43	0.20	0.190	-0.010 ± 0.002	-0.052 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.004	0.284 ± 0.008	0.190	-0.010	-4.96	0.0576	0.0584	1.014	94.456 ± 0.727	0.222 ± 0.001	92.944 ± 0.813
QML	beta44	0.11	0.102	-0.008 ± 0.002	-0.071 ± 0.020	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.419 ± 0.020	0.647 ± 0.018	0.100	-0.010	-9.54	0.0680	0.0688	0.998	93.750 ± 0.769	0.277 ± 0.002	31.855 ± 1.479
QML	beta45	0.11	0.110	0.000 ± 0.002	0.002 ± 0.017	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.302 ± 0.014	0.550 ± 0.015	0.113	0.003	2.60	0.0594	0.0595	1.005	95.262 ± 0.675	0.238 ± 0.001	45.665 ± 1.582
QML	beta46	0.08	0.078	-0.002 ± 0.002	-0.025 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.366 ± 0.017	0.605 ± 0.017	0.077	-0.003	-4.05	0.0483	0.0484	0.993	94.153 ± 0.745	0.188 ± 0.001	37.298 ± 1.535
QML	beta47	0.08	0.079	-0.001 ± 0.002	-0.014 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.360 ± 0.017	0.600 ± 0.017	0.078	-0.002	-3.00	0.0460	0.0460	0.984	93.246 ± 0.797	0.185 ± 0.001	38.810 ± 1.547
QML	beta51	0.16	0.177	0.017 ± 0.003	0.104 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.270 ± 0.012	0.519 ± 0.014	0.176	0.016	10.02	0.0787	0.0803	1.016	94.657 ± 0.714	0.324 ± 0.001	56.552 ± 1.574
QML	beta52	0.16	0.144	-0.016 ± 0.002	-0.103 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.183 ± 0.008	0.428 ± 0.011	0.143	-0.017	-10.69	0.0647	0.0669	1.045	95.060 ± 0.688	0.272 ± 0.001	54.435 ± 1.581
QML	beta53	0.16	0.136	-0.024 ± 0.002	-0.148 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.211 ± 0.010	0.459 ± 0.013	0.139	-0.021	-13.32	0.0682	0.0714	0.997	93.246 ± 0.797	0.272 ± 0.001	52.722 ± 1.585
QML	beta54	0.16	0.156	-0.004 ± 0.002	-0.024 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.006	0.387 ± 0.010	0.155	-0.005	-3.41	0.0630	0.0632	0.992	94.960 ± 0.695	0.241 ± 0.001	71.573 ± 1.432
QML	beta55	0.08	0.092	0.012 ± 0.003	0.154 ± 0.033	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	1.083 ± 0.049	1.041 ± 0.029	0.092	0.012	14.52	0.0803	0.0811	1.028	95.161 ± 0.681	0.332 ± 0.002	19.052 ± 1.247
QML	beta56	0.08	0.081	0.001 ± 0.003	0.011 ± 0.032	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.985 ± 0.046	0.993 ± 0.028	0.081	0.001	1.71	0.0787	0.0787	0.969	94.456 ± 0.727	0.302 ± 0.002	18.750 ± 1.239
QML	beta57	0.06	0.049	-0.011 ± 0.002	-0.190 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.765 ± 0.036	0.874 ± 0.024	0.049	-0.011	-18.68	0.0496	0.0508	0.993	93.145 ± 0.802	0.200 ± 0.002	17.440 ± 1.205
QML	beta58	0.06	0.055	-0.005 ± 0.001	-0.076 ± 0.023	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.536 ± 0.027	0.732 ± 0.022	0.054	-0.006	-9.33	0.0433	0.0437	0.981	95.363 ± 0.668	0.168 ± 0.001	26.411 ± 1.400
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.004	0.313 ± 0.009	0.199	-0.001	-0.32	0.0630	0.0630	0.979	94.254 ± 0.739	0.240 ± 0.001	91.129 ± 0.903
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.318 ± 0.009	0.200	0.000	0.24	0.0652	0.0652	0.999	95.565 ± 0.654	0.249 ± 0.001	90.020 ± 0.952
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.292 ± 0.008	0.200	0.000	0.19	0.0609	0.0609	1.017	95.363 ± 0.668	0.233 ± 0.001	93.649 ± 0.774
UPI	beta44	0.11	0.113	0.003 ± 0.003	0.031 ± 0.023	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.526 ± 0.026	0.725 ± 0.021	0.110	0.000	-0.42	0.0751	0.0751	1.020	94.859 ± 0.701	0.319 ± 0.003	29.839 ± 1.453
UPI	beta45	0.11	0.113	0.003 ± 0.002	0.030 ± 0.019	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.370 ± 0.017	0.609 ± 0.017	0.113	0.003	2.31	0.0642	0.0642	1.043	96.371 ± 0.594	0.273 ± 0.002	37.399 ± 1.536
UPI	beta46	0.08	0.082	0.002 ± 0.002	0.019 ± 0.024	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.558 ± 0.028	0.747 ± 0.022	0.079	-0.001	-1.59	0.0586	0.0586	0.973	93.952 ± 0.757	0.228 ± 0.002	30.141 ± 1.457

Table S147

Simulation 2 - Condition 3: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.084	0.004 ± 0.002	0.051 ± 0.023	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.523 ± 0.031	0.723 ± 0.024	0.079	-0.001	-1.70	0.0530	0.0530	1.002	94.960 ± 0.695	0.227 ± 0.002	29.234 ± 1.444
UPI	beta51	0.16	0.165	0.005 ± 0.003	0.029 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.244 ± 0.011	0.494 ± 0.013	0.164	0.004	2.42	0.0775	0.0776	1.012	95.363 ± 0.668	0.313 ± 0.001	54.435 ± 1.581
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.017 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.007	0.395 ± 0.011	0.158	-0.002	-1.11	0.0587	0.0588	1.017	95.464 ± 0.661	0.252 ± 0.001	72.984 ± 1.410
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.180 ± 0.008	0.425 ± 0.012	0.162	0.002	1.40	0.0675	0.0675	1.006	94.859 ± 0.701	0.268 ± 0.001	67.036 ± 1.493
UPI	beta54	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.158 ± 0.007	0.398 ± 0.011	0.157	-0.003	-2.09	0.0636	0.0636	0.995	94.355 ± 0.733	0.248 ± 0.001	71.169 ± 1.438
UPI	beta55	0.08	0.087	0.007 ± 0.003	0.091 ± 0.034	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	1.144 ± 0.060	1.070 ± 0.033	0.088	0.008	10.16	0.0777	0.0781	1.039	95.867 ± 0.632	0.347 ± 0.003	14.415 ± 1.115
UPI	beta56	0.08	0.081	0.001 ± 0.003	0.014 ± 0.033	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	1.105 ± 0.061	1.051 ± 0.034	0.080	0.000	0.00	0.0792	0.0792	0.974	95.262 ± 0.675	0.321 ± 0.003	16.935 ± 1.191
UPI	beta57	0.06	0.057	-0.003 ± 0.002	-0.045 ± 0.031	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.967 ± 0.052	0.984 ± 0.031	0.057	-0.003	-4.94	0.0554	0.0555	0.982	94.960 ± 0.695	0.227 ± 0.002	17.137 ± 1.196
UPI	beta58	0.06	0.063	0.003 ± 0.002	0.052 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.715 ± 0.035	0.845 ± 0.025	0.062	0.002	3.52	0.0501	0.0501	1.002	96.371 ± 0.594	0.199 ± 0.002	24.194 ± 1.360

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S148

Simulation 2 - Condition 4: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.202	0.002	1.02	0.0375	0.0376	0.997	95.200 ± 0.676	0.150 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.198	-0.002	-1.15	0.0384	0.0385	1.011	94.900 ± 0.696	0.153 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.202	0.002 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.201	0.001	0.58	0.0377	0.0377	0.964	94.200 ± 0.739	0.143 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.111	0.001 ± 0.001	0.013 ± 0.013	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.168 ± 0.007	0.410 ± 0.011	0.112	0.002	2.16	0.0465	0.0466	0.990	96.000 ± 0.620	0.175 ± 0.001	71.400 ± 1.429
LSAM	beta45	0.11	0.112	0.002 ± 0.001	0.019 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.133 ± 0.006	0.365 ± 0.010	0.112	0.002	2.03	0.0377	0.0378	0.980	94.700 ± 0.708	0.154 ± 0.001	80.800 ± 1.246
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.015 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.164 ± 0.008	0.405 ± 0.011	0.080	0.000	0.25	0.0321	0.0321	0.972	94.900 ± 0.696	0.123 ± 0.001	73.400 ± 1.397
LSAM	beta47	0.08	0.081	0.001 ± 0.001	0.011 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.155 ± 0.007	0.393 ± 0.011	0.082	0.002	1.94	0.0321	0.0321	0.988	94.300 ± 0.733	0.122 ± 0.001	74.600 ± 1.377
LSAM	beta51	0.16	0.160	0.000 ± 0.002	0.003 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.311 ± 0.009	0.160	0.000	-0.04	0.0494	0.0494	0.981	94.700 ± 0.708	0.192 ± 0.000	91.000 ± 0.905
LSAM	beta52	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.158	-0.002	-0.99	0.0388	0.0388	0.983	94.500 ± 0.721	0.154 ± 0.000	98.500 ± 0.384
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.161	0.001	0.76	0.0412	0.0412	1.012	95.200 ± 0.676	0.163 ± 0.000	97.100 ± 0.531
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.161	0.001	0.36	0.0378	0.0378	1.046	96.000 ± 0.620	0.153 ± 0.000	98.800 ± 0.344
LSAM	beta55	0.08	0.083	0.003 ± 0.002	0.035 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.404 ± 0.019	0.635 ± 0.018	0.081	0.001	1.75	0.0500	0.0501	0.974	95.700 ± 0.641	0.194 ± 0.001	39.300 ± 1.545
LSAM	beta56	0.08	0.078	-0.002 ± 0.002	-0.022 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.363 ± 0.016	0.602 ± 0.016	0.076	-0.004	-5.04	0.0487	0.0489	0.970	94.100 ± 0.745	0.183 ± 0.001	38.700 ± 1.540
LSAM	beta57	0.06	0.059	-0.001 ± 0.001	-0.016 ± 0.018	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.311 ± 0.015	0.558 ± 0.016	0.059	-0.001	-2.43	0.0333	0.0333	0.969	94.900 ± 0.696	0.127 ± 0.001	45.700 ± 1.575
LSAM	beta58	0.06	0.062	0.002 ± 0.001	0.033 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.229 ± 0.011	0.479 ± 0.014	0.063	0.003	4.30	0.0271	0.0273	0.988	94.600 ± 0.715	0.111 ± 0.000	61.800 ± 1.536
QML	beta41	0.20	0.193	-0.007 ± 0.001	-0.033 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.193	-0.007	-3.55	0.0355	0.0362	1.006	94.500 ± 0.721	0.146 ± 0.002	99.800 ± 0.141
QML	beta42	0.20	0.192	-0.008 ± 0.001	-0.039 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.192	-0.008	-4.13	0.0368	0.0377	1.018	93.700 ± 0.768	0.149 ± 0.001	99.800 ± 0.141
QML	beta43	0.20	0.191	-0.009 ± 0.001	-0.047 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.189	-0.011	-5.68	0.0358	0.0375	0.964	93.000 ± 0.807	0.139 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.105	-0.005 ± 0.001	-0.044 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.164 ± 0.007	0.405 ± 0.011	0.106	-0.004	-3.77	0.0452	0.0454	0.984	95.000 ± 0.689	0.171 ± 0.001	68.600 ± 1.468
QML	beta45	0.11	0.110	0.000 ± 0.001	0.003 ± 0.011	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.122 ± 0.006	0.349 ± 0.010	0.110	0.000	0.33	0.0366	0.0366	0.991	94.700 ± 0.708	0.149 ± 0.001	83.100 ± 1.185
QML	beta46	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.140 ± 0.006	0.374 ± 0.010	0.079	-0.001	-1.17	0.0312	0.0312	0.990	95.300 ± 0.669	0.116 ± 0.001	75.800 ± 1.354
QML	beta47	0.08	0.078	-0.002 ± 0.001	-0.025 ± 0.012	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.135 ± 0.006	0.368 ± 0.010	0.079	-0.001	-0.88	0.0287	0.0287	0.992	94.000 ± 0.751	0.114 ± 0.001	76.500 ± 1.341
QML	beta51	0.16	0.172	0.012 ± 0.002	0.078 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.328 ± 0.009	0.173	0.013	8.44	0.0506	0.0524	0.997	94.800 ± 0.702	0.199 ± 0.001	92.300 ± 0.843
QML	beta52	0.16	0.142	-0.018 ± 0.001	-0.114 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.291 ± 0.007	0.142	-0.018	-11.15	0.0430	0.0466	0.998	93.200 ± 0.796	0.167 ± 0.001	90.600 ± 0.923
QML	beta53	0.16	0.138	-0.022 ± 0.001	-0.137 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.139	-0.021	-13.33	0.0425	0.0475	1.030	92.800 ± 0.817	0.168 ± 0.000	90.700 ± 0.918
QML	beta54	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.160	0.000	0.21	0.0362	0.0362	1.049	95.500 ± 0.656	0.151 ± 0.000	98.800 ± 0.344
QML	beta55	0.08	0.091	0.011 ± 0.002	0.138 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.446 ± 0.021	0.668 ± 0.019	0.090	0.010	12.31	0.0511	0.0520	0.986	94.300 ± 0.733	0.202 ± 0.001	43.600 ± 1.568
QML	beta56	0.08	0.077	-0.003 ± 0.002	-0.035 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.367 ± 0.015	0.606 ± 0.016	0.076	-0.004	-4.45	0.0516	0.0517	0.983	95.100 ± 0.683	0.186 ± 0.001	37.700 ± 1.533
QML	beta57	0.06	0.050	-0.010 ± 0.001	-0.161 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.314 ± 0.014	0.560 ± 0.015	0.050	-0.010	-17.39	0.0322	0.0339	0.971	93.000 ± 0.807	0.123 ± 0.001	36.300 ± 1.521
QML	beta58	0.06	0.056	-0.004 ± 0.001	-0.060 ± 0.014	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.206 ± 0.010	0.454 ± 0.013	0.057	-0.003	-5.08	0.0260	0.0262	0.996	94.500 ± 0.721	0.105 ± 0.000	58.600 ± 1.558
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.201	0.001	0.72	0.0375	0.0376	0.967	94.200 ± 0.739	0.146 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.197	-0.003	-1.27	0.0390	0.0391	0.998	94.400 ± 0.727	0.151 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.202	0.002	0.86	0.0380	0.0380	0.965	94.100 ± 0.745	0.143 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.112	0.002 ± 0.001	0.016 ± 0.013	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.180 ± 0.009	0.424 ± 0.012	0.111	0.001	0.97	0.0463	0.0463	0.994	95.500 ± 0.656	0.182 ± 0.001	68.700 ± 1.466
UPI	beta45	0.11	0.113	0.003 ± 0.001	0.023 ± 0.012	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.143 ± 0.007	0.378 ± 0.011	0.113	0.003	3.16	0.0400	0.0401	0.992	94.900 ± 0.696	0.161 ± 0.001	79.400 ± 1.279
UPI	beta46	0.08	0.081	0.001 ± 0.001	0.009 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.174 ± 0.008	0.417 ± 0.012	0.080	0.000	-0.01	0.0349	0.0349	0.990	95.400 ± 0.662	0.129 ± 0.001	70.100 ± 1.448

Table S148

Simulation 2 - Condition 4: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.017 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.170 ± 0.008	0.412 ± 0.012	0.081	0.001	1.42	0.0329	0.0329	0.995	94.400 ± 0.727	0.128 ± 0.001	72.000 ± 1.420
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.161	0.001	0.70	0.0501	0.0501	0.979	94.500 ± 0.721	0.191 ± 0.000	91.400 ± 0.887
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.159	-0.001	-0.54	0.0387	0.0388	0.980	94.300 ± 0.733	0.153 ± 0.000	98.400 ± 0.397
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.161	0.001	0.86	0.0417	0.0418	1.023	95.400 ± 0.662	0.164 ± 0.000	97.400 ± 0.503
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.161	0.001	0.65	0.0379	0.0379	1.045	95.900 ± 0.627	0.153 ± 0.000	99.100 ± 0.299
UPI	beta55	0.08	0.082	0.002 ± 0.002	0.031 ± 0.020	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.416 ± 0.021	0.645 ± 0.019	0.080	0.000	-0.44	0.0495	0.0495	0.983	95.700 ± 0.641	0.199 ± 0.001	36.700 ± 1.524
UPI	beta56	0.08	0.078	-0.002 ± 0.002	-0.029 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.370 ± 0.016	0.608 ± 0.017	0.076	-0.004	-4.95	0.0475	0.0476	0.982	95.000 ± 0.689	0.187 ± 0.001	37.200 ± 1.528
UPI	beta57	0.06	0.059	-0.001 ± 0.001	-0.016 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.325 ± 0.016	0.570 ± 0.017	0.058	-0.002	-3.39	0.0332	0.0332	0.978	94.600 ± 0.715	0.131 ± 0.001	42.300 ± 1.562
UPI	beta58	0.06	0.062	0.002 ± 0.001	0.037 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.238 ± 0.012	0.488 ± 0.014	0.062	0.002	3.47	0.0287	0.0287	1.010	95.300 ± 0.669	0.116 ± 0.001	58.900 ± 1.556

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S149

Simulation 2 - Condition 5: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.198	-0.002	-1.07	0.0506	0.0506	0.982	94.600 ± 0.715	0.191 ± 0.000	98.600 ± 0.372
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.202	0.002	0.80	0.0509	0.0509	0.975	94.400 ± 0.727	0.194 ± 0.000	98.000 ± 0.443
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.202	0.002	0.93	0.0494	0.0495	0.961	93.300 ± 0.791	0.182 ± 0.000	98.600 ± 0.372
LSAM	beta44	0.11	0.114	0.004 ± 0.002	0.033 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.246 ± 0.011	0.496 ± 0.014	0.113	0.003	3.08	0.0540	0.0541	0.961	93.600 ± 0.774	0.205 ± 0.001	58.800 ± 1.556
LSAM	beta45	0.11	0.110	-0.000 ± 0.002	-0.004 ± 0.014	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.206 ± 0.010	0.454 ± 0.013	0.110	0.000	0.20	0.0497	0.0497	0.930	93.100 ± 0.801	0.182 ± 0.001	66.100 ± 1.497
LSAM	beta46	0.08	0.080	0.000 ± 0.001	0.004 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.224 ± 0.011	0.473 ± 0.014	0.080	0.000	0.33	0.0382	0.0382	0.968	94.400 ± 0.727	0.144 ± 0.001	60.800 ± 1.544
LSAM	beta47	0.08	0.080	0.000 ± 0.001	0.001 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.242 ± 0.012	0.492 ± 0.014	0.079	-0.001	-1.67	0.0396	0.0396	0.923	93.800 ± 0.763	0.143 ± 0.001	59.900 ± 1.550
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.393 ± 0.011	0.160	0.000	-0.12	0.0633	0.0633	0.976	93.800 ± 0.763	0.241 ± 0.001	73.700 ± 1.392
LSAM	beta52	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.008	0.159	-0.001	-0.90	0.0510	0.0510	1.024	94.700 ± 0.708	0.200 ± 0.001	88.000 ± 1.028
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.113 ± 0.005	0.336 ± 0.009	0.156	-0.004	-2.72	0.0528	0.0529	0.993	94.000 ± 0.751	0.210 ± 0.001	84.800 ± 1.135
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.003 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.160	0.000	-0.22	0.0520	0.0520	0.983	93.500 ± 0.780	0.199 ± 0.001	87.700 ± 1.039
LSAM	beta55	0.08	0.081	0.001 ± 0.002	0.010 ± 0.025	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.638 ± 0.030	0.799 ± 0.023	0.080	0.000	0.02	0.0628	0.0628	0.920	93.200 ± 0.796	0.231 ± 0.001	30.900 ± 1.461
LSAM	beta56	0.08	0.082	0.002 ± 0.002	0.022 ± 0.023	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.524 ± 0.025	0.724 ± 0.021	0.084	0.004	4.99	0.0558	0.0559	0.964	94.300 ± 0.733	0.219 ± 0.001	33.300 ± 1.490
LSAM	beta57	0.06	0.061	0.001 ± 0.001	0.023 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.469 ± 0.022	0.685 ± 0.019	0.061	0.001	1.68	0.0401	0.0401	0.938	92.800 ± 0.817	0.151 ± 0.001	36.800 ± 1.525
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.008 ± 0.019	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.375 ± 0.016	0.612 ± 0.017	0.061	0.001	1.01	0.0364	0.0364	0.939	93.000 ± 0.807	0.135 ± 0.001	43.500 ± 1.568
QML	beta41	0.20	0.195	-0.005 ± 0.002	-0.025 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.195	-0.005	-2.60	0.0484	0.0487	0.987	94.300 ± 0.733	0.188 ± 0.001	98.200 ± 0.420
QML	beta42	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.199	-0.001	-0.68	0.0500	0.0500	0.976	95.300 ± 0.669	0.190 ± 0.000	98.100 ± 0.432
QML	beta43	0.20	0.196	-0.004 ± 0.002	-0.019 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.240 ± 0.007	0.197	-0.003	-1.46	0.0482	0.0483	0.960	93.900 ± 0.757	0.180 ± 0.000	98.400 ± 0.397
QML	beta44	0.11	0.111	0.001 ± 0.002	0.007 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.236 ± 0.011	0.485 ± 0.013	0.110	0.000	-0.34	0.0530	0.0530	0.971	93.700 ± 0.768	0.203 ± 0.001	57.200 ± 1.565
QML	beta45	0.11	0.109	-0.001 ± 0.002	-0.010 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.196 ± 0.009	0.443 ± 0.012	0.109	-0.001	-0.70	0.0482	0.0482	0.943	93.300 ± 0.791	0.180 ± 0.001	67.300 ± 1.483
QML	beta46	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.208 ± 0.010	0.456 ± 0.013	0.079	-0.001	-0.87	0.0370	0.0370	0.978	94.800 ± 0.702	0.140 ± 0.001	61.100 ± 1.542
QML	beta47	0.08	0.079	-0.001 ± 0.001	-0.016 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.223 ± 0.011	0.473 ± 0.013	0.077	-0.003	-3.48	0.0382	0.0383	0.935	94.000 ± 0.751	0.139 ± 0.001	59.800 ± 1.550
QML	beta51	0.16	0.167	0.007 ± 0.002	0.042 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.007	0.399 ± 0.011	0.166	0.006	4.06	0.0635	0.0638	0.991	94.000 ± 0.751	0.247 ± 0.001	75.100 ± 1.367
QML	beta52	0.16	0.142	-0.018 ± 0.002	-0.111 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.125 ± 0.006	0.353 ± 0.009	0.142	-0.018	-11.45	0.0510	0.0542	1.017	93.500 ± 0.780	0.214 ± 0.001	75.100 ± 1.367
QML	beta53	0.16	0.137	-0.023 ± 0.002	-0.143 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.134	-0.026	-16.15	0.0521	0.0581	0.998	91.700 ± 0.872	0.219 ± 0.001	69.400 ± 1.457
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.320 ± 0.009	0.159	-0.001	-0.89	0.0511	0.0511	0.987	94.000 ± 0.751	0.198 ± 0.001	88.100 ± 1.024
QML	beta55	0.08	0.084	0.004 ± 0.002	0.052 ± 0.026	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.658 ± 0.031	0.811 ± 0.023	0.083	0.003	3.54	0.0635	0.0635	0.926	93.300 ± 0.791	0.235 ± 0.001	32.500 ± 1.481
QML	beta56	0.08	0.082	0.002 ± 0.002	0.024 ± 0.023	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.527 ± 0.025	0.726 ± 0.021	0.083	0.003	3.60	0.0561	0.0562	0.976	94.200 ± 0.739	0.222 ± 0.001	32.700 ± 1.483
QML	beta57	0.06	0.057	-0.003 ± 0.001	-0.046 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.441 ± 0.020	0.664 ± 0.019	0.056	-0.004	-6.22	0.0386	0.0388	0.949	92.900 ± 0.812	0.148 ± 0.001	33.000 ± 1.487
QML	beta58	0.06	0.058	-0.002 ± 0.001	-0.032 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.360 ± 0.016	0.600 ± 0.016	0.059	-0.001	-2.16	0.0359	0.0359	0.940	92.500 ± 0.833	0.132 ± 0.001	41.900 ± 1.560
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.199	-0.001	-0.71	0.0502	0.0502	0.940	93.500 ± 0.780	0.183 ± 0.000	98.900 ± 0.330
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.201	0.001	0.48	0.0507	0.0507	0.945	93.300 ± 0.791	0.189 ± 0.001	98.300 ± 0.409
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.242 ± 0.007	0.202	0.002	1.06	0.0487	0.0488	0.956	93.400 ± 0.785	0.181 ± 0.000	98.600 ± 0.372
UPI	beta44	0.11	0.114	0.004 ± 0.002	0.038 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.258 ± 0.012	0.508 ± 0.014	0.113	0.003	2.57	0.0548	0.0549	0.962	94.400 ± 0.727	0.210 ± 0.001	57.000 ± 1.566
UPI	beta45	0.11	0.110	0.000 ± 0.002	0.003 ± 0.014	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.209 ± 0.010	0.457 ± 0.013	0.111	0.001	1.01	0.0493	0.0493	0.949	92.900 ± 0.812	0.187 ± 0.001	64.600 ± 1.512
UPI	beta46	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.015	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.232 ± 0.011	0.482 ± 0.014	0.079	-0.001	-1.62	0.0380	0.0380	0.974	94.500 ± 0.721	0.147 ± 0.001	59.100 ± 1.555

Table S149

Simulation 2 - Condition 5: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.008 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.255 ± 0.012	0.505 ± 0.015	0.080	0.000	0.07	0.0407	0.0407	0.929	93.900 ± 0.757	0.147 ± 0.001	59.400 ± 1.553
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.007	0.392 ± 0.011	0.161	0.001	0.75	0.0650	0.0651	0.981	93.700 ± 0.768	0.241 ± 0.001	74.200 ± 1.384
UPI	beta52	0.16	0.160	0.000 ± 0.002	0.000 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.008	0.159	-0.001	-0.58	0.0505	0.0505	1.014	94.800 ± 0.702	0.198 ± 0.001	88.600 ± 1.005
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.113 ± 0.005	0.336 ± 0.009	0.157	-0.003	-1.61	0.0530	0.0531	0.994	94.000 ± 0.751	0.210 ± 0.001	84.400 ± 1.147
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.003 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.161	0.001	0.35	0.0520	0.0520	0.974	93.500 ± 0.780	0.198 ± 0.001	88.300 ± 1.016
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.008 ± 0.025	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.640 ± 0.030	0.800 ± 0.023	0.079	-0.001	-1.03	0.0606	0.0607	0.932	93.500 ± 0.780	0.234 ± 0.001	29.500 ± 1.442
UPI	beta56	0.08	0.082	0.002 ± 0.002	0.024 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.540 ± 0.026	0.735 ± 0.021	0.083	0.003	3.70	0.0553	0.0554	0.970	93.900 ± 0.757	0.224 ± 0.001	31.800 ± 1.473
UPI	beta57	0.06	0.061	0.001 ± 0.001	0.023 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.477 ± 0.022	0.691 ± 0.019	0.061	0.001	1.70	0.0415	0.0415	0.946	93.500 ± 0.780	0.154 ± 0.001	34.300 ± 1.501
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.007 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.383 ± 0.017	0.619 ± 0.017	0.059	-0.001	-1.46	0.0369	0.0369	0.948	93.200 ± 0.796	0.138 ± 0.001	42.700 ± 1.564

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S150

Simulation 2 - Condition 6: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.202	0.002	0.88	0.0307	0.0308	1.010	95.600 ± 0.649	0.120 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.202	0.002	0.84	0.0309	0.0310	0.985	94.200 ± 0.739	0.122 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.199	-0.001	-0.33	0.0290	0.0291	1.046	95.900 ± 0.627	0.114 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.021 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.113	0.003	2.63	0.0319	0.0321	0.986	93.800 ± 0.763	0.128 ± 0.000	91.700 ± 0.872
LSAM	beta45	0.11	0.110	-0.000 ± 0.001	-0.003 ± 0.008	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.072 ± 0.003	0.268 ± 0.007	0.108	-0.002	-1.55	0.0292	0.0292	0.985	94.800 ± 0.702	0.114 ± 0.000	96.600 ± 0.573
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.008 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.080	0.000	-0.32	0.0236	0.0236	0.960	93.900 ± 0.757	0.090 ± 0.000	93.100 ± 0.801
LSAM	beta47	0.08	0.080	0.000 ± 0.001	0.003 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.080	0.000	-0.25	0.0244	0.0244	0.954	94.300 ± 0.733	0.089 ± 0.000	92.900 ± 0.812
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.160	0.000	-0.10	0.0379	0.0379	1.000	94.000 ± 0.751	0.153 ± 0.000	98.000 ± 0.443
LSAM	beta52	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.159	-0.001	-0.70	0.0304	0.0304	1.048	96.100 ± 0.612	0.127 ± 0.000	100.000 ± 0.000
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.160	0.000	0.27	0.0318	0.0318	1.021	95.500 ± 0.656	0.132 ± 0.000	99.800 ± 0.141
LSAM	beta54	0.16	0.163	0.003 ± 0.001	0.016 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.199 ± 0.005	0.162	0.002	0.98	0.0330	0.0331	1.008	96.000 ± 0.620	0.125 ± 0.000	100.000 ± 0.000
LSAM	beta55	0.08	0.081	0.001 ± 0.001	0.008 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.202 ± 0.009	0.450 ± 0.012	0.080	0.000	0.06	0.0349	0.0349	1.023	95.400 ± 0.662	0.144 ± 0.000	58.900 ± 1.556
LSAM	beta56	0.08	0.078	-0.002 ± 0.001	-0.021 ± 0.014	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.209 ± 0.009	0.458 ± 0.013	0.080	0.000	0.28	0.0359	0.0359	0.959	93.000 ± 0.807	0.138 ± 0.000	60.900 ± 1.543
LSAM	beta57	0.06	0.060	0.000 ± 0.001	0.007 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.171 ± 0.008	0.413 ± 0.011	0.062	0.002	2.63	0.0250	0.0250	0.980	94.200 ± 0.739	0.095 ± 0.000	69.900 ± 1.451
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.006 ± 0.011	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.131 ± 0.006	0.362 ± 0.010	0.060	0.000	0.56	0.0219	0.0219	0.998	94.800 ± 0.702	0.085 ± 0.000	79.300 ± 1.281
QML	beta41	0.20	0.196	-0.004 ± 0.001	-0.019 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.198	-0.002	-1.20	0.0303	0.0304	1.021	95.200 ± 0.676	0.120 ± 0.001	99.900 ± 0.100
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.200	0.000	-0.03	0.0304	0.0304	0.992	94.200 ± 0.739	0.121 ± 0.001	99.900 ± 0.100
QML	beta43	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.194	-0.006	-2.91	0.0278	0.0284	1.049	95.800 ± 0.634	0.113 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.110	-0.000 ± 0.001	-0.001 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.089 ± 0.004	0.299 ± 0.008	0.111	0.001	0.85	0.0322	0.0322	0.990	93.800 ± 0.763	0.128 ± 0.000	91.400 ± 0.887
QML	beta45	0.11	0.109	-0.001 ± 0.001	-0.008 ± 0.008	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.108	-0.002	-2.05	0.0294	0.0295	0.989	94.700 ± 0.708	0.114 ± 0.000	96.500 ± 0.581
QML	beta46	0.08	0.080	0.000 ± 0.001	0.002 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.080	0.000	-0.02	0.0226	0.0226	0.958	93.700 ± 0.768	0.089 ± 0.000	93.400 ± 0.785
QML	beta47	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.009	0.001 ± 0.000	0.023 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.079	-0.001	-1.48	0.0244	0.0244	0.965	94.800 ± 0.702	0.087 ± 0.000	93.400 ± 0.785
QML	beta51	0.16	0.164	0.004 ± 0.001	0.024 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.164	0.004	2.76	0.0373	0.0376	1.011	94.600 ± 0.715	0.156 ± 0.001	98.300 ± 0.409
QML	beta52	0.16	0.142	-0.018 ± 0.001	-0.112 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.143	-0.017	-10.86	0.0329	0.0372	1.050	92.400 ± 0.838	0.136 ± 0.001	98.500 ± 0.384
QML	beta53	0.16	0.140	-0.020 ± 0.001	-0.123 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.006	0.141	-0.019	-12.14	0.0341	0.0392	1.016	90.900 ± 0.909	0.137 ± 0.000	98.600 ± 0.372
QML	beta54	0.16	0.162	0.002 ± 0.001	0.014 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.197 ± 0.005	0.161	0.001	0.71	0.0329	0.0329	1.012	95.900 ± 0.627	0.125 ± 0.000	100.000 ± 0.000
QML	beta55	0.08	0.084	0.004 ± 0.001	0.054 ± 0.014	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.212 ± 0.009	0.461 ± 0.013	0.083	0.003	4.03	0.0357	0.0359	1.030	95.100 ± 0.683	0.148 ± 0.000	61.600 ± 1.538
QML	beta56	0.08	0.079	-0.001 ± 0.001	-0.015 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.215 ± 0.010	0.463 ± 0.013	0.080	0.000	0.55	0.0357	0.0357	0.962	93.400 ± 0.785	0.140 ± 0.000	60.400 ± 1.547
QML	beta57	0.06	0.057	-0.003 ± 0.001	-0.055 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.170 ± 0.008	0.412 ± 0.011	0.058	-0.002	-3.61	0.0249	0.0250	0.981	94.200 ± 0.739	0.094 ± 0.000	66.600 ± 1.491
QML	beta58	0.06	0.058	-0.002 ± 0.001	-0.027 ± 0.011	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.125 ± 0.006	0.354 ± 0.010	0.058	-0.002	-2.65	0.0212	0.0212	1.006	94.700 ± 0.708	0.084 ± 0.000	78.200 ± 1.306
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.202	0.002	1.02	0.0307	0.0308	0.966	94.600 ± 0.715	0.115 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.008 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.202	0.002	0.82	0.0308	0.0309	0.959	93.500 ± 0.780	0.119 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.200	0.000	-0.22	0.0285	0.0285	1.040	95.400 ± 0.662	0.113 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.112	0.002 ± 0.001	0.023 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.092 ± 0.004	0.303 ± 0.009	0.113	0.003	2.69	0.0333	0.0334	0.991	93.800 ± 0.763	0.129 ± 0.000	92.000 ± 0.858
UPI	beta45	0.11	0.110	-0.000 ± 0.001	-0.001 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.074 ± 0.003	0.272 ± 0.007	0.108	-0.002	-1.38	0.0299	0.0300	0.985	95.200 ± 0.676	0.116 ± 0.000	96.900 ± 0.548
UPI	beta46	0.08	0.080	0.000 ± 0.001	0.005 ± 0.010	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.092 ± 0.004	0.303 ± 0.009	0.080	0.000	-0.16	0.0236	0.0236	0.964	93.600 ± 0.774	0.092 ± 0.000	92.500 ± 0.833

Table S150

Simulation 2 - Condition 6: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	0.000 ± 0.001	0.004 ± 0.010	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.093 ± 0.004	0.305 ± 0.008	0.079	-0.001	-1.18	0.0245	0.0245	0.943	94.100 ± 0.745	0.090 ± 0.000	93.600 ± 0.774
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.160	0.000	-0.23	0.0373	0.0373	0.999	93.800 ± 0.763	0.152 ± 0.000	98.200 ± 0.420
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.159	-0.001	-0.54	0.0307	0.0308	1.035	95.800 ± 0.634	0.125 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.161	0.001	0.39	0.0315	0.0315	1.017	95.600 ± 0.649	0.132 ± 0.000	99.800 ± 0.141
UPI	beta54	0.16	0.163	0.003 ± 0.001	0.017 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.199 ± 0.005	0.162	0.002	0.96	0.0332	0.0333	1.002	96.000 ± 0.620	0.124 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.007 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.202 ± 0.009	0.449 ± 0.012	0.080	0.000	0.14	0.0349	0.0349	1.030	95.300 ± 0.669	0.145 ± 0.000	58.500 ± 1.558
UPI	beta56	0.08	0.078	-0.002 ± 0.001	-0.023 ± 0.014	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.209 ± 0.009	0.457 ± 0.013	0.079	-0.001	-1.00	0.0354	0.0354	0.968	93.300 ± 0.791	0.139 ± 0.000	59.800 ± 1.550
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.007 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.171 ± 0.008	0.414 ± 0.011	0.061	0.001	2.08	0.0249	0.0249	0.985	94.600 ± 0.715	0.096 ± 0.000	70.500 ± 1.442
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.005 ± 0.012	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.133 ± 0.006	0.364 ± 0.010	0.060	0.000	0.58	0.0219	0.0219	1.002	95.300 ± 0.669	0.086 ± 0.000	79.100 ± 1.286

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S151

Simulation 2 - Condition 7: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.007	0.012 ± 0.034	0.030 ± 0.002	0.174 ± 0.005	0.030 ± 0.002	0.174 ± 0.006	0.758 ± 0.047	0.871 ± 0.032	0.201	0.001	0.52	0.1695	0.1695	1.015	97.633 ± 0.585	0.694 ± 0.012	25.296 ± 1.672
LSAM	beta42	0.20	0.203	0.003 ± 0.007	0.015 ± 0.033	0.030 ± 0.002	0.173 ± 0.005	0.030 ± 0.002	0.173 ± 0.006	0.750 ± 0.047	0.866 ± 0.032	0.208	0.008	4.24	0.1733	0.1735	1.051	97.633 ± 0.585	0.714 ± 0.011	24.852 ± 1.662
LSAM	beta43	0.20	0.197	-0.003 ± 0.003	-0.015 ± 0.017	0.008 ± 0.000	0.089 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	0.200 ± 0.012	0.447 ± 0.016	0.192	-0.008	-3.91	0.0893	0.0896	1.052	97.189 ± 0.636	0.369 ± 0.004	62.130 ± 1.866
LSAM	beta44	0.11	0.125	0.015 ± 0.006	0.132 ± 0.052	0.022 ± 0.002	0.149 ± 0.005	0.022 ± 0.002	0.149 ± 0.006	1.845 ± 0.134	1.358 ± 0.056	0.113	0.003	2.32	0.1285	0.1286	1.084	96.302 ± 0.726	0.632 ± 0.021	12.722 ± 1.282
LSAM	beta45	0.11	0.121	0.011 ± 0.005	0.098 ± 0.047	0.018 ± 0.001	0.134 ± 0.005	0.018 ± 0.001	0.134 ± 0.005	1.493 ± 0.101	1.222 ± 0.048	0.117	0.007	6.17	0.1258	0.1259	1.050	96.893 ± 0.667	0.552 ± 0.014	15.089 ± 1.377
LSAM	beta46	0.08	0.084	0.004 ± 0.003	0.047 ± 0.038	0.006 ± 0.000	0.080 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	0.999 ± 0.072	1.000 ± 0.041	0.083	0.003	4.29	0.0729	0.0730	1.007	95.562 ± 0.792	0.316 ± 0.009	25.000 ± 1.665
LSAM	beta47	0.08	0.085	0.005 ± 0.003	0.061 ± 0.036	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.075 ± 0.003	0.879 ± 0.065	0.937 ± 0.039	0.079	-0.001	-0.83	0.0662	0.0662	1.032	95.710 ± 0.779	0.303 ± 0.008	27.219 ± 1.712
LSAM	beta51	0.16	0.161	0.001 ± 0.005	0.007 ± 0.029	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.119 ± 0.004	0.550 ± 0.033	0.741 ± 0.026	0.162	0.002	1.47	0.1135	0.1135	0.991	95.266 ± 0.817	0.461 ± 0.003	28.254 ± 1.732
LSAM	beta52	0.16	0.157	-0.003 ± 0.007	-0.016 ± 0.043	0.031 ± 0.002	0.177 ± 0.006	0.031 ± 0.002	0.177 ± 0.007	1.225 ± 0.079	1.107 ± 0.042	0.152	-0.008	-5.04	0.1675	0.1677	1.045	97.041 ± 0.652	0.726 ± 0.013	15.385 ± 1.388
LSAM	beta53	0.16	0.161	0.001 ± 0.005	0.005 ± 0.033	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.137 ± 0.005	0.734 ± 0.044	0.856 ± 0.030	0.159	-0.001	-0.44	0.1323	0.1323	0.990	96.893 ± 0.667	0.532 ± 0.006	23.669 ± 1.635
LSAM	beta54	0.16	0.160	-0.000 ± 0.006	-0.000 ± 0.040	0.028 ± 0.002	0.168 ± 0.006	0.028 ± 0.002	0.167 ± 0.007	1.095 ± 0.081	1.046 ± 0.044	0.161	0.001	0.54	0.1575	0.1575	1.021	96.598 ± 0.697	0.671 ± 0.011	14.201 ± 1.343
LSAM	beta55	0.08	0.079	-0.001 ± 0.005	-0.015 ± 0.063	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	2.648 ± 0.179	1.627 ± 0.063	0.080	0.000	-0.33	0.1159	0.1159	1.048	96.746 ± 0.682	0.535 ± 0.012	9.763 ± 1.142
LSAM	beta56	0.08	0.086	0.006 ± 0.003	0.070 ± 0.037	0.006 ± 0.000	0.078 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	0.949 ± 0.072	0.974 ± 0.041	0.087	0.007	8.67	0.0734	0.0737	0.999	96.893 ± 0.667	0.305 ± 0.006	27.959 ± 1.726
LSAM	beta57	0.06	0.065	0.005 ± 0.005	0.088 ± 0.084	0.017 ± 0.001	0.131 ± 0.005	0.017 ± 0.001	0.131 ± 0.005	4.760 ± 0.346	2.182 ± 0.090	0.063	0.003	4.38	0.1167	0.1168	1.043	97.633 ± 0.585	0.535 ± 0.013	6.657 ± 0.959
LSAM	beta58	0.06	0.062	0.002 ± 0.002	0.032 ± 0.041	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.003	1.150 ± 0.084	1.072 ± 0.044	0.061	0.001	1.48	0.0556	0.0556	1.006	96.598 ± 0.697	0.254 ± 0.006	19.822 ± 1.533
QML	beta41	0.20	0.239	0.039 ± 0.004	0.195 ± 0.019	0.009 ± 0.001	0.097 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.271 ± 0.016	0.520 ± 0.017	0.238	0.038	19.08	0.0998	0.1069	0.958	94.822 ± 0.852	0.363 ± 0.003	72.781 ± 1.712
QML	beta42	0.20	0.237	0.037 ± 0.004	0.184 ± 0.019	0.009 ± 0.001	0.096 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	0.266 ± 0.016	0.515 ± 0.018	0.236	0.036	18.11	0.0975	0.1040	0.994	94.379 ± 0.886	0.375 ± 0.003	71.006 ± 1.745
QML	beta43	0.20	0.183	-0.017 ± 0.003	-0.086 ± 0.015	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.166 ± 0.009	0.408 ± 0.014	0.176	-0.024	-11.88	0.0809	0.0843	1.010	94.379 ± 0.886	0.316 ± 0.002	64.053 ± 1.846
QML	beta44	0.11	0.033	-0.077 ± 0.004	-0.702 ± 0.041	0.014 ± 0.001	0.116 ± 0.003	0.019 ± 0.001	0.140 ± 0.004	1.611 ± 0.093	1.269 ± 0.040	0.033	-0.077	-69.63	0.1294	0.1503	0.975	89.201 ± 1.194	0.445 ± 0.006	8.432 ± 1.069
QML	beta45	0.11	0.097	-0.013 ± 0.004	-0.122 ± 0.035	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.820 ± 0.044	0.906 ± 0.030	0.096	-0.014	-12.28	0.1002	0.1011	0.957	91.568 ± 1.069	0.371 ± 0.004	19.822 ± 1.533
QML	beta46	0.08	0.158	0.078 ± 0.003	0.975 ± 0.034	0.005 ± 0.000	0.071 ± 0.002	0.011 ± 0.001	0.105 ± 0.003	1.736 ± 0.089	1.318 ± 0.036	0.156	0.076	94.71	0.0710	0.1038	1.009	84.467 ± 1.393	0.281 ± 0.004	66.716 ± 1.812
QML	beta47	0.08	0.158	0.078 ± 0.003	0.969 ± 0.035	0.005 ± 0.000	0.072 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	1.752 ± 0.104	1.324 ± 0.041	0.152	0.072	90.11	0.0670	0.0985	0.955	85.059 ± 1.371	0.270 ± 0.005	71.302 ± 1.740
QML	beta51	0.16	0.187	0.027 ± 0.005	0.168 ± 0.032	0.018 ± 0.001	0.133 ± 0.004	0.018 ± 0.001	0.136 ± 0.004	0.718 ± 0.039	0.847 ± 0.028	0.180	0.020	12.63	0.1325	0.1340	1.012	94.231 ± 0.897	0.527 ± 0.006	30.769 ± 1.775
QML	beta52	0.16	0.158	-0.002 ± 0.005	-0.015 ± 0.030	0.015 ± 0.001	0.124 ± 0.004	0.015 ± 0.001	0.124 ± 0.005	0.603 ± 0.038	0.776 ± 0.029	0.155	-0.005	-2.99	0.1294	0.1295	0.987	94.822 ± 0.852	0.481 ± 0.006	26.775 ± 1.703
QML	beta53	0.16	0.108	-0.052 ± 0.004	-0.322 ± 0.028	0.014 ± 0.001	0.117 ± 0.003	0.016 ± 0.001	0.128 ± 0.004	0.635 ± 0.032	0.797 ± 0.024	0.102	-0.058	-36.55	0.1217	0.1351	0.983	92.751 ± 0.997	0.450 ± 0.004	17.899 ± 1.474
QML	beta54	0.16	0.178	0.018 ± 0.004	0.114 ± 0.023	0.009 ± 0.001	0.096 ± 0.003	0.010 ± 0.001	0.097 ± 0.003	0.371 ± 0.021	0.609 ± 0.021	0.177	0.017	10.43	0.0995	0.1009	1.011	94.231 ± 0.897	0.380 ± 0.003	45.414 ± 1.915
QML	beta55	0.08	0.077	-0.003 ± 0.006	-0.032 ± 0.069	0.021 ± 0.001	0.144 ± 0.005	0.021 ± 0.001	0.144 ± 0.006	3.230 ± 0.214	1.797 ± 0.069	0.078	-0.002	-2.91	0.1506	0.1507	0.906	89.941 ± 1.157	0.511 ± 0.010	17.899 ± 1.474
QML	beta56	0.08	0.119	0.039 ± 0.003	0.491 ± 0.039	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	1.291 ± 0.072	1.136 ± 0.036	0.117	0.037	45.90	0.0850	0.0926	0.945	90.976 ± 1.102	0.304 ± 0.005	40.089 ± 1.885
QML	beta57	0.06	0.132	0.072 ± 0.005	1.200 ± 0.080	0.015 ± 0.001	0.124 ± 0.004	0.021 ± 0.001	0.143 ± 0.005	5.713 ± 0.335	2.390 ± 0.078	0.130	0.070	116.22	0.1319	0.1492	0.965	87.130 ± 1.288	0.470 ± 0.010	27.663 ± 1.721
QML	beta58	0.06	0.108	0.048 ± 0.002	0.798 ± 0.037	0.003 ± 0.000	0.058 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	1.567 ± 0.091	1.252 ± 0.039	0.106	0.046	75.85	0.0523	0.0693	0.986	89.941 ± 1.157	0.224 ± 0.004	55.030 ± 1.913
UPI	beta41	0.20	0.218	0.018 ± 0.008	0.089 ± 0.039	0.041 ± 0.003	0.202 ± 0.007	0.041 ± 0.003	0.202 ± 0.008	1.022 ± 0.073	1.011 ± 0.041	0.219	0.019	9.54	0.1966	0.1975	1.309	99.112 ± 0.361	1.034 ± 0.030	18.195 ± 1.484
UPI	beta42	0.20	0.227	0.027 ± 0.008	0.134 ± 0.039	0.041 ± 0.003	0.204 ± 0.007	0.042 ± 0.003	0.205 ± 0.008	1.053 ± 0.072	1.026 ± 0.040	0.227	0.027	13.50	0.1877	0.1896	1.348	98.817 ± 0.416	1.076 ± 0.041	14.941 ± 1.371
UPI	beta43	0.20	0.195	-0.005 ± 0.004	-0.023 ± 0.019	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.236 ± 0.013	0.485 ± 0.017	0.192	-0.008	-4.14	0.1049	0.1052	1.243	97.337 ± 0.619	0.473 ± 0.016	49.408 ± 1.923
UPI	beta44	0.11	0.112	0.002 ± 0.007	0.019 ± 0.062	0.031 ± 0.003	0.177 ± 0.007	0.031 ± 0.003	0.176 ± 0.008	2.572 ± 0.208	1.604 ± 0.072	0.104	-0.006	-5.69	0.1579	0.1580	1.504	98.225 ± 0.508	1.041 ± 0.056	6.805 ± 0.969
UPI	beta45	0.11	0.126	0.016 ± 0.007	0.142 ± 0.066	0.036 ± 0.003	0.189 ± 0.008	0.036 ± 0.003	0.189 ± 0.008	2.964 ± 0.239	1.722 ± 0.077	0.098	-0.012	-11.11	0.1570	0.1574	1.416	96.154 ± 0.740	1.048 ± 0.049	6.657 ± 0.959
UPI	beta46	0.08	0.080	-0.000 ± 0.004	-0.006 ± 0.044	0.008 ± 0.001	0.092 ± 0.004	0.008 ± 0.001	0.092 ± 0.004	1.318 ± 0.103	1.148 ± 0.050	0.078	-0.002	-2.67	0.0818	0.0818	1.343	97.189 ± 0.636	0.484 ± 0.022	15.237 ± 1.382

Table S151

Simulation 2 - Condition 7: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.077	-0.003 ± 0.003	-0.038 ± 0.042	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.088 ± 0.004	1.219 ± 0.090	1.104 ± 0.046	0.071	-0.009	-10.92	0.0752	0.0757	1.342	96.302 ± 0.726	0.465 ± 0.023	17.012 ± 1.445
UPI	beta51	0.16	0.171	0.011 ± 0.005	0.069 ± 0.030	0.015 ± 0.001	0.123 ± 0.003	0.015 ± 0.001	0.123 ± 0.004	0.595 ± 0.033	0.772 ± 0.026	0.161	0.001	0.44	0.1284	0.1284	1.075	96.154 ± 0.740	0.519 ± 0.007	27.959 ± 1.726
UPI	beta52	0.16	0.180	0.020 ± 0.008	0.123 ± 0.048	0.040 ± 0.003	0.201 ± 0.008	0.041 ± 0.003	0.201 ± 0.009	1.586 ± 0.124	1.259 ± 0.055	0.177	0.017	10.88	0.2013	0.2021	1.232	97.633 ± 0.585	0.969 ± 0.036	15.089 ± 1.377
UPI	beta53	0.16	0.155	-0.005 ± 0.006	-0.028 ± 0.036	0.022 ± 0.001	0.148 ± 0.005	0.022 ± 0.001	0.148 ± 0.005	0.858 ± 0.052	0.926 ± 0.033	0.155	-0.005	-3.41	0.1520	0.1521	1.183	97.781 ± 0.567	0.687 ± 0.016	16.716 ± 1.435
UPI	beta54	0.16	0.169	0.009 ± 0.007	0.053 ± 0.045	0.035 ± 0.002	0.186 ± 0.006	0.035 ± 0.002	0.186 ± 0.007	1.359 ± 0.089	1.166 ± 0.044	0.167	0.007	4.55	0.1942	0.1943	1.311	97.781 ± 0.567	0.958 ± 0.035	11.391 ± 1.222
UPI	beta55	0.08	0.092	0.012 ± 0.005	0.152 ± 0.068	0.020 ± 0.001	0.141 ± 0.005	0.020 ± 0.001	0.142 ± 0.006	3.137 ± 0.222	1.771 ± 0.072	0.086	0.006	7.39	0.1389	0.1391	1.369	98.225 ± 0.508	0.758 ± 0.035	9.763 ± 1.142
UPI	beta56	0.08	0.083	0.003 ± 0.003	0.042 ± 0.040	0.007 ± 0.001	0.084 ± 0.003	0.007 ± 0.001	0.084 ± 0.004	1.101 ± 0.086	1.050 ± 0.046	0.082	0.002	2.19	0.0783	0.0783	1.223	97.929 ± 0.548	0.403 ± 0.016	18.935 ± 1.507
UPI	beta57	0.06	0.047	-0.013 ± 0.006	-0.211 ± 0.094	0.021 ± 0.002	0.146 ± 0.006	0.022 ± 0.002	0.147 ± 0.006	5.978 ± 0.475	2.445 ± 0.108	0.047	-0.013	-21.44	0.1301	0.1307	1.280	97.929 ± 0.548	0.734 ± 0.035	4.142 ± 0.766
UPI	beta58	0.06	0.059	-0.001 ± 0.003	-0.020 ± 0.050	0.006 ± 0.000	0.078 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	1.685 ± 0.132	1.298 ± 0.057	0.059	-0.001	-1.91	0.0672	0.0672	1.236	96.746 ± 0.682	0.378 ± 0.017	13.609 ± 1.319

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S152

Simulation 2 - Condition 8: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.003	0.012 ± 0.017	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.267 ± 0.014	0.517 ± 0.016	0.198	-0.002	-1.15	0.1021	0.1021	1.017	95.661 ± 0.663	0.412 ± 0.003	50.688 ± 1.626
LSAM	beta42	0.20	0.199	-0.001 ± 0.004	-0.005 ± 0.018	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.316 ± 0.016	0.562 ± 0.017	0.205	0.005	2.33	0.1083	0.1084	0.980	95.873 ± 0.647	0.432 ± 0.003	51.005 ± 1.626
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.004	0.275 ± 0.008	0.199	-0.001	-0.60	0.0549	0.0549	1.013	95.979 ± 0.639	0.218 ± 0.001	95.344 ± 0.685
LSAM	beta44	0.11	0.118	0.008 ± 0.003	0.072 ± 0.024	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.551 ± 0.029	0.742 ± 0.023	0.114	0.004	3.43	0.0804	0.0805	0.987	95.661 ± 0.663	0.315 ± 0.004	33.333 ± 1.533
LSAM	beta45	0.11	0.116	0.006 ± 0.002	0.054 ± 0.022	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.447 ± 0.024	0.668 ± 0.021	0.114	0.004	3.24	0.0683	0.0684	1.004	94.921 ± 0.714	0.288 ± 0.003	37.037 ± 1.571
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.012 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.307 ± 0.017	0.554 ± 0.018	0.080	0.000	0.62	0.0413	0.0413	0.926	93.757 ± 0.787	0.161 ± 0.002	55.979 ± 1.615
LSAM	beta47	0.08	0.082	0.002 ± 0.001	0.023 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.288 ± 0.015	0.536 ± 0.016	0.079	-0.001	-0.79	0.0404	0.0404	0.950	95.132 ± 0.700	0.160 ± 0.002	57.249 ± 1.609
LSAM	beta51	0.16	0.156	-0.004 ± 0.002	-0.027 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.204 ± 0.010	0.451 ± 0.013	0.157	-0.003	-2.07	0.0710	0.0711	1.007	95.873 ± 0.647	0.285 ± 0.001	60.529 ± 1.590
LSAM	beta52	0.16	0.165	0.005 ± 0.004	0.033 ± 0.023	0.013 ± 0.001	0.115 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.517 ± 0.028	0.719 ± 0.023	0.158	-0.002	-1.00	0.1089	0.1089	0.999	96.825 ± 0.570	0.450 ± 0.005	35.238 ± 1.554
LSAM	beta53	0.16	0.162	0.002 ± 0.003	0.011 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.264 ± 0.014	0.514 ± 0.016	0.164	0.004	2.52	0.0837	0.0838	0.997	95.767 ± 0.655	0.321 ± 0.002	53.545 ± 1.622
LSAM	beta54	0.16	0.161	0.001 ± 0.003	0.008 ± 0.021	0.011 ± 0.001	0.105 ± 0.002	0.011 ± 0.001	0.105 ± 0.003	0.433 ± 0.020	0.658 ± 0.019	0.165	0.005	3.41	0.1062	0.1063	0.992	95.873 ± 0.647	0.410 ± 0.003	36.190 ± 1.563
LSAM	beta55	0.08	0.081	0.001 ± 0.002	0.018 ± 0.031	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.885 ± 0.046	0.941 ± 0.029	0.083	0.003	3.31	0.0680	0.0681	0.982	96.296 ± 0.614	0.290 ± 0.005	26.138 ± 1.429
LSAM	beta56	0.08	0.084	0.004 ± 0.001	0.049 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.302 ± 0.016	0.549 ± 0.017	0.081	0.001	0.71	0.0389	0.0389	0.930	94.392 ± 0.748	0.160 ± 0.002	58.413 ± 1.603
LSAM	beta57	0.06	0.061	0.001 ± 0.003	0.017 ± 0.042	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	1.689 ± 0.095	1.300 ± 0.042	0.058	-0.002	-3.61	0.0681	0.0682	0.987	95.979 ± 0.639	0.302 ± 0.005	11.746 ± 1.047
LSAM	beta58	0.06	0.062	0.002 ± 0.001	0.035 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.384 ± 0.020	0.620 ± 0.019	0.061	0.001	1.14	0.0353	0.0353	0.962	94.603 ± 0.735	0.140 ± 0.002	48.254 ± 1.626
QML	beta41	0.20	0.235	0.035 ± 0.002	0.175 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.106 ± 0.005	0.325 ± 0.009	0.233	0.033	16.71	0.0534	0.0630	1.033	91.746 ± 0.895	0.222 ± 0.001	98.730 ± 0.364
QML	beta42	0.20	0.230	0.030 ± 0.002	0.152 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.106 ± 0.005	0.326 ± 0.009	0.232	0.032	15.82	0.0579	0.0660	1.006	93.122 ± 0.823	0.228 ± 0.001	98.413 ± 0.407
QML	beta43	0.20	0.183	-0.017 ± 0.002	-0.083 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.183	-0.017	-8.54	0.0484	0.0513	1.023	94.497 ± 0.742	0.194 ± 0.001	96.931 ± 0.561
QML	beta44	0.11	0.030	-0.080 ± 0.002	-0.728 ± 0.021	0.005 ± 0.000	0.072 ± 0.002	0.012 ± 0.000	0.108 ± 0.002	0.958 ± 0.038	0.979 ± 0.021	0.032	-0.078	-71.00	0.0669	0.1029	0.961	77.884 ± 1.350	0.271 ± 0.002	8.995 ± 0.931
QML	beta45	0.11	0.099	-0.011 ± 0.002	-0.103 ± 0.018	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.306 ± 0.016	0.553 ± 0.017	0.101	-0.009	-7.94	0.0542	0.0549	0.988	94.392 ± 0.748	0.231 ± 0.002	41.164 ± 1.601
QML	beta46	0.08	0.159	0.079 ± 0.001	0.984 ± 0.018	0.002 ± 0.000	0.043 ± 0.001	0.008 ± 0.000	0.090 ± 0.001	1.258 ± 0.041	1.122 ± 0.019	0.155	0.075	94.09	0.0408	0.0856	1.000	54.286 ± 1.621	0.169 ± 0.002	97.884 ± 0.468
QML	beta47	0.08	0.155	0.075 ± 0.001	0.935 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.007 ± 0.000	0.086 ± 0.001	1.157 ± 0.038	1.076 ± 0.018	0.152	0.072	89.48	0.0409	0.0825	0.954	53.651 ± 1.622	0.159 ± 0.001	97.037 ± 0.552
QML	beta51	0.16	0.175	0.015 ± 0.003	0.091 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.263 ± 0.012	0.512 ± 0.014	0.176	0.016	9.79	0.0802	0.0817	1.017	94.815 ± 0.721	0.322 ± 0.003	59.153 ± 1.599
QML	beta52	0.16	0.164	0.004 ± 0.003	0.023 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.232 ± 0.011	0.481 ± 0.014	0.164	0.004	2.51	0.0766	0.0767	0.988	94.074 ± 0.768	0.298 ± 0.003	62.646 ± 1.574
QML	beta53	0.16	0.107	-0.053 ± 0.002	-0.331 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.295 ± 0.013	0.543 ± 0.013	0.104	-0.056	-35.19	0.0674	0.0879	0.997	87.407 ± 1.079	0.269 ± 0.001	34.180 ± 1.543
QML	beta54	0.16	0.177	0.017 ± 0.002	0.109 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.007	0.381 ± 0.010	0.178	0.018	11.31	0.0610	0.0636	1.013	95.238 ± 0.693	0.232 ± 0.001	85.503 ± 1.145
QML	beta55	0.08	0.073	-0.007 ± 0.003	-0.084 ± 0.038	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	1.362 ± 0.066	1.167 ± 0.034	0.066	-0.014	-18.11	0.0893	0.0904	0.923	87.407 ± 1.079	0.337 ± 0.006	18.836 ± 1.272
QML	beta56	0.08	0.123	0.043 ± 0.002	0.540 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.004 ± 0.000	0.065 ± 0.001	0.664 ± 0.027	0.815 ± 0.018	0.123	0.043	53.56	0.0464	0.0631	0.962	81.058 ± 1.275	0.184 ± 0.002	74.921 ± 1.410
QML	beta57	0.06	0.139	0.079 ± 0.003	1.322 ± 0.045	0.007 ± 0.000	0.084 ± 0.002	0.013 ± 0.001	0.115 ± 0.002	3.700 ± 0.142	1.923 ± 0.041	0.141	0.081	135.47	0.0815	0.1151	0.967	73.862 ± 1.429	0.318 ± 0.005	51.323 ± 1.626
QML	beta58	0.06	0.113	0.053 ± 0.001	0.883 ± 0.019	0.001 ± 0.000	0.034 ± 0.001	0.004 ± 0.000	0.063 ± 0.001	1.106 ± 0.040	1.052 ± 0.020	0.112	0.052	86.14	0.0341	0.0619	0.991	66.561 ± 1.535	0.133 ± 0.001	93.439 ± 0.805
UPI	beta41	0.20	0.211	0.011 ± 0.003	0.057 ± 0.017	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.267 ± 0.013	0.517 ± 0.015	0.209	0.009	4.25	0.1000	0.1004	1.112	96.614 ± 0.588	0.448 ± 0.006	51.323 ± 1.626
UPI	beta42	0.20	0.205	0.005 ± 0.004	0.027 ± 0.018	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.307 ± 0.016	0.554 ± 0.017	0.207	0.007	3.34	0.0980	0.0982	1.070	96.508 ± 0.597	0.464 ± 0.005	47.302 ± 1.624
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.003	0.268 ± 0.008	0.199	-0.001	-0.59	0.0523	0.0523	1.045	95.979 ± 0.639	0.219 ± 0.002	93.968 ± 0.774
UPI	beta44	0.11	0.116	0.006 ± 0.003	0.057 ± 0.026	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.635 ± 0.037	0.797 ± 0.026	0.110	0.000	0.18	0.0804	0.0804	1.096	96.614 ± 0.588	0.376 ± 0.007	26.243 ± 1.431
UPI	beta45	0.11	0.113	0.003 ± 0.003	0.027 ± 0.025	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.573 ± 0.034	0.757 ± 0.026	0.108	-0.002	-1.53	0.0749	0.0749	1.170	95.979 ± 0.639	0.382 ± 0.011	24.233 ± 1.394
UPI	beta46	0.08	0.078	-0.002 ± 0.001	-0.025 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.320 ± 0.018	0.566 ± 0.018	0.075	-0.005	-6.06	0.0407	0.0410	1.008	95.026 ± 0.707	0.179 ± 0.003	48.254 ± 1.626

Table S152

Simulation 2 - Condition 8: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.018	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.291 ± 0.016	0.540 ± 0.017	0.077	-0.003	-4.19	0.0385	0.0386	1.028	95.132 ± 0.700	0.174 ± 0.003	51.111 ± 1.626
UPI	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.161	0.001	0.87	0.0691	0.0691	1.033	95.767 ± 0.655	0.281 ± 0.001	62.328 ± 1.576
UPI	beta52	0.16	0.177	0.017 ± 0.003	0.108 ± 0.022	0.011 ± 0.001	0.106 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	0.452 ± 0.024	0.672 ± 0.021	0.176	0.016	9.69	0.1008	0.1019	1.083	96.614 ± 0.588	0.451 ± 0.005	37.143 ± 1.572
UPI	beta53	0.16	0.161	0.001 ± 0.003	0.005 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.254 ± 0.013	0.504 ± 0.015	0.163	0.003	1.70	0.0820	0.0820	1.027	96.085 ± 0.631	0.325 ± 0.002	52.169 ± 1.625
UPI	beta54	0.16	0.159	-0.001 ± 0.003	-0.004 ± 0.021	0.011 ± 0.001	0.103 ± 0.002	0.011 ± 0.001	0.103 ± 0.003	0.415 ± 0.020	0.644 ± 0.019	0.163	0.003	1.90	0.1048	0.1048	1.068	97.354 ± 0.522	0.432 ± 0.006	33.333 ± 1.533
UPI	beta55	0.08	0.086	0.006 ± 0.002	0.074 ± 0.027	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.710 ± 0.038	0.842 ± 0.026	0.087	0.007	8.23	0.0623	0.0626	1.087	97.037 ± 0.552	0.286 ± 0.004	26.243 ± 1.431
UPI	beta56	0.08	0.082	0.002 ± 0.001	0.031 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.309 ± 0.017	0.555 ± 0.018	0.079	-0.001	-1.06	0.0396	0.0396	0.958	94.709 ± 0.728	0.167 ± 0.002	53.651 ± 1.622
UPI	beta57	0.06	0.053	-0.007 ± 0.002	-0.116 ± 0.036	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	1.232 ± 0.062	1.110 ± 0.033	0.051	-0.009	-14.57	0.0623	0.0629	1.132	96.720 ± 0.579	0.294 ± 0.005	9.418 ± 0.950
UPI	beta58	0.06	0.062	0.002 ± 0.001	0.038 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.398 ± 0.022	0.631 ± 0.020	0.060	0.000	0.59	0.0340	0.0340	1.014	95.450 ± 0.678	0.150 ± 0.003	44.339 ± 1.616

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S153

Simulation 2 - Condition 9: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.003	0.012 ± 0.017	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.287 ± 0.013	0.536 ± 0.015	0.201	0.001	0.62	0.1041	0.1041	1.025	95.988 ± 0.629	0.431 ± 0.003	48.663 ± 1.603
LSAM	beta42	0.20	0.200	-0.000 ± 0.004	-0.001 ± 0.018	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.305 ± 0.015	0.552 ± 0.016	0.203	0.003	1.74	0.1088	0.1089	1.026	96.399 ± 0.598	0.444 ± 0.003	46.605 ± 1.600
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.005	0.313 ± 0.010	0.201	0.001	0.70	0.0592	0.0592	0.989	93.930 ± 0.766	0.243 ± 0.001	89.712 ± 0.974
LSAM	beta44	0.11	0.115	0.005 ± 0.003	0.044 ± 0.025	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.596 ± 0.031	0.772 ± 0.023	0.112	0.002	1.61	0.0786	0.0787	0.950	94.547 ± 0.728	0.316 ± 0.003	32.305 ± 1.500
LSAM	beta45	0.11	0.123	0.013 ± 0.003	0.116 ± 0.023	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.546 ± 0.029	0.739 ± 0.023	0.121	0.011	10.24	0.0787	0.0795	0.925	94.033 ± 0.760	0.291 ± 0.003	42.078 ± 1.583
LSAM	beta46	0.08	0.082	0.002 ± 0.002	0.021 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.368 ± 0.020	0.607 ± 0.019	0.081	0.001	1.49	0.0439	0.0439	0.914	94.444 ± 0.735	0.174 ± 0.002	51.646 ± 1.603
LSAM	beta47	0.08	0.081	0.001 ± 0.001	0.015 ± 0.019	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.334 ± 0.018	0.578 ± 0.018	0.077	-0.003	-3.33	0.0417	0.0418	0.916	93.004 ± 0.818	0.166 ± 0.002	55.247 ± 1.595
LSAM	beta51	0.16	0.161	0.001 ± 0.003	0.005 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.257 ± 0.011	0.507 ± 0.014	0.162	0.002	1.41	0.0796	0.0796	1.000	94.650 ± 0.722	0.318 ± 0.001	51.955 ± 1.603
LSAM	beta52	0.16	0.169	0.009 ± 0.004	0.055 ± 0.024	0.014 ± 0.001	0.120 ± 0.003	0.014 ± 0.001	0.120 ± 0.003	0.562 ± 0.026	0.750 ± 0.021	0.170	0.010	5.97	0.1206	0.1210	0.989	96.193 ± 0.614	0.464 ± 0.003	33.745 ± 1.517
LSAM	beta53	0.16	0.157	-0.003 ± 0.003	-0.016 ± 0.018	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.327 ± 0.015	0.572 ± 0.016	0.155	-0.005	-2.96	0.0877	0.0878	0.989	94.444 ± 0.735	0.355 ± 0.002	43.004 ± 1.588
LSAM	beta54	0.16	0.160	-0.000 ± 0.004	-0.000 ± 0.023	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.003	0.523 ± 0.025	0.723 ± 0.021	0.164	0.004	2.67	0.1146	0.1147	0.960	94.136 ± 0.754	0.436 ± 0.003	32.099 ± 1.497
LSAM	beta55	0.08	0.083	0.003 ± 0.003	0.034 ± 0.033	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	1.067 ± 0.056	1.033 ± 0.032	0.084	0.004	5.36	0.0770	0.0771	0.926	93.827 ± 0.772	0.300 ± 0.003	23.971 ± 1.369
LSAM	beta56	0.08	0.080	0.000 ± 0.002	0.004 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.432 ± 0.024	0.657 ± 0.021	0.081	0.001	1.81	0.0464	0.0464	0.889	90.947 ± 0.920	0.183 ± 0.003	49.486 ± 1.604
LSAM	beta57	0.06	0.059	-0.001 ± 0.003	-0.016 ± 0.044	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	1.858 ± 0.103	1.363 ± 0.044	0.058	-0.002	-3.71	0.0716	0.0716	0.928	94.136 ± 0.754	0.298 ± 0.003	14.403 ± 1.126
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.023 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.556 ± 0.031	0.746 ± 0.024	0.062	0.002	2.61	0.0394	0.0395	0.888	93.313 ± 0.801	0.156 ± 0.002	43.004 ± 1.588
QML	beta41	0.20	0.217	0.017 ± 0.002	0.085 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.130 ± 0.006	0.361 ± 0.010	0.218	0.018	8.83	0.0716	0.0738	1.034	95.267 ± 0.681	0.285 ± 0.001	85.185 ± 1.139
QML	beta42	0.20	0.211	0.011 ± 0.002	0.057 ± 0.012	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.132 ± 0.006	0.363 ± 0.010	0.213	0.013	6.35	0.0729	0.0740	1.035	95.062 ± 0.695	0.291 ± 0.001	81.584 ± 1.243
QML	beta43	0.20	0.190	-0.010 ± 0.002	-0.051 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.294 ± 0.009	0.191	-0.009	-4.48	0.0554	0.0561	0.995	93.519 ± 0.790	0.226 ± 0.001	91.255 ± 0.906
QML	beta44	0.11	0.069	-0.041 ± 0.002	-0.370 ± 0.022	0.005 ± 0.000	0.074 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.588 ± 0.027	0.767 ± 0.020	0.070	-0.040	-36.43	0.0713	0.0817	0.961	89.712 ± 0.974	0.279 ± 0.002	19.650 ± 1.275
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.038 ± 0.019	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.368 ± 0.017	0.607 ± 0.017	0.106	-0.004	-3.58	0.0665	0.0666	0.949	93.930 ± 0.766	0.248 ± 0.002	42.181 ± 1.584
QML	beta46	0.08	0.115	0.035 ± 0.001	0.438 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.496 ± 0.024	0.704 ± 0.018	0.113	0.033	41.39	0.0407	0.0525	0.937	86.008 ± 1.113	0.162 ± 0.002	81.070 ± 1.257
QML	beta47	0.08	0.115	0.035 ± 0.001	0.439 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.454 ± 0.022	0.674 ± 0.017	0.114	0.034	41.90	0.0389	0.0514	0.949	86.008 ± 1.113	0.152 ± 0.002	84.156 ± 1.171
QML	beta51	0.16	0.178	0.018 ± 0.003	0.115 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.271 ± 0.012	0.521 ± 0.014	0.179	0.019	11.61	0.0800	0.0821	1.033	95.370 ± 0.674	0.329 ± 0.002	57.510 ± 1.586
QML	beta52	0.16	0.153	-0.007 ± 0.003	-0.047 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.248 ± 0.011	0.498 ± 0.014	0.154	-0.006	-3.74	0.0846	0.0848	1.011	95.473 ± 0.667	0.315 ± 0.002	49.177 ± 1.604
QML	beta53	0.16	0.122	-0.038 ± 0.003	-0.236 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.091 ± 0.002	0.321 ± 0.014	0.567 ± 0.015	0.121	-0.039	-24.21	0.0851	0.0935	0.986	92.593 ± 0.840	0.319 ± 0.001	33.848 ± 1.518
QML	beta54	0.16	0.165	0.005 ± 0.003	0.031 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.243 ± 0.011	0.492 ± 0.014	0.168	0.008	4.88	0.0810	0.0814	0.974	94.753 ± 0.715	0.300 ± 0.001	58.333 ± 1.581
QML	beta55	0.08	0.087	0.007 ± 0.003	0.089 ± 0.032	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	1.026 ± 0.051	1.013 ± 0.030	0.088	0.008	9.64	0.0761	0.0765	0.965	91.564 ± 0.891	0.305 ± 0.005	27.263 ± 1.428
QML	beta56	0.08	0.094	0.014 ± 0.002	0.180 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.453 ± 0.023	0.673 ± 0.020	0.096	0.016	19.64	0.0481	0.0506	0.893	89.198 ± 0.996	0.182 ± 0.002	57.922 ± 1.583
QML	beta57	0.06	0.084	0.024 ± 0.002	0.402 ± 0.038	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	1.560 ± 0.079	1.249 ± 0.036	0.084	0.024	39.42	0.0644	0.0686	1.003	90.947 ± 0.920	0.279 ± 0.004	27.984 ± 1.440
QML	beta58	0.06	0.084	0.024 ± 0.001	0.395 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.563 ± 0.026	0.750 ± 0.020	0.083	0.023	37.98	0.0356	0.0423	0.919	87.140 ± 1.074	0.138 ± 0.002	66.667 ± 1.512
UPI	beta41	0.20	0.212	0.012 ± 0.003	0.062 ± 0.017	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.285 ± 0.014	0.533 ± 0.016	0.216	0.016	7.90	0.1026	0.1038	1.046	96.605 ± 0.581	0.435 ± 0.003	51.337 ± 1.603
UPI	beta42	0.20	0.207	0.007 ± 0.004	0.036 ± 0.018	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.300 ± 0.014	0.548 ± 0.016	0.207	0.007	3.61	0.1101	0.1103	1.054	96.502 ± 0.589	0.452 ± 0.003	47.942 ± 1.602
UPI	beta43	0.20	0.196	-0.004 ± 0.002	-0.019 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.315 ± 0.009	0.198	-0.002	-0.94	0.0591	0.0591	0.972	93.827 ± 0.772	0.239 ± 0.001	87.757 ± 1.051
UPI	beta44	0.11	0.113	0.003 ± 0.003	0.024 ± 0.024	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.580 ± 0.033	0.762 ± 0.025	0.112	0.002	1.56	0.0784	0.0785	1.068	95.988 ± 0.629	0.351 ± 0.006	28.807 ± 1.453
UPI	beta45	0.11	0.119	0.009 ± 0.003	0.080 ± 0.026	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.644 ± 0.038	0.802 ± 0.027	0.114	0.004	3.20	0.0780	0.0781	0.987	94.856 ± 0.709	0.340 ± 0.006	32.202 ± 1.499
UPI	beta46	0.08	0.078	-0.002 ± 0.002	-0.021 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.363 ± 0.020	0.603 ± 0.019	0.078	-0.002	-2.82	0.0435	0.0435	0.953	94.856 ± 0.709	0.180 ± 0.002	47.634 ± 1.602

Table S153

Simulation 2 - Condition 9: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.016 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.314 ± 0.017	0.560 ± 0.017	0.077	-0.003	-3.91	0.0414	0.0415	0.974	93.210 ± 0.807	0.171 ± 0.003	52.263 ± 1.602
UPI	beta51	0.16	0.164	0.004 ± 0.003	0.027 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.249 ± 0.011	0.499 ± 0.014	0.164	0.004	2.20	0.0789	0.0790	1.000	94.753 ± 0.715	0.313 ± 0.001	55.350 ± 1.595
UPI	beta52	0.16	0.177	0.017 ± 0.004	0.106 ± 0.024	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.120 ± 0.003	0.562 ± 0.027	0.750 ± 0.022	0.179	0.019	11.73	0.1176	0.1190	0.971	94.444 ± 0.735	0.452 ± 0.003	37.243 ± 1.551
UPI	beta53	0.16	0.157	-0.003 ± 0.003	-0.021 ± 0.018	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.313 ± 0.015	0.559 ± 0.016	0.157	-0.003	-2.08	0.0855	0.0856	1.007	95.267 ± 0.681	0.353 ± 0.002	43.004 ± 1.588
UPI	beta54	0.16	0.161	0.001 ± 0.004	0.007 ± 0.023	0.013 ± 0.001	0.115 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.512 ± 0.025	0.716 ± 0.021	0.165	0.005	2.89	0.1148	0.1149	0.981	95.267 ± 0.681	0.441 ± 0.003	33.230 ± 1.511
UPI	beta55	0.08	0.085	0.005 ± 0.003	0.066 ± 0.033	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	1.059 ± 0.061	1.029 ± 0.034	0.085	0.005	5.93	0.0736	0.0737	0.929	93.519 ± 0.790	0.299 ± 0.004	23.971 ± 1.369
UPI	beta56	0.08	0.079	-0.001 ± 0.002	-0.007 ± 0.021	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	0.442 ± 0.025	0.665 ± 0.022	0.080	0.000	-0.29	0.0455	0.0455	0.893	92.387 ± 0.851	0.186 ± 0.003	47.325 ± 1.601
UPI	beta57	0.06	0.054	-0.006 ± 0.003	-0.093 ± 0.042	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	1.749 ± 0.097	1.323 ± 0.042	0.054	-0.006	-10.69	0.0723	0.0725	0.928	93.724 ± 0.778	0.288 ± 0.004	13.580 ± 1.099
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.010 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.555 ± 0.031	0.745 ± 0.024	0.060	0.000	0.43	0.0385	0.0385	0.910	92.798 ± 0.829	0.160 ± 0.002	41.461 ± 1.580

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S154

Simulation 2 - Condition 10: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.010 ± 0.011	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.113 ± 0.005	0.337 ± 0.010	0.199	-0.001	-0.44	0.0663	0.0663	0.966	94.444 ± 0.728	0.255 ± 0.001	86.768 ± 1.077
LSAM	beta42	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.011	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.111 ± 0.005	0.333 ± 0.010	0.198	-0.002	-0.91	0.0652	0.0653	1.010	96.364 ± 0.595	0.263 ± 0.001	83.737 ± 1.173
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.201	0.001	0.27	0.0369	0.0369	1.000	94.343 ± 0.734	0.149 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.014 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.183 ± 0.009	0.428 ± 0.013	0.110	0.000	0.45	0.0441	0.0441	0.963	93.838 ± 0.764	0.178 ± 0.001	69.091 ± 1.469
LSAM	beta45	0.11	0.113	0.003 ± 0.001	0.026 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.163 ± 0.008	0.404 ± 0.012	0.113	0.003	3.04	0.0448	0.0450	0.966	94.242 ± 0.740	0.168 ± 0.001	74.444 ± 1.386
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.013 ± 0.010	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.108 ± 0.005	0.328 ± 0.010	0.080	0.000	-0.14	0.0249	0.0249	0.933	93.939 ± 0.758	0.096 ± 0.001	89.091 ± 0.991
LSAM	beta47	0.08	0.081	0.001 ± 0.001	0.010 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.101 ± 0.005	0.318 ± 0.010	0.080	0.000	0.26	0.0245	0.0245	0.937	93.737 ± 0.770	0.093 ± 0.001	90.000 ± 0.953
LSAM	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.159	-0.001	-0.78	0.0464	0.0464	1.048	96.465 ± 0.587	0.197 ± 0.000	90.202 ± 0.945
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.004 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.192 ± 0.009	0.438 ± 0.012	0.161	0.001	0.32	0.0695	0.0695	0.999	94.646 ± 0.715	0.275 ± 0.001	65.152 ± 1.514
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.112 ± 0.006	0.335 ± 0.010	0.160	0.000	0.19	0.0521	0.0521	1.018	95.556 ± 0.655	0.214 ± 0.001	84.141 ± 1.161
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.190 ± 0.009	0.436 ± 0.013	0.162	0.002	1.03	0.0683	0.0683	0.952	94.343 ± 0.734	0.260 ± 0.001	67.273 ± 1.491
LSAM	beta55	0.08	0.080	0.000 ± 0.001	0.004 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.274 ± 0.013	0.523 ± 0.015	0.079	-0.001	-1.81	0.0400	0.0400	0.985	94.646 ± 0.715	0.162 ± 0.001	51.212 ± 1.589
LSAM	beta56	0.08	0.080	-0.000 ± 0.001	-0.005 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.111 ± 0.006	0.334 ± 0.010	0.080	0.000	-0.48	0.0240	0.0240	0.946	92.828 ± 0.820	0.099 ± 0.001	85.253 ± 1.127
LSAM	beta57	0.06	0.061	0.001 ± 0.001	0.018 ± 0.023	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.505 ± 0.026	0.711 ± 0.021	0.060	0.000	0.55	0.0402	0.0402	0.986	94.545 ± 0.722	0.165 ± 0.001	32.828 ± 1.492
LSAM	beta58	0.06	0.060	-0.000 ± 0.001	-0.005 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.152 ± 0.008	0.389 ± 0.012	0.059	-0.001	-0.89	0.0221	0.0222	0.964	95.152 ± 0.683	0.088 ± 0.001	76.970 ± 1.338
QML	beta41	0.20	0.217	0.017 ± 0.001	0.084 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.215	0.015	7.42	0.0463	0.0486	0.986	93.232 ± 0.798	0.177 ± 0.001	99.596 ± 0.202
QML	beta42	0.20	0.210	0.010 ± 0.001	0.050 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.003	0.230 ± 0.007	0.209	0.009	4.42	0.0436	0.0445	1.024	94.949 ± 0.696	0.180 ± 0.001	99.596 ± 0.202
QML	beta43	0.20	0.192	-0.008 ± 0.001	-0.040 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.183 ± 0.005	0.192	-0.008	-4.08	0.0358	0.0367	1.009	93.232 ± 0.798	0.142 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.072	-0.038 ± 0.001	-0.342 ± 0.013	0.002 ± 0.000	0.046 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.293 ± 0.012	0.541 ± 0.013	0.074	-0.036	-33.00	0.0447	0.0576	0.982	83.737 ± 1.173	0.178 ± 0.006	41.515 ± 1.566
QML	beta45	0.11	0.103	-0.007 ± 0.001	-0.065 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.139 ± 0.007	0.373 ± 0.011	0.104	-0.006	-5.75	0.0417	0.0421	0.989	93.636 ± 0.776	0.157 ± 0.002	72.525 ± 1.419
QML	beta46	0.08	0.115	0.035 ± 0.001	0.441 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.295 ± 0.011	0.543 ± 0.010	0.114	0.034	42.67	0.0237	0.0416	0.993	72.121 ± 1.425	0.099 ± 0.001	98.889 ± 0.333
QML	beta47	0.08	0.115	0.035 ± 0.001	0.435 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.287 ± 0.010	0.536 ± 0.010	0.113	0.033	41.70	0.0251	0.0417	0.961	68.081 ± 1.482	0.094 ± 0.002	99.192 ± 0.285
QML	beta51	0.16	0.175	0.015 ± 0.002	0.096 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.326 ± 0.009	0.176	0.016	9.91	0.0517	0.0541	1.051	95.455 ± 0.662	0.206 ± 0.001	92.525 ± 0.836
QML	beta52	0.16	0.147	-0.013 ± 0.002	-0.083 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.004	0.318 ± 0.008	0.147	-0.013	-8.30	0.0490	0.0507	1.001	94.141 ± 0.746	0.193 ± 0.001	83.838 ± 1.170
QML	beta53	0.16	0.124	-0.036 ± 0.002	-0.227 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.143 ± 0.006	0.378 ± 0.009	0.124	-0.036	-22.56	0.0479	0.0600	1.040	88.788 ± 1.003	0.197 ± 0.001	70.707 ± 1.446
QML	beta54	0.16	0.165	0.005 ± 0.002	0.030 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.317 ± 0.009	0.164	0.004	2.80	0.0484	0.0486	0.944	93.636 ± 0.776	0.187 ± 0.001	91.616 ± 0.881
QML	beta55	0.08	0.082	0.002 ± 0.002	0.024 ± 0.020	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.395 ± 0.017	0.628 ± 0.017	0.080	0.000	0.16	0.0508	0.0508	0.972	91.818 ± 0.871	0.191 ± 0.003	44.848 ± 1.581
QML	beta56	0.08	0.096	0.016 ± 0.001	0.200 ± 0.012	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.172 ± 0.008	0.415 ± 0.011	0.096	0.016	19.68	0.0286	0.0327	0.948	88.081 ± 1.030	0.108 ± 0.001	89.798 ± 0.962
QML	beta57	0.06	0.092	0.032 ± 0.001	0.532 ± 0.024	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.862 ± 0.038	0.929 ± 0.023	0.093	0.033	54.29	0.0427	0.0537	1.000	88.283 ± 1.022	0.179 ± 0.002	56.768 ± 1.574
QML	beta58	0.06	0.084	0.024 ± 0.001	0.399 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.295 ± 0.011	0.543 ± 0.011	0.084	0.024	39.84	0.0218	0.0324	0.971	77.879 ± 1.319	0.084 ± 0.001	95.657 ± 0.648
UPI	beta41	0.20	0.208	0.008 ± 0.002	0.038 ± 0.011	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.113 ± 0.005	0.337 ± 0.010	0.207	0.007	3.45	0.0655	0.0658	0.970	94.747 ± 0.709	0.255 ± 0.001	88.687 ± 1.007
UPI	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.107 ± 0.005	0.327 ± 0.009	0.197	-0.003	-1.25	0.0646	0.0647	1.026	95.859 ± 0.633	0.263 ± 0.001	84.545 ± 1.149
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.200	0.000	-0.12	0.0372	0.0372	0.984	93.535 ± 0.782	0.146 ± 0.001	100.000 ± 0.000
UPI	beta44	0.11	0.111	0.001 ± 0.002	0.010 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.197 ± 0.010	0.444 ± 0.013	0.109	-0.001	-0.71	0.0482	0.0482	0.966	94.040 ± 0.752	0.185 ± 0.002	66.970 ± 1.495
UPI	beta45	0.11	0.112	0.002 ± 0.001	0.021 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.183 ± 0.009	0.428 ± 0.013	0.111	0.001	1.19	0.0455	0.0455	0.998	94.848 ± 0.703	0.184 ± 0.002	70.808 ± 1.445
UPI	beta46	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.114 ± 0.006	0.337 ± 0.010	0.078	-0.002	-3.09	0.0244	0.0245	0.919	93.131 ± 0.804	0.097 ± 0.001	87.879 ± 1.037

Table S154

Simulation 2 - Condition 10: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	0.000 ± 0.001	0.002 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.101 ± 0.005	0.318 ± 0.010	0.080	0.000	0.11	0.0245	0.0245	0.942	92.626 ± 0.831	0.094 ± 0.001	90.505 ± 0.932
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.160	0.000	0.14	0.0475	0.0475	1.039	95.354 ± 0.669	0.195 ± 0.000	91.717 ± 0.876
UPI	beta52	0.16	0.165	0.005 ± 0.002	0.032 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.187 ± 0.008	0.432 ± 0.012	0.166	0.006	3.93	0.0682	0.0685	0.990	94.545 ± 0.722	0.268 ± 0.001	68.283 ± 1.479
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.110 ± 0.005	0.331 ± 0.010	0.158	-0.002	-1.37	0.0501	0.0501	1.017	95.455 ± 0.662	0.211 ± 0.001	85.354 ± 1.124
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.186 ± 0.009	0.432 ± 0.012	0.162	0.002	1.18	0.0672	0.0673	0.960	94.545 ± 0.722	0.260 ± 0.001	67.778 ± 1.485
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.019 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.271 ± 0.013	0.521 ± 0.015	0.080	0.000	0.19	0.0393	0.0393	0.979	93.838 ± 0.764	0.160 ± 0.001	52.424 ± 1.587
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.116 ± 0.006	0.340 ± 0.011	0.079	-0.001	-1.06	0.0246	0.0246	0.939	92.525 ± 0.836	0.100 ± 0.001	84.646 ± 1.146
UPI	beta57	0.06	0.059	-0.001 ± 0.001	-0.020 ± 0.022	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.501 ± 0.025	0.708 ± 0.021	0.057	-0.003	-4.58	0.0402	0.0403	0.957	94.141 ± 0.746	0.159 ± 0.001	31.313 ± 1.474
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.007 ± 0.012	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.154 ± 0.008	0.393 ± 0.012	0.059	-0.001	-0.84	0.0227	0.0227	0.968	94.545 ± 0.722	0.090 ± 0.001	76.364 ± 1.350

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S155

Simulation 2 - Condition 11: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.002	0.017 ± 0.012	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.141 ± 0.006	0.375 ± 0.010	0.203	0.003	1.45	0.0725	0.0725	1.040	96.061 ± 0.618	0.306 ± 0.001	75.455 ± 1.368
LSAM	beta42	0.20	0.201	0.001 ± 0.003	0.005 ± 0.013	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.167 ± 0.008	0.408 ± 0.011	0.205	0.005	2.46	0.0784	0.0785	0.988	94.444 ± 0.728	0.316 ± 0.001	70.808 ± 1.445
LSAM	beta43	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.199	-0.001	-0.41	0.0489	0.0489	0.978	94.444 ± 0.728	0.186 ± 0.001	97.677 ± 0.479
LSAM	beta44	0.11	0.114	0.004 ± 0.002	0.040 ± 0.017	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.285 ± 0.014	0.534 ± 0.016	0.115	0.005	4.15	0.0556	0.0558	0.897	92.121 ± 0.856	0.206 ± 0.002	59.495 ± 1.560
LSAM	beta45	0.11	0.112	0.002 ± 0.002	0.018 ± 0.016	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.264 ± 0.013	0.514 ± 0.015	0.110	0.000	0.19	0.0556	0.0556	0.881	91.717 ± 0.876	0.195 ± 0.002	59.394 ± 1.561
LSAM	beta46	0.08	0.080	-0.000 ± 0.001	-0.006 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.157 ± 0.008	0.396 ± 0.012	0.080	0.000	-0.61	0.0285	0.0285	0.936	93.030 ± 0.809	0.116 ± 0.001	75.960 ± 1.358
LSAM	beta47	0.08	0.080	-0.000 ± 0.001	-0.004 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.170 ± 0.009	0.413 ± 0.013	0.077	-0.003	-3.23	0.0301	0.0302	0.887	92.222 ± 0.851	0.115 ± 0.001	73.939 ± 1.395
LSAM	beta51	0.16	0.157	-0.003 ± 0.002	-0.019 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.007	0.399 ± 0.011	0.157	-0.003	-2.04	0.0639	0.0640	1.001	94.949 ± 0.696	0.250 ± 0.001	69.596 ± 1.462
LSAM	beta52	0.16	0.163	0.003 ± 0.003	0.020 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.255 ± 0.012	0.505 ± 0.014	0.162	0.002	1.46	0.0826	0.0826	1.043	96.263 ± 0.603	0.330 ± 0.001	51.616 ± 1.588
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.158	-0.002	-1.19	0.0700	0.0701	0.985	95.556 ± 0.655	0.268 ± 0.001	63.030 ± 1.534
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.241 ± 0.011	0.491 ± 0.013	0.156	-0.004	-2.48	0.0757	0.0759	1.031	95.354 ± 0.669	0.318 ± 0.001	50.505 ± 1.589
LSAM	beta55	0.08	0.079	-0.001 ± 0.002	-0.016 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.481 ± 0.023	0.694 ± 0.020	0.078	-0.002	-2.18	0.0562	0.0562	0.927	93.131 ± 0.804	0.202 ± 0.002	36.768 ± 1.532
LSAM	beta56	0.08	0.083	0.003 ± 0.001	0.037 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.212 ± 0.011	0.460 ± 0.014	0.081	0.001	1.54	0.0330	0.0331	0.931	92.929 ± 0.815	0.134 ± 0.002	66.970 ± 1.495
LSAM	beta57	0.06	0.061	0.001 ± 0.002	0.015 ± 0.027	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.748 ± 0.037	0.865 ± 0.025	0.061	0.001	1.63	0.0478	0.0478	0.949	93.434 ± 0.787	0.193 ± 0.002	26.768 ± 1.407
LSAM	beta58	0.06	0.062	0.002 ± 0.001	0.032 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.248 ± 0.013	0.498 ± 0.015	0.061	0.001	2.43	0.0273	0.0273	0.941	94.040 ± 0.752	0.110 ± 0.001	62.525 ± 1.538
QML	beta41	0.20	0.209	0.009 ± 0.002	0.044 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.004	0.313 ± 0.009	0.209	0.009	4.46	0.0610	0.0617	1.049	95.758 ± 0.641	0.255 ± 0.001	89.495 ± 0.974
QML	beta42	0.20	0.205	0.005 ± 0.002	0.024 ± 0.011	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.110 ± 0.005	0.332 ± 0.009	0.208	0.008	3.83	0.0660	0.0664	1.001	94.747 ± 0.709	0.260 ± 0.001	85.859 ± 1.107
QML	beta43	0.20	0.194	-0.006 ± 0.001	-0.028 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.194	-0.006	-3.11	0.0474	0.0478	0.986	95.354 ± 0.669	0.181 ± 0.001	97.677 ± 0.479
QML	beta44	0.11	0.098	-0.012 ± 0.002	-0.107 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.274 ± 0.013	0.524 ± 0.015	0.099	-0.011	-10.43	0.0551	0.0563	0.911	91.919 ± 0.866	0.202 ± 0.001	50.404 ± 1.589
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.036 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.235 ± 0.011	0.485 ± 0.013	0.106	-0.004	-3.66	0.0534	0.0535	0.896	91.414 ± 0.890	0.187 ± 0.001	57.980 ± 1.569
QML	beta46	0.08	0.091	0.011 ± 0.001	0.141 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.162 ± 0.008	0.402 ± 0.011	0.091	0.011	13.54	0.0270	0.0291	0.942	92.323 ± 0.846	0.111 ± 0.001	86.364 ± 1.091
QML	beta47	0.08	0.092	0.012 ± 0.001	0.149 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.168 ± 0.009	0.410 ± 0.012	0.090	0.010	12.01	0.0276	0.0292	0.906	91.212 ± 0.900	0.108 ± 0.001	87.273 ± 1.059
QML	beta51	0.16	0.165	0.005 ± 0.002	0.030 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.401 ± 0.011	0.164	0.004	2.80	0.0629	0.0631	1.005	95.657 ± 0.648	0.252 ± 0.001	72.727 ± 1.415
QML	beta52	0.16	0.148	-0.012 ± 0.002	-0.076 ± 0.013	0.004 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.180 ± 0.008	0.424 ± 0.011	0.147	-0.013	-7.97	0.0683	0.0695	1.032	95.354 ± 0.669	0.270 ± 0.001	58.081 ± 1.568
QML	beta53	0.16	0.133	-0.027 ± 0.002	-0.166 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.204 ± 0.009	0.451 ± 0.012	0.133	-0.027	-16.90	0.0672	0.0725	1.001	93.434 ± 0.787	0.264 ± 0.001	51.212 ± 1.589
QML	beta54	0.16	0.160	0.000 ± 0.002	0.001 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.008	0.413 ± 0.011	0.156	-0.004	-2.48	0.0647	0.0648	1.037	95.253 ± 0.676	0.268 ± 0.001	65.253 ± 1.513
QML	beta55	0.08	0.081	0.001 ± 0.002	0.009 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.483 ± 0.023	0.695 ± 0.020	0.080	0.000	-0.55	0.0548	0.0548	0.935	92.323 ± 0.846	0.204 ± 0.002	37.576 ± 1.539
QML	beta56	0.08	0.088	0.008 ± 0.001	0.105 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.220 ± 0.012	0.469 ± 0.014	0.086	0.006	7.67	0.0323	0.0329	0.937	91.919 ± 0.866	0.134 ± 0.002	72.222 ± 1.424
QML	beta57	0.06	0.070	0.010 ± 0.002	0.166 ± 0.026	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.687 ± 0.032	0.829 ± 0.023	0.071	0.011	18.77	0.0468	0.0481	0.973	93.333 ± 0.793	0.186 ± 0.002	36.263 ± 1.528
QML	beta58	0.06	0.071	0.011 ± 0.001	0.181 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.251 ± 0.012	0.501 ± 0.014	0.070	0.010	17.30	0.0265	0.0285	0.943	90.707 ± 0.923	0.104 ± 0.001	76.162 ± 1.354
UPI	beta41	0.20	0.209	0.009 ± 0.002	0.045 ± 0.012	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.142 ± 0.006	0.376 ± 0.010	0.209	0.009	4.47	0.0724	0.0730	1.023	95.354 ± 0.669	0.300 ± 0.001	77.475 ± 1.328
UPI	beta42	0.20	0.203	0.003 ± 0.003	0.013 ± 0.013	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.165 ± 0.007	0.406 ± 0.011	0.207	0.007	3.43	0.0832	0.0834	0.979	94.545 ± 0.722	0.312 ± 0.001	71.212 ± 1.439
UPI	beta43	0.20	0.197	-0.003 ± 0.002	-0.014 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.197	-0.003	-1.38	0.0485	0.0485	0.956	93.333 ± 0.793	0.182 ± 0.001	97.980 ± 0.447
UPI	beta44	0.11	0.112	0.002 ± 0.002	0.016 ± 0.017	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.289 ± 0.014	0.537 ± 0.016	0.111	0.001	0.99	0.0574	0.0574	0.905	92.121 ± 0.856	0.210 ± 0.002	57.980 ± 1.569
UPI	beta45	0.11	0.112	0.002 ± 0.002	0.017 ± 0.017	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.289 ± 0.014	0.538 ± 0.016	0.109	-0.001	-1.16	0.0562	0.0562	0.875	91.212 ± 0.900	0.203 ± 0.002	58.485 ± 1.566
UPI	beta46	0.08	0.078	-0.002 ± 0.001	-0.030 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.163 ± 0.008	0.403 ± 0.012	0.078	-0.002	-2.09	0.0298	0.0298	0.921	91.616 ± 0.881	0.116 ± 0.001	75.455 ± 1.368

Table S155

Simulation 2 - Condition 11: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.175 ± 0.009	0.419 ± 0.013	0.077	-0.003	-3.24	0.0306	0.0307	0.874	91.313 ± 0.895	0.115 ± 0.001	73.434 ± 1.404
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.158 ± 0.007	0.397 ± 0.011	0.158	-0.002	-1.40	0.0631	0.0632	0.992	94.747 ± 0.709	0.247 ± 0.001	70.606 ± 1.448
UPI	beta52	0.16	0.167	0.007 ± 0.003	0.041 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.252 ± 0.012	0.502 ± 0.014	0.167	0.007	4.53	0.0808	0.0811	1.033	96.162 ± 0.611	0.324 ± 0.001	55.556 ± 1.579
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.187 ± 0.009	0.433 ± 0.012	0.159	-0.001	-0.56	0.0690	0.0690	0.980	95.253 ± 0.676	0.266 ± 0.001	64.141 ± 1.524
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.238 ± 0.011	0.488 ± 0.013	0.158	-0.002	-1.45	0.0766	0.0766	1.034	95.051 ± 0.689	0.316 ± 0.001	51.010 ± 1.589
UPI	beta55	0.08	0.079	-0.001 ± 0.002	-0.010 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.494 ± 0.023	0.703 ± 0.020	0.077	-0.003	-3.69	0.0569	0.0570	0.914	92.525 ± 0.836	0.201 ± 0.002	36.364 ± 1.529
UPI	beta56	0.08	0.083	0.003 ± 0.001	0.032 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.215 ± 0.012	0.463 ± 0.014	0.080	0.000	0.59	0.0328	0.0328	0.927	93.030 ± 0.809	0.134 ± 0.002	67.677 ± 1.486
UPI	beta57	0.06	0.059	-0.001 ± 0.002	-0.010 ± 0.027	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.743 ± 0.036	0.862 ± 0.025	0.059	-0.001	-1.71	0.0491	0.0491	0.943	92.929 ± 0.815	0.191 ± 0.002	25.051 ± 1.377
UPI	beta58	0.06	0.062	0.002 ± 0.001	0.029 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.252 ± 0.013	0.502 ± 0.015	0.061	0.001	1.00	0.0271	0.0271	0.942	93.636 ± 0.776	0.111 ± 0.001	62.121 ± 1.542

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S156

Simulation 2 - Condition 12: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.200	0.000	-0.11	0.0447	0.0447	1.010	95.195 ± 0.677	0.186 ± 0.001	98.298 ± 0.409
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.252 ± 0.007	0.202	0.002	0.77	0.0494	0.0494	0.975	94.294 ± 0.734	0.193 ± 0.001	97.097 ± 0.531
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.198	-0.002	-0.82	0.0308	0.0308	0.973	94.595 ± 0.715	0.118 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.016 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.094 ± 0.005	0.307 ± 0.009	0.111	0.001	0.96	0.0337	0.0337	0.933	93.293 ± 0.791	0.123 ± 0.001	93.393 ± 0.786
LSAM	beta45	0.11	0.112	0.002 ± 0.001	0.014 ± 0.009	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.111	0.001	0.70	0.0310	0.0310	0.950	93.493 ± 0.780	0.119 ± 0.001	94.294 ± 0.734
LSAM	beta46	0.08	0.080	0.000 ± 0.001	0.001 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.051 ± 0.003	0.226 ± 0.007	0.080	0.000	-0.36	0.0175	0.0175	0.927	92.993 ± 0.808	0.066 ± 0.001	98.599 ± 0.372
LSAM	beta47	0.08	0.080	-0.000 ± 0.001	-0.001 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.051 ± 0.003	0.226 ± 0.007	0.080	0.000	-0.21	0.0170	0.0170	0.924	92.192 ± 0.849	0.065 ± 0.001	98.298 ± 0.409
LSAM	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.160	0.000	-0.22	0.0423	0.0423	0.988	95.495 ± 0.656	0.157 ± 0.000	97.798 ± 0.464
LSAM	beta52	0.16	0.157	-0.003 ± 0.002	-0.018 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.157	-0.003	-1.80	0.0514	0.0515	1.004	95.195 ± 0.677	0.203 ± 0.000	85.085 ± 1.127
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.010 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.276 ± 0.008	0.161	0.001	0.79	0.0446	0.0447	0.960	92.893 ± 0.813	0.166 ± 0.000	95.896 ± 0.628
LSAM	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.005	0.313 ± 0.009	0.160	0.000	0.13	0.0491	0.0491	0.997	94.494 ± 0.722	0.196 ± 0.001	88.589 ± 1.006
LSAM	beta55	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.153 ± 0.007	0.391 ± 0.011	0.079	-0.001	-1.66	0.0289	0.0289	0.927	92.893 ± 0.813	0.114 ± 0.001	74.875 ± 1.372
LSAM	beta56	0.08	0.080	0.000 ± 0.001	0.001 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.067 ± 0.003	0.258 ± 0.008	0.080	0.000	-0.58	0.0200	0.0200	0.914	92.793 ± 0.818	0.074 ± 0.001	95.696 ± 0.642
LSAM	beta57	0.06	0.062	0.002 ± 0.001	0.026 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.261 ± 0.012	0.511 ± 0.014	0.061	0.001	1.88	0.0315	0.0315	0.951	94.394 ± 0.728	0.114 ± 0.001	57.257 ± 1.565
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.012 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	0.084 ± 0.004	0.290 ± 0.009	0.060	0.000	0.69	0.0156	0.0157	0.954	92.693 ± 0.823	0.065 ± 0.001	92.292 ± 0.844
QML	beta41	0.20	0.205	0.005 ± 0.001	0.027 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.200 ± 0.006	0.205	0.005	2.74	0.0386	0.0389	1.019	94.494 ± 0.722	0.158 ± 0.000	99.900 ± 0.100
QML	beta42	0.20	0.205	0.005 ± 0.001	0.025 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.045 ± 0.002	0.211 ± 0.006	0.205	0.005	2.71	0.0416	0.0420	0.983	94.194 ± 0.740	0.162 ± 0.000	99.700 ± 0.173
QML	beta43	0.20	0.195	-0.005 ± 0.001	-0.026 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.153 ± 0.004	0.195	-0.005	-2.61	0.0299	0.0303	0.975	93.994 ± 0.752	0.116 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.098	-0.012 ± 0.001	-0.109 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.105 ± 0.005	0.323 ± 0.009	0.098	-0.012	-10.67	0.0337	0.0357	0.942	90.891 ± 0.910	0.124 ± 0.001	84.785 ± 1.136
QML	beta45	0.11	0.108	-0.002 ± 0.001	-0.017 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.079 ± 0.004	0.282 ± 0.008	0.109	-0.001	-1.34	0.0301	0.0301	0.963	93.894 ± 0.758	0.117 ± 0.001	93.994 ± 0.752
QML	beta46	0.08	0.092	0.012 ± 0.001	0.153 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.072 ± 0.003	0.269 ± 0.007	0.092	0.012	15.05	0.0169	0.0207	0.946	88.889 ± 0.994	0.066 ± 0.000	99.900 ± 0.100
QML	beta47	0.08	0.093	0.013 ± 0.001	0.157 ± 0.007	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.072 ± 0.003	0.269 ± 0.007	0.093	0.013	15.81	0.0162	0.0205	0.944	85.385 ± 1.118	0.064 ± 0.001	99.700 ± 0.173
QML	beta51	0.16	0.169	0.009 ± 0.001	0.054 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.167	0.007	4.52	0.0416	0.0423	0.990	94.494 ± 0.722	0.158 ± 0.000	98.799 ± 0.345
QML	beta52	0.16	0.141	-0.019 ± 0.001	-0.119 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.007	0.141	-0.019	-12.18	0.0424	0.0467	1.011	92.793 ± 0.818	0.168 ± 0.000	91.191 ± 0.897
QML	beta53	0.16	0.137	-0.023 ± 0.001	-0.147 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.307 ± 0.008	0.136	-0.024	-15.10	0.0421	0.0485	0.968	90.591 ± 0.924	0.164 ± 0.000	90.190 ± 0.941
QML	beta54	0.16	0.161	0.001 ± 0.001	0.004 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.268 ± 0.008	0.161	0.001	0.55	0.0413	0.0413	0.998	94.094 ± 0.746	0.168 ± 0.000	95.395 ± 0.663
QML	beta55	0.08	0.082	0.002 ± 0.001	0.029 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.167 ± 0.008	0.409 ± 0.011	0.082	0.002	3.00	0.0316	0.0317	0.922	92.693 ± 0.823	0.118 ± 0.001	74.775 ± 1.374
QML	beta56	0.08	0.086	0.006 ± 0.001	0.073 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.075 ± 0.004	0.273 ± 0.008	0.086	0.006	7.27	0.0198	0.0207	0.922	90.891 ± 0.910	0.076 ± 0.001	96.897 ± 0.549
QML	beta57	0.06	0.071	0.011 ± 0.001	0.177 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.282 ± 0.012	0.531 ± 0.014	0.070	0.010	17.36	0.0317	0.0334	0.962	93.193 ± 0.797	0.113 ± 0.001	68.368 ± 1.471
QML	beta58	0.06	0.070	0.010 ± 0.001	0.167 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.070	0.010	16.39	0.0155	0.0183	0.956	88.488 ± 1.010	0.063 ± 0.001	97.698 ± 0.475
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.011 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.202	0.002	0.98	0.0441	0.0441	1.000	94.595 ± 0.715	0.183 ± 0.001	98.799 ± 0.345
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.017 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.203	0.003	1.73	0.0488	0.0489	0.968	93.894 ± 0.758	0.190 ± 0.001	97.197 ± 0.522
UPI	beta43	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.197	-0.003	-1.40	0.0309	0.0311	0.962	93.994 ± 0.752	0.116 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.111	0.001 ± 0.001	0.009 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.110	0.000	-0.25	0.0346	0.0346	0.929	92.993 ± 0.808	0.124 ± 0.001	93.093 ± 0.802
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.013 ± 0.009	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.110	0.000	-0.03	0.0327	0.0327	0.962	93.794 ± 0.763	0.122 ± 0.001	93.994 ± 0.752
UPI	beta46	0.08	0.080	-0.000 ± 0.001	-0.006 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.051 ± 0.002	0.226 ± 0.007	0.080	0.000	-0.53	0.0180	0.0180	0.927	91.692 ± 0.873	0.066 ± 0.001	98.599 ± 0.372

Table S156

Simulation 2 - Condition 12: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.052 ± 0.003	0.227 ± 0.007	0.079	-0.001	-1.02	0.0168	0.0168	0.921	91.191 ± 0.897	0.066 ± 0.001	98.098 ± 0.432
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.252 ± 0.007	0.160	0.000	0.06	0.0409	0.0409	0.982	94.995 ± 0.690	0.155 ± 0.000	98.098 ± 0.432
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.159	-0.001	-0.68	0.0520	0.0520	0.990	94.895 ± 0.696	0.200 ± 0.000	86.787 ± 1.071
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.009 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.276 ± 0.008	0.161	0.001	0.87	0.0457	0.0457	0.953	92.993 ± 0.808	0.165 ± 0.000	96.396 ± 0.590
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.312 ± 0.009	0.161	0.001	0.50	0.0478	0.0478	0.996	94.795 ± 0.703	0.195 ± 0.001	88.488 ± 1.010
UPI	beta55	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.152 ± 0.007	0.390 ± 0.011	0.079	-0.001	-1.15	0.0299	0.0299	0.926	92.993 ± 0.808	0.113 ± 0.001	75.976 ± 1.352
UPI	beta56	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.067 ± 0.003	0.259 ± 0.008	0.079	-0.001	-0.94	0.0194	0.0194	0.913	92.492 ± 0.834	0.074 ± 0.001	95.395 ± 0.663
UPI	beta57	0.06	0.061	0.001 ± 0.001	0.016 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.263 ± 0.012	0.513 ± 0.014	0.061	0.001	1.11	0.0308	0.0308	0.940	93.694 ± 0.769	0.113 ± 0.001	55.255 ± 1.573
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.011 ± 0.009	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.085 ± 0.004	0.292 ± 0.009	0.060	0.000	0.33	0.0162	0.0162	0.954	92.492 ± 0.834	0.066 ± 0.001	92.192 ± 0.849

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S157

Simulation 2 - Condition 13: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.005	-0.008 ± 0.027	0.008 ± 0.001	0.088 ± 0.005	0.008 ± 0.001	0.088 ± 0.006	0.194 ± 0.021	0.440 ± 0.028	0.197	-0.003	-1.58	0.0863	0.0864	1.275	98.868 ± 0.650	0.441 ± 0.014	48.302 ± 3.070
LSAM	beta42	0.20	0.205	0.005 ± 0.006	0.025 ± 0.029	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.093 ± 0.005	0.218 ± 0.019	0.467 ± 0.025	0.203	0.003	1.57	0.0906	0.0906	1.206	97.358 ± 0.985	0.442 ± 0.010	48.679 ± 3.070
LSAM	beta43	0.20	0.204	0.004 ± 0.006	0.018 ± 0.028	0.008 ± 0.001	0.090 ± 0.005	0.008 ± 0.001	0.090 ± 0.006	0.200 ± 0.022	0.448 ± 0.028	0.198	-0.002	-0.97	0.0879	0.0879	1.097	96.604 ± 1.113	0.386 ± 0.007	58.868 ± 3.023
LSAM	beta44	0.11	0.130	0.020 ± 0.009	0.179 ± 0.079	0.020 ± 0.002	0.142 ± 0.008	0.020 ± 0.002	0.143 ± 0.009	1.681 ± 0.176	1.297 ± 0.079	0.116	0.006	5.11	0.1259	0.1261	1.214	99.245 ± 0.532	0.674 ± 0.029	10.189 ± 1.858
LSAM	beta45	0.11	0.113	0.003 ± 0.008	0.023 ± 0.071	0.016 ± 0.001	0.128 ± 0.006	0.016 ± 0.001	0.127 ± 0.007	1.340 ± 0.122	1.158 ± 0.064	0.101	-0.009	-8.11	0.1161	0.1165	1.070	96.981 ± 1.051	0.535 ± 0.023	13.208 ± 2.080
LSAM	beta46	0.08	0.079	-0.001 ± 0.011	-0.013 ± 0.140	0.033 ± 0.004	0.182 ± 0.010	0.033 ± 0.004	0.181 ± 0.011	5.147 ± 0.551	2.269 ± 0.141	0.084	0.004	5.02	0.1695	0.1696	1.278	98.868 ± 0.650	0.911 ± 0.054	1.132 ± 0.650
LSAM	beta47	0.08	0.071	-0.009 ± 0.011	-0.112 ± 0.142	0.034 ± 0.004	0.185 ± 0.011	0.034 ± 0.004	0.184 ± 0.012	5.318 ± 0.595	2.306 ± 0.148	0.072	-0.008	-10.22	0.1597	0.1599	1.163	97.736 ± 0.914	0.842 ± 0.039	1.132 ± 0.650
LSAM	beta51	0.16	0.160	-0.000 ± 0.008	-0.000 ± 0.048	0.016 ± 0.001	0.125 ± 0.006	0.016 ± 0.001	0.125 ± 0.007	0.607 ± 0.057	0.779 ± 0.044	0.154	-0.006	-3.57	0.1313	0.1314	1.030	97.358 ± 0.985	0.504 ± 0.008	26.038 ± 2.696
LSAM	beta52	0.16	0.158	-0.002 ± 0.006	-0.012 ± 0.041	0.011 ± 0.001	0.106 ± 0.007	0.011 ± 0.001	0.105 ± 0.007	0.434 ± 0.053	0.659 ± 0.046	0.160	0.000	-0.18	0.1041	0.1041	1.088	97.736 ± 0.914	0.450 ± 0.013	33.962 ± 2.909
LSAM	beta53	0.16	0.169	0.009 ± 0.006	0.055 ± 0.040	0.011 ± 0.001	0.104 ± 0.005	0.011 ± 0.001	0.104 ± 0.006	0.425 ± 0.042	0.652 ± 0.038	0.151	-0.009	-5.74	0.1048	0.1052	1.121	96.604 ± 1.113	0.457 ± 0.011	27.925 ± 2.756
LSAM	beta54	0.16	0.162	0.002 ± 0.006	0.013 ± 0.038	0.010 ± 0.001	0.098 ± 0.005	0.010 ± 0.001	0.098 ± 0.006	0.373 ± 0.035	0.611 ± 0.035	0.167	0.007	4.63	0.1015	0.1018	1.147	96.981 ± 1.051	0.440 ± 0.010	33.208 ± 2.893
LSAM	beta55	0.08	0.079	-0.001 ± 0.010	-0.009 ± 0.125	0.027 ± 0.003	0.163 ± 0.010	0.026 ± 0.003	0.163 ± 0.011	4.129 ± 0.477	2.032 ± 0.134	0.080	0.000	0.43	0.1503	0.1503	1.254	98.113 ± 0.836	0.801 ± 0.029	4.528 ± 1.277
LSAM	beta56	0.08	0.077	-0.003 ± 0.010	-0.034 ± 0.124	0.026 ± 0.003	0.161 ± 0.009	0.026 ± 0.003	0.161 ± 0.010	4.058 ± 0.457	2.015 ± 0.130	0.069	-0.011	-14.09	0.1455	0.1459	1.115	97.736 ± 0.914	0.705 ± 0.026	4.906 ± 1.327
LSAM	beta57	0.06	0.069	0.009 ± 0.013	0.146 ± 0.210	0.042 ± 0.005	0.205 ± 0.013	0.042 ± 0.005	0.205 ± 0.014	11.666 ± 1.464	3.416 ± 0.241	0.053	-0.007	-11.15	0.1806	0.1807	1.076	98.491 ± 0.749	0.865 ± 0.039	1.132 ± 0.650
LSAM	beta58	0.06	0.072	0.012 ± 0.011	0.200 ± 0.186	0.033 ± 0.004	0.182 ± 0.011	0.033 ± 0.004	0.182 ± 0.012	9.164 ± 1.079	3.027 ± 0.203	0.047	-0.013	-21.98	0.1470	0.1476	1.118	97.736 ± 0.914	0.796 ± 0.039	2.264 ± 0.914
QML	beta41	0.20	0.189	-0.011 ± 0.005	-0.057 ± 0.023	0.005 ± 0.001	0.074 ± 0.004	0.006 ± 0.001	0.074 ± 0.004	0.138 ± 0.013	0.372 ± 0.021	0.189	-0.011	-5.67	0.0760	0.0768	1.085	95.472 ± 1.277	0.313 ± 0.004	72.075 ± 2.756
QML	beta42	0.20	0.193	-0.007 ± 0.005	-0.035 ± 0.024	0.006 ± 0.001	0.078 ± 0.003	0.006 ± 0.001	0.078 ± 0.004	0.152 ± 0.013	0.389 ± 0.021	0.192	-0.008	-3.93	0.0814	0.0818	1.055	95.472 ± 1.277	0.321 ± 0.003	66.415 ± 2.901
QML	beta43	0.20	0.190	-0.010 ± 0.005	-0.049 ± 0.024	0.006 ± 0.001	0.078 ± 0.003	0.006 ± 0.001	0.078 ± 0.004	0.154 ± 0.013	0.392 ± 0.021	0.185	-0.015	-7.28	0.0832	0.0844	1.017	95.472 ± 1.277	0.311 ± 0.003	68.679 ± 2.849
QML	beta44	0.11	0.124	0.014 ± 0.007	0.128 ± 0.060	0.011 ± 0.001	0.107 ± 0.005	0.012 ± 0.001	0.108 ± 0.007	0.957 ± 0.100	0.978 ± 0.059	0.115	0.005	4.93	0.1018	0.1019	1.051	95.472 ± 1.277	0.441 ± 0.009	21.509 ± 2.524
QML	beta45	0.11	0.110	0.000 ± 0.006	0.003 ± 0.056	0.010 ± 0.001	0.100 ± 0.005	0.010 ± 0.001	0.100 ± 0.006	0.826 ± 0.076	0.909 ± 0.051	0.102	-0.008	-7.36	0.0983	0.0986	0.947	93.208 ± 1.546	0.372 ± 0.005	22.264 ± 2.556
QML	beta46	0.08	0.022	-0.058 ± 0.005	-0.726 ± 0.057	0.006 ± 0.001	0.074 ± 0.004	0.009 ± 0.001	0.094 ± 0.005	1.386 ± 0.127	1.177 ± 0.059	0.027	-0.053	-66.81	0.0704	0.0884	1.130	89.434 ± 1.888	0.329 ± 0.007	3.019 ± 1.051
QML	beta47	0.08	0.023	-0.057 ± 0.004	-0.719 ± 0.052	0.005 ± 0.000	0.068 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	1.233 ± 0.087	1.110 ± 0.044	0.024	-0.056	-69.52	0.0706	0.0899	1.159	87.170 ± 2.054	0.308 ± 0.007	4.151 ± 1.225
QML	beta51	0.16	0.184	0.024 ± 0.007	0.148 ± 0.044	0.013 ± 0.001	0.115 ± 0.005	0.014 ± 0.001	0.117 ± 0.006	0.539 ± 0.050	0.734 ± 0.040	0.173	0.013	7.94	0.1075	0.1082	1.083	98.113 ± 0.836	0.489 ± 0.007	29.434 ± 2.800
QML	beta52	0.16	0.152	-0.008 ± 0.005	-0.049 ± 0.032	0.007 ± 0.001	0.082 ± 0.004	0.007 ± 0.001	0.082 ± 0.005	0.266 ± 0.025	0.515 ± 0.029	0.154	-0.006	-3.74	0.0869	0.0871	1.144	97.358 ± 0.985	0.369 ± 0.006	38.868 ± 2.994
QML	beta53	0.16	0.151	-0.009 ± 0.005	-0.054 ± 0.034	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.089 ± 0.005	0.307 ± 0.028	0.554 ± 0.031	0.139	-0.021	-13.19	0.0785	0.0813	1.049	95.094 ± 1.327	0.364 ± 0.004	33.962 ± 2.909
QML	beta54	0.16	0.158	-0.002 ± 0.005	-0.013 ± 0.031	0.007 ± 0.001	0.082 ± 0.004	0.007 ± 0.001	0.082 ± 0.004	0.260 ± 0.022	0.510 ± 0.027	0.165	0.005	2.85	0.0864	0.0866	1.056	96.604 ± 1.113	0.338 ± 0.004	47.170 ± 3.067
QML	beta55	0.08	0.086	0.006 ± 0.008	0.072 ± 0.100	0.017 ± 0.002	0.130 ± 0.006	0.017 ± 0.002	0.130 ± 0.007	2.638 ± 0.242	1.624 ± 0.090	0.089	0.009	11.49	0.1385	0.1388	1.156	96.604 ± 1.113	0.589 ± 0.013	7.170 ± 1.585
QML	beta56	0.08	0.064	-0.016 ± 0.007	-0.202 ± 0.089	0.014 ± 0.001	0.116 ± 0.005	0.014 ± 0.001	0.117 ± 0.006	2.149 ± 0.193	1.466 ± 0.080	0.066	-0.014	-17.99	0.1264	0.1272	1.103	94.717 ± 1.374	0.503 ± 0.008	6.038 ± 1.463
QML	beta57	0.06	0.018	-0.042 ± 0.005	-0.693 ± 0.089	0.008 ± 0.001	0.087 ± 0.005	0.009 ± 0.001	0.096 ± 0.005	2.581 ± 0.239	1.607 ± 0.085	0.013	-0.047	-77.54	0.0837	0.0958	1.015	92.830 ± 1.585	0.347 ± 0.009	3.774 ± 1.171
QML	beta58	0.06	0.029	-0.031 ± 0.005	-0.524 ± 0.076	0.006 ± 0.000	0.075 ± 0.003	0.007 ± 0.001	0.081 ± 0.004	1.813 ± 0.161	1.346 ± 0.070	0.023	-0.037	-61.03	0.0730	0.0817	0.993	92.453 ± 1.623	0.290 ± 0.006	6.792 ± 1.546
UPI	beta41	0.20	0.196	-0.004 ± 0.007	-0.019 ± 0.037	0.015 ± 0.002	0.121 ± 0.008	0.015 ± 0.002	0.121 ± 0.009	0.363 ± 0.046	0.603 ± 0.043	0.196	-0.004	-1.82	0.1175	0.1176	1.650	99.245 ± 0.532	0.781 ± 0.076	32.830 ± 2.885
UPI	beta42	0.20	0.215	0.015 ± 0.008	0.073 ± 0.038	0.015 ± 0.002	0.124 ± 0.008	0.016 ± 0.002	0.125 ± 0.009	0.389 ± 0.051	0.624 ± 0.046	0.205	0.005	2.44	0.1229	0.1230	1.634	98.868 ± 0.650	0.795 ± 0.065	27.925 ± 2.756
UPI	beta43	0.20	0.202	0.002 ± 0.007	0.010 ± 0.033	0.011 ± 0.001	0.107 ± 0.006	0.011 ± 0.001	0.107 ± 0.007	0.286 ± 0.032	0.535 ± 0.035	0.195	-0.005	-2.37	0.1148	0.1149	1.580	98.113 ± 0.836	0.664 ± 0.068	37.736 ± 2.978
UPI	beta44	0.11	0.159	0.049 ± 0.014	0.447 ± 0.127	0.052 ± 0.008	0.228 ± 0.017	0.054 ± 0.008	0.233 ± 0.018	4.482 ± 0.641	2.117 ± 0.165	0.133	0.023	21.01	0.2283	0.2295	1.772	99.623 ± 0.377	1.584 ± 0.160	2.642 ± 0.985
UPI	beta45	0.11	0.117	0.007 ± 0.011	0.064 ± 0.099	0.032 ± 0.004	0.178 ± 0.011	0.032 ± 0.004	0.178 ± 0.012	2.612 ± 0.304	1.616 ± 0.107	0.116	0.006	5.23	0.1810	0.1811	1.702	99.245 ± 0.532	1.188 ± 0.086	3.019 ± 1.051
UPI	beta46	0.08	0.064	-0.016 ± 0.023	-0.203 ± 0.287	0.140 ± 0.020	0.374 ± 0.027	0.140 ± 0.020	0.374 ± 0.029	21.823 ± 3.088	4.672 ± 0.364	0.045	-0.035	-43.72	0.3212	0.3231	1.779	97.736 ± 0.914	2.609 ± 0.258	0.377 ± 0.377

Table S157

Simulation 2 - Condition 13: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.031	-0.049 ± 0.019	-0.608 ± 0.236	0.094 ± 0.012	0.307 ± 0.020	0.097 ± 0.012	0.311 ± 0.022	15.079 ± 1.888	3.883 ± 0.272	0.052	-0.028	-35.28	0.3069	0.3082	1.949	97.736 ± 0.914	2.348 ± 0.183	0.000 ± 0.000
UPI	beta51	0.16	0.173	0.013 ± 0.008	0.079 ± 0.048	0.016 ± 0.001	0.125 ± 0.006	0.016 ± 0.001	0.125 ± 0.007	0.614 ± 0.056	0.783 ± 0.044	0.163	0.003	1.84	0.1389	0.1389	1.538	99.623 ± 0.377	0.753 ± 0.050	18.868 ± 2.403
UPI	beta52	0.16	0.159	-0.001 ± 0.007	-0.008 ± 0.043	0.012 ± 0.001	0.111 ± 0.006	0.012 ± 0.001	0.111 ± 0.007	0.479 ± 0.048	0.692 ± 0.041	0.161	0.001	0.94	0.1162	0.1162	1.722	98.868 ± 0.650	0.749 ± 0.050	19.623 ± 2.440
UPI	beta53	0.16	0.164	0.004 ± 0.008	0.023 ± 0.048	0.016 ± 0.002	0.125 ± 0.007	0.016 ± 0.002	0.125 ± 0.008	0.613 ± 0.068	0.783 ± 0.050	0.142	-0.018	-11.29	0.1224	0.1237	1.597	98.491 ± 0.749	0.785 ± 0.055	17.358 ± 2.327
UPI	beta54	0.16	0.153	-0.007 ± 0.007	-0.047 ± 0.041	0.011 ± 0.001	0.107 ± 0.005	0.011 ± 0.001	0.107 ± 0.006	0.446 ± 0.046	0.668 ± 0.040	0.166	0.006	3.73	0.1176	0.1177	1.666	98.113 ± 0.836	0.698 ± 0.046	22.642 ± 2.571
UPI	beta55	0.08	0.108	0.028 ± 0.014	0.350 ± 0.175	0.052 ± 0.006	0.228 ± 0.013	0.053 ± 0.006	0.229 ± 0.015	8.216 ± 0.931	2.866 ± 0.186	0.082	0.002	2.46	0.2096	0.2096	2.055	100.000 ± 0.000	1.837 ± 0.150	1.132 ± 0.650
UPI	beta56	0.08	0.065	-0.015 ± 0.012	-0.185 ± 0.152	0.039 ± 0.004	0.198 ± 0.011	0.039 ± 0.004	0.198 ± 0.013	6.107 ± 0.681	2.471 ± 0.158	0.067	-0.013	-15.90	0.1868	0.1873	1.816	98.113 ± 0.836	1.406 ± 0.088	1.132 ± 0.650
UPI	beta57	0.06	0.025	-0.035 ± 0.018	-0.591 ± 0.305	0.089 ± 0.012	0.298 ± 0.021	0.090 ± 0.012	0.299 ± 0.023	24.912 ± 3.465	4.991 ± 0.383	0.021	-0.039	-64.49	0.2616	0.2645	2.054	96.981 ± 1.051	2.398 ± 0.222	0.000 ± 0.000
UPI	beta58	0.06	0.041	-0.019 ± 0.015	-0.311 ± 0.256	0.062 ± 0.008	0.250 ± 0.015	0.063 ± 0.008	0.250 ± 0.017	17.386 ± 2.126	4.170 ± 0.288	0.029	-0.031	-51.71	0.2026	0.2050	2.211	98.491 ± 0.749	2.166 ± 0.218	0.377 ± 0.377

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S158

Simulation 2 - Condition 14: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.020 ± 0.010	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.004	0.283 ± 0.009	0.205	0.005	2.40	0.0554	0.0556	1.029	95.342 ± 0.780	0.228 ± 0.002	95.753 ± 0.746
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.282 ± 0.009	0.200	0.000	0.15	0.0560	0.0560	1.061	97.123 ± 0.619	0.234 ± 0.002	93.836 ± 0.890
LSAM	beta43	0.20	0.203	0.003 ± 0.002	0.015 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.070 ± 0.004	0.264 ± 0.009	0.200	0.000	0.12	0.0505	0.0505	1.028	95.616 ± 0.758	0.213 ± 0.001	97.123 ± 0.619
LSAM	beta44	0.11	0.111	0.001 ± 0.003	0.010 ± 0.027	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.550 ± 0.033	0.742 ± 0.026	0.112	0.002	1.44	0.0793	0.0793	1.028	97.534 ± 0.574	0.329 ± 0.005	29.315 ± 1.685
LSAM	beta45	0.11	0.113	0.003 ± 0.002	0.031 ± 0.022	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.355 ± 0.021	0.596 ± 0.021	0.113	0.003	3.02	0.0620	0.0621	1.051	95.890 ± 0.735	0.270 ± 0.002	40.137 ± 1.814
LSAM	beta46	0.08	0.084	0.004 ± 0.004	0.049 ± 0.052	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	2.000 ± 0.134	1.414 ± 0.054	0.082	0.002	2.49	0.1052	0.1052	1.012	97.397 ± 0.589	0.449 ± 0.008	5.890 ± 0.871
LSAM	beta47	0.08	0.091	0.011 ± 0.004	0.133 ± 0.049	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	1.785 ± 0.104	1.336 ± 0.046	0.078	-0.002	-2.67	0.0993	0.0994	1.098	98.082 ± 0.508	0.458 ± 0.011	6.438 ± 0.908
LSAM	beta51	0.16	0.156	-0.004 ± 0.003	-0.026 ± 0.017	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.223 ± 0.012	0.472 ± 0.015	0.151	-0.009	-5.46	0.0763	0.0768	0.991	95.205 ± 0.791	0.293 ± 0.002	55.616 ± 1.839
LSAM	beta52	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.007	0.345 ± 0.012	0.159	-0.001	-0.44	0.0544	0.0544	1.069	95.616 ± 0.758	0.231 ± 0.001	79.726 ± 1.488
LSAM	beta53	0.16	0.163	0.003 ± 0.002	0.017 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.158 ± 0.009	0.398 ± 0.014	0.162	0.002	1.10	0.0611	0.0612	0.978	95.616 ± 0.758	0.244 ± 0.002	76.164 ± 1.577
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.006 ± 0.013	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.007	0.362 ± 0.012	0.161	0.001	0.69	0.0576	0.0576	1.010	95.205 ± 0.791	0.229 ± 0.001	80.274 ± 1.473
LSAM	beta55	0.08	0.083	0.003 ± 0.004	0.040 ± 0.046	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	1.542 ± 0.095	1.242 ± 0.045	0.078	-0.002	-3.10	0.0941	0.0941	1.012	96.986 ± 0.633	0.394 ± 0.004	13.288 ± 1.256
LSAM	beta56	0.08	0.087	0.007 ± 0.003	0.083 ± 0.041	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	1.211 ± 0.068	1.101 ± 0.037	0.085	0.005	6.62	0.0847	0.0849	1.047	96.712 ± 0.660	0.361 ± 0.003	13.836 ± 1.278
LSAM	beta57	0.06	0.071	0.011 ± 0.004	0.177 ± 0.068	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	3.397 ± 0.221	1.843 ± 0.069	0.065	0.005	8.67	0.1054	0.1056	1.038	97.945 ± 0.525	0.448 ± 0.007	5.342 ± 0.832
LSAM	beta58	0.06	0.067	0.007 ± 0.004	0.121 ± 0.061	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	2.720 ± 0.171	1.649 ± 0.060	0.064	0.004	7.24	0.0927	0.0928	1.017	98.082 ± 0.508	0.394 ± 0.006	7.397 ± 0.969
QML	beta41	0.20	0.190	-0.010 ± 0.002	-0.048 ± 0.009	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.063 ± 0.003	0.251 ± 0.008	0.190	-0.010	-4.92	0.0498	0.0508	1.021	92.055 ± 1.001	0.197 ± 0.007	97.945 ± 0.525
QML	beta42	0.20	0.195	-0.005 ± 0.002	-0.025 ± 0.009	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.062 ± 0.003	0.249 ± 0.008	0.191	-0.009	-4.51	0.0478	0.0486	1.026	95.753 ± 0.746	0.200 ± 0.003	98.356 ± 0.471
QML	beta43	0.20	0.188	-0.012 ± 0.002	-0.061 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.061 ± 0.003	0.248 ± 0.008	0.188	-0.012	-6.23	0.0471	0.0487	1.008	93.836 ± 0.890	0.190 ± 0.001	97.671 ± 0.558
QML	beta44	0.11	0.114	0.004 ± 0.003	0.039 ± 0.023	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.386 ± 0.021	0.621 ± 0.020	0.113	0.003	2.79	0.0644	0.0645	1.012	94.247 ± 0.862	0.271 ± 0.007	41.096 ± 1.821
QML	beta45	0.11	0.107	-0.003 ± 0.002	-0.025 ± 0.018	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.243 ± 0.013	0.493 ± 0.016	0.102	-0.008	-6.85	0.0544	0.0549	1.040	94.932 ± 0.812	0.221 ± 0.001	47.534 ± 1.848
QML	beta46	0.08	0.025	-0.055 ± 0.002	-0.683 ± 0.022	0.002 ± 0.000	0.048 ± 0.001	0.005 ± 0.000	0.072 ± 0.002	0.819 ± 0.035	0.905 ± 0.021	0.026	-0.054	-68.01	0.0497	0.0737	1.063	77.671 ± 1.541	0.198 ± 0.005	6.986 ± 0.943
QML	beta47	0.08	0.030	-0.050 ± 0.002	-0.631 ± 0.021	0.002 ± 0.000	0.045 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.708 ± 0.031	0.841 ± 0.020	0.029	-0.051	-64.29	0.0442	0.0678	1.142	79.452 ± 1.495	0.200 ± 0.010	7.534 ± 0.977
QML	beta51	0.16	0.180	0.020 ± 0.003	0.125 ± 0.017	0.005 ± 0.000	0.072 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.218 ± 0.012	0.467 ± 0.015	0.174	0.014	8.92	0.0726	0.0740	1.009	95.890 ± 0.735	0.284 ± 0.002	73.151 ± 1.640
QML	beta52	0.16	0.151	-0.009 ± 0.002	-0.059 ± 0.012	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.110 ± 0.006	0.332 ± 0.011	0.151	-0.009	-5.40	0.0496	0.0503	1.070	94.658 ± 0.832	0.219 ± 0.003	81.096 ± 1.449
QML	beta53	0.16	0.143	-0.017 ± 0.002	-0.104 ± 0.013	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.007	0.368 ± 0.012	0.143	-0.017	-10.49	0.0548	0.0573	0.978	94.795 ± 0.822	0.217 ± 0.001	75.479 ± 1.592
QML	beta54	0.16	0.157	-0.003 ± 0.002	-0.018 ± 0.012	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.108 ± 0.006	0.329 ± 0.011	0.159	-0.001	-0.90	0.0564	0.0564	1.025	93.699 ± 0.899	0.211 ± 0.008	86.849 ± 1.251
QML	beta55	0.08	0.085	0.005 ± 0.003	0.066 ± 0.038	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	1.077 ± 0.062	1.038 ± 0.036	0.082	0.002	2.91	0.0813	0.0813	1.070	96.301 ± 0.699	0.348 ± 0.005	15.205 ± 1.329
QML	beta56	0.08	0.074	-0.006 ± 0.003	-0.077 ± 0.034	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.854 ± 0.048	0.924 ± 0.031	0.075	-0.005	-5.91	0.0727	0.0729	1.031	94.110 ± 0.871	0.298 ± 0.003	15.890 ± 1.353
QML	beta57	0.06	0.021	-0.039 ± 0.002	-0.644 ± 0.032	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	1.139 ± 0.058	1.067 ± 0.030	0.022	-0.038	-62.62	0.0514	0.0637	1.059	88.767 ± 1.169	0.212 ± 0.005	6.301 ± 0.899
QML	beta58	0.06	0.030	-0.030 ± 0.002	-0.497 ± 0.027	0.002 ± 0.000	0.044 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.790 ± 0.037	0.889 ± 0.024	0.028	-0.032	-53.69	0.0443	0.0548	1.003	87.260 ± 1.234	0.174 ± 0.002	11.370 ± 1.175
UPI	beta41	0.20	0.205	0.005 ± 0.002	0.027 ± 0.011	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.005	0.294 ± 0.010	0.204	0.004	2.12	0.0585	0.0586	1.210	96.438 ± 0.686	0.278 ± 0.008	88.904 ± 1.162
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.011	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.005	0.294 ± 0.011	0.198	-0.002	-0.92	0.0600	0.0600	1.241	98.219 ± 0.489	0.286 ± 0.005	83.699 ± 1.367
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.007 ± 0.010	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.004	0.269 ± 0.009	0.201	0.001	0.53	0.0573	0.0573	1.197	97.397 ± 0.589	0.253 ± 0.005	90.822 ± 1.069
UPI	beta44	0.11	0.122	0.012 ± 0.003	0.109 ± 0.031	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.728 ± 0.050	0.853 ± 0.033	0.112	0.002	1.90	0.0935	0.0935	1.271	98.904 ± 0.385	0.464 ± 0.011	18.219 ± 1.429
UPI	beta45	0.11	0.114	0.004 ± 0.003	0.033 ± 0.024	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	0.427 ± 0.026	0.653 ± 0.023	0.112	0.002	1.57	0.0737	0.0737	1.322	98.630 ± 0.430	0.372 ± 0.010	20.137 ± 1.484
UPI	beta46	0.08	0.085	0.005 ± 0.005	0.068 ± 0.064	0.019 ± 0.001	0.138 ± 0.005	0.019 ± 0.001	0.138 ± 0.006	2.978 ± 0.214	1.726 ± 0.070	0.076	-0.004	-5.29	0.1234	0.1235	1.577	98.630 ± 0.430	0.853 ± 0.038	0.822 ± 0.334

Table S158

Simulation 2 - Condition 14: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.094	0.014 ± 0.005	0.172 ± 0.064	0.019 ± 0.001	0.137 ± 0.005	0.019 ± 0.001	0.138 ± 0.006	2.973 ± 0.219	1.724 ± 0.071	0.081	0.001	1.40	0.1177	0.1177	1.638	98.904 ± 0.385	0.882 ± 0.049	0.959 ± 0.361
UPI	beta51	0.16	0.159	-0.001 ± 0.003	-0.007 ± 0.017	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.216 ± 0.012	0.465 ± 0.015	0.157	-0.003	-2.06	0.0720	0.0721	1.093	96.575 ± 0.673	0.319 ± 0.005	54.384 ± 1.843
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.118 ± 0.006	0.344 ± 0.011	0.161	0.001	0.68	0.0537	0.0537	1.225	96.849 ± 0.647	0.264 ± 0.006	73.425 ± 1.635
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.009	0.401 ± 0.014	0.159	-0.001	-0.42	0.0630	0.0630	1.126	97.123 ± 0.619	0.283 ± 0.006	67.123 ± 1.739
UPI	beta54	0.16	0.160	0.000 ± 0.002	0.000 ± 0.014	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.007	0.367 ± 0.012	0.160	0.000	-0.30	0.0596	0.0596	1.143	96.575 ± 0.673	0.263 ± 0.005	73.014 ± 1.643
UPI	beta55	0.08	0.086	0.006 ± 0.004	0.079 ± 0.049	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	1.725 ± 0.112	1.313 ± 0.049	0.078	-0.002	-2.31	0.0983	0.0983	1.262	98.767 ± 0.408	0.519 ± 0.014	5.068 ± 0.812
UPI	beta56	0.08	0.086	0.006 ± 0.003	0.072 ± 0.042	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	1.320 ± 0.075	1.149 ± 0.039	0.082	0.002	2.17	0.0918	0.0918	1.285	98.493 ± 0.451	0.463 ± 0.009	5.753 ± 0.862
UPI	beta57	0.06	0.068	0.008 ± 0.005	0.134 ± 0.081	0.017 ± 0.001	0.131 ± 0.005	0.017 ± 0.001	0.131 ± 0.005	4.787 ± 0.358	2.188 ± 0.092	0.062	0.002	2.72	0.1185	0.1185	1.461	98.082 ± 0.508	0.751 ± 0.031	1.096 ± 0.385
UPI	beta58	0.06	0.062	0.002 ± 0.005	0.038 ± 0.077	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.125 ± 0.005	4.313 ± 0.333	2.077 ± 0.089	0.055	-0.005	-9.02	0.1037	0.1038	1.505	97.671 ± 0.558	0.736 ± 0.030	0.822 ± 0.334

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S159

Simulation 2 - Condition 15: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.004	0.306 ± 0.008	0.200	0.000	0.18	0.0622	0.0622	1.021	95.987 ± 0.638	0.245 ± 0.001	90.496 ± 0.953
LSAM	beta42	0.20	0.200	0.000 ± 0.002	0.001 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.005	0.307 ± 0.009	0.197	-0.003	-1.34	0.0624	0.0625	1.038	95.671 ± 0.661	0.250 ± 0.001	90.285 ± 0.962
LSAM	beta43	0.20	0.203	0.003 ± 0.002	0.016 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.008	0.201	0.001	0.29	0.0578	0.0578	1.039	95.987 ± 0.638	0.233 ± 0.001	95.248 ± 0.691
LSAM	beta44	0.11	0.111	0.001 ± 0.002	0.011 ± 0.022	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.461 ± 0.021	0.679 ± 0.019	0.110	0.000	0.06	0.0752	0.0752	0.997	96.199 ± 0.621	0.292 ± 0.002	34.636 ± 1.546
LSAM	beta45	0.11	0.110	0.000 ± 0.002	0.000 ± 0.019	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.348 ± 0.016	0.590 ± 0.017	0.107	-0.003	-2.49	0.0643	0.0643	1.017	96.410 ± 0.605	0.259 ± 0.001	40.760 ± 1.597
LSAM	beta46	0.08	0.083	0.003 ± 0.003	0.034 ± 0.037	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	1.316 ± 0.066	1.147 ± 0.034	0.085	0.005	6.16	0.0897	0.0898	1.044	96.727 ± 0.578	0.376 ± 0.003	10.243 ± 0.985
LSAM	beta47	0.08	0.078	-0.002 ± 0.003	-0.025 ± 0.038	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	1.391 ± 0.068	1.179 ± 0.035	0.078	-0.002	-2.14	0.0900	0.0900	0.995	94.720 ± 0.727	0.368 ± 0.003	10.454 ± 0.994
LSAM	beta51	0.16	0.161	0.001 ± 0.003	0.009 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.264 ± 0.012	0.514 ± 0.014	0.158	-0.002	-1.04	0.0797	0.0797	0.982	93.875 ± 0.779	0.317 ± 0.001	52.376 ± 1.623
LSAM	beta52	0.16	0.164	0.004 ± 0.002	0.023 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.009	0.414 ± 0.013	0.161	0.001	0.68	0.0625	0.0625	0.992	94.192 ± 0.760	0.257 ± 0.001	70.855 ± 1.477
LSAM	beta53	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.434 ± 0.012	0.158	-0.002	-1.29	0.0716	0.0716	0.974	94.509 ± 0.740	0.265 ± 0.001	64.625 ± 1.554
LSAM	beta54	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.013	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.007	0.389 ± 0.011	0.159	-0.001	-0.54	0.0602	0.0602	1.046	95.987 ± 0.638	0.255 ± 0.001	70.222 ± 1.486
LSAM	beta55	0.08	0.079	-0.001 ± 0.003	-0.012 ± 0.040	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	1.502 ± 0.087	1.226 ± 0.041	0.081	0.001	1.36	0.0968	0.0969	0.955	95.143 ± 0.699	0.367 ± 0.003	14.044 ± 1.129
LSAM	beta56	0.08	0.080	-0.000 ± 0.003	-0.002 ± 0.036	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	1.227 ± 0.057	1.108 ± 0.032	0.080	0.000	0.33	0.0876	0.0876	0.980	94.720 ± 0.727	0.341 ± 0.002	15.100 ± 1.164
LSAM	beta57	0.06	0.065	0.005 ± 0.003	0.077 ± 0.054	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	2.771 ± 0.138	1.665 ± 0.049	0.060	0.000	0.05	0.0917	0.0917	1.002	96.727 ± 0.578	0.392 ± 0.003	8.025 ± 0.883
LSAM	beta58	0.06	0.055	-0.005 ± 0.003	-0.084 ± 0.047	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	2.121 ± 0.099	1.456 ± 0.041	0.053	-0.007	-10.95	0.0861	0.0864	1.029	96.515 ± 0.596	0.352 ± 0.003	8.342 ± 0.899
QML	beta41	0.20	0.195	-0.005 ± 0.002	-0.025 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.288 ± 0.008	0.194	-0.006	-2.75	0.0589	0.0591	0.999	95.565 ± 0.669	0.225 ± 0.001	93.242 ± 0.816
QML	beta42	0.20	0.195	-0.005 ± 0.002	-0.026 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.289 ± 0.008	0.192	-0.008	-3.87	0.0583	0.0588	1.022	95.671 ± 0.661	0.231 ± 0.001	93.136 ± 0.822
QML	beta43	0.20	0.193	-0.007 ± 0.002	-0.033 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.075 ± 0.003	0.274 ± 0.007	0.193	-0.007	-3.65	0.0545	0.0550	1.039	95.671 ± 0.661	0.222 ± 0.001	93.981 ± 0.773
QML	beta44	0.11	0.115	0.005 ± 0.002	0.047 ± 0.021	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.406 ± 0.019	0.637 ± 0.018	0.113	0.003	2.70	0.0745	0.0746	0.998	96.093 ± 0.630	0.274 ± 0.002	38.226 ± 1.579
QML	beta45	0.11	0.109	-0.001 ± 0.002	-0.009 ± 0.018	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.295 ± 0.014	0.544 ± 0.015	0.108	-0.002	-1.64	0.0622	0.0623	1.030	96.938 ± 0.560	0.241 ± 0.001	43.611 ± 1.611
QML	beta46	0.08	0.041	-0.039 ± 0.002	-0.488 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.716 ± 0.033	0.846 ± 0.022	0.043	-0.037	-45.73	0.0552	0.0662	1.041	89.124 ± 1.012	0.226 ± 0.001	9.187 ± 0.939
QML	beta47	0.08	0.041	-0.039 ± 0.002	-0.485 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.005 ± 0.000	0.069 ± 0.002	0.734 ± 0.032	0.857 ± 0.021	0.042	-0.038	-47.09	0.0532	0.0652	1.004	88.912 ± 1.020	0.223 ± 0.001	11.404 ± 1.033
QML	beta51	0.16	0.176	0.016 ± 0.003	0.097 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.259 ± 0.012	0.509 ± 0.014	0.174	0.014	8.72	0.0773	0.0785	1.028	95.037 ± 0.706	0.323 ± 0.001	58.501 ± 1.601
QML	beta52	0.16	0.152	-0.008 ± 0.002	-0.047 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.008	0.414 ± 0.012	0.151	-0.009	-5.90	0.0619	0.0627	0.991	94.298 ± 0.754	0.256 ± 0.001	65.364 ± 1.546
QML	beta53	0.16	0.139	-0.021 ± 0.002	-0.132 ± 0.014	0.004 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.192 ± 0.008	0.438 ± 0.012	0.138	-0.022	-13.44	0.0706	0.0738	0.985	93.664 ± 0.792	0.258 ± 0.001	57.128 ± 1.608
QML	beta54	0.16	0.156	-0.004 ± 0.002	-0.023 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.006	0.369 ± 0.011	0.157	-0.003	-1.76	0.0577	0.0578	1.050	96.304 ± 0.613	0.242 ± 0.001	73.706 ± 1.431
QML	beta55	0.08	0.081	0.001 ± 0.003	0.015 ± 0.037	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	1.273 ± 0.066	1.128 ± 0.035	0.085	0.005	5.70	0.0875	0.0876	0.997	95.143 ± 0.699	0.353 ± 0.002	15.312 ± 1.170
QML	beta56	0.08	0.076	-0.004 ± 0.003	-0.046 ± 0.033	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	1.050 ± 0.048	1.025 ± 0.029	0.079	-0.001	-1.60	0.0820	0.0820	1.000	94.826 ± 0.720	0.321 ± 0.002	15.417 ± 1.173
QML	beta57	0.06	0.033	-0.027 ± 0.002	-0.455 ± 0.033	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	1.247 ± 0.056	1.117 ± 0.029	0.034	-0.026	-43.94	0.0594	0.0650	1.022	92.714 ± 0.845	0.245 ± 0.002	7.709 ± 0.867
QML	beta58	0.06	0.033	-0.027 ± 0.002	-0.450 ± 0.029	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.059 ± 0.001	0.974 ± 0.042	0.987 ± 0.025	0.034	-0.026	-43.97	0.0530	0.0592	1.052	91.975 ± 0.883	0.218 ± 0.001	8.976 ± 0.929
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.007 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.004	0.310 ± 0.009	0.200	0.000	-0.24	0.0654	0.0654	1.023	95.459 ± 0.677	0.249 ± 0.001	89.968 ± 0.976
UPI	beta42	0.20	0.200	0.000 ± 0.002	0.002 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.005	0.313 ± 0.009	0.195	-0.005	-2.30	0.0635	0.0637	1.057	96.199 ± 0.621	0.259 ± 0.002	87.962 ± 1.057
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.288 ± 0.008	0.201	0.001	0.45	0.0595	0.0595	1.087	96.727 ± 0.578	0.245 ± 0.002	92.714 ± 0.845
UPI	beta44	0.11	0.116	0.006 ± 0.003	0.051 ± 0.024	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.533 ± 0.028	0.730 ± 0.022	0.114	0.004	4.06	0.0804	0.0805	1.078	97.149 ± 0.541	0.339 ± 0.004	30.623 ± 1.498
UPI	beta45	0.11	0.113	0.003 ± 0.002	0.026 ± 0.021	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.403 ± 0.021	0.635 ± 0.019	0.110	0.000	0.45	0.0686	0.0686	1.069	97.466 ± 0.511	0.292 ± 0.003	33.052 ± 1.529
UPI	beta46	0.08	0.079	-0.001 ± 0.003	-0.016 ± 0.042	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	1.708 ± 0.092	1.307 ± 0.041	0.078	-0.002	-2.50	0.0997	0.0997	1.220	98.205 ± 0.431	0.500 ± 0.009	5.069 ± 0.713

Table S159

Simulation 2 - Condition 15: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.082	0.002 ± 0.004	0.024 ± 0.047	0.014 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.116 ± 0.004	2.112 ± 0.121	1.453 ± 0.048	0.075	-0.005	-6.33	0.1020	0.1021	1.169	97.677 ± 0.490	0.533 ± 0.011	4.963 ± 0.706
UPI	beta51	0.16	0.164	0.004 ± 0.003	0.023 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.251 ± 0.012	0.501 ± 0.014	0.162	0.002	1.17	0.0783	0.0783	1.029	95.248 ± 0.691	0.324 ± 0.002	52.270 ± 1.623
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.021 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.008	0.413 ± 0.012	0.161	0.001	0.87	0.0625	0.0625	1.008	94.509 ± 0.740	0.261 ± 0.001	69.694 ± 1.493
UPI	beta53	0.16	0.155	-0.005 ± 0.002	-0.029 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.434 ± 0.012	0.156	-0.004	-2.79	0.0712	0.0714	1.013	94.931 ± 0.713	0.275 ± 0.001	60.718 ± 1.587
UPI	beta54	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.013	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.388 ± 0.011	0.158	-0.002	-1.02	0.0617	0.0617	1.070	95.987 ± 0.638	0.260 ± 0.001	68.110 ± 1.514
UPI	beta55	0.08	0.081	0.001 ± 0.003	0.013 ± 0.039	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	1.471 ± 0.079	1.213 ± 0.038	0.082	0.002	2.79	0.0939	0.0939	1.051	96.410 ± 0.605	0.400 ± 0.004	9.609 ± 0.958
UPI	beta56	0.08	0.083	0.003 ± 0.003	0.036 ± 0.039	0.009 ± 0.000	0.095 ± 0.003	0.009 ± 0.000	0.095 ± 0.003	1.409 ± 0.075	1.187 ± 0.037	0.080	0.000	0.35	0.0871	0.0871	1.022	96.199 ± 0.621	0.380 ± 0.003	11.404 ± 1.033
UPI	beta57	0.06	0.060	-0.000 ± 0.004	-0.001 ± 0.059	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.003	3.283 ± 0.180	1.812 ± 0.058	0.057	-0.003	-5.65	0.1019	0.1020	1.134	97.677 ± 0.490	0.484 ± 0.008	3.696 ± 0.613
UPI	beta58	0.06	0.053	-0.007 ± 0.003	-0.123 ± 0.055	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	2.872 ± 0.153	1.695 ± 0.053	0.049	-0.011	-18.84	0.0927	0.0934	1.187	97.254 ± 0.531	0.472 ± 0.009	3.801 ± 0.621

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S160

Simulation 2 - Condition 16: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.199	-0.001	-0.66	0.0362	0.0363	1.023	95.596 ± 0.649	0.148 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.202	0.002 ± 0.001	0.011 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.202	0.002	1.02	0.0360	0.0361	1.027	95.495 ± 0.656	0.152 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.199	-0.001	-0.35	0.0367	0.0367	0.989	94.795 ± 0.703	0.143 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.015 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.162 ± 0.008	0.402 ± 0.012	0.113	0.003	3.08	0.0402	0.0403	1.002	94.394 ± 0.728	0.174 ± 0.001	73.273 ± 1.400
LSAM	beta45	0.11	0.110	0.000 ± 0.001	0.004 ± 0.011	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.131 ± 0.006	0.362 ± 0.010	0.110	0.000	-0.42	0.0404	0.0404	0.993	94.494 ± 0.722	0.155 ± 0.000	81.782 ± 1.221
LSAM	beta46	0.08	0.081	0.001 ± 0.002	0.016 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.495 ± 0.023	0.704 ± 0.020	0.081	0.001	1.74	0.0560	0.0560	0.990	95.495 ± 0.656	0.219 ± 0.001	29.930 ± 1.449
LSAM	beta47	0.08	0.080	-0.000 ± 0.002	-0.000 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.466 ± 0.021	0.683 ± 0.019	0.080	0.000	-0.54	0.0528	0.0528	1.017	95.395 ± 0.663	0.218 ± 0.001	28.829 ± 1.433
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.159	-0.001	-0.45	0.0507	0.0507	0.961	93.093 ± 0.802	0.196 ± 0.000	88.388 ± 1.014
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.003 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.161	0.001	0.49	0.0400	0.0400	1.000	94.595 ± 0.715	0.156 ± 0.000	98.498 ± 0.385
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.261 ± 0.007	0.160	0.000	0.27	0.0410	0.0410	0.998	95.195 ± 0.677	0.163 ± 0.000	97.798 ± 0.464
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.161	0.001	0.52	0.0391	0.0391	0.998	95.796 ± 0.635	0.155 ± 0.000	98.298 ± 0.409
LSAM	beta55	0.08	0.084	0.004 ± 0.002	0.049 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.443 ± 0.020	0.665 ± 0.019	0.083	0.003	4.23	0.0517	0.0518	1.049	95.996 ± 0.620	0.218 ± 0.001	33.333 ± 1.491
LSAM	beta56	0.08	0.075	-0.005 ± 0.002	-0.063 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.464 ± 0.021	0.681 ± 0.019	0.075	-0.005	-6.86	0.0509	0.0512	0.971	94.394 ± 0.728	0.207 ± 0.001	28.428 ± 1.427
LSAM	beta57	0.06	0.061	0.001 ± 0.002	0.022 ± 0.031	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.968 ± 0.046	0.984 ± 0.028	0.062	0.002	3.82	0.0565	0.0566	0.991	94.895 ± 0.696	0.229 ± 0.001	17.417 ± 1.200
LSAM	beta58	0.06	0.061	0.001 ± 0.002	0.013 ± 0.028	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.782 ± 0.035	0.884 ± 0.024	0.058	-0.002	-2.67	0.0533	0.0533	1.020	96.597 ± 0.574	0.212 ± 0.001	19.920 ± 1.264
QML	beta41	0.20	0.193	-0.007 ± 0.001	-0.037 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.194	-0.006	-2.91	0.0339	0.0344	1.013	94.294 ± 0.734	0.141 ± 0.001	99.700 ± 0.173
QML	beta42	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.001	0.183 ± 0.005	0.199	-0.001	-0.55	0.0370	0.0371	1.010	95.295 ± 0.670	0.145 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.190	-0.010 ± 0.001	-0.049 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.190	-0.010	-4.87	0.0351	0.0364	0.996	93.493 ± 0.780	0.138 ± 0.000	99.900 ± 0.100
QML	beta44	0.11	0.115	0.005 ± 0.001	0.046 ± 0.012	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.154 ± 0.007	0.392 ± 0.011	0.115	0.005	4.48	0.0392	0.0395	1.007	94.394 ± 0.728	0.169 ± 0.001	76.977 ± 1.332
QML	beta45	0.11	0.110	0.000 ± 0.001	0.004 ± 0.011	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.119 ± 0.005	0.345 ± 0.009	0.110	0.000	-0.18	0.0385	0.0385	1.000	94.995 ± 0.690	0.149 ± 0.000	84.084 ± 1.157
QML	beta46	0.08	0.042	-0.038 ± 0.001	-0.479 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.432 ± 0.016	0.657 ± 0.013	0.041	-0.039	-49.05	0.0362	0.0534	0.983	78.278 ± 1.305	0.139 ± 0.001	21.622 ± 1.302
QML	beta47	0.08	0.044	-0.036 ± 0.001	-0.452 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.391 ± 0.015	0.625 ± 0.013	0.044	-0.036	-45.00	0.0335	0.0492	1.016	82.082 ± 1.213	0.138 ± 0.001	23.724 ± 1.346
QML	beta51	0.16	0.172	0.012 ± 0.002	0.078 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.111 ± 0.005	0.333 ± 0.009	0.173	0.013	8.00	0.0508	0.0524	0.979	93.293 ± 0.791	0.199 ± 0.001	92.192 ± 0.849
QML	beta52	0.16	0.148	-0.012 ± 0.001	-0.074 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.256 ± 0.007	0.149	-0.011	-7.18	0.0395	0.0411	1.016	94.194 ± 0.740	0.157 ± 0.000	96.597 ± 0.574
QML	beta53	0.16	0.144	-0.016 ± 0.001	-0.101 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.274 ± 0.007	0.142	-0.018	-11.03	0.0402	0.0439	1.001	93.193 ± 0.797	0.160 ± 0.000	94.494 ± 0.722
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.002	0.240 ± 0.006	0.159	-0.001	-0.79	0.0380	0.0380	1.003	95.295 ± 0.670	0.151 ± 0.001	98.398 ± 0.397
QML	beta55	0.08	0.085	0.005 ± 0.002	0.067 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.415 ± 0.019	0.644 ± 0.018	0.088	0.008	9.63	0.0518	0.0524	1.077	96.396 ± 0.590	0.216 ± 0.001	34.334 ± 1.502
QML	beta56	0.08	0.071	-0.009 ± 0.002	-0.107 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.419 ± 0.019	0.648 ± 0.018	0.070	-0.010	-13.04	0.0506	0.0517	0.989	94.394 ± 0.728	0.198 ± 0.001	28.929 ± 1.435
QML	beta57	0.06	0.033	-0.027 ± 0.001	-0.458 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.615 ± 0.028	0.784 ± 0.020	0.034	-0.026	-43.52	0.0369	0.0452	1.004	87.588 ± 1.043	0.151 ± 0.001	12.412 ± 1.043
QML	beta58	0.06	0.037	-0.023 ± 0.001	-0.381 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.458 ± 0.019	0.677 ± 0.016	0.037	-0.023	-38.88	0.0337	0.0410	1.023	89.690 ± 0.962	0.135 ± 0.001	18.719 ± 1.234
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.200	0.000	-0.17	0.0367	0.0367	0.995	95.095 ± 0.683	0.145 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.012 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.202	0.002	0.95	0.0373	0.0374	1.010	94.795 ± 0.703	0.150 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.200	0.000	-0.02	0.0364	0.0364	0.990	95.095 ± 0.683	0.143 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.113	0.003 ± 0.001	0.030 ± 0.013	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.176 ± 0.008	0.419 ± 0.012	0.115	0.005	4.32	0.0429	0.0432	1.009	94.795 ± 0.703	0.182 ± 0.001	70.871 ± 1.438
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.009 ± 0.012	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.141 ± 0.006	0.375 ± 0.010	0.111	0.001	0.93	0.0416	0.0416	1.002	95.896 ± 0.628	0.162 ± 0.001	78.979 ± 1.289
UPI	beta46	0.08	0.082	0.002 ± 0.002	0.023 ± 0.024	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.559 ± 0.029	0.747 ± 0.022	0.079	-0.001	-1.51	0.0554	0.0554	1.033	96.897 ± 0.549	0.242 ± 0.002	23.123 ± 1.334

Table S160

Simulation 2 - Condition 16: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.082	0.002 ± 0.002	0.019 ± 0.023	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.533 ± 0.025	0.730 ± 0.021	0.079	-0.001	-1.73	0.0547	0.0548	1.054	95.996 ± 0.620	0.241 ± 0.002	23.323 ± 1.338
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.158	-0.002	-1.09	0.0525	0.0526	0.967	94.494 ± 0.722	0.197 ± 0.000	88.689 ± 1.002
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.161	0.001	0.82	0.0397	0.0398	0.997	94.194 ± 0.740	0.155 ± 0.000	98.298 ± 0.409
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.160	0.000	-0.04	0.0402	0.0402	1.008	95.796 ± 0.635	0.164 ± 0.000	97.397 ± 0.504
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.160	0.000	0.15	0.0404	0.0404	0.999	96.096 ± 0.613	0.155 ± 0.000	98.098 ± 0.432
UPI	beta55	0.08	0.084	0.004 ± 0.002	0.053 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.453 ± 0.021	0.673 ± 0.019	0.083	0.003	4.15	0.0536	0.0537	1.071	96.396 ± 0.590	0.226 ± 0.001	31.031 ± 1.464
UPI	beta56	0.08	0.076	-0.004 ± 0.002	-0.053 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.473 ± 0.023	0.688 ± 0.020	0.075	-0.005	-6.23	0.0519	0.0521	0.997	95.395 ± 0.663	0.215 ± 0.001	26.527 ± 1.397
UPI	beta57	0.06	0.061	0.001 ± 0.002	0.019 ± 0.032	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	1.039 ± 0.051	1.019 ± 0.030	0.062	0.002	3.66	0.0585	0.0586	1.031	95.896 ± 0.628	0.247 ± 0.002	12.813 ± 1.057
UPI	beta58	0.06	0.060	0.000 ± 0.002	0.001 ± 0.030	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.884 ± 0.041	0.940 ± 0.027	0.058	-0.002	-3.82	0.0559	0.0559	1.066	96.897 ± 0.549	0.236 ± 0.002	14.715 ± 1.121

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S161

Simulation 2 - Condition 17: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.012 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.197	-0.003	-1.39	0.0485	0.0486	1.003	95.796 ± 0.635	0.190 ± 0.000	98.599 ± 0.372
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.012 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.203	0.003	1.27	0.0506	0.0507	0.969	94.394 ± 0.728	0.193 ± 0.000	97.598 ± 0.484
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.200	0.000	-0.14	0.0479	0.0479	0.986	94.995 ± 0.690	0.181 ± 0.000	98.999 ± 0.315
LSAM	beta44	0.11	0.111	0.001 ± 0.002	0.010 ± 0.015	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.239 ± 0.011	0.489 ± 0.013	0.110	0.000	-0.38	0.0542	0.0542	0.961	93.994 ± 0.752	0.203 ± 0.001	57.858 ± 1.562
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.006 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.198 ± 0.009	0.445 ± 0.013	0.111	0.001	0.45	0.0473	0.0473	0.968	93.794 ± 0.763	0.186 ± 0.001	66.466 ± 1.494
LSAM	beta46	0.08	0.078	-0.002 ± 0.002	-0.027 ± 0.025	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.615 ± 0.026	0.784 ± 0.021	0.078	-0.002	-2.78	0.0635	0.0635	0.951	94.294 ± 0.734	0.234 ± 0.001	26.426 ± 1.395
LSAM	beta47	0.08	0.082	0.002 ± 0.002	0.021 ± 0.023	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.551 ± 0.027	0.742 ± 0.022	0.081	0.001	1.71	0.0590	0.0590	1.001	94.695 ± 0.709	0.233 ± 0.001	28.028 ± 1.421
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.007	0.408 ± 0.011	0.160	0.000	-0.25	0.0650	0.0650	0.957	94.294 ± 0.734	0.245 ± 0.001	73.874 ± 1.390
LSAM	beta52	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.160	0.000	-0.01	0.0520	0.0520	0.976	94.595 ± 0.715	0.201 ± 0.000	87.487 ± 1.047
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.111 ± 0.005	0.334 ± 0.009	0.159	-0.001	-0.82	0.0551	0.0551	0.993	94.494 ± 0.722	0.208 ± 0.001	84.785 ± 1.136
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.329 ± 0.009	0.162	0.002	0.98	0.0523	0.0523	0.974	94.294 ± 0.734	0.201 ± 0.000	86.987 ± 1.064
LSAM	beta55	0.08	0.084	0.004 ± 0.002	0.053 ± 0.026	0.004 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.703 ± 0.031	0.839 ± 0.023	0.087	0.007	8.94	0.0666	0.0670	0.968	93.894 ± 0.758	0.254 ± 0.001	26.226 ± 1.392
LSAM	beta56	0.08	0.079	-0.001 ± 0.002	-0.017 ± 0.026	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.655 ± 0.027	0.809 ± 0.021	0.081	0.001	1.73	0.0657	0.0658	0.962	94.294 ± 0.734	0.244 ± 0.001	26.426 ± 1.395
LSAM	beta57	0.06	0.057	-0.003 ± 0.002	-0.047 ± 0.034	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	1.134 ± 0.051	1.065 ± 0.029	0.057	-0.003	-4.33	0.0638	0.0638	1.000	95.596 ± 0.649	0.250 ± 0.001	14.214 ± 1.105
LSAM	beta58	0.06	0.056	-0.004 ± 0.002	-0.073 ± 0.032	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	1.033 ± 0.047	1.017 ± 0.028	0.054	-0.006	-9.52	0.0607	0.0610	0.982	93.994 ± 0.752	0.234 ± 0.001	16.817 ± 1.183
QML	beta41	0.20	0.195	-0.005 ± 0.001	-0.026 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.236 ± 0.007	0.195	-0.005	-2.64	0.0469	0.0472	1.003	95.596 ± 0.649	0.184 ± 0.000	98.599 ± 0.372
QML	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.202	0.002	0.88	0.0492	0.0492	0.969	94.194 ± 0.740	0.188 ± 0.000	98.098 ± 0.432
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.018 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.195	-0.005	-2.44	0.0472	0.0474	0.991	95.095 ± 0.683	0.179 ± 0.000	99.099 ± 0.299
QML	beta44	0.11	0.113	0.003 ± 0.002	0.025 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.236 ± 0.011	0.486 ± 0.014	0.111	0.001	0.57	0.0545	0.0545	0.963	93.694 ± 0.769	0.202 ± 0.001	58.358 ± 1.560
QML	beta45	0.11	0.111	0.001 ± 0.002	0.013 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.193 ± 0.009	0.439 ± 0.013	0.112	0.002	1.59	0.0467	0.0467	0.971	93.794 ± 0.763	0.184 ± 0.001	67.467 ± 1.482
QML	beta46	0.08	0.058	-0.022 ± 0.002	-0.276 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.477 ± 0.020	0.691 ± 0.017	0.058	-0.022	-27.61	0.0508	0.0554	0.948	90.791 ± 0.915	0.188 ± 0.001	24.625 ± 1.363
QML	beta47	0.08	0.062	-0.018 ± 0.001	-0.221 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.396 ± 0.018	0.629 ± 0.017	0.062	-0.018	-22.44	0.0458	0.0492	1.008	92.993 ± 0.808	0.186 ± 0.001	25.425 ± 1.378
QML	beta51	0.16	0.168	0.008 ± 0.002	0.048 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.008	0.414 ± 0.011	0.166	0.006	3.90	0.0646	0.0649	0.971	93.994 ± 0.752	0.250 ± 0.001	75.576 ± 1.359
QML	beta52	0.16	0.144	-0.016 ± 0.002	-0.099 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.125 ± 0.005	0.353 ± 0.009	0.145	-0.015	-9.54	0.0543	0.0564	0.973	92.993 ± 0.808	0.207 ± 0.001	77.277 ± 1.326
QML	beta53	0.16	0.141	-0.019 ± 0.002	-0.116 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.128 ± 0.005	0.358 ± 0.009	0.140	-0.020	-12.22	0.0552	0.0586	0.996	93.093 ± 0.802	0.212 ± 0.001	74.174 ± 1.385
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.325 ± 0.009	0.161	0.001	0.73	0.0516	0.0516	0.975	94.294 ± 0.734	0.199 ± 0.000	87.588 ± 1.043
QML	beta55	0.08	0.086	0.006 ± 0.002	0.075 ± 0.027	0.004 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.707 ± 0.031	0.841 ± 0.023	0.090	0.010	12.01	0.0672	0.0679	0.980	93.794 ± 0.763	0.258 ± 0.001	25.526 ± 1.379
QML	beta56	0.08	0.078	-0.002 ± 0.002	-0.023 ± 0.025	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.640 ± 0.027	0.800 ± 0.021	0.079	-0.001	-1.43	0.0654	0.0654	0.968	94.695 ± 0.709	0.243 ± 0.001	26.226 ± 1.392
QML	beta57	0.06	0.042	-0.018 ± 0.002	-0.307 ± 0.027	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.832 ± 0.037	0.912 ± 0.024	0.040	-0.020	-32.60	0.0523	0.0558	1.019	92.793 ± 0.818	0.206 ± 0.002	13.113 ± 1.068
QML	beta58	0.06	0.044	-0.016 ± 0.002	-0.268 ± 0.026	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.728 ± 0.032	0.853 ± 0.023	0.042	-0.018	-30.23	0.0481	0.0514	0.993	92.993 ± 0.808	0.189 ± 0.001	16.617 ± 1.178
UPI	beta41	0.20	0.198	-0.002 ± 0.002	-0.011 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.198	-0.002	-0.89	0.0491	0.0491	0.956	93.894 ± 0.758	0.181 ± 0.000	98.699 ± 0.359
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.203	0.003	1.62	0.0511	0.0512	0.941	93.794 ± 0.763	0.187 ± 0.000	97.998 ± 0.443
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.200	0.000	0.13	0.0483	0.0483	0.982	94.995 ± 0.690	0.180 ± 0.000	99.099 ± 0.299
UPI	beta44	0.11	0.112	0.002 ± 0.002	0.018 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.246 ± 0.011	0.496 ± 0.014	0.111	0.001	0.84	0.0554	0.0554	0.967	94.695 ± 0.709	0.207 ± 0.001	57.257 ± 1.565
UPI	beta45	0.11	0.111	0.001 ± 0.002	0.014 ± 0.014	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.208 ± 0.010	0.456 ± 0.013	0.111	0.001	1.16	0.0488	0.0488	0.963	94.194 ± 0.740	0.189 ± 0.001	66.066 ± 1.498
UPI	beta46	0.08	0.077	-0.003 ± 0.002	-0.037 ± 0.026	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.652 ± 0.029	0.808 ± 0.022	0.077	-0.003	-3.74	0.0644	0.0645	0.966	93.794 ± 0.763	0.244 ± 0.001	24.024 ± 1.352

Table S161

Simulation 2 - Condition 17: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.082	0.002 ± 0.002	0.021 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.600 ± 0.030	0.775 ± 0.023	0.080	0.000	0.29	0.0598	0.0598	1.010	95.395 ± 0.663	0.245 ± 0.001	24.525 ± 1.361
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.008 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.007	0.406 ± 0.011	0.160	0.000	-0.12	0.0653	0.0653	0.964	94.394 ± 0.728	0.245 ± 0.001	73.974 ± 1.388
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.329 ± 0.009	0.160	0.000	-0.01	0.0516	0.0516	0.960	93.794 ± 0.763	0.198 ± 0.000	87.487 ± 1.047
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.159	-0.001	-0.51	0.0550	0.0550	0.994	94.294 ± 0.734	0.209 ± 0.001	84.585 ± 1.142
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.161	0.001	0.85	0.0528	0.0528	0.964	93.794 ± 0.763	0.199 ± 0.000	87.788 ± 1.036
UPI	beta55	0.08	0.084	0.004 ± 0.002	0.047 ± 0.026	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.699 ± 0.031	0.836 ± 0.023	0.086	0.006	7.46	0.0651	0.0654	0.984	94.194 ± 0.740	0.258 ± 0.001	24.324 ± 1.357
UPI	beta56	0.08	0.078	-0.002 ± 0.002	-0.022 ± 0.026	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.668 ± 0.028	0.818 ± 0.022	0.079	-0.001	-1.57	0.0664	0.0664	0.971	94.695 ± 0.709	0.249 ± 0.001	25.425 ± 1.378
UPI	beta57	0.06	0.056	-0.004 ± 0.002	-0.069 ± 0.034	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	1.166 ± 0.054	1.080 ± 0.030	0.055	-0.005	-8.92	0.0630	0.0632	1.021	96.296 ± 0.598	0.259 ± 0.001	13.313 ± 1.075
UPI	beta58	0.06	0.056	-0.004 ± 0.002	-0.065 ± 0.033	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	1.100 ± 0.052	1.049 ± 0.030	0.054	-0.006	-9.75	0.0636	0.0639	0.997	95.796 ± 0.635	0.246 ± 0.001	14.515 ± 1.114

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S162

Simulation 2 - Condition 18: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.198	-0.002	-1.03	0.0296	0.0297	1.012	95.400 ± 0.662	0.119 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.202	0.002 ± 0.001	0.010 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.203	0.003	1.53	0.0291	0.0293	0.999	94.700 ± 0.708	0.120 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.201	0.001	0.29	0.0283	0.0283	0.991	95.200 ± 0.676	0.114 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.017 ± 0.009	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.114	0.004	3.33	0.0308	0.0310	1.012	94.400 ± 0.727	0.127 ± 0.000	93.500 ± 0.780
LSAM	beta45	0.11	0.110	0.000 ± 0.001	0.004 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.110	0.000	-0.11	0.0303	0.0303	0.949	92.800 ± 0.817	0.117 ± 0.000	95.000 ± 0.689
LSAM	beta46	0.08	0.082	0.002 ± 0.001	0.025 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.229 ± 0.010	0.478 ± 0.013	0.082	0.002	2.01	0.0384	0.0384	0.975	94.600 ± 0.715	0.146 ± 0.000	58.500 ± 1.558
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.015 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.224 ± 0.010	0.474 ± 0.013	0.078	-0.002	-2.09	0.0380	0.0380	0.980	94.100 ± 0.745	0.146 ± 0.000	55.900 ± 1.570
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.157	-0.003	-1.73	0.0372	0.0373	1.022	95.900 ± 0.627	0.156 ± 0.000	98.100 ± 0.432
LSAM	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.159	-0.001	-0.81	0.0329	0.0329	0.995	95.000 ± 0.689	0.127 ± 0.000	99.900 ± 0.100
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.162	0.002	1.29	0.0324	0.0325	1.019	95.800 ± 0.634	0.131 ± 0.000	99.800 ± 0.141
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.160	0.000	-0.29	0.0310	0.0310	1.048	96.400 ± 0.589	0.126 ± 0.000	100.000 ± 0.000
LSAM	beta55	0.08	0.080	0.000 ± 0.001	0.001 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.264 ± 0.012	0.514 ± 0.014	0.080	0.000	-0.46	0.0406	0.0406	0.996	95.200 ± 0.676	0.160 ± 0.000	49.700 ± 1.581
LSAM	beta56	0.08	0.081	0.001 ± 0.001	0.011 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.265 ± 0.011	0.515 ± 0.014	0.082	0.002	1.92	0.0405	0.0405	0.952	92.700 ± 0.823	0.154 ± 0.000	53.500 ± 1.577
LSAM	beta57	0.06	0.061	0.001 ± 0.001	0.025 ± 0.020	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.415 ± 0.019	0.644 ± 0.018	0.062	0.002	2.61	0.0374	0.0374	1.038	95.800 ± 0.634	0.157 ± 0.000	32.200 ± 1.478
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.383 ± 0.017	0.619 ± 0.017	0.060	0.000	-0.62	0.0370	0.0371	1.013	95.200 ± 0.676	0.148 ± 0.000	34.800 ± 1.506
QML	beta41	0.20	0.196	-0.004 ± 0.001	-0.022 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.195	-0.005	-2.56	0.0285	0.0290	1.015	95.500 ± 0.656	0.117 ± 0.000	99.900 ± 0.100
QML	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.201	0.001	0.69	0.0287	0.0287	0.998	94.200 ± 0.739	0.118 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.020 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.197	-0.003	-1.69	0.0284	0.0286	0.987	94.800 ± 0.702	0.113 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.114	0.004 ± 0.001	0.033 ± 0.009	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.115	0.005	4.33	0.0310	0.0314	1.013	94.700 ± 0.708	0.127 ± 0.000	94.500 ± 0.721
QML	beta45	0.11	0.111	0.001 ± 0.001	0.010 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.078 ± 0.003	0.280 ± 0.008	0.111	0.001	0.91	0.0297	0.0297	0.959	93.100 ± 0.801	0.116 ± 0.000	95.600 ± 0.649
QML	beta46	0.08	0.062	-0.018 ± 0.001	-0.230 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.201 ± 0.008	0.448 ± 0.011	0.062	-0.018	-22.95	0.0312	0.0362	0.979	88.700 ± 1.001	0.118 ± 0.000	53.500 ± 1.577
QML	beta47	0.08	0.061	-0.019 ± 0.001	-0.242 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.203 ± 0.009	0.450 ± 0.011	0.060	-0.020	-24.56	0.0302	0.0360	0.982	88.900 ± 0.993	0.117 ± 0.000	52.200 ± 1.580
QML	beta51	0.16	0.165	0.005 ± 0.001	0.033 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.165	0.005	2.84	0.0374	0.0376	1.032	95.600 ± 0.649	0.158 ± 0.000	98.700 ± 0.358
QML	beta52	0.16	0.145	-0.015 ± 0.001	-0.097 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.143	-0.017	-10.92	0.0341	0.0383	1.004	93.200 ± 0.796	0.130 ± 0.000	99.200 ± 0.282
QML	beta53	0.16	0.143	-0.017 ± 0.001	-0.108 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.236 ± 0.006	0.144	-0.016	-10.19	0.0343	0.0380	1.008	91.400 ± 0.887	0.133 ± 0.000	98.600 ± 0.372
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.159	-0.001	-0.79	0.0304	0.0304	1.049	96.500 ± 0.581	0.125 ± 0.000	100.000 ± 0.000
QML	beta55	0.08	0.081	0.001 ± 0.001	0.016 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.268 ± 0.012	0.517 ± 0.014	0.082	0.002	2.10	0.0416	0.0416	1.000	95.300 ± 0.669	0.162 ± 0.000	50.200 ± 1.581
QML	beta56	0.08	0.080	0.000 ± 0.001	0.005 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.261 ± 0.011	0.511 ± 0.014	0.080	0.000	0.49	0.0400	0.0400	0.956	93.000 ± 0.807	0.153 ± 0.000	52.800 ± 1.579
QML	beta57	0.06	0.046	-0.014 ± 0.001	-0.233 ± 0.017	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.328 ± 0.015	0.573 ± 0.015	0.046	-0.014	-23.20	0.0305	0.0335	1.048	94.100 ± 0.745	0.129 ± 0.000	28.400 ± 1.426
QML	beta58	0.06	0.048	-0.012 ± 0.001	-0.202 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.287 ± 0.013	0.536 ± 0.014	0.047	-0.013	-22.08	0.0287	0.0316	1.022	94.300 ± 0.733	0.119 ± 0.000	33.700 ± 1.495
UPI	beta41	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.198	-0.002	-1.08	0.0297	0.0298	0.965	94.300 ± 0.733	0.113 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.203	0.003	1.67	0.0290	0.0292	0.972	93.300 ± 0.791	0.117 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.201	0.001	0.29	0.0285	0.0285	0.985	94.400 ± 0.727	0.113 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.112	0.002 ± 0.001	0.020 ± 0.009	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.114	0.004	3.60	0.0315	0.0317	1.012	94.200 ± 0.739	0.128 ± 0.000	93.300 ± 0.791
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.005 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.109	-0.001	-0.79	0.0310	0.0310	0.955	93.000 ± 0.807	0.118 ± 0.000	95.000 ± 0.689
UPI	beta46	0.08	0.082	0.002 ± 0.001	0.022 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.236 ± 0.011	0.486 ± 0.013	0.081	0.001	1.44	0.0391	0.0391	0.976	94.800 ± 0.702	0.149 ± 0.000	56.800 ± 1.566

Table S162

Simulation 2 - Condition 18: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.015	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.232 ± 0.010	0.481 ± 0.013	0.078	-0.002	-2.91	0.0387	0.0387	0.984	94.600 ± 0.715	0.149 ± 0.000	54.500 ± 1.575
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.242 ± 0.007	0.158	-0.002	-1.54	0.0368	0.0369	1.024	95.900 ± 0.627	0.155 ± 0.000	98.100 ± 0.432
UPI	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.159	-0.001	-0.81	0.0330	0.0330	0.982	94.700 ± 0.708	0.125 ± 0.000	99.900 ± 0.100
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.162	0.002	1.23	0.0326	0.0327	1.017	95.700 ± 0.641	0.131 ± 0.000	99.800 ± 0.141
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.160	0.000	-0.21	0.0303	0.0303	1.042	96.100 ± 0.612	0.125 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.080	0.000 ± 0.001	0.000 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.265 ± 0.012	0.515 ± 0.014	0.080	0.000	0.27	0.0412	0.0412	1.000	94.500 ± 0.721	0.162 ± 0.000	49.300 ± 1.581
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.012 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.265 ± 0.011	0.515 ± 0.014	0.081	0.001	1.47	0.0407	0.0407	0.961	93.300 ± 0.791	0.155 ± 0.000	53.200 ± 1.578
UPI	beta57	0.06	0.061	0.001 ± 0.001	0.023 ± 0.020	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.419 ± 0.020	0.647 ± 0.018	0.062	0.002	2.59	0.0369	0.0369	1.048	95.600 ± 0.649	0.159 ± 0.000	30.800 ± 1.460
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.395 ± 0.018	0.629 ± 0.017	0.059	-0.001	-1.52	0.0376	0.0376	1.018	95.600 ± 0.649	0.151 ± 0.000	33.300 ± 1.490

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S163

Simulation 2 - Condition 19: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.194	-0.006 ± 0.010	-0.029 ± 0.050	0.010 ± 0.001	0.102 ± 0.007	0.010 ± 0.001	0.102 ± 0.009	0.260 ± 0.035	0.510 ± 0.044	0.190	-0.010	-4.88	0.1103	0.1107	1.075	97.115 ± 1.641	0.431 ± 0.010	41.346 ± 4.829
LSAM	beta42	0.20	0.206	0.006 ± 0.012	0.028 ± 0.058	0.014 ± 0.002	0.119 ± 0.008	0.014 ± 0.002	0.119 ± 0.010	0.352 ± 0.045	0.593 ± 0.049	0.201	0.001	0.31	0.1145	0.1145	1.017	97.115 ± 1.641	0.474 ± 0.017	44.231 ± 4.870
LSAM	beta43	0.20	0.203	0.003 ± 0.009	0.016 ± 0.045	0.008 ± 0.001	0.091 ± 0.005	0.008 ± 0.001	0.090 ± 0.007	0.205 ± 0.023	0.452 ± 0.034	0.206	0.006	3.08	0.0847	0.0850	0.985	98.077 ± 1.347	0.351 ± 0.008	61.538 ± 4.771
LSAM	beta44	0.11	0.160	0.050 ± 0.014	0.456 ± 0.127	0.020 ± 0.003	0.143 ± 0.012	0.023 ± 0.004	0.151 ± 0.015	1.872 ± 0.324	1.368 ± 0.136	0.164	0.054	48.71	0.1332	0.1436	1.014	96.154 ± 1.886	0.567 ± 0.028	24.038 ± 4.190
LSAM	beta45	0.11	0.122	0.012 ± 0.010	0.106 ± 0.095	0.011 ± 0.002	0.106 ± 0.008	0.011 ± 0.002	0.106 ± 0.009	0.931 ± 0.133	0.965 ± 0.085	0.124	0.014	12.44	0.1104	0.1113	1.043	98.077 ± 1.347	0.433 ± 0.015	19.231 ± 3.865
LSAM	beta46	0.08	0.037	-0.043 ± 0.009	-0.537 ± 0.112	0.008 ± 0.001	0.092 ± 0.008	0.010 ± 0.001	0.101 ± 0.007	1.587 ± 0.191	1.260 ± 0.092	0.025	-0.055	-68.55	0.0786	0.0958	0.964	89.423 ± 3.016	0.346 ± 0.016	5.769 ± 2.286
LSAM	beta47	0.08	0.028	-0.052 ± 0.008	-0.645 ± 0.104	0.007 ± 0.001	0.085 ± 0.007	0.010 ± 0.001	0.099 ± 0.008	1.539 ± 0.228	1.241 ± 0.104	0.026	-0.054	-67.54	0.0819	0.0982	1.015	88.462 ± 3.133	0.339 ± 0.013	3.846 ± 1.886
LSAM	beta51	0.16	0.187	0.027 ± 0.012	0.168 ± 0.078	0.016 ± 0.003	0.127 ± 0.011	0.017 ± 0.003	0.129 ± 0.014	0.649 ± 0.121	0.806 ± 0.086	0.181	0.021	13.20	0.1044	0.1065	0.953	93.269 ± 2.457	0.473 ± 0.010	31.731 ± 4.564
LSAM	beta52	0.16	0.147	-0.013 ± 0.010	-0.079 ± 0.063	0.010 ± 0.002	0.102 ± 0.008	0.011 ± 0.002	0.103 ± 0.009	0.412 ± 0.059	0.642 ± 0.056	0.143	-0.017	-10.87	0.1005	0.1020	1.022	96.154 ± 1.886	0.410 ± 0.022	29.808 ± 4.485
LSAM	beta53	0.16	0.152	-0.008 ± 0.010	-0.053 ± 0.063	0.011 ± 0.002	0.103 ± 0.009	0.011 ± 0.002	0.103 ± 0.010	0.413 ± 0.068	0.643 ± 0.063	0.160	0.000	-0.22	0.1085	0.1085	0.978	97.115 ± 1.641	0.395 ± 0.009	28.846 ± 4.442
LSAM	beta54	0.16	0.142	-0.018 ± 0.010	-0.111 ± 0.065	0.011 ± 0.002	0.106 ± 0.008	0.011 ± 0.002	0.107 ± 0.010	0.446 ± 0.071	0.668 ± 0.064	0.139	-0.021	-13.24	0.1109	0.1129	1.006	95.192 ± 2.098	0.418 ± 0.020	31.731 ± 4.564
LSAM	beta55	0.08	0.102	0.022 ± 0.018	0.272 ± 0.222	0.033 ± 0.005	0.181 ± 0.015	0.033 ± 0.005	0.182 ± 0.018	5.150 ± 0.840	2.269 ± 0.219	0.112	0.032	39.42	0.1609	0.1639	0.923	96.154 ± 1.886	0.655 ± 0.036	7.692 ± 2.613
LSAM	beta56	0.08	0.073	-0.007 ± 0.014	-0.081 ± 0.174	0.020 ± 0.003	0.142 ± 0.011	0.020 ± 0.003	0.141 ± 0.013	3.117 ± 0.486	1.766 ± 0.166	0.070	-0.010	-13.12	0.1348	0.1352	1.064	97.115 ± 1.641	0.591 ± 0.032	2.885 ± 1.641
LSAM	beta57	0.06	0.033	-0.027 ± 0.009	-0.442 ± 0.142	0.008 ± 0.001	0.087 ± 0.006	0.008 ± 0.001	0.090 ± 0.007	2.266 ± 0.308	1.505 ± 0.124	0.030	-0.030	-49.17	0.0902	0.0949	0.986	95.192 ± 2.098	0.335 ± 0.017	4.808 ± 2.098
LSAM	beta58	0.06	0.029	-0.031 ± 0.008	-0.524 ± 0.136	0.007 ± 0.001	0.083 ± 0.009	0.008 ± 0.001	0.089 ± 0.009	2.189 ± 0.384	1.480 ± 0.148	0.021	-0.039	-64.38	0.0655	0.0761	0.896	89.423 ± 3.016	0.293 ± 0.017	3.846 ± 1.886
QML	beta41	0.20	0.181	-0.019 ± 0.008	-0.096 ± 0.039	0.006 ± 0.001	0.079 ± 0.007	0.006 ± 0.001	0.081 ± 0.007	0.162 ± 0.024	0.403 ± 0.036	0.179	-0.021	-10.51	0.0850	0.0876	1.048	95.192 ± 2.098	0.323 ± 0.023	64.423 ± 4.694
QML	beta42	0.20	0.182	-0.018 ± 0.009	-0.091 ± 0.044	0.008 ± 0.001	0.090 ± 0.007	0.008 ± 0.001	0.091 ± 0.008	0.208 ± 0.030	0.456 ± 0.040	0.177	-0.023	-11.60	0.0816	0.0848	0.894	93.269 ± 2.457	0.315 ± 0.006	59.615 ± 4.811
QML	beta43	0.20	0.183	-0.017 ± 0.007	-0.083 ± 0.036	0.005 ± 0.001	0.073 ± 0.004	0.006 ± 0.001	0.074 ± 0.005	0.139 ± 0.014	0.372 ± 0.026	0.185	-0.015	-7.43	0.0824	0.0838	1.048	96.154 ± 1.886	0.300 ± 0.005	66.346 ± 4.633
QML	beta44	0.11	0.130	0.020 ± 0.011	0.182 ± 0.103	0.013 ± 0.002	0.115 ± 0.009	0.014 ± 0.002	0.116 ± 0.011	1.122 ± 0.173	1.059 ± 0.098	0.130	0.020	18.37	0.1008	0.1028	1.119	95.192 ± 2.098	0.506 ± 0.101	27.885 ± 4.397
QML	beta45	0.11	0.107	-0.003 ± 0.008	-0.028 ± 0.070	0.006 ± 0.001	0.079 ± 0.005	0.006 ± 0.001	0.079 ± 0.006	0.512 ± 0.063	0.716 ± 0.057	0.119	0.009	8.52	0.0901	0.0906	1.166	97.115 ± 1.641	0.361 ± 0.024	25.962 ± 4.299
QML	beta46	0.08	0.056	-0.024 ± 0.007	-0.298 ± 0.086	0.005 ± 0.001	0.070 ± 0.005	0.005 ± 0.001	0.074 ± 0.005	0.848 ± 0.099	0.921 ± 0.068	0.045	-0.035	-44.00	0.0716	0.0798	1.105	87.500 ± 3.243	0.303 ± 0.044	15.385 ± 3.538
QML	beta47	0.08	0.044	-0.036 ± 0.006	-0.450 ± 0.077	0.004 ± 0.001	0.063 ± 0.005	0.005 ± 0.001	0.072 ± 0.005	0.811 ± 0.103	0.901 ± 0.067	0.044	-0.036	-44.73	0.0621	0.0717	1.137	88.462 ± 3.133	0.279 ± 0.029	6.731 ± 2.457
QML	beta51	0.16	0.192	0.032 ± 0.013	0.198 ± 0.079	0.016 ± 0.002	0.128 ± 0.010	0.017 ± 0.003	0.131 ± 0.012	0.675 ± 0.103	0.822 ± 0.074	0.185	0.025	15.94	0.1203	0.1230	1.034	94.231 ± 2.286	0.520 ± 0.017	28.846 ± 4.442
QML	beta52	0.16	0.150	-0.010 ± 0.010	-0.060 ± 0.062	0.010 ± 0.002	0.100 ± 0.010	0.010 ± 0.002	0.100 ± 0.011	0.394 ± 0.071	0.628 ± 0.067	0.143	-0.017	-10.57	0.1093	0.1106	0.977	95.192 ± 2.098	0.385 ± 0.012	33.654 ± 4.633
QML	beta53	0.16	0.133	-0.027 ± 0.009	-0.172 ± 0.055	0.008 ± 0.001	0.090 ± 0.007	0.009 ± 0.001	0.094 ± 0.008	0.342 ± 0.046	0.585 ± 0.048	0.130	-0.030	-19.00	0.0979	0.1025	1.024	94.231 ± 2.286	0.361 ± 0.006	33.654 ± 4.633
QML	beta54	0.16	0.153	-0.007 ± 0.008	-0.045 ± 0.053	0.008 ± 0.001	0.087 ± 0.006	0.007 ± 0.001	0.086 ± 0.007	0.292 ± 0.038	0.541 ± 0.044	0.154	-0.006	-3.80	0.0893	0.0895	0.991	95.192 ± 2.098	0.337 ± 0.006	45.192 ± 4.880
QML	beta55	0.08	0.088	0.008 ± 0.014	0.097 ± 0.178	0.021 ± 0.003	0.145 ± 0.012	0.021 ± 0.003	0.145 ± 0.013	3.274 ± 0.491	1.809 ± 0.165	0.115	0.035	43.25	0.1478	0.1518	1.006	94.231 ± 2.286	0.573 ± 0.025	9.615 ± 2.891
QML	beta56	0.08	0.068	-0.012 ± 0.010	-0.152 ± 0.131	0.011 ± 0.002	0.107 ± 0.009	0.011 ± 0.002	0.107 ± 0.011	1.781 ± 0.302	1.334 ± 0.134	0.078	-0.002	-2.12	0.1133	0.1134	1.131	95.192 ± 2.098	0.472 ± 0.014	5.769 ± 2.286
QML	beta57	0.06	0.034	-0.026 ± 0.008	-0.435 ± 0.140	0.007 ± 0.001	0.086 ± 0.009	0.008 ± 0.001	0.089 ± 0.009	2.202 ± 0.383	1.484 ± 0.149	0.026	-0.034	-57.10	0.0812	0.0882	0.902	88.462 ± 3.133	0.302 ± 0.014	7.692 ± 2.613
QML	beta58	0.06	0.030	-0.030 ± 0.006	-0.494 ± 0.106	0.004 ± 0.001	0.065 ± 0.005	0.005 ± 0.001	0.071 ± 0.006	1.410 ± 0.199	1.187 ± 0.099	0.028	-0.032	-53.47	0.0595	0.0676	0.935	86.538 ± 3.347	0.239 ± 0.007	3.846 ± 1.886
UPI	beta41	0.20	0.204	0.004 ± 0.016	0.020 ± 0.079	0.026 ± 0.005	0.161 ± 0.016	0.026 ± 0.005	0.160 ± 0.018	0.639 ± 0.126	0.799 ± 0.090	0.222	0.022	11.13	0.1814	0.1828	1.698	99.038 ± 0.957	1.068 ± 0.190	30.769 ± 4.526
UPI	beta42	0.20	0.258	0.058 ± 0.020	0.289 ± 0.099	0.041 ± 0.008	0.202 ± 0.022	0.044 ± 0.010	0.209 ± 0.026	1.096 ± 0.247	1.047 ± 0.131	0.246	0.046	22.94	0.1761	0.1820	1.599	93.269 ± 2.457	1.267 ± 0.185	30.769 ± 4.526
UPI	beta43	0.20	0.168	-0.032 ± 0.011	-0.161 ± 0.054	0.012 ± 0.002	0.109 ± 0.008	0.013 ± 0.002	0.113 ± 0.009	0.322 ± 0.039	0.567 ± 0.043	0.155	-0.045	-22.61	0.1364	0.1437	1.272	84.615 ± 3.538	0.545 ± 0.041	25.962 ± 4.299
UPI	beta44	0.11	0.194	0.084 ± 0.021	0.766 ± 0.195	0.048 ± 0.008	0.219 ± 0.020	0.055 ± 0.010	0.233 ± 0.024	4.505 ± 0.836	2.122 ± 0.221	0.173	0.063	56.84	0.1919	0.2018	1.447	100.000 ± 0.000	1.241 ± 0.131	5.769 ± 2.286
UPI	beta45	0.11	0.108	-0.002 ± 0.014	-0.022 ± 0.124	0.019 ± 0.004	0.140 ± 0.015	0.019 ± 0.004	0.139 ± 0.016	1.595 ± 0.332	1.263 ± 0.150	0.116	0.006	5.80	0.1446	0.1447	1.594	99.038 ± 0.957	0.872 ± 0.116	5.769 ± 2.286
UPI	beta46	0.08	0.041	-0.039 ± 0.018	-0.491 ± 0.226	0.034 ± 0.007	0.184 ± 0.019	0.035 ± 0.007	0.188 ± 0.021	5.494 ± 1.083	2.344 ± 0.261	0.030	-0.050	-62.27	0.1648	0.1722	1.503	92.308 ± 2.613	1.085 ± 0.141	1.923 ± 1.347

Table S163

Simulation 2 - Condition 19: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.023	-0.103 ± 0.018	-1.293 ± 0.223	0.033 ± 0.006	0.182 ± 0.018	0.043 ± 0.009	0.208 ± 0.024	6.770 ± 1.415	2.602 ± 0.295	-0.006	-0.086	-107.51	0.1737	0.1938	1.797	92.308 ± 2.613	1.278 ± 0.187	0.962 ± 0.957
UPI	beta51	0.16	0.230	0.070 ± 0.012	0.436 ± 0.072	0.014 ± 0.002	0.118 ± 0.009	0.019 ± 0.003	0.137 ± 0.013	0.729 ± 0.120	0.854 ± 0.078	0.225	0.065	40.88	0.1316	0.1470	1.226	98.077 ± 1.347	0.567 ± 0.023	42.308 ± 4.845
UPI	beta52	0.16	0.158	-0.002 ± 0.013	-0.010 ± 0.080	0.017 ± 0.003	0.130 ± 0.013	0.017 ± 0.003	0.130 ± 0.014	0.656 ± 0.126	0.810 ± 0.090	0.159	-0.001	-0.64	0.1297	0.1297	1.214	94.231 ± 2.286	0.620 ± 0.056	26.923 ± 4.349
UPI	beta53	0.16	0.131	-0.029 ± 0.012	-0.180 ± 0.073	0.014 ± 0.002	0.120 ± 0.010	0.015 ± 0.002	0.122 ± 0.010	0.585 ± 0.081	0.765 ± 0.065	0.118	-0.042	-26.42	0.1229	0.1300	1.203	90.385 ± 2.891	0.564 ± 0.049	16.346 ± 3.626
UPI	beta54	0.16	0.140	-0.020 ± 0.012	-0.127 ± 0.076	0.015 ± 0.002	0.124 ± 0.010	0.016 ± 0.002	0.125 ± 0.012	0.607 ± 0.095	0.779 ± 0.073	0.108	-0.052	-32.31	0.1318	0.1416	1.583	88.462 ± 3.133	0.767 ± 0.084	14.423 ± 3.445
UPI	beta55	0.08	0.130	0.050 ± 0.019	0.624 ± 0.239	0.038 ± 0.007	0.195 ± 0.019	0.040 ± 0.008	0.201 ± 0.023	6.293 ± 1.256	2.509 ± 0.282	0.131	0.051	64.33	0.1586	0.1667	1.244	99.038 ± 0.957	0.952 ± 0.074	2.885 ± 1.641
UPI	beta56	0.08	0.062	-0.018 ± 0.016	-0.225 ± 0.206	0.028 ± 0.005	0.168 ± 0.015	0.028 ± 0.005	0.168 ± 0.017	4.419 ± 0.736	2.102 ± 0.207	0.053	-0.027	-33.63	0.1416	0.1442	1.299	99.038 ± 0.957	0.855 ± 0.072	1.923 ± 1.347
UPI	beta57	0.06	0.032	-0.028 ± 0.013	-0.469 ± 0.220	0.018 ± 0.004	0.135 ± 0.016	0.019 ± 0.004	0.137 ± 0.016	5.205 ± 1.098	2.281 ± 0.272	0.021	-0.039	-65.17	0.1296	0.1354	1.371	93.269 ± 2.457	0.723 ± 0.082	1.923 ± 1.347
UPI	beta58	0.06	0.000	-0.060 ± 0.015	-1.008 ± 0.250	0.023 ± 0.004	0.153 ± 0.015	0.027 ± 0.005	0.164 ± 0.017	7.447 ± 1.375	2.729 ± 0.283	0.009	-0.051	-85.62	0.1034	0.1155	1.512	85.577 ± 3.445	0.906 ± 0.111	2.885 ± 1.641

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S164

Simulation 2 - Condition 20: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.210	0.010 ± 0.006	0.052 ± 0.029	0.004 ± 0.001	0.065 ± 0.004	0.004 ± 0.001	0.066 ± 0.005	0.108 ± 0.014	0.329 ± 0.026	0.202	0.002	0.83	0.0649	0.0649	1.062	96.748 ± 1.599	0.272 ± 0.004	91.057 ± 2.573
LSAM	beta42	0.20	0.191	-0.009 ± 0.007	-0.047 ± 0.033	0.005 ± 0.001	0.073 ± 0.005	0.005 ± 0.001	0.073 ± 0.006	0.135 ± 0.019	0.367 ± 0.031	0.199	-0.001	-0.39	0.0786	0.0786	0.994	95.122 ± 1.942	0.285 ± 0.005	78.862 ± 3.681
LSAM	beta43	0.20	0.186	-0.014 ± 0.005	-0.071 ± 0.023	0.003 ± 0.000	0.052 ± 0.003	0.003 ± 0.000	0.053 ± 0.004	0.071 ± 0.008	0.266 ± 0.019	0.183	-0.017	-8.72	0.0536	0.0564	1.047	95.122 ± 1.942	0.212 ± 0.002	96.748 ± 1.599
LSAM	beta44	0.11	0.179	0.069 ± 0.008	0.627 ± 0.069	0.007 ± 0.001	0.085 ± 0.005	0.012 ± 0.001	0.109 ± 0.007	0.980 ± 0.122	0.990 ± 0.068	0.179	0.069	63.09	0.0938	0.1167	0.957	87.805 ± 2.951	0.318 ± 0.006	62.602 ± 4.363
LSAM	beta45	0.11	0.123	0.013 ± 0.006	0.117 ± 0.053	0.004 ± 0.000	0.065 ± 0.003	0.004 ± 0.000	0.066 ± 0.005	0.359 ± 0.038	0.599 ± 0.041	0.123	0.013	11.71	0.0677	0.0689	1.026	97.561 ± 1.391	0.261 ± 0.005	48.780 ± 4.507
LSAM	beta46	0.08	0.016	-0.064 ± 0.005	-0.799 ± 0.062	0.003 ± 0.000	0.055 ± 0.004	0.007 ± 0.001	0.084 ± 0.005	1.104 ± 0.117	1.051 ± 0.059	0.023	-0.057	-71.69	0.0550	0.0794	0.945	73.171 ± 3.995	0.203 ± 0.005	4.065 ± 1.781
LSAM	beta47	0.08	0.027	-0.053 ± 0.004	-0.665 ± 0.054	0.002 ± 0.000	0.048 ± 0.003	0.005 ± 0.001	0.071 ± 0.004	0.793 ± 0.082	0.890 ± 0.050	0.022	-0.058	-72.71	0.0414	0.0714	1.116	82.927 ± 3.393	0.208 ± 0.006	4.878 ± 1.942
LSAM	beta51	0.16	0.190	0.030 ± 0.005	0.186 ± 0.034	0.004 ± 0.000	0.061 ± 0.004	0.005 ± 0.001	0.067 ± 0.005	0.177 ± 0.022	0.420 ± 0.030	0.189	0.029	17.97	0.0628	0.0691	1.117	95.122 ± 1.942	0.265 ± 0.003	83.740 ± 3.327
LSAM	beta52	0.16	0.140	-0.020 ± 0.006	-0.125 ± 0.037	0.004 ± 0.001	0.065 ± 0.004	0.005 ± 0.001	0.068 ± 0.005	0.179 ± 0.022	0.423 ± 0.031	0.142	-0.018	-11.16	0.0668	0.0691	0.907	91.870 ± 2.464	0.231 ± 0.003	63.415 ± 4.343
LSAM	beta53	0.16	0.144	-0.016 ± 0.005	-0.101 ± 0.034	0.004 ± 0.000	0.060 ± 0.004	0.004 ± 0.000	0.062 ± 0.004	0.151 ± 0.016	0.389 ± 0.027	0.137	-0.023	-14.24	0.0556	0.0601	0.988	94.309 ± 2.089	0.233 ± 0.002	69.106 ± 4.166
LSAM	beta54	0.16	0.130	-0.030 ± 0.006	-0.189 ± 0.035	0.004 ± 0.000	0.061 ± 0.004	0.005 ± 0.001	0.068 ± 0.005	0.181 ± 0.024	0.426 ± 0.032	0.137	-0.023	-14.27	0.0546	0.0592	0.972	91.057 ± 2.573	0.234 ± 0.003	65.041 ± 4.300
LSAM	beta55	0.08	0.111	0.031 ± 0.007	0.393 ± 0.093	0.007 ± 0.001	0.082 ± 0.005	0.008 ± 0.001	0.088 ± 0.006	1.207 ± 0.135	1.099 ± 0.076	0.117	0.037	46.41	0.0923	0.0995	0.968	92.683 ± 2.348	0.313 ± 0.007	27.642 ± 4.033
LSAM	beta56	0.08	0.068	-0.012 ± 0.007	-0.152 ± 0.087	0.006 ± 0.001	0.077 ± 0.005	0.006 ± 0.001	0.078 ± 0.006	0.951 ± 0.124	0.975 ± 0.078	0.076	-0.004	-5.46	0.0777	0.0778	0.966	94.309 ± 2.089	0.293 ± 0.006	13.821 ± 3.112
LSAM	beta57	0.06	0.016	-0.044 ± 0.004	-0.741 ± 0.066	0.002 ± 0.000	0.044 ± 0.003	0.004 ± 0.000	0.063 ± 0.004	1.086 ± 0.123	1.042 ± 0.064	0.014	-0.046	-76.43	0.0456	0.0647	0.980	81.301 ± 3.516	0.170 ± 0.004	4.065 ± 1.781
LSAM	beta58	0.06	0.032	-0.028 ± 0.003	-0.471 ± 0.058	0.001 ± 0.000	0.039 ± 0.003	0.002 ± 0.000	0.048 ± 0.003	0.633 ± 0.081	0.795 ± 0.057	0.030	-0.030	-49.54	0.0331	0.0445	0.979	87.805 ± 2.951	0.148 ± 0.003	12.195 ± 2.951
QML	beta41	0.20	0.186	-0.014 ± 0.004	-0.072 ± 0.021	0.002 ± 0.000	0.046 ± 0.003	0.002 ± 0.000	0.048 ± 0.003	0.059 ± 0.007	0.242 ± 0.017	0.183	-0.017	-8.40	0.0469	0.0498	1.084	95.935 ± 1.781	0.197 ± 0.002	99.187 ± 0.810
QML	beta42	0.20	0.177	-0.023 ± 0.005	-0.117 ± 0.024	0.003 ± 0.000	0.054 ± 0.004	0.003 ± 0.000	0.059 ± 0.005	0.086 ± 0.011	0.294 ± 0.023	0.177	-0.023	-11.32	0.0525	0.0571	0.962	91.057 ± 2.573	0.204 ± 0.002	93.496 ± 2.223
QML	beta43	0.20	0.175	-0.025 ± 0.004	-0.127 ± 0.021	0.002 ± 0.000	0.047 ± 0.003	0.003 ± 0.000	0.053 ± 0.003	0.071 ± 0.008	0.266 ± 0.017	0.172	-0.028	-13.79	0.0477	0.0551	1.019	92.683 ± 2.348	0.187 ± 0.002	97.561 ± 1.391
QML	beta44	0.11	0.151	0.041 ± 0.007	0.373 ± 0.061	0.005 ± 0.001	0.074 ± 0.004	0.007 ± 0.001	0.084 ± 0.006	0.587 ± 0.069	0.766 ± 0.053	0.150	0.040	36.23	0.0730	0.0832	0.947	89.431 ± 2.772	0.274 ± 0.005	61.789 ± 4.381
QML	beta45	0.11	0.117	0.007 ± 0.005	0.060 ± 0.048	0.003 ± 0.000	0.058 ± 0.003	0.003 ± 0.000	0.058 ± 0.004	0.280 ± 0.026	0.529 ± 0.035	0.117	0.007	6.64	0.0634	0.0638	1.014	99.187 ± 0.810	0.231 ± 0.003	53.659 ± 4.496
QML	beta46	0.08	0.039	-0.041 ± 0.004	-0.514 ± 0.054	0.002 ± 0.000	0.048 ± 0.003	0.004 ± 0.000	0.063 ± 0.004	0.625 ± 0.062	0.791 ± 0.045	0.042	-0.038	-47.24	0.0448	0.0586	0.922	79.675 ± 3.628	0.174 ± 0.004	12.195 ± 2.951
QML	beta47	0.08	0.042	-0.038 ± 0.003	-0.477 ± 0.042	0.001 ± 0.000	0.037 ± 0.002	0.003 ± 0.000	0.053 ± 0.003	0.445 ± 0.048	0.667 ± 0.039	0.037	-0.043	-53.24	0.0415	0.0595	1.170	84.553 ± 3.259	0.172 ± 0.004	10.569 ± 2.772
QML	beta51	0.16	0.199	0.039 ± 0.006	0.244 ± 0.035	0.004 ± 0.000	0.061 ± 0.004	0.005 ± 0.001	0.072 ± 0.005	0.205 ± 0.027	0.453 ± 0.033	0.192	0.032	20.28	0.0543	0.0633	1.165	95.122 ± 1.942	0.280 ± 0.003	86.179 ± 3.112
QML	beta52	0.16	0.134	-0.026 ± 0.005	-0.163 ± 0.033	0.003 ± 0.000	0.059 ± 0.004	0.004 ± 0.000	0.064 ± 0.004	0.161 ± 0.018	0.402 ± 0.027	0.134	-0.026	-16.39	0.0664	0.0714	0.932	89.431 ± 2.772	0.215 ± 0.003	66.667 ± 4.251
QML	beta53	0.16	0.133	-0.027 ± 0.005	-0.171 ± 0.033	0.003 ± 0.000	0.059 ± 0.004	0.004 ± 0.000	0.065 ± 0.004	0.163 ± 0.016	0.404 ± 0.025	0.126	-0.034	-21.27	0.0591	0.0682	0.948	91.870 ± 2.464	0.219 ± 0.002	65.854 ± 4.276
QML	beta54	0.16	0.141	-0.019 ± 0.004	-0.119 ± 0.027	0.002 ± 0.000	0.049 ± 0.003	0.003 ± 0.000	0.052 ± 0.004	0.105 ± 0.014	0.325 ± 0.025	0.150	-0.010	-6.05	0.0439	0.0449	1.040	92.683 ± 2.348	0.198 ± 0.002	82.114 ± 3.456
QML	beta55	0.08	0.113	0.033 ± 0.007	0.415 ± 0.088	0.006 ± 0.001	0.078 ± 0.005	0.007 ± 0.001	0.085 ± 0.006	1.126 ± 0.127	1.061 ± 0.073	0.119	0.039	48.42	0.0794	0.0884	1.043	94.309 ± 2.089	0.321 ± 0.005	30.894 ± 4.166
QML	beta56	0.08	0.065	-0.015 ± 0.007	-0.184 ± 0.083	0.005 ± 0.001	0.073 ± 0.004	0.006 ± 0.001	0.074 ± 0.005	0.867 ± 0.098	0.931 ± 0.067	0.069	-0.011	-13.75	0.0726	0.0734	0.984	93.496 ± 2.223	0.283 ± 0.005	11.382 ± 2.864
QML	beta57	0.06	0.013	-0.047 ± 0.003	-0.784 ± 0.058	0.001 ± 0.000	0.039 ± 0.003	0.004 ± 0.000	0.061 ± 0.003	1.028 ± 0.093	1.014 ± 0.050	0.012	-0.048	-80.65	0.0410	0.0635	1.143	79.675 ± 3.628	0.173 ± 0.004	1.626 ± 1.140
QML	beta58	0.06	0.037	-0.023 ± 0.003	-0.380 ± 0.054	0.001 ± 0.000	0.036 ± 0.002	0.002 ± 0.000	0.042 ± 0.003	0.499 ± 0.060	0.706 ± 0.049	0.036	-0.024	-40.43	0.0334	0.0413	1.011	85.366 ± 3.187	0.142 ± 0.003	17.073 ± 3.393
UPI	beta41	0.20	0.233	0.033 ± 0.016	0.164 ± 0.078	0.030 ± 0.005	0.172 ± 0.015	0.031 ± 0.005	0.175 ± 0.018	0.763 ± 0.136	0.873 ± 0.089	0.234	0.034	17.05	0.1722	0.1756	1.201	91.870 ± 2.464	0.811 ± 0.068	43.902 ± 4.475
UPI	beta42	0.20	0.223	0.023 ± 0.016	0.114 ± 0.079	0.030 ± 0.005	0.175 ± 0.015	0.031 ± 0.005	0.175 ± 0.017	0.769 ± 0.129	0.877 ± 0.085	0.240	0.040	19.88	0.1904	0.1945	1.193	94.309 ± 2.089	0.816 ± 0.086	32.520 ± 4.224
UPI	beta43	0.20	0.150	-0.050 ± 0.008	-0.249 ± 0.042	0.009 ± 0.001	0.093 ± 0.006	0.011 ± 0.001	0.105 ± 0.008	0.275 ± 0.035	0.524 ± 0.039	0.132	-0.068	-33.92	0.1080	0.1275	1.078	85.366 ± 3.187	0.391 ± 0.035	48.780 ± 4.507
UPI	beta44	0.11	0.209	0.099 ± 0.013	0.903 ± 0.119	0.021 ± 0.004	0.146 ± 0.014	0.031 ± 0.005	0.176 ± 0.017	2.557 ± 0.451	1.599 ± 0.153	0.189	0.079	71.77	0.1504	0.1699	1.358	98.374 ± 1.140	0.776 ± 0.064	24.390 ± 3.872
UPI	beta45	0.11	0.121	0.011 ± 0.008	0.099 ± 0.070	0.007 ± 0.001	0.086 ± 0.007	0.007 ± 0.001	0.086 ± 0.008	0.610 ± 0.098	0.781 ± 0.073	0.126	0.016	14.95	0.0847	0.0863	1.338	100.000 ± 0.000	0.449 ± 0.022	17.073 ± 3.393
UPI	beta46	0.08	-0.049	-0.129 ± 0.019	-1.613 ± 0.242	0.046 ± 0.008	0.215 ± 0.019	0.062 ± 0.011	0.250 ± 0.024	9.749 ± 1.751	3.122 ± 0.305	-0.020	-0.100	-125.35	0.1688	0.1963	0.941	87.805 ± 2.951	0.792 ± 0.059	2.439 ± 1.391

Table S164

Simulation 2 - Condition 20: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.042	-0.122 ± 0.018	-1.522 ± 0.226	0.040 ± 0.008	0.200 ± 0.021	0.055 ± 0.010	0.234 ± 0.023	8.529 ± 1.575	2.920 ± 0.293	-0.048	-0.128	-160.26	0.1850	0.2251	1.196	91.870 ± 2.464	0.939 ± 0.106	4.065 ± 1.781
UPI	beta51	0.16	0.247	0.087 ± 0.006	0.545 ± 0.038	0.005 ± 0.000	0.068 ± 0.004	0.012 ± 0.001	0.110 ± 0.006	0.474 ± 0.049	0.689 ± 0.038	0.249	0.089	55.55	0.0724	0.1146	1.149	81.301 ± 3.516	0.305 ± 0.010	92.683 ± 2.348
UPI	beta52	0.16	0.155	-0.005 ± 0.008	-0.033 ± 0.053	0.009 ± 0.001	0.094 ± 0.008	0.009 ± 0.001	0.093 ± 0.009	0.341 ± 0.055	0.584 ± 0.055	0.157	-0.003	-1.59	0.1030	0.1030	1.144	91.057 ± 2.573	0.420 ± 0.027	32.520 ± 4.224
UPI	beta53	0.16	0.114	-0.046 ± 0.007	-0.286 ± 0.045	0.006 ± 0.001	0.080 ± 0.005	0.008 ± 0.001	0.092 ± 0.006	0.330 ± 0.033	0.575 ± 0.035	0.099	-0.061	-38.09	0.0916	0.1100	1.031	83.740 ± 3.327	0.324 ± 0.033	35.772 ± 4.322
UPI	beta54	0.16	0.100	-0.060 ± 0.008	-0.373 ± 0.052	0.008 ± 0.002	0.092 ± 0.009	0.012 ± 0.002	0.110 ± 0.010	0.469 ± 0.081	0.685 ± 0.065	0.110	-0.050	-31.45	0.0990	0.1110	1.316	82.114 ± 3.456	0.476 ± 0.062	26.016 ± 3.956
UPI	beta55	0.08	0.141	0.061 ± 0.008	0.759 ± 0.094	0.007 ± 0.001	0.084 ± 0.007	0.011 ± 0.002	0.103 ± 0.009	1.665 ± 0.265	1.290 ± 0.111	0.138	0.058	72.81	0.0931	0.1098	1.202	96.748 ± 1.599	0.395 ± 0.015	30.081 ± 4.135
UPI	beta56	0.08	0.070	-0.010 ± 0.008	-0.123 ± 0.100	0.008 ± 0.001	0.089 ± 0.007	0.008 ± 0.001	0.089 ± 0.008	1.231 ± 0.184	1.110 ± 0.098	0.073	-0.007	-8.90	0.0985	0.0988	1.154	97.561 ± 1.391	0.401 ± 0.020	9.756 ± 2.675
UPI	beta57	0.06	-0.035	-0.095 ± 0.012	-1.577 ± 0.199	0.017 ± 0.003	0.132 ± 0.013	0.026 ± 0.005	0.162 ± 0.016	7.307 ± 1.329	2.703 ± 0.264	-0.017	-0.077	-128.84	0.1084	0.1331	0.973	87.805 ± 2.951	0.504 ± 0.043	5.691 ± 2.089
UPI	beta58	0.06	-0.001	-0.061 ± 0.010	-1.016 ± 0.162	0.012 ± 0.002	0.108 ± 0.010	0.015 ± 0.002	0.123 ± 0.010	4.218 ± 0.627	2.054 ± 0.170	0.001	-0.059	-98.61	0.0943	0.1113	1.303	88.618 ± 2.864	0.549 ± 0.061	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S165

Simulation 2 - Condition 21: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.114 ± 0.005	0.338 ± 0.009	0.197	-0.003	-1.55	0.0685	0.0686	0.947	94.237 ± 0.741	0.251 ± 0.001	85.844 ± 1.108
LSAM	beta42	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.106 ± 0.005	0.325 ± 0.009	0.199	-0.001	-0.33	0.0645	0.0645	1.004	96.057 ± 0.619	0.256 ± 0.001	85.743 ± 1.112
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.301 ± 0.008	0.201	0.001	0.55	0.0580	0.0580	0.980	94.641 ± 0.716	0.232 ± 0.001	93.529 ± 0.782
LSAM	beta44	0.11	0.129	0.019 ± 0.003	0.169 ± 0.023	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.558 ± 0.028	0.747 ± 0.022	0.128	0.018	16.35	0.0747	0.0769	0.960	93.630 ± 0.777	0.301 ± 0.002	41.759 ± 1.568
LSAM	beta45	0.11	0.116	0.006 ± 0.002	0.054 ± 0.020	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.395 ± 0.020	0.628 ± 0.019	0.114	0.004	3.93	0.0672	0.0674	0.948	94.135 ± 0.747	0.256 ± 0.002	42.770 ± 1.573
LSAM	beta46	0.08	0.061	-0.019 ± 0.002	-0.235 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.545 ± 0.028	0.738 ± 0.022	0.062	-0.018	-23.12	0.0523	0.0555	0.956	91.608 ± 0.882	0.210 ± 0.002	23.660 ± 1.351
LSAM	beta47	0.08	0.064	-0.016 ± 0.002	-0.196 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.527 ± 0.025	0.726 ± 0.020	0.062	-0.018	-22.05	0.0563	0.0590	0.931	91.709 ± 0.877	0.204 ± 0.002	25.581 ± 1.387
LSAM	beta51	0.16	0.168	0.008 ± 0.003	0.051 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.263 ± 0.013	0.513 ± 0.015	0.166	0.006	3.97	0.0804	0.0806	0.953	94.439 ± 0.729	0.305 ± 0.001	58.645 ± 1.566
LSAM	beta52	0.16	0.151	-0.009 ± 0.002	-0.056 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.402 ± 0.011	0.153	-0.007	-4.64	0.0648	0.0652	1.001	95.248 ± 0.677	0.250 ± 0.001	67.139 ± 1.494
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.015 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.195 ± 0.009	0.441 ± 0.012	0.161	0.001	0.91	0.0724	0.0725	0.958	94.540 ± 0.722	0.265 ± 0.001	66.127 ± 1.505
LSAM	beta54	0.16	0.154	-0.006 ± 0.002	-0.034 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.008	0.410 ± 0.012	0.155	-0.005	-3.32	0.0643	0.0645	0.971	93.832 ± 0.765	0.249 ± 0.001	69.060 ± 1.470
LSAM	beta55	0.08	0.086	0.006 ± 0.003	0.071 ± 0.034	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	1.119 ± 0.056	1.058 ± 0.031	0.086	0.006	7.62	0.0764	0.0767	0.973	94.135 ± 0.747	0.322 ± 0.002	20.728 ± 1.289
LSAM	beta56	0.08	0.082	0.002 ± 0.003	0.031 ± 0.031	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.970 ± 0.047	0.985 ± 0.029	0.080	0.000	0.10	0.0750	0.0750	0.990	94.439 ± 0.729	0.306 ± 0.002	19.211 ± 1.253
LSAM	beta57	0.06	0.048	-0.012 ± 0.002	-0.206 ± 0.029	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.901 ± 0.041	0.949 ± 0.026	0.045	-0.015	-25.15	0.0546	0.0566	0.943	91.911 ± 0.867	0.206 ± 0.002	16.380 ± 1.177
LSAM	beta58	0.06	0.047	-0.013 ± 0.002	-0.213 ± 0.025	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.682 ± 0.031	0.826 ± 0.023	0.049	-0.011	-18.90	0.0472	0.0485	0.948	92.214 ± 0.852	0.178 ± 0.001	19.211 ± 1.253
QML	beta41	0.20	0.188	-0.012 ± 0.002	-0.061 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.102 ± 0.005	0.319 ± 0.009	0.190	-0.010	-5.15	0.0615	0.0623	0.946	93.124 ± 0.805	0.232 ± 0.001	87.664 ± 1.046
QML	beta42	0.20	0.195	-0.005 ± 0.002	-0.026 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.302 ± 0.008	0.194	-0.006	-2.89	0.0578	0.0581	1.005	94.944 ± 0.697	0.237 ± 0.001	88.777 ± 1.004
QML	beta43	0.20	0.191	-0.009 ± 0.002	-0.044 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.291 ± 0.008	0.192	-0.008	-4.14	0.0534	0.0540	0.986	93.428 ± 0.788	0.222 ± 0.001	93.225 ± 0.799
QML	beta44	0.11	0.119	0.009 ± 0.002	0.078 ± 0.021	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.443 ± 0.021	0.666 ± 0.019	0.118	0.008	6.90	0.0686	0.0690	0.994	93.933 ± 0.759	0.284 ± 0.002	39.838 ± 1.557
QML	beta45	0.11	0.111	0.001 ± 0.002	0.013 ± 0.018	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.334 ± 0.016	0.578 ± 0.017	0.112	0.002	1.51	0.0609	0.0610	0.974	94.135 ± 0.747	0.243 ± 0.001	43.579 ± 1.577
QML	beta46	0.08	0.065	-0.015 ± 0.002	-0.190 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.418 ± 0.020	0.647 ± 0.018	0.064	-0.016	-20.41	0.0477	0.0504	0.994	91.911 ± 0.867	0.193 ± 0.002	28.008 ± 1.428
QML	beta47	0.08	0.066	-0.014 ± 0.002	-0.181 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.412 ± 0.018	0.642 ± 0.017	0.064	-0.016	-19.86	0.0500	0.0525	0.964	92.518 ± 0.837	0.186 ± 0.001	28.109 ± 1.429
QML	beta51	0.16	0.177	0.017 ± 0.003	0.107 ± 0.016	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.278 ± 0.013	0.527 ± 0.015	0.176	0.016	10.07	0.0780	0.0797	0.988	94.944 ± 0.697	0.320 ± 0.001	60.263 ± 1.556
QML	beta52	0.16	0.140	-0.020 ± 0.002	-0.127 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.005 ± 0.000	0.069 ± 0.002	0.185 ± 0.008	0.430 ± 0.011	0.139	-0.021	-13.25	0.0666	0.0699	1.028	93.933 ± 0.759	0.265 ± 0.001	56.522 ± 1.576
QML	beta53	0.16	0.142	-0.018 ± 0.002	-0.110 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.198 ± 0.009	0.445 ± 0.012	0.143	-0.017	-10.37	0.0685	0.0705	0.990	93.630 ± 0.777	0.268 ± 0.001	55.308 ± 1.581
QML	beta54	0.16	0.156	-0.004 ± 0.002	-0.025 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.008	0.395 ± 0.012	0.157	-0.003	-1.93	0.0594	0.0595	0.974	94.135 ± 0.747	0.241 ± 0.001	72.194 ± 1.425
QML	beta55	0.08	0.089	0.009 ± 0.003	0.110 ± 0.033	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	1.099 ± 0.051	1.048 ± 0.029	0.089	0.009	11.55	0.0787	0.0793	1.017	94.742 ± 0.710	0.333 ± 0.002	19.919 ± 1.270
QML	beta56	0.08	0.080	0.000 ± 0.002	0.002 ± 0.030	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.902 ± 0.042	0.950 ± 0.027	0.076	-0.004	-4.64	0.0721	0.0722	1.017	95.551 ± 0.656	0.303 ± 0.002	18.402 ± 1.232
QML	beta57	0.06	0.042	-0.018 ± 0.002	-0.302 ± 0.028	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.846 ± 0.036	0.920 ± 0.023	0.040	-0.020	-33.54	0.0520	0.0557	0.977	91.102 ± 0.905	0.200 ± 0.002	14.156 ± 1.108
QML	beta58	0.06	0.045	-0.015 ± 0.001	-0.252 ± 0.023	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.606 ± 0.027	0.778 ± 0.021	0.044	-0.016	-25.86	0.0441	0.0467	0.966	91.608 ± 0.882	0.167 ± 0.001	19.717 ± 1.265
UPI	beta41	0.20	0.209	0.009 ± 0.002	0.043 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.117 ± 0.005	0.342 ± 0.010	0.211	0.011	5.37	0.0654	0.0663	0.951	94.034 ± 0.753	0.253 ± 0.001	87.968 ± 1.035
UPI	beta42	0.20	0.209	0.009 ± 0.002	0.046 ± 0.011	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.005	0.335 ± 0.009	0.208	0.008	4.19	0.0641	0.0646	1.009	95.450 ± 0.663	0.263 ± 0.002	87.765 ± 1.042
UPI	beta43	0.20	0.203	0.003 ± 0.002	0.014 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.004	0.307 ± 0.009	0.199	-0.001	-0.28	0.0598	0.0598	0.988	94.439 ± 0.729	0.237 ± 0.002	93.124 ± 0.805
UPI	beta44	0.11	0.126	0.016 ± 0.003	0.146 ± 0.024	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.569 ± 0.031	0.754 ± 0.024	0.124	0.014	12.78	0.0752	0.0765	1.014	95.450 ± 0.663	0.324 ± 0.003	34.277 ± 1.509
UPI	beta45	0.11	0.116	0.006 ± 0.002	0.054 ± 0.020	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.413 ± 0.021	0.643 ± 0.019	0.113	0.003	2.43	0.0666	0.0667	0.991	94.843 ± 0.703	0.274 ± 0.002	38.423 ± 1.547
UPI	beta46	0.08	0.072	-0.008 ± 0.002	-0.099 ± 0.024	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.558 ± 0.029	0.747 ± 0.023	0.073	-0.007	-8.84	0.0554	0.0558	1.044	95.046 ± 0.690	0.243 ± 0.003	22.447 ± 1.327

Table S165

Simulation 2 - Condition 21: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.072	-0.008 ± 0.002	-0.096 ± 0.024	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.565 ± 0.030	0.752 ± 0.023	0.070	-0.010	-11.94	0.0556	0.0565	1.015	94.439 ± 0.729	0.237 ± 0.003	24.065 ± 1.359
UPI	beta51	0.16	0.169	0.009 ± 0.003	0.056 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.257 ± 0.012	0.507 ± 0.014	0.168	0.008	4.75	0.0789	0.0792	0.983	95.551 ± 0.656	0.311 ± 0.003	57.735 ± 1.571
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.007 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.011	0.161	0.001	0.72	0.0642	0.0642	1.012	95.147 ± 0.683	0.256 ± 0.001	71.587 ± 1.434
UPI	beta53	0.16	0.164	0.004 ± 0.002	0.025 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.193 ± 0.009	0.440 ± 0.012	0.163	0.003	1.96	0.0713	0.0714	0.990	94.338 ± 0.735	0.273 ± 0.004	66.330 ± 1.503
UPI	beta54	0.16	0.160	0.000 ± 0.002	0.003 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.412 ± 0.012	0.161	0.001	0.58	0.0620	0.0620	1.057	94.540 ± 0.722	0.273 ± 0.014	69.666 ± 1.462
UPI	beta55	0.08	0.080	0.000 ± 0.003	0.003 ± 0.033	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	1.091 ± 0.054	1.045 ± 0.031	0.079	-0.001	-0.71	0.0761	0.0761	1.024	95.248 ± 0.677	0.336 ± 0.004	15.268 ± 1.144
UPI	beta56	0.08	0.084	0.004 ± 0.003	0.046 ± 0.032	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	1.007 ± 0.052	1.003 ± 0.030	0.079	-0.001	-1.09	0.0775	0.0775	1.028	96.057 ± 0.619	0.323 ± 0.006	17.088 ± 1.197
UPI	beta57	0.06	0.058	-0.002 ± 0.002	-0.039 ± 0.032	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.991 ± 0.049	0.995 ± 0.029	0.055	-0.005	-8.28	0.0562	0.0564	0.997	94.641 ± 0.716	0.233 ± 0.003	15.976 ± 1.165
UPI	beta58	0.06	0.053	-0.007 ± 0.002	-0.120 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.853 ± 0.045	0.924 ± 0.028	0.052	-0.008	-13.60	0.0534	0.0540	1.157	94.237 ± 0.741	0.249 ± 0.027	16.785 ± 1.188

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S166

Simulation 2 - Condition 22: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.195	-0.005	-2.51	0.0403	0.0406	0.954	93.888 ± 0.758	0.154 ± 0.000	99.800 ± 0.142
LSAM	beta42	0.20	0.197	-0.003 ± 0.001	-0.014 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.195	-0.005	-2.50	0.0413	0.0416	1.000	94.188 ± 0.741	0.158 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.180 ± 0.005	0.200	0.000	0.15	0.0368	0.0368	1.026	95.992 ± 0.621	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.129	0.019 ± 0.001	0.169 ± 0.013	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.203 ± 0.009	0.450 ± 0.011	0.127	0.017	15.87	0.0472	0.0503	0.998	93.788 ± 0.764	0.180 ± 0.001	80.461 ± 1.255
LSAM	beta45	0.11	0.114	0.004 ± 0.001	0.034 ± 0.012	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.140 ± 0.006	0.374 ± 0.010	0.114	0.004	3.70	0.0424	0.0426	0.968	94.790 ± 0.703	0.155 ± 0.001	80.661 ± 1.250
LSAM	beta46	0.08	0.062	-0.018 ± 0.001	-0.226 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.204 ± 0.010	0.451 ± 0.012	0.064	-0.016	-20.02	0.0304	0.0344	1.006	90.180 ± 0.942	0.123 ± 0.001	54.609 ± 1.576
LSAM	beta47	0.08	0.062	-0.018 ± 0.001	-0.222 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.202 ± 0.008	0.449 ± 0.011	0.062	-0.018	-22.32	0.0307	0.0355	1.002	89.880 ± 0.955	0.123 ± 0.001	53.307 ± 1.579
LSAM	beta51	0.16	0.172	0.012 ± 0.001	0.078 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.171	0.011	7.10	0.0438	0.0453	1.060	95.591 ± 0.650	0.190 ± 0.000	95.190 ± 0.677
LSAM	beta52	0.16	0.153	-0.007 ± 0.001	-0.045 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.153	-0.007	-4.41	0.0390	0.0397	1.005	94.890 ± 0.697	0.154 ± 0.000	97.595 ± 0.485
LSAM	beta53	0.16	0.156	-0.004 ± 0.001	-0.026 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.155	-0.005	-3.02	0.0406	0.0409	1.022	95.291 ± 0.671	0.164 ± 0.000	96.593 ± 0.574
LSAM	beta54	0.16	0.152	-0.008 ± 0.001	-0.047 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.154	-0.006	-4.06	0.0419	0.0424	1.004	93.888 ± 0.758	0.155 ± 0.000	96.293 ± 0.598
LSAM	beta55	0.08	0.088	0.008 ± 0.002	0.102 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.386 ± 0.019	0.621 ± 0.018	0.087	0.007	9.35	0.0457	0.0463	0.996	95.190 ± 0.677	0.191 ± 0.001	45.190 ± 1.575
LSAM	beta56	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.330 ± 0.016	0.575 ± 0.016	0.078	-0.002	-2.31	0.0441	0.0442	1.010	94.589 ± 0.716	0.182 ± 0.001	40.882 ± 1.556
LSAM	beta57	0.06	0.045	-0.015 ± 0.001	-0.253 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.348 ± 0.015	0.590 ± 0.015	0.044	-0.016	-27.28	0.0310	0.0351	0.974	90.381 ± 0.933	0.122 ± 0.001	30.261 ± 1.454
LSAM	beta58	0.06	0.051	-0.009 ± 0.001	-0.158 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.259 ± 0.012	0.509 ± 0.014	0.051	-0.009	-15.59	0.0274	0.0290	0.965	91.182 ± 0.898	0.110 ± 0.001	46.393 ± 1.579
QML	beta41	0.20	0.188	-0.012 ± 0.001	-0.059 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.189	-0.011	-5.57	0.0381	0.0397	0.948	91.784 ± 0.869	0.144 ± 0.000	99.800 ± 0.142
QML	beta42	0.20	0.193	-0.007 ± 0.001	-0.034 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.194 ± 0.005	0.193	-0.007	-3.67	0.0397	0.0404	0.989	94.188 ± 0.741	0.148 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.191	-0.009 ± 0.001	-0.046 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.190	-0.010	-4.88	0.0349	0.0363	1.021	94.389 ± 0.728	0.140 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.120	0.010 ± 0.001	0.095 ± 0.013	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.173 ± 0.007	0.416 ± 0.011	0.120	0.010	9.20	0.0467	0.0478	0.995	94.389 ± 0.728	0.174 ± 0.001	76.253 ± 1.347
QML	beta45	0.11	0.112	0.002 ± 0.001	0.015 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.133 ± 0.006	0.364 ± 0.010	0.112	0.002	2.08	0.0400	0.0400	0.958	94.088 ± 0.747	0.151 ± 0.001	81.062 ± 1.240
QML	beta46	0.08	0.065	-0.015 ± 0.001	-0.183 ± 0.011	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.162 ± 0.008	0.402 ± 0.011	0.066	-0.014	-17.56	0.0269	0.0304	1.035	92.385 ± 0.840	0.116 ± 0.001	62.725 ± 1.531
QML	beta47	0.08	0.064	-0.016 ± 0.001	-0.200 ± 0.011	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.172 ± 0.007	0.414 ± 0.010	0.064	-0.016	-20.20	0.0278	0.0322	1.010	90.180 ± 0.942	0.115 ± 0.001	60.922 ± 1.545
QML	beta51	0.16	0.181	0.021 ± 0.001	0.130 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.320 ± 0.008	0.180	0.020	12.63	0.0433	0.0478	1.068	94.088 ± 0.747	0.196 ± 0.000	95.892 ± 0.628
QML	beta52	0.16	0.141	-0.019 ± 0.001	-0.120 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.081 ± 0.004	0.284 ± 0.007	0.140	-0.020	-12.32	0.0404	0.0449	1.002	91.283 ± 0.893	0.162 ± 0.000	92.585 ± 0.829
QML	beta53	0.16	0.136	-0.024 ± 0.001	-0.148 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.007	0.134	-0.026	-16.15	0.0411	0.0486	1.019	92.084 ± 0.855	0.165 ± 0.000	90.481 ± 0.929
QML	beta54	0.16	0.155	-0.005 ± 0.001	-0.030 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.003	0.241 ± 0.006	0.155	-0.005	-2.96	0.0395	0.0398	1.009	94.689 ± 0.710	0.151 ± 0.000	97.996 ± 0.444
QML	beta55	0.08	0.092	0.012 ± 0.002	0.147 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.430 ± 0.020	0.656 ± 0.018	0.091	0.011	13.74	0.0481	0.0493	1.005	93.888 ± 0.758	0.201 ± 0.001	44.389 ± 1.573
QML	beta56	0.08	0.078	-0.002 ± 0.001	-0.031 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.330 ± 0.015	0.575 ± 0.016	0.076	-0.004	-5.29	0.0435	0.0437	1.025	95.090 ± 0.684	0.184 ± 0.001	37.575 ± 1.533
QML	beta57	0.06	0.041	-0.019 ± 0.001	-0.319 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.371 ± 0.015	0.609 ± 0.014	0.040	-0.020	-33.63	0.0309	0.0369	0.993	89.178 ± 0.983	0.121 ± 0.001	25.351 ± 1.377
QML	beta58	0.06	0.048	-0.012 ± 0.001	-0.196 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.256 ± 0.012	0.506 ± 0.014	0.048	-0.012	-19.28	0.0265	0.0289	0.962	89.880 ± 0.955	0.106 ± 0.000	44.990 ± 1.575
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.011 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.203	0.003	1.72	0.0394	0.0396	0.939	94.289 ± 0.735	0.152 ± 0.000	99.800 ± 0.142
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.012 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.005	0.202	0.002	1.10	0.0402	0.0403	0.998	94.689 ± 0.710	0.157 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.011 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.180 ± 0.005	0.202	0.002	1.18	0.0364	0.0364	1.028	96.192 ± 0.606	0.145 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.119	0.009 ± 0.001	0.083 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.190 ± 0.009	0.436 ± 0.012	0.118	0.008	7.49	0.0470	0.0477	0.995	95.291 ± 0.671	0.184 ± 0.001	73.146 ± 1.403
UPI	beta45	0.11	0.113	0.003 ± 0.001	0.030 ± 0.012	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.143 ± 0.007	0.378 ± 0.011	0.113	0.003	2.43	0.0408	0.0409	0.978	94.389 ± 0.728	0.159 ± 0.001	80.762 ± 1.248
UPI	beta46	0.08	0.074	-0.006 ± 0.001	-0.072 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.193 ± 0.010	0.440 ± 0.013	0.074	-0.006	-7.30	0.0347	0.0352	0.992	94.188 ± 0.741	0.135 ± 0.001	59.319 ± 1.555

Table S166

Simulation 2 - Condition 22: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.072	-0.008 ± 0.001	-0.103 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.192 ± 0.009	0.438 ± 0.012	0.070	-0.010	-12.23	0.0323	0.0338	1.008	94.188 ± 0.741	0.135 ± 0.001	57.615 ± 1.564
UPI	beta51	0.16	0.169	0.009 ± 0.001	0.057 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.168	0.008	5.19	0.0443	0.0451	1.068	95.892 ± 0.628	0.190 ± 0.000	95.190 ± 0.677
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.159	-0.001	-0.74	0.0389	0.0389	1.002	94.990 ± 0.691	0.155 ± 0.000	98.196 ± 0.421
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.256 ± 0.007	0.156	-0.004	-2.26	0.0406	0.0408	1.023	96.092 ± 0.613	0.164 ± 0.000	96.994 ± 0.541
UPI	beta54	0.16	0.158	-0.002 ± 0.001	-0.014 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.156	-0.004	-2.24	0.0410	0.0412	1.005	95.190 ± 0.677	0.156 ± 0.000	96.994 ± 0.541
UPI	beta55	0.08	0.084	0.004 ± 0.002	0.051 ± 0.020	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.387 ± 0.019	0.622 ± 0.018	0.084	0.004	4.99	0.0470	0.0472	0.994	94.890 ± 0.697	0.194 ± 0.001	41.283 ± 1.558
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.328 ± 0.016	0.573 ± 0.017	0.078	-0.002	-2.51	0.0427	0.0428	1.022	95.190 ± 0.677	0.184 ± 0.001	38.978 ± 1.544
UPI	beta57	0.06	0.054	-0.006 ± 0.001	-0.104 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.372 ± 0.019	0.610 ± 0.018	0.051	-0.009	-15.52	0.0339	0.0352	0.944	93.287 ± 0.792	0.133 ± 0.001	34.269 ± 1.502
UPI	beta58	0.06	0.056	-0.004 ± 0.001	-0.067 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.291 ± 0.014	0.539 ± 0.016	0.055	-0.005	-8.81	0.0308	0.0313	0.971	93.287 ± 0.792	0.122 ± 0.001	44.990 ± 1.575

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S167

Simulation 2 - Condition 23: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.064 ± 0.003	0.252 ± 0.007	0.200	0.000	0.09	0.0500	0.0500	0.968	94.400 ± 0.727	0.192 ± 0.000	98.100 ± 0.432
LSAM	beta42	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.197	-0.003	-1.61	0.0491	0.0492	0.993	94.800 ± 0.702	0.194 ± 0.001	98.100 ± 0.432
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.200	0.000	0.19	0.0484	0.0484	0.969	94.900 ± 0.696	0.182 ± 0.000	99.500 ± 0.223
LSAM	beta44	0.11	0.114	0.004 ± 0.002	0.037 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.251 ± 0.011	0.501 ± 0.014	0.116	0.006	5.31	0.0542	0.0545	0.944	92.800 ± 0.817	0.204 ± 0.001	60.200 ± 1.548
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.008 ± 0.014	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.206 ± 0.010	0.454 ± 0.013	0.112	0.002	1.45	0.0468	0.0468	0.933	93.200 ± 0.796	0.183 ± 0.001	65.600 ± 1.502
LSAM	beta46	0.08	0.078	-0.002 ± 0.001	-0.023 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.252 ± 0.011	0.502 ± 0.014	0.078	-0.002	-2.72	0.0391	0.0391	0.917	92.500 ± 0.833	0.144 ± 0.001	57.600 ± 1.563
LSAM	beta47	0.08	0.077	-0.003 ± 0.001	-0.040 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.221 ± 0.011	0.470 ± 0.014	0.076	-0.004	-4.70	0.0373	0.0375	0.968	93.000 ± 0.807	0.142 ± 0.001	57.100 ± 1.565
LSAM	beta51	0.16	0.163	0.003 ± 0.002	0.016 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.384 ± 0.011	0.165	0.005	3.03	0.0615	0.0617	1.008	95.200 ± 0.676	0.243 ± 0.001	74.600 ± 1.377
LSAM	beta52	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.004	0.317 ± 0.009	0.160	0.000	0.12	0.0502	0.0502	1.014	94.700 ± 0.708	0.202 ± 0.001	87.900 ± 1.031
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.161	0.001	0.79	0.0547	0.0547	1.016	95.800 ± 0.634	0.210 ± 0.001	85.500 ± 1.113
LSAM	beta54	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.323 ± 0.009	0.157	-0.003	-2.10	0.0526	0.0527	0.984	94.800 ± 0.702	0.200 ± 0.000	86.700 ± 1.074
LSAM	beta55	0.08	0.083	0.003 ± 0.002	0.035 ± 0.025	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.604 ± 0.027	0.777 ± 0.021	0.082	0.002	2.99	0.0623	0.0623	0.956	94.000 ± 0.751	0.233 ± 0.001	31.700 ± 1.471
LSAM	beta56	0.08	0.083	0.003 ± 0.002	0.039 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.513 ± 0.026	0.716 ± 0.021	0.082	0.002	2.70	0.0545	0.0545	0.990	94.700 ± 0.708	0.222 ± 0.001	32.700 ± 1.483
LSAM	beta57	0.06	0.059	-0.001 ± 0.001	-0.014 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.445 ± 0.019	0.667 ± 0.018	0.059	-0.001	-1.34	0.0406	0.0406	0.964	94.200 ± 0.739	0.151 ± 0.001	35.100 ± 1.509
LSAM	beta58	0.06	0.057	-0.003 ± 0.001	-0.051 ± 0.019	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.375 ± 0.018	0.612 ± 0.017	0.056	-0.004	-6.27	0.0367	0.0369	0.954	93.400 ± 0.785	0.137 ± 0.001	38.000 ± 1.535
QML	beta41	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.198	-0.002	-1.23	0.0485	0.0485	0.974	94.500 ± 0.721	0.189 ± 0.001	97.800 ± 0.464
QML	beta42	0.20	0.197	-0.003 ± 0.002	-0.017 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.244 ± 0.007	0.195	-0.005	-2.36	0.0482	0.0484	1.000	94.900 ± 0.696	0.191 ± 0.001	97.700 ± 0.474
QML	beta43	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.195	-0.005	-2.72	0.0482	0.0485	0.973	94.400 ± 0.727	0.180 ± 0.000	99.300 ± 0.264
QML	beta44	0.11	0.111	0.001 ± 0.002	0.012 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.241 ± 0.011	0.491 ± 0.013	0.113	0.003	3.02	0.0540	0.0541	0.953	93.400 ± 0.785	0.202 ± 0.001	59.300 ± 1.554
QML	beta45	0.11	0.110	-0.000 ± 0.002	-0.002 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.201 ± 0.010	0.448 ± 0.013	0.111	0.001	0.97	0.0466	0.0466	0.936	92.700 ± 0.823	0.181 ± 0.001	66.100 ± 1.497
QML	beta46	0.08	0.078	-0.002 ± 0.001	-0.028 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.236 ± 0.010	0.486 ± 0.013	0.077	-0.003	-3.40	0.0395	0.0396	0.924	92.800 ± 0.817	0.141 ± 0.001	59.700 ± 1.551
QML	beta47	0.08	0.076	-0.004 ± 0.001	-0.051 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.206 ± 0.010	0.454 ± 0.013	0.075	-0.005	-5.82	0.0353	0.0356	0.975	92.600 ± 0.828	0.138 ± 0.001	58.900 ± 1.556
QML	beta51	0.16	0.168	0.008 ± 0.002	0.049 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.170	0.010	6.46	0.0608	0.0617	1.028	95.400 ± 0.662	0.249 ± 0.001	75.100 ± 1.367
QML	beta52	0.16	0.142	-0.018 ± 0.002	-0.113 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.127 ± 0.006	0.357 ± 0.009	0.141	-0.019	-11.84	0.0551	0.0583	1.016	94.500 ± 0.721	0.216 ± 0.001	72.100 ± 1.418
QML	beta53	0.16	0.140	-0.020 ± 0.002	-0.123 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.006	0.363 ± 0.010	0.142	-0.018	-11.35	0.0576	0.0604	1.020	94.000 ± 0.751	0.219 ± 0.001	70.600 ± 1.441
QML	beta54	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.156	-0.004	-2.52	0.0519	0.0521	0.989	95.200 ± 0.676	0.199 ± 0.001	87.300 ± 1.053
QML	beta55	0.08	0.086	0.006 ± 0.002	0.073 ± 0.025	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.623 ± 0.028	0.789 ± 0.021	0.085	0.005	6.60	0.0626	0.0628	0.966	94.500 ± 0.721	0.238 ± 0.001	32.000 ± 1.475
QML	beta56	0.08	0.083	0.003 ± 0.002	0.039 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.510 ± 0.025	0.714 ± 0.021	0.084	0.004	4.73	0.0547	0.0549	1.004	95.400 ± 0.662	0.225 ± 0.001	31.900 ± 1.474
QML	beta57	0.06	0.055	-0.005 ± 0.001	-0.076 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.426 ± 0.018	0.653 ± 0.017	0.055	-0.005	-7.94	0.0395	0.0398	0.988	94.100 ± 0.745	0.151 ± 0.002	32.100 ± 1.476
QML	beta58	0.06	0.055	-0.005 ± 0.001	-0.088 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.361 ± 0.017	0.601 ± 0.017	0.054	-0.006	-9.76	0.0358	0.0363	0.958	93.700 ± 0.768	0.134 ± 0.001	36.800 ± 1.525
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.028 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.207	0.007	3.34	0.0485	0.0489	0.923	92.800 ± 0.817	0.183 ± 0.001	98.900 ± 0.330
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.016 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.200	0.000	0.23	0.0483	0.0483	0.968	94.000 ± 0.751	0.188 ± 0.001	98.600 ± 0.372
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.240 ± 0.006	0.201	0.001	0.59	0.0487	0.0487	0.960	94.600 ± 0.715	0.181 ± 0.000	99.700 ± 0.173
UPI	beta44	0.11	0.114	0.004 ± 0.002	0.035 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.256 ± 0.011	0.506 ± 0.014	0.116	0.006	5.56	0.0560	0.0563	0.946	93.700 ± 0.768	0.206 ± 0.001	59.500 ± 1.552
UPI	beta45	0.11	0.111	0.001 ± 0.002	0.008 ± 0.015	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.211 ± 0.010	0.459 ± 0.013	0.112	0.002	1.90	0.0468	0.0469	0.936	93.100 ± 0.801	0.185 ± 0.001	64.700 ± 1.511
UPI	beta46	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.262 ± 0.012	0.512 ± 0.014	0.079	-0.001	-1.48	0.0404	0.0404	0.914	92.700 ± 0.823	0.147 ± 0.001	56.600 ± 1.567

Table S167

Simulation 2 - Condition 23: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.225 ± 0.011	0.474 ± 0.014	0.079	-0.001	-1.47	0.0354	0.0354	0.977	94.000 ± 0.751	0.145 ± 0.001	57.100 ± 1.565
UPI	beta51	0.16	0.164	0.004 ± 0.002	0.026 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.383 ± 0.011	0.167	0.007	4.49	0.0601	0.0605	1.012	94.900 ± 0.696	0.242 ± 0.001	75.500 ± 1.360
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.021 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.004	0.317 ± 0.009	0.164	0.004	2.68	0.0504	0.0505	1.000	94.900 ± 0.696	0.199 ± 0.001	90.100 ± 0.944
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.329 ± 0.009	0.163	0.003	1.79	0.0537	0.0538	1.018	95.800 ± 0.634	0.210 ± 0.001	86.300 ± 1.087
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.158	-0.002	-0.96	0.0516	0.0516	0.977	94.700 ± 0.708	0.198 ± 0.001	88.200 ± 1.020
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.015 ± 0.024	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.593 ± 0.026	0.770 ± 0.021	0.081	0.001	0.67	0.0588	0.0588	0.964	93.700 ± 0.768	0.233 ± 0.001	30.700 ± 1.459
UPI	beta56	0.08	0.083	0.003 ± 0.002	0.042 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.512 ± 0.025	0.716 ± 0.021	0.082	0.002	2.22	0.0545	0.0546	0.998	94.600 ± 0.715	0.224 ± 0.001	32.200 ± 1.478
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.007 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.451 ± 0.021	0.672 ± 0.019	0.060	0.000	0.70	0.0410	0.0410	0.972	94.400 ± 0.727	0.154 ± 0.001	35.100 ± 1.509
UPI	beta58	0.06	0.059	-0.001 ± 0.001	-0.020 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.392 ± 0.018	0.626 ± 0.018	0.058	-0.002	-3.55	0.0383	0.0383	0.953	94.100 ± 0.745	0.140 ± 0.001	37.700 ± 1.533

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S168

Simulation 2 - Condition 24: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.197	-0.003	-1.32	0.0319	0.0320	1.015	95.700 ± 0.641	0.121 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.199	-0.001	-0.66	0.0329	0.0329	0.993	95.900 ± 0.627	0.122 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	-0.22	0.0280	0.0280	1.000	94.900 ± 0.696	0.115 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.020 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.097 ± 0.004	0.311 ± 0.008	0.113	0.003	2.88	0.0344	0.0345	0.964	94.000 ± 0.751	0.129 ± 0.000	90.600 ± 0.923
LSAM	beta45	0.11	0.112	0.002 ± 0.001	0.017 ± 0.008	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.072 ± 0.003	0.267 ± 0.008	0.113	0.003	2.67	0.0279	0.0281	0.995	95.500 ± 0.656	0.115 ± 0.000	97.100 ± 0.531
LSAM	beta46	0.08	0.076	-0.004 ± 0.001	-0.053 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.087 ± 0.004	0.296 ± 0.008	0.076	-0.004	-5.08	0.0235	0.0239	0.991	94.000 ± 0.751	0.090 ± 0.000	90.100 ± 0.944
LSAM	beta47	0.08	0.076	-0.004 ± 0.001	-0.049 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.076	-0.004	-5.45	0.0239	0.0243	0.978	94.300 ± 0.733	0.090 ± 0.000	91.100 ± 0.900
LSAM	beta51	0.16	0.165	0.005 ± 0.001	0.032 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.163	0.003	1.84	0.0409	0.0410	0.949	93.300 ± 0.791	0.153 ± 0.000	98.400 ± 0.397
LSAM	beta52	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.157	-0.003	-1.87	0.0332	0.0333	0.967	94.300 ± 0.733	0.126 ± 0.000	100.000 ± 0.000
LSAM	beta53	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.158	-0.002	-1.56	0.0347	0.0348	0.970	94.400 ± 0.727	0.133 ± 0.000	99.400 ± 0.244
LSAM	beta54	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.157	-0.003	-2.08	0.0324	0.0326	0.988	95.300 ± 0.669	0.126 ± 0.000	100.000 ± 0.000
LSAM	beta55	0.08	0.084	0.004 ± 0.001	0.049 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.244 ± 0.011	0.494 ± 0.014	0.084	0.004	4.67	0.0389	0.0391	0.936	92.500 ± 0.833	0.144 ± 0.000	61.100 ± 1.542
LSAM	beta56	0.08	0.080	0.000 ± 0.001	0.004 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.205 ± 0.009	0.453 ± 0.013	0.079	-0.001	-1.00	0.0358	0.0358	0.977	93.800 ± 0.763	0.139 ± 0.000	61.900 ± 1.536
LSAM	beta57	0.06	0.055	-0.005 ± 0.001	-0.077 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.176 ± 0.008	0.420 ± 0.011	0.056	-0.004	-6.45	0.0260	0.0263	0.971	93.600 ± 0.774	0.094 ± 0.000	62.900 ± 1.528
LSAM	beta58	0.06	0.056	-0.004 ± 0.001	-0.071 ± 0.011	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.134 ± 0.006	0.366 ± 0.010	0.055	-0.005	-7.80	0.0209	0.0215	1.013	95.400 ± 0.662	0.086 ± 0.000	73.000 ± 1.404
QML	beta41	0.20	0.194	-0.006 ± 0.001	-0.030 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.194	-0.006	-2.84	0.0314	0.0319	1.011	95.500 ± 0.656	0.118 ± 0.000	100.000 ± 0.000
QML	beta42	0.20	0.197	-0.003 ± 0.001	-0.017 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.197	-0.003	-1.42	0.0329	0.0330	0.990	95.100 ± 0.683	0.120 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.022 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.195	-0.005	-2.50	0.0276	0.0281	1.000	94.500 ± 0.721	0.113 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.110	-0.000 ± 0.001	-0.003 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.095 ± 0.004	0.309 ± 0.008	0.110	0.000	0.33	0.0343	0.0343	0.964	94.300 ± 0.733	0.128 ± 0.000	89.500 ± 0.969
QML	beta45	0.11	0.111	0.001 ± 0.001	0.012 ± 0.008	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.070 ± 0.003	0.264 ± 0.008	0.112	0.002	1.98	0.0282	0.0282	1.001	95.800 ± 0.634	0.114 ± 0.000	97.000 ± 0.539
QML	beta46	0.08	0.076	-0.004 ± 0.001	-0.053 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.076	-0.004	-4.53	0.0238	0.0240	0.993	94.000 ± 0.751	0.089 ± 0.000	91.700 ± 0.872
QML	beta47	0.08	0.076	-0.004 ± 0.001	-0.056 ± 0.009	0.001 ± 0.000	0.023 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.075	-0.005	-6.33	0.0232	0.0238	0.981	94.200 ± 0.739	0.088 ± 0.000	91.600 ± 0.877
QML	beta51	0.16	0.170	0.010 ± 0.001	0.061 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.268 ± 0.007	0.168	0.008	5.00	0.0416	0.0424	0.950	92.900 ± 0.812	0.155 ± 0.000	98.500 ± 0.384
QML	beta52	0.16	0.141	-0.019 ± 0.001	-0.120 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.252 ± 0.006	0.140	-0.020	-12.59	0.0342	0.0397	0.966	90.600 ± 0.923	0.134 ± 0.000	97.500 ± 0.494
QML	beta53	0.16	0.137	-0.023 ± 0.001	-0.141 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.137	-0.023	-14.33	0.0353	0.0421	0.971	89.100 ± 0.985	0.138 ± 0.000	96.900 ± 0.548
QML	beta54	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.156	-0.004	-2.19	0.0322	0.0324	0.987	95.400 ± 0.662	0.125 ± 0.000	100.000 ± 0.000
QML	beta55	0.08	0.087	0.007 ± 0.001	0.087 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.259 ± 0.012	0.509 ± 0.014	0.086	0.006	7.97	0.0406	0.0411	0.941	92.500 ± 0.833	0.148 ± 0.000	62.500 ± 1.531
QML	beta56	0.08	0.081	0.001 ± 0.001	0.009 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.212 ± 0.010	0.460 ± 0.013	0.080	0.000	0.48	0.0363	0.0363	0.977	93.900 ± 0.757	0.141 ± 0.000	60.800 ± 1.544
QML	beta57	0.06	0.052	-0.008 ± 0.001	-0.126 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.183 ± 0.008	0.427 ± 0.011	0.053	-0.007	-11.07	0.0249	0.0258	0.972	93.400 ± 0.785	0.093 ± 0.000	59.100 ± 1.555
QML	beta58	0.06	0.054	-0.006 ± 0.001	-0.099 ± 0.011	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.134 ± 0.006	0.366 ± 0.010	0.054	-0.006	-10.63	0.0209	0.0219	1.015	94.600 ± 0.715	0.084 ± 0.000	71.200 ± 1.432
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.201	0.001	0.47	0.0320	0.0321	0.967	94.500 ± 0.721	0.115 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.201	0.001	0.63	0.0328	0.0328	0.962	95.400 ± 0.662	0.119 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	0.25	0.0282	0.0282	0.991	94.500 ± 0.721	0.114 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.110	0.000 ± 0.001	0.004 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.098 ± 0.004	0.312 ± 0.008	0.111	0.001	0.66	0.0347	0.0347	0.958	93.100 ± 0.801	0.129 ± 0.000	90.300 ± 0.936
UPI	beta45	0.11	0.112	0.002 ± 0.001	0.018 ± 0.008	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.072 ± 0.004	0.268 ± 0.008	0.112	0.002	1.87	0.0291	0.0292	0.999	95.400 ± 0.662	0.115 ± 0.000	96.600 ± 0.573
UPI	beta46	0.08	0.078	-0.002 ± 0.001	-0.023 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.086 ± 0.004	0.294 ± 0.008	0.078	-0.002	-2.46	0.0232	0.0233	0.995	94.000 ± 0.751	0.091 ± 0.000	90.900 ± 0.909

Table S168

Simulation 2 - Condition 24: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.078	-0.002 ± 0.001	-0.021 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.078	-0.002	-3.00	0.0241	0.0242	0.965	94.500 ± 0.721	0.091 ± 0.000	91.500 ± 0.882
UPI	beta51	0.16	0.165	0.005 ± 0.001	0.030 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.163	0.003	1.98	0.0413	0.0414	0.944	93.100 ± 0.801	0.152 ± 0.000	98.400 ± 0.397
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.002 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.160	0.000	-0.28	0.0327	0.0327	0.956	93.500 ± 0.780	0.125 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.012 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.218 ± 0.006	0.158	-0.002	-1.28	0.0344	0.0345	0.967	94.600 ± 0.715	0.132 ± 0.000	99.500 ± 0.223
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.158	-0.002	-1.09	0.0318	0.0319	0.984	95.100 ± 0.683	0.125 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.083	0.003 ± 0.001	0.032 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.241 ± 0.011	0.491 ± 0.014	0.082	0.002	2.57	0.0380	0.0380	0.937	92.500 ± 0.833	0.144 ± 0.000	59.800 ± 1.550
UPI	beta56	0.08	0.080	0.000 ± 0.001	0.003 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.203 ± 0.009	0.451 ± 0.012	0.079	-0.001	-1.29	0.0363	0.0363	0.980	94.300 ± 0.733	0.139 ± 0.000	63.200 ± 1.525
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.040 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.174 ± 0.008	0.417 ± 0.011	0.058	-0.002	-3.67	0.0248	0.0249	0.976	94.500 ± 0.721	0.095 ± 0.000	64.600 ± 1.512
UPI	beta58	0.06	0.057	-0.003 ± 0.001	-0.045 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.135 ± 0.006	0.368 ± 0.010	0.057	-0.003	-4.49	0.0219	0.0221	1.009	95.600 ± 0.649	0.087 ± 0.000	73.000 ± 1.404

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S169

Simulation 2 - Condition 25: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.196	-0.004 ± 0.010	-0.021 ± 0.052	0.066 ± 0.005	0.257 ± 0.009	0.066 ± 0.005	0.256 ± 0.010	1.644 ± 0.115	1.282 ± 0.052	0.195	-0.005	-2.39	0.2342	0.2343	1.075	97.886 ± 0.580	1.082 ± 0.026	12.846 ± 1.349
LSAM	beta42	0.20	0.202	0.002 ± 0.010	0.008 ± 0.052	0.066 ± 0.004	0.257 ± 0.009	0.066 ± 0.004	0.256 ± 0.010	1.643 ± 0.109	1.282 ± 0.050	0.199	-0.001	-0.71	0.2361	0.2361	1.068	97.398 ± 0.642	1.074 ± 0.022	12.520 ± 1.335
LSAM	beta43	0.20	0.195	-0.005 ± 0.004	-0.025 ± 0.018	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	0.201 ± 0.012	0.449 ± 0.017	0.195	-0.005	-2.49	0.0878	0.0879	1.007	95.447 ± 0.841	0.354 ± 0.003	63.902 ± 1.937
LSAM	beta44	0.11	0.127	0.017 ± 0.005	0.153 ± 0.047	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.130 ± 0.006	1.402 ± 0.107	1.184 ± 0.051	0.126	0.016	14.65	0.1144	0.1156	1.032	97.073 ± 0.680	0.523 ± 0.011	16.585 ± 1.500
LSAM	beta45	0.11	0.104	-0.006 ± 0.005	-0.050 ± 0.043	0.013 ± 0.001	0.116 ± 0.004	0.013 ± 0.001	0.116 ± 0.004	1.113 ± 0.072	1.055 ± 0.040	0.097	-0.013	-11.72	0.1010	0.1018	0.983	94.959 ± 0.882	0.447 ± 0.009	13.496 ± 1.378
LSAM	beta46	0.08	0.065	-0.015 ± 0.004	-0.188 ± 0.055	0.012 ± 0.001	0.109 ± 0.004	0.012 ± 0.001	0.110 ± 0.005	1.879 ± 0.141	1.371 ± 0.058	0.064	-0.016	-20.24	0.0882	0.0897	1.056	94.959 ± 0.882	0.450 ± 0.014	13.333 ± 1.371
LSAM	beta47	0.08	0.060	-0.020 ± 0.004	-0.249 ± 0.052	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.105 ± 0.005	1.720 ± 0.134	1.312 ± 0.057	0.061	-0.019	-23.79	0.0877	0.0897	1.027	96.748 ± 0.715	0.415 ± 0.011	16.423 ± 1.494
LSAM	beta51	0.16	0.172	0.012 ± 0.004	0.073 ± 0.027	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	0.445 ± 0.026	0.667 ± 0.024	0.167	0.007	4.41	0.1010	0.1013	0.978	93.984 ± 0.959	0.407 ± 0.003	40.000 ± 1.975
LSAM	beta52	0.16	0.208	0.048 ± 0.012	0.297 ± 0.074	0.085 ± 0.006	0.292 ± 0.010	0.088 ± 0.006	0.296 ± 0.012	3.422 ± 0.237	1.850 ± 0.074	0.200	0.040	24.80	0.2507	0.2538	1.083	97.236 ± 0.661	1.241 ± 0.032	9.756 ± 1.196
LSAM	beta53	0.16	0.147	-0.013 ± 0.005	-0.080 ± 0.030	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	0.556 ± 0.037	0.745 ± 0.029	0.142	-0.018	-11.28	0.1208	0.1222	1.122	97.724 ± 0.601	0.522 ± 0.007	19.837 ± 1.608
LSAM	beta54	0.16	0.154	-0.006 ± 0.009	-0.039 ± 0.059	0.054 ± 0.004	0.233 ± 0.008	0.054 ± 0.004	0.233 ± 0.009	2.115 ± 0.147	1.454 ± 0.059	0.144	-0.016	-9.88	0.2070	0.2076	1.077	96.260 ± 0.765	0.983 ± 0.024	11.870 ± 1.304
LSAM	beta55	0.08	0.130	0.050 ± 0.005	0.628 ± 0.066	0.017 ± 0.001	0.131 ± 0.005	0.020 ± 0.001	0.141 ± 0.006	3.090 ± 0.219	1.758 ± 0.070	0.121	0.041	50.86	0.1111	0.1183	1.010	95.285 ± 0.855	0.520 ± 0.014	21.951 ± 1.669
LSAM	beta56	0.08	0.074	-0.006 ± 0.003	-0.079 ± 0.036	0.005 ± 0.000	0.072 ± 0.003	0.005 ± 0.000	0.072 ± 0.003	0.811 ± 0.060	0.900 ± 0.038	0.075	-0.005	-6.10	0.0635	0.0637	0.969	94.797 ± 0.896	0.273 ± 0.006	27.480 ± 1.800
LSAM	beta57	0.06	0.000	-0.060 ± 0.007	-1.001 ± 0.121	0.032 ± 0.003	0.180 ± 0.007	0.036 ± 0.003	0.190 ± 0.008	9.992 ± 0.751	3.161 ± 0.132	0.008	-0.052	-87.19	0.1432	0.1524	1.024	95.122 ± 0.869	0.723 ± 0.023	1.789 ± 0.534
LSAM	beta58	0.06	0.046	-0.014 ± 0.004	-0.241 ± 0.059	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	2.201 ± 0.171	1.484 ± 0.065	0.048	-0.012	-19.46	0.0745	0.0754	1.055	95.447 ± 0.841	0.364 ± 0.011	13.333 ± 1.371
QML	beta41	0.20	0.227	0.027 ± 0.004	0.136 ± 0.020	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	0.254 ± 0.015	0.504 ± 0.018	0.228	0.028	13.76	0.0996	0.1033	1.019	95.935 ± 0.796	0.388 ± 0.003	64.390 ± 1.931
QML	beta42	0.20	0.221	0.021 ± 0.004	0.105 ± 0.020	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	0.260 ± 0.015	0.510 ± 0.018	0.220	0.020	10.15	0.1028	0.1048	1.023	96.260 ± 0.765	0.401 ± 0.003	60.650 ± 1.970
QML	beta43	0.20	0.190	-0.010 ± 0.003	-0.048 ± 0.017	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	0.173 ± 0.011	0.415 ± 0.015	0.196	-0.004	-2.07	0.0818	0.0819	0.972	94.146 ± 0.947	0.315 ± 0.002	69.756 ± 1.852
QML	beta44	0.11	0.072	-0.038 ± 0.005	-0.349 ± 0.046	0.016 ± 0.001	0.126 ± 0.004	0.017 ± 0.001	0.132 ± 0.005	1.433 ± 0.093	1.197 ± 0.045	0.080	-0.030	-27.59	0.1212	0.1249	1.040	93.659 ± 0.983	0.514 ± 0.008	10.894 ± 1.256
QML	beta45	0.11	0.097	-0.013 ± 0.004	-0.122 ± 0.038	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.897 ± 0.057	0.947 ± 0.035	0.098	-0.012	-11.14	0.1068	0.1075	0.992	92.033 ± 1.092	0.402 ± 0.005	18.862 ± 1.577
QML	beta46	0.08	0.139	0.059 ± 0.003	0.737 ± 0.039	0.006 ± 0.000	0.077 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	1.474 ± 0.097	1.214 ± 0.043	0.132	0.052	64.90	0.0776	0.0933	1.003	92.033 ± 1.092	0.304 ± 0.005	47.154 ± 2.013
QML	beta47	0.08	0.132	0.052 ± 0.003	0.652 ± 0.038	0.006 ± 0.000	0.075 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	1.300 ± 0.097	1.140 ± 0.045	0.119	0.039	48.66	0.0659	0.0765	0.975	92.683 ± 1.050	0.286 ± 0.005	50.569 ± 2.016
QML	beta51	0.16	0.198	0.038 ± 0.006	0.236 ± 0.035	0.019 ± 0.001	0.137 ± 0.004	0.020 ± 0.001	0.142 ± 0.005	0.788 ± 0.048	0.888 ± 0.032	0.198	0.038	23.48	0.1400	0.1449	0.972	93.333 ± 1.006	0.522 ± 0.006	34.309 ± 1.914
QML	beta52	0.16	0.143	-0.017 ± 0.005	-0.105 ± 0.031	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.123 ± 0.005	0.588 ± 0.039	0.767 ± 0.029	0.146	-0.014	-8.70	0.1160	0.1168	0.979	94.634 ± 0.909	0.467 ± 0.006	23.415 ± 1.708
QML	beta53	0.16	0.118	-0.042 ± 0.005	-0.262 ± 0.028	0.013 ± 0.001	0.113 ± 0.004	0.014 ± 0.001	0.120 ± 0.004	0.565 ± 0.036	0.752 ± 0.027	0.114	-0.046	-29.00	0.1119	0.1211	1.020	95.772 ± 0.811	0.451 ± 0.004	16.911 ± 1.512
QML	beta54	0.16	0.177	0.017 ± 0.004	0.106 ± 0.026	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	0.442 ± 0.028	0.665 ± 0.025	0.172	0.012	7.33	0.1078	0.1084	0.985	95.285 ± 0.855	0.406 ± 0.003	40.488 ± 1.979
QML	beta55	0.08	0.110	0.030 ± 0.007	0.375 ± 0.085	0.028 ± 0.002	0.169 ± 0.005	0.029 ± 0.002	0.171 ± 0.006	4.586 ± 0.264	2.142 ± 0.075	0.093	0.013	16.61	0.1842	0.1847	0.840	83.902 ± 1.482	0.556 ± 0.012	25.366 ± 1.755
QML	beta56	0.08	0.109	0.029 ± 0.004	0.365 ± 0.045	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	1.368 ± 0.083	1.169 ± 0.042	0.114	0.034	42.95	0.0823	0.0892	0.907	90.244 ± 1.196	0.316 ± 0.005	35.447 ± 1.929
QML	beta57	0.06	0.098	0.038 ± 0.006	0.638 ± 0.104	0.024 ± 0.001	0.155 ± 0.005	0.025 ± 0.002	0.159 ± 0.006	7.066 ± 0.436	2.658 ± 0.097	0.099	0.039	65.09	0.1617	0.1663	0.841	82.927 ± 1.517	0.511 ± 0.012	22.114 ± 1.673
QML	beta58	0.06	0.096	0.036 ± 0.003	0.600 ± 0.042	0.004 ± 0.000	0.062 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	1.436 ± 0.095	1.198 ± 0.044	0.094	0.034	56.89	0.0606	0.0695	0.965	90.894 ± 1.160	0.236 ± 0.004	41.301 ± 1.985
UPI	beta41	0.20	0.291	0.091 ± 0.013	0.456 ± 0.066	0.108 ± 0.008	0.329 ± 0.012	0.117 ± 0.009	0.341 ± 0.014	2.913 ± 0.219	1.707 ± 0.072	0.280	0.080	39.87	0.3026	0.3130	1.349	98.374 ± 0.510	1.741 ± 0.084	10.894 ± 1.256
UPI	beta42	0.20	0.289	0.089 ± 0.012	0.447 ± 0.060	0.089 ± 0.007	0.299 ± 0.012	0.097 ± 0.008	0.312 ± 0.014	2.431 ± 0.190	1.559 ± 0.068	0.270	0.070	34.88	0.2863	0.2947	1.300	98.537 ± 0.484	1.523 ± 0.070	13.659 ± 1.385
UPI	beta43	0.20	0.170	-0.030 ± 0.004	-0.151 ± 0.022	0.012 ± 0.001	0.109 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	0.321 ± 0.022	0.567 ± 0.022	0.175	-0.025	-12.67	0.1082	0.1112	1.226	96.748 ± 0.715	0.526 ± 0.016	35.772 ± 1.933
UPI	beta44	0.11	0.165	0.055 ± 0.009	0.504 ± 0.079	0.046 ± 0.004	0.214 ± 0.009	0.049 ± 0.004	0.221 ± 0.010	4.043 ± 0.323	2.011 ± 0.089	0.156	0.046	41.37	0.1869	0.1923	1.492	99.024 ± 0.396	1.254 ± 0.060	4.228 ± 0.811
UPI	beta45	0.11	0.105	-0.005 ± 0.008	-0.044 ± 0.068	0.035 ± 0.003	0.187 ± 0.008	0.035 ± 0.003	0.187 ± 0.008	2.881 ± 0.232	1.697 ± 0.077	0.098	-0.012	-11.09	0.1611	0.1616	1.486	97.724 ± 0.601	1.088 ± 0.052	4.390 ± 0.826
UPI	beta46	0.08	0.027	-0.053 ± 0.006	-0.657 ± 0.072	0.020 ± 0.002	0.142 ± 0.006	0.023 ± 0.002	0.152 ± 0.006	3.598 ± 0.275	1.897 ± 0.080	0.036	-0.044	-54.77	0.1265	0.1339	1.363	97.398 ± 0.642	0.761 ± 0.035	7.317 ± 1.050

Table S169

Simulation 2 - Condition 25: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.028	-0.052 ± 0.005	-0.652 ± 0.061	0.015 ± 0.001	0.121 ± 0.005	0.017 ± 0.001	0.131 ± 0.006	2.700 ± 0.224	1.643 ± 0.074	0.036	-0.044	-54.38	0.1028	0.1116	1.303	96.911 ± 0.698	0.617 ± 0.030	8.130 ± 1.102
UPI	beta51	0.16	0.208	0.048 ± 0.005	0.301 ± 0.030	0.015 ± 0.001	0.121 ± 0.004	0.017 ± 0.001	0.130 ± 0.004	0.658 ± 0.039	0.811 ± 0.028	0.210	0.050	31.07	0.1240	0.1336	1.234	95.772 ± 0.811	0.584 ± 0.014	32.683 ± 1.891
UPI	beta52	0.16	0.245	0.085 ± 0.014	0.529 ± 0.088	0.123 ± 0.009	0.350 ± 0.013	0.130 ± 0.010	0.360 ± 0.016	5.069 ± 0.402	2.251 ± 0.100	0.210	0.050	31.18	0.3500	0.3536	1.561	99.350 ± 0.324	2.144 ± 0.109	6.829 ± 1.017
UPI	beta53	0.16	0.110	-0.050 ± 0.007	-0.313 ± 0.043	0.029 ± 0.002	0.171 ± 0.006	0.032 ± 0.002	0.178 ± 0.007	1.243 ± 0.089	1.115 ± 0.045	0.106	-0.054	-33.90	0.1660	0.1747	1.433	98.374 ± 0.510	0.963 ± 0.039	6.667 ± 1.006
UPI	beta54	0.16	0.259	0.099 ± 0.019	0.621 ± 0.121	0.232 ± 0.018	0.482 ± 0.019	0.241 ± 0.020	0.491 ± 0.022	9.431 ± 0.764	3.071 ± 0.138	0.235	0.075	46.89	0.4504	0.4566	1.529	99.187 ± 0.362	2.886 ± 0.142	2.764 ± 0.661
UPI	beta55	0.08	0.149	0.069 ± 0.008	0.861 ± 0.095	0.036 ± 0.003	0.189 ± 0.008	0.040 ± 0.003	0.201 ± 0.009	6.288 ± 0.506	2.508 ± 0.111	0.134	0.054	67.66	0.1670	0.1756	1.472	98.699 ± 0.457	1.088 ± 0.055	7.642 ± 1.071
UPI	beta56	0.08	0.073	-0.007 ± 0.004	-0.085 ± 0.049	0.010 ± 0.001	0.097 ± 0.004	0.010 ± 0.001	0.098 ± 0.004	1.489 ± 0.115	1.220 ± 0.053	0.074	-0.006	-8.05	0.0912	0.0914	1.428	97.236 ± 0.661	0.545 ± 0.020	13.008 ± 1.356
UPI	beta57	0.06	-0.022	-0.082 ± 0.009	-1.361 ± 0.143	0.045 ± 0.004	0.213 ± 0.009	0.052 ± 0.004	0.228 ± 0.010	14.418 ± 1.123	3.797 ± 0.163	-0.010	-0.070	-117.17	0.1805	0.1937	1.495	98.862 ± 0.428	1.247 ± 0.066	0.325 ± 0.230
UPI	beta58	0.06	0.001	-0.059 ± 0.007	-0.977 ± 0.117	0.030 ± 0.003	0.174 ± 0.007	0.034 ± 0.003	0.184 ± 0.008	9.372 ± 0.776	3.061 ± 0.139	0.014	-0.046	-76.56	0.1483	0.1553	1.498	98.862 ± 0.428	1.023 ± 0.050	3.577 ± 0.749

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S170

Simulation 2 - Condition 26: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.219	0.019 ± 0.007	0.093 ± 0.033	0.037 ± 0.002	0.191 ± 0.006	0.037 ± 0.002	0.192 ± 0.007	0.923 ± 0.055	0.961 ± 0.033	0.214	0.014	7.05	0.1745	0.1751	1.032	96.831 ± 0.600	0.774 ± 0.013	24.296 ± 1.469
LSAM	beta42	0.20	0.202	0.002 ± 0.007	0.011 ± 0.034	0.040 ± 0.002	0.200 ± 0.006	0.040 ± 0.002	0.200 ± 0.007	1.001 ± 0.056	1.001 ± 0.033	0.203	0.003	1.26	0.1883	0.1883	1.030	96.714 ± 0.611	0.808 ± 0.013	20.305 ± 1.378
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.010	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.292 ± 0.009	0.199	-0.001	-0.42	0.0579	0.0579	1.016	95.892 ± 0.680	0.233 ± 0.002	91.080 ± 0.977
LSAM	beta44	0.11	0.138	0.028 ± 0.003	0.258 ± 0.027	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.678 ± 0.040	0.823 ± 0.028	0.139	0.029	26.46	0.0817	0.0867	0.969	94.366 ± 0.790	0.327 ± 0.005	42.371 ± 1.693
LSAM	beta45	0.11	0.118	0.008 ± 0.003	0.069 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.485 ± 0.030	0.697 ± 0.024	0.112	0.002	1.93	0.0730	0.0731	0.959	93.427 ± 0.849	0.287 ± 0.004	37.559 ± 1.659
LSAM	beta46	0.08	0.058	-0.022 ± 0.003	-0.279 ± 0.032	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.930 ± 0.058	0.964 ± 0.034	0.060	-0.020	-25.12	0.0642	0.0672	0.992	94.601 ± 0.774	0.287 ± 0.006	21.244 ± 1.401
LSAM	beta47	0.08	0.061	-0.019 ± 0.003	-0.243 ± 0.032	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.920 ± 0.056	0.959 ± 0.033	0.059	-0.021	-26.12	0.0650	0.0682	0.984	94.014 ± 0.813	0.287 ± 0.006	21.831 ± 1.415
LSAM	beta51	0.16	0.174	0.014 ± 0.002	0.086 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.189 ± 0.009	0.434 ± 0.013	0.175	0.015	9.16	0.0668	0.0684	0.987	94.484 ± 0.782	0.264 ± 0.001	74.648 ± 1.490
LSAM	beta52	0.16	0.230	0.070 ± 0.008	0.439 ± 0.048	0.050 ± 0.003	0.223 ± 0.007	0.055 ± 0.003	0.234 ± 0.008	2.141 ± 0.133	1.463 ± 0.051	0.213	0.053	32.99	0.2014	0.2082	1.123	97.300 ± 0.555	0.983 ± 0.021	17.136 ± 1.291
LSAM	beta53	0.16	0.142	-0.018 ± 0.003	-0.112 ± 0.018	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.087 ± 0.003	0.293 ± 0.014	0.541 ± 0.016	0.143	-0.017	-10.68	0.0906	0.0922	1.062	96.479 ± 0.631	0.353 ± 0.003	37.911 ± 1.662
LSAM	beta54	0.16	0.140	-0.020 ± 0.006	-0.127 ± 0.037	0.030 ± 0.002	0.172 ± 0.005	0.030 ± 0.002	0.173 ± 0.006	1.171 ± 0.075	1.082 ± 0.039	0.142	-0.018	-11.27	0.1626	0.1635	1.085	97.066 ± 0.578	0.732 ± 0.016	16.080 ± 1.259
LSAM	beta55	0.08	0.138	0.058 ± 0.003	0.723 ± 0.038	0.008 ± 0.000	0.088 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	1.727 ± 0.095	1.314 ± 0.039	0.136	0.056	69.47	0.0772	0.0951	1.079	92.371 ± 0.909	0.371 ± 0.008	40.962 ± 1.685
LSAM	beta56	0.08	0.071	-0.009 ± 0.001	-0.116 ± 0.018	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.277 ± 0.016	0.526 ± 0.017	0.070	-0.010	-11.96	0.0406	0.0417	1.032	94.484 ± 0.782	0.166 ± 0.003	46.127 ± 1.708
LSAM	beta57	0.06	-0.008	-0.068 ± 0.004	-1.127 ± 0.071	0.015 ± 0.001	0.123 ± 0.004	0.020 ± 0.001	0.141 ± 0.005	5.501 ± 0.328	2.345 ± 0.077	-0.001	-0.061	-101.80	0.1071	0.1233	1.113	93.779 ± 0.827	0.539 ± 0.013	1.761 ± 0.451
LSAM	beta58	0.06	0.050	-0.010 ± 0.002	-0.164 ± 0.035	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	1.074 ± 0.067	1.036 ± 0.037	0.048	-0.012	-19.18	0.0531	0.0543	1.043	94.836 ± 0.758	0.251 ± 0.006	18.662 ± 1.335
QML	beta41	0.20	0.238	0.038 ± 0.002	0.192 ± 0.011	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.073 ± 0.002	0.132 ± 0.006	0.363 ± 0.009	0.239	0.039	19.56	0.0615	0.0729	0.994	90.258 ± 1.016	0.241 ± 0.001	96.244 ± 0.651
QML	beta42	0.20	0.220	0.020 ± 0.002	0.102 ± 0.011	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.006	0.335 ± 0.010	0.218	0.018	9.12	0.0652	0.0677	0.974	93.779 ± 0.827	0.244 ± 0.001	93.310 ± 0.856
QML	beta43	0.20	0.192	-0.008 ± 0.002	-0.038 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.192	-0.008	-4.14	0.0485	0.0492	1.008	94.131 ± 0.805	0.195 ± 0.001	96.714 ± 0.611
QML	beta44	0.11	0.076	-0.034 ± 0.003	-0.307 ± 0.027	0.007 ± 0.000	0.086 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	0.706 ± 0.035	0.840 ± 0.025	0.082	-0.028	-25.74	0.0863	0.0908	0.925	90.023 ± 1.027	0.312 ± 0.003	22.653 ± 1.434
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.036 ± 0.021	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.370 ± 0.020	0.608 ± 0.020	0.107	-0.003	-2.38	0.0676	0.0677	0.958	92.723 ± 0.890	0.251 ± 0.002	41.667 ± 1.689
QML	beta46	0.08	0.137	0.057 ± 0.002	0.708 ± 0.020	0.002 ± 0.000	0.047 ± 0.001	0.005 ± 0.000	0.074 ± 0.002	0.847 ± 0.035	0.920 ± 0.020	0.135	0.055	68.91	0.0473	0.0727	0.992	75.822 ± 1.467	0.183 ± 0.003	85.094 ± 1.220
QML	beta47	0.08	0.130	0.050 ± 0.002	0.630 ± 0.020	0.002 ± 0.000	0.046 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.726 ± 0.032	0.852 ± 0.020	0.125	0.045	56.44	0.0468	0.0650	0.942	78.169 ± 1.415	0.170 ± 0.002	85.915 ± 1.192
QML	beta51	0.16	0.185	0.025 ± 0.003	0.156 ± 0.018	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.312 ± 0.016	0.559 ± 0.017	0.184	0.024	14.70	0.0820	0.0853	0.981	92.606 ± 0.896	0.330 ± 0.004	61.150 ± 1.670
QML	beta52	0.16	0.151	-0.009 ± 0.003	-0.055 ± 0.016	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.228 ± 0.011	0.478 ± 0.014	0.152	-0.008	-5.01	0.0763	0.0767	0.996	93.779 ± 0.827	0.296 ± 0.003	53.991 ± 1.708
QML	beta53	0.16	0.117	-0.043 ± 0.002	-0.271 ± 0.015	0.005 ± 0.000	0.069 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.259 ± 0.011	0.509 ± 0.012	0.118	-0.042	-26.05	0.0708	0.0821	1.008	89.554 ± 1.048	0.273 ± 0.001	38.967 ± 1.671
QML	beta54	0.16	0.173	0.013 ± 0.002	0.081 ± 0.013	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.007	0.391 ± 0.011	0.172	0.012	7.38	0.0645	0.0656	1.046	95.188 ± 0.733	0.251 ± 0.001	78.169 ± 1.415
QML	beta55	0.08	0.086	0.006 ± 0.004	0.077 ± 0.055	0.016 ± 0.001	0.128 ± 0.003	0.016 ± 0.001	0.128 ± 0.004	2.549 ± 0.113	1.597 ± 0.045	0.075	-0.005	-6.39	0.1361	0.1362	0.831	78.638 ± 1.404	0.416 ± 0.011	26.526 ± 1.512
QML	beta56	0.08	0.117	0.037 ± 0.002	0.460 ± 0.023	0.003 ± 0.000	0.053 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.656 ± 0.032	0.810 ± 0.022	0.115	0.035	43.90	0.0519	0.0626	0.947	83.333 ± 1.277	0.198 ± 0.003	64.906 ± 1.635
QML	beta57	0.06	0.127	0.067 ± 0.004	1.118 ± 0.067	0.014 ± 0.001	0.118 ± 0.003	0.018 ± 0.001	0.136 ± 0.003	5.105 ± 0.213	2.259 ± 0.056	0.136	0.076	126.02	0.1210	0.1427	0.862	75.587 ± 1.472	0.398 ± 0.011	39.319 ± 1.673
QML	beta58	0.06	0.100	0.040 ± 0.001	0.667 ± 0.021	0.001 ± 0.000	0.038 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.837 ± 0.039	0.915 ± 0.023	0.100	0.040	66.43	0.0373	0.0546	0.975	79.930 ± 1.372	0.144 ± 0.001	77.113 ± 1.439
UPI	beta41	0.20	0.247	0.047 ± 0.006	0.237 ± 0.030	0.031 ± 0.002	0.176 ± 0.005	0.033 ± 0.002	0.182 ± 0.006	0.826 ± 0.047	0.909 ± 0.030	0.245	0.045	22.35	0.1658	0.1717	1.164	97.418 ± 0.543	0.801 ± 0.025	28.404 ± 1.545
UPI	beta42	0.20	0.238	0.038 ± 0.005	0.192 ± 0.027	0.025 ± 0.001	0.158 ± 0.004	0.027 ± 0.001	0.163 ± 0.005	0.663 ± 0.036	0.814 ± 0.026	0.241	0.041	20.56	0.1520	0.1575	1.135	97.183 ± 0.567	0.705 ± 0.013	31.221 ± 1.588
UPI	beta43	0.20	0.175	-0.025 ± 0.002	-0.125 ± 0.010	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.105 ± 0.005	0.324 ± 0.009	0.176	-0.024	-12.15	0.0584	0.0633	1.075	93.427 ± 0.849	0.252 ± 0.004	81.808 ± 1.322
UPI	beta44	0.11	0.147	0.037 ± 0.004	0.334 ± 0.034	0.012 ± 0.001	0.109 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	1.096 ± 0.067	1.047 ± 0.036	0.139	0.029	26.01	0.0998	0.1038	1.269	97.183 ± 0.567	0.543 ± 0.021	21.479 ± 1.407
UPI	beta45	0.11	0.130	0.020 ± 0.003	0.184 ± 0.032	0.010 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.891 ± 0.062	0.944 ± 0.037	0.120	0.010	9.15	0.0929	0.0934	1.224	96.009 ± 0.671	0.489 ± 0.033	22.183 ± 1.423
UPI	beta46	0.08	0.046	-0.034 ± 0.002	-0.429 ± 0.030	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.970 ± 0.059	0.985 ± 0.033	0.048	-0.032	-39.64	0.0643	0.0717	1.160	95.892 ± 0.680	0.323 ± 0.013	16.080 ± 1.259

Table S170

Simulation 2 - Condition 26: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.047	-0.033 ± 0.002	-0.413 ± 0.026	0.004 ± 0.000	0.060 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.728 ± 0.045	0.853 ± 0.029	0.052	-0.028	-35.28	0.0534	0.0604	1.154	93.192 ± 0.863	0.270 ± 0.008	17.606 ± 1.305
UPI	beta51	0.16	0.198	0.038 ± 0.002	0.240 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.260 ± 0.013	0.509 ± 0.014	0.200	0.040	25.22	0.0690	0.0799	1.078	93.192 ± 0.863	0.304 ± 0.003	74.178 ± 1.499
UPI	beta52	0.16	0.212	0.052 ± 0.008	0.324 ± 0.047	0.049 ± 0.003	0.221 ± 0.007	0.052 ± 0.003	0.227 ± 0.008	2.016 ± 0.131	1.420 ± 0.052	0.208	0.048	29.91	0.1930	0.1988	1.366	98.474 ± 0.420	1.185 ± 0.047	13.967 ± 1.188
UPI	beta53	0.16	0.101	-0.059 ± 0.004	-0.367 ± 0.022	0.011 ± 0.001	0.103 ± 0.003	0.014 ± 0.001	0.118 ± 0.003	0.548 ± 0.028	0.740 ± 0.021	0.106	-0.054	-33.66	0.1017	0.1151	1.200	93.310 ± 0.856	0.484 ± 0.013	21.009 ± 1.396
UPI	beta54	0.16	0.274	0.114 ± 0.012	0.710 ± 0.072	0.114 ± 0.007	0.338 ± 0.011	0.127 ± 0.008	0.356 ± 0.013	4.958 ± 0.322	2.227 ± 0.080	0.247	0.087	54.44	0.2718	0.2854	1.323	98.826 ± 0.369	1.752 ± 0.067	7.629 ± 0.909
UPI	beta55	0.08	0.132	0.052 ± 0.003	0.650 ± 0.042	0.009 ± 0.001	0.097 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	1.892 ± 0.103	1.376 ± 0.042	0.124	0.044	54.64	0.0872	0.0975	1.399	97.418 ± 0.543	0.532 ± 0.020	23.592 ± 1.455
UPI	beta56	0.08	0.079	-0.001 ± 0.002	-0.013 ± 0.022	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.428 ± 0.026	0.654 ± 0.023	0.078	-0.002	-2.21	0.0469	0.0469	1.237	96.009 ± 0.671	0.254 ± 0.009	37.089 ± 1.655
UPI	beta57	0.06	-0.003	-0.063 ± 0.004	-1.047 ± 0.065	0.013 ± 0.001	0.113 ± 0.004	0.017 ± 0.001	0.130 ± 0.004	4.670 ± 0.275	2.161 ± 0.070	0.007	-0.053	-88.89	0.0926	0.1068	1.402	96.714 ± 0.611	0.624 ± 0.025	0.469 ± 0.234
UPI	beta58	0.06	0.000	-0.060 ± 0.004	-1.000 ± 0.062	0.012 ± 0.001	0.108 ± 0.004	0.015 ± 0.001	0.124 ± 0.004	4.241 ± 0.266	2.059 ± 0.070	0.008	-0.052	-85.83	0.0878	0.1018	1.329	95.775 ± 0.689	0.563 ± 0.023	4.343 ± 0.698

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S171

Simulation 2 - Condition 27: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.210	0.010 ± 0.005	0.049 ± 0.024	0.020 ± 0.001	0.143 ± 0.004	0.020 ± 0.001	0.143 ± 0.004	0.510 ± 0.027	0.714 ± 0.022	0.209	0.009	4.46	0.1436	0.1438	0.970	95.842 ± 0.660	0.542 ± 0.006	37.965 ± 1.605
LSAM	beta42	0.20	0.199	-0.001 ± 0.005	-0.006 ± 0.024	0.021 ± 0.001	0.146 ± 0.004	0.021 ± 0.001	0.146 ± 0.005	0.531 ± 0.029	0.729 ± 0.024	0.201	0.001	0.33	0.1357	0.1357	0.979	95.842 ± 0.660	0.560 ± 0.006	34.026 ± 1.567
LSAM	beta43	0.20	0.204	0.004 ± 0.002	0.022 ± 0.011	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.105 ± 0.005	0.324 ± 0.009	0.204	0.004	2.15	0.0676	0.0677	0.987	95.405 ± 0.693	0.250 ± 0.001	89.387 ± 1.019
LSAM	beta44	0.11	0.131	0.021 ± 0.003	0.191 ± 0.028	0.009 ± 0.000	0.092 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.740 ± 0.044	0.860 ± 0.029	0.126	0.016	14.47	0.0867	0.0881	0.929	93.107 ± 0.838	0.336 ± 0.004	36.543 ± 1.593
LSAM	beta45	0.11	0.115	0.005 ± 0.003	0.044 ± 0.026	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.639 ± 0.038	0.800 ± 0.027	0.108	-0.002	-2.01	0.0766	0.0766	0.870	93.764 ± 0.800	0.300 ± 0.003	37.418 ± 1.601
LSAM	beta46	0.08	0.075	-0.005 ± 0.002	-0.068 ± 0.026	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.627 ± 0.038	0.792 ± 0.027	0.075	-0.005	-6.56	0.0564	0.0566	0.891	92.232 ± 0.885	0.221 ± 0.004	37.746 ± 1.603
LSAM	beta47	0.08	0.077	-0.003 ± 0.002	-0.042 ± 0.026	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.605 ± 0.036	0.778 ± 0.027	0.075	-0.005	-6.10	0.0550	0.0552	0.875	92.998 ± 0.844	0.213 ± 0.004	40.591 ± 1.624
LSAM	beta51	0.16	0.166	0.006 ± 0.003	0.036 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.276 ± 0.014	0.525 ± 0.016	0.168	0.008	4.84	0.0807	0.0811	0.975	94.748 ± 0.738	0.321 ± 0.001	54.814 ± 1.646
LSAM	beta52	0.16	0.165	0.005 ± 0.005	0.031 ± 0.033	0.025 ± 0.001	0.159 ± 0.004	0.025 ± 0.001	0.159 ± 0.005	0.989 ± 0.054	0.994 ± 0.032	0.162	0.002	1.22	0.1469	0.1470	0.966	95.624 ± 0.677	0.603 ± 0.006	21.554 ± 1.360
LSAM	beta53	0.16	0.161	0.001 ± 0.003	0.004 ± 0.020	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.380 ± 0.017	0.617 ± 0.017	0.165	0.005	3.41	0.1015	0.1017	0.963	94.530 ± 0.752	0.373 ± 0.002	42.232 ± 1.634
LSAM	beta54	0.16	0.157	-0.003 ± 0.004	-0.021 ± 0.026	0.016 ± 0.001	0.125 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	0.606 ± 0.030	0.778 ± 0.023	0.160	0.000	-0.04	0.1210	0.1210	1.058	96.937 ± 0.570	0.517 ± 0.004	22.429 ± 1.380
LSAM	beta55	0.08	0.096	0.016 ± 0.003	0.198 ± 0.038	0.008 ± 0.000	0.091 ± 0.003	0.009 ± 0.000	0.092 ± 0.003	1.335 ± 0.076	1.155 ± 0.038	0.095	0.015	18.53	0.0833	0.0846	0.937	93.435 ± 0.819	0.335 ± 0.005	27.243 ± 1.473
LSAM	beta56	0.08	0.078	-0.002 ± 0.002	-0.025 ± 0.022	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.459 ± 0.026	0.677 ± 0.022	0.077	-0.003	-3.88	0.0483	0.0484	0.892	93.107 ± 0.838	0.190 ± 0.003	42.013 ± 1.633
LSAM	beta57	0.06	0.049	-0.011 ± 0.004	-0.185 ± 0.059	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	3.179 ± 0.190	1.783 ± 0.061	0.046	-0.014	-23.82	0.0940	0.0951	0.905	94.967 ± 0.723	0.378 ± 0.006	7.549 ± 0.874
LSAM	beta58	0.06	0.059	-0.001 ± 0.002	-0.022 ± 0.027	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.662 ± 0.036	0.813 ± 0.026	0.057	-0.003	-5.28	0.0441	0.0443	0.956	93.654 ± 0.806	0.183 ± 0.003	33.042 ± 1.556
QML	beta41	0.20	0.220	0.020 ± 0.003	0.099 ± 0.013	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.161 ± 0.008	0.401 ± 0.011	0.221	0.021	10.67	0.0786	0.0815	0.974	94.201 ± 0.773	0.297 ± 0.001	81.072 ± 1.296
QML	beta42	0.20	0.211	0.011 ± 0.003	0.053 ± 0.013	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.213	0.013	6.52	0.0750	0.0761	0.992	94.530 ± 0.752	0.301 ± 0.001	78.556 ± 1.358
QML	beta43	0.20	0.195	-0.005 ± 0.002	-0.023 ± 0.010	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.293 ± 0.008	0.195	-0.005	-2.61	0.0591	0.0593	0.988	94.639 ± 0.745	0.226 ± 0.001	90.810 ± 0.956
QML	beta44	0.11	0.086	-0.024 ± 0.003	-0.222 ± 0.024	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.560 ± 0.027	0.749 ± 0.021	0.087	-0.023	-21.04	0.0779	0.0812	0.958	91.904 ± 0.902	0.295 ± 0.003	23.085 ± 1.394
QML	beta45	0.11	0.103	-0.007 ± 0.002	-0.062 ± 0.021	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.408 ± 0.020	0.639 ± 0.019	0.102	-0.008	-7.49	0.0692	0.0697	0.936	93.217 ± 0.832	0.257 ± 0.002	39.387 ± 1.616
QML	beta46	0.08	0.108	0.028 ± 0.002	0.350 ± 0.020	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.483 ± 0.026	0.695 ± 0.021	0.108	0.028	34.43	0.0462	0.0537	0.913	88.293 ± 1.063	0.172 ± 0.002	70.460 ± 1.509
QML	beta47	0.08	0.108	0.028 ± 0.002	0.353 ± 0.019	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.449 ± 0.024	0.670 ± 0.020	0.107	0.027	33.63	0.0427	0.0505	0.896	87.199 ± 1.105	0.160 ± 0.002	75.164 ± 1.429
QML	beta51	0.16	0.182	0.022 ± 0.003	0.138 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.298 ± 0.015	0.546 ± 0.016	0.181	0.021	13.36	0.0807	0.0835	1.005	94.311 ± 0.766	0.333 ± 0.002	57.877 ± 1.633
QML	beta52	0.16	0.144	-0.016 ± 0.003	-0.102 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.280 ± 0.013	0.530 ± 0.015	0.144	-0.016	-10.21	0.0818	0.0834	0.981	93.873 ± 0.793	0.320 ± 0.001	43.217 ± 1.639
QML	beta53	0.16	0.129	-0.031 ± 0.003	-0.194 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	0.315 ± 0.014	0.561 ± 0.015	0.131	-0.029	-18.39	0.0835	0.0885	0.974	92.560 ± 0.868	0.322 ± 0.002	36.105 ± 1.589
QML	beta54	0.16	0.162	0.002 ± 0.002	0.014 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.212 ± 0.010	0.461 ± 0.013	0.162	0.002	1.35	0.0763	0.0763	1.080	96.937 ± 0.570	0.312 ± 0.001	52.735 ± 1.651
QML	beta55	0.08	0.093	0.013 ± 0.003	0.159 ± 0.036	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	1.232 ± 0.063	1.110 ± 0.034	0.093	0.013	16.25	0.0802	0.0813	0.962	89.716 ± 1.005	0.331 ± 0.006	27.790 ± 1.482
QML	beta56	0.08	0.094	0.014 ± 0.002	0.169 ± 0.022	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.490 ± 0.026	0.700 ± 0.022	0.093	0.013	15.80	0.0503	0.0519	0.884	90.044 ± 0.990	0.188 ± 0.002	53.720 ± 1.649
QML	beta57	0.06	0.080	0.020 ± 0.003	0.333 ± 0.044	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	1.842 ± 0.097	1.357 ± 0.042	0.079	0.019	30.92	0.0763	0.0785	0.994	91.247 ± 0.935	0.308 ± 0.006	22.538 ± 1.382
QML	beta58	0.06	0.082	0.022 ± 0.001	0.361 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.547 ± 0.027	0.740 ± 0.020	0.080	0.020	32.81	0.0384	0.0431	0.942	89.934 ± 0.995	0.143 ± 0.002	63.020 ± 1.597
UPI	beta41	0.20	0.230	0.030 ± 0.005	0.149 ± 0.023	0.020 ± 0.001	0.141 ± 0.004	0.021 ± 0.001	0.144 ± 0.005	0.520 ± 0.030	0.721 ± 0.024	0.231	0.031	15.51	0.1321	0.1357	1.040	95.514 ± 0.685	0.575 ± 0.011	45.186 ± 1.646
UPI	beta42	0.20	0.223	0.023 ± 0.005	0.114 ± 0.023	0.020 ± 0.001	0.141 ± 0.003	0.020 ± 0.001	0.143 ± 0.004	0.512 ± 0.026	0.716 ± 0.021	0.221	0.021	10.51	0.1380	0.1396	1.060	96.171 ± 0.635	0.587 ± 0.009	38.074 ± 1.606
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.011	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.108 ± 0.005	0.328 ± 0.010	0.198	-0.002	-0.77	0.0657	0.0657	0.961	93.545 ± 0.813	0.247 ± 0.002	86.324 ± 1.137
UPI	beta44	0.11	0.123	0.013 ± 0.003	0.116 ± 0.029	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.759 ± 0.046	0.871 ± 0.030	0.118	0.008	7.44	0.0857	0.0861	1.014	94.530 ± 0.752	0.378 ± 0.008	28.009 ± 1.485
UPI	beta45	0.11	0.116	0.006 ± 0.003	0.054 ± 0.026	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.638 ± 0.039	0.799 ± 0.028	0.108	-0.002	-2.17	0.0784	0.0785	0.996	94.639 ± 0.745	0.342 ± 0.007	31.838 ± 1.541
UPI	beta46	0.08	0.069	-0.011 ± 0.002	-0.141 ± 0.027	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.665 ± 0.042	0.816 ± 0.029	0.069	-0.011	-13.98	0.0553	0.0564	0.959	92.451 ± 0.874	0.242 ± 0.007	33.370 ± 1.560

Table S171

Simulation 2 - Condition 27: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.069	-0.011 ± 0.002	-0.138 ± 0.025	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.572 ± 0.034	0.756 ± 0.026	0.073	-0.007	-8.89	0.0519	0.0524	0.979	93.654 ± 0.806	0.228 ± 0.005	36.543 ± 1.593
UPI	beta51	0.16	0.176	0.016 ± 0.003	0.099 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.276 ± 0.014	0.526 ± 0.016	0.176	0.016	9.88	0.0808	0.0823	0.974	94.967 ± 0.723	0.316 ± 0.001	59.847 ± 1.621
UPI	beta52	0.16	0.183	0.023 ± 0.005	0.144 ± 0.031	0.022 ± 0.001	0.150 ± 0.004	0.023 ± 0.001	0.151 ± 0.005	0.896 ± 0.047	0.946 ± 0.029	0.185	0.025	15.65	0.1493	0.1514	1.071	97.155 ± 0.550	0.629 ± 0.014	25.930 ± 1.450
UPI	beta53	0.16	0.156	-0.004 ± 0.003	-0.027 ± 0.020	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.367 ± 0.017	0.606 ± 0.017	0.154	-0.006	-3.49	0.0981	0.0983	0.979	94.748 ± 0.738	0.372 ± 0.003	41.357 ± 1.629
UPI	beta54	0.16	0.156	-0.004 ± 0.004	-0.023 ± 0.028	0.018 ± 0.001	0.135 ± 0.004	0.018 ± 0.001	0.135 ± 0.004	0.711 ± 0.039	0.843 ± 0.027	0.156	-0.004	-2.27	0.1296	0.1296	1.158	97.374 ± 0.529	0.612 ± 0.012	18.709 ± 1.290
UPI	beta55	0.08	0.100	0.020 ± 0.003	0.244 ± 0.038	0.008 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	1.384 ± 0.085	1.177 ± 0.041	0.097	0.017	21.83	0.0790	0.0809	1.019	95.842 ± 0.660	0.368 ± 0.009	23.195 ± 1.396
UPI	beta56	0.08	0.076	-0.004 ± 0.002	-0.051 ± 0.022	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.457 ± 0.026	0.676 ± 0.022	0.074	-0.006	-7.09	0.0474	0.0477	0.927	92.888 ± 0.850	0.196 ± 0.003	41.575 ± 1.630
UPI	beta57	0.06	0.039	-0.021 ± 0.003	-0.347 ± 0.056	0.010 ± 0.001	0.101 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	2.938 ± 0.171	1.714 ± 0.057	0.036	-0.024	-39.96	0.0892	0.0923	1.036	95.952 ± 0.652	0.409 ± 0.012	6.674 ± 0.826
UPI	beta58	0.06	0.058	-0.002 ± 0.002	-0.039 ± 0.030	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.800 ± 0.052	0.894 ± 0.032	0.058	-0.002	-3.10	0.0452	0.0452	1.048	94.967 ± 0.723	0.220 ± 0.006	31.182 ± 1.532

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S172

Simulation 2 - Condition 28: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.003	0.008 ± 0.013	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.163 ± 0.008	0.403 ± 0.012	0.206	0.006	3.06	0.0768	0.0770	0.970	94.960 ± 0.695	0.307 ± 0.002	73.387 ± 1.403
LSAM	beta42	0.20	0.200	-0.000 ± 0.003	-0.001 ± 0.013	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.167 ± 0.008	0.409 ± 0.011	0.199	-0.001	-0.32	0.0819	0.0819	0.994	95.665 ± 0.647	0.319 ± 0.002	71.472 ± 1.434
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.202	0.002	0.89	0.0373	0.0373	1.017	95.060 ± 0.688	0.153 ± 0.000	99.899 ± 0.101
LSAM	beta44	0.11	0.121	0.011 ± 0.002	0.103 ± 0.015	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.229 ± 0.012	0.478 ± 0.015	0.119	0.009	8.08	0.0484	0.0492	0.931	92.440 ± 0.839	0.188 ± 0.001	72.782 ± 1.413
LSAM	beta45	0.11	0.114	0.004 ± 0.002	0.039 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.196 ± 0.010	0.442 ± 0.013	0.115	0.005	4.81	0.0467	0.0470	0.915	93.145 ± 0.802	0.174 ± 0.001	73.589 ± 1.400
LSAM	beta46	0.08	0.076	-0.004 ± 0.001	-0.046 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.153 ± 0.007	0.391 ± 0.011	0.076	-0.004	-5.34	0.0289	0.0292	0.935	93.044 ± 0.808	0.114 ± 0.001	73.488 ± 1.401
LSAM	beta47	0.08	0.075	-0.005 ± 0.001	-0.060 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.156 ± 0.007	0.395 ± 0.011	0.074	-0.006	-7.06	0.0302	0.0308	0.909	92.440 ± 0.839	0.111 ± 0.001	73.185 ± 1.407
LSAM	beta51	0.16	0.164	0.004 ± 0.002	0.024 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.164	0.004	2.67	0.0522	0.0524	0.972	94.355 ± 0.733	0.195 ± 0.000	90.222 ± 0.943
LSAM	beta52	0.16	0.168	0.008 ± 0.003	0.049 ± 0.017	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.302 ± 0.014	0.549 ± 0.015	0.168	0.008	5.25	0.0864	0.0868	0.998	95.161 ± 0.681	0.342 ± 0.002	50.000 ± 1.588
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.134 ± 0.006	0.367 ± 0.010	0.160	0.000	-0.10	0.0571	0.0571	0.972	94.758 ± 0.708	0.224 ± 0.001	79.839 ± 1.274
LSAM	beta54	0.16	0.155	-0.005 ± 0.003	-0.031 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.260 ± 0.012	0.510 ± 0.015	0.153	-0.007	-4.52	0.0790	0.0793	0.958	94.456 ± 0.727	0.306 ± 0.002	51.411 ± 1.587
LSAM	beta55	0.08	0.097	0.017 ± 0.002	0.206 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.398 ± 0.019	0.631 ± 0.018	0.096	0.016	20.55	0.0467	0.0495	0.942	92.339 ± 0.844	0.176 ± 0.001	60.383 ± 1.553
LSAM	beta56	0.08	0.076	-0.004 ± 0.001	-0.044 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.128 ± 0.006	0.358 ± 0.010	0.076	-0.004	-4.69	0.0276	0.0278	0.904	91.230 ± 0.898	0.101 ± 0.001	81.351 ± 1.237
LSAM	beta57	0.06	0.044	-0.016 ± 0.002	-0.259 ± 0.029	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.872 ± 0.041	0.934 ± 0.026	0.042	-0.018	-29.53	0.0524	0.0553	0.952	92.137 ± 0.855	0.201 ± 0.002	16.734 ± 1.185
LSAM	beta58	0.06	0.058	-0.002 ± 0.001	-0.037 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.230 ± 0.012	0.480 ± 0.014	0.057	-0.003	-5.28	0.0257	0.0259	0.927	91.633 ± 0.879	0.104 ± 0.001	62.298 ± 1.539
QML	beta41	0.20	0.216	0.016 ± 0.002	0.082 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.217	0.017	8.35	0.0480	0.0509	0.988	94.052 ± 0.751	0.185 ± 0.001	99.194 ± 0.284
QML	beta42	0.20	0.209	0.009 ± 0.002	0.047 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.241 ± 0.006	0.210	0.010	5.04	0.0488	0.0498	1.019	95.161 ± 0.681	0.189 ± 0.001	98.286 ± 0.412
QML	beta43	0.20	0.192	-0.008 ± 0.001	-0.038 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.002	0.183 ± 0.005	0.194	-0.006	-3.07	0.0357	0.0363	1.012	93.851 ± 0.763	0.142 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.085	-0.025 ± 0.002	-0.230 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.243 ± 0.011	0.493 ± 0.013	0.085	-0.025	-22.90	0.0483	0.0544	0.960	90.726 ± 0.921	0.181 ± 0.001	46.976 ± 1.585
QML	beta45	0.11	0.106	-0.004 ± 0.001	-0.035 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.159 ± 0.008	0.398 ± 0.012	0.108	-0.002	-2.20	0.0418	0.0419	0.938	92.440 ± 0.839	0.161 ± 0.001	74.093 ± 1.391
QML	beta46	0.08	0.109	0.029 ± 0.001	0.365 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.244 ± 0.010	0.494 ± 0.011	0.108	0.028	35.04	0.0257	0.0380	0.965	79.637 ± 1.279	0.101 ± 0.001	98.891 ± 0.332
QML	beta47	0.08	0.109	0.029 ± 0.001	0.362 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.241 ± 0.010	0.491 ± 0.011	0.108	0.028	34.94	0.0257	0.0380	0.937	76.512 ± 1.346	0.097 ± 0.001	98.286 ± 0.412
QML	beta51	0.16	0.178	0.018 ± 0.002	0.115 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.121 ± 0.005	0.348 ± 0.009	0.179	0.019	11.96	0.0534	0.0567	0.993	94.052 ± 0.751	0.205 ± 0.001	92.036 ± 0.860
QML	beta52	0.16	0.141	-0.019 ± 0.002	-0.116 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.332 ± 0.009	0.141	-0.019	-11.77	0.0504	0.0538	1.001	93.649 ± 0.774	0.195 ± 0.001	80.746 ± 1.252
QML	beta53	0.16	0.128	-0.032 ± 0.002	-0.198 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.383 ± 0.010	0.129	-0.031	-19.52	0.0497	0.0587	0.965	88.306 ± 1.020	0.198 ± 0.001	72.883 ± 1.411
QML	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.160	0.000	-0.31	0.0487	0.0487	0.969	94.456 ± 0.727	0.195 ± 0.001	89.819 ± 0.960
QML	beta55	0.08	0.093	0.013 ± 0.002	0.161 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.476 ± 0.021	0.690 ± 0.019	0.096	0.016	19.61	0.0538	0.0561	0.977	89.718 ± 0.964	0.206 ± 0.004	50.000 ± 1.588
QML	beta56	0.08	0.094	0.014 ± 0.001	0.170 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.179 ± 0.009	0.423 ± 0.012	0.093	0.013	15.98	0.0294	0.0320	0.911	88.911 ± 0.997	0.111 ± 0.001	86.089 ± 1.099
QML	beta57	0.06	0.081	0.021 ± 0.002	0.354 ± 0.027	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.831 ± 0.038	0.911 ± 0.024	0.078	0.018	30.12	0.0487	0.0520	0.994	89.819 ± 0.960	0.197 ± 0.003	41.129 ± 1.562
QML	beta58	0.06	0.081	0.021 ± 0.001	0.358 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.285 ± 0.012	0.534 ± 0.012	0.081	0.021	34.33	0.0226	0.0306	0.946	83.367 ± 1.182	0.088 ± 0.001	93.044 ± 0.808
UPI	beta41	0.20	0.209	0.009 ± 0.002	0.045 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.151 ± 0.007	0.388 ± 0.010	0.210	0.010	5.07	0.0776	0.0782	1.023	96.371 ± 0.594	0.309 ± 0.002	74.597 ± 1.382
UPI	beta42	0.20	0.204	0.004 ± 0.003	0.022 ± 0.013	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.159 ± 0.008	0.398 ± 0.012	0.203	0.003	1.28	0.0727	0.0727	1.032	96.472 ± 0.586	0.322 ± 0.002	72.379 ± 1.420
UPI	beta43	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.197	-0.003	-1.31	0.0374	0.0375	0.991	94.052 ± 0.751	0.148 ± 0.001	99.899 ± 0.101
UPI	beta44	0.11	0.113	0.003 ± 0.002	0.028 ± 0.014	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.204 ± 0.010	0.451 ± 0.013	0.111	0.001	1.08	0.0498	0.0498	0.968	93.548 ± 0.780	0.188 ± 0.002	66.633 ± 1.497
UPI	beta45	0.11	0.114	0.004 ± 0.002	0.034 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.196 ± 0.010	0.443 ± 0.013	0.113	0.003	2.36	0.0459	0.0460	0.959	94.254 ± 0.739	0.183 ± 0.002	71.875 ± 1.428
UPI	beta46	0.08	0.075	-0.005 ± 0.001	-0.056 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.156 ± 0.007	0.395 ± 0.011	0.075	-0.005	-6.08	0.0302	0.0306	0.939	91.835 ± 0.869	0.115 ± 0.001	72.581 ± 1.416

Table S172

Simulation 2 - Condition 28: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.075	-0.005 ± 0.001	-0.064 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.149 ± 0.008	0.386 ± 0.012	0.075	-0.005	-6.28	0.0275	0.0280	0.949	91.331 ± 0.893	0.113 ± 0.001	74.194 ± 1.389
UPI	beta51	0.16	0.169	0.009 ± 0.002	0.053 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.320 ± 0.009	0.168	0.008	5.26	0.0511	0.0518	0.976	94.556 ± 0.720	0.193 ± 0.000	92.540 ± 0.834
UPI	beta52	0.16	0.173	0.013 ± 0.003	0.079 ± 0.017	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.087 ± 0.002	0.293 ± 0.014	0.541 ± 0.016	0.173	0.013	8.17	0.0853	0.0863	1.038	95.968 ± 0.625	0.349 ± 0.003	51.411 ± 1.587
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.126 ± 0.006	0.355 ± 0.010	0.156	-0.004	-2.22	0.0556	0.0557	0.979	94.859 ± 0.701	0.218 ± 0.001	79.839 ± 1.274
UPI	beta54	0.16	0.152	-0.008 ± 0.003	-0.047 ± 0.017	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.304 ± 0.014	0.551 ± 0.016	0.153	-0.007	-4.69	0.0856	0.0860	0.990	94.960 ± 0.695	0.341 ± 0.003	46.069 ± 1.583
UPI	beta55	0.08	0.093	0.013 ± 0.002	0.165 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.407 ± 0.019	0.638 ± 0.018	0.093	0.013	15.83	0.0451	0.0469	0.981	94.254 ± 0.739	0.190 ± 0.002	53.528 ± 1.584
UPI	beta56	0.08	0.077	-0.003 ± 0.001	-0.039 ± 0.012	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.133 ± 0.007	0.365 ± 0.011	0.077	-0.003	-4.15	0.0273	0.0275	0.909	91.028 ± 0.907	0.104 ± 0.001	80.242 ± 1.264
UPI	beta57	0.06	0.045	-0.015 ± 0.002	-0.253 ± 0.028	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.868 ± 0.043	0.932 ± 0.027	0.043	-0.017	-27.63	0.0513	0.0539	0.996	94.153 ± 0.745	0.210 ± 0.002	15.927 ± 1.162
UPI	beta58	0.06	0.058	-0.002 ± 0.001	-0.037 ± 0.017	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.271 ± 0.014	0.521 ± 0.016	0.058	-0.002	-4.16	0.0271	0.0272	0.955	93.448 ± 0.786	0.117 ± 0.002	57.056 ± 1.572

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S173

Simulation 2 - Condition 29: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.195	-0.005 ± 0.003	-0.027 ± 0.013	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.166 ± 0.007	0.408 ± 0.011	0.196	-0.004	-2.05	0.0804	0.0806	0.978	94.985 ± 0.691	0.312 ± 0.001	69.408 ± 1.459
LSAM	beta42	0.20	0.199	-0.001 ± 0.003	-0.005 ± 0.014	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.183 ± 0.008	0.428 ± 0.012	0.200	0.000	-0.00	0.0838	0.0838	0.967	94.283 ± 0.735	0.324 ± 0.001	67.603 ± 1.482
LSAM	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.202	0.002	0.83	0.0472	0.0473	1.015	95.286 ± 0.671	0.188 ± 0.001	98.696 ± 0.359
LSAM	beta44	0.11	0.111	0.001 ± 0.002	0.013 ± 0.017	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.279 ± 0.014	0.528 ± 0.016	0.112	0.002	1.89	0.0550	0.0551	0.907	92.678 ± 0.825	0.206 ± 0.002	58.475 ± 1.561
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.009 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.254 ± 0.012	0.504 ± 0.015	0.109	-0.001	-1.15	0.0537	0.0537	0.901	91.976 ± 0.860	0.196 ± 0.002	62.487 ± 1.533
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.013 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.194 ± 0.010	0.441 ± 0.013	0.081	0.001	1.69	0.0327	0.0327	0.848	90.973 ± 0.908	0.117 ± 0.001	75.627 ± 1.360
LSAM	beta47	0.08	0.081	0.001 ± 0.001	0.013 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.174 ± 0.009	0.417 ± 0.013	0.080	0.000	0.17	0.0302	0.0302	0.886	92.277 ± 0.845	0.116 ± 0.001	74.724 ± 1.376
LSAM	beta51	0.16	0.163	0.003 ± 0.002	0.016 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.008	0.413 ± 0.011	0.162	0.002	1.25	0.0632	0.0632	0.965	93.681 ± 0.771	0.250 ± 0.001	72.417 ± 1.415
LSAM	beta52	0.16	0.160	-0.000 ± 0.003	-0.001 ± 0.017	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.297 ± 0.014	0.545 ± 0.015	0.159	-0.001	-0.63	0.0830	0.0830	0.995	95.486 ± 0.657	0.340 ± 0.001	47.141 ± 1.581
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.192 ± 0.008	0.438 ± 0.012	0.163	0.003	1.77	0.0690	0.0690	0.981	94.784 ± 0.704	0.270 ± 0.001	64.193 ± 1.518
LSAM	beta54	0.16	0.153	-0.007 ± 0.003	-0.045 ± 0.018	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.311 ± 0.015	0.558 ± 0.016	0.154	-0.006	-3.73	0.0834	0.0836	0.941	92.778 ± 0.820	0.328 ± 0.001	46.239 ± 1.579
LSAM	beta55	0.08	0.084	0.004 ± 0.002	0.047 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.514 ± 0.025	0.717 ± 0.021	0.084	0.004	4.39	0.0535	0.0537	0.897	93.079 ± 0.804	0.201 ± 0.002	42.828 ± 1.567
LSAM	beta56	0.08	0.081	0.001 ± 0.001	0.009 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.235 ± 0.012	0.485 ± 0.015	0.080	0.000	0.34	0.0353	0.0353	0.852	89.870 ± 0.956	0.130 ± 0.001	67.703 ± 1.481
LSAM	beta57	0.06	0.058	-0.002 ± 0.002	-0.038 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.850 ± 0.040	0.922 ± 0.026	0.058	-0.002	-3.48	0.0540	0.0540	0.913	93.380 ± 0.787	0.198 ± 0.002	23.972 ± 1.352
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.015 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.314 ± 0.017	0.560 ± 0.018	0.060	0.000	0.80	0.0301	0.0301	0.871	89.669 ± 0.964	0.115 ± 0.001	56.971 ± 1.568
QML	beta41	0.20	0.201	0.001 ± 0.002	0.005 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.104 ± 0.005	0.323 ± 0.009	0.202	0.002	0.91	0.0661	0.0662	1.008	95.687 ± 0.643	0.255 ± 0.001	85.657 ± 1.110
QML	beta42	0.20	0.203	0.003 ± 0.002	0.015 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.114 ± 0.005	0.338 ± 0.009	0.203	0.003	1.47	0.0676	0.0677	0.987	94.885 ± 0.698	0.261 ± 0.001	85.657 ± 1.110
QML	beta43	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.198	-0.002	-0.80	0.0452	0.0452	1.011	94.885 ± 0.698	0.183 ± 0.001	98.897 ± 0.331
QML	beta44	0.11	0.096	-0.014 ± 0.002	-0.130 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.268 ± 0.013	0.517 ± 0.014	0.096	-0.014	-12.93	0.0563	0.0581	0.927	91.876 ± 0.865	0.200 ± 0.001	50.150 ± 1.584
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.037 ± 0.015	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.227 ± 0.010	0.477 ± 0.013	0.105	-0.005	-4.32	0.0514	0.0516	0.915	92.177 ± 0.850	0.188 ± 0.001	61.484 ± 1.541
QML	beta46	0.08	0.092	0.012 ± 0.001	0.154 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.182 ± 0.009	0.426 ± 0.012	0.091	0.011	13.92	0.0300	0.0320	0.887	90.070 ± 0.947	0.111 ± 0.001	85.757 ± 1.107
QML	beta47	0.08	0.093	0.013 ± 0.001	0.167 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.168 ± 0.009	0.409 ± 0.012	0.092	0.012	15.53	0.0285	0.0310	0.918	89.368 ± 0.976	0.108 ± 0.001	88.766 ± 1.000
QML	beta51	0.16	0.170	0.010 ± 0.002	0.064 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.177 ± 0.008	0.420 ± 0.012	0.169	0.009	5.57	0.0614	0.0620	0.964	93.480 ± 0.782	0.251 ± 0.001	76.329 ± 1.346
QML	beta52	0.16	0.143	-0.017 ± 0.002	-0.106 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.009	0.446 ± 0.012	0.141	-0.019	-11.90	0.0680	0.0706	1.003	95.386 ± 0.664	0.272 ± 0.001	54.463 ± 1.577
QML	beta53	0.16	0.136	-0.024 ± 0.002	-0.149 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.206 ± 0.009	0.454 ± 0.012	0.137	-0.023	-14.55	0.0668	0.0707	0.981	92.277 ± 0.845	0.264 ± 0.001	52.457 ± 1.582
QML	beta54	0.16	0.155	-0.005 ± 0.002	-0.033 ± 0.014	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.210 ± 0.010	0.458 ± 0.013	0.155	-0.005	-3.41	0.0724	0.0726	0.956	92.678 ± 0.825	0.274 ± 0.001	59.178 ± 1.557
QML	beta55	0.08	0.086	0.006 ± 0.002	0.075 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.503 ± 0.025	0.709 ± 0.021	0.085	0.005	6.32	0.0546	0.0549	0.911	92.578 ± 0.830	0.201 ± 0.002	43.731 ± 1.571
QML	beta56	0.08	0.086	0.006 ± 0.001	0.077 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.241 ± 0.012	0.491 ± 0.015	0.086	0.006	7.13	0.0352	0.0356	0.850	88.867 ± 0.996	0.129 ± 0.001	72.016 ± 1.422
QML	beta57	0.06	0.066	0.006 ± 0.002	0.103 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.731 ± 0.034	0.855 ± 0.024	0.067	0.007	12.38	0.0490	0.0495	0.929	93.581 ± 0.776	0.185 ± 0.002	32.798 ± 1.487
QML	beta58	0.06	0.069	0.009 ± 0.001	0.153 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.288 ± 0.016	0.536 ± 0.017	0.068	0.008	13.48	0.0279	0.0290	0.884	89.067 ± 0.988	0.107 ± 0.002	71.615 ± 1.428
UPI	beta41	0.20	0.205	0.005 ± 0.002	0.026 ± 0.012	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.156 ± 0.007	0.394 ± 0.011	0.207	0.007	3.27	0.0755	0.0758	0.975	94.985 ± 0.691	0.301 ± 0.001	77.131 ± 1.330
UPI	beta42	0.20	0.206	0.006 ± 0.003	0.032 ± 0.013	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.172 ± 0.008	0.415 ± 0.011	0.204	0.004	2.19	0.0837	0.0838	0.971	94.784 ± 0.704	0.315 ± 0.001	72.116 ± 1.420
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.201	0.001	0.67	0.0478	0.0478	1.001	94.483 ± 0.723	0.184 ± 0.001	98.796 ± 0.345
UPI	beta44	0.11	0.108	-0.002 ± 0.002	-0.014 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.262 ± 0.013	0.512 ± 0.015	0.109	-0.001	-0.68	0.0548	0.0548	0.928	92.578 ± 0.830	0.205 ± 0.002	56.971 ± 1.568
UPI	beta45	0.11	0.110	0.000 ± 0.002	0.003 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.245 ± 0.012	0.495 ± 0.015	0.108	-0.002	-1.86	0.0522	0.0522	0.926	93.180 ± 0.798	0.198 ± 0.002	60.682 ± 1.547
UPI	beta46	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.181 ± 0.009	0.425 ± 0.013	0.079	-0.001	-0.88	0.0318	0.0318	0.860	89.769 ± 0.960	0.115 ± 0.001	74.323 ± 1.384

Table S173

Simulation 2 - Condition 29: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.166 ± 0.009	0.407 ± 0.013	0.078	-0.002	-2.23	0.0303	0.0303	0.889	90.973 ± 0.908	0.113 ± 0.001	76.128 ± 1.350
UPI	beta51	0.16	0.167	0.007 ± 0.002	0.041 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.008	0.414 ± 0.011	0.164	0.004	2.75	0.0623	0.0624	0.954	93.480 ± 0.782	0.246 ± 0.001	75.627 ± 1.360
UPI	beta52	0.16	0.166	0.006 ± 0.003	0.040 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.281 ± 0.013	0.530 ± 0.015	0.163	0.003	1.56	0.0821	0.0821	0.987	94.784 ± 0.704	0.327 ± 0.001	50.953 ± 1.583
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.187 ± 0.008	0.432 ± 0.012	0.161	0.001	0.72	0.0660	0.0660	0.982	94.283 ± 0.735	0.266 ± 0.001	64.794 ± 1.513
UPI	beta54	0.16	0.155	-0.005 ± 0.003	-0.031 ± 0.017	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.303 ± 0.014	0.550 ± 0.015	0.156	-0.004	-2.81	0.0842	0.0843	0.942	92.879 ± 0.815	0.325 ± 0.001	47.944 ± 1.582
UPI	beta55	0.08	0.083	0.003 ± 0.002	0.031 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.493 ± 0.023	0.702 ± 0.020	0.082	0.002	2.04	0.0566	0.0566	0.899	92.377 ± 0.840	0.198 ± 0.002	42.026 ± 1.563
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.013 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.230 ± 0.012	0.479 ± 0.015	0.080	0.000	-0.38	0.0345	0.0345	0.854	88.867 ± 0.996	0.128 ± 0.002	68.104 ± 1.476
UPI	beta57	0.06	0.057	-0.003 ± 0.002	-0.056 ± 0.028	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.805 ± 0.038	0.897 ± 0.026	0.057	-0.003	-4.22	0.0518	0.0519	0.908	93.480 ± 0.782	0.191 ± 0.002	27.182 ± 1.409
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.007 ± 0.018	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.305 ± 0.016	0.553 ± 0.017	0.060	0.000	-0.58	0.0301	0.0301	0.874	90.371 ± 0.934	0.114 ± 0.001	57.573 ± 1.565

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S174

Simulation 2 - Condition 30: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.197	-0.003	-1.26	0.0494	0.0495	1.012	95.796 ± 0.635	0.191 ± 0.001	98.398 ± 0.397
LSAM	beta42	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.201	0.001	0.40	0.0519	0.0519	0.974	93.694 ± 0.769	0.199 ± 0.001	95.996 ± 0.620
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.200	0.000	0.20	0.0318	0.0318	0.988	94.995 ± 0.690	0.118 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.116	0.006 ± 0.001	0.054 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.096 ± 0.005	0.309 ± 0.009	0.114	0.004	3.88	0.0330	0.0333	0.957	93.293 ± 0.791	0.126 ± 0.001	93.093 ± 0.802
LSAM	beta45	0.11	0.110	-0.000 ± 0.001	-0.002 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.110	0.000	-0.11	0.0307	0.0307	0.985	93.894 ± 0.758	0.121 ± 0.001	93.093 ± 0.802
LSAM	beta46	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.053 ± 0.002	0.231 ± 0.007	0.079	-0.001	-0.67	0.0181	0.0181	0.935	93.594 ± 0.775	0.068 ± 0.001	98.098 ± 0.432
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.016 ± 0.007	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.079	-0.001	-0.98	0.0171	0.0171	0.922	91.391 ± 0.887	0.068 ± 0.001	97.397 ± 0.504
LSAM	beta51	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.160	0.000	-0.05	0.0402	0.0402	0.979	94.895 ± 0.696	0.157 ± 0.000	98.298 ± 0.409
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.120 ± 0.005	0.347 ± 0.009	0.158	-0.002	-1.06	0.0551	0.0551	0.964	94.595 ± 0.715	0.210 ± 0.001	83.784 ± 1.166
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.004	0.274 ± 0.008	0.161	0.001	0.53	0.0415	0.0415	0.977	93.493 ± 0.780	0.168 ± 0.000	94.795 ± 0.703
LSAM	beta54	0.16	0.156	-0.004 ± 0.002	-0.024 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.114 ± 0.005	0.338 ± 0.009	0.156	-0.004	-2.31	0.0544	0.0545	0.951	94.394 ± 0.728	0.201 ± 0.001	84.284 ± 1.151
LSAM	beta55	0.08	0.083	0.003 ± 0.001	0.038 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.159 ± 0.007	0.398 ± 0.011	0.082	0.002	2.33	0.0317	0.0317	0.939	92.092 ± 0.854	0.117 ± 0.001	76.877 ± 1.334
LSAM	beta56	0.08	0.080	-0.000 ± 0.001	-0.004 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.070 ± 0.003	0.265 ± 0.008	0.080	0.000	-0.17	0.0203	0.0203	0.905	92.793 ± 0.818	0.075 ± 0.001	95.195 ± 0.677
LSAM	beta57	0.06	0.057	-0.003 ± 0.001	-0.043 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.281 ± 0.014	0.530 ± 0.015	0.058	-0.002	-3.39	0.0304	0.0305	0.950	92.893 ± 0.813	0.118 ± 0.001	49.550 ± 1.582
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.013 ± 0.010	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.096 ± 0.005	0.309 ± 0.009	0.060	0.000	0.24	0.0174	0.0174	0.922	92.292 ± 0.844	0.067 ± 0.001	90.791 ± 0.915
QML	beta41	0.20	0.205	0.005 ± 0.001	0.026 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.005	0.205	0.005	2.28	0.0411	0.0413	1.041	94.995 ± 0.690	0.163 ± 0.002	99.600 ± 0.200
QML	beta42	0.20	0.203	0.003 ± 0.001	0.017 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.045 ± 0.002	0.213 ± 0.006	0.205	0.005	2.67	0.0433	0.0436	0.987	94.795 ± 0.703	0.165 ± 0.001	99.700 ± 0.173
QML	beta43	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.197	-0.003	-1.49	0.0310	0.0312	0.990	94.895 ± 0.696	0.116 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.103	-0.007 ± 0.001	-0.066 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.096 ± 0.004	0.310 ± 0.008	0.102	-0.008	-7.71	0.0320	0.0331	0.965	93.694 ± 0.769	0.126 ± 0.001	87.688 ± 1.040
QML	beta45	0.11	0.107	-0.003 ± 0.001	-0.030 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.078 ± 0.004	0.280 ± 0.008	0.107	-0.003	-3.02	0.0306	0.0308	0.990	93.493 ± 0.780	0.119 ± 0.001	92.893 ± 0.813
QML	beta46	0.08	0.091	0.011 ± 0.001	0.135 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.067 ± 0.003	0.259 ± 0.006	0.091	0.011	13.81	0.0178	0.0210	0.958	89.489 ± 0.970	0.066 ± 0.000	99.900 ± 0.100
QML	beta47	0.08	0.091	0.011 ± 0.001	0.141 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.068 ± 0.003	0.261 ± 0.006	0.091	0.011	14.20	0.0170	0.0204	0.960	88.288 ± 1.017	0.066 ± 0.000	99.600 ± 0.200
QML	beta51	0.16	0.169	0.009 ± 0.001	0.056 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.168	0.008	4.95	0.0401	0.0409	0.981	94.394 ± 0.728	0.158 ± 0.000	98.699 ± 0.359
QML	beta52	0.16	0.143	-0.017 ± 0.001	-0.108 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.142	-0.018	-11.21	0.0466	0.0499	0.962	92.092 ± 0.854	0.170 ± 0.000	89.089 ± 0.986
QML	beta53	0.16	0.134	-0.026 ± 0.001	-0.162 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.312 ± 0.008	0.135	-0.025	-15.47	0.0400	0.0470	0.985	88.789 ± 0.998	0.165 ± 0.001	88.088 ± 1.025
QML	beta54	0.16	0.157	-0.003 ± 0.001	-0.017 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.157	-0.003	-1.66	0.0461	0.0462	0.966	94.795 ± 0.703	0.172 ± 0.002	93.493 ± 0.780
QML	beta55	0.08	0.086	0.006 ± 0.001	0.070 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.177 ± 0.008	0.421 ± 0.011	0.085	0.005	6.02	0.0329	0.0333	0.945	92.192 ± 0.849	0.123 ± 0.001	74.775 ± 1.374
QML	beta56	0.08	0.085	0.005 ± 0.001	0.066 ± 0.009	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	0.079 ± 0.004	0.281 ± 0.008	0.086	0.006	7.48	0.0217	0.0225	0.906	90.791 ± 0.915	0.078 ± 0.001	96.396 ± 0.590
QML	beta57	0.06	0.067	0.007 ± 0.001	0.119 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.273 ± 0.012	0.522 ± 0.014	0.067	0.007	11.55	0.0307	0.0315	0.978	93.093 ± 0.802	0.117 ± 0.001	62.062 ± 1.535
QML	beta58	0.06	0.070	0.010 ± 0.001	0.165 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.070	0.010	16.29	0.0166	0.0193	0.932	87.688 ± 1.040	0.064 ± 0.000	96.897 ± 0.549
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.020 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.202	0.002	0.96	0.0496	0.0496	0.983	94.394 ± 0.728	0.186 ± 0.001	98.699 ± 0.359
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.205	0.005	2.40	0.0532	0.0535	0.956	93.594 ± 0.775	0.195 ± 0.001	96.697 ± 0.565
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.202	0.002	0.85	0.0315	0.0316	0.965	94.094 ± 0.746	0.116 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.113	0.003 ± 0.001	0.031 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.089 ± 0.004	0.298 ± 0.009	0.112	0.002	2.12	0.0313	0.0313	0.964	93.393 ± 0.786	0.123 ± 0.001	93.594 ± 0.775
UPI	beta45	0.11	0.109	-0.001 ± 0.001	-0.008 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.080 ± 0.004	0.283 ± 0.008	0.109	-0.001	-1.16	0.0295	0.0296	0.993	93.493 ± 0.780	0.121 ± 0.001	93.393 ± 0.786
UPI	beta46	0.08	0.079	-0.001 ± 0.001	-0.014 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.054 ± 0.003	0.231 ± 0.007	0.080	0.000	-0.59	0.0177	0.0177	0.918	92.192 ± 0.849	0.067 ± 0.001	97.898 ± 0.454

Table S174

Simulation 2 - Condition 30: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.016 ± 0.007	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.079	-0.001	-1.17	0.0176	0.0176	0.916	91.592 ± 0.878	0.067 ± 0.001	97.497 ± 0.494
UPI	beta51	0.16	0.163	0.003 ± 0.001	0.018 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.161	0.001	0.84	0.0400	0.0400	0.974	94.494 ± 0.722	0.155 ± 0.000	98.599 ± 0.372
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.018 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.005	0.341 ± 0.009	0.161	0.001	0.90	0.0556	0.0556	0.956	93.994 ± 0.752	0.205 ± 0.001	86.386 ± 1.085
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.073 ± 0.003	0.271 ± 0.008	0.161	0.001	0.61	0.0403	0.0403	0.977	93.594 ± 0.775	0.166 ± 0.000	95.195 ± 0.677
UPI	beta54	0.16	0.157	-0.003 ± 0.002	-0.019 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.157	-0.003	-1.68	0.0530	0.0531	0.955	94.494 ± 0.722	0.200 ± 0.001	84.985 ± 1.130
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.023 ± 0.012	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.156 ± 0.007	0.395 ± 0.011	0.080	0.000	0.55	0.0312	0.0312	0.936	93.193 ± 0.797	0.116 ± 0.001	76.076 ± 1.350
UPI	beta56	0.08	0.080	-0.000 ± 0.001	-0.005 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.071 ± 0.003	0.266 ± 0.008	0.080	0.000	-0.38	0.0208	0.0208	0.900	92.192 ± 0.849	0.075 ± 0.001	95.295 ± 0.670
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.037 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.272 ± 0.013	0.522 ± 0.015	0.059	-0.001	-1.01	0.0312	0.0312	0.945	93.093 ± 0.802	0.116 ± 0.001	52.252 ± 1.580
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.016 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.093 ± 0.005	0.305 ± 0.009	0.061	0.001	1.14	0.0168	0.0168	0.931	92.893 ± 0.813	0.067 ± 0.001	91.692 ± 0.873

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S175

Simulation 2 - Condition 31: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.206	0.006 ± 0.019	0.032 ± 0.097	0.006 ± 0.002	0.075 ± 0.013	0.005 ± 0.002	0.073 ± 0.017	0.133 ± 0.041	0.364 ± 0.083	0.203	0.003	1.70	0.0984	0.0984	1.884	100.000 ± 0.000	0.555 ± 0.064	13.333 ± 8.777
LSAM	beta42	0.20	0.210	0.010 ± 0.029	0.052 ± 0.143	0.012 ± 0.003	0.111 ± 0.017	0.012 ± 0.003	0.108 ± 0.022	0.291 ± 0.073	0.539 ± 0.108	0.210	0.010	4.81	0.1552	0.1555	1.428	100.000 ± 0.000	0.622 ± 0.081	40.000 ± 12.649
LSAM	beta43	0.20	0.172	-0.028 ± 0.028	-0.142 ± 0.138	0.011 ± 0.003	0.107 ± 0.017	0.011 ± 0.002	0.107 ± 0.019	0.287 ± 0.056	0.536 ± 0.093	0.163	-0.037	-18.30	0.1260	0.1312	0.880	93.333 ± 6.441	0.369 ± 0.025	40.000 ± 12.649
LSAM	beta44	0.11	0.235	0.125 ± 0.037	1.136 ± 0.335	0.020 ± 0.007	0.143 ± 0.032	0.035 ± 0.010	0.186 ± 0.031	2.859 ± 0.798	1.691 ± 0.285	0.217	0.107	96.84	0.1458	0.1806	1.079	93.333 ± 6.441	0.603 ± 0.070	40.000 ± 12.649
LSAM	beta45	0.11	0.077	-0.033 ± 0.028	-0.304 ± 0.258	0.012 ± 0.003	0.110 ± 0.018	0.012 ± 0.004	0.111 ± 0.025	1.021 ± 0.351	1.010 ± 0.232	0.104	-0.006	-5.01	0.0634	0.0636	1.024	86.667 ± 8.777	0.440 ± 0.032	13.333 ± 8.777
LSAM	beta46	0.08	-0.036	-0.116 ± 0.034	-1.452 ± 0.429	0.018 ± 0.007	0.133 ± 0.034	0.030 ± 0.010	0.173 ± 0.036	4.685 ± 1.626	2.164 ± 0.451	-0.051	-0.131	-164.12	0.1181	0.1766	1.024	80.000 ± 10.328	0.534 ± 0.078	0.000 ± 0.000
LSAM	beta47	0.08	-0.056	-0.136 ± 0.037	-1.697 ± 0.459	0.020 ± 0.005	0.142 ± 0.022	0.037 ± 0.009	0.193 ± 0.027	5.834 ± 1.389	2.415 ± 0.337	-0.028	-0.108	-135.58	0.1956	0.2236	1.185	86.667 ± 8.777	0.661 ± 0.091	0.000 ± 0.000
LSAM	beta51	0.16	0.199	0.039 ± 0.035	0.243 ± 0.221	0.019 ± 0.005	0.137 ± 0.024	0.019 ± 0.005	0.138 ± 0.029	0.744 ± 0.214	0.862 ± 0.181	0.229	0.069	43.07	0.1415	0.1574	0.909	86.667 ± 8.777	0.488 ± 0.030	33.333 ± 12.172
LSAM	beta52	0.16	0.134	-0.026 ± 0.029	-0.165 ± 0.184	0.013 ± 0.003	0.114 ± 0.018	0.013 ± 0.003	0.113 ± 0.022	0.502 ± 0.125	0.709 ± 0.137	0.162	0.002	1.51	0.1165	0.1165	1.112	93.333 ± 6.441	0.497 ± 0.036	20.000 ± 10.328
LSAM	beta53	0.16	0.115	-0.045 ± 0.024	-0.279 ± 0.152	0.009 ± 0.002	0.094 ± 0.012	0.010 ± 0.002	0.101 ± 0.015	0.400 ± 0.071	0.633 ± 0.095	0.092	-0.068	-42.34	0.1127	0.1315	1.137	100.000 ± 0.000	0.419 ± 0.033	20.000 ± 10.328
LSAM	beta54	0.16	0.200	0.040 ± 0.039	0.250 ± 0.246	0.023 ± 0.006	0.153 ± 0.027	0.023 ± 0.006	0.153 ± 0.030	0.911 ± 0.236	0.954 ± 0.189	0.185	0.025	15.74	0.1170	0.1197	0.825	100.000 ± 0.000	0.493 ± 0.041	46.667 ± 12.881
LSAM	beta55	0.08	0.104	0.024 ± 0.049	0.295 ± 0.612	0.036 ± 0.015	0.190 ± 0.052	0.034 ± 0.015	0.185 ± 0.056	5.328 ± 2.420	2.308 ± 0.704	0.095	0.015	18.32	0.1851	0.1857	0.976	93.333 ± 6.441	0.725 ± 0.063	6.667 ± 6.441
LSAM	beta56	0.08	0.072	-0.008 ± 0.046	-0.104 ± 0.570	0.031 ± 0.008	0.177 ± 0.030	0.029 ± 0.007	0.171 ± 0.035	4.564 ± 1.168	2.136 ± 0.441	0.068	-0.012	-15.09	0.2035	0.2039	1.017	93.333 ± 6.441	0.704 ± 0.094	6.667 ± 6.441
LSAM	beta57	0.06	0.013	-0.047 ± 0.034	-0.787 ± 0.569	0.017 ± 0.004	0.132 ± 0.022	0.019 ± 0.006	0.136 ± 0.030	5.151 ± 1.670	2.269 ± 0.495	0.037	-0.023	-38.01	0.1095	0.1118	0.908	93.333 ± 6.441	0.471 ± 0.052	0.000 ± 0.000
LSAM	beta58	0.06	-0.042	-0.102 ± 0.034	-1.692 ± 0.559	0.017 ± 0.005	0.130 ± 0.025	0.026 ± 0.009	0.161 ± 0.033	7.236 ± 2.463	2.690 ± 0.550	-0.048	-0.108	-179.38	0.0800	0.1341	0.936	73.333 ± 11.418	0.476 ± 0.057	0.000 ± 0.000
QML	beta41	0.20	0.186	-0.014 ± 0.013	-0.071 ± 0.063	0.002 ± 0.001	0.049 ± 0.008	0.002 ± 0.001	0.049 ± 0.010	0.060 ± 0.017	0.245 ± 0.049	0.180	-0.020	-10.09	0.0815	0.0839	1.616	100.000 ± 0.000	0.307 ± 0.013	66.667 ± 12.172
QML	beta42	0.20	0.179	-0.021 ± 0.015	-0.106 ± 0.077	0.004 ± 0.001	0.060 ± 0.008	0.004 ± 0.001	0.061 ± 0.011	0.094 ± 0.022	0.307 ± 0.056	0.198	-0.002	-1.22	0.0790	0.0790	1.369	100.000 ± 0.000	0.320 ± 0.012	60.000 ± 12.649
QML	beta43	0.20	0.179	-0.021 ± 0.021	-0.107 ± 0.106	0.007 ± 0.002	0.082 ± 0.012	0.007 ± 0.002	0.082 ± 0.016	0.168 ± 0.043	0.410 ± 0.080	0.178	-0.022	-10.97	0.0934	0.0959	0.917	93.333 ± 6.441	0.295 ± 0.013	60.000 ± 12.649
QML	beta44	0.11	0.196	0.086 ± 0.025	0.778 ± 0.231	0.010 ± 0.003	0.098 ± 0.020	0.016 ± 0.003	0.128 ± 0.016	1.351 ± 0.277	1.162 ± 0.150	0.195	0.085	77.34	0.1104	0.1394	1.112	93.333 ± 6.441	0.428 ± 0.029	53.333 ± 12.881
QML	beta45	0.11	0.073	-0.037 ± 0.027	-0.334 ± 0.244	0.011 ± 0.003	0.104 ± 0.017	0.011 ± 0.004	0.107 ± 0.025	0.946 ± 0.335	0.973 ± 0.226	0.105	-0.005	-4.31	0.1150	0.1151	0.901	86.667 ± 8.777	0.367 ± 0.035	6.667 ± 6.441
QML	beta46	0.08	-0.019	-0.099 ± 0.031	-1.244 ± 0.393	0.015 ± 0.007	0.122 ± 0.038	0.024 ± 0.006	0.154 ± 0.023	3.706 ± 0.877	1.925 ± 0.289	-0.012	-0.092	-114.72	0.1159	0.1479	0.618	53.333 ± 12.881	0.295 ± 0.032	26.667 ± 11.418
QML	beta47	0.08	-0.032	-0.112 ± 0.016	-1.398 ± 0.205	0.004 ± 0.001	0.063 ± 0.009	0.016 ± 0.003	0.128 ± 0.013	2.542 ± 0.510	1.594 ± 0.166	-0.040	-0.120	-149.87	0.0945	0.1526	1.145	66.667 ± 12.172	0.285 ± 0.025	0.000 ± 0.000
QML	beta51	0.16	0.194	0.034 ± 0.044	0.215 ± 0.272	0.028 ± 0.011	0.169 ± 0.042	0.028 ± 0.010	0.166 ± 0.041	1.082 ± 0.386	1.040 ± 0.257	0.234	0.074	45.99	0.1384	0.1568	0.713	80.000 ± 10.328	0.471 ± 0.025	33.333 ± 12.172
QML	beta52	0.16	0.177	0.017 ± 0.023	0.109 ± 0.143	0.008 ± 0.002	0.089 ± 0.014	0.008 ± 0.002	0.087 ± 0.018	0.298 ± 0.081	0.546 ± 0.112	0.192	0.032	20.25	0.1031	0.1080	1.034	93.333 ± 6.441	0.359 ± 0.017	46.667 ± 12.881
QML	beta53	0.16	0.116	-0.044 ± 0.020	-0.274 ± 0.128	0.006 ± 0.002	0.079 ± 0.013	0.008 ± 0.002	0.088 ± 0.013	0.304 ± 0.061	0.551 ± 0.083	0.116	-0.044	-27.54	0.0977	0.1072	1.116	100.000 ± 0.000	0.346 ± 0.013	20.000 ± 10.328
QML	beta54	0.16	0.178	0.018 ± 0.024	0.111 ± 0.151	0.009 ± 0.002	0.094 ± 0.013	0.008 ± 0.001	0.092 ± 0.016	0.332 ± 0.056	0.576 ± 0.098	0.172	0.012	7.34	0.0949	0.0956	0.864	93.333 ± 6.441	0.317 ± 0.011	53.333 ± 12.881
QML	beta55	0.08	0.116	0.036 ± 0.041	0.453 ± 0.506	0.025 ± 0.008	0.157 ± 0.035	0.024 ± 0.010	0.156 ± 0.043	3.794 ± 1.554	1.948 ± 0.533	0.051	-0.029	-35.76	0.1659	0.1684	0.816	86.667 ± 8.777	0.502 ± 0.030	13.333 ± 8.777
QML	beta56	0.08	0.049	-0.031 ± 0.030	-0.382 ± 0.379	0.014 ± 0.002	0.117 ± 0.014	0.014 ± 0.003	0.117 ± 0.022	2.157 ± 0.493	1.469 ± 0.272	0.030	-0.050	-62.07	0.1705	0.1776	1.049	93.333 ± 6.441	0.483 ± 0.026	0.000 ± 0.000
QML	beta57	0.06	0.015	-0.045 ± 0.026	-0.742 ± 0.432	0.010 ± 0.005	0.100 ± 0.036	0.011 ± 0.004	0.107 ± 0.026	3.164 ± 1.165	1.779 ± 0.431	0.017	-0.043	-72.34	0.0592	0.0734	0.624	73.333 ± 11.418	0.245 ± 0.020	13.333 ± 8.777
QML	beta58	0.06	-0.022	-0.082 ± 0.017	-1.372 ± 0.285	0.004 ± 0.001	0.066 ± 0.011	0.011 ± 0.003	0.104 ± 0.017	3.023 ± 0.904	1.739 ± 0.287	-0.018	-0.078	-129.36	0.0589	0.0975	0.899	73.333 ± 11.418	0.234 ± 0.016	6.667 ± 6.441
UPI	beta41	0.20	0.105	-0.095 ± 0.107	-0.476 ± 0.536	0.172 ± 0.099	0.415 ± 0.176	0.170 ± 0.112	0.412 ± 0.188	4.245 ± 2.808	2.060 ± 0.942	0.087	-0.113	-56.49	0.2773	0.2995	1.816	100.000 ± 0.000	2.954 ± 1.709	6.667 ± 6.441
UPI	beta42	0.20	0.075	-0.125 ± 0.063	-0.625 ± 0.316	0.060 ± 0.021	0.245 ± 0.056	0.072 ± 0.026	0.267 ± 0.061	1.788 ± 0.640	1.337 ± 0.304	0.175	-0.025	-12.52	0.3194	0.3203	1.258	93.333 ± 6.441	1.206 ± 0.190	0.000 ± 0.000
UPI	beta43	0.20	0.154	-0.046 ± 0.055	-0.228 ± 0.273	0.045 ± 0.021	0.212 ± 0.067	0.044 ± 0.017	0.210 ± 0.055	1.099 ± 0.414	1.048 ± 0.274	0.088	-0.112	-55.79	0.1119	0.1580	1.425	66.667 ± 12.172	1.183 ± 0.597	13.333 ± 8.777
UPI	beta44	0.11	0.194	0.084 ± 0.095	0.766 ± 0.866	0.136 ± 0.061	0.369 ± 0.116	0.134 ± 0.072	0.366 ± 0.132	11.089 ± 5.924	3.330 ± 1.197	0.189	0.079	72.12	0.2385	0.2514	1.482	100.000 ± 0.000	2.143 ± 0.459	6.667 ± 6.441
UPI	beta45	0.11	0.049	-0.061 ± 0.047	-0.553 ± 0.425	0.033 ± 0.010	0.181 ± 0.035	0.034 ± 0.012	0.185 ± 0.044	2.834 ± 1.032	1.683 ± 0.404	0.089	-0.021	-19.34	0.1923	0.1935	1.492	100.000 ± 0.000	1.058 ± 0.183	0.000 ± 0.000
UPI	beta46	0.08	0.161	0.081 ± 0.125	1.014 ± 1.568	0.236 ± 0.119	0.486 ± 0.168	0.227 ± 0.126	0.476 ± 0.174	35.435 ± 19.698	5.953 ± 2.178	0.116	0.036	44.96	0.3808	0.3825	2.000	86.667 ± 8.777	3.808 ± 1.557	0.000 ± 0.000

Table S175

Simulation 2 - Condition 31: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.038	-0.042 ± 0.109	-0.525 ± 1.358	0.177 ± 0.064	0.421 ± 0.105	0.167 ± 0.071	0.409 ± 0.118	26.078 ± 11.035	5.107 ± 1.481	0.072	-0.008	-10.33	0.5225	0.5225	1.912	100.000 ± 0.000	3.152 ± 0.916	0.000 ± 0.000
UPI	beta51	0.16	0.275	0.115 ± 0.020	0.721 ± 0.124	0.006 ± 0.002	0.077 ± 0.014	0.019 ± 0.006	0.137 ± 0.023	0.734 ± 0.221	0.857 ± 0.141	0.278	0.118	74.00	0.0972	0.1532	1.924	93.333 ± 6.441	0.579 ± 0.065	53.333 ± 12.881
UPI	beta52	0.16	0.087	-0.073 ± 0.052	-0.457 ± 0.326	0.041 ± 0.017	0.202 ± 0.059	0.043 ± 0.023	0.208 ± 0.072	1.695 ± 0.902	1.302 ± 0.452	0.116	-0.044	-27.62	0.2210	0.2254	1.247	100.000 ± 0.000	0.987 ± 0.238	13.333 ± 8.777
UPI	beta53	0.16	0.136	-0.024 ± 0.034	-0.150 ± 0.211	0.017 ± 0.005	0.131 ± 0.025	0.017 ± 0.004	0.129 ± 0.026	0.649 ± 0.157	0.805 ± 0.160	0.079	-0.081	-50.68	0.1419	0.1635	1.172	86.667 ± 8.777	0.602 ± 0.138	26.667 ± 11.418
UPI	beta54	0.16	0.211	0.051 ± 0.073	0.319 ± 0.457	0.080 ± 0.037	0.283 ± 0.092	0.077 ± 0.042	0.278 ± 0.103	3.027 ± 1.654	1.740 ± 0.641	0.150	-0.010	-5.95	0.2901	0.2902	1.252	73.333 ± 11.418	1.390 ± 0.342	0.000 ± 0.000
UPI	beta55	0.08	0.180	0.100 ± 0.053	1.245 ± 0.657	0.041 ± 0.015	0.204 ± 0.050	0.049 ± 0.022	0.221 ± 0.064	7.599 ± 3.482	2.757 ± 0.802	0.122	0.042	51.99	0.2449	0.2484	1.349	93.333 ± 6.441	1.077 ± 0.161	6.667 ± 6.441
UPI	beta56	0.08	0.093	0.013 ± 0.041	0.167 ± 0.506	0.025 ± 0.007	0.157 ± 0.029	0.023 ± 0.007	0.152 ± 0.034	3.618 ± 1.102	1.902 ± 0.426	0.048	-0.032	-40.54	0.2338	0.2361	1.666	93.333 ± 6.441	1.025 ± 0.151	0.000 ± 0.000
UPI	beta57	0.06	0.035	-0.025 ± 0.083	-0.417 ± 1.387	0.104 ± 0.045	0.322 ± 0.096	0.098 ± 0.047	0.312 ± 0.101	27.119 ± 12.966	5.208 ± 1.687	-0.010	-0.070	-117.05	0.3665	0.3732	1.619	100.000 ± 0.000	2.046 ± 0.568	0.000 ± 0.000
UPI	beta58	0.06	-0.161	-0.221 ± 0.075	-3.680 ± 1.249	0.084 ± 0.048	0.290 ± 0.124	0.127 ± 0.076	0.357 ± 0.139	35.377 ± 21.185	5.948 ± 2.313	-0.091	-0.151	-252.34	0.2045	0.2544	1.204	93.333 ± 6.441	1.370 ± 0.333	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S176

Simulation 2 - Condition 32: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.017	0.005 ± 0.083	0.003 ± 0.001	0.058 ± 0.011	0.003 ± 0.001	0.055 ± 0.013	0.076 ± 0.022	0.276 ± 0.066	0.189	-0.011	-5.58	0.0769	0.0777	1.186	100.000 ± 0.000	0.268 ± 0.013	91.667 ± 7.979
LSAM	beta42	0.20	0.212	0.012 ± 0.015	0.060 ± 0.074	0.003 ± 0.001	0.051 ± 0.017	0.003 ± 0.001	0.050 ± 0.014	0.063 ± 0.023	0.252 ± 0.069	0.230	0.030	14.85	0.0562	0.0636	1.410	91.667 ± 7.979	0.282 ± 0.014	91.667 ± 7.979
LSAM	beta43	0.20	0.162	-0.038 ± 0.016	-0.188 ± 0.082	0.003 ± 0.001	0.057 ± 0.018	0.004 ± 0.001	0.066 ± 0.012	0.109 ± 0.029	0.330 ± 0.060	0.160	-0.040	-20.17	0.0600	0.0723	0.875	83.333 ± 10.758	0.194 ± 0.005	100.000 ± 0.000
LSAM	beta44	0.11	0.185	0.075 ± 0.025	0.680 ± 0.230	0.008 ± 0.002	0.087 ± 0.014	0.013 ± 0.004	0.112 ± 0.020	1.042 ± 0.310	1.021 ± 0.182	0.203	0.093	84.78	0.0694	0.1162	0.772	75.000 ± 12.500	0.265 ± 0.017	75.000 ± 12.500
LSAM	beta45	0.11	0.123	0.013 ± 0.020	0.117 ± 0.180	0.005 ± 0.002	0.068 ± 0.024	0.004 ± 0.002	0.067 ± 0.020	0.369 ± 0.154	0.607 ± 0.185	0.127	0.017	15.80	0.0662	0.0684	0.946	91.667 ± 7.979	0.254 ± 0.009	58.333 ± 14.232
LSAM	beta46	0.08	-0.016	-0.096 ± 0.018	-1.202 ± 0.220	0.004 ± 0.001	0.061 ± 0.013	0.013 ± 0.003	0.112 ± 0.017	1.977 ± 0.538	1.406 ± 0.207	0.008	-0.072	-89.67	0.0827	0.1095	1.037	50.000 ± 14.434	0.248 ± 0.019	0.000 ± 0.000
LSAM	beta47	0.08	-0.041	-0.121 ± 0.016	-1.517 ± 0.195	0.003 ± 0.001	0.054 ± 0.018	0.017 ± 0.003	0.132 ± 0.014	2.721 ± 0.542	1.650 ± 0.169	-0.047	-0.127	-158.68	0.0364	0.1320	1.325	58.333 ± 14.232	0.281 ± 0.038	16.667 ± 10.758
LSAM	beta51	0.16	0.167	0.007 ± 0.025	0.047 ± 0.153	0.007 ± 0.002	0.085 ± 0.015	0.007 ± 0.002	0.082 ± 0.019	0.261 ± 0.073	0.511 ± 0.120	0.182	0.022	13.94	0.0687	0.0723	0.830	83.333 ± 10.758	0.277 ± 0.010	66.667 ± 13.608
LSAM	beta52	0.16	0.176	0.016 ± 0.015	0.099 ± 0.091	0.003 ± 0.000	0.050 ± 0.006	0.003 ± 0.001	0.051 ± 0.010	0.100 ± 0.020	0.317 ± 0.060	0.154	-0.006	-3.86	0.0789	0.0791	1.376	100.000 ± 0.000	0.271 ± 0.011	83.333 ± 10.758
LSAM	beta53	0.16	0.144	-0.016 ± 0.013	-0.101 ± 0.084	0.002 ± 0.001	0.046 ± 0.015	0.002 ± 0.001	0.047 ± 0.015	0.087 ± 0.040	0.295 ± 0.092	0.140	-0.020	-12.74	0.0362	0.0415	1.282	91.667 ± 7.979	0.233 ± 0.012	75.000 ± 12.500
LSAM	beta54	0.16	0.121	-0.039 ± 0.019	-0.244 ± 0.120	0.004 ± 0.002	0.066 ± 0.024	0.006 ± 0.003	0.075 ± 0.029	0.217 ± 0.125	0.466 ± 0.179	0.149	-0.011	-7.06	0.0639	0.0649	0.975	91.667 ± 7.979	0.253 ± 0.014	41.667 ± 14.232
LSAM	beta55	0.08	0.111	0.031 ± 0.017	0.384 ± 0.208	0.003 ± 0.001	0.058 ± 0.015	0.004 ± 0.002	0.063 ± 0.016	0.626 ± 0.239	0.791 ± 0.199	0.097	0.017	21.33	0.0780	0.0798	1.458	100.000 ± 0.000	0.330 ± 0.014	16.667 ± 10.758
LSAM	beta56	0.08	0.090	0.010 ± 0.026	0.122 ± 0.325	0.008 ± 0.002	0.090 ± 0.017	0.008 ± 0.002	0.087 ± 0.021	1.175 ± 0.360	1.084 ± 0.263	0.104	0.024	29.72	0.0799	0.0834	1.028	100.000 ± 0.000	0.363 ± 0.021	16.667 ± 10.758
LSAM	beta57	0.06	-0.033	-0.093 ± 0.018	-1.555 ± 0.306	0.004 ± 0.001	0.064 ± 0.012	0.012 ± 0.003	0.111 ± 0.016	3.451 ± 0.931	1.858 ± 0.270	-0.007	-0.067	-111.90	0.0756	0.1011	0.983	58.333 ± 14.232	0.245 ± 0.018	0.000 ± 0.000
LSAM	beta58	0.06	-0.002	-0.062 ± 0.012	-1.028 ± 0.201	0.002 ± 0.001	0.042 ± 0.009	0.005 ± 0.002	0.073 ± 0.012	1.498 ± 0.453	1.224 ± 0.203	-0.005	-0.065	-108.08	0.0646	0.0915	1.402	83.333 ± 10.758	0.229 ± 0.026	0.000 ± 0.000
QML	beta41	0.20	0.184	-0.016 ± 0.014	-0.081 ± 0.068	0.002 ± 0.001	0.047 ± 0.009	0.002 ± 0.001	0.048 ± 0.012	0.058 ± 0.020	0.241 ± 0.059	0.186	-0.014	-6.76	0.0568	0.0584	0.962	91.667 ± 7.979	0.179 ± 0.006	100.000 ± 0.000
QML	beta42	0.20	0.183	-0.017 ± 0.011	-0.084 ± 0.053	0.001 ± 0.000	0.037 ± 0.005	0.002 ± 0.000	0.039 ± 0.009	0.038 ± 0.012	0.195 ± 0.043	0.192	-0.008	-4.08	0.0440	0.0448	1.291	100.000 ± 0.000	0.186 ± 0.007	100.000 ± 0.000
QML	beta43	0.20	0.162	-0.038 ± 0.016	-0.189 ± 0.078	0.003 ± 0.001	0.054 ± 0.010	0.004 ± 0.001	0.064 ± 0.011	0.102 ± 0.026	0.320 ± 0.054	0.178	-0.022	-11.22	0.0719	0.0754	0.831	83.333 ± 10.758	0.176 ± 0.005	100.000 ± 0.000
QML	beta44	0.11	0.173	0.063 ± 0.024	0.571 ± 0.214	0.007 ± 0.002	0.082 ± 0.017	0.010 ± 0.003	0.100 ± 0.018	0.831 ± 0.229	0.912 ± 0.160	0.188	0.078	70.68	0.0783	0.1104	0.700	66.667 ± 13.608	0.224 ± 0.012	83.333 ± 10.758
QML	beta45	0.11	0.111	0.001 ± 0.018	0.007 ± 0.164	0.004 ± 0.001	0.063 ± 0.012	0.004 ± 0.001	0.060 ± 0.014	0.296 ± 0.089	0.544 ± 0.132	0.108	-0.002	-1.54	0.0724	0.0724	0.845	91.667 ± 7.979	0.207 ± 0.008	66.667 ± 13.608
QML	beta46	0.08	-0.029	-0.109 ± 0.013	-1.361 ± 0.161	0.002 ± 0.001	0.045 ± 0.012	0.014 ± 0.003	0.117 ± 0.015	2.138 ± 0.521	1.462 ± 0.191	-0.023	-0.103	-128.43	0.0354	0.1087	0.937	8.333 ± 7.979	0.164 ± 0.008	16.667 ± 10.758
QML	beta47	0.08	-0.042	-0.122 ± 0.010	-1.520 ± 0.130	0.001 ± 0.000	0.036 ± 0.010	0.016 ± 0.002	0.126 ± 0.010	2.499 ± 0.375	1.581 ± 0.119	-0.037	-0.117	-146.25	0.0366	0.1226	1.134	25.000 ± 12.500	0.160 ± 0.015	16.667 ± 10.758
QML	beta51	0.16	0.185	0.025 ± 0.026	0.154 ± 0.163	0.008 ± 0.001	0.090 ± 0.012	0.008 ± 0.002	0.090 ± 0.018	0.316 ± 0.071	0.562 ± 0.112	0.192	0.032	20.18	0.1142	0.1187	0.777	91.667 ± 7.979	0.275 ± 0.011	66.667 ± 13.608
QML	beta52	0.16	0.166	0.006 ± 0.008	0.035 ± 0.050	0.001 ± 0.000	0.028 ± 0.003	0.001 ± 0.000	0.027 ± 0.006	0.029 ± 0.007	0.169 ± 0.035	0.156	-0.004	-2.24	0.0509	0.0510	1.843	100.000 ± 0.000	0.199 ± 0.005	100.000 ± 0.000
QML	beta53	0.16	0.155	-0.005 ± 0.014	-0.032 ± 0.091	0.003 ± 0.001	0.050 ± 0.018	0.002 ± 0.001	0.048 ± 0.017	0.091 ± 0.045	0.302 ± 0.104	0.143	-0.017	-10.42	0.0424	0.0456	1.071	100.000 ± 0.000	0.211 ± 0.008	91.667 ± 7.979
QML	beta54	0.16	0.117	-0.043 ± 0.014	-0.271 ± 0.090	0.002 ± 0.001	0.050 ± 0.014	0.004 ± 0.002	0.065 ± 0.017	0.163 ± 0.071	0.403 ± 0.109	0.135	-0.025	-15.58	0.0539	0.0594	0.960	91.667 ± 7.979	0.188 ± 0.004	75.000 ± 12.500
QML	beta55	0.08	0.103	0.023 ± 0.013	0.287 ± 0.163	0.002 ± 0.001	0.045 ± 0.012	0.002 ± 0.001	0.049 ± 0.015	0.375 ± 0.176	0.612 ± 0.190	0.093	0.013	16.23	0.0552	0.0567	1.731	100.000 ± 0.000	0.306 ± 0.013	8.333 ± 7.979
QML	beta56	0.08	0.092	0.012 ± 0.022	0.144 ± 0.269	0.006 ± 0.001	0.074 ± 0.011	0.005 ± 0.001	0.072 ± 0.015	0.816 ± 0.172	0.903 ± 0.184	0.093	0.013	16.28	0.0934	0.0943	1.078	100.000 ± 0.000	0.315 ± 0.015	16.667 ± 10.758
QML	beta57	0.06	-0.014	-0.074 ± 0.010	-1.225 ± 0.170	0.001 ± 0.000	0.035 ± 0.005	0.007 ± 0.002	0.081 ± 0.010	1.820 ± 0.429	1.349 ± 0.166	-0.001	-0.061	-101.18	0.0419	0.0738	1.145	50.000 ± 14.434	0.159 ± 0.011	0.000 ± 0.000
QML	beta58	0.06	0.009	-0.051 ± 0.008	-0.854 ± 0.139	0.001 ± 0.000	0.029 ± 0.005	0.003 ± 0.001	0.058 ± 0.009	0.942 ± 0.268	0.971 ± 0.149	0.001	-0.059	-98.75	0.0528	0.0794	1.358	75.000 ± 12.500	0.154 ± 0.009	0.000 ± 0.000
UPI	beta41	0.20	0.169	-0.031 ± 0.077	-0.157 ± 0.384	0.071 ± 0.029	0.266 ± 0.076	0.066 ± 0.029	0.256 ± 0.080	1.644 ± 0.734	1.282 ± 0.401	0.233	0.033	16.29	0.3441	0.3457	1.299	91.667 ± 7.979	1.354 ± 0.368	25.000 ± 12.500
UPI	beta42	0.20	0.342	0.142 ± 0.073	0.709 ± 0.366	0.064 ± 0.016	0.254 ± 0.047	0.079 ± 0.032	0.281 ± 0.073	1.978 ± 0.812	1.407 ± 0.365	0.290	0.090	45.02	0.4414	0.4505	2.987	100.000 ± 0.000	2.970 ± 1.177	33.333 ± 13.608
UPI	beta43	0.20	0.113	-0.087 ± 0.038	-0.437 ± 0.188	0.017 ± 0.005	0.130 ± 0.027	0.023 ± 0.006	0.152 ± 0.026	0.578 ± 0.143	0.760 ± 0.128	0.023	-0.177	-88.63	0.2560	0.3114	0.957	58.333 ± 14.232	0.488 ± 0.177	33.333 ± 13.608
UPI	beta44	0.11	0.182	0.072 ± 0.070	0.654 ± 0.637	0.059 ± 0.022	0.243 ± 0.062	0.059 ± 0.028	0.243 ± 0.075	4.886 ± 2.280	2.210 ± 0.684	0.148	0.038	34.25	0.1571	0.1615	1.399	100.000 ± 0.000	1.331 ± 0.495	41.667 ± 14.232
UPI	beta45	0.11	0.058	-0.052 ± 0.065	-0.474 ± 0.589	0.050 ± 0.020	0.224 ± 0.063	0.049 ± 0.021	0.221 ± 0.064	4.040 ± 1.717	2.010 ± 0.585	0.112	0.002	2.22	0.1526	0.1526	1.445	100.000 ± 0.000	1.271 ± 0.494	16.667 ± 10.758
UPI	beta46	0.08	-0.023	-0.103 ± 0.081	-1.290 ± 1.009	0.078 ± 0.019	0.280 ± 0.049	0.082 ± 0.029	0.287 ± 0.071	12.864 ± 4.537	3.587 ± 0.883	-0.176	-0.256	-320.08	0.6024	0.6545	1.408	83.333 ± 10.758	1.543 ± 0.421	0.000 ± 0.000

Table S176

Simulation 2 - Condition 32: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.415	-0.495 ± 0.160	-6.182 ± 2.005	0.309 ± 0.109	0.556 ± 0.140	0.528 ± 0.250	0.726 ± 0.205	82.432 ± 39.060	9.079 ± 2.558	-0.232	-0.312	-390.51	0.5287	0.6141	1.711	91.667 ± 7.979	3.725 ± 1.015	0.000 ± 0.000
UPI	beta51	0.16	0.273	0.113 ± 0.034	0.706 ± 0.212	0.014 ± 0.005	0.118 ± 0.029	0.025 ± 0.008	0.160 ± 0.028	0.995 ± 0.296	0.998 ± 0.176	0.258	0.098	61.18	0.1276	0.1608	1.103	58.333 ± 14.232	0.509 ± 0.193	75.000 ± 12.500
UPI	beta52	0.16	0.096	-0.064 ± 0.032	-0.401 ± 0.201	0.012 ± 0.004	0.111 ± 0.028	0.015 ± 0.008	0.124 ± 0.039	0.603 ± 0.294	0.777 ± 0.242	0.089	-0.071	-44.18	0.1480	0.1640	1.196	100.000 ± 0.000	0.521 ± 0.097	25.000 ± 12.500
UPI	beta53	0.16	0.127	-0.033 ± 0.034	-0.206 ± 0.213	0.014 ± 0.006	0.118 ± 0.037	0.014 ± 0.005	0.118 ± 0.030	0.542 ± 0.186	0.736 ± 0.190	0.063	-0.097	-60.35	0.0964	0.1364	0.549	66.667 ± 13.608	0.254 ± 0.046	41.667 ± 14.232
UPI	beta54	0.16	0.035	-0.125 ± 0.018	-0.779 ± 0.112	0.004 ± 0.001	0.062 ± 0.014	0.019 ± 0.005	0.138 ± 0.017	0.744 ± 0.178	0.863 ± 0.109	0.038	-0.122	-76.53	0.0818	0.1472	1.761	41.667 ± 14.232	0.428 ± 0.211	16.667 ± 10.758
UPI	beta55	0.08	0.122	0.042 ± 0.023	0.527 ± 0.291	0.007 ± 0.002	0.081 ± 0.015	0.008 ± 0.003	0.088 ± 0.021	1.211 ± 0.431	1.100 ± 0.256	0.128	0.048	60.61	0.0975	0.1089	1.369	100.000 ± 0.000	0.433 ± 0.076	16.667 ± 10.758
UPI	beta56	0.08	0.111	0.031 ± 0.030	0.387 ± 0.378	0.011 ± 0.005	0.105 ± 0.038	0.011 ± 0.006	0.105 ± 0.043	1.722 ± 1.011	1.312 ± 0.536	0.095	0.015	18.42	0.0769	0.0783	1.059	91.667 ± 7.979	0.435 ± 0.050	0.000 ± 0.000
UPI	beta57	0.06	0.050	-0.010 ± 0.063	-0.163 ± 1.057	0.048 ± 0.016	0.220 ± 0.050	0.044 ± 0.015	0.211 ± 0.055	12.323 ± 4.296	3.510 ± 0.923	0.005	-0.055	-91.47	0.2278	0.2343	0.878	91.667 ± 7.979	0.756 ± 0.234	0.000 ± 0.000
UPI	beta58	0.06	0.009	-0.051 ± 0.019	-0.848 ± 0.308	0.004 ± 0.002	0.064 ± 0.026	0.006 ± 0.004	0.080 ± 0.029	1.765 ± 0.983	1.329 ± 0.483	0.014	-0.046	-76.88	0.1043	0.1141	1.617	75.000 ± 12.500	0.406 ± 0.143	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S177

Simulation 2 - Condition 33: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.003	-0.009 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.314 ± 0.011	0.200	0.000	0.09	0.0647	0.0647	1.022	96.284 ± 0.777	0.252 ± 0.001	88.682 ± 1.302
LSAM	beta42	0.20	0.198	-0.002 ± 0.003	-0.012 ± 0.014	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.003	0.111 ± 0.007	0.333 ± 0.013	0.199	-0.001	-0.46	0.0663	0.0663	0.998	95.101 ± 0.887	0.261 ± 0.001	84.966 ± 1.469
LSAM	beta43	0.20	0.195	-0.005 ± 0.003	-0.024 ± 0.013	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.005	0.310 ± 0.011	0.193	-0.007	-3.41	0.0579	0.0583	0.961	92.399 ± 1.089	0.233 ± 0.001	91.554 ± 1.143
LSAM	beta44	0.11	0.132	0.022 ± 0.003	0.199 ± 0.026	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.453 ± 0.030	0.673 ± 0.026	0.130	0.020	18.20	0.0667	0.0696	1.067	95.946 ± 0.811	0.296 ± 0.003	42.399 ± 2.031
LSAM	beta45	0.11	0.115	0.005 ± 0.003	0.050 ± 0.026	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.403 ± 0.026	0.635 ± 0.024	0.116	0.006	5.22	0.0662	0.0665	0.953	93.919 ± 0.982	0.260 ± 0.002	41.892 ± 2.028
LSAM	beta46	0.08	0.046	-0.034 ± 0.004	-0.428 ± 0.044	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	1.326 ± 0.073	1.151 ± 0.038	0.039	-0.041	-51.86	0.0784	0.0887	1.037	92.230 ± 1.100	0.348 ± 0.004	4.561 ± 0.857
LSAM	beta47	0.08	0.049	-0.031 ± 0.004	-0.383 ± 0.048	0.009 ± 0.001	0.093 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	1.504 ± 0.097	1.226 ± 0.046	0.044	-0.036	-44.45	0.0817	0.0891	0.974	92.230 ± 1.100	0.356 ± 0.004	6.926 ± 1.043
LSAM	beta51	0.16	0.174	0.014 ± 0.003	0.087 ± 0.020	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.233 ± 0.015	0.482 ± 0.018	0.168	0.008	5.04	0.0723	0.0728	1.037	94.257 ± 0.956	0.309 ± 0.002	60.304 ± 2.011
LSAM	beta52	0.16	0.157	-0.003 ± 0.003	-0.021 ± 0.017	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.176 ± 0.011	0.420 ± 0.015	0.152	-0.008	-5.26	0.0607	0.0613	0.982	95.101 ± 0.887	0.259 ± 0.001	66.047 ± 1.946
LSAM	beta53	0.16	0.151	-0.009 ± 0.003	-0.056 ± 0.018	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.189 ± 0.012	0.435 ± 0.016	0.151	-0.009	-5.67	0.0669	0.0675	0.984	94.088 ± 0.969	0.266 ± 0.001	60.135 ± 2.012
LSAM	beta54	0.16	0.156	-0.004 ± 0.003	-0.026 ± 0.017	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.172 ± 0.010	0.415 ± 0.014	0.155	-0.005	-3.33	0.0678	0.0680	0.991	94.764 ± 0.916	0.258 ± 0.001	65.541 ± 1.953
LSAM	beta55	0.08	0.096	0.016 ± 0.004	0.196 ± 0.050	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	1.493 ± 0.095	1.222 ± 0.046	0.093	0.013	15.81	0.0931	0.0940	0.957	93.919 ± 0.982	0.362 ± 0.003	18.581 ± 1.599
LSAM	beta56	0.08	0.079	-0.001 ± 0.004	-0.014 ± 0.047	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.000	0.091 ± 0.003	1.284 ± 0.072	1.133 ± 0.040	0.075	-0.005	-5.75	0.0931	0.0932	0.956	94.426 ± 0.943	0.340 ± 0.003	17.230 ± 1.552
LSAM	beta57	0.06	0.029	-0.031 ± 0.004	-0.512 ± 0.061	0.008 ± 0.000	0.089 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	2.466 ± 0.143	1.570 ± 0.054	0.028	-0.032	-52.82	0.0840	0.0898	1.012	93.581 ± 1.007	0.354 ± 0.004	4.561 ± 0.857
LSAM	beta58	0.06	0.036	-0.024 ± 0.003	-0.405 ± 0.056	0.007 ± 0.000	0.081 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	1.994 ± 0.127	1.412 ± 0.053	0.034	-0.026	-44.16	0.0761	0.0806	1.051	94.595 ± 0.929	0.335 ± 0.004	4.561 ± 0.857
QML	beta41	0.20	0.194	-0.006 ± 0.002	-0.030 ± 0.012	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.083 ± 0.004	0.287 ± 0.009	0.193	-0.007	-3.55	0.0605	0.0609	1.013	95.439 ± 0.857	0.227 ± 0.002	92.230 ± 1.100
QML	beta42	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.012	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.006	0.304 ± 0.011	0.197	-0.003	-1.31	0.0598	0.0599	0.981	94.764 ± 0.916	0.234 ± 0.001	92.399 ± 1.089
QML	beta43	0.20	0.187	-0.013 ± 0.002	-0.063 ± 0.012	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.092 ± 0.005	0.304 ± 0.010	0.184	-0.016	-8.25	0.0573	0.0597	0.949	92.736 ± 1.067	0.221 ± 0.001	93.074 ± 1.043
QML	beta44	0.11	0.129	0.019 ± 0.003	0.177 ± 0.025	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.390 ± 0.026	0.624 ± 0.024	0.127	0.017	15.72	0.0637	0.0660	1.073	94.932 ± 0.901	0.277 ± 0.002	46.622 ± 2.050
QML	beta45	0.11	0.114	0.004 ± 0.003	0.038 ± 0.024	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.345 ± 0.022	0.588 ± 0.022	0.116	0.006	5.01	0.0641	0.0644	0.968	94.595 ± 0.929	0.245 ± 0.002	45.439 ± 2.046
QML	beta46	0.08	0.025	-0.055 ± 0.002	-0.693 ± 0.027	0.003 ± 0.000	0.053 ± 0.001	0.006 ± 0.000	0.077 ± 0.002	0.916 ± 0.041	0.957 ± 0.023	0.020	-0.060	-75.57	0.0532	0.0805	1.087	82.432 ± 1.564	0.225 ± 0.002	4.392 ± 0.842
QML	beta47	0.08	0.029	-0.051 ± 0.002	-0.632 ± 0.030	0.003 ± 0.000	0.058 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.933 ± 0.050	0.966 ± 0.028	0.025	-0.055	-68.66	0.0541	0.0771	0.986	83.953 ± 1.509	0.226 ± 0.002	7.770 ± 1.100
QML	beta51	0.16	0.185	0.025 ± 0.003	0.158 ± 0.019	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.249 ± 0.015	0.499 ± 0.018	0.183	0.023	14.12	0.0739	0.0773	1.067	94.257 ± 0.956	0.317 ± 0.002	65.541 ± 1.953
QML	beta52	0.16	0.148	-0.012 ± 0.003	-0.074 ± 0.017	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.010	0.410 ± 0.014	0.144	-0.016	-10.13	0.0623	0.0644	0.987	94.595 ± 0.929	0.250 ± 0.001	64.189 ± 1.971
QML	beta53	0.16	0.138	-0.022 ± 0.003	-0.138 ± 0.017	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.011	0.437 ± 0.015	0.136	-0.024	-14.77	0.0677	0.0717	0.988	94.088 ± 0.969	0.257 ± 0.001	56.081 ± 2.040
QML	beta54	0.16	0.157	-0.003 ± 0.003	-0.021 ± 0.016	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.009	0.395 ± 0.014	0.152	-0.008	-4.87	0.0625	0.0630	0.971	95.439 ± 0.857	0.240 ± 0.001	71.622 ± 1.853
QML	beta55	0.08	0.095	0.015 ± 0.004	0.182 ± 0.047	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.001	0.092 ± 0.003	1.322 ± 0.080	1.150 ± 0.042	0.091	0.011	13.60	0.0943	0.0949	0.991	93.919 ± 0.982	0.353 ± 0.003	18.750 ± 1.604
QML	beta56	0.08	0.074	-0.006 ± 0.003	-0.076 ± 0.043	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	1.087 ± 0.057	1.043 ± 0.035	0.068	-0.012	-14.59	0.0897	0.0904	0.982	95.101 ± 0.887	0.321 ± 0.002	16.723 ± 1.534
QML	beta57	0.06	0.016	-0.044 ± 0.003	-0.741 ± 0.042	0.004 ± 0.000	0.061 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	1.577 ± 0.085	1.256 ± 0.038	0.012	-0.048	-79.18	0.0571	0.0743	1.021	86.149 ± 1.420	0.243 ± 0.003	6.081 ± 0.982
QML	beta58	0.06	0.025	-0.035 ± 0.002	-0.586 ± 0.035	0.003 ± 0.000	0.051 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	1.064 ± 0.059	1.031 ± 0.032	0.020	-0.040	-67.36	0.0502	0.0645	1.086	89.189 ± 1.276	0.217 ± 0.004	5.068 ± 0.901
UPI	beta41	0.20	0.211	0.011 ± 0.003	0.056 ± 0.015	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	0.134 ± 0.010	0.367 ± 0.015	0.208	0.008	4.16	0.0743	0.0748	1.120	96.791 ± 0.724	0.318 ± 0.008	79.392 ± 1.662
UPI	beta42	0.20	0.213	0.013 ± 0.003	0.063 ± 0.016	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	0.159 ± 0.011	0.399 ± 0.016	0.213	0.013	6.44	0.0751	0.0762	1.201	96.791 ± 0.724	0.371 ± 0.010	72.297 ± 1.839
UPI	beta43	0.20	0.190	-0.010 ± 0.003	-0.050 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.007	0.335 ± 0.012	0.188	-0.012	-5.98	0.0698	0.0708	1.063	93.750 ± 0.995	0.276 ± 0.004	81.419 ± 1.599
UPI	beta44	0.11	0.141	0.031 ± 0.003	0.283 ± 0.029	0.006 ± 0.000	0.076 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	0.562 ± 0.039	0.750 ± 0.029	0.143	0.033	30.14	0.0744	0.0814	1.258	96.959 ± 0.706	0.377 ± 0.013	34.797 ± 1.958
UPI	beta45	0.11	0.118	0.008 ± 0.003	0.076 ± 0.027	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	0.444 ± 0.029	0.666 ± 0.026	0.118	0.008	7.67	0.0722	0.0727	1.081	96.622 ± 0.743	0.309 ± 0.005	35.642 ± 1.968
UPI	beta46	0.08	0.067	-0.013 ± 0.006	-0.165 ± 0.071	0.019 ± 0.002	0.139 ± 0.006	0.019 ± 0.002	0.139 ± 0.006	3.039 ± 0.244	1.743 ± 0.079	0.053	-0.027	-33.68	0.1281	0.1309	1.172	95.439 ± 0.857	0.638 ± 0.025	4.561 ± 0.857

Table S177

Simulation 2 - Condition 33: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.052	-0.028 ± 0.006	-0.355 ± 0.081	0.025 ± 0.002	0.158 ± 0.006	0.026 ± 0.002	0.160 ± 0.007	4.023 ± 0.303	2.006 ± 0.086	0.044	-0.036	-45.07	0.1411	0.1456	1.213	94.257 ± 0.956	0.752 ± 0.037	4.392 ± 0.842
UPI	beta51	0.16	0.180	0.020 ± 0.003	0.126 ± 0.019	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.235 ± 0.014	0.485 ± 0.018	0.177	0.017	10.89	0.0771	0.0791	1.099	95.777 ± 0.827	0.323 ± 0.003	59.797 ± 2.015
UPI	beta52	0.16	0.163	0.003 ± 0.003	0.022 ± 0.018	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.193 ± 0.012	0.440 ± 0.017	0.158	-0.002	-0.98	0.0662	0.0662	1.113	95.777 ± 0.827	0.307 ± 0.006	61.824 ± 1.997
UPI	beta53	0.16	0.147	-0.013 ± 0.003	-0.079 ± 0.018	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.194 ± 0.012	0.440 ± 0.016	0.146	-0.014	-9.04	0.0686	0.0701	1.106	95.439 ± 0.857	0.301 ± 0.004	50.507 ± 2.055
UPI	beta54	0.16	0.160	-0.000 ± 0.003	-0.001 ± 0.019	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	0.208 ± 0.013	0.456 ± 0.017	0.159	-0.001	-0.37	0.0689	0.0689	1.185	96.791 ± 0.724	0.339 ± 0.010	54.730 ± 2.046
UPI	beta55	0.08	0.099	0.019 ± 0.004	0.233 ± 0.048	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	1.420 ± 0.094	1.192 ± 0.046	0.091	0.011	14.12	0.0884	0.0891	1.114	96.115 ± 0.794	0.409 ± 0.007	14.865 ± 1.462
UPI	beta56	0.08	0.078	-0.002 ± 0.004	-0.022 ± 0.048	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	1.369 ± 0.080	1.170 ± 0.042	0.073	-0.007	-8.52	0.0954	0.0956	1.044	97.128 ± 0.686	0.384 ± 0.005	11.486 ± 1.311
UPI	beta57	0.06	0.048	-0.012 ± 0.005	-0.192 ± 0.084	0.015 ± 0.001	0.123 ± 0.004	0.015 ± 0.001	0.123 ± 0.005	4.199 ± 0.298	2.049 ± 0.084	0.043	-0.017	-28.90	0.1139	0.1153	1.155	94.595 ± 0.929	0.555 ± 0.019	5.236 ± 0.916
UPI	beta58	0.06	0.040	-0.020 ± 0.005	-0.326 ± 0.087	0.016 ± 0.001	0.127 ± 0.005	0.017 ± 0.001	0.129 ± 0.006	4.600 ± 0.361	2.145 ± 0.095	0.032	-0.028	-45.84	0.1036	0.1072	1.313	94.932 ± 0.901	0.655 ± 0.031	2.365 ± 0.625

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S178

Simulation 2 - Condition 34: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.011 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.007	0.198	-0.002	-0.97	0.0391	0.0392	0.997	95.502 ± 0.802	0.155 ± 0.000	99.550 ± 0.259
LSAM	beta42	0.20	0.194	-0.006 ± 0.002	-0.032 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.209 ± 0.007	0.194	-0.006	-3.17	0.0397	0.0402	0.979	94.453 ± 0.886	0.159 ± 0.000	99.850 ± 0.150
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.006	0.197	-0.003	-1.39	0.0383	0.0384	0.990	95.202 ± 0.828	0.145 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.127	0.017 ± 0.002	0.156 ± 0.016	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.206 ± 0.012	0.454 ± 0.015	0.125	0.015	13.50	0.0465	0.0488	0.965	91.904 ± 1.056	0.177 ± 0.001	79.160 ± 1.573
LSAM	beta45	0.11	0.117	0.007 ± 0.002	0.064 ± 0.014	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.140 ± 0.008	0.374 ± 0.012	0.116	0.006	5.13	0.0402	0.0406	1.000	95.202 ± 0.828	0.159 ± 0.001	83.358 ± 1.442
LSAM	beta46	0.08	0.043	-0.037 ± 0.002	-0.462 ± 0.026	0.003 ± 0.000	0.054 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.676 ± 0.033	0.822 ± 0.023	0.040	-0.040	-49.50	0.0515	0.0650	1.007	86.807 ± 1.310	0.215 ± 0.002	10.945 ± 1.209
LSAM	beta47	0.08	0.045	-0.035 ± 0.002	-0.440 ± 0.026	0.003 ± 0.000	0.054 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.651 ± 0.033	0.807 ± 0.023	0.043	-0.037	-46.20	0.0541	0.0656	0.981	85.307 ± 1.371	0.208 ± 0.002	11.094 ± 1.216
LSAM	beta51	0.16	0.171	0.011 ± 0.002	0.071 ± 0.011	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.092 ± 0.005	0.303 ± 0.010	0.172	0.012	7.72	0.0472	0.0488	1.047	95.952 ± 0.763	0.193 ± 0.001	94.303 ± 0.897
LSAM	beta52	0.16	0.154	-0.006 ± 0.002	-0.038 ± 0.010	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.004	0.260 ± 0.009	0.152	-0.008	-5.00	0.0399	0.0407	0.985	94.603 ± 0.875	0.159 ± 0.000	96.252 ± 0.735
LSAM	beta53	0.16	0.155	-0.005 ± 0.002	-0.033 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.070 ± 0.004	0.265 ± 0.008	0.154	-0.006	-3.74	0.0436	0.0440	0.995	95.202 ± 0.828	0.164 ± 0.000	95.802 ± 0.776
LSAM	beta54	0.16	0.155	-0.005 ± 0.002	-0.032 ± 0.010	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.004	0.256 ± 0.009	0.155	-0.005	-3.34	0.0419	0.0422	0.996	95.202 ± 0.828	0.158 ± 0.000	96.552 ± 0.707
LSAM	beta55	0.08	0.094	0.014 ± 0.002	0.180 ± 0.027	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.534 ± 0.030	0.731 ± 0.024	0.090	0.010	13.07	0.0593	0.0602	0.985	94.603 ± 0.875	0.219 ± 0.001	40.930 ± 1.904
LSAM	beta56	0.08	0.082	0.002 ± 0.002	0.019 ± 0.026	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.456 ± 0.025	0.675 ± 0.022	0.082	0.002	2.69	0.0546	0.0547	0.976	94.903 ± 0.852	0.207 ± 0.001	32.984 ± 1.820
LSAM	beta57	0.06	0.031	-0.029 ± 0.002	-0.485 ± 0.036	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	1.107 ± 0.056	1.052 ± 0.031	0.030	-0.030	-49.70	0.0579	0.0651	0.986	89.505 ± 1.187	0.217 ± 0.001	7.046 ± 0.991
LSAM	beta58	0.06	0.034	-0.026 ± 0.002	-0.426 ± 0.035	0.003 ± 0.000	0.054 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.984 ± 0.049	0.992 ± 0.030	0.030	-0.030	-50.30	0.0510	0.0592	0.946	88.456 ± 1.237	0.199 ± 0.001	10.045 ± 1.164
QML	beta41	0.20	0.194	-0.006 ± 0.001	-0.028 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.184 ± 0.006	0.193	-0.007	-3.25	0.0368	0.0374	0.989	95.802 ± 0.776	0.141 ± 0.000	100.000 ± 0.000
QML	beta42	0.20	0.192	-0.008 ± 0.001	-0.041 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.196 ± 0.006	0.190	-0.010	-5.13	0.0366	0.0381	0.962	92.504 ± 1.020	0.145 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.191	-0.009 ± 0.001	-0.044 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.002	0.183 ± 0.006	0.190	-0.010	-4.84	0.0370	0.0382	0.997	94.603 ± 0.875	0.139 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.125	0.015 ± 0.002	0.139 ± 0.016	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.181 ± 0.010	0.425 ± 0.014	0.124	0.014	12.83	0.0443	0.0465	0.990	92.204 ± 1.038	0.172 ± 0.001	80.660 ± 1.529
QML	beta45	0.11	0.115	0.005 ± 0.001	0.049 ± 0.014	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.124 ± 0.007	0.353 ± 0.012	0.114	0.004	3.56	0.0389	0.0391	1.016	95.802 ± 0.776	0.153 ± 0.001	85.307 ± 1.371
QML	beta46	0.08	0.023	-0.057 ± 0.001	-0.709 ± 0.017	0.001 ± 0.000	0.035 ± 0.001	0.004 ± 0.000	0.066 ± 0.001	0.690 ± 0.025	0.830 ± 0.016	0.023	-0.057	-71.64	0.0338	0.0666	1.041	64.168 ± 1.857	0.141 ± 0.001	8.096 ± 1.056
QML	beta47	0.08	0.027	-0.053 ± 0.001	-0.663 ± 0.017	0.001 ± 0.000	0.036 ± 0.001	0.004 ± 0.000	0.064 ± 0.001	0.639 ± 0.025	0.799 ± 0.016	0.026	-0.054	-67.95	0.0360	0.0652	0.972	64.318 ± 1.855	0.136 ± 0.001	12.144 ± 1.265
QML	beta51	0.16	0.181	0.021 ± 0.002	0.134 ± 0.011	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.106 ± 0.005	0.325 ± 0.010	0.182	0.022	13.68	0.0483	0.0530	1.052	94.753 ± 0.863	0.196 ± 0.001	96.252 ± 0.735
QML	beta52	0.16	0.146	-0.014 ± 0.002	-0.085 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.071 ± 0.004	0.266 ± 0.008	0.147	-0.013	-8.12	0.0387	0.0408	0.974	91.904 ± 1.056	0.154 ± 0.000	96.402 ± 0.721
QML	beta53	0.16	0.142	-0.018 ± 0.002	-0.115 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.077 ± 0.004	0.277 ± 0.008	0.141	-0.019	-11.76	0.0421	0.0461	0.996	92.654 ± 1.010	0.158 ± 0.000	94.453 ± 0.886
QML	beta54	0.16	0.157	-0.003 ± 0.001	-0.022 ± 0.009	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.239 ± 0.008	0.155	-0.005	-2.95	0.0380	0.0383	1.009	95.652 ± 0.790	0.151 ± 0.000	98.351 ± 0.493
QML	beta55	0.08	0.094	0.014 ± 0.002	0.175 ± 0.027	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.504 ± 0.027	0.710 ± 0.023	0.089	0.009	11.27	0.0590	0.0597	1.010	95.202 ± 0.828	0.218 ± 0.001	40.930 ± 1.904
QML	beta56	0.08	0.077	-0.003 ± 0.002	-0.038 ± 0.025	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.418 ± 0.023	0.646 ± 0.022	0.077	-0.003	-4.02	0.0513	0.0514	0.976	94.603 ± 0.875	0.198 ± 0.001	32.984 ± 1.820
QML	beta57	0.06	0.015	-0.045 ± 0.001	-0.742 ± 0.025	0.001 ± 0.000	0.038 ± 0.001	0.003 ± 0.000	0.059 ± 0.001	0.951 ± 0.041	0.975 ± 0.023	0.015	-0.045	-74.17	0.0397	0.0596	1.013	76.762 ± 1.635	0.151 ± 0.001	5.847 ± 0.908
QML	beta58	0.06	0.024	-0.036 ± 0.001	-0.602 ± 0.023	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.711 ± 0.032	0.843 ± 0.020	0.021	-0.039	-64.71	0.0347	0.0521	0.960	78.261 ± 1.597	0.133 ± 0.001	11.244 ± 1.223
UPI	beta41	0.20	0.205	0.005 ± 0.002	0.025 ± 0.009	0.002 ± 0.000	0.047 ± 0.002	0.002 ± 0.000	0.047 ± 0.002	0.055 ± 0.004	0.235 ± 0.009	0.205	0.005	2.73	0.0447	0.0450	1.187	95.652 ± 0.790	0.217 ± 0.009	94.603 ± 0.875
UPI	beta42	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.010	0.002 ± 0.000	0.049 ± 0.002	0.002 ± 0.000	0.049 ± 0.002	0.060 ± 0.004	0.246 ± 0.010	0.199	-0.001	-0.63	0.0454	0.0455	1.193	96.702 ± 0.692	0.230 ± 0.009	92.504 ± 1.020
UPI	beta43	0.20	0.195	-0.005 ± 0.002	-0.025 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.200 ± 0.007	0.193	-0.007	-3.72	0.0414	0.0421	1.062	95.652 ± 0.790	0.165 ± 0.003	98.801 ± 0.422
UPI	beta44	0.11	0.128	0.018 ± 0.002	0.166 ± 0.017	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.223 ± 0.013	0.473 ± 0.016	0.126	0.016	14.30	0.0498	0.0522	1.140	94.903 ± 0.852	0.218 ± 0.004	68.066 ± 1.805
UPI	beta45	0.11	0.119	0.009 ± 0.002	0.083 ± 0.015	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.148 ± 0.008	0.385 ± 0.012	0.118	0.008	6.91	0.0433	0.0440	1.075	96.402 ± 0.721	0.174 ± 0.002	78.111 ± 1.601
UPI	beta46	0.08	0.057	-0.023 ± 0.003	-0.286 ± 0.043	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	1.300 ± 0.083	1.140 ± 0.042	0.052	-0.028	-35.09	0.0800	0.0848	1.170	94.003 ± 0.919	0.405 ± 0.026	10.045 ± 1.164

Table S178

Simulation 2 - Condition 34: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.046	-0.034 ± 0.003	-0.421 ± 0.042	0.007 ± 0.001	0.086 ± 0.003	0.009 ± 0.001	0.092 ± 0.003	1.333 ± 0.087	1.155 ± 0.043	0.042	-0.038	-47.06	0.0794	0.0879	1.144	93.703 ± 0.941	0.386 ± 0.014	7.946 ± 1.047
UPI	beta51	0.16	0.174	0.014 ± 0.002	0.089 ± 0.011	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.094 ± 0.005	0.307 ± 0.009	0.175	0.015	9.62	0.0478	0.0502	1.084	96.702 ± 0.692	0.200 ± 0.001	94.003 ± 0.919
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.011	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.002	0.077 ± 0.004	0.277 ± 0.009	0.161	0.001	0.34	0.0458	0.0458	1.121	96.702 ± 0.692	0.195 ± 0.004	92.204 ± 1.038
UPI	beta53	0.16	0.150	-0.010 ± 0.002	-0.061 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.004	0.276 ± 0.009	0.148	-0.012	-7.75	0.0465	0.0482	1.085	96.702 ± 0.692	0.183 ± 0.002	89.505 ± 1.187
UPI	beta54	0.16	0.157	-0.003 ± 0.002	-0.020 ± 0.012	0.002 ± 0.000	0.048 ± 0.002	0.002 ± 0.000	0.048 ± 0.002	0.089 ± 0.006	0.299 ± 0.011	0.155	-0.005	-2.93	0.0478	0.0480	1.107	96.702 ± 0.692	0.207 ± 0.004	87.256 ± 1.291
UPI	beta55	0.08	0.096	0.016 ± 0.002	0.203 ± 0.029	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.589 ± 0.036	0.768 ± 0.027	0.092	0.012	15.31	0.0595	0.0608	1.027	95.502 ± 0.802	0.239 ± 0.002	37.331 ± 1.873
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.015 ± 0.027	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.472 ± 0.025	0.687 ± 0.023	0.083	0.003	3.51	0.0549	0.0550	1.019	96.252 ± 0.735	0.220 ± 0.001	30.435 ± 1.782
UPI	beta57	0.06	0.040	-0.020 ± 0.003	-0.341 ± 0.051	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	1.830 ± 0.112	1.353 ± 0.048	0.041	-0.019	-32.25	0.0727	0.0752	1.114	94.003 ± 0.919	0.343 ± 0.011	7.046 ± 0.991
UPI	beta58	0.06	0.033	-0.027 ± 0.003	-0.453 ± 0.056	0.007 ± 0.001	0.086 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	2.257 ± 0.167	1.502 ± 0.062	0.026	-0.034	-57.06	0.0727	0.0804	1.064	93.553 ± 0.951	0.359 ± 0.011	6.897 ± 0.981

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S179

Simulation 2 - Condition 35: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.012 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.241 ± 0.006	0.199	-0.001	-0.72	0.0478	0.0478	1.011	95.095 ± 0.683	0.191 ± 0.000	98.398 ± 0.397
LSAM	beta42	0.20	0.200	0.000 ± 0.002	0.000 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.198	-0.002	-1.14	0.0495	0.0495	1.015	95.195 ± 0.677	0.193 ± 0.000	98.699 ± 0.359
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.202	0.002	0.81	0.0458	0.0458	1.009	95.195 ± 0.677	0.181 ± 0.000	99.199 ± 0.282
LSAM	beta44	0.11	0.114	0.004 ± 0.002	0.036 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.244 ± 0.011	0.494 ± 0.014	0.114	0.004	3.91	0.0522	0.0524	0.962	94.094 ± 0.746	0.204 ± 0.001	60.060 ± 1.550
LSAM	beta45	0.11	0.112	0.002 ± 0.002	0.022 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.199 ± 0.009	0.446 ± 0.013	0.109	-0.001	-0.93	0.0486	0.0486	0.968	94.394 ± 0.728	0.186 ± 0.001	64.965 ± 1.509
LSAM	beta46	0.08	0.071	-0.009 ± 0.002	-0.119 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.551 ± 0.025	0.742 ± 0.021	0.070	-0.010	-12.47	0.0579	0.0587	1.023	94.695 ± 0.709	0.235 ± 0.001	19.920 ± 1.264
LSAM	beta47	0.08	0.071	-0.009 ± 0.002	-0.113 ± 0.024	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.564 ± 0.026	0.751 ± 0.021	0.069	-0.011	-14.04	0.0591	0.0602	1.000	94.595 ± 0.715	0.233 ± 0.001	21.522 ± 1.300
LSAM	beta51	0.16	0.166	0.006 ± 0.002	0.038 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.007	0.405 ± 0.011	0.169	0.009	5.55	0.0650	0.0656	0.970	94.995 ± 0.690	0.245 ± 0.001	74.775 ± 1.374
LSAM	beta52	0.16	0.160	0.000 ± 0.002	0.000 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.004	0.326 ± 0.009	0.162	0.002	1.55	0.0536	0.0536	0.989	95.195 ± 0.677	0.202 ± 0.000	86.086 ± 1.095
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.120 ± 0.005	0.347 ± 0.009	0.157	-0.003	-1.75	0.0554	0.0554	0.960	93.894 ± 0.758	0.209 ± 0.001	81.782 ± 1.221
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.159	-0.001	-0.73	0.0522	0.0522	0.994	94.595 ± 0.715	0.202 ± 0.001	87.087 ± 1.061
LSAM	beta55	0.08	0.083	0.003 ± 0.002	0.039 ± 0.027	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.705 ± 0.030	0.840 ± 0.022	0.080	0.000	0.52	0.0712	0.0712	0.973	93.594 ± 0.775	0.256 ± 0.001	25.826 ± 1.385
LSAM	beta56	0.08	0.081	0.001 ± 0.002	0.018 ± 0.025	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.645 ± 0.028	0.803 ± 0.022	0.081	0.001	0.99	0.0644	0.0644	0.968	94.094 ± 0.746	0.244 ± 0.001	26.026 ± 1.388
LSAM	beta57	0.06	0.052	-0.008 ± 0.002	-0.133 ± 0.035	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	1.222 ± 0.060	1.105 ± 0.032	0.051	-0.009	-15.64	0.0610	0.0618	0.972	93.894 ± 0.758	0.251 ± 0.001	13.113 ± 1.068
LSAM	beta58	0.06	0.055	-0.005 ± 0.002	-0.080 ± 0.031	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.977 ± 0.047	0.988 ± 0.028	0.057	-0.003	-5.08	0.0575	0.0576	1.010	94.494 ± 0.722	0.234 ± 0.001	13.914 ± 1.095
QML	beta41	0.20	0.196	-0.004 ± 0.001	-0.021 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.236 ± 0.006	0.196	-0.004	-2.02	0.0458	0.0460	1.010	95.095 ± 0.683	0.186 ± 0.001	98.398 ± 0.397
QML	beta42	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.197	-0.003	-1.67	0.0494	0.0495	1.009	95.295 ± 0.670	0.188 ± 0.000	99.099 ± 0.299
QML	beta43	0.20	0.197	-0.003 ± 0.001	-0.016 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.197	-0.003	-1.42	0.0443	0.0444	1.013	94.995 ± 0.690	0.179 ± 0.000	98.999 ± 0.315
QML	beta44	0.11	0.115	0.005 ± 0.002	0.049 ± 0.015	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.241 ± 0.011	0.491 ± 0.014	0.115	0.005	4.49	0.0523	0.0525	0.966	93.994 ± 0.752	0.204 ± 0.001	61.261 ± 1.541
QML	beta45	0.11	0.113	0.003 ± 0.002	0.025 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.195 ± 0.009	0.441 ± 0.013	0.109	-0.001	-0.99	0.0475	0.0475	0.969	93.794 ± 0.763	0.184 ± 0.001	66.166 ± 1.497
QML	beta46	0.08	0.053	-0.027 ± 0.001	-0.332 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.455 ± 0.020	0.675 ± 0.017	0.053	-0.027	-33.51	0.0454	0.0527	1.034	90.891 ± 0.910	0.191 ± 0.001	18.819 ± 1.237
QML	beta47	0.08	0.055	-0.025 ± 0.002	-0.317 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.455 ± 0.019	0.674 ± 0.017	0.054	-0.026	-32.94	0.0476	0.0544	1.003	90.290 ± 0.937	0.187 ± 0.001	22.022 ± 1.311
QML	beta51	0.16	0.172	0.012 ± 0.002	0.074 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.007	0.412 ± 0.011	0.174	0.014	8.99	0.0677	0.0692	0.984	94.695 ± 0.709	0.250 ± 0.001	76.176 ± 1.348
QML	beta52	0.16	0.147	-0.013 ± 0.002	-0.083 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.113 ± 0.005	0.336 ± 0.009	0.147	-0.013	-8.40	0.0529	0.0546	1.010	94.795 ± 0.703	0.206 ± 0.001	78.679 ± 1.296
QML	beta53	0.16	0.141	-0.019 ± 0.002	-0.120 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.001	0.135 ± 0.006	0.367 ± 0.009	0.141	-0.019	-12.17	0.0559	0.0592	0.968	92.192 ± 0.849	0.211 ± 0.001	73.774 ± 1.392
QML	beta54	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.319 ± 0.009	0.159	-0.001	-0.51	0.0512	0.0512	0.993	94.494 ± 0.722	0.198 ± 0.000	87.888 ± 1.032
QML	beta55	0.08	0.084	0.004 ± 0.002	0.047 ± 0.026	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.690 ± 0.029	0.831 ± 0.022	0.081	0.001	1.20	0.0691	0.0691	0.991	94.094 ± 0.746	0.258 ± 0.001	26.727 ± 1.400
QML	beta56	0.08	0.081	0.001 ± 0.002	0.007 ± 0.025	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.625 ± 0.028	0.791 ± 0.021	0.080	0.000	-0.62	0.0628	0.0628	0.977	94.895 ± 0.696	0.242 ± 0.001	25.526 ± 1.379
QML	beta57	0.06	0.039	-0.021 ± 0.002	-0.356 ± 0.028	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.925 ± 0.044	0.962 ± 0.026	0.038	-0.022	-36.00	0.0507	0.0551	0.977	92.693 ± 0.823	0.205 ± 0.001	11.712 ± 1.017
QML	beta58	0.06	0.044	-0.016 ± 0.002	-0.265 ± 0.025	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.696 ± 0.033	0.834 ± 0.023	0.045	-0.015	-24.18	0.0472	0.0494	1.013	92.693 ± 0.823	0.189 ± 0.001	13.413 ± 1.078
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.018 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.006	0.204	0.004	1.92	0.0496	0.0497	0.960	93.393 ± 0.786	0.182 ± 0.000	98.999 ± 0.315
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.023 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.006	0.203	0.003	1.51	0.0498	0.0498	0.981	94.895 ± 0.696	0.188 ± 0.000	99.399 ± 0.244
UPI	beta43	0.20	0.203	0.003 ± 0.001	0.016 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.204	0.004	2.05	0.0454	0.0456	1.009	94.995 ± 0.690	0.180 ± 0.000	99.399 ± 0.244
UPI	beta44	0.11	0.113	0.003 ± 0.002	0.031 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.253 ± 0.012	0.503 ± 0.014	0.112	0.002	1.49	0.0551	0.0551	0.952	94.094 ± 0.746	0.206 ± 0.001	57.758 ± 1.563
UPI	beta45	0.11	0.113	0.003 ± 0.002	0.023 ± 0.014	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.209 ± 0.010	0.457 ± 0.013	0.109	-0.001	-0.49	0.0481	0.0481	0.953	94.394 ± 0.728	0.187 ± 0.001	63.964 ± 1.519
UPI	beta46	0.08	0.076	-0.004 ± 0.002	-0.055 ± 0.025	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.622 ± 0.030	0.788 ± 0.023	0.073	-0.007	-8.60	0.0612	0.0616	1.007	94.494 ± 0.722	0.249 ± 0.001	21.421 ± 1.298

Table S179

Simulation 2 - Condition 35: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.076	-0.004 ± 0.002	-0.053 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.588 ± 0.028	0.767 ± 0.022	0.075	-0.005	-6.41	0.0602	0.0604	1.027	94.795 ± 0.703	0.247 ± 0.001	21.822 ± 1.307
UPI	beta51	0.16	0.167	0.007 ± 0.002	0.046 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.007	0.404 ± 0.011	0.168	0.008	5.31	0.0670	0.0676	0.975	95.095 ± 0.683	0.245 ± 0.001	75.375 ± 1.363
UPI	beta52	0.16	0.164	0.004 ± 0.002	0.025 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.004	0.326 ± 0.009	0.165	0.005	3.13	0.0538	0.0540	0.976	94.595 ± 0.715	0.199 ± 0.000	88.088 ± 1.025
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.120 ± 0.005	0.347 ± 0.009	0.158	-0.002	-1.08	0.0557	0.0558	0.959	93.894 ± 0.758	0.209 ± 0.001	82.482 ± 1.203
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.162	0.002	1.24	0.0524	0.0524	0.983	94.094 ± 0.746	0.200 ± 0.001	88.889 ± 0.994
UPI	beta55	0.08	0.082	0.002 ± 0.002	0.022 ± 0.027	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.705 ± 0.030	0.840 ± 0.023	0.080	0.000	-0.38	0.0701	0.0702	0.977	93.994 ± 0.752	0.257 ± 0.001	25.325 ± 1.376
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.015 ± 0.026	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.654 ± 0.029	0.808 ± 0.022	0.081	0.001	1.43	0.0646	0.0646	0.970	94.294 ± 0.734	0.246 ± 0.001	24.725 ± 1.365
UPI	beta57	0.06	0.057	-0.003 ± 0.002	-0.057 ± 0.036	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	1.299 ± 0.066	1.140 ± 0.034	0.055	-0.005	-7.57	0.0605	0.0607	0.977	94.795 ± 0.703	0.262 ± 0.001	12.913 ± 1.061
UPI	beta58	0.06	0.060	-0.000 ± 0.002	-0.008 ± 0.033	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	1.073 ± 0.051	1.036 ± 0.030	0.062	0.002	2.61	0.0611	0.0612	1.031	95.896 ± 0.628	0.251 ± 0.001	12.212 ± 1.036

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S180

Simulation 2 - Condition 36: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.198	-0.002	-0.76	0.0296	0.0297	1.030	95.600 ± 0.649	0.120 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.198	-0.002	-0.81	0.0318	0.0318	0.992	95.700 ± 0.641	0.121 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.000 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.201	0.001	0.47	0.0282	0.0282	1.003	95.600 ± 0.649	0.114 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.115	0.005 ± 0.001	0.048 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.092 ± 0.004	0.304 ± 0.008	0.114	0.004	4.01	0.0342	0.0345	0.989	94.300 ± 0.733	0.128 ± 0.000	93.800 ± 0.763
LSAM	beta45	0.11	0.113	0.003 ± 0.001	0.024 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.073 ± 0.003	0.271 ± 0.007	0.113	0.003	2.87	0.0301	0.0303	1.006	95.500 ± 0.656	0.117 ± 0.000	96.700 ± 0.565
LSAM	beta46	0.08	0.072	-0.008 ± 0.001	-0.101 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.226 ± 0.010	0.476 ± 0.013	0.070	-0.010	-13.11	0.0363	0.0378	1.012	94.500 ± 0.721	0.148 ± 0.000	46.200 ± 1.577
LSAM	beta47	0.08	0.072	-0.008 ± 0.001	-0.103 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.215 ± 0.010	0.463 ± 0.013	0.071	-0.009	-11.26	0.0369	0.0380	1.035	95.600 ± 0.649	0.147 ± 0.000	47.100 ± 1.578
LSAM	beta51	0.16	0.162	0.002 ± 0.001	0.015 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.256 ± 0.007	0.161	0.001	0.65	0.0398	0.0398	0.967	94.900 ± 0.696	0.155 ± 0.000	98.500 ± 0.384
LSAM	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.160	0.000	0.29	0.0316	0.0316	0.995	94.600 ± 0.715	0.127 ± 0.000	99.700 ± 0.173
LSAM	beta53	0.16	0.158	-0.002 ± 0.001	-0.014 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.045 ± 0.002	0.213 ± 0.006	0.158	-0.002	-1.53	0.0340	0.0341	0.983	94.700 ± 0.708	0.131 ± 0.000	99.900 ± 0.100
LSAM	beta54	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.158	-0.002	-1.24	0.0326	0.0327	0.981	94.600 ± 0.715	0.126 ± 0.000	99.800 ± 0.141
LSAM	beta55	0.08	0.082	0.002 ± 0.001	0.028 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.251 ± 0.012	0.501 ± 0.014	0.082	0.002	2.51	0.0394	0.0394	1.019	94.900 ± 0.696	0.160 ± 0.000	52.700 ± 1.579
LSAM	beta56	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.245 ± 0.012	0.495 ± 0.014	0.079	-0.001	-1.39	0.0383	0.0383	0.987	95.000 ± 0.689	0.153 ± 0.000	53.100 ± 1.578
LSAM	beta57	0.06	0.052	-0.008 ± 0.001	-0.125 ± 0.021	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.474 ± 0.023	0.689 ± 0.020	0.053	-0.007	-11.17	0.0413	0.0418	0.983	94.300 ± 0.733	0.157 ± 0.000	26.200 ± 1.391
LSAM	beta58	0.06	0.056	-0.004 ± 0.001	-0.069 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.403 ± 0.018	0.635 ± 0.017	0.056	-0.004	-7.49	0.0381	0.0383	0.987	94.700 ± 0.708	0.147 ± 0.000	31.900 ± 1.474
QML	beta41	0.20	0.197	-0.003 ± 0.001	-0.017 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.197	-0.003	-1.54	0.0297	0.0299	1.367	95.600 ± 0.649	0.156 ± 0.035	99.800 ± 0.141
QML	beta42	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.197	-0.003	-1.57	0.0314	0.0315	1.273	95.500 ± 0.656	0.152 ± 0.028	99.800 ± 0.141
QML	beta43	0.20	0.195	-0.005 ± 0.001	-0.023 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.196	-0.004	-2.03	0.0285	0.0288	1.185	94.900 ± 0.696	0.134 ± 0.018	99.800 ± 0.141
QML	beta44	0.11	0.117	0.007 ± 0.001	0.060 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.093 ± 0.004	0.305 ± 0.008	0.116	0.006	5.08	0.0337	0.0341	0.990	94.600 ± 0.715	0.128 ± 0.000	94.500 ± 0.721
QML	beta45	0.11	0.113	0.003 ± 0.001	0.027 ± 0.008	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.072 ± 0.003	0.269 ± 0.007	0.113	0.003	2.72	0.0304	0.0305	1.006	95.400 ± 0.662	0.116 ± 0.000	96.900 ± 0.548
QML	beta46	0.08	0.055	-0.025 ± 0.001	-0.317 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.240 ± 0.009	0.490 ± 0.011	0.054	-0.026	-33.12	0.0288	0.0391	1.020	86.500 ± 1.081	0.120 ± 0.000	43.000 ± 1.566
QML	beta47	0.08	0.056	-0.024 ± 0.001	-0.302 ± 0.012	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.225 ± 0.010	0.474 ± 0.011	0.056	-0.024	-30.05	0.0305	0.0388	1.033	87.600 ± 1.042	0.119 ± 0.001	45.200 ± 1.574
QML	beta51	0.16	0.168	0.008 ± 0.001	0.053 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.167	0.007	4.38	0.0418	0.0424	1.112	94.000 ± 0.751	0.180 ± 0.019	98.600 ± 0.372
QML	beta52	0.16	0.146	-0.014 ± 0.001	-0.087 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.223 ± 0.006	0.147	-0.013	-8.17	0.0324	0.0349	1.160	93.100 ± 0.801	0.150 ± 0.016	98.800 ± 0.344
QML	beta53	0.16	0.141	-0.019 ± 0.001	-0.122 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.002	0.244 ± 0.006	0.139	-0.021	-13.12	0.0342	0.0402	1.009	90.700 ± 0.918	0.134 ± 0.002	98.500 ± 0.384
QML	beta54	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.204 ± 0.006	0.157	-0.003	-1.67	0.0321	0.0322	0.998	94.500 ± 0.721	0.127 ± 0.003	99.700 ± 0.173
QML	beta55	0.08	0.083	0.003 ± 0.001	0.041 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.252 ± 0.012	0.502 ± 0.014	0.083	0.003	3.93	0.0388	0.0390	1.030	95.100 ± 0.683	0.162 ± 0.001	53.100 ± 1.578
QML	beta56	0.08	0.079	-0.001 ± 0.001	-0.013 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.241 ± 0.011	0.491 ± 0.014	0.078	-0.002	-3.00	0.0377	0.0378	0.994	95.200 ± 0.676	0.153 ± 0.001	52.200 ± 1.580
QML	beta57	0.06	0.039	-0.021 ± 0.001	-0.345 ± 0.018	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.427 ± 0.020	0.653 ± 0.017	0.040	-0.020	-33.93	0.0328	0.0386	0.991	90.100 ± 0.944	0.129 ± 0.001	22.000 ± 1.310
QML	beta58	0.06	0.045	-0.015 ± 0.001	-0.251 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.323 ± 0.013	0.569 ± 0.014	0.044	-0.016	-27.09	0.0310	0.0350	0.993	91.200 ± 0.896	0.119 ± 0.001	31.500 ± 1.469
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.009 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.202	0.002	1.14	0.0310	0.0311	0.984	94.300 ± 0.733	0.114 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.200	0.000	0.20	0.0323	0.0323	0.960	94.600 ± 0.715	0.118 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.201	0.001	0.69	0.0283	0.0283	0.992	95.300 ± 0.669	0.113 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.113	0.003 ± 0.001	0.029 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.112	0.002	2.22	0.0345	0.0346	0.989	94.600 ± 0.715	0.128 ± 0.000	93.100 ± 0.801
UPI	beta45	0.11	0.112	0.002 ± 0.001	0.020 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.074 ± 0.003	0.271 ± 0.007	0.113	0.003	2.28	0.0308	0.0309	1.004	95.900 ± 0.627	0.117 ± 0.000	96.700 ± 0.565
UPI	beta46	0.08	0.077	-0.003 ± 0.001	-0.040 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.232 ± 0.011	0.482 ± 0.014	0.076	-0.004	-5.13	0.0367	0.0369	1.005	94.400 ± 0.727	0.151 ± 0.000	51.000 ± 1.581

Table S180

Simulation 2 - Condition 36: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.077	-0.003 ± 0.001	-0.039 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.217 ± 0.010	0.466 ± 0.013	0.076	-0.004	-4.84	0.0372	0.0374	1.034	96.100 ± 0.612	0.151 ± 0.000	51.300 ± 1.581
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.014 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.256 ± 0.007	0.161	0.001	0.44	0.0406	0.0406	0.964	94.500 ± 0.721	0.154 ± 0.000	98.500 ± 0.384
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.041 ± 0.002	0.204 ± 0.006	0.162	0.002	1.55	0.0314	0.0315	0.979	94.100 ± 0.745	0.125 ± 0.000	99.900 ± 0.100
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.045 ± 0.002	0.212 ± 0.006	0.158	-0.002	-1.17	0.0339	0.0340	0.981	95.100 ± 0.683	0.131 ± 0.000	99.900 ± 0.100
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.160	0.000	-0.23	0.0324	0.0324	0.968	94.500 ± 0.721	0.125 ± 0.000	99.900 ± 0.100
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.012 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.251 ± 0.012	0.501 ± 0.014	0.081	0.001	1.33	0.0392	0.0392	1.017	95.400 ± 0.662	0.160 ± 0.000	51.900 ± 1.580
UPI	beta56	0.08	0.080	-0.000 ± 0.001	-0.006 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.240 ± 0.011	0.489 ± 0.014	0.078	-0.002	-2.28	0.0384	0.0385	0.998	95.100 ± 0.683	0.153 ± 0.000	52.800 ± 1.579
UPI	beta57	0.06	0.056	-0.004 ± 0.001	-0.059 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.480 ± 0.023	0.693 ± 0.020	0.057	-0.003	-5.35	0.0406	0.0407	0.984	94.300 ± 0.733	0.160 ± 0.000	29.200 ± 1.438
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.000 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.422 ± 0.019	0.650 ± 0.018	0.058	-0.002	-2.77	0.0395	0.0395	0.993	94.800 ± 0.702	0.152 ± 0.001	34.100 ± 1.499

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S181

Simulation 2 - Condition 37: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.003	0.011 ± 0.017	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.192 ± 0.011	0.438 ± 0.015	0.201	0.001	0.47	0.0854	0.0854	1.098	96.154 ± 0.754	0.377 ± 0.004	62.154 ± 1.902
LSAM	beta42	0.20	0.205	0.005 ± 0.004	0.025 ± 0.019	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.235 ± 0.014	0.485 ± 0.017	0.202	0.002	1.18	0.0979	0.0979	1.033	97.692 ± 0.589	0.393 ± 0.004	57.385 ± 1.940
LSAM	beta43	0.20	0.206	0.006 ± 0.004	0.030 ± 0.018	0.008 ± 0.001	0.092 ± 0.003	0.008 ± 0.001	0.092 ± 0.003	0.210 ± 0.013	0.459 ± 0.017	0.201	0.001	0.63	0.0834	0.0834	0.991	94.615 ± 0.885	0.356 ± 0.003	67.692 ± 1.834
LSAM	beta44	0.11	0.114	0.004 ± 0.006	0.034 ± 0.051	0.021 ± 0.002	0.144 ± 0.005	0.021 ± 0.002	0.144 ± 0.006	1.718 ± 0.128	1.311 ± 0.055	0.107	-0.003	-2.86	0.1360	0.1361	1.032	98.000 ± 0.549	0.584 ± 0.012	12.154 ± 1.282
LSAM	beta45	0.11	0.111	0.001 ± 0.005	0.009 ± 0.041	0.013 ± 0.001	0.115 ± 0.004	0.013 ± 0.001	0.115 ± 0.005	1.093 ± 0.074	1.045 ± 0.041	0.111	0.001	1.29	0.1044	0.1044	1.030	97.846 ± 0.569	0.465 ± 0.008	16.462 ± 1.455
LSAM	beta46	0.08	0.089	0.009 ± 0.004	0.115 ± 0.049	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	1.548 ± 0.099	1.244 ± 0.047	0.087	0.007	9.31	0.0990	0.0992	1.043	97.077 ± 0.661	0.406 ± 0.009	10.308 ± 1.193
LSAM	beta47	0.08	0.086	0.006 ± 0.004	0.075 ± 0.048	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	1.503 ± 0.106	1.226 ± 0.050	0.079	-0.001	-0.86	0.0854	0.0854	1.024	97.846 ± 0.569	0.393 ± 0.008	10.615 ± 1.208
LSAM	beta51	0.16	0.162	0.002 ± 0.005	0.011 ± 0.032	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	0.657 ± 0.040	0.810 ± 0.029	0.164	0.004	2.44	0.1224	0.1225	0.953	94.615 ± 0.885	0.485 ± 0.005	27.846 ± 1.758
LSAM	beta52	0.16	0.159	-0.001 ± 0.004	-0.006 ± 0.023	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.003	0.338 ± 0.020	0.582 ± 0.021	0.157	-0.003	-2.10	0.0923	0.0923	1.040	96.769 ± 0.694	0.380 ± 0.004	39.385 ± 1.916
LSAM	beta53	0.16	0.165	0.005 ± 0.004	0.034 ± 0.025	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.407 ± 0.023	0.638 ± 0.022	0.162	0.002	1.12	0.1044	0.1044	1.010	94.615 ± 0.885	0.404 ± 0.004	37.385 ± 1.898
LSAM	beta54	0.16	0.160	0.000 ± 0.004	0.001 ± 0.025	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.392 ± 0.025	0.626 ± 0.023	0.156	-0.004	-2.49	0.0947	0.0947	0.972	95.692 ± 0.796	0.382 ± 0.004	39.385 ± 1.916
LSAM	beta55	0.08	0.096	0.016 ± 0.007	0.201 ± 0.086	0.031 ± 0.002	0.175 ± 0.006	0.031 ± 0.002	0.175 ± 0.007	4.799 ± 0.330	2.191 ± 0.087	0.092	0.012	14.42	0.1593	0.1598	0.989	96.769 ± 0.694	0.677 ± 0.015	7.385 ± 1.026
LSAM	beta56	0.08	0.082	0.002 ± 0.006	0.029 ± 0.075	0.023 ± 0.002	0.152 ± 0.005	0.023 ± 0.002	0.152 ± 0.006	3.607 ± 0.253	1.899 ± 0.077	0.082	0.002	3.08	0.1356	0.1357	1.029	96.769 ± 0.694	0.613 ± 0.013	6.462 ± 0.964
LSAM	beta57	0.06	0.060	0.000 ± 0.004	0.005 ± 0.061	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	2.423 ± 0.171	1.557 ± 0.063	0.058	-0.002	-3.13	0.0911	0.0911	1.094	98.000 ± 0.549	0.401 ± 0.008	4.154 ± 0.783
LSAM	beta58	0.06	0.064	0.004 ± 0.003	0.060 ± 0.051	0.006 ± 0.000	0.078 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	1.707 ± 0.121	1.307 ± 0.053	0.056	-0.004	-6.52	0.0732	0.0733	1.065	97.538 ± 0.608	0.327 ± 0.007	7.538 ± 1.036
QML	beta41	0.20	0.184	-0.016 ± 0.003	-0.079 ± 0.015	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.159 ± 0.009	0.399 ± 0.014	0.183	-0.017	-8.36	0.0766	0.0784	1.055	94.154 ± 0.920	0.324 ± 0.003	64.462 ± 1.877
QML	beta42	0.20	0.190	-0.010 ± 0.003	-0.051 ± 0.016	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.176 ± 0.009	0.420 ± 0.014	0.190	-0.010	-4.85	0.0830	0.0836	1.028	94.154 ± 0.920	0.336 ± 0.002	62.000 ± 1.904
QML	beta43	0.20	0.186	-0.014 ± 0.003	-0.069 ± 0.016	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	0.174 ± 0.010	0.417 ± 0.015	0.186	-0.014	-6.78	0.0771	0.0783	0.973	91.692 ± 1.083	0.314 ± 0.002	67.077 ± 1.843
QML	beta44	0.11	0.088	-0.022 ± 0.005	-0.201 ± 0.042	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	1.164 ± 0.078	1.079 ± 0.042	0.091	-0.019	-16.83	0.1197	0.1211	1.030	96.308 ± 0.740	0.471 ± 0.008	15.538 ± 1.421
QML	beta45	0.11	0.103	-0.007 ± 0.004	-0.061 ± 0.033	0.008 ± 0.000	0.092 ± 0.003	0.009 ± 0.000	0.092 ± 0.003	0.705 ± 0.040	0.840 ± 0.029	0.102	-0.008	-7.16	0.0934	0.0937	1.061	96.923 ± 0.677	0.384 ± 0.004	19.231 ± 1.546
QML	beta46	0.08	0.088	0.008 ± 0.003	0.102 ± 0.039	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	0.989 ± 0.069	0.995 ± 0.040	0.085	0.005	6.04	0.0821	0.0822	1.024	95.231 ± 0.836	0.318 ± 0.006	13.538 ± 1.342
QML	beta47	0.08	0.083	0.003 ± 0.003	0.043 ± 0.038	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.918 ± 0.058	0.958 ± 0.036	0.076	-0.004	-5.58	0.0715	0.0717	1.030	93.692 ± 0.954	0.309 ± 0.005	14.308 ± 1.373
QML	beta51	0.16	0.183	0.023 ± 0.005	0.144 ± 0.032	0.017 ± 0.001	0.130 ± 0.005	0.017 ± 0.001	0.132 ± 0.005	0.681 ± 0.047	0.825 ± 0.033	0.182	0.022	13.65	0.1229	0.1248	1.093	97.692 ± 0.589	0.557 ± 0.012	27.692 ± 1.755
QML	beta52	0.16	0.140	-0.020 ± 0.004	-0.128 ± 0.026	0.011 ± 0.001	0.105 ± 0.004	0.012 ± 0.001	0.107 ± 0.004	0.450 ± 0.029	0.671 ± 0.025	0.140	-0.020	-12.73	0.0991	0.1012	1.067	95.077 ± 0.849	0.441 ± 0.010	26.615 ± 1.733
QML	beta53	0.16	0.141	-0.019 ± 0.004	-0.116 ± 0.024	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.377 ± 0.022	0.614 ± 0.021	0.134	-0.026	-16.45	0.0994	0.1028	1.054	95.385 ± 0.823	0.399 ± 0.004	28.615 ± 1.773
QML	beta54	0.16	0.157	-0.003 ± 0.004	-0.017 ± 0.023	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.092 ± 0.003	0.333 ± 0.020	0.577 ± 0.021	0.155	-0.005	-3.02	0.0897	0.0898	0.980	95.077 ± 0.849	0.355 ± 0.003	42.615 ± 1.940
QML	beta55	0.08	0.102	0.022 ± 0.006	0.277 ± 0.072	0.022 ± 0.001	0.147 ± 0.005	0.022 ± 0.001	0.148 ± 0.006	3.441 ± 0.228	1.855 ± 0.071	0.105	0.025	30.80	0.1430	0.1451	1.033	95.846 ± 0.783	0.595 ± 0.014	11.692 ± 1.260
QML	beta56	0.08	0.071	-0.009 ± 0.005	-0.110 ± 0.064	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	2.645 ± 0.174	1.626 ± 0.062	0.069	-0.011	-13.25	0.1246	0.1251	1.030	94.615 ± 0.885	0.525 ± 0.008	6.923 ± 0.996
QML	beta57	0.06	0.044	-0.016 ± 0.003	-0.265 ± 0.053	0.007 ± 0.000	0.081 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	1.884 ± 0.123	1.373 ± 0.052	0.038	-0.022	-36.86	0.0775	0.0806	1.075	92.615 ± 1.026	0.341 ± 0.009	3.846 ± 0.754
QML	beta58	0.06	0.054	-0.006 ± 0.003	-0.095 ± 0.043	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.190 ± 0.077	1.091 ± 0.041	0.047	-0.013	-21.39	0.0620	0.0633	1.049	94.769 ± 0.873	0.268 ± 0.004	5.846 ± 0.920
UPI	beta41	0.20	0.204	0.004 ± 0.004	0.018 ± 0.019	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.231 ± 0.014	0.480 ± 0.018	0.204	0.004	1.93	0.1030	0.1031	1.383	96.923 ± 0.677	0.520 ± 0.015	46.000 ± 1.955
UPI	beta42	0.20	0.207	0.007 ± 0.004	0.037 ± 0.020	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	0.257 ± 0.016	0.507 ± 0.018	0.204	0.004	1.91	0.1083	0.1084	1.319	98.462 ± 0.483	0.523 ± 0.012	40.923 ± 1.929
UPI	beta43	0.20	0.205	0.005 ± 0.004	0.024 ± 0.019	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.246 ± 0.016	0.496 ± 0.019	0.201	0.001	0.64	0.0999	0.0999	1.167	96.462 ± 0.725	0.454 ± 0.010	50.154 ± 1.961
UPI	beta44	0.11	0.129	0.019 ± 0.008	0.174 ± 0.070	0.039 ± 0.003	0.197 ± 0.008	0.039 ± 0.003	0.198 ± 0.009	3.224 ± 0.275	1.796 ± 0.085	0.120	0.010	9.39	0.1877	0.1880	1.506	99.077 ± 0.375	1.161 ± 0.060	2.769 ± 0.644
UPI	beta45	0.11	0.114	0.004 ± 0.006	0.034 ± 0.053	0.022 ± 0.002	0.148 ± 0.006	0.022 ± 0.002	0.148 ± 0.006	1.803 ± 0.137	1.343 ± 0.058	0.106	-0.004	-3.32	0.1457	0.1457	1.478	99.077 ± 0.375	0.856 ± 0.034	4.769 ± 0.836
UPI	beta46	0.08	0.092	0.012 ± 0.005	0.156 ± 0.067	0.019 ± 0.002	0.136 ± 0.006	0.019 ± 0.002	0.137 ± 0.007	2.913 ± 0.252	1.707 ± 0.081	0.085	0.005	6.78	0.1241	0.1242	1.484	98.923 ± 0.405	0.792 ± 0.042	2.769 ± 0.644

Table S181

Simulation 2 - Condition 37: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.087	0.007 ± 0.005	0.090 ± 0.066	0.018 ± 0.001	0.134 ± 0.005	0.018 ± 0.001	0.134 ± 0.006	2.819 ± 0.227	1.679 ± 0.075	0.077	-0.003	-3.69	0.1178	0.1178	1.470	98.154 ± 0.528	0.774 ± 0.032	2.769 ± 0.644
UPI	beta51	0.16	0.171	0.011 ± 0.005	0.066 ± 0.030	0.015 ± 0.001	0.124 ± 0.004	0.015 ± 0.001	0.124 ± 0.004	0.604 ± 0.036	0.777 ± 0.028	0.169	0.009	5.84	0.1173	0.1176	1.142	97.846 ± 0.569	0.555 ± 0.012	24.923 ± 1.697
UPI	beta52	0.16	0.157	-0.003 ± 0.004	-0.018 ± 0.023	0.008 ± 0.001	0.092 ± 0.003	0.008 ± 0.001	0.092 ± 0.003	0.329 ± 0.021	0.574 ± 0.021	0.161	0.001	0.44	0.0987	0.0987	1.252	98.308 ± 0.506	0.451 ± 0.009	30.769 ± 1.810
UPI	beta53	0.16	0.163	0.003 ± 0.004	0.020 ± 0.026	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.425 ± 0.023	0.652 ± 0.022	0.157	-0.003	-1.61	0.1062	0.1062	1.161	97.077 ± 0.661	0.474 ± 0.010	29.077 ± 1.781
UPI	beta54	0.16	0.159	-0.001 ± 0.004	-0.006 ± 0.025	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.103 ± 0.004	0.417 ± 0.029	0.646 ± 0.026	0.156	-0.004	-2.19	0.1011	0.1012	1.141	97.538 ± 0.608	0.463 ± 0.012	29.077 ± 1.781
UPI	beta55	0.08	0.099	0.019 ± 0.008	0.242 ± 0.099	0.041 ± 0.003	0.203 ± 0.008	0.041 ± 0.003	0.203 ± 0.009	6.467 ± 0.525	2.543 ± 0.115	0.093	0.013	16.32	0.1705	0.1710	1.329	98.154 ± 0.528	1.056 ± 0.042	1.385 ± 0.458
UPI	beta56	0.08	0.083	0.003 ± 0.007	0.040 ± 0.087	0.031 ± 0.002	0.177 ± 0.007	0.031 ± 0.002	0.177 ± 0.008	4.911 ± 0.378	2.216 ± 0.096	0.070	-0.010	-12.36	0.1454	0.1458	1.375	98.308 ± 0.506	0.956 ± 0.039	2.462 ± 0.608
UPI	beta57	0.06	0.058	-0.002 ± 0.004	-0.037 ± 0.074	0.013 ± 0.001	0.113 ± 0.005	0.013 ± 0.001	0.113 ± 0.005	3.546 ± 0.289	1.883 ± 0.085	0.046	-0.014	-23.70	0.0947	0.0958	1.382	97.692 ± 0.589	0.613 ± 0.024	1.077 ± 0.405
UPI	beta58	0.06	0.061	0.001 ± 0.004	0.017 ± 0.063	0.009 ± 0.001	0.096 ± 0.004	0.009 ± 0.001	0.096 ± 0.004	2.560 ± 0.202	1.600 ± 0.071	0.053	-0.007	-11.14	0.0802	0.0805	1.380	97.538 ± 0.608	0.520 ± 0.021	1.692 ± 0.506

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S182

Simulation 2 - Condition 38: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.009 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.066 ± 0.003	0.258 ± 0.007	0.201	0.001	0.59	0.0502	0.0502	1.069	97.322 ± 0.518	0.216 ± 0.001	96.601 ± 0.581
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.021 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.009	0.201	0.001	0.59	0.0552	0.0552	0.997	95.881 ± 0.638	0.223 ± 0.001	96.704 ± 0.573
LSAM	beta43	0.20	0.203	0.003 ± 0.002	0.014 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.069 ± 0.003	0.262 ± 0.007	0.200	0.000	0.09	0.0528	0.0528	0.999	95.572 ± 0.660	0.205 ± 0.001	98.249 ± 0.421
LSAM	beta44	0.11	0.112	0.002 ± 0.002	0.022 ± 0.022	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.473 ± 0.023	0.687 ± 0.020	0.111	0.001	1.27	0.0757	0.0757	1.007	96.704 ± 0.573	0.299 ± 0.002	36.972 ± 1.549
LSAM	beta45	0.11	0.114	0.004 ± 0.002	0.033 ± 0.019	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.347 ± 0.018	0.589 ± 0.018	0.113	0.003	2.28	0.0588	0.0589	0.989	95.881 ± 0.638	0.251 ± 0.002	44.387 ± 1.594
LSAM	beta46	0.08	0.084	0.004 ± 0.002	0.053 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.473 ± 0.026	0.688 ± 0.022	0.082	0.002	3.03	0.0523	0.0523	0.964	94.748 ± 0.716	0.207 ± 0.002	35.530 ± 1.536
LSAM	beta47	0.08	0.082	0.002 ± 0.002	0.025 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.419 ± 0.022	0.648 ± 0.020	0.079	-0.001	-1.69	0.0488	0.0488	1.015	96.189 ± 0.614	0.206 ± 0.002	34.089 ± 1.521
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.204 ± 0.010	0.452 ± 0.013	0.163	0.003	1.79	0.0693	0.0694	0.983	94.645 ± 0.722	0.279 ± 0.001	64.470 ± 1.536
LSAM	beta52	0.16	0.160	0.000 ± 0.002	0.000 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.120 ± 0.006	0.347 ± 0.010	0.159	-0.001	-0.34	0.0552	0.0552	0.994	94.645 ± 0.722	0.216 ± 0.001	83.625 ± 1.188
LSAM	beta53	0.16	0.164	0.004 ± 0.002	0.024 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.006	0.361 ± 0.011	0.163	0.003	2.11	0.0559	0.0560	1.025	95.057 ± 0.696	0.232 ± 0.001	81.977 ± 1.234
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.006	0.342 ± 0.010	0.158	-0.002	-1.04	0.0549	0.0550	1.002	94.645 ± 0.722	0.215 ± 0.001	84.964 ± 1.147
LSAM	beta55	0.08	0.081	0.001 ± 0.003	0.009 ± 0.034	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	1.117 ± 0.057	1.057 ± 0.032	0.080	0.000	0.39	0.0799	0.0799	1.006	95.778 ± 0.645	0.334 ± 0.003	16.478 ± 1.191
LSAM	beta56	0.08	0.084	0.004 ± 0.002	0.050 ± 0.031	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.949 ± 0.045	0.974 ± 0.028	0.085	0.005	6.09	0.0759	0.0761	1.006	95.469 ± 0.667	0.307 ± 0.003	19.773 ± 1.278
LSAM	beta57	0.06	0.061	0.001 ± 0.002	0.024 ± 0.028	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.780 ± 0.042	0.883 ± 0.028	0.060	0.000	-0.32	0.0503	0.0503	1.005	95.469 ± 0.667	0.209 ± 0.002	20.082 ± 1.286
LSAM	beta58	0.06	0.062	0.002 ± 0.001	0.035 ± 0.024	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.540 ± 0.026	0.735 ± 0.021	0.061	0.001	1.53	0.0421	0.0421	1.006	96.292 ± 0.606	0.174 ± 0.001	27.291 ± 1.430
QML	beta41	0.20	0.186	-0.014 ± 0.002	-0.072 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.186	-0.014	-7.03	0.0494	0.0513	1.028	94.233 ± 0.748	0.199 ± 0.001	96.601 ± 0.581
QML	beta42	0.20	0.192	-0.008 ± 0.002	-0.041 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.003	0.270 ± 0.008	0.190	-0.010	-5.15	0.0533	0.0542	0.987	94.336 ± 0.742	0.206 ± 0.001	97.116 ± 0.537
QML	beta43	0.20	0.183	-0.017 ± 0.002	-0.084 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.261 ± 0.007	0.181	-0.019	-9.25	0.0501	0.0534	0.984	93.203 ± 0.808	0.191 ± 0.001	97.116 ± 0.537
QML	beta44	0.11	0.097	-0.013 ± 0.002	-0.118 ± 0.020	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.419 ± 0.021	0.648 ± 0.019	0.100	-0.010	-9.31	0.0676	0.0683	1.007	94.954 ± 0.702	0.277 ± 0.002	32.853 ± 1.507
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.038 ± 0.017	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.287 ± 0.013	0.536 ± 0.015	0.105	-0.005	-4.36	0.0587	0.0589	1.008	95.160 ± 0.689	0.232 ± 0.001	45.005 ± 1.597
QML	beta46	0.08	0.084	0.004 ± 0.002	0.047 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.352 ± 0.018	0.593 ± 0.018	0.081	0.001	0.86	0.0463	0.0463	0.999	94.542 ± 0.729	0.185 ± 0.002	40.886 ± 1.578
QML	beta47	0.08	0.078	-0.002 ± 0.001	-0.022 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.320 ± 0.016	0.565 ± 0.016	0.076	-0.004	-5.13	0.0430	0.0432	1.016	94.851 ± 0.709	0.180 ± 0.002	38.002 ± 1.558
QML	beta51	0.16	0.180	0.020 ± 0.002	0.123 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.229 ± 0.010	0.478 ± 0.013	0.181	0.021	13.08	0.0727	0.0757	1.034	96.292 ± 0.606	0.300 ± 0.002	69.310 ± 1.480
QML	beta52	0.16	0.147	-0.013 ± 0.002	-0.083 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.146	-0.014	-8.70	0.0609	0.0625	1.009	94.130 ± 0.754	0.248 ± 0.002	66.838 ± 1.511
QML	beta53	0.16	0.137	-0.023 ± 0.002	-0.141 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.148 ± 0.007	0.385 ± 0.010	0.138	-0.022	-13.73	0.0561	0.0602	1.014	92.276 ± 0.857	0.228 ± 0.001	67.868 ± 1.499
QML	beta54	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.112 ± 0.005	0.335 ± 0.009	0.157	-0.003	-2.19	0.0549	0.0551	0.998	95.057 ± 0.696	0.210 ± 0.001	86.612 ± 1.093
QML	beta55	0.08	0.090	0.010 ± 0.003	0.121 ± 0.034	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	1.118 ± 0.053	1.057 ± 0.030	0.087	0.007	8.61	0.0845	0.0847	1.032	95.366 ± 0.675	0.340 ± 0.003	19.361 ± 1.268
QML	beta56	0.08	0.078	-0.002 ± 0.002	-0.028 ± 0.030	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.848 ± 0.038	0.921 ± 0.025	0.079	-0.001	-1.61	0.0733	0.0733	1.018	94.439 ± 0.735	0.294 ± 0.002	18.023 ± 1.234
QML	beta57	0.06	0.049	-0.011 ± 0.002	-0.178 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.764 ± 0.039	0.874 ± 0.026	0.047	-0.013	-21.65	0.0485	0.0503	0.997	91.658 ± 0.887	0.201 ± 0.002	11.020 ± 1.005
QML	beta58	0.06	0.053	-0.007 ± 0.001	-0.121 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.472 ± 0.020	0.687 ± 0.018	0.051	-0.009	-15.54	0.0382	0.0393	1.010	93.615 ± 0.785	0.161 ± 0.001	21.215 ± 1.312
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.201	0.001	0.42	0.0529	0.0529	1.103	97.631 ± 0.488	0.228 ± 0.001	95.057 ± 0.696
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.023 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.288 ± 0.009	0.202	0.002	1.07	0.0565	0.0565	1.050	95.778 ± 0.645	0.236 ± 0.002	94.439 ± 0.735
UPI	beta43	0.20	0.203	0.003 ± 0.002	0.016 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.003	0.270 ± 0.008	0.200	0.000	-0.16	0.0559	0.0559	1.028	96.395 ± 0.598	0.217 ± 0.002	97.219 ± 0.528
UPI	beta44	0.11	0.113	0.003 ± 0.003	0.032 ± 0.025	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.587 ± 0.030	0.766 ± 0.023	0.112	0.002	2.16	0.0804	0.0804	1.159	98.352 ± 0.409	0.383 ± 0.006	25.129 ± 1.392
UPI	beta45	0.11	0.117	0.007 ± 0.002	0.066 ± 0.022	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.454 ± 0.027	0.674 ± 0.023	0.110	0.000	0.16	0.0620	0.0620	1.071	98.043 ± 0.444	0.310 ± 0.003	31.205 ± 1.487
UPI	beta46	0.08	0.088	0.008 ± 0.002	0.102 ± 0.025	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.624 ± 0.036	0.790 ± 0.026	0.083	0.003	3.72	0.0556	0.0557	1.108	97.425 ± 0.508	0.272 ± 0.005	21.627 ± 1.321

Table S182

Simulation 2 - Condition 38: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.083	0.003 ± 0.002	0.039 ± 0.024	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.572 ± 0.033	0.756 ± 0.025	0.079	-0.001	-1.15	0.0533	0.0533	1.114	96.704 ± 0.573	0.264 ± 0.004	20.494 ± 1.295
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.020 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.009	0.437 ± 0.012	0.165	0.005	3.14	0.0657	0.0659	1.033	96.601 ± 0.581	0.283 ± 0.002	66.117 ± 1.519
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.006	0.345 ± 0.010	0.159	-0.001	-0.75	0.0556	0.0556	1.020	95.263 ± 0.682	0.221 ± 0.001	82.698 ± 1.214
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.019 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.129 ± 0.006	0.360 ± 0.010	0.163	0.003	1.69	0.0571	0.0572	1.059	95.881 ± 0.638	0.239 ± 0.002	79.712 ± 1.291
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.116 ± 0.006	0.341 ± 0.010	0.157	-0.003	-1.72	0.0535	0.0536	1.037	94.748 ± 0.716	0.222 ± 0.003	84.140 ± 1.172
UPI	beta55	0.08	0.082	0.002 ± 0.003	0.019 ± 0.034	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	1.117 ± 0.061	1.057 ± 0.034	0.080	0.000	-0.35	0.0781	0.0781	1.131	97.219 ± 0.528	0.375 ± 0.005	11.740 ± 1.033
UPI	beta56	0.08	0.080	-0.000 ± 0.003	-0.002 ± 0.032	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.973 ± 0.048	0.986 ± 0.029	0.080	0.000	0.41	0.0792	0.0792	1.132	96.189 ± 0.614	0.350 ± 0.005	12.461 ± 1.060
UPI	beta57	0.06	0.062	0.002 ± 0.002	0.030 ± 0.030	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.850 ± 0.048	0.922 ± 0.030	0.061	0.001	0.88	0.0521	0.0521	1.101	96.704 ± 0.573	0.239 ± 0.003	12.049 ± 1.045
UPI	beta58	0.06	0.063	0.003 ± 0.002	0.051 ± 0.026	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.658 ± 0.038	0.811 ± 0.027	0.059	-0.001	-0.85	0.0466	0.0466	1.124	96.498 ± 0.590	0.214 ± 0.011	18.435 ± 1.244

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S183

Simulation 2 - Condition 39: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.002	0.013 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.004	0.301 ± 0.009	0.201	0.001	0.32	0.0566	0.0566	1.026	95.951 ± 0.627	0.242 ± 0.001	91.296 ± 0.897
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.022 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.005	0.321 ± 0.009	0.204	0.004	2.19	0.0637	0.0638	0.983	96.154 ± 0.612	0.247 ± 0.001	90.283 ± 0.942
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.092 ± 0.004	0.303 ± 0.008	0.197	-0.003	-1.41	0.0623	0.0623	0.964	93.927 ± 0.760	0.229 ± 0.001	92.409 ± 0.843
LSAM	beta44	0.11	0.109	-0.001 ± 0.002	-0.010 ± 0.022	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.473 ± 0.025	0.688 ± 0.021	0.105	-0.005	-4.88	0.0707	0.0709	0.984	95.142 ± 0.684	0.292 ± 0.002	35.324 ± 1.521
LSAM	beta45	0.11	0.115	0.005 ± 0.002	0.043 ± 0.020	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.388 ± 0.018	0.623 ± 0.017	0.114	0.004	3.60	0.0683	0.0684	0.931	92.510 ± 0.837	0.250 ± 0.002	45.142 ± 1.583
LSAM	beta46	0.08	0.084	0.004 ± 0.002	0.051 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.437 ± 0.022	0.661 ± 0.020	0.082	0.002	2.24	0.0482	0.0482	0.993	95.445 ± 0.663	0.205 ± 0.002	35.628 ± 1.524
LSAM	beta47	0.08	0.084	0.004 ± 0.002	0.053 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.444 ± 0.021	0.667 ± 0.019	0.081	0.001	0.84	0.0520	0.0520	0.961	94.636 ± 0.717	0.200 ± 0.002	37.247 ± 1.538
LSAM	beta51	0.16	0.159	-0.001 ± 0.003	-0.007 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.283 ± 0.013	0.532 ± 0.015	0.157	-0.003	-1.84	0.0824	0.0824	0.934	92.713 ± 0.827	0.312 ± 0.002	52.227 ± 1.589
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.007	0.413 ± 0.011	0.158	-0.002	-0.95	0.0698	0.0699	0.958	95.445 ± 0.663	0.248 ± 0.001	71.559 ± 1.435
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.008	0.437 ± 0.012	0.157	-0.003	-1.67	0.0678	0.0678	0.952	94.231 ± 0.742	0.261 ± 0.001	65.891 ± 1.508
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.007	0.406 ± 0.011	0.163	0.003	1.88	0.0678	0.0678	0.959	94.737 ± 0.710	0.244 ± 0.001	73.482 ± 1.404
LSAM	beta55	0.08	0.084	0.004 ± 0.003	0.052 ± 0.036	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	1.258 ± 0.064	1.122 ± 0.034	0.083	0.003	3.68	0.0843	0.0843	0.949	94.737 ± 0.710	0.334 ± 0.003	19.231 ± 1.254
LSAM	beta56	0.08	0.081	0.001 ± 0.003	0.014 ± 0.031	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.969 ± 0.048	0.984 ± 0.029	0.079	-0.001	-1.62	0.0758	0.0758	0.992	95.040 ± 0.691	0.306 ± 0.003	20.445 ± 1.283
LSAM	beta57	0.06	0.063	0.003 ± 0.002	0.051 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.844 ± 0.041	0.919 ± 0.027	0.063	0.003	5.19	0.0522	0.0523	0.980	94.838 ± 0.704	0.212 ± 0.002	21.053 ± 1.297
LSAM	beta58	0.06	0.060	-0.000 ± 0.001	-0.002 ± 0.025	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.596 ± 0.031	0.772 ± 0.023	0.058	-0.002	-3.50	0.0435	0.0435	0.979	94.636 ± 0.717	0.178 ± 0.001	23.583 ± 1.351
QML	beta41	0.20	0.194	-0.006 ± 0.002	-0.032 ± 0.009	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.009	0.192	-0.008	-4.15	0.0569	0.0575	1.016	95.547 ± 0.656	0.235 ± 0.001	90.789 ± 0.920
QML	beta42	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.004	0.313 ± 0.009	0.198	-0.002	-0.75	0.0625	0.0625	0.985	95.951 ± 0.627	0.242 ± 0.001	90.081 ± 0.951
QML	beta43	0.20	0.188	-0.012 ± 0.002	-0.062 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.294 ± 0.008	0.185	-0.015	-7.28	0.0579	0.0597	0.984	93.725 ± 0.772	0.221 ± 0.001	91.802 ± 0.873
QML	beta44	0.11	0.098	-0.012 ± 0.002	-0.113 ± 0.022	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.473 ± 0.025	0.688 ± 0.021	0.099	-0.011	-9.80	0.0692	0.0701	1.004	95.648 ± 0.649	0.294 ± 0.003	32.895 ± 1.495
QML	beta45	0.11	0.111	0.001 ± 0.002	0.007 ± 0.019	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.358 ± 0.017	0.599 ± 0.017	0.110	0.000	0.31	0.0664	0.0664	0.958	93.725 ± 0.772	0.247 ± 0.002	43.320 ± 1.576
QML	beta46	0.08	0.085	0.005 ± 0.002	0.058 ± 0.020	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.396 ± 0.020	0.629 ± 0.019	0.081	0.001	1.57	0.0466	0.0467	1.014	95.243 ± 0.677	0.199 ± 0.002	36.640 ± 1.533
QML	beta47	0.08	0.084	0.004 ± 0.002	0.050 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.410 ± 0.021	0.640 ± 0.019	0.079	-0.001	-1.35	0.0508	0.0508	0.983	95.040 ± 0.691	0.197 ± 0.002	34.919 ± 1.517
QML	beta51	0.16	0.170	0.010 ± 0.003	0.063 ± 0.017	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.298 ± 0.014	0.546 ± 0.016	0.170	0.010	6.02	0.0826	0.0832	0.993	94.534 ± 0.723	0.338 ± 0.003	51.822 ± 1.590
QML	beta52	0.16	0.144	-0.016 ± 0.002	-0.103 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.210 ± 0.009	0.458 ± 0.012	0.139	-0.021	-13.19	0.0739	0.0769	1.008	93.219 ± 0.800	0.282 ± 0.002	53.543 ± 1.587
QML	beta53	0.16	0.134	-0.026 ± 0.002	-0.161 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.215 ± 0.009	0.464 ± 0.012	0.135	-0.025	-15.54	0.0707	0.0749	0.980	93.623 ± 0.777	0.268 ± 0.002	51.518 ± 1.590
QML	beta54	0.16	0.163	0.003 ± 0.002	0.017 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.007	0.404 ± 0.011	0.162	0.002	1.11	0.0660	0.0660	0.968	94.737 ± 0.710	0.246 ± 0.001	73.785 ± 1.399
QML	beta55	0.08	0.089	0.009 ± 0.003	0.111 ± 0.035	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	1.215 ± 0.057	1.102 ± 0.031	0.091	0.011	13.93	0.0860	0.0867	1.001	95.344 ± 0.670	0.344 ± 0.003	19.534 ± 1.261
QML	beta56	0.08	0.081	0.001 ± 0.002	0.009 ± 0.031	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.920 ± 0.046	0.959 ± 0.028	0.077	-0.003	-4.33	0.0721	0.0721	1.031	95.445 ± 0.663	0.310 ± 0.003	17.713 ± 1.215
QML	beta57	0.06	0.056	-0.004 ± 0.002	-0.072 ± 0.028	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.791 ± 0.037	0.889 ± 0.025	0.052	-0.008	-13.65	0.0522	0.0528	1.017	93.320 ± 0.794	0.212 ± 0.003	13.866 ± 1.099
QML	beta58	0.06	0.056	-0.004 ± 0.001	-0.063 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.578 ± 0.031	0.760 ± 0.024	0.052	-0.008	-12.85	0.0411	0.0418	0.979	94.332 ± 0.736	0.175 ± 0.002	19.231 ± 1.254
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.016 ± 0.010	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.005	0.306 ± 0.009	0.201	0.001	0.70	0.0579	0.0579	1.015	95.547 ± 0.656	0.244 ± 0.001	90.789 ± 0.920
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.025 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.104 ± 0.005	0.323 ± 0.009	0.203	0.003	1.52	0.0627	0.0628	0.995	96.356 ± 0.596	0.251 ± 0.001	89.271 ± 0.985
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.004	0.306 ± 0.008	0.197	-0.003	-1.69	0.0639	0.0640	0.977	94.332 ± 0.736	0.235 ± 0.001	91.700 ± 0.878
UPI	beta44	0.11	0.112	0.002 ± 0.003	0.018 ± 0.024	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.567 ± 0.031	0.753 ± 0.024	0.109	-0.001	-0.50	0.0749	0.0749	1.032	97.166 ± 0.528	0.335 ± 0.003	27.126 ± 1.414
UPI	beta45	0.11	0.115	0.005 ± 0.002	0.046 ± 0.021	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.446 ± 0.022	0.668 ± 0.020	0.115	0.005	4.32	0.0674	0.0676	0.981	94.130 ± 0.748	0.282 ± 0.002	38.765 ± 1.550
UPI	beta46	0.08	0.084	0.004 ± 0.002	0.048 ± 0.023	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.542 ± 0.028	0.736 ± 0.023	0.081	0.001	0.68	0.0552	0.0552	1.037	96.559 ± 0.580	0.239 ± 0.003	27.834 ± 1.426

Table S183

Simulation 2 - Condition 39: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.086	0.006 ± 0.002	0.079 ± 0.023	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.542 ± 0.029	0.736 ± 0.023	0.082	0.002	1.96	0.0557	0.0557	1.021	94.939 ± 0.697	0.235 ± 0.003	27.126 ± 1.414
UPI	beta51	0.16	0.160	0.000 ± 0.003	0.001 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.265 ± 0.012	0.515 ± 0.015	0.159	-0.001	-0.68	0.0785	0.0785	0.977	93.623 ± 0.777	0.316 ± 0.002	51.012 ± 1.590
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.016 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.007	0.411 ± 0.011	0.161	0.001	0.76	0.0687	0.0687	0.962	95.547 ± 0.656	0.248 ± 0.001	72.976 ± 1.413
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.008	0.437 ± 0.012	0.158	-0.002	-1.47	0.0667	0.0667	0.965	94.028 ± 0.754	0.264 ± 0.001	64.777 ± 1.520
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.021 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.162 ± 0.007	0.402 ± 0.011	0.162	0.002	1.23	0.0658	0.0659	0.978	94.433 ± 0.729	0.246 ± 0.001	73.178 ± 1.409
UPI	beta55	0.08	0.081	0.001 ± 0.003	0.010 ± 0.036	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	1.277 ± 0.064	1.130 ± 0.034	0.082	0.002	3.02	0.0846	0.0846	1.002	96.761 ± 0.563	0.355 ± 0.004	16.802 ± 1.189
UPI	beta56	0.08	0.081	0.001 ± 0.003	0.010 ± 0.032	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	1.030 ± 0.052	1.015 ± 0.030	0.078	-0.002	-2.47	0.0721	0.0722	1.040	95.850 ± 0.634	0.331 ± 0.004	13.968 ± 1.103
UPI	beta57	0.06	0.063	0.003 ± 0.002	0.058 ± 0.030	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.885 ± 0.046	0.941 ± 0.029	0.064	0.004	6.14	0.0541	0.0542	1.027	95.344 ± 0.670	0.227 ± 0.002	16.599 ± 1.184
UPI	beta58	0.06	0.061	0.001 ± 0.002	0.009 ± 0.026	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.676 ± 0.035	0.822 ± 0.025	0.055	-0.005	-7.51	0.0451	0.0453	1.009	94.737 ± 0.710	0.195 ± 0.002	18.826 ± 1.244

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S184

Simulation 2 - Condition 40: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.199	-0.001	-0.65	0.0375	0.0376	1.001	95.300 ± 0.669	0.149 ± 0.000	99.800 ± 0.141
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.202	0.002	1.09	0.0378	0.0379	1.018	95.900 ± 0.627	0.153 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.200	0.000	-0.01	0.0348	0.0348	0.980	93.800 ± 0.763	0.143 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.109	-0.001 ± 0.001	-0.006 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.162 ± 0.007	0.402 ± 0.011	0.108	-0.002	-1.54	0.0437	0.0437	1.009	95.000 ± 0.689	0.175 ± 0.001	68.900 ± 1.464
LSAM	beta45	0.11	0.110	0.000 ± 0.001	0.003 ± 0.011	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.124 ± 0.005	0.353 ± 0.009	0.111	0.001	0.91	0.0402	0.0402	1.003	95.900 ± 0.627	0.153 ± 0.001	81.400 ± 1.230
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.009 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.161 ± 0.007	0.401 ± 0.011	0.079	-0.001	-1.59	0.0318	0.0319	0.975	94.500 ± 0.721	0.123 ± 0.001	74.800 ± 1.373
LSAM	beta47	0.08	0.082	0.002 ± 0.001	0.020 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.158 ± 0.008	0.397 ± 0.012	0.081	0.001	1.55	0.0306	0.0307	0.978	94.000 ± 0.751	0.122 ± 0.001	77.700 ± 1.316
LSAM	beta51	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.314 ± 0.009	0.161	0.001	0.55	0.0532	0.0532	0.979	94.900 ± 0.696	0.193 ± 0.001	90.000 ± 0.949
LSAM	beta52	0.16	0.162	0.002 ± 0.001	0.014 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.163	0.003	1.57	0.0376	0.0377	0.985	94.200 ± 0.739	0.155 ± 0.000	98.200 ± 0.420
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.274 ± 0.007	0.161	0.001	0.92	0.0443	0.0443	0.945	93.900 ± 0.757	0.163 ± 0.000	96.300 ± 0.597
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.160	0.000	-0.26	0.0379	0.0379	1.022	95.800 ± 0.634	0.153 ± 0.000	98.700 ± 0.358
LSAM	beta55	0.08	0.080	-0.000 ± 0.002	-0.001 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.400 ± 0.018	0.633 ± 0.017	0.079	-0.001	-1.78	0.0506	0.0506	0.983	95.300 ± 0.669	0.195 ± 0.001	37.100 ± 1.528
LSAM	beta56	0.08	0.082	0.002 ± 0.002	0.020 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.370 ± 0.017	0.608 ± 0.017	0.081	0.001	1.84	0.0468	0.0468	0.966	95.000 ± 0.689	0.184 ± 0.001	43.400 ± 1.567
LSAM	beta57	0.06	0.060	0.000 ± 0.001	0.003 ± 0.018	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.309 ± 0.014	0.556 ± 0.016	0.059	-0.001	-1.90	0.0324	0.0324	0.970	94.000 ± 0.751	0.127 ± 0.001	47.300 ± 1.579
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.018 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.225 ± 0.010	0.475 ± 0.013	0.060	0.000	-0.49	0.0288	0.0288	0.981	95.200 ± 0.676	0.110 ± 0.001	59.300 ± 1.554
QML	beta41	0.20	0.190	-0.010 ± 0.001	-0.050 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.190	-0.010	-5.10	0.0365	0.0379	0.996	93.200 ± 0.796	0.148 ± 0.001	99.700 ± 0.173
QML	beta42	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.196	-0.004	-1.97	0.0374	0.0376	1.020	95.100 ± 0.683	0.151 ± 0.001	99.900 ± 0.100
QML	beta43	0.20	0.190	-0.010 ± 0.001	-0.049 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.188	-0.012	-5.81	0.0341	0.0360	0.982	92.800 ± 0.817	0.139 ± 0.000	99.900 ± 0.100
QML	beta44	0.11	0.102	-0.008 ± 0.001	-0.074 ± 0.013	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.171 ± 0.008	0.413 ± 0.011	0.102	-0.008	-7.08	0.0446	0.0452	1.025	95.500 ± 0.656	0.180 ± 0.001	60.900 ± 1.543
QML	beta45	0.11	0.108	-0.002 ± 0.001	-0.014 ± 0.011	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.127 ± 0.006	0.357 ± 0.010	0.108	-0.002	-1.40	0.0397	0.0397	1.008	95.700 ± 0.641	0.155 ± 0.001	78.100 ± 1.308
QML	beta46	0.08	0.081	0.001 ± 0.001	0.007 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.161 ± 0.008	0.402 ± 0.012	0.077	-0.003	-3.49	0.0316	0.0317	0.970	94.100 ± 0.745	0.122 ± 0.001	77.400 ± 1.323
QML	beta47	0.08	0.080	-0.000 ± 0.001	-0.000 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.150 ± 0.007	0.387 ± 0.011	0.079	-0.001	-1.87	0.0292	0.0292	0.987	93.500 ± 0.780	0.120 ± 0.001	79.700 ± 1.272
QML	beta51	0.16	0.171	0.011 ± 0.002	0.069 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.332 ± 0.009	0.171	0.011	6.68	0.0540	0.0551	0.994	94.400 ± 0.727	0.202 ± 0.001	91.300 ± 0.891
QML	beta52	0.16	0.146	-0.014 ± 0.001	-0.089 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.145	-0.015	-9.58	0.0407	0.0435	0.980	92.200 ± 0.848	0.172 ± 0.001	92.000 ± 0.858
QML	beta53	0.16	0.137	-0.023 ± 0.001	-0.144 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.008	0.139	-0.021	-13.39	0.0445	0.0494	0.951	89.600 ± 0.965	0.164 ± 0.000	89.200 ± 0.982
QML	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.242 ± 0.007	0.160	0.000	-0.23	0.0375	0.0375	1.021	95.900 ± 0.627	0.155 ± 0.000	98.300 ± 0.409
QML	beta55	0.08	0.086	0.006 ± 0.002	0.081 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.446 ± 0.020	0.668 ± 0.019	0.085	0.005	5.97	0.0523	0.0525	0.991	95.100 ± 0.683	0.206 ± 0.001	39.700 ± 1.547
QML	beta56	0.08	0.082	0.002 ± 0.002	0.020 ± 0.020	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.389 ± 0.018	0.624 ± 0.017	0.082	0.002	2.96	0.0493	0.0493	0.966	93.900 ± 0.757	0.189 ± 0.001	41.100 ± 1.556
QML	beta57	0.06	0.053	-0.007 ± 0.001	-0.116 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.325 ± 0.014	0.570 ± 0.015	0.051	-0.009	-15.81	0.0321	0.0335	0.977	92.200 ± 0.848	0.128 ± 0.001	34.800 ± 1.506
QML	beta58	0.06	0.057	-0.003 ± 0.001	-0.051 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.232 ± 0.010	0.482 ± 0.013	0.055	-0.005	-8.09	0.0296	0.0300	0.978	93.300 ± 0.791	0.110 ± 0.001	52.100 ± 1.580
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.199	-0.001	-0.56	0.0372	0.0372	0.977	94.700 ± 0.708	0.147 ± 0.000	99.800 ± 0.141
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.201	0.001	0.42	0.0389	0.0389	1.008	95.800 ± 0.634	0.151 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.187 ± 0.005	0.199	-0.001	-0.36	0.0357	0.0357	0.979	93.800 ± 0.763	0.144 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.109	-0.001 ± 0.001	-0.006 ± 0.013	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.180 ± 0.008	0.424 ± 0.012	0.108	-0.002	-1.75	0.0467	0.0467	1.016	95.700 ± 0.641	0.186 ± 0.001	64.600 ± 1.512
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.007 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.135 ± 0.006	0.368 ± 0.010	0.111	0.001	0.48	0.0389	0.0389	1.015	95.700 ± 0.641	0.161 ± 0.001	78.500 ± 1.299
UPI	beta46	0.08	0.081	0.001 ± 0.001	0.010 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.176 ± 0.008	0.419 ± 0.011	0.079	-0.001	-1.68	0.0342	0.0342	0.994	95.200 ± 0.676	0.131 ± 0.001	70.400 ± 1.444

Table S184

Simulation 2 - Condition 40: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.082	0.002 ± 0.001	0.024 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.174 ± 0.009	0.417 ± 0.012	0.081	0.001	1.52	0.0321	0.0321	0.995	94.700 ± 0.708	0.130 ± 0.001	73.600 ± 1.394
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.008	0.160	0.000	0.26	0.0528	0.0528	0.988	94.800 ± 0.702	0.193 ± 0.001	90.600 ± 0.923
UPI	beta52	0.16	0.163	0.003 ± 0.001	0.017 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.163	0.003	1.73	0.0374	0.0375	0.979	94.100 ± 0.745	0.154 ± 0.000	98.200 ± 0.420
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.274 ± 0.007	0.161	0.001	0.89	0.0448	0.0448	0.948	94.000 ± 0.751	0.163 ± 0.000	96.100 ± 0.612
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.004 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.160	0.000	0.05	0.0382	0.0382	1.017	95.600 ± 0.649	0.153 ± 0.000	98.600 ± 0.372
UPI	beta55	0.08	0.080	0.000 ± 0.002	0.001 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.414 ± 0.018	0.643 ± 0.018	0.080	0.000	-0.49	0.0494	0.0494	1.001	96.900 ± 0.548	0.202 ± 0.001	35.200 ± 1.510
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.013 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.377 ± 0.017	0.614 ± 0.017	0.080	0.000	0.38	0.0473	0.0473	0.992	95.000 ± 0.689	0.191 ± 0.001	38.700 ± 1.540
UPI	beta57	0.06	0.060	-0.000 ± 0.001	-0.007 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.322 ± 0.016	0.567 ± 0.017	0.058	-0.002	-2.91	0.0336	0.0336	0.976	94.700 ± 0.708	0.130 ± 0.001	43.300 ± 1.567
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.016 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.231 ± 0.011	0.481 ± 0.014	0.060	0.000	0.22	0.0285	0.0285	1.008	95.800 ± 0.634	0.114 ± 0.001	57.000 ± 1.566

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S185

Simulation 2 - Condition 41: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.196	-0.004 ± 0.002	-0.018 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.197	-0.003	-1.70	0.0484	0.0485	0.996	95.892 ± 0.628	0.191 ± 0.001	96.894 ± 0.549
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.198	-0.002	-0.82	0.0503	0.0503	0.987	93.888 ± 0.758	0.192 ± 0.001	98.297 ± 0.410
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.200	0.000	0.11	0.0478	0.0478	0.969	93.387 ± 0.787	0.180 ± 0.001	98.998 ± 0.315
LSAM	beta44	0.11	0.108	-0.002 ± 0.002	-0.018 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.260 ± 0.012	0.510 ± 0.014	0.108	-0.002	-2.14	0.0543	0.0544	0.924	93.287 ± 0.792	0.203 ± 0.001	57.014 ± 1.567
LSAM	beta45	0.11	0.112	0.002 ± 0.002	0.021 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.201 ± 0.009	0.448 ± 0.012	0.112	0.002	2.25	0.0500	0.0500	0.935	93.788 ± 0.764	0.181 ± 0.001	68.337 ± 1.472
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.014 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.229 ± 0.010	0.478 ± 0.013	0.079	-0.001	-0.98	0.0374	0.0374	0.948	93.687 ± 0.770	0.142 ± 0.001	62.926 ± 1.529
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.218 ± 0.010	0.467 ± 0.013	0.078	-0.002	-1.99	0.0372	0.0372	0.966	93.687 ± 0.770	0.142 ± 0.001	60.621 ± 1.547
LSAM	beta51	0.16	0.160	0.000 ± 0.002	0.003 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.007	0.413 ± 0.011	0.161	0.001	0.38	0.0692	0.0692	0.929	93.487 ± 0.781	0.241 ± 0.001	73.547 ± 1.396
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.004 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.116 ± 0.005	0.340 ± 0.009	0.160	0.000	-0.20	0.0527	0.0527	0.943	93.387 ± 0.787	0.201 ± 0.001	84.669 ± 1.140
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.003 ± 0.011	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.129 ± 0.006	0.359 ± 0.010	0.160	0.000	-0.11	0.0575	0.0575	0.933	93.687 ± 0.770	0.211 ± 0.001	82.766 ± 1.196
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.158	-0.002	-1.48	0.0550	0.0550	0.929	93.387 ± 0.787	0.199 ± 0.001	86.072 ± 1.096
LSAM	beta55	0.08	0.081	0.001 ± 0.002	0.015 ± 0.026	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.694 ± 0.032	0.833 ± 0.024	0.081	0.001	1.14	0.0647	0.0648	0.906	93.687 ± 0.770	0.237 ± 0.002	31.964 ± 1.476
LSAM	beta56	0.08	0.078	-0.002 ± 0.002	-0.024 ± 0.024	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.594 ± 0.028	0.771 ± 0.022	0.077	-0.003	-3.34	0.0574	0.0574	0.924	91.583 ± 0.879	0.224 ± 0.002	29.960 ± 1.450
LSAM	beta57	0.06	0.060	-0.000 ± 0.001	-0.000 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.464 ± 0.023	0.681 ± 0.020	0.059	-0.001	-2.10	0.0424	0.0424	0.955	94.188 ± 0.741	0.153 ± 0.001	34.870 ± 1.509
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.001 ± 0.019	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.370 ± 0.016	0.609 ± 0.016	0.059	-0.001	-2.38	0.0369	0.0369	0.941	92.385 ± 0.840	0.135 ± 0.001	41.483 ± 1.560
QML	beta41	0.20	0.192	-0.008 ± 0.002	-0.038 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.193	-0.007	-3.34	0.0492	0.0497	1.008	95.491 ± 0.657	0.194 ± 0.003	95.892 ± 0.628
QML	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.196	-0.004	-1.91	0.0503	0.0505	0.991	94.389 ± 0.728	0.193 ± 0.001	97.796 ± 0.465
QML	beta43	0.20	0.197	-0.003 ± 0.001	-0.017 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.195	-0.005	-2.28	0.0481	0.0483	0.967	93.687 ± 0.770	0.178 ± 0.001	99.098 ± 0.299
QML	beta44	0.11	0.104	-0.006 ± 0.002	-0.058 ± 0.017	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.276 ± 0.013	0.525 ± 0.015	0.103	-0.007	-5.99	0.0545	0.0549	0.928	93.287 ± 0.792	0.209 ± 0.002	52.505 ± 1.581
QML	beta45	0.11	0.112	0.002 ± 0.002	0.015 ± 0.014	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.207 ± 0.009	0.455 ± 0.012	0.111	0.001	0.73	0.0505	0.0505	0.938	93.687 ± 0.770	0.184 ± 0.001	67.635 ± 1.481
QML	beta46	0.08	0.082	0.002 ± 0.001	0.021 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.230 ± 0.011	0.480 ± 0.014	0.079	-0.001	-1.20	0.0366	0.0366	0.959	93.587 ± 0.775	0.144 ± 0.001	62.124 ± 1.535
QML	beta47	0.08	0.079	-0.001 ± 0.001	-0.006 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.221 ± 0.010	0.470 ± 0.013	0.079	-0.001	-1.67	0.0378	0.0378	0.968	92.786 ± 0.819	0.143 ± 0.001	59.920 ± 1.551
QML	beta51	0.16	0.164	0.004 ± 0.002	0.027 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.172 ± 0.007	0.415 ± 0.011	0.164	0.004	2.59	0.0694	0.0695	0.966	95.090 ± 0.684	0.251 ± 0.001	72.345 ± 1.416
QML	beta52	0.16	0.144	-0.016 ± 0.002	-0.103 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.148 ± 0.007	0.385 ± 0.011	0.143	-0.017	-10.57	0.0586	0.0610	0.945	92.285 ± 0.845	0.220 ± 0.001	72.345 ± 1.416
QML	beta53	0.16	0.140	-0.020 ± 0.002	-0.123 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.006	0.387 ± 0.010	0.139	-0.021	-12.92	0.0598	0.0633	0.946	91.683 ± 0.874	0.218 ± 0.001	70.040 ± 1.450
QML	beta54	0.16	0.158	-0.002 ± 0.002	-0.009 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.117 ± 0.005	0.343 ± 0.009	0.157	-0.003	-1.71	0.0548	0.0549	0.944	93.186 ± 0.798	0.203 ± 0.002	85.070 ± 1.128
QML	beta55	0.08	0.084	0.004 ± 0.002	0.048 ± 0.027	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.704 ± 0.033	0.839 ± 0.024	0.083	0.003	4.15	0.0670	0.0670	0.928	93.888 ± 0.758	0.244 ± 0.002	32.265 ± 1.480
QML	beta56	0.08	0.079	-0.001 ± 0.002	-0.018 ± 0.024	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.591 ± 0.028	0.769 ± 0.022	0.078	-0.002	-2.51	0.0557	0.0557	0.945	92.786 ± 0.819	0.228 ± 0.002	28.557 ± 1.430
QML	beta57	0.06	0.057	-0.003 ± 0.001	-0.044 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.472 ± 0.023	0.687 ± 0.020	0.056	-0.004	-6.74	0.0416	0.0418	0.963	94.088 ± 0.747	0.155 ± 0.001	29.459 ± 1.443
QML	beta58	0.06	0.059	-0.001 ± 0.001	-0.023 ± 0.019	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.379 ± 0.017	0.615 ± 0.017	0.056	-0.004	-5.98	0.0357	0.0359	0.943	91.984 ± 0.860	0.136 ± 0.001	37.876 ± 1.535
UPI	beta41	0.20	0.196	-0.004 ± 0.002	-0.018 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.197	-0.003	-1.73	0.0480	0.0481	0.952	95.391 ± 0.664	0.184 ± 0.001	96.794 ± 0.558
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.198	-0.002	-0.84	0.0498	0.0498	0.962	93.788 ± 0.764	0.188 ± 0.001	97.996 ± 0.444
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.200	0.000	-0.18	0.0490	0.0490	0.963	93.387 ± 0.787	0.180 ± 0.001	99.098 ± 0.299
UPI	beta44	0.11	0.109	-0.001 ± 0.002	-0.012 ± 0.017	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.293 ± 0.015	0.541 ± 0.016	0.108	-0.002	-2.14	0.0567	0.0568	0.908	92.786 ± 0.819	0.212 ± 0.001	54.509 ± 1.576
UPI	beta45	0.11	0.113	0.003 ± 0.002	0.025 ± 0.015	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.214 ± 0.010	0.462 ± 0.013	0.111	0.001	1.23	0.0520	0.0520	0.939	94.489 ± 0.722	0.187 ± 0.001	66.132 ± 1.498
UPI	beta46	0.08	0.081	0.001 ± 0.001	0.014 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.246 ± 0.011	0.496 ± 0.014	0.079	-0.001	-0.84	0.0373	0.0373	0.955	93.888 ± 0.758	0.149 ± 0.001	58.818 ± 1.558

Table S185

Simulation 2 - Condition 41: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.239 ± 0.012	0.489 ± 0.014	0.078	-0.002	-2.72	0.0383	0.0384	0.967	93.988 ± 0.752	0.148 ± 0.001	56.814 ± 1.568
UPI	beta51	0.16	0.160	0.000 ± 0.002	0.003 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.007	0.408 ± 0.011	0.161	0.001	0.37	0.0690	0.0690	0.949	94.689 ± 0.710	0.243 ± 0.001	73.347 ± 1.400
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.006 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.005	0.340 ± 0.009	0.161	0.001	0.57	0.0528	0.0528	0.934	93.387 ± 0.787	0.199 ± 0.001	86.273 ± 1.089
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.159	-0.001	-0.48	0.0566	0.0566	0.938	93.788 ± 0.764	0.211 ± 0.001	83.768 ± 1.167
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.157	-0.003	-1.81	0.0544	0.0544	0.925	93.086 ± 0.803	0.198 ± 0.001	86.373 ± 1.086
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.010 ± 0.027	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.711 ± 0.035	0.843 ± 0.025	0.080	0.000	-0.14	0.0684	0.0684	0.917	93.888 ± 0.758	0.243 ± 0.002	29.860 ± 1.449
UPI	beta56	0.08	0.078	-0.002 ± 0.002	-0.020 ± 0.024	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.597 ± 0.028	0.773 ± 0.022	0.077	-0.003	-4.00	0.0594	0.0595	0.947	93.287 ± 0.792	0.230 ± 0.002	28.557 ± 1.430
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.001 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.467 ± 0.022	0.683 ± 0.019	0.059	-0.001	-1.60	0.0416	0.0416	0.968	93.788 ± 0.764	0.156 ± 0.001	33.367 ± 1.493
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.011 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.384 ± 0.017	0.620 ± 0.017	0.059	-0.001	-1.72	0.0348	0.0349	0.954	93.186 ± 0.798	0.139 ± 0.001	39.579 ± 1.548

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S186

Simulation 2 - Condition 42: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.200	0.000	0.05	0.0294	0.0294	0.998	94.800 ± 0.702	0.120 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.201	0.001	0.31	0.0309	0.0309	1.010	95.300 ± 0.669	0.121 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	0.01	0.0299	0.0299	0.993	94.800 ± 0.702	0.114 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.110	0.000 ± 0.001	0.000 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.111	0.001	0.65	0.0320	0.0321	0.982	93.500 ± 0.780	0.126 ± 0.001	90.600 ± 0.923
LSAM	beta45	0.11	0.111	0.001 ± 0.001	0.011 ± 0.008	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.111	0.001	1.27	0.0295	0.0296	0.999	94.900 ± 0.696	0.114 ± 0.000	96.300 ± 0.597
LSAM	beta46	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.009	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.077 ± 0.004	0.278 ± 0.008	0.079	-0.001	-0.83	0.0217	0.0217	1.021	95.400 ± 0.662	0.089 ± 0.000	95.400 ± 0.662
LSAM	beta47	0.08	0.080	0.000 ± 0.001	0.005 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.082 ± 0.004	0.287 ± 0.008	0.080	0.000	-0.30	0.0214	0.0214	0.981	93.000 ± 0.807	0.088 ± 0.000	95.000 ± 0.689
LSAM	beta51	0.16	0.162	0.002 ± 0.001	0.010 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.262 ± 0.007	0.160	0.000	-0.02	0.0412	0.0412	0.932	92.100 ± 0.853	0.153 ± 0.001	98.200 ± 0.420
LSAM	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.039 ± 0.002	0.196 ± 0.005	0.159	-0.001	-0.67	0.0303	0.0304	1.023	95.800 ± 0.634	0.126 ± 0.000	99.800 ± 0.141
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.160	0.000	0.07	0.0336	0.0336	1.004	95.200 ± 0.676	0.132 ± 0.000	99.600 ± 0.200
LSAM	beta54	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.211 ± 0.006	0.160	0.000	-0.07	0.0345	0.0345	0.948	93.500 ± 0.780	0.125 ± 0.000	99.800 ± 0.141
LSAM	beta55	0.08	0.082	0.002 ± 0.001	0.025 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.231 ± 0.011	0.480 ± 0.014	0.082	0.002	2.49	0.0384	0.0384	0.958	93.300 ± 0.791	0.144 ± 0.001	60.300 ± 1.547
LSAM	beta56	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.216 ± 0.009	0.465 ± 0.013	0.079	-0.001	-0.81	0.0370	0.0370	0.942	92.800 ± 0.817	0.137 ± 0.001	61.000 ± 1.542
LSAM	beta57	0.06	0.060	-0.000 ± 0.001	-0.006 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.167 ± 0.008	0.409 ± 0.012	0.059	-0.001	-1.01	0.0231	0.0231	0.977	94.200 ± 0.739	0.094 ± 0.000	71.600 ± 1.426
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.004 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.132 ± 0.006	0.364 ± 0.011	0.059	-0.001	-1.10	0.0223	0.0224	0.993	94.300 ± 0.733	0.085 ± 0.000	82.200 ± 1.210
QML	beta41	0.20	0.196	-0.004 ± 0.001	-0.020 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.196	-0.004	-1.79	0.0303	0.0305	1.043	94.400 ± 0.727	0.125 ± 0.004	99.800 ± 0.141
QML	beta42	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.198	-0.002	-1.01	0.0309	0.0310	1.050	95.300 ± 0.669	0.127 ± 0.004	99.800 ± 0.141
QML	beta43	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.195	-0.005	-2.26	0.0297	0.0300	0.997	94.800 ± 0.702	0.114 ± 0.001	99.900 ± 0.100
QML	beta44	0.11	0.107	-0.003 ± 0.001	-0.027 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.095 ± 0.004	0.308 ± 0.008	0.108	-0.002	-2.13	0.0331	0.0332	0.984	94.100 ± 0.745	0.130 ± 0.001	86.900 ± 1.067
QML	beta45	0.11	0.111	0.001 ± 0.001	0.005 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.073 ± 0.003	0.269 ± 0.007	0.111	0.001	0.92	0.0306	0.0306	1.002	95.500 ± 0.656	0.116 ± 0.001	95.800 ± 0.634
QML	beta46	0.08	0.080	0.000 ± 0.001	0.000 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.080	0.000	-0.60	0.0220	0.0220	1.011	94.800 ± 0.702	0.090 ± 0.000	95.500 ± 0.656
QML	beta47	0.08	0.080	-0.000 ± 0.001	-0.001 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.079	-0.001	-1.27	0.0215	0.0215	0.987	93.100 ± 0.801	0.090 ± 0.000	95.600 ± 0.649
QML	beta51	0.16	0.166	0.006 ± 0.001	0.040 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.166	0.006	3.66	0.0409	0.0413	0.965	93.400 ± 0.785	0.159 ± 0.003	98.400 ± 0.397
QML	beta52	0.16	0.142	-0.018 ± 0.001	-0.111 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.141	-0.019	-11.85	0.0313	0.0366	1.044	92.300 ± 0.843	0.139 ± 0.002	98.700 ± 0.358
QML	beta53	0.16	0.139	-0.021 ± 0.001	-0.132 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.006	0.139	-0.021	-13.05	0.0347	0.0405	1.001	90.300 ± 0.936	0.137 ± 0.001	97.100 ± 0.531
QML	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.045 ± 0.002	0.213 ± 0.006	0.160	0.000	-0.28	0.0346	0.0346	0.958	93.400 ± 0.785	0.128 ± 0.001	99.500 ± 0.223
QML	beta55	0.08	0.085	0.005 ± 0.001	0.064 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.242 ± 0.011	0.492 ± 0.014	0.085	0.005	6.71	0.0383	0.0387	0.966	93.500 ± 0.780	0.148 ± 0.001	62.000 ± 1.535
QML	beta56	0.08	0.080	-0.000 ± 0.001	-0.004 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.224 ± 0.010	0.473 ± 0.013	0.080	0.000	0.07	0.0385	0.0385	0.948	93.100 ± 0.801	0.141 ± 0.001	60.900 ± 1.543
QML	beta57	0.06	0.057	-0.003 ± 0.001	-0.056 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.173 ± 0.008	0.416 ± 0.012	0.057	-0.003	-5.09	0.0232	0.0234	0.979	93.700 ± 0.768	0.095 ± 0.000	66.200 ± 1.496
QML	beta58	0.06	0.059	-0.001 ± 0.001	-0.023 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.137 ± 0.007	0.370 ± 0.011	0.058	-0.002	-3.28	0.0216	0.0217	0.994	93.800 ± 0.763	0.086 ± 0.000	79.400 ± 1.279
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.000 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.200	0.000	0.09	0.0289	0.0289	0.958	94.000 ± 0.751	0.115 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.201	0.001	0.29	0.0307	0.0307	0.983	94.500 ± 0.721	0.118 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	-0.18	0.0297	0.0297	0.987	94.600 ± 0.715	0.113 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.110	0.000 ± 0.001	0.003 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.110	0.000	-0.10	0.0323	0.0323	0.978	94.300 ± 0.733	0.129 ± 0.001	89.800 ± 0.957
UPI	beta45	0.11	0.112	0.002 ± 0.001	0.015 ± 0.008	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.071 ± 0.003	0.267 ± 0.007	0.111	0.001	1.24	0.0305	0.0305	1.006	95.400 ± 0.662	0.116 ± 0.000	96.400 ± 0.589
UPI	beta46	0.08	0.080	0.000 ± 0.001	0.001 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.083 ± 0.004	0.287 ± 0.008	0.080	0.000	-0.25	0.0227	0.0227	1.010	95.000 ± 0.689	0.091 ± 0.000	94.800 ± 0.702

Table S186

Simulation 2 - Condition 42: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.007 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.080	0.000	-0.42	0.0217	0.0217	0.993	93.500 ± 0.780	0.090 ± 0.000	94.300 ± 0.733
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.010 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.161	0.001	0.36	0.0417	0.0417	0.936	92.700 ± 0.823	0.153 ± 0.001	98.400 ± 0.397
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.039 ± 0.002	0.196 ± 0.005	0.159	-0.001	-0.55	0.0299	0.0299	1.011	95.400 ± 0.662	0.125 ± 0.000	99.800 ± 0.141
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.210 ± 0.006	0.160	0.000	0.06	0.0334	0.0334	1.003	95.100 ± 0.683	0.132 ± 0.000	99.600 ± 0.200
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.211 ± 0.006	0.159	-0.001	-0.38	0.0347	0.0347	0.941	93.200 ± 0.796	0.124 ± 0.000	99.800 ± 0.141
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.024 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.237 ± 0.011	0.487 ± 0.014	0.081	0.001	1.57	0.0379	0.0379	0.955	93.800 ± 0.763	0.146 ± 0.001	59.200 ± 1.554
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.214 ± 0.009	0.463 ± 0.013	0.080	0.000	-0.21	0.0366	0.0366	0.959	93.400 ± 0.785	0.139 ± 0.001	60.600 ± 1.545
UPI	beta57	0.06	0.060	-0.000 ± 0.001	-0.004 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.171 ± 0.009	0.414 ± 0.012	0.059	-0.001	-2.16	0.0238	0.0239	0.977	94.000 ± 0.751	0.095 ± 0.000	71.000 ± 1.435
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.007 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.136 ± 0.007	0.369 ± 0.011	0.060	0.000	-0.12	0.0215	0.0215	0.992	94.400 ± 0.727	0.086 ± 0.000	81.900 ± 1.218

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S187

Simulation 2 - Condition 43: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.209	0.009 ± 0.006	0.044 ± 0.031	0.024 ± 0.002	0.156 ± 0.005	0.024 ± 0.002	0.156 ± 0.006	0.611 ± 0.039	0.782 ± 0.029	0.211	0.011	5.35	0.1581	0.1584	1.100	97.981 ± 0.554	0.674 ± 0.011	28.261 ± 1.774
LSAM	beta42	0.20	0.197	-0.003 ± 0.007	-0.014 ± 0.034	0.030 ± 0.002	0.173 ± 0.006	0.030 ± 0.002	0.173 ± 0.007	0.750 ± 0.049	0.866 ± 0.033	0.203	0.003	1.60	0.1672	0.1673	1.037	96.584 ± 0.716	0.705 ± 0.012	24.534 ± 1.696
LSAM	beta43	0.20	0.203	0.003 ± 0.004	0.013 ± 0.018	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.000	0.091 ± 0.003	0.205 ± 0.012	0.453 ± 0.016	0.203	0.003	1.32	0.0844	0.0845	1.003	96.429 ± 0.731	0.356 ± 0.004	64.752 ± 1.883
LSAM	beta44	0.11	0.130	0.020 ± 0.005	0.185 ± 0.049	0.018 ± 0.001	0.135 ± 0.005	0.019 ± 0.001	0.137 ± 0.005	1.549 ± 0.105	1.245 ± 0.049	0.115	0.005	4.47	0.1231	0.1232	1.107	97.826 ± 0.575	0.588 ± 0.015	15.683 ± 1.433
LSAM	beta45	0.11	0.113	0.003 ± 0.005	0.031 ± 0.046	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.005	1.377 ± 0.102	1.173 ± 0.049	0.110	0.000	-0.13	0.1159	0.1159	1.026	95.186 ± 0.843	0.519 ± 0.012	13.975 ± 1.366
LSAM	beta46	0.08	0.077	-0.003 ± 0.003	-0.034 ± 0.036	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	0.844 ± 0.061	0.919 ± 0.038	0.078	-0.002	-2.81	0.0714	0.0714	1.027	95.497 ± 0.817	0.296 ± 0.007	23.602 ± 1.673
LSAM	beta47	0.08	0.082	0.002 ± 0.003	0.020 ± 0.039	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	0.963 ± 0.074	0.981 ± 0.043	0.077	-0.003	-3.26	0.0688	0.0689	0.956	96.894 ± 0.684	0.294 ± 0.007	27.950 ± 1.768
LSAM	beta51	0.16	0.145	-0.015 ± 0.005	-0.095 ± 0.031	0.015 ± 0.001	0.124 ± 0.004	0.016 ± 0.001	0.125 ± 0.005	0.610 ± 0.038	0.781 ± 0.029	0.148	-0.012	-7.55	0.1137	0.1143	0.963	94.720 ± 0.881	0.468 ± 0.004	24.379 ± 1.692
LSAM	beta52	0.16	0.156	-0.004 ± 0.007	-0.025 ± 0.042	0.029 ± 0.002	0.170 ± 0.005	0.029 ± 0.002	0.170 ± 0.006	1.126 ± 0.071	1.061 ± 0.040	0.161	0.001	0.77	0.1660	0.1660	1.075	98.758 ± 0.436	0.716 ± 0.012	15.528 ± 1.427
LSAM	beta53	0.16	0.162	0.002 ± 0.005	0.013 ± 0.032	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	0.659 ± 0.042	0.812 ± 0.031	0.164	0.004	2.70	0.1313	0.1314	1.019	96.429 ± 0.731	0.519 ± 0.006	23.758 ± 1.677
LSAM	beta54	0.16	0.175	0.015 ± 0.006	0.094 ± 0.040	0.026 ± 0.002	0.161 ± 0.005	0.026 ± 0.002	0.162 ± 0.006	1.025 ± 0.065	1.012 ± 0.038	0.173	0.013	7.85	0.1592	0.1597	1.040	96.739 ± 0.700	0.658 ± 0.010	19.720 ± 1.568
LSAM	beta55	0.08	0.076	-0.004 ± 0.005	-0.054 ± 0.066	0.018 ± 0.001	0.134 ± 0.005	0.018 ± 0.001	0.134 ± 0.006	2.785 ± 0.214	1.669 ± 0.072	0.073	-0.007	-8.13	0.1224	0.1226	1.027	97.050 ± 0.667	0.537 ± 0.012	10.559 ± 1.211
LSAM	beta56	0.08	0.087	0.007 ± 0.003	0.087 ± 0.038	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	0.913 ± 0.061	0.955 ± 0.037	0.086	0.006	6.98	0.0697	0.0699	1.012	95.497 ± 0.817	0.302 ± 0.006	26.087 ± 1.730
LSAM	beta57	0.06	0.071	0.011 ± 0.005	0.188 ± 0.084	0.016 ± 0.001	0.128 ± 0.005	0.016 ± 0.001	0.128 ± 0.006	4.574 ± 0.374	2.139 ± 0.097	0.066	0.006	9.42	0.1192	0.1194	1.041	98.292 ± 0.511	0.522 ± 0.012	5.124 ± 0.869
LSAM	beta58	0.06	0.060	-0.000 ± 0.002	-0.000 ± 0.040	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	1.014 ± 0.070	1.007 ± 0.040	0.057	-0.003	-4.88	0.0560	0.0561	1.039	95.031 ± 0.856	0.246 ± 0.006	20.031 ± 1.577
QML	beta41	0.20	0.237	0.037 ± 0.004	0.183 ± 0.018	0.008 ± 0.000	0.092 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	0.246 ± 0.014	0.496 ± 0.016	0.240	0.040	19.80	0.0945	0.1025	1.064	95.031 ± 0.856	0.385 ± 0.006	71.429 ± 1.780
QML	beta42	0.20	0.231	0.031 ± 0.004	0.153 ± 0.019	0.009 ± 0.001	0.095 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.247 ± 0.015	0.497 ± 0.018	0.228	0.028	13.77	0.0960	0.0999	1.049	94.720 ± 0.881	0.389 ± 0.004	68.478 ± 1.831
QML	beta43	0.20	0.191	-0.009 ± 0.003	-0.047 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.172 ± 0.010	0.414 ± 0.014	0.190	-0.010	-5.20	0.0823	0.0830	0.974	93.944 ± 0.940	0.315 ± 0.003	68.478 ± 1.831
QML	beta44	0.11	0.043	-0.067 ± 0.005	-0.612 ± 0.044	0.015 ± 0.001	0.124 ± 0.004	0.020 ± 0.001	0.141 ± 0.005	1.634 ± 0.097	1.278 ± 0.043	0.040	-0.070	-64.07	0.1230	0.1418	1.058	91.460 ± 1.101	0.512 ± 0.024	8.696 ± 1.110
QML	beta45	0.11	0.090	-0.020 ± 0.004	-0.186 ± 0.037	0.010 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.898 ± 0.052	0.948 ± 0.033	0.088	-0.022	-20.20	0.1030	0.1053	0.999	93.012 ± 1.005	0.401 ± 0.010	15.994 ± 1.444
QML	beta46	0.08	0.160	0.080 ± 0.003	0.999 ± 0.038	0.006 ± 0.000	0.076 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	1.905 ± 0.114	1.380 ± 0.043	0.156	0.076	94.51	0.0734	0.1054	1.043	90.528 ± 1.154	0.312 ± 0.008	62.112 ± 1.912
QML	beta47	0.08	0.154	0.074 ± 0.003	0.925 ± 0.035	0.005 ± 0.000	0.071 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	1.638 ± 0.092	1.280 ± 0.038	0.148	0.068	85.52	0.0691	0.0972	1.049	89.752 ± 1.195	0.291 ± 0.008	65.062 ± 1.879
QML	beta51	0.16	0.164	0.004 ± 0.006	0.025 ± 0.035	0.020 ± 0.001	0.142 ± 0.005	0.020 ± 0.001	0.142 ± 0.005	0.787 ± 0.052	0.887 ± 0.034	0.166	0.006	3.60	0.1423	0.1424	1.016	95.807 ± 0.790	0.565 ± 0.008	22.516 ± 1.646
QML	beta52	0.16	0.164	0.004 ± 0.005	0.024 ± 0.034	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.137 ± 0.005	0.734 ± 0.043	0.857 ± 0.030	0.154	-0.006	-4.04	0.1415	0.1417	0.939	95.031 ± 0.856	0.505 ± 0.007	28.106 ± 1.771
QML	beta53	0.16	0.114	-0.046 ± 0.004	-0.285 ± 0.028	0.013 ± 0.001	0.114 ± 0.004	0.015 ± 0.001	0.122 ± 0.004	0.584 ± 0.034	0.764 ± 0.026	0.113	-0.047	-29.23	0.1154	0.1245	1.012	92.857 ± 1.015	0.450 ± 0.004	17.081 ± 1.483
QML	beta54	0.16	0.186	0.026 ± 0.004	0.161 ± 0.025	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	0.442 ± 0.025	0.665 ± 0.023	0.184	0.024	15.23	0.1053	0.1081	0.996	94.720 ± 0.881	0.403 ± 0.005	49.224 ± 1.970
QML	beta55	0.08	0.070	-0.010 ± 0.006	-0.130 ± 0.076	0.024 ± 0.002	0.154 ± 0.005	0.024 ± 0.002	0.154 ± 0.006	3.709 ± 0.244	1.926 ± 0.074	0.065	-0.015	-18.19	0.1446	0.1454	0.866	87.578 ± 1.300	0.522 ± 0.013	18.478 ± 1.529
QML	beta56	0.08	0.122	0.042 ± 0.003	0.527 ± 0.044	0.008 ± 0.001	0.089 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	1.503 ± 0.096	1.226 ± 0.044	0.123	0.043	53.35	0.0845	0.0947	0.915	89.441 ± 1.211	0.318 ± 0.006	42.236 ± 1.946
QML	beta57	0.06	0.142	0.082 ± 0.005	1.360 ± 0.091	0.019 ± 0.001	0.139 ± 0.004	0.026 ± 0.002	0.161 ± 0.006	7.178 ± 0.449	2.679 ± 0.093	0.136	0.076	126.38	0.1335	0.1535	0.911	88.975 ± 1.234	0.495 ± 0.012	23.758 ± 1.677
QML	beta58	0.06	0.113	0.053 ± 0.002	0.883 ± 0.041	0.004 ± 0.000	0.063 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	1.868 ± 0.126	1.367 ± 0.049	0.104	0.044	73.16	0.0572	0.0721	0.996	92.081 ± 1.064	0.245 ± 0.005	49.068 ± 1.970
UPI	beta41	0.20	0.230	0.030 ± 0.008	0.151 ± 0.039	0.039 ± 0.003	0.198 ± 0.008	0.040 ± 0.003	0.201 ± 0.009	1.006 ± 0.082	1.003 ± 0.045	0.229	0.029	14.64	0.2086	0.2107	1.416	98.913 ± 0.409	1.101 ± 0.042	19.720 ± 1.568
UPI	beta42	0.20	0.207	0.007 ± 0.007	0.035 ± 0.037	0.036 ± 0.003	0.189 ± 0.007	0.036 ± 0.003	0.189 ± 0.008	0.894 ± 0.063	0.946 ± 0.038	0.199	-0.001	-0.36	0.1984	0.1984	1.440	98.602 ± 0.463	1.067 ± 0.034	15.528 ± 1.427
UPI	beta43	0.20	0.204	0.004 ± 0.004	0.020 ± 0.019	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	0.221 ± 0.013	0.470 ± 0.017	0.208	0.008	4.18	0.1027	0.1031	1.265	97.205 ± 0.650	0.466 ± 0.012	50.776 ± 1.970
UPI	beta44	0.11	0.142	0.032 ± 0.008	0.287 ± 0.070	0.038 ± 0.003	0.195 ± 0.009	0.039 ± 0.003	0.198 ± 0.010	3.230 ± 0.284	1.797 ± 0.087	0.127	0.017	15.64	0.1742	0.1751	1.419	98.447 ± 0.487	1.087 ± 0.047	7.143 ± 1.015
UPI	beta45	0.11	0.113	0.003 ± 0.007	0.024 ± 0.065	0.033 ± 0.003	0.181 ± 0.007	0.033 ± 0.003	0.181 ± 0.008	2.713 ± 0.223	1.647 ± 0.075	0.102	-0.008	-7.44	0.1625	0.1627	1.559	96.273 ± 0.746	1.108 ± 0.061	4.658 ± 0.830
UPI	beta46	0.08	0.075	-0.005 ± 0.004	-0.062 ± 0.049	0.010 ± 0.001	0.100 ± 0.004	0.010 ± 0.001	0.100 ± 0.005	1.554 ± 0.128	1.246 ± 0.057	0.071	-0.009	-11.16	0.0966	0.0970	1.329	97.205 ± 0.650	0.519 ± 0.023	14.130 ± 1.373

Table S187

Simulation 2 - Condition 43: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.077	-0.003 ± 0.004	-0.035 ± 0.045	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.091 ± 0.004	1.284 ± 0.109	1.133 ± 0.053	0.080	0.000	0.21	0.0813	0.0813	1.300	97.981 ± 0.554	0.462 ± 0.019	19.255 ± 1.554
UPI	beta51	0.16	0.154	-0.006 ± 0.005	-0.036 ± 0.032	0.016 ± 0.001	0.128 ± 0.004	0.016 ± 0.001	0.128 ± 0.005	0.642 ± 0.039	0.801 ± 0.029	0.160	0.000	0.18	0.1250	0.1250	1.061	96.429 ± 0.731	0.533 ± 0.007	21.118 ± 1.608
UPI	beta52	0.16	0.178	0.018 ± 0.008	0.115 ± 0.050	0.041 ± 0.003	0.202 ± 0.007	0.041 ± 0.003	0.203 ± 0.008	1.610 ± 0.114	1.269 ± 0.051	0.182	0.022	13.71	0.1951	0.1964	1.216	99.068 ± 0.379	0.964 ± 0.027	12.422 ± 1.300
UPI	beta53	0.16	0.158	-0.002 ± 0.006	-0.010 ± 0.034	0.020 ± 0.001	0.140 ± 0.005	0.019 ± 0.001	0.140 ± 0.005	0.762 ± 0.050	0.873 ± 0.033	0.159	-0.001	-0.56	0.1482	0.1482	1.214	98.447 ± 0.487	0.665 ± 0.016	17.702 ± 1.504
UPI	beta54	0.16	0.182	0.022 ± 0.008	0.138 ± 0.047	0.037 ± 0.003	0.192 ± 0.007	0.037 ± 0.003	0.193 ± 0.008	1.459 ± 0.102	1.208 ± 0.048	0.177	0.017	10.81	0.1854	0.1862	1.228	98.447 ± 0.487	0.925 ± 0.024	12.267 ± 1.293
UPI	beta55	0.08	0.078	-0.002 ± 0.006	-0.021 ± 0.075	0.023 ± 0.002	0.151 ± 0.006	0.023 ± 0.002	0.151 ± 0.007	3.572 ± 0.276	1.890 ± 0.082	0.076	-0.004	-5.36	0.1391	0.1391	1.265	97.981 ± 0.554	0.750 ± 0.029	7.609 ± 1.045
UPI	beta56	0.08	0.085	0.005 ± 0.003	0.057 ± 0.042	0.007 ± 0.001	0.085 ± 0.003	0.007 ± 0.001	0.085 ± 0.004	1.130 ± 0.086	1.063 ± 0.046	0.083	0.003	3.13	0.0828	0.0829	1.215	97.205 ± 0.650	0.405 ± 0.018	18.323 ± 1.524
UPI	beta57	0.06	0.061	0.001 ± 0.006	0.017 ± 0.092	0.020 ± 0.002	0.140 ± 0.005	0.020 ± 0.002	0.140 ± 0.006	5.474 ± 0.422	2.340 ± 0.102	0.056	-0.004	-6.29	0.1298	0.1298	1.310	99.224 ± 0.346	0.721 ± 0.027	2.950 ± 0.667
UPI	beta58	0.06	0.057	-0.003 ± 0.003	-0.043 ± 0.047	0.005 ± 0.000	0.072 ± 0.003	0.005 ± 0.000	0.072 ± 0.003	1.421 ± 0.102	1.192 ± 0.049	0.055	-0.005	-8.50	0.0638	0.0640	1.255	97.360 ± 0.632	0.352 ± 0.011	13.043 ± 1.327

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S188

Simulation 2 - Condition 44: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.004	-0.002 ± 0.018	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.300 ± 0.016	0.548 ± 0.017	0.196	-0.004	-2.24	0.1068	0.1069	0.946	94.212 ± 0.764	0.407 ± 0.003	51.554 ± 1.636
LSAM	beta42	0.20	0.200	-0.000 ± 0.004	-0.002 ± 0.018	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.310 ± 0.017	0.556 ± 0.017	0.199	-0.001	-0.34	0.1035	0.1035	0.992	95.927 ± 0.647	0.433 ± 0.004	50.911 ± 1.637
LSAM	beta43	0.20	0.203	0.003 ± 0.002	0.013 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.004	0.275 ± 0.008	0.202	0.002	0.87	0.0529	0.0529	1.018	95.498 ± 0.679	0.219 ± 0.001	95.284 ± 0.694
LSAM	beta44	0.11	0.119	0.009 ± 0.003	0.079 ± 0.024	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.561 ± 0.032	0.749 ± 0.025	0.117	0.007	6.48	0.0768	0.0771	0.993	94.534 ± 0.744	0.319 ± 0.004	32.262 ± 1.530
LSAM	beta45	0.11	0.118	0.008 ± 0.003	0.071 ± 0.023	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.510 ± 0.030	0.714 ± 0.024	0.113	0.003	2.94	0.0713	0.0714	0.965	95.820 ± 0.655	0.296 ± 0.003	38.478 ± 1.593
LSAM	beta46	0.08	0.082	0.002 ± 0.001	0.025 ± 0.018	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.292 ± 0.017	0.541 ± 0.018	0.081	0.001	0.87	0.0389	0.0389	0.935	93.355 ± 0.815	0.158 ± 0.002	57.342 ± 1.619
LSAM	beta47	0.08	0.083	0.003 ± 0.001	0.036 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.299 ± 0.018	0.546 ± 0.018	0.082	0.002	2.10	0.0367	0.0367	0.939	95.284 ± 0.694	0.161 ± 0.002	60.021 ± 1.604
LSAM	beta51	0.16	0.158	-0.002 ± 0.003	-0.015 ± 0.016	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.228 ± 0.011	0.477 ± 0.014	0.160	0.000	-0.28	0.0722	0.0722	0.972	94.641 ± 0.737	0.291 ± 0.001	56.592 ± 1.623
LSAM	beta52	0.16	0.164	0.004 ± 0.004	0.022 ± 0.023	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.491 ± 0.024	0.701 ± 0.020	0.160	0.000	-0.22	0.1095	0.1095	1.013	96.785 ± 0.578	0.445 ± 0.004	34.834 ± 1.560
LSAM	beta53	0.16	0.161	0.001 ± 0.003	0.007 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.267 ± 0.012	0.517 ± 0.015	0.160	0.000	-0.25	0.0811	0.0811	0.994	95.606 ± 0.671	0.323 ± 0.002	51.661 ± 1.636
LSAM	beta54	0.16	0.165	0.005 ± 0.003	0.032 ± 0.021	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.397 ± 0.021	0.630 ± 0.020	0.162	0.002	1.21	0.0994	0.0994	1.048	97.106 ± 0.549	0.414 ± 0.003	36.013 ± 1.572
LSAM	beta55	0.08	0.084	0.004 ± 0.003	0.044 ± 0.033	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.997 ± 0.056	0.999 ± 0.033	0.081	0.001	1.24	0.0729	0.0729	0.937	95.498 ± 0.679	0.293 ± 0.004	27.974 ± 1.470
LSAM	beta56	0.08	0.083	0.003 ± 0.001	0.036 ± 0.018	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.292 ± 0.017	0.540 ± 0.018	0.083	0.003	3.36	0.0385	0.0386	0.942	93.676 ± 0.797	0.159 ± 0.002	58.628 ± 1.612
LSAM	beta57	0.06	0.061	0.001 ± 0.003	0.019 ± 0.044	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	1.789 ± 0.099	1.338 ± 0.043	0.059	-0.001	-2.07	0.0688	0.0688	0.955	95.284 ± 0.694	0.301 ± 0.004	12.004 ± 1.064
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.021 ± 0.020	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.363 ± 0.020	0.602 ± 0.019	0.061	0.001	1.35	0.0320	0.0321	1.018	95.391 ± 0.686	0.144 ± 0.002	45.230 ± 1.629
QML	beta41	0.20	0.234	0.034 ± 0.002	0.170 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.121 ± 0.006	0.348 ± 0.009	0.235	0.035	17.34	0.0581	0.0677	0.970	90.032 ± 0.981	0.231 ± 0.001	97.535 ± 0.508
QML	beta42	0.20	0.230	0.030 ± 0.002	0.148 ± 0.010	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.110 ± 0.005	0.332 ± 0.009	0.230	0.030	15.09	0.0564	0.0640	1.023	92.926 ± 0.839	0.238 ± 0.001	95.391 ± 0.686
QML	beta43	0.20	0.186	-0.014 ± 0.002	-0.069 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.185	-0.015	-7.69	0.0448	0.0474	1.019	93.462 ± 0.809	0.193 ± 0.001	96.463 ± 0.605
QML	beta44	0.11	0.029	-0.081 ± 0.002	-0.738 ± 0.022	0.005 ± 0.000	0.074 ± 0.002	0.012 ± 0.001	0.110 ± 0.002	0.997 ± 0.042	0.999 ± 0.023	0.032	-0.078	-70.78	0.0711	0.1054	1.011	81.029 ± 1.284	0.294 ± 0.003	9.003 ± 0.937
QML	beta45	0.11	0.097	-0.013 ± 0.002	-0.119 ± 0.019	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.353 ± 0.018	0.594 ± 0.017	0.096	-0.014	-12.70	0.0606	0.0622	0.960	92.390 ± 0.868	0.241 ± 0.002	37.085 ± 1.581
QML	beta46	0.08	0.159	0.079 ± 0.001	0.990 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.008 ± 0.000	0.091 ± 0.002	1.283 ± 0.043	1.133 ± 0.019	0.155	0.075	94.27	0.0440	0.0873	1.038	61.415 ± 1.594	0.179 ± 0.002	96.463 ± 0.605
QML	beta47	0.08	0.159	0.079 ± 0.001	0.989 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.008 ± 0.000	0.091 ± 0.002	1.283 ± 0.043	1.133 ± 0.019	0.155	0.075	93.66	0.0427	0.0862	0.979	55.949 ± 1.625	0.169 ± 0.002	97.964 ± 0.462
QML	beta51	0.16	0.175	0.015 ± 0.003	0.091 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.289 ± 0.014	0.537 ± 0.016	0.175	0.015	9.51	0.0845	0.0859	1.008	94.534 ± 0.744	0.335 ± 0.004	57.020 ± 1.621
QML	beta52	0.16	0.165	0.005 ± 0.003	0.029 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.252 ± 0.013	0.502 ± 0.016	0.161	0.001	0.40	0.0778	0.0778	1.017	95.284 ± 0.694	0.320 ± 0.004	56.270 ± 1.624
QML	beta53	0.16	0.107	-0.053 ± 0.002	-0.332 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.292 ± 0.012	0.540 ± 0.012	0.107	-0.053	-33.37	0.0689	0.0871	0.994	84.995 ± 1.169	0.266 ± 0.001	35.691 ± 1.568
QML	beta54	0.16	0.179	0.019 ± 0.002	0.120 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.007	0.386 ± 0.011	0.178	0.018	11.17	0.0592	0.0619	1.053	94.748 ± 0.730	0.242 ± 0.001	82.637 ± 1.240
QML	beta55	0.08	0.070	-0.010 ± 0.003	-0.129 ± 0.041	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	1.609 ± 0.081	1.268 ± 0.038	0.069	-0.011	-13.59	0.0921	0.0928	0.910	86.710 ± 1.111	0.360 ± 0.008	21.865 ± 1.353
QML	beta56	0.08	0.124	0.044 ± 0.002	0.545 ± 0.021	0.003 ± 0.000	0.051 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.703 ± 0.034	0.839 ± 0.022	0.124	0.044	54.85	0.0493	0.0660	0.939	83.280 ± 1.222	0.188 ± 0.002	75.456 ± 1.409
QML	beta57	0.06	0.144	0.084 ± 0.003	1.394 ± 0.051	0.009 ± 0.000	0.093 ± 0.002	0.016 ± 0.001	0.125 ± 0.003	4.337 ± 0.189	2.083 ± 0.049	0.144	0.084	139.34	0.0905	0.1232	0.947	78.242 ± 1.351	0.345 ± 0.008	45.552 ± 1.630
QML	beta58	0.06	0.114	0.054 ± 0.001	0.897 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.004 ± 0.000	0.065 ± 0.001	1.157 ± 0.043	1.075 ± 0.021	0.112	0.052	87.41	0.0336	0.0623	1.026	73.419 ± 1.446	0.143 ± 0.002	92.283 ± 0.874
UPI	beta41	0.20	0.209	0.009 ± 0.003	0.045 ± 0.016	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.253 ± 0.014	0.503 ± 0.016	0.206	0.006	2.87	0.0995	0.0996	1.097	96.999 ± 0.559	0.431 ± 0.005	52.947 ± 1.634
UPI	beta42	0.20	0.205	0.005 ± 0.003	0.024 ± 0.017	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.264 ± 0.014	0.513 ± 0.016	0.204	0.004	2.25	0.0924	0.0925	1.131	97.320 ± 0.529	0.455 ± 0.007	52.197 ± 1.635
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.004	0.276 ± 0.008	0.202	0.002	0.93	0.0552	0.0552	1.018	94.427 ± 0.751	0.221 ± 0.002	94.105 ± 0.771
UPI	beta44	0.11	0.119	0.009 ± 0.003	0.078 ± 0.027	0.008 ± 0.001	0.092 ± 0.003	0.008 ± 0.001	0.092 ± 0.003	0.702 ± 0.043	0.838 ± 0.029	0.110	0.000	-0.18	0.0807	0.0807	1.148	96.892 ± 0.568	0.413 ± 0.013	22.615 ± 1.370
UPI	beta45	0.11	0.116	0.006 ± 0.003	0.055 ± 0.026	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.639 ± 0.039	0.799 ± 0.028	0.110	0.000	-0.38	0.0759	0.0759	1.145	95.927 ± 0.647	0.394 ± 0.010	21.651 ± 1.348
UPI	beta46	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.283 ± 0.014	0.532 ± 0.016	0.078	-0.002	-3.04	0.0407	0.0408	1.072	95.070 ± 0.709	0.179 ± 0.003	48.767 ± 1.636

Table S188

Simulation 2 - Condition 44: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.010 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.279 ± 0.016	0.528 ± 0.018	0.079	-0.001	-1.14	0.0380	0.0380	1.062	95.070 ± 0.709	0.176 ± 0.004	53.269 ± 1.633
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.020 ± 0.015	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.215 ± 0.010	0.464 ± 0.014	0.162	0.002	1.41	0.0724	0.0724	0.984	94.855 ± 0.723	0.286 ± 0.001	61.629 ± 1.592
UPI	beta52	0.16	0.178	0.018 ± 0.003	0.111 ± 0.022	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.444 ± 0.022	0.667 ± 0.020	0.176	0.016	9.98	0.1041	0.1053	1.067	97.106 ± 0.549	0.440 ± 0.004	39.657 ± 1.602
UPI	beta53	0.16	0.157	-0.003 ± 0.003	-0.018 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.252 ± 0.011	0.502 ± 0.014	0.157	-0.003	-1.59	0.0817	0.0817	1.027	96.249 ± 0.622	0.323 ± 0.002	50.804 ± 1.637
UPI	beta54	0.16	0.164	0.004 ± 0.003	0.027 ± 0.020	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.389 ± 0.020	0.624 ± 0.019	0.162	0.002	0.98	0.0995	0.0995	1.098	97.535 ± 0.508	0.429 ± 0.004	33.441 ± 1.545
UPI	beta55	0.08	0.089	0.009 ± 0.002	0.115 ± 0.030	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.879 ± 0.049	0.937 ± 0.030	0.088	0.008	10.24	0.0677	0.0682	0.986	96.463 ± 0.605	0.288 ± 0.004	28.939 ± 1.485
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.010 ± 0.018	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.291 ± 0.017	0.540 ± 0.018	0.079	-0.001	-1.33	0.0399	0.0399	0.969	94.105 ± 0.771	0.164 ± 0.003	54.555 ± 1.630
UPI	beta57	0.06	0.052	-0.008 ± 0.002	-0.141 ± 0.040	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	1.478 ± 0.083	1.216 ± 0.039	0.052	-0.008	-14.10	0.0642	0.0647	1.025	95.927 ± 0.647	0.291 ± 0.004	9.539 ± 0.962
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.008 ± 0.020	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.369 ± 0.020	0.607 ± 0.019	0.061	0.001	1.06	0.0329	0.0330	1.055	95.177 ± 0.701	0.151 ± 0.002	41.693 ± 1.614

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S189

Simulation 2 - Condition 45: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.210	0.010 ± 0.004	0.050 ± 0.018	0.012 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.315 ± 0.015	0.561 ± 0.016	0.210	0.010	4.95	0.1112	0.1116	0.969	95.173 ± 0.694	0.425 ± 0.003	51.941 ± 1.618
LSAM	beta42	0.20	0.199	-0.001 ± 0.004	-0.005 ± 0.019	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	0.344 ± 0.016	0.587 ± 0.017	0.197	-0.003	-1.31	0.1160	0.1160	0.971	95.488 ± 0.672	0.447 ± 0.003	47.849 ± 1.618
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.315 ± 0.009	0.201	0.001	0.74	0.0601	0.0601	0.975	94.544 ± 0.736	0.241 ± 0.001	90.241 ± 0.961
LSAM	beta44	0.11	0.120	0.010 ± 0.003	0.088 ± 0.026	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.628 ± 0.035	0.792 ± 0.025	0.114	0.004	3.23	0.0817	0.0818	0.912	94.124 ± 0.762	0.310 ± 0.003	35.992 ± 1.555
LSAM	beta45	0.11	0.113	0.003 ± 0.003	0.032 ± 0.023	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.493 ± 0.026	0.702 ± 0.022	0.108	-0.002	-1.41	0.0746	0.0746	0.934	92.760 ± 0.839	0.283 ± 0.003	37.461 ± 1.568
LSAM	beta46	0.08	0.077	-0.003 ± 0.002	-0.037 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.354 ± 0.020	0.595 ± 0.019	0.075	-0.005	-6.22	0.0455	0.0457	0.898	92.235 ± 0.867	0.167 ± 0.002	51.312 ± 1.619
LSAM	beta47	0.08	0.082	0.002 ± 0.002	0.020 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.382 ± 0.022	0.618 ± 0.021	0.080	0.000	0.56	0.0440	0.0440	0.873	92.865 ± 0.834	0.169 ± 0.002	55.614 ± 1.609
LSAM	beta51	0.16	0.155	-0.005 ± 0.003	-0.029 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.280 ± 0.015	0.530 ± 0.017	0.155	-0.005	-3.37	0.0791	0.0793	0.964	93.075 ± 0.822	0.320 ± 0.002	48.688 ± 1.619
LSAM	beta52	0.16	0.164	0.004 ± 0.004	0.023 ± 0.023	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.506 ± 0.025	0.712 ± 0.021	0.170	0.010	5.99	0.1011	0.1015	1.011	95.908 ± 0.642	0.451 ± 0.003	32.634 ± 1.519
LSAM	beta53	0.16	0.158	-0.002 ± 0.003	-0.010 ± 0.019	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.361 ± 0.016	0.601 ± 0.017	0.151	-0.009	-5.34	0.0992	0.0996	0.937	94.124 ± 0.762	0.353 ± 0.002	42.183 ± 1.600
LSAM	beta54	0.16	0.162	0.002 ± 0.004	0.015 ± 0.024	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	0.536 ± 0.025	0.732 ± 0.021	0.162	0.002	1.09	0.1100	0.1100	0.940	94.439 ± 0.742	0.432 ± 0.003	36.201 ± 1.557
LSAM	beta55	0.08	0.083	0.003 ± 0.003	0.032 ± 0.034	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	1.074 ± 0.058	1.036 ± 0.033	0.080	0.000	0.35	0.0806	0.0806	0.928	94.858 ± 0.715	0.302 ± 0.004	26.128 ± 1.423
LSAM	beta56	0.08	0.081	0.001 ± 0.002	0.017 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.428 ± 0.023	0.654 ± 0.020	0.081	0.001	0.66	0.0493	0.0493	0.890	92.235 ± 0.867	0.183 ± 0.003	48.688 ± 1.619
LSAM	beta57	0.06	0.060	-0.000 ± 0.003	-0.006 ± 0.042	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	1.696 ± 0.096	1.302 ± 0.043	0.058	-0.002	-3.77	0.0679	0.0679	0.938	95.698 ± 0.657	0.287 ± 0.004	12.802 ± 1.082
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.010 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.557 ± 0.032	0.746 ± 0.024	0.058	-0.002	-2.91	0.0398	0.0398	0.860	92.550 ± 0.851	0.151 ± 0.002	42.602 ± 1.602
QML	beta41	0.20	0.221	0.021 ± 0.002	0.103 ± 0.012	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.150 ± 0.007	0.387 ± 0.010	0.217	0.017	8.72	0.0763	0.0783	0.993	94.439 ± 0.742	0.290 ± 0.002	82.371 ± 1.234
QML	beta42	0.20	0.211	0.011 ± 0.002	0.056 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.149 ± 0.007	0.386 ± 0.011	0.215	0.015	7.32	0.0779	0.0792	0.998	95.698 ± 0.657	0.299 ± 0.002	77.650 ± 1.349
QML	beta43	0.20	0.192	-0.008 ± 0.002	-0.038 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.293 ± 0.008	0.191	-0.009	-4.48	0.0534	0.0542	0.978	93.809 ± 0.781	0.223 ± 0.001	91.815 ± 0.888
QML	beta44	0.11	0.072	-0.038 ± 0.003	-0.342 ± 0.023	0.006 ± 0.000	0.078 ± 0.002	0.007 ± 0.000	0.087 ± 0.002	0.619 ± 0.030	0.787 ± 0.022	0.072	-0.038	-34.42	0.0751	0.0841	0.953	90.556 ± 0.947	0.291 ± 0.003	21.406 ± 1.329
QML	beta45	0.11	0.099	-0.011 ± 0.002	-0.102 ± 0.020	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.396 ± 0.021	0.629 ± 0.020	0.100	-0.010	-9.44	0.0685	0.0693	0.948	93.389 ± 0.805	0.254 ± 0.003	37.146 ± 1.565
QML	beta46	0.08	0.111	0.031 ± 0.001	0.394 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.440 ± 0.022	0.663 ± 0.018	0.109	0.029	36.18	0.0402	0.0496	0.993	90.661 ± 0.943	0.166 ± 0.002	77.964 ± 1.343
QML	beta47	0.08	0.117	0.037 ± 0.001	0.466 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.536 ± 0.029	0.732 ± 0.021	0.113	0.033	41.16	0.0418	0.0532	0.920	89.717 ± 0.984	0.163 ± 0.002	84.050 ± 1.186
QML	beta51	0.16	0.170	0.010 ± 0.003	0.063 ± 0.018	0.007 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.296 ± 0.015	0.544 ± 0.017	0.170	0.010	6.13	0.0805	0.0811	1.008	93.599 ± 0.793	0.342 ± 0.003	52.466 ± 1.618
QML	beta52	0.16	0.151	-0.009 ± 0.003	-0.055 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.254 ± 0.012	0.504 ± 0.014	0.154	-0.006	-3.68	0.0796	0.0798	1.023	95.488 ± 0.672	0.322 ± 0.002	47.324 ± 1.617
QML	beta53	0.16	0.124	-0.036 ± 0.003	-0.227 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.092 ± 0.002	0.329 ± 0.015	0.574 ± 0.015	0.118	-0.042	-26.25	0.0848	0.0946	0.950	90.556 ± 0.947	0.314 ± 0.002	34.627 ± 1.541
QML	beta54	0.16	0.166	0.006 ± 0.003	0.035 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.081 ± 0.002	0.253 ± 0.012	0.503 ± 0.014	0.166	0.006	3.98	0.0797	0.0799	0.964	94.229 ± 0.755	0.304 ± 0.002	58.762 ± 1.595
QML	beta55	0.08	0.078	-0.002 ± 0.003	-0.024 ± 0.035	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	1.190 ± 0.066	1.091 ± 0.035	0.081	0.001	1.31	0.0809	0.0810	0.952	91.501 ± 0.903	0.326 ± 0.008	27.702 ± 1.450
QML	beta56	0.08	0.098	0.018 ± 0.002	0.224 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.471 ± 0.025	0.687 ± 0.021	0.099	0.019	23.13	0.0477	0.0511	0.917	91.081 ± 0.923	0.187 ± 0.003	58.342 ± 1.597
QML	beta57	0.06	0.092	0.032 ± 0.003	0.537 ± 0.043	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	2.012 ± 0.131	1.418 ± 0.050	0.086	0.026	43.15	0.0665	0.0714	0.969	93.914 ± 0.774	0.299 ± 0.007	23.820 ± 1.380
QML	beta58	0.06	0.084	0.024 ± 0.001	0.408 ± 0.022	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.608 ± 0.033	0.780 ± 0.023	0.082	0.022	36.61	0.0368	0.0429	0.893	90.556 ± 0.947	0.140 ± 0.002	69.465 ± 1.492
UPI	beta41	0.20	0.217	0.017 ± 0.003	0.087 ± 0.017	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.270 ± 0.013	0.520 ± 0.015	0.221	0.021	10.62	0.1043	0.1065	1.041	96.118 ± 0.626	0.419 ± 0.005	59.811 ± 1.588
UPI	beta42	0.20	0.206	0.006 ± 0.004	0.029 ± 0.018	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.306 ± 0.016	0.553 ± 0.017	0.203	0.003	1.57	0.1035	0.1035	1.018	96.118 ± 0.626	0.441 ± 0.006	50.682 ± 1.620
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.010	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.319 ± 0.009	0.198	-0.002	-0.86	0.0633	0.0634	0.967	94.019 ± 0.768	0.242 ± 0.002	89.297 ± 1.001
UPI	beta44	0.11	0.118	0.008 ± 0.003	0.068 ± 0.028	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	0.725 ± 0.040	0.851 ± 0.027	0.108	-0.002	-1.40	0.0866	0.0866	1.001	95.593 ± 0.665	0.367 ± 0.007	27.387 ± 1.445
UPI	beta45	0.11	0.112	0.002 ± 0.003	0.017 ± 0.025	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.577 ± 0.033	0.760 ± 0.025	0.108	-0.002	-2.19	0.0792	0.0793	1.070	94.229 ± 0.755	0.351 ± 0.009	28.227 ± 1.458
UPI	beta46	0.08	0.075	-0.005 ± 0.002	-0.057 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.362 ± 0.018	0.602 ± 0.018	0.073	-0.007	-8.43	0.0448	0.0453	0.957	93.075 ± 0.822	0.180 ± 0.003	44.806 ± 1.611

Table S189

Simulation 2 - Condition 45: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.002	-0.002 ± 0.020	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.396 ± 0.023	0.629 ± 0.021	0.076	-0.004	-4.68	0.0428	0.0430	0.916	92.550 ± 0.851	0.181 ± 0.004	51.626 ± 1.619
UPI	beta51	0.16	0.160	-0.000 ± 0.003	-0.003 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.269 ± 0.014	0.519 ± 0.016	0.159	-0.001	-0.92	0.0786	0.0786	0.970	94.019 ± 0.768	0.316 ± 0.002	52.571 ± 1.618
UPI	beta52	0.16	0.174	0.014 ± 0.004	0.085 ± 0.022	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.477 ± 0.023	0.690 ± 0.020	0.178	0.018	11.49	0.1037	0.1053	1.025	95.278 ± 0.687	0.441 ± 0.003	37.880 ± 1.571
UPI	beta53	0.16	0.157	-0.003 ± 0.003	-0.017 ± 0.019	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.356 ± 0.017	0.597 ± 0.017	0.151	-0.009	-5.62	0.0967	0.0971	0.936	93.914 ± 0.774	0.350 ± 0.002	42.812 ± 1.603
UPI	beta54	0.16	0.163	0.003 ± 0.004	0.019 ± 0.023	0.013 ± 0.001	0.115 ± 0.003	0.013 ± 0.001	0.115 ± 0.003	0.514 ± 0.024	0.717 ± 0.020	0.162	0.002	1.17	0.1126	0.1126	0.971	95.488 ± 0.672	0.436 ± 0.003	35.152 ± 1.547
UPI	beta55	0.08	0.085	0.005 ± 0.002	0.065 ± 0.031	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.926 ± 0.048	0.962 ± 0.030	0.080	0.000	0.00	0.0734	0.0734	0.991	95.278 ± 0.687	0.299 ± 0.004	27.072 ± 1.439
UPI	beta56	0.08	0.080	-0.000 ± 0.002	-0.006 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.425 ± 0.024	0.652 ± 0.021	0.077	-0.003	-3.34	0.0474	0.0475	0.907	92.235 ± 0.867	0.185 ± 0.003	45.960 ± 1.614
UPI	beta57	0.06	0.053	-0.007 ± 0.002	-0.110 ± 0.037	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	1.351 ± 0.072	1.162 ± 0.036	0.053	-0.007	-11.74	0.0626	0.0630	1.022	94.858 ± 0.715	0.278 ± 0.004	10.913 ± 1.010
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.004 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.551 ± 0.032	0.742 ± 0.025	0.057	-0.003	-5.12	0.0406	0.0407	0.882	92.340 ± 0.862	0.154 ± 0.002	40.504 ± 1.590

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S190

Simulation 2 - Condition 46: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.005	0.316 ± 0.009	0.201	0.001	0.28	0.0613	0.0613	1.031	94.885 ± 0.698	0.256 ± 0.001	86.259 ± 1.090
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.009 ± 0.011	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.113 ± 0.005	0.336 ± 0.010	0.203	0.003	1.36	0.0646	0.0647	1.006	94.383 ± 0.729	0.265 ± 0.001	85.055 ± 1.129
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.198	-0.002	-1.14	0.0370	0.0371	1.022	94.985 ± 0.691	0.149 ± 0.001	100.000 ± 0.000
LSAM	beta44	0.11	0.113	0.003 ± 0.001	0.026 ± 0.013	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.182 ± 0.008	0.426 ± 0.012	0.113	0.003	2.45	0.0476	0.0477	0.955	94.183 ± 0.741	0.175 ± 0.001	71.013 ± 1.437
LSAM	beta45	0.11	0.113	0.003 ± 0.001	0.030 ± 0.013	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.171 ± 0.008	0.414 ± 0.011	0.110	0.000	0.43	0.0447	0.0447	0.939	93.681 ± 0.771	0.167 ± 0.001	76.028 ± 1.352
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.008 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.100 ± 0.005	0.317 ± 0.010	0.080	0.000	0.30	0.0249	0.0249	0.959	93.781 ± 0.765	0.095 ± 0.001	90.471 ± 0.930
LSAM	beta47	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.096 ± 0.005	0.311 ± 0.009	0.080	0.000	-0.34	0.0227	0.0227	0.962	93.982 ± 0.753	0.094 ± 0.001	89.769 ± 0.960
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.009	0.162	0.002	1.42	0.0519	0.0519	1.017	95.888 ± 0.629	0.199 ± 0.001	89.970 ± 0.951
LSAM	beta52	0.16	0.164	0.004 ± 0.002	0.026 ± 0.014	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.206 ± 0.009	0.454 ± 0.012	0.167	0.007	4.36	0.0738	0.0741	0.966	94.684 ± 0.711	0.275 ± 0.001	64.393 ± 1.516
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.113 ± 0.005	0.337 ± 0.010	0.158	-0.002	-1.19	0.0531	0.0531	1.016	95.687 ± 0.643	0.215 ± 0.001	83.551 ± 1.174
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.173 ± 0.008	0.415 ± 0.011	0.161	0.001	0.59	0.0645	0.0645	0.996	94.985 ± 0.691	0.260 ± 0.001	67.904 ± 1.479
LSAM	beta55	0.08	0.081	0.001 ± 0.001	0.009 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.322 ± 0.015	0.568 ± 0.016	0.080	0.000	0.51	0.0452	0.0452	0.923	93.180 ± 0.798	0.164 ± 0.001	52.859 ± 1.581
LSAM	beta56	0.08	0.081	0.001 ± 0.001	0.012 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.120 ± 0.006	0.347 ± 0.010	0.080	0.000	0.27	0.0259	0.0259	0.910	91.374 ± 0.889	0.099 ± 0.001	85.657 ± 1.110
LSAM	beta57	0.06	0.060	-0.000 ± 0.001	-0.000 ± 0.023	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.535 ± 0.025	0.731 ± 0.021	0.059	-0.001	-2.22	0.0443	0.0443	0.963	94.183 ± 0.741	0.166 ± 0.001	30.491 ± 1.458
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.162 ± 0.008	0.402 ± 0.012	0.061	0.001	1.67	0.0225	0.0225	0.921	93.380 ± 0.787	0.087 ± 0.001	77.934 ± 1.313
QML	beta41	0.20	0.215	0.015 ± 0.001	0.073 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.215	0.015	7.53	0.0435	0.0461	1.038	94.183 ± 0.741	0.184 ± 0.001	99.097 ± 0.300
QML	beta42	0.20	0.213	0.013 ± 0.001	0.064 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.213	0.013	6.47	0.0470	0.0488	1.012	93.280 ± 0.793	0.186 ± 0.001	98.796 ± 0.345
QML	beta43	0.20	0.189	-0.011 ± 0.001	-0.053 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.189	-0.011	-5.48	0.0347	0.0364	1.030	93.581 ± 0.776	0.141 ± 0.001	100.000 ± 0.000
QML	beta44	0.11	0.074	-0.036 ± 0.002	-0.330 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.306 ± 0.013	0.553 ± 0.013	0.076	-0.034	-30.52	0.0472	0.0579	0.945	85.657 ± 1.110	0.181 ± 0.001	39.920 ± 1.551
QML	beta45	0.11	0.103	-0.007 ± 0.001	-0.066 ± 0.013	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.160 ± 0.007	0.400 ± 0.011	0.103	-0.007	-6.50	0.0417	0.0423	0.947	93.380 ± 0.787	0.161 ± 0.001	71.113 ± 1.435
QML	beta46	0.08	0.116	0.036 ± 0.001	0.449 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.320 ± 0.013	0.566 ± 0.012	0.114	0.034	43.11	0.0240	0.0420	0.941	74.524 ± 1.380	0.102 ± 0.001	99.398 ± 0.245
QML	beta47	0.08	0.115	0.035 ± 0.001	0.437 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.289 ± 0.011	0.538 ± 0.010	0.113	0.033	41.39	0.0227	0.0401	0.983	73.019 ± 1.406	0.097 ± 0.001	99.900 ± 0.100
QML	beta51	0.16	0.174	0.014 ± 0.002	0.089 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.115 ± 0.005	0.339 ± 0.009	0.176	0.016	10.26	0.0533	0.0558	1.022	95.286 ± 0.671	0.209 ± 0.001	90.873 ± 0.912
QML	beta52	0.16	0.149	-0.011 ± 0.002	-0.066 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.118 ± 0.005	0.344 ± 0.009	0.151	-0.009	-5.32	0.0555	0.0561	0.940	92.979 ± 0.809	0.199 ± 0.001	81.143 ± 1.239
QML	beta53	0.16	0.124	-0.036 ± 0.002	-0.224 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.143 ± 0.006	0.379 ± 0.009	0.122	-0.038	-23.48	0.0469	0.0601	1.008	88.365 ± 1.015	0.193 ± 0.001	72.016 ± 1.422
QML	beta54	0.16	0.165	0.005 ± 0.002	0.034 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.166	0.006	3.48	0.0468	0.0472	1.005	94.584 ± 0.717	0.190 ± 0.001	90.572 ± 0.925
QML	beta55	0.08	0.078	-0.002 ± 0.002	-0.024 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.484 ± 0.023	0.695 ± 0.020	0.080	0.000	-0.38	0.0522	0.0522	0.906	91.274 ± 0.894	0.198 ± 0.003	43.029 ± 1.568
QML	beta56	0.08	0.098	0.018 ± 0.001	0.228 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.197 ± 0.009	0.444 ± 0.012	0.097	0.017	20.93	0.0296	0.0340	0.922	87.763 ± 1.038	0.110 ± 0.001	91.675 ± 0.875
QML	beta57	0.06	0.096	0.036 ± 0.002	0.594 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	1.067 ± 0.054	1.033 ± 0.029	0.091	0.031	51.79	0.0478	0.0570	0.951	90.371 ± 0.934	0.189 ± 0.002	53.059 ± 1.581
QML	beta58	0.06	0.085	0.025 ± 0.001	0.417 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.337 ± 0.015	0.581 ± 0.013	0.084	0.024	40.54	0.0219	0.0327	0.919	82.146 ± 1.213	0.087 ± 0.001	96.489 ± 0.583
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.032 ± 0.009	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.296 ± 0.008	0.205	0.005	2.27	0.0615	0.0617	1.013	95.286 ± 0.671	0.234 ± 0.001	91.474 ± 0.884
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.016 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.092 ± 0.004	0.303 ± 0.008	0.204	0.004	1.77	0.0607	0.0608	1.021	95.186 ± 0.678	0.243 ± 0.001	89.368 ± 0.976
UPI	beta43	0.20	0.197	-0.003 ± 0.001	-0.014 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.001	0.186 ± 0.005	0.196	-0.004	-1.98	0.0374	0.0376	1.007	95.085 ± 0.685	0.146 ± 0.001	100.000 ± 0.000
UPI	beta44	0.11	0.111	0.001 ± 0.002	0.008 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.198 ± 0.010	0.445 ± 0.013	0.109	-0.001	-1.27	0.0471	0.0472	0.961	93.982 ± 0.753	0.184 ± 0.002	67.603 ± 1.482
UPI	beta45	0.11	0.113	0.003 ± 0.002	0.023 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.186 ± 0.009	0.432 ± 0.012	0.111	0.001	0.74	0.0466	0.0466	0.979	93.882 ± 0.759	0.182 ± 0.002	70.913 ± 1.438
UPI	beta46	0.08	0.078	-0.002 ± 0.001	-0.023 ± 0.010	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.110 ± 0.006	0.331 ± 0.010	0.077	-0.003	-3.62	0.0260	0.0261	0.919	90.471 ± 0.930	0.095 ± 0.001	88.465 ± 1.012

Table S190

Simulation 2 - Condition 46: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.099 ± 0.005	0.314 ± 0.010	0.078	-0.002	-2.02	0.0230	0.0230	0.942	92.979 ± 0.809	0.093 ± 0.001	89.769 ± 0.960
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.018 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.308 ± 0.008	0.164	0.004	2.64	0.0502	0.0504	1.021	95.687 ± 0.643	0.197 ± 0.001	90.672 ± 0.921
UPI	beta52	0.16	0.169	0.009 ± 0.002	0.056 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.200 ± 0.009	0.447 ± 0.012	0.168	0.008	5.01	0.0717	0.0721	0.962	94.483 ± 0.723	0.268 ± 0.001	67.904 ± 1.479
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.328 ± 0.009	0.158	-0.002	-1.30	0.0516	0.0516	1.025	95.286 ± 0.671	0.211 ± 0.001	85.657 ± 1.110
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.008	0.413 ± 0.011	0.162	0.002	1.53	0.0630	0.0631	1.000	94.885 ± 0.698	0.259 ± 0.001	68.806 ± 1.467
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.014 ± 0.017	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.296 ± 0.015	0.544 ± 0.016	0.081	0.001	1.18	0.0414	0.0415	0.946	93.681 ± 0.771	0.161 ± 0.001	53.159 ± 1.580
UPI	beta56	0.08	0.080	0.000 ± 0.001	0.000 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.120 ± 0.006	0.347 ± 0.010	0.079	-0.001	-0.66	0.0264	0.0264	0.918	92.076 ± 0.855	0.100 ± 0.001	85.055 ± 1.129
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.039 ± 0.022	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.496 ± 0.024	0.704 ± 0.020	0.057	-0.003	-4.27	0.0417	0.0418	0.967	94.183 ± 0.741	0.160 ± 0.001	30.090 ± 1.453
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.002 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.163 ± 0.008	0.404 ± 0.012	0.060	0.000	0.80	0.0227	0.0227	0.923	92.778 ± 0.820	0.088 ± 0.001	76.931 ± 1.334

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S191

Simulation 2 - Condition 47: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.003	0.004 ± 0.013	0.006 ± 0.000	0.081 ± 0.002	0.006 ± 0.000	0.081 ± 0.002	0.162 ± 0.007	0.403 ± 0.011	0.201	0.001	0.36	0.0816	0.0816	0.961	93.598 ± 0.780	0.303 ± 0.001	73.171 ± 1.412
LSAM	beta42	0.20	0.203	0.003 ± 0.003	0.013 ± 0.013	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.168 ± 0.008	0.410 ± 0.011	0.203	0.003	1.53	0.0773	0.0774	0.975	94.512 ± 0.726	0.313 ± 0.001	72.459 ± 1.424
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.202	0.002	0.79	0.0509	0.0509	0.961	94.411 ± 0.732	0.185 ± 0.001	98.577 ± 0.378
LSAM	beta44	0.11	0.112	0.002 ± 0.002	0.022 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.262 ± 0.013	0.512 ± 0.015	0.110	0.000	0.36	0.0548	0.0548	0.918	92.480 ± 0.841	0.203 ± 0.002	59.959 ± 1.562
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.012 ± 0.015	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.226 ± 0.011	0.476 ± 0.014	0.111	0.001	1.21	0.0489	0.0489	0.933	93.699 ± 0.775	0.191 ± 0.002	64.939 ± 1.521
LSAM	beta46	0.08	0.079	-0.001 ± 0.001	-0.007 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.162 ± 0.009	0.403 ± 0.013	0.078	-0.002	-2.67	0.0294	0.0295	0.897	90.854 ± 0.919	0.113 ± 0.001	76.118 ± 1.359
LSAM	beta47	0.08	0.080	-0.000 ± 0.001	-0.004 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.158 ± 0.009	0.397 ± 0.013	0.078	-0.002	-2.02	0.0282	0.0282	0.902	92.378 ± 0.846	0.112 ± 0.001	78.455 ± 1.311
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.003 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.008	0.410 ± 0.012	0.159	-0.001	-0.42	0.0663	0.0663	0.977	94.512 ± 0.726	0.252 ± 0.001	69.512 ± 1.468
LSAM	beta52	0.16	0.164	0.004 ± 0.003	0.028 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.278 ± 0.013	0.527 ± 0.015	0.164	0.004	2.60	0.0841	0.0842	0.989	95.020 ± 0.693	0.327 ± 0.001	52.236 ± 1.592
LSAM	beta53	0.16	0.156	-0.004 ± 0.002	-0.024 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.200 ± 0.009	0.447 ± 0.013	0.157	-0.003	-2.03	0.0703	0.0703	0.961	94.411 ± 0.732	0.269 ± 0.001	63.211 ± 1.537
LSAM	beta54	0.16	0.159	-0.001 ± 0.003	-0.009 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.270 ± 0.012	0.520 ± 0.014	0.160	0.000	0.10	0.0832	0.0832	0.982	94.715 ± 0.713	0.320 ± 0.002	52.134 ± 1.592
LSAM	beta55	0.08	0.080	0.000 ± 0.002	0.002 ± 0.024	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.548 ± 0.027	0.740 ± 0.022	0.080	0.000	0.10	0.0552	0.0552	0.860	91.159 ± 0.905	0.200 ± 0.002	41.057 ± 1.568
LSAM	beta56	0.08	0.083	0.003 ± 0.001	0.036 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.237 ± 0.013	0.487 ± 0.016	0.081	0.001	1.46	0.0337	0.0337	0.885	92.378 ± 0.846	0.135 ± 0.002	67.988 ± 1.487
LSAM	beta57	0.06	0.059	-0.001 ± 0.002	-0.013 ± 0.028	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.791 ± 0.038	0.889 ± 0.026	0.057	-0.003	-4.63	0.0509	0.0510	0.895	91.667 ± 0.881	0.187 ± 0.002	28.557 ± 1.440
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.017 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.285 ± 0.014	0.534 ± 0.016	0.060	0.000	0.07	0.0288	0.0288	0.894	92.175 ± 0.856	0.112 ± 0.001	61.280 ± 1.553
QML	beta41	0.20	0.206	0.006 ± 0.002	0.028 ± 0.011	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.111 ± 0.005	0.334 ± 0.009	0.206	0.006	3.05	0.0665	0.0668	0.977	94.106 ± 0.751	0.255 ± 0.002	85.264 ± 1.130
QML	beta42	0.20	0.206	0.006 ± 0.002	0.028 ± 0.011	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.113 ± 0.005	0.335 ± 0.009	0.208	0.008	3.82	0.0648	0.0652	0.990	94.512 ± 0.726	0.259 ± 0.002	85.366 ± 1.127
QML	beta43	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.240 ± 0.007	0.198	-0.002	-0.75	0.0498	0.0499	0.963	93.598 ± 0.780	0.180 ± 0.001	98.679 ± 0.364
QML	beta44	0.11	0.096	-0.014 ± 0.002	-0.130 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.267 ± 0.013	0.517 ± 0.015	0.093	-0.017	-15.00	0.0544	0.0568	0.942	92.378 ± 0.846	0.203 ± 0.002	50.203 ± 1.594
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.037 ± 0.015	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.211 ± 0.011	0.460 ± 0.014	0.107	-0.003	-3.16	0.0469	0.0470	0.948	93.293 ± 0.797	0.187 ± 0.002	62.907 ± 1.540
QML	beta46	0.08	0.091	0.011 ± 0.001	0.143 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.166 ± 0.010	0.407 ± 0.013	0.090	0.010	12.26	0.0273	0.0290	0.925	91.667 ± 0.881	0.111 ± 0.001	88.821 ± 1.005
QML	beta47	0.08	0.093	0.013 ± 0.001	0.161 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.170 ± 0.009	0.412 ± 0.013	0.091	0.011	13.28	0.0288	0.0307	0.915	92.276 ± 0.851	0.109 ± 0.001	90.752 ± 0.924
QML	beta51	0.16	0.167	0.007 ± 0.002	0.047 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.008	0.412 ± 0.012	0.167	0.007	4.44	0.0672	0.0675	0.993	95.122 ± 0.687	0.255 ± 0.001	72.967 ± 1.416
QML	beta52	0.16	0.148	-0.012 ± 0.002	-0.075 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.186 ± 0.008	0.431 ± 0.012	0.146	-0.014	-8.92	0.0669	0.0684	1.011	95.224 ± 0.680	0.269 ± 0.001	57.622 ± 1.575
QML	beta53	0.16	0.132	-0.028 ± 0.002	-0.174 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.212 ± 0.010	0.461 ± 0.012	0.131	-0.029	-18.31	0.0666	0.0728	0.973	91.463 ± 0.891	0.261 ± 0.001	51.118 ± 1.594
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.189 ± 0.008	0.434 ± 0.012	0.162	0.002	0.95	0.0710	0.0710	0.997	95.020 ± 0.693	0.272 ± 0.001	65.244 ± 1.518
QML	beta55	0.08	0.080	-0.000 ± 0.002	-0.002 ± 0.024	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.551 ± 0.028	0.742 ± 0.022	0.080	0.000	-0.16	0.0562	0.0562	0.886	92.480 ± 0.841	0.206 ± 0.003	41.159 ± 1.569
QML	beta56	0.08	0.089	0.009 ± 0.001	0.108 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.242 ± 0.013	0.492 ± 0.015	0.087	0.007	8.28	0.0352	0.0358	0.900	92.073 ± 0.861	0.135 ± 0.002	72.154 ± 1.429
QML	beta57	0.06	0.070	0.010 ± 0.002	0.170 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.740 ± 0.038	0.860 ± 0.026	0.068	0.008	13.49	0.0469	0.0476	0.933	93.191 ± 0.803	0.185 ± 0.002	34.654 ± 1.517
QML	beta58	0.06	0.070	0.010 ± 0.001	0.172 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.289 ± 0.016	0.537 ± 0.017	0.069	0.009	14.89	0.0269	0.0284	0.896	90.752 ± 0.924	0.107 ± 0.001	72.663 ± 1.421
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.028 ± 0.012	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.135 ± 0.006	0.367 ± 0.010	0.205	0.005	2.38	0.0728	0.0729	0.962	93.801 ± 0.769	0.276 ± 0.001	82.012 ± 1.224
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.023 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.146 ± 0.007	0.382 ± 0.011	0.206	0.006	2.85	0.0736	0.0738	0.959	93.801 ± 0.769	0.287 ± 0.001	79.776 ± 1.280
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.199	-0.001	-0.29	0.0502	0.0502	0.942	93.496 ± 0.786	0.183 ± 0.001	98.577 ± 0.378
UPI	beta44	0.11	0.112	0.002 ± 0.002	0.018 ± 0.017	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.282 ± 0.015	0.531 ± 0.017	0.107	-0.003	-2.36	0.0543	0.0544	0.919	92.683 ± 0.830	0.210 ± 0.002	56.911 ± 1.579
UPI	beta45	0.11	0.110	0.000 ± 0.002	0.003 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.239 ± 0.013	0.489 ± 0.015	0.111	0.001	0.55	0.0510	0.0510	0.953	94.004 ± 0.757	0.201 ± 0.002	62.297 ± 1.545
UPI	beta46	0.08	0.078	-0.002 ± 0.001	-0.026 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.165 ± 0.009	0.406 ± 0.013	0.077	-0.003	-3.73	0.0293	0.0294	0.879	89.431 ± 0.980	0.112 ± 0.001	76.321 ± 1.355

Table S191

Simulation 2 - Condition 47: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.013 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.163 ± 0.009	0.404 ± 0.013	0.077	-0.003	-3.87	0.0294	0.0296	0.877	90.346 ± 0.941	0.111 ± 0.001	77.642 ± 1.328
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.012	0.159	-0.001	-0.42	0.0653	0.0653	0.984	94.817 ± 0.707	0.249 ± 0.001	71.951 ± 1.432
UPI	beta52	0.16	0.169	0.009 ± 0.003	0.055 ± 0.016	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.270 ± 0.012	0.520 ± 0.014	0.168	0.008	5.03	0.0819	0.0823	0.989	95.020 ± 0.693	0.321 ± 0.001	54.472 ± 1.588
UPI	beta53	0.16	0.155	-0.005 ± 0.002	-0.029 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.194 ± 0.009	0.441 ± 0.012	0.157	-0.003	-1.62	0.0713	0.0714	0.967	94.512 ± 0.726	0.267 ± 0.001	64.736 ± 1.523
UPI	beta54	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.271 ± 0.012	0.520 ± 0.014	0.160	0.000	-0.10	0.0844	0.0844	0.978	94.817 ± 0.707	0.319 ± 0.002	51.931 ± 1.593
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.013 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.536 ± 0.027	0.732 ± 0.022	0.079	-0.001	-1.01	0.0540	0.0540	0.872	92.175 ± 0.856	0.200 ± 0.002	41.159 ± 1.569
UPI	beta56	0.08	0.082	0.002 ± 0.001	0.030 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.236 ± 0.013	0.486 ± 0.015	0.081	0.001	0.72	0.0343	0.0343	0.891	91.463 ± 0.891	0.136 ± 0.002	67.683 ± 1.491
UPI	beta57	0.06	0.057	-0.003 ± 0.002	-0.046 ± 0.028	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.752 ± 0.035	0.867 ± 0.025	0.058	-0.002	-3.52	0.0498	0.0498	0.915	92.785 ± 0.825	0.186 ± 0.002	27.033 ± 1.416
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.012 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.288 ± 0.015	0.537 ± 0.016	0.059	-0.001	-1.69	0.0293	0.0293	0.896	91.260 ± 0.900	0.113 ± 0.001	60.671 ± 1.557

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S192

Simulation 2 - Condition 48: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.201	0.001	0.59	0.0475	0.0475	0.985	94.076 ± 0.748	0.186 ± 0.001	98.394 ± 0.398
LSAM	beta42	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.200	0.000	-0.10	0.0496	0.0496	0.961	93.675 ± 0.771	0.192 ± 0.001	96.285 ± 0.599
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.199	-0.001	-0.29	0.0310	0.0310	0.970	94.076 ± 0.748	0.117 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.109	-0.001 ± 0.001	-0.011 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.108	-0.002	-1.38	0.0330	0.0331	0.948	93.876 ± 0.760	0.123 ± 0.001	90.462 ± 0.931
LSAM	beta45	0.11	0.111	0.001 ± 0.001	0.007 ± 0.009	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.083 ± 0.004	0.287 ± 0.008	0.110	0.000	0.34	0.0294	0.0294	0.951	93.474 ± 0.783	0.118 ± 0.001	93.574 ± 0.777
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.008 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.053 ± 0.003	0.230 ± 0.007	0.080	0.000	0.23	0.0176	0.0176	0.911	91.165 ± 0.899	0.066 ± 0.001	98.795 ± 0.346
LSAM	beta47	0.08	0.080	0.000 ± 0.001	0.001 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.052 ± 0.003	0.227 ± 0.007	0.080	0.000	-0.56	0.0166	0.0166	0.919	91.968 ± 0.861	0.065 ± 0.001	99.498 ± 0.224
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.158	-0.002	-1.51	0.0383	0.0384	0.993	94.177 ± 0.742	0.157 ± 0.001	98.193 ± 0.422
LSAM	beta52	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.161	0.001	0.33	0.0508	0.0508	0.977	94.880 ± 0.698	0.202 ± 0.001	85.743 ± 1.108
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.005 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.004	0.273 ± 0.008	0.160	0.000	-0.26	0.0422	0.0422	0.973	94.177 ± 0.742	0.167 ± 0.001	96.285 ± 0.599
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.004	0.320 ± 0.009	0.164	0.004	2.25	0.0526	0.0528	0.974	94.378 ± 0.730	0.196 ± 0.001	88.353 ± 1.016
LSAM	beta55	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.156 ± 0.008	0.395 ± 0.012	0.080	0.000	-0.12	0.0301	0.0301	0.928	92.972 ± 0.810	0.115 ± 0.001	75.402 ± 1.365
LSAM	beta56	0.08	0.081	0.001 ± 0.001	0.015 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.071 ± 0.004	0.266 ± 0.008	0.081	0.001	1.35	0.0196	0.0196	0.893	90.562 ± 0.926	0.074 ± 0.001	96.687 ± 0.567
LSAM	beta57	0.06	0.060	0.000 ± 0.001	0.007 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.260 ± 0.012	0.510 ± 0.015	0.060	0.000	-0.25	0.0306	0.0306	0.951	94.177 ± 0.742	0.114 ± 0.001	56.627 ± 1.570
LSAM	beta58	0.06	0.060	-0.000 ± 0.001	-0.003 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.093 ± 0.005	0.305 ± 0.009	0.059	-0.001	-2.13	0.0173	0.0173	0.899	92.771 ± 0.821	0.064 ± 0.001	92.972 ± 0.810
QML	beta41	0.20	0.206	0.006 ± 0.001	0.030 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.207	0.007	3.36	0.0401	0.0406	1.005	95.181 ± 0.679	0.160 ± 0.002	99.598 ± 0.200
QML	beta42	0.20	0.202	0.002 ± 0.001	0.012 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.045 ± 0.002	0.213 ± 0.006	0.203	0.003	1.66	0.0410	0.0411	0.984	93.574 ± 0.777	0.164 ± 0.002	99.297 ± 0.265
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.020 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.195	-0.005	-2.35	0.0300	0.0304	0.980	93.574 ± 0.777	0.114 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.095	-0.015 ± 0.001	-0.140 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.094	-0.016	-14.10	0.0334	0.0368	0.955	91.165 ± 0.899	0.128 ± 0.001	82.129 ± 1.214
QML	beta45	0.11	0.107	-0.003 ± 0.001	-0.024 ± 0.009	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.084 ± 0.004	0.291 ± 0.008	0.107	-0.003	-2.69	0.0310	0.0312	0.950	92.871 ± 0.815	0.119 ± 0.001	92.570 ± 0.831
QML	beta46	0.08	0.093	0.013 ± 0.001	0.165 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	0.080 ± 0.004	0.283 ± 0.007	0.093	0.013	16.09	0.0185	0.0225	0.936	88.153 ± 1.024	0.067 ± 0.001	99.900 ± 0.100
QML	beta47	0.08	0.093	0.013 ± 0.001	0.162 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.077 ± 0.004	0.277 ± 0.007	0.093	0.013	15.63	0.0162	0.0205	0.937	90.462 ± 0.931	0.066 ± 0.001	99.900 ± 0.100
QML	beta51	0.16	0.167	0.007 ± 0.001	0.043 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.166	0.006	3.74	0.0387	0.0392	1.004	94.679 ± 0.711	0.160 ± 0.001	98.494 ± 0.386
QML	beta52	0.16	0.143	-0.017 ± 0.001	-0.107 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.294 ± 0.008	0.143	-0.017	-10.53	0.0439	0.0470	0.983	93.474 ± 0.783	0.169 ± 0.001	89.558 ± 0.969
QML	beta53	0.16	0.136	-0.024 ± 0.001	-0.151 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.307 ± 0.008	0.134	-0.026	-16.00	0.0415	0.0488	0.967	89.357 ± 0.977	0.162 ± 0.001	90.161 ± 0.944
QML	beta54	0.16	0.162	0.002 ± 0.001	0.010 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.275 ± 0.007	0.164	0.004	2.51	0.0439	0.0440	0.979	94.679 ± 0.711	0.169 ± 0.001	95.984 ± 0.622
QML	beta55	0.08	0.082	0.002 ± 0.001	0.025 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.176 ± 0.009	0.419 ± 0.012	0.082	0.002	2.54	0.0320	0.0320	0.939	93.373 ± 0.788	0.123 ± 0.001	72.992 ± 1.407
QML	beta56	0.08	0.087	0.007 ± 0.001	0.090 ± 0.009	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.084 ± 0.004	0.289 ± 0.009	0.087	0.007	8.52	0.0206	0.0217	0.899	91.265 ± 0.895	0.077 ± 0.001	97.892 ± 0.455
QML	beta57	0.06	0.070	0.010 ± 0.001	0.172 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.292 ± 0.015	0.540 ± 0.016	0.070	0.010	15.87	0.0303	0.0318	0.974	94.880 ± 0.698	0.117 ± 0.001	69.578 ± 1.458
QML	beta58	0.06	0.069	0.009 ± 0.001	0.157 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.116 ± 0.006	0.340 ± 0.010	0.068	0.008	13.60	0.0176	0.0194	0.899	89.458 ± 0.973	0.064 ± 0.001	97.791 ± 0.466
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.016 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.204	0.004	1.90	0.0426	0.0428	0.971	93.976 ± 0.754	0.167 ± 0.001	99.398 ± 0.245
UPI	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.003	0.231 ± 0.007	0.201	0.001	0.69	0.0449	0.0449	0.953	93.574 ± 0.777	0.173 ± 0.001	98.594 ± 0.373
UPI	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.198	-0.002	-1.02	0.0305	0.0306	0.954	93.775 ± 0.766	0.115 ± 0.001	100.000 ± 0.000
UPI	beta44	0.11	0.108	-0.002 ± 0.001	-0.022 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.094 ± 0.004	0.306 ± 0.009	0.105	-0.005	-4.14	0.0341	0.0344	0.949	93.976 ± 0.754	0.125 ± 0.001	90.261 ± 0.939
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.006 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.088 ± 0.004	0.296 ± 0.009	0.110	0.000	0.35	0.0305	0.0305	0.948	92.771 ± 0.821	0.121 ± 0.001	93.273 ± 0.794
UPI	beta46	0.08	0.080	-0.000 ± 0.001	-0.001 ± 0.007	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.054 ± 0.003	0.232 ± 0.007	0.079	-0.001	-1.02	0.0174	0.0174	0.885	89.960 ± 0.952	0.065 ± 0.001	98.996 ± 0.316

Table S192

Simulation 2 - Condition 48: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.005 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.051 ± 0.003	0.225 ± 0.007	0.078	-0.002	-2.09	0.0169	0.0170	0.912	91.265 ± 0.895	0.064 ± 0.001	99.598 ± 0.200
UPI	beta51	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.252 ± 0.007	0.158	-0.002	-1.07	0.0389	0.0389	0.988	94.478 ± 0.724	0.156 ± 0.001	98.293 ± 0.410
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.162	0.002	1.33	0.0499	0.0500	0.977	94.378 ± 0.730	0.199 ± 0.001	87.751 ± 1.039
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.002 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.269 ± 0.008	0.160	0.000	-0.27	0.0415	0.0415	0.981	94.578 ± 0.718	0.166 ± 0.001	96.988 ± 0.542
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.004	0.319 ± 0.009	0.164	0.004	2.30	0.0507	0.0509	0.973	94.478 ± 0.724	0.195 ± 0.001	88.755 ± 1.001
UPI	beta55	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.153 ± 0.007	0.392 ± 0.011	0.079	-0.001	-0.63	0.0301	0.0301	0.932	93.574 ± 0.777	0.115 ± 0.001	75.201 ± 1.368
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.013 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.072 ± 0.004	0.268 ± 0.008	0.081	0.001	0.66	0.0191	0.0191	0.890	90.964 ± 0.908	0.075 ± 0.001	96.285 ± 0.599
UPI	beta57	0.06	0.060	-0.000 ± 0.001	-0.006 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.255 ± 0.013	0.505 ± 0.015	0.059	-0.001	-1.28	0.0291	0.0291	0.951	94.277 ± 0.736	0.113 ± 0.001	58.333 ± 1.562
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.007 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.093 ± 0.005	0.306 ± 0.009	0.058	-0.002	-2.85	0.0176	0.0177	0.901	91.667 ± 0.876	0.065 ± 0.001	92.570 ± 0.831

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S193

Simulation 2 - Condition 49: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.192	-0.008 ± 0.006	-0.041 ± 0.028	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.089 ± 0.005	0.200 ± 0.018	0.447 ± 0.025	0.196	-0.004	-1.82	0.0887	0.0888	1.137	96.875 ± 1.087	0.398 ± 0.008	57.031 ± 3.094
LSAM	beta42	0.20	0.204	0.004 ± 0.005	0.018 ± 0.027	0.007 ± 0.001	0.086 ± 0.004	0.007 ± 0.001	0.086 ± 0.005	0.185 ± 0.019	0.430 ± 0.026	0.204	0.004	1.91	0.0860	0.0860	1.234	97.656 ± 0.946	0.417 ± 0.009	53.125 ± 3.119
LSAM	beta43	0.20	0.204	0.004 ± 0.005	0.020 ± 0.027	0.008 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.087 ± 0.005	0.190 ± 0.016	0.436 ± 0.023	0.207	0.007	3.61	0.0930	0.0933	1.078	96.484 ± 1.151	0.369 ± 0.007	61.328 ± 3.044
LSAM	beta44	0.11	0.113	0.003 ± 0.009	0.026 ± 0.080	0.020 ± 0.002	0.141 ± 0.009	0.020 ± 0.002	0.140 ± 0.010	1.629 ± 0.195	1.276 ± 0.087	0.100	-0.010	-9.22	0.1104	0.1109	1.123	98.828 ± 0.673	0.619 ± 0.025	10.547 ± 1.920
LSAM	beta45	0.11	0.113	0.003 ± 0.007	0.031 ± 0.067	0.014 ± 0.001	0.117 ± 0.006	0.014 ± 0.001	0.117 ± 0.007	1.135 ± 0.110	1.065 ± 0.062	0.111	0.001	1.18	0.1013	0.1013	1.085	98.047 ± 0.865	0.499 ± 0.015	14.062 ± 2.173
LSAM	beta46	0.08	0.067	-0.013 ± 0.011	-0.163 ± 0.137	0.031 ± 0.004	0.175 ± 0.011	0.031 ± 0.004	0.175 ± 0.013	4.811 ± 0.613	2.193 ± 0.157	0.071	-0.009	-10.71	0.1591	0.1593	1.208	98.828 ± 0.673	0.830 ± 0.041	1.562 ± 0.775
LSAM	beta47	0.08	0.093	0.013 ± 0.011	0.159 ± 0.141	0.033 ± 0.004	0.180 ± 0.010	0.033 ± 0.004	0.181 ± 0.012	5.092 ± 0.564	2.256 ± 0.145	0.081	0.001	1.11	0.1426	0.1426	1.128	98.828 ± 0.673	0.798 ± 0.034	1.953 ± 0.865
LSAM	beta51	0.16	0.164	0.004 ± 0.009	0.026 ± 0.054	0.019 ± 0.002	0.139 ± 0.006	0.019 ± 0.002	0.139 ± 0.008	0.756 ± 0.069	0.870 ± 0.049	0.163	0.003	1.94	0.1376	0.1376	0.943	95.312 ± 1.321	0.515 ± 0.008	28.516 ± 2.822
LSAM	beta52	0.16	0.162	0.002 ± 0.006	0.015 ± 0.040	0.010 ± 0.001	0.102 ± 0.005	0.010 ± 0.001	0.102 ± 0.006	0.405 ± 0.039	0.636 ± 0.037	0.164	0.004	2.69	0.0902	0.0903	1.045	98.047 ± 0.865	0.418 ± 0.009	35.938 ± 2.999
LSAM	beta53	0.16	0.159	-0.001 ± 0.006	-0.009 ± 0.039	0.010 ± 0.001	0.101 ± 0.005	0.010 ± 0.001	0.100 ± 0.006	0.393 ± 0.040	0.627 ± 0.038	0.160	0.000	0.05	0.1035	0.1035	1.100	95.703 ± 1.267	0.434 ± 0.008	30.469 ± 2.877
LSAM	beta54	0.16	0.168	0.008 ± 0.006	0.052 ± 0.038	0.009 ± 0.001	0.097 ± 0.005	0.009 ± 0.001	0.097 ± 0.006	0.366 ± 0.038	0.605 ± 0.037	0.161	0.001	0.36	0.0931	0.0931	1.112	97.266 ± 1.019	0.421 ± 0.009	36.719 ± 3.013
LSAM	beta55	0.08	0.089	0.009 ± 0.012	0.115 ± 0.145	0.034 ± 0.003	0.186 ± 0.009	0.034 ± 0.003	0.186 ± 0.011	5.381 ± 0.534	2.320 ± 0.137	0.087	0.007	8.62	0.1684	0.1686	1.125	98.828 ± 0.673	0.819 ± 0.031	6.250 ± 1.513
LSAM	beta56	0.08	0.074	-0.006 ± 0.010	-0.076 ± 0.130	0.028 ± 0.003	0.166 ± 0.009	0.028 ± 0.003	0.166 ± 0.010	4.308 ± 0.441	2.076 ± 0.126	0.081	0.001	1.14	0.1632	0.1632	1.072	98.828 ± 0.673	0.698 ± 0.020	3.125 ± 1.087
LSAM	beta57	0.06	0.062	0.002 ± 0.011	0.037 ± 0.177	0.029 ± 0.003	0.170 ± 0.010	0.029 ± 0.003	0.169 ± 0.012	7.953 ± 0.965	2.820 ± 0.195	0.065	0.005	8.70	0.1667	0.1667	1.271	99.219 ± 0.550	0.845 ± 0.046	1.172 ± 0.673
LSAM	beta58	0.06	0.085	0.025 ± 0.010	0.411 ± 0.174	0.028 ± 0.003	0.167 ± 0.010	0.028 ± 0.003	0.169 ± 0.011	7.895 ± 0.934	2.810 ± 0.189	0.070	0.010	17.38	0.1530	0.1534	1.131	98.438 ± 0.775	0.741 ± 0.028	2.344 ± 0.946
QML	beta41	0.20	0.186	-0.014 ± 0.005	-0.068 ± 0.025	0.006 ± 0.001	0.079 ± 0.004	0.006 ± 0.001	0.080 ± 0.005	0.160 ± 0.016	0.400 ± 0.024	0.184	-0.016	-7.99	0.0821	0.0837	1.017	94.922 ± 1.372	0.315 ± 0.004	69.531 ± 2.877
QML	beta42	0.20	0.195	-0.005 ± 0.005	-0.027 ± 0.024	0.006 ± 0.001	0.076 ± 0.005	0.006 ± 0.001	0.076 ± 0.005	0.145 ± 0.017	0.381 ± 0.025	0.194	-0.006	-3.12	0.0793	0.0796	1.105	96.094 ± 1.211	0.330 ± 0.004	69.531 ± 2.877
QML	beta43	0.20	0.190	-0.010 ± 0.005	-0.051 ± 0.025	0.006 ± 0.000	0.080 ± 0.003	0.006 ± 0.000	0.081 ± 0.004	0.162 ± 0.012	0.403 ± 0.020	0.192	-0.008	-4.07	0.0905	0.0909	0.983	95.312 ± 1.321	0.308 ± 0.003	73.047 ± 2.773
QML	beta44	0.11	0.114	0.004 ± 0.007	0.033 ± 0.063	0.012 ± 0.001	0.111 ± 0.006	0.012 ± 0.001	0.111 ± 0.007	1.010 ± 0.110	1.005 ± 0.064	0.109	-0.001	-0.91	0.1086	0.1086	1.073	94.922 ± 1.372	0.466 ± 0.015	17.188 ± 2.358
QML	beta45	0.11	0.109	-0.001 ± 0.006	-0.012 ± 0.054	0.009 ± 0.001	0.096 ± 0.004	0.009 ± 0.001	0.096 ± 0.005	0.755 ± 0.070	0.869 ± 0.049	0.112	0.002	2.18	0.0946	0.0946	1.002	96.484 ± 1.151	0.376 ± 0.007	20.703 ± 2.532
QML	beta46	0.08	0.027	-0.053 ± 0.005	-0.666 ± 0.062	0.006 ± 0.001	0.079 ± 0.005	0.009 ± 0.001	0.095 ± 0.005	1.412 ± 0.133	1.188 ± 0.062	0.024	-0.056	-69.56	0.0737	0.0923	1.151	88.281 ± 2.010	0.356 ± 0.012	2.734 ± 1.019
QML	beta47	0.08	0.041	-0.039 ± 0.005	-0.492 ± 0.062	0.006 ± 0.001	0.079 ± 0.004	0.008 ± 0.001	0.088 ± 0.004	1.213 ± 0.095	1.101 ± 0.051	0.032	-0.048	-59.96	0.0794	0.0928	1.079	88.672 ± 1.981	0.334 ± 0.011	2.734 ± 1.019
QML	beta51	0.16	0.187	0.027 ± 0.008	0.167 ± 0.051	0.017 ± 0.002	0.130 ± 0.007	0.017 ± 0.002	0.132 ± 0.009	0.682 ± 0.077	0.826 ± 0.053	0.181	0.021	13.40	0.1334	0.1351	1.098	98.438 ± 0.775	0.558 ± 0.015	28.516 ± 2.822
QML	beta52	0.16	0.158	-0.002 ± 0.006	-0.011 ± 0.039	0.010 ± 0.001	0.099 ± 0.005	0.010 ± 0.001	0.099 ± 0.006	0.385 ± 0.042	0.620 ± 0.039	0.165	0.005	2.99	0.1019	0.1021	1.089	96.094 ± 1.211	0.425 ± 0.015	35.547 ± 2.992
QML	beta53	0.16	0.143	-0.017 ± 0.006	-0.107 ± 0.036	0.008 ± 0.001	0.091 ± 0.004	0.009 ± 0.001	0.093 ± 0.005	0.336 ± 0.030	0.580 ± 0.031	0.141	-0.019	-11.88	0.0922	0.0941	1.058	96.484 ± 1.151	0.379 ± 0.008	31.641 ± 2.907
QML	beta54	0.16	0.168	0.008 ± 0.005	0.052 ± 0.034	0.007 ± 0.001	0.086 ± 0.004	0.008 ± 0.001	0.087 ± 0.005	0.293 ± 0.029	0.542 ± 0.031	0.162	0.002	1.26	0.0856	0.0856	1.050	96.094 ± 1.211	0.356 ± 0.005	48.047 ± 3.123
QML	beta55	0.08	0.082	0.002 ± 0.010	0.031 ± 0.124	0.025 ± 0.003	0.159 ± 0.008	0.025 ± 0.003	0.158 ± 0.010	3.913 ± 0.410	1.978 ± 0.122	0.068	-0.012	-15.19	0.1379	0.1384	1.081	96.094 ± 1.211	0.672 ± 0.023	9.375 ± 1.822
QML	beta56	0.08	0.057	-0.023 ± 0.008	-0.285 ± 0.099	0.016 ± 0.002	0.126 ± 0.006	0.016 ± 0.002	0.128 ± 0.008	2.570 ± 0.260	1.603 ± 0.096	0.068	-0.012	-15.36	0.1445	0.1450	1.176	96.484 ± 1.151	0.583 ± 0.035	3.516 ± 1.151
QML	beta57	0.06	0.026	-0.034 ± 0.006	-0.575 ± 0.095	0.008 ± 0.001	0.091 ± 0.006	0.009 ± 0.001	0.097 ± 0.006	2.611 ± 0.263	1.616 ± 0.093	0.030	-0.030	-50.67	0.0904	0.0953	1.094	89.062 ± 1.951	0.389 ± 0.013	1.172 ± 0.673
QML	beta58	0.06	0.046	-0.014 ± 0.005	-0.239 ± 0.080	0.006 ± 0.001	0.077 ± 0.004	0.006 ± 0.001	0.078 ± 0.004	1.689 ± 0.153	1.300 ± 0.071	0.035	-0.025	-42.22	0.0778	0.0818	1.021	91.797 ± 1.715	0.307 ± 0.007	7.031 ± 1.598
UPI	beta41	0.20	0.195	-0.005 ± 0.007	-0.027 ± 0.037	0.014 ± 0.002	0.119 ± 0.008	0.014 ± 0.002	0.119 ± 0.009	0.356 ± 0.048	0.596 ± 0.045	0.195	-0.005	-2.26	0.1325	0.1326	1.868	100.000 ± 0.000	0.874 ± 0.070	28.125 ± 2.810
UPI	beta42	0.20	0.202	0.002 ± 0.008	0.010 ± 0.039	0.015 ± 0.002	0.124 ± 0.008	0.015 ± 0.002	0.124 ± 0.009	0.383 ± 0.050	0.619 ± 0.046	0.210	0.010	5.13	0.1364	0.1368	1.801	99.219 ± 0.550	0.875 ± 0.052	24.219 ± 2.678
UPI	beta43	0.20	0.202	0.002 ± 0.007	0.009 ± 0.037	0.014 ± 0.002	0.118 ± 0.008	0.014 ± 0.002	0.117 ± 0.009	0.345 ± 0.047	0.587 ± 0.045	0.201	0.001	0.64	0.1398	0.1398	1.538	98.438 ± 0.775	0.710 ± 0.040	33.203 ± 2.943
UPI	beta44	0.11	0.143	0.033 ± 0.017	0.297 ± 0.158	0.077 ± 0.009	0.278 ± 0.016	0.078 ± 0.009	0.279 ± 0.019	6.442 ± 0.758	2.538 ± 0.170	0.125	0.015	13.49	0.2373	0.2377	1.639	99.609 ± 0.390	1.785 ± 0.120	1.172 ± 0.673
UPI	beta45	0.11	0.106	-0.004 ± 0.014	-0.037 ± 0.130	0.053 ± 0.007	0.229 ± 0.017	0.052 ± 0.007	0.229 ± 0.018	4.336 ± 0.614	2.082 ± 0.163	0.103	-0.007	-6.71	0.1874	0.1875	1.721	99.609 ± 0.390	1.548 ± 0.120	3.906 ± 1.211
UPI	beta46	0.08	0.109	0.029 ± 0.026	0.363 ± 0.320	0.168 ± 0.025	0.410 ± 0.030	0.168 ± 0.025	0.410 ± 0.034	26.303 ± 3.927	5.129 ± 0.419	0.067	-0.013	-16.04	0.3778	0.3781	1.783	97.266 ± 1.019	2.866 ± 0.260	0.000 ± 0.000

Table S193

Simulation 2 - Condition 49: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.155	0.075 ± 0.026	0.936 ± 0.331	0.179 ± 0.024	0.424 ± 0.029	0.184 ± 0.026	0.429 ± 0.033	28.805 ± 4.052	5.367 ± 0.415	0.066	-0.014	-17.57	0.3230	0.3233	1.850	98.438 ± 0.775	3.071 ± 0.283	0.391 ± 0.390
UPI	beta51	0.16	0.172	0.012 ± 0.009	0.077 ± 0.054	0.019 ± 0.002	0.139 ± 0.008	0.019 ± 0.002	0.139 ± 0.009	0.754 ± 0.089	0.868 ± 0.058	0.163	0.003	1.58	0.1433	0.1433	1.535	100.000 ± 0.000	0.834 ± 0.045	16.406 ± 2.315
UPI	beta52	0.16	0.164	0.004 ± 0.006	0.025 ± 0.040	0.011 ± 0.001	0.103 ± 0.005	0.011 ± 0.001	0.103 ± 0.006	0.417 ± 0.040	0.646 ± 0.038	0.169	0.009	5.89	0.1068	0.1072	1.820	98.828 ± 0.673	0.738 ± 0.056	21.875 ± 2.584
UPI	beta53	0.16	0.152	-0.008 ± 0.007	-0.049 ± 0.047	0.014 ± 0.002	0.119 ± 0.007	0.014 ± 0.002	0.119 ± 0.008	0.556 ± 0.061	0.746 ± 0.047	0.157	-0.003	-1.77	0.1237	0.1237	1.560	98.047 ± 0.865	0.730 ± 0.043	14.453 ± 2.198
UPI	beta54	0.16	0.161	0.001 ± 0.007	0.004 ± 0.043	0.012 ± 0.001	0.109 ± 0.007	0.012 ± 0.001	0.109 ± 0.007	0.461 ± 0.055	0.679 ± 0.046	0.150	-0.010	-5.97	0.1057	0.1062	1.548	98.828 ± 0.673	0.660 ± 0.038	19.141 ± 2.459
UPI	beta55	0.08	0.103	0.023 ± 0.015	0.294 ± 0.188	0.058 ± 0.007	0.240 ± 0.015	0.058 ± 0.007	0.241 ± 0.016	9.051 ± 1.089	3.009 ± 0.205	0.079	-0.001	-0.94	0.2306	0.2306	2.015	99.609 ± 0.390	1.896 ± 0.169	0.000 ± 0.000
UPI	beta56	0.08	0.078	-0.002 ± 0.014	-0.023 ± 0.172	0.049 ± 0.006	0.220 ± 0.013	0.048 ± 0.006	0.220 ± 0.015	7.559 ± 0.898	2.749 ± 0.186	0.071	-0.009	-11.13	0.1971	0.1973	1.938	99.219 ± 0.550	1.674 ± 0.126	0.391 ± 0.390
UPI	beta57	0.06	0.060	0.000 ± 0.019	0.003 ± 0.321	0.095 ± 0.014	0.308 ± 0.023	0.095 ± 0.014	0.308 ± 0.025	26.327 ± 3.812	5.131 ± 0.409	0.038	-0.022	-36.41	0.2268	0.2279	1.835	99.219 ± 0.550	2.218 ± 0.228	0.000 ± 0.000
UPI	beta58	0.06	0.079	0.019 ± 0.015	0.323 ± 0.246	0.056 ± 0.007	0.236 ± 0.016	0.056 ± 0.008	0.237 ± 0.018	15.563 ± 2.085	3.945 ± 0.294	0.061	0.001	1.84	0.1919	0.1919	2.047	98.047 ± 0.865	1.897 ± 0.145	0.781 ± 0.550

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S194

Simulation 2 - Condition 50: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.008 ± 0.011	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.077 ± 0.004	0.277 ± 0.009	0.200	0.000	-0.18	0.0565	0.0565	1.028	96.398 ± 0.707	0.224 ± 0.001	94.957 ± 0.831
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.019 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.004	0.268 ± 0.009	0.206	0.006	2.83	0.0603	0.0606	1.091	97.406 ± 0.603	0.229 ± 0.001	95.245 ± 0.808
LSAM	beta43	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.010	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	0.070 ± 0.004	0.264 ± 0.009	0.199	-0.001	-0.39	0.0486	0.0486	1.010	94.957 ± 0.831	0.209 ± 0.001	96.974 ± 0.650
LSAM	beta44	0.11	0.112	0.002 ± 0.003	0.017 ± 0.027	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.514 ± 0.030	0.717 ± 0.025	0.110	0.000	-0.24	0.0781	0.0781	1.021	97.406 ± 0.603	0.316 ± 0.004	33.429 ± 1.791
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.010 ± 0.023	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.356 ± 0.021	0.597 ± 0.021	0.113	0.003	2.87	0.0631	0.0632	1.027	95.965 ± 0.747	0.264 ± 0.002	41.066 ± 1.867
LSAM	beta46	0.08	0.090	0.010 ± 0.004	0.120 ± 0.052	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	1.897 ± 0.129	1.377 ± 0.054	0.087	0.007	8.34	0.1016	0.1018	1.017	97.118 ± 0.635	0.438 ± 0.007	8.934 ± 1.083
LSAM	beta47	0.08	0.075	-0.005 ± 0.004	-0.067 ± 0.049	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	1.645 ± 0.107	1.283 ± 0.048	0.072	-0.008	-10.56	0.0979	0.0982	1.049	97.695 ± 0.570	0.422 ± 0.007	5.043 ± 0.831
LSAM	beta51	0.16	0.161	0.001 ± 0.003	0.007 ± 0.019	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	0.258 ± 0.016	0.508 ± 0.018	0.157	-0.003	-2.11	0.0784	0.0785	0.939	93.660 ± 0.925	0.300 ± 0.002	56.340 ± 1.883
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.014	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.007	0.369 ± 0.012	0.161	0.001	0.36	0.0642	0.0642	1.006	96.686 ± 0.679	0.233 ± 0.002	79.971 ± 1.519
LSAM	beta53	0.16	0.165	0.005 ± 0.002	0.030 ± 0.015	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.008	0.387 ± 0.013	0.167	0.007	4.08	0.0633	0.0636	1.007	95.677 ± 0.772	0.244 ± 0.001	76.081 ± 1.619
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.014	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.126 ± 0.007	0.356 ± 0.012	0.159	-0.001	-0.36	0.0587	0.0587	1.025	96.398 ± 0.707	0.229 ± 0.001	79.683 ± 1.527
LSAM	beta55	0.08	0.081	0.001 ± 0.004	0.017 ± 0.052	0.012 ± 0.001	0.109 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	1.852 ± 0.129	1.361 ± 0.054	0.084	0.004	4.76	0.0968	0.0969	0.974	96.830 ± 0.665	0.416 ± 0.006	11.816 ± 1.225
LSAM	beta56	0.08	0.087	0.007 ± 0.004	0.084 ± 0.046	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	1.474 ± 0.094	1.214 ± 0.045	0.079	-0.001	-1.08	0.1009	0.1009	0.990	96.542 ± 0.694	0.376 ± 0.004	13.112 ± 1.281
LSAM	beta57	0.06	0.066	0.006 ± 0.005	0.102 ± 0.076	0.014 ± 0.001	0.120 ± 0.004	0.014 ± 0.001	0.120 ± 0.005	3.997 ± 0.270	1.999 ± 0.078	0.063	0.003	5.40	0.1074	0.1075	0.969	96.830 ± 0.665	0.455 ± 0.008	3.314 ± 0.679
LSAM	beta58	0.06	0.064	0.004 ± 0.004	0.068 ± 0.059	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.000	0.093 ± 0.003	2.408 ± 0.137	1.552 ± 0.053	0.063	0.003	4.83	0.0910	0.0910	1.055	96.830 ± 0.665	0.385 ± 0.005	6.196 ± 0.915
QML	beta41	0.20	0.193	-0.007 ± 0.002	-0.037 ± 0.009	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.063 ± 0.003	0.250 ± 0.008	0.190	-0.010	-4.75	0.0517	0.0526	0.996	94.524 ± 0.864	0.193 ± 0.001	97.550 ± 0.587
QML	beta42	0.20	0.195	-0.005 ± 0.002	-0.024 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.059 ± 0.003	0.242 ± 0.008	0.196	-0.004	-1.87	0.0533	0.0534	1.048	95.677 ± 0.772	0.198 ± 0.001	97.406 ± 0.603
QML	beta43	0.20	0.183	-0.017 ± 0.002	-0.083 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.065 ± 0.004	0.256 ± 0.008	0.185	-0.015	-7.71	0.0478	0.0503	0.987	92.795 ± 0.981	0.187 ± 0.001	97.406 ± 0.603
QML	beta44	0.11	0.110	-0.000 ± 0.002	-0.002 ± 0.022	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.344 ± 0.018	0.587 ± 0.019	0.109	-0.001	-0.54	0.0708	0.0708	1.058	97.550 ± 0.587	0.268 ± 0.003	41.210 ± 1.868
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.034 ± 0.020	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.274 ± 0.015	0.523 ± 0.017	0.106	-0.004	-3.57	0.0561	0.0562	1.008	94.380 ± 0.874	0.227 ± 0.001	45.245 ± 1.889
QML	beta46	0.08	0.032	-0.048 ± 0.002	-0.600 ± 0.024	0.003 ± 0.000	0.050 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.751 ± 0.034	0.866 ± 0.022	0.031	-0.049	-61.85	0.0467	0.0681	1.042	80.692 ± 1.498	0.205 ± 0.004	7.061 ± 0.972
QML	beta47	0.08	0.028	-0.052 ± 0.002	-0.648 ± 0.022	0.002 ± 0.000	0.047 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.761 ± 0.032	0.872 ± 0.020	0.027	-0.053	-65.82	0.0472	0.0707	1.056	75.648 ± 1.629	0.194 ± 0.003	4.611 ± 0.796
QML	beta51	0.16	0.181	0.021 ± 0.003	0.132 ± 0.018	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.251 ± 0.015	0.501 ± 0.018	0.181	0.021	13.04	0.0725	0.0754	0.978	93.804 ± 0.915	0.297 ± 0.002	70.173 ± 1.737
QML	beta52	0.16	0.155	-0.005 ± 0.002	-0.033 ± 0.013	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.007	0.352 ± 0.012	0.151	-0.009	-5.76	0.0578	0.0585	1.024	95.533 ± 0.784	0.226 ± 0.003	80.692 ± 1.498
QML	beta53	0.16	0.148	-0.012 ± 0.002	-0.077 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.122 ± 0.006	0.349 ± 0.011	0.147	-0.013	-8.02	0.0565	0.0579	1.010	93.948 ± 0.905	0.216 ± 0.001	77.233 ± 1.592
QML	beta54	0.16	0.157	-0.003 ± 0.002	-0.020 ± 0.013	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.109 ± 0.005	0.330 ± 0.010	0.156	-0.004	-2.74	0.0547	0.0549	1.008	95.389 ± 0.796	0.208 ± 0.002	84.582 ± 1.371
QML	beta55	0.08	0.078	-0.002 ± 0.003	-0.027 ± 0.042	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	1.205 ± 0.067	1.098 ± 0.037	0.078	-0.002	-2.41	0.0849	0.0849	1.062	96.110 ± 0.734	0.366 ± 0.008	13.545 ± 1.299
QML	beta56	0.08	0.073	-0.007 ± 0.003	-0.087 ± 0.037	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.950 ± 0.049	0.975 ± 0.031	0.071	-0.009	-10.84	0.0833	0.0838	1.007	95.389 ± 0.796	0.307 ± 0.003	14.986 ± 1.355
QML	beta57	0.06	0.026	-0.034 ± 0.002	-0.560 ± 0.035	0.003 ± 0.000	0.056 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.173 ± 0.057	1.083 ± 0.030	0.025	-0.035	-58.66	0.0541	0.0646	1.085	84.726 ± 1.366	0.237 ± 0.013	4.467 ± 0.784
QML	beta58	0.06	0.032	-0.028 ± 0.002	-0.471 ± 0.029	0.002 ± 0.000	0.045 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.786 ± 0.035	0.887 ± 0.023	0.029	-0.031	-50.91	0.0481	0.0570	1.103	86.023 ± 1.316	0.195 ± 0.012	6.484 ± 0.935
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.014 ± 0.011	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.005	0.295 ± 0.010	0.208	0.008	4.01	0.0706	0.0710	1.287	97.550 ± 0.587	0.298 ± 0.008	82.853 ± 1.431
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.016 ± 0.011	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.006	0.298 ± 0.011	0.202	0.002	0.99	0.0653	0.0654	1.364	98.415 ± 0.474	0.318 ± 0.012	81.124 ± 1.485
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.011	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.005	0.283 ± 0.011	0.199	-0.001	-0.37	0.0617	0.0617	1.194	96.830 ± 0.665	0.265 ± 0.006	89.337 ± 1.172
UPI	beta44	0.11	0.116	0.006 ± 0.004	0.058 ± 0.033	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	0.739 ± 0.054	0.860 ± 0.036	0.113	0.003	3.05	0.0938	0.0938	1.415	99.568 ± 0.249	0.523 ± 0.021	16.571 ± 1.411
UPI	beta45	0.11	0.112	0.002 ± 0.003	0.018 ± 0.027	0.006 ± 0.000	0.077 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	0.494 ± 0.034	0.703 ± 0.028	0.109	-0.001	-0.83	0.0822	0.0822	1.363	98.559 ± 0.452	0.413 ± 0.024	23.055 ± 1.599
UPI	beta46	0.08	0.102	0.022 ± 0.006	0.270 ± 0.079	0.028 ± 0.002	0.166 ± 0.007	0.028 ± 0.002	0.168 ± 0.008	4.385 ± 0.362	2.094 ± 0.095	0.088	0.008	10.15	0.1428	0.1431	1.672	98.559 ± 0.452	1.089 ± 0.065	1.441 ± 0.452

Table S194

Simulation 2 - Condition 50: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.084	0.004 ± 0.006	0.048 ± 0.077	0.026 ± 0.002	0.161 ± 0.007	0.026 ± 0.002	0.161 ± 0.007	4.060 ± 0.327	2.015 ± 0.090	0.073	-0.007	-8.94	0.1408	0.1409	1.561	98.271 ± 0.495	0.986 ± 0.045	0.288 ± 0.203
UPI	beta51	0.16	0.164	0.004 ± 0.003	0.023 ± 0.018	0.006 ± 0.000	0.078 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	0.237 ± 0.015	0.487 ± 0.018	0.161	0.001	0.68	0.0763	0.0763	1.096	96.110 ± 0.734	0.334 ± 0.005	55.331 ± 1.887
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.014	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.008	0.370 ± 0.013	0.160	0.000	-0.10	0.0611	0.0611	1.200	97.695 ± 0.570	0.279 ± 0.007	71.326 ± 1.717
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.017 ± 0.015	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.008	0.391 ± 0.013	0.162	0.002	0.94	0.0694	0.0695	1.211	97.839 ± 0.552	0.297 ± 0.006	64.409 ± 1.817
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.014	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.007	0.358 ± 0.012	0.159	-0.001	-0.60	0.0583	0.0583	1.174	97.550 ± 0.587	0.263 ± 0.004	71.902 ± 1.706
UPI	beta55	0.08	0.083	0.003 ± 0.004	0.041 ± 0.055	0.013 ± 0.001	0.115 ± 0.004	0.013 ± 0.001	0.115 ± 0.005	2.060 ± 0.151	1.435 ± 0.060	0.086	0.006	7.09	0.1033	0.1035	1.288	98.847 ± 0.405	0.580 ± 0.020	4.611 ± 0.796
UPI	beta56	0.08	0.088	0.008 ± 0.004	0.097 ± 0.052	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.005	1.902 ± 0.140	1.379 ± 0.057	0.077	-0.003	-3.38	0.1011	0.1011	1.242	98.991 ± 0.379	0.536 ± 0.014	5.043 ± 0.831
UPI	beta57	0.06	0.068	0.008 ± 0.006	0.136 ± 0.092	0.021 ± 0.002	0.146 ± 0.005	0.021 ± 0.002	0.146 ± 0.006	5.894 ± 0.434	2.428 ± 0.101	0.059	-0.001	-1.93	0.1262	0.1262	1.514	98.847 ± 0.405	0.864 ± 0.048	0.576 ± 0.287
UPI	beta58	0.06	0.059	-0.001 ± 0.004	-0.019 ± 0.074	0.014 ± 0.001	0.116 ± 0.004	0.014 ± 0.001	0.116 ± 0.005	3.764 ± 0.264	1.940 ± 0.078	0.050	-0.010	-16.12	0.0955	0.0960	1.602	97.550 ± 0.587	0.732 ± 0.042	1.009 ± 0.379

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S195

Simulation 2 - Condition 51: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.004 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.004	0.306 ± 0.008	0.195	-0.005	-2.38	0.0649	0.0650	1.011	96.237 ± 0.624	0.243 ± 0.001	90.753 ± 0.950
LSAM	beta42	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.005	0.313 ± 0.009	0.197	-0.003	-1.38	0.0597	0.0597	1.009	95.914 ± 0.649	0.248 ± 0.001	88.387 ± 1.051
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.010	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.004	0.294 ± 0.009	0.201	0.001	0.73	0.0565	0.0565	0.997	94.086 ± 0.774	0.230 ± 0.001	93.011 ± 0.836
LSAM	beta44	0.11	0.108	-0.002 ± 0.002	-0.020 ± 0.022	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.462 ± 0.023	0.680 ± 0.020	0.113	0.003	2.65	0.0738	0.0739	0.984	95.376 ± 0.689	0.289 ± 0.002	35.806 ± 1.572
LSAM	beta45	0.11	0.114	0.004 ± 0.002	0.038 ± 0.020	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.358 ± 0.017	0.598 ± 0.017	0.114	0.004	3.49	0.0642	0.0643	0.996	96.559 ± 0.598	0.257 ± 0.002	42.688 ± 1.622
LSAM	beta46	0.08	0.078	-0.002 ± 0.003	-0.020 ± 0.039	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	1.414 ± 0.070	1.189 ± 0.035	0.080	0.000	-0.33	0.0949	0.0949	0.979	94.946 ± 0.718	0.365 ± 0.003	12.043 ± 1.067
LSAM	beta47	0.08	0.085	0.005 ± 0.003	0.066 ± 0.038	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	1.377 ± 0.073	1.174 ± 0.037	0.082	0.002	2.51	0.0869	0.0870	0.988	95.484 ± 0.681	0.363 ± 0.003	12.366 ± 1.079
LSAM	beta51	0.16	0.160	0.000 ± 0.003	0.003 ± 0.019	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.325 ± 0.017	0.570 ± 0.018	0.154	-0.006	-3.66	0.0884	0.0886	0.904	91.935 ± 0.893	0.323 ± 0.002	49.570 ± 1.640
LSAM	beta52	0.16	0.164	0.004 ± 0.002	0.026 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.434 ± 0.012	0.161	0.001	0.54	0.0657	0.0657	0.952	93.978 ± 0.780	0.259 ± 0.001	72.043 ± 1.472
LSAM	beta53	0.16	0.163	0.003 ± 0.002	0.019 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.201 ± 0.010	0.448 ± 0.013	0.161	0.001	0.59	0.0724	0.0724	0.950	93.226 ± 0.824	0.267 ± 0.001	67.312 ± 1.538
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.018 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.008	0.408 ± 0.012	0.164	0.004	2.19	0.0596	0.0598	0.999	94.731 ± 0.733	0.255 ± 0.001	73.118 ± 1.454
LSAM	beta55	0.08	0.083	0.003 ± 0.003	0.032 ± 0.043	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	1.696 ± 0.085	1.302 ± 0.039	0.083	0.003	4.33	0.1021	0.1022	0.937	94.946 ± 0.718	0.383 ± 0.004	16.559 ± 1.219
LSAM	beta56	0.08	0.079	-0.001 ± 0.003	-0.018 ± 0.041	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	1.537 ± 0.075	1.240 ± 0.036	0.077	-0.003	-3.83	0.0965	0.0966	0.920	93.548 ± 0.806	0.358 ± 0.003	16.774 ± 1.225
LSAM	beta57	0.06	0.061	0.001 ± 0.003	0.022 ± 0.056	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	2.905 ± 0.153	1.704 ± 0.053	0.060	0.000	-0.48	0.0984	0.0984	0.970	95.269 ± 0.696	0.389 ± 0.004	7.957 ± 0.887
LSAM	beta58	0.06	0.067	0.007 ± 0.003	0.115 ± 0.049	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	2.223 ± 0.117	1.491 ± 0.046	0.063	0.003	4.41	0.0867	0.0867	1.010	96.022 ± 0.641	0.353 ± 0.003	8.602 ± 0.919
QML	beta41	0.20	0.194	-0.006 ± 0.002	-0.032 ± 0.010	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.008	0.190	-0.010	-5.10	0.0642	0.0650	0.988	96.344 ± 0.615	0.229 ± 0.001	91.828 ± 0.898
QML	beta42	0.20	0.195	-0.005 ± 0.002	-0.025 ± 0.010	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.004	0.298 ± 0.009	0.193	-0.007	-3.49	0.0588	0.0592	1.007	95.699 ± 0.665	0.234 ± 0.001	91.183 ± 0.930
QML	beta43	0.20	0.192	-0.008 ± 0.002	-0.038 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.004	0.285 ± 0.008	0.191	-0.009	-4.55	0.0555	0.0562	0.993	93.871 ± 0.787	0.220 ± 0.001	92.473 ± 0.865
QML	beta44	0.11	0.106	-0.004 ± 0.002	-0.036 ± 0.022	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.449 ± 0.024	0.670 ± 0.021	0.111	0.001	0.80	0.0732	0.0733	0.994	96.237 ± 0.624	0.287 ± 0.004	38.602 ± 1.596
QML	beta45	0.11	0.112	0.002 ± 0.002	0.023 ± 0.019	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.334 ± 0.016	0.578 ± 0.017	0.109	-0.001	-0.80	0.0639	0.0639	1.000	95.591 ± 0.673	0.249 ± 0.002	43.871 ± 1.627
QML	beta46	0.08	0.047	-0.033 ± 0.002	-0.418 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.776 ± 0.033	0.881 ± 0.022	0.045	-0.035	-43.96	0.0625	0.0717	1.005	87.312 ± 1.091	0.244 ± 0.004	8.172 ± 0.898
QML	beta47	0.08	0.051	-0.029 ± 0.002	-0.367 ± 0.025	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.697 ± 0.031	0.835 ± 0.021	0.047	-0.033	-41.69	0.0566	0.0657	0.991	87.742 ± 1.075	0.233 ± 0.002	10.000 ± 0.984
QML	beta51	0.16	0.174	0.014 ± 0.003	0.085 ± 0.018	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.322 ± 0.017	0.568 ± 0.017	0.166	0.006	3.89	0.0858	0.0860	0.981	93.656 ± 0.799	0.346 ± 0.004	51.183 ± 1.639
QML	beta52	0.16	0.152	-0.008 ± 0.002	-0.049 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.205 ± 0.010	0.453 ± 0.013	0.150	-0.010	-6.56	0.0689	0.0697	0.972	93.871 ± 0.787	0.275 ± 0.006	62.043 ± 1.591
QML	beta53	0.16	0.144	-0.016 ± 0.002	-0.099 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.189 ± 0.009	0.435 ± 0.012	0.143	-0.017	-10.71	0.0663	0.0685	0.981	93.441 ± 0.812	0.260 ± 0.002	59.247 ± 1.611
QML	beta54	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.008	0.399 ± 0.012	0.162	0.002	1.17	0.0617	0.0617	1.007	95.269 ± 0.696	0.252 ± 0.002	74.086 ± 1.437
QML	beta55	0.08	0.081	0.001 ± 0.003	0.007 ± 0.041	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	1.547 ± 0.075	1.244 ± 0.036	0.083	0.003	4.25	0.0986	0.0987	0.982	95.054 ± 0.711	0.383 ± 0.007	17.204 ± 1.238
QML	beta56	0.08	0.074	-0.006 ± 0.003	-0.076 ± 0.037	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	1.286 ± 0.063	1.134 ± 0.034	0.071	-0.009	-11.71	0.0908	0.0913	0.956	93.333 ± 0.818	0.339 ± 0.003	15.269 ± 1.179
QML	beta57	0.06	0.037	-0.023 ± 0.002	-0.384 ± 0.037	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	1.425 ± 0.066	1.194 ± 0.033	0.032	-0.028	-46.51	0.0650	0.0708	1.007	88.817 ± 1.033	0.268 ± 0.006	4.946 ± 0.711
QML	beta58	0.06	0.045	-0.015 ± 0.002	-0.251 ± 0.032	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	1.013 ± 0.045	1.006 ± 0.027	0.039	-0.021	-35.23	0.0565	0.0603	1.007	92.043 ± 0.887	0.231 ± 0.003	6.774 ± 0.824
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.005	0.316 ± 0.010	0.198	-0.002	-0.79	0.0635	0.0636	1.046	97.527 ± 0.509	0.259 ± 0.002	88.280 ± 1.055
UPI	beta42	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.010	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.005	0.312 ± 0.009	0.197	-0.003	-1.33	0.0622	0.0623	1.096	96.452 ± 0.607	0.268 ± 0.003	85.484 ± 1.155
UPI	beta43	0.20	0.203	0.003 ± 0.002	0.016 ± 0.010	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.005	0.305 ± 0.010	0.204	0.004	2.19	0.0605	0.0607	1.057	95.806 ± 0.657	0.252 ± 0.002	90.215 ± 0.974
UPI	beta44	0.11	0.111	0.001 ± 0.003	0.012 ± 0.025	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.599 ± 0.031	0.774 ± 0.024	0.113	0.003	2.96	0.0847	0.0848	1.112	97.097 ± 0.551	0.371 ± 0.006	26.344 ± 1.444
UPI	beta45	0.11	0.116	0.006 ± 0.002	0.058 ± 0.022	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.438 ± 0.024	0.662 ± 0.021	0.114	0.004	3.85	0.0730	0.0731	1.095	97.634 ± 0.498	0.311 ± 0.004	32.903 ± 1.541
UPI	beta46	0.08	0.087	0.007 ± 0.004	0.085 ± 0.052	0.016 ± 0.001	0.126 ± 0.004	0.016 ± 0.001	0.126 ± 0.005	2.489 ± 0.157	1.578 ± 0.056	0.075	-0.005	-5.95	0.1097	0.1098	1.233	97.419 ± 0.520	0.610 ± 0.016	3.978 ± 0.641

Table S195

Simulation 2 - Condition 51: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.094	0.014 ± 0.004	0.176 ± 0.052	0.016 ± 0.001	0.126 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	2.523 ± 0.160	1.588 ± 0.057	0.082	0.002	2.32	0.1107	0.1107	1.255	98.172 ± 0.439	0.621 ± 0.019	3.226 ± 0.579
UPI	beta51	0.16	0.161	0.001 ± 0.003	0.007 ± 0.018	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.306 ± 0.016	0.553 ± 0.017	0.155	-0.005	-3.03	0.0849	0.0851	0.974	94.194 ± 0.767	0.338 ± 0.003	47.097 ± 1.637
UPI	beta52	0.16	0.164	0.004 ± 0.002	0.027 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.189 ± 0.009	0.435 ± 0.012	0.162	0.002	1.15	0.0698	0.0698	0.972	94.839 ± 0.725	0.265 ± 0.002	69.677 ± 1.507
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.016 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.201 ± 0.009	0.449 ± 0.013	0.161	0.001	0.63	0.0711	0.0711	0.998	94.194 ± 0.767	0.281 ± 0.002	63.871 ± 1.575
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.021 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.409 ± 0.012	0.162	0.002	1.46	0.0620	0.0621	1.024	95.269 ± 0.696	0.262 ± 0.001	71.183 ± 1.485
UPI	beta55	0.08	0.083	0.003 ± 0.004	0.038 ± 0.045	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.003	1.843 ± 0.099	1.357 ± 0.043	0.083	0.003	3.55	0.1035	0.1035	1.016	96.882 ± 0.570	0.433 ± 0.005	12.581 ± 1.087
UPI	beta56	0.08	0.078	-0.002 ± 0.003	-0.022 ± 0.042	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	1.626 ± 0.086	1.275 ± 0.040	0.073	-0.007	-8.34	0.0942	0.0944	1.020	96.452 ± 0.607	0.408 ± 0.005	11.613 ± 1.051
UPI	beta57	0.06	0.068	0.008 ± 0.004	0.142 ± 0.065	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.120 ± 0.004	3.968 ± 0.236	1.992 ± 0.068	0.058	-0.002	-2.93	0.1040	0.1041	1.110	97.957 ± 0.464	0.519 ± 0.010	3.871 ± 0.633
UPI	beta58	0.06	0.066	0.006 ± 0.003	0.102 ± 0.054	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	2.745 ± 0.146	1.657 ± 0.052	0.060	0.000	0.82	0.0935	0.0935	1.239	97.312 ± 0.530	0.482 ± 0.010	3.763 ± 0.624

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S196

Simulation 2 - Condition 52: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.187 ± 0.005	0.200	0.000	-0.03	0.0368	0.0368	1.014	95.281 ± 0.672	0.148 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.198	-0.002	-0.89	0.0359	0.0360	1.040	96.386 ± 0.591	0.152 ± 0.000	99.900 ± 0.100
LSAM	beta43	0.20	0.202	0.002 ± 0.001	0.011 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.002	0.178 ± 0.005	0.202	0.002	1.15	0.0352	0.0353	1.023	94.578 ± 0.718	0.143 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.109	-0.001 ± 0.001	-0.006 ± 0.013	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.180 ± 0.009	0.424 ± 0.012	0.110	0.000	0.39	0.0450	0.0450	0.944	94.679 ± 0.711	0.173 ± 0.001	70.080 ± 1.451
LSAM	beta45	0.11	0.109	-0.001 ± 0.001	-0.008 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.135 ± 0.006	0.367 ± 0.010	0.108	-0.002	-2.09	0.0394	0.0395	0.979	94.378 ± 0.730	0.155 ± 0.001	78.514 ± 1.301
LSAM	beta46	0.08	0.084	0.004 ± 0.002	0.045 ± 0.023	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.533 ± 0.026	0.730 ± 0.021	0.083	0.003	3.42	0.0589	0.0589	0.964	94.277 ± 0.736	0.220 ± 0.001	33.333 ± 1.494
LSAM	beta47	0.08	0.084	0.004 ± 0.002	0.046 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.466 ± 0.021	0.683 ± 0.019	0.082	0.002	1.90	0.0533	0.0534	1.017	96.586 ± 0.575	0.217 ± 0.001	30.622 ± 1.460
LSAM	beta51	0.16	0.160	0.000 ± 0.002	0.003 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.115 ± 0.005	0.339 ± 0.009	0.161	0.001	0.75	0.0551	0.0551	0.925	92.771 ± 0.821	0.197 ± 0.001	89.056 ± 0.989
LSAM	beta52	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.159	-0.001	-0.77	0.0381	0.0381	1.003	95.080 ± 0.685	0.155 ± 0.000	97.892 ± 0.455
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.013 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.161	0.001	0.32	0.0407	0.0407	1.007	94.980 ± 0.692	0.163 ± 0.000	97.992 ± 0.444
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.161	0.001	0.44	0.0399	0.0399	1.002	94.880 ± 0.698	0.155 ± 0.000	98.193 ± 0.422
LSAM	beta55	0.08	0.078	-0.002 ± 0.002	-0.023 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.535 ± 0.024	0.732 ± 0.020	0.079	-0.001	-1.54	0.0584	0.0585	0.962	93.876 ± 0.760	0.221 ± 0.001	30.321 ± 1.456
LSAM	beta56	0.08	0.082	0.002 ± 0.002	0.029 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.476 ± 0.022	0.690 ± 0.020	0.081	0.001	1.60	0.0509	0.0509	0.963	93.675 ± 0.771	0.208 ± 0.001	33.936 ± 1.500
LSAM	beta57	0.06	0.064	0.004 ± 0.002	0.066 ± 0.032	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	1.042 ± 0.048	1.021 ± 0.028	0.062	0.002	4.02	0.0639	0.0640	0.967	94.679 ± 0.711	0.232 ± 0.001	18.775 ± 1.237
LSAM	beta58	0.06	0.061	0.001 ± 0.002	0.020 ± 0.029	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.811 ± 0.038	0.900 ± 0.026	0.061	0.001	1.93	0.0532	0.0532	0.992	95.281 ± 0.672	0.210 ± 0.001	19.076 ± 1.245
QML	beta41	0.20	0.195	-0.005 ± 0.001	-0.023 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.195	-0.005	-2.57	0.0352	0.0355	1.013	94.378 ± 0.730	0.142 ± 0.000	100.000 ± 0.000
QML	beta42	0.20	0.195	-0.005 ± 0.001	-0.025 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.194	-0.006	-2.92	0.0365	0.0369	1.019	95.582 ± 0.651	0.146 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.193	-0.007 ± 0.001	-0.035 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.177 ± 0.005	0.193	-0.007	-3.55	0.0335	0.0343	1.012	94.076 ± 0.748	0.138 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.111	0.001 ± 0.001	0.013 ± 0.013	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.173 ± 0.008	0.415 ± 0.012	0.113	0.003	2.78	0.0444	0.0445	0.968	94.578 ± 0.718	0.173 ± 0.001	70.884 ± 1.440
QML	beta45	0.11	0.109	-0.001 ± 0.001	-0.011 ± 0.011	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.126 ± 0.006	0.355 ± 0.010	0.109	-0.001	-0.80	0.0388	0.0388	1.005	94.880 ± 0.698	0.154 ± 0.001	80.522 ± 1.255
QML	beta46	0.08	0.046	-0.034 ± 0.001	-0.430 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.408 ± 0.016	0.639 ± 0.014	0.045	-0.035	-43.88	0.0377	0.0515	0.992	79.618 ± 1.276	0.147 ± 0.002	20.382 ± 1.276
QML	beta47	0.08	0.049	-0.031 ± 0.001	-0.391 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.349 ± 0.013	0.591 ± 0.012	0.046	-0.034	-42.33	0.0349	0.0486	1.046	82.329 ± 1.209	0.146 ± 0.002	21.888 ± 1.310
QML	beta51	0.16	0.174	0.014 ± 0.002	0.085 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.121 ± 0.005	0.348 ± 0.009	0.172	0.012	7.50	0.0546	0.0559	0.950	92.972 ± 0.810	0.201 ± 0.001	92.570 ± 0.831
QML	beta52	0.16	0.148	-0.012 ± 0.001	-0.078 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.267 ± 0.007	0.147	-0.013	-8.39	0.0397	0.0419	0.994	92.671 ± 0.826	0.159 ± 0.001	95.683 ± 0.644
QML	beta53	0.16	0.143	-0.017 ± 0.001	-0.108 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.074 ± 0.003	0.272 ± 0.007	0.142	-0.018	-11.32	0.0408	0.0446	1.011	92.871 ± 0.815	0.158 ± 0.001	95.080 ± 0.685
QML	beta54	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.158	-0.002	-0.98	0.0390	0.0390	0.992	94.779 ± 0.705	0.153 ± 0.000	98.594 ± 0.373
QML	beta55	0.08	0.078	-0.002 ± 0.002	-0.021 ± 0.022	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.500 ± 0.022	0.707 ± 0.019	0.079	-0.001	-1.65	0.0573	0.0573	0.988	94.478 ± 0.724	0.219 ± 0.001	30.823 ± 1.463
QML	beta56	0.08	0.079	-0.001 ± 0.002	-0.018 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.438 ± 0.020	0.662 ± 0.019	0.077	-0.003	-3.36	0.0511	0.0512	0.971	94.478 ± 0.724	0.202 ± 0.001	33.534 ± 1.496
QML	beta57	0.06	0.036	-0.024 ± 0.001	-0.407 ± 0.021	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.624 ± 0.025	0.790 ± 0.018	0.033	-0.027	-44.57	0.0410	0.0489	0.986	86.647 ± 1.078	0.157 ± 0.001	10.341 ± 0.965
QML	beta58	0.06	0.039	-0.021 ± 0.001	-0.345 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.479 ± 0.019	0.692 ± 0.016	0.038	-0.022	-36.91	0.0349	0.0414	0.991	86.446 ± 1.085	0.140 ± 0.001	14.357 ± 1.111
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.201	0.001	0.72	0.0372	0.0372	0.996	94.880 ± 0.698	0.145 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.199	-0.001	-0.72	0.0365	0.0365	1.033	96.084 ± 0.615	0.149 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.203	0.003 ± 0.001	0.014 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.002	0.180 ± 0.005	0.204	0.004	1.93	0.0358	0.0360	1.028	94.478 ± 0.724	0.144 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.110	-0.000 ± 0.002	-0.002 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.199 ± 0.010	0.446 ± 0.013	0.111	0.001	1.16	0.0479	0.0479	0.967	94.880 ± 0.698	0.186 ± 0.001	66.165 ± 1.499
UPI	beta45	0.11	0.109	-0.001 ± 0.001	-0.005 ± 0.012	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.146 ± 0.007	0.382 ± 0.011	0.108	-0.002	-1.68	0.0398	0.0398	0.996	95.080 ± 0.685	0.164 ± 0.001	74.398 ± 1.383
UPI	beta46	0.08	0.087	0.007 ± 0.002	0.088 ± 0.026	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.670 ± 0.033	0.818 ± 0.024	0.083	0.003	3.44	0.0644	0.0645	1.030	95.984 ± 0.622	0.263 ± 0.002	22.590 ± 1.325

Table S196

Simulation 2 - Condition 52: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.088	0.008 ± 0.002	0.102 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.609 ± 0.034	0.780 ± 0.025	0.084	0.004	5.17	0.0562	0.0564	1.076	97.390 ± 0.505	0.261 ± 0.003	22.390 ± 1.321
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.004 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.113 ± 0.005	0.336 ± 0.009	0.160	0.000	-0.01	0.0546	0.0546	0.937	93.273 ± 0.794	0.198 ± 0.001	89.056 ± 0.989
UPI	beta52	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.159	-0.001	-0.54	0.0379	0.0379	0.998	94.578 ± 0.718	0.154 ± 0.000	97.892 ± 0.455
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.012 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.160	0.000	0.13	0.0415	0.0415	1.015	95.382 ± 0.665	0.164 ± 0.000	97.892 ± 0.455
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.161	0.001	0.45	0.0396	0.0396	1.001	94.679 ± 0.711	0.155 ± 0.000	98.494 ± 0.386
UPI	beta55	0.08	0.078	-0.002 ± 0.002	-0.030 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.546 ± 0.025	0.739 ± 0.021	0.077	-0.003	-3.85	0.0582	0.0583	0.985	95.482 ± 0.658	0.228 ± 0.001	27.510 ± 1.415
UPI	beta56	0.08	0.083	0.003 ± 0.002	0.034 ± 0.022	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.502 ± 0.025	0.709 ± 0.021	0.080	0.000	-0.60	0.0521	0.0521	0.981	94.679 ± 0.711	0.218 ± 0.001	31.124 ± 1.467
UPI	beta57	0.06	0.065	0.005 ± 0.002	0.080 ± 0.034	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.159 ± 0.058	1.077 ± 0.032	0.063	0.003	5.48	0.0639	0.0640	0.999	96.185 ± 0.607	0.252 ± 0.002	13.454 ± 1.081
UPI	beta58	0.06	0.062	0.002 ± 0.002	0.033 ± 0.031	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.970 ± 0.052	0.985 ± 0.031	0.058	-0.002	-2.93	0.0551	0.0551	1.006	96.084 ± 0.615	0.233 ± 0.002	14.357 ± 1.111

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S197

Simulation 2 - Condition 53: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.199	-0.001	-0.26	0.0486	0.0487	1.024	96.000 ± 0.620	0.190 ± 0.001	98.400 ± 0.397
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.201	0.001	0.61	0.0494	0.0495	0.942	93.300 ± 0.791	0.192 ± 0.001	96.600 ± 0.573
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.202	0.002	0.92	0.0456	0.0457	1.013	95.400 ± 0.662	0.181 ± 0.001	98.600 ± 0.372
LSAM	beta44	0.11	0.108	-0.002 ± 0.002	-0.017 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.235 ± 0.011	0.484 ± 0.014	0.111	0.001	0.94	0.0548	0.0549	0.972	94.900 ± 0.696	0.203 ± 0.001	57.500 ± 1.563
LSAM	beta45	0.11	0.113	0.003 ± 0.001	0.027 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.183 ± 0.009	0.428 ± 0.012	0.113	0.003	2.65	0.0438	0.0439	1.008	95.100 ± 0.683	0.186 ± 0.001	67.700 ± 1.479
LSAM	beta46	0.08	0.084	0.004 ± 0.002	0.049 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.548 ± 0.026	0.740 ± 0.021	0.082	0.002	2.72	0.0573	0.0574	1.016	95.600 ± 0.649	0.235 ± 0.001	26.300 ± 1.392
LSAM	beta47	0.08	0.080	-0.000 ± 0.002	-0.003 ± 0.024	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.593 ± 0.026	0.770 ± 0.021	0.078	-0.002	-2.65	0.0621	0.0622	0.968	94.500 ± 0.721	0.234 ± 0.001	25.200 ± 1.373
LSAM	beta51	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.007	0.408 ± 0.011	0.157	-0.003	-1.84	0.0680	0.0680	0.958	94.600 ± 0.715	0.245 ± 0.001	71.200 ± 1.432
LSAM	beta52	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.327 ± 0.009	0.158	-0.002	-0.95	0.0532	0.0532	0.991	94.900 ± 0.696	0.203 ± 0.001	87.300 ± 1.053
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.005	0.344 ± 0.010	0.161	0.001	0.42	0.0535	0.0535	0.968	93.300 ± 0.791	0.209 ± 0.001	85.900 ± 1.101
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.015 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.322 ± 0.009	0.162	0.002	1.33	0.0488	0.0489	1.005	94.700 ± 0.708	0.203 ± 0.001	87.400 ± 1.049
LSAM	beta55	0.08	0.079	-0.001 ± 0.002	-0.010 ± 0.029	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.829 ± 0.042	0.911 ± 0.027	0.081	0.001	1.74	0.0695	0.0695	0.900	93.600 ± 0.774	0.257 ± 0.002	27.800 ± 1.417
LSAM	beta56	0.08	0.083	0.003 ± 0.002	0.033 ± 0.027	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.754 ± 0.036	0.868 ± 0.025	0.082	0.002	2.28	0.0673	0.0674	0.903	93.300 ± 0.791	0.246 ± 0.002	28.900 ± 1.433
LSAM	beta57	0.06	0.064	0.004 ± 0.002	0.064 ± 0.035	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	1.256 ± 0.060	1.121 ± 0.032	0.061	0.001	1.56	0.0625	0.0625	0.971	93.300 ± 0.791	0.256 ± 0.001	15.200 ± 1.135
LSAM	beta58	0.06	0.057	-0.003 ± 0.002	-0.049 ± 0.032	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	1.034 ± 0.047	1.017 ± 0.028	0.057	-0.003	-5.00	0.0600	0.0600	0.979	94.500 ± 0.721	0.234 ± 0.001	15.000 ± 1.129
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.197	-0.003	-1.36	0.0467	0.0468	1.021	96.000 ± 0.620	0.188 ± 0.001	98.800 ± 0.344
QML	beta42	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.199	-0.001	-0.39	0.0494	0.0494	0.937	93.200 ± 0.796	0.190 ± 0.001	97.100 ± 0.531
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.019 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.197	-0.003	-1.40	0.0443	0.0444	1.013	95.100 ± 0.683	0.179 ± 0.001	98.600 ± 0.372
QML	beta44	0.11	0.109	-0.001 ± 0.002	-0.011 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.243 ± 0.011	0.493 ± 0.014	0.112	0.002	1.78	0.0549	0.0550	0.979	95.300 ± 0.669	0.208 ± 0.001	57.000 ± 1.566
QML	beta45	0.11	0.113	0.003 ± 0.001	0.025 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.186 ± 0.009	0.431 ± 0.012	0.112	0.002	2.06	0.0433	0.0433	1.015	95.300 ± 0.669	0.189 ± 0.001	66.400 ± 1.494
QML	beta46	0.08	0.065	-0.015 ± 0.002	-0.184 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.414 ± 0.019	0.644 ± 0.017	0.063	-0.017	-21.70	0.0464	0.0496	1.021	92.300 ± 0.843	0.198 ± 0.001	20.200 ± 1.270
QML	beta47	0.08	0.064	-0.016 ± 0.002	-0.204 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.449 ± 0.018	0.670 ± 0.017	0.061	-0.019	-24.33	0.0521	0.0556	0.971	91.300 ± 0.891	0.194 ± 0.001	21.600 ± 1.301
QML	beta51	0.16	0.164	0.004 ± 0.002	0.023 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.007	0.408 ± 0.011	0.163	0.003	1.88	0.0691	0.0692	0.991	95.600 ± 0.649	0.254 ± 0.001	71.600 ± 1.426
QML	beta52	0.16	0.145	-0.015 ± 0.002	-0.096 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.120 ± 0.005	0.347 ± 0.009	0.143	-0.017	-10.65	0.0552	0.0577	1.013	94.500 ± 0.721	0.212 ± 0.001	76.700 ± 1.337
QML	beta53	0.16	0.143	-0.017 ± 0.002	-0.106 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.128 ± 0.006	0.357 ± 0.009	0.142	-0.018	-11.01	0.0541	0.0569	0.989	93.900 ± 0.757	0.212 ± 0.001	75.100 ± 1.367
QML	beta54	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.161	0.001	0.90	0.0499	0.0499	1.003	94.900 ± 0.696	0.203 ± 0.001	87.600 ± 1.042
QML	beta55	0.08	0.080	-0.000 ± 0.002	-0.003 ± 0.029	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.812 ± 0.041	0.901 ± 0.027	0.080	0.000	0.59	0.0687	0.0687	0.927	94.300 ± 0.733	0.262 ± 0.002	26.200 ± 1.391
QML	beta56	0.08	0.082	0.002 ± 0.002	0.020 ± 0.027	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.724 ± 0.035	0.851 ± 0.024	0.082	0.002	2.75	0.0665	0.0665	0.926	93.900 ± 0.757	0.247 ± 0.002	26.400 ± 1.394
QML	beta57	0.06	0.049	-0.011 ± 0.002	-0.179 ± 0.029	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.893 ± 0.040	0.945 ± 0.025	0.045	-0.015	-25.31	0.0523	0.0545	0.982	92.300 ± 0.843	0.214 ± 0.001	11.100 ± 0.993
QML	beta58	0.06	0.047	-0.013 ± 0.002	-0.214 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.754 ± 0.033	0.868 ± 0.023	0.046	-0.014	-24.13	0.0509	0.0530	0.982	92.200 ± 0.848	0.195 ± 0.001	12.900 ± 1.060
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.199	-0.001	-0.32	0.0463	0.0463	0.981	95.300 ± 0.669	0.180 ± 0.001	98.800 ± 0.344
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.003	0.259 ± 0.008	0.202	0.002	0.83	0.0494	0.0494	0.914	92.400 ± 0.838	0.185 ± 0.001	97.300 ± 0.513
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.202	0.002	0.99	0.0456	0.0457	1.005	95.200 ± 0.676	0.181 ± 0.001	98.700 ± 0.358
UPI	beta44	0.11	0.109	-0.001 ± 0.002	-0.013 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.249 ± 0.012	0.499 ± 0.014	0.112	0.002	1.50	0.0547	0.0548	0.977	95.600 ± 0.649	0.210 ± 0.001	56.500 ± 1.568
UPI	beta45	0.11	0.114	0.004 ± 0.002	0.035 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.191 ± 0.009	0.436 ± 0.012	0.113	0.003	2.90	0.0461	0.0462	1.021	95.400 ± 0.662	0.192 ± 0.001	65.800 ± 1.500
UPI	beta46	0.08	0.085	0.005 ± 0.002	0.061 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.612 ± 0.030	0.782 ± 0.023	0.082	0.002	3.01	0.0572	0.0573	1.051	95.600 ± 0.649	0.257 ± 0.002	21.800 ± 1.306

Table S197

Simulation 2 - Condition 53: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.002	0.016 ± 0.026	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.698 ± 0.033	0.835 ± 0.024	0.078	-0.002	-2.59	0.0685	0.0685	0.982	94.900 ± 0.696	0.257 ± 0.002	20.600 ± 1.279
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.402 ± 0.011	0.157	-0.003	-2.02	0.0690	0.0690	0.979	95.400 ± 0.662	0.247 ± 0.001	71.000 ± 1.435
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.158	-0.002	-1.14	0.0525	0.0525	0.977	94.500 ± 0.721	0.200 ± 0.001	87.800 ± 1.035
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.118 ± 0.005	0.344 ± 0.010	0.161	0.001	0.35	0.0539	0.0539	0.970	93.500 ± 0.780	0.209 ± 0.001	85.500 ± 1.113
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.015 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.162	0.002	1.22	0.0489	0.0489	0.999	94.700 ± 0.708	0.201 ± 0.001	88.500 ± 1.009
UPI	beta55	0.08	0.080	-0.000 ± 0.002	-0.004 ± 0.029	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.853 ± 0.045	0.923 ± 0.028	0.081	0.001	1.75	0.0700	0.0700	0.909	94.400 ± 0.727	0.263 ± 0.002	26.000 ± 1.387
UPI	beta56	0.08	0.082	0.002 ± 0.002	0.030 ± 0.028	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.771 ± 0.037	0.878 ± 0.025	0.083	0.003	3.45	0.0671	0.0671	0.919	94.600 ± 0.715	0.253 ± 0.002	27.000 ± 1.404
UPI	beta57	0.06	0.064	0.004 ± 0.002	0.060 ± 0.036	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	1.307 ± 0.064	1.143 ± 0.033	0.060	0.000	-0.64	0.0627	0.0627	0.988	93.800 ± 0.763	0.265 ± 0.002	14.000 ± 1.097
UPI	beta58	0.06	0.057	-0.003 ± 0.002	-0.043 ± 0.033	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	1.112 ± 0.052	1.055 ± 0.030	0.058	-0.002	-3.80	0.0627	0.0627	0.991	94.600 ± 0.715	0.246 ± 0.001	12.800 ± 1.056

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S198

Simulation 2 - Condition 54: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.200	0.000	-0.22	0.0292	0.0292	1.003	94.995 ± 0.690	0.119 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.200	0.000	-0.06	0.0300	0.0300	1.033	95.295 ± 0.670	0.120 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.199	-0.001	-0.38	0.0292	0.0292	0.994	95.095 ± 0.683	0.113 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.110	0.000 ± 0.001	0.004 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.111	0.001	0.68	0.0320	0.0320	0.994	93.994 ± 0.752	0.127 ± 0.000	90.490 ± 0.928
LSAM	beta45	0.11	0.110	0.000 ± 0.001	0.002 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.079 ± 0.004	0.281 ± 0.008	0.111	0.001	1.21	0.0302	0.0302	0.961	93.493 ± 0.780	0.116 ± 0.000	94.494 ± 0.722
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.013 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.216 ± 0.010	0.465 ± 0.013	0.082	0.002	2.49	0.0360	0.0361	1.000	94.795 ± 0.703	0.146 ± 0.000	59.560 ± 1.553
LSAM	beta47	0.08	0.077	-0.003 ± 0.001	-0.035 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.221 ± 0.009	0.470 ± 0.012	0.075	-0.005	-5.93	0.0382	0.0385	0.988	94.695 ± 0.709	0.145 ± 0.000	53.053 ± 1.579
LSAM	beta51	0.16	0.162	0.002 ± 0.001	0.014 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.162	0.002	1.01	0.0407	0.0407	0.956	93.994 ± 0.752	0.155 ± 0.001	98.599 ± 0.372
LSAM	beta52	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.195 ± 0.005	0.160	0.000	0.15	0.0317	0.0317	1.034	95.295 ± 0.670	0.126 ± 0.000	99.900 ± 0.100
LSAM	beta53	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.161	0.001	0.84	0.0324	0.0324	1.010	96.096 ± 0.613	0.130 ± 0.000	99.600 ± 0.200
LSAM	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.161	0.001	0.49	0.0323	0.0323	1.006	94.895 ± 0.696	0.126 ± 0.000	99.800 ± 0.141
LSAM	beta55	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.017	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.296 ± 0.013	0.544 ± 0.015	0.080	0.000	-0.09	0.0440	0.0440	0.937	94.494 ± 0.722	0.160 ± 0.001	49.950 ± 1.582
LSAM	beta56	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.254 ± 0.011	0.504 ± 0.014	0.078	-0.002	-2.07	0.0416	0.0416	0.969	95.095 ± 0.683	0.153 ± 0.001	52.753 ± 1.580
LSAM	beta57	0.06	0.060	0.000 ± 0.001	0.004 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.451 ± 0.021	0.671 ± 0.019	0.060	0.000	-0.59	0.0398	0.0398	0.991	94.995 ± 0.690	0.157 ± 0.001	31.632 ± 1.471
LSAM	beta58	0.06	0.060	-0.000 ± 0.001	-0.006 ± 0.019	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.375 ± 0.016	0.612 ± 0.016	0.061	0.001	1.62	0.0375	0.0375	1.017	95.195 ± 0.677	0.147 ± 0.000	35.736 ± 1.516
QML	beta41	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.198	-0.002	-1.22	0.0293	0.0294	1.013	94.494 ± 0.722	0.119 ± 0.001	99.800 ± 0.141
QML	beta42	0.20	0.197	-0.003 ± 0.001	-0.015 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.198	-0.002	-1.07	0.0297	0.0298	1.034	95.796 ± 0.635	0.119 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.022 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.195	-0.005	-2.55	0.0291	0.0295	0.996	94.595 ± 0.715	0.112 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.111	0.001 ± 0.001	0.012 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.090 ± 0.004	0.301 ± 0.008	0.112	0.002	1.57	0.0325	0.0325	0.999	94.494 ± 0.722	0.130 ± 0.001	89.089 ± 0.986
QML	beta45	0.11	0.111	0.001 ± 0.001	0.006 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.080 ± 0.004	0.283 ± 0.008	0.111	0.001	1.17	0.0307	0.0307	0.969	94.294 ± 0.734	0.118 ± 0.000	94.494 ± 0.722
QML	beta46	0.08	0.063	-0.017 ± 0.001	-0.218 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.201 ± 0.009	0.448 ± 0.011	0.062	-0.018	-22.40	0.0310	0.0358	1.001	89.189 ± 0.982	0.123 ± 0.000	51.151 ± 1.582
QML	beta47	0.08	0.060	-0.020 ± 0.001	-0.248 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.216 ± 0.008	0.464 ± 0.010	0.058	-0.022	-27.54	0.0309	0.0380	0.980	86.787 ± 1.071	0.121 ± 0.000	47.147 ± 1.579
QML	beta51	0.16	0.168	0.008 ± 0.001	0.052 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.167	0.007	4.27	0.0403	0.0409	0.976	94.895 ± 0.696	0.158 ± 0.001	98.899 ± 0.330
QML	beta52	0.16	0.143	-0.017 ± 0.001	-0.103 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.145	-0.015	-9.56	0.0310	0.0345	1.044	92.192 ± 0.849	0.130 ± 0.000	99.199 ± 0.282
QML	beta53	0.16	0.142	-0.018 ± 0.001	-0.113 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.144	-0.016	-10.25	0.0321	0.0360	1.013	89.990 ± 0.950	0.131 ± 0.000	98.699 ± 0.359
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.006	0.160	0.000	0.17	0.0320	0.0320	1.013	94.895 ± 0.696	0.127 ± 0.001	99.600 ± 0.200
QML	beta55	0.08	0.080	0.000 ± 0.001	0.003 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.293 ± 0.013	0.541 ± 0.015	0.080	0.000	0.30	0.0443	0.0443	0.954	94.094 ± 0.746	0.162 ± 0.001	50.551 ± 1.582
QML	beta56	0.08	0.078	-0.002 ± 0.001	-0.019 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.250 ± 0.011	0.500 ± 0.014	0.077	-0.003	-3.59	0.0407	0.0408	0.976	95.095 ± 0.683	0.153 ± 0.001	52.553 ± 1.580
QML	beta57	0.06	0.046	-0.014 ± 0.001	-0.237 ± 0.018	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.367 ± 0.016	0.606 ± 0.015	0.045	-0.015	-24.44	0.0322	0.0353	0.998	90.190 ± 0.941	0.131 ± 0.001	24.324 ± 1.357
QML	beta58	0.06	0.048	-0.012 ± 0.001	-0.196 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.299 ± 0.013	0.547 ± 0.014	0.048	-0.012	-19.81	0.0314	0.0336	1.016	92.192 ± 0.849	0.122 ± 0.000	31.231 ± 1.466
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.200	0.000	-0.06	0.0289	0.0289	0.959	93.994 ± 0.752	0.113 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.200	0.000	-0.06	0.0293	0.0293	1.005	95.095 ± 0.683	0.116 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.199	-0.001	-0.47	0.0288	0.0288	0.991	95.195 ± 0.677	0.113 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.111	0.001 ± 0.001	0.005 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.110	0.000	0.22	0.0323	0.0323	1.006	93.894 ± 0.758	0.129 ± 0.000	90.490 ± 0.928
UPI	beta45	0.11	0.110	0.000 ± 0.001	0.004 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.080 ± 0.004	0.282 ± 0.008	0.112	0.002	1.48	0.0299	0.0300	0.969	93.894 ± 0.758	0.118 ± 0.000	94.394 ± 0.728
UPI	beta46	0.08	0.082	0.002 ± 0.001	0.023 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.229 ± 0.011	0.478 ± 0.014	0.081	0.001	0.96	0.0384	0.0384	1.014	94.995 ± 0.690	0.152 ± 0.001	57.257 ± 1.565

Table S198

Simulation 2 - Condition 54: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.078	-0.002 ± 0.001	-0.028 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.230 ± 0.010	0.480 ± 0.013	0.076	-0.004	-4.76	0.0386	0.0388	1.008	95.495 ± 0.656	0.151 ± 0.001	51.251 ± 1.581
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.014 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.161	0.001	0.75	0.0393	0.0393	0.962	94.394 ± 0.728	0.155 ± 0.001	98.599 ± 0.372
UPI	beta52	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.195 ± 0.005	0.160	0.000	-0.20	0.0319	0.0319	1.021	95.095 ± 0.683	0.124 ± 0.000	99.900 ± 0.100
UPI	beta53	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.161	0.001	0.87	0.0324	0.0324	1.007	96.196 ± 0.605	0.130 ± 0.000	99.600 ± 0.200
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.161	0.001	0.63	0.0324	0.0324	0.999	94.595 ± 0.715	0.125 ± 0.000	99.800 ± 0.141
UPI	beta55	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.017	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.297 ± 0.013	0.545 ± 0.015	0.080	0.000	-0.10	0.0440	0.0440	0.945	93.794 ± 0.763	0.162 ± 0.001	51.051 ± 1.582
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.256 ± 0.011	0.506 ± 0.014	0.077	-0.003	-3.17	0.0407	0.0408	0.979	95.195 ± 0.677	0.155 ± 0.001	51.652 ± 1.581
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.005 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.473 ± 0.023	0.688 ± 0.020	0.060	0.000	-0.22	0.0403	0.0403	0.985	95.596 ± 0.649	0.159 ± 0.001	31.331 ± 1.468
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.005 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.383 ± 0.016	0.619 ± 0.016	0.062	0.002	2.68	0.0381	0.0381	1.029	95.495 ± 0.656	0.150 ± 0.001	34.034 ± 1.499

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S199

Simulation 2 - Condition 55: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.194	-0.006 ± 0.012	-0.031 ± 0.062	0.015 ± 0.004	0.124 ± 0.015	0.015 ± 0.004	0.124 ± 0.016	0.381 ± 0.089	0.618 ± 0.082	0.206	0.006	2.84	0.1069	0.1071	0.970	92.929 ± 2.576	0.471 ± 0.019	38.384 ± 4.888
LSAM	beta42	0.20	0.215	0.015 ± 0.012	0.077 ± 0.062	0.015 ± 0.002	0.122 ± 0.010	0.015 ± 0.003	0.123 ± 0.012	0.377 ± 0.063	0.614 ± 0.062	0.196	-0.004	-1.99	0.1287	0.1287	1.036	96.970 ± 1.723	0.497 ± 0.018	42.424 ± 4.967
LSAM	beta43	0.20	0.189	-0.011 ± 0.009	-0.054 ± 0.046	0.008 ± 0.001	0.092 ± 0.006	0.008 ± 0.001	0.092 ± 0.007	0.212 ± 0.026	0.460 ± 0.037	0.181	-0.019	-9.34	0.0956	0.0974	0.975	96.970 ± 1.723	0.351 ± 0.006	52.525 ± 5.019
LSAM	beta44	0.11	0.174	0.064 ± 0.014	0.582 ± 0.128	0.020 ± 0.004	0.140 ± 0.013	0.023 ± 0.004	0.153 ± 0.016	1.940 ± 0.360	1.393 ± 0.144	0.160	0.050	45.75	0.1164	0.1268	1.063	96.970 ± 1.723	0.583 ± 0.026	17.172 ± 3.790
LSAM	beta45	0.11	0.129	0.019 ± 0.012	0.169 ± 0.112	0.015 ± 0.002	0.122 ± 0.010	0.015 ± 0.002	0.123 ± 0.012	1.248 ± 0.204	1.117 ± 0.109	0.113	0.003	2.99	0.0869	0.0870	0.966	94.949 ± 2.201	0.462 ± 0.026	19.192 ± 3.958
LSAM	beta46	0.08	0.036	-0.044 ± 0.010	-0.547 ± 0.121	0.009 ± 0.002	0.096 ± 0.010	0.011 ± 0.002	0.105 ± 0.010	1.727 ± 0.300	1.314 ± 0.130	0.032	-0.048	-59.92	0.0888	0.1009	1.030	92.929 ± 2.576	0.388 ± 0.025	2.020 ± 1.414
LSAM	beta47	0.08	0.023	-0.057 ± 0.009	-0.707 ± 0.114	0.008 ± 0.001	0.091 ± 0.006	0.011 ± 0.001	0.107 ± 0.008	1.773 ± 0.219	1.332 ± 0.097	0.027	-0.053	-65.63	0.0857	0.1005	1.023	86.869 ± 3.394	0.364 ± 0.018	3.030 ± 1.723
LSAM	beta51	0.16	0.204	0.044 ± 0.012	0.275 ± 0.076	0.015 ± 0.002	0.121 ± 0.008	0.016 ± 0.002	0.128 ± 0.010	0.642 ± 0.086	0.801 ± 0.066	0.194	0.034	21.11	0.1269	0.1313	1.035	97.980 ± 1.414	0.491 ± 0.013	40.404 ± 4.932
LSAM	beta52	0.16	0.134	-0.026 ± 0.010	-0.165 ± 0.065	0.011 ± 0.002	0.103 ± 0.008	0.011 ± 0.001	0.106 ± 0.009	0.435 ± 0.058	0.660 ± 0.055	0.133	-0.027	-17.18	0.1139	0.1172	0.966	92.929 ± 2.576	0.389 ± 0.010	27.273 ± 4.476
LSAM	beta53	0.16	0.157	-0.003 ± 0.010	-0.017 ± 0.064	0.011 ± 0.001	0.103 ± 0.008	0.010 ± 0.001	0.102 ± 0.009	0.407 ± 0.058	0.638 ± 0.057	0.164	0.004	2.49	0.0950	0.0951	1.045	94.949 ± 2.201	0.420 ± 0.013	36.364 ± 4.835
LSAM	beta54	0.16	0.145	-0.015 ± 0.010	-0.095 ± 0.065	0.011 ± 0.002	0.104 ± 0.009	0.011 ± 0.002	0.104 ± 0.010	0.424 ± 0.070	0.651 ± 0.064	0.144	-0.016	-10.26	0.0890	0.0905	1.030	95.960 ± 1.979	0.418 ± 0.012	23.232 ± 4.244
LSAM	beta55	0.08	0.085	0.005 ± 0.015	0.062 ± 0.190	0.023 ± 0.004	0.151 ± 0.014	0.023 ± 0.004	0.150 ± 0.016	3.538 ± 0.653	1.881 ± 0.203	0.084	0.004	5.23	0.1417	0.1418	0.990	95.960 ± 1.979	0.587 ± 0.031	9.091 ± 2.889
LSAM	beta56	0.08	0.098	0.018 ± 0.014	0.219 ± 0.171	0.019 ± 0.003	0.136 ± 0.010	0.019 ± 0.003	0.137 ± 0.013	2.919 ± 0.453	1.709 ± 0.160	0.084	0.004	5.16	0.1322	0.1323	1.090	98.990 ± 1.005	0.582 ± 0.030	8.081 ± 2.739
LSAM	beta57	0.06	0.040	-0.020 ± 0.007	-0.333 ± 0.120	0.005 ± 0.001	0.071 ± 0.006	0.005 ± 0.001	0.074 ± 0.007	1.512 ± 0.230	1.230 ± 0.112	0.033	-0.027	-44.82	0.0771	0.0817	1.118	92.929 ± 2.576	0.313 ± 0.012	2.020 ± 1.414
LSAM	beta58	0.06	0.024	-0.036 ± 0.006	-0.601 ± 0.107	0.004 ± 0.001	0.064 ± 0.006	0.005 ± 0.001	0.073 ± 0.008	1.492 ± 0.278	1.222 ± 0.126	0.030	-0.030	-49.89	0.0554	0.0630	1.042	88.889 ± 3.159	0.262 ± 0.010	2.020 ± 1.414
QML	beta41	0.20	0.173	-0.027 ± 0.008	-0.134 ± 0.041	0.007 ± 0.001	0.081 ± 0.005	0.007 ± 0.001	0.085 ± 0.007	0.182 ± 0.025	0.427 ± 0.035	0.185	-0.015	-7.63	0.0850	0.0863	1.029	92.929 ± 2.576	0.328 ± 0.007	59.596 ± 4.932
QML	beta42	0.20	0.185	-0.015 ± 0.009	-0.074 ± 0.043	0.007 ± 0.001	0.085 ± 0.006	0.007 ± 0.001	0.086 ± 0.007	0.183 ± 0.025	0.428 ± 0.037	0.180	-0.020	-9.80	0.0875	0.0897	1.018	95.960 ± 1.979	0.338 ± 0.006	59.596 ± 4.932
QML	beta43	0.20	0.176	-0.024 ± 0.008	-0.122 ± 0.040	0.006 ± 0.001	0.080 ± 0.006	0.007 ± 0.001	0.083 ± 0.007	0.173 ± 0.022	0.416 ± 0.034	0.173	-0.027	-13.57	0.0807	0.0852	0.988	92.929 ± 2.576	0.310 ± 0.006	62.626 ± 4.862
QML	beta44	0.11	0.122	0.012 ± 0.011	0.111 ± 0.102	0.013 ± 0.002	0.112 ± 0.009	0.013 ± 0.002	0.112 ± 0.011	1.040 ± 0.163	1.020 ± 0.096	0.124	0.014	12.63	0.1129	0.1138	1.104	94.949 ± 2.201	0.485 ± 0.020	22.222 ± 4.178
QML	beta45	0.11	0.121	0.011 ± 0.010	0.097 ± 0.089	0.010 ± 0.001	0.098 ± 0.007	0.010 ± 0.001	0.098 ± 0.008	0.792 ± 0.109	0.890 ± 0.077	0.118	0.008	7.28	0.0974	0.0977	1.015	97.980 ± 1.414	0.389 ± 0.013	23.232 ± 4.244
QML	beta46	0.08	0.063	-0.017 ± 0.006	-0.213 ± 0.079	0.004 ± 0.000	0.063 ± 0.004	0.004 ± 0.001	0.065 ± 0.005	0.659 ± 0.082	0.812 ± 0.064	0.059	-0.021	-26.72	0.0749	0.0779	1.269	90.909 ± 2.889	0.313 ± 0.016	6.061 ± 2.398
QML	beta47	0.08	0.049	-0.031 ± 0.008	-0.388 ± 0.098	0.006 ± 0.001	0.078 ± 0.006	0.007 ± 0.001	0.084 ± 0.007	1.090 ± 0.153	1.044 ± 0.087	0.046	-0.034	-42.85	0.0692	0.0772	1.026	83.838 ± 3.700	0.313 ± 0.015	5.051 ± 2.201
QML	beta51	0.16	0.204	0.044 ± 0.013	0.274 ± 0.083	0.017 ± 0.003	0.131 ± 0.010	0.019 ± 0.003	0.138 ± 0.013	0.743 ± 0.116	0.862 ± 0.079	0.190	0.030	18.52	0.1380	0.1412	1.091	96.970 ± 1.723	0.562 ± 0.022	33.333 ± 4.738
QML	beta52	0.16	0.141	-0.019 ± 0.010	-0.116 ± 0.065	0.011 ± 0.002	0.104 ± 0.008	0.011 ± 0.001	0.105 ± 0.009	0.429 ± 0.057	0.655 ± 0.055	0.135	-0.025	-15.33	0.1050	0.1078	1.008	93.939 ± 2.398	0.410 ± 0.016	35.354 ± 4.805
QML	beta53	0.16	0.140	-0.020 ± 0.009	-0.123 ± 0.057	0.008 ± 0.001	0.091 ± 0.007	0.009 ± 0.001	0.093 ± 0.009	0.337 ± 0.053	0.580 ± 0.055	0.162	0.002	0.95	0.0935	0.0935	1.098	95.960 ± 1.979	0.393 ± 0.010	32.323 ± 4.701
QML	beta54	0.16	0.152	-0.008 ± 0.007	-0.047 ± 0.046	0.005 ± 0.001	0.074 ± 0.005	0.005 ± 0.001	0.074 ± 0.006	0.212 ± 0.029	0.460 ± 0.040	0.145	-0.015	-9.48	0.0819	0.0833	1.196	96.970 ± 1.723	0.345 ± 0.008	42.424 ± 4.967
QML	beta55	0.08	0.077	-0.003 ± 0.013	-0.043 ± 0.164	0.017 ± 0.002	0.131 ± 0.009	0.017 ± 0.002	0.130 ± 0.011	2.648 ± 0.333	1.627 ± 0.133	0.085	0.005	6.60	0.1640	0.1641	1.222	96.970 ± 1.723	0.627 ± 0.037	10.101 ± 3.029
QML	beta56	0.08	0.083	0.003 ± 0.011	0.036 ± 0.143	0.013 ± 0.002	0.114 ± 0.009	0.013 ± 0.002	0.113 ± 0.011	2.004 ± 0.319	1.416 ± 0.136	0.067	-0.013	-15.80	0.1210	0.1216	1.152	93.939 ± 2.398	0.514 ± 0.019	10.101 ± 3.029
QML	beta57	0.06	0.044	-0.016 ± 0.008	-0.274 ± 0.127	0.006 ± 0.001	0.076 ± 0.008	0.006 ± 0.001	0.077 ± 0.008	1.660 ± 0.293	1.289 ± 0.133	0.023	-0.037	-62.33	0.0615	0.0719	1.204	95.960 ± 1.979	0.358 ± 0.037	2.020 ± 1.414
QML	beta58	0.06	0.030	-0.030 ± 0.006	-0.495 ± 0.103	0.004 ± 0.001	0.061 ± 0.005	0.005 ± 0.001	0.068 ± 0.006	1.281 ± 0.182	1.132 ± 0.094	0.036	-0.024	-39.94	0.0646	0.0689	1.112	87.879 ± 3.280	0.268 ± 0.019	2.020 ± 1.414
UPI	beta41	0.20	0.211	0.011 ± 0.019	0.054 ± 0.096	0.036 ± 0.009	0.191 ± 0.026	0.036 ± 0.009	0.190 ± 0.027	0.904 ± 0.230	0.951 ± 0.135	0.221	0.021	10.62	0.1938	0.1950	1.788	92.929 ± 2.576	1.337 ± 0.369	37.374 ± 4.862
UPI	beta42	0.20	0.251	0.051 ± 0.021	0.254 ± 0.106	0.045 ± 0.010	0.211 ± 0.025	0.047 ± 0.010	0.216 ± 0.026	1.169 ± 0.251	1.081 ± 0.131	0.233	0.033	16.51	0.1979	0.2006	1.540	96.970 ± 1.723	1.275 ± 0.201	28.283 ± 4.526
UPI	beta43	0.20	0.175	-0.025 ± 0.013	-0.126 ± 0.065	0.017 ± 0.002	0.129 ± 0.009	0.017 ± 0.002	0.130 ± 0.011	0.425 ± 0.055	0.652 ± 0.054	0.163	-0.037	-18.53	0.1572	0.1615	1.158	85.859 ± 3.502	0.584 ± 0.046	26.263 ± 4.423
UPI	beta44	0.11	0.165	0.055 ± 0.021	0.503 ± 0.195	0.046 ± 0.011	0.214 ± 0.026	0.048 ± 0.011	0.220 ± 0.028	3.990 ± 0.919	1.998 ± 0.257	0.159	0.049	44.70	0.2059	0.2117	1.533	100.000 ± 0.000	1.285 ± 0.156	7.071 ± 2.576
UPI	beta45	0.11	0.138	0.028 ± 0.016	0.253 ± 0.145	0.025 ± 0.005	0.158 ± 0.018	0.026 ± 0.006	0.160 ± 0.020	2.114 ± 0.470	1.454 ± 0.182	0.117	0.007	6.35	0.1413	0.1414	1.434	97.980 ± 1.414	0.890 ± 0.135	5.051 ± 2.201
UPI	beta46	0.08	0.034	-0.046 ± 0.019	-0.580 ± 0.242	0.037 ± 0.010	0.193 ± 0.029	0.039 ± 0.010	0.197 ± 0.029	6.069 ± 1.637	2.464 ± 0.368	0.025	-0.055	-68.20	0.2008	0.2081	1.544	88.889 ± 3.159	1.165 ± 0.219	3.030 ± 1.723

Table S199

Simulation 2 - Condition 55: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.027	-0.107 ± 0.024	-1.340 ± 0.306	0.059 ± 0.010	0.243 ± 0.021	0.070 ± 0.012	0.265 ± 0.025	10.962 ± 1.808	3.311 ± 0.312	-0.013	-0.093	-116.59	0.2481	0.2651	1.348	90.909 ± 2.889	1.287 ± 0.145	2.020 ± 1.414
UPI	beta51	0.16	0.250	0.090 ± 0.012	0.560 ± 0.073	0.014 ± 0.002	0.116 ± 0.009	0.021 ± 0.003	0.146 ± 0.012	0.836 ± 0.125	0.914 ± 0.075	0.244	0.084	52.80	0.1272	0.1527	1.322	92.929 ± 2.576	0.602 ± 0.074	50.505 ± 5.025
UPI	beta52	0.16	0.148	-0.012 ± 0.012	-0.072 ± 0.077	0.015 ± 0.003	0.123 ± 0.013	0.015 ± 0.003	0.123 ± 0.015	0.586 ± 0.124	0.766 ± 0.093	0.153	-0.007	-4.63	0.1347	0.1349	1.291	92.929 ± 2.576	0.620 ± 0.094	25.253 ± 4.366
UPI	beta53	0.16	0.147	-0.013 ± 0.010	-0.083 ± 0.065	0.011 ± 0.002	0.104 ± 0.008	0.011 ± 0.002	0.104 ± 0.009	0.422 ± 0.059	0.650 ± 0.057	0.123	-0.037	-23.19	0.1153	0.1211	1.409	95.960 ± 1.979	0.572 ± 0.068	21.212 ± 4.109
UPI	beta54	0.16	0.142	-0.018 ± 0.012	-0.113 ± 0.073	0.013 ± 0.003	0.116 ± 0.016	0.014 ± 0.004	0.117 ± 0.017	0.532 ± 0.138	0.730 ± 0.107	0.133	-0.027	-16.79	0.1244	0.1273	1.565	95.960 ± 1.979	0.711 ± 0.087	19.192 ± 3.958
UPI	beta55	0.08	0.112	0.032 ± 0.016	0.401 ± 0.204	0.026 ± 0.005	0.162 ± 0.016	0.027 ± 0.005	0.165 ± 0.019	4.242 ± 0.847	2.060 ± 0.234	0.087	0.007	8.26	0.1588	0.1589	1.246	98.990 ± 1.005	0.794 ± 0.057	6.061 ± 2.398
UPI	beta56	0.08	0.079	-0.001 ± 0.021	-0.011 ± 0.260	0.043 ± 0.010	0.207 ± 0.025	0.042 ± 0.010	0.206 ± 0.027	6.638 ± 1.535	2.576 ± 0.335	0.080	0.000	0.27	0.1627	0.1627	1.106	96.970 ± 1.723	0.898 ± 0.101	3.030 ± 1.723
UPI	beta57	0.06	0.026	-0.034 ± 0.011	-0.559 ± 0.177	0.011 ± 0.002	0.106 ± 0.011	0.012 ± 0.002	0.110 ± 0.013	3.382 ± 0.690	1.839 ± 0.213	0.009	-0.051	-84.47	0.1001	0.1122	1.433	91.919 ± 2.739	0.594 ± 0.085	1.010 ± 1.005
UPI	beta58	0.06	-0.009	-0.069 ± 0.013	-1.152 ± 0.210	0.016 ± 0.003	0.126 ± 0.013	0.020 ± 0.004	0.143 ± 0.016	5.663 ± 1.117	2.380 ± 0.259	-0.002	-0.062	-103.76	0.1140	0.1299	1.643	83.838 ± 3.700	0.809 ± 0.156	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S200

Simulation 2 - Condition 56: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.208	0.008 ± 0.006	0.039 ± 0.029	0.004 ± 0.001	0.067 ± 0.005	0.005 ± 0.001	0.067 ± 0.006	0.113 ± 0.016	0.336 ± 0.028	0.213	0.013	6.53	0.0701	0.0713	1.022	91.538 ± 2.441	0.268 ± 0.004	90.769 ± 2.539
LSAM	beta42	0.20	0.201	0.001 ± 0.006	0.007 ± 0.031	0.005 ± 0.001	0.071 ± 0.005	0.005 ± 0.001	0.071 ± 0.006	0.125 ± 0.017	0.353 ± 0.029	0.196	-0.004	-1.98	0.0700	0.0701	1.003	94.615 ± 1.980	0.279 ± 0.004	80.000 ± 3.508
LSAM	beta43	0.20	0.190	-0.010 ± 0.005	-0.048 ± 0.025	0.003 ± 0.000	0.057 ± 0.004	0.003 ± 0.000	0.058 ± 0.004	0.083 ± 0.010	0.289 ± 0.022	0.193	-0.007	-3.61	0.0560	0.0565	0.933	94.615 ± 1.980	0.209 ± 0.002	95.385 ± 1.840
LSAM	beta44	0.11	0.191	0.081 ± 0.007	0.737 ± 0.064	0.006 ± 0.001	0.080 ± 0.006	0.013 ± 0.001	0.114 ± 0.007	1.073 ± 0.123	1.036 ± 0.064	0.185	0.075	67.96	0.0758	0.1065	1.018	87.692 ± 2.881	0.321 ± 0.006	69.231 ± 4.048
LSAM	beta45	0.11	0.114	0.004 ± 0.006	0.039 ± 0.050	0.004 ± 0.000	0.063 ± 0.004	0.004 ± 0.000	0.063 ± 0.004	0.325 ± 0.036	0.570 ± 0.041	0.122	0.012	10.70	0.0662	0.0673	1.038	97.692 ± 1.317	0.255 ± 0.004	45.385 ± 4.367
LSAM	beta46	0.08	0.021	-0.059 ± 0.005	-0.738 ± 0.063	0.003 ± 0.000	0.057 ± 0.004	0.007 ± 0.001	0.082 ± 0.005	1.050 ± 0.118	1.025 ± 0.062	0.017	-0.063	-79.37	0.0530	0.0827	0.915	73.846 ± 3.854	0.205 ± 0.005	5.385 ± 1.980
LSAM	beta47	0.08	0.019	-0.061 ± 0.005	-0.768 ± 0.059	0.003 ± 0.000	0.054 ± 0.004	0.007 ± 0.001	0.082 ± 0.004	1.038 ± 0.105	1.019 ± 0.055	0.021	-0.059	-73.17	0.0487	0.0762	0.938	68.462 ± 4.075	0.198 ± 0.005	5.385 ± 1.980
LSAM	beta51	0.16	0.192	0.032 ± 0.007	0.202 ± 0.045	0.007 ± 0.001	0.082 ± 0.005	0.008 ± 0.001	0.088 ± 0.007	0.303 ± 0.041	0.551 ± 0.044	0.195	0.035	22.12	0.0885	0.0953	0.854	91.538 ± 2.441	0.275 ± 0.003	76.154 ± 3.738
LSAM	beta52	0.16	0.154	-0.006 ± 0.006	-0.039 ± 0.036	0.004 ± 0.000	0.066 ± 0.004	0.004 ± 0.000	0.066 ± 0.005	0.168 ± 0.019	0.410 ± 0.030	0.154	-0.006	-3.73	0.0672	0.0675	0.923	93.077 ± 2.226	0.237 ± 0.003	71.538 ± 3.958
LSAM	beta53	0.16	0.148	-0.012 ± 0.005	-0.074 ± 0.034	0.004 ± 0.000	0.062 ± 0.004	0.004 ± 0.000	0.063 ± 0.005	0.157 ± 0.019	0.396 ± 0.030	0.140	-0.020	-12.32	0.0620	0.0650	0.978	94.615 ± 1.980	0.239 ± 0.002	68.462 ± 4.075
LSAM	beta54	0.16	0.133	-0.027 ± 0.005	-0.172 ± 0.032	0.003 ± 0.000	0.059 ± 0.003	0.004 ± 0.000	0.065 ± 0.004	0.164 ± 0.019	0.406 ± 0.028	0.141	-0.019	-11.85	0.0633	0.0661	1.031	92.308 ± 2.337	0.239 ± 0.003	57.692 ± 4.333
LSAM	beta55	0.08	0.108	0.028 ± 0.008	0.349 ± 0.099	0.008 ± 0.001	0.090 ± 0.007	0.009 ± 0.001	0.094 ± 0.008	1.375 ± 0.192	1.172 ± 0.096	0.108	0.028	34.98	0.0847	0.0892	0.953	96.923 ± 1.515	0.336 ± 0.008	23.077 ± 3.695
LSAM	beta56	0.08	0.072	-0.008 ± 0.006	-0.098 ± 0.080	0.005 ± 0.001	0.073 ± 0.005	0.005 ± 0.001	0.073 ± 0.006	0.832 ± 0.111	0.912 ± 0.074	0.077	-0.003	-3.84	0.0703	0.0704	1.121	96.154 ± 1.687	0.320 ± 0.009	8.462 ± 2.441
LSAM	beta57	0.06	0.014	-0.046 ± 0.004	-0.759 ± 0.072	0.002 ± 0.000	0.049 ± 0.004	0.004 ± 0.000	0.067 ± 0.004	1.247 ± 0.137	1.117 ± 0.068	0.013	-0.047	-77.64	0.0511	0.0691	0.932	82.308 ± 3.347	0.180 ± 0.004	6.154 ± 2.108
LSAM	beta58	0.06	0.031	-0.029 ± 0.003	-0.476 ± 0.056	0.001 ± 0.000	0.039 ± 0.002	0.002 ± 0.000	0.048 ± 0.003	0.637 ± 0.062	0.798 ± 0.045	0.029	-0.031	-51.09	0.0423	0.0522	1.023	80.000 ± 3.508	0.155 ± 0.003	6.923 ± 2.226
QML	beta41	0.20	0.184	-0.016 ± 0.004	-0.078 ± 0.022	0.003 ± 0.000	0.051 ± 0.004	0.003 ± 0.000	0.053 ± 0.004	0.071 ± 0.009	0.267 ± 0.021	0.182	-0.018	-8.76	0.0486	0.0517	0.984	92.308 ± 2.337	0.197 ± 0.002	95.385 ± 1.840
QML	beta42	0.20	0.182	-0.018 ± 0.005	-0.088 ± 0.023	0.003 ± 0.000	0.053 ± 0.004	0.003 ± 0.000	0.055 ± 0.004	0.076 ± 0.010	0.276 ± 0.021	0.175	-0.025	-12.45	0.0486	0.0546	0.982	93.077 ± 2.226	0.202 ± 0.002	96.154 ± 1.687
QML	beta43	0.20	0.181	-0.019 ± 0.004	-0.097 ± 0.022	0.003 ± 0.000	0.051 ± 0.003	0.003 ± 0.000	0.054 ± 0.004	0.073 ± 0.008	0.270 ± 0.019	0.182	-0.018	-8.99	0.0527	0.0557	0.947	89.231 ± 2.719	0.188 ± 0.002	96.154 ± 1.687
QML	beta44	0.11	0.155	0.045 ± 0.006	0.412 ± 0.053	0.004 ± 0.001	0.067 ± 0.005	0.007 ± 0.001	0.081 ± 0.005	0.539 ± 0.053	0.734 ± 0.043	0.152	0.042	38.21	0.0644	0.0769	1.030	92.308 ± 2.337	0.271 ± 0.006	65.385 ± 4.173
QML	beta45	0.11	0.108	-0.002 ± 0.006	-0.020 ± 0.051	0.004 ± 0.000	0.064 ± 0.004	0.004 ± 0.000	0.064 ± 0.005	0.340 ± 0.037	0.583 ± 0.041	0.114	0.004	4.06	0.0675	0.0676	0.922	93.846 ± 2.108	0.233 ± 0.004	50.000 ± 4.385
QML	beta46	0.08	0.048	-0.032 ± 0.004	-0.406 ± 0.054	0.002 ± 0.000	0.049 ± 0.003	0.003 ± 0.000	0.059 ± 0.003	0.542 ± 0.054	0.736 ± 0.043	0.045	-0.035	-43.98	0.0529	0.0635	0.942	82.308 ± 3.347	0.182 ± 0.009	16.923 ± 3.289
QML	beta47	0.08	0.039	-0.041 ± 0.004	-0.515 ± 0.051	0.002 ± 0.000	0.046 ± 0.003	0.004 ± 0.000	0.062 ± 0.003	0.600 ± 0.059	0.774 ± 0.043	0.036	-0.044	-54.46	0.0452	0.0628	0.931	72.308 ± 3.925	0.170 ± 0.004	10.769 ± 2.719
QML	beta51	0.16	0.198	0.038 ± 0.007	0.240 ± 0.045	0.007 ± 0.001	0.083 ± 0.005	0.008 ± 0.001	0.091 ± 0.007	0.323 ± 0.043	0.568 ± 0.043	0.195	0.035	21.83	0.0848	0.0917	0.936	93.077 ± 2.226	0.304 ± 0.005	73.077 ± 3.890
QML	beta52	0.16	0.148	-0.012 ± 0.005	-0.077 ± 0.033	0.004 ± 0.001	0.061 ± 0.004	0.004 ± 0.000	0.062 ± 0.005	0.148 ± 0.019	0.385 ± 0.030	0.147	-0.013	-8.26	0.0591	0.0606	0.972	91.538 ± 2.441	0.231 ± 0.005	73.077 ± 3.890
QML	beta53	0.16	0.136	-0.024 ± 0.005	-0.148 ± 0.029	0.003 ± 0.000	0.054 ± 0.003	0.003 ± 0.000	0.058 ± 0.004	0.133 ± 0.017	0.365 ± 0.027	0.129	-0.031	-19.37	0.0582	0.0660	1.060	93.077 ± 2.226	0.223 ± 0.002	66.923 ± 4.126
QML	beta54	0.16	0.144	-0.016 ± 0.005	-0.101 ± 0.029	0.003 ± 0.000	0.054 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	0.122 ± 0.015	0.349 ± 0.026	0.143	-0.017	-10.32	0.0567	0.0591	0.984	91.538 ± 2.441	0.207 ± 0.002	76.154 ± 3.738
QML	beta55	0.08	0.103	0.023 ± 0.008	0.290 ± 0.098	0.008 ± 0.001	0.090 ± 0.006	0.009 ± 0.001	0.092 ± 0.007	1.334 ± 0.161	1.155 ± 0.085	0.112	0.032	39.99	0.0978	0.1029	1.005	93.846 ± 2.108	0.354 ± 0.009	21.538 ± 3.605
QML	beta56	0.08	0.071	-0.009 ± 0.006	-0.108 ± 0.071	0.004 ± 0.000	0.065 ± 0.004	0.004 ± 0.000	0.065 ± 0.005	0.662 ± 0.077	0.814 ± 0.060	0.075	-0.005	-6.56	0.0724	0.0726	1.155	96.923 ± 1.515	0.293 ± 0.005	8.462 ± 2.441
QML	beta57	0.06	0.022	-0.038 ± 0.004	-0.626 ± 0.073	0.003 ± 0.000	0.050 ± 0.004	0.004 ± 0.000	0.062 ± 0.003	1.084 ± 0.101	1.041 ± 0.057	0.013	-0.047	-78.50	0.0549	0.0723	0.984	76.923 ± 3.695	0.193 ± 0.007	3.077 ± 1.515
QML	beta58	0.06	0.035	-0.025 ± 0.003	-0.417 ± 0.055	0.001 ± 0.000	0.038 ± 0.002	0.002 ± 0.000	0.045 ± 0.003	0.565 ± 0.053	0.752 ± 0.042	0.030	-0.030	-50.08	0.0344	0.0457	1.035	81.538 ± 3.403	0.153 ± 0.004	7.692 ± 2.337
UPI	beta41	0.20	0.219	0.019 ± 0.016	0.096 ± 0.081	0.034 ± 0.006	0.184 ± 0.018	0.034 ± 0.007	0.184 ± 0.020	0.849 ± 0.168	0.921 ± 0.102	0.220	0.020	9.77	0.1793	0.1804	1.235	92.308 ± 2.337	0.890 ± 0.099	29.231 ± 3.989
UPI	beta42	0.20	0.240	0.040 ± 0.018	0.198 ± 0.088	0.041 ± 0.008	0.202 ± 0.020	0.042 ± 0.008	0.205 ± 0.022	1.048 ± 0.200	1.024 ± 0.109	0.238	0.038	18.78	0.1677	0.1719	1.090	93.077 ± 2.226	0.862 ± 0.068	36.923 ± 4.233
UPI	beta43	0.20	0.149	-0.051 ± 0.008	-0.253 ± 0.041	0.009 ± 0.001	0.094 ± 0.005	0.011 ± 0.001	0.106 ± 0.008	0.283 ± 0.035	0.532 ± 0.038	0.144	-0.056	-27.79	0.1385	0.1492	0.926	82.308 ± 3.347	0.341 ± 0.020	49.231 ± 4.385
UPI	beta44	0.11	0.210	0.100 ± 0.013	0.910 ± 0.116	0.021 ± 0.004	0.145 ± 0.015	0.031 ± 0.005	0.176 ± 0.015	2.551 ± 0.403	1.597 ± 0.137	0.196	0.086	78.62	0.1308	0.1568	1.376	100.000 ± 0.000	0.782 ± 0.056	24.615 ± 3.778
UPI	beta45	0.11	0.124	0.014 ± 0.008	0.124 ± 0.073	0.008 ± 0.001	0.091 ± 0.008	0.008 ± 0.002	0.092 ± 0.010	0.700 ± 0.128	0.836 ± 0.087	0.125	0.015	13.73	0.1003	0.1015	1.214	99.231 ± 0.766	0.435 ± 0.026	20.769 ± 3.558
UPI	beta46	0.08	-0.031	-0.111 ± 0.017	-1.387 ± 0.208	0.036 ± 0.007	0.190 ± 0.019	0.048 ± 0.009	0.220 ± 0.023	7.530 ± 1.438	2.744 ± 0.284	-0.017	-0.097	-121.14	0.1795	0.2040	1.222	90.000 ± 2.631	0.911 ± 0.099	2.308 ± 1.317

Table S200

Simulation 2 - Condition 56: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.039	-0.119 ± 0.016	-1.491 ± 0.196	0.032 ± 0.006	0.179 ± 0.017	0.046 ± 0.007	0.215 ± 0.017	7.204 ± 1.070	2.684 ± 0.219	-0.023	-0.103	-128.37	0.1560	0.1868	1.092	89.231 ± 2.719	0.767 ± 0.061	4.615 ± 1.840
UPI	beta51	0.16	0.249	0.089 ± 0.007	0.553 ± 0.044	0.007 ± 0.001	0.081 ± 0.005	0.014 ± 0.002	0.120 ± 0.007	0.559 ± 0.063	0.747 ± 0.045	0.245	0.085	52.85	0.0793	0.1159	1.000	78.462 ± 3.605	0.316 ± 0.009	86.923 ± 2.957
UPI	beta52	0.16	0.153	-0.007 ± 0.009	-0.045 ± 0.053	0.010 ± 0.001	0.098 ± 0.008	0.009 ± 0.001	0.097 ± 0.008	0.371 ± 0.054	0.609 ± 0.053	0.161	0.001	0.69	0.1012	0.1012	1.182	91.538 ± 2.441	0.452 ± 0.030	30.000 ± 4.019
UPI	beta53	0.16	0.130	-0.030 ± 0.007	-0.190 ± 0.043	0.006 ± 0.001	0.078 ± 0.005	0.007 ± 0.001	0.084 ± 0.006	0.273 ± 0.034	0.522 ± 0.038	0.123	-0.037	-23.14	0.0945	0.1015	0.974	89.231 ± 2.719	0.298 ± 0.012	46.154 ± 4.372
UPI	beta54	0.16	0.110	-0.050 ± 0.008	-0.314 ± 0.053	0.009 ± 0.001	0.096 ± 0.007	0.012 ± 0.002	0.108 ± 0.008	0.456 ± 0.059	0.676 ± 0.050	0.108	-0.052	-32.50	0.1000	0.1127	1.034	81.538 ± 3.403	0.389 ± 0.023	20.769 ± 3.558
UPI	beta55	0.08	0.129	0.049 ± 0.009	0.615 ± 0.108	0.010 ± 0.002	0.099 ± 0.009	0.012 ± 0.002	0.110 ± 0.009	1.894 ± 0.290	1.376 ± 0.119	0.131	0.051	63.33	0.1002	0.1123	1.208	94.615 ± 1.980	0.468 ± 0.022	20.769 ± 3.558
UPI	beta56	0.08	0.074	-0.006 ± 0.008	-0.081 ± 0.096	0.008 ± 0.001	0.088 ± 0.007	0.008 ± 0.001	0.088 ± 0.008	1.206 ± 0.188	1.098 ± 0.100	0.074	-0.006	-7.48	0.0784	0.0786	1.178	96.923 ± 1.515	0.406 ± 0.015	4.615 ± 1.840
UPI	beta57	0.06	-0.036	-0.096 ± 0.011	-1.599 ± 0.188	0.017 ± 0.003	0.129 ± 0.012	0.026 ± 0.005	0.160 ± 0.016	7.135 ± 1.298	2.671 ± 0.259	-0.007	-0.067	-111.22	0.1034	0.1231	1.009	85.385 ± 3.098	0.510 ± 0.041	0.769 ± 0.766
UPI	beta58	0.06	0.006	-0.054 ± 0.008	-0.899 ± 0.140	0.009 ± 0.001	0.096 ± 0.008	0.012 ± 0.002	0.110 ± 0.008	3.348 ± 0.458	1.830 ± 0.141	0.004	-0.056	-93.55	0.0831	0.1003	1.060	83.846 ± 3.228	0.399 ± 0.033	0.769 ± 0.766

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S201

Simulation 2 - Condition 57: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.002	-0.014 ± 0.010	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.106 ± 0.005	0.325 ± 0.009	0.198	-0.002	-1.01	0.0650	0.0650	0.978	93.615 ± 0.785	0.249 ± 0.001	87.951 ± 1.045
LSAM	beta42	0.20	0.196	-0.004 ± 0.002	-0.018 ± 0.010	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.005	0.321 ± 0.009	0.196	-0.004	-1.86	0.0611	0.0612	1.014	95.469 ± 0.667	0.255 ± 0.001	85.788 ± 1.121
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.004	0.310 ± 0.009	0.197	-0.003	-1.58	0.0651	0.0652	0.956	94.336 ± 0.742	0.232 ± 0.001	93.203 ± 0.808
LSAM	beta44	0.11	0.132	0.022 ± 0.002	0.204 ± 0.023	0.006 ± 0.000	0.078 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.542 ± 0.026	0.736 ± 0.021	0.133	0.023	20.89	0.0774	0.0807	0.974	93.821 ± 0.773	0.297 ± 0.002	44.902 ± 1.596
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.012 ± 0.020	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.375 ± 0.019	0.613 ± 0.019	0.108	-0.002	-1.37	0.0676	0.0676	0.964	94.645 ± 0.722	0.255 ± 0.002	41.916 ± 1.583
LSAM	beta46	0.08	0.062	-0.018 ± 0.002	-0.227 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.564 ± 0.026	0.751 ± 0.021	0.060	-0.020	-24.53	0.0548	0.0582	0.925	90.422 ± 0.944	0.208 ± 0.002	22.657 ± 1.343
LSAM	beta47	0.08	0.059	-0.021 ± 0.002	-0.257 ± 0.022	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.524 ± 0.024	0.724 ± 0.019	0.055	-0.025	-31.10	0.0503	0.0561	0.951	89.186 ± 0.997	0.202 ± 0.002	20.494 ± 1.295
LSAM	beta51	0.16	0.175	0.015 ± 0.003	0.093 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.282 ± 0.014	0.531 ± 0.015	0.174	0.014	8.94	0.0801	0.0814	0.937	93.409 ± 0.796	0.307 ± 0.002	63.234 ± 1.547
LSAM	beta52	0.16	0.149	-0.011 ± 0.002	-0.067 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.403 ± 0.012	0.150	-0.010	-6.50	0.0620	0.0628	0.996	94.439 ± 0.735	0.248 ± 0.001	67.250 ± 1.506
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.004 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.181 ± 0.009	0.425 ± 0.012	0.160	0.000	-0.11	0.0686	0.0686	0.991	94.027 ± 0.761	0.264 ± 0.001	66.632 ± 1.513
LSAM	beta54	0.16	0.153	-0.007 ± 0.002	-0.047 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.172 ± 0.008	0.415 ± 0.012	0.154	-0.006	-3.58	0.0647	0.0649	0.967	93.512 ± 0.790	0.250 ± 0.001	66.838 ± 1.511
LSAM	beta55	0.08	0.092	0.012 ± 0.003	0.156 ± 0.036	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	1.273 ± 0.066	1.128 ± 0.034	0.094	0.014	17.52	0.0803	0.0816	0.952	94.439 ± 0.735	0.334 ± 0.003	21.112 ± 1.310
LSAM	beta56	0.08	0.080	0.000 ± 0.003	0.002 ± 0.034	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	1.110 ± 0.056	1.054 ± 0.031	0.076	-0.004	-4.79	0.0779	0.0780	0.950	94.542 ± 0.729	0.314 ± 0.003	17.508 ± 1.220
LSAM	beta57	0.06	0.045	-0.015 ± 0.002	-0.243 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.903 ± 0.042	0.950 ± 0.026	0.042	-0.018	-30.52	0.0540	0.0570	0.953	91.967 ± 0.872	0.206 ± 0.002	13.697 ± 1.103
LSAM	beta58	0.06	0.052	-0.008 ± 0.002	-0.141 ± 0.026	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.700 ± 0.034	0.837 ± 0.024	0.049	-0.011	-17.86	0.0474	0.0486	0.924	90.834 ± 0.926	0.179 ± 0.001	19.156 ± 1.263
QML	beta41	0.20	0.190	-0.010 ± 0.002	-0.051 ± 0.010	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.005	0.313 ± 0.009	0.191	-0.009	-4.71	0.0630	0.0637	0.981	93.924 ± 0.767	0.238 ± 0.001	88.568 ± 1.021
QML	beta42	0.20	0.192	-0.008 ± 0.002	-0.040 ± 0.010	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.005	0.312 ± 0.009	0.191	-0.009	-4.56	0.0602	0.0609	1.002	94.954 ± 0.702	0.243 ± 0.001	87.642 ± 1.056
QML	beta43	0.20	0.190	-0.010 ± 0.002	-0.048 ± 0.010	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.004	0.300 ± 0.008	0.188	-0.012	-6.19	0.0622	0.0634	0.966	94.233 ± 0.748	0.225 ± 0.001	91.761 ± 0.882
QML	beta44	0.11	0.119	0.009 ± 0.002	0.080 ± 0.022	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.496 ± 0.025	0.704 ± 0.021	0.121	0.011	10.09	0.0761	0.0769	0.973	94.542 ± 0.729	0.294 ± 0.003	42.225 ± 1.585
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.033 ± 0.019	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.362 ± 0.019	0.601 ± 0.018	0.106	-0.004	-3.75	0.0633	0.0635	0.979	94.851 ± 0.709	0.254 ± 0.002	40.165 ± 1.573
QML	beta46	0.08	0.069	-0.011 ± 0.002	-0.140 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.457 ± 0.021	0.676 ± 0.019	0.067	-0.013	-16.60	0.0517	0.0534	0.966	90.628 ± 0.935	0.200 ± 0.002	24.717 ± 1.384
QML	beta47	0.08	0.065	-0.015 ± 0.002	-0.186 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.460 ± 0.020	0.678 ± 0.018	0.059	-0.021	-25.96	0.0505	0.0546	0.957	88.363 ± 1.029	0.196 ± 0.002	21.421 ± 1.317
QML	beta51	0.16	0.181	0.021 ± 0.003	0.134 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.302 ± 0.015	0.550 ± 0.016	0.179	0.019	11.65	0.0830	0.0851	0.987	94.027 ± 0.761	0.330 ± 0.002	60.144 ± 1.571
QML	beta52	0.16	0.138	-0.022 ± 0.002	-0.140 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.195 ± 0.009	0.442 ± 0.012	0.137	-0.023	-14.51	0.0618	0.0660	1.030	94.336 ± 0.742	0.271 ± 0.002	53.244 ± 1.601
QML	beta53	0.16	0.141	-0.019 ± 0.002	-0.117 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.138	-0.022	-13.53	0.0650	0.0685	1.011	93.615 ± 0.785	0.265 ± 0.001	55.304 ± 1.596
QML	beta54	0.16	0.156	-0.004 ± 0.002	-0.025 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.409 ± 0.012	0.156	-0.004	-2.67	0.0632	0.0634	0.970	94.130 ± 0.754	0.249 ± 0.001	70.649 ± 1.461
QML	beta55	0.08	0.091	0.011 ± 0.003	0.141 ± 0.035	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	1.215 ± 0.058	1.102 ± 0.032	0.097	0.017	20.71	0.0840	0.0856	0.998	94.748 ± 0.716	0.342 ± 0.003	19.773 ± 1.278
QML	beta56	0.08	0.079	-0.001 ± 0.003	-0.017 ± 0.033	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	1.034 ± 0.051	1.017 ± 0.030	0.074	-0.006	-7.59	0.0787	0.0790	0.978	94.645 ± 0.722	0.312 ± 0.003	15.860 ± 1.172
QML	beta57	0.06	0.043	-0.017 ± 0.002	-0.285 ± 0.029	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.875 ± 0.038	0.936 ± 0.024	0.038	-0.022	-36.85	0.0505	0.0552	0.987	90.319 ± 0.949	0.207 ± 0.002	10.093 ± 0.967
QML	beta58	0.06	0.051	-0.009 ± 0.002	-0.147 ± 0.026	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.655 ± 0.031	0.810 ± 0.023	0.047	-0.013	-21.71	0.0459	0.0477	0.954	89.907 ± 0.967	0.179 ± 0.002	15.963 ± 1.175
UPI	beta41	0.20	0.210	0.010 ± 0.002	0.048 ± 0.011	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.115 ± 0.006	0.339 ± 0.010	0.210	0.010	5.09	0.0683	0.0691	0.969	94.748 ± 0.716	0.255 ± 0.002	88.877 ± 1.009
UPI	beta42	0.20	0.206	0.006 ± 0.002	0.029 ± 0.010	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.104 ± 0.005	0.322 ± 0.009	0.204	0.004	2.01	0.0604	0.0605	1.048	95.984 ± 0.630	0.263 ± 0.002	87.333 ± 1.067
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.011 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.318 ± 0.009	0.199	-0.001	-0.32	0.0647	0.0647	0.967	94.645 ± 0.722	0.241 ± 0.001	91.658 ± 0.887
UPI	beta44	0.11	0.129	0.019 ± 0.003	0.170 ± 0.024	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.591 ± 0.029	0.769 ± 0.022	0.127	0.017	15.34	0.0833	0.0850	1.044	96.498 ± 0.590	0.338 ± 0.004	35.942 ± 1.540
UPI	beta45	0.11	0.112	0.002 ± 0.002	0.021 ± 0.021	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.428 ± 0.023	0.654 ± 0.020	0.110	0.000	-0.16	0.0696	0.0696	0.989	95.778 ± 0.645	0.279 ± 0.003	36.869 ± 1.548
UPI	beta46	0.08	0.074	-0.006 ± 0.002	-0.077 ± 0.026	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.668 ± 0.035	0.817 ± 0.025	0.071	-0.009	-11.11	0.0600	0.0607	0.986	94.439 ± 0.735	0.252 ± 0.003	20.803 ± 1.303

Table S201

Simulation 2 - Condition 57: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.069	-0.011 ± 0.002	-0.132 ± 0.026	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.658 ± 0.033	0.811 ± 0.024	0.064	-0.016	-19.89	0.0604	0.0624	1.010	93.306 ± 0.802	0.254 ± 0.004	19.258 ± 1.265
UPI	beta51	0.16	0.175	0.015 ± 0.003	0.095 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.271 ± 0.013	0.521 ± 0.015	0.172	0.012	7.57	0.0799	0.0808	0.972	94.130 ± 0.754	0.312 ± 0.002	60.865 ± 1.566
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.162 ± 0.008	0.403 ± 0.012	0.159	-0.001	-0.62	0.0650	0.0650	1.005	95.057 ± 0.696	0.254 ± 0.001	69.619 ± 1.476
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.016 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.179 ± 0.009	0.423 ± 0.012	0.161	0.001	0.70	0.0683	0.0683	1.011	94.645 ± 0.722	0.269 ± 0.001	67.868 ± 1.499
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.173 ± 0.008	0.416 ± 0.012	0.159	-0.001	-0.32	0.0659	0.0659	0.989	93.924 ± 0.767	0.258 ± 0.001	68.486 ± 1.491
UPI	beta55	0.08	0.086	0.006 ± 0.003	0.080 ± 0.035	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	1.213 ± 0.062	1.101 ± 0.033	0.089	0.009	11.70	0.0839	0.0844	1.005	95.778 ± 0.645	0.346 ± 0.004	17.096 ± 1.208
UPI	beta56	0.08	0.080	0.000 ± 0.003	0.005 ± 0.035	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	1.158 ± 0.066	1.076 ± 0.035	0.074	-0.006	-8.03	0.0797	0.0800	0.979	95.263 ± 0.682	0.331 ± 0.004	13.285 ± 1.089
UPI	beta57	0.06	0.056	-0.004 ± 0.002	-0.071 ± 0.031	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.944 ± 0.049	0.972 ± 0.030	0.053	-0.007	-11.83	0.0576	0.0580	1.018	94.027 ± 0.761	0.232 ± 0.003	13.800 ± 1.107
UPI	beta58	0.06	0.058	-0.002 ± 0.002	-0.030 ± 0.030	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.887 ± 0.048	0.942 ± 0.030	0.055	-0.005	-8.74	0.0530	0.0532	0.989	93.718 ± 0.779	0.219 ± 0.004	18.538 ± 1.247

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S202

Simulation 2 - Condition 58: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.192	-0.008 ± 0.001	-0.039 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.005	0.191	-0.009	-4.36	0.0383	0.0393	1.001	94.584 ± 0.717	0.155 ± 0.000	99.799 ± 0.142
LSAM	beta42	0.20	0.195	-0.005 ± 0.001	-0.026 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.193	-0.007	-3.27	0.0427	0.0432	0.977	94.684 ± 0.711	0.158 ± 0.000	99.498 ± 0.224
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.200	0.000	-0.17	0.0352	0.0352	0.986	93.581 ± 0.776	0.144 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.127	0.017 ± 0.001	0.151 ± 0.013	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.203 ± 0.009	0.450 ± 0.012	0.127	0.017	15.58	0.0470	0.0500	0.978	93.280 ± 0.793	0.179 ± 0.001	78.134 ± 1.309
LSAM	beta45	0.11	0.113	0.003 ± 0.001	0.029 ± 0.011	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.129 ± 0.006	0.359 ± 0.010	0.112	0.002	2.21	0.0415	0.0415	1.015	95.687 ± 0.643	0.157 ± 0.001	80.742 ± 1.249
LSAM	beta46	0.08	0.064	-0.016 ± 0.001	-0.205 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.205 ± 0.009	0.453 ± 0.011	0.064	-0.016	-19.96	0.0308	0.0347	0.984	89.970 ± 0.951	0.125 ± 0.001	51.454 ± 1.583
LSAM	beta47	0.08	0.062	-0.018 ± 0.001	-0.220 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.206 ± 0.010	0.454 ± 0.012	0.062	-0.018	-22.81	0.0307	0.0357	0.985	89.368 ± 0.976	0.123 ± 0.001	53.260 ± 1.580
LSAM	beta51	0.16	0.170	0.010 ± 0.002	0.065 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.329 ± 0.009	0.170	0.010	6.14	0.0517	0.0526	0.949	93.280 ± 0.793	0.192 ± 0.001	93.480 ± 0.782
LSAM	beta52	0.16	0.153	-0.007 ± 0.001	-0.045 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.153	-0.007	-4.44	0.0399	0.0405	1.000	94.684 ± 0.711	0.155 ± 0.000	97.292 ± 0.514
LSAM	beta53	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.157	-0.003	-2.10	0.0404	0.0406	1.011	94.584 ± 0.717	0.163 ± 0.000	96.189 ± 0.606
LSAM	beta54	0.16	0.151	-0.009 ± 0.001	-0.056 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.150	-0.010	-6.34	0.0390	0.0403	0.973	94.584 ± 0.717	0.154 ± 0.000	97.192 ± 0.523
LSAM	beta55	0.08	0.094	0.014 ± 0.002	0.169 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.430 ± 0.020	0.656 ± 0.018	0.093	0.013	16.61	0.0489	0.0507	0.979	94.483 ± 0.723	0.195 ± 0.001	48.144 ± 1.582
LSAM	beta56	0.08	0.076	-0.004 ± 0.002	-0.044 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.400 ± 0.018	0.633 ± 0.018	0.075	-0.005	-6.29	0.0487	0.0489	0.929	92.879 ± 0.815	0.184 ± 0.001	37.412 ± 1.533
LSAM	beta57	0.06	0.046	-0.014 ± 0.001	-0.229 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.335 ± 0.015	0.579 ± 0.015	0.046	-0.014	-23.03	0.0339	0.0366	0.989	90.672 ± 0.921	0.124 ± 0.001	31.695 ± 1.474
LSAM	beta58	0.06	0.049	-0.011 ± 0.001	-0.188 ± 0.014	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.243 ± 0.011	0.493 ± 0.013	0.048	-0.012	-20.09	0.0261	0.0288	1.017	90.873 ± 0.912	0.109 ± 0.001	40.120 ± 1.552
QML	beta41	0.20	0.186	-0.014 ± 0.001	-0.072 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.205 ± 0.005	0.185	-0.015	-7.72	0.0379	0.0409	1.000	92.879 ± 0.815	0.151 ± 0.001	99.298 ± 0.264
QML	beta42	0.20	0.191	-0.009 ± 0.001	-0.044 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.191	-0.009	-4.60	0.0408	0.0418	0.980	94.283 ± 0.735	0.152 ± 0.001	99.398 ± 0.245
QML	beta43	0.20	0.192	-0.008 ± 0.001	-0.040 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.190	-0.010	-5.03	0.0359	0.0372	0.983	93.079 ± 0.804	0.140 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.118	0.008 ± 0.001	0.068 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.188 ± 0.008	0.434 ± 0.012	0.118	0.008	7.60	0.0473	0.0480	0.984	93.882 ± 0.759	0.182 ± 0.001	71.916 ± 1.423
QML	beta45	0.11	0.110	0.000 ± 0.001	0.001 ± 0.011	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.130 ± 0.006	0.360 ± 0.010	0.110	0.000	-0.05	0.0402	0.0402	1.018	94.985 ± 0.691	0.158 ± 0.001	77.934 ± 1.313
QML	beta46	0.08	0.068	-0.012 ± 0.001	-0.146 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.180 ± 0.008	0.425 ± 0.011	0.067	-0.013	-16.42	0.0300	0.0328	0.992	90.171 ± 0.943	0.124 ± 0.001	59.378 ± 1.555
QML	beta47	0.08	0.066	-0.014 ± 0.001	-0.174 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.182 ± 0.008	0.427 ± 0.011	0.065	-0.015	-19.32	0.0306	0.0343	0.985	89.067 ± 0.988	0.120 ± 0.001	59.378 ± 1.555
QML	beta51	0.16	0.178	0.018 ± 0.002	0.114 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.119 ± 0.005	0.346 ± 0.009	0.177	0.017	10.43	0.0527	0.0553	0.982	93.781 ± 0.765	0.201 ± 0.001	94.383 ± 0.729
QML	beta52	0.16	0.140	-0.020 ± 0.001	-0.125 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.291 ± 0.007	0.139	-0.021	-13.04	0.0412	0.0462	1.010	91.575 ± 0.880	0.166 ± 0.001	92.277 ± 0.845
QML	beta53	0.16	0.138	-0.022 ± 0.001	-0.139 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.137	-0.023	-14.62	0.0401	0.0464	1.006	91.675 ± 0.875	0.163 ± 0.000	90.271 ± 0.939
QML	beta54	0.16	0.154	-0.006 ± 0.001	-0.039 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.153	-0.007	-4.20	0.0392	0.0398	0.978	94.885 ± 0.698	0.155 ± 0.001	96.891 ± 0.550
QML	beta55	0.08	0.096	0.016 ± 0.002	0.202 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.479 ± 0.021	0.692 ± 0.018	0.097	0.017	21.11	0.0543	0.0568	0.997	95.085 ± 0.685	0.207 ± 0.001	47.342 ± 1.581
QML	beta56	0.08	0.076	-0.004 ± 0.002	-0.049 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.415 ± 0.019	0.645 ± 0.018	0.075	-0.005	-5.66	0.0512	0.0514	0.932	92.377 ± 0.840	0.188 ± 0.001	35.707 ± 1.517
QML	beta57	0.06	0.044	-0.016 ± 0.001	-0.272 ± 0.017	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.374 ± 0.016	0.611 ± 0.015	0.042	-0.018	-30.35	0.0332	0.0378	0.991	88.064 ± 1.027	0.128 ± 0.001	24.373 ± 1.360
QML	beta58	0.06	0.047	-0.013 ± 0.001	-0.209 ± 0.014	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.248 ± 0.010	0.498 ± 0.012	0.046	-0.014	-23.60	0.0264	0.0300	1.035	90.572 ± 0.925	0.110 ± 0.001	36.510 ± 1.525
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.006	0.199	-0.001	-0.30	0.0380	0.0380	0.976	94.383 ± 0.729	0.154 ± 0.001	99.699 ± 0.173
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.200	0.000	-0.19	0.0411	0.0411	0.984	95.386 ± 0.664	0.157 ± 0.001	99.599 ± 0.200
UPI	beta43	0.20	0.203	0.003 ± 0.001	0.014 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.201	0.001	0.63	0.0361	0.0361	0.982	94.183 ± 0.741	0.145 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.119	0.009 ± 0.002	0.079 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.196 ± 0.009	0.443 ± 0.013	0.119	0.009	8.02	0.0458	0.0467	0.993	94.684 ± 0.711	0.187 ± 0.001	72.116 ± 1.420
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.009 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.132 ± 0.006	0.364 ± 0.010	0.110	0.000	-0.05	0.0390	0.0390	1.030	95.988 ± 0.622	0.162 ± 0.001	79.338 ± 1.282
UPI	beta46	0.08	0.075	-0.005 ± 0.001	-0.068 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.195 ± 0.009	0.441 ± 0.013	0.073	-0.007	-8.16	0.0339	0.0345	1.017	94.082 ± 0.747	0.139 ± 0.001	58.375 ± 1.561

Table S202

Simulation 2 - Condition 58: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.073	-0.007 ± 0.001	-0.089 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.197 ± 0.009	0.444 ± 0.012	0.071	-0.009	-10.91	0.0336	0.0348	1.016	94.082 ± 0.747	0.139 ± 0.001	55.868 ± 1.573
UPI	beta51	0.16	0.167	0.007 ± 0.002	0.046 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.319 ± 0.009	0.167	0.007	4.29	0.0514	0.0519	0.971	94.183 ± 0.741	0.192 ± 0.001	92.578 ± 0.830
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.160	0.000	-0.17	0.0386	0.0386	0.999	94.183 ± 0.741	0.156 ± 0.000	97.994 ± 0.444
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.158	-0.002	-1.13	0.0402	0.0403	1.010	94.784 ± 0.704	0.164 ± 0.000	96.991 ± 0.541
UPI	beta54	0.16	0.156	-0.004 ± 0.001	-0.023 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.155	-0.005	-2.86	0.0392	0.0395	0.982	95.085 ± 0.685	0.156 ± 0.000	97.492 ± 0.495
UPI	beta55	0.08	0.088	0.008 ± 0.002	0.100 ± 0.020	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.403 ± 0.017	0.635 ± 0.017	0.087	0.007	8.31	0.0525	0.0530	1.004	95.587 ± 0.650	0.198 ± 0.001	44.132 ± 1.573
UPI	beta56	0.08	0.076	-0.004 ± 0.002	-0.048 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.414 ± 0.019	0.643 ± 0.018	0.074	-0.006	-7.64	0.0501	0.0505	0.929	92.277 ± 0.845	0.187 ± 0.001	35.707 ± 1.517
UPI	beta57	0.06	0.055	-0.005 ± 0.001	-0.081 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.321 ± 0.014	0.567 ± 0.015	0.055	-0.005	-8.33	0.0339	0.0343	1.018	94.183 ± 0.741	0.134 ± 0.001	36.710 ± 1.527
UPI	beta58	0.06	0.054	-0.006 ± 0.001	-0.096 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.269 ± 0.013	0.519 ± 0.015	0.053	-0.007	-11.79	0.0287	0.0296	1.014	93.982 ± 0.753	0.122 ± 0.001	42.227 ± 1.564

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S203

Simulation 2 - Condition 59: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.254 ± 0.007	0.200	0.000	0.09	0.0499	0.0499	0.960	94.076 ± 0.748	0.191 ± 0.001	96.787 ± 0.559
LSAM	beta42	0.20	0.197	-0.003 ± 0.002	-0.014 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.197	-0.003	-1.70	0.0489	0.0490	0.987	93.976 ± 0.754	0.194 ± 0.001	96.787 ± 0.559
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.202	0.002	0.97	0.0487	0.0487	0.970	94.378 ± 0.730	0.180 ± 0.001	98.594 ± 0.373
LSAM	beta44	0.11	0.117	0.007 ± 0.002	0.061 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.245 ± 0.011	0.495 ± 0.014	0.115	0.005	4.91	0.0538	0.0541	0.962	93.876 ± 0.760	0.204 ± 0.001	61.847 ± 1.539
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.009 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.193 ± 0.009	0.439 ± 0.013	0.111	0.001	1.12	0.0471	0.0472	0.965	93.675 ± 0.771	0.183 ± 0.001	66.165 ± 1.499
LSAM	beta46	0.08	0.077	-0.003 ± 0.001	-0.032 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.245 ± 0.012	0.495 ± 0.014	0.077	-0.003	-4.17	0.0369	0.0370	0.926	91.466 ± 0.885	0.144 ± 0.001	56.225 ± 1.572
LSAM	beta47	0.08	0.076	-0.004 ± 0.001	-0.046 ± 0.015	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.234 ± 0.011	0.484 ± 0.014	0.076	-0.004	-5.56	0.0385	0.0388	0.938	93.173 ± 0.799	0.142 ± 0.001	57.430 ± 1.567
LSAM	beta51	0.16	0.160	0.000 ± 0.002	0.002 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.008	0.417 ± 0.012	0.159	-0.001	-0.62	0.0652	0.0652	0.920	92.169 ± 0.851	0.240 ± 0.001	75.301 ± 1.366
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.323 ± 0.009	0.161	0.001	0.40	0.0500	0.0500	0.994	95.281 ± 0.672	0.202 ± 0.001	88.655 ± 1.005
LSAM	beta53	0.16	0.156	-0.004 ± 0.002	-0.023 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.157	-0.003	-2.00	0.0520	0.0521	1.006	95.181 ± 0.679	0.210 ± 0.001	82.631 ± 1.200
LSAM	beta54	0.16	0.156	-0.004 ± 0.002	-0.023 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.156	-0.004	-2.43	0.0572	0.0574	0.966	94.880 ± 0.698	0.199 ± 0.001	85.241 ± 1.124
LSAM	beta55	0.08	0.082	0.002 ± 0.002	0.023 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.602 ± 0.029	0.776 ± 0.022	0.081	0.001	1.81	0.0617	0.0617	0.959	94.177 ± 0.742	0.233 ± 0.002	31.526 ± 1.472
LSAM	beta56	0.08	0.082	0.002 ± 0.002	0.020 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.608 ± 0.030	0.780 ± 0.023	0.077	-0.003	-3.50	0.0588	0.0589	0.910	91.767 ± 0.871	0.223 ± 0.002	30.321 ± 1.456
LSAM	beta57	0.06	0.058	-0.002 ± 0.001	-0.033 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.437 ± 0.020	0.661 ± 0.018	0.058	-0.002	-2.68	0.0393	0.0394	0.984	93.976 ± 0.754	0.153 ± 0.001	31.124 ± 1.467
LSAM	beta58	0.06	0.057	-0.003 ± 0.001	-0.044 ± 0.019	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.348 ± 0.016	0.590 ± 0.017	0.056	-0.004	-7.33	0.0347	0.0350	0.966	93.373 ± 0.788	0.134 ± 0.001	38.956 ± 1.545
QML	beta41	0.20	0.195	-0.005 ± 0.002	-0.025 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.197	-0.003	-1.75	0.0502	0.0503	0.969	93.775 ± 0.766	0.192 ± 0.001	96.285 ± 0.599
QML	beta42	0.20	0.195	-0.005 ± 0.002	-0.024 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.252 ± 0.007	0.195	-0.005	-2.46	0.0492	0.0494	0.989	94.277 ± 0.736	0.194 ± 0.001	96.486 ± 0.583
QML	beta43	0.20	0.197	-0.003 ± 0.001	-0.015 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.198	-0.002	-1.18	0.0475	0.0476	0.973	94.378 ± 0.730	0.179 ± 0.001	98.594 ± 0.373
QML	beta44	0.11	0.112	0.002 ± 0.002	0.021 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.247 ± 0.011	0.497 ± 0.014	0.112	0.002	2.03	0.0549	0.0549	0.973	94.277 ± 0.736	0.208 ± 0.001	58.133 ± 1.563
QML	beta45	0.11	0.110	-0.000 ± 0.002	-0.000 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.197 ± 0.010	0.444 ± 0.013	0.111	0.001	0.55	0.0460	0.0460	0.972	94.177 ± 0.742	0.186 ± 0.001	65.462 ± 1.507
QML	beta46	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.242 ± 0.012	0.492 ± 0.014	0.077	-0.003	-3.93	0.0357	0.0359	0.937	92.068 ± 0.856	0.145 ± 0.001	57.932 ± 1.564
QML	beta47	0.08	0.077	-0.003 ± 0.001	-0.035 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.240 ± 0.012	0.490 ± 0.014	0.075	-0.005	-6.61	0.0376	0.0380	0.931	92.871 ± 0.815	0.143 ± 0.001	57.932 ± 1.564
QML	beta51	0.16	0.164	0.004 ± 0.002	0.024 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.009	0.419 ± 0.012	0.163	0.003	2.08	0.0641	0.0642	0.951	93.675 ± 0.771	0.250 ± 0.001	75.502 ± 1.363
QML	beta52	0.16	0.145	-0.015 ± 0.002	-0.096 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.129 ± 0.006	0.360 ± 0.010	0.143	-0.017	-10.45	0.0552	0.0576	1.002	94.478 ± 0.724	0.218 ± 0.001	75.201 ± 1.368
QML	beta53	0.16	0.136	-0.024 ± 0.002	-0.150 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.135	-0.025	-15.39	0.0550	0.0602	1.001	92.470 ± 0.836	0.217 ± 0.001	69.478 ± 1.459
QML	beta54	0.16	0.157	-0.003 ± 0.002	-0.021 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.157	-0.003	-2.14	0.0571	0.0572	0.973	95.482 ± 0.658	0.201 ± 0.001	84.438 ± 1.149
QML	beta55	0.08	0.084	0.004 ± 0.002	0.047 ± 0.025	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.617 ± 0.030	0.785 ± 0.023	0.085	0.005	6.61	0.0638	0.0640	0.972	94.578 ± 0.718	0.239 ± 0.002	32.932 ± 1.489
QML	beta56	0.08	0.082	0.002 ± 0.002	0.024 ± 0.025	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.612 ± 0.030	0.782 ± 0.023	0.079	-0.001	-1.39	0.0574	0.0574	0.923	91.867 ± 0.866	0.226 ± 0.002	28.514 ± 1.431
QML	beta57	0.06	0.056	-0.004 ± 0.001	-0.068 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.454 ± 0.022	0.674 ± 0.019	0.055	-0.005	-8.77	0.0374	0.0378	0.978	92.671 ± 0.826	0.154 ± 0.001	26.104 ± 1.392
QML	beta58	0.06	0.056	-0.004 ± 0.001	-0.062 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.363 ± 0.017	0.602 ± 0.017	0.054	-0.006	-10.72	0.0347	0.0353	0.959	92.369 ± 0.841	0.135 ± 0.001	35.442 ± 1.516
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.022 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.256 ± 0.007	0.204	0.004	2.15	0.0503	0.0505	0.918	92.369 ± 0.841	0.183 ± 0.001	97.490 ± 0.496
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.201	0.001	0.48	0.0491	0.0491	0.964	93.675 ± 0.771	0.189 ± 0.001	97.189 ± 0.524
UPI	beta43	0.20	0.203	0.003 ± 0.002	0.016 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.204	0.004	1.83	0.0485	0.0486	0.964	94.478 ± 0.724	0.180 ± 0.001	98.996 ± 0.316
UPI	beta44	0.11	0.115	0.005 ± 0.002	0.047 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.257 ± 0.011	0.507 ± 0.014	0.115	0.005	4.79	0.0565	0.0568	0.955	94.177 ± 0.742	0.208 ± 0.001	58.735 ± 1.560
UPI	beta45	0.11	0.111	0.001 ± 0.002	0.013 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.199 ± 0.009	0.446 ± 0.013	0.112	0.002	2.07	0.0454	0.0454	0.976	94.478 ± 0.724	0.188 ± 0.001	66.265 ± 1.498
UPI	beta46	0.08	0.080	0.000 ± 0.001	0.006 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.259 ± 0.013	0.509 ± 0.015	0.077	-0.003	-3.22	0.0383	0.0384	0.930	93.273 ± 0.794	0.149 ± 0.001	57.028 ± 1.569

Table S203

Simulation 2 - Condition 59: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.254 ± 0.012	0.504 ± 0.015	0.078	-0.002	-3.11	0.0382	0.0383	0.933	93.474 ± 0.783	0.148 ± 0.001	56.928 ± 1.569
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.411 ± 0.012	0.160	0.000	0.30	0.0642	0.0642	0.937	92.771 ± 0.821	0.242 ± 0.001	76.406 ± 1.345
UPI	beta52	0.16	0.165	0.005 ± 0.002	0.034 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.165	0.005	3.42	0.0488	0.0491	0.985	94.980 ± 0.692	0.199 ± 0.001	90.361 ± 0.935
UPI	beta53	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.111 ± 0.005	0.333 ± 0.009	0.157	-0.003	-1.64	0.0531	0.0531	1.005	95.582 ± 0.651	0.210 ± 0.001	83.233 ± 1.184
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.004	0.327 ± 0.009	0.159	-0.001	-0.87	0.0564	0.0564	0.962	94.980 ± 0.692	0.197 ± 0.001	86.044 ± 1.098
UPI	beta55	0.08	0.080	0.000 ± 0.002	0.004 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.602 ± 0.030	0.776 ± 0.023	0.081	0.001	1.01	0.0615	0.0615	0.964	95.582 ± 0.651	0.235 ± 0.002	30.422 ± 1.458
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.007 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.605 ± 0.030	0.778 ± 0.023	0.078	-0.002	-2.64	0.0577	0.0577	0.919	92.671 ± 0.826	0.224 ± 0.002	29.518 ± 1.445
UPI	beta57	0.06	0.060	-0.000 ± 0.001	-0.001 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.449 ± 0.021	0.670 ± 0.019	0.059	-0.001	-1.45	0.0393	0.0393	0.982	94.076 ± 0.748	0.155 ± 0.001	32.229 ± 1.481
UPI	beta58	0.06	0.059	-0.001 ± 0.001	-0.011 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.370 ± 0.018	0.608 ± 0.018	0.057	-0.003	-4.21	0.0349	0.0350	0.962	93.072 ± 0.805	0.138 ± 0.001	39.960 ± 1.552

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S204

Simulation 2 - Condition 60: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.001	-0.016 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.197	-0.003	-1.63	0.0318	0.0320	0.996	95.200 ± 0.676	0.121 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.198	-0.002	-1.05	0.0313	0.0313	1.024	95.300 ± 0.669	0.122 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.201	0.001	0.34	0.0293	0.0293	0.998	95.000 ± 0.689	0.115 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.113	0.003 ± 0.001	0.028 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.097 ± 0.005	0.311 ± 0.009	0.113	0.003	2.43	0.0340	0.0341	0.958	94.600 ± 0.715	0.128 ± 0.001	90.700 ± 0.918
LSAM	beta45	0.11	0.111	0.001 ± 0.001	0.007 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.077 ± 0.004	0.278 ± 0.008	0.112	0.002	1.58	0.0302	0.0302	0.956	93.100 ± 0.801	0.114 ± 0.000	95.700 ± 0.641
LSAM	beta46	0.08	0.076	-0.004 ± 0.001	-0.052 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.075	-0.005	-5.66	0.0240	0.0244	0.958	93.300 ± 0.791	0.090 ± 0.000	92.600 ± 0.828
LSAM	beta47	0.08	0.077	-0.003 ± 0.001	-0.036 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.077	-0.003	-3.63	0.0232	0.0234	0.971	93.000 ± 0.807	0.089 ± 0.000	93.500 ± 0.780
LSAM	beta51	0.16	0.162	0.002 ± 0.001	0.013 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.242 ± 0.006	0.162	0.002	1.26	0.0405	0.0406	1.007	95.700 ± 0.641	0.153 ± 0.001	99.000 ± 0.315
LSAM	beta52	0.16	0.158	-0.002 ± 0.001	-0.012 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.158	-0.002	-1.13	0.0337	0.0338	0.966	94.500 ± 0.721	0.126 ± 0.000	99.900 ± 0.100
LSAM	beta53	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.005	0.158	-0.002	-1.15	0.0351	0.0351	1.020	95.800 ± 0.634	0.132 ± 0.000	99.600 ± 0.200
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.158	-0.002	-1.48	0.0352	0.0353	0.986	95.100 ± 0.683	0.126 ± 0.000	99.900 ± 0.100
LSAM	beta55	0.08	0.083	0.003 ± 0.001	0.040 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.215 ± 0.010	0.464 ± 0.013	0.082	0.002	2.67	0.0369	0.0370	1.003	95.000 ± 0.689	0.145 ± 0.001	61.300 ± 1.540
LSAM	beta56	0.08	0.081	0.001 ± 0.001	0.009 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.203 ± 0.010	0.451 ± 0.013	0.079	-0.001	-1.07	0.0348	0.0348	0.983	94.300 ± 0.733	0.139 ± 0.001	63.900 ± 1.519
LSAM	beta57	0.06	0.057	-0.003 ± 0.001	-0.056 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.171 ± 0.008	0.414 ± 0.011	0.057	-0.003	-5.57	0.0245	0.0247	0.979	92.800 ± 0.817	0.094 ± 0.000	67.300 ± 1.483
LSAM	beta58	0.06	0.058	-0.002 ± 0.001	-0.037 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.137 ± 0.006	0.370 ± 0.010	0.057	-0.003	-4.24	0.0229	0.0230	0.977	94.000 ± 0.751	0.085 ± 0.000	77.100 ± 1.329
QML	beta41	0.20	0.193	-0.007 ± 0.001	-0.034 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.193	-0.007	-3.42	0.0315	0.0322	1.002	94.900 ± 0.696	0.121 ± 0.001	99.800 ± 0.141
QML	beta42	0.20	0.196	-0.004 ± 0.001	-0.018 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.196	-0.004	-1.79	0.0300	0.0302	1.025	95.100 ± 0.683	0.123 ± 0.001	99.900 ± 0.100
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.020 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.196	-0.004	-1.78	0.0285	0.0287	0.996	94.600 ± 0.715	0.114 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.110	-0.000 ± 0.001	-0.000 ± 0.010	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.110	0.000	0.21	0.0361	0.0361	0.958	94.600 ± 0.715	0.131 ± 0.001	87.200 ± 1.056
QML	beta45	0.11	0.110	0.000 ± 0.001	0.000 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.080 ± 0.004	0.284 ± 0.008	0.110	0.000	0.39	0.0304	0.0304	0.958	93.700 ± 0.768	0.117 ± 0.000	94.600 ± 0.715
QML	beta46	0.08	0.077	-0.003 ± 0.001	-0.043 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.076	-0.004	-5.52	0.0238	0.0242	0.967	92.900 ± 0.812	0.091 ± 0.000	94.100 ± 0.745
QML	beta47	0.08	0.077	-0.003 ± 0.001	-0.032 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.076	-0.004	-4.46	0.0234	0.0237	0.971	93.200 ± 0.796	0.090 ± 0.001	94.700 ± 0.708
QML	beta51	0.16	0.166	0.006 ± 0.001	0.040 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.248 ± 0.007	0.166	0.006	4.02	0.0407	0.0412	1.021	96.100 ± 0.612	0.157 ± 0.001	99.400 ± 0.244
QML	beta52	0.16	0.141	-0.019 ± 0.001	-0.118 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.006	0.140	-0.020	-12.57	0.0365	0.0417	0.977	91.300 ± 0.891	0.136 ± 0.000	97.900 ± 0.453
QML	beta53	0.16	0.138	-0.022 ± 0.001	-0.136 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.006	0.138	-0.022	-13.60	0.0371	0.0430	1.007	91.100 ± 0.900	0.136 ± 0.000	97.800 ± 0.464
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.204 ± 0.005	0.157	-0.003	-1.68	0.0352	0.0353	0.989	95.300 ± 0.669	0.127 ± 0.000	100.000 ± 0.000
QML	beta55	0.08	0.086	0.006 ± 0.001	0.073 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.228 ± 0.010	0.478 ± 0.013	0.085	0.005	5.69	0.0373	0.0376	1.008	94.700 ± 0.708	0.149 ± 0.001	61.100 ± 1.542
QML	beta56	0.08	0.082	0.002 ± 0.001	0.019 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.208 ± 0.010	0.456 ± 0.013	0.080	0.000	0.42	0.0349	0.0349	0.996	94.200 ± 0.739	0.142 ± 0.001	62.600 ± 1.530
QML	beta57	0.06	0.054	-0.006 ± 0.001	-0.099 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.181 ± 0.008	0.426 ± 0.011	0.053	-0.007	-11.22	0.0247	0.0256	0.981	92.000 ± 0.858	0.096 ± 0.001	62.900 ± 1.528
QML	beta58	0.06	0.056	-0.004 ± 0.001	-0.058 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.144 ± 0.007	0.380 ± 0.010	0.055	-0.005	-7.80	0.0227	0.0231	0.974	94.100 ± 0.745	0.086 ± 0.000	74.800 ± 1.373
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.000 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.200	0.000	-0.05	0.0313	0.0313	0.956	94.900 ± 0.696	0.116 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.202	0.002	0.77	0.0300	0.0300	1.002	94.600 ± 0.715	0.119 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.201	0.001	0.54	0.0297	0.0297	0.987	94.800 ± 0.702	0.114 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.112	0.002 ± 0.001	0.015 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.097 ± 0.005	0.311 ± 0.009	0.112	0.002	1.43	0.0355	0.0356	0.963	94.600 ± 0.715	0.129 ± 0.001	90.000 ± 0.949
UPI	beta45	0.11	0.110	0.000 ± 0.001	0.002 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.080 ± 0.004	0.283 ± 0.008	0.110	0.000	-0.22	0.0304	0.0304	0.947	93.000 ± 0.807	0.116 ± 0.000	94.500 ± 0.721
UPI	beta46	0.08	0.078	-0.002 ± 0.001	-0.020 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.078	-0.002	-2.50	0.0238	0.0239	0.970	94.400 ± 0.727	0.091 ± 0.000	94.800 ± 0.702

Table S204

Simulation 2 - Condition 60: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.089 ± 0.004	0.299 ± 0.008	0.079	-0.001	-1.55	0.0245	0.0245	0.977	93.500 ± 0.780	0.092 ± 0.000	94.200 ± 0.739
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.010 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.002	0.241 ± 0.006	0.162	0.002	1.32	0.0404	0.0405	1.009	96.200 ± 0.605	0.153 ± 0.001	99.000 ± 0.315
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.160	0.000	0.31	0.0336	0.0336	0.955	93.900 ± 0.757	0.125 ± 0.000	99.900 ± 0.100
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.005	0.159	-0.001	-0.55	0.0346	0.0347	1.017	95.600 ± 0.649	0.132 ± 0.000	99.600 ± 0.200
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.160	0.000	-0.20	0.0354	0.0354	0.979	95.100 ± 0.683	0.125 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.018 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.216 ± 0.010	0.465 ± 0.013	0.080	0.000	0.31	0.0363	0.0363	0.998	94.200 ± 0.739	0.146 ± 0.001	59.600 ± 1.552
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.011 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.201 ± 0.010	0.448 ± 0.013	0.079	-0.001	-0.95	0.0337	0.0337	0.990	94.300 ± 0.733	0.139 ± 0.001	64.300 ± 1.515
UPI	beta57	0.06	0.059	-0.001 ± 0.001	-0.017 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.173 ± 0.008	0.416 ± 0.012	0.059	-0.001	-2.43	0.0247	0.0248	0.973	92.800 ± 0.817	0.095 ± 0.000	69.800 ± 1.452
UPI	beta58	0.06	0.059	-0.001 ± 0.001	-0.010 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.138 ± 0.007	0.372 ± 0.011	0.058	-0.002	-2.73	0.0227	0.0228	0.978	93.700 ± 0.768	0.086 ± 0.000	79.400 ± 1.279

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S205

Simulation 2 - Condition 61: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.220	0.020 ± 0.009	0.098 ± 0.047	0.054 ± 0.003	0.233 ± 0.007	0.054 ± 0.003	0.233 ± 0.009	1.359 ± 0.087	1.166 ± 0.044	0.216	0.016	8.15	0.2256	0.2262	1.085	97.749 ± 0.595	0.989 ± 0.023	16.720 ± 1.496
LSAM	beta42	0.20	0.198	-0.002 ± 0.010	-0.008 ± 0.048	0.058 ± 0.004	0.241 ± 0.008	0.058 ± 0.004	0.241 ± 0.009	1.449 ± 0.097	1.204 ± 0.047	0.197	-0.003	-1.61	0.2261	0.2261	1.069	97.910 ± 0.574	1.010 ± 0.022	14.952 ± 1.430
LSAM	beta43	0.20	0.192	-0.008 ± 0.003	-0.040 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.174 ± 0.010	0.417 ± 0.015	0.195	-0.005	-2.30	0.0859	0.0860	1.078	95.981 ± 0.788	0.351 ± 0.004	63.666 ± 1.928
LSAM	beta44	0.11	0.127	0.017 ± 0.005	0.154 ± 0.048	0.017 ± 0.001	0.131 ± 0.005	0.017 ± 0.001	0.132 ± 0.005	1.433 ± 0.103	1.197 ± 0.049	0.122	0.012	10.87	0.1256	0.1262	0.993	96.141 ± 0.772	0.509 ± 0.010	19.132 ± 1.577
LSAM	beta45	0.11	0.115	0.005 ± 0.005	0.048 ± 0.042	0.013 ± 0.001	0.115 ± 0.004	0.013 ± 0.001	0.116 ± 0.005	1.103 ± 0.077	1.050 ± 0.043	0.115	0.005	4.67	0.1107	0.1108	0.988	93.408 ± 0.995	0.447 ± 0.009	16.559 ± 1.490
LSAM	beta46	0.08	0.056	-0.024 ± 0.004	-0.306 ± 0.051	0.010 ± 0.001	0.101 ± 0.004	0.011 ± 0.001	0.104 ± 0.004	1.694 ± 0.126	1.301 ± 0.054	0.057	-0.023	-28.52	0.0887	0.0916	1.012	95.016 ± 0.873	0.402 ± 0.012	14.309 ± 1.404
LSAM	beta47	0.08	0.065	-0.015 ± 0.004	-0.193 ± 0.048	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	1.452 ± 0.104	1.205 ± 0.049	0.065	-0.015	-18.17	0.0816	0.0828	1.050	94.855 ± 0.886	0.392 ± 0.011	16.077 ± 1.473
LSAM	beta51	0.16	0.171	0.011 ± 0.004	0.070 ± 0.026	0.011 ± 0.001	0.105 ± 0.004	0.011 ± 0.001	0.106 ± 0.004	0.438 ± 0.029	0.661 ± 0.026	0.172	0.012	7.44	0.1074	0.1081	0.984	95.820 ± 0.802	0.406 ± 0.004	41.158 ± 1.973
LSAM	beta52	0.16	0.217	0.057 ± 0.011	0.353 ± 0.066	0.069 ± 0.005	0.262 ± 0.010	0.072 ± 0.006	0.268 ± 0.012	2.811 ± 0.218	1.677 ± 0.073	0.194	0.034	21.22	0.2320	0.2345	1.142	98.232 ± 0.528	1.175 ± 0.031	9.325 ± 1.166
LSAM	beta53	0.16	0.144	-0.016 ± 0.005	-0.102 ± 0.029	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.118 ± 0.004	0.542 ± 0.034	0.736 ± 0.027	0.147	-0.013	-8.15	0.1143	0.1150	1.110	96.463 ± 0.741	0.508 ± 0.007	20.096 ± 1.607
LSAM	beta54	0.16	0.140	-0.020 ± 0.009	-0.123 ± 0.057	0.053 ± 0.004	0.229 ± 0.008	0.053 ± 0.004	0.230 ± 0.009	2.063 ± 0.142	1.436 ± 0.057	0.145	-0.015	-9.11	0.1950	0.1955	1.088	97.106 ± 0.672	0.978 ± 0.026	8.521 ± 1.119
LSAM	beta55	0.08	0.139	0.059 ± 0.005	0.739 ± 0.059	0.014 ± 0.001	0.119 ± 0.004	0.018 ± 0.001	0.132 ± 0.005	2.736 ± 0.189	1.654 ± 0.063	0.135	0.055	69.07	0.1067	0.1202	1.108	95.659 ± 0.817	0.515 ± 0.015	27.010 ± 1.780
LSAM	beta56	0.08	0.068	-0.012 ± 0.003	-0.153 ± 0.034	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	0.733 ± 0.054	0.856 ± 0.036	0.071	-0.009	-11.68	0.0576	0.0584	1.010	94.855 ± 0.886	0.267 ± 0.007	23.955 ± 1.711
LSAM	beta57	0.06	-0.004	-0.064 ± 0.006	-1.070 ± 0.100	0.022 ± 0.002	0.149 ± 0.006	0.026 ± 0.002	0.162 ± 0.007	7.316 ± 0.578	2.705 ± 0.117	0.008	-0.052	-86.01	0.1224	0.1328	1.149	96.302 ± 0.757	0.672 ± 0.022	1.608 ± 0.504
LSAM	beta58	0.06	0.048	-0.012 ± 0.004	-0.193 ± 0.061	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.092 ± 0.004	2.334 ± 0.179	1.528 ± 0.066	0.046	-0.014	-23.06	0.0677	0.0691	1.047	94.855 ± 0.886	0.374 ± 0.013	11.254 ± 1.267
QML	beta41	0.20	0.234	0.034 ± 0.004	0.172 ± 0.020	0.010 ± 0.001	0.098 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.269 ± 0.016	0.519 ± 0.018	0.233	0.033	16.38	0.0999	0.1051	1.040	95.016 ± 0.873	0.399 ± 0.004	65.916 ± 1.901
QML	beta42	0.20	0.224	0.024 ± 0.004	0.120 ± 0.021	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	0.283 ± 0.017	0.532 ± 0.019	0.220	0.020	9.91	0.1017	0.1036	1.012	95.177 ± 0.859	0.412 ± 0.004	60.611 ± 1.959
QML	beta43	0.20	0.191	-0.009 ± 0.003	-0.044 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.146 ± 0.008	0.382 ± 0.013	0.196	-0.004	-2.09	0.0796	0.0797	1.060	95.659 ± 0.817	0.316 ± 0.002	67.685 ± 1.875
QML	beta44	0.11	0.065	-0.045 ± 0.006	-0.408 ± 0.051	0.020 ± 0.001	0.141 ± 0.005	0.022 ± 0.001	0.148 ± 0.005	1.799 ± 0.117	1.341 ± 0.050	0.067	-0.043	-39.40	0.1390	0.1456	0.980	93.248 ± 1.006	0.540 ± 0.011	13.666 ± 1.377
QML	beta45	0.11	0.108	-0.002 ± 0.004	-0.020 ± 0.041	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	1.020 ± 0.057	1.010 ± 0.035	0.110	0.000	0.05	0.1135	0.1135	0.964	91.801 ± 1.100	0.420 ± 0.006	18.810 ± 1.567
QML	beta46	0.08	0.138	0.058 ± 0.003	0.723 ± 0.042	0.007 ± 0.001	0.084 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	1.629 ± 0.118	1.276 ± 0.050	0.129	0.049	61.47	0.0804	0.0943	0.967	92.765 ± 1.039	0.319 ± 0.008	46.785 ± 2.001
QML	beta47	0.08	0.133	0.053 ± 0.003	0.660 ± 0.038	0.006 ± 0.000	0.077 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	1.356 ± 0.095	1.164 ± 0.044	0.127	0.047	58.25	0.0710	0.0849	0.999	93.730 ± 0.972	0.301 ± 0.007	47.588 ± 2.002
QML	beta51	0.16	0.184	0.024 ± 0.006	0.148 ± 0.037	0.022 ± 0.001	0.149 ± 0.005	0.023 ± 0.001	0.151 ± 0.006	0.889 ± 0.055	0.943 ± 0.035	0.182	0.022	13.49	0.1406	0.1422	0.963	94.373 ± 0.924	0.563 ± 0.013	33.280 ± 1.889
QML	beta52	0.16	0.154	-0.006 ± 0.005	-0.035 ± 0.033	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	0.660 ± 0.043	0.813 ± 0.031	0.146	-0.014	-9.00	0.1250	0.1258	0.991	94.051 ± 0.948	0.505 ± 0.011	21.865 ± 1.657
QML	beta53	0.16	0.125	-0.035 ± 0.004	-0.219 ± 0.028	0.013 ± 0.001	0.112 ± 0.004	0.014 ± 0.001	0.117 ± 0.004	0.537 ± 0.032	0.733 ± 0.026	0.127	-0.033	-20.49	0.1072	0.1121	1.026	92.765 ± 1.039	0.450 ± 0.006	18.650 ± 1.562
QML	beta54	0.16	0.168	0.008 ± 0.004	0.050 ± 0.026	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.423 ± 0.024	0.651 ± 0.023	0.173	0.013	8.21	0.1063	0.1071	1.034	95.981 ± 0.788	0.421 ± 0.004	38.585 ± 1.952
QML	beta55	0.08	0.096	0.016 ± 0.007	0.202 ± 0.088	0.031 ± 0.002	0.175 ± 0.005	0.031 ± 0.002	0.175 ± 0.006	4.808 ± 0.291	2.193 ± 0.080	0.099	0.019	23.72	0.1828	0.1838	0.897	85.370 ± 1.417	0.615 ± 0.024	24.277 ± 1.719
QML	beta56	0.08	0.105	0.025 ± 0.003	0.312 ± 0.043	0.007 ± 0.000	0.085 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	1.236 ± 0.084	1.112 ± 0.043	0.106	0.026	31.95	0.0814	0.0853	0.973	92.283 ± 1.070	0.326 ± 0.008	32.797 ± 1.882
QML	beta57	0.06	0.118	0.058 ± 0.006	0.965 ± 0.108	0.026 ± 0.002	0.162 ± 0.005	0.030 ± 0.002	0.172 ± 0.007	8.208 ± 0.561	2.865 ± 0.111	0.110	0.050	83.16	0.1615	0.1691	0.902	85.370 ± 1.417	0.573 ± 0.021	19.775 ± 1.597
QML	beta58	0.06	0.102	0.042 ± 0.003	0.706 ± 0.047	0.005 ± 0.000	0.070 ± 0.003	0.007 ± 0.001	0.082 ± 0.003	1.858 ± 0.144	1.363 ± 0.057	0.101	0.041	68.51	0.0642	0.0762	0.977	94.051 ± 0.948	0.268 ± 0.007	35.691 ± 1.921
UPI	beta41	0.20	0.294	0.094 ± 0.014	0.470 ± 0.068	0.116 ± 0.009	0.341 ± 0.014	0.125 ± 0.009	0.353 ± 0.015	3.118 ± 0.236	1.766 ± 0.075	0.278	0.078	39.20	0.3322	0.3414	1.288	98.232 ± 0.528	1.720 ± 0.069	11.736 ± 1.291
UPI	beta42	0.20	0.318	0.118 ± 0.013	0.588 ± 0.065	0.105 ± 0.008	0.324 ± 0.012	0.119 ± 0.009	0.344 ± 0.015	2.963 ± 0.234	1.721 ± 0.075	0.292	0.092	46.06	0.3179	0.3310	1.174	98.232 ± 0.528	1.490 ± 0.047	12.058 ± 1.306
UPI	beta43	0.20	0.164	-0.036 ± 0.004	-0.182 ± 0.022	0.012 ± 0.001	0.109 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	0.327 ± 0.022	0.572 ± 0.022	0.163	-0.037	-18.63	0.1149	0.1208	1.279	96.624 ± 0.724	0.544 ± 0.019	31.511 ± 1.863
UPI	beta44	0.11	0.188	0.078 ± 0.010	0.707 ± 0.094	0.067 ± 0.005	0.258 ± 0.010	0.073 ± 0.006	0.270 ± 0.012	6.004 ± 0.480	2.450 ± 0.108	0.165	0.055	50.32	0.2334	0.2399	1.355	98.553 ± 0.479	1.372 ± 0.072	5.627 ± 0.924

Table S205

Simulation 2 - Condition 61: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.126	0.016 ± 0.008	0.149 ± 0.075	0.042 ± 0.004	0.205 ± 0.009	0.042 ± 0.004	0.205 ± 0.010	3.482 ± 0.307	1.866 ± 0.091	0.116	0.006	5.46	0.1955	0.1956	1.308	95.498 ± 0.831	1.050 ± 0.046	5.466 ± 0.911
UPI	beta46	0.08	0.022	-0.058 ± 0.006	-0.731 ± 0.078	0.024 ± 0.002	0.156 ± 0.007	0.028 ± 0.002	0.166 ± 0.008	4.324 ± 0.363	2.079 ± 0.095	0.032	-0.048	-59.88	0.1327	0.1411	1.263	96.302 ± 0.757	0.771 ± 0.035	5.949 ± 0.948
UPI	beta47	0.08	0.022	-0.058 ± 0.005	-0.722 ± 0.066	0.017 ± 0.001	0.131 ± 0.005	0.021 ± 0.002	0.143 ± 0.006	3.204 ± 0.242	1.790 ± 0.074	0.034	-0.046	-58.04	0.1239	0.1323	1.162	97.106 ± 0.672	0.598 ± 0.020	7.235 ± 1.039
UPI	beta51	0.16	0.209	0.049 ± 0.005	0.307 ± 0.032	0.016 ± 0.001	0.127 ± 0.004	0.018 ± 0.001	0.136 ± 0.005	0.722 ± 0.045	0.850 ± 0.030	0.208	0.048	30.24	0.1318	0.1404	1.203	96.141 ± 0.772	0.598 ± 0.014	33.923 ± 1.898
UPI	beta52	0.16	0.247	0.087 ± 0.014	0.543 ± 0.085	0.115 ± 0.009	0.339 ± 0.013	0.123 ± 0.009	0.350 ± 0.015	4.788 ± 0.363	2.188 ± 0.093	0.233	0.073	45.62	0.3291	0.3371	1.561	98.392 ± 0.504	2.077 ± 0.120	4.662 ± 0.845
UPI	beta53	0.16	0.107	-0.053 ± 0.007	-0.332 ± 0.042	0.029 ± 0.002	0.169 ± 0.007	0.031 ± 0.002	0.177 ± 0.007	1.226 ± 0.090	1.107 ± 0.046	0.100	-0.060	-37.31	0.1606	0.1714	1.370	97.428 ± 0.635	0.908 ± 0.035	6.913 ± 1.017
UPI	beta54	0.16	0.244	0.084 ± 0.020	0.526 ± 0.127	0.257 ± 0.022	0.507 ± 0.022	0.264 ± 0.022	0.514 ± 0.024	10.312 ± 0.859	3.211 ± 0.148	0.218	0.058	36.17	0.4133	0.4174	1.473	98.875 ± 0.423	2.930 ± 0.143	3.537 ± 0.741
UPI	beta55	0.08	0.155	0.075 ± 0.008	0.937 ± 0.096	0.037 ± 0.003	0.192 ± 0.008	0.042 ± 0.004	0.206 ± 0.009	6.611 ± 0.550	2.571 ± 0.116	0.146	0.066	82.76	0.1702	0.1826	1.433	98.714 ± 0.452	1.077 ± 0.054	7.556 ± 1.060
UPI	beta56	0.08	0.074	-0.006 ± 0.004	-0.073 ± 0.050	0.010 ± 0.001	0.100 ± 0.004	0.010 ± 0.001	0.100 ± 0.004	1.577 ± 0.117	1.256 ± 0.053	0.069	-0.011	-13.31	0.0917	0.0924	1.294	96.945 ± 0.690	0.509 ± 0.018	12.862 ± 1.342
UPI	beta57	0.06	-0.020	-0.080 ± 0.008	-1.335 ± 0.132	0.039 ± 0.003	0.198 ± 0.008	0.045 ± 0.003	0.213 ± 0.009	12.606 ± 0.972	3.550 ± 0.150	-0.008	-0.068	-112.86	0.1844	0.1964	1.523	98.392 ± 0.504	1.179 ± 0.069	0.643 ± 0.321
UPI	beta58	0.06	0.003	-0.057 ± 0.007	-0.942 ± 0.122	0.033 ± 0.003	0.182 ± 0.008	0.036 ± 0.003	0.191 ± 0.008	10.086 ± 0.800	3.176 ± 0.139	0.014	-0.046	-76.96	0.1430	0.1503	1.544	96.624 ± 0.724	1.102 ± 0.071	3.215 ± 0.707

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S206

Simulation 2 - Condition 62: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.221	0.021 ± 0.006	0.103 ± 0.031	0.033 ± 0.002	0.182 ± 0.005	0.033 ± 0.002	0.183 ± 0.006	0.834 ± 0.049	0.913 ± 0.031	0.207	0.007	3.75	0.1658	0.1660	1.035	96.479 ± 0.631	0.737 ± 0.012	26.174 ± 1.506
LSAM	beta42	0.20	0.203	0.003 ± 0.006	0.014 ± 0.032	0.035 ± 0.002	0.187 ± 0.005	0.035 ± 0.002	0.187 ± 0.006	0.872 ± 0.050	0.934 ± 0.031	0.200	0.000	0.05	0.1682	0.1682	1.057	96.831 ± 0.600	0.774 ± 0.012	21.831 ± 1.415
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.074 ± 0.004	0.273 ± 0.008	0.197	-0.003	-1.71	0.0527	0.0528	1.059	95.657 ± 0.698	0.226 ± 0.001	94.014 ± 0.813
LSAM	beta44	0.11	0.137	0.027 ± 0.003	0.250 ± 0.025	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.600 ± 0.032	0.775 ± 0.024	0.133	0.023	20.49	0.0808	0.0839	0.991	95.423 ± 0.716	0.314 ± 0.004	43.192 ± 1.697
LSAM	beta45	0.11	0.110	-0.000 ± 0.003	-0.003 ± 0.023	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.451 ± 0.024	0.672 ± 0.022	0.107	-0.003	-2.90	0.0697	0.0698	0.946	94.601 ± 0.774	0.274 ± 0.003	37.089 ± 1.655
LSAM	beta46	0.08	0.056	-0.024 ± 0.002	-0.299 ± 0.030	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.849 ± 0.050	0.922 ± 0.031	0.060	-0.020	-24.49	0.0646	0.0675	0.985	93.662 ± 0.835	0.270 ± 0.005	24.178 ± 1.467
LSAM	beta47	0.08	0.062	-0.018 ± 0.002	-0.219 ± 0.029	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	0.781 ± 0.049	0.884 ± 0.031	0.063	-0.017	-21.67	0.0615	0.0639	1.015	94.601 ± 0.774	0.273 ± 0.005	22.887 ± 1.439
LSAM	beta51	0.16	0.172	0.012 ± 0.002	0.074 ± 0.015	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.196 ± 0.010	0.442 ± 0.014	0.171	0.011	7.06	0.0706	0.0715	0.972	94.484 ± 0.782	0.266 ± 0.001	73.122 ± 1.519
LSAM	beta52	0.16	0.224	0.064 ± 0.008	0.401 ± 0.048	0.050 ± 0.003	0.224 ± 0.007	0.054 ± 0.003	0.233 ± 0.008	2.112 ± 0.130	1.453 ± 0.050	0.220	0.060	37.71	0.1962	0.2053	1.063	98.239 ± 0.451	0.932 ± 0.018	17.488 ± 1.301
LSAM	beta53	0.16	0.147	-0.013 ± 0.003	-0.081 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.266 ± 0.014	0.516 ± 0.016	0.146	-0.014	-9.03	0.0811	0.0824	1.101	96.714 ± 0.611	0.352 ± 0.003	40.258 ± 1.680
LSAM	beta54	0.16	0.142	-0.018 ± 0.006	-0.115 ± 0.037	0.030 ± 0.002	0.172 ± 0.005	0.030 ± 0.002	0.173 ± 0.006	1.170 ± 0.070	1.082 ± 0.037	0.141	-0.019	-12.14	0.1497	0.1509	1.067	97.535 ± 0.531	0.720 ± 0.017	15.141 ± 1.228
LSAM	beta55	0.08	0.135	0.055 ± 0.003	0.682 ± 0.038	0.008 ± 0.000	0.088 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	1.664 ± 0.093	1.290 ± 0.040	0.130	0.050	62.62	0.0806	0.0949	1.025	91.901 ± 0.935	0.352 ± 0.007	40.610 ± 1.682
LSAM	beta56	0.08	0.068	-0.012 ± 0.001	-0.150 ± 0.018	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.287 ± 0.017	0.535 ± 0.018	0.071	-0.009	-11.74	0.0379	0.0391	0.973	93.662 ± 0.835	0.157 ± 0.002	49.883 ± 1.713
LSAM	beta57	0.06	-0.004	-0.064 ± 0.004	-1.065 ± 0.070	0.015 ± 0.001	0.122 ± 0.004	0.019 ± 0.001	0.138 ± 0.005	5.283 ± 0.313	2.299 ± 0.075	0.000	-0.060	-100.52	0.1060	0.1219	1.056	94.601 ± 0.774	0.506 ± 0.011	1.291 ± 0.387
LSAM	beta58	0.06	0.050	-0.010 ± 0.002	-0.160 ± 0.034	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	1.013 ± 0.065	1.007 ± 0.036	0.053	-0.007	-11.65	0.0486	0.0491	1.064	96.479 ± 0.631	0.249 ± 0.007	20.423 ± 1.381
QML	beta41	0.20	0.238	0.038 ± 0.002	0.189 ± 0.011	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.133 ± 0.006	0.365 ± 0.010	0.237	0.037	18.27	0.0605	0.0707	1.015	90.610 ± 0.999	0.249 ± 0.002	95.188 ± 0.733
QML	beta42	0.20	0.223	0.023 ± 0.002	0.113 ± 0.011	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.110 ± 0.005	0.331 ± 0.009	0.223	0.023	11.44	0.0619	0.0660	1.028	94.014 ± 0.813	0.251 ± 0.001	92.254 ± 0.916
QML	beta43	0.20	0.194	-0.006 ± 0.002	-0.031 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.192	-0.008	-4.08	0.0472	0.0479	1.006	94.131 ± 0.805	0.194 ± 0.001	98.592 ± 0.404
QML	beta44	0.11	0.074	-0.036 ± 0.003	-0.331 ± 0.028	0.008 ± 0.000	0.090 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.778 ± 0.039	0.882 ± 0.026	0.078	-0.032	-29.04	0.0919	0.0973	0.944	91.667 ± 0.947	0.333 ± 0.004	20.188 ± 1.375
QML	beta45	0.11	0.104	-0.006 ± 0.002	-0.058 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.367 ± 0.017	0.605 ± 0.017	0.102	-0.008	-7.22	0.0673	0.0677	0.990	94.953 ± 0.750	0.257 ± 0.002	39.906 ± 1.678
QML	beta46	0.08	0.137	0.057 ± 0.002	0.716 ± 0.020	0.002 ± 0.000	0.048 ± 0.001	0.006 ± 0.000	0.074 ± 0.002	0.866 ± 0.037	0.931 ± 0.021	0.136	0.056	69.63	0.0460	0.0722	1.034	81.221 ± 1.338	0.193 ± 0.003	84.155 ± 1.251
QML	beta47	0.08	0.134	0.054 ± 0.002	0.680 ± 0.021	0.002 ± 0.000	0.048 ± 0.001	0.005 ± 0.000	0.073 ± 0.002	0.821 ± 0.037	0.906 ± 0.022	0.131	0.051	63.28	0.0452	0.0679	0.946	80.399 ± 1.360	0.178 ± 0.002	86.502 ± 1.171
QML	beta51	0.16	0.185	0.025 ± 0.003	0.154 ± 0.019	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.345 ± 0.018	0.587 ± 0.018	0.185	0.025	15.38	0.0889	0.0922	0.960	91.549 ± 0.953	0.341 ± 0.005	59.742 ± 1.680
QML	beta52	0.16	0.150	-0.010 ± 0.003	-0.064 ± 0.017	0.006 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.258 ± 0.013	0.507 ± 0.015	0.147	-0.013	-8.34	0.0777	0.0788	0.985	93.075 ± 0.870	0.311 ± 0.005	49.061 ± 1.713
QML	beta53	0.16	0.120	-0.040 ± 0.002	-0.247 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.244 ± 0.012	0.494 ± 0.014	0.119	-0.041	-25.41	0.0647	0.0764	1.005	90.141 ± 1.021	0.269 ± 0.002	42.371 ± 1.693
QML	beta54	0.16	0.172	0.012 ± 0.002	0.077 ± 0.014	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.008	0.412 ± 0.012	0.170	0.010	6.06	0.0648	0.0655	1.005	94.249 ± 0.798	0.256 ± 0.001	76.056 ± 1.462
QML	beta55	0.08	0.083	0.003 ± 0.005	0.035 ± 0.056	0.017 ± 0.001	0.132 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	2.713 ± 0.118	1.647 ± 0.046	0.073	-0.007	-8.89	0.1463	0.1465	0.815	78.756 ± 1.401	0.421 ± 0.011	27.465 ± 1.529
QML	beta56	0.08	0.114	0.034 ± 0.002	0.420 ± 0.023	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.631 ± 0.030	0.794 ± 0.021	0.114	0.034	42.73	0.0559	0.0656	0.928	84.859 ± 1.228	0.196 ± 0.003	63.028 ± 1.654
QML	beta57	0.06	0.130	0.070 ± 0.004	1.164 ± 0.070	0.015 ± 0.001	0.123 ± 0.003	0.020 ± 0.001	0.141 ± 0.004	5.561 ± 0.247	2.358 ± 0.061	0.136	0.076	127.28	0.1387	0.1584	0.847	75.235 ± 1.479	0.409 ± 0.011	37.441 ± 1.658
QML	beta58	0.06	0.102	0.042 ± 0.001	0.707 ± 0.021	0.001 ± 0.000	0.037 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.876 ± 0.039	0.936 ± 0.022	0.101	0.041	67.78	0.0374	0.0552	1.045	84.507 ± 1.240	0.151 ± 0.002	80.282 ± 1.363
UPI	beta41	0.20	0.240	0.040 ± 0.006	0.202 ± 0.031	0.033 ± 0.002	0.183 ± 0.006	0.035 ± 0.002	0.187 ± 0.007	0.875 ± 0.056	0.935 ± 0.034	0.235	0.035	17.35	0.1655	0.1691	1.196	98.357 ± 0.436	0.857 ± 0.048	26.878 ± 1.519
UPI	beta42	0.20	0.245	0.045 ± 0.006	0.224 ± 0.029	0.028 ± 0.002	0.168 ± 0.005	0.030 ± 0.002	0.174 ± 0.006	0.755 ± 0.043	0.869 ± 0.028	0.246	0.046	22.93	0.1630	0.1694	1.129	96.596 ± 0.621	0.744 ± 0.025	32.864 ± 1.609
UPI	beta43	0.20	0.177	-0.023 ± 0.002	-0.115 ± 0.010	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.107 ± 0.005	0.326 ± 0.010	0.174	-0.026	-13.12	0.0596	0.0651	1.058	93.310 ± 0.856	0.254 ± 0.004	81.573 ± 1.328
UPI	beta44	0.11	0.151	0.041 ± 0.004	0.370 ± 0.037	0.014 ± 0.001	0.119 ± 0.004	0.016 ± 0.001	0.126 ± 0.005	1.313 ± 0.087	1.146 ± 0.042	0.138	0.028	25.40	0.1063	0.1099	1.138	97.183 ± 0.567	0.532 ± 0.018	19.836 ± 1.366

Table S206

Simulation 2 - Condition 62: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.121	0.011 ± 0.003	0.099 ± 0.029	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.735 ± 0.050	0.857 ± 0.033	0.119	0.009	7.98	0.0875	0.0879	1.250	97.066 ± 0.578	0.459 ± 0.021	21.127 ± 1.398
UPI	beta46	0.08	0.048	-0.032 ± 0.003	-0.394 ± 0.032	0.006 ± 0.000	0.074 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	1.020 ± 0.064	1.010 ± 0.035	0.054	-0.026	-32.07	0.0642	0.0691	1.170	95.305 ± 0.725	0.341 ± 0.017	15.258 ± 1.232
UPI	beta47	0.08	0.048	-0.032 ± 0.002	-0.402 ± 0.028	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.838 ± 0.051	0.915 ± 0.031	0.050	-0.030	-38.04	0.0589	0.0663	1.088	94.249 ± 0.798	0.281 ± 0.007	19.131 ± 1.348
UPI	beta51	0.16	0.199	0.039 ± 0.002	0.246 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.264 ± 0.013	0.514 ± 0.014	0.199	0.039	24.62	0.0728	0.0828	1.086	94.249 ± 0.798	0.307 ± 0.003	76.643 ± 1.450
UPI	beta52	0.16	0.216	0.056 ± 0.007	0.349 ± 0.043	0.040 ± 0.003	0.201 ± 0.006	0.043 ± 0.003	0.208 ± 0.007	1.698 ± 0.109	1.303 ± 0.047	0.205	0.045	28.11	0.1859	0.1913	1.440	98.239 ± 0.451	1.135 ± 0.052	14.319 ± 1.200
UPI	beta53	0.16	0.109	-0.051 ± 0.003	-0.317 ± 0.021	0.009 ± 0.001	0.097 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.469 ± 0.027	0.685 ± 0.022	0.110	-0.050	-31.07	0.0900	0.1028	1.224	94.836 ± 0.758	0.467 ± 0.011	21.244 ± 1.401
UPI	beta54	0.16	0.251	0.091 ± 0.010	0.571 ± 0.065	0.092 ± 0.006	0.304 ± 0.010	0.100 ± 0.007	0.317 ± 0.012	3.920 ± 0.257	1.980 ± 0.072	0.217	0.057	35.77	0.2647	0.2708	1.416	98.592 ± 0.404	1.684 ± 0.072	8.216 ± 0.941
UPI	beta55	0.08	0.131	0.051 ± 0.003	0.637 ± 0.039	0.008 ± 0.000	0.091 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	1.692 ± 0.099	1.301 ± 0.042	0.128	0.048	59.60	0.0871	0.0993	1.425	97.183 ± 0.567	0.507 ± 0.020	24.883 ± 1.481
UPI	beta56	0.08	0.075	-0.005 ± 0.002	-0.069 ± 0.021	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.387 ± 0.021	0.622 ± 0.020	0.075	-0.005	-6.68	0.0455	0.0458	1.208	95.423 ± 0.716	0.234 ± 0.008	36.854 ± 1.653
UPI	beta57	0.06	-0.003	-0.063 ± 0.004	-1.051 ± 0.060	0.011 ± 0.001	0.105 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	4.159 ± 0.254	2.039 ± 0.068	0.001	-0.059	-98.50	0.0930	0.1102	1.450	96.596 ± 0.621	0.596 ± 0.027	0.822 ± 0.309
UPI	beta58	0.06	0.009	-0.051 ± 0.003	-0.855 ± 0.056	0.009 ± 0.001	0.097 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	3.359 ± 0.209	1.833 ± 0.062	0.019	-0.041	-67.97	0.0893	0.0982	1.383	96.009 ± 0.671	0.528 ± 0.021	4.577 ± 0.716

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S207

Simulation 2 - Condition 63: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.196	-0.004 ± 0.004	-0.018 ± 0.022	0.018 ± 0.001	0.135 ± 0.003	0.018 ± 0.001	0.135 ± 0.004	0.453 ± 0.023	0.673 ± 0.020	0.194	-0.006	-3.04	0.1391	0.1392	0.998	96.354 ± 0.623	0.527 ± 0.005	35.138 ± 1.587
LSAM	beta42	0.20	0.205	0.005 ± 0.005	0.026 ± 0.023	0.020 ± 0.001	0.141 ± 0.003	0.020 ± 0.001	0.141 ± 0.004	0.496 ± 0.024	0.704 ± 0.021	0.212	0.012	5.78	0.1388	0.1393	1.001	96.685 ± 0.595	0.553 ± 0.006	35.249 ± 1.588
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.007 ± 0.011	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.109 ± 0.005	0.330 ± 0.010	0.201	0.001	0.44	0.0661	0.0661	0.967	94.696 ± 0.745	0.250 ± 0.002	89.171 ± 1.033
LSAM	beta44	0.11	0.125	0.015 ± 0.003	0.134 ± 0.028	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	0.703 ± 0.035	0.839 ± 0.025	0.122	0.012	11.20	0.0903	0.0912	0.930	93.702 ± 0.808	0.332 ± 0.004	36.796 ± 1.603
LSAM	beta45	0.11	0.116	0.006 ± 0.003	0.054 ± 0.026	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.637 ± 0.037	0.798 ± 0.027	0.111	0.001	1.34	0.0785	0.0785	0.888	93.039 ± 0.846	0.305 ± 0.004	36.906 ± 1.604
LSAM	beta46	0.08	0.078	-0.002 ± 0.002	-0.029 ± 0.025	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.558 ± 0.032	0.747 ± 0.025	0.076	-0.004	-4.67	0.0563	0.0564	0.895	94.144 ± 0.781	0.210 ± 0.003	40.000 ± 1.628
LSAM	beta47	0.08	0.077	-0.003 ± 0.002	-0.042 ± 0.024	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.520 ± 0.029	0.721 ± 0.023	0.076	-0.004	-5.35	0.0559	0.0560	0.920	93.370 ± 0.827	0.208 ± 0.003	38.895 ± 1.621
LSAM	beta51	0.16	0.165	0.005 ± 0.003	0.031 ± 0.018	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.303 ± 0.014	0.551 ± 0.016	0.165	0.005	3.16	0.0884	0.0885	0.935	94.475 ± 0.759	0.323 ± 0.002	53.260 ± 1.659
LSAM	beta52	0.16	0.171	0.011 ± 0.005	0.070 ± 0.032	0.024 ± 0.001	0.154 ± 0.004	0.024 ± 0.001	0.154 ± 0.005	0.924 ± 0.047	0.961 ± 0.029	0.170	0.010	6.22	0.1512	0.1515	0.981	95.359 ± 0.699	0.590 ± 0.006	24.641 ± 1.432
LSAM	beta53	0.16	0.155	-0.005 ± 0.003	-0.032 ± 0.020	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.375 ± 0.018	0.612 ± 0.018	0.153	-0.007	-4.33	0.0969	0.0971	0.971	94.917 ± 0.730	0.373 ± 0.002	39.006 ± 1.621
LSAM	beta54	0.16	0.155	-0.005 ± 0.004	-0.028 ± 0.027	0.017 ± 0.001	0.130 ± 0.003	0.017 ± 0.001	0.130 ± 0.004	0.662 ± 0.035	0.814 ± 0.026	0.155	-0.005	-3.39	0.1270	0.1271	1.012	95.801 ± 0.667	0.517 ± 0.004	24.862 ± 1.437
LSAM	beta55	0.08	0.097	0.017 ± 0.003	0.209 ± 0.038	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	1.382 ± 0.078	1.176 ± 0.038	0.098	0.018	22.02	0.0794	0.0813	0.906	93.370 ± 0.827	0.329 ± 0.005	29.061 ± 1.509
LSAM	beta56	0.08	0.078	-0.002 ± 0.002	-0.024 ± 0.022	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.432 ± 0.024	0.657 ± 0.021	0.076	-0.004	-4.59	0.0484	0.0485	0.908	91.823 ± 0.911	0.187 ± 0.003	43.646 ± 1.649
LSAM	beta57	0.06	0.043	-0.017 ± 0.003	-0.280 ± 0.056	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	2.863 ± 0.163	1.692 ± 0.056	0.040	-0.020	-32.72	0.0897	0.0918	0.925	93.370 ± 0.827	0.363 ± 0.005	8.398 ± 0.922
LSAM	beta58	0.06	0.057	-0.003 ± 0.002	-0.048 ± 0.028	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.703 ± 0.041	0.839 ± 0.028	0.057	-0.003	-5.29	0.0444	0.0446	0.944	94.033 ± 0.787	0.186 ± 0.003	31.492 ± 1.544
QML	beta41	0.20	0.211	0.011 ± 0.003	0.054 ± 0.013	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.209	0.009	4.69	0.0778	0.0784	1.010	95.359 ± 0.699	0.301 ± 0.002	77.459 ± 1.389
QML	beta42	0.20	0.212	0.012 ± 0.003	0.059 ± 0.013	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.214	0.014	7.02	0.0747	0.0760	1.011	95.580 ± 0.683	0.307 ± 0.002	76.685 ± 1.406
QML	beta43	0.20	0.192	-0.008 ± 0.002	-0.039 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.004	0.301 ± 0.009	0.193	-0.007	-3.49	0.0596	0.0600	0.962	93.260 ± 0.833	0.225 ± 0.001	91.492 ± 0.927
QML	beta44	0.11	0.081	-0.029 ± 0.003	-0.260 ± 0.025	0.007 ± 0.000	0.083 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.641 ± 0.030	0.800 ± 0.022	0.085	-0.025	-23.10	0.0862	0.0899	0.947	92.597 ± 0.870	0.309 ± 0.003	24.641 ± 1.432
QML	beta45	0.11	0.102	-0.008 ± 0.002	-0.070 ± 0.022	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.459 ± 0.023	0.677 ± 0.020	0.104	-0.006	-5.66	0.0679	0.0682	0.922	91.823 ± 0.911	0.268 ± 0.003	36.243 ± 1.598
QML	beta46	0.08	0.111	0.031 ± 0.002	0.382 ± 0.020	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.503 ± 0.026	0.709 ± 0.020	0.107	0.027	33.63	0.0481	0.0551	0.933	90.608 ± 0.970	0.175 ± 0.003	72.597 ± 1.483
QML	beta47	0.08	0.112	0.032 ± 0.002	0.404 ± 0.020	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.509 ± 0.030	0.714 ± 0.022	0.107	0.027	34.00	0.0440	0.0518	0.907	89.945 ± 1.000	0.167 ± 0.002	78.232 ± 1.372
QML	beta51	0.16	0.176	0.016 ± 0.003	0.103 ± 0.019	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.323 ± 0.016	0.568 ± 0.017	0.174	0.014	8.92	0.0885	0.0896	0.973	95.470 ± 0.691	0.341 ± 0.003	54.033 ± 1.657
QML	beta52	0.16	0.145	-0.015 ± 0.003	-0.094 ± 0.018	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.305 ± 0.014	0.552 ± 0.016	0.144	-0.016	-10.14	0.0889	0.0903	0.961	93.591 ± 0.814	0.328 ± 0.002	42.210 ± 1.642
QML	beta53	0.16	0.125	-0.035 ± 0.003	-0.219 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.324 ± 0.015	0.569 ± 0.016	0.126	-0.034	-21.33	0.0865	0.0930	0.961	91.602 ± 0.922	0.317 ± 0.002	35.580 ± 1.591
QML	beta54	0.16	0.162	0.002 ± 0.003	0.011 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.244 ± 0.012	0.494 ± 0.015	0.164	0.004	2.75	0.0778	0.0780	1.019	95.138 ± 0.715	0.315 ± 0.002	54.254 ± 1.656
QML	beta55	0.08	0.084	0.004 ± 0.003	0.044 ± 0.041	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	1.490 ± 0.082	1.221 ± 0.039	0.090	0.010	11.91	0.0874	0.0880	0.878	89.613 ± 1.014	0.336 ± 0.006	30.939 ± 1.537
QML	beta56	0.08	0.096	0.016 ± 0.002	0.195 ± 0.023	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.497 ± 0.030	0.705 ± 0.024	0.093	0.013	15.75	0.0468	0.0485	0.892	89.945 ± 1.000	0.189 ± 0.003	54.475 ± 1.655
QML	beta57	0.06	0.086	0.026 ± 0.003	0.430 ± 0.047	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	2.203 ± 0.128	1.484 ± 0.049	0.077	0.017	28.75	0.0780	0.0799	0.938	91.713 ± 0.916	0.314 ± 0.006	17.569 ± 1.265
QML	beta58	0.06	0.081	0.021 ± 0.001	0.349 ± 0.023	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.597 ± 0.033	0.773 ± 0.024	0.078	0.018	30.64	0.0392	0.0433	0.924	92.265 ± 0.888	0.150 ± 0.002	59.448 ± 1.632
UPI	beta41	0.20	0.225	0.025 ± 0.005	0.123 ± 0.023	0.018 ± 0.001	0.136 ± 0.004	0.019 ± 0.001	0.138 ± 0.005	0.475 ± 0.030	0.689 ± 0.025	0.218	0.018	9.06	0.1236	0.1249	1.040	96.354 ± 0.623	0.553 ± 0.010	45.414 ± 1.655
UPI	beta42	0.20	0.216	0.016 ± 0.004	0.080 ± 0.022	0.017 ± 0.001	0.132 ± 0.004	0.018 ± 0.001	0.133 ± 0.004	0.441 ± 0.025	0.664 ± 0.022	0.223	0.023	11.55	0.1277	0.1297	1.097	97.790 ± 0.489	0.568 ± 0.011	44.420 ± 1.652
UPI	beta43	0.20	0.198	-0.002 ± 0.002	-0.011 ± 0.011	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.005	0.321 ± 0.010	0.197	-0.003	-1.51	0.0632	0.0633	1.001	94.696 ± 0.745	0.252 ± 0.002	89.061 ± 1.038
UPI	beta44	0.11	0.118	0.008 ± 0.003	0.075 ± 0.029	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.770 ± 0.046	0.878 ± 0.030	0.117	0.007	6.22	0.0841	0.0844	1.017	95.912 ± 0.658	0.383 ± 0.008	29.282 ± 1.513

Table S207

Simulation 2 - Condition 63: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.111	0.001 ± 0.003	0.006 ± 0.027	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	0.676 ± 0.038	0.822 ± 0.027	0.109	-0.001	-1.16	0.0814	0.0814	0.989	93.923 ± 0.794	0.351 ± 0.007	29.171 ± 1.511
UPI	beta46	0.08	0.069	-0.011 ± 0.002	-0.140 ± 0.027	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.674 ± 0.043	0.821 ± 0.030	0.069	-0.011	-13.88	0.0553	0.0564	0.946	94.254 ± 0.774	0.240 ± 0.006	33.039 ± 1.564
UPI	beta47	0.08	0.073	-0.007 ± 0.002	-0.082 ± 0.025	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.561 ± 0.036	0.749 ± 0.027	0.071	-0.009	-10.84	0.0509	0.0517	0.979	92.707 ± 0.864	0.229 ± 0.006	36.354 ± 1.599
UPI	beta51	0.16	0.175	0.015 ± 0.003	0.092 ± 0.018	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.288 ± 0.014	0.536 ± 0.015	0.175	0.015	9.16	0.0839	0.0851	0.958	95.470 ± 0.691	0.318 ± 0.002	60.110 ± 1.628
UPI	beta52	0.16	0.182	0.022 ± 0.005	0.140 ± 0.029	0.019 ± 0.001	0.138 ± 0.004	0.019 ± 0.001	0.140 ± 0.004	0.761 ± 0.039	0.872 ± 0.027	0.186	0.026	16.26	0.1359	0.1384	1.117	95.470 ± 0.691	0.603 ± 0.011	26.961 ± 1.475
UPI	beta53	0.16	0.150	-0.010 ± 0.003	-0.060 ± 0.019	0.009 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.337 ± 0.016	0.581 ± 0.017	0.148	-0.012	-7.22	0.0901	0.0908	1.022	94.586 ± 0.752	0.370 ± 0.003	36.243 ± 1.598
UPI	beta54	0.16	0.160	-0.000 ± 0.005	-0.002 ± 0.029	0.020 ± 0.001	0.141 ± 0.004	0.020 ± 0.001	0.141 ± 0.004	0.774 ± 0.040	0.880 ± 0.027	0.160	0.000	0.22	0.1376	0.1376	1.111	97.238 ± 0.545	0.613 ± 0.011	20.442 ± 1.341
UPI	beta55	0.08	0.094	0.014 ± 0.003	0.175 ± 0.037	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	1.253 ± 0.068	1.120 ± 0.035	0.094	0.014	17.53	0.0831	0.0843	1.008	94.365 ± 0.767	0.350 ± 0.006	25.304 ± 1.445
UPI	beta56	0.08	0.077	-0.003 ± 0.002	-0.033 ± 0.022	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.421 ± 0.024	0.649 ± 0.022	0.077	-0.003	-3.41	0.0482	0.0483	0.964	91.934 ± 0.905	0.196 ± 0.003	41.436 ± 1.637
UPI	beta57	0.06	0.041	-0.019 ± 0.003	-0.321 ± 0.049	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	2.308 ± 0.123	1.519 ± 0.047	0.035	-0.025	-41.98	0.0824	0.0862	1.074	95.028 ± 0.723	0.375 ± 0.009	6.851 ± 0.840
UPI	beta58	0.06	0.055	-0.005 ± 0.002	-0.089 ± 0.031	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.853 ± 0.051	0.924 ± 0.032	0.055	-0.005	-7.92	0.0462	0.0465	1.044	95.580 ± 0.683	0.226 ± 0.006	26.961 ± 1.475

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S208

Simulation 2 - Condition 64: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.003	-0.016 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.156 ± 0.007	0.395 ± 0.011	0.193	-0.007	-3.54	0.0812	0.0815	0.968	95.082 ± 0.692	0.300 ± 0.002	72.131 ± 1.435
LSAM	beta42	0.20	0.204	0.004 ± 0.003	0.022 ± 0.013	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.175 ± 0.009	0.418 ± 0.012	0.202	0.002	1.20	0.0815	0.0815	0.970	94.160 ± 0.751	0.317 ± 0.002	70.492 ± 1.460
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.201	0.001	0.69	0.0400	0.0400	0.965	95.594 ± 0.657	0.153 ± 0.001	99.795 ± 0.145
LSAM	beta44	0.11	0.124	0.014 ± 0.002	0.128 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.245 ± 0.012	0.495 ± 0.014	0.123	0.013	11.82	0.0491	0.0508	0.896	93.033 ± 0.815	0.185 ± 0.001	75.307 ± 1.380
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.013 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.193 ± 0.010	0.439 ± 0.013	0.112	0.002	2.17	0.0447	0.0448	0.919	93.033 ± 0.815	0.174 ± 0.001	71.311 ± 1.448
LSAM	beta46	0.08	0.076	-0.004 ± 0.001	-0.048 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.149 ± 0.007	0.386 ± 0.011	0.077	-0.003	-4.30	0.0299	0.0301	0.902	91.086 ± 0.912	0.108 ± 0.001	76.025 ± 1.367
LSAM	beta47	0.08	0.074	-0.006 ± 0.001	-0.074 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.155 ± 0.008	0.394 ± 0.012	0.074	-0.006	-7.13	0.0270	0.0276	0.916	92.213 ± 0.858	0.111 ± 0.001	74.898 ± 1.388
LSAM	beta51	0.16	0.167	0.007 ± 0.002	0.045 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.167	0.007	4.41	0.0529	0.0534	0.990	95.697 ± 0.650	0.197 ± 0.001	91.701 ± 0.883
LSAM	beta52	0.16	0.168	0.008 ± 0.003	0.049 ± 0.018	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.301 ± 0.014	0.549 ± 0.016	0.169	0.009	5.34	0.0839	0.0843	0.980	94.160 ± 0.751	0.336 ± 0.002	51.127 ± 1.600
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.016 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.156	-0.004	-2.39	0.0595	0.0596	0.992	94.672 ± 0.719	0.223 ± 0.001	78.996 ± 1.304
LSAM	beta54	0.16	0.156	-0.004 ± 0.003	-0.027 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.258 ± 0.013	0.508 ± 0.015	0.155	-0.005	-2.96	0.0839	0.0840	0.958	94.570 ± 0.725	0.305 ± 0.002	54.918 ± 1.593
LSAM	beta55	0.08	0.096	0.016 ± 0.001	0.204 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.372 ± 0.017	0.610 ± 0.016	0.099	0.019	23.41	0.0445	0.0483	0.975	92.930 ± 0.820	0.176 ± 0.001	60.246 ± 1.566
LSAM	beta56	0.08	0.076	-0.004 ± 0.001	-0.049 ± 0.012	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.138 ± 0.007	0.371 ± 0.011	0.076	-0.004	-5.14	0.0271	0.0274	0.866	90.369 ± 0.944	0.100 ± 0.001	80.943 ± 1.257
LSAM	beta57	0.06	0.045	-0.015 ± 0.002	-0.247 ± 0.028	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.802 ± 0.036	0.895 ± 0.024	0.042	-0.018	-30.37	0.0493	0.0526	0.968	92.418 ± 0.847	0.196 ± 0.002	15.061 ± 1.145
LSAM	beta58	0.06	0.057	-0.003 ± 0.001	-0.055 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.217 ± 0.012	0.466 ± 0.015	0.057	-0.003	-4.74	0.0265	0.0267	0.951	94.365 ± 0.738	0.103 ± 0.001	60.553 ± 1.564
QML	beta41	0.20	0.212	0.012 ± 0.002	0.061 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.210	0.010	5.05	0.0485	0.0495	0.994	92.930 ± 0.820	0.188 ± 0.001	98.566 ± 0.381
QML	beta42	0.20	0.212	0.012 ± 0.002	0.060 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.211	0.011	5.50	0.0477	0.0490	1.012	94.570 ± 0.725	0.192 ± 0.001	98.770 ± 0.353
QML	beta43	0.20	0.193	-0.007 ± 0.001	-0.034 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.194	-0.006	-3.23	0.0357	0.0363	0.969	93.648 ± 0.781	0.142 ± 0.001	99.898 ± 0.102
QML	beta44	0.11	0.087	-0.023 ± 0.002	-0.208 ± 0.015	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.257 ± 0.012	0.507 ± 0.014	0.089	-0.021	-19.47	0.0496	0.0541	0.949	90.574 ± 0.935	0.189 ± 0.001	46.721 ± 1.597
QML	beta45	0.11	0.103	-0.007 ± 0.001	-0.063 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.167 ± 0.008	0.409 ± 0.012	0.103	-0.007	-6.81	0.0437	0.0443	0.963	92.725 ± 0.831	0.168 ± 0.001	68.135 ± 1.491
QML	beta46	0.08	0.109	0.029 ± 0.001	0.358 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.245 ± 0.011	0.495 ± 0.012	0.107	0.027	34.14	0.0263	0.0379	0.958	82.172 ± 1.225	0.103 ± 0.001	98.053 ± 0.442
QML	beta47	0.08	0.109	0.029 ± 0.001	0.358 ± 0.010	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.231 ± 0.009	0.481 ± 0.010	0.107	0.027	34.30	0.0264	0.0381	0.989	80.635 ± 1.265	0.100 ± 0.001	99.180 ± 0.289
QML	beta51	0.16	0.181	0.021 ± 0.002	0.130 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.122 ± 0.005	0.349 ± 0.009	0.181	0.021	13.04	0.0555	0.0593	1.036	94.980 ± 0.699	0.211 ± 0.001	92.316 ± 0.853
QML	beta52	0.16	0.142	-0.018 ± 0.002	-0.114 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.119 ± 0.005	0.345 ± 0.009	0.141	-0.019	-12.11	0.0515	0.0550	0.971	92.008 ± 0.868	0.198 ± 0.001	79.201 ± 1.299
QML	beta53	0.16	0.126	-0.034 ± 0.002	-0.214 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.143 ± 0.006	0.378 ± 0.009	0.124	-0.036	-22.20	0.0498	0.0612	0.991	88.320 ± 1.028	0.194 ± 0.001	71.516 ± 1.445
QML	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.162	0.002	0.99	0.0527	0.0528	0.972	94.980 ± 0.699	0.198 ± 0.001	86.578 ± 1.091
QML	beta55	0.08	0.092	0.012 ± 0.002	0.149 ± 0.023	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.522 ± 0.027	0.723 ± 0.022	0.093	0.013	16.30	0.0556	0.0571	0.969	89.242 ± 0.992	0.215 ± 0.004	49.385 ± 1.600
QML	beta56	0.08	0.094	0.014 ± 0.001	0.169 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.190 ± 0.009	0.435 ± 0.012	0.092	0.012	15.52	0.0300	0.0325	0.889	88.422 ± 1.024	0.112 ± 0.001	87.193 ± 1.070
QML	beta57	0.06	0.083	0.023 ± 0.002	0.376 ± 0.029	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.935 ± 0.052	0.967 ± 0.030	0.077	0.017	27.91	0.0510	0.0537	0.982	92.623 ± 0.837	0.206 ± 0.004	37.295 ± 1.548
QML	beta58	0.06	0.081	0.021 ± 0.001	0.353 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.281 ± 0.013	0.530 ± 0.013	0.081	0.021	34.62	0.0231	0.0311	0.987	86.988 ± 1.077	0.092 ± 0.001	94.672 ± 0.719
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.029 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.124 ± 0.006	0.351 ± 0.010	0.205	0.005	2.35	0.0714	0.0716	1.034	96.414 ± 0.595	0.284 ± 0.003	81.967 ± 1.231
UPI	beta42	0.20	0.209	0.009 ± 0.002	0.046 ± 0.012	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.143 ± 0.008	0.378 ± 0.012	0.209	0.009	4.30	0.0734	0.0739	1.027	96.107 ± 0.619	0.302 ± 0.003	78.586 ± 1.313
UPI	beta43	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.198	-0.002	-0.90	0.0386	0.0387	0.942	94.262 ± 0.744	0.150 ± 0.001	99.693 ± 0.177
UPI	beta44	0.11	0.118	0.008 ± 0.002	0.069 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.233 ± 0.012	0.482 ± 0.015	0.116	0.006	5.29	0.0489	0.0493	0.919	94.365 ± 0.738	0.189 ± 0.002	71.824 ± 1.440

Table S208

Simulation 2 - Condition 64: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.110	0.000 ± 0.002	0.004 ± 0.015	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.207 ± 0.010	0.455 ± 0.013	0.110	0.000	-0.29	0.0471	0.0471	0.940	93.135 ± 0.809	0.185 ± 0.002	66.803 ± 1.507
UPI	beta46	0.08	0.074	-0.006 ± 0.001	-0.073 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.154 ± 0.008	0.393 ± 0.012	0.075	-0.005	-6.13	0.0299	0.0303	0.940	90.984 ± 0.917	0.114 ± 0.002	72.746 ± 1.425
UPI	beta47	0.08	0.073	-0.007 ± 0.001	-0.089 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.160 ± 0.010	0.400 ± 0.013	0.072	-0.008	-9.53	0.0271	0.0281	0.955	92.316 ± 0.853	0.117 ± 0.002	72.336 ± 1.432
UPI	beta51	0.16	0.172	0.012 ± 0.002	0.078 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.319 ± 0.009	0.171	0.011	7.10	0.0522	0.0535	1.003	95.902 ± 0.635	0.195 ± 0.001	94.980 ± 0.699
UPI	beta52	0.16	0.173	0.013 ± 0.003	0.082 ± 0.018	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.310 ± 0.016	0.557 ± 0.017	0.175	0.015	9.13	0.0797	0.0810	0.992	95.902 ± 0.635	0.343 ± 0.003	54.713 ± 1.593
UPI	beta53	0.16	0.154	-0.006 ± 0.002	-0.035 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.126 ± 0.006	0.355 ± 0.010	0.153	-0.007	-4.48	0.0573	0.0578	0.978	94.262 ± 0.744	0.217 ± 0.001	79.816 ± 1.285
UPI	beta54	0.16	0.153	-0.007 ± 0.003	-0.045 ± 0.018	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.302 ± 0.015	0.550 ± 0.016	0.153	-0.007	-4.35	0.0869	0.0872	0.989	95.492 ± 0.664	0.340 ± 0.003	47.029 ± 1.598
UPI	beta55	0.08	0.092	0.012 ± 0.002	0.152 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.376 ± 0.020	0.613 ± 0.019	0.092	0.012	15.56	0.0451	0.0468	1.004	94.365 ± 0.738	0.187 ± 0.002	53.279 ± 1.597
UPI	beta56	0.08	0.076	-0.004 ± 0.001	-0.044 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.139 ± 0.007	0.372 ± 0.011	0.076	-0.004	-4.61	0.0270	0.0272	0.880	90.574 ± 0.935	0.102 ± 0.001	79.918 ± 1.282
UPI	beta57	0.06	0.046	-0.014 ± 0.002	-0.238 ± 0.028	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.843 ± 0.043	0.918 ± 0.027	0.044	-0.016	-26.67	0.0488	0.0514	0.981	93.238 ± 0.804	0.205 ± 0.002	15.471 ± 1.158
UPI	beta58	0.06	0.057	-0.003 ± 0.001	-0.053 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.253 ± 0.014	0.503 ± 0.016	0.056	-0.004	-6.34	0.0275	0.0277	0.980	94.775 ± 0.712	0.115 ± 0.002	55.225 ± 1.592

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S209

Simulation 2 - Condition 65: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.003	-0.012 ± 0.013	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.161 ± 0.007	0.401 ± 0.011	0.197	-0.003	-1.63	0.0798	0.0799	0.998	95.547 ± 0.656	0.314 ± 0.001	70.648 ± 1.449
LSAM	beta42	0.20	0.199	-0.001 ± 0.003	-0.007 ± 0.014	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.189 ± 0.009	0.435 ± 0.012	0.197	-0.003	-1.44	0.0820	0.0821	0.960	94.028 ± 0.754	0.328 ± 0.002	66.194 ± 1.505
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.200	0.000	0.09	0.0498	0.0498	0.976	93.320 ± 0.794	0.189 ± 0.001	98.785 ± 0.348
LSAM	beta44	0.11	0.115	0.005 ± 0.002	0.044 ± 0.017	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.286 ± 0.014	0.534 ± 0.016	0.114	0.004	3.88	0.0556	0.0557	0.901	91.397 ± 0.892	0.207 ± 0.002	59.919 ± 1.559
LSAM	beta45	0.11	0.112	0.002 ± 0.002	0.021 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.258 ± 0.014	0.508 ± 0.016	0.109	-0.001	-1.00	0.0524	0.0524	0.895	93.219 ± 0.800	0.196 ± 0.002	62.652 ± 1.539
LSAM	beta46	0.08	0.082	0.002 ± 0.001	0.022 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.186 ± 0.010	0.431 ± 0.013	0.082	0.002	1.92	0.0315	0.0315	0.859	90.587 ± 0.929	0.116 ± 0.001	77.632 ± 1.326
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.189 ± 0.010	0.435 ± 0.014	0.078	-0.002	-2.46	0.0327	0.0327	0.863	90.891 ± 0.915	0.118 ± 0.001	74.899 ± 1.379
LSAM	beta51	0.16	0.163	0.003 ± 0.002	0.017 ± 0.013	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.176 ± 0.008	0.419 ± 0.012	0.162	0.002	1.14	0.0656	0.0656	0.946	93.522 ± 0.783	0.249 ± 0.001	73.279 ± 1.408
LSAM	beta52	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.018	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.307 ± 0.013	0.554 ± 0.015	0.156	-0.004	-2.20	0.0873	0.0873	0.980	94.636 ± 0.717	0.341 ± 0.001	46.356 ± 1.586
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.009	0.450 ± 0.013	0.160	0.000	0.16	0.0717	0.0717	0.955	94.838 ± 0.704	0.269 ± 0.001	64.676 ± 1.521
LSAM	beta54	0.16	0.157	-0.003 ± 0.003	-0.020 ± 0.017	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.296 ± 0.013	0.544 ± 0.015	0.158	-0.002	-1.02	0.0903	0.0904	0.967	94.838 ± 0.704	0.330 ± 0.002	48.077 ± 1.590
LSAM	beta55	0.08	0.083	0.003 ± 0.002	0.036 ± 0.024	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.573 ± 0.029	0.757 ± 0.023	0.084	0.004	4.39	0.0571	0.0572	0.853	90.789 ± 0.920	0.202 ± 0.002	42.814 ± 1.574
LSAM	beta56	0.08	0.081	0.001 ± 0.001	0.012 ± 0.015	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.232 ± 0.012	0.482 ± 0.015	0.079	-0.001	-0.84	0.0348	0.0348	0.878	92.105 ± 0.858	0.133 ± 0.002	67.713 ± 1.488
LSAM	beta57	0.06	0.059	-0.001 ± 0.002	-0.023 ± 0.030	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.887 ± 0.041	0.942 ± 0.027	0.059	-0.001	-1.79	0.0556	0.0556	0.888	90.789 ± 0.920	0.197 ± 0.002	26.518 ± 1.404
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.010 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.316 ± 0.017	0.562 ± 0.018	0.058	-0.002	-2.88	0.0289	0.0290	0.891	92.510 ± 0.837	0.118 ± 0.001	57.287 ± 1.574
QML	beta41	0.20	0.203	0.003 ± 0.002	0.014 ± 0.010	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.108 ± 0.005	0.329 ± 0.009	0.204	0.004	2.10	0.0662	0.0663	1.009	96.255 ± 0.604	0.260 ± 0.001	84.615 ± 1.148
QML	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.123 ± 0.006	0.350 ± 0.010	0.200	0.000	-0.02	0.0677	0.0677	0.965	93.522 ± 0.783	0.265 ± 0.001	82.389 ± 1.212
QML	beta43	0.20	0.198	-0.002 ± 0.002	-0.012 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.197	-0.003	-1.71	0.0466	0.0468	0.983	93.219 ± 0.800	0.183 ± 0.001	98.887 ± 0.334
QML	beta44	0.11	0.099	-0.011 ± 0.002	-0.099 ± 0.017	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.299 ± 0.015	0.547 ± 0.016	0.099	-0.011	-10.23	0.0563	0.0574	0.906	91.903 ± 0.868	0.210 ± 0.002	48.178 ± 1.590
QML	beta45	0.11	0.107	-0.003 ± 0.002	-0.027 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.242 ± 0.012	0.492 ± 0.014	0.106	-0.004	-4.04	0.0482	0.0484	0.915	92.814 ± 0.822	0.194 ± 0.002	60.020 ± 1.558
QML	beta46	0.08	0.094	0.014 ± 0.001	0.175 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.199 ± 0.011	0.446 ± 0.014	0.093	0.013	15.93	0.0313	0.0338	0.874	90.182 ± 0.947	0.113 ± 0.001	87.449 ± 1.054
QML	beta47	0.08	0.092	0.012 ± 0.001	0.155 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.188 ± 0.010	0.433 ± 0.013	0.090	0.010	11.90	0.0302	0.0317	0.880	90.789 ± 0.920	0.112 ± 0.001	87.146 ± 1.065
QML	beta51	0.16	0.170	0.010 ± 0.002	0.060 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.183 ± 0.008	0.428 ± 0.012	0.171	0.011	6.59	0.0670	0.0678	0.956	93.522 ± 0.783	0.254 ± 0.002	75.304 ± 1.372
QML	beta52	0.16	0.141	-0.019 ± 0.002	-0.116 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.209 ± 0.009	0.457 ± 0.012	0.144	-0.016	-10.02	0.0714	0.0732	0.989	93.421 ± 0.789	0.274 ± 0.001	53.138 ± 1.588
QML	beta53	0.16	0.135	-0.025 ± 0.002	-0.157 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.205 ± 0.009	0.453 ± 0.012	0.135	-0.025	-15.56	0.0670	0.0715	0.977	91.397 ± 0.892	0.260 ± 0.001	54.352 ± 1.585
QML	beta54	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.009	0.449 ± 0.012	0.160	0.000	-0.19	0.0731	0.0731	0.978	94.838 ± 0.704	0.276 ± 0.001	61.640 ± 1.547
QML	beta55	0.08	0.083	0.003 ± 0.002	0.039 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.604 ± 0.033	0.777 ± 0.024	0.084	0.004	5.26	0.0560	0.0561	0.878	91.700 ± 0.878	0.214 ± 0.004	44.332 ± 1.580
QML	beta56	0.08	0.087	0.007 ± 0.001	0.084 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.246 ± 0.013	0.496 ± 0.015	0.085	0.005	5.83	0.0353	0.0356	0.886	91.700 ± 0.878	0.136 ± 0.002	71.457 ± 1.437
QML	beta57	0.06	0.070	0.010 ± 0.002	0.158 ± 0.029	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.869 ± 0.051	0.932 ± 0.031	0.066	0.006	10.78	0.0533	0.0537	0.907	93.117 ± 0.805	0.196 ± 0.004	31.883 ± 1.483
QML	beta58	0.06	0.070	0.010 ± 0.001	0.162 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.310 ± 0.018	0.557 ± 0.018	0.066	0.006	10.49	0.0272	0.0280	0.886	92.004 ± 0.863	0.111 ± 0.001	70.648 ± 1.449
UPI	beta41	0.20	0.208	0.008 ± 0.002	0.040 ± 0.011	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.131 ± 0.006	0.362 ± 0.010	0.209	0.009	4.42	0.0706	0.0712	0.991	94.636 ± 0.717	0.280 ± 0.001	82.692 ± 1.204
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.023 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.202	0.002	1.23	0.0776	0.0777	0.964	94.534 ± 0.723	0.295 ± 0.002	76.822 ± 1.342
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.201	0.001	0.69	0.0489	0.0490	0.970	92.814 ± 0.822	0.186 ± 0.001	98.988 ± 0.318
UPI	beta44	0.11	0.111	0.001 ± 0.002	0.012 ± 0.017	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.282 ± 0.014	0.531 ± 0.016	0.110	0.000	-0.04	0.0546	0.0546	0.916	91.599 ± 0.883	0.210 ± 0.002	57.389 ± 1.573

Table S209

Simulation 2 - Condition 65: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.111	0.001 ± 0.002	0.014 ± 0.016	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.265 ± 0.013	0.515 ± 0.015	0.110	0.000	0.22	0.0527	0.0527	0.906	92.713 ± 0.827	0.201 ± 0.002	61.134 ± 1.551
UPI	beta46	0.08	0.081	0.001 ± 0.001	0.008 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.173 ± 0.009	0.416 ± 0.013	0.080	0.000	-0.26	0.0313	0.0313	0.856	89.879 ± 0.960	0.112 ± 0.001	77.227 ± 1.334
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.182 ± 0.010	0.426 ± 0.014	0.077	-0.003	-4.09	0.0301	0.0303	0.852	89.676 ± 0.968	0.114 ± 0.002	75.506 ± 1.368
UPI	beta51	0.16	0.167	0.007 ± 0.002	0.042 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.173 ± 0.008	0.416 ± 0.011	0.166	0.006	3.98	0.0658	0.0661	0.947	94.130 ± 0.748	0.246 ± 0.001	75.911 ± 1.360
UPI	beta52	0.16	0.165	0.005 ± 0.003	0.033 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.281 ± 0.012	0.530 ± 0.014	0.166	0.006	3.47	0.0839	0.0841	0.985	94.939 ± 0.697	0.327 ± 0.001	51.113 ± 1.590
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.009	0.446 ± 0.012	0.160	0.000	0.19	0.0726	0.0726	0.951	93.927 ± 0.760	0.266 ± 0.001	66.093 ± 1.506
UPI	beta54	0.16	0.158	-0.002 ± 0.003	-0.011 ± 0.017	0.008 ± 0.000	0.087 ± 0.002	0.007 ± 0.000	0.087 ± 0.002	0.293 ± 0.012	0.541 ± 0.014	0.162	0.002	1.03	0.0880	0.0880	0.962	94.433 ± 0.729	0.327 ± 0.002	49.696 ± 1.591
UPI	beta55	0.08	0.082	0.002 ± 0.002	0.019 ± 0.024	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.548 ± 0.028	0.740 ± 0.022	0.081	0.001	1.70	0.0565	0.0565	0.858	90.789 ± 0.920	0.199 ± 0.002	43.320 ± 1.576
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.009 ± 0.015	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.233 ± 0.012	0.483 ± 0.015	0.078	-0.002	-2.05	0.0338	0.0339	0.866	90.182 ± 0.947	0.131 ± 0.002	69.028 ± 1.471
UPI	beta57	0.06	0.058	-0.002 ± 0.002	-0.036 ± 0.029	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.807 ± 0.040	0.898 ± 0.026	0.057	-0.003	-4.97	0.0549	0.0550	0.901	92.713 ± 0.827	0.190 ± 0.002	25.810 ± 1.392
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.008 ± 0.018	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.310 ± 0.017	0.557 ± 0.017	0.058	-0.002	-3.02	0.0291	0.0291	0.892	91.903 ± 0.868	0.117 ± 0.001	56.984 ± 1.575

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S210

Simulation 2 - Condition 66: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.198	-0.002	-1.03	0.0471	0.0471	0.980	94.277 ± 0.736	0.191 ± 0.001	97.390 ± 0.505
LSAM	beta42	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.075 ± 0.003	0.275 ± 0.008	0.200	0.000	-0.10	0.0566	0.0566	0.922	92.771 ± 0.821	0.199 ± 0.001	96.084 ± 0.615
LSAM	beta43	0.20	0.202	0.002 ± 0.001	0.012 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.202	0.002	1.18	0.0287	0.0288	1.028	95.683 ± 0.644	0.117 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.021 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.093 ± 0.004	0.304 ± 0.008	0.111	0.001	1.33	0.0327	0.0328	0.951	93.976 ± 0.754	0.125 ± 0.001	92.972 ± 0.810
LSAM	beta45	0.11	0.112	0.002 ± 0.001	0.018 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.082 ± 0.004	0.286 ± 0.008	0.112	0.002	1.58	0.0323	0.0324	0.968	93.675 ± 0.771	0.119 ± 0.001	94.277 ± 0.736
LSAM	beta46	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.008	0.000 ± 0.000	0.019 ± 0.001	0.000 ± 0.000	0.019 ± 0.001	0.059 ± 0.003	0.242 ± 0.008	0.078	-0.002	-2.43	0.0183	0.0184	0.902	91.667 ± 0.876	0.069 ± 0.001	97.892 ± 0.455
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.015 ± 0.008	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.079	-0.001	-1.69	0.0180	0.0180	0.881	90.361 ± 0.935	0.068 ± 0.001	97.590 ± 0.486
LSAM	beta51	0.16	0.163	0.003 ± 0.001	0.017 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.162	0.002	1.16	0.0403	0.0403	1.002	94.578 ± 0.718	0.158 ± 0.001	98.494 ± 0.386
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.020 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.164	0.004	2.56	0.0553	0.0554	0.998	95.281 ± 0.672	0.209 ± 0.001	84.839 ± 1.136
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.162	0.002	1.24	0.0432	0.0433	1.011	95.683 ± 0.644	0.168 ± 0.001	96.888 ± 0.550
LSAM	beta54	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.103 ± 0.005	0.321 ± 0.009	0.155	-0.005	-2.86	0.0451	0.0454	0.998	94.478 ± 0.724	0.200 ± 0.001	87.349 ± 1.053
LSAM	beta55	0.08	0.083	0.003 ± 0.001	0.040 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.153 ± 0.007	0.392 ± 0.011	0.083	0.003	3.54	0.0308	0.0309	0.966	95.080 ± 0.685	0.118 ± 0.001	77.410 ± 1.325
LSAM	beta56	0.08	0.079	-0.001 ± 0.001	-0.007 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.070 ± 0.004	0.264 ± 0.008	0.078	-0.002	-2.77	0.0190	0.0191	0.930	92.570 ± 0.831	0.077 ± 0.001	96.185 ± 0.607
LSAM	beta57	0.06	0.056	-0.004 ± 0.001	-0.071 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.263 ± 0.011	0.513 ± 0.014	0.055	-0.005	-8.67	0.0299	0.0303	0.987	93.976 ± 0.754	0.118 ± 0.001	45.080 ± 1.577
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.010	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.096 ± 0.005	0.310 ± 0.009	0.060	0.000	-0.21	0.0173	0.0173	0.919	93.273 ± 0.794	0.067 ± 0.001	90.964 ± 0.908
QML	beta41	0.20	0.204	0.004 ± 0.001	0.018 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.203	0.003	1.58	0.0396	0.0398	1.002	94.378 ± 0.730	0.160 ± 0.000	99.699 ± 0.174
QML	beta42	0.20	0.203	0.003 ± 0.001	0.014 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.203	0.003	1.57	0.0443	0.0444	0.930	92.771 ± 0.821	0.164 ± 0.000	99.297 ± 0.265
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.198	-0.002	-0.89	0.0280	0.0280	1.028	95.482 ± 0.658	0.114 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.099	-0.011 ± 0.001	-0.100 ± 0.010	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.099	-0.011	-10.17	0.0361	0.0378	0.954	92.671 ± 0.826	0.129 ± 0.001	83.032 ± 1.189
QML	beta45	0.11	0.109	-0.001 ± 0.001	-0.012 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.078 ± 0.004	0.279 ± 0.008	0.110	0.000	-0.25	0.0303	0.0303	0.998	94.378 ± 0.730	0.120 ± 0.001	93.574 ± 0.777
QML	beta46	0.08	0.091	0.011 ± 0.001	0.141 ± 0.008	0.000 ± 0.000	0.019 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.076 ± 0.004	0.276 ± 0.008	0.090	0.010	12.82	0.0178	0.0205	0.920	90.964 ± 0.908	0.069 ± 0.001	99.900 ± 0.100
QML	beta47	0.08	0.091	0.011 ± 0.001	0.143 ± 0.007	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.075 ± 0.003	0.275 ± 0.007	0.090	0.010	13.00	0.0178	0.0206	0.913	88.353 ± 1.016	0.067 ± 0.001	99.900 ± 0.100
QML	beta51	0.16	0.170	0.010 ± 0.001	0.064 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.169	0.009	5.74	0.0412	0.0423	0.996	94.578 ± 0.718	0.159 ± 0.001	99.197 ± 0.283
QML	beta52	0.16	0.145	-0.015 ± 0.001	-0.096 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.146	-0.014	-9.05	0.0444	0.0467	0.990	93.173 ± 0.799	0.170 ± 0.000	89.960 ± 0.952
QML	beta53	0.16	0.136	-0.024 ± 0.001	-0.147 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.088 ± 0.004	0.296 ± 0.007	0.136	-0.024	-14.85	0.0446	0.0505	1.008	90.863 ± 0.913	0.162 ± 0.001	91.365 ± 0.890
QML	beta54	0.16	0.158	-0.002 ± 0.001	-0.011 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.074 ± 0.004	0.273 ± 0.008	0.158	-0.002	-1.41	0.0394	0.0395	0.996	94.679 ± 0.711	0.170 ± 0.000	94.378 ± 0.730
QML	beta55	0.08	0.085	0.005 ± 0.001	0.066 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.180 ± 0.009	0.424 ± 0.012	0.085	0.005	6.36	0.0316	0.0320	0.966	94.076 ± 0.748	0.127 ± 0.001	73.996 ± 1.390
QML	beta56	0.08	0.085	0.005 ± 0.001	0.068 ± 0.009	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.081 ± 0.004	0.284 ± 0.009	0.084	0.004	5.10	0.0206	0.0210	0.921	93.876 ± 0.760	0.080 ± 0.001	97.791 ± 0.466
QML	beta57	0.06	0.066	0.006 ± 0.001	0.099 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.270 ± 0.014	0.519 ± 0.015	0.064	0.004	7.05	0.0302	0.0305	0.999	95.382 ± 0.665	0.120 ± 0.001	59.237 ± 1.557
QML	beta58	0.06	0.069	0.009 ± 0.001	0.158 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.118 ± 0.006	0.343 ± 0.010	0.068	0.008	14.15	0.0175	0.0195	0.910	91.165 ± 0.899	0.065 ± 0.001	97.892 ± 0.455
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.203	0.003	1.43	0.0433	0.0434	0.961	93.273 ± 0.794	0.169 ± 0.001	98.695 ± 0.360
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.014 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.201	0.001	0.45	0.0493	0.0493	0.922	92.671 ± 0.826	0.176 ± 0.001	98.494 ± 0.386
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.012 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.203	0.003	1.27	0.0288	0.0289	1.013	95.382 ± 0.665	0.115 ± 0.001	100.000 ± 0.000
UPI	beta44	0.11	0.110	0.000 ± 0.001	0.002 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.111	0.001	0.50	0.0337	0.0337	0.951	94.277 ± 0.736	0.124 ± 0.001	92.068 ± 0.856

Table S210

Simulation 2 - Condition 66: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.011 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.082 ± 0.004	0.286 ± 0.008	0.112	0.002	1.74	0.0326	0.0327	0.971	93.775 ± 0.766	0.120 ± 0.001	94.076 ± 0.748
UPI	beta46	0.08	0.079	-0.001 ± 0.001	-0.016 ± 0.008	0.000 ± 0.000	0.019 ± 0.001	0.000 ± 0.000	0.019 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.078	-0.002	-2.37	0.0175	0.0176	0.888	91.265 ± 0.895	0.066 ± 0.001	98.695 ± 0.360
UPI	beta47	0.08	0.078	-0.002 ± 0.001	-0.021 ± 0.008	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.078	-0.002	-2.24	0.0177	0.0178	0.866	88.755 ± 1.001	0.066 ± 0.001	98.494 ± 0.386
UPI	beta51	0.16	0.165	0.005 ± 0.001	0.029 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.164	0.004	2.39	0.0405	0.0407	0.989	94.177 ± 0.742	0.156 ± 0.001	98.695 ± 0.360
UPI	beta52	0.16	0.166	0.006 ± 0.002	0.037 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.167	0.007	4.63	0.0516	0.0521	0.989	95.080 ± 0.685	0.204 ± 0.001	87.550 ± 1.046
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.160	0.000	0.12	0.0437	0.0437	1.004	95.683 ± 0.644	0.166 ± 0.001	97.490 ± 0.496
UPI	beta54	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.101 ± 0.005	0.317 ± 0.009	0.156	-0.004	-2.50	0.0464	0.0466	1.001	94.880 ± 0.698	0.199 ± 0.001	87.249 ± 1.057
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.025 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.147 ± 0.007	0.383 ± 0.011	0.081	0.001	1.83	0.0301	0.0302	0.973	95.683 ± 0.644	0.117 ± 0.001	77.108 ± 1.331
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.069 ± 0.003	0.262 ± 0.008	0.078	-0.002	-2.39	0.0189	0.0190	0.929	92.269 ± 0.846	0.076 ± 0.001	96.386 ± 0.591
UPI	beta57	0.06	0.056	-0.004 ± 0.001	-0.070 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.253 ± 0.011	0.503 ± 0.014	0.054	-0.006	-10.04	0.0293	0.0299	0.983	94.880 ± 0.698	0.115 ± 0.001	49.096 ± 1.584
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.004 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.095 ± 0.005	0.308 ± 0.009	0.060	0.000	-0.48	0.0174	0.0174	0.924	93.474 ± 0.783	0.067 ± 0.001	91.265 ± 0.895

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S211

Simulation 2 - Condition 67: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.295	0.095 ± 0.092	0.477 ± 0.461	0.043 ± 0.009	0.206 ± 0.078	0.043 ± 0.027	0.208 ± 0.106	1.079 ± 0.669	1.039 ± 0.528	0.233	0.033	16.33	0.1402	0.1439	1.738	100.000 ± 0.000	1.406 ± 0.728	20.000 ± 17.889
LSAM	beta42	0.20	0.133	-0.067 ± 0.067	-0.337 ± 0.337	0.023 ± 0.007	0.151 ± 0.070	0.023 ± 0.015	0.151 ± 0.087	0.567 ± 0.383	0.753 ± 0.433	0.214	0.014	7.08	0.1286	0.1294	2.123	100.000 ± 0.000	1.253 ± 0.598	20.000 ± 17.889
LSAM	beta43	0.20	0.238	0.038 ± 0.030	0.188 ± 0.150	0.004 ± 0.001	0.067 ± 0.025	0.005 ± 0.001	0.071 ± 0.019	0.125 ± 0.033	0.354 ± 0.094	0.172	-0.028	-14.00	0.0915	0.0957	1.328	100.000 ± 0.000	0.349 ± 0.026	80.000 ± 17.889
LSAM	beta44	0.11	0.192	0.082 ± 0.058	0.748 ± 0.524	0.017 ± NaN	0.129 ± 0.031	0.020 ± 0.010	0.142 ± 0.047	1.656 ± 0.790	1.287 ± 0.430	0.164	0.054	48.66	0.1831	0.1907	1.183	100.000 ± 0.000	0.597 ± 0.186	20.000 ± 17.889
LSAM	beta45	0.11	0.151	0.041 ± 0.023	0.375 ± 0.209	0.003 ± NaN	0.051 ± 0.008	0.004 ± 0.002	0.062 ± 0.021	0.316 ± 0.174	0.562 ± 0.193	0.119	0.009	8.44	0.1047	0.1051	2.548	100.000 ± 0.000	0.513 ± 0.091	0.000 ± 0.000
LSAM	beta46	0.08	-0.062	-0.142 ± 0.029	-1.776 ± 0.364	0.004 ± 0.002	0.065 ± 0.040	0.024 ± 0.010	0.154 ± 0.040	3.683 ± 1.548	1.919 ± 0.499	-0.019	-0.099	-124.26	0.1057	0.1451	2.940	100.000 ± 0.000	0.750 ± 0.335	0.000 ± 0.000
LSAM	beta47	0.08	0.026	-0.054 ± 0.087	-0.678 ± 1.083	0.038 ± 0.015	0.194 ± 0.113	0.033 ± 0.008	0.182 ± 0.072	5.153 ± 1.291	2.270 ± 0.894	-0.025	-0.105	-130.84	0.1630	0.1937	2.043	100.000 ± 0.000	1.551 ± 0.841	0.000 ± 0.000
LSAM	beta51	0.16	0.178	0.018 ± 0.029	0.115 ± 0.183	0.004 ± 0.001	0.066 ± 0.023	0.004 ± 0.001	0.061 ± 0.022	0.147 ± 0.033	0.384 ± 0.137	0.208	0.048	29.88	0.1606	0.1676	1.827	100.000 ± 0.000	0.469 ± 0.030	40.000 ± 21.909
LSAM	beta52	0.16	0.110	-0.050 ± 0.050	-0.313 ± 0.313	0.013 ± 0.002	0.112 ± 0.037	0.013 ± 0.005	0.112 ± 0.038	0.489 ± 0.180	0.700 ± 0.241	0.157	-0.003	-1.71	0.1267	0.1267	1.074	100.000 ± 0.000	0.471 ± 0.041	20.000 ± 17.889
LSAM	beta53	0.16	0.136	-0.024 ± 0.061	-0.149 ± 0.379	0.018 ± 0.001	0.135 ± 0.036	0.015 ± 0.004	0.123 ± 0.047	0.595 ± 0.158	0.771 ± 0.296	0.162	0.002	1.01	0.1299	0.1299	0.804	100.000 ± 0.000	0.427 ± 0.032	40.000 ± 21.909
LSAM	beta54	0.16	0.196	0.036 ± 0.086	0.225 ± 0.537	0.037 ± 0.004	0.192 ± 0.053	0.031 ± 0.006	0.175 ± 0.063	1.202 ± 0.234	1.097 ± 0.395	0.188	0.028	17.26	0.1278	0.1307	0.604	80.000 ± 17.889	0.455 ± 0.077	40.000 ± 21.909
LSAM	beta55	0.08	0.054	-0.026 ± 0.021	-0.324 ± 0.264	0.002 ± NaN	0.047 ± 0.010	0.002 ± 0.001	0.049 ± 0.018	0.383 ± 0.183	0.619 ± 0.224	0.073	-0.007	-9.35	0.1083	0.1086	3.444	100.000 ± 0.000	0.637 ± 0.039	0.000 ± 0.000
LSAM	beta56	0.08	0.139	0.059 ± 0.044	0.738 ± 0.547	0.010 ± 0.003	0.098 ± 0.050	0.011 ± 0.003	0.106 ± 0.029	1.741 ± 0.513	1.320 ± 0.368	0.125	0.045	55.99	0.1648	0.1707	2.177	100.000 ± 0.000	0.835 ± 0.093	0.000 ± 0.000
LSAM	beta57	0.06	-0.062	-0.122 ± 0.018	-2.031 ± 0.296	0.002 ± 0.000	0.040 ± 0.016	0.016 ± 0.005	0.127 ± 0.020	4.478 ± 1.292	2.116 ± 0.337	-0.041	-0.101	-167.83	0.0779	0.1273	2.005	80.000 ± 17.889	0.312 ± 0.018	0.000 ± 0.000
LSAM	beta58	0.06	-0.062	-0.122 ± 0.069	-2.032 ± 1.153	0.024 ± 0.007	0.155 ± 0.063	0.034 ± 0.022	0.184 ± 0.085	9.448 ± 6.009	3.074 ± 1.416	-0.022	-0.082	-137.06	0.0949	0.1255	0.683	80.000 ± 17.889	0.414 ± 0.144	0.000 ± 0.000
QML	beta41	0.20	0.211	0.011 ± 0.037	0.053 ± 0.186	0.007 ± 0.002	0.083 ± 0.044	0.006 ± 0.003	0.075 ± 0.043	0.141 ± 0.079	0.376 ± 0.214	0.230	0.030	15.14	0.1120	0.1160	1.064	100.000 ± 0.000	0.347 ± 0.046	80.000 ± 17.889
QML	beta42	0.20	0.155	-0.045 ± 0.029	-0.226 ± 0.145	0.004 ± NaN	0.065 ± 0.011	0.005 ± 0.003	0.074 ± 0.026	0.135 ± 0.072	0.368 ± 0.128	0.191	-0.009	-4.74	0.0604	0.0611	1.347	100.000 ± 0.000	0.342 ± 0.019	60.000 ± 21.909
QML	beta43	0.20	0.247	0.047 ± 0.035	0.237 ± 0.176	0.006 ± 0.002	0.079 ± 0.031	0.007 ± 0.002	0.085 ± 0.023	0.180 ± 0.057	0.424 ± 0.117	0.190	-0.010	-5.04	0.0981	0.0986	1.091	100.000 ± 0.000	0.337 ± 0.018	80.000 ± 17.889
QML	beta44	0.11	0.163	0.053 ± 0.055	0.484 ± 0.499	0.015 ± 0.005	0.123 ± 0.064	0.015 ± 0.011	0.122 ± 0.076	1.230 ± 0.881	1.109 ± 0.694	0.144	0.034	30.52	0.1112	0.1162	0.865	80.000 ± 17.889	0.416 ± 0.045	40.000 ± 21.909
QML	beta45	0.11	0.170	0.060 ± 0.043	0.543 ± 0.393	0.009 ± 0.004	0.097 ± 0.060	0.011 ± 0.009	0.105 ± 0.072	0.914 ± 0.723	0.956 ± 0.655	0.126	0.016	14.26	0.0760	0.0776	1.110	100.000 ± 0.000	0.421 ± 0.053	20.000 ± 17.889
QML	beta46	0.08	-0.034	-0.114 ± 0.045	-1.422 ± 0.565	0.010 ± 0.003	0.101 ± 0.046	0.021 ± 0.009	0.145 ± 0.038	3.300 ± 1.433	1.817 ± 0.472	-0.031	-0.111	-138.36	0.0887	0.1419	0.869	60.000 ± 21.909	0.345 ± 0.058	0.000 ± 0.000
QML	beta47	0.08	-0.036	-0.116 ± 0.034	-1.448 ± 0.421	0.006 ± 0.001	0.075 ± 0.027	0.018 ± 0.007	0.134 ± 0.024	2.806 ± 1.043	1.675 ± 0.304	-0.008	-0.088	-109.58	0.0855	0.1225	0.950	60.000 ± 21.909	0.281 ± 0.024	0.000 ± 0.000
QML	beta51	0.16	0.201	0.041 ± 0.033	0.254 ± 0.207	0.006 ± 0.001	0.074 ± 0.028	0.006 ± 0.003	0.078 ± 0.032	0.237 ± 0.123	0.486 ± 0.202	0.188	0.028	17.50	0.0826	0.0872	1.655	100.000 ± 0.000	0.482 ± 0.037	40.000 ± 21.909
QML	beta52	0.16	0.097	-0.063 ± 0.043	-0.393 ± 0.268	0.009 ± 0.002	0.096 ± 0.039	0.011 ± 0.004	0.106 ± 0.029	0.442 ± 0.157	0.664 ± 0.183	0.141	-0.019	-11.77	0.0731	0.0755	0.881	100.000 ± 0.000	0.331 ± 0.020	20.000 ± 17.889
QML	beta53	0.16	0.176	0.016 ± 0.049	0.098 ± 0.306	0.012 ± 0.002	0.109 ± 0.036	0.010 ± 0.002	0.099 ± 0.039	0.384 ± 0.096	0.619 ± 0.243	0.132	-0.028	-17.31	0.0827	0.0872	0.901	100.000 ± 0.000	0.387 ± 0.030	60.000 ± 21.909
QML	beta54	0.16	0.158	-0.002 ± 0.061	-0.012 ± 0.383	0.019 ± 0.005	0.137 ± 0.057	0.015 ± 0.005	0.123 ± 0.056	0.588 ± 0.209	0.767 ± 0.351	0.193	0.033	20.34	0.0819	0.0881	0.609	80.000 ± 17.889	0.327 ± 0.012	60.000 ± 21.909
QML	beta55	0.08	0.060	-0.020 ± 0.030	-0.252 ± 0.371	0.004 ± 0.002	0.066 ± 0.041	0.004 ± 0.001	0.063 ± 0.024	0.614 ± 0.123	0.783 ± 0.297	0.122	0.042	52.41	0.1426	0.1486	1.988	100.000 ± 0.000	0.517 ± 0.037	0.000 ± 0.000
QML	beta56	0.08	0.076	-0.004 ± 0.030	-0.056 ± 0.373	0.004 ± 0.001	0.067 ± 0.021	0.004 ± 0.001	0.060 ± 0.025	0.560 ± 0.146	0.748 ± 0.306	0.097	0.017	21.44	0.0970	0.0985	2.357	100.000 ± 0.000	0.617 ± 0.069	0.000 ± 0.000
QML	beta57	0.06	-0.061	-0.121 ± 0.018	-2.010 ± 0.307	0.002 ± 0.001	0.041 ± 0.024	0.016 ± 0.005	0.126 ± 0.023	4.417 ± 1.398	2.102 ± 0.388	-0.040	-0.100	-167.48	0.0416	0.1088	1.217	20.000 ± 17.889	0.196 ± 0.018	20.000 ± 17.889
QML	beta58	0.06	-0.003	-0.063 ± 0.025	-1.052 ± 0.411	0.003 ± 0.001	0.055 ± 0.020	0.006 ± 0.003	0.080 ± 0.020	1.782 ± 0.790	1.335 ± 0.341	-0.028	-0.088	-147.06	0.0752	0.1159	0.974	80.000 ± 17.889	0.210 ± 0.032	0.000 ± 0.000
UPI	beta41	0.20	0.676	0.476 ± 0.248	2.382 ± 1.241	0.308 ± 0.095	0.555 ± 0.253	0.474 ± 0.316	0.688 ± 0.337	11.838 ± 7.904	3.441 ± 1.685	0.270	0.070	35.15	0.6435	0.6473	1.536	100.000 ± 0.000	3.342 ± 1.871	0.000 ± 0.000
UPI	beta42	0.20	-0.009	-0.209 ± 0.238	-1.044 ± 1.189	0.283 ± 0.105	0.532 ± 0.294	0.270 ± 0.197	0.519 ± 0.342	6.743 ± 4.933	2.597 ± 1.709	0.163	-0.037	-18.32	0.2374	0.2402	1.034	100.000 ± 0.000	2.156 ± 1.090	20.000 ± 17.889
UPI	beta43	0.20	0.082	-0.118 ± 0.082	-0.592 ± 0.409	0.033 ± 0.006	0.183 ± 0.058	0.041 ± 0.023	0.202 ± 0.084	1.019 ± 0.573	1.009 ± 0.419	0.103	-0.097	-48.47	0.1641	0.1906	2.046	80.000 ± 17.889	1.466 ± 0.715	20.000 ± 17.889
UPI	beta44	0.11	0.010	-0.100 ± 0.157	-0.912 ± 1.429	0.124 ± 0.057	0.352 ± 0.291	0.109 ± 0.089	0.330 ± 0.282	9.004 ± 7.335	3.001 ± 2.559	0.115	0.005	4.95	0.2934	0.2935	1.351	100.000 ± 0.000	1.862 ± 0.934	0.000 ± 0.000
UPI	beta45	0.11	0.146	0.036 ± 0.067	0.328 ± 0.613	0.023 ± NaN	0.151 ± 0.027	0.019 ± 0.004	0.140 ± 0.047	1.611 ± 0.319	1.269 ± 0.430	0.081	-0.029	-26.37	0.1851	0.1874	2.156	100.000 ± 0.000	1.274 ± 0.443	0.000 ± 0.000
UPI	beta46	0.08	-0.791	-0.871 ± 0.448	-10.884 ± 5.594	1.001 ± 0.401	1.001 ± 0.623	1.559 ± 1.169	1.249 ± 0.741	243.648 ± 182.693	15.609 ± 9.265	-0.193	-0.273	-341.68	1.0418	1.0771	1.187	100.000 ± 0.000	4.656 ± 3.348	0.000 ± 0.000

Table S211

Simulation 2 - Condition 67: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.159	-0.239 ± 0.090	-2.983 ± 1.122	0.040 ± NaN	0.201 ± 0.032	0.089 ± 0.040	0.299 ± 0.073	13.936 ± 6.274	3.733 ± 0.909	-0.022	-0.102	-126.89	0.4588	0.4699	1.684	100.000 ± 0.000	1.325 ± 0.374	0.000 ± 0.000
UPI	beta51	0.16	0.190	0.030 ± 0.062	0.187 ± 0.388	0.019 ± 0.006	0.139 ± 0.062	0.016 ± 0.003	0.128 ± 0.049	0.636 ± 0.126	0.798 ± 0.304	0.236	0.076	47.80	0.1359	0.1560	1.138	100.000 ± 0.000	0.619 ± 0.061	40.000 ± 21.909
UPI	beta52	0.16	0.289	0.129 ± 0.101	0.805 ± 0.630	0.051 ± 0.016	0.225 ± 0.103	0.057 ± 0.039	0.239 ± 0.133	2.236 ± 1.533	1.495 ± 0.829	0.303	0.143	89.26	0.4404	0.4629	1.137	100.000 ± 0.000	1.004 ± 0.141	20.000 ± 17.889
UPI	beta53	0.16	0.090	-0.070 ± 0.061	-0.436 ± 0.381	0.019 ± 0.007	0.136 ± 0.080	0.020 ± 0.015	0.141 ± 0.097	0.771 ± 0.604	0.878 ± 0.607	0.100	-0.060	-37.25	0.1342	0.1468	0.912	80.000 ± 17.889	0.488 ± 0.037	0.000 ± 0.000
UPI	beta54	0.16	0.124	-0.036 ± 0.090	-0.225 ± 0.562	0.040 ± 0.013	0.201 ± 0.085	0.034 ± 0.013	0.183 ± 0.080	1.312 ± 0.502	1.146 ± 0.500	0.137	-0.023	-14.32	0.2333	0.2344	0.935	100.000 ± 0.000	0.736 ± 0.183	0.000 ± 0.000
UPI	beta55	0.08	0.297	0.217 ± 0.233	2.711 ± 2.912	0.271 ± 0.113	0.521 ± 0.344	0.264 ± 0.213	0.514 ± 0.391	41.258 ± 33.205	6.423 ± 4.893	0.074	-0.006	-6.92	0.2790	0.2790	0.693	100.000 ± 0.000	1.416 ± 0.661	0.000 ± 0.000
UPI	beta56	0.08	-0.070	-0.150 ± 0.146	-1.872 ± 1.831	0.107 ± 0.047	0.328 ± 0.235	0.108 ± 0.091	0.329 ± 0.268	16.917 ± 14.150	4.113 ± 3.349	0.047	-0.033	-41.50	0.2194	0.2219	0.979	100.000 ± 0.000	1.257 ± 0.463	0.000 ± 0.000
UPI	beta57	0.06	-0.202	-0.262 ± 0.057	-4.361 ± 0.949	0.016 ± NaN	0.127 ± 0.028	0.081 ± 0.031	0.285 ± 0.059	22.617 ± 8.625	4.756 ± 0.980	-0.136	-0.196	-327.26	0.4251	0.4683	2.067	100.000 ± 0.000	1.032 ± 0.198	0.000 ± 0.000
UPI	beta58	0.06	0.060	-0.000 ± 0.106	-0.008 ± 1.759	0.056 ± 0.023	0.236 ± 0.148	0.045 ± 0.023	0.211 ± 0.119	12.370 ± 6.301	3.517 ± 1.985	-0.058	-0.118	-196.41	0.2700	0.2946	0.862	80.000 ± 17.889	0.797 ± 0.302	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S212

Simulation 2 - Condition 68: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.181	-0.019 ± 0.027	-0.096 ± 0.133	0.008 ± 0.001	0.088 ± 0.010	0.007 ± 0.002	0.086 ± 0.018	0.186 ± 0.042	0.431 ± 0.092	0.192	-0.008	-3.77	0.0882	0.0885	0.892	81.818 ± 11.629	0.308 ± 0.025	63.636 ± 14.504
LSAM	beta42	0.20	0.225	0.025 ± 0.024	0.125 ± 0.118	0.006 ± 0.002	0.078 ± 0.014	0.006 ± 0.002	0.079 ± 0.019	0.155 ± 0.050	0.394 ± 0.095	0.253	0.053	26.57	0.1056	0.1182	1.008	100.000 ± 0.000	0.309 ± 0.023	90.909 ± 8.668
LSAM	beta43	0.20	0.168	-0.032 ± 0.014	-0.162 ± 0.069	0.002 ± 0.001	0.046 ± 0.010	0.003 ± 0.001	0.054 ± 0.012	0.073 ± 0.027	0.271 ± 0.062	0.166	-0.034	-16.88	0.0485	0.0591	1.190	100.000 ± 0.000	0.213 ± 0.008	81.818 ± 11.629
LSAM	beta44	0.11	0.150	0.040 ± 0.022	0.361 ± 0.201	0.005 ± 0.001	0.073 ± 0.011	0.006 ± 0.002	0.081 ± 0.018	0.536 ± 0.181	0.732 ± 0.164	0.169	0.059	53.94	0.0984	0.1149	0.990	100.000 ± 0.000	0.285 ± 0.024	63.636 ± 14.504
LSAM	beta45	0.11	0.125	0.015 ± 0.020	0.137 ± 0.185	0.005 ± 0.001	0.067 ± 0.013	0.004 ± 0.001	0.066 ± 0.015	0.361 ± 0.099	0.601 ± 0.139	0.135	0.025	22.96	0.0632	0.0681	0.956	90.909 ± 8.668	0.253 ± 0.009	45.455 ± 15.013
LSAM	beta46	0.08	-0.001	-0.081 ± 0.028	-1.008 ± 0.351	0.009 ± 0.002	0.093 ± 0.013	0.014 ± 0.004	0.120 ± 0.022	2.245 ± 0.684	1.498 ± 0.273	-0.024	-0.104	-130.06	0.0834	0.1334	0.938	63.636 ± 14.504	0.342 ± 0.056	0.000 ± 0.000
LSAM	beta47	0.08	-0.029	-0.109 ± 0.016	-1.366 ± 0.198	0.003 ± 0.001	0.053 ± 0.014	0.014 ± 0.003	0.120 ± 0.013	2.258 ± 0.472	1.503 ± 0.160	-0.054	-0.134	-166.91	0.0645	0.1483	1.417	72.727 ± 13.428	0.292 ± 0.052	9.091 ± 8.668
LSAM	beta51	0.16	0.167	0.007 ± 0.016	0.041 ± 0.102	0.003 ± 0.001	0.054 ± 0.018	0.003 ± 0.001	0.052 ± 0.019	0.106 ± 0.055	0.326 ± 0.121	0.158	-0.002	-1.11	0.0595	0.0595	1.379	100.000 ± 0.000	0.293 ± 0.010	54.545 ± 15.013
LSAM	beta52	0.16	0.186	0.026 ± 0.029	0.165 ± 0.182	0.009 ± 0.002	0.096 ± 0.015	0.009 ± 0.002	0.096 ± 0.021	0.357 ± 0.090	0.597 ± 0.129	0.170	0.010	6.55	0.0787	0.0794	0.810	100.000 ± 0.000	0.306 ± 0.019	63.636 ± 14.504
LSAM	beta53	0.16	0.135	-0.025 ± 0.029	-0.156 ± 0.182	0.009 ± 0.004	0.097 ± 0.026	0.009 ± 0.004	0.096 ± 0.029	0.357 ± 0.155	0.597 ± 0.183	0.139	-0.021	-12.98	0.0637	0.0670	0.654	72.727 ± 13.428	0.248 ± 0.011	63.636 ± 14.504
LSAM	beta54	0.16	0.158	-0.002 ± 0.013	-0.015 ± 0.080	0.002 ± 0.000	0.042 ± 0.006	0.002 ± 0.000	0.040 ± 0.009	0.064 ± 0.014	0.252 ± 0.056	0.156	-0.004	-2.45	0.0557	0.0559	1.735	100.000 ± 0.000	0.288 ± 0.011	63.636 ± 14.504
LSAM	beta55	0.08	0.149	0.069 ± 0.044	0.862 ± 0.555	0.022 ± 0.011	0.147 ± 0.057	0.024 ± 0.015	0.156 ± 0.067	3.822 ± 2.368	1.955 ± 0.838	0.112	0.032	40.61	0.1008	0.1059	0.719	100.000 ± 0.000	0.415 ± 0.058	27.273 ± 13.428
LSAM	beta56	0.08	0.058	-0.022 ± 0.027	-0.271 ± 0.333	0.008 ± 0.002	0.088 ± 0.016	0.008 ± 0.001	0.087 ± 0.017	1.184 ± 0.217	1.088 ± 0.218	0.077	-0.003	-4.00	0.0926	0.0927	0.978	90.909 ± 8.668	0.339 ± 0.031	0.000 ± 0.000
LSAM	beta57	0.06	-0.057	-0.117 ± 0.027	-1.945 ± 0.451	0.008 ± 0.003	0.090 ± 0.021	0.021 ± 0.008	0.145 ± 0.031	5.814 ± 2.184	2.411 ± 0.513	-0.036	-0.096	-159.27	0.0531	0.1093	0.894	63.636 ± 14.504	0.315 ± 0.030	18.182 ± 11.629
LSAM	beta58	0.06	0.001	-0.059 ± 0.008	-0.980 ± 0.129	0.001 ± 0.000	0.026 ± 0.008	0.004 ± 0.001	0.064 ± 0.010	1.127 ± 0.314	1.061 ± 0.163	0.005	-0.055	-92.00	0.0487	0.0736	2.658	81.818 ± 11.629	0.268 ± 0.020	0.000 ± 0.000
QML	beta41	0.20	0.179	-0.021 ± 0.017	-0.103 ± 0.086	0.003 ± 0.001	0.057 ± 0.009	0.003 ± 0.001	0.058 ± 0.013	0.084 ± 0.024	0.291 ± 0.064	0.190	-0.010	-4.87	0.0525	0.0534	0.855	90.909 ± 8.668	0.191 ± 0.006	90.909 ± 8.668
QML	beta42	0.20	0.204	0.004 ± 0.020	0.022 ± 0.098	0.004 ± 0.001	0.065 ± 0.013	0.004 ± 0.001	0.062 ± 0.017	0.096 ± 0.032	0.310 ± 0.083	0.205	0.005	2.60	0.0730	0.0732	0.787	90.909 ± 8.668	0.200 ± 0.008	100.000 ± 0.000
QML	beta43	0.20	0.174	-0.026 ± 0.011	-0.129 ± 0.057	0.001 ± 0.000	0.038 ± 0.008	0.002 ± 0.001	0.044 ± 0.009	0.049 ± 0.016	0.221 ± 0.046	0.174	-0.026	-13.01	0.0509	0.0572	1.286	100.000 ± 0.000	0.190 ± 0.006	100.000 ± 0.000
QML	beta44	0.11	0.151	0.041 ± 0.022	0.370 ± 0.200	0.005 ± 0.001	0.073 ± 0.014	0.006 ± 0.003	0.081 ± 0.021	0.536 ± 0.221	0.732 ± 0.195	0.157	0.047	42.51	0.0942	0.1051	0.934	81.818 ± 11.629	0.267 ± 0.015	63.636 ± 14.504
QML	beta45	0.11	0.105	-0.005 ± 0.021	-0.048 ± 0.190	0.005 ± 0.001	0.069 ± 0.014	0.004 ± 0.001	0.066 ± 0.017	0.365 ± 0.107	0.604 ± 0.151	0.111	0.001	0.59	0.0844	0.0844	0.879	90.909 ± 8.668	0.239 ± 0.011	36.364 ± 14.504
QML	beta46	0.08	0.001	-0.079 ± 0.016	-0.984 ± 0.205	0.003 ± 0.001	0.054 ± 0.009	0.009 ± 0.003	0.094 ± 0.015	1.388 ± 0.398	1.178 ± 0.182	-0.006	-0.086	-107.02	0.0643	0.1071	0.875	54.545 ± 15.013	0.186 ± 0.020	9.091 ± 8.668
QML	beta47	0.08	-0.011	-0.091 ± 0.013	-1.143 ± 0.158	0.002 ± 0.001	0.042 ± 0.010	0.010 ± 0.002	0.100 ± 0.012	1.555 ± 0.362	1.247 ± 0.152	-0.035	-0.115	-143.75	0.0566	0.1282	1.106	45.455 ± 15.013	0.181 ± 0.025	18.182 ± 11.629
QML	beta51	0.16	0.184	0.024 ± 0.021	0.147 ± 0.134	0.005 ± 0.002	0.071 ± 0.027	0.005 ± 0.003	0.072 ± 0.031	0.201 ± 0.122	0.448 ± 0.191	0.180	0.020	12.51	0.0673	0.0702	1.051	90.909 ± 8.668	0.293 ± 0.009	81.818 ± 11.629
QML	beta52	0.16	0.164	0.004 ± 0.021	0.026 ± 0.130	0.005 ± 0.001	0.069 ± 0.012	0.004 ± 0.001	0.066 ± 0.016	0.170 ± 0.044	0.413 ± 0.097	0.166	0.006	3.96	0.0807	0.0809	0.872	100.000 ± 0.000	0.236 ± 0.025	81.818 ± 11.629
QML	beta53	0.16	0.146	-0.014 ± 0.026	-0.086 ± 0.162	0.007 ± 0.003	0.086 ± 0.022	0.007 ± 0.003	0.083 ± 0.025	0.269 ± 0.112	0.519 ± 0.158	0.160	0.000	0.20	0.0792	0.0792	0.724	81.818 ± 11.629	0.244 ± 0.023	63.636 ± 14.504
QML	beta54	0.16	0.162	0.002 ± 0.009	0.011 ± 0.059	0.001 ± 0.000	0.031 ± 0.006	0.001 ± 0.000	0.030 ± 0.008	0.035 ± 0.011	0.187 ± 0.049	0.155	-0.005	-3.42	0.0514	0.0516	1.992	100.000 ± 0.000	0.245 ± 0.041	90.909 ± 8.668
QML	beta55	0.08	0.126	0.046 ± 0.036	0.579 ± 0.453	0.014 ± 0.007	0.120 ± 0.042	0.015 ± 0.009	0.124 ± 0.049	2.387 ± 1.352	1.545 ± 0.608	0.125	0.045	55.67	0.1168	0.1250	0.892	100.000 ± 0.000	0.420 ± 0.059	36.364 ± 14.504
QML	beta56	0.08	0.052	-0.028 ± 0.023	-0.348 ± 0.283	0.006 ± 0.001	0.075 ± 0.010	0.006 ± 0.002	0.077 ± 0.016	0.920 ± 0.240	0.959 ± 0.200	0.052	-0.028	-34.65	0.0680	0.0734	1.323	72.727 ± 13.428	0.389 ± 0.104	0.000 ± 0.000
QML	beta57	0.06	-0.039	-0.099 ± 0.015	-1.644 ± 0.242	0.002 ± 0.001	0.048 ± 0.009	0.012 ± 0.003	0.109 ± 0.012	3.287 ± 0.736	1.813 ± 0.208	-0.029	-0.089	-148.08	0.0475	0.1008	0.976	36.364 ± 14.504	0.184 ± 0.015	27.273 ± 13.428
QML	beta58	0.06	0.005	-0.055 ± 0.007	-0.912 ± 0.124	0.001 ± 0.000	0.025 ± 0.008	0.004 ± 0.001	0.060 ± 0.009	0.986 ± 0.283	0.993 ± 0.158	0.011	-0.049	-82.10	0.0364	0.0612	5.370	81.818 ± 11.629	0.520 ± 0.335	0.000 ± 0.000
UPI	beta41	0.20	0.851	0.651 ± 0.199	3.255 ± 0.994	0.435 ± 0.151	0.660 ± 0.166	0.819 ± 0.269	0.905 ± 0.177	20.478 ± 6.729	4.525 ± 0.883	0.641	0.441	220.42	0.8605	0.9668	1.588	90.909 ± 8.668	4.107 ± 1.192	0.000 ± 0.000
UPI	beta42	0.20	0.310	0.110 ± 0.140	0.550 ± 0.699	0.215 ± 0.081	0.464 ± 0.128	0.208 ± 0.110	0.456 ± 0.162	5.190 ± 2.753	2.278 ± 0.809	0.291	0.091	45.55	0.5024	0.5106	1.132	90.909 ± 8.668	2.058 ± 0.991	9.091 ± 8.668
UPI	beta43	0.20	-0.041	-0.241 ± 0.063	-1.204 ± 0.315	0.044 ± 0.009	0.209 ± 0.033	0.098 ± 0.034	0.313 ± 0.062	2.442 ± 0.859	1.563 ± 0.308	0.002	-0.198	-98.97	0.2345	0.3069	1.863	90.909 ± 8.668	1.525 ± 0.592	18.182 ± 11.629
UPI	beta44	0.11	0.194	0.084 ± 0.050	0.760 ± 0.453	0.027 ± 0.011	0.165 ± 0.049	0.032 ± 0.016	0.178 ± 0.061	2.626 ± 1.320	1.621 ± 0.551	0.215	0.105	95.60	0.1622	0.1933	1.692	100.000 ± 0.000	1.095 ± 0.209	9.091 ± 8.668
UPI	beta45	0.11	0.226	0.116 ± 0.067	1.051 ± 0.611	0.050 ± 0.017	0.223 ± 0.056	0.059 ± 0.026	0.242 ± 0.071	4.843 ± 2.156	2.201 ± 0.642	0.137	0.027	24.80	0.1618	0.1641	1.411	100.000 ± 0.000	1.234 ± 0.380	0.000 ± 0.000
UPI	beta46	0.08	-0.684	-0.764 ± 0.231	-9.546 ± 2.891	0.588 ± 0.235	0.767 ± 0.234	1.118 ± 0.296	1.057 ± 0.167	174.698 ± 46.327	13.217 ± 2.089	-0.538	-0.618	-772.02	1.1865	1.3376	2.378	100.000 ± 0.000	7.150 ± 1.825	0.000 ± 0.000

Table S212

Simulation 2 - Condition 68: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.309	-0.389 ± 0.135	-4.867 ± 1.693	0.202 ± 0.051	0.449 ± 0.085	0.335 ± 0.152	0.579 ± 0.154	52.350 ± 23.683	7.235 ± 1.920	-0.124	-0.204	-255.40	0.4169	0.4643	1.653	100.000 ± 0.000	2.910 ± 1.353	9.091 ± 8.668
UPI	beta51	0.16	0.160	0.000 ± 0.030	0.002 ± 0.187	0.010 ± 0.005	0.099 ± 0.041	0.009 ± 0.005	0.094 ± 0.039	0.348 ± 0.193	0.590 ± 0.241	0.208	0.048	29.77	0.0950	0.1063	1.336	100.000 ± 0.000	0.519 ± 0.096	54.545 ± 15.013
UPI	beta52	0.16	0.469	0.309 ± 0.125	1.929 ± 0.784	0.173 ± 0.062	0.416 ± 0.113	0.253 ± 0.125	0.503 ± 0.156	9.869 ± 4.871	3.141 ± 0.975	0.368	0.208	130.25	0.3828	0.4358	0.971	100.000 ± 0.000	1.584 ± 0.369	0.000 ± 0.000
UPI	beta53	0.16	0.064	-0.096 ± 0.034	-0.598 ± 0.211	0.013 ± 0.004	0.112 ± 0.029	0.021 ± 0.009	0.143 ± 0.040	0.803 ± 0.362	0.896 ± 0.251	0.046	-0.114	-71.06	0.1665	0.2016	0.904	72.727 ± 13.428	0.397 ± 0.073	27.273 ± 13.428
UPI	beta54	0.16	0.076	-0.084 ± 0.060	-0.527 ± 0.376	0.040 ± 0.026	0.200 ± 0.116	0.043 ± 0.018	0.208 ± 0.064	1.695 ± 0.714	1.302 ± 0.403	0.054	-0.106	-66.28	0.1318	0.1691	0.890	90.909 ± 8.668	0.697 ± 0.247	0.000 ± 0.000
UPI	beta55	0.08	0.164	0.084 ± 0.040	1.055 ± 0.506	0.018 ± 0.008	0.134 ± 0.042	0.024 ± 0.014	0.153 ± 0.058	3.674 ± 2.224	1.917 ± 0.727	0.105	0.025	31.50	0.1200	0.1227	1.176	100.000 ± 0.000	0.619 ± 0.115	0.000 ± 0.000
UPI	beta56	0.08	0.045	-0.035 ± 0.037	-0.442 ± 0.463	0.015 ± 0.005	0.123 ± 0.033	0.015 ± 0.007	0.122 ± 0.039	2.341 ± 1.044	1.530 ± 0.485	0.115	0.035	44.25	0.1468	0.1510	1.394	90.909 ± 8.668	0.672 ± 0.117	0.000 ± 0.000
UPI	beta57	0.06	-0.435	-0.495 ± 0.167	-8.252 ± 2.785	0.307 ± 0.154	0.554 ± 0.225	0.524 ± 0.296	0.724 ± 0.265	145.659 ± 82.336	12.069 ± 4.417	-0.291	-0.351	-585.78	0.5914	0.6879	1.434	100.000 ± 0.000	3.116 ± 1.246	0.000 ± 0.000
UPI	beta58	0.06	0.012	-0.048 ± 0.078	-0.792 ± 1.295	0.066 ± 0.048	0.258 ± 0.186	0.063 ± 0.055	0.250 ± 0.186	17.405 ± 15.195	4.172 ± 3.101	0.031	-0.029	-47.75	0.1361	0.1391	1.497	100.000 ± 0.000	1.513 ± 0.928	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S213

Simulation 2 - Condition 69: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.003	0.011 ± 0.013	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.006	0.300 ± 0.012	0.198	-0.002	-0.98	0.0603	0.0604	1.064	96.514 ± 0.786	0.251 ± 0.001	89.541 ± 1.311
LSAM	beta42	0.20	0.195	-0.005 ± 0.003	-0.025 ± 0.014	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.006	0.320 ± 0.012	0.200	0.000	0.07	0.0637	0.0637	1.033	95.413 ± 0.896	0.259 ± 0.002	84.954 ± 1.531
LSAM	beta43	0.20	0.199	-0.001 ± 0.003	-0.006 ± 0.013	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.006	0.310 ± 0.011	0.199	-0.001	-0.33	0.0643	0.0643	0.948	93.761 ± 1.036	0.231 ± 0.001	91.743 ± 1.179
LSAM	beta44	0.11	0.133	0.023 ± 0.003	0.205 ± 0.030	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.518 ± 0.035	0.720 ± 0.028	0.133	0.023	20.94	0.0751	0.0786	0.992	94.495 ± 0.977	0.295 ± 0.003	45.872 ± 2.134
LSAM	beta45	0.11	0.118	0.008 ± 0.003	0.075 ± 0.026	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.385 ± 0.023	0.621 ± 0.023	0.120	0.010	8.94	0.0661	0.0668	0.963	95.229 ± 0.913	0.256 ± 0.002	49.174 ± 2.141
LSAM	beta46	0.08	0.040	-0.040 ± 0.004	-0.497 ± 0.049	0.008 ± 0.001	0.091 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	1.542 ± 0.097	1.242 ± 0.045	0.042	-0.038	-47.83	0.0888	0.0967	1.000	88.807 ± 1.350	0.357 ± 0.005	5.138 ± 0.946
LSAM	beta47	0.08	0.052	-0.028 ± 0.004	-0.346 ± 0.051	0.009 ± 0.001	0.095 ± 0.004	0.010 ± 0.001	0.099 ± 0.004	1.537 ± 0.103	1.240 ± 0.048	0.043	-0.037	-46.72	0.0838	0.0918	0.938	89.908 ± 1.290	0.351 ± 0.004	6.055 ± 1.022
LSAM	beta51	0.16	0.170	0.010 ± 0.004	0.065 ± 0.025	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.347 ± 0.025	0.589 ± 0.025	0.165	0.005	2.96	0.0877	0.0879	0.866	90.826 ± 1.236	0.318 ± 0.002	55.229 ± 2.130
LSAM	beta52	0.16	0.155	-0.005 ± 0.003	-0.032 ± 0.017	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.009	0.392 ± 0.014	0.155	-0.005	-3.34	0.0663	0.0665	1.072	97.248 ± 0.701	0.263 ± 0.002	65.505 ± 2.036
LSAM	beta53	0.16	0.151	-0.009 ± 0.003	-0.054 ± 0.018	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	0.183 ± 0.012	0.428 ± 0.017	0.151	-0.009	-5.86	0.0637	0.0644	1.008	94.128 ± 1.007	0.269 ± 0.002	61.468 ± 2.085
LSAM	beta54	0.16	0.155	-0.005 ± 0.003	-0.029 ± 0.018	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.011	0.419 ± 0.016	0.155	-0.005	-3.35	0.0636	0.0638	0.991	95.229 ± 0.913	0.260 ± 0.002	65.138 ± 2.041
LSAM	beta55	0.08	0.094	0.014 ± 0.005	0.173 ± 0.058	0.012 ± 0.001	0.108 ± 0.004	0.012 ± 0.001	0.109 ± 0.005	1.858 ± 0.138	1.363 ± 0.058	0.089	0.009	11.06	0.1016	0.1020	0.905	94.128 ± 1.007	0.384 ± 0.005	19.450 ± 1.695
LSAM	beta56	0.08	0.086	0.006 ± 0.004	0.077 ± 0.051	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	1.397 ± 0.098	1.182 ± 0.049	0.085	0.005	6.82	0.0921	0.0922	0.957	95.046 ± 0.930	0.354 ± 0.004	17.248 ± 1.618
LSAM	beta57	0.06	0.034	-0.026 ± 0.004	-0.432 ± 0.071	0.010 ± 0.001	0.099 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	2.928 ± 0.181	1.711 ± 0.063	0.028	-0.032	-53.70	0.0931	0.0985	0.943	91.376 ± 1.202	0.367 ± 0.005	4.954 ± 0.930
LSAM	beta58	0.06	0.032	-0.028 ± 0.004	-0.459 ± 0.060	0.007 ± 0.001	0.084 ± 0.003	0.008 ± 0.001	0.088 ± 0.003	2.145 ± 0.148	1.465 ± 0.058	0.032	-0.028	-46.34	0.0792	0.0839	1.005	92.844 ± 1.104	0.329 ± 0.004	4.587 ± 0.896
QML	beta41	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.012	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.005	0.270 ± 0.010	0.193	-0.007	-3.38	0.0559	0.0563	1.080	95.596 ± 0.879	0.228 ± 0.001	92.110 ± 1.155
QML	beta42	0.20	0.194	-0.006 ± 0.002	-0.032 ± 0.012	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.005	0.291 ± 0.011	0.197	-0.003	-1.47	0.0589	0.0589	1.031	95.596 ± 0.879	0.234 ± 0.001	90.826 ± 1.236
QML	beta43	0.20	0.192	-0.008 ± 0.003	-0.041 ± 0.013	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.005	0.295 ± 0.011	0.193	-0.007	-3.57	0.0602	0.0606	0.956	94.312 ± 0.992	0.219 ± 0.001	93.394 ± 1.064
QML	beta44	0.11	0.126	0.016 ± 0.003	0.147 ± 0.028	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	0.443 ± 0.031	0.666 ± 0.027	0.127	0.017	15.85	0.0684	0.0706	1.008	94.862 ± 0.946	0.282 ± 0.003	47.156 ± 2.138
QML	beta45	0.11	0.116	0.006 ± 0.003	0.055 ± 0.025	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	0.356 ± 0.023	0.597 ± 0.023	0.115	0.005	4.77	0.0643	0.0645	0.978	95.413 ± 0.896	0.251 ± 0.002	47.156 ± 2.138
QML	beta46	0.08	0.030	-0.050 ± 0.003	-0.629 ± 0.032	0.004 ± 0.000	0.060 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.956 ± 0.048	0.978 ± 0.028	0.027	-0.053	-65.79	0.0595	0.0794	1.012	78.349 ± 1.764	0.238 ± 0.003	5.321 ± 0.961
QML	beta47	0.08	0.035	-0.045 ± 0.003	-0.557 ± 0.033	0.004 ± 0.000	0.061 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.887 ± 0.044	0.942 ± 0.027	0.032	-0.048	-59.41	0.0576	0.0747	0.969	79.266 ± 1.737	0.231 ± 0.003	5.688 ± 0.992
QML	beta51	0.16	0.180	0.020 ± 0.004	0.124 ± 0.025	0.008 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.346 ± 0.025	0.588 ± 0.024	0.176	0.016	9.98	0.0852	0.0866	0.936	93.394 ± 1.064	0.338 ± 0.003	56.697 ± 2.122
QML	beta52	0.16	0.148	-0.012 ± 0.003	-0.076 ± 0.017	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.009	0.406 ± 0.014	0.146	-0.014	-9.00	0.0673	0.0689	1.045	96.514 ± 0.786	0.261 ± 0.003	61.101 ± 2.088
QML	beta53	0.16	0.138	-0.022 ± 0.003	-0.135 ± 0.017	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.179 ± 0.011	0.423 ± 0.016	0.138	-0.022	-13.64	0.0649	0.0685	1.020	94.495 ± 0.977	0.256 ± 0.002	57.431 ± 2.118
QML	beta54	0.16	0.156	-0.004 ± 0.003	-0.025 ± 0.017	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.009	0.389 ± 0.014	0.153	-0.007	-4.37	0.0600	0.0604	1.006	95.046 ± 0.930	0.245 ± 0.001	72.110 ± 1.921
QML	beta55	0.08	0.090	0.010 ± 0.004	0.123 ± 0.053	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	1.532 ± 0.102	1.238 ± 0.049	0.088	0.008	9.49	0.0937	0.0940	0.976	94.495 ± 0.977	0.377 ± 0.005	19.817 ± 1.707
QML	beta56	0.08	0.078	-0.002 ± 0.004	-0.024 ± 0.045	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	1.085 ± 0.069	1.042 ± 0.040	0.078	-0.002	-3.10	0.0807	0.0807	1.012	94.679 ± 0.961	0.331 ± 0.003	14.312 ± 1.500
QML	beta57	0.06	0.021	-0.039 ± 0.003	-0.643 ± 0.049	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	1.694 ± 0.090	1.301 ± 0.041	0.017	-0.043	-72.23	0.0646	0.0778	0.976	83.486 ± 1.590	0.260 ± 0.005	4.404 ± 0.879
QML	beta58	0.06	0.026	-0.034 ± 0.002	-0.571 ± 0.038	0.003 ± 0.000	0.054 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	1.122 ± 0.059	1.059 ± 0.032	0.022	-0.038	-62.92	0.0502	0.0628	1.051	85.321 ± 1.516	0.221 ± 0.003	4.587 ± 0.896
UPI	beta41	0.20	0.218	0.018 ± 0.003	0.091 ± 0.015	0.005 ± 0.000	0.070 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	0.132 ± 0.010	0.363 ± 0.016	0.212	0.012	6.21	0.0723	0.0734	1.203	97.431 ± 0.678	0.332 ± 0.009	79.817 ± 1.719
UPI	beta42	0.20	0.206	0.006 ± 0.003	0.030 ± 0.016	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.075 ± 0.004	0.142 ± 0.012	0.377 ± 0.018	0.210	0.010	4.91	0.0757	0.0763	1.263	97.064 ± 0.723	0.372 ± 0.012	71.560 ± 1.932
UPI	beta43	0.20	0.196	-0.004 ± 0.003	-0.022 ± 0.014	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.007	0.334 ± 0.012	0.193	-0.007	-3.54	0.0740	0.0743	1.085	94.862 ± 0.946	0.284 ± 0.006	81.101 ± 1.677
UPI	beta44	0.11	0.139	0.029 ± 0.004	0.265 ± 0.033	0.007 ± 0.001	0.085 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	0.670 ± 0.048	0.819 ± 0.033	0.137	0.027	24.62	0.0842	0.0885	1.174	96.881 ± 0.745	0.392 ± 0.010	34.312 ± 2.034
UPI	beta45	0.11	0.122	0.012 ± 0.003	0.107 ± 0.029	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.468 ± 0.030	0.684 ± 0.027	0.123	0.013	12.07	0.0754	0.0766	1.075	96.514 ± 0.786	0.313 ± 0.005	38.165 ± 2.081
UPI	beta46	0.08	0.060	-0.020 ± 0.006	-0.248 ± 0.077	0.020 ± 0.002	0.143 ± 0.006	0.021 ± 0.002	0.144 ± 0.007	3.256 ± 0.276	1.805 ± 0.086	0.051	-0.029	-36.53	0.1309	0.1342	1.202	96.147 ± 0.824	0.675 ± 0.022	3.670 ± 0.805

Table S213

Simulation 2 - Condition 69: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.067	-0.013 ± 0.007	-0.161 ± 0.083	0.024 ± 0.002	0.155 ± 0.007	0.024 ± 0.002	0.156 ± 0.007	3.791 ± 0.323	1.947 ± 0.093	0.059	-0.021	-25.74	0.1409	0.1424	1.259	95.596 ± 0.879	0.767 ± 0.030	3.486 ± 0.786
UPI	beta51	0.16	0.176	0.016 ± 0.004	0.102 ± 0.024	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	0.316 ± 0.023	0.562 ± 0.024	0.172	0.012	7.36	0.0833	0.0842	0.985	93.578 ± 1.050	0.342 ± 0.005	55.596 ± 2.128
UPI	beta52	0.16	0.166	0.006 ± 0.003	0.035 ± 0.019	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.188 ± 0.014	0.433 ± 0.018	0.165	0.005	2.82	0.0727	0.0729	1.211	98.165 ± 0.575	0.328 ± 0.010	60.183 ± 2.097
UPI	beta53	0.16	0.146	-0.014 ± 0.003	-0.090 ± 0.019	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	0.196 ± 0.013	0.442 ± 0.017	0.148	-0.012	-7.68	0.0724	0.0735	1.130	94.679 ± 0.961	0.307 ± 0.006	52.110 ± 2.140
UPI	beta54	0.16	0.160	-0.000 ± 0.003	-0.000 ± 0.020	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	0.228 ± 0.017	0.477 ± 0.021	0.160	0.000	-0.29	0.0780	0.0780	1.224	97.431 ± 0.678	0.367 ± 0.014	52.661 ± 2.139
UPI	beta55	0.08	0.094	0.014 ± 0.005	0.176 ± 0.057	0.011 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	1.811 ± 0.111	1.346 ± 0.050	0.093	0.013	16.87	0.1096	0.1104	1.086	96.330 ± 0.805	0.455 ± 0.012	14.679 ± 1.516
UPI	beta56	0.08	0.086	0.006 ± 0.004	0.071 ± 0.053	0.010 ± 0.001	0.099 ± 0.004	0.010 ± 0.001	0.099 ± 0.004	1.540 ± 0.109	1.241 ± 0.051	0.084	0.004	4.93	0.0938	0.0939	1.087	97.064 ± 0.723	0.423 ± 0.010	10.459 ± 1.311
UPI	beta57	0.06	0.047	-0.013 ± 0.006	-0.215 ± 0.094	0.017 ± 0.001	0.132 ± 0.005	0.017 ± 0.001	0.132 ± 0.006	4.850 ± 0.365	2.202 ± 0.096	0.049	-0.011	-18.30	0.1192	0.1197	1.170	94.128 ± 1.007	0.604 ± 0.024	3.853 ± 0.824
UPI	beta58	0.06	0.034	-0.026 ± 0.006	-0.434 ± 0.097	0.018 ± 0.002	0.136 ± 0.006	0.019 ± 0.002	0.138 ± 0.007	5.290 ± 0.450	2.300 ± 0.109	0.030	-0.030	-49.78	0.1103	0.1142	1.275	95.413 ± 0.896	0.678 ± 0.036	3.303 ± 0.766

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S214

Simulation 2 - Condition 70: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.002	0.183 ± 0.006	0.196	-0.004	-1.77	0.0359	0.0361	1.078	96.682 ± 0.712	0.154 ± 0.000	99.842 ± 0.158
LSAM	beta42	0.20	0.196	-0.004 ± 0.002	-0.022 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.039 ± 0.002	0.196 ± 0.007	0.198	-0.002	-0.97	0.0381	0.0381	1.036	95.419 ± 0.831	0.159 ± 0.001	100.000 ± 0.000
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.006	0.200	0.000	0.24	0.0382	0.0382	1.017	95.893 ± 0.789	0.144 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.130	0.020 ± 0.002	0.180 ± 0.016	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.195 ± 0.011	0.441 ± 0.015	0.130	0.020	17.95	0.0428	0.0471	1.016	92.733 ± 1.032	0.176 ± 0.001	81.675 ± 1.538
LSAM	beta45	0.11	0.114	0.004 ± 0.002	0.038 ± 0.015	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.136 ± 0.007	0.369 ± 0.012	0.114	0.004	3.41	0.0430	0.0431	0.991	96.209 ± 0.759	0.157 ± 0.001	80.411 ± 1.577
LSAM	beta46	0.08	0.041	-0.039 ± 0.002	-0.485 ± 0.027	0.003 ± 0.000	0.055 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.708 ± 0.036	0.842 ± 0.024	0.041	-0.039	-49.22	0.0549	0.0676	0.981	85.624 ± 1.394	0.212 ± 0.002	8.847 ± 1.129
LSAM	beta47	0.08	0.046	-0.034 ± 0.002	-0.431 ± 0.027	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.646 ± 0.031	0.804 ± 0.023	0.041	-0.039	-48.59	0.0547	0.0671	0.999	86.256 ± 1.369	0.213 ± 0.002	9.479 ± 1.164
LSAM	beta51	0.16	0.171	0.011 ± 0.002	0.067 ± 0.013	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.103 ± 0.006	0.321 ± 0.011	0.170	0.010	6.34	0.0497	0.0507	0.985	94.945 ± 0.871	0.194 ± 0.001	93.839 ± 0.956
LSAM	beta52	0.16	0.151	-0.009 ± 0.002	-0.055 ± 0.010	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.250 ± 0.008	0.152	-0.008	-5.31	0.0406	0.0415	1.038	95.103 ± 0.858	0.159 ± 0.001	96.367 ± 0.744
LSAM	beta53	0.16	0.155	-0.005 ± 0.002	-0.029 ± 0.011	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.004	0.268 ± 0.009	0.154	-0.006	-3.72	0.0425	0.0429	0.983	93.681 ± 0.967	0.165 ± 0.001	96.051 ± 0.774
LSAM	beta54	0.16	0.154	-0.006 ± 0.002	-0.035 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.004	0.255 ± 0.009	0.152	-0.008	-5.23	0.0414	0.0422	1.000	95.419 ± 0.831	0.159 ± 0.001	96.051 ± 0.774
LSAM	beta55	0.08	0.091	0.011 ± 0.002	0.140 ± 0.028	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.522 ± 0.031	0.723 ± 0.025	0.091	0.011	13.49	0.0562	0.0573	0.995	94.945 ± 0.871	0.221 ± 0.002	37.283 ± 1.922
LSAM	beta56	0.08	0.080	-0.000 ± 0.002	-0.003 ± 0.030	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.569 ± 0.034	0.754 ± 0.027	0.082	0.002	2.92	0.0557	0.0558	0.887	93.049 ± 1.011	0.210 ± 0.001	35.071 ± 1.897
LSAM	beta57	0.06	0.033	-0.027 ± 0.002	-0.449 ± 0.037	0.003 ± 0.000	0.055 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	1.054 ± 0.054	1.026 ± 0.031	0.027	-0.033	-54.20	0.0560	0.0647	0.993	90.047 ± 1.190	0.216 ± 0.002	8.215 ± 1.091
LSAM	beta58	0.06	0.034	-0.026 ± 0.002	-0.427 ± 0.032	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.831 ± 0.041	0.911 ± 0.027	0.031	-0.029	-47.94	0.0496	0.0574	1.043	92.259 ± 1.062	0.198 ± 0.001	7.267 ± 1.032
QML	beta41	0.20	0.194	-0.006 ± 0.001	-0.029 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.002	0.172 ± 0.006	0.194	-0.006	-3.07	0.0344	0.0349	1.089	96.051 ± 0.774	0.145 ± 0.003	99.842 ± 0.158
QML	beta42	0.20	0.194	-0.006 ± 0.001	-0.031 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.002	0.183 ± 0.006	0.194	-0.006	-3.04	0.0334	0.0339	1.034	95.261 ± 0.845	0.146 ± 0.001	100.000 ± 0.000
QML	beta43	0.20	0.193	-0.007 ± 0.001	-0.036 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.002	0.177 ± 0.006	0.194	-0.006	-3.22	0.0356	0.0362	1.022	95.419 ± 0.831	0.139 ± 0.000	99.842 ± 0.158
QML	beta44	0.11	0.127	0.017 ± 0.002	0.159 ± 0.015	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.174 ± 0.009	0.417 ± 0.013	0.128	0.018	16.00	0.0422	0.0457	1.044	92.891 ± 1.021	0.174 ± 0.001	81.517 ± 1.543
QML	beta45	0.11	0.112	0.002 ± 0.002	0.021 ± 0.014	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.127 ± 0.007	0.356 ± 0.012	0.110	0.000	0.36	0.0436	0.0436	1.011	96.682 ± 0.712	0.155 ± 0.001	81.675 ± 1.538
QML	beta46	0.08	0.024	-0.056 ± 0.001	-0.703 ± 0.018	0.001 ± 0.000	0.036 ± 0.001	0.004 ± 0.000	0.067 ± 0.001	0.696 ± 0.026	0.834 ± 0.016	0.024	-0.056	-70.19	0.0361	0.0668	1.032	61.453 ± 1.934	0.145 ± 0.002	5.845 ± 0.932
QML	beta47	0.08	0.028	-0.052 ± 0.001	-0.647 ± 0.018	0.001 ± 0.000	0.036 ± 0.001	0.004 ± 0.000	0.063 ± 0.001	0.618 ± 0.024	0.786 ± 0.016	0.025	-0.055	-68.95	0.0364	0.0661	1.019	65.087 ± 1.895	0.143 ± 0.001	8.215 ± 1.091
QML	beta51	0.16	0.180	0.020 ± 0.002	0.128 ± 0.012	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.007	0.339 ± 0.011	0.180	0.020	12.49	0.0491	0.0530	1.015	93.997 ± 0.944	0.200 ± 0.001	94.787 ± 0.884
QML	beta52	0.16	0.144	-0.016 ± 0.002	-0.099 ± 0.009	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.004	0.257 ± 0.008	0.145	-0.015	-9.47	0.0390	0.0419	1.041	93.207 ± 1.000	0.155 ± 0.001	96.524 ± 0.728
QML	beta53	0.16	0.143	-0.017 ± 0.002	-0.109 ± 0.010	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.004	0.276 ± 0.009	0.142	-0.018	-11.43	0.0410	0.0449	0.991	92.575 ± 1.042	0.158 ± 0.000	94.313 ± 0.921
QML	beta54	0.16	0.156	-0.004 ± 0.002	-0.023 ± 0.010	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.008	0.156	-0.004	-2.43	0.0399	0.0401	1.004	95.893 ± 0.789	0.152 ± 0.001	98.104 ± 0.542
QML	beta55	0.08	0.089	0.009 ± 0.002	0.111 ± 0.027	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.486 ± 0.027	0.697 ± 0.024	0.087	0.007	8.68	0.0536	0.0541	1.031	95.419 ± 0.831	0.223 ± 0.002	35.703 ± 1.904
QML	beta56	0.08	0.076	-0.004 ± 0.002	-0.056 ± 0.028	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.502 ± 0.028	0.709 ± 0.024	0.076	-0.004	-5.59	0.0532	0.0534	0.913	93.049 ± 1.011	0.202 ± 0.001	32.701 ± 1.865
QML	beta57	0.06	0.020	-0.040 ± 0.002	-0.669 ± 0.026	0.002 ± 0.000	0.040 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.889 ± 0.038	0.943 ± 0.022	0.016	-0.044	-72.81	0.0398	0.0591	0.997	74.092 ± 1.741	0.156 ± 0.002	4.423 ± 0.817
QML	beta58	0.06	0.026	-0.034 ± 0.001	-0.573 ± 0.022	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.644 ± 0.028	0.803 ± 0.019	0.024	-0.036	-60.44	0.0348	0.0503	1.045	78.199 ± 1.641	0.138 ± 0.001	5.371 ± 0.896
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.017 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.204 ± 0.007	0.203	0.003	1.61	0.0438	0.0440	1.357	97.314 ± 0.643	0.217 ± 0.007	93.681 ± 0.967
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.002 ± 0.000	0.045 ± 0.002	0.002 ± 0.000	0.045 ± 0.002	0.050 ± 0.003	0.223 ± 0.009	0.199	-0.001	-0.36	0.0441	0.0441	1.313	97.314 ± 0.643	0.230 ± 0.009	92.417 ± 1.052
UPI	beta43	0.20	0.197	-0.003 ± 0.002	-0.015 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.037 ± 0.002	0.191 ± 0.007	0.196	-0.004	-1.87	0.0439	0.0440	1.122	96.524 ± 0.728	0.168 ± 0.004	98.262 ± 0.519
UPI	beta44	0.11	0.132	0.022 ± 0.002	0.196 ± 0.018	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.235 ± 0.014	0.485 ± 0.017	0.134	0.024	21.98	0.0478	0.0535	1.159	95.893 ± 0.789	0.222 ± 0.004	69.668 ± 1.827
UPI	beta45	0.11	0.115	0.005 ± 0.002	0.044 ± 0.015	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.146 ± 0.008	0.383 ± 0.013	0.114	0.004	3.42	0.0460	0.0462	1.074	96.367 ± 0.744	0.176 ± 0.002	73.776 ± 1.748
UPI	beta46	0.08	0.052	-0.028 ± 0.003	-0.348 ± 0.043	0.008 ± 0.001	0.087 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	1.301 ± 0.091	1.141 ± 0.045	0.048	-0.032	-39.99	0.0817	0.0878	1.264	95.577 ± 0.817	0.431 ± 0.019	6.477 ± 0.978

Table S214

Simulation 2 - Condition 70: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.050	-0.030 ± 0.004	-0.371 ± 0.044	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	1.380 ± 0.090	1.175 ± 0.044	0.040	-0.040	-49.88	0.0856	0.0945	1.194	93.523 ± 0.978	0.418 ± 0.018	7.267 ± 1.032
UPI	beta51	0.16	0.173	0.013 ± 0.002	0.083 ± 0.012	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.105 ± 0.006	0.323 ± 0.011	0.173	0.013	7.96	0.0522	0.0538	1.029	94.471 ± 0.908	0.202 ± 0.002	92.891 ± 1.021
UPI	beta52	0.16	0.155	-0.005 ± 0.002	-0.032 ± 0.011	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.073 ± 0.004	0.270 ± 0.009	0.155	-0.005	-3.20	0.0439	0.0442	1.177	96.367 ± 0.744	0.198 ± 0.003	88.626 ± 1.262
UPI	beta53	0.16	0.152	-0.008 ± 0.002	-0.048 ± 0.011	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.004	0.276 ± 0.009	0.147	-0.013	-8.12	0.0446	0.0464	1.081	95.419 ± 0.831	0.185 ± 0.002	92.101 ± 1.072
UPI	beta54	0.16	0.156	-0.004 ± 0.002	-0.026 ± 0.012	0.002 ± 0.000	0.047 ± 0.002	0.002 ± 0.000	0.047 ± 0.002	0.087 ± 0.006	0.294 ± 0.011	0.156	-0.004	-2.63	0.0473	0.0475	1.165	96.998 ± 0.678	0.214 ± 0.006	87.204 ± 1.328
UPI	beta55	0.08	0.090	0.010 ± 0.002	0.127 ± 0.029	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.537 ± 0.031	0.732 ± 0.026	0.088	0.008	10.44	0.0566	0.0572	1.071	96.682 ± 0.712	0.243 ± 0.002	31.912 ± 1.853
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.018 ± 0.030	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.574 ± 0.031	0.757 ± 0.026	0.082	0.002	3.11	0.0610	0.0610	0.951	93.681 ± 0.967	0.226 ± 0.002	29.542 ± 1.813
UPI	beta57	0.06	0.044	-0.016 ± 0.003	-0.266 ± 0.049	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	1.588 ± 0.097	1.260 ± 0.045	0.041	-0.019	-31.79	0.0777	0.0800	1.182	95.577 ± 0.817	0.343 ± 0.009	5.845 ± 0.932
UPI	beta58	0.06	0.037	-0.023 ± 0.003	-0.381 ± 0.053	0.006 ± 0.000	0.080 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	1.919 ± 0.131	1.385 ± 0.054	0.027	-0.033	-54.23	0.0702	0.0773	1.221	94.787 ± 0.884	0.383 ± 0.014	5.529 ± 0.908

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S215

Simulation 2 - Condition 71: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based								Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.240 ± 0.006	0.199	-0.001	-0.66	0.0512	0.0513	1.006	95.696 ± 0.642	0.189 ± 0.001	98.198 ± 0.421
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.203	0.003	1.62	0.0498	0.0499	0.976	94.194 ± 0.740	0.192 ± 0.001	97.698 ± 0.475
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.010 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.197	-0.003	-1.65	0.0470	0.0471	0.974	94.895 ± 0.696	0.180 ± 0.001	98.699 ± 0.359
LSAM	beta44	0.11	0.112	0.002 ± 0.002	0.016 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.230 ± 0.011	0.480 ± 0.013	0.114	0.004	4.02	0.0518	0.0520	0.975	94.294 ± 0.734	0.202 ± 0.001	60.360 ± 1.548
LSAM	beta45	0.11	0.113	0.003 ± 0.002	0.031 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.191 ± 0.008	0.437 ± 0.012	0.114	0.004	4.06	0.0506	0.0508	0.974	95.195 ± 0.677	0.183 ± 0.001	68.068 ± 1.475
LSAM	beta46	0.08	0.073	-0.007 ± 0.002	-0.084 ± 0.024	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.563 ± 0.025	0.751 ± 0.020	0.069	-0.011	-14.09	0.0588	0.0599	0.997	93.994 ± 0.752	0.233 ± 0.001	22.322 ± 1.317
LSAM	beta47	0.08	0.073	-0.007 ± 0.002	-0.083 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.578 ± 0.026	0.760 ± 0.021	0.071	-0.009	-10.65	0.0594	0.0600	0.977	93.794 ± 0.763	0.232 ± 0.001	22.823 ± 1.328
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.008 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.008	0.418 ± 0.012	0.159	-0.001	-0.61	0.0649	0.0649	0.923	93.393 ± 0.786	0.242 ± 0.001	73.473 ± 1.397
LSAM	beta52	0.16	0.156	-0.004 ± 0.002	-0.024 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.111 ± 0.005	0.333 ± 0.009	0.155	-0.005	-2.84	0.0537	0.0539	0.961	94.995 ± 0.690	0.200 ± 0.001	85.085 ± 1.127
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.159	-0.001	-0.40	0.0532	0.0532	1.004	96.096 ± 0.613	0.206 ± 0.001	86.086 ± 1.095
LSAM	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.159	-0.001	-0.31	0.0503	0.0503	1.024	95.295 ± 0.670	0.200 ± 0.001	88.889 ± 0.994
LSAM	beta55	0.08	0.081	0.001 ± 0.002	0.013 ± 0.027	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.715 ± 0.033	0.846 ± 0.024	0.082	0.002	2.53	0.0641	0.0642	0.957	93.493 ± 0.780	0.254 ± 0.002	27.528 ± 1.413
LSAM	beta56	0.08	0.083	0.003 ± 0.002	0.036 ± 0.026	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.698 ± 0.034	0.835 ± 0.024	0.082	0.002	2.32	0.0649	0.0649	0.931	93.493 ± 0.780	0.244 ± 0.002	28.028 ± 1.421
LSAM	beta57	0.06	0.054	-0.006 ± 0.002	-0.096 ± 0.035	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	1.232 ± 0.060	1.110 ± 0.032	0.052	-0.008	-12.71	0.0673	0.0677	0.950	93.794 ± 0.763	0.247 ± 0.001	14.314 ± 1.108
LSAM	beta58	0.06	0.057	-0.003 ± 0.002	-0.044 ± 0.032	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	1.031 ± 0.047	1.016 ± 0.028	0.054	-0.006	-9.21	0.0608	0.0611	0.969	94.394 ± 0.728	0.231 ± 0.001	14.915 ± 1.127
QML	beta41	0.20	0.196	-0.004 ± 0.001	-0.018 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.056 ± 0.002	0.238 ± 0.006	0.196	-0.004	-1.85	0.0493	0.0494	1.002	95.295 ± 0.670	0.186 ± 0.001	97.598 ± 0.484
QML	beta42	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.200	0.000	-0.02	0.0487	0.0487	0.974	94.294 ± 0.734	0.189 ± 0.001	97.798 ± 0.464
QML	beta43	0.20	0.194	-0.006 ± 0.001	-0.031 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.193	-0.007	-3.38	0.0458	0.0463	0.974	94.895 ± 0.696	0.177 ± 0.001	98.699 ± 0.359
QML	beta44	0.11	0.111	0.001 ± 0.002	0.012 ± 0.015	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.239 ± 0.011	0.489 ± 0.014	0.116	0.006	5.29	0.0542	0.0545	0.982	94.795 ± 0.703	0.207 ± 0.001	59.359 ± 1.554
QML	beta45	0.11	0.113	0.003 ± 0.002	0.030 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.194 ± 0.008	0.441 ± 0.012	0.114	0.004	3.72	0.0504	0.0505	0.981	94.895 ± 0.696	0.186 ± 0.001	66.667 ± 1.491
QML	beta46	0.08	0.059	-0.021 ± 0.002	-0.265 ± 0.020	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.454 ± 0.019	0.674 ± 0.016	0.054	-0.026	-31.96	0.0494	0.0556	1.010	90.591 ± 0.924	0.196 ± 0.001	17.718 ± 1.208
QML	beta47	0.08	0.060	-0.020 ± 0.002	-0.254 ± 0.020	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.450 ± 0.019	0.671 ± 0.017	0.057	-0.023	-28.48	0.0473	0.0525	0.989	91.091 ± 0.901	0.193 ± 0.001	19.319 ± 1.249
QML	beta51	0.16	0.167	0.007 ± 0.002	0.042 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.178 ± 0.008	0.422 ± 0.012	0.164	0.004	2.31	0.0678	0.0679	0.954	93.794 ± 0.763	0.252 ± 0.001	73.874 ± 1.390
QML	beta52	0.16	0.142	-0.018 ± 0.002	-0.109 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.143	-0.017	-10.89	0.0551	0.0578	0.970	92.893 ± 0.813	0.207 ± 0.001	76.276 ± 1.346
QML	beta53	0.16	0.144	-0.016 ± 0.002	-0.100 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.142	-0.018	-11.01	0.0495	0.0525	1.018	94.695 ± 0.709	0.208 ± 0.001	78.378 ± 1.302
QML	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.160	0.000	-0.31	0.0505	0.0505	1.027	95.696 ± 0.642	0.199 ± 0.001	89.089 ± 0.986
QML	beta55	0.08	0.081	0.001 ± 0.002	0.006 ± 0.027	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.721 ± 0.034	0.849 ± 0.024	0.083	0.003	3.13	0.0645	0.0645	0.976	93.894 ± 0.758	0.260 ± 0.002	26.727 ± 1.400
QML	beta56	0.08	0.082	0.002 ± 0.002	0.026 ± 0.026	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.681 ± 0.033	0.825 ± 0.024	0.081	0.001	0.92	0.0633	0.0633	0.944	93.293 ± 0.791	0.244 ± 0.002	25.826 ± 1.385
QML	beta57	0.06	0.042	-0.018 ± 0.002	-0.294 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.940 ± 0.043	0.970 ± 0.026	0.040	-0.020	-32.90	0.0546	0.0580	0.972	91.091 ± 0.901	0.211 ± 0.003	11.111 ± 0.994
QML	beta58	0.06	0.048	-0.012 ± 0.002	-0.201 ± 0.027	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.748 ± 0.034	0.865 ± 0.024	0.046	-0.014	-24.15	0.0502	0.0522	0.978	91.892 ± 0.864	0.193 ± 0.001	12.913 ± 1.061
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.022 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.002	0.238 ± 0.006	0.204	0.004	2.15	0.0495	0.0497	0.964	94.895 ± 0.696	0.179 ± 0.001	99.399 ± 0.244
UPI	beta42	0.20	0.207	0.007 ± 0.002	0.033 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.206	0.006	3.05	0.0499	0.0502	0.950	93.694 ± 0.769	0.185 ± 0.001	98.198 ± 0.421
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.000 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.199	-0.001	-0.55	0.0475	0.0475	0.973	94.995 ± 0.690	0.179 ± 0.001	98.699 ± 0.359
UPI	beta44	0.11	0.110	-0.000 ± 0.002	-0.001 ± 0.015	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.239 ± 0.011	0.489 ± 0.014	0.113	0.003	2.60	0.0522	0.0522	0.979	94.194 ± 0.740	0.207 ± 0.001	57.658 ± 1.563
UPI	beta45	0.11	0.114	0.004 ± 0.002	0.036 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.198 ± 0.008	0.445 ± 0.012	0.115	0.005	4.91	0.0517	0.0520	0.979	94.995 ± 0.690	0.187 ± 0.001	66.066 ± 1.498
UPI	beta46	0.08	0.081	0.001 ± 0.002	0.009 ± 0.025	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.607 ± 0.026	0.779 ± 0.021	0.081	0.001	1.06	0.0615	0.0615	1.037	96.096 ± 0.613	0.254 ± 0.002	20.621 ± 1.280

Table S215

Simulation 2 - Condition 71: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.002	-0.003 ± 0.025	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.648 ± 0.029	0.805 ± 0.022	0.077	-0.003	-3.82	0.0623	0.0624	1.023	96.096 ± 0.613	0.259 ± 0.002	22.022 ± 1.311
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.017 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.008	0.413 ± 0.012	0.160	0.000	-0.28	0.0658	0.0658	0.943	93.393 ± 0.786	0.244 ± 0.001	74.174 ± 1.385
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.332 ± 0.009	0.159	-0.001	-0.50	0.0551	0.0551	0.946	94.494 ± 0.722	0.197 ± 0.001	87.988 ± 1.029
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.018 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.328 ± 0.009	0.161	0.001	0.48	0.0525	0.0525	1.004	95.696 ± 0.642	0.206 ± 0.001	86.687 ± 1.075
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.008	0.162	0.002	1.44	0.0510	0.0511	1.019	95.395 ± 0.663	0.198 ± 0.001	90.090 ± 0.945
UPI	beta55	0.08	0.080	-0.000 ± 0.002	-0.005 ± 0.027	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.713 ± 0.034	0.844 ± 0.024	0.080	0.000	-0.39	0.0629	0.0629	0.968	93.694 ± 0.769	0.256 ± 0.002	24.925 ± 1.369
UPI	beta56	0.08	0.083	0.003 ± 0.002	0.032 ± 0.027	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.703 ± 0.036	0.838 ± 0.025	0.081	0.001	1.42	0.0618	0.0618	0.938	92.993 ± 0.808	0.247 ± 0.002	27.127 ± 1.407
UPI	beta57	0.06	0.059	-0.001 ± 0.002	-0.020 ± 0.036	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	1.294 ± 0.064	1.138 ± 0.033	0.058	-0.002	-4.00	0.0644	0.0644	0.967	93.894 ± 0.758	0.259 ± 0.002	13.714 ± 1.088
UPI	beta58	0.06	0.062	0.002 ± 0.002	0.027 ± 0.033	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	1.093 ± 0.053	1.045 ± 0.030	0.059	-0.001	-1.96	0.0617	0.0618	1.005	94.494 ± 0.722	0.247 ± 0.002	14.915 ± 1.127

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S216

Simulation 2 - Condition 72: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.201	0.001	0.53	0.0309	0.0309	0.999	95.500 ± 0.656	0.119 ± 0.000	100.000 ± 0.000
LSAM	beta42	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.199	-0.001	-0.62	0.0296	0.0297	1.001	95.200 ± 0.676	0.121 ± 0.000	100.000 ± 0.000
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.200	0.000	-0.06	0.0282	0.0282	1.035	95.300 ± 0.669	0.114 ± 0.000	100.000 ± 0.000
LSAM	beta44	0.11	0.116	0.006 ± 0.001	0.052 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.093 ± 0.004	0.305 ± 0.008	0.115	0.005	4.61	0.0317	0.0321	0.983	94.200 ± 0.739	0.127 ± 0.000	92.800 ± 0.817
LSAM	beta45	0.11	0.112	0.002 ± 0.001	0.021 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.078 ± 0.003	0.279 ± 0.008	0.112	0.002	2.15	0.0303	0.0304	0.972	93.500 ± 0.780	0.117 ± 0.000	95.700 ± 0.641
LSAM	beta46	0.08	0.069	-0.011 ± 0.001	-0.135 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.245 ± 0.011	0.495 ± 0.013	0.068	-0.012	-15.49	0.0384	0.0404	0.985	93.000 ± 0.807	0.147 ± 0.000	43.900 ± 1.569
LSAM	beta47	0.08	0.071	-0.009 ± 0.001	-0.114 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.241 ± 0.010	0.491 ± 0.013	0.070	-0.010	-12.68	0.0392	0.0405	0.975	93.200 ± 0.796	0.146 ± 0.000	47.800 ± 1.580
LSAM	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.159	-0.001	-0.75	0.0383	0.0383	0.950	92.800 ± 0.817	0.154 ± 0.001	97.700 ± 0.474
LSAM	beta52	0.16	0.160	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.162	0.002	0.97	0.0317	0.0318	1.013	95.800 ± 0.634	0.126 ± 0.000	99.900 ± 0.100
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.210 ± 0.006	0.158	-0.002	-1.05	0.0330	0.0330	0.992	94.700 ± 0.708	0.131 ± 0.000	100.000 ± 0.000
LSAM	beta54	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.157	-0.003	-1.58	0.0325	0.0326	0.991	95.100 ± 0.683	0.126 ± 0.000	99.900 ± 0.100
LSAM	beta55	0.08	0.082	0.002 ± 0.001	0.024 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.279 ± 0.013	0.529 ± 0.015	0.082	0.002	2.97	0.0412	0.0413	0.969	94.900 ± 0.696	0.161 ± 0.001	53.200 ± 1.578
LSAM	beta56	0.08	0.083	0.003 ± 0.001	0.033 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.254 ± 0.011	0.504 ± 0.014	0.083	0.003	4.35	0.0417	0.0418	0.972	94.800 ± 0.702	0.153 ± 0.001	57.100 ± 1.565
LSAM	beta57	0.06	0.052	-0.008 ± 0.001	-0.129 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.482 ± 0.021	0.694 ± 0.018	0.051	-0.009	-15.58	0.0404	0.0415	0.975	93.500 ± 0.780	0.156 ± 0.001	24.500 ± 1.360
LSAM	beta58	0.06	0.054	-0.006 ± 0.001	-0.100 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.451 ± 0.020	0.671 ± 0.018	0.052	-0.008	-13.75	0.0403	0.0411	0.938	93.000 ± 0.807	0.147 ± 0.000	29.700 ± 1.445
QML	beta41	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.199	-0.001	-0.66	0.0297	0.0297	0.999	95.000 ± 0.689	0.117 ± 0.000	100.000 ± 0.000
QML	beta42	0.20	0.197	-0.003 ± 0.001	-0.014 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.197	-0.003	-1.41	0.0293	0.0294	0.996	95.500 ± 0.656	0.119 ± 0.000	100.000 ± 0.000
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.020 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.195	-0.005	-2.41	0.0283	0.0287	1.030	95.700 ± 0.641	0.112 ± 0.000	100.000 ± 0.000
QML	beta44	0.11	0.116	0.006 ± 0.001	0.058 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.116	0.006	5.18	0.0330	0.0335	0.982	93.600 ± 0.774	0.129 ± 0.001	91.500 ± 0.882
QML	beta45	0.11	0.113	0.003 ± 0.001	0.024 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.079 ± 0.003	0.281 ± 0.008	0.113	0.003	2.56	0.0298	0.0300	0.980	94.000 ± 0.751	0.118 ± 0.000	95.500 ± 0.656
QML	beta46	0.08	0.053	-0.027 ± 0.001	-0.332 ± 0.012	0.001 ± 0.000	0.032 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.266 ± 0.010	0.516 ± 0.011	0.053	-0.027	-34.09	0.0314	0.0416	0.988	82.300 ± 1.207	0.122 ± 0.001	37.400 ± 1.530
QML	beta47	0.08	0.056	-0.024 ± 0.001	-0.301 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.243 ± 0.010	0.492 ± 0.011	0.055	-0.025	-31.42	0.0326	0.0411	0.988	84.400 ± 1.147	0.121 ± 0.000	43.500 ± 1.568
QML	beta51	0.16	0.166	0.006 ± 0.001	0.038 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.262 ± 0.008	0.164	0.004	2.43	0.0389	0.0390	0.965	93.700 ± 0.768	0.157 ± 0.001	97.700 ± 0.474
QML	beta52	0.16	0.147	-0.013 ± 0.001	-0.082 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.147	-0.013	-8.33	0.0321	0.0348	1.019	93.000 ± 0.807	0.130 ± 0.000	99.800 ± 0.141
QML	beta53	0.16	0.142	-0.018 ± 0.001	-0.114 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.142	-0.018	-11.50	0.0334	0.0381	0.996	91.800 ± 0.868	0.131 ± 0.000	98.900 ± 0.330
QML	beta54	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.158	-0.002	-1.34	0.0323	0.0324	0.992	94.700 ± 0.708	0.126 ± 0.000	99.900 ± 0.100
QML	beta55	0.08	0.082	0.002 ± 0.001	0.031 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.279 ± 0.013	0.528 ± 0.015	0.083	0.003	4.27	0.0423	0.0424	0.982	95.200 ± 0.676	0.162 ± 0.001	52.100 ± 1.580
QML	beta56	0.08	0.081	0.001 ± 0.001	0.018 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.247 ± 0.011	0.497 ± 0.013	0.082	0.002	2.83	0.0403	0.0404	0.985	94.700 ± 0.708	0.153 ± 0.001	55.800 ± 1.570
QML	beta57	0.06	0.040	-0.020 ± 0.001	-0.337 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.440 ± 0.017	0.663 ± 0.015	0.039	-0.021	-35.76	0.0334	0.0397	0.972	87.400 ± 1.049	0.131 ± 0.001	19.800 ± 1.260
QML	beta58	0.06	0.044	-0.016 ± 0.001	-0.266 ± 0.017	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.374 ± 0.015	0.611 ± 0.015	0.042	-0.018	-29.80	0.0325	0.0371	0.939	88.800 ± 0.997	0.122 ± 0.000	27.200 ± 1.407
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.204	0.004	1.92	0.0306	0.0309	0.953	94.200 ± 0.739	0.113 ± 0.000	100.000 ± 0.000
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.201	0.001	0.31	0.0299	0.0300	0.976	94.400 ± 0.727	0.116 ± 0.000	100.000 ± 0.000
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.201	0.001	0.35	0.0279	0.0279	1.031	95.600 ± 0.649	0.113 ± 0.000	100.000 ± 0.000
UPI	beta44	0.11	0.113	0.003 ± 0.001	0.029 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.091 ± 0.004	0.302 ± 0.009	0.113	0.003	3.08	0.0310	0.0312	0.991	94.300 ± 0.733	0.128 ± 0.000	91.700 ± 0.872
UPI	beta45	0.11	0.112	0.002 ± 0.001	0.019 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.079 ± 0.004	0.281 ± 0.008	0.112	0.002	1.75	0.0294	0.0295	0.969	93.800 ± 0.763	0.117 ± 0.000	95.400 ± 0.662
UPI	beta46	0.08	0.075	-0.005 ± 0.001	-0.059 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.255 ± 0.011	0.505 ± 0.014	0.073	-0.007	-8.44	0.0401	0.0407	0.982	94.500 ± 0.721	0.154 ± 0.001	47.000 ± 1.578

Table S216

Simulation 2 - Condition 72: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.076	-0.004 ± 0.001	-0.044 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.252 ± 0.011	0.502 ± 0.014	0.076	-0.004	-4.84	0.0408	0.0410	0.980	94.400 ± 0.727	0.154 ± 0.001	50.100 ± 1.581
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.258 ± 0.007	0.159	-0.001	-0.87	0.0389	0.0389	0.954	92.600 ± 0.828	0.154 ± 0.001	97.400 ± 0.503
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.015 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.005	0.164	0.004	2.32	0.0320	0.0322	0.997	95.300 ± 0.669	0.125 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.002 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.210 ± 0.006	0.159	-0.001	-0.58	0.0328	0.0328	0.992	94.800 ± 0.702	0.130 ± 0.000	100.000 ± 0.000
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.160	0.000	0.01	0.0324	0.0324	0.985	94.900 ± 0.696	0.125 ± 0.000	99.900 ± 0.100
UPI	beta55	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.277 ± 0.013	0.526 ± 0.015	0.080	0.000	0.47	0.0416	0.0416	0.973	94.800 ± 0.702	0.161 ± 0.001	52.000 ± 1.580
UPI	beta56	0.08	0.083	0.003 ± 0.001	0.032 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.253 ± 0.011	0.503 ± 0.014	0.083	0.003	3.46	0.0431	0.0431	0.976	94.800 ± 0.702	0.154 ± 0.001	57.500 ± 1.563
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.041 ± 0.022	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.484 ± 0.022	0.696 ± 0.019	0.057	-0.003	-5.76	0.0418	0.0419	0.984	94.200 ± 0.739	0.161 ± 0.001	27.400 ± 1.410
UPI	beta58	0.06	0.058	-0.002 ± 0.001	-0.037 ± 0.021	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.461 ± 0.020	0.679 ± 0.018	0.056	-0.004	-6.58	0.0415	0.0417	0.951	94.300 ± 0.733	0.152 ± 0.001	31.800 ± 1.473

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S217

Simulation 2 - Condition 73: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.003	-0.001 ± 0.017	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.190 ± 0.011	0.436 ± 0.015	0.199	-0.001	-0.31	0.0888	0.0888	1.032	95.376 ± 0.798	0.353 ± 0.003	—
LSAM	beta42	0.20	0.203	0.003 ± 0.003	0.017 ± 0.016	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.186 ± 0.010	0.432 ± 0.015	0.200	0.000	0.23	0.0861	0.0861	1.069	96.965 ± 0.652	0.362 ± 0.003	—
LSAM	beta43	0.20	0.202	0.002 ± 0.003	0.011 ± 0.016	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.177 ± 0.010	0.421 ± 0.014	0.201	0.001	0.40	0.0830	0.0830	1.000	94.220 ± 0.887	0.330 ± 0.003	—
LSAM	beta44	0.00	0.001	0.001 ± 0.005	—	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	—	—	0.001	0.001	—	0.1198	0.1198	1.038	98.266 ± 0.496	0.533 ± 0.010	1.734 ± 0.496
LSAM	beta45	0.00	0.001	0.001 ± 0.004	—	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	—	—	0.003	0.003	—	0.0937	0.0938	0.985	96.243 ± 0.723	0.428 ± 0.007	3.757 ± 0.723
LSAM	beta46	0.00	0.000	-0.000 ± 0.004	—	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	—	—	0.004	0.004	—	0.0788	0.0789	1.032	97.543 ± 0.588	0.373 ± 0.008	2.457 ± 0.588
LSAM	beta47	0.00	0.002	0.002 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	—	—	0.002	0.002	—	0.0755	0.0755	1.069	98.121 ± 0.516	0.353 ± 0.006	1.879 ± 0.516
LSAM	beta51	0.16	0.173	0.013 ± 0.005	0.080 ± 0.032	0.018 ± 0.001	0.133 ± 0.004	0.018 ± 0.001	0.134 ± 0.005	0.699 ± 0.041	0.836 ± 0.029	0.170	0.010	6.19	0.1291	0.1295	0.924	94.509 ± 0.866	0.483 ± 0.004	—
LSAM	beta52	0.16	0.164	0.004 ± 0.004	0.026 ± 0.023	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.357 ± 0.022	0.597 ± 0.021	0.163	0.003	1.66	0.0905	0.0906	1.007	94.942 ± 0.833	0.377 ± 0.003	—
LSAM	beta53	0.16	0.153	-0.007 ± 0.004	-0.045 ± 0.022	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	0.342 ± 0.019	0.585 ± 0.020	0.148	-0.012	-7.64	0.0898	0.0907	1.041	96.098 ± 0.736	0.381 ± 0.003	—
LSAM	beta54	0.16	0.160	0.000 ± 0.004	0.002 ± 0.023	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.351 ± 0.021	0.592 ± 0.021	0.158	-0.002	-1.48	0.0931	0.0931	0.990	95.087 ± 0.822	0.368 ± 0.003	—
LSAM	beta55	0.00	-0.013	-0.013 ± 0.007	—	0.036 ± 0.002	0.189 ± 0.006	0.036 ± 0.002	0.189 ± 0.007	—	—	-0.015	-0.015	—	0.1763	0.1770	1.001	96.965 ± 0.652	0.741 ± 0.015	3.035 ± 0.652
LSAM	beta56	0.00	-0.004	-0.004 ± 0.006	—	0.022 ± 0.001	0.150 ± 0.005	0.022 ± 0.001	0.150 ± 0.005	—	—	-0.006	-0.006	—	0.1443	0.1444	1.026	98.121 ± 0.516	0.602 ± 0.011	1.879 ± 0.516
LSAM	beta57	0.00	0.011	0.011 ± 0.004	—	0.011 ± 0.001	0.107 ± 0.004	0.012 ± 0.001	0.107 ± 0.004	—	—	0.011	0.011	—	0.0960	0.0966	0.979	97.688 ± 0.571	0.410 ± 0.008	2.312 ± 0.571
LSAM	beta58	0.00	0.002	0.002 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	—	—	0.003	0.003	—	0.0663	0.0664	1.031	97.543 ± 0.588	0.316 ± 0.005	2.457 ± 0.588
QML	beta41	0.20	0.196	-0.004 ± 0.003	-0.022 ± 0.014	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.140 ± 0.007	0.374 ± 0.012	0.194	-0.006	-3.00	0.0791	0.0793	1.060	94.942 ± 0.833	0.310 ± 0.002	—
QML	beta42	0.20	0.199	-0.001 ± 0.003	-0.004 ± 0.015	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.150 ± 0.008	0.387 ± 0.013	0.197	-0.003	-1.67	0.0796	0.0797	1.054	95.231 ± 0.810	0.320 ± 0.002	—
QML	beta43	0.20	0.197	-0.003 ± 0.003	-0.015 ± 0.015	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.151 ± 0.008	0.388 ± 0.013	0.193	-0.007	-3.41	0.0766	0.0769	0.999	94.075 ± 0.897	0.304 ± 0.002	—
QML	beta44	0.00	-0.015	-0.015 ± 0.004	—	0.011 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.107 ± 0.004	—	—	-0.016	-0.016	—	0.1020	0.1033	1.064	96.387 ± 0.709	0.444 ± 0.007	3.613 ± 0.709
QML	beta45	0.00	-0.014	-0.014 ± 0.003	—	0.008 ± 0.001	0.091 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	—	—	-0.014	-0.014	—	0.0852	0.0864	0.972	94.653 ± 0.855	0.348 ± 0.004	5.347 ± 0.855
QML	beta46	0.00	-0.001	-0.001 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	—	—	-0.001	-0.001	—	0.0652	0.0652	1.092	96.387 ± 0.709	0.298 ± 0.005	3.613 ± 0.709
QML	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.002	-0.002	—	0.0612	0.0613	1.082	95.954 ± 0.749	0.279 ± 0.004	4.046 ± 0.749
QML	beta51	0.16	0.176	0.016 ± 0.005	0.102 ± 0.030	0.016 ± 0.001	0.126 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	0.635 ± 0.039	0.797 ± 0.029	0.171	0.011	7.02	0.1298	0.1303	1.037	96.821 ± 0.667	0.514 ± 0.005	—
QML	beta52	0.16	0.160	-0.000 ± 0.003	-0.000 ± 0.021	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.303 ± 0.018	0.551 ± 0.019	0.159	-0.001	-0.50	0.0858	0.0858	1.067	95.665 ± 0.774	0.369 ± 0.004	—
QML	beta53	0.16	0.150	-0.010 ± 0.003	-0.062 ± 0.020	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.294 ± 0.017	0.542 ± 0.018	0.146	-0.014	-9.02	0.0862	0.0874	1.069	96.387 ± 0.709	0.361 ± 0.003	—
QML	beta54	0.16	0.156	-0.004 ± 0.003	-0.023 ± 0.020	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.276 ± 0.016	0.525 ± 0.018	0.154	-0.006	-3.91	0.0844	0.0846	1.022	94.509 ± 0.866	0.337 ± 0.002	—
QML	beta55	0.00	-0.012	-0.012 ± 0.006	—	0.021 ± 0.001	0.146 ± 0.004	0.021 ± 0.001	0.146 ± 0.005	—	—	-0.011	-0.011	—	0.1530	0.1533	1.082	95.520 ± 0.786	0.619 ± 0.010	4.480 ± 0.786
QML	beta56	0.00	-0.018	-0.018 ± 0.005	—	0.014 ± 0.001	0.119 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	—	—	-0.016	-0.016	—	0.1178	0.1189	1.038	96.098 ± 0.736	0.485 ± 0.006	3.902 ± 0.736
QML	beta57	0.00	0.004	0.004 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	0.003	0.003	—	0.0784	0.0785	1.025	94.942 ± 0.833	0.329 ± 0.007	5.058 ± 0.833
QML	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	—	0.0565	0.0565	1.017	96.387 ± 0.709	0.245 ± 0.003	3.613 ± 0.709
UPI	beta41	0.20	0.203	0.003 ± 0.003	0.015 ± 0.017	0.008 ± 0.000	0.090 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	0.202 ± 0.012	0.450 ± 0.016	0.205	0.005	2.68	0.0980	0.0982	1.254	97.977 ± 0.535	0.442 ± 0.008	—
UPI	beta42	0.20	0.205	0.005 ± 0.003	0.026 ± 0.017	0.008 ± 0.000	0.090 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	0.205 ± 0.012	0.452 ± 0.016	0.200	0.000	0.18	0.0976	0.0976	1.297	98.555 ± 0.454	0.460 ± 0.011	—
UPI	beta43	0.20	0.201	0.001 ± 0.003	0.005 ± 0.016	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.182 ± 0.010	0.427 ± 0.015	0.196	-0.004	-1.91	0.0880	0.0881	1.199	97.110 ± 0.637	0.401 ± 0.009	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.006	—	0.026 ± 0.002	0.161 ± 0.006	0.026 ± 0.002	0.161 ± 0.007	—	—	-0.007	-0.007	—	0.1425	0.1426	1.520	99.711 ± 0.204	0.962 ± 0.043	0.289 ± 0.204
UPI	beta45	0.00	-0.001	-0.001 ± 0.005	—	0.018 ± 0.001	0.134 ± 0.005	0.018 ± 0.001	0.134 ± 0.006	—	—	0.005	0.005	—	0.1172	0.1173	1.469	99.566 ± 0.250	0.774 ± 0.041	0.434 ± 0.250
UPI	beta46	0.00	-0.002	-0.002 ± 0.004	—	0.013 ± 0.001	0.115 ± 0.005	0.013 ± 0.001	0.115 ± 0.005	—	—	0.002	0.002	—	0.1020	0.1020	1.472	99.711 ± 0.204	0.662 ± 0.025	0.289 ± 0.204

Table S217

Simulation 2 - Condition 73: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.006	0.006 ± 0.004	—	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.103 ± 0.004	—	—	0.006	0.006	—	0.0925	0.0927	1.586	99.566 ± 0.250	0.639 ± 0.031	0.434 ± 0.250
UPI	beta51	0.16	0.172	0.012 ± 0.005	0.077 ± 0.029	0.015 ± 0.001	0.124 ± 0.004	0.016 ± 0.001	0.125 ± 0.004	0.606 ± 0.036	0.778 ± 0.027	0.172	0.012	7.52	0.1234	0.1240	1.071	97.543 ± 0.588	0.521 ± 0.006	—
UPI	beta52	0.16	0.164	0.004 ± 0.004	0.028 ± 0.023	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.365 ± 0.023	0.604 ± 0.022	0.159	-0.001	-0.72	0.0942	0.0942	1.159	96.532 ± 0.696	0.439 ± 0.008	—
UPI	beta53	0.16	0.154	-0.006 ± 0.004	-0.038 ± 0.022	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.095 ± 0.003	0.350 ± 0.019	0.591 ± 0.020	0.147	-0.013	-8.20	0.0940	0.0949	1.200	97.688 ± 0.571	0.444 ± 0.008	—
UPI	beta54	0.16	0.162	0.002 ± 0.004	0.013 ± 0.023	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.351 ± 0.022	0.593 ± 0.022	0.160	0.000	0.26	0.0959	0.0959	1.143	96.821 ± 0.667	0.425 ± 0.007	—
UPI	beta55	0.00	-0.010	-0.010 ± 0.008	—	0.045 ± 0.003	0.212 ± 0.008	0.045 ± 0.003	0.212 ± 0.009	—	—	-0.010	-0.010	—	0.1801	0.1804	1.363	99.277 ± 0.322	1.132 ± 0.045	0.723 ± 0.322
UPI	beta56	0.00	-0.005	-0.005 ± 0.006	—	0.028 ± 0.002	0.168 ± 0.006	0.028 ± 0.002	0.168 ± 0.007	—	—	0.000	0.000	—	0.1423	0.1423	1.455	99.711 ± 0.204	0.961 ± 0.042	0.289 ± 0.204
UPI	beta57	0.00	0.008	0.008 ± 0.005	—	0.015 ± 0.001	0.123 ± 0.005	0.015 ± 0.001	0.123 ± 0.005	—	—	0.007	0.007	—	0.1052	0.1054	1.334	99.711 ± 0.204	0.641 ± 0.026	0.289 ± 0.204
UPI	beta58	0.00	0.000	-0.000 ± 0.003	—	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	—	—	0.002	0.002	—	0.0776	0.0776	1.398	99.133 ± 0.352	0.497 ± 0.017	0.867 ± 0.352

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S218

Simulation 2 - Condition 74: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.201	0.001	0.49	0.0507	0.0507	1.016	95.723 ± 0.646	0.205 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.004	0.270 ± 0.008	0.202	0.002	0.87	0.0512	0.0513	1.005	95.214 ± 0.681	0.213 ± 0.001	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.197	-0.003	-1.59	0.0532	0.0533	0.971	93.585 ± 0.782	0.197 ± 0.001	—
LSAM	beta44	0.00	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.002	0.002	—	0.0676	0.0677	0.989	96.436 ± 0.592	0.282 ± 0.002	3.564 ± 0.592
LSAM	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	—	0.0558	0.0558	0.990	96.130 ± 0.615	0.232 ± 0.002	3.870 ± 0.615
LSAM	beta46	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.001	-0.001	—	0.0479	0.0479	0.973	96.130 ± 0.615	0.196 ± 0.002	3.870 ± 0.615
LSAM	beta47	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	—	0.0477	0.0477	1.026	96.640 ± 0.575	0.191 ± 0.002	3.360 ± 0.575
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.009 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.209 ± 0.010	0.457 ± 0.013	0.160	0.000	0.07	0.0721	0.0721	0.975	94.705 ± 0.715	0.280 ± 0.001	—
LSAM	beta52	0.16	0.160	0.000 ± 0.002	0.002 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.118 ± 0.005	0.343 ± 0.009	0.160	0.000	-0.18	0.0562	0.0562	0.998	95.825 ± 0.638	0.215 ± 0.001	—
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.125 ± 0.006	0.354 ± 0.010	0.159	-0.001	-0.60	0.0571	0.0571	0.997	95.214 ± 0.681	0.221 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.118 ± 0.005	0.344 ± 0.010	0.162	0.002	1.33	0.0563	0.0563	0.974	95.010 ± 0.695	0.210 ± 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.003	—	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	0.001	0.001	—	0.0913	0.0913	0.981	96.232 ± 0.608	0.374 ± 0.003	3.768 ± 0.608
LSAM	beta56	0.00	0.005	0.005 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	0.005	0.005	—	0.0741	0.0743	0.995	95.825 ± 0.638	0.309 ± 0.002	4.175 ± 0.638
LSAM	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.001	-0.001	—	0.0524	0.0525	0.984	96.029 ± 0.623	0.215 ± 0.002	3.971 ± 0.623
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0429	0.0429	0.968	94.705 ± 0.715	0.171 ± 0.001	5.295 ± 0.715
QML	beta41	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.197	-0.003	-1.54	0.0474	0.0475	1.021	95.316 ± 0.674	0.191 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.199	-0.001	-0.35	0.0496	0.0496	1.000	94.807 ± 0.708	0.198 ± 0.001	—
QML	beta43	0.20	0.196	-0.004 ± 0.002	-0.018 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.194	-0.006	-2.93	0.0512	0.0516	0.987	94.196 ± 0.746	0.190 ± 0.001	—
QML	beta44	0.00	-0.014	-0.014 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.015	-0.015	—	0.0688	0.0704	1.050	94.705 ± 0.715	0.272 ± 0.008	5.295 ± 0.715
QML	beta45	0.00	-0.015	-0.015 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.015	-0.015	—	0.0533	0.0553	1.011	94.501 ± 0.727	0.216 ± 0.001	5.499 ± 0.727
QML	beta46	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.002	-0.002	—	0.0427	0.0427	1.033	95.927 ± 0.631	0.175 ± 0.002	4.073 ± 0.631
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.003	-0.003	—	0.0407	0.0408	1.108	95.316 ± 0.674	0.177 ± 0.009	4.684 ± 0.674
QML	beta51	0.16	0.163	0.003 ± 0.002	0.019 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.204 ± 0.010	0.452 ± 0.013	0.163	0.003	1.64	0.0696	0.0697	1.013	95.621 ± 0.653	0.287 ± 0.001	—
QML	beta52	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.158	-0.002	-1.51	0.0536	0.0536	1.028	95.418 ± 0.667	0.212 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.005	0.339 ± 0.009	0.156	-0.004	-2.19	0.0535	0.0536	1.009	95.519 ± 0.660	0.215 ± 0.001	—
QML	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.329 ± 0.009	0.159	-0.001	-0.59	0.0535	0.0535	0.983	95.112 ± 0.688	0.203 ± 0.001	—
QML	beta55	0.00	-0.004	-0.004 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	—	—	-0.004	-0.004	—	0.0851	0.0852	1.047	95.519 ± 0.660	0.354 ± 0.003	4.481 ± 0.660
QML	beta56	0.00	-0.008	-0.008 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.007	-0.007	—	0.0682	0.0686	1.011	95.316 ± 0.674	0.279 ± 0.002	4.684 ± 0.674
QML	beta57	0.00	-0.006	-0.006 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.004	-0.004	—	0.0476	0.0478	1.036	94.603 ± 0.721	0.196 ± 0.002	5.397 ± 0.721
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.004	-0.004	—	0.0380	0.0383	1.016	94.196 ± 0.746	0.153 ± 0.003	5.804 ± 0.746
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.001 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.200	0.000	-0.08	0.0514	0.0514	1.058	96.640 ± 0.575	0.214 ± 0.001	—
UPI	beta42	0.20	0.204	0.004 ± 0.002	0.021 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.075 ± 0.004	0.274 ± 0.008	0.202	0.002	1.19	0.0527	0.0527	1.038	95.316 ± 0.674	0.222 ± 0.001	—
UPI	beta43	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.197	-0.003	-1.70	0.0531	0.0532	0.999	94.399 ± 0.734	0.203 ± 0.001	—
UPI	beta44	0.00	0.000	-0.000 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	-0.001	-0.001	—	0.0745	0.0745	1.061	98.371 ± 0.404	0.333 ± 0.005	1.629 ± 0.404
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0574	0.0574	1.076	98.167 ± 0.428	0.267 ± 0.003	1.833 ± 0.428
UPI	beta46	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	0.000	0.000	—	0.0505	0.0505	1.053	97.862 ± 0.462	0.238 ± 0.003	2.138 ± 0.462

Table S218

Simulation 2 - Condition 74: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0476	0.0476	1.102	97.862 ± 0.462	0.227 ± 0.003	2.138 ± 0.462
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.005 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.200 ± 0.009	0.447 ± 0.013	0.161	0.001	0.60	0.0694	0.0694	1.006	95.519 ± 0.660	0.282 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.008 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.005	0.342 ± 0.009	0.162	0.002	1.23	0.0558	0.0558	1.029	96.741 ± 0.567	0.221 ± 0.001	—
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.005	0.352 ± 0.010	0.161	0.001	0.72	0.0584	0.0584	1.027	96.334 ± 0.600	0.227 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.020 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.118 ± 0.005	0.343 ± 0.010	0.163	0.003	1.92	0.0556	0.0557	1.000	95.621 ± 0.653	0.215 ± 0.001	—
UPI	beta55	0.00	0.000	0.000 ± 0.003	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	—	—	0.001	0.001	—	0.0904	0.0904	1.114	98.167 ± 0.428	0.432 ± 0.010	1.833 ± 0.428
UPI	beta56	0.00	0.006	0.006 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	0.004	0.004	—	0.0731	0.0732	1.084	97.760 ± 0.472	0.347 ± 0.005	2.240 ± 0.472
UPI	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0547	0.0548	1.060	97.556 ± 0.493	0.250 ± 0.005	2.444 ± 0.493
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.001	-0.001	—	0.0431	0.0431	1.079	97.352 ± 0.512	0.198 ± 0.002	2.648 ± 0.512

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S219

Simulation 2 - Condition 75: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.009	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.008	0.201	0.001	0.73	0.0591	0.0591	0.982	94.260 ± 0.738	0.229 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.293 ± 0.008	0.200	0.000	-0.15	0.0576	0.0576	1.032	96.576 ± 0.577	0.237 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.011 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.008	0.201	0.001	0.26	0.0592	0.0592	0.998	94.763 ± 0.707	0.224 ± 0.001	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.005	-0.005	—	0.0706	0.0708	0.982	94.965 ± 0.694	0.278 ± 0.002	5.035 ± 0.694
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.002	0.002	—	0.0635	0.0635	0.915	92.850 ± 0.818	0.237 ± 0.001	7.150 ± 0.818
LSAM	beta46	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	0.001	0.001	—	0.0479	0.0479	0.997	95.368 ± 0.667	0.195 ± 0.001	4.632 ± 0.667
LSAM	beta47	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	—	0.0492	0.0493	0.944	94.058 ± 0.750	0.192 ± 0.001	5.942 ± 0.750
LSAM	beta51	0.16	0.162	0.002 ± 0.003	0.010 ± 0.016	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.270 ± 0.012	0.519 ± 0.015	0.162	0.002	1.07	0.0843	0.0843	0.971	94.663 ± 0.713	0.316 ± 0.001	—
LSAM	beta52	0.16	0.160	0.000 ± 0.002	0.002 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.411 ± 0.012	0.161	0.001	0.35	0.0631	0.0631	0.951	93.152 ± 0.801	0.245 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.412 ± 0.011	0.156	-0.004	-2.45	0.0661	0.0662	0.985	94.965 ± 0.694	0.254 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.008	0.399 ± 0.011	0.160	0.000	0.23	0.0587	0.0587	0.972	93.857 ± 0.762	0.244 ± 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.003	—	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	—	—	-0.003	-0.003	—	0.0954	0.0954	0.955	94.159 ± 0.744	0.362 ± 0.003	5.841 ± 0.744
LSAM	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	—	—	0.000	0.000	—	0.0741	0.0741	0.974	94.562 ± 0.720	0.311 ± 0.002	5.438 ± 0.720
LSAM	beta57	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.001	-0.001	—	0.0574	0.0574	0.952	93.656 ± 0.774	0.212 ± 0.002	6.344 ± 0.774
LSAM	beta58	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.002	-0.002	—	0.0447	0.0447	0.961	94.965 ± 0.694	0.178 ± 0.001	5.035 ± 0.694
QML	beta41	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.004	0.283 ± 0.008	0.198	-0.002	-1.19	0.0553	0.0554	0.994	94.562 ± 0.720	0.221 ± 0.001	—
QML	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.199	-0.001	-0.34	0.0553	0.0553	1.043	96.475 ± 0.585	0.229 ± 0.001	—
QML	beta43	0.20	0.200	0.000 ± 0.002	0.001 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.077 ± 0.003	0.277 ± 0.007	0.199	-0.001	-0.48	0.0575	0.0575	1.011	94.965 ± 0.694	0.219 ± 0.001	—
QML	beta44	0.00	-0.011	-0.011 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.013	-0.013	—	0.0675	0.0686	1.007	94.965 ± 0.694	0.269 ± 0.002	5.035 ± 0.694
QML	beta45	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.002	-0.002	—	0.0615	0.0615	0.930	93.152 ± 0.801	0.230 ± 0.001	6.848 ± 0.801
QML	beta46	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0437	0.0437	1.016	95.166 ± 0.681	0.182 ± 0.001	4.834 ± 0.681
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.004	-0.004	—	0.0448	0.0449	0.964	93.756 ± 0.768	0.177 ± 0.001	6.244 ± 0.768
QML	beta51	0.16	0.163	0.003 ± 0.003	0.016 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.260 ± 0.012	0.510 ± 0.014	0.162	0.002	1.09	0.0827	0.0827	1.017	96.475 ± 0.585	0.325 ± 0.001	—
QML	beta52	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.012	0.158	-0.002	-1.16	0.0626	0.0626	0.967	93.656 ± 0.774	0.245 ± 0.001	—
QML	beta53	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.402 ± 0.011	0.156	-0.004	-2.74	0.0650	0.0652	1.001	95.770 ± 0.639	0.252 ± 0.001	—
QML	beta54	0.16	0.160	0.000 ± 0.002	0.000 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.160	0.000	-0.15	0.0570	0.0570	0.980	93.756 ± 0.768	0.240 ± 0.001	—
QML	beta55	0.00	-0.003	-0.003 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	—	—	-0.005	-0.005	—	0.0887	0.0888	1.005	94.965 ± 0.694	0.351 ± 0.002	5.035 ± 0.694
QML	beta56	0.00	-0.004	-0.004 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.005	-0.005	—	0.0702	0.0704	1.017	94.663 ± 0.713	0.298 ± 0.002	5.337 ± 0.713
QML	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.002	-0.002	—	0.0519	0.0520	0.981	93.656 ± 0.774	0.199 ± 0.002	6.344 ± 0.774
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.005	-0.005	—	0.0416	0.0418	0.988	95.166 ± 0.681	0.166 ± 0.001	4.834 ± 0.681
UPI	beta41	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.009	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.004	0.299 ± 0.008	0.202	0.002	0.87	0.0600	0.0600	0.994	95.065 ± 0.687	0.233 ± 0.001	—
UPI	beta42	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.294 ± 0.008	0.201	0.001	0.35	0.0567	0.0567	1.049	97.684 ± 0.477	0.241 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.287 ± 0.008	0.200	0.000	-0.11	0.0600	0.0600	1.015	95.166 ± 0.681	0.228 ± 0.001	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.004	-0.004	—	0.0704	0.0704	1.026	96.274 ± 0.601	0.303 ± 0.003	3.726 ± 0.601
UPI	beta45	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.004	0.004	—	0.0646	0.0647	0.946	94.058 ± 0.750	0.258 ± 0.002	5.942 ± 0.750
UPI	beta46	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.001	0.001	—	0.0472	0.0473	1.049	95.972 ± 0.624	0.215 ± 0.002	4.028 ± 0.624

Table S219

Simulation 2 - Condition 75: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.000	0.000	—	0.0512	0.0512	0.991	95.267 ± 0.674	0.211 ± 0.002	4.733 ± 0.674
UPI	beta51	0.16	0.162	0.002 ± 0.003	0.011 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.257 ± 0.011	0.507 ± 0.014	0.162	0.002	1.22	0.0824	0.0824	1.010	95.468 ± 0.660	0.321 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.002	0.001 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.412 ± 0.012	0.159	-0.001	-0.39	0.0633	0.0633	0.962	93.656 ± 0.774	0.248 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.008	0.410 ± 0.011	0.155	-0.005	-2.94	0.0646	0.0647	1.006	95.972 ± 0.624	0.259 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.008	0.399 ± 0.011	0.159	-0.001	-0.90	0.0575	0.0576	0.987	93.756 ± 0.768	0.247 ± 0.001	—
UPI	beta55	0.00	0.000	0.000 ± 0.003	—	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	—	—	0.001	0.001	—	0.0934	0.0934	1.005	96.777 ± 0.560	0.387 ± 0.004	3.223 ± 0.560
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.000	0.000	—	0.0777	0.0777	1.019	95.972 ± 0.624	0.331 ± 0.003	4.028 ± 0.624
UPI	beta57	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0561	0.0562	0.976	95.166 ± 0.681	0.226 ± 0.002	4.834 ± 0.681
UPI	beta58	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.002	-0.002	—	0.0454	0.0454	1.035	96.576 ± 0.577	0.197 ± 0.004	3.424 ± 0.577

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S220

Simulation 2 - Condition 76: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.002	0.174 ± 0.005	0.199	-0.001	-0.61	0.0329	0.0330	1.037	96.100 ± 0.612	0.141 ± 0.000	—
LSAM	beta42	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.200	0.000	-0.02	0.0357	0.0357	1.021	95.100 ± 0.683	0.146 ± 0.000	—
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.177 ± 0.005	0.201	0.001	0.45	0.0347	0.0347	0.999	94.300 ± 0.733	0.139 ± 0.000	—
LSAM	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0436	0.0436	0.974	93.700 ± 0.768	0.165 ± 0.001	6.300 ± 0.768
LSAM	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0343	0.0343	1.022	95.000 ± 0.689	0.145 ± 0.001	5.000 ± 0.689
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0304	0.0304	0.979	95.000 ± 0.689	0.116 ± 0.001	5.000 ± 0.689
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0295	0.0295	1.009	95.300 ± 0.669	0.116 ± 0.001	4.700 ± 0.669
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.315 ± 0.009	0.161	0.001	0.85	0.0499	0.0499	1.004	95.400 ± 0.662	0.198 ± 0.000	—
LSAM	beta52	0.16	0.160	0.000 ± 0.001	0.003 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.160	0.000	0.04	0.0374	0.0374	1.008	95.200 ± 0.676	0.152 ± 0.000	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.159	-0.001	-0.89	0.0395	0.0395	0.993	94.800 ± 0.702	0.158 ± 0.000	—
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.240 ± 0.007	0.161	0.001	0.75	0.0387	0.0387	1.006	95.900 ± 0.627	0.151 ± 0.000	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	—	0.0561	0.0561	0.985	95.300 ± 0.669	0.219 ± 0.001	4.700 ± 0.669
LSAM	beta56	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.002	0.002	—	0.0526	0.0526	0.946	93.900 ± 0.757	0.189 ± 0.001	6.100 ± 0.757
LSAM	beta57	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.000	0.000	—	0.0332	0.0332	0.991	95.200 ± 0.676	0.129 ± 0.001	4.800 ± 0.676
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0284	0.0284	0.967	95.200 ± 0.676	0.110 ± 0.001	4.800 ± 0.676
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.168 ± 0.005	0.197	-0.003	-1.37	0.0326	0.0327	1.043	96.100 ± 0.612	0.137 ± 0.000	—
QML	beta42	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.199	-0.001	-0.56	0.0349	0.0350	1.021	95.000 ± 0.689	0.142 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.174 ± 0.005	0.200	0.000	-0.22	0.0337	0.0337	1.002	94.100 ± 0.745	0.137 ± 0.000	—
QML	beta44	0.00	-0.009	-0.009 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.009	-0.009	—	0.0435	0.0445	0.984	93.500 ± 0.780	0.166 ± 0.001	6.500 ± 0.780
QML	beta45	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.008	-0.008	—	0.0340	0.0349	1.014	94.800 ± 0.702	0.142 ± 0.001	5.200 ± 0.702
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0281	0.0281	0.991	95.400 ± 0.662	0.111 ± 0.001	4.600 ± 0.662
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.001	-0.001	—	0.0277	0.0277	1.023	95.500 ± 0.656	0.111 ± 0.001	4.500 ± 0.656
QML	beta51	0.16	0.163	0.003 ± 0.002	0.017 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.314 ± 0.009	0.163	0.003	1.70	0.0501	0.0502	1.018	95.500 ± 0.656	0.200 ± 0.000	—
QML	beta52	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.159	-0.001	-0.50	0.0373	0.0374	1.019	95.400 ± 0.662	0.151 ± 0.000	—
QML	beta53	0.16	0.158	-0.002 ± 0.001	-0.011 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.157	-0.003	-1.83	0.0387	0.0389	0.996	94.400 ± 0.727	0.156 ± 0.000	—
QML	beta54	0.16	0.160	0.000 ± 0.001	0.000 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.236 ± 0.006	0.160	0.000	0.14	0.0387	0.0387	1.009	96.200 ± 0.605	0.150 ± 0.000	—
QML	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.002	-0.002	—	0.0550	0.0550	0.997	95.500 ± 0.656	0.217 ± 0.001	4.500 ± 0.656
QML	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.004	-0.004	—	0.0488	0.0489	0.951	93.300 ± 0.791	0.184 ± 0.001	6.700 ± 0.791
QML	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.002	-0.002	—	0.0318	0.0318	1.003	95.000 ± 0.689	0.125 ± 0.001	5.000 ± 0.689
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.003	-0.003	—	0.0271	0.0273	0.982	94.700 ± 0.708	0.105 ± 0.000	5.300 ± 0.708
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.002	0.174 ± 0.005	0.200	0.000	-0.11	0.0340	0.0340	1.042	96.300 ± 0.597	0.142 ± 0.000	—
UPI	beta42	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.200	0.000	0.01	0.0355	0.0355	1.026	95.200 ± 0.676	0.147 ± 0.000	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.177 ± 0.005	0.201	0.001	0.42	0.0346	0.0346	1.008	94.500 ± 0.721	0.140 ± 0.000	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	—	0.0424	0.0424	0.992	95.100 ± 0.683	0.170 ± 0.001	4.900 ± 0.683
UPI	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0340	0.0340	1.033	95.600 ± 0.649	0.148 ± 0.001	4.400 ± 0.649
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0307	0.0307	0.981	95.800 ± 0.634	0.119 ± 0.001	4.200 ± 0.634

Table S220

Simulation 2 - Condition 76: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.002	0.002	—	0.0291	0.0292	1.022	96.300 ± 0.597	0.119 ± 0.001	3.700 ± 0.597
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.009	0.162	0.002	1.18	0.0498	0.0499	1.014	95.300 ± 0.669	0.199 ± 0.000	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.161	0.001	0.43	0.0370	0.0370	1.012	95.200 ± 0.676	0.153 ± 0.000	—
UPI	beta53	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.159	-0.001	-0.83	0.0394	0.0395	0.997	95.200 ± 0.676	0.159 ± 0.000	—
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.161	0.001	0.76	0.0389	0.0389	1.009	96.300 ± 0.597	0.152 ± 0.000	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0532	0.0532	1.010	95.800 ± 0.634	0.223 ± 0.001	4.200 ± 0.634
UPI	beta56	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.003	0.003	—	0.0518	0.0519	0.960	94.600 ± 0.715	0.193 ± 0.001	5.400 ± 0.715
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.000	0.000	—	0.0337	0.0337	1.001	95.800 ± 0.634	0.131 ± 0.001	4.200 ± 0.634
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0279	0.0280	1.000	95.500 ± 0.656	0.113 ± 0.001	4.500 ± 0.656

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S221

Simulation 2 - Condition 77: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.201	0.001	0.48	0.0473	0.0473	0.981	94.700 ± 0.708	0.181 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.204	0.004	2.22	0.0471	0.0473	0.976	93.900 ± 0.757	0.186 ± 0.000	—
LSAM	beta43	0.20	0.203	0.003 ± 0.001	0.015 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.205	0.005	2.63	0.0456	0.0459	0.989	94.400 ± 0.727	0.178 ± 0.000	—
LSAM	beta44	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0516	0.0516	0.946	93.600 ± 0.774	0.200 ± 0.001	6.400 ± 0.774
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.003	0.003	—	0.0485	0.0486	0.929	92.200 ± 0.848	0.178 ± 0.001	7.800 ± 0.848
LSAM	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0346	0.0347	0.957	93.600 ± 0.774	0.140 ± 0.001	6.400 ± 0.774
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0363	0.0363	0.950	92.800 ± 0.817	0.139 ± 0.001	7.200 ± 0.817
LSAM	beta51	0.16	0.158	-0.002 ± 0.002	-0.015 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.008	0.412 ± 0.011	0.157	-0.003	-1.68	0.0673	0.0674	0.967	94.200 ± 0.739	0.250 ± 0.001	—
LSAM	beta52	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.005	0.314 ± 0.009	0.160	0.000	0.03	0.0487	0.0487	1.004	94.700 ± 0.708	0.198 ± 0.001	—
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.115 ± 0.005	0.339 ± 0.009	0.158	-0.002	-1.53	0.0539	0.0540	0.963	93.800 ± 0.763	0.205 ± 0.001	—
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.315 ± 0.009	0.161	0.001	0.90	0.0485	0.0486	1.004	94.000 ± 0.751	0.198 ± 0.001	—
LSAM	beta55	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.005	0.005	—	0.0681	0.0683	0.946	93.500 ± 0.780	0.259 ± 0.001	6.500 ± 0.780
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.002	-0.002	—	0.0596	0.0596	0.958	94.200 ± 0.739	0.228 ± 0.001	5.800 ± 0.739
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.003	-0.003	—	0.0400	0.0401	0.948	93.400 ± 0.785	0.155 ± 0.001	6.600 ± 0.785
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0371	0.0371	0.949	94.300 ± 0.733	0.136 ± 0.001	5.700 ± 0.733
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.199	-0.001	-0.26	0.0464	0.0464	0.982	94.600 ± 0.715	0.179 ± 0.000	—
QML	beta42	0.20	0.200	0.000 ± 0.002	0.001 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.203	0.003	1.43	0.0468	0.0469	0.981	94.100 ± 0.745	0.184 ± 0.000	—
QML	beta43	0.20	0.202	0.002 ± 0.001	0.011 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.204	0.004	2.16	0.0457	0.0459	0.996	94.500 ± 0.721	0.177 ± 0.000	—
QML	beta44	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.005	-0.005	—	0.0517	0.0520	0.960	93.800 ± 0.763	0.200 ± 0.001	6.200 ± 0.763
QML	beta45	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.000	0.000	—	0.0463	0.0463	0.939	92.700 ± 0.823	0.177 ± 0.001	7.300 ± 0.823
QML	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0340	0.0340	0.972	94.000 ± 0.751	0.137 ± 0.001	6.000 ± 0.751
QML	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0356	0.0357	0.959	92.600 ± 0.828	0.135 ± 0.001	7.400 ± 0.828
QML	beta51	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.007	0.411 ± 0.011	0.157	-0.003	-1.72	0.0658	0.0658	0.982	94.400 ± 0.727	0.253 ± 0.001	—
QML	beta52	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.314 ± 0.009	0.159	-0.001	-0.44	0.0475	0.0475	1.010	94.800 ± 0.702	0.199 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.113 ± 0.005	0.336 ± 0.009	0.157	-0.003	-2.06	0.0531	0.0532	0.969	94.000 ± 0.751	0.204 ± 0.001	—
QML	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.008	0.161	0.001	0.62	0.0488	0.0488	1.005	94.400 ± 0.727	0.197 ± 0.001	—
QML	beta55	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.005	0.005	—	0.0671	0.0673	0.957	93.800 ± 0.763	0.258 ± 0.001	6.200 ± 0.763
QML	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.004	-0.004	—	0.0601	0.0603	0.968	94.300 ± 0.733	0.228 ± 0.001	5.700 ± 0.733
QML	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.003	-0.003	—	0.0397	0.0398	0.958	93.500 ± 0.780	0.152 ± 0.001	6.500 ± 0.780
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.003	-0.003	—	0.0361	0.0362	0.954	94.500 ± 0.721	0.133 ± 0.001	5.500 ± 0.721
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.201	0.001	0.47	0.0475	0.0475	0.978	94.900 ± 0.696	0.181 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.204	0.004	1.99	0.0478	0.0480	0.975	94.200 ± 0.739	0.186 ± 0.001	—
UPI	beta43	0.20	0.203	0.003 ± 0.001	0.015 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.205	0.005	2.43	0.0464	0.0466	0.994	94.900 ± 0.696	0.179 ± 0.000	—
UPI	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.003	-0.003	—	0.0521	0.0522	0.955	94.100 ± 0.745	0.204 ± 0.001	5.900 ± 0.745
UPI	beta45	0.00	0.003	0.003 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.002	0.002	—	0.0497	0.0497	0.938	92.900 ± 0.812	0.181 ± 0.001	7.100 ± 0.812
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0356	0.0356	0.973	94.200 ± 0.739	0.143 ± 0.001	5.800 ± 0.739

Table S221

Simulation 2 - Condition 77: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0374	0.0374	0.955	93.100 ± 0.801	0.142 ± 0.001	6.900 ± 0.801
UPI	beta51	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.007	0.410 ± 0.011	0.157	-0.003	-2.12	0.0664	0.0665	0.979	94.100 ± 0.745	0.252 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.002	0.003 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.005	0.314 ± 0.009	0.160	0.000	0.10	0.0483	0.0483	1.007	94.500 ± 0.721	0.198 ± 0.001	—
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.115 ± 0.005	0.339 ± 0.009	0.158	-0.002	-1.27	0.0549	0.0550	0.966	93.700 ± 0.768	0.206 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.315 ± 0.009	0.161	0.001	0.50	0.0480	0.0480	1.007	94.200 ± 0.739	0.199 ± 0.001	—
UPI	beta55	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.003	0.003	—	0.0686	0.0687	0.958	93.800 ± 0.763	0.262 ± 0.001	6.200 ± 0.763
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.003	-0.003	—	0.0581	0.0582	0.972	94.700 ± 0.708	0.233 ± 0.001	5.300 ± 0.708
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0394	0.0395	0.956	93.500 ± 0.780	0.157 ± 0.001	6.500 ± 0.780
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.002	-0.002	—	0.0369	0.0370	0.958	94.800 ± 0.702	0.138 ± 0.001	5.200 ± 0.702

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S222

Simulation 2 - Condition 78: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.200	0.000	0.16	0.0301	0.0301	0.976	93.500 ± 0.780	0.114 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.200	0.000	-0.12	0.0288	0.0288	0.990	94.800 ± 0.702	0.117 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.199	-0.001	-0.39	0.0294	0.0294	0.986	95.000 ± 0.689	0.112 ± 0.000	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.002	-0.002	—	0.0327	0.0328	0.969	94.500 ± 0.721	0.124 ± 0.000	5.500 ± 0.721
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0314	0.0314	0.950	94.800 ± 0.702	0.111 ± 0.000	5.200 ± 0.702
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.001	0.001	—	0.0223	0.0223	0.976	94.400 ± 0.727	0.087 ± 0.000	5.600 ± 0.727
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.001	0.001	—	0.0226	0.0227	0.975	93.400 ± 0.785	0.087 ± 0.000	6.600 ± 0.785
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.159	-0.001	-0.43	0.0415	0.0415	0.989	94.500 ± 0.721	0.158 ± 0.000	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.161	0.001	0.54	0.0304	0.0304	0.999	94.800 ± 0.702	0.125 ± 0.000	—
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.014 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.162	0.002	1.09	0.0330	0.0330	1.008	94.900 ± 0.696	0.129 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.197 ± 0.005	0.158	-0.002	-1.40	0.0314	0.0315	1.007	94.900 ± 0.696	0.124 ± 0.000	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	—	0.0424	0.0425	1.008	95.300 ± 0.669	0.163 ± 0.001	4.700 ± 0.669
LSAM	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0353	0.0353	0.995	93.800 ± 0.763	0.144 ± 0.000	6.200 ± 0.763
LSAM	beta57	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.000	0.000	—	0.0260	0.0260	0.979	93.900 ± 0.757	0.097 ± 0.000	6.100 ± 0.757
LSAM	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.001	0.001	—	0.0213	0.0213	0.966	92.900 ± 0.812	0.085 ± 0.000	7.100 ± 0.812
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.199	-0.001	-0.52	0.0298	0.0299	0.981	93.500 ± 0.780	0.112 ± 0.000	—
QML	beta42	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.199	-0.001	-0.33	0.0286	0.0286	0.991	94.500 ± 0.721	0.116 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.198	-0.002	-0.94	0.0295	0.0296	0.987	95.100 ± 0.683	0.111 ± 0.000	—
QML	beta44	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.005	-0.005	—	0.0321	0.0325	0.974	93.800 ± 0.763	0.125 ± 0.000	6.200 ± 0.763
QML	beta45	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.003	-0.003	—	0.0313	0.0315	0.956	94.500 ± 0.721	0.112 ± 0.000	5.500 ± 0.721
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	—	0.0220	0.0220	0.983	94.000 ± 0.751	0.086 ± 0.000	6.000 ± 0.751
QML	beta47	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0221	0.0221	0.981	93.800 ± 0.763	0.085 ± 0.000	6.200 ± 0.763
QML	beta51	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.159	-0.001	-0.33	0.0415	0.0415	0.994	94.500 ± 0.721	0.159 ± 0.000	—
QML	beta52	0.16	0.160	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.161	0.001	0.31	0.0302	0.0302	1.000	94.900 ± 0.696	0.125 ± 0.000	—
QML	beta53	0.16	0.162	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.161	0.001	0.68	0.0326	0.0326	1.010	94.600 ± 0.715	0.129 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.158	-0.002	-1.51	0.0311	0.0312	1.011	95.000 ± 0.689	0.124 ± 0.000	—
QML	beta55	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0424	0.0425	1.015	95.900 ± 0.627	0.163 ± 0.001	4.100 ± 0.627
QML	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.002	-0.002	—	0.0359	0.0360	1.004	93.600 ± 0.774	0.144 ± 0.000	6.400 ± 0.774
QML	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	-0.001	-0.001	—	0.0259	0.0259	0.988	93.800 ± 0.763	0.096 ± 0.000	6.200 ± 0.763
QML	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	—	0.0210	0.0210	0.986	93.400 ± 0.785	0.085 ± 0.001	6.600 ± 0.785
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.200	0.000	0.03	0.0301	0.0301	0.976	93.600 ± 0.774	0.114 ± 0.000	—
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.200	0.000	0.01	0.0288	0.0288	0.991	94.500 ± 0.721	0.117 ± 0.000	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.200	0.000	-0.23	0.0296	0.0296	0.987	94.700 ± 0.708	0.112 ± 0.000	—
UPI	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.002	-0.002	—	0.0335	0.0336	0.970	94.800 ± 0.702	0.125 ± 0.000	5.200 ± 0.702
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0311	0.0311	0.953	94.300 ± 0.733	0.112 ± 0.000	5.700 ± 0.733
UPI	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.001	0.001	—	0.0223	0.0223	0.978	94.400 ± 0.727	0.088 ± 0.000	5.600 ± 0.727

Table S222

Simulation 2 - Condition 78: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.001	0.001	—	0.0224	0.0224	0.984	93.900 ± 0.757	0.088 ± 0.000	6.100 ± 0.757
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.159	-0.001	-0.50	0.0417	0.0417	0.993	94.500 ± 0.721	0.159 ± 0.000	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.161	0.001	0.60	0.0303	0.0303	1.000	94.900 ± 0.696	0.125 ± 0.000	—
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.014 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.204 ± 0.006	0.162	0.002	0.96	0.0335	0.0336	1.010	95.200 ± 0.676	0.129 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.197 ± 0.005	0.158	-0.002	-1.22	0.0319	0.0320	1.007	94.900 ± 0.696	0.125 ± 0.000	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0429	0.0429	1.017	95.500 ± 0.656	0.164 ± 0.001	4.500 ± 0.656
UPI	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0353	0.0353	0.998	94.300 ± 0.733	0.145 ± 0.000	5.700 ± 0.733
UPI	beta57	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.000	0.000	—	0.0256	0.0256	0.980	93.900 ± 0.757	0.098 ± 0.000	6.100 ± 0.757
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.002	0.002	—	0.0214	0.0215	0.974	93.400 ± 0.785	0.086 ± 0.000	6.600 ± 0.785

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S223

Simulation 2 - Condition 79: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.007	0.013 ± 0.035	0.027 ± 0.002	0.165 ± 0.006	0.027 ± 0.002	0.165 ± 0.007	0.679 ± 0.049	0.824 ± 0.034	0.198	-0.002	-1.12	0.1672	0.1672	1.089	97.317 ± 0.683	0.704 ± 0.015	—
LSAM	beta42	0.20	0.198	-0.002 ± 0.008	-0.010 ± 0.038	0.032 ± 0.002	0.178 ± 0.006	0.032 ± 0.002	0.178 ± 0.007	0.794 ± 0.055	0.891 ± 0.036	0.200	0.000	-0.05	0.1788	0.1788	1.068	97.138 ± 0.705	0.747 ± 0.015	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.004	-0.002 ± 0.019	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.088 ± 0.004	0.196 ± 0.014	0.442 ± 0.019	0.196	-0.004	-2.23	0.0860	0.0861	1.051	96.601 ± 0.766	0.365 ± 0.005	—
LSAM	beta44	0.00	0.008	0.008 ± 0.006	—	0.019 ± 0.002	0.137 ± 0.006	0.019 ± 0.002	0.137 ± 0.006	—	—	0.011	0.011	—	0.1208	0.1212	1.087	98.032 ± 0.587	0.583 ± 0.016	1.968 ± 0.587
LSAM	beta45	0.00	0.003	0.003 ± 0.005	—	0.015 ± 0.001	0.123 ± 0.005	0.015 ± 0.001	0.123 ± 0.005	—	—	0.001	0.001	—	0.1099	0.1099	1.050	97.674 ± 0.637	0.506 ± 0.013	2.326 ± 0.637
LSAM	beta46	0.00	0.002	0.002 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.071 ± 0.003	—	—	-0.002	-0.002	—	0.0648	0.0649	1.065	97.853 ± 0.613	0.295 ± 0.009	2.147 ± 0.613
LSAM	beta47	0.00	0.001	0.001 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.002	-0.002	—	0.0653	0.0654	1.007	97.853 ± 0.613	0.297 ± 0.009	2.147 ± 0.613
LSAM	beta51	0.16	0.169	0.009 ± 0.005	0.053 ± 0.033	0.016 ± 0.001	0.126 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	0.627 ± 0.042	0.792 ± 0.031	0.166	0.006	3.87	0.1224	0.1225	0.952	95.170 ± 0.907	0.472 ± 0.004	—
LSAM	beta52	0.16	0.153	-0.007 ± 0.007	-0.042 ± 0.044	0.027 ± 0.002	0.166 ± 0.005	0.027 ± 0.002	0.166 ± 0.006	1.073 ± 0.068	1.036 ± 0.040	0.152	-0.008	-5.09	0.1706	0.1708	1.087	97.317 ± 0.683	0.706 ± 0.011	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.005	-0.018 ± 0.034	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.005	0.653 ± 0.048	0.808 ± 0.034	0.158	-0.002	-1.54	0.1258	0.1259	1.046	96.601 ± 0.766	0.531 ± 0.008	—
LSAM	beta54	0.16	0.162	0.002 ± 0.007	0.013 ± 0.044	0.028 ± 0.002	0.168 ± 0.005	0.028 ± 0.002	0.168 ± 0.006	1.102 ± 0.070	1.050 ± 0.040	0.154	-0.006	-3.89	0.1676	0.1677	1.099	96.243 ± 0.804	0.724 ± 0.015	—
LSAM	beta55	0.00	-0.006	-0.006 ± 0.008	—	0.032 ± 0.003	0.180 ± 0.007	0.032 ± 0.003	0.179 ± 0.008	—	—	-0.009	-0.009	—	0.1554	0.1556	1.155	98.390 ± 0.532	0.813 ± 0.022	1.610 ± 0.532
LSAM	beta56	0.00	0.008	0.008 ± 0.006	—	0.022 ± 0.002	0.149 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	0.005	0.005	—	0.1298	0.1299	1.055	97.853 ± 0.613	0.615 ± 0.015	2.147 ± 0.613
LSAM	beta57	0.00	0.006	0.006 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	—	—	0.005	0.005	—	0.0734	0.0735	1.132	98.390 ± 0.532	0.351 ± 0.009	1.610 ± 0.532
LSAM	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	—	—	0.000	0.000	—	0.0574	0.0574	1.055	98.032 ± 0.587	0.274 ± 0.008	1.968 ± 0.587
QML	beta41	0.20	0.175	-0.025 ± 0.004	-0.126 ± 0.018	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.197 ± 0.011	0.443 ± 0.015	0.173	-0.027	-13.26	0.0903	0.0941	1.041	94.454 ± 0.968	0.347 ± 0.003	—
QML	beta42	0.20	0.173	-0.027 ± 0.004	-0.137 ± 0.020	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.233 ± 0.015	0.483 ± 0.018	0.173	-0.027	-13.31	0.0950	0.0987	1.016	91.771 ± 1.162	0.369 ± 0.004	—
QML	beta43	0.20	0.203	0.003 ± 0.003	0.013 ± 0.016	0.006 ± 0.000	0.077 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	0.149 ± 0.010	0.386 ± 0.015	0.202	0.002	0.85	0.0795	0.0795	1.027	95.528 ± 0.874	0.311 ± 0.003	—
QML	beta44	0.00	-0.020	-0.020 ± 0.005	—	0.012 ± 0.001	0.109 ± 0.004	0.012 ± 0.001	0.111 ± 0.005	—	—	-0.013	-0.013	—	0.1060	0.1068	1.014	95.707 ± 0.857	0.433 ± 0.009	4.293 ± 0.857
QML	beta45	0.00	0.005	0.005 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	0.008	0.008	—	0.0881	0.0885	0.983	95.707 ± 0.857	0.346 ± 0.005	4.293 ± 0.857
QML	beta46	0.00	0.043	0.043 ± 0.003	—	0.004 ± 0.000	0.062 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	0.037	0.037	—	0.0601	0.0707	0.992	91.592 ± 1.174	0.241 ± 0.005	8.408 ± 1.174
QML	beta47	0.00	0.044	0.044 ± 0.003	—	0.004 ± 0.000	0.062 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.037	0.037	—	0.0592	0.0695	0.989	88.909 ± 1.328	0.242 ± 0.007	11.091 ± 1.328
QML	beta51	0.16	0.180	0.020 ± 0.005	0.127 ± 0.034	0.016 ± 0.001	0.127 ± 0.005	0.017 ± 0.001	0.129 ± 0.005	0.646 ± 0.047	0.804 ± 0.034	0.175	0.015	9.55	0.1206	0.1216	1.011	96.601 ± 0.766	0.504 ± 0.006	—
QML	beta52	0.16	0.135	-0.025 ± 0.004	-0.157 ± 0.025	0.009 ± 0.001	0.096 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.382 ± 0.026	0.618 ± 0.024	0.141	-0.019	-11.89	0.1009	0.1026	1.064	95.528 ± 0.874	0.399 ± 0.005	—
QML	beta53	0.16	0.156	-0.004 ± 0.004	-0.024 ± 0.026	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.368 ± 0.023	0.607 ± 0.023	0.156	-0.004	-2.36	0.0932	0.0933	1.053	95.707 ± 0.857	0.401 ± 0.004	—
QML	beta54	0.16	0.130	-0.030 ± 0.004	-0.186 ± 0.025	0.009 ± 0.001	0.094 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	0.376 ± 0.023	0.613 ± 0.022	0.131	-0.029	-17.84	0.0955	0.0997	1.047	95.170 ± 0.907	0.384 ± 0.004	—
QML	beta55	0.00	-0.020	-0.020 ± 0.006	—	0.019 ± 0.001	0.137 ± 0.005	0.019 ± 0.001	0.139 ± 0.006	—	—	-0.022	-0.022	—	0.1380	0.1398	1.094	95.349 ± 0.891	0.589 ± 0.013	4.651 ± 0.891
QML	beta56	0.00	0.027	0.027 ± 0.005	—	0.012 ± 0.001	0.111 ± 0.004	0.013 ± 0.001	0.115 ± 0.005	—	—	0.030	0.030	—	0.1059	0.1100	0.983	94.812 ± 0.938	0.430 ± 0.006	5.188 ± 0.938
QML	beta57	0.00	0.034	0.034 ± 0.003	—	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.034	0.034	—	0.0673	0.0752	1.103	94.097 ± 0.997	0.296 ± 0.010	5.903 ± 0.997
QML	beta58	0.00	0.029	0.029 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.027	0.027	—	0.0455	0.0527	0.996	93.023 ± 1.077	0.205 ± 0.004	6.977 ± 1.077
UPI	beta41	0.20	0.203	0.003 ± 0.008	0.015 ± 0.040	0.036 ± 0.003	0.191 ± 0.008	0.036 ± 0.003	0.191 ± 0.009	0.908 ± 0.073	0.953 ± 0.043	0.210	0.010	4.89	0.2003	0.2005	1.424	98.390 ± 0.532	1.065 ± 0.046	—
UPI	beta42	0.20	0.203	0.003 ± 0.009	0.013 ± 0.044	0.043 ± 0.003	0.208 ± 0.008	0.043 ± 0.003	0.208 ± 0.009	1.081 ± 0.086	1.040 ± 0.047	0.208	0.008	3.87	0.2001	0.2002	1.398	98.390 ± 0.532	1.140 ± 0.050	—
UPI	beta43	0.20	0.202	0.002 ± 0.004	0.010 ± 0.019	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.088 ± 0.004	0.195 ± 0.015	0.441 ± 0.020	0.202	0.002	1.01	0.0923	0.0923	1.313	96.959 ± 0.726	0.455 ± 0.014	—
UPI	beta44	0.00	0.011	0.011 ± 0.007	—	0.029 ± 0.003	0.170 ± 0.008	0.029 ± 0.003	0.170 ± 0.009	—	—	0.007	0.007	—	0.1517	0.1518	1.589	99.106 ± 0.398	1.058 ± 0.051	0.894 ± 0.398
UPI	beta45	0.00	-0.007	-0.007 ± 0.006	—	0.021 ± 0.002	0.144 ± 0.007	0.021 ± 0.002	0.144 ± 0.007	—	—	-0.003	-0.003	—	0.1336	0.1337	1.543	99.284 ± 0.356	0.871 ± 0.039	0.716 ± 0.356
UPI	beta46	0.00	0.001	0.001 ± 0.004	—	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.084 ± 0.004	—	—	-0.003	-0.003	—	0.0831	0.0832	1.424	98.748 ± 0.470	0.468 ± 0.022	1.252 ± 0.470

Table S223

Simulation 2 - Condition 79: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.004	—	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.088 ± 0.004	—	—	-0.002	-0.002	—	0.0759	0.0759	1.326	99.284 ± 0.356	0.457 ± 0.022	0.716 ± 0.356
UPI	beta51	0.16	0.167	0.007 ± 0.005	0.043 ± 0.033	0.016 ± 0.001	0.126 ± 0.004	0.016 ± 0.001	0.126 ± 0.005	0.616 ± 0.039	0.785 ± 0.030	0.165	0.005	2.98	0.1269	0.1269	1.058	97.138 ± 0.705	0.521 ± 0.006	—
UPI	beta52	0.16	0.164	0.004 ± 0.007	0.028 ± 0.046	0.030 ± 0.002	0.173 ± 0.006	0.030 ± 0.002	0.173 ± 0.007	1.169 ± 0.078	1.081 ± 0.043	0.164	0.004	2.79	0.1927	0.1928	1.344	98.569 ± 0.502	0.912 ± 0.033	—
UPI	beta53	0.16	0.165	0.005 ± 0.006	0.029 ± 0.036	0.018 ± 0.001	0.136 ± 0.005	0.018 ± 0.001	0.136 ± 0.006	0.720 ± 0.050	0.848 ± 0.035	0.158	-0.002	-1.50	0.1361	0.1361	1.246	97.674 ± 0.637	0.663 ± 0.023	—
UPI	beta54	0.16	0.154	-0.006 ± 0.007	-0.039 ± 0.045	0.029 ± 0.002	0.169 ± 0.005	0.029 ± 0.002	0.169 ± 0.006	1.114 ± 0.069	1.056 ± 0.040	0.155	-0.005	-2.96	0.1749	0.1750	1.362	97.674 ± 0.637	0.902 ± 0.026	—
UPI	beta55	0.00	-0.009	-0.009 ± 0.009	—	0.041 ± 0.003	0.203 ± 0.008	0.041 ± 0.003	0.203 ± 0.009	—	—	-0.006	-0.006	—	0.1791	0.1791	1.574	99.821 ± 0.179	1.253 ± 0.062	0.179 ± 0.179
UPI	beta56	0.00	0.004	0.004 ± 0.007	—	0.026 ± 0.002	0.162 ± 0.006	0.026 ± 0.002	0.161 ± 0.007	—	—	-0.001	-0.001	—	0.1439	0.1439	1.389	99.106 ± 0.398	0.880 ± 0.036	0.894 ± 0.398
UPI	beta57	0.00	0.001	0.001 ± 0.004	—	0.007 ± 0.001	0.086 ± 0.004	0.007 ± 0.001	0.086 ± 0.004	—	—	0.002	0.002	—	0.0745	0.0746	1.434	98.569 ± 0.502	0.481 ± 0.024	1.431 ± 0.502
UPI	beta58	0.00	0.001	0.001 ± 0.003	—	0.005 ± 0.000	0.068 ± 0.003	0.005 ± 0.000	0.068 ± 0.003	—	—	0.000	0.000	—	0.0620	0.0620	1.299	98.927 ± 0.436	0.345 ± 0.013	1.073 ± 0.436

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S224

Simulation 2 - Condition 80: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.003	-0.016 ± 0.016	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.240 ± 0.012	0.490 ± 0.014	0.194	-0.006	-2.86	0.1011	0.1013	1.018	96.467 ± 0.604	0.391 ± 0.003	—
LSAM	beta42	0.20	0.211	0.011 ± 0.003	0.056 ± 0.017	0.010 ± 0.000	0.101 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	0.258 ± 0.013	0.508 ± 0.015	0.210	0.010	4.93	0.1007	0.1012	1.054	97.109 ± 0.548	0.417 ± 0.003	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.002	-0.012 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.070 ± 0.003	0.264 ± 0.008	0.199	-0.001	-0.55	0.0513	0.0513	1.016	94.861 ± 0.722	0.210 ± 0.001	—
LSAM	beta44	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0616	0.0616	1.032	96.360 ± 0.613	0.266 ± 0.003	3.640 ± 0.613
LSAM	beta45	0.00	0.006	0.006 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.004	0.004	—	0.0576	0.0578	0.929	95.503 ± 0.678	0.246 ± 0.003	4.497 ± 0.678
LSAM	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0344	0.0344	1.011	97.216 ± 0.538	0.142 ± 0.002	2.784 ± 0.538
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0326	0.0327	1.011	96.360 ± 0.613	0.142 ± 0.002	3.640 ± 0.613
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.014 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.214 ± 0.010	0.462 ± 0.013	0.165	0.005	2.97	0.0760	0.0762	0.971	95.182 ± 0.701	0.282 ± 0.001	—
LSAM	beta52	0.16	0.155	-0.005 ± 0.003	-0.030 ± 0.021	0.010 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.410 ± 0.022	0.641 ± 0.020	0.159	-0.001	-0.67	0.0916	0.0916	0.997	95.503 ± 0.678	0.400 ± 0.003	—
LSAM	beta53	0.16	0.164	0.004 ± 0.003	0.026 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.237 ± 0.011	0.487 ± 0.014	0.164	0.004	2.26	0.0766	0.0767	1.020	96.360 ± 0.613	0.311 ± 0.002	—
LSAM	beta54	0.16	0.162	0.002 ± 0.003	0.011 ± 0.021	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.407 ± 0.021	0.638 ± 0.019	0.160	0.000	-0.31	0.0941	0.0941	0.987	94.968 ± 0.715	0.395 ± 0.003	—
LSAM	beta55	0.00	0.000	-0.000 ± 0.003	—	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	—	—	-0.004	-0.004	—	0.0967	0.0968	0.994	97.645 ± 0.496	0.393 ± 0.005	2.355 ± 0.496
LSAM	beta56	0.00	-0.003	-0.003 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.004	-0.004	—	0.0744	0.0745	1.027	96.788 ± 0.577	0.317 ± 0.003	3.212 ± 0.577
LSAM	beta57	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	0.000	0.000	—	0.0401	0.0401	0.950	95.717 ± 0.662	0.174 ± 0.002	4.283 ± 0.662
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.000	0.000	—	0.0319	0.0319	0.960	96.039 ± 0.638	0.133 ± 0.002	3.961 ± 0.638
QML	beta41	0.20	0.171	-0.029 ± 0.002	-0.145 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.004	0.311 ± 0.008	0.169	-0.031	-15.36	0.0559	0.0638	1.001	88.972 ± 1.025	0.216 ± 0.002	—
QML	beta42	0.20	0.178	-0.022 ± 0.002	-0.108 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.004	0.295 ± 0.008	0.178	-0.022	-11.11	0.0541	0.0585	1.045	94.325 ± 0.757	0.225 ± 0.001	—
QML	beta43	0.20	0.203	0.003 ± 0.002	0.013 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.240 ± 0.007	0.203	0.003	1.51	0.0459	0.0460	1.031	94.861 ± 0.722	0.194 ± 0.001	—
QML	beta44	0.00	-0.024	-0.024 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.024	-0.024	—	0.0600	0.0647	1.072	94.111 ± 0.770	0.253 ± 0.005	5.889 ± 0.770
QML	beta45	0.00	0.010	0.010 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.010	0.010	—	0.0529	0.0538	0.971	94.111 ± 0.770	0.213 ± 0.002	5.889 ± 0.770
QML	beta46	0.00	0.039	0.039 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.037	0.037	—	0.0342	0.0502	1.013	78.587 ± 1.342	0.138 ± 0.002	21.413 ± 1.342
QML	beta47	0.00	0.040	0.040 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.038	0.038	—	0.0338	0.0507	0.980	79.443 ± 1.322	0.136 ± 0.001	20.557 ± 1.322
QML	beta51	0.16	0.168	0.008 ± 0.002	0.048 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.204 ± 0.010	0.452 ± 0.013	0.171	0.011	7.03	0.0727	0.0735	1.018	96.253 ± 0.621	0.287 ± 0.001	—
QML	beta52	0.16	0.138	-0.022 ± 0.002	-0.138 ± 0.012	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.007	0.378 ± 0.010	0.138	-0.022	-13.95	0.0581	0.0622	1.052	93.148 ± 0.827	0.232 ± 0.002	—
QML	beta53	0.16	0.157	-0.003 ± 0.002	-0.018 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.360 ± 0.010	0.156	-0.004	-2.57	0.0573	0.0574	1.080	96.039 ± 0.638	0.244 ± 0.005	—
QML	beta54	0.16	0.136	-0.024 ± 0.002	-0.147 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.157 ± 0.007	0.396 ± 0.011	0.137	-0.023	-14.69	0.0554	0.0601	1.002	91.649 ± 0.905	0.231 ± 0.002	—
QML	beta55	0.00	-0.026	-0.026 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	—	—	-0.026	-0.026	—	0.0854	0.0891	1.060	94.004 ± 0.777	0.354 ± 0.012	5.996 ± 0.777
QML	beta56	0.00	0.025	0.025 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	—	—	0.027	0.027	—	0.0584	0.0643	1.118	94.325 ± 0.757	0.270 ± 0.021	5.675 ± 0.757
QML	beta57	0.00	0.035	0.035 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	—	—	0.035	0.035	—	0.0404	0.0534	1.041	87.366 ± 1.087	0.174 ± 0.003	12.634 ± 1.087
QML	beta58	0.00	0.029	0.029 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.028	0.028	—	0.0299	0.0408	1.006	83.619 ± 1.211	0.121 ± 0.002	16.381 ± 1.211
UPI	beta41	0.20	0.196	-0.004 ± 0.003	-0.021 ± 0.016	0.010 ± 0.000	0.101 ± 0.002	0.010 ± 0.000	0.101 ± 0.003	0.254 ± 0.012	0.504 ± 0.015	0.195	-0.005	-2.48	0.0985	0.0986	1.132	97.216 ± 0.538	0.447 ± 0.009	—
UPI	beta42	0.20	0.215	0.015 ± 0.003	0.073 ± 0.017	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.271 ± 0.014	0.521 ± 0.016	0.211	0.011	5.30	0.1018	0.1024	1.181	97.537 ± 0.507	0.478 ± 0.015	—
UPI	beta43	0.20	0.198	-0.002 ± 0.002	-0.012 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.197	-0.003	-1.57	0.0499	0.0500	1.053	95.396 ± 0.686	0.209 ± 0.002	—
UPI	beta44	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.001	0.001	—	0.0615	0.0615	1.235	98.501 ± 0.398	0.336 ± 0.011	1.499 ± 0.398
UPI	beta45	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.001	0.001	—	0.0596	0.0596	1.172	98.608 ± 0.383	0.317 ± 0.008	1.392 ± 0.383
UPI	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0339	0.0339	1.146	98.287 ± 0.425	0.166 ± 0.004	1.713 ± 0.425

Table S224

Simulation 2 - Condition 80: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0316	0.0318	1.152	97.537 ± 0.507	0.166 ± 0.007	2.463 ± 0.507
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.015	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.197 ± 0.009	0.444 ± 0.013	0.163	0.003	2.01	0.0725	0.0725	1.018	96.360 ± 0.613	0.284 ± 0.001	—
UPI	beta52	0.16	0.158	-0.002 ± 0.003	-0.011 ± 0.020	0.010 ± 0.000	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	0.383 ± 0.020	0.619 ± 0.019	0.162	0.002	1.24	0.0926	0.0926	1.079	96.895 ± 0.568	0.419 ± 0.007	—
UPI	beta53	0.16	0.163	0.003 ± 0.003	0.016 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.230 ± 0.011	0.479 ± 0.014	0.164	0.004	2.45	0.0795	0.0796	1.065	96.360 ± 0.613	0.320 ± 0.003	—
UPI	beta54	0.16	0.162	0.002 ± 0.003	0.013 ± 0.021	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.393 ± 0.020	0.627 ± 0.019	0.159	-0.001	-0.42	0.0959	0.0959	1.038	95.717 ± 0.662	0.408 ± 0.004	—
UPI	beta55	0.00	-0.004	-0.004 ± 0.003	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	—	—	-0.002	-0.002	—	0.0897	0.0897	1.148	98.394 ± 0.411	0.460 ± 0.011	1.606 ± 0.411
UPI	beta56	0.00	-0.003	-0.003 ± 0.003	—	0.006 ± 0.000	0.081 ± 0.002	0.006 ± 0.000	0.081 ± 0.003	—	—	-0.005	-0.005	—	0.0757	0.0759	1.148	97.859 ± 0.474	0.363 ± 0.006	2.141 ± 0.474
UPI	beta57	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	—	0.0392	0.0392	1.041	97.216 ± 0.538	0.185 ± 0.004	2.784 ± 0.538
UPI	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	—	0.0306	0.0306	0.999	96.788 ± 0.577	0.138 ± 0.002	3.212 ± 0.577

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S225

Simulation 2 - Condition 81: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.209	0.009 ± 0.003	0.043 ± 0.017	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	0.277 ± 0.013	0.526 ± 0.015	0.209	0.009	4.54	0.1025	0.1029	1.019	96.230 ± 0.616	0.419 ± 0.003	—
LSAM	beta42	0.20	0.197	-0.003 ± 0.004	-0.017 ± 0.018	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.315 ± 0.015	0.561 ± 0.016	0.192	-0.008	-3.76	0.1142	0.1145	1.000	95.288 ± 0.686	0.440 ± 0.003	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.009 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.004	0.307 ± 0.008	0.202	0.002	1.12	0.0643	0.0643	0.992	95.288 ± 0.686	0.239 ± 0.001	—
LSAM	beta44	0.00	0.007	0.007 ± 0.002	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	0.005	0.005	—	0.0737	0.0739	0.948	94.764 ± 0.721	0.286 ± 0.003	5.236 ± 0.721
LSAM	beta45	0.00	0.000	-0.000 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.001	0.001	—	0.0739	0.0739	0.924	93.717 ± 0.785	0.272 ± 0.003	6.283 ± 0.785
LSAM	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	—	0.0401	0.0401	0.907	93.613 ± 0.791	0.161 ± 0.002	6.387 ± 0.791
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.001	0.001	—	0.0379	0.0379	0.944	94.869 ± 0.714	0.159 ± 0.002	5.131 ± 0.714
LSAM	beta51	0.16	0.158	-0.002 ± 0.003	-0.011 ± 0.018	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.298 ± 0.014	0.546 ± 0.016	0.159	-0.001	-0.83	0.0849	0.0849	0.929	92.984 ± 0.826	0.318 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.004	0.021 ± 0.023	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.510 ± 0.025	0.714 ± 0.021	0.161	0.001	0.78	0.1143	0.1143	0.977	95.288 ± 0.686	0.438 ± 0.003	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.003	-0.010 ± 0.017	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.290 ± 0.014	0.539 ± 0.016	0.161	0.001	0.42	0.0845	0.0845	0.979	94.660 ± 0.728	0.331 ± 0.002	—
LSAM	beta54	0.16	0.169	0.009 ± 0.004	0.054 ± 0.023	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.503 ± 0.024	0.710 ± 0.020	0.163	0.003	1.79	0.1108	0.1109	0.983	95.183 ± 0.693	0.437 ± 0.003	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.003	—	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	—	—	0.002	0.002	—	0.0977	0.0977	0.943	94.764 ± 0.721	0.399 ± 0.004	5.236 ± 0.721
LSAM	beta56	0.00	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	—	—	0.001	0.001	—	0.0856	0.0856	0.927	95.183 ± 0.693	0.328 ± 0.003	4.817 ± 0.693
LSAM	beta57	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.002	-0.002	—	0.0464	0.0464	0.928	94.974 ± 0.707	0.190 ± 0.002	5.026 ± 0.707
LSAM	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.003	-0.003	—	0.0388	0.0389	0.932	95.288 ± 0.686	0.157 ± 0.002	4.712 ± 0.686
QML	beta41	0.20	0.185	-0.015 ± 0.002	-0.076 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.122 ± 0.005	0.349 ± 0.009	0.186	-0.014	-7.14	0.0678	0.0693	1.040	94.450 ± 0.741	0.278 ± 0.001	—
QML	beta42	0.20	0.178	-0.022 ± 0.002	-0.111 ± 0.012	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.145 ± 0.006	0.380 ± 0.010	0.176	-0.024	-12.22	0.0735	0.0774	1.010	93.613 ± 0.791	0.288 ± 0.001	—
QML	beta43	0.20	0.205	0.005 ± 0.002	0.027 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.287 ± 0.008	0.206	0.006	2.96	0.0593	0.0596	1.004	94.974 ± 0.707	0.225 ± 0.001	—
QML	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.006	-0.006	—	0.0628	0.0631	0.970	94.555 ± 0.734	0.257 ± 0.002	5.445 ± 0.734
QML	beta45	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.001	0.001	—	0.0627	0.0627	0.942	93.194 ± 0.815	0.236 ± 0.002	6.806 ± 0.815
QML	beta46	0.00	0.018	0.018 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.015	0.015	—	0.0361	0.0392	0.905	89.529 ± 0.991	0.140 ± 0.002	10.471 ± 0.991
QML	beta47	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.020	0.020	—	0.0356	0.0406	0.930	88.586 ± 1.029	0.138 ± 0.002	11.414 ± 1.029
QML	beta51	0.16	0.161	0.001 ± 0.003	0.009 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.285 ± 0.014	0.534 ± 0.015	0.164	0.004	2.21	0.0841	0.0841	0.967	94.031 ± 0.767	0.324 ± 0.001	—
QML	beta52	0.16	0.147	-0.013 ± 0.003	-0.080 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.248 ± 0.011	0.498 ± 0.014	0.151	-0.009	-5.91	0.0791	0.0797	0.979	94.450 ± 0.741	0.302 ± 0.001	—
QML	beta53	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.009	0.447 ± 0.012	0.158	-0.002	-0.98	0.0706	0.0706	1.006	94.346 ± 0.747	0.282 ± 0.001	—
QML	beta54	0.16	0.148	-0.012 ± 0.003	-0.073 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.242 ± 0.011	0.492 ± 0.013	0.146	-0.014	-8.71	0.0801	0.0813	0.984	94.346 ± 0.747	0.300 ± 0.001	—
QML	beta55	0.00	-0.011	-0.011 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	-0.008	-0.008	—	0.0893	0.0896	0.967	93.298 ± 0.809	0.355 ± 0.003	6.702 ± 0.809
QML	beta56	0.00	0.016	0.016 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	0.015	0.015	—	0.0756	0.0770	0.940	92.670 ± 0.843	0.279 ± 0.002	7.330 ± 0.843
QML	beta57	0.00	0.016	0.016 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.016	0.016	—	0.0432	0.0461	0.965	93.194 ± 0.815	0.172 ± 0.002	6.806 ± 0.815
QML	beta58	0.00	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.011	0.011	—	0.0341	0.0357	0.952	92.461 ± 0.854	0.134 ± 0.002	7.539 ± 0.854
UPI	beta41	0.20	0.209	0.009 ± 0.003	0.045 ± 0.017	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	0.281 ± 0.013	0.531 ± 0.015	0.209	0.009	4.65	0.1032	0.1036	1.072	96.649 ± 0.582	0.444 ± 0.004	—
UPI	beta42	0.20	0.196	-0.004 ± 0.004	-0.018 ± 0.018	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.320 ± 0.015	0.566 ± 0.016	0.189	-0.011	-5.42	0.1115	0.1120	1.055	96.021 ± 0.633	0.468 ± 0.006	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.009 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.301 ± 0.008	0.201	0.001	0.46	0.0620	0.0620	1.002	95.079 ± 0.700	0.237 ± 0.002	—
UPI	beta44	0.00	0.005	0.005 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	0.005	0.005	—	0.0741	0.0742	1.072	96.440 ± 0.600	0.340 ± 0.007	3.560 ± 0.600
UPI	beta45	0.00	-0.001	-0.001 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	—	—	0.001	0.001	—	0.0707	0.0707	1.076	96.335 ± 0.608	0.333 ± 0.009	3.665 ± 0.608
UPI	beta46	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	-0.002	-0.002	—	0.0389	0.0390	0.968	95.079 ± 0.700	0.176 ± 0.003	4.921 ± 0.700

Table S225

Simulation 2 - Condition 81: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.001	0.001	—	0.0394	0.0394	1.003	95.916 ± 0.640	0.173 ± 0.003	4.084 ± 0.640
UPI	beta51	0.16	0.158	-0.002 ± 0.003	-0.013 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.285 ± 0.014	0.534 ± 0.015	0.159	-0.001	-0.34	0.0848	0.0848	0.959	93.822 ± 0.779	0.321 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.004	0.007 ± 0.023	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.502 ± 0.025	0.708 ± 0.021	0.162	0.002	1.39	0.1135	0.1135	0.986	95.183 ± 0.693	0.438 ± 0.003	—
UPI	beta53	0.16	0.160	-0.000 ± 0.003	-0.001 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.270 ± 0.013	0.520 ± 0.015	0.161	0.001	0.86	0.0798	0.0798	1.015	95.393 ± 0.678	0.331 ± 0.002	—
UPI	beta54	0.16	0.168	0.008 ± 0.004	0.050 ± 0.023	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.492 ± 0.023	0.702 ± 0.020	0.161	0.001	0.68	0.1128	0.1128	1.000	95.393 ± 0.678	0.439 ± 0.003	—
UPI	beta55	0.00	-0.003	-0.003 ± 0.003	—	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	—	—	0.000	0.000	—	0.0940	0.0940	1.044	95.602 ± 0.664	0.438 ± 0.006	4.398 ± 0.664
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	—	—	0.000	0.000	—	0.0821	0.0821	0.995	95.602 ± 0.664	0.357 ± 0.005	4.398 ± 0.664
UPI	beta57	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.001	-0.001	—	0.0454	0.0454	0.970	95.079 ± 0.700	0.194 ± 0.003	4.921 ± 0.700
UPI	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.003	-0.003	—	0.0374	0.0374	0.958	95.602 ± 0.664	0.159 ± 0.002	4.398 ± 0.664

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S226

Simulation 2 - Condition 82: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.001 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.005	0.310 ± 0.009	0.198	-0.002	-1.24	0.0608	0.0608	1.007	95.281 ± 0.672	0.245 ± 0.001	—
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.102 ± 0.005	0.319 ± 0.009	0.203	0.003	1.29	0.0608	0.0608	1.034	95.783 ± 0.637	0.259 ± 0.001	—
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.201	0.001	0.26	0.0403	0.0403	0.965	94.378 ± 0.730	0.145 ± 0.000	—
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.002	0.002	—	0.0388	0.0389	0.960	94.277 ± 0.736	0.154 ± 0.001	5.723 ± 0.736
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0373	0.0373	0.980	94.980 ± 0.692	0.148 ± 0.001	5.020 ± 0.692
LSAM	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0199	0.0199	0.957	93.675 ± 0.771	0.083 ± 0.001	6.325 ± 0.771
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0208	0.0208	0.976	94.378 ± 0.730	0.086 ± 0.001	5.622 ± 0.730
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.016 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.320 ± 0.009	0.162	0.002	1.45	0.0502	0.0503	0.987	94.679 ± 0.711	0.198 ± 0.000	—
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.177 ± 0.008	0.421 ± 0.012	0.162	0.002	0.95	0.0657	0.0658	0.970	94.880 ± 0.698	0.256 ± 0.001	—
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.004 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.317 ± 0.009	0.162	0.002	1.08	0.0506	0.0506	1.009	95.281 ± 0.672	0.201 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.012 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.007	0.407 ± 0.011	0.160	0.000	0.16	0.0673	0.0673	1.002	94.880 ± 0.698	0.255 ± 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0547	0.0547	0.982	94.880 ± 0.698	0.222 ± 0.001	5.120 ± 0.698
LSAM	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.002	-0.002	—	0.0494	0.0494	0.988	94.277 ± 0.736	0.190 ± 0.001	5.723 ± 0.736
LSAM	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.001	-0.001	—	0.0247	0.0248	0.963	94.277 ± 0.736	0.100 ± 0.001	5.723 ± 0.736
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.000	0.000	—	0.0202	0.0202	0.922	92.972 ± 0.810	0.084 ± 0.001	7.028 ± 0.810
QML	beta41	0.20	0.181	-0.019 ± 0.001	-0.097 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.237 ± 0.006	0.179	-0.021	-10.31	0.0436	0.0482	1.047	92.169 ± 0.851	0.178 ± 0.007	—
QML	beta42	0.20	0.182	-0.018 ± 0.001	-0.092 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.183	-0.017	-8.30	0.0413	0.0446	1.056	93.173 ± 0.799	0.179 ± 0.002	—
QML	beta43	0.20	0.205	0.005 ± 0.001	0.023 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.001	0.185 ± 0.005	0.205	0.005	2.34	0.0389	0.0391	0.967	94.378 ± 0.730	0.139 ± 0.000	—
QML	beta44	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.007	-0.007	—	0.0402	0.0408	0.965	93.876 ± 0.760	0.155 ± 0.001	6.124 ± 0.760
QML	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	—	0.0351	0.0351	1.012	95.181 ± 0.679	0.142 ± 0.001	4.819 ± 0.679
QML	beta46	0.00	0.018	0.018 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.017	0.017	—	0.0204	0.0264	0.943	85.542 ± 1.114	0.081 ± 0.001	14.458 ± 1.114
QML	beta47	0.00	0.018	0.018 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.028 ± 0.001	—	—	0.017	0.017	—	0.0223	0.0283	0.950	84.839 ± 1.136	0.081 ± 0.001	15.161 ± 1.136
QML	beta51	0.16	0.164	0.004 ± 0.002	0.028 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.320 ± 0.009	0.165	0.005	3.00	0.0498	0.0500	1.017	94.880 ± 0.698	0.203 ± 0.004	—
QML	beta52	0.16	0.144	-0.016 ± 0.002	-0.098 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.316 ± 0.009	0.145	-0.015	-9.52	0.0479	0.0502	0.983	91.767 ± 0.871	0.185 ± 0.001	—
QML	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.276 ± 0.008	0.161	0.001	0.77	0.0440	0.0440	1.015	95.582 ± 0.651	0.176 ± 0.001	—
QML	beta54	0.16	0.145	-0.015 ± 0.001	-0.096 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.143	-0.017	-10.81	0.0490	0.0520	1.020	94.980 ± 0.692	0.186 ± 0.001	—
QML	beta55	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.011	-0.011	—	0.0542	0.0553	1.014	94.779 ± 0.705	0.215 ± 0.001	5.221 ± 0.705
QML	beta56	0.00	0.011	0.011 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.011	0.011	—	0.0434	0.0447	1.000	93.775 ± 0.766	0.170 ± 0.001	6.225 ± 0.766
QML	beta57	0.00	0.015	0.015 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.014	0.014	—	0.0249	0.0288	1.006	91.064 ± 0.904	0.101 ± 0.001	8.936 ± 0.904
QML	beta58	0.00	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.013	0.013	—	0.0191	0.0231	0.924	89.056 ± 0.989	0.078 ± 0.001	10.944 ± 0.989
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.095 ± 0.004	0.308 ± 0.009	0.198	-0.002	-1.13	0.0623	0.0624	1.024	96.185 ± 0.607	0.247 ± 0.001	—
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.318 ± 0.009	0.202	0.002	0.81	0.0608	0.0608	1.045	96.185 ± 0.607	0.260 ± 0.001	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.201	0.001	0.42	0.0382	0.0382	0.962	93.976 ± 0.754	0.140 ± 0.001	—
UPI	beta44	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.002	0.002	—	0.0384	0.0384	0.999	95.783 ± 0.637	0.160 ± 0.001	4.217 ± 0.637
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0386	0.0386	1.019	95.683 ± 0.644	0.157 ± 0.001	4.317 ± 0.644
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0201	0.0201	0.977	94.880 ± 0.698	0.085 ± 0.001	5.120 ± 0.698

Table S226

Simulation 2 - Condition 82: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0205	0.0205	0.986	94.578 ± 0.718	0.087 ± 0.001	5.422 ± 0.718
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.317 ± 0.009	0.163	0.003	1.78	0.0501	0.0501	1.001	95.181 ± 0.679	0.199 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.008 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.008	0.417 ± 0.012	0.160	0.000	0.06	0.0653	0.0653	0.973	94.880 ± 0.698	0.254 ± 0.001	—
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.005	0.313 ± 0.009	0.162	0.002	1.31	0.0501	0.0501	1.010	95.281 ± 0.672	0.198 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.007	0.406 ± 0.011	0.161	0.001	0.35	0.0654	0.0654	1.002	95.181 ± 0.679	0.255 ± 0.001	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.002	-0.002	—	0.0537	0.0538	1.018	96.084 ± 0.615	0.227 ± 0.002	3.916 ± 0.615
UPI	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.002	-0.002	—	0.0500	0.0500	1.006	94.980 ± 0.692	0.195 ± 0.002	5.020 ± 0.692
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.000	0.000	—	0.0240	0.0240	0.978	95.382 ± 0.665	0.100 ± 0.001	4.618 ± 0.665
UPI	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.000	0.000	—	0.0200	0.0200	0.923	92.972 ± 0.810	0.084 ± 0.001	7.028 ± 0.810

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S227

Simulation 2 - Condition 83: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.152 ± 0.008	0.390 ± 0.012	0.198	-0.002	-0.79	0.0732	0.0732	0.968	94.165 ± 0.743	0.296 ± 0.001	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.153 ± 0.007	0.392 ± 0.011	0.200	0.000	-0.01	0.0773	0.0773	1.004	95.573 ± 0.652	0.308 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.203	0.003	1.33	0.0481	0.0482	0.960	93.159 ± 0.801	0.183 ± 0.001	—
LSAM	beta44	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0481	0.0481	0.942	92.555 ± 0.833	0.194 ± 0.002	7.445 ± 0.833
LSAM	beta45	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.000	0.000	—	0.0526	0.0526	0.903	92.757 ± 0.822	0.184 ± 0.002	7.243 ± 0.822
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.001	0.001	—	0.0315	0.0315	0.851	92.254 ± 0.848	0.110 ± 0.001	7.746 ± 0.848
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	—	0.0258	0.0258	0.894	92.958 ± 0.812	0.108 ± 0.001	7.042 ± 0.812
LSAM	beta51	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.177 ± 0.008	0.421 ± 0.012	0.159	-0.001	-0.40	0.0685	0.0685	0.950	94.266 ± 0.737	0.251 ± 0.001	—
LSAM	beta52	0.16	0.160	-0.000 ± 0.003	-0.001 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.284 ± 0.016	0.533 ± 0.017	0.159	-0.001	-0.51	0.0811	0.0811	0.953	94.366 ± 0.731	0.318 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.013	0.005 ± 0.000	0.068 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.179 ± 0.008	0.423 ± 0.011	0.159	-0.001	-0.87	0.0711	0.0711	0.931	92.555 ± 0.833	0.247 ± 0.001	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.003	-0.007 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.255 ± 0.012	0.505 ± 0.014	0.156	-0.004	-2.22	0.0817	0.0818	0.999	94.970 ± 0.693	0.316 ± 0.001	—
LSAM	beta55	0.00	0.001	0.001 ± 0.002	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	-0.002	-0.002	—	0.0729	0.0729	0.926	93.360 ± 0.790	0.270 ± 0.002	6.640 ± 0.790
LSAM	beta56	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.001	0.001	—	0.0603	0.0603	0.947	93.461 ± 0.784	0.233 ± 0.002	6.539 ± 0.784
LSAM	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0368	0.0368	0.905	92.656 ± 0.827	0.131 ± 0.001	7.344 ± 0.827
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0270	0.0270	0.925	92.153 ± 0.853	0.109 ± 0.001	7.847 ± 0.853
QML	beta41	0.20	0.188	-0.012 ± 0.002	-0.060 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.104 ± 0.005	0.322 ± 0.009	0.188	-0.012	-6.15	0.0600	0.0613	0.988	94.266 ± 0.737	0.245 ± 0.001	—
QML	beta42	0.20	0.190	-0.010 ± 0.002	-0.051 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.005	0.321 ± 0.009	0.189	-0.011	-5.38	0.0633	0.0642	1.020	95.070 ± 0.687	0.253 ± 0.001	—
QML	beta43	0.20	0.204	0.004 ± 0.002	0.022 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.205	0.005	2.61	0.0462	0.0465	0.964	93.159 ± 0.801	0.180 ± 0.001	—
QML	beta44	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.002	-0.002	—	0.0474	0.0475	0.948	92.757 ± 0.822	0.189 ± 0.001	7.243 ± 0.822
QML	beta45	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.001	-0.001	—	0.0507	0.0507	0.912	93.260 ± 0.795	0.177 ± 0.001	6.740 ± 0.795
QML	beta46	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.008	0.008	—	0.0297	0.0307	0.851	89.437 ± 0.975	0.102 ± 0.001	10.563 ± 0.975
QML	beta47	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.007	0.007	—	0.0250	0.0258	0.895	91.549 ± 0.882	0.100 ± 0.001	8.451 ± 0.882
QML	beta51	0.16	0.160	0.000 ± 0.002	0.000 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.008	0.418 ± 0.012	0.160	0.000	0.08	0.0686	0.0686	0.967	94.567 ± 0.719	0.254 ± 0.001	—
QML	beta52	0.16	0.151	-0.009 ± 0.002	-0.053 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.011	0.449 ± 0.014	0.150	-0.010	-6.02	0.0688	0.0695	0.959	94.567 ± 0.719	0.269 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.007	0.391 ± 0.011	0.159	-0.001	-0.74	0.0647	0.0647	0.943	93.561 ± 0.778	0.232 ± 0.001	—
QML	beta54	0.16	0.151	-0.009 ± 0.002	-0.059 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.185 ± 0.008	0.431 ± 0.012	0.148	-0.012	-7.44	0.0692	0.0702	1.002	95.372 ± 0.666	0.268 ± 0.001	—
QML	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.004	-0.004	—	0.0704	0.0705	0.988	93.662 ± 0.773	0.274 ± 0.013	6.338 ± 0.773
QML	beta56	0.00	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	0.006	0.006	—	0.0597	0.0600	0.959	93.360 ± 0.790	0.222 ± 0.002	6.640 ± 0.790
QML	beta57	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.005	0.005	—	0.0354	0.0358	0.933	92.052 ± 0.858	0.127 ± 0.003	7.948 ± 0.858
QML	beta58	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.006	0.006	—	0.0262	0.0270	0.926	91.247 ± 0.896	0.102 ± 0.001	8.753 ± 0.896
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.150 ± 0.008	0.388 ± 0.012	0.198	-0.002	-1.18	0.0719	0.0719	0.978	94.366 ± 0.731	0.297 ± 0.001	—
UPI	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.153 ± 0.007	0.391 ± 0.011	0.201	0.001	0.30	0.0744	0.0744	1.011	95.976 ± 0.623	0.310 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.204	0.004	1.88	0.0480	0.0482	0.954	93.159 ± 0.801	0.181 ± 0.001	—
UPI	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.001	0.001	—	0.0485	0.0486	0.963	93.461 ± 0.784	0.200 ± 0.002	6.539 ± 0.784
UPI	beta45	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.001	-0.001	—	0.0525	0.0525	0.927	93.561 ± 0.778	0.192 ± 0.002	6.439 ± 0.778
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.001	0.001	—	0.0314	0.0314	0.865	92.455 ± 0.838	0.111 ± 0.001	7.545 ± 0.838

Table S227

Simulation 2 - Condition 83: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0261	0.0262	0.902	93.360 ± 0.790	0.110 ± 0.001	6.640 ± 0.790
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.008	0.419 ± 0.012	0.159	-0.001	-0.75	0.0690	0.0690	0.961	94.668 ± 0.713	0.252 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.003	0.002 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.276 ± 0.016	0.526 ± 0.017	0.158	-0.002	-1.29	0.0802	0.0802	0.960	94.668 ± 0.713	0.317 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.008	0.418 ± 0.011	0.161	0.001	0.68	0.0685	0.0685	0.934	92.958 ± 0.812	0.245 ± 0.001	—
UPI	beta54	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.253 ± 0.012	0.503 ± 0.014	0.157	-0.003	-1.86	0.0777	0.0778	1.000	94.869 ± 0.700	0.316 ± 0.001	—
UPI	beta55	0.00	0.002	0.002 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.001	-0.001	—	0.0734	0.0734	0.924	93.260 ± 0.795	0.276 ± 0.002	6.740 ± 0.795
UPI	beta56	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.000	0.000	—	0.0620	0.0620	0.951	93.964 ± 0.755	0.238 ± 0.002	6.036 ± 0.755
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0357	0.0357	0.914	92.555 ± 0.833	0.132 ± 0.001	7.445 ± 0.833
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0265	0.0265	0.921	92.254 ± 0.848	0.109 ± 0.001	7.746 ± 0.848

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S228

Simulation 2 - Condition 84: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.001	-0.010 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.198	-0.002	-0.89	0.0466	0.0466	0.981	94.494 ± 0.722	0.181 ± 0.001	—
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.003 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.061 ± 0.003	0.248 ± 0.007	0.201	0.001	0.41	0.0475	0.0475	0.968	93.894 ± 0.758	0.188 ± 0.001	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.199	-0.001	-0.41	0.0289	0.0289	0.987	94.795 ± 0.703	0.115 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0290	0.0290	0.981	94.294 ± 0.734	0.115 ± 0.001	5.706 ± 0.734
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0307	0.0307	0.946	93.694 ± 0.769	0.111 ± 0.001	6.306 ± 0.769
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.002	0.002	—	0.0169	0.0170	0.927	92.993 ± 0.808	0.062 ± 0.001	7.007 ± 0.808
LSAM	beta47	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.000	0.000	—	0.0157	0.0157	0.885	92.192 ± 0.849	0.061 ± 0.001	7.808 ± 0.849
LSAM	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.252 ± 0.007	0.161	0.001	0.45	0.0394	0.0394	1.001	94.895 ± 0.696	0.158 ± 0.000	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.314 ± 0.008	0.161	0.001	0.72	0.0515	0.0515	0.985	95.195 ± 0.677	0.194 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.158	-0.002	-1.25	0.0410	0.0411	0.960	94.394 ± 0.728	0.155 ± 0.000	—
LSAM	beta54	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.302 ± 0.008	0.158	-0.002	-1.36	0.0488	0.0489	1.019	95.095 ± 0.683	0.193 ± 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.002	-0.002	—	0.0430	0.0430	0.988	95.195 ± 0.677	0.163 ± 0.001	4.805 ± 0.677
LSAM	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.003	0.003	—	0.0376	0.0376	0.953	93.894 ± 0.758	0.140 ± 0.001	6.106 ± 0.758
LSAM	beta57	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0201	0.0201	0.940	93.393 ± 0.786	0.075 ± 0.001	6.607 ± 0.786
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.000	0.000	—	0.0159	0.0159	0.961	94.194 ± 0.740	0.064 ± 0.001	5.806 ± 0.740
QML	beta41	0.20	0.189	-0.011 ± 0.001	-0.057 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.189	-0.011	-5.27	0.0391	0.0405	0.986	93.994 ± 0.752	0.153 ± 0.000	—
QML	beta42	0.20	0.191	-0.009 ± 0.001	-0.046 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.211 ± 0.006	0.192	-0.008	-4.11	0.0405	0.0413	0.975	93.493 ± 0.780	0.157 ± 0.000	—
QML	beta43	0.20	0.202	0.002 ± 0.001	0.010 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.202	0.002	0.96	0.0282	0.0283	0.988	94.094 ± 0.746	0.113 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0290	0.0290	0.992	93.994 ± 0.752	0.116 ± 0.001	6.006 ± 0.752
QML	beta45	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0295	0.0295	0.961	93.894 ± 0.758	0.110 ± 0.001	6.106 ± 0.758
QML	beta46	0.00	0.008	0.008 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.018 ± 0.000	—	—	0.008	0.008	—	0.0169	0.0187	0.927	90.390 ± 0.932	0.060 ± 0.001	9.610 ± 0.932
QML	beta47	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.006	0.006	—	0.0153	0.0164	0.880	89.690 ± 0.962	0.059 ± 0.000	10.310 ± 0.962
QML	beta51	0.16	0.161	0.001 ± 0.001	0.004 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.252 ± 0.007	0.161	0.001	0.57	0.0396	0.0396	1.008	95.095 ± 0.683	0.159 ± 0.000	—
QML	beta52	0.16	0.154	-0.006 ± 0.001	-0.040 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.273 ± 0.007	0.153	-0.007	-4.15	0.0445	0.0450	0.984	94.194 ± 0.740	0.167 ± 0.000	—
QML	beta53	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.158	-0.002	-1.19	0.0381	0.0381	0.966	95.395 ± 0.663	0.146 ± 0.000	—
QML	beta54	0.16	0.151	-0.009 ± 0.001	-0.059 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.151	-0.009	-5.43	0.0420	0.0429	1.033	94.695 ± 0.709	0.168 ± 0.002	—
QML	beta55	0.00	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.003	-0.003	—	0.0417	0.0418	0.999	95.095 ± 0.683	0.162 ± 0.001	4.905 ± 0.683
QML	beta56	0.00	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.007	0.007	—	0.0365	0.0372	0.959	93.493 ± 0.780	0.135 ± 0.001	6.507 ± 0.780
QML	beta57	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	0.006	0.006	—	0.0203	0.0212	0.941	92.092 ± 0.854	0.074 ± 0.001	7.908 ± 0.854
QML	beta58	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.005	0.005	—	0.0161	0.0169	0.981	91.592 ± 0.878	0.063 ± 0.001	8.408 ± 0.878
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.199	-0.001	-0.52	0.0470	0.0470	0.986	94.294 ± 0.734	0.181 ± 0.001	—
UPI	beta42	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.200	0.000	-0.00	0.0480	0.0480	0.976	94.194 ± 0.740	0.188 ± 0.001	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.200	0.000	-0.07	0.0286	0.0286	0.979	94.394 ± 0.728	0.114 ± 0.000	—
UPI	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0301	0.0301	0.991	94.194 ± 0.740	0.116 ± 0.001	5.806 ± 0.740
UPI	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0303	0.0303	0.960	94.494 ± 0.722	0.112 ± 0.001	5.506 ± 0.722
UPI	beta46	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.002	0.002	—	0.0166	0.0167	0.928	93.193 ± 0.797	0.062 ± 0.001	6.807 ± 0.797

Table S228

Simulation 2 - Condition 84: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.001	-0.001	—	0.0159	0.0159	0.891	92.492 ± 0.834	0.061 ± 0.001	7.508 ± 0.834
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.252 ± 0.007	0.160	0.000	0.23	0.0399	0.0399	1.006	95.095 ± 0.683	0.159 ± 0.000	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.008	0.160	0.000	0.22	0.0502	0.0502	0.988	95.596 ± 0.649	0.194 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.159	-0.001	-0.80	0.0406	0.0406	0.957	94.094 ± 0.746	0.155 ± 0.000	—
UPI	beta54	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.158	-0.002	-1.03	0.0471	0.0471	1.020	94.695 ± 0.709	0.193 ± 0.001	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0426	0.0426	0.994	94.995 ± 0.690	0.165 ± 0.001	5.005 ± 0.690
UPI	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0370	0.0371	0.954	93.794 ± 0.763	0.141 ± 0.001	6.206 ± 0.763
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0198	0.0198	0.947	94.294 ± 0.734	0.075 ± 0.001	5.706 ± 0.734
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.000	0.000	—	0.0158	0.0158	0.960	93.794 ± 0.763	0.064 ± 0.001	6.206 ± 0.763

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S229

Simulation 2 - Condition 85: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.006	0.002 ± 0.028	0.009 ± 0.001	0.096 ± 0.005	0.009 ± 0.001	0.095 ± 0.005	0.228 ± 0.022	0.477 ± 0.027	0.202	0.002	0.84	0.0938	0.0938	1.084	96.321 ± 1.089	0.406 ± 0.010	—
LSAM	beta42	0.20	0.189	-0.011 ± 0.005	-0.053 ± 0.027	0.008 ± 0.001	0.092 ± 0.004	0.009 ± 0.001	0.092 ± 0.005	0.213 ± 0.017	0.462 ± 0.023	0.196	-0.004	-1.94	0.0904	0.0905	1.177	96.990 ± 0.988	0.424 ± 0.012	—
LSAM	beta43	0.20	0.200	0.000 ± 0.005	0.000 ± 0.026	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.089 ± 0.005	0.199 ± 0.017	0.446 ± 0.023	0.198	-0.002	-1.17	0.0936	0.0937	1.099	97.324 ± 0.933	0.385 ± 0.008	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.008	—	0.019 ± 0.002	0.136 ± 0.007	0.018 ± 0.002	0.136 ± 0.008	—	—	0.004	0.004	—	0.1322	0.1322	1.214	98.328 ± 0.742	0.648 ± 0.026	1.672 ± 0.742
LSAM	beta45	0.00	0.005	0.005 ± 0.007	—	0.015 ± 0.002	0.123 ± 0.007	0.015 ± 0.002	0.123 ± 0.008	—	—	0.010	0.010	—	0.1207	0.1211	1.054	98.662 ± 0.664	0.507 ± 0.016	1.338 ± 0.664
LSAM	beta46	0.00	-0.004	-0.004 ± 0.010	—	0.031 ± 0.003	0.176 ± 0.010	0.031 ± 0.003	0.176 ± 0.011	—	—	0.000	0.000	—	0.1628	0.1628	1.197	99.666 ± 0.334	0.825 ± 0.034	0.334 ± 0.334
LSAM	beta47	0.00	0.001	0.001 ± 0.010	—	0.030 ± 0.003	0.173 ± 0.009	0.030 ± 0.003	0.173 ± 0.010	—	—	0.009	0.009	—	0.1523	0.1525	1.249	99.666 ± 0.334	0.849 ± 0.051	0.334 ± 0.334
LSAM	beta51	0.16	0.169	0.009 ± 0.008	0.054 ± 0.047	0.017 ± 0.002	0.131 ± 0.006	0.017 ± 0.002	0.131 ± 0.008	0.675 ± 0.066	0.821 ± 0.047	0.165	0.005	3.05	0.1326	0.1326	0.968	96.990 ± 0.988	0.499 ± 0.007	—
LSAM	beta52	0.16	0.163	0.003 ± 0.006	0.018 ± 0.037	0.011 ± 0.001	0.104 ± 0.005	0.011 ± 0.001	0.104 ± 0.006	0.419 ± 0.037	0.647 ± 0.034	0.158	-0.002	-0.97	0.0977	0.0978	1.066	95.987 ± 1.135	0.433 ± 0.012	—
LSAM	beta53	0.16	0.155	-0.005 ± 0.006	-0.033 ± 0.037	0.011 ± 0.001	0.103 ± 0.006	0.011 ± 0.001	0.103 ± 0.007	0.412 ± 0.046	0.642 ± 0.041	0.153	-0.007	-4.44	0.0973	0.0975	1.044	96.990 ± 0.988	0.420 ± 0.009	—
LSAM	beta54	0.16	0.160	0.000 ± 0.006	0.001 ± 0.037	0.011 ± 0.001	0.103 ± 0.006	0.010 ± 0.001	0.102 ± 0.006	0.410 ± 0.043	0.640 ± 0.039	0.154	-0.006	-3.61	0.1010	0.1011	1.099	97.324 ± 0.933	0.442 ± 0.012	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.010	—	0.032 ± 0.003	0.179 ± 0.010	0.032 ± 0.003	0.179 ± 0.011	—	—	0.006	0.006	—	0.1729	0.1730	1.152	98.328 ± 0.742	0.808 ± 0.033	1.672 ± 0.742
LSAM	beta56	0.00	-0.003	-0.003 ± 0.010	—	0.028 ± 0.003	0.167 ± 0.008	0.028 ± 0.003	0.167 ± 0.010	—	—	0.001	0.001	—	0.1576	0.1576	1.087	97.993 ± 0.811	0.711 ± 0.026	2.007 ± 0.811
LSAM	beta57	0.00	-0.009	-0.009 ± 0.011	—	0.039 ± 0.005	0.197 ± 0.012	0.039 ± 0.005	0.197 ± 0.013	—	—	-0.020	-0.020	—	0.1670	0.1681	1.102	100.000 ± 0.000	0.850 ± 0.037	0.000 ± 0.000
LSAM	beta58	0.00	0.013	0.013 ± 0.011	—	0.033 ± 0.004	0.182 ± 0.010	0.033 ± 0.004	0.182 ± 0.011	—	—	0.001	0.001	—	0.1535	0.1535	1.162	98.662 ± 0.664	0.828 ± 0.037	1.338 ± 0.664
QML	beta41	0.20	0.197	-0.003 ± 0.005	-0.014 ± 0.024	0.007 ± 0.001	0.083 ± 0.004	0.007 ± 0.001	0.083 ± 0.004	0.171 ± 0.014	0.413 ± 0.021	0.199	-0.001	-0.33	0.0879	0.0879	0.955	93.311 ± 1.445	0.309 ± 0.003	—
QML	beta42	0.20	0.193	-0.007 ± 0.004	-0.035 ± 0.022	0.006 ± 0.000	0.077 ± 0.003	0.006 ± 0.000	0.078 ± 0.004	0.150 ± 0.012	0.388 ± 0.020	0.195	-0.005	-2.48	0.0826	0.0827	1.074	96.656 ± 1.040	0.326 ± 0.003	—
QML	beta43	0.20	0.198	-0.002 ± 0.004	-0.009 ± 0.022	0.006 ± 0.001	0.077 ± 0.003	0.006 ± 0.001	0.077 ± 0.004	0.148 ± 0.013	0.385 ± 0.020	0.199	-0.001	-0.56	0.0769	0.0769	1.039	96.656 ± 1.040	0.314 ± 0.003	—
QML	beta44	0.00	-0.012	-0.012 ± 0.006	—	0.011 ± 0.001	0.103 ± 0.005	0.011 ± 0.001	0.103 ± 0.006	—	—	-0.004	-0.004	—	0.1097	0.1098	1.117	96.990 ± 0.988	0.449 ± 0.008	3.010 ± 0.988
QML	beta45	0.00	-0.006	-0.006 ± 0.006	—	0.009 ± 0.001	0.096 ± 0.004	0.009 ± 0.001	0.096 ± 0.005	—	—	0.002	0.002	—	0.1045	0.1045	0.992	95.987 ± 1.135	0.373 ± 0.006	4.013 ± 1.135
QML	beta46	0.00	0.001	0.001 ± 0.005	—	0.006 ± 0.001	0.080 ± 0.004	0.006 ± 0.001	0.080 ± 0.004	—	—	-0.002	-0.002	—	0.0829	0.0829	1.071	96.321 ± 1.089	0.338 ± 0.008	3.679 ± 1.089
QML	beta47	0.00	0.002	0.002 ± 0.004	—	0.006 ± 0.001	0.077 ± 0.003	0.006 ± 0.001	0.077 ± 0.004	—	—	-0.002	-0.002	—	0.0805	0.0806	1.078	98.328 ± 0.742	0.327 ± 0.008	1.672 ± 0.742
QML	beta51	0.16	0.172	0.012 ± 0.007	0.076 ± 0.044	0.015 ± 0.001	0.122 ± 0.005	0.015 ± 0.001	0.122 ± 0.006	0.585 ± 0.051	0.765 ± 0.040	0.162	0.002	1.43	0.1265	0.1266	1.052	97.324 ± 0.933	0.503 ± 0.007	—
QML	beta52	0.16	0.162	0.002 ± 0.005	0.012 ± 0.034	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.094 ± 0.005	0.347 ± 0.033	0.589 ± 0.033	0.159	-0.001	-0.31	0.0935	0.0935	0.994	94.649 ± 1.302	0.368 ± 0.006	—
QML	beta53	0.16	0.155	-0.005 ± 0.005	-0.029 ± 0.033	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.091 ± 0.005	0.321 ± 0.027	0.567 ± 0.029	0.158	-0.002	-1.26	0.0907	0.0907	1.025	95.318 ± 1.222	0.364 ± 0.004	—
QML	beta54	0.16	0.159	-0.001 ± 0.005	-0.006 ± 0.030	0.007 ± 0.001	0.084 ± 0.003	0.007 ± 0.001	0.084 ± 0.004	0.273 ± 0.022	0.523 ± 0.026	0.151	-0.009	-5.79	0.0849	0.0854	1.040	96.656 ± 1.040	0.341 ± 0.004	—
QML	beta55	0.00	0.000	-0.000 ± 0.008	—	0.020 ± 0.002	0.140 ± 0.007	0.020 ± 0.002	0.140 ± 0.008	—	—	0.007	0.007	—	0.1388	0.1390	1.098	96.990 ± 0.988	0.603 ± 0.015	3.010 ± 0.988
QML	beta56	0.00	-0.011	-0.011 ± 0.007	—	0.016 ± 0.001	0.127 ± 0.006	0.016 ± 0.002	0.127 ± 0.007	—	—	-0.015	-0.015	—	0.1268	0.1277	1.003	95.318 ± 1.222	0.498 ± 0.008	4.682 ± 1.222
QML	beta57	0.00	-0.005	-0.005 ± 0.005	—	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.084 ± 0.004	—	—	-0.008	-0.008	—	0.0876	0.0880	1.082	97.993 ± 0.811	0.356 ± 0.012	2.007 ± 0.811
QML	beta58	0.00	0.001	0.001 ± 0.004	—	0.005 ± 0.000	0.070 ± 0.003	0.005 ± 0.000	0.070 ± 0.004	—	—	-0.003	-0.003	—	0.0699	0.0700	1.074	96.990 ± 0.988	0.294 ± 0.005	3.010 ± 0.988
UPI	beta41	0.20	0.208	0.008 ± 0.007	0.040 ± 0.034	0.014 ± 0.002	0.118 ± 0.007	0.014 ± 0.002	0.118 ± 0.008	0.348 ± 0.040	0.590 ± 0.038	0.205	0.005	2.68	0.1204	0.1205	1.502	97.993 ± 0.811	0.694 ± 0.051	—
UPI	beta42	0.20	0.192	-0.008 ± 0.006	-0.039 ± 0.032	0.012 ± 0.001	0.110 ± 0.005	0.012 ± 0.001	0.110 ± 0.006	0.305 ± 0.030	0.552 ± 0.032	0.203	0.003	1.36	0.1156	0.1156	1.533	98.328 ± 0.742	0.663 ± 0.034	—
UPI	beta43	0.20	0.195	-0.005 ± 0.006	-0.023 ± 0.029	0.010 ± 0.001	0.101 ± 0.005	0.010 ± 0.001	0.100 ± 0.005	0.252 ± 0.023	0.502 ± 0.027	0.195	-0.005	-2.48	0.1058	0.1059	1.589	99.331 ± 0.471	0.626 ± 0.039	—
UPI	beta44	0.00	0.012	0.012 ± 0.012	—	0.041 ± 0.005	0.203 ± 0.013	0.041 ± 0.005	0.203 ± 0.014	—	—	0.001	0.001	—	0.1717	0.1717	1.706	100.000 ± 0.000	1.355 ± 0.111	0.000 ± 0.000
UPI	beta45	0.00	-0.020	-0.020 ± 0.010	—	0.029 ± 0.003	0.171 ± 0.010	0.029 ± 0.003	0.172 ± 0.011	—	—	-0.011	-0.011	—	0.1635	0.1639	1.680	99.666 ± 0.334	1.126 ± 0.091	0.334 ± 0.334
UPI	beta46	0.00	0.012	0.012 ± 0.017	—	0.090 ± 0.011	0.300 ± 0.019	0.090 ± 0.011	0.299 ± 0.020	—	—	0.011	0.011	—	0.2440	0.2443	1.829	100.000 ± 0.000	2.148 ± 0.156	0.000 ± 0.000

Table S229

Simulation 2 - Condition 85: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.003	0.003 ± 0.019	—	0.103 ± 0.012	0.321 ± 0.020	0.103 ± 0.012	0.321 ± 0.022	—	—	-0.004	-0.004	—	0.2448	0.2448	2.073	100.000 ± 0.000	2.608 ± 0.255	0.000 ± 0.000
UPI	beta51	0.16	0.166	0.006 ± 0.007	0.039 ± 0.046	0.016 ± 0.001	0.127 ± 0.006	0.016 ± 0.001	0.127 ± 0.007	0.629 ± 0.056	0.793 ± 0.042	0.151	-0.009	-5.45	0.1366	0.1369	1.324	97.993 ± 0.811	0.659 ± 0.024	—
UPI	beta52	0.16	0.166	0.006 ± 0.007	0.034 ± 0.041	0.013 ± 0.001	0.114 ± 0.006	0.013 ± 0.001	0.114 ± 0.006	0.507 ± 0.048	0.712 ± 0.040	0.166	0.006	3.57	0.1176	0.1178	1.430	97.659 ± 0.874	0.639 ± 0.034	—
UPI	beta53	0.16	0.149	-0.011 ± 0.006	-0.066 ± 0.039	0.012 ± 0.001	0.107 ± 0.005	0.012 ± 0.001	0.108 ± 0.006	0.453 ± 0.038	0.673 ± 0.034	0.145	-0.015	-9.33	0.1147	0.1156	1.632	98.328 ± 0.742	0.686 ± 0.045	—
UPI	beta54	0.16	0.166	0.006 ± 0.006	0.035 ± 0.040	0.012 ± 0.001	0.110 ± 0.006	0.012 ± 0.001	0.110 ± 0.007	0.469 ± 0.048	0.685 ± 0.041	0.163	0.003	2.18	0.1117	0.1118	1.547	98.662 ± 0.664	0.665 ± 0.036	—
UPI	beta55	0.00	-0.019	-0.019 ± 0.012	—	0.045 ± 0.005	0.213 ± 0.013	0.045 ± 0.005	0.213 ± 0.014	—	—	-0.007	-0.007	—	0.1956	0.1957	1.909	99.666 ± 0.334	1.592 ± 0.146	0.334 ± 0.334
UPI	beta56	0.00	0.005	0.005 ± 0.013	—	0.047 ± 0.005	0.217 ± 0.012	0.047 ± 0.005	0.217 ± 0.013	—	—	-0.005	-0.005	—	0.1865	0.1866	1.733	100.000 ± 0.000	1.477 ± 0.094	0.000 ± 0.000
UPI	beta57	0.00	-0.038	-0.038 ± 0.017	—	0.084 ± 0.010	0.290 ± 0.017	0.086 ± 0.010	0.293 ± 0.019	—	—	-0.012	-0.012	—	0.2149	0.2153	1.648	100.000 ± 0.000	1.876 ± 0.142	0.000 ± 0.000
UPI	beta58	0.00	0.012	0.012 ± 0.017	—	0.086 ± 0.011	0.293 ± 0.019	0.086 ± 0.011	0.292 ± 0.020	—	—	0.000	0.000	—	0.2214	0.2214	1.794	100.000 ± 0.000	2.058 ± 0.168	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S230

Simulation 2 - Condition 86: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.004 ± 0.010	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.051 ± 0.002	0.066 ± 0.004	0.256 ± 0.009	0.199	-0.001	-0.29	0.0510	0.0510	1.058	95.742 ± 0.748	0.213 ± 0.001	—
LSAM	beta42	0.20	0.205	0.005 ± 0.002	0.025 ± 0.010	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.004	0.272 ± 0.009	0.206	0.006	2.76	0.0517	0.0520	1.055	95.467 ± 0.771	0.224 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.003	0.258 ± 0.008	0.199	-0.001	-0.41	0.0554	0.0554	1.001	95.055 ± 0.804	0.203 ± 0.001	—
LSAM	beta44	0.00	0.003	0.003 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.002	0.002	—	0.0714	0.0714	1.054	97.253 ± 0.606	0.301 ± 0.003	2.747 ± 0.606
LSAM	beta45	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.001	0.001	—	0.0645	0.0645	1.013	96.016 ± 0.725	0.247 ± 0.002	3.984 ± 0.725
LSAM	beta46	0.00	-0.001	-0.001 ± 0.004	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	—	—	-0.003	-0.003	—	0.0901	0.0901	1.055	98.489 ± 0.452	0.408 ± 0.006	1.511 ± 0.452
LSAM	beta47	0.00	-0.005	-0.005 ± 0.004	—	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	—	—	-0.005	-0.005	—	0.0923	0.0925	1.069	97.665 ± 0.560	0.418 ± 0.006	2.335 ± 0.560
LSAM	beta51	0.16	0.165	0.005 ± 0.003	0.030 ± 0.018	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.243 ± 0.014	0.493 ± 0.017	0.163	0.003	2.15	0.0762	0.0763	0.939	94.368 ± 0.854	0.290 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.007	0.342 ± 0.012	0.160	0.000	-0.24	0.0526	0.0526	1.041	96.703 ± 0.662	0.223 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.014	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.008	0.378 ± 0.012	0.158	-0.002	-1.46	0.0600	0.0601	0.975	95.467 ± 0.771	0.231 ± 0.001	—
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.012	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.112 ± 0.006	0.334 ± 0.010	0.160	0.000	-0.07	0.0547	0.0547	1.055	97.527 ± 0.576	0.221 ± 0.001	—
LSAM	beta55	0.00	0.002	0.002 ± 0.004	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	—	—	-0.003	-0.003	—	0.0924	0.0924	1.004	96.016 ± 0.725	0.400 ± 0.006	3.984 ± 0.725
LSAM	beta56	0.00	0.002	0.002 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.002	0.002	—	0.0891	0.0891	1.014	96.291 ± 0.700	0.355 ± 0.004	3.709 ± 0.700
LSAM	beta57	0.00	-0.003	-0.003 ± 0.004	—	0.012 ± 0.001	0.107 ± 0.004	0.012 ± 0.001	0.107 ± 0.004	—	—	0.000	0.000	—	0.0935	0.0935	1.034	97.527 ± 0.576	0.435 ± 0.008	2.473 ± 0.576
LSAM	beta58	0.00	-0.001	-0.001 ± 0.004	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	—	—	0.002	0.002	—	0.0859	0.0859	1.024	97.802 ± 0.543	0.384 ± 0.006	2.198 ± 0.543
QML	beta41	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.056 ± 0.003	0.236 ± 0.008	0.196	-0.004	-1.79	0.0475	0.0476	1.021	95.055 ± 0.804	0.189 ± 0.002	—
QML	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.009	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.063 ± 0.004	0.251 ± 0.009	0.203	0.003	1.55	0.0455	0.0456	1.008	94.780 ± 0.824	0.198 ± 0.001	—
QML	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.198	-0.002	-0.77	0.0507	0.0508	1.001	94.918 ± 0.814	0.188 ± 0.001	—
QML	beta44	0.00	-0.007	-0.007 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.006	-0.006	—	0.0674	0.0676	1.060	95.742 ± 0.748	0.268 ± 0.008	4.258 ± 0.748
QML	beta45	0.00	-0.006	-0.006 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.003	-0.003	—	0.0547	0.0548	1.039	94.918 ± 0.814	0.221 ± 0.005	5.082 ± 0.814
QML	beta46	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	-0.003	-0.003	—	0.0465	0.0466	1.114	96.566 ± 0.675	0.207 ± 0.012	3.434 ± 0.675
QML	beta47	0.00	-0.007	-0.007 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	-0.010	-0.010	—	0.0461	0.0472	1.047	95.879 ± 0.737	0.192 ± 0.002	4.121 ± 0.737
QML	beta51	0.16	0.167	0.007 ± 0.003	0.041 ± 0.018	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.225 ± 0.012	0.474 ± 0.016	0.164	0.004	2.20	0.0768	0.0769	0.978	94.918 ± 0.814	0.290 ± 0.001	—
QML	beta52	0.16	0.160	0.000 ± 0.002	0.001 ± 0.012	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.106 ± 0.006	0.326 ± 0.011	0.159	-0.001	-0.49	0.0517	0.0517	1.027	95.742 ± 0.748	0.210 ± 0.001	—
QML	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.013	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.007	0.352 ± 0.012	0.157	-0.003	-1.77	0.0591	0.0591	0.978	94.643 ± 0.835	0.216 ± 0.001	—
QML	beta54	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.100 ± 0.005	0.316 ± 0.010	0.158	-0.002	-1.14	0.0516	0.0516	1.027	96.566 ± 0.675	0.204 ± 0.001	—
QML	beta55	0.00	0.000	-0.000 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.000	0.091 ± 0.003	—	—	-0.006	-0.006	—	0.0845	0.0848	0.990	95.192 ± 0.793	0.353 ± 0.004	4.808 ± 0.793
QML	beta56	0.00	-0.007	-0.007 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.009	-0.009	—	0.0755	0.0761	0.995	95.742 ± 0.748	0.295 ± 0.002	4.258 ± 0.748
QML	beta57	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.003	-0.003	—	0.0521	0.0522	1.044	95.192 ± 0.793	0.213 ± 0.003	4.808 ± 0.793
QML	beta58	0.00	-0.005	-0.005 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.004	-0.004	—	0.0430	0.0432	1.024	95.467 ± 0.771	0.173 ± 0.001	4.533 ± 0.771
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	0.071 ± 0.004	0.267 ± 0.009	0.202	0.002	1.00	0.0546	0.0546	1.254	97.253 ± 0.606	0.263 ± 0.006	—
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.025 ± 0.010	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.279 ± 0.009	0.205	0.005	2.72	0.0568	0.0571	1.264	96.016 ± 0.725	0.276 ± 0.007	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.068 ± 0.003	0.261 ± 0.008	0.200	0.000	0.14	0.0580	0.0580	1.167	97.390 ± 0.591	0.238 ± 0.004	—
UPI	beta44	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	0.003	0.003	—	0.0822	0.0823	1.405	99.038 ± 0.362	0.451 ± 0.016	0.962 ± 0.362
UPI	beta45	0.00	0.000	-0.000 ± 0.003	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	—	—	0.000	0.000	—	0.0723	0.0723	1.238	98.489 ± 0.452	0.336 ± 0.009	1.511 ± 0.452
UPI	beta46	0.00	0.000	0.000 ± 0.005	—	0.016 ± 0.001	0.126 ± 0.005	0.016 ± 0.001	0.125 ± 0.005	—	—	0.000	0.000	—	0.1099	0.1099	1.541	99.725 ± 0.194	0.758 ± 0.035	0.275 ± 0.194

Table S230

Simulation 2 - Condition 86: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.005	-0.005 ± 0.005	—	0.018 ± 0.001	0.133 ± 0.005	0.018 ± 0.001	0.133 ± 0.005	—	—	-0.005	-0.005	—	0.1159	0.1161	1.663	99.725 ± 0.194	0.864 ± 0.042	0.275 ± 0.194
UPI	beta51	0.16	0.165	0.005 ± 0.003	0.028 ± 0.018	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.225 ± 0.012	0.474 ± 0.016	0.163	0.003	1.81	0.0773	0.0774	1.028	95.742 ± 0.748	0.305 ± 0.003	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.121 ± 0.008	0.348 ± 0.013	0.160	0.000	-0.19	0.0523	0.0523	1.161	97.253 ± 0.606	0.253 ± 0.005	—
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.009 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.008	0.376 ± 0.013	0.157	-0.003	-1.95	0.0593	0.0593	1.080	96.291 ± 0.700	0.255 ± 0.003	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.111 ± 0.005	0.332 ± 0.010	0.160	0.000	-0.22	0.0540	0.0540	1.169	97.390 ± 0.591	0.244 ± 0.004	—
UPI	beta55	0.00	0.001	0.001 ± 0.004	—	0.011 ± 0.001	0.106 ± 0.004	0.011 ± 0.001	0.106 ± 0.004	—	—	-0.002	-0.002	—	0.0998	0.0998	1.242	98.626 ± 0.431	0.516 ± 0.019	1.374 ± 0.431
UPI	beta56	0.00	0.000	0.000 ± 0.004	—	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	0.007	0.007	—	0.0955	0.0957	1.207	98.489 ± 0.452	0.458 ± 0.010	1.511 ± 0.452
UPI	beta57	0.00	-0.005	-0.005 ± 0.005	—	0.015 ± 0.001	0.124 ± 0.005	0.015 ± 0.001	0.124 ± 0.005	—	—	0.000	0.000	—	0.1049	0.1049	1.407	100.000 ± 0.000	0.682 ± 0.028	0.000 ± 0.000
UPI	beta58	0.00	-0.007	-0.007 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.113 ± 0.005	—	—	-0.001	-0.001	—	0.0917	0.0917	1.563	99.313 ± 0.306	0.694 ± 0.047	0.687 ± 0.306

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S231

Simulation 2 - Condition 87: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.290 ± 0.008	0.202	0.002	0.82	0.0573	0.0574	1.020	95.734 ± 0.652	0.232 ± 0.001	—
LSAM	beta42	0.20	0.200	0.000 ± 0.002	0.000 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.302 ± 0.009	0.200	0.000	-0.20	0.0597	0.0597	1.015	95.109 ± 0.696	0.241 ± 0.001	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.009	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.293 ± 0.008	0.199	-0.001	-0.69	0.0610	0.0610	0.983	94.589 ± 0.730	0.226 ± 0.001	—
LSAM	beta44	0.00	0.003	0.003 ± 0.002	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	0.004	0.004	—	0.0757	0.0758	0.963	95.630 ± 0.659	0.280 ± 0.002	4.370 ± 0.659
LSAM	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.003	-0.003	—	0.0652	0.0652	0.973	95.317 ± 0.682	0.247 ± 0.001	4.683 ± 0.682
LSAM	beta46	0.00	-0.002	-0.002 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	0.002	0.002	—	0.0862	0.0863	0.983	96.046 ± 0.629	0.355 ± 0.003	3.954 ± 0.629
LSAM	beta47	0.00	-0.002	-0.002 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.001	-0.001	—	0.0869	0.0869	0.982	95.109 ± 0.696	0.359 ± 0.003	4.891 ± 0.696
LSAM	beta51	0.16	0.160	-0.000 ± 0.003	-0.002 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.282 ± 0.013	0.531 ± 0.015	0.161	0.001	0.55	0.0881	0.0881	0.966	94.277 ± 0.749	0.322 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.009 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.007	0.399 ± 0.011	0.162	0.002	1.46	0.0654	0.0654	1.003	95.838 ± 0.644	0.251 ± 0.001	—
LSAM	beta53	0.16	0.163	0.003 ± 0.002	0.019 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.179 ± 0.009	0.423 ± 0.013	0.161	0.001	0.58	0.0677	0.0677	0.974	94.485 ± 0.736	0.258 ± 0.001	—
LSAM	beta54	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.008	0.407 ± 0.012	0.158	-0.002	-1.04	0.0633	0.0633	0.984	94.901 ± 0.710	0.251 ± 0.001	—
LSAM	beta55	0.00	0.003	0.003 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	-0.001	-0.001	—	0.0900	0.0901	1.007	95.317 ± 0.682	0.370 ± 0.002	4.683 ± 0.682
LSAM	beta56	0.00	-0.001	-0.001 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	—	—	0.000	0.000	—	0.0885	0.0885	0.992	95.213 ± 0.689	0.344 ± 0.002	4.787 ± 0.689
LSAM	beta57	0.00	0.000	0.000 ± 0.003	—	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	—	—	0.003	0.003	—	0.0946	0.0946	0.958	95.005 ± 0.703	0.381 ± 0.003	4.995 ± 0.703
LSAM	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	—	—	0.000	0.000	—	0.0901	0.0901	0.931	94.381 ± 0.743	0.350 ± 0.003	5.619 ± 0.743
QML	beta41	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.004	0.276 ± 0.008	0.199	-0.001	-0.38	0.0530	0.0530	1.010	95.213 ± 0.689	0.219 ± 0.001	—
QML	beta42	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.004	0.283 ± 0.008	0.198	-0.002	-0.93	0.0572	0.0573	1.024	95.109 ± 0.696	0.227 ± 0.001	—
QML	beta43	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.280 ± 0.008	0.198	-0.002	-0.94	0.0599	0.0599	0.993	94.797 ± 0.716	0.218 ± 0.001	—
QML	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.001	-0.001	—	0.0704	0.0704	0.994	95.525 ± 0.667	0.269 ± 0.002	4.475 ± 0.667
QML	beta45	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.006	-0.006	—	0.0624	0.0627	0.987	94.901 ± 0.710	0.236 ± 0.001	5.099 ± 0.710
QML	beta46	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.001	-0.001	—	0.0554	0.0554	1.024	96.046 ± 0.629	0.222 ± 0.002	3.954 ± 0.629
QML	beta47	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.002	-0.002	—	0.0528	0.0528	1.017	94.797 ± 0.716	0.221 ± 0.002	5.203 ± 0.716
QML	beta51	0.16	0.160	0.000 ± 0.003	0.002 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.272 ± 0.012	0.521 ± 0.014	0.164	0.004	2.71	0.0841	0.0842	1.005	95.525 ± 0.667	0.329 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.162	0.002	1.11	0.0642	0.0642	1.007	95.942 ± 0.637	0.248 ± 0.001	—
QML	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.172 ± 0.009	0.415 ± 0.013	0.159	-0.001	-0.50	0.0650	0.0650	0.973	94.277 ± 0.749	0.253 ± 0.001	—
QML	beta54	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.393 ± 0.011	0.157	-0.003	-2.14	0.0640	0.0641	0.981	95.005 ± 0.703	0.242 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	—	—	-0.003	-0.003	—	0.0846	0.0846	1.052	95.838 ± 0.644	0.360 ± 0.002	4.162 ± 0.644
QML	beta56	0.00	-0.005	-0.005 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.004	-0.004	—	0.0821	0.0822	1.037	96.462 ± 0.596	0.322 ± 0.002	3.538 ± 0.596
QML	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.001	-0.001	—	0.0600	0.0600	0.990	93.965 ± 0.768	0.247 ± 0.002	6.035 ± 0.768
QML	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.005	-0.005	—	0.0573	0.0575	0.964	93.444 ± 0.798	0.218 ± 0.001	6.556 ± 0.798
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.017 ± 0.010	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.009	0.202	0.002	1.18	0.0579	0.0579	1.053	95.942 ± 0.637	0.245 ± 0.003	—
UPI	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.095 ± 0.005	0.308 ± 0.009	0.200	0.000	-0.22	0.0607	0.0607	1.053	95.005 ± 0.703	0.254 ± 0.002	—
UPI	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.008	0.200	0.000	0.21	0.0625	0.0625	1.017	95.317 ± 0.682	0.237 ± 0.001	—
UPI	beta44	0.00	0.003	0.003 ± 0.002	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	0.004	0.004	—	0.0751	0.0752	1.052	96.462 ± 0.596	0.319 ± 0.004	3.538 ± 0.596
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.002	-0.002	—	0.0675	0.0676	1.045	96.358 ± 0.604	0.276 ± 0.002	3.642 ± 0.604
UPI	beta46	0.00	-0.003	-0.003 ± 0.003	—	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	—	—	0.000	0.000	—	0.0957	0.0957	1.165	98.751 ± 0.358	0.481 ± 0.008	1.249 ± 0.358

Table S231

Simulation 2 - Condition 87: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.004	—	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	—	—	0.000	0.000	—	0.0955	0.0955	1.152	98.231 ± 0.425	0.513 ± 0.011	1.769 ± 0.425
UPI	beta51	0.16	0.159	-0.001 ± 0.003	-0.004 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.268 ± 0.012	0.518 ± 0.014	0.161	0.001	0.49	0.0836	0.0836	1.013	95.525 ± 0.667	0.329 ± 0.002	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.007	0.403 ± 0.011	0.161	0.001	0.73	0.0669	0.0670	1.023	95.942 ± 0.637	0.259 ± 0.002	—
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.019 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.182 ± 0.009	0.427 ± 0.013	0.161	0.001	0.60	0.0670	0.0670	1.001	95.005 ± 0.703	0.268 ± 0.001	—
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.409 ± 0.012	0.161	0.001	0.89	0.0635	0.0635	1.007	95.421 ± 0.674	0.258 ± 0.001	—
UPI	beta55	0.00	0.001	0.001 ± 0.003	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	—	—	0.000	0.000	—	0.0916	0.0916	1.050	96.982 ± 0.552	0.409 ± 0.004	3.018 ± 0.552
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	0.001	0.001	—	0.0912	0.0912	1.046	97.607 ± 0.493	0.383 ± 0.003	2.393 ± 0.493
UPI	beta57	0.00	0.001	0.001 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	—	—	0.002	0.002	—	0.0986	0.0986	1.101	96.254 ± 0.613	0.487 ± 0.010	3.746 ± 0.613
UPI	beta58	0.00	0.001	0.001 ± 0.004	—	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	—	—	0.000	0.000	—	0.0934	0.0934	1.124	98.023 ± 0.449	0.489 ± 0.011	1.977 ± 0.449

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S232

Simulation 2 - Condition 88: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.187 ± 0.005	0.199	-0.001	-0.63	0.0360	0.0360	0.961	94.394 ± 0.728	0.141 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.197	-0.003	-1.44	0.0381	0.0382	0.984	94.995 ± 0.690	0.146 ± 0.000	—
LSAM	beta43	0.20	0.203	0.003 ± 0.001	0.017 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.204	0.004	1.77	0.0362	0.0364	1.001	96.196 ± 0.605	0.139 ± 0.000	—
LSAM	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.002	-0.002	—	0.0414	0.0414	1.020	95.295 ± 0.670	0.168 ± 0.001	4.705 ± 0.670
LSAM	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0377	0.0377	0.989	94.595 ± 0.715	0.150 ± 0.000	5.405 ± 0.715
LSAM	beta46	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.001	0.001	—	0.0525	0.0525	0.991	94.995 ± 0.690	0.212 ± 0.001	5.005 ± 0.690
LSAM	beta47	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.001	-0.001	—	0.0557	0.0557	0.979	95.195 ± 0.677	0.211 ± 0.001	4.805 ± 0.677
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.004 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.008	0.161	0.001	0.37	0.0500	0.0500	1.020	95.796 ± 0.635	0.198 ± 0.000	—
LSAM	beta52	0.16	0.163	0.003 ± 0.001	0.016 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.164	0.004	2.29	0.0405	0.0406	0.992	95.295 ± 0.670	0.153 ± 0.000	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.256 ± 0.007	0.156	-0.004	-2.36	0.0408	0.0409	0.986	94.394 ± 0.728	0.158 ± 0.000	—
LSAM	beta54	0.16	0.162	0.002 ± 0.001	0.013 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.160	0.000	0.22	0.0380	0.0380	1.012	95.796 ± 0.635	0.153 ± 0.000	—
LSAM	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.003	-0.003	—	0.0544	0.0544	1.017	95.195 ± 0.677	0.221 ± 0.001	4.805 ± 0.677
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0559	0.0559	0.957	93.994 ± 0.752	0.207 ± 0.001	6.006 ± 0.752
LSAM	beta57	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.001	0.001	—	0.0602	0.0602	0.977	94.595 ± 0.715	0.225 ± 0.001	5.405 ± 0.715
LSAM	beta58	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.002	0.002	—	0.0559	0.0560	0.958	94.695 ± 0.709	0.210 ± 0.001	5.305 ± 0.709
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.198	-0.002	-0.98	0.0353	0.0353	0.974	94.094 ± 0.746	0.139 ± 0.002	—
QML	beta42	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.197	-0.003	-1.28	0.0367	0.0368	0.994	94.895 ± 0.696	0.142 ± 0.000	—
QML	beta43	0.20	0.202	0.002 ± 0.001	0.012 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.174 ± 0.005	0.202	0.002	1.00	0.0359	0.0360	1.004	96.096 ± 0.613	0.137 ± 0.000	—
QML	beta44	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.006	-0.006	—	0.0413	0.0417	1.025	95.095 ± 0.683	0.168 ± 0.001	4.905 ± 0.683
QML	beta45	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	—	0.0370	0.0373	1.002	94.194 ± 0.740	0.148 ± 0.001	5.806 ± 0.740
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.002	-0.002	—	0.0343	0.0343	1.007	95.395 ± 0.663	0.139 ± 0.001	4.605 ± 0.663
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.003	-0.003	—	0.0367	0.0369	0.990	94.595 ± 0.715	0.137 ± 0.001	5.405 ± 0.715
QML	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.004	0.306 ± 0.008	0.160	0.000	0.24	0.0491	0.0491	1.045	96.196 ± 0.605	0.200 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.001	0.013 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.163	0.003	2.09	0.0386	0.0388	1.006	95.395 ± 0.663	0.152 ± 0.000	—
QML	beta53	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.156	-0.004	-2.30	0.0401	0.0403	0.999	94.895 ± 0.696	0.156 ± 0.000	—
QML	beta54	0.16	0.161	0.001 ± 0.001	0.009 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.160	0.000	0.07	0.0370	0.0370	1.036	95.596 ± 0.649	0.154 ± 0.003	—
QML	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.001	-0.001	—	0.0529	0.0529	1.041	95.495 ± 0.656	0.221 ± 0.001	4.505 ± 0.656
QML	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.002	-0.002	—	0.0532	0.0532	0.964	94.294 ± 0.734	0.197 ± 0.001	5.706 ± 0.734
QML	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0403	0.0403	1.017	95.295 ± 0.670	0.153 ± 0.002	4.705 ± 0.670
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	—	0.0354	0.0354	0.975	94.194 ± 0.740	0.135 ± 0.001	5.806 ± 0.740
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.199	-0.001	-0.72	0.0362	0.0362	0.959	94.094 ± 0.746	0.142 ± 0.000	—
UPI	beta42	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.197	-0.003	-1.34	0.0377	0.0378	0.989	94.695 ± 0.709	0.147 ± 0.000	—
UPI	beta43	0.20	0.203	0.003 ± 0.001	0.017 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.179 ± 0.005	0.203	0.003	1.54	0.0365	0.0367	1.005	96.196 ± 0.605	0.141 ± 0.000	—
UPI	beta44	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.001	-0.001	—	0.0425	0.0425	1.027	95.596 ± 0.649	0.175 ± 0.001	4.404 ± 0.649
UPI	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0377	0.0377	1.000	95.596 ± 0.649	0.155 ± 0.001	4.404 ± 0.649
UPI	beta46	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.000	0.000	—	0.0538	0.0538	1.051	96.597 ± 0.574	0.233 ± 0.002	3.403 ± 0.574

Table S232

Simulation 2 - Condition 88: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.002	-0.002	—	0.0580	0.0580	1.022	96.496 ± 0.582	0.232 ± 0.002	3.504 ± 0.582
UPI	beta51	0.16	0.160	0.000 ± 0.002	0.003 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.160	0.000	-0.11	0.0494	0.0494	1.038	96.096 ± 0.613	0.199 ± 0.000	—
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.015 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.163	0.003	2.03	0.0403	0.0405	0.994	95.295 ± 0.670	0.153 ± 0.000	—
UPI	beta53	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.256 ± 0.007	0.157	-0.003	-2.07	0.0398	0.0399	0.996	95.095 ± 0.683	0.160 ± 0.000	—
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.014 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.242 ± 0.007	0.161	0.001	0.50	0.0381	0.0381	1.014	95.796 ± 0.635	0.154 ± 0.000	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.000	0.000	—	0.0546	0.0546	1.022	95.996 ± 0.620	0.227 ± 0.001	4.004 ± 0.620
UPI	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.002	0.002	—	0.0555	0.0555	0.961	94.795 ± 0.703	0.213 ± 0.001	5.205 ± 0.703
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.002	0.002	—	0.0595	0.0596	1.013	96.396 ± 0.590	0.241 ± 0.002	3.604 ± 0.590
UPI	beta58	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	0.001	0.001	—	0.0535	0.0535	1.007	96.697 ± 0.565	0.232 ± 0.002	3.303 ± 0.565

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S233

Simulation 2 - Condition 89: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.229 ± 0.006	0.201	0.001	0.57	0.0448	0.0448	1.000	94.600 ± 0.715	0.180 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.202	0.002	0.94	0.0485	0.0485	1.000	94.900 ± 0.696	0.185 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.200	0.000	-0.02	0.0436	0.0436	0.996	94.800 ± 0.702	0.178 ± 0.000	—
LSAM	beta44	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.001	0.001	—	0.0532	0.0533	0.946	93.700 ± 0.768	0.200 ± 0.001	6.300 ± 0.768
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.001	-0.001	—	0.0466	0.0466	0.984	93.500 ± 0.780	0.182 ± 0.001	6.500 ± 0.780
LSAM	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0566	0.0567	1.013	95.300 ± 0.669	0.230 ± 0.001	4.700 ± 0.669
LSAM	beta47	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	0.003	0.003	—	0.0569	0.0570	0.986	94.200 ± 0.739	0.229 ± 0.001	5.800 ± 0.739
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.008	0.407 ± 0.012	0.158	-0.002	-1.54	0.0635	0.0635	0.980	94.600 ± 0.715	0.250 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.325 ± 0.009	0.161	0.001	0.33	0.0516	0.0516	0.974	94.700 ± 0.708	0.198 ± 0.000	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.160	0.000	-0.18	0.0512	0.0512	1.015	95.300 ± 0.669	0.204 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.160	0.000	-0.21	0.0509	0.0509	0.971	93.500 ± 0.780	0.199 ± 0.000	—
LSAM	beta55	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.003	-0.003	—	0.0650	0.0651	0.985	93.900 ± 0.757	0.259 ± 0.001	6.100 ± 0.757
LSAM	beta56	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.003	-0.003	—	0.0621	0.0622	0.967	93.600 ± 0.774	0.245 ± 0.001	6.400 ± 0.774
LSAM	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0666	0.0666	0.958	93.800 ± 0.763	0.248 ± 0.001	6.200 ± 0.763
LSAM	beta58	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.001	0.001	—	0.0623	0.0623	0.985	94.900 ± 0.696	0.233 ± 0.001	5.100 ± 0.696
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.201	0.001	0.35	0.0448	0.0448	1.004	94.300 ± 0.733	0.177 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.201	0.001	0.68	0.0472	0.0473	1.000	94.500 ± 0.721	0.182 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.200	0.000	-0.15	0.0430	0.0430	0.999	94.800 ± 0.702	0.176 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0533	0.0534	0.954	94.000 ± 0.751	0.200 ± 0.001	6.000 ± 0.751
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.002	-0.002	—	0.0473	0.0473	0.990	94.100 ± 0.745	0.181 ± 0.001	5.900 ± 0.745
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.002	-0.002	—	0.0457	0.0457	1.028	95.200 ± 0.676	0.186 ± 0.001	4.800 ± 0.676
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	—	0.0450	0.0450	0.998	94.200 ± 0.739	0.185 ± 0.001	5.800 ± 0.739
QML	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.008	0.405 ± 0.012	0.158	-0.002	-1.11	0.0633	0.0633	0.997	95.000 ± 0.689	0.254 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.160	0.000	0.07	0.0519	0.0519	0.979	95.000 ± 0.689	0.199 ± 0.000	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.319 ± 0.009	0.160	0.000	-0.18	0.0505	0.0505	1.021	95.300 ± 0.669	0.204 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.159	-0.001	-0.78	0.0512	0.0512	0.973	93.700 ± 0.768	0.197 ± 0.000	—
QML	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.003	-0.003	—	0.0644	0.0645	0.999	94.400 ± 0.727	0.259 ± 0.001	5.600 ± 0.727
QML	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.003	-0.003	—	0.0604	0.0605	0.976	93.600 ± 0.774	0.241 ± 0.001	6.400 ± 0.774
QML	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.002	-0.002	—	0.0551	0.0551	0.971	93.900 ± 0.757	0.203 ± 0.001	6.100 ± 0.757
QML	beta58	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.001	-0.001	—	0.0488	0.0488	0.993	94.900 ± 0.696	0.189 ± 0.001	5.100 ± 0.696
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.201	0.001	0.71	0.0450	0.0450	0.998	94.600 ± 0.715	0.179 ± 0.000	—
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.202	0.002	1.02	0.0475	0.0475	0.999	95.000 ± 0.689	0.185 ± 0.000	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.200	0.000	0.06	0.0436	0.0436	1.000	94.600 ± 0.715	0.179 ± 0.000	—
UPI	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.001	0.001	—	0.0540	0.0541	0.949	93.400 ± 0.785	0.204 ± 0.001	6.600 ± 0.785
UPI	beta45	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.001	0.001	—	0.0468	0.0468	0.988	93.900 ± 0.757	0.185 ± 0.001	6.100 ± 0.757
UPI	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0581	0.0581	1.023	95.600 ± 0.649	0.241 ± 0.001	4.400 ± 0.649

Table S233

Simulation 2 - Condition 89: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.001	0.001	—	0.0559	0.0559	1.017	94.500 ± 0.721	0.240 ± 0.001	5.500 ± 0.721
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.012	0.158	-0.002	-1.35	0.0638	0.0638	0.995	95.000 ± 0.689	0.252 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.161	0.001	0.68	0.0519	0.0519	0.970	94.600 ± 0.715	0.198 ± 0.000	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.159	-0.001	-0.60	0.0521	0.0521	1.017	95.500 ± 0.656	0.205 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.160	0.000	-0.25	0.0520	0.0520	0.971	93.200 ± 0.796	0.198 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.003	-0.003	—	0.0667	0.0668	0.996	94.600 ± 0.715	0.263 ± 0.001	5.400 ± 0.715
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.003	-0.003	—	0.0611	0.0612	0.976	94.000 ± 0.751	0.249 ± 0.001	6.000 ± 0.751
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.001	-0.001	—	0.0639	0.0639	0.973	94.400 ± 0.727	0.257 ± 0.001	5.600 ± 0.727
UPI	beta58	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.001	0.001	—	0.0601	0.0601	1.012	96.300 ± 0.597	0.245 ± 0.001	3.700 ± 0.597

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S234

Simulation 2 - Condition 90: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.201	0.001	0.56	0.0284	0.0284	1.008	94.900 ± 0.696	0.113 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.201	0.001	0.64	0.0313	0.0314	0.999	94.500 ± 0.721	0.116 ± 0.000	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.140 ± 0.004	0.200	0.000	0.09	0.0278	0.0278	1.023	96.100 ± 0.612	0.112 ± 0.000	—
LSAM	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.002	0.002	—	0.0324	0.0324	0.980	94.600 ± 0.715	0.126 ± 0.000	5.400 ± 0.715
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0296	0.0296	1.005	95.200 ± 0.676	0.115 ± 0.000	4.800 ± 0.676
LSAM	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.002	-0.002	—	0.0365	0.0366	0.996	95.500 ± 0.656	0.144 ± 0.000	4.500 ± 0.656
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0364	0.0364	0.994	95.500 ± 0.656	0.143 ± 0.000	4.500 ± 0.656
LSAM	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.258 ± 0.007	0.160	0.000	-0.18	0.0409	0.0409	0.981	93.800 ± 0.763	0.159 ± 0.000	—
LSAM	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.159	-0.001	-0.44	0.0328	0.0328	0.971	93.900 ± 0.757	0.125 ± 0.000	—
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.005	0.163	0.003	1.65	0.0347	0.0348	0.988	95.400 ± 0.662	0.129 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.039 ± 0.002	0.196 ± 0.005	0.160	0.000	-0.23	0.0325	0.0325	1.012	94.600 ± 0.715	0.125 ± 0.000	—
LSAM	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.001	0.001	—	0.0439	0.0440	0.985	95.000 ± 0.689	0.163 ± 0.000	5.000 ± 0.689
LSAM	beta56	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0393	0.0393	1.012	95.900 ± 0.627	0.154 ± 0.000	4.100 ± 0.627
LSAM	beta57	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	—	0.0403	0.0403	0.972	94.700 ± 0.708	0.156 ± 0.000	5.300 ± 0.708
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0379	0.0380	0.988	95.900 ± 0.627	0.147 ± 0.000	4.100 ± 0.627
QML	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.201	0.001	0.45	0.0277	0.0277	1.006	95.100 ± 0.683	0.112 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	0.21	0.0311	0.0311	1.003	94.500 ± 0.721	0.115 ± 0.000	—
QML	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.200	0.000	-0.02	0.0281	0.0281	1.025	96.100 ± 0.612	0.111 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	—	0.0325	0.0325	0.991	94.400 ± 0.727	0.127 ± 0.001	5.600 ± 0.727
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0299	0.0299	1.013	95.000 ± 0.689	0.115 ± 0.001	5.000 ± 0.689
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.002	-0.002	—	0.0294	0.0295	1.006	94.800 ± 0.702	0.118 ± 0.000	5.200 ± 0.702
QML	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.002	-0.002	—	0.0303	0.0303	1.016	95.600 ± 0.649	0.117 ± 0.001	4.400 ± 0.649
QML	beta51	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.160	0.000	-0.04	0.0404	0.0404	0.990	94.000 ± 0.751	0.159 ± 0.000	—
QML	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.041 ± 0.002	0.204 ± 0.006	0.159	-0.001	-0.40	0.0320	0.0320	0.977	94.300 ± 0.733	0.125 ± 0.000	—
QML	beta53	0.16	0.162	0.002 ± 0.001	0.012 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.005	0.162	0.002	1.31	0.0339	0.0339	0.991	95.800 ± 0.634	0.128 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.195 ± 0.005	0.159	-0.001	-0.35	0.0319	0.0319	1.015	95.100 ± 0.683	0.124 ± 0.000	—
QML	beta55	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0435	0.0435	0.994	95.000 ± 0.689	0.164 ± 0.001	5.000 ± 0.689
QML	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0383	0.0383	1.025	96.000 ± 0.620	0.154 ± 0.002	4.000 ± 0.620
QML	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.003	-0.003	—	0.0323	0.0325	0.981	94.500 ± 0.721	0.128 ± 0.000	5.500 ± 0.721
QML	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0306	0.0307	0.993	95.700 ± 0.641	0.119 ± 0.000	4.300 ± 0.641
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.201	0.001	0.53	0.0284	0.0284	1.004	94.900 ± 0.696	0.113 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.006 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.201	0.001	0.68	0.0317	0.0317	0.997	94.500 ± 0.721	0.116 ± 0.000	—
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.200	0.000	0.20	0.0280	0.0280	1.023	96.200 ± 0.605	0.112 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.001	0.001	—	0.0332	0.0332	0.978	94.200 ± 0.739	0.126 ± 0.000	5.800 ± 0.739
UPI	beta45	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0305	0.0305	1.009	95.400 ± 0.662	0.115 ± 0.000	4.600 ± 0.662
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0370	0.0370	0.993	95.200 ± 0.676	0.147 ± 0.000	4.800 ± 0.676

Table S234

Simulation 2 - Condition 90: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0374	0.0374	1.001	95.900 ± 0.627	0.146 ± 0.000	4.100 ± 0.627
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.160	0.000	-0.27	0.0404	0.0404	0.988	93.900 ± 0.757	0.159 ± 0.000	—
UPI	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.159	-0.001	-0.48	0.0327	0.0327	0.968	93.500 ± 0.780	0.124 ± 0.000	—
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.005	0.163	0.003	1.60	0.0349	0.0350	0.990	95.700 ± 0.641	0.129 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.039 ± 0.002	0.196 ± 0.005	0.160	0.000	-0.28	0.0326	0.0326	1.011	94.800 ± 0.702	0.125 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	—	0.0445	0.0445	0.989	94.700 ± 0.708	0.164 ± 0.000	5.300 ± 0.708
UPI	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0390	0.0390	1.015	95.700 ± 0.641	0.155 ± 0.000	4.300 ± 0.641
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0403	0.0404	0.979	94.600 ± 0.715	0.158 ± 0.000	5.400 ± 0.715
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0383	0.0384	0.990	95.900 ± 0.627	0.149 ± 0.000	4.100 ± 0.627

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S235

Simulation 2 - Condition 91: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.222	0.022 ± 0.011	0.108 ± 0.057	0.014 ± 0.002	0.119 ± 0.009	0.015 ± 0.002	0.121 ± 0.011	0.365 ± 0.058	0.604 ± 0.057	0.221	0.021	10.45	0.1152	0.1170	0.993	97.273 ± 1.553	0.465 ± 0.017	—
LSAM	beta42	0.20	0.240	0.040 ± 0.012	0.200 ± 0.061	0.016 ± 0.003	0.127 ± 0.013	0.018 ± 0.004	0.133 ± 0.015	0.442 ± 0.089	0.665 ± 0.075	0.227	0.027	13.25	0.1175	0.1204	1.040	99.091 ± 0.905	0.519 ± 0.021	—
LSAM	beta43	0.20	0.167	-0.033 ± 0.007	-0.165 ± 0.037	0.006 ± 0.001	0.078 ± 0.005	0.007 ± 0.001	0.085 ± 0.006	0.180 ± 0.022	0.424 ± 0.031	0.174	-0.026	-13.07	0.0856	0.0895	1.136	95.455 ± 1.986	0.349 ± 0.009	—
LSAM	beta44	0.00	0.075	0.075 ± 0.016	—	0.029 ± 0.005	0.170 ± 0.016	0.034 ± 0.006	0.185 ± 0.018	—	—	0.061	0.061	—	0.1326	0.1461	0.964	97.273 ± 1.553	0.643 ± 0.045	2.727 ± 1.553
LSAM	beta45	0.00	-0.015	-0.015 ± 0.011	—	0.014 ± 0.002	0.117 ± 0.010	0.014 ± 0.002	0.118 ± 0.011	—	—	-0.003	-0.003	—	0.1199	0.1200	0.981	96.364 ± 1.785	0.452 ± 0.023	3.636 ± 1.785
LSAM	beta46	0.00	-0.048	-0.048 ± 0.011	—	0.014 ± 0.003	0.117 ± 0.012	0.016 ± 0.003	0.126 ± 0.013	—	—	-0.054	-0.054	—	0.0924	0.1071	0.853	93.636 ± 2.327	0.391 ± 0.024	6.364 ± 2.327
LSAM	beta47	0.00	-0.061	-0.061 ± 0.009	—	0.009 ± 0.002	0.096 ± 0.009	0.013 ± 0.002	0.113 ± 0.012	—	—	-0.047	-0.047	—	0.0845	0.0966	1.126	97.273 ± 1.553	0.424 ± 0.029	2.727 ± 1.553
LSAM	beta51	0.16	0.155	-0.005 ± 0.013	-0.029 ± 0.083	0.019 ± 0.003	0.139 ± 0.012	0.019 ± 0.003	0.138 ± 0.013	0.748 ± 0.121	0.865 ± 0.083	0.156	-0.004	-2.64	0.1341	0.1342	0.889	93.636 ± 2.327	0.484 ± 0.009	—
LSAM	beta52	0.16	0.178	0.018 ± 0.011	0.113 ± 0.066	0.012 ± 0.002	0.112 ± 0.007	0.013 ± 0.002	0.112 ± 0.009	0.494 ± 0.062	0.703 ± 0.055	0.186	0.026	16.17	0.1119	0.1148	1.008	97.273 ± 1.553	0.441 ± 0.012	—
LSAM	beta53	0.16	0.141	-0.019 ± 0.011	-0.121 ± 0.069	0.013 ± 0.002	0.115 ± 0.009	0.014 ± 0.002	0.116 ± 0.011	0.530 ± 0.083	0.728 ± 0.068	0.135	-0.025	-15.74	0.1009	0.1040	0.840	89.091 ± 2.972	0.380 ± 0.007	—
LSAM	beta54	0.16	0.169	0.009 ± 0.009	0.053 ± 0.055	0.009 ± 0.001	0.093 ± 0.007	0.009 ± 0.001	0.093 ± 0.008	0.335 ± 0.050	0.579 ± 0.052	0.171	0.011	6.75	0.1070	0.1076	1.076	97.273 ± 1.553	0.391 ± 0.008	—
LSAM	beta55	0.00	0.043	0.043 ± 0.017	—	0.031 ± 0.004	0.175 ± 0.013	0.032 ± 0.005	0.179 ± 0.016	—	—	0.030	0.030	—	0.1537	0.1566	1.042	95.455 ± 1.986	0.714 ± 0.032	4.545 ± 1.986
LSAM	beta56	0.00	0.007	0.007 ± 0.011	—	0.014 ± 0.002	0.119 ± 0.009	0.014 ± 0.002	0.119 ± 0.011	—	—	0.013	0.013	—	0.1122	0.1130	1.234	99.091 ± 0.905	0.577 ± 0.025	0.909 ± 0.905
LSAM	beta57	0.00	-0.031	-0.031 ± 0.009	—	0.008 ± 0.001	0.090 ± 0.008	0.009 ± 0.001	0.095 ± 0.008	—	—	-0.031	-0.031	—	0.0841	0.0898	1.063	96.364 ± 1.785	0.376 ± 0.020	3.636 ± 1.785
LSAM	beta58	0.00	-0.029	-0.029 ± 0.006	—	0.004 ± 0.001	0.063 ± 0.005	0.005 ± 0.001	0.069 ± 0.006	—	—	-0.033	-0.033	—	0.0603	0.0687	1.010	97.273 ± 1.553	0.250 ± 0.010	2.727 ± 1.553
QML	beta41	0.20	0.205	0.005 ± 0.007	0.023 ± 0.037	0.006 ± 0.001	0.077 ± 0.005	0.006 ± 0.001	0.077 ± 0.007	0.149 ± 0.021	0.386 ± 0.033	0.202	0.002	0.76	0.0787	0.0787	1.001	94.545 ± 2.165	0.304 ± 0.005	—
QML	beta42	0.20	0.215	0.015 ± 0.008	0.076 ± 0.040	0.007 ± 0.001	0.084 ± 0.006	0.007 ± 0.001	0.085 ± 0.007	0.180 ± 0.024	0.424 ± 0.035	0.215	0.015	7.60	0.0899	0.0912	0.992	93.636 ± 2.327	0.326 ± 0.006	—
QML	beta43	0.20	0.178	-0.022 ± 0.007	-0.109 ± 0.033	0.005 ± 0.001	0.070 ± 0.005	0.005 ± 0.001	0.073 ± 0.006	0.133 ± 0.017	0.365 ± 0.029	0.185	-0.015	-7.58	0.0794	0.0808	1.066	94.545 ± 2.165	0.292 ± 0.005	—
QML	beta44	0.00	0.022	0.022 ± 0.010	—	0.012 ± 0.002	0.109 ± 0.008	0.012 ± 0.002	0.110 ± 0.009	—	—	0.019	0.019	—	0.1215	0.1231	1.055	96.364 ± 1.785	0.449 ± 0.025	3.636 ± 1.785
QML	beta45	0.00	-0.028	-0.028 ± 0.008	—	0.007 ± 0.001	0.082 ± 0.006	0.008 ± 0.001	0.087 ± 0.007	—	—	-0.026	-0.026	—	0.0975	0.1010	1.040	92.727 ± 2.476	0.336 ± 0.009	7.273 ± 2.476
QML	beta46	0.00	-0.025	-0.025 ± 0.007	—	0.005 ± 0.001	0.069 ± 0.005	0.005 ± 0.001	0.073 ± 0.006	—	—	-0.030	-0.030	—	0.0753	0.0810	1.011	91.818 ± 2.613	0.273 ± 0.014	8.182 ± 2.613
QML	beta47	0.00	-0.037	-0.037 ± 0.006	—	0.004 ± 0.000	0.060 ± 0.004	0.005 ± 0.001	0.070 ± 0.005	—	—	-0.039	-0.039	—	0.0657	0.0763	1.178	93.636 ± 2.327	0.277 ± 0.016	6.364 ± 2.327
QML	beta51	0.16	0.164	0.004 ± 0.013	0.026 ± 0.080	0.018 ± 0.003	0.134 ± 0.012	0.018 ± 0.003	0.133 ± 0.013	0.694 ± 0.118	0.833 ± 0.083	0.157	-0.003	-1.98	0.1263	0.1263	0.979	97.273 ± 1.553	0.513 ± 0.014	—
QML	beta52	0.16	0.162	0.002 ± 0.009	0.011 ± 0.056	0.009 ± 0.001	0.093 ± 0.007	0.009 ± 0.001	0.093 ± 0.008	0.338 ± 0.046	0.581 ± 0.049	0.170	0.010	6.16	0.0924	0.0930	1.023	93.636 ± 2.327	0.375 ± 0.011	—
QML	beta53	0.16	0.145	-0.015 ± 0.010	-0.095 ± 0.064	0.012 ± 0.002	0.107 ± 0.009	0.012 ± 0.002	0.108 ± 0.010	0.456 ± 0.073	0.675 ± 0.064	0.137	-0.023	-14.06	0.1112	0.1135	0.857	92.727 ± 2.476	0.361 ± 0.009	—
QML	beta54	0.16	0.155	-0.005 ± 0.007	-0.030 ± 0.043	0.005 ± 0.001	0.072 ± 0.005	0.005 ± 0.001	0.072 ± 0.006	0.203 ± 0.027	0.451 ± 0.037	0.163	0.003	1.99	0.0837	0.0837	1.129	96.364 ± 1.785	0.320 ± 0.006	—
QML	beta55	0.00	0.024	0.024 ± 0.015	—	0.024 ± 0.004	0.154 ± 0.012	0.024 ± 0.004	0.155 ± 0.014	—	—	0.026	0.026	—	0.1501	0.1523	1.043	95.455 ± 1.986	0.628 ± 0.031	4.545 ± 1.986
QML	beta56	0.00	-0.007	-0.007 ± 0.009	—	0.009 ± 0.001	0.094 ± 0.008	0.009 ± 0.001	0.094 ± 0.009	—	—	-0.012	-0.012	—	0.0921	0.0929	1.292	100.000 ± 0.000	0.477 ± 0.014	0.000 ± 0.000
QML	beta57	0.00	-0.022	-0.022 ± 0.007	—	0.006 ± 0.001	0.078 ± 0.006	0.006 ± 0.001	0.080 ± 0.007	—	—	-0.031	-0.031	—	0.0745	0.0807	1.039	93.636 ± 2.327	0.317 ± 0.019	6.364 ± 2.327
QML	beta58	0.00	-0.022	-0.022 ± 0.005	—	0.003 ± 0.000	0.054 ± 0.004	0.003 ± 0.000	0.058 ± 0.004	—	—	-0.031	-0.031	—	0.0569	0.0651	1.000	91.818 ± 2.613	0.212 ± 0.008	8.182 ± 2.613
UPI	beta41	0.20	0.195	-0.005 ± 0.015	-0.027 ± 0.075	0.025 ± 0.004	0.157 ± 0.014	0.024 ± 0.004	0.156 ± 0.016	0.609 ± 0.107	0.780 ± 0.080	0.187	-0.013	-6.40	0.1730	0.1735	1.476	96.364 ± 1.785	0.907 ± 0.082	—
UPI	beta42	0.20	0.311	0.111 ± 0.022	0.553 ± 0.109	0.052 ± 0.012	0.229 ± 0.027	0.064 ± 0.015	0.253 ± 0.033	1.605 ± 0.379	1.267 ± 0.163	0.289	0.089	44.33	0.2356	0.2517	1.406	92.727 ± 2.476	1.262 ± 0.168	—
UPI	beta43	0.20	0.132	-0.068 ± 0.011	-0.339 ± 0.053	0.012 ± 0.002	0.111 ± 0.011	0.017 ± 0.003	0.129 ± 0.011	0.419 ± 0.064	0.647 ± 0.056	0.128	-0.072	-36.11	0.1299	0.1486	1.331	89.091 ± 2.972	0.578 ± 0.058	—
UPI	beta44	0.00	0.065	0.065 ± 0.023	—	0.056 ± 0.011	0.236 ± 0.025	0.059 ± 0.013	0.244 ± 0.029	—	—	0.071	0.071	—	0.1922	0.2048	1.342	100.000 ± 0.000	1.242 ± 0.115	0.000 ± 0.000
UPI	beta45	0.00	-0.045	-0.045 ± 0.014	—	0.021 ± 0.004	0.146 ± 0.015	0.023 ± 0.004	0.153 ± 0.016	—	—	-0.037	-0.037	—	0.1378	0.1427	1.587	99.091 ± 0.905	0.911 ± 0.107	0.909 ± 0.905
UPI	beta46	0.00	-0.031	-0.031 ± 0.016	—	0.028 ± 0.006	0.166 ± 0.017	0.028 ± 0.006	0.168 ± 0.020	—	—	-0.037	-0.037	—	0.1532	0.1576	1.413	99.091 ± 0.905	0.919 ± 0.096	0.909 ± 0.905

Table S235

Simulation 2 - Condition 91: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.122	-0.122 ± 0.026	—	0.074 ± 0.015	0.272 ± 0.028	0.088 ± 0.019	0.297 ± 0.035	—	—	-0.064	-0.064	—	0.1804	0.1914	1.492	98.182 ± 1.274	1.591 ± 0.250	1.818 ± 1.274
UPI	beta51	0.16	0.203	0.043 ± 0.013	0.266 ± 0.082	0.019 ± 0.003	0.138 ± 0.011	0.021 ± 0.003	0.144 ± 0.012	0.805 ± 0.118	0.897 ± 0.078	0.204	0.044	27.28	0.1495	0.1557	1.057	96.364 ± 1.785	0.571 ± 0.020	—
UPI	beta52	0.16	0.161	0.001 ± 0.012	0.006 ± 0.077	0.017 ± 0.003	0.129 ± 0.012	0.016 ± 0.003	0.128 ± 0.013	0.644 ± 0.113	0.803 ± 0.082	0.165	0.005	3.43	0.1454	0.1455	1.296	95.455 ± 1.986	0.656 ± 0.043	—
UPI	beta53	0.16	0.144	-0.016 ± 0.011	-0.100 ± 0.071	0.014 ± 0.002	0.119 ± 0.008	0.014 ± 0.002	0.119 ± 0.010	0.554 ± 0.072	0.744 ± 0.061	0.130	-0.030	-18.61	0.1296	0.1329	0.952	92.727 ± 2.476	0.443 ± 0.016	—
UPI	beta54	0.16	0.142	-0.018 ± 0.011	-0.110 ± 0.066	0.012 ± 0.002	0.111 ± 0.010	0.012 ± 0.002	0.112 ± 0.011	0.487 ± 0.087	0.698 ± 0.072	0.132	-0.028	-17.77	0.1200	0.1233	1.380	87.273 ± 3.178	0.599 ± 0.040	—
UPI	beta55	0.00	0.060	0.060 ± 0.019	—	0.038 ± 0.006	0.196 ± 0.015	0.042 ± 0.007	0.204 ± 0.019	—	—	0.033	0.033	—	0.1913	0.1940	1.453	99.091 ± 0.905	1.117 ± 0.101	0.909 ± 0.905
UPI	beta56	0.00	-0.021	-0.021 ± 0.015	—	0.026 ± 0.005	0.161 ± 0.015	0.026 ± 0.005	0.162 ± 0.017	—	—	0.003	0.003	—	0.1514	0.1514	1.402	100.000 ± 0.000	0.886 ± 0.067	0.000 ± 0.000
UPI	beta57	0.00	-0.042	-0.042 ± 0.013	—	0.019 ± 0.004	0.139 ± 0.016	0.021 ± 0.005	0.145 ± 0.018	—	—	-0.030	-0.030	—	0.1199	0.1237	1.231	100.000 ± 0.000	0.672 ± 0.065	0.000 ± 0.000
UPI	beta58	0.00	-0.049	-0.049 ± 0.012	—	0.016 ± 0.003	0.126 ± 0.010	0.018 ± 0.003	0.135 ± 0.013	—	—	-0.025	-0.025	—	0.0881	0.0915	1.413	100.000 ± 0.000	0.698 ± 0.081	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S236

Simulation 2 - Condition 92: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.215	0.015 ± 0.006	0.074 ± 0.028	0.005 ± 0.001	0.068 ± 0.005	0.005 ± 0.001	0.070 ± 0.005	0.122 ± 0.016	0.349 ± 0.027	0.218	0.018	9.02	0.0625	0.0650	0.917	92.667 ± 2.128	0.246 ± 0.003	—
LSAM	beta42	0.20	0.231	0.031 ± 0.007	0.154 ± 0.033	0.007 ± 0.001	0.081 ± 0.005	0.007 ± 0.001	0.086 ± 0.007	0.186 ± 0.025	0.432 ± 0.034	0.230	0.030	14.77	0.0799	0.0852	0.845	92.000 ± 2.215	0.268 ± 0.005	—
LSAM	beta43	0.20	0.173	-0.027 ± 0.004	-0.137 ± 0.019	0.002 ± 0.000	0.047 ± 0.003	0.003 ± 0.000	0.055 ± 0.003	0.075 ± 0.007	0.273 ± 0.016	0.175	-0.025	-12.63	0.0465	0.0529	1.088	94.000 ± 1.939	0.203 ± 0.002	—
LSAM	beta44	0.00	0.044	0.044 ± 0.005	—	0.004 ± 0.000	0.065 ± 0.004	0.006 ± 0.001	0.079 ± 0.005	—	—	0.047	0.047	—	0.0669	0.0818	1.112	94.000 ± 1.939	0.286 ± 0.006	6.000 ± 1.939
LSAM	beta45	0.00	0.008	0.008 ± 0.005	—	0.004 ± 0.000	0.060 ± 0.004	0.004 ± 0.000	0.060 ± 0.005	—	—	0.014	0.014	—	0.0611	0.0626	0.973	96.000 ± 1.600	0.227 ± 0.004	4.000 ± 1.600
LSAM	beta46	0.00	-0.040	-0.040 ± 0.004	—	0.002 ± 0.000	0.045 ± 0.003	0.004 ± 0.000	0.060 ± 0.003	—	—	-0.044	-0.044	—	0.0403	0.0600	1.008	84.000 ± 2.993	0.177 ± 0.004	16.000 ± 2.993
LSAM	beta47	0.00	-0.050	-0.050 ± 0.005	—	0.003 ± 0.001	0.058 ± 0.005	0.006 ± 0.001	0.077 ± 0.006	—	—	-0.048	-0.048	—	0.0484	0.0683	0.835	79.333 ± 3.306	0.189 ± 0.005	20.667 ± 3.306
LSAM	beta51	0.16	0.148	-0.012 ± 0.006	-0.075 ± 0.040	0.006 ± 0.001	0.078 ± 0.004	0.006 ± 0.001	0.079 ± 0.005	0.241 ± 0.024	0.491 ± 0.031	0.149	-0.011	-6.85	0.0821	0.0829	0.928	95.333 ± 1.722	0.283 ± 0.003	—
LSAM	beta52	0.16	0.180	0.020 ± 0.005	0.122 ± 0.033	0.004 ± 0.000	0.064 ± 0.004	0.004 ± 0.001	0.067 ± 0.005	0.174 ± 0.021	0.417 ± 0.030	0.176	0.016	9.87	0.0665	0.0683	0.991	94.667 ± 1.835	0.248 ± 0.003	—
LSAM	beta53	0.16	0.142	-0.018 ± 0.005	-0.109 ± 0.029	0.003 ± 0.000	0.058 ± 0.004	0.004 ± 0.000	0.060 ± 0.004	0.142 ± 0.016	0.376 ± 0.026	0.138	-0.022	-14.05	0.0557	0.0600	0.991	95.333 ± 1.722	0.225 ± 0.002	—
LSAM	beta54	0.16	0.175	0.015 ± 0.005	0.096 ± 0.030	0.004 ± 0.000	0.059 ± 0.003	0.004 ± 0.000	0.061 ± 0.004	0.146 ± 0.015	0.382 ± 0.025	0.179	0.019	11.62	0.0602	0.0630	1.020	96.667 ± 1.466	0.238 ± 0.003	—
LSAM	beta55	0.00	0.039	0.039 ± 0.008	—	0.010 ± 0.001	0.099 ± 0.007	0.011 ± 0.001	0.107 ± 0.007	—	—	0.032	0.032	—	0.0915	0.0968	0.943	94.667 ± 1.835	0.367 ± 0.009	5.333 ± 1.835
LSAM	beta56	0.00	0.006	0.006 ± 0.006	—	0.005 ± 0.001	0.072 ± 0.005	0.005 ± 0.001	0.072 ± 0.006	—	—	0.008	0.008	—	0.0648	0.0652	1.079	96.000 ± 1.600	0.305 ± 0.006	4.000 ± 1.600
LSAM	beta57	0.00	-0.038	-0.038 ± 0.005	—	0.003 ± 0.000	0.055 ± 0.004	0.004 ± 0.001	0.067 ± 0.004	—	—	-0.037	-0.037	—	0.0445	0.0582	0.873	86.667 ± 2.776	0.189 ± 0.004	13.333 ± 2.776
LSAM	beta58	0.00	-0.029	-0.029 ± 0.003	—	0.001 ± 0.000	0.037 ± 0.002	0.002 ± 0.000	0.047 ± 0.003	—	—	-0.028	-0.028	—	0.0405	0.0492	1.036	89.333 ± 2.520	0.152 ± 0.003	10.667 ± 2.520
QML	beta41	0.20	0.202	0.002 ± 0.004	0.008 ± 0.021	0.003 ± 0.000	0.052 ± 0.004	0.003 ± 0.000	0.052 ± 0.004	0.067 ± 0.009	0.259 ± 0.020	0.205	0.005	2.29	0.0535	0.0537	0.918	93.333 ± 2.037	0.187 ± 0.002	—
QML	beta42	0.20	0.208	0.008 ± 0.005	0.040 ± 0.023	0.003 ± 0.000	0.057 ± 0.003	0.003 ± 0.000	0.057 ± 0.004	0.081 ± 0.009	0.285 ± 0.020	0.211	0.011	5.32	0.0574	0.0584	0.879	92.667 ± 2.128	0.195 ± 0.002	—
QML	beta43	0.20	0.185	-0.015 ± 0.004	-0.077 ± 0.018	0.002 ± 0.000	0.045 ± 0.003	0.002 ± 0.000	0.047 ± 0.003	0.056 ± 0.006	0.237 ± 0.015	0.184	-0.016	-8.01	0.0446	0.0474	1.047	94.000 ± 1.939	0.184 ± 0.002	—
QML	beta44	0.00	0.016	0.016 ± 0.005	—	0.004 ± 0.000	0.060 ± 0.003	0.004 ± 0.000	0.062 ± 0.004	—	—	0.021	0.021	—	0.0589	0.0624	1.103	94.667 ± 1.835	0.260 ± 0.005	5.333 ± 1.835
QML	beta45	0.00	-0.010	-0.010 ± 0.004	—	0.003 ± 0.000	0.052 ± 0.003	0.003 ± 0.000	0.053 ± 0.004	—	—	-0.009	-0.009	—	0.0551	0.0559	1.022	96.667 ± 1.466	0.210 ± 0.003	3.333 ± 1.466
QML	beta46	0.00	-0.031	-0.031 ± 0.003	—	0.001 ± 0.000	0.038 ± 0.002	0.002 ± 0.000	0.049 ± 0.002	—	—	-0.037	-0.037	—	0.0409	0.0553	1.080	88.000 ± 2.653	0.163 ± 0.004	12.000 ± 2.653
QML	beta47	0.00	-0.038	-0.038 ± 0.004	—	0.002 ± 0.000	0.045 ± 0.003	0.003 ± 0.000	0.059 ± 0.003	—	—	-0.039	-0.039	—	0.0410	0.0568	0.899	79.333 ± 3.306	0.158 ± 0.004	20.667 ± 3.306
QML	beta51	0.16	0.158	-0.002 ± 0.006	-0.012 ± 0.038	0.005 ± 0.001	0.074 ± 0.004	0.005 ± 0.001	0.073 ± 0.005	0.211 ± 0.020	0.459 ± 0.029	0.163	0.003	1.94	0.0801	0.0802	0.990	96.667 ± 1.466	0.286 ± 0.003	—
QML	beta52	0.16	0.165	0.005 ± 0.004	0.033 ± 0.025	0.002 ± 0.000	0.048 ± 0.003	0.002 ± 0.000	0.048 ± 0.003	0.092 ± 0.010	0.303 ± 0.021	0.161	0.001	0.52	0.0556	0.0556	1.115	96.667 ± 1.466	0.211 ± 0.002	—
QML	beta53	0.16	0.149	-0.011 ± 0.004	-0.068 ± 0.027	0.003 ± 0.000	0.052 ± 0.003	0.003 ± 0.000	0.053 ± 0.004	0.112 ± 0.013	0.334 ± 0.024	0.146	-0.014	-8.93	0.0536	0.0555	1.026	96.667 ± 1.466	0.211 ± 0.002	—
QML	beta54	0.16	0.162	0.002 ± 0.004	0.015 ± 0.025	0.002 ± 0.000	0.048 ± 0.003	0.002 ± 0.000	0.048 ± 0.003	0.091 ± 0.010	0.302 ± 0.021	0.165	0.005	3.27	0.0508	0.0511	1.046	96.000 ± 1.600	0.199 ± 0.002	—
QML	beta55	0.00	0.031	0.031 ± 0.007	—	0.008 ± 0.001	0.087 ± 0.005	0.008 ± 0.001	0.092 ± 0.006	—	—	0.032	0.032	—	0.0910	0.0965	1.005	95.333 ± 1.722	0.342 ± 0.006	4.667 ± 1.722
QML	beta56	0.00	-0.013	-0.013 ± 0.005	—	0.004 ± 0.001	0.064 ± 0.004	0.004 ± 0.000	0.065 ± 0.005	—	—	-0.013	-0.013	—	0.0675	0.0687	1.085	97.333 ± 1.315	0.272 ± 0.005	2.667 ± 1.315
QML	beta57	0.00	-0.033	-0.033 ± 0.004	—	0.002 ± 0.000	0.046 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	—	—	-0.033	-0.033	—	0.0471	0.0577	0.962	88.000 ± 2.653	0.172 ± 0.004	12.000 ± 2.653
QML	beta58	0.00	-0.024	-0.024 ± 0.003	—	0.001 ± 0.000	0.032 ± 0.002	0.002 ± 0.000	0.040 ± 0.002	—	—	-0.024	-0.024	—	0.0335	0.0413	1.077	88.000 ± 2.653	0.137 ± 0.003	12.000 ± 2.653
UPI	beta41	0.20	0.250	0.050 ± 0.014	0.249 ± 0.072	0.031 ± 0.005	0.177 ± 0.016	0.033 ± 0.006	0.183 ± 0.018	0.836 ± 0.148	0.914 ± 0.090	0.252	0.052	25.84	0.1801	0.1874	1.113	90.667 ± 2.375	0.771 ± 0.060	—
UPI	beta42	0.20	0.291	0.091 ± 0.020	0.457 ± 0.099	0.059 ± 0.010	0.242 ± 0.022	0.067 ± 0.013	0.258 ± 0.027	1.666 ± 0.321	1.291 ± 0.136	0.280	0.080	39.76	0.2515	0.2638	0.972	89.333 ± 2.520	0.923 ± 0.086	—
UPI	beta43	0.20	0.148	-0.052 ± 0.008	-0.259 ± 0.041	0.010 ± 0.001	0.100 ± 0.005	0.013 ± 0.001	0.113 ± 0.007	0.317 ± 0.035	0.563 ± 0.036	0.128	-0.072	-36.22	0.1249	0.1444	0.951	81.333 ± 3.181	0.374 ± 0.030	—
UPI	beta44	0.00	0.064	0.064 ± 0.013	—	0.026 ± 0.005	0.160 ± 0.016	0.030 ± 0.006	0.172 ± 0.019	—	—	0.048	0.048	—	0.1417	0.1498	1.085	98.000 ± 1.143	0.681 ± 0.056	2.000 ± 1.143
UPI	beta45	0.00	-0.008	-0.008 ± 0.006	—	0.006 ± 0.001	0.078 ± 0.006	0.006 ± 0.001	0.078 ± 0.007	—	—	0.000	0.000	—	0.0820	0.0820	1.225	98.000 ± 1.143	0.375 ± 0.019	2.000 ± 1.143
UPI	beta46	0.00	-0.117	-0.117 ± 0.016	—	0.036 ± 0.008	0.191 ± 0.022	0.050 ± 0.010	0.224 ± 0.023	—	—	-0.098	-0.098	—	0.1551	0.1835	1.354	99.333 ± 0.664	1.014 ± 0.152	0.667 ± 0.664

Table S236

Simulation 2 - Condition 92: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.156	-0.156 ± 0.018	—	0.050 ± 0.008	0.223 ± 0.019	0.074 ± 0.012	0.272 ± 0.023	—	—	-0.124	-0.124	—	0.1807	0.2190	1.187	96.667 ± 1.466	1.038 ± 0.087	3.333 ± 1.466
UPI	beta51	0.16	0.211	0.051 ± 0.008	0.321 ± 0.048	0.009 ± 0.001	0.094 ± 0.005	0.011 ± 0.001	0.107 ± 0.007	0.446 ± 0.047	0.668 ± 0.041	0.206	0.046	28.53	0.1010	0.1109	0.986	87.333 ± 2.716	0.364 ± 0.014	—
UPI	beta52	0.16	0.205	0.045 ± 0.011	0.280 ± 0.071	0.019 ± 0.002	0.139 ± 0.009	0.021 ± 0.003	0.146 ± 0.012	0.830 ± 0.118	0.911 ± 0.074	0.184	0.024	15.07	0.1489	0.1509	1.045	90.000 ± 2.449	0.570 ± 0.038	—
UPI	beta53	0.16	0.132	-0.028 ± 0.007	-0.174 ± 0.043	0.007 ± 0.001	0.083 ± 0.006	0.008 ± 0.001	0.088 ± 0.007	0.301 ± 0.039	0.548 ± 0.041	0.119	-0.041	-25.49	0.0910	0.0997	1.010	90.000 ± 2.449	0.330 ± 0.022	—
UPI	beta54	0.16	0.135	-0.025 ± 0.008	-0.153 ± 0.051	0.010 ± 0.001	0.100 ± 0.007	0.010 ± 0.001	0.102 ± 0.008	0.410 ± 0.057	0.640 ± 0.052	0.133	-0.027	-16.96	0.0988	0.1025	1.237	87.333 ± 2.716	0.484 ± 0.040	—
UPI	beta55	0.00	0.057	0.057 ± 0.012	—	0.020 ± 0.003	0.141 ± 0.012	0.023 ± 0.004	0.152 ± 0.015	—	—	0.033	0.033	—	0.1203	0.1247	1.102	98.000 ± 1.143	0.609 ± 0.043	2.000 ± 1.143
UPI	beta56	0.00	-0.018	-0.018 ± 0.007	—	0.008 ± 0.001	0.089 ± 0.006	0.008 ± 0.001	0.091 ± 0.007	—	—	-0.019	-0.019	—	0.0955	0.0973	1.396	99.333 ± 0.664	0.489 ± 0.031	0.667 ± 0.664
UPI	beta57	0.00	-0.108	-0.108 ± 0.013	—	0.024 ± 0.003	0.154 ± 0.011	0.035 ± 0.005	0.188 ± 0.015	—	—	-0.082	-0.082	—	0.1348	0.1576	1.298	100.000 ± 0.000	0.784 ± 0.117	0.000 ± 0.000
UPI	beta58	0.00	-0.066	-0.066 ± 0.011	—	0.018 ± 0.003	0.136 ± 0.012	0.023 ± 0.004	0.151 ± 0.015	—	—	-0.044	-0.044	—	0.1063	0.1151	1.218	99.333 ± 0.664	0.648 ± 0.073	0.667 ± 0.664

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S237

Simulation 2 - Condition 93: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.206	0.006 ± 0.002	0.029 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.095 ± 0.005	0.308 ± 0.009	0.201	0.001	0.70	0.0563	0.0563	0.987	94.540 ± 0.722	0.237 ± 0.001	—
LSAM	beta42	0.20	0.205	0.005 ± 0.002	0.023 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.005	0.311 ± 0.009	0.204	0.004	1.98	0.0621	0.0622	1.014	95.450 ± 0.663	0.247 ± 0.001	—
LSAM	beta43	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.196	-0.004	-1.79	0.0572	0.0573	1.024	94.944 ± 0.697	0.224 ± 0.001	—
LSAM	beta44	0.00	0.015	0.015 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	0.013	0.013	—	0.0721	0.0733	0.956	93.731 ± 0.771	0.282 ± 0.002	6.269 ± 0.771
LSAM	beta45	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.003	0.003	—	0.0660	0.0661	0.946	94.439 ± 0.729	0.240 ± 0.002	5.561 ± 0.729
LSAM	beta46	0.00	-0.014	-0.014 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.016	-0.016	—	0.0472	0.0499	0.995	94.135 ± 0.747	0.195 ± 0.002	5.865 ± 0.747
LSAM	beta47	0.00	-0.016	-0.016 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.015	-0.015	—	0.0468	0.0492	0.988	93.529 ± 0.782	0.192 ± 0.002	6.471 ± 0.782
LSAM	beta51	0.16	0.163	0.003 ± 0.003	0.020 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.271 ± 0.013	0.521 ± 0.015	0.159	-0.001	-0.62	0.0811	0.0811	0.959	93.124 ± 0.805	0.313 ± 0.001	—
LSAM	beta52	0.16	0.165	0.005 ± 0.002	0.034 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.402 ± 0.011	0.166	0.006	3.79	0.0656	0.0659	1.001	95.551 ± 0.656	0.251 ± 0.001	—
LSAM	beta53	0.16	0.153	-0.007 ± 0.002	-0.043 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.179 ± 0.008	0.423 ± 0.012	0.154	-0.006	-3.91	0.0680	0.0683	0.969	94.338 ± 0.735	0.256 ± 0.001	—
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.401 ± 0.011	0.159	-0.001	-0.69	0.0634	0.0634	0.981	95.147 ± 0.683	0.247 ± 0.001	—
LSAM	beta55	0.00	0.013	0.013 ± 0.003	—	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	—	—	0.017	0.017	—	0.0948	0.0964	0.954	94.135 ± 0.747	0.360 ± 0.003	5.865 ± 0.747
LSAM	beta56	0.00	0.002	0.002 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.003	-0.003	—	0.0822	0.0822	0.954	93.630 ± 0.777	0.312 ± 0.002	6.370 ± 0.777
LSAM	beta57	0.00	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.014	-0.014	—	0.0517	0.0535	0.988	93.832 ± 0.765	0.210 ± 0.002	6.168 ± 0.765
LSAM	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.009	-0.009	—	0.0464	0.0473	0.962	94.641 ± 0.716	0.175 ± 0.001	5.359 ± 0.716
QML	beta41	0.20	0.202	0.002 ± 0.002	0.009 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.008	0.198	-0.002	-0.90	0.0546	0.0547	0.993	94.742 ± 0.710	0.222 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.291 ± 0.008	0.200	0.000	0.07	0.0570	0.0570	1.008	95.248 ± 0.677	0.230 ± 0.001	—
QML	beta43	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.074 ± 0.003	0.273 ± 0.008	0.197	-0.003	-1.42	0.0547	0.0548	1.025	95.248 ± 0.677	0.219 ± 0.001	—
QML	beta44	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.002	0.002	—	0.0696	0.0696	0.992	94.439 ± 0.729	0.272 ± 0.002	5.561 ± 0.729
QML	beta45	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.004	-0.004	—	0.0636	0.0637	0.965	94.641 ± 0.716	0.231 ± 0.001	5.359 ± 0.716
QML	beta46	0.00	-0.011	-0.011 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.013	-0.013	—	0.0442	0.0460	1.006	93.832 ± 0.765	0.183 ± 0.002	6.168 ± 0.765
QML	beta47	0.00	-0.015	-0.015 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.015	-0.015	—	0.0445	0.0468	1.013	93.225 ± 0.799	0.178 ± 0.001	6.775 ± 0.799
QML	beta51	0.16	0.164	0.004 ± 0.003	0.028 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.266 ± 0.013	0.515 ± 0.015	0.160	0.000	-0.05	0.0805	0.0805	0.995	94.540 ± 0.722	0.321 ± 0.001	—
QML	beta52	0.16	0.163	0.003 ± 0.002	0.019 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.006	0.389 ± 0.010	0.164	0.004	2.65	0.0642	0.0644	1.018	96.158 ± 0.611	0.248 ± 0.001	—
QML	beta53	0.16	0.153	-0.007 ± 0.002	-0.042 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.007	0.411 ± 0.011	0.153	-0.007	-4.60	0.0659	0.0663	0.979	94.540 ± 0.722	0.251 ± 0.001	—
QML	beta54	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.007	0.386 ± 0.011	0.156	-0.004	-2.32	0.0602	0.0603	0.986	94.843 ± 0.703	0.239 ± 0.001	—
QML	beta55	0.00	0.010	0.010 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	—	—	0.012	0.012	—	0.0930	0.0937	0.979	93.832 ± 0.765	0.352 ± 0.002	6.168 ± 0.765
QML	beta56	0.00	-0.006	-0.006 ± 0.002	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.007	-0.007	—	0.0763	0.0766	0.973	94.641 ± 0.716	0.298 ± 0.002	5.359 ± 0.716
QML	beta57	0.00	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.015	-0.015	—	0.0479	0.0502	1.012	93.327 ± 0.794	0.201 ± 0.002	6.673 ± 0.794
QML	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.010	-0.010	—	0.0435	0.0447	0.973	93.529 ± 0.782	0.165 ± 0.001	6.471 ± 0.782
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.028 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.005	0.310 ± 0.009	0.202	0.002	0.76	0.0589	0.0589	1.013	95.046 ± 0.690	0.245 ± 0.001	—
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.026 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.005	0.313 ± 0.009	0.204	0.004	2.18	0.0633	0.0635	1.045	95.652 ± 0.648	0.256 ± 0.001	—
UPI	beta43	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.080 ± 0.004	0.282 ± 0.008	0.198	-0.002	-1.02	0.0563	0.0563	1.042	95.450 ± 0.663	0.230 ± 0.001	—
UPI	beta44	0.00	0.011	0.011 ± 0.002	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.008	0.008	—	0.0739	0.0743	1.000	95.450 ± 0.663	0.305 ± 0.003	4.550 ± 0.663
UPI	beta45	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.002	0.002	—	0.0629	0.0629	1.017	96.158 ± 0.611	0.258 ± 0.002	3.842 ± 0.611
UPI	beta46	0.00	-0.008	-0.008 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.009	-0.009	—	0.0553	0.0561	1.018	94.944 ± 0.697	0.230 ± 0.003	5.056 ± 0.697

Table S237

Simulation 2 - Condition 93: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.011	-0.011 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.008	-0.008	—	0.0525	0.0532	1.038	96.461 ± 0.588	0.230 ± 0.004	3.539 ± 0.588
UPI	beta51	0.16	0.163	0.003 ± 0.003	0.017 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.260 ± 0.012	0.510 ± 0.015	0.159	-0.001	-0.66	0.0791	0.0791	0.994	94.338 ± 0.735	0.318 ± 0.001	—
UPI	beta52	0.16	0.165	0.005 ± 0.002	0.033 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.007	0.399 ± 0.011	0.165	0.005	3.01	0.0634	0.0636	1.028	95.652 ± 0.648	0.256 ± 0.001	—
UPI	beta53	0.16	0.155	-0.005 ± 0.002	-0.034 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.177 ± 0.008	0.421 ± 0.012	0.155	-0.005	-3.32	0.0673	0.0675	0.988	94.843 ± 0.703	0.260 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.007	0.405 ± 0.011	0.162	0.002	0.95	0.0636	0.0636	1.004	95.248 ± 0.677	0.255 ± 0.001	—
UPI	beta55	0.00	0.012	0.012 ± 0.003	—	0.009 ± 0.000	0.097 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	—	—	0.013	0.013	—	0.0893	0.0903	0.999	95.248 ± 0.677	0.379 ± 0.004	4.752 ± 0.677
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.002	-0.002	—	0.0794	0.0794	1.002	95.652 ± 0.648	0.327 ± 0.003	4.348 ± 0.648
UPI	beta57	0.00	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.007	-0.007	—	0.0564	0.0569	1.027	96.461 ± 0.588	0.233 ± 0.003	3.539 ± 0.588
UPI	beta58	0.00	-0.006	-0.006 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.005	-0.005	—	0.0481	0.0484	1.043	95.956 ± 0.626	0.210 ± 0.003	4.044 ± 0.626

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S238

Simulation 2 - Condition 94: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.203	0.003	1.41	0.0370	0.0372	0.989	94.494 ± 0.722	0.146 ± 0.000	—
LSAM	beta42	0.20	0.205	0.005 ± 0.001	0.026 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.204	0.004	1.88	0.0392	0.0393	0.963	94.094 ± 0.746	0.152 ± 0.000	—
LSAM	beta43	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.001	0.185 ± 0.005	0.195	-0.005	-2.62	0.0375	0.0379	0.964	93.694 ± 0.769	0.139 ± 0.000	—
LSAM	beta44	0.00	0.015	0.015 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.013	0.013	—	0.0436	0.0456	0.952	93.493 ± 0.780	0.167 ± 0.001	6.507 ± 0.780
LSAM	beta45	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.006	0.006	—	0.0382	0.0386	0.982	94.595 ± 0.715	0.144 ± 0.001	5.405 ± 0.715
LSAM	beta46	0.00	-0.013	-0.013 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.014	-0.014	—	0.0292	0.0322	0.960	91.992 ± 0.859	0.115 ± 0.001	8.008 ± 0.859
LSAM	beta47	0.00	-0.016	-0.016 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.016	-0.016	—	0.0301	0.0342	0.960	90.190 ± 0.941	0.114 ± 0.001	9.810 ± 0.941
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.161	0.001	0.91	0.0489	0.0489	1.018	96.096 ± 0.613	0.198 ± 0.000	—
LSAM	beta52	0.16	0.164	0.004 ± 0.001	0.024 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.252 ± 0.007	0.164	0.004	2.31	0.0385	0.0386	0.989	94.194 ± 0.740	0.155 ± 0.000	—
LSAM	beta53	0.16	0.156	-0.004 ± 0.001	-0.023 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.158	-0.002	-1.48	0.0398	0.0399	1.015	94.895 ± 0.696	0.159 ± 0.000	—
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.247 ± 0.006	0.160	0.000	-0.19	0.0403	0.0403	0.990	95.395 ± 0.663	0.154 ± 0.000	—
LSAM	beta55	0.00	0.013	0.013 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.014	0.014	—	0.0565	0.0581	0.980	94.695 ± 0.709	0.219 ± 0.001	5.305 ± 0.709
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	—	0.0459	0.0459	0.991	94.494 ± 0.722	0.190 ± 0.001	5.506 ± 0.722
LSAM	beta57	0.00	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.014	-0.014	—	0.0311	0.0342	0.974	91.592 ± 0.878	0.125 ± 0.001	8.408 ± 0.878
LSAM	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.010	-0.010	—	0.0284	0.0300	0.916	92.893 ± 0.813	0.107 ± 0.001	7.107 ± 0.813
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.200	0.000	-0.13	0.0351	0.0351	0.991	94.795 ± 0.703	0.139 ± 0.001	—
QML	beta42	0.20	0.202	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.201	0.001	0.50	0.0370	0.0371	0.964	93.193 ± 0.797	0.143 ± 0.000	—
QML	beta43	0.20	0.195	-0.005 ± 0.001	-0.023 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.180 ± 0.005	0.195	-0.005	-2.34	0.0366	0.0369	0.969	94.194 ± 0.740	0.136 ± 0.000	—
QML	beta44	0.00	0.005	0.005 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.004	0.004	—	0.0436	0.0438	0.957	95.195 ± 0.677	0.165 ± 0.001	4.805 ± 0.677
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0372	0.0372	0.982	94.294 ± 0.734	0.142 ± 0.001	5.706 ± 0.734
QML	beta46	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.011	-0.011	—	0.0286	0.0308	0.970	92.693 ± 0.823	0.111 ± 0.001	7.307 ± 0.823
QML	beta47	0.00	-0.016	-0.016 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.016	-0.016	—	0.0286	0.0329	0.968	89.690 ± 0.962	0.108 ± 0.001	10.310 ± 0.962
QML	beta51	0.16	0.163	0.003 ± 0.002	0.017 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.309 ± 0.009	0.163	0.003	1.60	0.0489	0.0490	1.033	96.597 ± 0.574	0.200 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.001	0.015 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.162	0.002	1.27	0.0384	0.0384	0.990	94.494 ± 0.722	0.152 ± 0.000	—
QML	beta53	0.16	0.157	-0.003 ± 0.001	-0.021 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.157	-0.003	-1.67	0.0382	0.0383	1.020	95.495 ± 0.656	0.156 ± 0.000	—
QML	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.160	0.000	-0.28	0.0384	0.0384	0.999	95.395 ± 0.663	0.150 ± 0.000	—
QML	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.011	0.011	—	0.0562	0.0572	0.993	94.494 ± 0.722	0.217 ± 0.001	5.506 ± 0.722
QML	beta56	0.00	-0.005	-0.005 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.005	-0.005	—	0.0455	0.0458	1.000	94.895 ± 0.696	0.186 ± 0.001	5.105 ± 0.696
QML	beta57	0.00	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.014	-0.014	—	0.0296	0.0329	0.987	91.792 ± 0.868	0.122 ± 0.001	8.208 ± 0.868
QML	beta58	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.011	-0.011	—	0.0273	0.0296	0.934	91.692 ± 0.873	0.104 ± 0.001	8.308 ± 0.873
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.202	0.002	1.19	0.0372	0.0372	0.997	95.395 ± 0.663	0.147 ± 0.000	—
UPI	beta42	0.20	0.205	0.005 ± 0.001	0.027 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.205	0.005	2.39	0.0390	0.0393	0.975	94.494 ± 0.722	0.153 ± 0.000	—
UPI	beta43	0.20	0.196	-0.004 ± 0.001	-0.022 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.001	0.186 ± 0.005	0.195	-0.005	-2.42	0.0376	0.0379	0.965	93.594 ± 0.775	0.140 ± 0.000	—
UPI	beta44	0.00	0.011	0.011 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.010	0.010	—	0.0448	0.0460	0.958	94.394 ± 0.728	0.174 ± 0.001	5.606 ± 0.728
UPI	beta45	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.005	0.005	—	0.0366	0.0370	0.999	95.195 ± 0.677	0.148 ± 0.001	4.805 ± 0.677
UPI	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.009	-0.009	—	0.0316	0.0328	0.987	95.095 ± 0.683	0.128 ± 0.001	4.905 ± 0.683

Table S238

Simulation 2 - Condition 94: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.012	-0.012 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.012	-0.012	—	0.0325	0.0347	0.973	93.594 ± 0.775	0.126 ± 0.001	6.406 ± 0.775
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.016 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.308 ± 0.008	0.162	0.002	1.54	0.0478	0.0478	1.032	96.196 ± 0.605	0.199 ± 0.000	—
UPI	beta52	0.16	0.163	0.003 ± 0.001	0.018 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.162	0.002	1.19	0.0390	0.0391	1.002	94.695 ± 0.709	0.156 ± 0.000	—
UPI	beta53	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.158	-0.002	-1.39	0.0402	0.0403	1.018	94.494 ± 0.722	0.160 ± 0.000	—
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.246 ± 0.006	0.160	0.000	-0.25	0.0392	0.0392	1.008	95.596 ± 0.649	0.156 ± 0.000	—
UPI	beta55	0.00	0.010	0.010 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	0.010	0.010	—	0.0545	0.0555	0.975	94.695 ± 0.709	0.223 ± 0.001	5.305 ± 0.709
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.002	0.002	—	0.0450	0.0450	0.998	95.596 ± 0.649	0.192 ± 0.001	4.404 ± 0.649
UPI	beta57	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.009	-0.009	—	0.0331	0.0342	0.994	93.694 ± 0.769	0.135 ± 0.001	6.306 ± 0.769
UPI	beta58	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.008	-0.008	—	0.0308	0.0317	0.933	93.794 ± 0.763	0.119 ± 0.001	6.206 ± 0.763

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S239

Simulation 2 - Condition 95: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.201	0.001	0.43	0.0510	0.0510	0.944	94.000 ± 0.751	0.182 ± 0.000	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.020 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.204	0.004	1.92	0.0509	0.0510	0.975	94.500 ± 0.721	0.187 ± 0.001	—
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.201	0.001	0.34	0.0439	0.0440	0.986	94.200 ± 0.739	0.179 ± 0.000	—
LSAM	beta44	0.00	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.003	0.003	—	0.0492	0.0492	0.980	94.100 ± 0.745	0.200 ± 0.001	5.900 ± 0.745
LSAM	beta45	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.000	0.000	—	0.0477	0.0477	0.951	94.800 ± 0.702	0.178 ± 0.001	5.200 ± 0.702
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0353	0.0353	0.936	92.700 ± 0.823	0.140 ± 0.001	7.300 ± 0.823
LSAM	beta47	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.006	-0.006	—	0.0376	0.0381	0.954	93.500 ± 0.780	0.139 ± 0.001	6.500 ± 0.780
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.008 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.007	0.404 ± 0.011	0.161	0.001	0.89	0.0620	0.0620	0.987	94.400 ± 0.727	0.250 ± 0.001	—
LSAM	beta52	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.009	0.160	0.000	-0.19	0.0480	0.0480	1.014	94.700 ± 0.708	0.199 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.004	0.319 ± 0.009	0.157	-0.003	-1.62	0.0518	0.0518	1.027	96.000 ± 0.620	0.205 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.018 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.008	0.162	0.002	0.97	0.0517	0.0517	1.014	95.800 ± 0.634	0.199 ± 0.001	—
LSAM	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.005	-0.005	—	0.0698	0.0700	0.960	94.000 ± 0.751	0.258 ± 0.001	6.000 ± 0.751
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.002	0.002	—	0.0556	0.0556	0.960	93.700 ± 0.768	0.229 ± 0.001	6.300 ± 0.768
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.001	-0.001	—	0.0403	0.0403	0.932	92.100 ± 0.853	0.155 ± 0.001	7.900 ± 0.853
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.003	-0.003	—	0.0343	0.0344	0.998	94.800 ± 0.702	0.136 ± 0.001	5.200 ± 0.702
QML	beta41	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.241 ± 0.006	0.200	0.000	-0.13	0.0504	0.0504	0.946	94.200 ± 0.739	0.179 ± 0.000	—
QML	beta42	0.20	0.203	0.003 ± 0.002	0.016 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.241 ± 0.006	0.203	0.003	1.62	0.0505	0.0506	0.976	94.700 ± 0.708	0.184 ± 0.001	—
QML	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.200	0.000	0.00	0.0435	0.0435	0.989	94.000 ± 0.751	0.177 ± 0.000	—
QML	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.000	0.000	—	0.0483	0.0483	0.985	94.500 ± 0.721	0.199 ± 0.001	5.500 ± 0.721
QML	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.003	-0.003	—	0.0454	0.0455	0.958	94.600 ± 0.715	0.177 ± 0.001	5.400 ± 0.715
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0342	0.0342	0.942	93.200 ± 0.796	0.137 ± 0.001	6.800 ± 0.796
QML	beta47	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.006	-0.006	—	0.0369	0.0373	0.959	93.400 ± 0.785	0.135 ± 0.001	6.600 ± 0.785
QML	beta51	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.402 ± 0.011	0.162	0.002	1.29	0.0628	0.0628	1.003	94.900 ± 0.696	0.253 ± 0.001	—
QML	beta52	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.009	0.159	-0.001	-0.82	0.0483	0.0483	1.017	94.900 ± 0.696	0.199 ± 0.001	—
QML	beta53	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.004	0.316 ± 0.009	0.157	-0.003	-1.93	0.0510	0.0511	1.033	95.700 ± 0.641	0.205 ± 0.001	—
QML	beta54	0.16	0.162	0.002 ± 0.002	0.015 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.309 ± 0.008	0.161	0.001	0.93	0.0506	0.0506	1.020	95.900 ± 0.627	0.197 ± 0.001	—
QML	beta55	0.00	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.006	-0.006	—	0.0691	0.0694	0.974	94.600 ± 0.715	0.258 ± 0.001	5.400 ± 0.715
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	—	0.0553	0.0553	0.971	94.000 ± 0.751	0.228 ± 0.001	6.000 ± 0.751
QML	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0390	0.0391	0.948	92.200 ± 0.848	0.153 ± 0.001	7.800 ± 0.848
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.003	-0.003	—	0.0327	0.0329	1.009	94.900 ± 0.696	0.133 ± 0.001	5.100 ± 0.696
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.246 ± 0.006	0.201	0.001	0.53	0.0509	0.0509	0.942	93.700 ± 0.768	0.181 ± 0.000	—
UPI	beta42	0.20	0.204	0.004 ± 0.002	0.020 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.244 ± 0.006	0.203	0.003	1.68	0.0516	0.0517	0.979	94.500 ± 0.721	0.187 ± 0.001	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.201	0.001	0.34	0.0439	0.0439	0.988	94.000 ± 0.751	0.179 ± 0.000	—
UPI	beta44	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.002	0.002	—	0.0499	0.0499	0.976	94.200 ± 0.739	0.202 ± 0.001	5.800 ± 0.739
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.001	-0.001	—	0.0464	0.0464	0.942	93.600 ± 0.774	0.181 ± 0.001	6.400 ± 0.774
UPI	beta46	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0355	0.0355	0.939	92.400 ± 0.838	0.143 ± 0.001	7.600 ± 0.838

Table S239

Simulation 2 - Condition 95: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	—	0.0387	0.0389	0.957	93.900 ± 0.757	0.143 ± 0.001	6.100 ± 0.757
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.160	0.000	0.29	0.0611	0.0611	1.001	94.800 ± 0.702	0.251 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.159	-0.001	-0.37	0.0471	0.0471	1.016	95.200 ± 0.676	0.198 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.004	0.318 ± 0.009	0.158	-0.002	-1.46	0.0517	0.0517	1.030	96.200 ± 0.605	0.206 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.008	0.161	0.001	0.68	0.0516	0.0516	1.016	95.500 ± 0.656	0.199 ± 0.001	—
UPI	beta55	0.00	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.006	-0.006	—	0.0688	0.0691	0.959	93.900 ± 0.757	0.260 ± 0.001	6.100 ± 0.757
UPI	beta56	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.003	0.003	—	0.0566	0.0566	0.963	94.000 ± 0.751	0.230 ± 0.001	6.000 ± 0.751
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.001	0.001	—	0.0416	0.0417	0.936	92.500 ± 0.833	0.157 ± 0.001	7.500 ± 0.833
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.000	0.000	—	0.0344	0.0344	0.998	95.200 ± 0.676	0.138 ± 0.001	4.800 ± 0.676

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S240

Simulation 2 - Condition 96: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.201	0.001	0.61	0.0298	0.0298	0.970	94.300 ± 0.733	0.114 ± 0.000	—
LSAM	beta42	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.200	0.000	0.22	0.0280	0.0280	1.035	95.800 ± 0.634	0.118 ± 0.000	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.200	0.000	0.13	0.0303	0.0303	0.940	93.400 ± 0.785	0.112 ± 0.000	—
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.003	0.003	—	0.0309	0.0310	0.993	94.500 ± 0.721	0.125 ± 0.000	5.500 ± 0.721
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0281	0.0282	1.026	95.900 ± 0.627	0.112 ± 0.000	4.100 ± 0.627
LSAM	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.003	-0.003	—	0.0225	0.0228	0.976	95.200 ± 0.676	0.087 ± 0.000	4.800 ± 0.676
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0221	0.0223	0.988	94.200 ± 0.739	0.087 ± 0.000	5.800 ± 0.739
LSAM	beta51	0.16	0.162	0.002 ± 0.001	0.012 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.161	0.001	0.69	0.0404	0.0405	0.996	93.500 ± 0.780	0.158 ± 0.000	—
LSAM	beta52	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.157	-0.003	-1.91	0.0320	0.0321	0.975	94.100 ± 0.745	0.125 ± 0.000	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.159	-0.001	-0.62	0.0324	0.0325	1.000	94.700 ± 0.708	0.129 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.158	-0.002	-1.03	0.0311	0.0311	0.978	94.000 ± 0.751	0.125 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.001	0.001	—	0.0403	0.0403	0.977	93.400 ± 0.785	0.163 ± 0.000	6.600 ± 0.785
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0353	0.0353	1.000	95.200 ± 0.676	0.143 ± 0.000	4.800 ± 0.676
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.002	-0.002	—	0.0254	0.0255	0.956	94.300 ± 0.733	0.097 ± 0.000	5.700 ± 0.733
LSAM	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.002	-0.002	—	0.0234	0.0235	0.955	93.400 ± 0.785	0.085 ± 0.000	6.600 ± 0.785
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.200	0.000	-0.16	0.0289	0.0289	0.972	94.500 ± 0.721	0.113 ± 0.000	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.200	0.000	-0.21	0.0281	0.0281	1.037	95.900 ± 0.627	0.116 ± 0.000	—
QML	beta43	0.20	0.201	0.001 ± 0.001	0.003 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.200	0.000	-0.14	0.0308	0.0308	0.941	93.500 ± 0.780	0.112 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0309	0.0309	1.000	95.500 ± 0.656	0.125 ± 0.001	4.500 ± 0.656
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.002	-0.002	—	0.0281	0.0281	1.031	95.800 ± 0.634	0.112 ± 0.000	4.200 ± 0.634
QML	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.003	-0.003	—	0.0223	0.0225	0.987	95.100 ± 0.683	0.087 ± 0.001	4.900 ± 0.683
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.004	-0.004	—	0.0215	0.0218	0.996	94.200 ± 0.739	0.085 ± 0.001	5.800 ± 0.739
QML	beta51	0.16	0.162	0.002 ± 0.001	0.013 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.161	0.001	0.82	0.0403	0.0403	1.001	94.100 ± 0.745	0.159 ± 0.000	—
QML	beta52	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.157	-0.003	-1.95	0.0321	0.0323	0.976	94.200 ± 0.739	0.125 ± 0.000	—
QML	beta53	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.159	-0.001	-0.77	0.0328	0.0328	1.001	94.700 ± 0.708	0.128 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.158	-0.002	-1.20	0.0311	0.0312	0.981	94.000 ± 0.751	0.124 ± 0.000	—
QML	beta55	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	—	0.0406	0.0406	0.982	93.200 ± 0.796	0.163 ± 0.001	6.800 ± 0.796
QML	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0347	0.0348	1.004	95.000 ± 0.689	0.142 ± 0.000	5.000 ± 0.689
QML	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.003	-0.003	—	0.0251	0.0252	0.964	94.200 ± 0.739	0.096 ± 0.000	5.800 ± 0.739
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0227	0.0229	0.964	93.600 ± 0.774	0.084 ± 0.000	6.400 ± 0.774
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.200	0.000	0.18	0.0297	0.0297	0.969	94.800 ± 0.702	0.114 ± 0.000	—
UPI	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.200	0.000	0.14	0.0285	0.0285	1.038	96.100 ± 0.612	0.117 ± 0.000	—
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.201	0.001	0.35	0.0307	0.0307	0.940	93.200 ± 0.796	0.112 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.001	0.001	—	0.0315	0.0315	0.994	95.600 ± 0.649	0.125 ± 0.000	4.400 ± 0.649
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0282	0.0282	1.033	96.000 ± 0.620	0.112 ± 0.000	4.000 ± 0.620
UPI	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.002	-0.002	—	0.0236	0.0236	0.978	94.900 ± 0.696	0.088 ± 0.000	5.100 ± 0.696

Table S240

Simulation 2 - Condition 96: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.001	-0.001	—	0.0227	0.0227	0.982	93.800 ± 0.763	0.088 ± 0.000	6.200 ± 0.763
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.012 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.161	0.001	0.74	0.0401	0.0401	1.000	94.000 ± 0.751	0.158 ± 0.000	—
UPI	beta52	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.157	-0.003	-1.74	0.0322	0.0324	0.975	93.800 ± 0.763	0.125 ± 0.000	—
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.159	-0.001	-0.71	0.0324	0.0325	0.999	94.900 ± 0.696	0.129 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.159	-0.001	-0.80	0.0311	0.0312	0.979	94.000 ± 0.751	0.125 ± 0.000	—
UPI	beta55	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.001	0.001	—	0.0408	0.0409	0.975	93.400 ± 0.785	0.163 ± 0.001	6.600 ± 0.785
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	—	0.0349	0.0350	1.005	94.800 ± 0.702	0.144 ± 0.000	5.200 ± 0.702
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.000	0.000	—	0.0246	0.0246	0.953	94.400 ± 0.727	0.097 ± 0.000	5.600 ± 0.727
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.001	-0.001	—	0.0229	0.0230	0.968	94.400 ± 0.727	0.086 ± 0.000	5.600 ± 0.727

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S241

Simulation 2 - Condition 97: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.251	0.051 ± 0.014	0.256 ± 0.071	0.095 ± 0.008	0.308 ± 0.014	0.098 ± 0.008	0.312 ± 0.015	2.440 ± 0.210	1.562 ± 0.076	0.229	0.029	14.74	0.2728	0.2744	1.261	98.309 ± 0.593	1.525 ± 0.053	—
LSAM	beta42	0.20	0.261	0.061 ± 0.015	0.304 ± 0.077	0.112 ± 0.009	0.335 ± 0.014	0.116 ± 0.010	0.340 ± 0.017	2.888 ± 0.257	1.699 ± 0.085	0.240	0.040	20.12	0.2905	0.2933	1.232	99.366 ± 0.365	1.617 ± 0.054	—
LSAM	beta43	0.20	0.174	-0.026 ± 0.004	-0.129 ± 0.022	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	0.255 ± 0.018	0.505 ± 0.021	0.172	-0.028	-14.06	0.0961	0.1001	1.079	95.772 ± 0.925	0.414 ± 0.008	—
LSAM	beta44	0.00	0.033	0.033 ± 0.006	—	0.017 ± 0.001	0.129 ± 0.006	0.018 ± 0.002	0.132 ± 0.007	—	—	0.032	0.032	—	0.1187	0.1230	1.179	98.097 ± 0.628	0.594 ± 0.018	1.903 ± 0.628
LSAM	beta45	0.00	0.013	0.013 ± 0.005	—	0.013 ± 0.001	0.115 ± 0.005	0.013 ± 0.001	0.115 ± 0.005	—	—	0.013	0.013	—	0.1044	0.1051	1.133	98.097 ± 0.628	0.510 ± 0.015	1.903 ± 0.628
LSAM	beta46	0.00	-0.033	-0.033 ± 0.006	—	0.016 ± 0.001	0.126 ± 0.006	0.017 ± 0.002	0.130 ± 0.007	—	—	-0.025	-0.025	—	0.1044	0.1073	1.243	98.732 ± 0.515	0.612 ± 0.025	1.268 ± 0.515
LSAM	beta47	0.00	-0.035	-0.035 ± 0.006	—	0.017 ± 0.001	0.129 ± 0.006	0.018 ± 0.002	0.134 ± 0.007	—	—	-0.026	-0.026	—	0.1011	0.1045	1.212	99.577 ± 0.298	0.613 ± 0.023	0.423 ± 0.298
LSAM	beta51	0.16	0.147	-0.013 ± 0.006	-0.079 ± 0.036	0.015 ± 0.001	0.124 ± 0.004	0.016 ± 0.001	0.125 ± 0.005	0.607 ± 0.040	0.779 ± 0.032	0.152	-0.008	-5.14	0.1171	0.1174	0.942	93.446 ± 1.138	0.458 ± 0.005	—
LSAM	beta52	0.16	0.243	0.083 ± 0.013	0.517 ± 0.081	0.079 ± 0.006	0.282 ± 0.010	0.086 ± 0.007	0.293 ± 0.013	3.363 ± 0.264	1.834 ± 0.082	0.224	0.064	39.79	0.2714	0.2787	1.153	98.943 ± 0.470	1.274 ± 0.039	—
LSAM	beta53	0.16	0.142	-0.018 ± 0.007	-0.114 ± 0.041	0.020 ± 0.001	0.141 ± 0.005	0.020 ± 0.001	0.142 ± 0.006	0.793 ± 0.053	0.890 ± 0.036	0.132	-0.028	-17.39	0.1389	0.1417	1.023	94.926 ± 1.009	0.567 ± 0.011	—
LSAM	beta54	0.16	0.181	0.021 ± 0.013	0.129 ± 0.079	0.076 ± 0.006	0.276 ± 0.011	0.076 ± 0.006	0.276 ± 0.013	2.986 ± 0.246	1.728 ± 0.082	0.188	0.028	17.31	0.2450	0.2466	1.110	96.829 ± 0.806	1.201 ± 0.038	—
LSAM	beta55	0.00	0.042	0.042 ± 0.008	—	0.027 ± 0.002	0.166 ± 0.007	0.029 ± 0.003	0.171 ± 0.008	—	—	0.032	0.032	—	0.1429	0.1465	1.082	99.789 ± 0.211	0.702 ± 0.020	0.211 ± 0.211
LSAM	beta56	0.00	-0.003	-0.003 ± 0.006	—	0.018 ± 0.001	0.133 ± 0.005	0.018 ± 0.001	0.132 ± 0.006	—	—	-0.001	-0.001	—	0.1145	0.1145	1.055	98.732 ± 0.515	0.548 ± 0.013	1.268 ± 0.515
LSAM	beta57	0.00	-0.045	-0.045 ± 0.006	—	0.016 ± 0.001	0.127 ± 0.006	0.018 ± 0.002	0.135 ± 0.007	—	—	-0.034	-0.034	—	0.0967	0.1026	1.131	98.732 ± 0.515	0.563 ± 0.022	1.268 ± 0.515
LSAM	beta58	0.00	-0.015	-0.015 ± 0.005	—	0.011 ± 0.001	0.103 ± 0.005	0.011 ± 0.001	0.104 ± 0.005	—	—	-0.016	-0.016	—	0.0842	0.0858	1.104	98.309 ± 0.593	0.445 ± 0.018	1.691 ± 0.593
QML	beta41	0.20	0.188	-0.012 ± 0.004	-0.060 ± 0.021	0.008 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	0.213 ± 0.016	0.461 ± 0.020	0.182	-0.018	-8.79	0.0932	0.0949	1.052	95.349 ± 0.968	0.377 ± 0.004	—
QML	beta42	0.20	0.179	-0.021 ± 0.004	-0.104 ± 0.022	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.231 ± 0.015	0.481 ± 0.019	0.179	-0.021	-10.68	0.1001	0.1023	1.068	96.195 ± 0.880	0.393 ± 0.005	—
QML	beta43	0.20	0.200	-0.000 ± 0.004	-0.000 ± 0.020	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	0.180 ± 0.012	0.424 ± 0.017	0.197	-0.003	-1.45	0.0830	0.0831	0.945	93.869 ± 1.103	0.314 ± 0.003	—
QML	beta44	0.00	-0.006	-0.006 ± 0.005	—	0.012 ± 0.001	0.111 ± 0.005	0.012 ± 0.001	0.111 ± 0.005	—	—	0.002	0.002	—	0.1088	0.1088	1.052	95.983 ± 0.903	0.460 ± 0.009	4.017 ± 0.903
QML	beta45	0.00	0.011	0.011 ± 0.004	—	0.009 ± 0.001	0.096 ± 0.004	0.009 ± 0.001	0.097 ± 0.005	—	—	0.011	0.011	—	0.0922	0.0929	0.991	95.772 ± 0.925	0.373 ± 0.007	4.228 ± 0.925
QML	beta46	0.00	0.024	0.024 ± 0.003	—	0.004 ± 0.000	0.061 ± 0.003	0.004 ± 0.000	0.066 ± 0.003	—	—	0.018	0.018	—	0.0626	0.0653	1.069	95.772 ± 0.925	0.256 ± 0.006	4.228 ± 0.925
QML	beta47	0.00	0.027	0.027 ± 0.003	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.067 ± 0.003	—	—	0.020	0.020	—	0.0592	0.0626	1.052	95.772 ± 0.925	0.252 ± 0.006	4.228 ± 0.925
QML	beta51	0.16	0.172	0.012 ± 0.006	0.077 ± 0.036	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	0.630 ± 0.043	0.794 ± 0.032	0.169	0.009	5.38	0.1172	0.1175	1.016	96.195 ± 0.880	0.504 ± 0.005	—
QML	beta52	0.16	0.153	-0.007 ± 0.005	-0.043 ± 0.029	0.010 ± 0.001	0.102 ± 0.004	0.010 ± 0.001	0.102 ± 0.005	0.408 ± 0.031	0.639 ± 0.029	0.151	-0.009	-5.89	0.1031	0.1035	1.026	96.829 ± 0.806	0.411 ± 0.004	—
QML	beta53	0.16	0.158	-0.002 ± 0.005	-0.011 ± 0.030	0.011 ± 0.001	0.106 ± 0.004	0.011 ± 0.001	0.106 ± 0.005	0.437 ± 0.034	0.661 ± 0.030	0.150	-0.010	-6.35	0.1008	0.1013	0.970	94.080 ± 1.085	0.402 ± 0.005	—
QML	beta54	0.16	0.138	-0.022 ± 0.005	-0.139 ± 0.031	0.011 ± 0.001	0.107 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	0.466 ± 0.031	0.683 ± 0.027	0.137	-0.023	-14.34	0.1003	0.1029	0.979	94.292 ± 1.067	0.411 ± 0.005	—
QML	beta55	0.00	0.012	0.012 ± 0.007	—	0.022 ± 0.002	0.148 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	0.011	0.011	—	0.1449	0.1453	1.058	96.829 ± 0.806	0.616 ± 0.013	3.171 ± 0.806
QML	beta56	0.00	0.018	0.018 ± 0.005	—	0.012 ± 0.001	0.111 ± 0.004	0.013 ± 0.001	0.113 ± 0.005	—	—	0.019	0.019	—	0.1084	0.1100	1.052	97.040 ± 0.779	0.458 ± 0.009	2.960 ± 0.779
QML	beta57	0.00	0.010	0.010 ± 0.003	—	0.006 ± 0.001	0.076 ± 0.003	0.006 ± 0.001	0.077 ± 0.004	—	—	0.006	0.006	—	0.0706	0.0708	1.044	97.040 ± 0.779	0.311 ± 0.009	2.960 ± 0.779
QML	beta58	0.00	0.020	0.020 ± 0.003	—	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.059 ± 0.003	—	—	0.015	0.015	—	0.0528	0.0548	0.980	94.926 ± 1.009	0.212 ± 0.005	5.074 ± 1.009
UPI	beta41	0.20	0.368	0.168 ± 0.029	0.840 ± 0.144	0.393 ± 0.036	0.627 ± 0.029	0.420 ± 0.038	0.648 ± 0.033	10.500 ± 0.955	3.240 ± 0.164	0.358	0.158	78.96	0.5547	0.5768	1.371	99.366 ± 0.365	3.368 ± 0.183	—
UPI	beta42	0.20	0.321	0.121 ± 0.023	0.604 ± 0.114	0.247 ± 0.022	0.497 ± 0.023	0.261 ± 0.025	0.511 ± 0.027	6.534 ± 0.614	2.556 ± 0.133	0.272	0.072	36.00	0.4934	0.4987	1.444	99.366 ± 0.365	2.815 ± 0.142	—
UPI	beta43	0.20	0.154	-0.046 ± 0.006	-0.231 ± 0.029	0.015 ± 0.001	0.124 ± 0.005	0.018 ± 0.001	0.132 ± 0.006	0.438 ± 0.033	0.662 ± 0.029	0.148	-0.052	-25.82	0.1243	0.1346	1.407	96.829 ± 0.806	0.685 ± 0.025	—
UPI	beta44	0.00	0.056	0.056 ± 0.012	—	0.070 ± 0.007	0.264 ± 0.013	0.073 ± 0.007	0.270 ± 0.014	—	—	0.036	0.036	—	0.2324	0.2353	1.674	99.577 ± 0.298	1.736 ± 0.099	0.423 ± 0.298
UPI	beta45	0.00	0.026	0.026 ± 0.010	—	0.044 ± 0.004	0.209 ± 0.010	0.044 ± 0.004	0.211 ± 0.011	—	—	0.025	0.025	—	0.1776	0.1794	1.632	99.577 ± 0.298	1.339 ± 0.083	0.423 ± 0.298
UPI	beta46	0.00	-0.076	-0.076 ± 0.012	—	0.064 ± 0.006	0.252 ± 0.012	0.069 ± 0.007	0.263 ± 0.014	—	—	-0.057	-0.057	—	0.2210	0.2284	1.382	99.789 ± 0.211	1.366 ± 0.077	0.211 ± 0.211

Table S241

Simulation 2 - Condition 97: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.056	-0.056 ± 0.008	—	0.031 ± 0.003	0.177 ± 0.008	0.034 ± 0.003	0.186 ± 0.009	—	—	-0.047	-0.047	—	0.1628	0.1694	1.512	99.366 ± 0.365	1.049 ± 0.053	0.634 ± 0.365
UPI	beta51	0.16	0.162	0.002 ± 0.006	0.011 ± 0.040	0.019 ± 0.001	0.138 ± 0.005	0.019 ± 0.001	0.138 ± 0.006	0.747 ± 0.052	0.864 ± 0.036	0.154	-0.006	-3.91	0.1482	0.1483	1.287	98.097 ± 0.628	0.698 ± 0.017	—
UPI	beta52	0.16	0.220	0.060 ± 0.018	0.373 ± 0.114	0.158 ± 0.014	0.397 ± 0.017	0.161 ± 0.014	0.401 ± 0.020	6.279 ± 0.548	2.506 ± 0.124	0.196	0.036	22.66	0.4074	0.4090	1.643	98.732 ± 0.515	2.556 ± 0.133	—
UPI	beta53	0.16	0.136	-0.024 ± 0.008	-0.153 ± 0.049	0.029 ± 0.002	0.171 ± 0.007	0.030 ± 0.002	0.172 ± 0.008	1.161 ± 0.094	1.078 ± 0.050	0.120	-0.040	-25.22	0.1602	0.1652	1.487	97.886 ± 0.661	0.996 ± 0.049	—
UPI	beta54	0.16	0.237	0.077 ± 0.021	0.480 ± 0.129	0.201 ± 0.019	0.448 ± 0.021	0.206 ± 0.020	0.454 ± 0.024	8.050 ± 0.778	2.837 ± 0.152	0.174	0.014	9.02	0.4306	0.4308	1.558	99.366 ± 0.365	2.735 ± 0.141	—
UPI	beta55	0.00	0.027	0.027 ± 0.012	—	0.070 ± 0.007	0.264 ± 0.013	0.070 ± 0.007	0.265 ± 0.014	—	—	0.021	0.021	—	0.2245	0.2254	1.644	100.000 ± 0.000	1.701 ± 0.081	0.000 ± 0.000
UPI	beta56	0.00	-0.003	-0.003 ± 0.009	—	0.039 ± 0.004	0.198 ± 0.010	0.039 ± 0.004	0.198 ± 0.011	—	—	0.002	0.002	—	0.1688	0.1688	1.560	100.000 ± 0.000	1.210 ± 0.064	0.000 ± 0.000
UPI	beta57	0.00	-0.025	-0.025 ± 0.008	—	0.030 ± 0.003	0.174 ± 0.009	0.031 ± 0.003	0.176 ± 0.009	—	—	-0.028	-0.028	—	0.1526	0.1552	1.590	100.000 ± 0.000	1.086 ± 0.059	0.000 ± 0.000
UPI	beta58	0.00	-0.029	-0.029 ± 0.007	—	0.024 ± 0.002	0.155 ± 0.007	0.025 ± 0.002	0.158 ± 0.008	—	—	-0.010	-0.010	—	0.1406	0.1410	1.582	100.000 ± 0.000	0.962 ± 0.052	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S242

Simulation 2 - Condition 98: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.252	0.052 ± 0.011	0.262 ± 0.056	0.091 ± 0.006	0.302 ± 0.010	0.094 ± 0.006	0.306 ± 0.012	2.338 ± 0.161	1.529 ± 0.060	0.255	0.055	27.45	0.2580	0.2637	1.299	99.450 ± 0.274	1.536 ± 0.055	—
LSAM	beta42	0.20	0.307	0.107 ± 0.012	0.535 ± 0.062	0.112 ± 0.007	0.334 ± 0.010	0.123 ± 0.008	0.351 ± 0.013	3.075 ± 0.196	1.754 ± 0.063	0.284	0.084	41.99	0.2887	0.3007	1.266	99.862 ± 0.137	1.659 ± 0.053	—
LSAM	beta43	0.20	0.175	-0.025 ± 0.002	-0.123 ± 0.012	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.124 ± 0.007	0.352 ± 0.011	0.174	-0.026	-12.80	0.0673	0.0720	1.173	95.873 ± 0.738	0.304 ± 0.005	—
LSAM	beta44	0.00	0.036	0.036 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.036	0.036	—	0.0784	0.0861	1.233	97.249 ± 0.607	0.395 ± 0.011	2.751 ± 0.607
LSAM	beta45	0.00	0.015	0.015 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	—	—	0.015	0.015	—	0.0669	0.0686	1.204	97.937 ± 0.527	0.332 ± 0.007	2.063 ± 0.527
LSAM	beta46	0.00	-0.033	-0.033 ± 0.004	—	0.011 ± 0.001	0.105 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	—	—	-0.032	-0.032	—	0.0832	0.0891	1.307	99.037 ± 0.362	0.536 ± 0.022	0.963 ± 0.362
LSAM	beta47	0.00	-0.052	-0.052 ± 0.004	—	0.014 ± 0.001	0.118 ± 0.004	0.017 ± 0.001	0.129 ± 0.005	—	—	-0.040	-0.040	—	0.0906	0.0989	1.227	99.037 ± 0.362	0.569 ± 0.020	0.963 ± 0.362
LSAM	beta51	0.16	0.149	-0.011 ± 0.003	-0.068 ± 0.018	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.227 ± 0.012	0.477 ± 0.015	0.145	-0.015	-9.57	0.0814	0.0828	1.017	95.323 ± 0.783	0.301 ± 0.002	—
LSAM	beta52	0.16	0.242	0.082 ± 0.009	0.513 ± 0.054	0.053 ± 0.004	0.231 ± 0.008	0.060 ± 0.004	0.245 ± 0.009	2.350 ± 0.155	1.533 ± 0.057	0.229	0.069	43.11	0.2271	0.2373	1.143	98.074 ± 0.510	1.036 ± 0.030	—
LSAM	beta53	0.16	0.130	-0.030 ± 0.003	-0.187 ± 0.021	0.008 ± 0.000	0.091 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.360 ± 0.020	0.600 ± 0.020	0.124	-0.036	-22.37	0.0890	0.0959	1.091	96.424 ± 0.689	0.391 ± 0.005	—
LSAM	beta54	0.16	0.230	0.070 ± 0.008	0.436 ± 0.049	0.044 ± 0.003	0.209 ± 0.007	0.049 ± 0.003	0.221 ± 0.008	1.899 ± 0.125	1.378 ± 0.051	0.221	0.061	37.99	0.1851	0.1948	1.160	98.349 ± 0.473	0.952 ± 0.026	—
LSAM	beta55	0.00	0.040	0.040 ± 0.004	—	0.013 ± 0.001	0.115 ± 0.004	0.015 ± 0.001	0.122 ± 0.004	—	—	0.032	0.032	—	0.1100	0.1145	1.090	96.699 ± 0.663	0.492 ± 0.011	3.301 ± 0.663
LSAM	beta56	0.00	0.007	0.007 ± 0.004	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	—	—	0.006	0.006	—	0.0874	0.0876	0.968	96.561 ± 0.676	0.364 ± 0.006	3.439 ± 0.676
LSAM	beta57	0.00	-0.045	-0.045 ± 0.004	—	0.010 ± 0.001	0.099 ± 0.004	0.012 ± 0.001	0.108 ± 0.004	—	—	-0.035	-0.035	—	0.0804	0.0877	1.121	98.212 ± 0.491	0.434 ± 0.015	1.788 ± 0.491
LSAM	beta58	0.00	-0.032	-0.032 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.026	-0.026	—	0.0589	0.0643	1.122	98.487 ± 0.453	0.334 ± 0.011	1.513 ± 0.453
QML	beta41	0.20	0.183	-0.017 ± 0.002	-0.086 ± 0.010	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.293 ± 0.009	0.182	-0.018	-9.23	0.0576	0.0605	1.040	94.635 ± 0.836	0.228 ± 0.001	—
QML	beta42	0.20	0.182	-0.018 ± 0.002	-0.089 ± 0.011	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.092 ± 0.005	0.303 ± 0.009	0.182	-0.018	-9.19	0.0622	0.0648	1.037	94.773 ± 0.825	0.235 ± 0.001	—
QML	beta43	0.20	0.206	0.006 ± 0.002	0.032 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.058 ± 0.003	0.241 ± 0.008	0.202	0.002	1.18	0.0481	0.0481	1.029	95.048 ± 0.805	0.193 ± 0.001	—
QML	beta44	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0631	0.0631	1.014	94.911 ± 0.815	0.259 ± 0.003	5.089 ± 0.815
QML	beta45	0.00	0.014	0.014 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.011	0.011	—	0.0556	0.0567	1.037	94.911 ± 0.815	0.221 ± 0.002	5.089 ± 0.815
QML	beta46	0.00	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.016	0.016	—	0.0328	0.0368	1.059	94.911 ± 0.815	0.140 ± 0.002	5.089 ± 0.815
QML	beta47	0.00	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.016	0.016	—	0.0324	0.0361	1.007	93.810 ± 0.894	0.141 ± 0.002	6.190 ± 0.894
QML	beta51	0.16	0.165	0.005 ± 0.003	0.032 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.192 ± 0.010	0.438 ± 0.014	0.162	0.002	1.19	0.0707	0.0707	1.042	96.149 ± 0.714	0.286 ± 0.002	—
QML	beta52	0.16	0.144	-0.016 ± 0.002	-0.103 ± 0.014	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.155 ± 0.007	0.393 ± 0.012	0.143	-0.017	-10.86	0.0655	0.0678	1.026	94.911 ± 0.815	0.244 ± 0.002	—
QML	beta53	0.16	0.157	-0.003 ± 0.002	-0.021 ± 0.013	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.007	0.353 ± 0.012	0.155	-0.005	-2.93	0.0574	0.0576	1.048	94.911 ± 0.815	0.231 ± 0.001	—
QML	beta54	0.16	0.144	-0.016 ± 0.002	-0.100 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.133 ± 0.007	0.365 ± 0.011	0.149	-0.011	-6.98	0.0601	0.0612	1.100	96.149 ± 0.714	0.243 ± 0.001	—
QML	beta55	0.00	0.003	0.003 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	-0.001	-0.001	—	0.0885	0.0885	1.030	94.223 ± 0.865	0.354 ± 0.005	5.777 ± 0.865
QML	beta56	0.00	0.024	0.024 ± 0.003	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.020	0.020	—	0.0673	0.0702	0.946	92.022 ± 1.005	0.256 ± 0.002	7.978 ± 1.005
QML	beta57	0.00	0.015	0.015 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	—	—	0.015	0.015	—	0.0435	0.0459	1.065	94.773 ± 0.825	0.179 ± 0.004	5.227 ± 0.825
QML	beta58	0.00	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.012	0.012	—	0.0311	0.0331	0.964	93.535 ± 0.912	0.125 ± 0.002	6.465 ± 0.912
UPI	beta41	0.20	0.421	0.221 ± 0.014	1.104 ± 0.070	0.142 ± 0.009	0.377 ± 0.012	0.190 ± 0.012	0.436 ± 0.015	4.762 ± 0.291	2.182 ± 0.073	0.416	0.216	108.08	0.3974	0.4524	1.379	96.699 ± 0.663	2.036 ± 0.086	—
UPI	beta42	0.20	0.295	0.095 ± 0.012	0.475 ± 0.060	0.105 ± 0.008	0.324 ± 0.012	0.114 ± 0.008	0.337 ± 0.014	2.841 ± 0.211	1.685 ± 0.069	0.269	0.069	34.52	0.3064	0.3141	1.310	98.074 ± 0.510	1.662 ± 0.064	—
UPI	beta43	0.20	0.149	-0.051 ± 0.002	-0.253 ± 0.012	0.005 ± 0.000	0.067 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.177 ± 0.009	0.421 ± 0.012	0.144	-0.056	-27.94	0.0700	0.0896	1.242	92.022 ± 1.005	0.328 ± 0.008	—
UPI	beta44	0.00	0.058	0.058 ± 0.005	—	0.015 ± 0.001	0.123 ± 0.004	0.018 ± 0.001	0.136 ± 0.005	—	—	0.051	0.051	—	0.1119	0.1228	1.436	99.450 ± 0.274	0.694 ± 0.028	0.550 ± 0.274
UPI	beta45	0.00	0.030	0.030 ± 0.003	—	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	—	—	0.021	0.021	—	0.0800	0.0827	1.505	99.312 ± 0.307	0.545 ± 0.021	0.688 ± 0.307
UPI	beta46	0.00	-0.090	-0.090 ± 0.005	—	0.017 ± 0.001	0.129 ± 0.004	0.025 ± 0.001	0.157 ± 0.005	—	—	-0.088	-0.088	—	0.1354	0.1614	1.418	97.524 ± 0.576	0.717 ± 0.031	2.476 ± 0.576

Table S242

Simulation 2 - Condition 98: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.050	-0.050 ± 0.004	—	0.011 ± 0.001	0.105 ± 0.004	0.013 ± 0.001	0.116 ± 0.005	—	—	-0.039	-0.039	—	0.0945	0.1023	1.325	98.762 ± 0.410	0.545 ± 0.019	1.238 ± 0.410
UPI	beta51	0.16	0.154	-0.006 ± 0.003	-0.038 ± 0.018	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.246 ± 0.013	0.496 ± 0.016	0.153	-0.007	-4.59	0.0812	0.0816	1.235	97.524 ± 0.576	0.383 ± 0.007	—
UPI	beta52	0.16	0.283	0.123 ± 0.014	0.766 ± 0.085	0.133 ± 0.010	0.365 ± 0.013	0.148 ± 0.011	0.385 ± 0.016	5.781 ± 0.424	2.404 ± 0.097	0.225	0.065	40.48	0.3082	0.3150	1.430	99.312 ± 0.307	2.046 ± 0.093	—
UPI	beta53	0.16	0.113	-0.047 ± 0.004	-0.296 ± 0.022	0.009 ± 0.001	0.096 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	0.447 ± 0.029	0.668 ± 0.024	0.106	-0.054	-33.76	0.0911	0.1059	1.262	95.736 ± 0.749	0.474 ± 0.013	—
UPI	beta54	0.16	0.265	0.105 ± 0.015	0.659 ± 0.094	0.163 ± 0.011	0.404 ± 0.014	0.174 ± 0.013	0.417 ± 0.017	6.797 ± 0.504	2.607 ± 0.107	0.220	0.060	37.43	0.3444	0.3495	1.369	97.937 ± 0.527	2.168 ± 0.103	—
UPI	beta55	0.00	0.057	0.057 ± 0.006	—	0.024 ± 0.002	0.154 ± 0.006	0.027 ± 0.002	0.164 ± 0.007	—	—	0.044	0.044	—	0.1425	0.1493	1.423	99.587 ± 0.238	0.857 ± 0.030	0.413 ± 0.238
UPI	beta56	0.00	0.008	0.008 ± 0.004	—	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	—	—	0.008	0.008	—	0.0993	0.0997	1.295	99.725 ± 0.194	0.530 ± 0.015	0.275 ± 0.194
UPI	beta57	0.00	-0.057	-0.057 ± 0.005	—	0.019 ± 0.001	0.137 ± 0.005	0.022 ± 0.002	0.148 ± 0.006	—	—	-0.038	-0.038	—	0.1135	0.1197	1.437	99.175 ± 0.336	0.771 ± 0.034	0.825 ± 0.336
UPI	beta58	0.00	-0.040	-0.040 ± 0.005	—	0.016 ± 0.001	0.125 ± 0.004	0.017 ± 0.001	0.132 ± 0.005	—	—	-0.027	-0.027	—	0.1032	0.1067	1.386	99.175 ± 0.336	0.682 ± 0.032	0.825 ± 0.336

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S243

Simulation 2 - Condition 99: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.211	0.011 ± 0.005	0.054 ± 0.023	0.018 ± 0.001	0.136 ± 0.003	0.018 ± 0.001	0.136 ± 0.004	0.462 ± 0.024	0.680 ± 0.021	0.211	0.011	5.75	0.1309	0.1315	0.968	95.362 ± 0.707	0.514 ± 0.005	—
LSAM	beta42	0.20	0.224	0.024 ± 0.005	0.120 ± 0.024	0.020 ± 0.001	0.142 ± 0.004	0.021 ± 0.001	0.144 ± 0.005	0.517 ± 0.031	0.719 ± 0.025	0.219	0.019	9.33	0.1335	0.1348	0.986	96.154 ± 0.647	0.549 ± 0.006	—
LSAM	beta43	0.20	0.195	-0.005 ± 0.002	-0.026 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.004	0.305 ± 0.009	0.192	-0.008	-3.88	0.0590	0.0595	1.018	95.136 ± 0.724	0.243 ± 0.001	—
LSAM	beta44	0.00	0.009	0.009 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	—	—	0.008	0.008	—	0.0683	0.0687	0.959	94.005 ± 0.798	0.292 ± 0.004	5.995 ± 0.798
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.001	0.001	—	0.0672	0.0673	0.973	94.457 ± 0.770	0.273 ± 0.003	5.543 ± 0.770
LSAM	beta46	0.00	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.008	-0.008	—	0.0470	0.0477	0.900	95.362 ± 0.707	0.194 ± 0.003	4.638 ± 0.707
LSAM	beta47	0.00	-0.011	-0.011 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.008	-0.008	—	0.0474	0.0481	0.969	95.023 ± 0.731	0.201 ± 0.004	4.977 ± 0.731
LSAM	beta51	0.16	0.162	0.002 ± 0.003	0.010 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.267 ± 0.014	0.517 ± 0.016	0.159	-0.001	-0.55	0.0831	0.0831	0.993	94.910 ± 0.739	0.322 ± 0.001	—
LSAM	beta52	0.16	0.178	0.018 ± 0.005	0.112 ± 0.029	0.019 ± 0.001	0.137 ± 0.003	0.019 ± 0.001	0.138 ± 0.004	0.743 ± 0.039	0.862 ± 0.027	0.177	0.017	10.31	0.1331	0.1341	0.950	93.100 ± 0.852	0.509 ± 0.004	—
LSAM	beta53	0.16	0.155	-0.005 ± 0.003	-0.031 ± 0.018	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.302 ± 0.017	0.549 ± 0.018	0.154	-0.006	-3.91	0.0882	0.0884	0.991	94.796 ± 0.747	0.341 ± 0.002	—
LSAM	beta54	0.16	0.181	0.021 ± 0.004	0.132 ± 0.027	0.017 ± 0.001	0.130 ± 0.003	0.017 ± 0.001	0.131 ± 0.004	0.675 ± 0.034	0.822 ± 0.025	0.181	0.021	13.41	0.1275	0.1293	0.994	95.023 ± 0.731	0.506 ± 0.004	—
LSAM	beta55	0.00	0.015	0.015 ± 0.004	—	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	—	—	0.012	0.012	—	0.0968	0.0976	0.961	95.249 ± 0.715	0.400 ± 0.004	4.751 ± 0.715
LSAM	beta56	0.00	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	—	—	0.000	0.000	—	0.0795	0.0795	0.929	94.344 ± 0.777	0.334 ± 0.003	5.656 ± 0.777
LSAM	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.009	-0.009	—	0.0570	0.0578	0.897	94.457 ± 0.770	0.217 ± 0.003	5.543 ± 0.770
LSAM	beta58	0.00	-0.009	-0.009 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.010	-0.010	—	0.0455	0.0465	0.936	93.778 ± 0.812	0.182 ± 0.003	6.222 ± 0.812
QML	beta41	0.20	0.183	-0.017 ± 0.002	-0.083 ± 0.012	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.144 ± 0.007	0.380 ± 0.011	0.187	-0.013	-6.57	0.0785	0.0796	0.982	94.683 ± 0.755	0.286 ± 0.001	—
QML	beta42	0.20	0.189	-0.011 ± 0.003	-0.053 ± 0.013	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.151 ± 0.007	0.388 ± 0.011	0.187	-0.013	-6.66	0.0773	0.0784	0.977	94.118 ± 0.791	0.295 ± 0.001	—
QML	beta43	0.20	0.203	0.003 ± 0.002	0.015 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.279 ± 0.008	0.200	0.000	0.17	0.0554	0.0554	1.015	95.475 ± 0.699	0.222 ± 0.001	—
QML	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.004	-0.004	—	0.0629	0.0630	0.969	93.326 ± 0.839	0.257 ± 0.003	6.674 ± 0.839
QML	beta45	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	—	0.0621	0.0621	0.984	93.326 ± 0.839	0.232 ± 0.002	6.674 ± 0.839
QML	beta46	0.00	0.015	0.015 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.013	0.013	—	0.0375	0.0395	0.882	91.176 ± 0.954	0.140 ± 0.002	8.824 ± 0.954
QML	beta47	0.00	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.011	0.011	—	0.0363	0.0381	0.947	93.439 ± 0.833	0.141 ± 0.002	6.561 ± 0.833
QML	beta51	0.16	0.166	0.006 ± 0.003	0.035 ± 0.017	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.255 ± 0.013	0.505 ± 0.016	0.161	0.001	0.84	0.0776	0.0776	1.030	95.814 ± 0.674	0.326 ± 0.002	—
QML	beta52	0.16	0.149	-0.011 ± 0.003	-0.071 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.266 ± 0.013	0.516 ± 0.015	0.152	-0.008	-4.84	0.0800	0.0803	0.959	92.421 ± 0.890	0.307 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.015	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.198 ± 0.010	0.445 ± 0.014	0.159	-0.001	-0.52	0.0714	0.0714	0.999	94.910 ± 0.739	0.279 ± 0.001	—
QML	beta54	0.16	0.152	-0.008 ± 0.003	-0.053 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.242 ± 0.011	0.492 ± 0.014	0.153	-0.007	-4.29	0.0777	0.0780	1.003	95.023 ± 0.731	0.308 ± 0.001	—
QML	beta55	0.00	0.003	0.003 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	—	—	0.003	0.003	—	0.0906	0.0906	0.996	95.362 ± 0.707	0.354 ± 0.004	4.638 ± 0.707
QML	beta56	0.00	0.011	0.011 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	0.009	0.009	—	0.0708	0.0714	0.948	93.439 ± 0.833	0.282 ± 0.003	6.561 ± 0.833
QML	beta57	0.00	0.007	0.007 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	0.007	0.007	—	0.0436	0.0441	0.939	93.100 ± 0.852	0.172 ± 0.003	6.900 ± 0.852
QML	beta58	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.007	0.007	—	0.0364	0.0369	0.908	90.385 ± 0.992	0.135 ± 0.002	9.615 ± 0.992
UPI	beta41	0.20	0.216	0.016 ± 0.005	0.081 ± 0.023	0.019 ± 0.001	0.138 ± 0.004	0.019 ± 0.001	0.139 ± 0.004	0.484 ± 0.026	0.695 ± 0.022	0.218	0.018	9.06	0.1403	0.1414	1.173	97.851 ± 0.488	0.635 ± 0.015	—
UPI	beta42	0.20	0.242	0.042 ± 0.005	0.210 ± 0.026	0.023 ± 0.001	0.153 ± 0.004	0.025 ± 0.002	0.158 ± 0.005	0.627 ± 0.038	0.792 ± 0.027	0.235	0.035	17.74	0.1508	0.1549	1.174	97.172 ± 0.558	0.703 ± 0.021	—
UPI	beta43	0.20	0.188	-0.012 ± 0.002	-0.061 ± 0.010	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.301 ± 0.009	0.183	-0.017	-8.41	0.0626	0.0648	1.061	95.362 ± 0.707	0.246 ± 0.003	—
UPI	beta44	0.00	0.014	0.014 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.081 ± 0.003	—	—	0.014	0.014	—	0.0745	0.0758	1.166	96.041 ± 0.656	0.363 ± 0.016	3.959 ± 0.656
UPI	beta45	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	—	—	0.001	0.001	—	0.0680	0.0680	1.128	97.059 ± 0.568	0.320 ± 0.007	2.941 ± 0.568
UPI	beta46	0.00	-0.011	-0.011 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.009	-0.009	—	0.0501	0.0510	1.134	96.833 ± 0.589	0.250 ± 0.009	3.167 ± 0.589

Table S243

Simulation 2 - Condition 99: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.020	-0.020 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.013	-0.013	—	0.0515	0.0532	1.167	96.946 ± 0.579	0.267 ± 0.011	3.054 ± 0.579
UPI	beta51	0.16	0.163	0.003 ± 0.003	0.017 ± 0.017	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.251 ± 0.013	0.501 ± 0.015	0.158	-0.002	-1.21	0.0759	0.0759	1.033	95.588 ± 0.691	0.325 ± 0.001	—
UPI	beta52	0.16	0.178	0.018 ± 0.005	0.115 ± 0.028	0.018 ± 0.001	0.135 ± 0.003	0.019 ± 0.001	0.137 ± 0.004	0.728 ± 0.038	0.853 ± 0.026	0.175	0.015	9.43	0.1379	0.1387	1.022	95.249 ± 0.715	0.542 ± 0.007	—
UPI	beta53	0.16	0.151	-0.009 ± 0.003	-0.058 ± 0.018	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.275 ± 0.015	0.525 ± 0.017	0.149	-0.011	-7.10	0.0829	0.0837	1.038	95.475 ± 0.699	0.340 ± 0.003	—
UPI	beta54	0.16	0.190	0.030 ± 0.005	0.189 ± 0.029	0.019 ± 0.001	0.139 ± 0.004	0.020 ± 0.001	0.142 ± 0.005	0.790 ± 0.044	0.889 ± 0.029	0.196	0.036	22.74	0.1356	0.1404	1.082	96.833 ± 0.589	0.589 ± 0.010	—
UPI	beta55	0.00	0.019	0.019 ± 0.004	—	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.112 ± 0.004	—	—	0.015	0.015	—	0.0968	0.0979	1.056	96.719 ± 0.599	0.456 ± 0.008	3.281 ± 0.599
UPI	beta56	0.00	0.000	-0.000 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	-0.002	-0.002	—	0.0777	0.0778	1.045	96.380 ± 0.628	0.352 ± 0.005	3.620 ± 0.628
UPI	beta57	0.00	-0.014	-0.014 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.011	-0.011	—	0.0547	0.0559	0.983	96.154 ± 0.647	0.242 ± 0.005	3.846 ± 0.647
UPI	beta58	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.014	-0.014	—	0.0458	0.0477	1.032	94.796 ± 0.747	0.217 ± 0.005	5.204 ± 0.747

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S244

Simulation 2 - Condition 100: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.218	0.018 ± 0.002	0.089 ± 0.011	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.136 ± 0.006	0.369 ± 0.010	0.222	0.022	11.20	0.0702	0.0736	1.016	94.969 ± 0.700	0.285 ± 0.002	—
LSAM	beta42	0.20	0.208	0.008 ± 0.002	0.041 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.205	0.005	2.45	0.0753	0.0755	1.020	95.688 ± 0.651	0.305 ± 0.002	—
LSAM	beta43	0.20	0.195	-0.005 ± 0.001	-0.023 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.194	-0.006	-2.86	0.0349	0.0354	1.002	94.456 ± 0.733	0.147 ± 0.000	—
LSAM	beta44	0.00	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.009	0.009	—	0.0380	0.0389	0.977	94.764 ± 0.714	0.154 ± 0.001	5.236 ± 0.714
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.001	0.001	—	0.0378	0.0379	0.968	93.737 ± 0.776	0.150 ± 0.001	6.263 ± 0.776
LSAM	beta46	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.008	-0.008	—	0.0238	0.0252	0.985	93.532 ± 0.788	0.097 ± 0.001	6.468 ± 0.788
LSAM	beta47	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0253	0.0261	0.959	93.121 ± 0.811	0.101 ± 0.001	6.879 ± 0.811
LSAM	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.011	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.328 ± 0.009	0.160	0.000	-0.02	0.0508	0.0508	0.965	93.840 ± 0.770	0.198 ± 0.000	—
LSAM	beta52	0.16	0.174	0.014 ± 0.002	0.088 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.221 ± 0.010	0.470 ± 0.013	0.174	0.014	8.81	0.0701	0.0715	1.004	95.380 ± 0.673	0.291 ± 0.001	—
LSAM	beta53	0.16	0.151	-0.009 ± 0.002	-0.054 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.123 ± 0.006	0.351 ± 0.010	0.152	-0.008	-5.10	0.0571	0.0577	0.942	92.916 ± 0.822	0.205 ± 0.001	—
LSAM	beta54	0.16	0.176	0.016 ± 0.002	0.101 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.208 ± 0.010	0.457 ± 0.013	0.176	0.016	10.31	0.0692	0.0711	1.035	95.893 ± 0.636	0.289 ± 0.001	—
LSAM	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.012	0.012	—	0.0567	0.0579	0.985	94.559 ± 0.727	0.223 ± 0.001	5.441 ± 0.727
LSAM	beta56	0.00	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	0.003	0.003	—	0.0467	0.0469	0.968	93.943 ± 0.764	0.189 ± 0.001	6.057 ± 0.764
LSAM	beta57	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.008	-0.008	—	0.0277	0.0289	0.956	93.634 ± 0.782	0.112 ± 0.001	6.366 ± 0.782
LSAM	beta58	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.006	-0.006	—	0.0223	0.0230	0.988	95.072 ± 0.694	0.096 ± 0.001	4.928 ± 0.694
QML	beta41	0.20	0.188	-0.012 ± 0.001	-0.060 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.192	-0.008	-4.24	0.0426	0.0435	1.020	94.045 ± 0.758	0.176 ± 0.000	—
QML	beta42	0.20	0.182	-0.018 ± 0.001	-0.089 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.006	0.182	-0.018	-9.04	0.0475	0.0508	1.020	93.532 ± 0.788	0.183 ± 0.001	—
QML	beta43	0.20	0.202	0.002 ± 0.001	0.012 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.002	0.176 ± 0.005	0.202	0.002	0.79	0.0333	0.0333	1.005	95.380 ± 0.673	0.139 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0396	0.0396	0.971	94.969 ± 0.700	0.155 ± 0.001	5.031 ± 0.700
QML	beta45	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0365	0.0366	0.967	93.634 ± 0.782	0.143 ± 0.001	6.366 ± 0.782
QML	beta46	0.00	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	—	0.0217	0.0243	0.970	91.992 ± 0.870	0.082 ± 0.001	8.008 ± 0.870
QML	beta47	0.00	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.012	0.012	—	0.0225	0.0256	0.940	88.912 ± 1.006	0.083 ± 0.001	11.088 ± 1.006
QML	beta51	0.16	0.162	0.002 ± 0.002	0.015 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.325 ± 0.009	0.163	0.003	1.84	0.0519	0.0520	0.978	94.456 ± 0.733	0.199 ± 0.001	—
QML	beta52	0.16	0.149	-0.011 ± 0.002	-0.068 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.150	-0.010	-6.24	0.0464	0.0474	1.012	94.353 ± 0.740	0.190 ± 0.000	—
QML	beta53	0.16	0.155	-0.005 ± 0.002	-0.031 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.296 ± 0.008	0.155	-0.005	-2.99	0.0466	0.0468	0.936	92.197 ± 0.859	0.173 ± 0.001	—
QML	beta54	0.16	0.150	-0.010 ± 0.001	-0.061 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.150	-0.010	-6.01	0.0466	0.0476	1.052	96.304 ± 0.605	0.191 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.003	0.003	—	0.0554	0.0555	0.989	94.045 ± 0.758	0.213 ± 0.001	5.955 ± 0.758
QML	beta56	0.00	0.014	0.014 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.014	0.014	—	0.0425	0.0447	0.981	93.224 ± 0.805	0.170 ± 0.001	6.776 ± 0.805
QML	beta57	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.008	0.008	—	0.0257	0.0269	0.975	92.197 ± 0.859	0.100 ± 0.001	7.803 ± 0.859
QML	beta58	0.00	0.008	0.008 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.008	0.008	—	0.0191	0.0208	0.993	94.045 ± 0.758	0.080 ± 0.001	5.955 ± 0.758
UPI	beta41	0.20	0.222	0.022 ± 0.003	0.109 ± 0.013	0.006 ± 0.000	0.078 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.166 ± 0.008	0.407 ± 0.011	0.223	0.023	11.45	0.0747	0.0782	1.049	96.509 ± 0.588	0.323 ± 0.003	—
UPI	beta42	0.20	0.219	0.019 ± 0.003	0.093 ± 0.013	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.177 ± 0.009	0.421 ± 0.012	0.216	0.016	8.11	0.0784	0.0801	1.077	96.201 ± 0.613	0.347 ± 0.003	—
UPI	beta43	0.20	0.190	-0.010 ± 0.001	-0.048 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.189	-0.011	-5.29	0.0348	0.0364	0.998	93.224 ± 0.805	0.143 ± 0.001	—
UPI	beta44	0.00	0.009	0.009 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.008	0.008	—	0.0391	0.0399	1.010	95.791 ± 0.643	0.166 ± 0.002	4.209 ± 0.643

Table S244

Simulation 2 - Condition 100: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.002	0.002	—	0.0395	0.0395	0.998	94.559 ± 0.727	0.158 ± 0.002	5.441 ± 0.727
UPI	beta46	0.00	-0.012	-0.012 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.009	-0.009	—	0.0244	0.0261	0.990	95.380 ± 0.673	0.111 ± 0.002	4.620 ± 0.673
UPI	beta47	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.009	-0.009	—	0.0273	0.0287	1.007	94.764 ± 0.714	0.118 ± 0.002	5.236 ± 0.714
UPI	beta51	0.16	0.160	0.000 ± 0.002	0.001 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.160	0.000	-0.22	0.0515	0.0515	0.977	94.251 ± 0.746	0.199 ± 0.001	—
UPI	beta52	0.16	0.177	0.017 ± 0.002	0.105 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.245 ± 0.011	0.495 ± 0.014	0.175	0.015	9.34	0.0756	0.0771	1.020	96.099 ± 0.620	0.309 ± 0.002	—
UPI	beta53	0.16	0.148	-0.012 ± 0.002	-0.075 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.005	0.346 ± 0.009	0.148	-0.012	-7.20	0.0537	0.0549	0.954	93.737 ± 0.776	0.202 ± 0.001	—
UPI	beta54	0.16	0.183	0.023 ± 0.003	0.141 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.265 ± 0.014	0.515 ± 0.016	0.177	0.017	10.80	0.0731	0.0751	1.034	95.791 ± 0.643	0.321 ± 0.003	—
UPI	beta55	0.00	0.012	0.012 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.013	0.013	—	0.0589	0.0603	1.003	94.867 ± 0.707	0.236 ± 0.002	5.133 ± 0.707
UPI	beta56	0.00	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	—	—	0.005	0.005	—	0.0464	0.0466	0.994	94.764 ± 0.714	0.193 ± 0.002	5.236 ± 0.714
UPI	beta57	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.009	-0.009	—	0.0294	0.0308	0.966	93.634 ± 0.782	0.122 ± 0.002	6.366 ± 0.782
UPI	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.008	-0.008	—	0.0228	0.0241	0.977	96.099 ± 0.620	0.108 ± 0.002	3.901 ± 0.620

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S245

Simulation 2 - Condition 101: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.003	0.008 ± 0.013	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.203	0.003	1.41	0.0821	0.0822	0.975	94.630 ± 0.718	0.306 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.003	0.022 ± 0.013	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.164 ± 0.008	0.405 ± 0.011	0.204	0.004	2.03	0.0807	0.0808	1.017	95.846 ± 0.635	0.322 ± 0.001	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.199	-0.001	-0.53	0.0486	0.0486	0.953	94.326 ± 0.736	0.184 ± 0.001	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0520	0.0520	0.908	92.199 ± 0.854	0.196 ± 0.002	7.801 ± 0.854
LSAM	beta45	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	0.001	0.001	—	0.0460	0.0460	0.957	93.212 ± 0.801	0.188 ± 0.002	6.788 ± 0.801
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.002	-0.002	—	0.0274	0.0274	0.917	92.705 ± 0.828	0.112 ± 0.001	7.295 ± 0.828
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0298	0.0299	0.905	91.692 ± 0.879	0.114 ± 0.001	8.308 ± 0.879
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.012 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.007	0.413 ± 0.011	0.162	0.002	1.03	0.0664	0.0664	0.965	94.428 ± 0.730	0.250 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.003	0.018 ± 0.017	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.294 ± 0.014	0.542 ± 0.015	0.162	0.002	1.16	0.0850	0.0850	0.960	94.124 ± 0.749	0.326 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.007	0.410 ± 0.011	0.160	0.000	-0.13	0.0651	0.0651	0.972	94.326 ± 0.736	0.250 ± 0.001	—
LSAM	beta54	0.16	0.161	0.001 ± 0.003	0.009 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.285 ± 0.013	0.534 ± 0.015	0.158	-0.002	-1.48	0.0879	0.0879	0.966	95.137 ± 0.685	0.323 ± 0.001	—
LSAM	beta55	0.00	0.004	0.004 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.003	0.003	—	0.0736	0.0737	0.920	92.300 ± 0.849	0.269 ± 0.002	7.700 ± 0.849
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.001	0.001	—	0.0609	0.0609	0.929	91.996 ± 0.864	0.234 ± 0.002	8.004 ± 0.864
LSAM	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0338	0.0339	0.897	92.705 ± 0.828	0.132 ± 0.001	7.295 ± 0.828
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0288	0.0288	0.880	92.705 ± 0.828	0.111 ± 0.001	7.295 ± 0.828
QML	beta41	0.20	0.190	-0.010 ± 0.002	-0.048 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.106 ± 0.005	0.326 ± 0.009	0.191	-0.009	-4.74	0.0654	0.0661	0.983	94.124 ± 0.749	0.249 ± 0.001	—
QML	beta42	0.20	0.193	-0.007 ± 0.002	-0.034 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.005	0.321 ± 0.009	0.194	-0.006	-3.23	0.0650	0.0653	1.033	95.339 ± 0.671	0.258 ± 0.001	—
QML	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.201	0.001	0.72	0.0493	0.0493	0.951	94.428 ± 0.730	0.180 ± 0.001	—
QML	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.002	-0.002	—	0.0506	0.0507	0.913	91.895 ± 0.869	0.190 ± 0.001	8.105 ± 0.869
QML	beta45	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0447	0.0447	0.972	93.516 ± 0.784	0.179 ± 0.001	6.484 ± 0.784
QML	beta46	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.006	0.006	—	0.0245	0.0251	0.917	92.199 ± 0.854	0.102 ± 0.001	7.801 ± 0.854
QML	beta47	0.00	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.006	0.006	—	0.0277	0.0284	0.894	90.476 ± 0.934	0.103 ± 0.001	9.524 ± 0.934
QML	beta51	0.16	0.162	0.002 ± 0.002	0.015 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.007	0.412 ± 0.011	0.162	0.002	1.40	0.0664	0.0665	0.978	94.833 ± 0.705	0.253 ± 0.001	—
QML	beta52	0.16	0.153	-0.007 ± 0.002	-0.043 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.009	0.449 ± 0.012	0.153	-0.007	-4.09	0.0709	0.0712	0.966	94.124 ± 0.749	0.271 ± 0.001	—
QML	beta53	0.16	0.160	0.000 ± 0.002	0.000 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.376 ± 0.010	0.161	0.001	0.38	0.0608	0.0608	0.983	94.428 ± 0.730	0.232 ± 0.001	—
QML	beta54	0.16	0.152	-0.008 ± 0.002	-0.050 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.197 ± 0.009	0.444 ± 0.012	0.150	-0.010	-6.15	0.0740	0.0746	0.977	95.339 ± 0.671	0.270 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.000	0.000	—	0.0714	0.0714	0.932	92.806 ± 0.822	0.259 ± 0.002	7.194 ± 0.822
QML	beta56	0.00	0.005	0.005 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.005	0.005	—	0.0565	0.0568	0.937	91.591 ± 0.883	0.220 ± 0.002	8.409 ± 0.883
QML	beta57	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.004	0.004	—	0.0316	0.0318	0.912	93.009 ± 0.812	0.124 ± 0.001	6.991 ± 0.812
QML	beta58	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.004	0.004	—	0.0273	0.0276	0.894	91.692 ± 0.879	0.102 ± 0.001	8.308 ± 0.879
UPI	beta41	0.20	0.202	0.002 ± 0.003	0.009 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.155 ± 0.007	0.394 ± 0.011	0.203	0.003	1.51	0.0785	0.0786	0.978	94.732 ± 0.711	0.302 ± 0.001	—
UPI	beta42	0.20	0.206	0.006 ± 0.003	0.030 ± 0.013	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.161 ± 0.007	0.401 ± 0.011	0.204	0.004	2.06	0.0797	0.0798	1.021	96.049 ± 0.620	0.320 ± 0.002	—
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.198	-0.002	-0.98	0.0495	0.0496	0.940	94.225 ± 0.743	0.181 ± 0.001	—
UPI	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.001	-0.001	—	0.0517	0.0517	0.930	92.806 ± 0.822	0.199 ± 0.002	7.194 ± 0.822
UPI	beta45	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.001	-0.001	—	0.0453	0.0453	0.983	94.022 ± 0.755	0.190 ± 0.002	5.978 ± 0.755
UPI	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.002	-0.002	—	0.0285	0.0285	0.930	93.414 ± 0.789	0.111 ± 0.001	6.586 ± 0.789

Table S245

Simulation 2 - Condition 101: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.002	-0.002	—	0.0292	0.0293	0.914	92.300 ± 0.849	0.114 ± 0.001	7.700 ± 0.849
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.007	0.413 ± 0.011	0.162	0.002	1.49	0.0654	0.0655	0.972	94.630 ± 0.718	0.252 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.003	0.012 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.277 ± 0.013	0.526 ± 0.015	0.161	0.001	0.69	0.0823	0.0823	0.968	94.732 ± 0.711	0.320 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.007	0.404 ± 0.011	0.160	0.000	-0.17	0.0650	0.0650	0.974	94.225 ± 0.743	0.247 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.003	0.008 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.276 ± 0.013	0.525 ± 0.015	0.158	-0.002	-1.46	0.0862	0.0862	0.975	95.137 ± 0.685	0.321 ± 0.001	—
UPI	beta55	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	0.005	0.005	—	0.0716	0.0717	0.931	92.908 ± 0.817	0.269 ± 0.002	7.092 ± 0.817
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.001	-0.001	—	0.0606	0.0607	0.937	92.300 ± 0.849	0.234 ± 0.002	7.700 ± 0.849
UPI	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0331	0.0332	0.913	92.604 ± 0.833	0.130 ± 0.001	7.396 ± 0.833
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0290	0.0290	0.881	92.401 ± 0.843	0.110 ± 0.001	7.599 ± 0.843

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S246

Simulation 2 - Condition 102: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.018 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.244 ± 0.007	0.202	0.002	0.82	0.0467	0.0467	0.971	94.795 ± 0.703	0.185 ± 0.001	—
LSAM	beta42	0.20	0.203	0.003 ± 0.002	0.017 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.203	0.003	1.35	0.0525	0.0526	0.971	94.695 ± 0.709	0.194 ± 0.001	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.197	-0.003	-1.31	0.0296	0.0297	1.001	95.395 ± 0.663	0.115 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.001	0.001	—	0.0293	0.0293	0.944	92.793 ± 0.818	0.114 ± 0.001	7.207 ± 0.818
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.002	0.002	—	0.0280	0.0281	0.988	94.595 ± 0.715	0.113 ± 0.001	5.405 ± 0.715
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.002	-0.002	—	0.0153	0.0154	0.959	94.294 ± 0.734	0.064 ± 0.001	5.706 ± 0.734
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0169	0.0170	0.904	93.093 ± 0.802	0.064 ± 0.001	6.907 ± 0.802
LSAM	beta51	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.159	-0.001	-0.54	0.0397	0.0398	0.998	94.995 ± 0.690	0.159 ± 0.000	—
LSAM	beta52	0.16	0.164	0.004 ± 0.002	0.027 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.166	0.006	3.49	0.0518	0.0521	0.987	94.895 ± 0.696	0.198 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.242 ± 0.006	0.158	-0.002	-1.40	0.0385	0.0386	1.031	96.096 ± 0.613	0.157 ± 0.000	—
LSAM	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.096 ± 0.004	0.309 ± 0.009	0.162	0.002	1.24	0.0466	0.0466	1.018	94.995 ± 0.690	0.198 ± 0.001	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.005	0.005	—	0.0408	0.0411	0.968	93.493 ± 0.780	0.161 ± 0.001	6.507 ± 0.780
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0355	0.0356	0.997	94.795 ± 0.703	0.142 ± 0.001	5.205 ± 0.703
LSAM	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0187	0.0190	0.943	93.794 ± 0.763	0.076 ± 0.001	6.206 ± 0.763
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.001	-0.001	—	0.0179	0.0179	0.920	92.492 ± 0.834	0.065 ± 0.001	7.508 ± 0.834
QML	beta41	0.20	0.193	-0.007 ± 0.001	-0.037 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.192	-0.008	-4.03	0.0389	0.0397	0.979	93.894 ± 0.758	0.155 ± 0.001	—
QML	beta42	0.20	0.193	-0.007 ± 0.001	-0.036 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.045 ± 0.002	0.212 ± 0.006	0.193	-0.007	-3.41	0.0425	0.0430	0.975	94.494 ± 0.722	0.159 ± 0.000	—
QML	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.200	0.000	-0.02	0.0289	0.0289	1.002	95.495 ± 0.656	0.113 ± 0.000	—
QML	beta44	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0299	0.0300	0.948	93.193 ± 0.797	0.116 ± 0.001	6.807 ± 0.797
QML	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0279	0.0279	0.989	94.795 ± 0.703	0.111 ± 0.001	5.205 ± 0.703
QML	beta46	0.00	0.005	0.005 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.005	0.005	—	0.0157	0.0165	0.975	93.093 ± 0.802	0.061 ± 0.001	6.907 ± 0.802
QML	beta47	0.00	0.005	0.005 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.000	—	—	0.005	0.005	—	0.0158	0.0165	0.908	91.491 ± 0.883	0.060 ± 0.001	8.509 ± 0.883
QML	beta51	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.159	-0.001	-0.65	0.0401	0.0401	1.001	94.995 ± 0.690	0.159 ± 0.000	—
QML	beta52	0.16	0.155	-0.005 ± 0.001	-0.032 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.274 ± 0.008	0.156	-0.004	-2.53	0.0441	0.0443	0.985	93.994 ± 0.752	0.168 ± 0.000	—
QML	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.158	-0.002	-0.95	0.0357	0.0357	1.037	95.796 ± 0.635	0.146 ± 0.000	—
QML	beta54	0.16	0.152	-0.008 ± 0.001	-0.053 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.153	-0.007	-4.40	0.0397	0.0403	1.025	94.394 ± 0.728	0.168 ± 0.000	—
QML	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.002	0.002	—	0.0408	0.0409	0.976	94.194 ± 0.740	0.160 ± 0.001	5.806 ± 0.740
QML	beta56	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.006	0.006	—	0.0333	0.0339	0.998	94.294 ± 0.734	0.136 ± 0.001	5.706 ± 0.734
QML	beta57	0.00	0.004	0.004 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.002	0.002	—	0.0179	0.0181	0.947	93.093 ± 0.802	0.074 ± 0.001	6.907 ± 0.802
QML	beta58	0.00	0.005	0.005 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.000	—	—	0.004	0.004	—	0.0172	0.0178	0.918	91.592 ± 0.878	0.062 ± 0.001	8.408 ± 0.878
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.016 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.202	0.002	1.00	0.0462	0.0462	0.969	95.095 ± 0.683	0.184 ± 0.001	—
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.015 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.064 ± 0.003	0.252 ± 0.007	0.202	0.002	1.05	0.0520	0.0520	0.975	94.595 ± 0.715	0.193 ± 0.001	—
UPI	beta43	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.197	-0.003	-1.47	0.0297	0.0299	1.001	94.294 ± 0.734	0.114 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.001	0.001	—	0.0295	0.0295	0.942	93.393 ± 0.786	0.115 ± 0.001	6.607 ± 0.786

Table S246

Simulation 2 - Condition 102: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0272	0.0273	0.994	95.095 ± 0.683	0.113 ± 0.001	4.905 ± 0.683
UPI	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0159	0.0159	0.956	93.994 ± 0.752	0.063 ± 0.001	6.006 ± 0.752
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.001	-0.001	—	0.0169	0.0170	0.909	93.193 ± 0.797	0.063 ± 0.001	6.807 ± 0.797
UPI	beta51	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.159	-0.001	-0.87	0.0403	0.0403	1.000	95.195 ± 0.677	0.159 ± 0.000	—
UPI	beta52	0.16	0.164	0.004 ± 0.002	0.024 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.165	0.005	3.32	0.0504	0.0507	0.988	95.395 ± 0.663	0.196 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.158	-0.002	-1.10	0.0381	0.0381	1.023	95.696 ± 0.642	0.155 ± 0.000	—
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.309 ± 0.009	0.161	0.001	0.79	0.0475	0.0475	1.017	94.394 ± 0.728	0.197 ± 0.001	—
UPI	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.004	0.004	—	0.0413	0.0416	0.966	94.094 ± 0.746	0.162 ± 0.001	5.906 ± 0.746
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	—	0.0357	0.0357	0.993	94.795 ± 0.703	0.142 ± 0.001	5.205 ± 0.703
UPI	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0196	0.0198	0.938	94.494 ± 0.722	0.076 ± 0.001	5.506 ± 0.722
UPI	beta58	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.000	0.000	—	0.0177	0.0177	0.918	92.793 ± 0.818	0.065 ± 0.001	7.207 ± 0.818

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S247

Simulation 2 - Condition 103: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.254	0.054 ± 0.028	0.271 ± 0.140	0.010 ± 0.004	0.101 ± 0.027	0.012 ± 0.006	0.111 ± 0.035	0.309 ± 0.152	0.556 ± 0.175	0.241	0.041	20.70	0.1221	0.1289	1.335	100.000 ± 0.000	0.529 ± 0.058	—
LSAM	beta42	0.20	0.199	-0.001 ± 0.033	-0.007 ± 0.166	0.014 ± 0.004	0.120 ± 0.021	0.013 ± 0.004	0.115 ± 0.026	0.331 ± 0.091	0.575 ± 0.130	0.192	-0.008	-4.09	0.1576	0.1578	1.121	100.000 ± 0.000	0.526 ± 0.038	—
LSAM	beta43	0.20	0.145	-0.055 ± 0.022	-0.275 ± 0.108	0.006 ± 0.002	0.078 ± 0.013	0.009 ± 0.003	0.093 ± 0.019	0.217 ± 0.073	0.466 ± 0.097	0.147	-0.053	-26.56	0.0621	0.0818	0.995	84.615 ± 10.007	0.305 ± 0.015	—
LSAM	beta44	0.00	0.036	0.036 ± 0.038	—	0.019 ± 0.006	0.136 ± 0.032	0.018 ± 0.005	0.136 ± 0.031	—	—	0.047	0.047	—	0.1046	0.1146	0.923	92.308 ± 7.391	0.493 ± 0.050	7.692 ± 7.391
LSAM	beta45	0.00	0.002	0.002 ± 0.024	—	0.007 ± 0.002	0.085 ± 0.015	0.007 ± 0.002	0.082 ± 0.019	—	—	0.000	0.000	—	0.0740	0.0740	1.066	100.000 ± 0.000	0.356 ± 0.027	0.000 ± 0.000
LSAM	beta46	0.00	-0.096	-0.096 ± 0.036	—	0.017 ± 0.006	0.129 ± 0.034	0.024 ± 0.010	0.156 ± 0.040	—	—	-0.074	-0.074	—	0.1290	0.1486	1.025	100.000 ± 0.000	0.518 ± 0.080	0.000 ± 0.000
LSAM	beta47	0.00	-0.046	-0.046 ± 0.039	—	0.020 ± 0.008	0.142 ± 0.039	0.021 ± 0.007	0.144 ± 0.036	—	—	-0.027	-0.027	—	0.1546	0.1570	0.997	84.615 ± 10.007	0.555 ± 0.051	15.385 ± 10.007
LSAM	beta51	0.16	0.092	-0.068 ± 0.038	-0.423 ± 0.235	0.018 ± 0.006	0.136 ± 0.030	0.022 ± 0.008	0.147 ± 0.036	0.842 ± 0.317	0.918 ± 0.227	0.102	-0.058	-36.47	0.1281	0.1407	0.934	92.308 ± 7.391	0.497 ± 0.026	—
LSAM	beta52	0.16	0.171	0.011 ± 0.031	0.069 ± 0.193	0.012 ± 0.002	0.111 ± 0.015	0.012 ± 0.003	0.108 ± 0.022	0.452 ± 0.103	0.672 ± 0.137	0.163	0.003	1.98	0.0977	0.0977	1.358	100.000 ± 0.000	0.593 ± 0.114	—
LSAM	beta53	0.16	0.186	0.026 ± 0.027	0.164 ± 0.166	0.009 ± 0.002	0.096 ± 0.015	0.009 ± 0.003	0.096 ± 0.021	0.356 ± 0.101	0.597 ± 0.129	0.169	0.009	5.91	0.1016	0.1020	1.073	100.000 ± 0.000	0.402 ± 0.018	—
LSAM	beta54	0.16	0.229	0.069 ± 0.031	0.433 ± 0.194	0.013 ± 0.003	0.112 ± 0.016	0.016 ± 0.005	0.128 ± 0.026	0.640 ± 0.202	0.800 ± 0.160	0.190	0.030	18.62	0.1509	0.1538	1.261	100.000 ± 0.000	0.554 ± 0.062	—
LSAM	beta55	0.00	0.124	0.124 ± 0.055	—	0.040 ± 0.009	0.199 ± 0.033	0.052 ± 0.020	0.228 ± 0.054	—	—	0.108	0.108	—	0.1584	0.1917	1.153	100.000 ± 0.000	0.900 ± 0.189	0.000 ± 0.000
LSAM	beta56	0.00	-0.069	-0.069 ± 0.041	—	0.021 ± 0.006	0.147 ± 0.027	0.025 ± 0.006	0.157 ± 0.028	—	—	-0.065	-0.065	—	0.1353	0.1501	1.306	100.000 ± 0.000	0.750 ± 0.081	0.000 ± 0.000
LSAM	beta57	0.00	-0.076	-0.076 ± 0.029	—	0.011 ± 0.004	0.105 ± 0.024	0.016 ± 0.007	0.126 ± 0.035	—	—	-0.055	-0.055	—	0.1077	0.1207	1.567	100.000 ± 0.000	0.648 ± 0.159	0.000 ± 0.000
LSAM	beta58	0.00	-0.075	-0.075 ± 0.029	—	0.011 ± 0.004	0.105 ± 0.028	0.016 ± 0.008	0.126 ± 0.038	—	—	-0.052	-0.052	—	0.0974	0.1105	1.133	100.000 ± 0.000	0.467 ± 0.060	0.000 ± 0.000
QML	beta41	0.20	0.203	0.003 ± 0.016	0.017 ± 0.082	0.003 ± 0.001	0.059 ± 0.014	0.003 ± 0.001	0.057 ± 0.016	0.081 ± 0.031	0.284 ± 0.078	0.195	-0.005	-2.59	0.0506	0.0508	1.179	100.000 ± 0.000	0.273 ± 0.011	—
QML	beta42	0.20	0.189	-0.011 ± 0.019	-0.056 ± 0.093	0.004 ± 0.001	0.067 ± 0.012	0.004 ± 0.001	0.065 ± 0.015	0.107 ± 0.031	0.326 ± 0.074	0.200	0.000	0.02	0.0707	0.0707	1.134	100.000 ± 0.000	0.298 ± 0.010	—
QML	beta43	0.20	0.155	-0.045 ± 0.020	-0.223 ± 0.100	0.005 ± 0.002	0.072 ± 0.016	0.007 ± 0.003	0.082 ± 0.021	0.169 ± 0.069	0.411 ± 0.106	0.156	-0.044	-21.97	0.0708	0.0834	0.985	84.615 ± 10.007	0.277 ± 0.014	—
QML	beta44	0.00	0.021	0.021 ± 0.032	—	0.013 ± 0.004	0.115 ± 0.023	0.013 ± 0.003	0.112 ± 0.025	—	—	0.027	0.027	—	0.1037	0.1072	0.786	92.308 ± 7.391	0.354 ± 0.021	7.692 ± 7.391
QML	beta45	0.00	-0.016	-0.016 ± 0.021	—	0.006 ± 0.001	0.077 ± 0.013	0.006 ± 0.002	0.076 ± 0.018	—	—	0.001	0.001	—	0.0684	0.0684	0.971	92.308 ± 7.391	0.294 ± 0.013	7.692 ± 7.391
QML	beta46	0.00	-0.037	-0.037 ± 0.022	—	0.006 ± 0.002	0.080 ± 0.019	0.007 ± 0.002	0.085 ± 0.015	—	—	-0.051	-0.051	—	0.0516	0.0724	0.768	76.923 ± 11.685	0.240 ± 0.017	23.077 ± 11.685
QML	beta47	0.00	-0.046	-0.046 ± 0.019	—	0.005 ± 0.001	0.070 ± 0.014	0.007 ± 0.003	0.082 ± 0.020	—	—	-0.045	-0.045	—	0.0895	0.1002	0.936	84.615 ± 10.007	0.257 ± 0.022	15.385 ± 10.007
QML	beta51	0.16	0.132	-0.028 ± 0.033	-0.178 ± 0.209	0.014 ± 0.006	0.120 ± 0.034	0.014 ± 0.006	0.119 ± 0.037	0.554 ± 0.250	0.745 ± 0.230	0.126	-0.034	-21.15	0.1030	0.1084	1.014	92.308 ± 7.391	0.479 ± 0.031	—
QML	beta52	0.16	0.134	-0.026 ± 0.021	-0.165 ± 0.129	0.006 ± 0.001	0.074 ± 0.013	0.006 ± 0.002	0.076 ± 0.015	0.226 ± 0.059	0.475 ± 0.095	0.153	-0.007	-4.25	0.1073	0.1076	1.188	100.000 ± 0.000	0.346 ± 0.024	—
QML	beta53	0.16	0.192	0.032 ± 0.029	0.201 ± 0.179	0.011 ± 0.003	0.103 ± 0.018	0.011 ± 0.004	0.104 ± 0.024	0.425 ± 0.137	0.652 ± 0.152	0.185	0.025	15.68	0.1155	0.1182	0.855	92.308 ± 7.391	0.346 ± 0.010	—
QML	beta54	0.16	0.183	0.023 ± 0.023	0.143 ± 0.144	0.007 ± 0.003	0.083 ± 0.028	0.007 ± 0.004	0.083 ± 0.032	0.269 ± 0.152	0.518 ± 0.200	0.173	0.013	8.01	0.0774	0.0784	0.957	92.308 ± 7.391	0.311 ± 0.013	—
QML	beta55	0.00	0.053	0.053 ± 0.035	—	0.016 ± 0.005	0.126 ± 0.025	0.017 ± 0.007	0.132 ± 0.035	—	—	0.095	0.095	—	0.1633	0.1891	1.114	100.000 ± 0.000	0.550 ± 0.061	0.000 ± 0.000
QML	beta56	0.00	-0.043	-0.043 ± 0.031	—	0.012 ± 0.003	0.110 ± 0.019	0.013 ± 0.003	0.115 ± 0.022	—	—	-0.030	-0.030	—	0.1360	0.1392	1.146	100.000 ± 0.000	0.496 ± 0.038	0.000 ± 0.000
QML	beta57	0.00	-0.044	-0.044 ± 0.014	—	0.002 ± 0.001	0.050 ± 0.013	0.004 ± 0.002	0.065 ± 0.017	—	—	-0.051	-0.051	—	0.0713	0.0874	1.342	92.308 ± 7.391	0.261 ± 0.022	7.692 ± 7.391
QML	beta58	0.00	-0.025	-0.025 ± 0.018	—	0.004 ± 0.001	0.064 ± 0.014	0.004 ± 0.001	0.066 ± 0.013	—	—	-0.034	-0.034	—	0.0500	0.0605	0.913	100.000 ± 0.000	0.230 ± 0.024	0.000 ± 0.000
UPI	beta41	0.20	0.193	-0.007 ± 0.074	-0.037 ± 0.372	0.072 ± 0.025	0.268 ± 0.065	0.067 ± 0.025	0.258 ± 0.069	1.664 ± 0.614	1.290 ± 0.346	0.182	-0.018	-8.78	0.1985	0.1993	1.510	84.615 ± 10.007	1.589 ± 0.622	—
UPI	beta42	0.20	0.382	0.182 ± 0.120	0.910 ± 0.602	0.188 ± 0.061	0.434 ± 0.099	0.207 ± 0.096	0.455 ± 0.136	5.176 ± 2.391	2.275 ± 0.681	0.268	0.068	34.19	0.4271	0.4325	1.915	92.308 ± 7.391	3.258 ± 1.432	—
UPI	beta43	0.20	0.101	-0.099 ± 0.024	-0.495 ± 0.121	0.008 ± 0.001	0.087 ± 0.009	0.017 ± 0.005	0.130 ± 0.020	0.420 ± 0.120	0.648 ± 0.101	0.037	-0.163	-81.34	0.1481	0.2200	1.739	61.538 ± 13.493	0.593 ± 0.207	—
UPI	beta44	0.00	-0.068	-0.068 ± 0.054	—	0.039 ± 0.018	0.196 ± 0.066	0.040 ± 0.023	0.201 ± 0.077	—	—	0.019	0.019	—	0.2080	0.2089	3.050	100.000 ± 0.000	2.348 ± 1.499	0.000 ± 0.000
UPI	beta45	0.00	0.015	0.015 ± 0.032	—	0.013 ± 0.003	0.114 ± 0.021	0.012 ± 0.004	0.110 ± 0.025	—	—	-0.022	-0.022	—	0.1690	0.1705	2.752	100.000 ± 0.000	1.227 ± 0.620	0.000 ± 0.000
UPI	beta46	0.00	-0.129	-0.129 ± 0.112	—	0.163 ± 0.074	0.404 ± 0.130	0.167 ± 0.069	0.409 ± 0.114	—	—	-0.146	-0.146	—	0.4549	0.4778	1.774	100.000 ± 0.000	2.807 ± 1.207	0.000 ± 0.000

Table S247

Simulation 2 - Condition 103: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.513	-0.513 ± 0.254	—	0.840 ± 0.252	0.917 ± 0.193	1.039 ± 0.488	1.019 ± 0.295	—	—	-0.048	-0.048	—	0.3680	0.3711	2.434	100.000 ± 0.000	8.747 ± 3.291	0.000 ± 0.000
UPI	beta51	0.16	0.133	-0.027 ± 0.055	-0.170 ± 0.344	0.039 ± 0.019	0.198 ± 0.070	0.037 ± 0.021	0.193 ± 0.075	1.448 ± 0.805	1.203 ± 0.468	0.146	-0.014	-9.03	0.1252	0.1261	1.294	100.000 ± 0.000	1.006 ± 0.300	—
UPI	beta52	0.16	0.273	0.113 ± 0.063	0.709 ± 0.392	0.051 ± 0.016	0.226 ± 0.048	0.060 ± 0.024	0.245 ± 0.064	2.343 ± 0.933	1.531 ± 0.402	0.234	0.074	46.13	0.3079	0.3166	1.922	100.000 ± 0.000	1.702 ± 0.543	—
UPI	beta53	0.16	0.094	-0.066 ± 0.040	-0.412 ± 0.250	0.021 ± 0.006	0.144 ± 0.030	0.024 ± 0.009	0.153 ± 0.038	0.919 ± 0.339	0.959 ± 0.236	0.093	-0.067	-42.05	0.1298	0.1462	2.562	92.308 ± 7.391	1.447 ± 0.540	—
UPI	beta54	0.16	0.317	0.157 ± 0.135	0.982 ± 0.845	0.238 ± 0.088	0.487 ± 0.125	0.244 ± 0.126	0.494 ± 0.164	9.530 ± 4.939	3.087 ± 1.025	0.118	-0.042	-26.06	0.1930	0.1974	1.506	84.615 ± 10.007	2.877 ± 1.223	—
UPI	beta55	0.00	0.080	0.080 ± 0.090	—	0.105 ± 0.035	0.324 ± 0.075	0.103 ± 0.039	0.321 ± 0.085	—	—	0.064	0.064	—	0.2387	0.2471	1.622	100.000 ± 0.000	2.057 ± 0.598	0.000 ± 0.000
UPI	beta56	0.00	-0.124	-0.124 ± 0.061	—	0.048 ± 0.026	0.220 ± 0.087	0.060 ± 0.035	0.245 ± 0.097	—	—	-0.095	-0.095	—	0.2309	0.2498	2.865	100.000 ± 0.000	2.467 ± 0.922	0.000 ± 0.000
UPI	beta57	0.00	-0.199	-0.199 ± 0.097	—	0.122 ± 0.055	0.349 ± 0.109	0.152 ± 0.062	0.390 ± 0.103	—	—	-0.049	-0.049	—	0.4194	0.4223	2.572	100.000 ± 0.000	3.520 ± 1.237	0.000 ± 0.000
UPI	beta58	0.00	-0.227	-0.227 ± 0.126	—	0.206 ± 0.090	0.454 ± 0.146	0.242 ± 0.138	0.492 ± 0.184	—	—	-0.010	-0.010	—	0.3856	0.3857	2.322	100.000 ± 0.000	4.134 ± 1.585	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S248

Simulation 2 - Condition 104: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.020	0.015 ± 0.102	0.004 ± 0.001	0.064 ± 0.009	0.004 ± 0.001	0.061 ± 0.013	0.093 ± 0.016	0.305 ± 0.067	0.226	0.026	13.00	0.0651	0.0701	1.173	100.000 ± 0.000	0.295 ± 0.020	—
LSAM	beta42	0.20	0.246	0.046 ± 0.024	0.232 ± 0.121	0.006 ± 0.002	0.076 ± 0.025	0.007 ± 0.004	0.086 ± 0.031	0.185 ± 0.101	0.430 ± 0.157	0.240	0.040	20.08	0.0699	0.0806	0.994	100.000 ± 0.000	0.298 ± 0.032	—
LSAM	beta43	0.20	0.175	-0.025 ± 0.019	-0.127 ± 0.097	0.004 ± 0.001	0.061 ± 0.009	0.004 ± 0.001	0.064 ± 0.015	0.101 ± 0.034	0.318 ± 0.077	0.197	-0.003	-1.58	0.0463	0.0464	0.859	80.000 ± 12.649	0.207 ± 0.013	—
LSAM	beta44	0.00	0.042	0.042 ± 0.024	—	0.006 ± 0.001	0.076 ± 0.013	0.007 ± 0.003	0.084 ± 0.023	—	—	0.074	0.074	—	0.0917	0.1180	1.093	100.000 ± 0.000	0.327 ± 0.029	0.000 ± 0.000
LSAM	beta45	0.00	0.014	0.014 ± 0.014	—	0.002 ± 0.000	0.046 ± 0.008	0.002 ± 0.001	0.046 ± 0.012	—	—	0.010	0.010	—	0.0678	0.0685	1.367	100.000 ± 0.000	0.244 ± 0.015	0.000 ± 0.000
LSAM	beta46	0.00	-0.099	-0.099 ± 0.026	—	0.007 ± 0.003	0.083 ± 0.026	0.016 ± 0.006	0.127 ± 0.027	—	—	-0.094	-0.094	—	0.0600	0.1118	0.996	90.000 ± 9.487	0.324 ± 0.065	10.000 ± 9.487
LSAM	beta47	0.00	-0.057	-0.057 ± 0.025	—	0.006 ± 0.003	0.078 ± 0.031	0.009 ± 0.005	0.094 ± 0.038	—	—	-0.075	-0.075	—	0.0806	0.1098	0.969	90.000 ± 9.487	0.298 ± 0.041	10.000 ± 9.487
LSAM	beta51	0.16	0.166	0.006 ± 0.015	0.038 ± 0.096	0.002 ± 0.001	0.048 ± 0.014	0.002 ± 0.001	0.046 ± 0.014	0.084 ± 0.031	0.289 ± 0.086	0.172	0.012	7.62	0.0491	0.0506	1.546	100.000 ± 0.000	0.293 ± 0.010	—
LSAM	beta52	0.16	0.206	0.046 ± 0.017	0.285 ± 0.107	0.003 ± 0.001	0.054 ± 0.018	0.005 ± 0.002	0.069 ± 0.023	0.183 ± 0.096	0.428 ± 0.145	0.203	0.043	27.00	0.0686	0.0810	1.360	90.000 ± 9.487	0.287 ± 0.016	—
LSAM	beta53	0.16	0.109	-0.051 ± 0.014	-0.316 ± 0.087	0.002 ± 0.001	0.044 ± 0.012	0.004 ± 0.002	0.066 ± 0.014	0.168 ± 0.062	0.410 ± 0.088	0.123	-0.037	-23.25	0.0480	0.0607	1.250	90.000 ± 9.487	0.216 ± 0.006	—
LSAM	beta54	0.16	0.172	0.012 ± 0.020	0.077 ± 0.125	0.004 ± 0.001	0.063 ± 0.014	0.004 ± 0.001	0.061 ± 0.016	0.146 ± 0.046	0.383 ± 0.099	0.178	0.018	11.46	0.0734	0.0756	1.128	100.000 ± 0.000	0.280 ± 0.018	—
LSAM	beta55	0.00	0.056	0.056 ± 0.021	—	0.005 ± 0.001	0.068 ± 0.015	0.007 ± 0.003	0.085 ± 0.022	—	—	0.052	0.052	—	0.0642	0.0825	1.482	100.000 ± 0.000	0.394 ± 0.025	0.000 ± 0.000
LSAM	beta56	0.00	-0.020	-0.020 ± 0.025	—	0.006 ± 0.001	0.078 ± 0.013	0.006 ± 0.001	0.076 ± 0.017	—	—	-0.019	-0.019	—	0.0689	0.0716	1.092	100.000 ± 0.000	0.334 ± 0.020	0.000 ± 0.000
LSAM	beta57	0.00	-0.068	-0.068 ± 0.010	—	0.001 ± 0.000	0.032 ± 0.008	0.006 ± 0.002	0.075 ± 0.012	—	—	-0.062	-0.062	—	0.0386	0.0727	2.266	90.000 ± 9.487	0.286 ± 0.046	10.000 ± 9.487
LSAM	beta58	0.00	-0.028	-0.028 ± 0.031	—	0.010 ± 0.003	0.098 ± 0.022	0.009 ± 0.004	0.097 ± 0.027	—	—	-0.054	-0.054	—	0.0979	0.1117	0.790	80.000 ± 12.649	0.304 ± 0.042	20.000 ± 12.649
QML	beta41	0.20	0.172	-0.028 ± 0.017	-0.141 ± 0.083	0.003 ± 0.001	0.052 ± 0.012	0.003 ± 0.001	0.057 ± 0.014	0.082 ± 0.028	0.286 ± 0.069	0.172	-0.028	-14.16	0.0683	0.0739	0.929	90.000 ± 9.487	0.191 ± 0.006	—
QML	beta42	0.20	0.225	0.025 ± 0.014	0.123 ± 0.072	0.002 ± 0.001	0.046 ± 0.011	0.002 ± 0.001	0.050 ± 0.012	0.062 ± 0.022	0.250 ± 0.060	0.214	0.014	7.04	0.0499	0.0519	1.085	100.000 ± 0.000	0.195 ± 0.007	—
QML	beta43	0.20	0.194	-0.006 ± 0.022	-0.031 ± 0.111	0.005 ± 0.001	0.070 ± 0.014	0.005 ± 0.001	0.067 ± 0.017	0.113 ± 0.032	0.336 ± 0.086	0.207	0.007	3.72	0.0590	0.0594	0.675	80.000 ± 12.649	0.187 ± 0.006	—
QML	beta44	0.00	0.012	0.012 ± 0.014	—	0.002 ± 0.001	0.045 ± 0.013	0.002 ± 0.001	0.045 ± 0.016	—	—	0.030	0.030	—	0.0605	0.0674	1.432	100.000 ± 0.000	0.255 ± 0.014	0.000 ± 0.000
QML	beta45	0.00	0.002	0.002 ± 0.010	—	0.001 ± 0.000	0.030 ± 0.006	0.001 ± 0.000	0.029 ± 0.007	—	—	0.014	0.014	—	0.0383	0.0409	1.758	100.000 ± 0.000	0.208 ± 0.008	0.000 ± 0.000
QML	beta46	0.00	-0.059	-0.059 ± 0.009	—	0.001 ± 0.000	0.030 ± 0.008	0.004 ± 0.001	0.065 ± 0.008	—	—	-0.055	-0.055	—	0.0478	0.0726	1.476	70.000 ± 14.491	0.173 ± 0.015	30.000 ± 14.491
QML	beta47	0.00	-0.021	-0.021 ± 0.013	—	0.002 ± 0.000	0.040 ± 0.008	0.002 ± 0.001	0.043 ± 0.012	—	—	-0.050	-0.050	—	0.0555	0.0746	1.112	100.000 ± 0.000	0.173 ± 0.008	0.000 ± 0.000
QML	beta51	0.16	0.171	0.011 ± 0.015	0.071 ± 0.093	0.002 ± 0.001	0.047 ± 0.012	0.002 ± 0.001	0.046 ± 0.012	0.083 ± 0.024	0.288 ± 0.072	0.179	0.019	11.90	0.0434	0.0474	1.573	100.000 ± 0.000	0.291 ± 0.009	—
QML	beta52	0.16	0.181	0.021 ± 0.014	0.130 ± 0.086	0.002 ± 0.001	0.043 ± 0.015	0.002 ± 0.001	0.046 ± 0.019	0.083 ± 0.048	0.288 ± 0.116	0.166	0.006	4.05	0.0570	0.0574	1.218	90.000 ± 9.487	0.207 ± 0.008	—
QML	beta53	0.16	0.120	-0.040 ± 0.012	-0.249 ± 0.077	0.002 ± 0.000	0.039 ± 0.007	0.003 ± 0.001	0.054 ± 0.012	0.116 ± 0.043	0.340 ± 0.073	0.140	-0.020	-12.52	0.0497	0.0536	1.361	100.000 ± 0.000	0.208 ± 0.006	—
QML	beta54	0.16	0.164	0.004 ± 0.012	0.025 ± 0.075	0.001 ± 0.001	0.038 ± 0.013	0.001 ± 0.001	0.036 ± 0.012	0.051 ± 0.023	0.226 ± 0.077	0.165	0.005	2.83	0.0509	0.0511	1.347	100.000 ± 0.000	0.200 ± 0.007	—
QML	beta55	0.00	0.044	0.044 ± 0.018	—	0.003 ± 0.001	0.055 ± 0.020	0.005 ± 0.003	0.069 ± 0.025	—	—	0.048	0.048	—	0.0909	0.1028	1.563	90.000 ± 9.487	0.339 ± 0.018	10.000 ± 9.487
QML	beta56	0.00	-0.025	-0.025 ± 0.023	—	0.005 ± 0.002	0.074 ± 0.021	0.006 ± 0.002	0.074 ± 0.019	—	—	-0.033	-0.033	—	0.0649	0.0726	1.031	100.000 ± 0.000	0.298 ± 0.015	0.000 ± 0.000
QML	beta57	0.00	-0.036	-0.036 ± 0.008	—	0.001 ± 0.000	0.026 ± 0.004	0.002 ± 0.001	0.043 ± 0.008	—	—	-0.042	-0.042	—	0.0388	0.0571	1.648	100.000 ± 0.000	0.167 ± 0.011	0.000 ± 0.000
QML	beta58	0.00	-0.019	-0.019 ± 0.018	—	0.003 ± 0.001	0.056 ± 0.010	0.003 ± 0.001	0.056 ± 0.014	—	—	-0.051	-0.051	—	0.0659	0.0831	0.725	80.000 ± 12.649	0.159 ± 0.013	20.000 ± 12.649
UPI	beta41	0.20	0.269	0.069 ± 0.100	0.343 ± 0.500	0.100 ± 0.038	0.316 ± 0.093	0.095 ± 0.049	0.308 ± 0.112	2.369 ± 1.213	1.539 ± 0.560	0.294	0.094	46.86	0.4217	0.4320	1.023	100.000 ± 0.000	1.269 ± 0.236	—
UPI	beta42	0.20	0.375	0.175 ± 0.128	0.873 ± 0.642	0.165 ± 0.057	0.406 ± 0.106	0.179 ± 0.097	0.423 ± 0.150	4.471 ± 2.432	2.115 ± 0.749	0.163	-0.037	-18.52	0.3637	0.3656	0.801	90.000 ± 9.487	1.274 ± 0.320	—
UPI	beta43	0.20	0.152	-0.048 ± 0.059	-0.238 ± 0.296	0.035 ± 0.011	0.187 ± 0.044	0.034 ± 0.010	0.184 ± 0.047	0.846 ± 0.261	0.920 ± 0.234	0.018	-0.182	-91.22	0.1779	0.2548	0.725	70.000 ± 14.491	0.532 ± 0.106	—
UPI	beta44	0.00	0.024	0.024 ± 0.048	—	0.023 ± 0.008	0.153 ± 0.038	0.022 ± 0.008	0.147 ± 0.043	—	—	0.038	0.038	—	0.2207	0.2240	1.229	100.000 ± 0.000	0.738 ± 0.098	0.000 ± 0.000
UPI	beta45	0.00	0.019	0.019 ± 0.026	—	0.007 ± 0.003	0.082 ± 0.024	0.006 ± 0.002	0.080 ± 0.022	—	—	-0.012	-0.012	—	0.1564	0.1569	1.942	100.000 ± 0.000	0.623 ± 0.081	0.000 ± 0.000
UPI	beta46	0.00	-0.495	-0.495 ± 0.222	—	0.495 ± 0.304	0.703 ± 0.400	0.690 ± 0.509	0.831 ± 0.445	—	—	-0.409	-0.409	—	0.5811	0.7106	1.384	100.000 ± 0.000	3.814 ± 2.501	0.000 ± 0.000

Table S248

Simulation 2 - Condition 104: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.216	-0.216 ± 0.099	—	0.098 ± 0.021	0.314 ± 0.054	0.135 ± 0.055	0.368 ± 0.092	—	—	-0.064	-0.064	—	0.3255	0.3317	1.603	100.000 ± 0.000	1.970 ± 0.817	0.000 ± 0.000
UPI	beta51	0.16	0.229	0.069 ± 0.033	0.431 ± 0.207	0.011 ± 0.003	0.105 ± 0.020	0.015 ± 0.003	0.121 ± 0.021	0.572 ± 0.134	0.756 ± 0.130	0.279	0.119	74.55	0.1734	0.2105	1.505	70.000 ± 14.491	0.619 ± 0.164	—
UPI	beta52	0.16	0.236	0.076 ± 0.068	0.472 ± 0.428	0.047 ± 0.022	0.217 ± 0.085	0.048 ± 0.030	0.219 ± 0.099	1.873 ± 1.177	1.368 ± 0.621	0.211	0.051	31.96	0.2525	0.2576	1.092	100.000 ± 0.000	0.928 ± 0.176	—
UPI	beta53	0.16	0.094	-0.066 ± 0.019	-0.411 ± 0.121	0.004 ± 0.001	0.061 ± 0.014	0.008 ± 0.002	0.088 ± 0.013	0.302 ± 0.080	0.549 ± 0.084	0.090	-0.070	-43.71	0.0886	0.1128	1.386	70.000 ± 14.491	0.334 ± 0.068	—
UPI	beta54	0.16	0.170	0.010 ± 0.058	0.063 ± 0.364	0.034 ± 0.014	0.184 ± 0.059	0.031 ± 0.015	0.175 ± 0.063	1.200 ± 0.573	1.095 ± 0.391	0.070	-0.090	-56.38	0.2674	0.2822	1.051	90.000 ± 9.487	0.760 ± 0.348	—
UPI	beta55	0.00	0.024	0.024 ± 0.033	—	0.011 ± 0.005	0.103 ± 0.043	0.010 ± 0.005	0.100 ± 0.034	—	—	0.027	0.027	—	0.1287	0.1316	1.729	100.000 ± 0.000	0.697 ± 0.121	0.000 ± 0.000
UPI	beta56	0.00	-0.018	-0.018 ± 0.032	—	0.010 ± 0.004	0.101 ± 0.026	0.010 ± 0.004	0.098 ± 0.028	—	—	-0.005	-0.005	—	0.1856	0.1857	1.749	90.000 ± 9.487	0.693 ± 0.131	10.000 ± 9.487
UPI	beta57	0.00	-0.377	-0.377 ± 0.233	—	0.541 ± 0.335	0.736 ± 0.424	0.629 ± 0.498	0.793 ± 0.480	—	—	-0.333	-0.333	—	0.7266	0.7994	1.246	100.000 ± 0.000	3.592 ± 2.511	0.000 ± 0.000
UPI	beta58	0.00	-0.057	-0.057 ± 0.042	—	0.018 ± 0.006	0.133 ± 0.038	0.019 ± 0.010	0.139 ± 0.051	—	—	0.006	0.006	—	0.1953	0.1953	2.288	100.000 ± 0.000	1.194 ± 0.480	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S249

Simulation 2 - Condition 105: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.003	0.010 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.314 ± 0.011	0.200	0.000	0.18	0.0658	0.0658	0.988	95.126 ± 0.883	0.244 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.003	0.021 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	0.109 ± 0.007	0.330 ± 0.013	0.204	0.004	1.79	0.0661	0.0662	0.992	94.790 ± 0.911	0.256 ± 0.002	—
LSAM	beta43	0.20	0.193	-0.007 ± 0.002	-0.037 ± 0.011	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.010	0.193	-0.007	-3.70	0.0597	0.0602	1.045	96.303 ± 0.774	0.227 ± 0.001	—
LSAM	beta44	0.00	0.015	0.015 ± 0.003	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.013	0.013	—	0.0701	0.0713	0.970	94.790 ± 0.911	0.282 ± 0.003	5.210 ± 0.911
LSAM	beta45	0.00	0.007	0.007 ± 0.003	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.005	0.005	—	0.0654	0.0657	0.981	94.622 ± 0.925	0.246 ± 0.002	5.378 ± 0.925
LSAM	beta46	0.00	-0.029	-0.029 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	—	—	-0.029	-0.029	—	0.0832	0.0881	0.971	93.613 ± 1.002	0.340 ± 0.004	6.387 ± 1.002
LSAM	beta47	0.00	-0.029	-0.029 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	—	—	-0.029	-0.029	—	0.0789	0.0839	0.969	93.782 ± 0.990	0.339 ± 0.004	6.218 ± 0.990
LSAM	beta51	0.16	0.159	-0.001 ± 0.004	-0.003 ± 0.023	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.303 ± 0.019	0.550 ± 0.021	0.156	-0.004	-2.44	0.0831	0.0832	0.916	93.109 ± 1.038	0.317 ± 0.002	—
LSAM	beta52	0.16	0.163	0.003 ± 0.003	0.020 ± 0.017	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.172 ± 0.010	0.415 ± 0.015	0.162	0.002	1.20	0.0658	0.0658	0.994	95.630 ± 0.838	0.259 ± 0.001	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.003	-0.019 ± 0.018	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	0.199 ± 0.013	0.446 ± 0.017	0.159	-0.001	-0.55	0.0712	0.0712	0.931	92.773 ± 1.062	0.260 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.003	0.011 ± 0.017	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.162 ± 0.010	0.403 ± 0.015	0.159	-0.001	-0.39	0.0630	0.0630	1.016	95.630 ± 0.838	0.257 ± 0.001	—
LSAM	beta55	0.00	0.009	0.009 ± 0.004	—	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	—	—	0.008	0.008	—	0.0934	0.0937	0.959	93.782 ± 0.990	0.368 ± 0.003	6.218 ± 0.990
LSAM	beta56	0.00	-0.002	-0.002 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	0.000	0.000	—	0.0787	0.0787	1.059	95.630 ± 0.838	0.344 ± 0.003	4.370 ± 0.838
LSAM	beta57	0.00	-0.025	-0.025 ± 0.004	—	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	—	—	-0.026	-0.026	—	0.0946	0.0980	0.930	93.613 ± 1.002	0.355 ± 0.004	6.387 ± 1.002
LSAM	beta58	0.00	-0.022	-0.022 ± 0.004	—	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	—	—	-0.025	-0.025	—	0.0851	0.0887	0.941	94.622 ± 0.925	0.326 ± 0.004	5.378 ± 0.925
QML	beta41	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.012	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.005	0.288 ± 0.010	0.198	-0.002	-1.09	0.0553	0.0554	1.020	94.454 ± 0.938	0.230 ± 0.011	—
QML	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.012	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.005	0.292 ± 0.011	0.201	0.001	0.45	0.0568	0.0568	1.014	94.790 ± 0.911	0.232 ± 0.003	—
QML	beta43	0.20	0.194	-0.006 ± 0.002	-0.028 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.070 ± 0.004	0.265 ± 0.009	0.194	-0.006	-3.20	0.0564	0.0568	1.051	96.807 ± 0.721	0.218 ± 0.001	—
QML	beta44	0.00	0.007	0.007 ± 0.003	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	—	—	0.006	0.006	—	0.0703	0.0705	1.000	95.294 ± 0.868	0.269 ± 0.002	4.706 ± 0.868
QML	beta45	0.00	0.002	0.002 ± 0.003	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	—	0.0605	0.0605	0.978	94.118 ± 0.965	0.234 ± 0.001	5.882 ± 0.965
QML	beta46	0.00	-0.019	-0.019 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.020	-0.020	—	0.0573	0.0607	1.005	91.933 ± 1.116	0.219 ± 0.002	8.067 ± 1.116
QML	beta47	0.00	-0.021	-0.021 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	-0.022	-0.022	—	0.0539	0.0583	0.999	93.277 ± 1.027	0.218 ± 0.002	6.723 ± 1.027
QML	beta51	0.16	0.161	0.001 ± 0.004	0.007 ± 0.022	0.008 ± 0.001	0.087 ± 0.003	0.008 ± 0.001	0.087 ± 0.003	0.293 ± 0.020	0.542 ± 0.021	0.159	-0.001	-0.93	0.0818	0.0818	0.957	94.454 ± 0.938	0.325 ± 0.003	—
QML	beta52	0.16	0.160	0.000 ± 0.003	0.003 ± 0.016	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.009	0.391 ± 0.014	0.159	-0.001	-0.75	0.0626	0.0627	1.034	95.630 ± 0.838	0.254 ± 0.005	—
QML	beta53	0.16	0.160	0.000 ± 0.003	0.000 ± 0.017	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	0.181 ± 0.012	0.425 ± 0.016	0.161	0.001	0.35	0.0686	0.0686	0.966	93.782 ± 0.990	0.258 ± 0.004	—
QML	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.015	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.008	0.370 ± 0.014	0.158	-0.002	-0.96	0.0578	0.0578	1.133	96.471 ± 0.756	0.263 ± 0.018	—
QML	beta55	0.00	0.008	0.008 ± 0.004	—	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	—	—	0.009	0.009	—	0.0919	0.0923	1.063	92.941 ± 1.050	0.389 ± 0.033	7.059 ± 1.050
QML	beta56	0.00	-0.006	-0.006 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	-0.009	-0.009	—	0.0754	0.0760	1.121	95.798 ± 0.822	0.334 ± 0.013	4.202 ± 0.822
QML	beta57	0.00	-0.017	-0.017 ± 0.003	—	0.004 ± 0.000	0.065 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.020	-0.020	—	0.0646	0.0675	0.970	91.765 ± 1.127	0.248 ± 0.012	8.235 ± 1.127
QML	beta58	0.00	-0.015	-0.015 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.017	-0.017	—	0.0544	0.0569	0.986	93.782 ± 0.990	0.210 ± 0.002	6.218 ± 0.990
UPI	beta41	0.20	0.204	0.004 ± 0.003	0.020 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	0.125 ± 0.008	0.354 ± 0.014	0.201	0.001	0.65	0.0700	0.0700	1.229	97.647 ± 0.621	0.340 ± 0.009	—
UPI	beta42	0.20	0.220	0.020 ± 0.004	0.098 ± 0.018	0.008 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.089 ± 0.004	0.198 ± 0.016	0.445 ± 0.020	0.216	0.016	7.96	0.0783	0.0799	1.196	97.143 ± 0.683	0.407 ± 0.014	—
UPI	beta43	0.20	0.184	-0.016 ± 0.003	-0.078 ± 0.013	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.006	0.315 ± 0.011	0.178	-0.022	-10.94	0.0619	0.0657	1.162	96.303 ± 0.774	0.278 ± 0.004	—
UPI	beta44	0.00	0.021	0.021 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	—	—	0.019	0.019	—	0.0805	0.0827	1.160	96.807 ± 0.721	0.379 ± 0.009	3.193 ± 0.721
UPI	beta45	0.00	0.007	0.007 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	—	—	0.004	0.004	—	0.0690	0.0691	1.101	95.630 ± 0.838	0.301 ± 0.005	4.370 ± 0.838
UPI	beta46	0.00	-0.023	-0.023 ± 0.005	—	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.131 ± 0.006	—	—	-0.023	-0.023	—	0.1231	0.1252	1.288	98.319 ± 0.527	0.651 ± 0.024	1.681 ± 0.527

Table S249

Simulation 2 - Condition 105: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.041	-0.041 ± 0.006	—	0.024 ± 0.002	0.154 ± 0.006	0.025 ± 0.002	0.159 ± 0.007	—	—	-0.042	-0.042	—	0.1239	0.1307	1.319	96.975 ± 0.702	0.798 ± 0.037	3.025 ± 0.702
UPI	beta51	0.16	0.158	-0.002 ± 0.004	-0.014 ± 0.022	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.299 ± 0.019	0.547 ± 0.021	0.153	-0.007	-4.55	0.0882	0.0885	0.980	94.622 ± 0.925	0.337 ± 0.003	—
UPI	beta52	0.16	0.168	0.008 ± 0.003	0.049 ± 0.018	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	0.205 ± 0.014	0.453 ± 0.018	0.165	0.005	3.11	0.0739	0.0741	1.171	96.975 ± 0.702	0.331 ± 0.008	—
UPI	beta53	0.16	0.154	-0.006 ± 0.003	-0.037 ± 0.019	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	0.210 ± 0.014	0.458 ± 0.018	0.156	-0.004	-2.76	0.0758	0.0759	1.079	95.126 ± 0.883	0.309 ± 0.006	—
UPI	beta54	0.16	0.168	0.008 ± 0.003	0.047 ± 0.020	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	0.228 ± 0.016	0.478 ± 0.020	0.166	0.006	3.74	0.0709	0.0711	1.278	97.479 ± 0.643	0.381 ± 0.017	—
UPI	beta55	0.00	0.013	0.013 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	—	—	0.014	0.014	—	0.0943	0.0953	1.102	96.471 ± 0.756	0.435 ± 0.008	3.529 ± 0.756
UPI	beta56	0.00	0.002	0.002 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	—	—	0.003	0.003	—	0.0855	0.0855	1.156	97.647 ± 0.621	0.402 ± 0.007	2.353 ± 0.621
UPI	beta57	0.00	-0.023	-0.023 ± 0.006	—	0.018 ± 0.001	0.136 ± 0.005	0.019 ± 0.002	0.138 ± 0.006	—	—	-0.017	-0.017	—	0.1183	0.1195	1.133	96.134 ± 0.790	0.603 ± 0.024	3.866 ± 0.790
UPI	beta58	0.00	-0.023	-0.023 ± 0.006	—	0.019 ± 0.001	0.138 ± 0.005	0.020 ± 0.001	0.140 ± 0.006	—	—	-0.023	-0.023	—	0.1180	0.1203	1.301	97.983 ± 0.576	0.704 ± 0.038	2.017 ± 0.576

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S250

Simulation 2 - Condition 106: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.187 ± 0.006	0.203	0.003	1.45	0.0366	0.0368	1.014	95.582 ± 0.789	0.148 ± 0.000	—
LSAM	beta42	0.20	0.205	0.005 ± 0.002	0.025 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.007	0.206	0.006	2.78	0.0396	0.0400	1.005	94.993 ± 0.837	0.155 ± 0.000	—
LSAM	beta43	0.20	0.196	-0.004 ± 0.001	-0.019 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.184 ± 0.006	0.194	-0.006	-2.83	0.0365	0.0369	0.974	94.109 ± 0.904	0.140 ± 0.000	—
LSAM	beta44	0.00	0.016	0.016 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.015	0.015	—	0.0418	0.0444	1.000	93.373 ± 0.955	0.166 ± 0.001	6.627 ± 0.955
LSAM	beta45	0.00	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.007	0.007	—	0.0407	0.0413	1.017	96.024 ± 0.750	0.150 ± 0.001	3.976 ± 0.750
LSAM	beta46	0.00	-0.028	-0.028 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.030	-0.030	—	0.0526	0.0606	0.947	89.691 ± 1.167	0.200 ± 0.001	10.309 ± 1.167
LSAM	beta47	0.00	-0.034	-0.034 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.034	-0.034	—	0.0485	0.0591	1.022	89.838 ± 1.160	0.199 ± 0.001	10.162 ± 1.160
LSAM	beta51	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.012	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.100 ± 0.005	0.316 ± 0.011	0.160	0.000	-0.13	0.0488	0.0488	1.005	94.404 ± 0.882	0.199 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.001	0.014 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.008	0.161	0.001	0.80	0.0395	0.0395	1.038	94.993 ± 0.837	0.158 ± 0.000	—
LSAM	beta53	0.16	0.155	-0.005 ± 0.002	-0.030 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.004	0.251 ± 0.009	0.155	-0.005	-3.12	0.0372	0.0375	1.020	94.404 ± 0.882	0.160 ± 0.000	—
LSAM	beta54	0.16	0.166	0.006 ± 0.002	0.036 ± 0.010	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.004	0.259 ± 0.009	0.166	0.006	3.61	0.0398	0.0402	0.977	94.256 ± 0.893	0.157 ± 0.000	—
LSAM	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	0.012	0.012	—	0.0567	0.0579	0.964	94.256 ± 0.893	0.221 ± 0.001	5.744 ± 0.893
LSAM	beta56	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	0.003	0.003	—	0.0498	0.0498	0.993	94.109 ± 0.904	0.206 ± 0.001	5.891 ± 0.904
LSAM	beta57	0.00	-0.023	-0.023 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.026	-0.026	—	0.0550	0.0607	0.996	94.109 ± 0.904	0.211 ± 0.002	5.891 ± 0.904
LSAM	beta58	0.00	-0.024	-0.024 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.025	-0.025	—	0.0525	0.0581	0.971	90.574 ± 1.121	0.195 ± 0.001	9.426 ± 1.121
QML	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.002	0.173 ± 0.006	0.201	0.001	0.56	0.0341	0.0341	1.007	94.109 ± 0.904	0.137 ± 0.000	—
QML	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.180 ± 0.006	0.201	0.001	0.68	0.0372	0.0373	1.007	94.845 ± 0.849	0.142 ± 0.001	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.002	0.178 ± 0.006	0.197	-0.003	-1.36	0.0351	0.0352	0.972	94.551 ± 0.871	0.136 ± 0.000	—
QML	beta44	0.00	0.010	0.010 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.009	0.009	—	0.0421	0.0431	1.024	93.667 ± 0.935	0.168 ± 0.003	6.333 ± 0.935
QML	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	—	0.0396	0.0397	1.035	96.024 ± 0.750	0.148 ± 0.001	3.976 ± 0.750
QML	beta46	0.00	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.020	-0.020	—	0.0359	0.0410	0.965	88.954 ± 1.203	0.135 ± 0.001	11.046 ± 1.203
QML	beta47	0.00	-0.025	-0.025 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.025	-0.025	—	0.0318	0.0405	1.034	88.513 ± 1.224	0.134 ± 0.001	11.487 ± 1.224
QML	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.097 ± 0.005	0.312 ± 0.010	0.161	0.001	0.75	0.0483	0.0483	1.028	94.698 ± 0.860	0.201 ± 0.001	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.009	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.003	0.234 ± 0.008	0.161	0.001	0.59	0.0364	0.0364	1.041	94.845 ± 0.849	0.153 ± 0.001	—
QML	beta53	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.004	0.244 ± 0.009	0.158	-0.002	-1.13	0.0365	0.0365	1.020	94.551 ± 0.871	0.156 ± 0.000	—
QML	beta54	0.16	0.163	0.003 ± 0.001	0.020 ± 0.009	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.242 ± 0.008	0.163	0.003	1.66	0.0373	0.0374	1.003	94.256 ± 0.893	0.152 ± 0.002	—
QML	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.012	0.012	—	0.0570	0.0583	0.986	93.962 ± 0.914	0.218 ± 0.001	6.038 ± 0.914
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.002	-0.002	—	0.0475	0.0475	0.998	94.404 ± 0.882	0.197 ± 0.001	5.596 ± 0.882
QML	beta57	0.00	-0.017	-0.017 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.019	-0.019	—	0.0377	0.0422	1.008	92.489 ± 1.011	0.145 ± 0.001	7.511 ± 1.011
QML	beta58	0.00	-0.017	-0.017 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.018	-0.018	—	0.0338	0.0381	0.991	90.722 ± 1.113	0.130 ± 0.001	9.278 ± 1.113
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.029 ± 0.009	0.002 ± 0.000	0.046 ± 0.002	0.002 ± 0.000	0.046 ± 0.002	0.053 ± 0.004	0.230 ± 0.009	0.204	0.004	1.93	0.0466	0.0467	1.293	97.054 ± 0.649	0.231 ± 0.011	—
UPI	beta42	0.20	0.216	0.016 ± 0.002	0.081 ± 0.010	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.005	0.269 ± 0.010	0.214	0.014	6.94	0.0510	0.0528	1.260	97.202 ± 0.633	0.254 ± 0.011	—
UPI	beta43	0.20	0.189	-0.011 ± 0.001	-0.057 ± 0.007	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.185	-0.015	-7.56	0.0415	0.0441	1.098	95.140 ± 0.825	0.167 ± 0.004	—
UPI	beta44	0.00	0.022	0.022 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.022	0.022	—	0.0495	0.0541	1.136	93.373 ± 0.955	0.204 ± 0.004	6.627 ± 0.955
UPI	beta45	0.00	0.008	0.008 ± 0.002	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.009	0.009	—	0.0418	0.0427	1.109	96.465 ± 0.709	0.170 ± 0.003	3.535 ± 0.709
UPI	beta46	0.00	-0.030	-0.030 ± 0.003	—	0.007 ± 0.001	0.086 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	—	—	-0.030	-0.030	—	0.0787	0.0842	1.217	95.140 ± 0.825	0.409 ± 0.022	4.860 ± 0.825

Table S250

Simulation 2 - Condition 106: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.050	-0.050 ± 0.003	—	0.007 ± 0.001	0.085 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	—	—	-0.048	-0.048	—	0.0747	0.0888	1.236	94.845 ± 0.849	0.414 ± 0.019	5.155 ± 0.849
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.012	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.100 ± 0.005	0.317 ± 0.010	0.158	-0.002	-1.40	0.0522	0.0523	1.046	95.140 ± 0.825	0.208 ± 0.001	—
UPI	beta52	0.16	0.166	0.006 ± 0.002	0.040 ± 0.011	0.002 ± 0.000	0.046 ± 0.002	0.002 ± 0.000	0.047 ± 0.002	0.085 ± 0.006	0.291 ± 0.011	0.166	0.006	3.62	0.0444	0.0448	1.134	96.318 ± 0.723	0.205 ± 0.004	—
UPI	beta53	0.16	0.149	-0.011 ± 0.002	-0.067 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.044 ± 0.002	0.074 ± 0.005	0.272 ± 0.010	0.147	-0.013	-7.83	0.0429	0.0447	1.111	95.140 ± 0.825	0.184 ± 0.004	—
UPI	beta54	0.16	0.174	0.014 ± 0.002	0.089 ± 0.012	0.003 ± 0.000	0.050 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.107 ± 0.007	0.327 ± 0.012	0.173	0.013	7.83	0.0483	0.0499	1.233	95.876 ± 0.763	0.243 ± 0.017	—
UPI	beta55	0.00	0.014	0.014 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.014	0.014	—	0.0619	0.0634	1.035	93.962 ± 0.914	0.248 ± 0.005	6.038 ± 0.914
UPI	beta56	0.00	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	0.005	0.005	—	0.0521	0.0523	1.092	96.318 ± 0.723	0.227 ± 0.006	3.682 ± 0.723
UPI	beta57	0.00	-0.021	-0.021 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.003	0.007 ± 0.000	0.081 ± 0.003	—	—	-0.024	-0.024	—	0.0739	0.0778	1.104	96.171 ± 0.736	0.340 ± 0.009	3.829 ± 0.736
UPI	beta58	0.00	-0.035	-0.035 ± 0.003	—	0.007 ± 0.001	0.084 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	—	—	-0.032	-0.032	—	0.0721	0.0791	1.264	96.024 ± 0.750	0.416 ± 0.032	3.976 ± 0.750

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S251

Simulation 2 - Condition 107: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.198	-0.002	-1.05	0.0478	0.0479	0.968	94.100 ± 0.745	0.180 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.240 ± 0.006	0.199	-0.001	-0.74	0.0486	0.0486	0.988	94.900 ± 0.696	0.186 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.200	0.000	-0.09	0.0479	0.0479	0.976	95.100 ± 0.683	0.177 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.002	0.002	—	0.0502	0.0503	0.980	94.000 ± 0.751	0.199 ± 0.001	6.000 ± 0.751
LSAM	beta45	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	—	0.0464	0.0464	0.987	94.900 ± 0.696	0.181 ± 0.001	5.100 ± 0.696
LSAM	beta46	0.00	-0.006	-0.006 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.008	-0.008	—	0.0577	0.0582	0.994	95.400 ± 0.662	0.228 ± 0.001	4.600 ± 0.662
LSAM	beta47	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.006	-0.006	—	0.0576	0.0579	0.986	94.300 ± 0.733	0.226 ± 0.001	5.700 ± 0.733
LSAM	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.013	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.176 ± 0.008	0.419 ± 0.011	0.156	-0.004	-2.26	0.0650	0.0651	0.955	93.400 ± 0.785	0.251 ± 0.001	—
LSAM	beta52	0.16	0.164	0.004 ± 0.002	0.023 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.317 ± 0.009	0.165	0.005	2.95	0.0500	0.0502	1.006	94.900 ± 0.696	0.199 ± 0.000	—
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.114 ± 0.005	0.337 ± 0.009	0.158	-0.002	-1.26	0.0554	0.0555	0.965	93.800 ± 0.763	0.204 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.008	0.160	0.000	0.12	0.0487	0.0487	1.019	94.400 ± 0.727	0.199 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.003	0.003	—	0.0712	0.0712	0.928	93.100 ± 0.801	0.261 ± 0.001	6.900 ± 0.801
LSAM	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.002	0.002	—	0.0636	0.0636	0.944	92.900 ± 0.812	0.247 ± 0.001	7.100 ± 0.812
LSAM	beta57	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.004	-0.004	—	0.0636	0.0637	0.979	94.200 ± 0.739	0.247 ± 0.001	5.800 ± 0.739
LSAM	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.006	-0.006	—	0.0624	0.0627	0.968	93.900 ± 0.757	0.233 ± 0.001	6.100 ± 0.757
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.198	-0.002	-0.85	0.0472	0.0472	0.970	94.300 ± 0.733	0.176 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.198	-0.002	-0.78	0.0471	0.0471	0.988	94.500 ± 0.721	0.182 ± 0.000	—
QML	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.200	0.000	0.05	0.0480	0.0480	0.977	94.600 ± 0.715	0.175 ± 0.000	—
QML	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.000	0.000	—	0.0511	0.0511	0.985	93.900 ± 0.757	0.199 ± 0.001	6.100 ± 0.757
QML	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0459	0.0459	0.994	94.800 ± 0.702	0.180 ± 0.001	5.200 ± 0.702
QML	beta46	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.006	-0.006	—	0.0466	0.0470	1.009	95.400 ± 0.662	0.185 ± 0.001	4.600 ± 0.662
QML	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.005	-0.005	—	0.0450	0.0453	0.992	94.200 ± 0.739	0.182 ± 0.001	5.800 ± 0.739
QML	beta51	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.013	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.008	0.417 ± 0.011	0.158	-0.002	-1.15	0.0632	0.0632	0.970	94.400 ± 0.727	0.254 ± 0.001	—
QML	beta52	0.16	0.164	0.004 ± 0.002	0.023 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.316 ± 0.009	0.165	0.005	2.93	0.0506	0.0508	1.007	94.800 ± 0.702	0.199 ± 0.000	—
QML	beta53	0.16	0.160	0.000 ± 0.002	0.001 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.112 ± 0.005	0.335 ± 0.009	0.158	-0.002	-1.22	0.0541	0.0541	0.968	94.000 ± 0.751	0.204 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.308 ± 0.008	0.159	-0.001	-0.77	0.0480	0.0480	1.019	94.400 ± 0.727	0.197 ± 0.000	—
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.001	0.001	—	0.0700	0.0700	0.938	93.200 ± 0.796	0.260 ± 0.001	6.800 ± 0.796
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0618	0.0618	0.950	92.800 ± 0.817	0.243 ± 0.001	7.200 ± 0.817
QML	beta57	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.004	-0.004	—	0.0518	0.0519	0.986	94.300 ± 0.733	0.202 ± 0.001	5.700 ± 0.733
QML	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.005	-0.005	—	0.0496	0.0498	0.979	94.300 ± 0.733	0.188 ± 0.001	5.700 ± 0.733
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.199	-0.001	-0.71	0.0474	0.0474	0.968	93.500 ± 0.780	0.178 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.199	-0.001	-0.36	0.0485	0.0485	0.984	94.600 ± 0.715	0.184 ± 0.000	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.000 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.200	0.000	0.06	0.0476	0.0476	0.979	95.100 ± 0.683	0.178 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.000	0.000	—	0.0519	0.0519	0.978	93.900 ± 0.757	0.201 ± 0.001	6.100 ± 0.757
UPI	beta45	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.000	0.000	—	0.0473	0.0473	0.977	94.100 ± 0.745	0.183 ± 0.001	5.900 ± 0.745
UPI	beta46	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.002	-0.002	—	0.0594	0.0594	1.016	94.700 ± 0.708	0.242 ± 0.001	5.300 ± 0.708

Table S251

Simulation 2 - Condition 107: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.003	-0.003	—	0.0605	0.0606	1.010	94.800 ± 0.702	0.241 ± 0.001	5.200 ± 0.702
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.013	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.173 ± 0.008	0.416 ± 0.011	0.158	-0.002	-1.17	0.0634	0.0635	0.967	94.300 ± 0.733	0.252 ± 0.001	—
UPI	beta52	0.16	0.164	0.004 ± 0.002	0.022 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.316 ± 0.009	0.164	0.004	2.46	0.0488	0.0489	0.999	94.200 ± 0.739	0.198 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.003 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.114 ± 0.005	0.337 ± 0.009	0.158	-0.002	-1.14	0.0561	0.0561	0.967	93.900 ± 0.757	0.205 ± 0.000	—
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.008	0.160	0.000	0.00	0.0497	0.0497	1.014	94.500 ± 0.721	0.199 ± 0.000	—
UPI	beta55	0.00	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.002	0.002	—	0.0729	0.0729	0.940	93.300 ± 0.791	0.263 ± 0.001	6.700 ± 0.791
UPI	beta56	0.00	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.002	0.002	—	0.0643	0.0643	0.949	93.000 ± 0.807	0.250 ± 0.001	7.000 ± 0.807
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0647	0.0647	0.997	94.100 ± 0.745	0.257 ± 0.001	5.900 ± 0.745
UPI	beta58	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	—	0.0627	0.0627	1.018	95.500 ± 0.656	0.248 ± 0.001	4.500 ± 0.656

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S252

Simulation 2 - Condition 108: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.143 ± 0.004	0.199	-0.001	-0.45	0.0279	0.0279	1.004	95.300 ± 0.669	0.113 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.200	0.000	-0.20	0.0293	0.0293	0.987	94.900 ± 0.696	0.117 ± 0.000	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.199	-0.001	-0.52	0.0269	0.0269	1.014	95.100 ± 0.683	0.112 ± 0.000	—
LSAM	beta44	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.005	0.005	—	0.0315	0.0319	0.995	94.400 ± 0.727	0.125 ± 0.000	5.600 ± 0.727
LSAM	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0293	0.0293	0.998	95.000 ± 0.689	0.114 ± 0.000	5.000 ± 0.689
LSAM	beta46	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.010	-0.010	—	0.0376	0.0389	0.973	93.500 ± 0.780	0.142 ± 0.000	6.500 ± 0.780
LSAM	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.005	-0.005	—	0.0369	0.0373	0.970	93.900 ± 0.757	0.142 ± 0.000	6.100 ± 0.757
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.256 ± 0.007	0.159	-0.001	-0.93	0.0405	0.0406	0.985	94.700 ± 0.708	0.158 ± 0.000	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.199 ± 0.005	0.161	0.001	0.70	0.0321	0.0322	1.002	95.400 ± 0.662	0.125 ± 0.000	—
LSAM	beta53	0.16	0.160	0.000 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.160	0.000	-0.05	0.0330	0.0330	0.988	94.400 ± 0.727	0.129 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.160	0.000	0.22	0.0320	0.0320	0.970	93.200 ± 0.796	0.125 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.002	0.002	—	0.0393	0.0393	1.017	95.100 ± 0.683	0.163 ± 0.000	4.900 ± 0.683
LSAM	beta56	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.001	0.001	—	0.0400	0.0400	0.990	94.700 ± 0.708	0.154 ± 0.000	5.300 ± 0.708
LSAM	beta57	0.00	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.008	-0.008	—	0.0422	0.0429	0.974	93.700 ± 0.768	0.154 ± 0.000	6.300 ± 0.768
LSAM	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.005	-0.005	—	0.0379	0.0383	1.007	96.000 ± 0.620	0.145 ± 0.000	4.000 ± 0.620
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.198	-0.002	-0.76	0.0273	0.0274	1.040	94.900 ± 0.696	0.116 ± 0.004	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.199	-0.001	-0.32	0.0286	0.0286	0.994	94.800 ± 0.702	0.115 ± 0.001	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.140 ± 0.004	0.199	-0.001	-0.54	0.0270	0.0270	1.026	95.000 ± 0.689	0.112 ± 0.002	—
QML	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.003	0.003	—	0.0311	0.0312	1.004	94.500 ± 0.721	0.126 ± 0.001	5.500 ± 0.721
QML	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.003	-0.003	—	0.0291	0.0293	0.998	94.900 ± 0.696	0.114 ± 0.000	5.100 ± 0.696
QML	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.009	-0.009	—	0.0301	0.0315	0.987	93.300 ± 0.791	0.117 ± 0.001	6.700 ± 0.791
QML	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.006	-0.006	—	0.0295	0.0301	0.989	93.700 ± 0.768	0.117 ± 0.001	6.300 ± 0.768
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.256 ± 0.007	0.158	-0.002	-0.94	0.0402	0.0403	1.012	94.800 ± 0.702	0.162 ± 0.003	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.197 ± 0.005	0.161	0.001	0.68	0.0321	0.0321	1.033	95.500 ± 0.656	0.128 ± 0.003	—
QML	beta53	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.160	0.000	0.02	0.0325	0.0325	0.996	94.500 ± 0.721	0.129 ± 0.001	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.160	0.000	-0.04	0.0317	0.0317	1.101	93.100 ± 0.801	0.140 ± 0.016	—
QML	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.001	0.001	—	0.0389	0.0389	1.034	95.400 ± 0.662	0.165 ± 0.001	4.600 ± 0.662
QML	beta56	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0396	0.0396	1.013	95.500 ± 0.656	0.155 ± 0.002	4.500 ± 0.656
QML	beta57	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.008	-0.008	—	0.0348	0.0357	0.986	93.000 ± 0.807	0.128 ± 0.001	7.000 ± 0.807
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.005	-0.005	—	0.0306	0.0310	1.051	95.900 ± 0.627	0.121 ± 0.003	4.100 ± 0.627
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.199	-0.001	-0.54	0.0275	0.0275	1.001	95.300 ± 0.669	0.112 ± 0.000	—
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.199	-0.001	-0.32	0.0290	0.0290	0.989	95.000 ± 0.689	0.116 ± 0.000	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.199	-0.001	-0.29	0.0270	0.0270	1.015	95.100 ± 0.683	0.112 ± 0.000	—
UPI	beta44	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.003	0.003	—	0.0314	0.0316	0.993	94.000 ± 0.751	0.125 ± 0.000	6.000 ± 0.751
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0302	0.0302	0.987	94.300 ± 0.733	0.114 ± 0.000	5.700 ± 0.733
UPI	beta46	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.006	-0.006	—	0.0381	0.0387	0.987	94.400 ± 0.727	0.146 ± 0.000	5.600 ± 0.727

Table S252

Simulation 2 - Condition 108: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0372	0.0373	0.976	94.700 ± 0.708	0.146 ± 0.000	5.300 ± 0.708
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.256 ± 0.007	0.158	-0.002	-1.09	0.0403	0.0403	0.991	94.900 ± 0.696	0.159 ± 0.000	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.161	0.001	0.74	0.0318	0.0319	1.001	95.200 ± 0.676	0.124 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.160	0.000	0.22	0.0322	0.0322	0.986	94.200 ± 0.739	0.129 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.160	0.000	0.06	0.0319	0.0319	0.967	93.200 ± 0.796	0.124 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.001	0.001	—	0.0373	0.0373	1.013	94.400 ± 0.727	0.164 ± 0.000	5.600 ± 0.727
UPI	beta56	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.001	0.001	—	0.0400	0.0400	0.991	95.300 ± 0.669	0.154 ± 0.000	4.700 ± 0.669
UPI	beta57	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.005	-0.005	—	0.0439	0.0442	0.960	93.700 ± 0.768	0.157 ± 0.001	6.300 ± 0.768
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0385	0.0386	1.016	96.400 ± 0.589	0.150 ± 0.000	3.600 ± 0.589

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S253

Simulation 2 - Condition 109: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.210	0.010 ± 0.004	0.050 ± 0.018	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.001	0.088 ± 0.003	0.193 ± 0.013	0.440 ± 0.017	0.210	0.010	4.99	0.0901	0.0907	1.031	95.876 ± 0.824	0.353 ± 0.004	—
LSAM	beta42	0.20	0.199	-0.001 ± 0.004	-0.003 ± 0.019	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	0.217 ± 0.014	0.466 ± 0.018	0.193	-0.007	-3.56	0.0899	0.0902	1.010	96.392 ± 0.773	0.369 ± 0.004	—
LSAM	beta43	0.20	0.207	0.007 ± 0.003	0.035 ± 0.017	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	0.169 ± 0.010	0.411 ± 0.015	0.206	0.006	2.90	0.0841	0.0843	1.043	96.735 ± 0.737	0.335 ± 0.003	—
LSAM	beta44	0.00	0.003	0.003 ± 0.005	—	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.125 ± 0.005	—	—	-0.003	-0.003	—	0.1194	0.1194	1.076	97.938 ± 0.589	0.526 ± 0.010	2.062 ± 0.589
LSAM	beta45	0.00	-0.003	-0.003 ± 0.005	—	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.005	—	—	0.002	0.002	—	0.1124	0.1124	0.988	97.938 ± 0.589	0.427 ± 0.007	2.062 ± 0.589
LSAM	beta46	0.00	0.003	0.003 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	—	—	0.002	0.002	—	0.0830	0.0830	1.045	98.454 ± 0.511	0.364 ± 0.007	1.546 ± 0.511
LSAM	beta47	0.00	0.001	0.001 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	0.003	0.003	—	0.0819	0.0820	1.018	98.110 ± 0.564	0.361 ± 0.007	1.890 ± 0.564
LSAM	beta51	0.16	0.165	0.005 ± 0.006	0.030 ± 0.036	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.137 ± 0.005	0.738 ± 0.046	0.859 ± 0.032	0.164	0.004	2.73	0.1424	0.1425	0.901	93.127 ± 1.049	0.485 ± 0.005	—
LSAM	beta52	0.16	0.162	0.002 ± 0.004	0.012 ± 0.025	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.360 ± 0.024	0.600 ± 0.023	0.159	-0.001	-0.42	0.0929	0.0929	1.006	95.704 ± 0.840	0.379 ± 0.004	—
LSAM	beta53	0.16	0.161	0.001 ± 0.004	0.003 ± 0.025	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	0.350 ± 0.023	0.592 ± 0.023	0.161	0.001	0.51	0.0933	0.0933	1.029	96.392 ± 0.773	0.382 ± 0.003	—
LSAM	beta54	0.16	0.162	0.002 ± 0.004	0.015 ± 0.025	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	0.355 ± 0.022	0.596 ± 0.023	0.163	0.003	1.87	0.0974	0.0974	0.983	95.361 ± 0.872	0.367 ± 0.003	—
LSAM	beta55	0.00	-0.002	-0.002 ± 0.008	—	0.038 ± 0.003	0.195 ± 0.007	0.038 ± 0.003	0.195 ± 0.008	—	—	-0.005	-0.005	—	0.1772	0.1773	1.027	98.282 ± 0.539	0.786 ± 0.020	1.718 ± 0.539
LSAM	beta56	0.00	0.003	0.003 ± 0.007	—	0.025 ± 0.002	0.157 ± 0.006	0.025 ± 0.002	0.157 ± 0.007	—	—	-0.004	-0.004	—	0.1496	0.1496	1.023	97.079 ± 0.698	0.630 ± 0.014	2.921 ± 0.698
LSAM	beta57	0.00	-0.005	-0.005 ± 0.005	—	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.005	—	—	-0.001	-0.001	—	0.0991	0.0991	1.010	98.110 ± 0.564	0.434 ± 0.012	1.890 ± 0.564
LSAM	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	—	—	-0.001	-0.001	—	0.0752	0.0752	1.020	96.735 ± 0.737	0.317 ± 0.006	3.265 ± 0.737
QML	beta41	0.20	0.205	0.005 ± 0.003	0.025 ± 0.016	0.006 ± 0.000	0.078 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	0.151 ± 0.010	0.389 ± 0.015	0.205	0.005	2.26	0.0801	0.0802	1.044	96.220 ± 0.791	0.318 ± 0.003	—
QML	beta42	0.20	0.196	-0.004 ± 0.003	-0.018 ± 0.017	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	0.163 ± 0.010	0.403 ± 0.015	0.192	-0.008	-4.03	0.0780	0.0784	1.043	95.876 ± 0.824	0.330 ± 0.003	—
QML	beta43	0.20	0.203	0.003 ± 0.003	0.016 ± 0.016	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.140 ± 0.008	0.374 ± 0.014	0.202	0.002	1.17	0.0785	0.0786	1.064	97.595 ± 0.635	0.312 ± 0.002	—
QML	beta44	0.00	-0.020	-0.020 ± 0.004	—	0.012 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	—	—	-0.021	-0.021	—	0.1039	0.1060	1.088	97.251 ± 0.678	0.458 ± 0.007	2.749 ± 0.678
QML	beta45	0.00	-0.015	-0.015 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	—	—	-0.009	-0.009	—	0.1047	0.1050	1.003	96.220 ± 0.791	0.369 ± 0.005	3.780 ± 0.791
QML	beta46	0.00	0.009	0.009 ± 0.003	—	0.005 ± 0.000	0.074 ± 0.003	0.006 ± 0.000	0.074 ± 0.003	—	—	0.007	0.007	—	0.0752	0.0756	1.064	97.595 ± 0.635	0.307 ± 0.006	2.405 ± 0.635
QML	beta47	0.00	0.000	0.000 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	—	—	-0.001	-0.001	—	0.0684	0.0684	1.027	96.735 ± 0.737	0.294 ± 0.005	3.265 ± 0.737
QML	beta51	0.16	0.170	0.010 ± 0.006	0.060 ± 0.035	0.018 ± 0.001	0.135 ± 0.005	0.018 ± 0.001	0.135 ± 0.005	0.714 ± 0.049	0.845 ± 0.034	0.163	0.003	1.60	0.1280	0.1280	0.999	96.048 ± 0.808	0.529 ± 0.007	—
QML	beta52	0.16	0.158	-0.002 ± 0.004	-0.015 ± 0.024	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	0.324 ± 0.020	0.569 ± 0.021	0.157	-0.003	-1.60	0.0927	0.0928	1.058	97.251 ± 0.678	0.378 ± 0.005	—
QML	beta53	0.16	0.160	-0.000 ± 0.004	-0.003 ± 0.023	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.305 ± 0.019	0.552 ± 0.021	0.159	-0.001	-0.57	0.0873	0.0873	1.056	96.220 ± 0.791	0.366 ± 0.003	—
QML	beta54	0.16	0.158	-0.002 ± 0.004	-0.014 ± 0.023	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	0.309 ± 0.020	0.556 ± 0.021	0.160	0.000	0.05	0.0890	0.0890	0.980	95.361 ± 0.872	0.342 ± 0.003	—
QML	beta55	0.00	-0.013	-0.013 ± 0.006	—	0.022 ± 0.001	0.149 ± 0.005	0.022 ± 0.001	0.150 ± 0.006	—	—	-0.019	-0.019	—	0.1520	0.1532	1.081	97.251 ± 0.678	0.632 ± 0.011	2.749 ± 0.678
QML	beta56	0.00	-0.011	-0.011 ± 0.005	—	0.014 ± 0.001	0.120 ± 0.004	0.015 ± 0.001	0.120 ± 0.004	—	—	-0.012	-0.012	—	0.1221	0.1227	1.051	96.735 ± 0.737	0.494 ± 0.007	3.265 ± 0.737
QML	beta57	0.00	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.003	0.007 ± 0.000	0.081 ± 0.003	—	—	-0.002	-0.002	—	0.0807	0.0807	1.059	95.876 ± 0.824	0.337 ± 0.007	4.124 ± 0.824
QML	beta58	0.00	-0.003	-0.003 ± 0.003	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.005	-0.005	—	0.0603	0.0605	1.027	94.674 ± 0.931	0.256 ± 0.005	5.326 ± 0.931
UPI	beta41	0.20	0.214	0.014 ± 0.004	0.068 ± 0.020	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.249 ± 0.017	0.499 ± 0.020	0.215	0.015	7.60	0.1135	0.1145	1.227	98.625 ± 0.483	0.476 ± 0.012	—
UPI	beta42	0.20	0.198	-0.002 ± 0.004	-0.012 ± 0.021	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	0.258 ± 0.015	0.508 ± 0.018	0.189	-0.011	-5.52	0.1199	0.1204	1.261	98.282 ± 0.539	0.503 ± 0.011	—
UPI	beta43	0.20	0.211	0.011 ± 0.004	0.055 ± 0.018	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	0.196 ± 0.014	0.443 ± 0.019	0.211	0.011	5.52	0.1024	0.1030	1.279	98.625 ± 0.483	0.441 ± 0.010	—
UPI	beta44	0.00	0.011	0.011 ± 0.007	—	0.030 ± 0.003	0.174 ± 0.008	0.030 ± 0.003	0.174 ± 0.008	—	—	-0.006	-0.006	—	0.1762	0.1763	1.515	100.000 ± 0.000	1.033 ± 0.043	0.000 ± 0.000
UPI	beta45	0.00	-0.004	-0.004 ± 0.006	—	0.023 ± 0.002	0.152 ± 0.006	0.023 ± 0.002	0.152 ± 0.007	—	—	0.005	0.005	—	0.1562	0.1562	1.376	99.485 ± 0.297	0.819 ± 0.034	0.515 ± 0.297
UPI	beta46	0.00	0.001	0.001 ± 0.005	—	0.015 ± 0.001	0.122 ± 0.005	0.015 ± 0.001	0.122 ± 0.006	—	—	0.003	0.003	—	0.1199	0.1199	1.537	99.656 ± 0.243	0.734 ± 0.031	0.344 ± 0.243

Table S253

Simulation 2 - Condition 109: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.005	0.005 ± 0.005	—	0.017 ± 0.001	0.132 ± 0.005	0.017 ± 0.001	0.132 ± 0.006	—	—	0.008	0.008	—	0.1132	0.1135	1.449	99.828 ± 0.172	0.747 ± 0.035	0.172 ± 0.172
UPI	beta51	0.16	0.163	0.003 ± 0.005	0.019 ± 0.034	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	0.673 ± 0.046	0.820 ± 0.033	0.163	0.003	1.87	0.1332	0.1333	1.039	95.876 ± 0.824	0.535 ± 0.007	—
UPI	beta52	0.16	0.163	0.003 ± 0.004	0.019 ± 0.024	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.343 ± 0.024	0.586 ± 0.024	0.157	-0.003	-1.92	0.1002	0.1003	1.227	98.454 ± 0.511	0.451 ± 0.009	—
UPI	beta53	0.16	0.161	0.001 ± 0.004	0.005 ± 0.025	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	0.356 ± 0.025	0.596 ± 0.024	0.161	0.001	0.72	0.0980	0.0980	1.222	97.423 ± 0.657	0.458 ± 0.009	—
UPI	beta54	0.16	0.164	0.004 ± 0.004	0.026 ± 0.025	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.370 ± 0.024	0.608 ± 0.024	0.163	0.003	2.18	0.1035	0.1035	1.147	96.564 ± 0.755	0.437 ± 0.008	—
UPI	beta55	0.00	0.002	0.002 ± 0.009	—	0.043 ± 0.004	0.208 ± 0.009	0.043 ± 0.004	0.208 ± 0.010	—	—	-0.007	-0.007	—	0.1839	0.1840	1.590	99.828 ± 0.172	1.296 ± 0.066	0.172 ± 0.172
UPI	beta56	0.00	-0.002	-0.002 ± 0.007	—	0.031 ± 0.003	0.175 ± 0.008	0.031 ± 0.003	0.175 ± 0.009	—	—	-0.009	-0.009	—	0.1556	0.1558	1.489	99.313 ± 0.342	1.022 ± 0.054	0.687 ± 0.342
UPI	beta57	0.00	-0.006	-0.006 ± 0.005	—	0.016 ± 0.001	0.126 ± 0.006	0.016 ± 0.001	0.126 ± 0.006	—	—	0.000	0.000	—	0.1138	0.1138	1.412	99.141 ± 0.383	0.695 ± 0.030	0.859 ± 0.383
UPI	beta58	0.00	-0.002	-0.002 ± 0.004	—	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.091 ± 0.004	—	—	-0.004	-0.004	—	0.0799	0.0801	1.433	99.141 ± 0.383	0.513 ± 0.020	0.859 ± 0.383

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S254

Simulation 2 - Condition 110: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.203	0.003	1.38	0.0533	0.0533	1.014	95.036 ± 0.698	0.203 ± 0.001	—
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.077 ± 0.004	0.278 ± 0.008	0.200	0.000	-0.23	0.0552	0.0552	0.966	94.416 ± 0.738	0.210 ± 0.001	—
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.199	-0.001	-0.31	0.0483	0.0483	1.022	96.174 ± 0.617	0.196 ± 0.001	—
LSAM	beta44	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.003	0.003	—	0.0677	0.0677	1.034	97.311 ± 0.520	0.278 ± 0.002	2.689 ± 0.520
LSAM	beta45	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	—	0.0586	0.0587	0.983	96.174 ± 0.617	0.230 ± 0.002	3.826 ± 0.617
LSAM	beta46	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.005	-0.005	—	0.0480	0.0483	0.982	96.174 ± 0.617	0.195 ± 0.002	3.826 ± 0.617
LSAM	beta47	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0466	0.0466	1.010	96.174 ± 0.617	0.188 ± 0.002	3.826 ± 0.617
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.004 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.208 ± 0.010	0.456 ± 0.014	0.159	-0.001	-0.50	0.0746	0.0746	0.994	95.450 ± 0.670	0.284 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.016 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.123 ± 0.006	0.351 ± 0.010	0.161	0.001	0.43	0.0536	0.0536	0.976	94.726 ± 0.719	0.215 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.010 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.111 ± 0.005	0.334 ± 0.009	0.162	0.002	1.21	0.0515	0.0515	1.055	96.898 ± 0.558	0.221 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.021 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.109 ± 0.005	0.330 ± 0.010	0.164	0.004	2.61	0.0524	0.0525	1.023	95.863 ± 0.640	0.211 ± 0.001	—
LSAM	beta55	0.00	0.004	0.004 ± 0.003	—	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	—	—	0.006	0.006	—	0.0963	0.0964	0.954	97.311 ± 0.520	0.394 ± 0.005	2.689 ± 0.520
LSAM	beta56	0.00	-0.002	-0.002 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.006	-0.006	—	0.0762	0.0764	0.980	96.484 ± 0.592	0.319 ± 0.003	3.516 ± 0.592
LSAM	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.003	-0.003	—	0.0527	0.0528	0.970	96.691 ± 0.575	0.218 ± 0.002	3.309 ± 0.575
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0407	0.0407	1.026	95.760 ± 0.648	0.170 ± 0.001	4.240 ± 0.648
QML	beta41	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.007	0.200	0.000	-0.23	0.0484	0.0484	1.043	95.036 ± 0.698	0.193 ± 0.001	—
QML	beta42	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.066 ± 0.003	0.258 ± 0.007	0.197	-0.003	-1.69	0.0525	0.0526	0.982	94.829 ± 0.712	0.198 ± 0.001	—
QML	beta43	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.196	-0.004	-2.23	0.0470	0.0472	1.042	96.277 ± 0.609	0.191 ± 0.001	—
QML	beta44	0.00	-0.016	-0.016 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.013	-0.013	—	0.0653	0.0665	1.042	97.518 ± 0.500	0.268 ± 0.002	2.482 ± 0.500
QML	beta45	0.00	-0.015	-0.015 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.016	-0.016	—	0.0563	0.0587	0.988	94.312 ± 0.745	0.220 ± 0.002	5.688 ± 0.745
QML	beta46	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	—	0.0425	0.0425	1.009	95.450 ± 0.670	0.180 ± 0.002	4.550 ± 0.670
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.002	-0.002	—	0.0406	0.0407	1.027	95.346 ± 0.677	0.170 ± 0.002	4.654 ± 0.677
QML	beta51	0.16	0.163	0.003 ± 0.002	0.021 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.196 ± 0.010	0.443 ± 0.013	0.160	0.000	0.04	0.0698	0.0698	1.057	96.381 ± 0.601	0.293 ± 0.002	—
QML	beta52	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.006	0.340 ± 0.010	0.160	0.000	0.15	0.0525	0.0525	1.003	95.553 ± 0.663	0.214 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.320 ± 0.009	0.158	-0.002	-1.51	0.0491	0.0491	1.067	96.174 ± 0.617	0.214 ± 0.001	—
QML	beta54	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.005	0.315 ± 0.009	0.161	0.001	0.42	0.0497	0.0497	1.040	96.070 ± 0.625	0.206 ± 0.001	—
QML	beta55	0.00	-0.004	-0.004 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	—	—	-0.004	-0.004	—	0.0913	0.0914	1.048	96.277 ± 0.609	0.368 ± 0.006	3.723 ± 0.609
QML	beta56	0.00	-0.014	-0.014 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.017	-0.017	—	0.0700	0.0720	1.007	94.209 ± 0.751	0.285 ± 0.002	5.791 ± 0.751
QML	beta57	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.007	-0.007	—	0.0484	0.0489	1.029	94.933 ± 0.705	0.204 ± 0.004	5.067 ± 0.705
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.006	-0.006	—	0.0361	0.0367	1.048	94.312 ± 0.745	0.153 ± 0.001	5.688 ± 0.745
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.014 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.205	0.005	2.63	0.0521	0.0524	1.062	95.967 ± 0.633	0.215 ± 0.001	—
UPI	beta42	0.20	0.200	0.000 ± 0.002	0.001 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.200	0.000	-0.06	0.0564	0.0564	1.017	95.450 ± 0.670	0.223 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.198	-0.002	-0.90	0.0492	0.0492	1.055	95.967 ± 0.633	0.207 ± 0.001	—
UPI	beta44	0.00	0.003	0.003 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.001	0.001	—	0.0721	0.0721	1.160	99.069 ± 0.309	0.363 ± 0.007	0.931 ± 0.309
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.003	-0.003	—	0.0610	0.0611	1.085	97.622 ± 0.490	0.282 ± 0.004	2.378 ± 0.490
UPI	beta46	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.004	-0.004	—	0.0509	0.0511	1.128	98.656 ± 0.370	0.253 ± 0.005	1.344 ± 0.370

Table S254

Simulation 2 - Condition 110: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	0.001	0.001	—	0.0490	0.0490	1.143	98.552 ± 0.384	0.240 ± 0.005	1.448 ± 0.384
UPI	beta51	0.16	0.160	0.000 ± 0.002	0.001 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.009	0.436 ± 0.013	0.157	-0.003	-1.95	0.0691	0.0691	1.043	96.381 ± 0.601	0.286 ± 0.001	—
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.016 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.006	0.345 ± 0.010	0.161	0.001	0.32	0.0539	0.0539	1.022	95.243 ± 0.684	0.221 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.112 ± 0.005	0.334 ± 0.009	0.161	0.001	0.37	0.0510	0.0510	1.077	97.208 ± 0.530	0.226 ± 0.001	—
UPI	beta54	0.16	0.164	0.004 ± 0.002	0.023 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.108 ± 0.005	0.329 ± 0.010	0.165	0.005	3.04	0.0514	0.0516	1.049	96.277 ± 0.609	0.216 ± 0.001	—
UPI	beta55	0.00	0.003	0.003 ± 0.003	—	0.012 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.107 ± 0.004	—	—	0.005	0.005	—	0.0960	0.0961	1.088	98.552 ± 0.384	0.458 ± 0.010	1.448 ± 0.384
UPI	beta56	0.00	-0.001	-0.001 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.002	-0.002	—	0.0770	0.0771	1.128	98.759 ± 0.356	0.363 ± 0.006	1.241 ± 0.356
UPI	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0523	0.0523	1.089	98.345 ± 0.410	0.245 ± 0.004	1.655 ± 0.410
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0416	0.0416	1.089	97.932 ± 0.458	0.194 ± 0.002	2.068 ± 0.458

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S255

Simulation 2 - Condition 111: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.009 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.287 ± 0.008	0.200	0.000	0.23	0.0542	0.0542	1.013	94.366 ± 0.731	0.228 ± 0.001	—
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.009 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.004	0.313 ± 0.008	0.200	0.000	0.12	0.0641	0.0641	0.956	94.668 ± 0.713	0.235 ± 0.001	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.009	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.293 ± 0.008	0.198	-0.002	-1.06	0.0584	0.0584	0.968	94.064 ± 0.749	0.223 ± 0.001	—
LSAM	beta44	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	0.005	0.005	—	0.0724	0.0726	0.958	94.970 ± 0.693	0.278 ± 0.002	5.030 ± 0.693
LSAM	beta45	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.002	0.002	—	0.0619	0.0619	0.976	95.070 ± 0.687	0.239 ± 0.002	4.930 ± 0.687
LSAM	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.003	-0.003	—	0.0469	0.0470	1.018	95.674 ± 0.645	0.196 ± 0.002	4.326 ± 0.645
LSAM	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.001	-0.001	—	0.0449	0.0450	1.003	95.573 ± 0.652	0.190 ± 0.002	4.427 ± 0.652
LSAM	beta51	0.16	0.163	0.003 ± 0.003	0.016 ± 0.017	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.299 ± 0.014	0.547 ± 0.016	0.164	0.004	2.40	0.0851	0.0851	0.930	92.656 ± 0.827	0.319 ± 0.002	—
LSAM	beta52	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.387 ± 0.010	0.158	-0.002	-1.43	0.0620	0.0620	1.016	96.177 ± 0.608	0.247 ± 0.001	—
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.007 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.008	0.418 ± 0.012	0.160	0.000	0.15	0.0665	0.0665	0.967	93.863 ± 0.761	0.254 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.018 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.008	0.399 ± 0.012	0.161	0.001	0.40	0.0591	0.0591	0.977	94.567 ± 0.719	0.245 ± 0.001	—
LSAM	beta55	0.00	-0.003	-0.003 ± 0.003	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	—	—	-0.001	-0.001	—	0.0942	0.0942	0.954	94.165 ± 0.743	0.382 ± 0.004	5.835 ± 0.743
LSAM	beta56	0.00	0.007	0.007 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	—	—	0.005	0.005	—	0.0839	0.0840	0.908	92.757 ± 0.822	0.324 ± 0.003	7.243 ± 0.822
LSAM	beta57	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0566	0.0566	0.959	94.769 ± 0.706	0.219 ± 0.002	5.231 ± 0.706
LSAM	beta58	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.001	-0.001	—	0.0468	0.0468	0.971	94.769 ± 0.706	0.182 ± 0.001	5.231 ± 0.706
QML	beta41	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.077 ± 0.004	0.277 ± 0.008	0.198	-0.002	-1.09	0.0549	0.0550	1.025	94.869 ± 0.700	0.223 ± 0.001	—
QML	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.302 ± 0.008	0.197	-0.003	-1.46	0.0613	0.0614	0.965	94.366 ± 0.731	0.228 ± 0.001	—
QML	beta43	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.287 ± 0.008	0.197	-0.003	-1.31	0.0560	0.0560	0.980	94.165 ± 0.743	0.221 ± 0.001	—
QML	beta44	0.00	-0.009	-0.009 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.007	-0.007	—	0.0659	0.0663	0.996	94.769 ± 0.706	0.280 ± 0.003	5.231 ± 0.706
QML	beta45	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.007	-0.007	—	0.0600	0.0604	1.005	94.970 ± 0.693	0.239 ± 0.003	5.030 ± 0.693
QML	beta46	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0443	0.0444	1.064	95.674 ± 0.645	0.190 ± 0.002	4.326 ± 0.645
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	—	0.0431	0.0431	1.041	95.372 ± 0.666	0.182 ± 0.002	4.628 ± 0.666
QML	beta51	0.16	0.163	0.003 ± 0.003	0.022 ± 0.017	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.290 ± 0.014	0.538 ± 0.015	0.166	0.006	3.99	0.0845	0.0847	0.990	94.970 ± 0.693	0.334 ± 0.002	—
QML	beta52	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.006	0.384 ± 0.010	0.155	-0.005	-2.85	0.0628	0.0630	1.042	96.479 ± 0.585	0.251 ± 0.002	—
QML	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.008	0.410 ± 0.012	0.159	-0.001	-0.47	0.0633	0.0633	0.979	94.266 ± 0.737	0.252 ± 0.001	—
QML	beta54	0.16	0.162	0.002 ± 0.002	0.010 ± 0.012	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.008	0.391 ± 0.012	0.160	0.000	-0.02	0.0569	0.0569	0.988	94.567 ± 0.719	0.242 ± 0.001	—
QML	beta55	0.00	-0.009	-0.009 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	—	—	-0.007	-0.007	—	0.0881	0.0884	1.033	94.869 ± 0.700	0.373 ± 0.004	5.131 ± 0.700
QML	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	—	—	-0.001	-0.001	—	0.0795	0.0796	0.936	94.064 ± 0.749	0.308 ± 0.003	5.936 ± 0.749
QML	beta57	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0523	0.0523	1.014	94.467 ± 0.725	0.212 ± 0.003	5.533 ± 0.725
QML	beta58	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.003	-0.003	—	0.0411	0.0413	0.985	94.366 ± 0.731	0.173 ± 0.002	5.634 ± 0.731
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.010 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.288 ± 0.008	0.202	0.002	0.88	0.0524	0.0525	1.042	94.869 ± 0.700	0.235 ± 0.001	—
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.004	0.311 ± 0.008	0.200	0.000	0.20	0.0649	0.0649	0.990	95.171 ± 0.680	0.242 ± 0.001	—
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.008	0.198	-0.002	-0.90	0.0575	0.0575	0.991	94.668 ± 0.713	0.231 ± 0.001	—
UPI	beta44	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.003	-0.003	—	0.0720	0.0721	1.000	95.976 ± 0.623	0.324 ± 0.004	4.024 ± 0.623
UPI	beta45	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.004	0.004	—	0.0630	0.0631	1.014	96.177 ± 0.608	0.269 ± 0.003	3.823 ± 0.608
UPI	beta46	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0513	0.0513	1.069	96.479 ± 0.585	0.232 ± 0.003	3.521 ± 0.585

Table S255

Simulation 2 - Condition 111: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0495	0.0495	1.083	97.485 ± 0.497	0.225 ± 0.003	2.515 ± 0.497
UPI	beta51	0.16	0.162	0.002 ± 0.003	0.015 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.280 ± 0.013	0.530 ± 0.015	0.164	0.004	2.73	0.0810	0.0811	0.976	93.964 ± 0.755	0.324 ± 0.002	—
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.007	0.388 ± 0.010	0.156	-0.004	-2.27	0.0636	0.0637	1.026	96.278 ± 0.600	0.250 ± 0.001	—
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.007 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.008	0.418 ± 0.012	0.161	0.001	0.36	0.0649	0.0649	0.982	94.567 ± 0.719	0.258 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.008	0.399 ± 0.012	0.162	0.002	1.30	0.0606	0.0607	0.991	94.668 ± 0.713	0.248 ± 0.001	—
UPI	beta55	0.00	-0.004	-0.004 ± 0.003	—	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	—	—	0.000	0.000	—	0.0922	0.0922	1.005	95.875 ± 0.631	0.406 ± 0.005	4.125 ± 0.631
UPI	beta56	0.00	0.006	0.006 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	0.006	0.006	—	0.0853	0.0856	0.957	95.976 ± 0.623	0.352 ± 0.004	4.024 ± 0.623
UPI	beta57	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.002	-0.002	—	0.0555	0.0555	1.012	96.076 ± 0.616	0.234 ± 0.002	3.924 ± 0.616
UPI	beta58	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	0.000	0.000	—	0.0465	0.0465	1.012	96.076 ± 0.616	0.198 ± 0.002	3.924 ± 0.616

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S256

Simulation 2 - Condition 112: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.201	0.001	0.64	0.0340	0.0340	1.015	93.982 ± 0.753	0.141 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.201	0.001	0.42	0.0382	0.0382	0.995	95.186 ± 0.678	0.147 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.001	0.187 ± 0.005	0.199	-0.001	-0.30	0.0377	0.0377	0.945	93.079 ± 0.804	0.139 ± 0.000	—
LSAM	beta44	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.001	0.001	—	0.0410	0.0410	1.002	94.584 ± 0.717	0.165 ± 0.001	5.416 ± 0.717
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0370	0.0370	0.979	94.885 ± 0.698	0.144 ± 0.001	5.115 ± 0.698
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0268	0.0268	1.009	95.186 ± 0.678	0.116 ± 0.001	4.814 ± 0.678
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0288	0.0288	1.024	95.787 ± 0.636	0.116 ± 0.001	4.213 ± 0.636
LSAM	beta51	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.005	0.315 ± 0.009	0.156	-0.004	-2.22	0.0482	0.0484	0.997	94.784 ± 0.704	0.197 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.001	0.015 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.161	0.001	0.36	0.0397	0.0397	0.986	94.584 ± 0.717	0.152 ± 0.000	—
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.264 ± 0.007	0.158	-0.002	-1.44	0.0411	0.0412	0.952	93.982 ± 0.753	0.157 ± 0.000	—
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.000 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.159	-0.001	-0.80	0.0398	0.0399	0.970	94.584 ± 0.717	0.152 ± 0.000	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0543	0.0543	0.999	94.483 ± 0.723	0.220 ± 0.001	5.517 ± 0.723
LSAM	beta56	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	—	0.0471	0.0471	0.982	94.784 ± 0.704	0.189 ± 0.001	5.216 ± 0.704
LSAM	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.001	0.001	—	0.0312	0.0312	1.009	94.885 ± 0.698	0.128 ± 0.001	5.115 ± 0.698
LSAM	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.000	0.000	—	0.0258	0.0258	1.039	95.186 ± 0.678	0.110 ± 0.001	4.814 ± 0.678
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.173 ± 0.005	0.199	-0.001	-0.36	0.0326	0.0326	1.027	94.985 ± 0.691	0.139 ± 0.001	—
QML	beta42	0.20	0.200	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.002	0.183 ± 0.005	0.200	0.000	0.04	0.0368	0.0368	0.999	95.587 ± 0.650	0.143 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.001	0.184 ± 0.005	0.199	-0.001	-0.65	0.0376	0.0376	0.957	92.979 ± 0.809	0.138 ± 0.001	—
QML	beta44	0.00	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.008	-0.008	—	0.0420	0.0428	1.008	95.286 ± 0.671	0.170 ± 0.001	4.714 ± 0.671
QML	beta45	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.005	-0.005	—	0.0364	0.0367	0.993	93.681 ± 0.771	0.144 ± 0.001	6.319 ± 0.771
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0268	0.0268	1.018	94.985 ± 0.691	0.114 ± 0.001	5.015 ± 0.691
QML	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0273	0.0274	1.050	95.687 ± 0.643	0.114 ± 0.001	4.313 ± 0.643
QML	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.312 ± 0.009	0.158	-0.002	-1.43	0.0468	0.0469	1.027	95.186 ± 0.678	0.201 ± 0.001	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.160	0.000	-0.27	0.0397	0.0397	1.002	94.784 ± 0.704	0.153 ± 0.001	—
QML	beta53	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.157	-0.003	-1.89	0.0404	0.0405	0.960	94.383 ± 0.729	0.156 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.158	-0.002	-1.55	0.0393	0.0394	0.980	94.784 ± 0.704	0.150 ± 0.000	—
QML	beta55	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.003	-0.003	—	0.0534	0.0535	1.023	95.186 ± 0.678	0.220 ± 0.002	4.814 ± 0.678
QML	beta56	0.00	-0.006	-0.006 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.005	-0.005	—	0.0454	0.0457	0.991	94.684 ± 0.711	0.185 ± 0.001	5.316 ± 0.711
QML	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0300	0.0300	1.032	94.985 ± 0.691	0.127 ± 0.001	5.015 ± 0.691
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.002	-0.002	—	0.0239	0.0240	1.054	95.186 ± 0.678	0.106 ± 0.001	4.814 ± 0.678
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.202	0.002	0.78	0.0341	0.0341	1.022	94.684 ± 0.711	0.142 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.200	0.000	0.14	0.0374	0.0374	1.007	95.888 ± 0.629	0.147 ± 0.000	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.200	0.000	-0.25	0.0382	0.0382	0.952	93.380 ± 0.787	0.140 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.002	0.002	—	0.0423	0.0423	1.019	94.985 ± 0.691	0.175 ± 0.001	5.015 ± 0.691
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0372	0.0373	1.005	95.486 ± 0.657	0.151 ± 0.001	4.514 ± 0.657
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0287	0.0287	1.029	95.888 ± 0.629	0.123 ± 0.001	4.112 ± 0.629

Table S256

Simulation 2 - Condition 112: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.002	0.002	—	0.0297	0.0298	1.057	96.289 ± 0.599	0.123 ± 0.001	3.711 ± 0.599
UPI	beta51	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.311 ± 0.009	0.157	-0.003	-1.99	0.0480	0.0481	1.019	94.985 ± 0.691	0.198 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.015 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.161	0.001	0.43	0.0397	0.0397	0.993	94.784 ± 0.704	0.152 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.157	-0.003	-1.69	0.0409	0.0410	0.958	94.082 ± 0.747	0.158 ± 0.000	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.159	-0.001	-0.85	0.0400	0.0401	0.971	94.483 ± 0.723	0.153 ± 0.000	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0545	0.0545	1.017	95.386 ± 0.664	0.225 ± 0.002	4.614 ± 0.664
UPI	beta56	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.001	0.001	—	0.0473	0.0474	1.005	95.587 ± 0.650	0.193 ± 0.001	4.413 ± 0.650
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.000	0.000	—	0.0320	0.0320	1.017	96.088 ± 0.614	0.131 ± 0.001	3.912 ± 0.614
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.000	0.000	—	0.0258	0.0258	1.052	95.988 ± 0.622	0.113 ± 0.001	4.012 ± 0.622

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S257

Simulation 2 - Condition 113: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.198	-0.002	-0.99	0.0483	0.0483	0.946	93.373 ± 0.788	0.180 ± 0.001	—
LSAM	beta42	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.199	-0.001	-0.36	0.0470	0.0470	0.996	94.679 ± 0.711	0.185 ± 0.001	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.201	0.001	0.37	0.0416	0.0416	0.999	94.880 ± 0.698	0.178 ± 0.001	—
LSAM	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.003	0.003	—	0.0508	0.0509	0.966	93.273 ± 0.794	0.198 ± 0.001	6.727 ± 0.794
LSAM	beta45	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.003	-0.003	—	0.0483	0.0484	0.912	91.667 ± 0.876	0.177 ± 0.001	8.333 ± 0.876
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0387	0.0387	0.937	93.775 ± 0.766	0.139 ± 0.001	6.225 ± 0.766
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0332	0.0333	0.969	94.076 ± 0.748	0.136 ± 0.001	5.924 ± 0.748
LSAM	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.009	0.408 ± 0.012	0.155	-0.005	-2.88	0.0627	0.0629	0.965	94.277 ± 0.736	0.247 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.331 ± 0.009	0.161	0.001	0.54	0.0493	0.0494	0.954	93.876 ± 0.760	0.198 ± 0.001	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.002	-0.019 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.157	-0.003	-1.78	0.0492	0.0492	0.996	94.980 ± 0.692	0.204 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.016 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.162	0.002	1.19	0.0496	0.0496	0.991	95.783 ± 0.637	0.197 ± 0.001	—
LSAM	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.000	0.000	—	0.0709	0.0709	0.919	93.574 ± 0.777	0.262 ± 0.002	6.426 ± 0.777
LSAM	beta56	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.001	0.001	—	0.0591	0.0592	0.930	93.273 ± 0.794	0.228 ± 0.002	6.727 ± 0.794
LSAM	beta57	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0402	0.0402	0.952	93.976 ± 0.754	0.157 ± 0.001	6.024 ± 0.754
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0354	0.0355	0.932	93.574 ± 0.777	0.136 ± 0.001	6.426 ± 0.777
QML	beta41	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.197	-0.003	-1.38	0.0476	0.0477	0.961	93.876 ± 0.760	0.181 ± 0.002	—
QML	beta42	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.234 ± 0.007	0.199	-0.001	-0.71	0.0460	0.0461	1.002	94.679 ± 0.711	0.184 ± 0.001	—
QML	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.201	0.001	0.34	0.0431	0.0431	1.005	95.281 ± 0.672	0.178 ± 0.001	—
QML	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.001	-0.001	—	0.0496	0.0496	0.988	93.976 ± 0.754	0.203 ± 0.004	6.024 ± 0.754
QML	beta45	0.00	-0.005	-0.005 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.006	-0.006	—	0.0490	0.0493	0.987	91.265 ± 0.895	0.191 ± 0.013	8.735 ± 0.895
QML	beta46	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0375	0.0376	0.943	94.277 ± 0.736	0.138 ± 0.001	5.723 ± 0.736
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.003	-0.003	—	0.0330	0.0332	0.971	94.277 ± 0.736	0.135 ± 0.001	5.723 ± 0.736
QML	beta51	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.008	0.406 ± 0.012	0.155	-0.005	-3.27	0.0622	0.0624	0.997	94.679 ± 0.711	0.254 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.161	0.001	0.70	0.0501	0.0501	0.963	93.876 ± 0.760	0.200 ± 0.001	—
QML	beta53	0.16	0.157	-0.003 ± 0.002	-0.022 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.157	-0.003	-1.72	0.0499	0.0500	1.004	94.980 ± 0.692	0.203 ± 0.001	—
QML	beta54	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.005	0.315 ± 0.009	0.162	0.002	1.02	0.0497	0.0497	0.998	95.884 ± 0.630	0.197 ± 0.001	—
QML	beta55	0.00	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.003	-0.003	—	0.0699	0.0700	0.944	93.072 ± 0.805	0.262 ± 0.002	6.928 ± 0.805
QML	beta56	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	—	0.0579	0.0579	0.947	93.976 ± 0.754	0.228 ± 0.002	6.024 ± 0.754
QML	beta57	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0382	0.0383	0.965	93.574 ± 0.777	0.157 ± 0.002	6.426 ± 0.777
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.003	-0.003	—	0.0345	0.0347	0.942	92.470 ± 0.836	0.134 ± 0.001	7.530 ± 0.836
UPI	beta41	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.198	-0.002	-0.87	0.0494	0.0494	0.950	93.072 ± 0.805	0.181 ± 0.001	—
UPI	beta42	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.200	0.000	-0.22	0.0467	0.0467	1.003	95.080 ± 0.685	0.185 ± 0.001	—
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.201	0.001	0.39	0.0419	0.0419	1.006	95.181 ± 0.679	0.179 ± 0.001	—
UPI	beta44	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.002	0.002	—	0.0515	0.0515	0.972	93.373 ± 0.788	0.207 ± 0.001	6.627 ± 0.788
UPI	beta45	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.003	-0.003	—	0.0495	0.0496	0.915	91.968 ± 0.861	0.182 ± 0.001	8.032 ± 0.861
UPI	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0397	0.0397	0.937	94.076 ± 0.748	0.144 ± 0.001	5.924 ± 0.748

Table S257

Simulation 2 - Condition 113: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0336	0.0337	0.978	94.880 ± 0.698	0.143 ± 0.001	5.120 ± 0.698
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.012	0.156	-0.004	-2.45	0.0608	0.0610	0.991	94.478 ± 0.724	0.251 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.331 ± 0.009	0.161	0.001	0.59	0.0494	0.0494	0.956	93.373 ± 0.788	0.198 ± 0.001	—
UPI	beta53	0.16	0.157	-0.003 ± 0.002	-0.019 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.327 ± 0.009	0.157	-0.003	-1.73	0.0489	0.0490	0.999	94.779 ± 0.705	0.204 ± 0.001	—
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.015 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.162	0.002	1.55	0.0501	0.0501	0.993	95.582 ± 0.651	0.198 ± 0.001	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.000	0.000	—	0.0699	0.0699	0.943	94.277 ± 0.736	0.268 ± 0.002	5.723 ± 0.736
UPI	beta56	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.000	0.000	—	0.0583	0.0583	0.944	94.578 ± 0.718	0.233 ± 0.002	5.422 ± 0.718
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0403	0.0403	0.963	94.378 ± 0.730	0.159 ± 0.001	5.622 ± 0.730
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.002	-0.002	—	0.0362	0.0363	0.949	93.976 ± 0.754	0.138 ± 0.001	6.024 ± 0.754

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S258

Simulation 2 - Condition 114: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.200	0.000	-0.03	0.0299	0.0299	0.955	93.900 ± 0.757	0.113 ± 0.000	—
LSAM	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.200	0.000	-0.08	0.0299	0.0299	0.952	93.200 ± 0.796	0.117 ± 0.000	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.010 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.199	-0.001	-0.69	0.0289	0.0289	1.005	94.100 ± 0.745	0.112 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.002	0.002	—	0.0333	0.0334	0.943	93.400 ± 0.785	0.124 ± 0.001	6.600 ± 0.785
LSAM	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0290	0.0290	0.977	94.000 ± 0.751	0.112 ± 0.000	6.000 ± 0.751
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.001	-0.001	—	0.0231	0.0231	0.966	93.900 ± 0.757	0.087 ± 0.000	6.100 ± 0.757
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.002	-0.002	—	0.0231	0.0232	0.947	93.700 ± 0.768	0.086 ± 0.000	6.300 ± 0.768
LSAM	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.161	0.001	0.33	0.0407	0.0407	0.969	94.100 ± 0.745	0.158 ± 0.001	—
LSAM	beta52	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.201 ± 0.006	0.159	-0.001	-0.36	0.0300	0.0300	0.984	94.500 ± 0.721	0.124 ± 0.000	—
LSAM	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.159	-0.001	-0.40	0.0340	0.0340	0.996	94.400 ± 0.727	0.129 ± 0.000	—
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.005	0.161	0.001	0.76	0.0327	0.0327	0.992	95.500 ± 0.656	0.124 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0427	0.0427	0.966	94.400 ± 0.727	0.164 ± 0.001	5.600 ± 0.727
LSAM	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0375	0.0375	0.955	94.800 ± 0.702	0.144 ± 0.001	5.200 ± 0.702
LSAM	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	-0.001	-0.001	—	0.0243	0.0244	1.005	95.400 ± 0.662	0.097 ± 0.000	4.600 ± 0.662
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.001	-0.001	—	0.0226	0.0227	0.961	94.200 ± 0.739	0.085 ± 0.000	5.800 ± 0.739
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.199	-0.001	-0.62	0.0299	0.0299	0.959	94.000 ± 0.751	0.112 ± 0.000	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.199	-0.001	-0.42	0.0296	0.0296	0.956	93.500 ± 0.780	0.115 ± 0.000	—
QML	beta43	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.198	-0.002	-0.83	0.0292	0.0292	1.007	94.200 ± 0.739	0.112 ± 0.000	—
QML	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.001	-0.001	—	0.0340	0.0340	0.947	93.800 ± 0.763	0.125 ± 0.001	6.200 ± 0.763
QML	beta45	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0289	0.0290	0.979	94.200 ± 0.739	0.112 ± 0.001	5.800 ± 0.739
QML	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.001	-0.001	—	0.0229	0.0229	0.974	94.000 ± 0.751	0.086 ± 0.000	6.000 ± 0.751
QML	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.002	-0.002	—	0.0230	0.0231	0.951	92.000 ± 0.858	0.085 ± 0.000	8.000 ± 0.858
QML	beta51	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.161	0.001	0.44	0.0408	0.0408	0.983	94.300 ± 0.733	0.160 ± 0.001	—
QML	beta52	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.006	0.159	-0.001	-0.64	0.0298	0.0298	0.989	94.400 ± 0.727	0.124 ± 0.000	—
QML	beta53	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.204 ± 0.006	0.159	-0.001	-0.72	0.0333	0.0333	1.000	94.700 ± 0.708	0.128 ± 0.000	—
QML	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.161	0.001	0.56	0.0326	0.0326	0.996	95.200 ± 0.676	0.124 ± 0.000	—
QML	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.001	-0.001	—	0.0419	0.0419	0.979	95.100 ± 0.683	0.165 ± 0.001	4.900 ± 0.683
QML	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0372	0.0372	0.971	94.700 ± 0.708	0.144 ± 0.001	5.300 ± 0.708
QML	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.001	-0.001	—	0.0238	0.0239	1.017	94.700 ± 0.708	0.096 ± 0.001	5.300 ± 0.708
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.001	-0.001	—	0.0219	0.0220	0.966	93.800 ± 0.763	0.084 ± 0.000	6.200 ± 0.763
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.200	0.000	-0.02	0.0300	0.0300	0.956	94.100 ± 0.745	0.113 ± 0.000	—
UPI	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.200	0.000	-0.03	0.0296	0.0296	0.953	93.700 ± 0.768	0.116 ± 0.000	—
UPI	beta43	0.20	0.198	-0.002 ± 0.001	-0.010 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.199	-0.001	-0.72	0.0288	0.0288	1.008	94.500 ± 0.721	0.113 ± 0.000	—
UPI	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.002	0.002	—	0.0341	0.0341	0.944	93.900 ± 0.757	0.126 ± 0.001	6.100 ± 0.757
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0286	0.0286	0.982	94.300 ± 0.733	0.113 ± 0.001	5.700 ± 0.733
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.002	-0.002	—	0.0240	0.0240	0.978	93.900 ± 0.757	0.088 ± 0.000	6.100 ± 0.757

Table S258

Simulation 2 - Condition 114: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.002	-0.002	—	0.0239	0.0240	0.954	92.900 ± 0.812	0.088 ± 0.000	7.100 ± 0.812
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.160	0.000	0.09	0.0409	0.0409	0.981	94.200 ± 0.739	0.159 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.201 ± 0.006	0.159	-0.001	-0.53	0.0302	0.0302	0.986	94.300 ± 0.733	0.124 ± 0.000	—
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.159	-0.001	-0.41	0.0338	0.0338	0.998	94.500 ± 0.721	0.129 ± 0.000	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.005	0.161	0.001	0.83	0.0328	0.0328	0.995	95.500 ± 0.656	0.124 ± 0.000	—
UPI	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.001	0.001	—	0.0414	0.0414	0.976	94.300 ± 0.733	0.165 ± 0.001	5.700 ± 0.733
UPI	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0386	0.0387	0.967	94.900 ± 0.696	0.145 ± 0.001	5.100 ± 0.696
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.000	0.000	—	0.0251	0.0251	1.007	95.100 ± 0.683	0.097 ± 0.000	4.900 ± 0.683
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.000	0.000	—	0.0226	0.0226	0.960	93.900 ± 0.757	0.085 ± 0.000	6.100 ± 0.757

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S259

Simulation 2 - Condition 115: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.007	-0.013 ± 0.036	0.025 ± 0.002	0.158 ± 0.006	0.025 ± 0.002	0.158 ± 0.007	0.623 ± 0.044	0.789 ± 0.033	0.201	0.001	0.65	0.1518	0.1518	1.077	97.546 ± 0.700	0.667 ± 0.011	—
LSAM	beta42	0.20	0.215	0.015 ± 0.008	0.074 ± 0.038	0.029 ± 0.002	0.169 ± 0.006	0.029 ± 0.002	0.170 ± 0.007	0.721 ± 0.054	0.849 ± 0.037	0.203	0.003	1.62	0.1603	0.1603	1.064	97.955 ± 0.640	0.706 ± 0.014	—
LSAM	beta43	0.20	0.197	-0.003 ± 0.004	-0.016 ± 0.019	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	0.177 ± 0.012	0.420 ± 0.017	0.199	-0.001	-0.63	0.0883	0.0883	1.053	95.501 ± 0.937	0.347 ± 0.004	—
LSAM	beta44	0.00	0.005	0.005 ± 0.006	—	0.018 ± 0.001	0.134 ± 0.006	0.018 ± 0.001	0.134 ± 0.006	—	—	0.005	0.005	—	0.1176	0.1178	1.049	97.955 ± 0.640	0.550 ± 0.017	2.045 ± 0.640
LSAM	beta45	0.00	0.005	0.005 ± 0.005	—	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.005	—	—	0.003	0.003	—	0.1012	0.1013	1.060	98.364 ± 0.574	0.461 ± 0.011	1.636 ± 0.574
LSAM	beta46	0.00	0.000	0.000 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.071 ± 0.003	—	—	-0.006	-0.006	—	0.0648	0.0651	1.009	98.569 ± 0.537	0.283 ± 0.007	1.431 ± 0.537
LSAM	beta47	0.00	-0.004	-0.004 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.003	0.004 ± 0.000	0.066 ± 0.003	—	—	-0.004	-0.004	—	0.0590	0.0592	1.075	98.160 ± 0.608	0.278 ± 0.008	1.840 ± 0.608
LSAM	beta51	0.16	0.160	-0.000 ± 0.006	-0.001 ± 0.037	0.017 ± 0.001	0.131 ± 0.005	0.017 ± 0.001	0.131 ± 0.006	0.671 ± 0.049	0.819 ± 0.035	0.158	-0.002	-1.52	0.1226	0.1227	0.928	93.865 ± 1.085	0.477 ± 0.005	—
LSAM	beta52	0.16	0.149	-0.011 ± 0.007	-0.069 ± 0.046	0.027 ± 0.002	0.163 ± 0.006	0.027 ± 0.002	0.164 ± 0.007	1.047 ± 0.070	1.023 ± 0.042	0.150	-0.010	-6.07	0.1656	0.1659	1.098	97.546 ± 0.700	0.704 ± 0.013	—
LSAM	beta53	0.16	0.161	0.001 ± 0.006	0.005 ± 0.035	0.016 ± 0.001	0.125 ± 0.004	0.016 ± 0.001	0.125 ± 0.005	0.612 ± 0.044	0.782 ± 0.033	0.167	0.007	4.44	0.1208	0.1210	1.054	97.751 ± 0.671	0.517 ± 0.008	—
LSAM	beta54	0.16	0.163	0.003 ± 0.007	0.020 ± 0.047	0.027 ± 0.002	0.165 ± 0.006	0.027 ± 0.002	0.165 ± 0.007	1.066 ± 0.080	1.032 ± 0.045	0.157	-0.003	-1.85	0.1673	0.1673	1.052	96.933 ± 0.780	0.681 ± 0.011	—
LSAM	beta55	0.00	0.017	0.017 ± 0.009	—	0.042 ± 0.003	0.205 ± 0.009	0.042 ± 0.004	0.205 ± 0.010	—	—	0.014	0.014	—	0.1808	0.1813	1.062	98.364 ± 0.574	0.853 ± 0.030	1.636 ± 0.574
LSAM	beta56	0.00	-0.002	-0.002 ± 0.007	—	0.021 ± 0.002	0.146 ± 0.006	0.021 ± 0.002	0.146 ± 0.006	—	—	0.000	0.000	—	0.1368	0.1368	1.097	98.160 ± 0.608	0.630 ± 0.016	1.840 ± 0.608
LSAM	beta57	0.00	-0.001	-0.001 ± 0.004	—	0.008 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.087 ± 0.004	—	—	0.002	0.002	—	0.0751	0.0751	1.064	97.751 ± 0.671	0.363 ± 0.011	2.249 ± 0.671
LSAM	beta58	0.00	0.000	0.000 ± 0.003	—	0.004 ± 0.000	0.063 ± 0.003	0.004 ± 0.000	0.063 ± 0.003	—	—	0.000	0.000	—	0.0544	0.0544	1.043	97.137 ± 0.754	0.256 ± 0.007	2.863 ± 0.754
QML	beta41	0.20	0.170	-0.030 ± 0.004	-0.150 ± 0.019	0.007 ± 0.000	0.084 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	0.198 ± 0.013	0.444 ± 0.017	0.174	-0.026	-13.12	0.0865	0.0904	1.094	94.888 ± 0.996	0.359 ± 0.003	—
QML	beta42	0.20	0.180	-0.020 ± 0.004	-0.099 ± 0.020	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.092 ± 0.004	0.211 ± 0.014	0.459 ± 0.019	0.173	-0.027	-13.62	0.0881	0.0922	1.050	94.888 ± 0.996	0.370 ± 0.004	—
QML	beta43	0.20	0.203	0.003 ± 0.003	0.016 ± 0.017	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.075 ± 0.003	0.141 ± 0.010	0.375 ± 0.015	0.206	0.006	3.09	0.0781	0.0784	1.056	94.888 ± 0.996	0.311 ± 0.003	—
QML	beta44	0.00	-0.029	-0.029 ± 0.005	—	0.012 ± 0.001	0.109 ± 0.004	0.013 ± 0.001	0.113 ± 0.005	—	—	-0.024	-0.024	—	0.1080	0.1107	1.098	95.706 ± 0.917	0.468 ± 0.015	4.294 ± 0.917
QML	beta45	0.00	0.005	0.005 ± 0.004	—	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.088 ± 0.004	—	—	0.001	0.001	—	0.0871	0.0871	1.058	98.364 ± 0.574	0.364 ± 0.008	1.636 ± 0.574
QML	beta46	0.00	0.047	0.047 ± 0.003	—	0.005 ± 0.000	0.069 ± 0.003	0.007 ± 0.001	0.083 ± 0.004	—	—	0.038	0.038	—	0.0641	0.0745	1.033	93.456 ± 1.118	0.281 ± 0.013	6.544 ± 1.118
QML	beta47	0.00	0.043	0.043 ± 0.003	—	0.004 ± 0.000	0.059 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	—	—	0.040	0.040	—	0.0580	0.0703	1.084	95.706 ± 0.917	0.253 ± 0.008	4.294 ± 0.917
QML	beta51	0.16	0.169	0.009 ± 0.006	0.057 ± 0.037	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	0.657 ± 0.042	0.811 ± 0.032	0.169	0.009	5.43	0.1396	0.1399	1.054	95.501 ± 0.937	0.535 ± 0.011	—
QML	beta52	0.16	0.134	-0.026 ± 0.004	-0.164 ± 0.027	0.009 ± 0.001	0.095 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.381 ± 0.025	0.617 ± 0.024	0.133	-0.027	-17.14	0.0950	0.0988	1.119	96.524 ± 0.828	0.418 ± 0.007	—
QML	beta53	0.16	0.154	-0.006 ± 0.004	-0.040 ± 0.027	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.360 ± 0.025	0.600 ± 0.025	0.157	-0.003	-1.88	0.1016	0.1017	1.050	96.933 ± 0.780	0.395 ± 0.005	—
QML	beta54	0.16	0.135	-0.025 ± 0.004	-0.155 ± 0.027	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.392 ± 0.027	0.626 ± 0.026	0.131	-0.029	-17.87	0.0966	0.1007	1.003	95.706 ± 0.917	0.382 ± 0.004	—
QML	beta55	0.00	-0.020	-0.020 ± 0.007	—	0.021 ± 0.001	0.146 ± 0.005	0.022 ± 0.001	0.147 ± 0.006	—	—	-0.022	-0.022	—	0.1484	0.1500	1.116	95.706 ± 0.917	0.638 ± 0.022	4.294 ± 0.917
QML	beta56	0.00	0.027	0.027 ± 0.005	—	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.107 ± 0.004	—	—	0.020	0.020	—	0.1074	0.1093	1.093	95.910 ± 0.896	0.442 ± 0.009	4.090 ± 0.896
QML	beta57	0.00	0.036	0.036 ± 0.003	—	0.006 ± 0.000	0.074 ± 0.003	0.007 ± 0.001	0.083 ± 0.004	—	—	0.029	0.029	—	0.0667	0.0728	1.131	95.910 ± 0.896	0.329 ± 0.014	4.090 ± 0.896
QML	beta58	0.00	0.034	0.034 ± 0.003	—	0.003 ± 0.000	0.056 ± 0.003	0.004 ± 0.000	0.065 ± 0.003	—	—	0.029	0.029	—	0.0503	0.0579	0.958	93.252 ± 1.134	0.210 ± 0.005	6.748 ± 1.134
UPI	beta41	0.20	0.205	0.005 ± 0.009	0.025 ± 0.047	0.043 ± 0.004	0.208 ± 0.009	0.043 ± 0.004	0.208 ± 0.010	1.085 ± 0.096	1.042 ± 0.052	0.210	0.010	4.99	0.2238	0.2240	1.508	99.591 ± 0.289	1.233 ± 0.073	—
UPI	beta42	0.20	0.213	0.013 ± 0.010	0.063 ± 0.048	0.045 ± 0.004	0.212 ± 0.009	0.045 ± 0.004	0.212 ± 0.011	1.127 ± 0.099	1.061 ± 0.053	0.203	0.003	1.32	0.2371	0.2371	1.510	99.387 ± 0.353	1.256 ± 0.059	—
UPI	beta43	0.20	0.201	0.001 ± 0.004	0.006 ± 0.020	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.088 ± 0.004	0.196 ± 0.014	0.442 ± 0.019	0.199	-0.001	-0.41	0.1025	0.1025	1.349	96.524 ± 0.828	0.468 ± 0.023	—
UPI	beta44	0.00	0.011	0.011 ± 0.008	—	0.030 ± 0.003	0.174 ± 0.009	0.030 ± 0.003	0.174 ± 0.010	—	—	0.004	0.004	—	0.1577	0.1578	1.660	99.591 ± 0.289	1.134 ± 0.068	0.409 ± 0.289
UPI	beta45	0.00	0.003	0.003 ± 0.007	—	0.022 ± 0.002	0.149 ± 0.008	0.022 ± 0.002	0.149 ± 0.009	—	—	0.007	0.007	—	0.1374	0.1376	1.731	99.796 ± 0.204	1.013 ± 0.073	0.204 ± 0.204
UPI	beta46	0.00	-0.001	-0.001 ± 0.004	—	0.009 ± 0.001	0.095 ± 0.005	0.009 ± 0.001	0.095 ± 0.005	—	—	-0.005	-0.005	—	0.0956	0.0958	1.510	99.591 ± 0.289	0.561 ± 0.032	0.409 ± 0.289

Table S259

Simulation 2 - Condition 115: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.004	—	0.008 ± 0.001	0.087 ± 0.005	0.008 ± 0.001	0.087 ± 0.005	—	—	-0.002	-0.002	—	0.0865	0.0865	1.503	98.978 ± 0.455	0.515 ± 0.027	1.022 ± 0.455
UPI	beta51	0.16	0.159	-0.001 ± 0.006	-0.004 ± 0.036	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	0.628 ± 0.039	0.792 ± 0.030	0.154	-0.006	-3.65	0.1389	0.1390	1.066	96.319 ± 0.851	0.530 ± 0.009	—
UPI	beta52	0.16	0.149	-0.011 ± 0.008	-0.067 ± 0.051	0.033 ± 0.003	0.182 ± 0.008	0.033 ± 0.003	0.182 ± 0.009	1.293 ± 0.113	1.137 ± 0.056	0.150	-0.010	-6.32	0.1837	0.1840	1.330	97.546 ± 0.700	0.948 ± 0.038	—
UPI	beta53	0.16	0.160	-0.000 ± 0.006	-0.000 ± 0.036	0.016 ± 0.001	0.127 ± 0.005	0.016 ± 0.001	0.126 ± 0.006	0.625 ± 0.049	0.790 ± 0.036	0.167	0.007	4.31	0.1360	0.1361	1.301	98.773 ± 0.498	0.646 ± 0.023	—
UPI	beta54	0.16	0.170	0.010 ± 0.008	0.064 ± 0.052	0.034 ± 0.003	0.184 ± 0.008	0.034 ± 0.003	0.184 ± 0.009	1.319 ± 0.110	1.148 ± 0.055	0.156	-0.004	-2.37	0.1806	0.1806	1.226	98.773 ± 0.498	0.882 ± 0.026	—
UPI	beta55	0.00	0.010	0.010 ± 0.009	—	0.042 ± 0.004	0.205 ± 0.010	0.042 ± 0.004	0.205 ± 0.011	—	—	0.003	0.003	—	0.1773	0.1773	1.454	99.182 ± 0.407	1.170 ± 0.056	0.818 ± 0.407
UPI	beta56	0.00	-0.001	-0.001 ± 0.007	—	0.025 ± 0.002	0.157 ± 0.007	0.025 ± 0.002	0.157 ± 0.008	—	—	-0.002	-0.002	—	0.1447	0.1448	1.479	98.773 ± 0.498	0.913 ± 0.045	1.227 ± 0.498
UPI	beta57	0.00	0.002	0.002 ± 0.004	—	0.008 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.087 ± 0.004	—	—	0.004	0.004	—	0.0819	0.0820	1.397	98.160 ± 0.608	0.475 ± 0.020	1.840 ± 0.608
UPI	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.003	0.005 ± 0.000	0.070 ± 0.004	—	—	-0.001	-0.001	—	0.0606	0.0606	1.217	98.160 ± 0.608	0.335 ± 0.012	1.840 ± 0.608

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S260

Simulation 2 - Condition 116: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.003	-0.017 ± 0.017	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.248 ± 0.012	0.498 ± 0.014	0.198	-0.002	-1.00	0.0993	0.0993	1.001	94.934 ± 0.728	0.391 ± 0.003	—
LSAM	beta42	0.20	0.206	0.006 ± 0.004	0.029 ± 0.018	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.286 ± 0.015	0.535 ± 0.016	0.205	0.005	2.53	0.1049	0.1051	0.990	96.366 ± 0.621	0.415 ± 0.003	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.004	0.271 ± 0.008	0.200	0.000	0.10	0.0517	0.0517	0.984	94.824 ± 0.735	0.209 ± 0.001	—
LSAM	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.003	-0.003	—	0.0663	0.0664	1.001	97.026 ± 0.564	0.265 ± 0.003	2.974 ± 0.564
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.001	0.001	—	0.0607	0.0607	0.984	96.035 ± 0.648	0.246 ± 0.003	3.965 ± 0.648
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0329	0.0329	1.003	96.476 ± 0.612	0.141 ± 0.002	3.524 ± 0.612
LSAM	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0315	0.0315	0.952	96.586 ± 0.603	0.143 ± 0.002	3.414 ± 0.603
LSAM	beta51	0.16	0.166	0.006 ± 0.003	0.035 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.231 ± 0.011	0.481 ± 0.014	0.165	0.005	3.40	0.0764	0.0766	0.956	94.163 ± 0.778	0.288 ± 0.001	—
LSAM	beta52	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.022	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.422 ± 0.021	0.649 ± 0.020	0.161	0.001	0.40	0.1031	0.1031	1.001	96.806 ± 0.584	0.408 ± 0.003	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.003	-0.009 ± 0.016	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.223 ± 0.011	0.472 ± 0.014	0.156	-0.004	-2.41	0.0733	0.0734	1.052	96.586 ± 0.603	0.312 ± 0.002	—
LSAM	beta54	0.16	0.167	0.007 ± 0.003	0.047 ± 0.022	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.425 ± 0.022	0.652 ± 0.020	0.159	-0.001	-0.53	0.0980	0.0980	0.988	96.256 ± 0.630	0.403 ± 0.003	—
LSAM	beta55	0.00	-0.002	-0.002 ± 0.004	—	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	—	—	-0.005	-0.005	—	0.0979	0.0980	1.007	97.577 ± 0.510	0.428 ± 0.007	2.423 ± 0.510
LSAM	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	—	—	0.000	0.000	—	0.0762	0.0762	1.019	97.357 ± 0.532	0.336 ± 0.004	2.643 ± 0.532
LSAM	beta57	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	—	—	-0.001	-0.001	—	0.0421	0.0421	1.005	96.366 ± 0.621	0.180 ± 0.003	3.634 ± 0.621
LSAM	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0332	0.0332	0.961	95.815 ± 0.665	0.137 ± 0.002	4.185 ± 0.665
QML	beta41	0.20	0.169	-0.031 ± 0.002	-0.153 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.315 ± 0.008	0.169	-0.031	-15.72	0.0560	0.0642	0.997	91.189 ± 0.941	0.215 ± 0.001	—
QML	beta42	0.20	0.175	-0.025 ± 0.002	-0.123 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.005	0.311 ± 0.009	0.176	-0.024	-12.04	0.0550	0.0600	1.003	92.511 ± 0.874	0.224 ± 0.001	—
QML	beta43	0.20	0.204	0.004 ± 0.002	0.021 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.206	0.006	3.18	0.0486	0.0490	0.986	94.493 ± 0.757	0.192 ± 0.001	—
QML	beta44	0.00	-0.030	-0.030 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.029	-0.029	—	0.0632	0.0695	1.020	94.273 ± 0.771	0.264 ± 0.003	5.727 ± 0.771
QML	beta45	0.00	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.007	0.007	—	0.0547	0.0551	1.002	94.824 ± 0.735	0.219 ± 0.002	5.176 ± 0.735
QML	beta46	0.00	0.043	0.043 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	—	—	0.042	0.042	—	0.0364	0.0553	1.012	82.599 ± 1.258	0.144 ± 0.002	17.401 ± 1.258
QML	beta47	0.00	0.044	0.044 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	—	—	0.041	0.041	—	0.0328	0.0525	0.969	81.498 ± 1.289	0.143 ± 0.002	18.502 ± 1.289
QML	beta51	0.16	0.172	0.012 ± 0.002	0.078 ± 0.016	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.225 ± 0.011	0.474 ± 0.014	0.174	0.014	8.87	0.0731	0.0745	1.016	95.925 ± 0.656	0.298 ± 0.003	—
QML	beta52	0.16	0.140	-0.020 ± 0.002	-0.126 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.007	0.395 ± 0.011	0.138	-0.022	-13.47	0.0596	0.0634	1.006	94.934 ± 0.728	0.236 ± 0.001	—
QML	beta53	0.16	0.153	-0.007 ± 0.002	-0.042 ± 0.012	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.360 ± 0.011	0.151	-0.009	-5.43	0.0591	0.0597	1.056	96.035 ± 0.648	0.237 ± 0.004	—
QML	beta54	0.16	0.137	-0.023 ± 0.002	-0.145 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.157 ± 0.007	0.396 ± 0.011	0.133	-0.027	-16.60	0.0595	0.0652	1.001	94.383 ± 0.764	0.231 ± 0.001	—
QML	beta55	0.00	-0.029	-0.029 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	—	—	-0.032	-0.032	—	0.0854	0.0912	1.039	95.485 ± 0.689	0.367 ± 0.006	4.515 ± 0.689
QML	beta56	0.00	0.028	0.028 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.029	0.029	—	0.0610	0.0678	1.057	94.934 ± 0.728	0.261 ± 0.008	5.066 ± 0.728
QML	beta57	0.00	0.038	0.038 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	0.033	0.033	—	0.0438	0.0550	1.041	91.630 ± 0.919	0.186 ± 0.003	8.370 ± 0.919
QML	beta58	0.00	0.028	0.028 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.026	0.026	—	0.0305	0.0398	0.968	88.877 ± 1.043	0.125 ± 0.002	11.123 ± 1.043
UPI	beta41	0.20	0.200	0.000 ± 0.004	0.001 ± 0.018	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.284 ± 0.015	0.533 ± 0.017	0.203	0.003	1.25	0.1030	0.1031	1.132	97.577 ± 0.510	0.473 ± 0.010	—
UPI	beta42	0.20	0.204	0.004 ± 0.004	0.022 ± 0.018	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.310 ± 0.016	0.557 ± 0.017	0.199	-0.001	-0.64	0.1066	0.1066	1.140	97.687 ± 0.499	0.497 ± 0.012	—
UPI	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.069 ± 0.003	0.263 ± 0.008	0.200	0.000	-0.03	0.0510	0.0510	1.024	95.044 ± 0.720	0.211 ± 0.002	—
UPI	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.005	-0.005	—	0.0653	0.0655	1.346	98.458 ± 0.409	0.397 ± 0.023	1.542 ± 0.409

Table S260

Simulation 2 - Condition 116: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.001	0.001	—	0.0577	0.0577	1.275	97.907 ± 0.475	0.345 ± 0.011	2.093 ± 0.475
UPI	beta46	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0337	0.0337	1.155	98.018 ± 0.463	0.176 ± 0.004	1.982 ± 0.463
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0335	0.0335	1.124	98.238 ± 0.437	0.178 ± 0.006	1.762 ± 0.437
UPI	beta51	0.16	0.165	0.005 ± 0.002	0.033 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.213 ± 0.010	0.462 ± 0.014	0.166	0.006	3.66	0.0725	0.0727	1.001	95.485 ± 0.689	0.289 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.003	0.012 ± 0.022	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.420 ± 0.021	0.648 ± 0.020	0.161	0.001	0.73	0.1019	0.1019	1.038	97.357 ± 0.532	0.422 ± 0.004	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.208 ± 0.011	0.457 ± 0.014	0.157	-0.003	-1.95	0.0712	0.0713	1.109	96.916 ± 0.574	0.318 ± 0.003	—
UPI	beta54	0.16	0.167	0.007 ± 0.003	0.041 ± 0.021	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.405 ± 0.021	0.637 ± 0.019	0.160	0.000	-0.17	0.0954	0.0954	1.043	96.916 ± 0.574	0.416 ± 0.004	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	—	—	-0.004	-0.004	—	0.0981	0.0982	1.092	98.568 ± 0.394	0.485 ± 0.011	1.432 ± 0.394
UPI	beta56	0.00	-0.003	-0.003 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	—	—	0.000	0.000	—	0.0731	0.0731	1.148	98.568 ± 0.394	0.376 ± 0.008	1.432 ± 0.394
UPI	beta57	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	—	—	-0.002	-0.002	—	0.0417	0.0417	1.058	97.467 ± 0.521	0.190 ± 0.003	2.533 ± 0.521
UPI	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0326	0.0327	1.018	96.806 ± 0.584	0.142 ± 0.002	3.194 ± 0.584

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S261

Simulation 2 - Condition 117: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.003	0.015 ± 0.017	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.001	0.100 ± 0.003	0.252 ± 0.013	0.502 ± 0.015	0.201	0.001	0.69	0.0997	0.0997	1.054	96.208 ± 0.629	0.415 ± 0.003	—
LSAM	beta42	0.20	0.201	0.001 ± 0.004	0.003 ± 0.018	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.285 ± 0.013	0.534 ± 0.015	0.200	0.000	0.07	0.1066	0.1066	1.033	95.450 ± 0.686	0.432 ± 0.003	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.010	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.005	0.305 ± 0.009	0.197	-0.003	-1.36	0.0598	0.0599	0.980	94.150 ± 0.773	0.234 ± 0.001	—
LSAM	beta44	0.00	0.002	0.002 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.003	0.003	—	0.0697	0.0697	0.955	95.341 ± 0.694	0.281 ± 0.003	4.659 ± 0.694
LSAM	beta45	0.00	0.006	0.006 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.008	0.008	—	0.0663	0.0667	0.935	95.341 ± 0.694	0.265 ± 0.003	4.659 ± 0.694
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0356	0.0356	0.993	95.883 ± 0.654	0.159 ± 0.002	4.117 ± 0.654
LSAM	beta47	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0369	0.0370	0.983	95.558 ± 0.678	0.154 ± 0.002	4.442 ± 0.678
LSAM	beta51	0.16	0.162	0.002 ± 0.003	0.014 ± 0.018	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.287 ± 0.014	0.536 ± 0.015	0.163	0.003	2.15	0.0827	0.0827	0.952	93.716 ± 0.799	0.320 ± 0.002	—
LSAM	beta52	0.16	0.156	-0.004 ± 0.004	-0.026 ± 0.024	0.013 ± 0.001	0.115 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.520 ± 0.027	0.721 ± 0.022	0.154	-0.006	-4.04	0.1147	0.1148	0.962	94.908 ± 0.724	0.435 ± 0.003	—
LSAM	beta53	0.16	0.162	0.002 ± 0.003	0.011 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.279 ± 0.013	0.528 ± 0.015	0.158	-0.002	-1.41	0.0845	0.0845	1.004	95.341 ± 0.694	0.333 ± 0.002	—
LSAM	beta54	0.16	0.163	0.003 ± 0.004	0.021 ± 0.023	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.491 ± 0.023	0.701 ± 0.020	0.169	0.009	5.35	0.1111	0.1114	0.974	94.366 ± 0.759	0.428 ± 0.003	—
LSAM	beta55	0.00	0.001	0.001 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.112 ± 0.004	—	—	-0.003	-0.003	—	0.1039	0.1039	0.940	95.883 ± 0.654	0.415 ± 0.005	4.117 ± 0.654
LSAM	beta56	0.00	-0.004	-0.004 ± 0.003	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	—	—	-0.002	-0.002	—	0.0916	0.0916	0.919	93.283 ± 0.824	0.347 ± 0.004	6.717 ± 0.824
LSAM	beta57	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.000	0.000	—	0.0460	0.0460	0.929	95.016 ± 0.716	0.188 ± 0.003	4.984 ± 0.716
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	—	0.0385	0.0385	0.939	95.125 ± 0.709	0.152 ± 0.002	4.875 ± 0.709
QML	beta41	0.20	0.179	-0.021 ± 0.002	-0.105 ± 0.011	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.120 ± 0.006	0.346 ± 0.010	0.179	-0.021	-10.55	0.0656	0.0689	1.066	94.908 ± 0.724	0.276 ± 0.001	—
QML	beta42	0.20	0.179	-0.021 ± 0.002	-0.105 ± 0.012	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.136 ± 0.007	0.368 ± 0.011	0.180	-0.020	-9.81	0.0691	0.0719	1.028	95.125 ± 0.709	0.285 ± 0.002	—
QML	beta43	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.287 ± 0.009	0.203	0.003	1.61	0.0558	0.0559	0.986	94.475 ± 0.752	0.221 ± 0.001	—
QML	beta44	0.00	-0.012	-0.012 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.011	-0.011	—	0.0656	0.0664	0.998	94.258 ± 0.766	0.265 ± 0.003	5.742 ± 0.766
QML	beta45	0.00	0.005	0.005 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.004	0.004	—	0.0581	0.0582	0.966	95.016 ± 0.716	0.237 ± 0.003	4.984 ± 0.716
QML	beta46	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.018	0.018	—	0.0338	0.0381	1.015	93.608 ± 0.805	0.145 ± 0.002	6.392 ± 0.805
QML	beta47	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.016	0.016	—	0.0329	0.0367	0.947	93.283 ± 0.824	0.139 ± 0.002	6.717 ± 0.824
QML	beta51	0.16	0.164	0.004 ± 0.003	0.027 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.267 ± 0.013	0.517 ± 0.015	0.165	0.005	2.97	0.0812	0.0814	1.025	95.558 ± 0.678	0.332 ± 0.003	—
QML	beta52	0.16	0.140	-0.020 ± 0.003	-0.126 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.247 ± 0.012	0.497 ± 0.015	0.139	-0.021	-13.24	0.0757	0.0786	0.993	94.583 ± 0.745	0.300 ± 0.003	—
QML	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.185 ± 0.009	0.430 ± 0.012	0.157	-0.003	-1.79	0.0686	0.0687	1.027	95.883 ± 0.654	0.277 ± 0.002	—
QML	beta54	0.16	0.144	-0.016 ± 0.003	-0.103 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.243 ± 0.012	0.493 ± 0.014	0.147	-0.013	-7.87	0.0761	0.0771	0.974	93.824 ± 0.792	0.295 ± 0.001	—
QML	beta55	0.00	-0.014	-0.014 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.095 ± 0.003	—	—	-0.014	-0.014	—	0.0897	0.0908	1.019	95.450 ± 0.686	0.377 ± 0.007	4.550 ± 0.686
QML	beta56	0.00	0.010	0.010 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	—	—	0.009	0.009	—	0.0732	0.0737	0.939	92.849 ± 0.848	0.283 ± 0.003	7.151 ± 0.848
QML	beta57	0.00	0.020	0.020 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	—	—	0.015	0.015	—	0.0431	0.0456	1.014	94.691 ± 0.738	0.182 ± 0.004	5.309 ± 0.738
QML	beta58	0.00	0.015	0.015 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.014	0.014	—	0.0337	0.0363	0.954	94.041 ± 0.779	0.130 ± 0.002	5.959 ± 0.779
UPI	beta41	0.20	0.206	0.006 ± 0.003	0.029 ± 0.017	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.270 ± 0.014	0.520 ± 0.016	0.202	0.002	1.14	0.0981	0.0981	1.157	97.508 ± 0.513	0.471 ± 0.010	—
UPI	beta42	0.20	0.200	0.000 ± 0.004	0.002 ± 0.018	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.003	0.299 ± 0.015	0.546 ± 0.017	0.200	0.000	-0.24	0.1065	0.1065	1.127	97.183 ± 0.545	0.483 ± 0.007	—
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.010	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.005	0.300 ± 0.009	0.196	-0.004	-2.08	0.0587	0.0589	1.005	94.908 ± 0.724	0.236 ± 0.002	—
UPI	beta44	0.00	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.000	0.000	—	0.0709	0.0709	1.117	96.533 ± 0.602	0.362 ± 0.008	3.467 ± 0.602
UPI	beta45	0.00	0.007	0.007 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.007	0.007	—	0.0686	0.0690	1.133	97.508 ± 0.513	0.353 ± 0.020	2.492 ± 0.513
UPI	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.002	-0.002	—	0.0393	0.0393	1.125	97.508 ± 0.513	0.192 ± 0.006	2.492 ± 0.513

Table S261

Simulation 2 - Condition 117: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0373	0.0373	1.092	97.183 ± 0.545	0.177 ± 0.003	2.817 ± 0.545
UPI	beta51	0.16	0.161	0.001 ± 0.003	0.004 ± 0.017	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.257 ± 0.012	0.507 ± 0.015	0.159	-0.001	-0.39	0.0797	0.0797	1.017	95.775 ± 0.662	0.323 ± 0.002	—
UPI	beta52	0.16	0.158	-0.002 ± 0.004	-0.011 ± 0.023	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.498 ± 0.026	0.706 ± 0.022	0.159	-0.001	-0.82	0.1128	0.1128	0.987	95.666 ± 0.670	0.437 ± 0.003	—
UPI	beta53	0.16	0.163	0.003 ± 0.003	0.016 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.263 ± 0.012	0.513 ± 0.015	0.157	-0.003	-1.56	0.0859	0.0860	1.035	96.316 ± 0.620	0.333 ± 0.003	—
UPI	beta54	0.16	0.162	0.002 ± 0.004	0.011 ± 0.023	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.471 ± 0.021	0.686 ± 0.019	0.165	0.005	3.24	0.1129	0.1130	1.005	94.908 ± 0.724	0.433 ± 0.003	—
UPI	beta55	0.00	0.004	0.004 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	—	—	-0.005	-0.005	—	0.0959	0.0960	1.060	97.075 ± 0.555	0.471 ± 0.010	2.925 ± 0.555
UPI	beta56	0.00	-0.006	-0.006 ± 0.003	—	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	-0.003	-0.003	—	0.0861	0.0862	1.014	95.125 ± 0.709	0.384 ± 0.008	4.875 ± 0.709
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	—	0.0472	0.0472	0.955	95.233 ± 0.701	0.193 ± 0.003	4.767 ± 0.701
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0367	0.0367	0.980	95.558 ± 0.678	0.155 ± 0.002	4.442 ± 0.678

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S262

Simulation 2 - Condition 118: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.001 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.005	0.311 ± 0.009	0.199	-0.001	-0.29	0.0615	0.0615	1.006	96.053 ± 0.619	0.245 ± 0.001	—
LSAM	beta42	0.20	0.197	-0.003 ± 0.002	-0.017 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.104 ± 0.005	0.323 ± 0.009	0.196	-0.004	-2.06	0.0673	0.0674	1.026	96.457 ± 0.588	0.260 ± 0.001	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.187 ± 0.005	0.201	0.001	0.39	0.0380	0.0380	0.986	93.725 ± 0.772	0.145 ± 0.001	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	—	0.0384	0.0384	0.979	95.243 ± 0.677	0.152 ± 0.001	4.757 ± 0.677
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0377	0.0378	0.948	95.142 ± 0.684	0.148 ± 0.001	4.858 ± 0.684
LSAM	beta46	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.002	-0.002	—	0.0190	0.0190	0.971	95.142 ± 0.684	0.085 ± 0.001	4.858 ± 0.684
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.002	0.002	—	0.0212	0.0212	0.975	95.344 ± 0.670	0.087 ± 0.001	4.656 ± 0.670
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.012 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.161	0.001	0.54	0.0541	0.0542	0.946	93.725 ± 0.772	0.198 ± 0.001	—
LSAM	beta52	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.172 ± 0.008	0.415 ± 0.012	0.158	-0.002	-1.37	0.0661	0.0661	0.986	95.142 ± 0.684	0.256 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.332 ± 0.009	0.161	0.001	0.54	0.0533	0.0533	0.969	94.231 ± 0.742	0.202 ± 0.001	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.007	0.399 ± 0.011	0.159	-0.001	-0.82	0.0610	0.0611	1.023	95.648 ± 0.649	0.256 ± 0.001	—
LSAM	beta55	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.002	0.002	—	0.0567	0.0567	0.944	94.332 ± 0.736	0.227 ± 0.002	5.668 ± 0.736
LSAM	beta56	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.000	0.000	—	0.0488	0.0488	0.954	94.838 ± 0.704	0.194 ± 0.002	5.162 ± 0.704
LSAM	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0253	0.0254	0.932	94.028 ± 0.754	0.101 ± 0.001	5.972 ± 0.754
LSAM	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.000	0.000	—	0.0209	0.0209	0.952	94.130 ± 0.748	0.085 ± 0.001	5.870 ± 0.748
QML	beta41	0.20	0.179	-0.021 ± 0.001	-0.104 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.238 ± 0.006	0.180	-0.020	-10.03	0.0435	0.0479	1.013	94.130 ± 0.748	0.170 ± 0.001	—
QML	beta42	0.20	0.177	-0.023 ± 0.001	-0.114 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.006	0.177	-0.023	-11.43	0.0446	0.0501	1.030	94.332 ± 0.736	0.177 ± 0.001	—
QML	beta43	0.20	0.206	0.006 ± 0.001	0.028 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.205	0.005	2.33	0.0358	0.0361	0.990	94.737 ± 0.710	0.139 ± 0.001	—
QML	beta44	0.00	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.011	-0.011	—	0.0410	0.0424	0.971	94.433 ± 0.729	0.162 ± 0.001	5.567 ± 0.729
QML	beta45	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0371	0.0372	0.965	95.142 ± 0.684	0.143 ± 0.001	4.858 ± 0.684
QML	beta46	0.00	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.017	0.017	—	0.0202	0.0266	0.965	89.069 ± 0.993	0.085 ± 0.001	10.931 ± 0.993
QML	beta47	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.019	0.019	—	0.0218	0.0290	0.943	87.146 ± 1.065	0.085 ± 0.001	12.854 ± 1.065
QML	beta51	0.16	0.164	0.004 ± 0.002	0.027 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.331 ± 0.009	0.163	0.003	1.91	0.0533	0.0534	0.975	94.939 ± 0.697	0.202 ± 0.001	—
QML	beta52	0.16	0.143	-0.017 ± 0.001	-0.108 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.312 ± 0.009	0.142	-0.018	-11.09	0.0459	0.0492	0.989	92.510 ± 0.837	0.182 ± 0.001	—
QML	beta53	0.16	0.160	0.000 ± 0.001	0.003 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.080 ± 0.004	0.282 ± 0.008	0.159	-0.001	-0.48	0.0458	0.0458	0.975	94.231 ± 0.742	0.173 ± 0.001	—
QML	beta54	0.16	0.142	-0.018 ± 0.001	-0.113 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.308 ± 0.008	0.142	-0.018	-11.28	0.0441	0.0477	1.024	93.421 ± 0.789	0.184 ± 0.001	—
QML	beta55	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.009	-0.009	—	0.0538	0.0545	1.016	94.028 ± 0.754	0.231 ± 0.006	5.972 ± 0.754
QML	beta56	0.00	0.014	0.014 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.013	0.013	—	0.0453	0.0471	0.982	94.737 ± 0.710	0.175 ± 0.002	5.263 ± 0.710
QML	beta57	0.00	0.017	0.017 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.016	0.016	—	0.0266	0.0310	0.960	92.510 ± 0.837	0.107 ± 0.002	7.490 ± 0.837
QML	beta58	0.00	0.015	0.015 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.013	0.013	—	0.0209	0.0247	0.929	92.105 ± 0.858	0.081 ± 0.001	7.895 ± 0.858
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.005 ± 0.010	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.005	0.310 ± 0.009	0.200	0.000	0.22	0.0600	0.0600	1.025	96.053 ± 0.619	0.249 ± 0.001	—
UPI	beta42	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.315 ± 0.009	0.196	-0.004	-2.06	0.0632	0.0633	1.064	96.862 ± 0.555	0.263 ± 0.001	—
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.002	0.183 ± 0.005	0.201	0.001	0.33	0.0373	0.0373	0.987	94.231 ± 0.742	0.141 ± 0.001	—
UPI	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.002	-0.002	—	0.0392	0.0392	1.002	95.749 ± 0.642	0.164 ± 0.002	4.251 ± 0.642

Table S262

Simulation 2 - Condition 118: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0369	0.0370	0.971	95.648 ± 0.649	0.159 ± 0.002	4.352 ± 0.649
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.001	-0.001	—	0.0204	0.0205	0.977	95.142 ± 0.684	0.088 ± 0.001	4.858 ± 0.684
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.001	0.001	—	0.0213	0.0213	0.998	95.445 ± 0.663	0.089 ± 0.001	4.555 ± 0.663
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.328 ± 0.009	0.161	0.001	0.32	0.0526	0.0527	0.971	95.142 ± 0.684	0.200 ± 0.001	—
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.408 ± 0.012	0.157	-0.003	-1.73	0.0661	0.0661	0.998	94.838 ± 0.704	0.256 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.162	0.002	1.29	0.0519	0.0519	0.990	94.939 ± 0.697	0.200 ± 0.001	—
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.007	0.395 ± 0.011	0.158	-0.002	-1.00	0.0618	0.0618	1.034	96.356 ± 0.596	0.256 ± 0.001	—
UPI	beta55	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	—	0.0571	0.0571	0.994	95.951 ± 0.627	0.233 ± 0.002	4.049 ± 0.627
UPI	beta56	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.001	0.001	—	0.0475	0.0475	1.015	95.850 ± 0.634	0.199 ± 0.002	4.150 ± 0.634
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.002	0.002	—	0.0260	0.0260	0.954	94.939 ± 0.697	0.102 ± 0.001	5.061 ± 0.697
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0210	0.0210	0.967	94.838 ± 0.704	0.085 ± 0.001	5.162 ± 0.704

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S263

Simulation 2 - Condition 119: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.012 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.144 ± 0.006	0.380 ± 0.010	0.198	-0.002	-1.11	0.0779	0.0779	0.990	95.194 ± 0.684	0.295 ± 0.001	—
LSAM	beta42	0.20	0.201	0.001 ± 0.003	0.007 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.155 ± 0.007	0.394 ± 0.011	0.200	0.000	-0.01	0.0774	0.0774	0.993	96.012 ± 0.626	0.306 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.009 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.200	0.000	-0.06	0.0467	0.0467	0.965	93.865 ± 0.767	0.183 ± 0.001	—
LSAM	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.001	-0.001	—	0.0483	0.0483	0.920	93.661 ± 0.779	0.191 ± 0.002	6.339 ± 0.779
LSAM	beta45	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	—	—	0.002	0.002	—	0.0474	0.0475	0.946	94.172 ± 0.749	0.185 ± 0.002	5.828 ± 0.749
LSAM	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0274	0.0274	0.900	93.047 ± 0.813	0.107 ± 0.001	6.953 ± 0.813
LSAM	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0273	0.0274	0.924	94.070 ± 0.755	0.107 ± 0.001	5.930 ± 0.755
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.009 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.178 ± 0.009	0.422 ± 0.012	0.162	0.002	1.06	0.0681	0.0681	0.937	93.252 ± 0.802	0.248 ± 0.001	—
LSAM	beta52	0.16	0.164	0.004 ± 0.003	0.022 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.284 ± 0.013	0.533 ± 0.015	0.163	0.003	1.74	0.0869	0.0870	0.948	92.536 ± 0.840	0.317 ± 0.002	—
LSAM	beta53	0.16	0.160	0.000 ± 0.002	0.001 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.008	0.414 ± 0.011	0.160	0.000	0.12	0.0668	0.0668	0.950	93.149 ± 0.808	0.247 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.003	0.014 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.257 ± 0.012	0.507 ± 0.015	0.158	-0.002	-1.06	0.0783	0.0783	0.981	94.172 ± 0.749	0.312 ± 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.001	-0.001	—	0.0750	0.0750	0.928	93.763 ± 0.773	0.272 ± 0.003	6.237 ± 0.773
LSAM	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.004	-0.004	—	0.0595	0.0596	0.881	90.798 ± 0.924	0.229 ± 0.002	9.202 ± 0.924
LSAM	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0344	0.0344	0.872	92.945 ± 0.819	0.127 ± 0.001	7.055 ± 0.819
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0261	0.0262	0.935	93.661 ± 0.779	0.107 ± 0.001	6.339 ± 0.779
QML	beta41	0.20	0.187	-0.013 ± 0.002	-0.066 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.004	0.316 ± 0.009	0.187	-0.013	-6.29	0.0634	0.0646	1.008	94.990 ± 0.698	0.245 ± 0.002	—
QML	beta42	0.20	0.190	-0.010 ± 0.002	-0.048 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.102 ± 0.005	0.319 ± 0.009	0.189	-0.011	-5.33	0.0624	0.0633	1.007	95.706 ± 0.648	0.249 ± 0.001	—
QML	beta43	0.20	0.204	0.004 ± 0.002	0.021 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.202	0.002	1.19	0.0450	0.0451	0.968	93.252 ± 0.802	0.179 ± 0.001	—
QML	beta44	0.00	-0.006	-0.006 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.006	-0.006	—	0.0474	0.0477	0.927	93.456 ± 0.791	0.189 ± 0.002	6.544 ± 0.791
QML	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.001	0.001	—	0.0445	0.0445	0.949	93.865 ± 0.767	0.178 ± 0.002	6.135 ± 0.767
QML	beta46	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.007	0.007	—	0.0252	0.0262	0.908	92.536 ± 0.840	0.102 ± 0.002	7.464 ± 0.840
QML	beta47	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.007	0.007	—	0.0253	0.0263	0.918	93.865 ± 0.767	0.100 ± 0.001	6.135 ± 0.767
QML	beta51	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.008	0.417 ± 0.012	0.162	0.002	1.44	0.0681	0.0682	0.972	94.479 ± 0.730	0.254 ± 0.001	—
QML	beta52	0.16	0.154	-0.006 ± 0.002	-0.038 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.198 ± 0.009	0.444 ± 0.012	0.154	-0.006	-4.00	0.0699	0.0702	0.953	93.558 ± 0.785	0.265 ± 0.001	—
QML	beta53	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.144 ± 0.007	0.380 ± 0.011	0.160	0.000	0.00	0.0620	0.0620	0.956	93.558 ± 0.785	0.228 ± 0.001	—
QML	beta54	0.16	0.152	-0.008 ± 0.002	-0.049 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.180 ± 0.008	0.424 ± 0.012	0.151	-0.009	-5.32	0.0663	0.0669	0.991	94.888 ± 0.704	0.262 ± 0.001	—
QML	beta55	0.00	-0.005	-0.005 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.003	-0.003	—	0.0699	0.0699	0.948	94.479 ± 0.730	0.262 ± 0.003	5.521 ± 0.730
QML	beta56	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	—	0.0576	0.0576	0.891	91.002 ± 0.915	0.213 ± 0.002	8.998 ± 0.915
QML	beta57	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.005	0.005	—	0.0339	0.0343	0.888	92.536 ± 0.840	0.122 ± 0.001	7.464 ± 0.840
QML	beta58	0.00	0.005	0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.003	0.003	—	0.0238	0.0240	0.943	94.581 ± 0.724	0.099 ± 0.001	5.419 ± 0.724
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.012	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.140 ± 0.006	0.374 ± 0.010	0.195	-0.005	-2.37	0.0775	0.0776	1.011	95.706 ± 0.648	0.296 ± 0.002	—
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.148 ± 0.007	0.384 ± 0.011	0.199	-0.001	-0.43	0.0748	0.0748	1.026	96.626 ± 0.577	0.309 ± 0.002	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.200	0.000	-0.01	0.0466	0.0466	0.966	93.865 ± 0.767	0.182 ± 0.001	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.002	-0.002	—	0.0507	0.0508	0.961	94.683 ± 0.717	0.200 ± 0.002	5.317 ± 0.717
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.000	0.000	—	0.0448	0.0448	0.976	95.092 ± 0.691	0.194 ± 0.002	4.908 ± 0.691
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0270	0.0270	0.919	94.376 ± 0.737	0.110 ± 0.001	5.624 ± 0.737

Table S263

Simulation 2 - Condition 119: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0270	0.0270	0.953	94.581 ± 0.724	0.110 ± 0.001	5.419 ± 0.724
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.173 ± 0.008	0.415 ± 0.012	0.161	0.001	0.45	0.0674	0.0674	0.965	94.479 ± 0.730	0.252 ± 0.001	—
UPI	beta52	0.16	0.164	0.004 ± 0.003	0.024 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.283 ± 0.013	0.532 ± 0.015	0.163	0.003	1.91	0.0836	0.0836	0.945	92.843 ± 0.824	0.315 ± 0.002	—
UPI	beta53	0.16	0.160	0.000 ± 0.002	0.000 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.007	0.408 ± 0.011	0.160	0.000	-0.01	0.0654	0.0654	0.955	93.047 ± 0.813	0.244 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.003	0.016 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.251 ± 0.012	0.501 ± 0.014	0.159	-0.001	-0.60	0.0791	0.0791	0.990	94.785 ± 0.711	0.311 ± 0.001	—
UPI	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	-0.003	-0.003	—	0.0715	0.0716	0.960	93.967 ± 0.761	0.278 ± 0.003	6.033 ± 0.761
UPI	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.006	-0.006	—	0.0600	0.0603	0.904	91.411 ± 0.896	0.233 ± 0.002	8.589 ± 0.896
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0342	0.0342	0.879	93.252 ± 0.802	0.128 ± 0.001	6.748 ± 0.802
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0256	0.0257	0.946	94.785 ± 0.711	0.107 ± 0.001	5.215 ± 0.711

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S264

Simulation 2 - Condition 120: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.200	0.000	0.03	0.0477	0.0477	0.975	94.990 ± 0.691	0.180 ± 0.001	—
LSAM	beta42	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.199	-0.001	-0.34	0.0481	0.0481	1.015	95.591 ± 0.650	0.189 ± 0.001	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.201	0.001	0.46	0.0292	0.0292	0.978	94.489 ± 0.722	0.115 ± 0.000	—
LSAM	beta44	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	—	0.0297	0.0297	0.954	94.289 ± 0.735	0.114 ± 0.001	5.711 ± 0.735
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0289	0.0289	0.936	93.487 ± 0.781	0.112 ± 0.001	6.513 ± 0.781
LSAM	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0155	0.0156	0.928	93.587 ± 0.775	0.061 ± 0.001	6.413 ± 0.775
LSAM	beta47	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.000	0.000	—	0.0163	0.0163	0.947	93.988 ± 0.752	0.062 ± 0.001	6.012 ± 0.752
LSAM	beta51	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.159	-0.001	-0.44	0.0405	0.0405	0.969	93.888 ± 0.758	0.158 ± 0.001	—
LSAM	beta52	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.157	-0.003	-1.89	0.0507	0.0508	0.952	94.389 ± 0.728	0.193 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.012 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.161	0.001	0.38	0.0393	0.0393	0.986	94.589 ± 0.716	0.156 ± 0.001	—
LSAM	beta54	0.16	0.160	0.000 ± 0.002	0.001 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.315 ± 0.009	0.161	0.001	0.43	0.0520	0.0520	0.978	94.489 ± 0.722	0.193 ± 0.001	—
LSAM	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.002	0.002	—	0.0391	0.0392	0.952	95.291 ± 0.671	0.162 ± 0.001	4.709 ± 0.671
LSAM	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0390	0.0391	0.914	94.188 ± 0.741	0.140 ± 0.001	5.812 ± 0.741
LSAM	beta57	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.001	-0.001	—	0.0195	0.0196	0.887	91.884 ± 0.864	0.074 ± 0.001	8.116 ± 0.864
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.000	0.000	—	0.0162	0.0162	0.900	92.685 ± 0.824	0.063 ± 0.001	7.315 ± 0.824
QML	beta41	0.20	0.191	-0.009 ± 0.001	-0.046 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.201 ± 0.005	0.190	-0.010	-4.79	0.0387	0.0399	1.015	94.589 ± 0.716	0.156 ± 0.006	—
QML	beta42	0.20	0.189	-0.011 ± 0.001	-0.056 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.190	-0.010	-5.01	0.0397	0.0410	1.028	94.890 ± 0.697	0.156 ± 0.001	—
QML	beta43	0.20	0.203	0.003 ± 0.001	0.016 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.203	0.003	1.65	0.0287	0.0289	0.985	94.689 ± 0.710	0.113 ± 0.001	—
QML	beta44	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0306	0.0308	0.974	94.088 ± 0.747	0.118 ± 0.001	5.912 ± 0.747
QML	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0289	0.0290	0.946	93.587 ± 0.775	0.111 ± 0.001	6.413 ± 0.775
QML	beta46	0.00	0.007	0.007 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.006	0.006	—	0.0156	0.0166	0.973	92.285 ± 0.845	0.062 ± 0.003	7.715 ± 0.845
QML	beta47	0.00	0.007	0.007 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.007	0.007	—	0.0158	0.0172	0.931	93.186 ± 0.798	0.060 ± 0.001	6.814 ± 0.798
QML	beta51	0.16	0.161	0.001 ± 0.001	0.004 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.159	-0.001	-0.41	0.0405	0.0405	0.984	94.589 ± 0.716	0.159 ± 0.001	—
QML	beta52	0.16	0.150	-0.010 ± 0.001	-0.062 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.004	0.281 ± 0.008	0.149	-0.011	-6.79	0.0432	0.0445	0.960	93.888 ± 0.758	0.165 ± 0.001	—
QML	beta53	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.161	0.001	0.80	0.0366	0.0366	0.987	94.689 ± 0.710	0.145 ± 0.000	—
QML	beta54	0.16	0.152	-0.008 ± 0.001	-0.052 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.273 ± 0.007	0.153	-0.007	-4.55	0.0446	0.0452	0.980	94.188 ± 0.741	0.165 ± 0.000	—
QML	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	—	0.0383	0.0383	0.972	95.391 ± 0.664	0.161 ± 0.002	4.609 ± 0.664
QML	beta56	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.004	0.004	—	0.0380	0.0381	0.909	93.186 ± 0.798	0.134 ± 0.001	6.814 ± 0.798
QML	beta57	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.005	0.005	—	0.0196	0.0202	0.898	92.786 ± 0.819	0.074 ± 0.001	7.214 ± 0.819
QML	beta58	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.004	0.004	—	0.0162	0.0167	0.893	92.786 ± 0.819	0.060 ± 0.001	7.214 ± 0.819
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.231 ± 0.006	0.200	0.000	0.08	0.0462	0.0462	0.985	95.491 ± 0.657	0.179 ± 0.001	—
UPI	beta42	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.200	0.000	0.05	0.0472	0.0472	1.024	96.393 ± 0.590	0.188 ± 0.001	—
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.200	0.000	0.12	0.0294	0.0294	0.978	94.289 ± 0.735	0.114 ± 0.000	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0297	0.0297	0.947	94.289 ± 0.735	0.117 ± 0.001	5.711 ± 0.735

Table S264

Simulation 2 - Condition 120: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.001	0.001	—	0.0294	0.0294	0.947	93.788 ± 0.764	0.113 ± 0.001	6.212 ± 0.764
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0153	0.0153	0.930	93.988 ± 0.752	0.061 ± 0.001	6.012 ± 0.752
UPI	beta47	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	0.000	0.000	—	0.0162	0.0162	0.954	94.088 ± 0.747	0.063 ± 0.001	5.912 ± 0.747
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.000 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.159	-0.001	-0.69	0.0404	0.0405	0.981	94.389 ± 0.728	0.159 ± 0.001	—
UPI	beta52	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.156	-0.004	-2.39	0.0507	0.0508	0.956	93.788 ± 0.764	0.193 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.013 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.160	0.000	0.08	0.0394	0.0394	0.989	94.790 ± 0.703	0.155 ± 0.001	—
UPI	beta54	0.16	0.160	0.000 ± 0.002	0.001 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.100 ± 0.004	0.316 ± 0.009	0.160	0.000	0.09	0.0515	0.0515	0.975	94.188 ± 0.741	0.193 ± 0.001	—
UPI	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.002	0.002	—	0.0403	0.0404	0.968	95.591 ± 0.650	0.164 ± 0.001	4.409 ± 0.650
UPI	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0391	0.0392	0.926	94.088 ± 0.747	0.142 ± 0.001	5.912 ± 0.747
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.001	-0.001	—	0.0192	0.0192	0.893	92.084 ± 0.855	0.074 ± 0.001	7.916 ± 0.855
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.001	-0.001	—	0.0161	0.0161	0.897	93.186 ± 0.798	0.063 ± 0.001	6.814 ± 0.798

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S265

Simulation 2 - Condition 121: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.206	0.006 ± 0.006	0.030 ± 0.030	0.009 ± 0.001	0.094 ± 0.005	0.009 ± 0.001	0.094 ± 0.006	0.220 ± 0.023	0.469 ± 0.029	0.197	-0.003	-1.35	0.0879	0.0880	1.083	95.951 ± 1.254	0.398 ± 0.011	—
LSAM	beta42	0.20	0.204	0.004 ± 0.006	0.022 ± 0.030	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.094 ± 0.005	0.219 ± 0.020	0.468 ± 0.026	0.196	-0.004	-2.14	0.0883	0.0884	1.133	94.737 ± 1.421	0.416 ± 0.010	—
LSAM	beta43	0.20	0.200	0.000 ± 0.005	0.002 ± 0.027	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.084 ± 0.005	0.175 ± 0.015	0.419 ± 0.023	0.198	-0.002	-0.85	0.0939	0.0939	1.113	97.976 ± 0.896	0.366 ± 0.008	—
LSAM	beta44	0.00	-0.007	-0.007 ± 0.008	—	0.014 ± 0.002	0.120 ± 0.007	0.014 ± 0.002	0.120 ± 0.008	—	—	-0.007	-0.007	—	0.1139	0.1141	1.360	99.190 ± 0.570	0.640 ± 0.025	0.810 ± 0.570
LSAM	beta45	0.00	0.005	0.005 ± 0.007	—	0.011 ± 0.001	0.103 ± 0.005	0.011 ± 0.001	0.103 ± 0.006	—	—	0.013	0.013	—	0.1002	0.1011	1.176	98.381 ± 0.803	0.476 ± 0.013	1.619 ± 0.803
LSAM	beta46	0.00	0.009	0.009 ± 0.010	—	0.027 ± 0.003	0.165 ± 0.010	0.027 ± 0.003	0.165 ± 0.012	—	—	0.010	0.010	—	0.1487	0.1490	1.274	100.000 ± 0.000	0.822 ± 0.042	0.000 ± 0.000
LSAM	beta47	0.00	-0.011	-0.011 ± 0.010	—	0.026 ± 0.003	0.162 ± 0.008	0.026 ± 0.003	0.162 ± 0.009	—	—	-0.004	-0.004	—	0.1494	0.1494	1.243	99.595 ± 0.404	0.791 ± 0.032	0.405 ± 0.404
LSAM	beta51	0.16	0.174	0.014 ± 0.009	0.086 ± 0.058	0.021 ± 0.002	0.146 ± 0.008	0.022 ± 0.002	0.147 ± 0.010	0.842 ± 0.095	0.917 ± 0.060	0.168	0.008	4.88	0.1420	0.1422	0.891	93.117 ± 1.611	0.511 ± 0.009	—
LSAM	beta52	0.16	0.154	-0.006 ± 0.006	-0.037 ± 0.035	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.088 ± 0.005	0.305 ± 0.026	0.552 ± 0.030	0.152	-0.008	-5.25	0.0888	0.0892	1.185	99.190 ± 0.570	0.410 ± 0.008	—
LSAM	beta53	0.16	0.161	0.001 ± 0.006	0.003 ± 0.038	0.009 ± 0.001	0.096 ± 0.005	0.009 ± 0.001	0.096 ± 0.006	0.357 ± 0.034	0.598 ± 0.035	0.168	0.008	4.76	0.0965	0.0968	1.098	97.976 ± 0.896	0.412 ± 0.009	—
LSAM	beta54	0.16	0.161	0.001 ± 0.007	0.005 ± 0.042	0.011 ± 0.001	0.105 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	0.428 ± 0.042	0.654 ± 0.038	0.153	-0.007	-4.30	0.0921	0.0923	1.010	93.927 ± 1.520	0.415 ± 0.008	—
LSAM	beta55	0.00	-0.032	-0.032 ± 0.013	—	0.040 ± 0.004	0.201 ± 0.011	0.041 ± 0.005	0.203 ± 0.013	—	—	-0.023	-0.023	—	0.1773	0.1788	1.048	99.595 ± 0.404	0.824 ± 0.031	0.405 ± 0.404
LSAM	beta56	0.00	0.044	0.044 ± 0.012	—	0.035 ± 0.004	0.186 ± 0.011	0.036 ± 0.004	0.191 ± 0.012	—	—	0.035	0.035	—	0.1702	0.1738	1.008	97.166 ± 1.056	0.734 ± 0.027	2.834 ± 1.056
LSAM	beta57	0.00	0.012	0.012 ± 0.011	—	0.032 ± 0.004	0.178 ± 0.010	0.032 ± 0.004	0.178 ± 0.012	—	—	0.007	0.007	—	0.1573	0.1575	1.114	98.381 ± 0.803	0.778 ± 0.029	1.619 ± 0.803
LSAM	beta58	0.00	-0.002	-0.002 ± 0.010	—	0.027 ± 0.003	0.163 ± 0.009	0.027 ± 0.003	0.163 ± 0.011	—	—	-0.003	-0.003	—	0.1557	0.1557	1.141	99.595 ± 0.404	0.730 ± 0.028	0.405 ± 0.404
QML	beta41	0.20	0.209	0.009 ± 0.005	0.046 ± 0.027	0.007 ± 0.001	0.084 ± 0.005	0.007 ± 0.001	0.084 ± 0.006	0.177 ± 0.021	0.420 ± 0.029	0.201	0.001	0.56	0.0779	0.0780	0.984	95.951 ± 1.254	0.323 ± 0.005	—
QML	beta42	0.20	0.201	0.001 ± 0.005	0.004 ± 0.026	0.007 ± 0.001	0.082 ± 0.004	0.007 ± 0.001	0.082 ± 0.005	0.166 ± 0.017	0.408 ± 0.025	0.197	-0.003	-1.54	0.0812	0.0813	1.019	95.142 ± 1.368	0.327 ± 0.004	—
QML	beta43	0.20	0.199	-0.001 ± 0.005	-0.005 ± 0.024	0.006 ± 0.001	0.075 ± 0.003	0.006 ± 0.001	0.075 ± 0.004	0.140 ± 0.013	0.374 ± 0.021	0.197	-0.003	-1.66	0.0822	0.0822	1.053	95.951 ± 1.254	0.310 ± 0.003	—
QML	beta44	0.00	-0.018	-0.018 ± 0.006	—	0.009 ± 0.001	0.097 ± 0.005	0.010 ± 0.001	0.098 ± 0.006	—	—	-0.014	-0.014	—	0.1076	0.1085	1.211	96.761 ± 1.126	0.460 ± 0.011	3.239 ± 1.126
QML	beta45	0.00	-0.011	-0.011 ± 0.006	—	0.008 ± 0.001	0.090 ± 0.006	0.008 ± 0.001	0.091 ± 0.007	—	—	-0.004	-0.004	—	0.0901	0.0902	1.089	97.166 ± 1.056	0.385 ± 0.010	2.834 ± 1.056
QML	beta46	0.00	0.013	0.013 ± 0.005	—	0.006 ± 0.001	0.080 ± 0.005	0.007 ± 0.001	0.081 ± 0.005	—	—	0.005	0.005	—	0.0757	0.0759	1.141	99.190 ± 0.570	0.359 ± 0.012	0.810 ± 0.570
QML	beta47	0.00	0.002	0.002 ± 0.005	—	0.006 ± 0.001	0.079 ± 0.004	0.006 ± 0.001	0.079 ± 0.005	—	—	-0.002	-0.002	—	0.0810	0.0810	1.103	98.785 ± 0.697	0.343 ± 0.011	1.215 ± 0.697
QML	beta51	0.16	0.171	0.011 ± 0.008	0.067 ± 0.052	0.017 ± 0.002	0.132 ± 0.007	0.017 ± 0.002	0.132 ± 0.008	0.678 ± 0.070	0.823 ± 0.050	0.159	-0.001	-0.66	0.1205	0.1206	1.094	96.356 ± 1.192	0.564 ± 0.013	—
QML	beta52	0.16	0.155	-0.005 ± 0.006	-0.033 ± 0.038	0.009 ± 0.001	0.095 ± 0.006	0.009 ± 0.001	0.095 ± 0.006	0.353 ± 0.041	0.594 ± 0.039	0.161	0.001	0.78	0.0916	0.0916	1.140	98.785 ± 0.697	0.425 ± 0.015	—
QML	beta53	0.16	0.159	-0.001 ± 0.006	-0.004 ± 0.035	0.008 ± 0.001	0.089 ± 0.005	0.008 ± 0.001	0.089 ± 0.005	0.307 ± 0.031	0.554 ± 0.034	0.161	0.001	0.63	0.0941	0.0941	1.086	97.976 ± 0.896	0.378 ± 0.006	—
QML	beta54	0.16	0.161	0.001 ± 0.006	0.006 ± 0.038	0.009 ± 0.001	0.095 ± 0.004	0.009 ± 0.001	0.095 ± 0.005	0.354 ± 0.032	0.595 ± 0.033	0.156	-0.004	-2.62	0.0835	0.0836	0.945	93.927 ± 1.520	0.353 ± 0.004	—
QML	beta55	0.00	-0.036	-0.036 ± 0.011	—	0.029 ± 0.004	0.170 ± 0.011	0.030 ± 0.004	0.174 ± 0.012	—	—	-0.025	-0.025	—	0.1727	0.1744	1.079	98.381 ± 0.803	0.721 ± 0.029	1.619 ± 0.803
QML	beta56	0.00	0.022	0.022 ± 0.008	—	0.016 ± 0.001	0.126 ± 0.006	0.016 ± 0.001	0.128 ± 0.007	—	—	0.012	0.012	—	0.1310	0.1316	1.076	96.356 ± 1.192	0.532 ± 0.011	3.644 ± 1.192
QML	beta57	0.00	0.015	0.015 ± 0.007	—	0.011 ± 0.001	0.106 ± 0.006	0.011 ± 0.001	0.107 ± 0.007	—	—	0.003	0.003	—	0.0904	0.0905	1.032	97.166 ± 1.056	0.429 ± 0.019	2.834 ± 1.056
QML	beta58	0.00	0.000	0.000 ± 0.005	—	0.005 ± 0.001	0.074 ± 0.004	0.005 ± 0.001	0.073 ± 0.004	—	—	-0.005	-0.005	—	0.0675	0.0677	1.050	95.951 ± 1.254	0.302 ± 0.007	4.049 ± 1.254
UPI	beta41	0.20	0.211	0.011 ± 0.008	0.053 ± 0.039	0.015 ± 0.002	0.122 ± 0.007	0.015 ± 0.002	0.122 ± 0.008	0.374 ± 0.042	0.611 ± 0.040	0.202	0.002	0.89	0.1326	0.1326	1.499	97.571 ± 0.980	0.717 ± 0.041	—
UPI	beta42	0.20	0.204	0.004 ± 0.008	0.022 ± 0.038	0.014 ± 0.002	0.118 ± 0.007	0.014 ± 0.002	0.118 ± 0.008	0.350 ± 0.041	0.592 ± 0.039	0.204	0.004	2.18	0.1282	0.1283	1.635	98.785 ± 0.697	0.759 ± 0.042	—
UPI	beta43	0.20	0.202	0.002 ± 0.007	0.009 ± 0.034	0.011 ± 0.001	0.106 ± 0.006	0.011 ± 0.001	0.105 ± 0.007	0.278 ± 0.031	0.527 ± 0.035	0.194	-0.006	-3.08	0.1169	0.1170	1.590	98.785 ± 0.697	0.658 ± 0.033	—
UPI	beta44	0.00	-0.010	-0.010 ± 0.015	—	0.052 ± 0.007	0.228 ± 0.017	0.052 ± 0.007	0.228 ± 0.018	—	—	-0.017	-0.017	—	0.2011	0.2019	1.771	99.595 ± 0.404	1.586 ± 0.108	0.405 ± 0.404
UPI	beta45	0.00	-0.007	-0.007 ± 0.013	—	0.040 ± 0.006	0.200 ± 0.016	0.040 ± 0.006	0.200 ± 0.017	—	—	-0.008	-0.008	—	0.1782	0.1784	1.604	100.000 ± 0.000	1.258 ± 0.104	0.000 ± 0.000
UPI	beta46	0.00	0.021	0.021 ± 0.022	—	0.116 ± 0.016	0.341 ± 0.024	0.116 ± 0.016	0.341 ± 0.026	—	—	0.021	0.021	—	0.2799	0.2807	1.946	100.000 ± 0.000	2.598 ± 0.238	0.000 ± 0.000

Table S265

Simulation 2 - Condition 121: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.019	-0.019 ± 0.024	—	0.137 ± 0.019	0.370 ± 0.027	0.137 ± 0.019	0.370 ± 0.029	—	—	-0.003	-0.003	—	0.2938	0.2938	1.699	100.000 ± 0.000	2.466 ± 0.241	0.000 ± 0.000
UPI	beta51	0.16	0.168	0.008 ± 0.009	0.053 ± 0.058	0.021 ± 0.003	0.147 ± 0.009	0.021 ± 0.003	0.147 ± 0.010	0.839 ± 0.099	0.916 ± 0.062	0.165	0.005	3.12	0.1407	0.1408	1.328	97.166 ± 1.056	0.763 ± 0.051	—
UPI	beta52	0.16	0.153	-0.007 ± 0.007	-0.042 ± 0.044	0.012 ± 0.001	0.110 ± 0.007	0.012 ± 0.001	0.110 ± 0.008	0.469 ± 0.059	0.685 ± 0.049	0.149	-0.011	-7.12	0.1091	0.1097	1.499	98.785 ± 0.697	0.644 ± 0.036	—
UPI	beta53	0.16	0.164	0.004 ± 0.007	0.022 ± 0.042	0.011 ± 0.001	0.104 ± 0.006	0.011 ± 0.001	0.104 ± 0.007	0.424 ± 0.051	0.652 ± 0.044	0.171	0.011	6.63	0.1074	0.1079	1.529	99.190 ± 0.570	0.626 ± 0.034	—
UPI	beta54	0.16	0.164	0.004 ± 0.007	0.022 ± 0.044	0.012 ± 0.001	0.112 ± 0.005	0.012 ± 0.001	0.111 ± 0.006	0.486 ± 0.043	0.697 ± 0.038	0.160	0.000	-0.24	0.1137	0.1137	1.480	98.785 ± 0.697	0.648 ± 0.034	—
UPI	beta55	0.00	-0.020	-0.020 ± 0.016	—	0.066 ± 0.009	0.256 ± 0.018	0.066 ± 0.009	0.257 ± 0.019	—	—	-0.023	-0.023	—	0.2106	0.2118	1.680	100.000 ± 0.000	1.688 ± 0.129	0.000 ± 0.000
UPI	beta56	0.00	0.039	0.039 ± 0.016	—	0.064 ± 0.009	0.252 ± 0.018	0.065 ± 0.009	0.255 ± 0.020	—	—	0.021	0.021	—	0.1982	0.1993	1.552	100.000 ± 0.000	1.535 ± 0.108	0.000 ± 0.000
UPI	beta57	0.00	-0.008	-0.008 ± 0.018	—	0.081 ± 0.010	0.285 ± 0.017	0.081 ± 0.010	0.284 ± 0.020	—	—	0.001	0.001	—	0.2155	0.2155	1.790	99.595 ± 0.404	1.996 ± 0.175	0.405 ± 0.404
UPI	beta58	0.00	0.008	0.008 ± 0.018	—	0.078 ± 0.010	0.279 ± 0.019	0.077 ± 0.010	0.278 ± 0.021	—	—	0.005	0.005	—	0.2028	0.2029	1.993	100.000 ± 0.000	2.177 ± 0.212	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S266

Simulation 2 - Condition 122: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.002	0.017 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.070 ± 0.004	0.265 ± 0.009	0.202	0.002	1.15	0.0535	0.0535	1.026	96.657 ± 0.701	0.213 ± 0.002	—
LSAM	beta42	0.20	0.206	0.006 ± 0.002	0.032 ± 0.011	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.077 ± 0.004	0.278 ± 0.009	0.210	0.010	4.95	0.0552	0.0561	1.022	96.049 ± 0.759	0.221 ± 0.001	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.062 ± 0.003	0.250 ± 0.009	0.197	-0.003	-1.66	0.0499	0.0500	1.030	96.201 ± 0.745	0.202 ± 0.001	—
LSAM	beta44	0.00	0.007	0.007 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	0.006	0.006	—	0.0723	0.0726	1.037	97.264 ± 0.636	0.305 ± 0.004	2.736 ± 0.636
LSAM	beta45	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	0.000	0.000	—	0.0559	0.0559	1.076	98.176 ± 0.522	0.251 ± 0.002	1.824 ± 0.522
LSAM	beta46	0.00	-0.005	-0.005 ± 0.004	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	—	—	-0.003	-0.003	—	0.0931	0.0931	1.013	97.568 ± 0.600	0.404 ± 0.007	2.432 ± 0.600
LSAM	beta47	0.00	-0.003	-0.003 ± 0.004	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	—	—	-0.003	-0.003	—	0.0974	0.0975	1.001	97.416 ± 0.618	0.400 ± 0.006	2.584 ± 0.618
LSAM	beta51	0.16	0.157	-0.003 ± 0.003	-0.016 ± 0.019	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.233 ± 0.014	0.483 ± 0.018	0.155	-0.005	-3.04	0.0774	0.0776	0.955	94.225 ± 0.909	0.289 ± 0.002	—
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.018 ± 0.015	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.008	0.377 ± 0.013	0.159	-0.001	-0.60	0.0626	0.0627	0.954	94.377 ± 0.898	0.225 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.014	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.007	0.361 ± 0.012	0.160	0.000	-0.07	0.0601	0.0601	1.013	95.441 ± 0.813	0.229 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.010 ± 0.014	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.122 ± 0.007	0.350 ± 0.012	0.157	-0.003	-1.71	0.0561	0.0562	1.006	94.073 ± 0.921	0.221 ± 0.001	—
LSAM	beta55	0.00	-0.007	-0.007 ± 0.004	—	0.012 ± 0.001	0.107 ± 0.004	0.012 ± 0.001	0.107 ± 0.004	—	—	-0.004	-0.004	—	0.1058	0.1058	0.991	97.112 ± 0.653	0.417 ± 0.006	2.888 ± 0.653
LSAM	beta56	0.00	0.003	0.003 ± 0.003	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	—	—	-0.002	-0.002	—	0.0818	0.0818	1.055	96.505 ± 0.716	0.368 ± 0.004	3.495 ± 0.716
LSAM	beta57	0.00	0.006	0.006 ± 0.004	—	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	—	—	0.005	0.005	—	0.0979	0.0981	0.978	97.568 ± 0.600	0.423 ± 0.006	2.432 ± 0.600
LSAM	beta58	0.00	0.000	-0.000 ± 0.004	—	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	-0.004	-0.004	—	0.0884	0.0885	1.036	98.632 ± 0.453	0.378 ± 0.006	1.368 ± 0.453
QML	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.057 ± 0.003	0.239 ± 0.008	0.203	0.003	1.47	0.0485	0.0485	1.026	95.441 ± 0.813	0.192 ± 0.001	—
QML	beta42	0.20	0.205	0.005 ± 0.002	0.025 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.063 ± 0.003	0.252 ± 0.008	0.206	0.006	2.77	0.0518	0.0521	1.013	96.201 ± 0.745	0.199 ± 0.001	—
QML	beta43	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.055 ± 0.003	0.234 ± 0.008	0.196	-0.004	-2.13	0.0471	0.0473	1.032	95.593 ± 0.800	0.189 ± 0.001	—
QML	beta44	0.00	-0.004	-0.004 ± 0.003	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.003	-0.003	—	0.0613	0.0614	1.040	96.353 ± 0.731	0.266 ± 0.003	3.647 ± 0.731
QML	beta45	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.010	-0.010	—	0.0549	0.0557	1.062	96.657 ± 0.701	0.222 ± 0.002	3.343 ± 0.701
QML	beta46	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.000	0.000	—	0.0506	0.0506	1.024	96.049 ± 0.759	0.199 ± 0.002	3.951 ± 0.759
QML	beta47	0.00	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	—	—	-0.003	-0.003	—	0.0491	0.0491	1.029	95.137 ± 0.839	0.194 ± 0.002	4.863 ± 0.839
QML	beta51	0.16	0.158	-0.002 ± 0.003	-0.014 ± 0.018	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.212 ± 0.013	0.461 ± 0.017	0.156	-0.004	-2.63	0.0712	0.0713	1.035	95.137 ± 0.839	0.299 ± 0.004	—
QML	beta52	0.16	0.163	0.003 ± 0.002	0.018 ± 0.014	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.129 ± 0.007	0.359 ± 0.012	0.159	-0.001	-0.37	0.0620	0.0620	0.993	94.529 ± 0.887	0.224 ± 0.006	—
QML	beta53	0.16	0.160	0.000 ± 0.002	0.002 ± 0.013	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.006	0.339 ± 0.011	0.159	-0.001	-0.73	0.0563	0.0564	1.050	94.377 ± 0.898	0.223 ± 0.009	—
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.013	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.110 ± 0.006	0.331 ± 0.011	0.157	-0.003	-2.08	0.0540	0.0541	0.989	94.529 ± 0.887	0.206 ± 0.002	—
QML	beta55	0.00	-0.013	-0.013 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	—	—	-0.013	-0.013	—	0.0938	0.0947	1.060	96.049 ± 0.759	0.389 ± 0.020	3.951 ± 0.759
QML	beta56	0.00	-0.005	-0.005 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	—	—	-0.008	-0.008	—	0.0673	0.0678	1.234	96.353 ± 0.731	0.342 ± 0.043	3.647 ± 0.731
QML	beta57	0.00	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	0.000	0.000	—	0.0555	0.0555	1.004	95.745 ± 0.787	0.226 ± 0.004	4.255 ± 0.787
QML	beta58	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.005	-0.005	—	0.0416	0.0419	1.132	96.353 ± 0.731	0.191 ± 0.013	3.647 ± 0.731
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.022 ± 0.011	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.005	0.285 ± 0.011	0.203	0.003	1.25	0.0626	0.0627	1.236	98.328 ± 0.500	0.276 ± 0.006	—
UPI	beta42	0.20	0.207	0.007 ± 0.002	0.033 ± 0.012	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.006	0.298 ± 0.011	0.210	0.010	5.21	0.0613	0.0622	1.262	97.872 ± 0.563	0.293 ± 0.009	—
UPI	beta43	0.20	0.200	0.000 ± 0.002	0.000 ± 0.010	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.069 ± 0.004	0.262 ± 0.010	0.198	-0.002	-0.94	0.0581	0.0581	1.214	98.176 ± 0.522	0.249 ± 0.005	—
UPI	beta44	0.00	0.005	0.005 ± 0.004	—	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	0.005	0.005	—	0.0898	0.0899	1.405	99.088 ± 0.371	0.512 ± 0.017	0.912 ± 0.371
UPI	beta45	0.00	0.000	0.000 ± 0.003	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	—	—	-0.003	-0.003	—	0.0731	0.0731	1.382	99.088 ± 0.371	0.372 ± 0.009	0.912 ± 0.371
UPI	beta46	0.00	-0.004	-0.004 ± 0.006	—	0.022 ± 0.002	0.149 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	-0.003	-0.003	—	0.1301	0.1301	1.591	100.000 ± 0.000	0.930 ± 0.038	0.000 ± 0.000

Table S266

Simulation 2 - Condition 122: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.003	0.003 ± 0.006	—	0.022 ± 0.002	0.149 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	0.004	0.004	—	0.1269	0.1270	1.555	99.848 ± 0.152	0.911 ± 0.038	0.152 ± 0.152
UPI	beta51	0.16	0.157	-0.003 ± 0.003	-0.019 ± 0.018	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.213 ± 0.013	0.461 ± 0.017	0.157	-0.003	-1.95	0.0731	0.0732	1.062	95.745 ± 0.787	0.307 ± 0.003	—
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.019 ± 0.014	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.008	0.372 ± 0.013	0.160	0.000	0.21	0.0635	0.0635	1.079	94.985 ± 0.851	0.251 ± 0.004	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.010 ± 0.014	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.007	0.360 ± 0.012	0.160	0.000	0.27	0.0609	0.0609	1.139	97.112 ± 0.653	0.257 ± 0.004	—
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.015 ± 0.014	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.007	0.353 ± 0.012	0.158	-0.002	-1.05	0.0556	0.0556	1.121	95.745 ± 0.787	0.248 ± 0.004	—
UPI	beta55	0.00	-0.009	-0.009 ± 0.004	—	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	—	—	-0.008	-0.008	—	0.1023	0.1026	1.305	98.784 ± 0.427	0.560 ± 0.018	1.216 ± 0.427
UPI	beta56	0.00	0.001	0.001 ± 0.004	—	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	—	—	-0.003	-0.003	—	0.0940	0.0940	1.368	99.544 ± 0.263	0.521 ± 0.019	0.456 ± 0.263
UPI	beta57	0.00	0.011	0.011 ± 0.005	—	0.018 ± 0.001	0.132 ± 0.005	0.018 ± 0.001	0.133 ± 0.006	—	—	0.010	0.010	—	0.1101	0.1106	1.441	99.696 ± 0.215	0.749 ± 0.034	0.304 ± 0.215
UPI	beta58	0.00	0.005	0.005 ± 0.004	—	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.005	—	—	0.000	0.000	—	0.0951	0.0951	1.529	100.000 ± 0.000	0.665 ± 0.024	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S267

Simulation 2 - Condition 123: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.009 ± 0.010	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.290 ± 0.008	0.202	0.002	0.93	0.0607	0.0608	1.015	95.991 ± 0.646	0.231 ± 0.001	—
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.010	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.294 ± 0.008	0.201	0.001	0.59	0.0592	0.0592	1.035	96.533 ± 0.602	0.238 ± 0.001	—
LSAM	beta43	0.20	0.204	0.004 ± 0.002	0.020 ± 0.010	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.291 ± 0.008	0.203	0.003	1.37	0.0590	0.0591	0.990	95.233 ± 0.701	0.225 ± 0.001	—
LSAM	beta44	0.00	0.000	0.000 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	0.000	0.000	—	0.0728	0.0728	0.937	94.583 ± 0.745	0.278 ± 0.002	5.417 ± 0.745
LSAM	beta45	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.003	0.003	—	0.0598	0.0599	0.973	93.608 ± 0.805	0.245 ± 0.001	6.392 ± 0.805
LSAM	beta46	0.00	0.002	0.002 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	—	—	0.000	0.000	—	0.0912	0.0912	0.980	95.558 ± 0.678	0.354 ± 0.003	4.442 ± 0.678
LSAM	beta47	0.00	0.000	-0.000 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	-0.003	-0.003	—	0.0843	0.0843	0.964	94.908 ± 0.724	0.348 ± 0.003	5.092 ± 0.724
LSAM	beta51	0.16	0.160	0.000 ± 0.003	0.000 ± 0.018	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.315 ± 0.016	0.561 ± 0.017	0.161	0.001	0.45	0.0881	0.0881	0.923	93.933 ± 0.786	0.325 ± 0.002	—
LSAM	beta52	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.008	0.400 ± 0.012	0.161	0.001	0.40	0.0638	0.0638	1.008	94.800 ± 0.731	0.253 ± 0.001	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.014	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.173 ± 0.008	0.416 ± 0.012	0.159	-0.001	-0.39	0.0659	0.0659	0.989	95.450 ± 0.686	0.258 ± 0.001	—
LSAM	beta54	0.16	0.165	0.005 ± 0.002	0.029 ± 0.013	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.165	0.005	2.91	0.0604	0.0606	1.029	96.316 ± 0.620	0.251 ± 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.004	—	0.012 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	—	—	-0.001	-0.001	—	0.1007	0.1007	0.928	94.475 ± 0.752	0.390 ± 0.004	5.525 ± 0.752
LSAM	beta56	0.00	0.001	0.001 ± 0.003	—	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	0.005	0.005	—	0.0943	0.0944	0.946	95.125 ± 0.709	0.360 ± 0.003	4.875 ± 0.709
LSAM	beta57	0.00	0.000	0.000 ± 0.003	—	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	—	—	-0.001	-0.001	—	0.0903	0.0903	0.973	96.100 ± 0.637	0.386 ± 0.004	3.900 ± 0.637
LSAM	beta58	0.00	-0.004	-0.004 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	—	—	-0.002	-0.002	—	0.0913	0.0914	0.980	95.016 ± 0.716	0.350 ± 0.003	4.984 ± 0.716
QML	beta41	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.077 ± 0.003	0.278 ± 0.008	0.202	0.002	1.03	0.0565	0.0565	1.024	95.775 ± 0.662	0.223 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.200	0.000	-0.24	0.0568	0.0568	1.040	95.558 ± 0.678	0.228 ± 0.001	—
QML	beta43	0.20	0.203	0.003 ± 0.002	0.016 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.280 ± 0.008	0.202	0.002	1.20	0.0582	0.0583	1.002	95.341 ± 0.694	0.220 ± 0.001	—
QML	beta44	0.00	-0.007	-0.007 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.003	-0.003	—	0.0698	0.0699	0.973	95.450 ± 0.686	0.277 ± 0.003	4.550 ± 0.686
QML	beta45	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.003	-0.003	—	0.0602	0.0603	0.998	94.475 ± 0.752	0.242 ± 0.002	5.525 ± 0.752
QML	beta46	0.00	0.005	0.005 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	—	0.0569	0.0569	1.012	95.666 ± 0.670	0.236 ± 0.003	4.334 ± 0.670
QML	beta47	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.003	-0.003	—	0.0551	0.0552	1.008	93.716 ± 0.799	0.229 ± 0.003	6.284 ± 0.799
QML	beta51	0.16	0.161	0.001 ± 0.003	0.005 ± 0.018	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.303 ± 0.015	0.550 ± 0.016	0.163	0.003	2.05	0.0889	0.0889	0.991	95.558 ± 0.678	0.342 ± 0.003	—
QML	beta52	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.008	0.405 ± 0.012	0.160	0.000	0.09	0.0645	0.0645	1.018	95.233 ± 0.701	0.259 ± 0.003	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.008	0.400 ± 0.012	0.160	0.000	-0.24	0.0619	0.0619	1.012	95.016 ± 0.716	0.254 ± 0.001	—
QML	beta54	0.16	0.164	0.004 ± 0.002	0.022 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.165	0.005	2.86	0.0610	0.0611	1.031	96.858 ± 0.574	0.243 ± 0.001	—
QML	beta55	0.00	-0.008	-0.008 ± 0.003	—	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	—	—	-0.008	-0.008	—	0.0903	0.0907	1.004	94.691 ± 0.738	0.385 ± 0.007	5.309 ± 0.738
QML	beta56	0.00	-0.003	-0.003 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	-0.003	-0.003	—	0.0835	0.0836	0.999	95.883 ± 0.654	0.331 ± 0.003	4.117 ± 0.654
QML	beta57	0.00	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0631	0.0631	1.017	94.908 ± 0.724	0.262 ± 0.005	5.092 ± 0.724
QML	beta58	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.004	-0.004	—	0.0557	0.0558	1.012	94.691 ± 0.738	0.219 ± 0.002	5.309 ± 0.738
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.013 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.004	0.299 ± 0.008	0.204	0.004	1.95	0.0636	0.0637	1.088	97.183 ± 0.545	0.255 ± 0.003	—
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.009 ± 0.010	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.008	0.202	0.002	1.17	0.0624	0.0624	1.122	96.858 ± 0.574	0.262 ± 0.003	—
UPI	beta43	0.20	0.205	0.005 ± 0.002	0.023 ± 0.010	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.009	0.202	0.002	1.24	0.0605	0.0606	1.063	95.775 ± 0.662	0.247 ± 0.002	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.001	0.001	—	0.0810	0.0810	1.081	96.966 ± 0.565	0.352 ± 0.006	3.034 ± 0.565
UPI	beta45	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.001	0.001	—	0.0660	0.0660	1.075	96.208 ± 0.629	0.294 ± 0.004	3.792 ± 0.629
UPI	beta46	0.00	0.005	0.005 ± 0.004	—	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.121 ± 0.004	—	—	0.004	0.004	—	0.1050	0.1050	1.303	98.917 ± 0.341	0.618 ± 0.024	1.083 ± 0.341

Table S267

Simulation 2 - Condition 123: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.004	0.004 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	—	—	0.005	0.005	—	0.0992	0.0993	1.293	98.917 ± 0.341	0.573 ± 0.015	1.083 ± 0.341
UPI	beta51	0.16	0.160	0.000 ± 0.003	0.000 ± 0.018	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.290 ± 0.014	0.539 ± 0.016	0.159	-0.001	-0.59	0.0854	0.0854	0.995	95.125 ± 0.709	0.336 ± 0.002	—
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.008	0.401 ± 0.012	0.161	0.001	0.94	0.0640	0.0641	1.036	95.450 ± 0.686	0.261 ± 0.001	—
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.172 ± 0.008	0.415 ± 0.012	0.160	0.000	-0.03	0.0655	0.0655	1.036	96.316 ± 0.620	0.270 ± 0.002	—
UPI	beta54	0.16	0.165	0.005 ± 0.002	0.032 ± 0.013	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.007	0.389 ± 0.011	0.165	0.005	3.00	0.0627	0.0629	1.077	96.533 ± 0.602	0.262 ± 0.003	—
UPI	beta55	0.00	0.001	0.001 ± 0.004	—	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	—	—	0.003	0.003	—	0.0972	0.0973	1.060	96.100 ± 0.637	0.443 ± 0.007	3.900 ± 0.637
UPI	beta56	0.00	-0.002	-0.002 ± 0.003	—	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	—	—	-0.001	-0.001	—	0.0938	0.0938	1.047	97.291 ± 0.534	0.412 ± 0.006	2.709 ± 0.534
UPI	beta57	0.00	0.000	0.000 ± 0.004	—	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	—	—	-0.002	-0.002	—	0.1021	0.1021	1.215	98.700 ± 0.373	0.530 ± 0.018	1.300 ± 0.373
UPI	beta58	0.00	-0.004	-0.004 ± 0.004	—	0.012 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.107 ± 0.004	—	—	-0.002	-0.002	—	0.0967	0.0967	1.196	97.833 ± 0.479	0.503 ± 0.012	2.167 ± 0.479

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S268

Simulation 2 - Condition 124: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.205	0.005 ± 0.001	0.024 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.204	0.004	1.86	0.0355	0.0357	0.966	94.088 ± 0.747	0.142 ± 0.000	—
LSAM	beta42	0.20	0.202	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.001	0.180 ± 0.005	0.202	0.002	0.94	0.0368	0.0369	1.041	95.892 ± 0.628	0.147 ± 0.000	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.176 ± 0.005	0.199	-0.001	-0.26	0.0351	0.0351	1.010	94.890 ± 0.697	0.139 ± 0.000	—
LSAM	beta44	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0474	0.0474	0.944	94.188 ± 0.741	0.168 ± 0.001	5.812 ± 0.741
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0389	0.0389	0.966	93.687 ± 0.770	0.149 ± 0.001	6.313 ± 0.770
LSAM	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0519	0.0519	0.997	94.188 ± 0.741	0.212 ± 0.001	5.812 ± 0.741
LSAM	beta47	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.003	-0.003	—	0.0538	0.0539	0.970	93.988 ± 0.752	0.212 ± 0.001	6.012 ± 0.752
LSAM	beta51	0.16	0.164	0.004 ± 0.002	0.023 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.329 ± 0.009	0.163	0.003	1.79	0.0522	0.0523	0.966	95.090 ± 0.684	0.199 ± 0.001	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.161	0.001	0.66	0.0399	0.0399	0.976	95.591 ± 0.650	0.153 ± 0.000	—
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.160	0.000	0.05	0.0418	0.0418	0.999	95.691 ± 0.643	0.159 ± 0.000	—
LSAM	beta54	0.16	0.162	0.002 ± 0.001	0.010 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.160	0.000	0.23	0.0378	0.0378	1.022	94.890 ± 0.697	0.153 ± 0.000	—
LSAM	beta55	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	-0.001	-0.001	—	0.0572	0.0572	0.956	94.188 ± 0.741	0.222 ± 0.001	5.812 ± 0.741
LSAM	beta56	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.001	-0.001	—	0.0559	0.0559	0.965	94.489 ± 0.722	0.208 ± 0.001	5.511 ± 0.722
LSAM	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.003	-0.003	—	0.0562	0.0563	1.004	95.291 ± 0.671	0.226 ± 0.001	4.709 ± 0.671
LSAM	beta58	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.000	0.000	—	0.0533	0.0533	0.989	94.890 ± 0.697	0.208 ± 0.001	5.110 ± 0.697
QML	beta41	0.20	0.204	0.004 ± 0.001	0.019 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.033 ± 0.002	0.183 ± 0.005	0.202	0.002	1.24	0.0351	0.0352	0.970	94.790 ± 0.703	0.138 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.175 ± 0.005	0.201	0.001	0.54	0.0346	0.0346	1.038	95.591 ± 0.650	0.143 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.173 ± 0.005	0.199	-0.001	-0.56	0.0338	0.0339	1.010	94.990 ± 0.691	0.137 ± 0.000	—
QML	beta44	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.005	-0.005	—	0.0476	0.0478	0.958	94.489 ± 0.722	0.170 ± 0.001	5.511 ± 0.722
QML	beta45	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.003	-0.003	—	0.0382	0.0384	0.975	94.289 ± 0.735	0.148 ± 0.001	5.711 ± 0.735
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	—	0.0333	0.0333	1.029	94.890 ± 0.697	0.142 ± 0.001	5.110 ± 0.697
QML	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0351	0.0353	0.967	94.790 ± 0.703	0.140 ± 0.001	5.210 ± 0.703
QML	beta51	0.16	0.164	0.004 ± 0.002	0.023 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.163	0.003	1.96	0.0522	0.0522	0.995	95.190 ± 0.677	0.202 ± 0.001	—
QML	beta52	0.16	0.160	0.000 ± 0.001	0.003 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.161	0.001	0.57	0.0407	0.0407	0.984	95.391 ± 0.664	0.153 ± 0.000	—
QML	beta53	0.16	0.161	0.001 ± 0.001	0.007 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.160	0.000	0.25	0.0410	0.0410	1.010	95.792 ± 0.636	0.156 ± 0.000	—
QML	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.160	0.000	-0.09	0.0371	0.0371	1.023	95.190 ± 0.677	0.150 ± 0.000	—
QML	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0567	0.0567	0.975	94.389 ± 0.728	0.222 ± 0.001	5.611 ± 0.728
QML	beta56	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.003	-0.003	—	0.0533	0.0534	0.986	95.591 ± 0.650	0.198 ± 0.001	4.409 ± 0.650
QML	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.004	-0.004	—	0.0378	0.0380	1.012	94.289 ± 0.735	0.154 ± 0.001	5.711 ± 0.735
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.003	-0.003	—	0.0348	0.0349	0.999	94.389 ± 0.728	0.136 ± 0.001	5.611 ± 0.728
UPI	beta41	0.20	0.205	0.005 ± 0.001	0.026 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.204	0.004	2.19	0.0357	0.0359	0.974	94.289 ± 0.735	0.143 ± 0.000	—
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.202	0.002	0.99	0.0357	0.0357	1.058	96.092 ± 0.613	0.148 ± 0.000	—
UPI	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.177 ± 0.005	0.199	-0.001	-0.28	0.0343	0.0343	1.021	95.190 ± 0.677	0.142 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0475	0.0475	0.969	95.591 ± 0.650	0.180 ± 0.001	4.409 ± 0.650
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	—	0.0396	0.0397	0.987	94.990 ± 0.691	0.156 ± 0.001	5.010 ± 0.691
UPI	beta46	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.001	-0.001	—	0.0565	0.0566	1.084	96.493 ± 0.582	0.249 ± 0.002	3.507 ± 0.582

Table S268

Simulation 2 - Condition 124: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.001	-0.001	—	0.0597	0.0598	1.042	96.593 ± 0.574	0.253 ± 0.002	3.407 ± 0.574
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.020 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.004	0.322 ± 0.009	0.162	0.002	1.48	0.0519	0.0519	0.991	95.291 ± 0.671	0.200 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.161	0.001	0.72	0.0411	0.0411	0.986	95.491 ± 0.657	0.154 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.160	0.000	0.07	0.0420	0.0420	1.010	95.691 ± 0.643	0.160 ± 0.000	—
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.011 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.160	0.000	-0.08	0.0372	0.0372	1.026	95.591 ± 0.650	0.153 ± 0.000	—
UPI	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	—	0.0561	0.0561	0.973	94.689 ± 0.710	0.228 ± 0.001	5.311 ± 0.710
UPI	beta56	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.002	-0.002	—	0.0557	0.0557	0.978	95.491 ± 0.657	0.215 ± 0.001	4.509 ± 0.657
UPI	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.004	-0.004	—	0.0584	0.0585	1.043	95.792 ± 0.636	0.244 ± 0.002	4.208 ± 0.636
UPI	beta58	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.001	-0.001	—	0.0538	0.0538	1.031	96.794 ± 0.558	0.230 ± 0.002	3.206 ± 0.558

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S269

Simulation 2 - Condition 125: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.199	-0.001	-0.56	0.0465	0.0465	1.022	96.196 ± 0.605	0.180 ± 0.001	—
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.056 ± 0.003	0.238 ± 0.007	0.202	0.002	0.82	0.0483	0.0483	0.995	93.994 ± 0.752	0.185 ± 0.001	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.056 ± 0.003	0.238 ± 0.006	0.200	0.000	-0.22	0.0454	0.0454	0.952	92.893 ± 0.813	0.177 ± 0.001	—
LSAM	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.001	0.001	—	0.0504	0.0504	0.954	93.994 ± 0.752	0.199 ± 0.001	6.006 ± 0.752
LSAM	beta45	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.002	0.002	—	0.0466	0.0467	0.977	95.095 ± 0.683	0.182 ± 0.001	4.905 ± 0.683
LSAM	beta46	0.00	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.004	0.004	—	0.0626	0.0628	0.928	93.994 ± 0.752	0.230 ± 0.001	6.006 ± 0.752
LSAM	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.006	-0.006	—	0.0595	0.0598	0.972	94.895 ± 0.696	0.228 ± 0.001	5.105 ± 0.696
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.013 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.181 ± 0.009	0.425 ± 0.013	0.159	-0.001	-0.42	0.0652	0.0652	0.924	92.993 ± 0.808	0.246 ± 0.001	—
LSAM	beta52	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.320 ± 0.009	0.158	-0.002	-1.19	0.0515	0.0515	0.987	94.695 ± 0.709	0.198 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.164	0.004	2.28	0.0498	0.0500	0.991	93.794 ± 0.763	0.203 ± 0.001	—
LSAM	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.161	0.001	0.38	0.0496	0.0496	0.987	94.494 ± 0.722	0.197 ± 0.001	—
LSAM	beta55	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.002	0.002	—	0.0676	0.0676	0.952	93.794 ± 0.763	0.258 ± 0.002	6.206 ± 0.763
LSAM	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.003	-0.003	—	0.0631	0.0632	0.954	94.795 ± 0.703	0.245 ± 0.002	5.205 ± 0.703
LSAM	beta57	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0639	0.0639	0.952	94.494 ± 0.722	0.248 ± 0.001	5.506 ± 0.722
LSAM	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.003	-0.003	—	0.0581	0.0582	0.973	93.493 ± 0.780	0.231 ± 0.001	6.507 ± 0.780
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.049 ± 0.002	0.222 ± 0.006	0.199	-0.001	-0.49	0.0460	0.0460	1.027	95.796 ± 0.635	0.179 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.002	0.233 ± 0.007	0.202	0.002	0.84	0.0474	0.0475	1.001	94.294 ± 0.734	0.183 ± 0.001	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.199	-0.001	-0.45	0.0453	0.0453	0.958	93.694 ± 0.769	0.177 ± 0.001	—
QML	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0504	0.0504	0.959	94.394 ± 0.728	0.203 ± 0.001	5.606 ± 0.728
QML	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0478	0.0478	0.996	95.596 ± 0.649	0.184 ± 0.001	4.404 ± 0.649
QML	beta46	0.00	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.004	0.004	—	0.0511	0.0513	0.940	93.994 ± 0.752	0.191 ± 0.001	6.006 ± 0.752
QML	beta47	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.005	-0.005	—	0.0489	0.0492	0.972	94.695 ± 0.709	0.186 ± 0.001	5.305 ± 0.709
QML	beta51	0.16	0.162	0.002 ± 0.002	0.014 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.179 ± 0.009	0.423 ± 0.013	0.160	0.000	-0.16	0.0663	0.0663	0.956	93.994 ± 0.752	0.253 ± 0.001	—
QML	beta52	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.159	-0.001	-0.79	0.0511	0.0511	0.994	94.595 ± 0.715	0.200 ± 0.001	—
QML	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.163	0.003	1.94	0.0503	0.0504	0.997	93.894 ± 0.758	0.203 ± 0.001	—
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.005	0.315 ± 0.009	0.160	0.000	-0.01	0.0495	0.0495	0.990	94.595 ± 0.715	0.195 ± 0.001	—
QML	beta55	0.00	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.001	0.001	—	0.0657	0.0657	0.971	94.595 ± 0.715	0.261 ± 0.002	5.405 ± 0.715
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.003	-0.003	—	0.0631	0.0632	0.973	95.295 ± 0.670	0.243 ± 0.002	4.705 ± 0.670
QML	beta57	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.002	-0.002	—	0.0519	0.0519	0.962	94.995 ± 0.690	0.206 ± 0.001	5.005 ± 0.690
QML	beta58	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.003	-0.003	—	0.0470	0.0471	0.986	93.393 ± 0.786	0.187 ± 0.001	6.607 ± 0.786
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.199	-0.001	-0.46	0.0455	0.0456	1.021	96.196 ± 0.605	0.180 ± 0.001	—
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.202	0.002	0.75	0.0482	0.0482	0.997	94.194 ± 0.740	0.185 ± 0.001	—
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.201	0.001	0.27	0.0457	0.0457	0.959	93.293 ± 0.791	0.179 ± 0.001	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.002	0.002	—	0.0521	0.0521	0.975	94.094 ± 0.746	0.207 ± 0.001	5.906 ± 0.746
UPI	beta45	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	—	0.0499	0.0499	0.983	94.595 ± 0.715	0.187 ± 0.001	5.405 ± 0.715
UPI	beta46	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.005	0.005	—	0.0646	0.0647	0.953	94.795 ± 0.703	0.251 ± 0.002	5.205 ± 0.703

Table S269

Simulation 2 - Condition 125: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.008	-0.008	—	0.0616	0.0621	1.011	96.196 ± 0.605	0.251 ± 0.002	3.804 ± 0.605
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.176 ± 0.009	0.420 ± 0.013	0.160	0.000	-0.31	0.0652	0.0652	0.951	93.794 ± 0.763	0.250 ± 0.001	—
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.320 ± 0.009	0.158	-0.002	-0.99	0.0509	0.0510	0.985	94.795 ± 0.703	0.197 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.326 ± 0.009	0.163	0.003	2.13	0.0495	0.0496	0.996	93.794 ± 0.763	0.204 ± 0.001	—
UPI	beta54	0.16	0.160	0.000 ± 0.002	0.003 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.317 ± 0.009	0.161	0.001	0.32	0.0496	0.0496	0.987	94.595 ± 0.715	0.196 ± 0.001	—
UPI	beta55	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.002	0.002	—	0.0650	0.0650	0.970	94.695 ± 0.709	0.263 ± 0.002	5.305 ± 0.709
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.004	-0.004	—	0.0646	0.0647	0.983	94.895 ± 0.696	0.252 ± 0.002	5.105 ± 0.696
UPI	beta57	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.001	-0.001	—	0.0658	0.0658	0.958	94.995 ± 0.690	0.257 ± 0.001	5.005 ± 0.690
UPI	beta58	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	—	0.0603	0.0603	0.999	94.294 ± 0.734	0.242 ± 0.001	5.706 ± 0.734

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S270

Simulation 2 - Condition 126: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.200	0.000	0.09	0.0290	0.0290	1.015	95.295 ± 0.670	0.113 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.199	-0.001	-0.26	0.0301	0.0301	0.994	95.596 ± 0.649	0.116 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.200	0.000	-0.12	0.0301	0.0301	0.970	93.894 ± 0.758	0.112 ± 0.000	—
LSAM	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0321	0.0322	1.022	95.696 ± 0.642	0.125 ± 0.000	4.304 ± 0.642
LSAM	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0313	0.0313	0.971	94.394 ± 0.728	0.114 ± 0.000	5.606 ± 0.728
LSAM	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0390	0.0390	0.975	94.394 ± 0.728	0.144 ± 0.000	5.606 ± 0.728
LSAM	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0374	0.0374	0.990	94.995 ± 0.690	0.144 ± 0.000	5.005 ± 0.690
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.159	-0.001	-0.68	0.0404	0.0404	1.008	94.995 ± 0.690	0.157 ± 0.001	—
LSAM	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.160	0.000	-0.13	0.0335	0.0335	0.951	93.694 ± 0.769	0.124 ± 0.000	—
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.211 ± 0.005	0.162	0.002	1.04	0.0342	0.0342	0.975	94.895 ± 0.696	0.129 ± 0.000	—
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.006	0.161	0.001	0.50	0.0305	0.0305	1.015	94.895 ± 0.696	0.125 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.001	0.001	—	0.0390	0.0391	1.002	94.995 ± 0.690	0.162 ± 0.001	5.005 ± 0.690
LSAM	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.000	0.000	—	0.0374	0.0374	0.982	93.994 ± 0.752	0.153 ± 0.001	6.006 ± 0.752
LSAM	beta57	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0390	0.0390	0.977	94.595 ± 0.715	0.156 ± 0.000	5.405 ± 0.715
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0394	0.0394	0.955	93.493 ± 0.780	0.147 ± 0.000	6.507 ± 0.780
QML	beta41	0.20	0.200	0.000 ± 0.001	0.000 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.200	0.000	-0.14	0.0285	0.0285	1.017	95.495 ± 0.656	0.112 ± 0.000	—
QML	beta42	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.199	-0.001	-0.59	0.0301	0.0301	0.995	95.696 ± 0.642	0.115 ± 0.000	—
QML	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.199	-0.001	-0.28	0.0300	0.0300	0.977	93.994 ± 0.752	0.111 ± 0.000	—
QML	beta44	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.003	-0.003	—	0.0323	0.0325	1.029	95.796 ± 0.635	0.127 ± 0.001	4.204 ± 0.635
QML	beta45	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.003	-0.003	—	0.0308	0.0310	0.981	94.494 ± 0.722	0.115 ± 0.000	5.506 ± 0.722
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	—	0.0320	0.0320	0.977	94.494 ± 0.722	0.118 ± 0.001	5.506 ± 0.722
QML	beta47	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0313	0.0313	1.006	95.095 ± 0.683	0.118 ± 0.001	4.905 ± 0.683
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.159	-0.001	-0.72	0.0405	0.0405	1.026	95.395 ± 0.663	0.159 ± 0.001	—
QML	beta52	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.160	0.000	-0.23	0.0323	0.0323	0.958	94.194 ± 0.740	0.125 ± 0.000	—
QML	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.044 ± 0.002	0.209 ± 0.005	0.161	0.001	0.66	0.0337	0.0337	0.979	94.595 ± 0.715	0.128 ± 0.000	—
QML	beta54	0.16	0.161	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.195 ± 0.006	0.161	0.001	0.39	0.0301	0.0302	1.017	95.095 ± 0.683	0.124 ± 0.000	—
QML	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0392	0.0392	1.010	95.696 ± 0.642	0.163 ± 0.001	4.304 ± 0.642
QML	beta56	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0365	0.0365	0.995	94.294 ± 0.734	0.151 ± 0.001	5.706 ± 0.734
QML	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	—	0.0329	0.0329	0.984	95.095 ± 0.683	0.129 ± 0.001	4.905 ± 0.683
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.002	-0.002	—	0.0327	0.0328	0.966	92.993 ± 0.808	0.120 ± 0.001	7.007 ± 0.808
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.201	0.001	0.30	0.0291	0.0291	1.015	95.495 ± 0.656	0.112 ± 0.000	—
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.200	0.000	-0.21	0.0295	0.0295	0.996	95.696 ± 0.642	0.116 ± 0.000	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.200	0.000	-0.21	0.0306	0.0306	0.973	93.694 ± 0.769	0.112 ± 0.000	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0327	0.0327	1.025	95.495 ± 0.656	0.127 ± 0.000	4.505 ± 0.656
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.002	-0.002	—	0.0316	0.0316	0.971	94.695 ± 0.709	0.116 ± 0.000	5.305 ± 0.709
UPI	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0386	0.0386	0.977	95.195 ± 0.677	0.150 ± 0.001	4.805 ± 0.677

Table S270

Simulation 2 - Condition 126: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0397	0.0398	1.006	95.596 ± 0.649	0.149 ± 0.001	4.404 ± 0.649
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.159	-0.001	-0.88	0.0400	0.0400	1.023	95.395 ± 0.663	0.158 ± 0.001	—
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.159	-0.001	-0.40	0.0333	0.0333	0.951	93.994 ± 0.752	0.124 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.044 ± 0.002	0.211 ± 0.005	0.161	0.001	0.72	0.0343	0.0343	0.976	94.795 ± 0.703	0.129 ± 0.000	—
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.006	0.161	0.001	0.61	0.0305	0.0305	1.015	94.995 ± 0.690	0.125 ± 0.000	—
UPI	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.001	0.001	—	0.0392	0.0392	1.008	95.495 ± 0.656	0.163 ± 0.001	4.505 ± 0.656
UPI	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.001	0.001	—	0.0385	0.0385	0.984	94.294 ± 0.734	0.155 ± 0.001	5.706 ± 0.734
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.001	0.001	—	0.0412	0.0413	0.980	94.995 ± 0.690	0.158 ± 0.001	5.005 ± 0.690
UPI	beta58	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0395	0.0395	0.967	94.194 ± 0.740	0.149 ± 0.001	5.806 ± 0.740

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S271

Simulation 2 - Condition 127: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.221	0.021 ± 0.010	0.105 ± 0.051	0.010 ± 0.001	0.100 ± 0.007	0.010 ± 0.001	0.101 ± 0.009	0.256 ± 0.035	0.506 ± 0.044	0.213	0.013	6.43	0.1008	0.1016	1.136	95.745 ± 2.082	0.443 ± 0.015	—
LSAM	beta42	0.20	0.223	0.023 ± 0.010	0.117 ± 0.051	0.010 ± 0.002	0.099 ± 0.011	0.010 ± 0.002	0.101 ± 0.013	0.257 ± 0.055	0.507 ± 0.063	0.217	0.017	8.36	0.0893	0.0908	1.230	97.872 ± 1.488	0.479 ± 0.016	—
LSAM	beta43	0.20	0.190	-0.010 ± 0.008	-0.048 ± 0.040	0.006 ± 0.001	0.078 ± 0.006	0.006 ± 0.001	0.078 ± 0.007	0.153 ± 0.023	0.392 ± 0.036	0.181	-0.019	-9.66	0.0645	0.0674	1.135	95.745 ± 2.082	0.348 ± 0.009	—
LSAM	beta44	0.00	0.038	0.038 ± 0.014	—	0.018 ± 0.003	0.134 ± 0.011	0.019 ± 0.003	0.139 ± 0.014	—	—	0.034	0.034	—	0.1344	0.1387	1.077	94.681 ± 2.315	0.567 ± 0.030	5.319 ± 2.315
LSAM	beta45	0.00	0.036	0.036 ± 0.011	—	0.012 ± 0.002	0.108 ± 0.009	0.013 ± 0.002	0.113 ± 0.011	—	—	0.027	0.027	—	0.1014	0.1049	0.990	97.872 ± 1.488	0.418 ± 0.016	2.128 ± 1.488
LSAM	beta46	0.00	-0.043	-0.043 ± 0.009	—	0.007 ± 0.001	0.086 ± 0.006	0.009 ± 0.001	0.095 ± 0.008	—	—	-0.043	-0.043	—	0.0776	0.0889	1.110	96.809 ± 1.813	0.372 ± 0.022	3.191 ± 1.813
LSAM	beta47	0.00	-0.045	-0.045 ± 0.009	—	0.008 ± 0.002	0.087 ± 0.010	0.009 ± 0.002	0.097 ± 0.012	—	—	-0.036	-0.036	—	0.0768	0.0848	1.068	92.553 ± 2.708	0.364 ± 0.019	7.447 ± 2.708
LSAM	beta51	0.16	0.149	-0.011 ± 0.012	-0.067 ± 0.076	0.014 ± 0.003	0.118 ± 0.012	0.014 ± 0.003	0.118 ± 0.013	0.542 ± 0.100	0.737 ± 0.080	0.135	-0.025	-15.88	0.1183	0.1210	1.026	96.809 ± 1.813	0.475 ± 0.012	—
LSAM	beta52	0.16	0.178	0.018 ± 0.013	0.115 ± 0.081	0.016 ± 0.002	0.126 ± 0.010	0.016 ± 0.002	0.127 ± 0.012	0.630 ± 0.096	0.793 ± 0.074	0.162	0.002	1.31	0.1050	0.1050	0.919	94.681 ± 2.315	0.455 ± 0.014	—
LSAM	beta53	0.16	0.154	-0.006 ± 0.010	-0.035 ± 0.061	0.009 ± 0.002	0.095 ± 0.010	0.009 ± 0.002	0.094 ± 0.011	0.347 ± 0.067	0.589 ± 0.067	0.141	-0.019	-11.83	0.0922	0.0942	1.075	94.681 ± 2.315	0.398 ± 0.012	—
LSAM	beta54	0.16	0.181	0.021 ± 0.011	0.131 ± 0.071	0.012 ± 0.002	0.109 ± 0.010	0.012 ± 0.002	0.111 ± 0.012	0.479 ± 0.088	0.692 ± 0.074	0.195	0.035	21.70	0.1029	0.1086	0.978	97.872 ± 1.488	0.419 ± 0.013	—
LSAM	beta55	0.00	0.041	0.041 ± 0.017	—	0.026 ± 0.004	0.161 ± 0.014	0.027 ± 0.005	0.166 ± 0.017	—	—	0.048	0.048	—	0.1671	0.1740	1.157	98.936 ± 1.058	0.731 ± 0.044	1.064 ± 1.058
LSAM	beta56	0.00	0.006	0.006 ± 0.015	—	0.022 ± 0.004	0.150 ± 0.014	0.022 ± 0.004	0.149 ± 0.016	—	—	0.002	0.002	—	0.1372	0.1372	1.065	96.809 ± 1.813	0.625 ± 0.043	3.191 ± 1.813
LSAM	beta57	0.00	-0.035	-0.035 ± 0.010	—	0.010 ± 0.002	0.099 ± 0.010	0.011 ± 0.002	0.105 ± 0.012	—	—	-0.029	-0.029	—	0.0923	0.0967	0.942	94.681 ± 2.315	0.366 ± 0.019	5.319 ± 2.315
LSAM	beta58	0.00	-0.029	-0.029 ± 0.008	—	0.006 ± 0.001	0.077 ± 0.007	0.007 ± 0.001	0.082 ± 0.008	—	—	-0.032	-0.032	—	0.0603	0.0684	0.954	90.426 ± 3.035	0.287 ± 0.014	9.574 ± 3.035
QML	beta41	0.20	0.203	0.003 ± 0.008	0.015 ± 0.039	0.006 ± 0.001	0.076 ± 0.006	0.006 ± 0.001	0.076 ± 0.007	0.144 ± 0.021	0.380 ± 0.035	0.196	-0.004	-2.11	0.0789	0.0790	1.089	97.872 ± 1.488	0.326 ± 0.008	—
QML	beta42	0.20	0.210	0.010 ± 0.008	0.050 ± 0.038	0.005 ± 0.001	0.073 ± 0.008	0.005 ± 0.001	0.073 ± 0.009	0.133 ± 0.028	0.365 ± 0.044	0.207	0.007	3.46	0.0673	0.0676	1.127	97.872 ± 1.488	0.321 ± 0.006	—
QML	beta43	0.20	0.197	-0.003 ± 0.008	-0.014 ± 0.040	0.006 ± 0.001	0.078 ± 0.006	0.006 ± 0.001	0.078 ± 0.007	0.153 ± 0.021	0.391 ± 0.034	0.185	-0.015	-7.72	0.0714	0.0731	1.000	95.745 ± 2.082	0.308 ± 0.006	—
QML	beta44	0.00	0.004	0.004 ± 0.012	—	0.013 ± 0.002	0.116 ± 0.008	0.013 ± 0.002	0.115 ± 0.010	—	—	0.007	0.007	—	0.1309	0.1311	1.012	92.553 ± 2.708	0.458 ± 0.018	7.447 ± 2.708
QML	beta45	0.00	0.006	0.006 ± 0.010	—	0.009 ± 0.002	0.096 ± 0.008	0.009 ± 0.002	0.095 ± 0.010	—	—	-0.001	-0.001	—	0.0965	0.0965	0.990	100.000 ± 0.000	0.371 ± 0.014	0.000 ± 0.000
QML	beta46	0.00	-0.019	-0.019 ± 0.008	—	0.007 ± 0.001	0.082 ± 0.008	0.007 ± 0.001	0.084 ± 0.008	—	—	-0.027	-0.027	—	0.0683	0.0734	0.970	93.617 ± 2.521	0.311 ± 0.018	6.383 ± 2.521
QML	beta47	0.00	-0.030	-0.030 ± 0.008	—	0.005 ± 0.001	0.073 ± 0.008	0.006 ± 0.001	0.079 ± 0.009	—	—	-0.029	-0.029	—	0.0772	0.0826	0.961	86.170 ± 3.561	0.277 ± 0.012	13.830 ± 3.561
QML	beta51	0.16	0.177	0.017 ± 0.012	0.106 ± 0.073	0.013 ± 0.002	0.114 ± 0.010	0.013 ± 0.002	0.115 ± 0.013	0.513 ± 0.097	0.716 ± 0.079	0.150	-0.010	-6.43	0.1191	0.1195	1.163	100.000 ± 0.000	0.519 ± 0.019	—
QML	beta52	0.16	0.154	-0.006 ± 0.010	-0.039 ± 0.064	0.010 ± 0.002	0.100 ± 0.008	0.010 ± 0.002	0.099 ± 0.010	0.386 ± 0.062	0.621 ± 0.061	0.163	0.003	1.59	0.0972	0.0973	0.994	97.872 ± 1.488	0.389 ± 0.015	—
QML	beta53	0.16	0.162	0.002 ± 0.009	0.012 ± 0.055	0.007 ± 0.001	0.086 ± 0.007	0.007 ± 0.001	0.085 ± 0.009	0.285 ± 0.047	0.534 ± 0.053	0.152	-0.008	-5.06	0.0895	0.0899	1.077	96.809 ± 1.813	0.363 ± 0.008	—
QML	beta54	0.16	0.161	0.001 ± 0.008	0.009 ± 0.050	0.006 ± 0.001	0.077 ± 0.005	0.006 ± 0.001	0.077 ± 0.007	0.231 ± 0.031	0.480 ± 0.042	0.171	0.011	6.91	0.0885	0.0892	1.087	97.872 ± 1.488	0.329 ± 0.007	—
QML	beta55	0.00	0.018	0.018 ± 0.015	—	0.021 ± 0.004	0.144 ± 0.013	0.021 ± 0.004	0.145 ± 0.015	—	—	0.027	0.027	—	0.1422	0.1447	1.087	97.872 ± 1.488	0.615 ± 0.032	2.128 ± 1.488
QML	beta56	0.00	-0.008	-0.008 ± 0.012	—	0.013 ± 0.002	0.113 ± 0.010	0.013 ± 0.002	0.113 ± 0.011	—	—	-0.012	-0.012	—	0.1263	0.1268	1.051	94.681 ± 2.315	0.466 ± 0.016	5.319 ± 2.315
QML	beta57	0.00	-0.024	-0.024 ± 0.008	—	0.006 ± 0.001	0.079 ± 0.009	0.007 ± 0.001	0.082 ± 0.009	—	—	-0.033	-0.033	—	0.0740	0.0812	1.037	92.553 ± 2.708	0.322 ± 0.020	7.447 ± 2.708
QML	beta58	0.00	-0.020	-0.020 ± 0.006	—	0.003 ± 0.001	0.057 ± 0.006	0.004 ± 0.001	0.060 ± 0.006	—	—	-0.024	-0.024	—	0.0527	0.0580	1.059	93.617 ± 2.521	0.236 ± 0.011	6.383 ± 2.521
UPI	beta41	0.20	0.233	0.033 ± 0.025	0.164 ± 0.127	0.061 ± 0.015	0.247 ± 0.033	0.061 ± 0.016	0.248 ± 0.036	1.537 ± 0.404	1.240 ± 0.182	0.206	0.006	3.01	0.2782	0.2782	1.613	95.745 ± 2.082	1.563 ± 0.239	—
UPI	beta42	0.20	0.282	0.082 ± 0.027	0.411 ± 0.135	0.069 ± 0.015	0.262 ± 0.030	0.075 ± 0.016	0.274 ± 0.033	1.873 ± 0.405	1.369 ± 0.166	0.286	0.086	42.97	0.2638	0.2775	1.632	96.809 ± 1.813	1.679 ± 0.201	—
UPI	beta43	0.20	0.145	-0.055 ± 0.014	-0.274 ± 0.069	0.018 ± 0.004	0.134 ± 0.016	0.021 ± 0.005	0.144 ± 0.018	0.521 ± 0.115	0.721 ± 0.089	0.127	-0.073	-36.72	0.1466	0.1639	1.468	87.234 ± 3.442	0.772 ± 0.090	—
UPI	beta44	0.00	0.071	0.071 ± 0.030	—	0.084 ± 0.021	0.289 ± 0.038	0.088 ± 0.023	0.296 ± 0.043	—	—	0.056	0.056	—	0.2268	0.2335	1.718	98.936 ± 1.058	1.948 ± 0.317	1.064 ± 1.058
UPI	beta45	0.00	0.015	0.015 ± 0.016	—	0.025 ± 0.006	0.159 ± 0.018	0.025 ± 0.005	0.159 ± 0.020	—	—	0.009	0.009	—	0.1462	0.1465	1.699	100.000 ± 0.000	1.060 ± 0.123	0.000 ± 0.000
UPI	beta46	0.00	-0.080	-0.080 ± 0.025	—	0.060 ± 0.014	0.245 ± 0.029	0.066 ± 0.016	0.256 ± 0.034	—	—	-0.040	-0.040	—	0.2285	0.2320	1.515	100.000 ± 0.000	1.452 ± 0.195	0.000 ± 0.000

Table S271

Simulation 2 - Condition 127: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.105	-0.105 ± 0.033	—	0.101 ± 0.028	0.318 ± 0.047	0.111 ± 0.030	0.333 ± 0.050	—	—	-0.130	-0.130	—	0.2547	0.2862	1.507	97.872 ± 1.488	1.879 ± 0.296	2.128 ± 1.488
UPI	beta51	0.16	0.198	0.038 ± 0.012	0.237 ± 0.077	0.014 ± 0.003	0.119 ± 0.014	0.015 ± 0.003	0.124 ± 0.016	0.604 ± 0.136	0.777 ± 0.098	0.185	0.025	15.43	0.1298	0.1321	1.311	98.936 ± 1.058	0.612 ± 0.029	—
UPI	beta52	0.16	0.174	0.014 ± 0.014	0.088 ± 0.089	0.019 ± 0.003	0.138 ± 0.012	0.019 ± 0.003	0.138 ± 0.014	0.743 ± 0.127	0.862 ± 0.088	0.190	0.030	18.96	0.1471	0.1502	1.533	96.809 ± 1.813	0.829 ± 0.088	—
UPI	beta53	0.16	0.155	-0.005 ± 0.011	-0.033 ± 0.066	0.010 ± 0.001	0.102 ± 0.007	0.010 ± 0.001	0.102 ± 0.009	0.405 ± 0.057	0.637 ± 0.056	0.122	-0.038	-23.49	0.1372	0.1422	1.321	91.489 ± 2.878	0.530 ± 0.032	—
UPI	beta54	0.16	0.150	-0.010 ± 0.013	-0.060 ± 0.084	0.017 ± 0.004	0.131 ± 0.015	0.017 ± 0.004	0.130 ± 0.016	0.663 ± 0.140	0.814 ± 0.099	0.156	-0.004	-2.81	0.1453	0.1454	1.487	93.617 ± 2.521	0.761 ± 0.081	—
UPI	beta55	0.00	0.009	0.009 ± 0.019	—	0.035 ± 0.009	0.187 ± 0.026	0.035 ± 0.009	0.186 ± 0.027	—	—	0.035	0.035	—	0.1658	0.1694	1.492	100.000 ± 0.000	1.095 ± 0.123	0.000 ± 0.000
UPI	beta56	0.00	0.001	0.001 ± 0.015	—	0.022 ± 0.006	0.150 ± 0.019	0.022 ± 0.006	0.149 ± 0.021	—	—	0.006	0.006	—	0.1402	0.1403	1.927	100.000 ± 0.000	1.132 ± 0.264	0.000 ± 0.000
UPI	beta57	0.00	-0.034	-0.034 ± 0.016	—	0.026 ± 0.006	0.160 ± 0.021	0.026 ± 0.007	0.162 ± 0.024	—	—	-0.043	-0.043	—	0.1190	0.1266	1.247	100.000 ± 0.000	0.781 ± 0.087	0.000 ± 0.000
UPI	beta58	0.00	-0.036	-0.036 ± 0.011	—	0.011 ± 0.002	0.102 ± 0.008	0.012 ± 0.002	0.108 ± 0.011	—	—	-0.031	-0.031	—	0.0865	0.0919	2.137	97.872 ± 1.488	0.858 ± 0.129	2.128 ± 1.488

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S272

Simulation 2 - Condition 128: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.211	0.011 ± 0.005	0.053 ± 0.026	0.003 ± 0.000	0.054 ± 0.004	0.003 ± 0.000	0.055 ± 0.005	0.075 ± 0.011	0.274 ± 0.024	0.215	0.015	7.70	0.0515	0.0537	1.179	96.429 ± 1.754	0.250 ± 0.004	—
LSAM	beta42	0.20	0.222	0.022 ± 0.006	0.110 ± 0.030	0.004 ± 0.000	0.063 ± 0.004	0.004 ± 0.001	0.067 ± 0.005	0.111 ± 0.014	0.334 ± 0.026	0.218	0.018	9.12	0.0613	0.0639	1.063	93.750 ± 2.287	0.264 ± 0.004	—
LSAM	beta43	0.20	0.182	-0.018 ± 0.004	-0.090 ± 0.022	0.002 ± 0.000	0.047 ± 0.003	0.002 ± 0.000	0.050 ± 0.004	0.062 ± 0.008	0.250 ± 0.019	0.182	-0.018	-8.92	0.0511	0.0541	1.071	94.643 ± 2.128	0.197 ± 0.002	—
LSAM	beta44	0.00	0.050	0.050 ± 0.007	—	0.006 ± 0.001	0.078 ± 0.006	0.009 ± 0.001	0.092 ± 0.007	—	—	0.048	0.048	—	0.0725	0.0867	0.915	89.286 ± 2.923	0.281 ± 0.007	10.714 ± 2.923
LSAM	beta45	0.00	0.016	0.016 ± 0.005	—	0.003 ± 0.000	0.054 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	—	—	0.019	0.019	—	0.0552	0.0585	1.070	97.321 ± 1.526	0.225 ± 0.004	2.679 ± 1.526
LSAM	beta46	0.00	-0.045	-0.045 ± 0.004	—	0.002 ± 0.000	0.041 ± 0.003	0.004 ± 0.000	0.061 ± 0.004	—	—	-0.046	-0.046	—	0.0352	0.0577	1.150	87.500 ± 3.125	0.186 ± 0.005	12.500 ± 3.125
LSAM	beta47	0.00	-0.045	-0.045 ± 0.004	—	0.002 ± 0.000	0.044 ± 0.003	0.004 ± 0.000	0.063 ± 0.004	—	—	-0.041	-0.041	—	0.0531	0.0671	1.059	84.821 ± 3.390	0.184 ± 0.006	15.179 ± 3.390
LSAM	beta51	0.16	0.158	-0.002 ± 0.006	-0.013 ± 0.039	0.004 ± 0.001	0.066 ± 0.004	0.004 ± 0.001	0.066 ± 0.005	0.170 ± 0.020	0.413 ± 0.032	0.164	0.004	2.31	0.0742	0.0743	1.091	98.214 ± 1.251	0.284 ± 0.003	—
LSAM	beta52	0.16	0.179	0.019 ± 0.005	0.121 ± 0.034	0.003 ± 0.000	0.058 ± 0.003	0.004 ± 0.000	0.061 ± 0.005	0.144 ± 0.019	0.379 ± 0.030	0.176	0.016	10.12	0.0606	0.0627	1.110	97.321 ± 1.526	0.251 ± 0.004	—
LSAM	beta53	0.16	0.136	-0.024 ± 0.006	-0.151 ± 0.035	0.004 ± 0.000	0.059 ± 0.004	0.004 ± 0.001	0.064 ± 0.005	0.159 ± 0.021	0.399 ± 0.032	0.138	-0.022	-13.84	0.0526	0.0570	0.948	88.393 ± 3.027	0.221 ± 0.002	—
LSAM	beta54	0.16	0.169	0.009 ± 0.005	0.056 ± 0.032	0.003 ± 0.000	0.054 ± 0.003	0.003 ± 0.000	0.054 ± 0.004	0.114 ± 0.015	0.338 ± 0.027	0.174	0.014	8.72	0.0539	0.0557	1.113	97.321 ± 1.526	0.234 ± 0.003	—
LSAM	beta55	0.00	0.045	0.045 ± 0.010	—	0.011 ± 0.002	0.107 ± 0.009	0.013 ± 0.002	0.115 ± 0.011	—	—	0.042	0.042	—	0.0917	0.1010	0.901	91.964 ± 2.569	0.377 ± 0.012	8.036 ± 2.569
LSAM	beta56	0.00	0.001	0.001 ± 0.008	—	0.007 ± 0.001	0.082 ± 0.007	0.007 ± 0.001	0.081 ± 0.008	—	—	0.005	0.005	—	0.0720	0.0722	1.000	96.429 ± 1.754	0.320 ± 0.008	3.571 ± 1.754
LSAM	beta57	0.00	-0.041	-0.041 ± 0.004	—	0.002 ± 0.000	0.045 ± 0.004	0.004 ± 0.000	0.061 ± 0.004	—	—	-0.046	-0.046	—	0.0452	0.0642	1.079	90.179 ± 2.812	0.191 ± 0.005	9.821 ± 2.812
LSAM	beta58	0.00	-0.021	-0.021 ± 0.003	—	0.001 ± 0.000	0.037 ± 0.003	0.002 ± 0.000	0.042 ± 0.003	—	—	-0.022	-0.022	—	0.0381	0.0441	1.025	91.964 ± 2.569	0.147 ± 0.003	8.036 ± 2.569
QML	beta41	0.20	0.194	-0.006 ± 0.004	-0.031 ± 0.020	0.002 ± 0.000	0.041 ± 0.003	0.002 ± 0.000	0.042 ± 0.004	0.043 ± 0.006	0.208 ± 0.018	0.197	-0.003	-1.62	0.0394	0.0395	1.128	95.536 ± 1.951	0.183 ± 0.002	—
QML	beta42	0.20	0.203	0.003 ± 0.004	0.015 ± 0.022	0.002 ± 0.000	0.046 ± 0.003	0.002 ± 0.000	0.046 ± 0.004	0.052 ± 0.007	0.228 ± 0.019	0.202	0.002	0.98	0.0481	0.0482	1.061	95.536 ± 1.951	0.190 ± 0.002	—
QML	beta43	0.20	0.191	-0.009 ± 0.004	-0.046 ± 0.021	0.002 ± 0.000	0.044 ± 0.003	0.002 ± 0.000	0.044 ± 0.004	0.049 ± 0.006	0.222 ± 0.018	0.190	-0.010	-4.88	0.0493	0.0502	1.069	96.429 ± 1.754	0.183 ± 0.002	—
QML	beta44	0.00	0.015	0.015 ± 0.007	—	0.005 ± 0.001	0.070 ± 0.006	0.005 ± 0.001	0.071 ± 0.006	—	—	0.020	0.020	—	0.0619	0.0651	0.903	93.750 ± 2.287	0.247 ± 0.006	6.250 ± 2.287
QML	beta45	0.00	-0.003	-0.003 ± 0.005	—	0.003 ± 0.000	0.051 ± 0.003	0.003 ± 0.000	0.051 ± 0.004	—	—	-0.001	-0.001	—	0.0570	0.0570	1.044	96.429 ± 1.754	0.207 ± 0.004	3.571 ± 1.754
QML	beta46	0.00	-0.030	-0.030 ± 0.004	—	0.001 ± 0.000	0.038 ± 0.003	0.002 ± 0.000	0.049 ± 0.003	—	—	-0.034	-0.034	—	0.0369	0.0500	1.079	86.607 ± 3.218	0.162 ± 0.005	13.393 ± 3.218
QML	beta47	0.00	-0.035	-0.035 ± 0.003	—	0.001 ± 0.000	0.036 ± 0.003	0.003 ± 0.000	0.050 ± 0.003	—	—	-0.038	-0.038	—	0.0352	0.0518	1.076	81.250 ± 3.688	0.151 ± 0.004	18.750 ± 3.688
QML	beta51	0.16	0.164	0.004 ± 0.006	0.023 ± 0.038	0.004 ± 0.000	0.064 ± 0.004	0.004 ± 0.000	0.064 ± 0.005	0.158 ± 0.019	0.397 ± 0.031	0.163	0.003	1.93	0.0756	0.0757	1.150	99.107 ± 0.889	0.287 ± 0.004	—
QML	beta52	0.16	0.164	0.004 ± 0.004	0.026 ± 0.027	0.002 ± 0.000	0.045 ± 0.003	0.002 ± 0.000	0.045 ± 0.004	0.079 ± 0.011	0.281 ± 0.023	0.160	0.000	-0.16	0.0504	0.0504	1.189	97.321 ± 1.526	0.210 ± 0.003	—
QML	beta53	0.16	0.143	-0.017 ± 0.005	-0.109 ± 0.034	0.003 ± 0.000	0.057 ± 0.004	0.004 ± 0.000	0.059 ± 0.005	0.137 ± 0.019	0.370 ± 0.031	0.146	-0.014	-8.63	0.0519	0.0537	0.938	91.071 ± 2.694	0.209 ± 0.002	—
QML	beta54	0.16	0.161	0.001 ± 0.004	0.004 ± 0.027	0.002 ± 0.000	0.046 ± 0.003	0.002 ± 0.000	0.046 ± 0.003	0.081 ± 0.009	0.285 ± 0.021	0.163	0.003	1.88	0.0467	0.0468	1.104	97.321 ± 1.526	0.198 ± 0.002	—
QML	beta55	0.00	0.034	0.034 ± 0.009	—	0.008 ± 0.001	0.092 ± 0.007	0.010 ± 0.001	0.097 ± 0.008	—	—	0.028	0.028	—	0.0847	0.0891	0.937	91.964 ± 2.569	0.338 ± 0.009	8.036 ± 2.569
QML	beta56	0.00	-0.014	-0.014 ± 0.007	—	0.005 ± 0.001	0.072 ± 0.005	0.005 ± 0.001	0.073 ± 0.006	—	—	-0.013	-0.013	—	0.0679	0.0691	0.992	91.964 ± 2.569	0.279 ± 0.006	8.036 ± 2.569
QML	beta57	0.00	-0.035	-0.035 ± 0.004	—	0.001 ± 0.000	0.038 ± 0.003	0.003 ± 0.000	0.052 ± 0.004	—	—	-0.038	-0.038	—	0.0388	0.0541	1.123	83.036 ± 3.546	0.166 ± 0.006	16.964 ± 3.546
QML	beta58	0.00	-0.018	-0.018 ± 0.003	—	0.001 ± 0.000	0.033 ± 0.003	0.001 ± 0.000	0.037 ± 0.003	—	—	-0.021	-0.021	—	0.0360	0.0414	1.082	89.286 ± 2.923	0.139 ± 0.004	10.714 ± 2.923
UPI	beta41	0.20	0.235	0.035 ± 0.031	0.177 ± 0.154	0.106 ± 0.023	0.326 ± 0.037	0.106 ± 0.024	0.326 ± 0.041	2.659 ± 0.598	1.631 ± 0.203	0.228	0.028	13.80	0.2556	0.2571	1.372	91.071 ± 2.694	1.752 ± 0.300	—
UPI	beta42	0.20	0.294	0.094 ± 0.029	0.471 ± 0.143	0.091 ± 0.018	0.302 ± 0.031	0.099 ± 0.020	0.315 ± 0.035	2.487 ± 0.491	1.577 ± 0.174	0.285	0.085	42.73	0.2668	0.2802	1.606	90.179 ± 2.812	1.903 ± 0.414	—
UPI	beta43	0.20	0.127	-0.073 ± 0.011	-0.364 ± 0.055	0.013 ± 0.003	0.115 ± 0.012	0.018 ± 0.003	0.136 ± 0.014	0.462 ± 0.085	0.680 ± 0.069	0.103	-0.097	-48.36	0.1341	0.1653	1.638	80.357 ± 3.754	0.741 ± 0.201	—
UPI	beta44	0.00	0.111	0.111 ± 0.021	—	0.051 ± 0.009	0.227 ± 0.022	0.063 ± 0.013	0.251 ± 0.028	—	—	0.071	0.071	—	0.1800	0.1933	1.439	98.214 ± 1.251	1.278 ± 0.194	1.786 ± 1.251

Table S272

Simulation 2 - Condition 128: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.021	0.021 ± 0.011	—	0.012 ± 0.003	0.112 ± 0.012	0.013 ± 0.003	0.113 ± 0.014	—	—	0.021	0.021	—	0.0895	0.0920	1.547	99.107 ± 0.889	0.677 ± 0.097	0.893 ± 0.889
UPI	beta46	0.00	-0.100	-0.100 ± 0.028	—	0.088 ± 0.016	0.297 ± 0.029	0.098 ± 0.018	0.312 ± 0.032	—	—	-0.096	-0.096	—	0.2322	0.2513	1.295	99.107 ± 0.889	1.509 ± 0.202	0.893 ± 0.889
UPI	beta47	0.00	-0.161	-0.161 ± 0.023	—	0.059 ± 0.011	0.243 ± 0.023	0.085 ± 0.016	0.291 ± 0.029	—	—	-0.116	-0.116	—	0.2068	0.2370	1.520	96.429 ± 1.754	1.450 ± 0.228	3.571 ± 1.754
UPI	beta51	0.16	0.211	0.051 ± 0.008	0.316 ± 0.053	0.008 ± 0.002	0.090 ± 0.010	0.011 ± 0.002	0.103 ± 0.009	0.412 ± 0.068	0.642 ± 0.059	0.211	0.051	32.17	0.0759	0.0917	1.062	91.964 ± 2.569	0.374 ± 0.020	—
UPI	beta52	0.16	0.197	0.037 ± 0.012	0.232 ± 0.075	0.016 ± 0.003	0.128 ± 0.012	0.017 ± 0.003	0.132 ± 0.014	0.684 ± 0.126	0.827 ± 0.087	0.197	0.037	23.09	0.1636	0.1677	1.227	98.214 ± 1.251	0.613 ± 0.067	—
UPI	beta53	0.16	0.126	-0.034 ± 0.008	-0.212 ± 0.048	0.006 ± 0.001	0.080 ± 0.005	0.008 ± 0.001	0.087 ± 0.006	0.296 ± 0.030	0.544 ± 0.036	0.117	-0.043	-26.73	0.0976	0.1065	0.924	83.929 ± 3.470	0.292 ± 0.017	—
UPI	beta54	0.16	0.135	-0.025 ± 0.011	-0.154 ± 0.068	0.013 ± 0.003	0.116 ± 0.015	0.014 ± 0.003	0.118 ± 0.015	0.544 ± 0.121	0.738 ± 0.092	0.140	-0.020	-12.32	0.1249	0.1265	1.039	83.929 ± 3.470	0.472 ± 0.055	—
UPI	beta55	0.00	0.051	0.051 ± 0.012	—	0.017 ± 0.004	0.132 ± 0.015	0.020 ± 0.004	0.141 ± 0.017	—	—	0.048	0.048	—	0.1366	0.1449	1.281	99.107 ± 0.889	0.661 ± 0.058	0.893 ± 0.889
UPI	beta56	0.00	-0.012	-0.012 ± 0.010	—	0.011 ± 0.002	0.106 ± 0.008	0.011 ± 0.002	0.106 ± 0.010	—	—	0.003	0.003	—	0.1088	0.1088	1.242	96.429 ± 1.754	0.517 ± 0.036	3.571 ± 1.754
UPI	beta57	0.00	-0.087	-0.087 ± 0.012	—	0.016 ± 0.003	0.125 ± 0.011	0.023 ± 0.004	0.152 ± 0.014	—	—	-0.093	-0.093	—	0.1267	0.1569	1.459	100.000 ± 0.000	0.716 ± 0.065	0.000 ± 0.000
UPI	beta58	0.00	-0.058	-0.058 ± 0.013	—	0.020 ± 0.004	0.142 ± 0.013	0.023 ± 0.004	0.153 ± 0.016	—	—	-0.039	-0.039	—	0.1187	0.1248	1.143	97.321 ± 1.526	0.636 ± 0.082	2.679 ± 1.526

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S273

Simulation 2 - Condition 129: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.019 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.302 ± 0.009	0.202	0.002	1.00	0.0582	0.0582	0.996	95.124 ± 0.694	0.235 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.022 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.318 ± 0.009	0.203	0.003	1.56	0.0629	0.0630	0.983	94.917 ± 0.707	0.244 ± 0.001	—
LSAM	beta43	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.004	0.298 ± 0.008	0.197	-0.003	-1.66	0.0574	0.0575	0.960	94.502 ± 0.734	0.224 ± 0.001	—
LSAM	beta44	0.00	0.014	0.014 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.013	0.013	—	0.0782	0.0792	0.962	93.880 ± 0.772	0.277 ± 0.002	6.120 ± 0.772
LSAM	beta45	0.00	0.007	0.007 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.007	0.007	—	0.0610	0.0615	0.957	94.398 ± 0.741	0.240 ± 0.002	5.602 ± 0.741
LSAM	beta46	0.00	-0.016	-0.016 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.017	-0.017	—	0.0473	0.0504	0.950	91.598 ± 0.894	0.192 ± 0.002	8.402 ± 0.894
LSAM	beta47	0.00	-0.016	-0.016 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.017	-0.017	—	0.0494	0.0522	0.947	91.909 ± 0.878	0.189 ± 0.002	8.091 ± 0.878
LSAM	beta51	0.16	0.170	0.010 ± 0.003	0.060 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.288 ± 0.015	0.537 ± 0.016	0.167	0.007	4.13	0.0811	0.0813	0.950	93.361 ± 0.802	0.318 ± 0.002	—
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.020 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.008	0.402 ± 0.012	0.162	0.002	1.17	0.0625	0.0625	1.005	95.124 ± 0.694	0.253 ± 0.001	—
LSAM	beta53	0.16	0.153	-0.007 ± 0.002	-0.043 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.408 ± 0.012	0.150	-0.010	-6.21	0.0650	0.0658	0.997	94.398 ± 0.741	0.254 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.016 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.168 ± 0.008	0.409 ± 0.012	0.163	0.003	2.16	0.0665	0.0666	0.968	94.813 ± 0.714	0.248 ± 0.001	—
LSAM	beta55	0.00	0.010	0.010 ± 0.003	—	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	—	—	0.009	0.009	—	0.1013	0.1017	0.911	93.465 ± 0.796	0.374 ± 0.004	6.535 ± 0.796
LSAM	beta56	0.00	0.005	0.005 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.005	0.005	—	0.0843	0.0845	0.926	93.361 ± 0.802	0.319 ± 0.003	6.639 ± 0.802
LSAM	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.013	-0.013	—	0.0509	0.0524	0.993	94.191 ± 0.753	0.210 ± 0.002	5.809 ± 0.753
LSAM	beta58	0.00	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.014	-0.014	—	0.0432	0.0453	0.971	93.154 ± 0.813	0.177 ± 0.001	6.846 ± 0.813
QML	beta41	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.281 ± 0.008	0.198	-0.002	-0.79	0.0547	0.0547	1.016	95.124 ± 0.694	0.224 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.291 ± 0.008	0.200	0.000	-0.22	0.0571	0.0571	1.001	95.332 ± 0.679	0.229 ± 0.001	—
QML	beta43	0.20	0.198	-0.002 ± 0.002	-0.011 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.083 ± 0.004	0.287 ± 0.008	0.196	-0.004	-1.79	0.0558	0.0559	0.977	94.502 ± 0.734	0.220 ± 0.001	—
QML	beta44	0.00	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.001	0.001	—	0.0724	0.0725	0.989	94.606 ± 0.728	0.276 ± 0.003	5.394 ± 0.728
QML	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.001	-0.001	—	0.0581	0.0581	0.978	95.021 ± 0.701	0.236 ± 0.002	4.979 ± 0.701
QML	beta46	0.00	-0.008	-0.008 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.012	-0.012	—	0.0456	0.0471	0.992	92.220 ± 0.863	0.186 ± 0.002	7.780 ± 0.863
QML	beta47	0.00	-0.013	-0.013 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.015	-0.015	—	0.0454	0.0479	0.983	90.456 ± 0.946	0.179 ± 0.002	9.544 ± 0.946
QML	beta51	0.16	0.172	0.012 ± 0.003	0.072 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.281 ± 0.015	0.530 ± 0.016	0.166	0.006	3.71	0.0803	0.0806	1.013	95.124 ± 0.694	0.334 ± 0.002	—
QML	beta52	0.16	0.160	0.000 ± 0.002	0.000 ± 0.012	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.388 ± 0.011	0.160	0.000	-0.28	0.0597	0.0597	1.036	95.851 ± 0.642	0.252 ± 0.002	—
QML	beta53	0.16	0.154	-0.006 ± 0.002	-0.038 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.008	0.401 ± 0.012	0.152	-0.008	-5.08	0.0635	0.0640	1.008	94.606 ± 0.728	0.253 ± 0.002	—
QML	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.161	0.001	0.91	0.0618	0.0618	0.977	95.021 ± 0.701	0.241 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.003	—	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	—	—	0.003	0.003	—	0.0956	0.0957	0.975	95.851 ± 0.642	0.376 ± 0.008	4.149 ± 0.642
QML	beta56	0.00	-0.001	-0.001 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	—	—	-0.001	-0.001	—	0.0782	0.0782	0.956	94.398 ± 0.741	0.308 ± 0.003	5.602 ± 0.741
QML	beta57	0.00	-0.009	-0.009 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	-0.012	-0.012	—	0.0464	0.0479	1.095	93.568 ± 0.790	0.213 ± 0.008	6.432 ± 0.790
QML	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.013	-0.013	—	0.0406	0.0427	0.986	92.531 ± 0.847	0.169 ± 0.002	7.469 ± 0.847
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.021 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.004	0.299 ± 0.009	0.203	0.003	1.50	0.0601	0.0602	1.052	95.747 ± 0.650	0.246 ± 0.001	—
UPI	beta42	0.20	0.206	0.006 ± 0.002	0.030 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.318 ± 0.010	0.204	0.004	2.10	0.0631	0.0632	1.031	95.954 ± 0.635	0.256 ± 0.002	—
UPI	beta43	0.20	0.197	-0.003 ± 0.002	-0.015 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.004	0.301 ± 0.009	0.196	-0.004	-2.25	0.0595	0.0597	0.997	94.710 ± 0.721	0.235 ± 0.001	—
UPI	beta44	0.00	0.011	0.011 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	—	—	0.009	0.009	—	0.0763	0.0768	1.022	95.954 ± 0.635	0.325 ± 0.004	4.046 ± 0.635
UPI	beta45	0.00	0.007	0.007 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.008	0.008	—	0.0664	0.0669	1.005	95.539 ± 0.665	0.265 ± 0.002	4.461 ± 0.665
UPI	beta46	0.00	-0.011	-0.011 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.012	-0.012	—	0.0592	0.0604	1.014	95.124 ± 0.694	0.242 ± 0.003	4.876 ± 0.694

Table S273

Simulation 2 - Condition 129: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.014	-0.014 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.014	-0.014	—	0.0558	0.0575	1.040	95.539 ± 0.665	0.245 ± 0.004	4.461 ± 0.665
UPI	beta51	0.16	0.169	0.009 ± 0.003	0.056 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.268 ± 0.014	0.517 ± 0.015	0.165	0.005	3.04	0.0793	0.0795	1.004	95.124 ± 0.694	0.324 ± 0.002	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.015 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.007	0.395 ± 0.011	0.165	0.005	2.95	0.0603	0.0605	1.035	95.851 ± 0.642	0.256 ± 0.001	—
UPI	beta53	0.16	0.154	-0.006 ± 0.002	-0.037 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.409 ± 0.012	0.152	-0.008	-4.87	0.0648	0.0653	1.011	95.332 ± 0.679	0.258 ± 0.001	—
UPI	beta54	0.16	0.164	0.004 ± 0.002	0.025 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.409 ± 0.012	0.165	0.005	3.18	0.0647	0.0649	1.007	95.851 ± 0.642	0.258 ± 0.001	—
UPI	beta55	0.00	0.003	0.003 ± 0.003	—	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	—	—	0.004	0.004	—	0.1037	0.1037	0.954	95.747 ± 0.650	0.397 ± 0.005	4.253 ± 0.650
UPI	beta56	0.00	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.000	0.000	—	0.0840	0.0840	0.978	95.539 ± 0.665	0.339 ± 0.004	4.461 ± 0.665
UPI	beta57	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.005	-0.005	—	0.0570	0.0572	1.025	95.851 ± 0.642	0.236 ± 0.003	4.149 ± 0.642
UPI	beta58	0.00	-0.008	-0.008 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.007	-0.007	—	0.0457	0.0462	1.058	96.888 ± 0.559	0.214 ± 0.003	3.112 ± 0.559

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S274

Simulation 2 - Condition 130: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.001	0.018 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.194 ± 0.005	0.202	0.002	1.16	0.0386	0.0387	0.965	94.289 ± 0.735	0.146 ± 0.000	—
LSAM	beta42	0.20	0.203	0.003 ± 0.001	0.016 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.195 ± 0.005	0.203	0.003	1.49	0.0389	0.0390	0.996	95.190 ± 0.677	0.152 ± 0.000	—
LSAM	beta43	0.20	0.197	-0.003 ± 0.001	-0.017 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.174 ± 0.005	0.194	-0.006	-2.99	0.0340	0.0345	1.034	95.792 ± 0.636	0.140 ± 0.000	—
LSAM	beta44	0.00	0.014	0.014 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.015	0.015	—	0.0427	0.0452	1.007	93.086 ± 0.803	0.167 ± 0.001	6.914 ± 0.803
LSAM	beta45	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.004	0.004	—	0.0373	0.0375	0.967	94.489 ± 0.722	0.146 ± 0.001	5.511 ± 0.722
LSAM	beta46	0.00	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.013	-0.013	—	0.0299	0.0327	0.988	90.982 ± 0.907	0.115 ± 0.001	9.018 ± 0.907
LSAM	beta47	0.00	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.014	-0.014	—	0.0291	0.0324	0.983	91.383 ± 0.888	0.114 ± 0.001	8.617 ± 0.888
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.158	-0.002	-1.17	0.0499	0.0499	0.975	94.489 ± 0.722	0.197 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.001	0.012 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.161	0.001	0.35	0.0375	0.0375	1.026	94.990 ± 0.691	0.155 ± 0.000	—
LSAM	beta53	0.16	0.153	-0.007 ± 0.001	-0.045 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.152	-0.008	-4.91	0.0390	0.0398	0.990	93.988 ± 0.752	0.158 ± 0.000	—
LSAM	beta54	0.16	0.162	0.002 ± 0.001	0.013 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.162	0.002	1.39	0.0369	0.0370	0.998	94.489 ± 0.722	0.154 ± 0.000	—
LSAM	beta55	0.00	0.013	0.013 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.013	0.013	—	0.0546	0.0561	0.966	94.389 ± 0.728	0.218 ± 0.001	5.611 ± 0.728
LSAM	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	—	0.0481	0.0481	0.929	93.788 ± 0.764	0.189 ± 0.001	6.212 ± 0.764
LSAM	beta57	0.00	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.014	-0.014	—	0.0320	0.0350	0.956	89.780 ± 0.959	0.125 ± 0.001	10.220 ± 0.959
LSAM	beta58	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.012	-0.012	—	0.0265	0.0291	0.985	91.683 ± 0.874	0.108 ± 0.001	8.317 ± 0.874
QML	beta41	0.20	0.200	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.001	0.181 ± 0.005	0.199	-0.001	-0.58	0.0352	0.0352	0.991	95.291 ± 0.671	0.140 ± 0.002	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.182 ± 0.005	0.200	0.000	0.14	0.0362	0.0362	1.015	95.591 ± 0.650	0.145 ± 0.002	—
QML	beta43	0.20	0.197	-0.003 ± 0.001	-0.016 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.170 ± 0.005	0.195	-0.005	-2.72	0.0327	0.0331	1.041	95.591 ± 0.650	0.138 ± 0.000	—
QML	beta44	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.004	0.004	—	0.0427	0.0428	1.018	95.090 ± 0.684	0.167 ± 0.001	4.910 ± 0.684
QML	beta45	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	—	0.0368	0.0370	0.970	93.687 ± 0.770	0.146 ± 0.001	6.313 ± 0.770
QML	beta46	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.010	-0.010	—	0.0285	0.0301	1.008	91.383 ± 0.888	0.112 ± 0.001	8.617 ± 0.888
QML	beta47	0.00	-0.013	-0.013 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.014	-0.014	—	0.0273	0.0306	0.994	90.581 ± 0.925	0.109 ± 0.001	9.419 ± 0.925
QML	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.320 ± 0.009	0.159	-0.001	-0.88	0.0492	0.0493	1.000	94.990 ± 0.691	0.201 ± 0.001	—
QML	beta52	0.16	0.160	0.000 ± 0.001	0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.159	-0.001	-0.64	0.0356	0.0356	1.047	95.591 ± 0.650	0.151 ± 0.001	—
QML	beta53	0.16	0.154	-0.006 ± 0.001	-0.040 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.153	-0.007	-4.11	0.0392	0.0398	0.995	94.389 ± 0.728	0.156 ± 0.001	—
QML	beta54	0.16	0.161	0.001 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.161	0.001	0.71	0.0356	0.0356	1.010	94.890 ± 0.697	0.149 ± 0.000	—
QML	beta55	0.00	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.008	0.008	—	0.0550	0.0556	1.000	94.890 ± 0.697	0.217 ± 0.001	5.110 ± 0.697
QML	beta56	0.00	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.007	-0.007	—	0.0469	0.0474	0.938	93.587 ± 0.775	0.185 ± 0.001	6.413 ± 0.775
QML	beta57	0.00	-0.013	-0.013 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.015	-0.015	—	0.0297	0.0331	0.982	88.778 ± 0.999	0.122 ± 0.001	11.222 ± 0.999
QML	beta58	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.012	-0.012	—	0.0258	0.0285	0.998	90.581 ± 0.925	0.104 ± 0.001	9.419 ± 0.925
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.203	0.003	1.32	0.0385	0.0385	0.984	95.090 ± 0.684	0.146 ± 0.000	—
UPI	beta42	0.20	0.204	0.004 ± 0.001	0.019 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.204	0.004	2.21	0.0380	0.0383	1.017	96.092 ± 0.613	0.151 ± 0.001	—
UPI	beta43	0.20	0.197	-0.003 ± 0.001	-0.016 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.175 ± 0.005	0.195	-0.005	-2.39	0.0338	0.0341	1.039	95.591 ± 0.650	0.142 ± 0.000	—
UPI	beta44	0.00	0.010	0.010 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.011	0.011	—	0.0441	0.0454	1.013	93.888 ± 0.758	0.179 ± 0.001	6.112 ± 0.758

Table S274

Simulation 2 - Condition 130: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.003	0.003	—	0.0374	0.0375	0.999	95.291 ± 0.671	0.152 ± 0.001	4.709 ± 0.671
UPI	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.007	-0.007	—	0.0329	0.0338	0.999	95.491 ± 0.657	0.132 ± 0.001	4.509 ± 0.657
UPI	beta47	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.011	-0.011	—	0.0318	0.0335	1.002	93.988 ± 0.752	0.131 ± 0.001	6.012 ± 0.752
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.319 ± 0.009	0.158	-0.002	-1.01	0.0492	0.0492	0.988	94.589 ± 0.716	0.198 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.007 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.160	0.000	-0.22	0.0372	0.0372	1.040	95.591 ± 0.650	0.155 ± 0.000	—
UPI	beta53	0.16	0.153	-0.007 ± 0.001	-0.041 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.153	-0.007	-4.28	0.0391	0.0397	0.995	94.088 ± 0.747	0.159 ± 0.000	—
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.015 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.162	0.002	1.41	0.0371	0.0372	1.010	94.890 ± 0.697	0.155 ± 0.000	—
UPI	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.010	0.010	—	0.0556	0.0565	0.990	95.591 ± 0.650	0.225 ± 0.002	4.409 ± 0.650
UPI	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.003	-0.003	—	0.0481	0.0482	0.945	94.088 ± 0.747	0.191 ± 0.001	5.912 ± 0.747
UPI	beta57	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.010	-0.010	—	0.0335	0.0349	0.959	93.287 ± 0.792	0.135 ± 0.001	6.713 ± 0.792
UPI	beta58	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.009	-0.009	—	0.0279	0.0293	1.013	94.389 ± 0.728	0.119 ± 0.001	5.611 ± 0.728

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S275

Simulation 2 - Condition 131: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.200	0.000	-0.18	0.0455	0.0455	0.986	94.900 ± 0.696	0.180 ± 0.001	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.201	0.001	0.40	0.0462	0.0462	1.002	94.400 ± 0.727	0.186 ± 0.001	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.010 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.197	-0.003	-1.27	0.0465	0.0466	0.952	93.000 ± 0.807	0.176 ± 0.001	—
LSAM	beta44	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.000	0.000	—	0.0511	0.0511	0.980	95.000 ± 0.689	0.199 ± 0.001	5.000 ± 0.689
LSAM	beta45	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0447	0.0447	0.962	94.100 ± 0.745	0.175 ± 0.001	5.900 ± 0.745
LSAM	beta46	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.005	-0.005	—	0.0358	0.0361	0.985	93.500 ± 0.780	0.139 ± 0.001	6.500 ± 0.780
LSAM	beta47	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.004	-0.004	—	0.0366	0.0368	0.970	94.200 ± 0.739	0.138 ± 0.001	5.800 ± 0.739
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.009 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.180 ± 0.008	0.425 ± 0.012	0.160	0.000	0.22	0.0675	0.0675	0.936	94.100 ± 0.745	0.249 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.016 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.005	0.314 ± 0.009	0.164	0.004	2.48	0.0475	0.0477	1.000	95.200 ± 0.676	0.197 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.113 ± 0.005	0.336 ± 0.009	0.158	-0.002	-1.12	0.0528	0.0528	0.971	94.400 ± 0.727	0.205 ± 0.001	—
LSAM	beta54	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.160	0.000	-0.06	0.0508	0.0508	0.953	93.500 ± 0.780	0.197 ± 0.001	—
LSAM	beta55	0.00	0.005	0.005 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	0.007	0.007	—	0.0694	0.0697	0.907	93.400 ± 0.785	0.261 ± 0.002	6.600 ± 0.785
LSAM	beta56	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.004	0.004	—	0.0547	0.0548	0.973	94.600 ± 0.715	0.230 ± 0.002	5.400 ± 0.715
LSAM	beta57	0.00	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.006	-0.006	—	0.0402	0.0406	0.992	93.500 ± 0.780	0.153 ± 0.001	6.500 ± 0.780
LSAM	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.003	-0.003	—	0.0364	0.0365	0.958	93.100 ± 0.801	0.134 ± 0.001	6.900 ± 0.801
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.229 ± 0.006	0.198	-0.002	-0.88	0.0448	0.0449	0.998	95.200 ± 0.676	0.179 ± 0.001	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.200	0.000	-0.04	0.0455	0.0455	1.005	94.900 ± 0.696	0.183 ± 0.001	—
QML	beta43	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.198	-0.002	-1.24	0.0464	0.0465	0.957	93.300 ± 0.791	0.175 ± 0.001	—
QML	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.004	-0.004	—	0.0515	0.0516	0.996	95.400 ± 0.662	0.203 ± 0.002	4.600 ± 0.662
QML	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.002	-0.002	—	0.0453	0.0453	0.999	94.500 ± 0.721	0.181 ± 0.006	5.500 ± 0.721
QML	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.004	-0.004	—	0.0341	0.0343	0.998	93.400 ± 0.785	0.139 ± 0.001	6.600 ± 0.785
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.004	-0.004	—	0.0354	0.0356	0.976	94.300 ± 0.733	0.136 ± 0.001	5.700 ± 0.733
QML	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.178 ± 0.008	0.422 ± 0.012	0.161	0.001	0.44	0.0667	0.0667	0.970	94.900 ± 0.696	0.257 ± 0.002	—
QML	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.312 ± 0.009	0.163	0.003	1.78	0.0476	0.0477	1.010	95.500 ± 0.656	0.198 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.111 ± 0.005	0.333 ± 0.009	0.157	-0.003	-1.72	0.0514	0.0515	0.976	94.600 ± 0.715	0.204 ± 0.001	—
QML	beta54	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.326 ± 0.009	0.160	0.000	0.00	0.0500	0.0500	0.960	93.900 ± 0.757	0.196 ± 0.001	—
QML	beta55	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.004	0.004	—	0.0685	0.0686	0.937	94.200 ± 0.739	0.264 ± 0.002	5.800 ± 0.739
QML	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.000	0.000	—	0.0544	0.0545	0.989	95.000 ± 0.689	0.229 ± 0.002	5.000 ± 0.689
QML	beta57	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.006	-0.006	—	0.0390	0.0394	1.020	92.900 ± 0.812	0.153 ± 0.002	7.100 ± 0.812
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.003	-0.003	—	0.0352	0.0354	0.960	93.000 ± 0.807	0.133 ± 0.001	7.000 ± 0.807
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.199	-0.001	-0.51	0.0453	0.0453	0.991	95.000 ± 0.689	0.180 ± 0.001	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.201	0.001	0.26	0.0466	0.0466	1.003	94.400 ± 0.727	0.185 ± 0.001	—
UPI	beta43	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.198	-0.002	-1.02	0.0469	0.0469	0.956	93.300 ± 0.791	0.177 ± 0.001	—
UPI	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.000	0.000	—	0.0531	0.0531	0.984	94.900 ± 0.696	0.205 ± 0.001	5.100 ± 0.696
UPI	beta45	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0431	0.0431	0.976	93.800 ± 0.763	0.179 ± 0.001	6.200 ± 0.763
UPI	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0379	0.0381	0.990	94.000 ± 0.751	0.146 ± 0.001	6.000 ± 0.751

Table S275

Simulation 2 - Condition 131: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	—	0.0367	0.0370	0.974	94.300 ± 0.733	0.144 ± 0.001	5.700 ± 0.733
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.008	0.419 ± 0.012	0.161	0.001	0.69	0.0666	0.0666	0.966	94.400 ± 0.727	0.254 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.015 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.312 ± 0.009	0.163	0.003	2.08	0.0466	0.0468	1.006	95.600 ± 0.649	0.197 ± 0.001	—
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.112 ± 0.005	0.335 ± 0.009	0.158	-0.002	-1.04	0.0520	0.0521	0.977	94.400 ± 0.727	0.205 ± 0.001	—
UPI	beta54	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.160	0.000	0.18	0.0503	0.0503	0.956	93.600 ± 0.774	0.198 ± 0.001	—
UPI	beta55	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.003	0.003	—	0.0689	0.0690	0.945	94.000 ± 0.751	0.265 ± 0.002	6.000 ± 0.751
UPI	beta56	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.001	0.001	—	0.0537	0.0537	0.990	95.200 ± 0.676	0.233 ± 0.002	4.800 ± 0.676
UPI	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.004	-0.004	—	0.0392	0.0394	1.011	94.900 ± 0.696	0.156 ± 0.001	5.100 ± 0.696
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0368	0.0369	0.968	93.500 ± 0.780	0.138 ± 0.001	6.500 ± 0.780

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S276

Simulation 2 - Condition 132: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.201	0.001	0.49	0.0288	0.0288	1.025	96.200 ± 0.605	0.114 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.200	0.000	0.18	0.0299	0.0299	0.971	94.500 ± 0.721	0.117 ± 0.000	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.199	-0.001	-0.26	0.0304	0.0304	0.972	94.800 ± 0.702	0.112 ± 0.000	—
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.003	0.003	—	0.0350	0.0351	0.965	94.000 ± 0.751	0.125 ± 0.001	6.000 ± 0.751
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0288	0.0288	0.961	94.300 ± 0.733	0.111 ± 0.000	5.700 ± 0.733
LSAM	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.004	-0.004	—	0.0218	0.0221	0.978	93.800 ± 0.763	0.087 ± 0.000	6.200 ± 0.763
LSAM	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0211	0.0214	0.998	93.100 ± 0.801	0.087 ± 0.000	6.900 ± 0.801
LSAM	beta51	0.16	0.161	0.001 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.162	0.002	1.38	0.0410	0.0411	0.957	94.600 ± 0.715	0.158 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.006	0.162	0.002	1.40	0.0306	0.0307	0.995	95.000 ± 0.689	0.125 ± 0.000	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.001	-0.017 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.046 ± 0.002	0.214 ± 0.006	0.157	-0.003	-1.64	0.0331	0.0332	0.961	92.600 ± 0.828	0.129 ± 0.000	—
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.161	0.001	0.65	0.0309	0.0309	0.960	94.100 ± 0.745	0.125 ± 0.000	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.002	0.002	—	0.0416	0.0417	0.989	94.600 ± 0.715	0.163 ± 0.001	5.400 ± 0.715
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0373	0.0373	0.972	94.600 ± 0.715	0.143 ± 0.001	5.400 ± 0.715
LSAM	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.004	-0.004	—	0.0261	0.0264	0.977	93.800 ± 0.763	0.097 ± 0.000	6.200 ± 0.763
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.004	-0.004	—	0.0218	0.0221	0.979	93.800 ± 0.763	0.085 ± 0.000	6.200 ± 0.763
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.140 ± 0.004	0.200	0.000	0.09	0.0288	0.0288	1.028	96.800 ± 0.557	0.113 ± 0.000	—
QML	beta42	0.20	0.200	0.000 ± 0.001	0.000 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.199	-0.001	-0.43	0.0291	0.0291	0.975	94.600 ± 0.715	0.116 ± 0.000	—
QML	beta43	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.199	-0.001	-0.55	0.0301	0.0301	0.972	95.000 ± 0.689	0.112 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	—	0.0352	0.0352	0.971	95.000 ± 0.689	0.127 ± 0.001	5.000 ± 0.689
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0289	0.0289	0.965	93.800 ± 0.763	0.111 ± 0.000	6.200 ± 0.763
QML	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0210	0.0213	0.985	93.500 ± 0.780	0.087 ± 0.001	6.500 ± 0.780
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.004	-0.004	—	0.0210	0.0214	1.014	93.100 ± 0.801	0.086 ± 0.001	6.900 ± 0.801
QML	beta51	0.16	0.162	0.002 ± 0.001	0.010 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.162	0.002	1.04	0.0412	0.0412	0.969	94.600 ± 0.715	0.160 ± 0.001	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.006	0.162	0.002	0.97	0.0309	0.0309	1.000	95.100 ± 0.683	0.125 ± 0.000	—
QML	beta53	0.16	0.157	-0.003 ± 0.001	-0.019 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.045 ± 0.002	0.213 ± 0.006	0.157	-0.003	-2.07	0.0326	0.0327	0.961	92.700 ± 0.823	0.128 ± 0.000	—
QML	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.161	0.001	0.60	0.0307	0.0307	0.964	94.700 ± 0.708	0.124 ± 0.000	—
QML	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.001	0.001	—	0.0417	0.0417	0.997	94.600 ± 0.715	0.163 ± 0.001	5.400 ± 0.715
QML	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0367	0.0368	0.983	94.800 ± 0.702	0.143 ± 0.001	5.200 ± 0.702
QML	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	-0.005	-0.005	—	0.0250	0.0255	0.983	93.300 ± 0.791	0.096 ± 0.001	6.700 ± 0.791
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.004	-0.004	—	0.0215	0.0219	0.983	93.200 ± 0.796	0.084 ± 0.000	6.800 ± 0.796
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.202	0.002	0.77	0.0286	0.0286	1.025	96.300 ± 0.597	0.114 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.003 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.200	0.000	-0.03	0.0299	0.0299	0.972	94.500 ± 0.721	0.117 ± 0.000	—
UPI	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.199	-0.001	-0.26	0.0304	0.0304	0.971	94.900 ± 0.696	0.112 ± 0.000	—
UPI	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.003	0.003	—	0.0351	0.0352	0.968	94.500 ± 0.721	0.127 ± 0.001	5.500 ± 0.721

Table S276

Simulation 2 - Condition 132: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0299	0.0299	0.964	94.900 ± 0.696	0.112 ± 0.000	5.100 ± 0.696
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.002	-0.002	—	0.0221	0.0222	0.989	94.700 ± 0.708	0.089 ± 0.000	5.300 ± 0.708
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.002	-0.002	—	0.0220	0.0221	1.009	94.500 ± 0.721	0.089 ± 0.000	5.500 ± 0.721
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.009 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.162	0.002	1.33	0.0409	0.0409	0.966	94.500 ± 0.721	0.159 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.006	0.162	0.002	1.23	0.0305	0.0306	0.997	94.900 ± 0.696	0.125 ± 0.000	—
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.014 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.045 ± 0.002	0.213 ± 0.006	0.158	-0.002	-1.22	0.0324	0.0325	0.964	92.500 ± 0.833	0.129 ± 0.000	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.161	0.001	0.78	0.0309	0.0310	0.964	94.500 ± 0.721	0.125 ± 0.000	—
UPI	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.002	0.002	—	0.0432	0.0432	0.990	94.500 ± 0.721	0.164 ± 0.001	5.500 ± 0.721
UPI	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0365	0.0365	0.984	94.200 ± 0.739	0.144 ± 0.001	5.800 ± 0.739
UPI	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.002	-0.002	—	0.0269	0.0270	0.968	93.700 ± 0.768	0.098 ± 0.001	6.300 ± 0.768
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.002	-0.002	—	0.0222	0.0223	0.985	94.400 ± 0.727	0.086 ± 0.000	5.600 ± 0.727

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S277

Simulation 2 - Condition 133: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.248	0.048 ± 0.014	0.238 ± 0.070	0.088 ± 0.008	0.297 ± 0.013	0.090 ± 0.008	0.301 ± 0.015	2.262 ± 0.201	1.504 ± 0.076	0.233	0.033	16.40	0.2659	0.2679	1.188	99.101 ± 0.447	1.385 ± 0.048	—
LSAM	beta42	0.20	0.257	0.057 ± 0.014	0.284 ± 0.072	0.092 ± 0.007	0.304 ± 0.012	0.095 ± 0.008	0.309 ± 0.014	2.380 ± 0.189	1.543 ± 0.071	0.251	0.051	25.74	0.2746	0.2794	1.194	99.326 ± 0.388	1.421 ± 0.044	—
LSAM	beta43	0.20	0.171	-0.029 ± 0.004	-0.143 ± 0.022	0.009 ± 0.001	0.094 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.243 ± 0.018	0.493 ± 0.021	0.180	-0.020	-10.22	0.0920	0.0942	1.067	94.382 ± 1.092	0.395 ± 0.007	—
LSAM	beta44	0.00	0.035	0.035 ± 0.005	—	0.013 ± 0.001	0.115 ± 0.005	0.014 ± 0.001	0.120 ± 0.006	—	—	0.026	0.026	—	0.1113	0.1143	1.276	98.876 ± 0.500	0.576 ± 0.019	1.124 ± 0.500
LSAM	beta45	0.00	0.007	0.007 ± 0.005	—	0.012 ± 0.001	0.111 ± 0.005	0.012 ± 0.001	0.111 ± 0.006	—	—	0.012	0.012	—	0.1036	0.1043	1.137	98.427 ± 0.590	0.496 ± 0.017	1.573 ± 0.590
LSAM	beta46	0.00	-0.033	-0.033 ± 0.005	—	0.013 ± 0.001	0.116 ± 0.005	0.014 ± 0.001	0.120 ± 0.006	—	—	-0.025	-0.025	—	0.0953	0.0985	1.183	99.775 ± 0.224	0.538 ± 0.021	0.225 ± 0.224
LSAM	beta47	0.00	-0.030	-0.030 ± 0.005	—	0.013 ± 0.001	0.113 ± 0.005	0.014 ± 0.001	0.117 ± 0.006	—	—	-0.026	-0.026	—	0.0959	0.0995	1.185	98.202 ± 0.630	0.527 ± 0.019	1.798 ± 0.630
LSAM	beta51	0.16	0.168	0.008 ± 0.006	0.050 ± 0.038	0.017 ± 0.001	0.130 ± 0.006	0.017 ± 0.001	0.130 ± 0.006	0.659 ± 0.056	0.812 ± 0.040	0.155	-0.005	-3.40	0.1328	0.1329	0.905	96.629 ± 0.856	0.460 ± 0.007	—
LSAM	beta52	0.16	0.227	0.067 ± 0.014	0.418 ± 0.087	0.086 ± 0.009	0.294 ± 0.015	0.091 ± 0.009	0.301 ± 0.017	3.543 ± 0.356	1.882 ± 0.104	0.219	0.059	36.71	0.2599	0.2664	1.095	98.652 ± 0.547	1.261 ± 0.046	—
LSAM	beta53	0.16	0.130	-0.030 ± 0.006	-0.187 ± 0.039	0.017 ± 0.001	0.131 ± 0.005	0.018 ± 0.002	0.134 ± 0.006	0.699 ± 0.059	0.836 ± 0.040	0.123	-0.037	-23.37	0.1200	0.1257	1.094	96.854 ± 0.827	0.560 ± 0.013	—
LSAM	beta54	0.16	0.185	0.025 ± 0.013	0.154 ± 0.081	0.074 ± 0.007	0.272 ± 0.012	0.075 ± 0.007	0.273 ± 0.014	2.912 ± 0.264	1.707 ± 0.088	0.193	0.033	20.35	0.2319	0.2341	1.126	99.101 ± 0.447	1.202 ± 0.040	—
LSAM	beta55	0.00	0.044	0.044 ± 0.009	—	0.034 ± 0.003	0.185 ± 0.008	0.036 ± 0.003	0.190 ± 0.010	—	—	0.030	0.030	—	0.1545	0.1573	1.072	97.978 ± 0.667	0.776 ± 0.026	2.022 ± 0.667
LSAM	beta56	0.00	0.011	0.011 ± 0.007	—	0.019 ± 0.002	0.138 ± 0.006	0.019 ± 0.002	0.138 ± 0.007	—	—	0.007	0.007	—	0.1234	0.1236	1.073	97.978 ± 0.667	0.580 ± 0.019	2.022 ± 0.667
LSAM	beta57	0.00	-0.044	-0.044 ± 0.006	—	0.017 ± 0.002	0.129 ± 0.007	0.018 ± 0.002	0.136 ± 0.008	—	—	-0.033	-0.033	—	0.1042	0.1092	1.101	99.101 ± 0.447	0.556 ± 0.023	0.899 ± 0.447
LSAM	beta58	0.00	-0.020	-0.020 ± 0.005	—	0.011 ± 0.001	0.104 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	—	—	-0.017	-0.017	—	0.0796	0.0813	1.119	98.876 ± 0.500	0.455 ± 0.018	1.124 ± 0.500
QML	beta41	0.20	0.175	-0.025 ± 0.004	-0.123 ± 0.022	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.229 ± 0.017	0.479 ± 0.021	0.178	-0.022	-11.00	0.0929	0.0954	1.085	96.180 ± 0.909	0.394 ± 0.013	—
QML	beta42	0.20	0.185	-0.015 ± 0.005	-0.074 ± 0.023	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.232 ± 0.016	0.482 ± 0.020	0.184	-0.016	-7.84	0.0935	0.0948	1.133	97.303 ± 0.768	0.423 ± 0.029	—
QML	beta43	0.20	0.199	-0.001 ± 0.004	-0.005 ± 0.019	0.007 ± 0.000	0.081 ± 0.003	0.007 ± 0.000	0.081 ± 0.003	0.165 ± 0.011	0.406 ± 0.017	0.205	0.005	2.37	0.0813	0.0814	1.006	94.607 ± 1.071	0.320 ± 0.005	—
QML	beta44	0.00	-0.006	-0.006 ± 0.005	—	0.012 ± 0.001	0.109 ± 0.004	0.012 ± 0.001	0.109 ± 0.005	—	—	-0.001	-0.001	—	0.1114	0.1114	1.082	97.528 ± 0.736	0.463 ± 0.012	2.472 ± 0.736
QML	beta45	0.00	0.005	0.005 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.094 ± 0.004	—	—	0.010	0.010	—	0.0916	0.0922	1.001	96.854 ± 0.827	0.370 ± 0.007	3.146 ± 0.827
QML	beta46	0.00	0.025	0.025 ± 0.003	—	0.004 ± 0.000	0.063 ± 0.003	0.005 ± 0.000	0.067 ± 0.003	—	—	0.015	0.015	—	0.0578	0.0597	1.044	96.854 ± 0.827	0.257 ± 0.007	3.146 ± 0.827
QML	beta47	0.00	0.031	0.031 ± 0.003	—	0.004 ± 0.000	0.062 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	—	—	0.024	0.024	—	0.0603	0.0651	1.098	95.506 ± 0.982	0.267 ± 0.013	4.494 ± 0.982
QML	beta51	0.16	0.182	0.022 ± 0.006	0.140 ± 0.039	0.017 ± 0.001	0.131 ± 0.005	0.018 ± 0.001	0.132 ± 0.006	0.684 ± 0.050	0.827 ± 0.036	0.168	0.008	5.00	0.1337	0.1340	1.031	97.978 ± 0.667	0.528 ± 0.008	—
QML	beta52	0.16	0.142	-0.018 ± 0.005	-0.116 ± 0.031	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.105 ± 0.005	0.429 ± 0.033	0.655 ± 0.030	0.151	-0.009	-5.92	0.1029	0.1034	1.041	96.180 ± 0.909	0.421 ± 0.007	—
QML	beta53	0.16	0.152	-0.008 ± 0.005	-0.047 ± 0.029	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.387 ± 0.026	0.622 ± 0.026	0.152	-0.008	-4.76	0.1014	0.1017	1.017	95.955 ± 0.934	0.396 ± 0.005	—
QML	beta54	0.16	0.135	-0.025 ± 0.004	-0.156 ± 0.027	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.358 ± 0.026	0.598 ± 0.026	0.144	-0.016	-10.29	0.1010	0.1024	1.109	96.854 ± 0.827	0.402 ± 0.004	—
QML	beta55	0.00	0.005	0.005 ± 0.008	—	0.027 ± 0.002	0.166 ± 0.007	0.027 ± 0.002	0.166 ± 0.008	—	—	0.004	0.004	—	0.1435	0.1436	0.978	94.607 ± 1.071	0.636 ± 0.017	5.393 ± 1.071
QML	beta56	0.00	0.022	0.022 ± 0.006	—	0.014 ± 0.001	0.118 ± 0.005	0.014 ± 0.001	0.120 ± 0.006	—	—	0.017	0.017	—	0.1041	0.1055	0.976	94.607 ± 1.071	0.451 ± 0.009	5.393 ± 1.071
QML	beta57	0.00	0.016	0.016 ± 0.004	—	0.006 ± 0.001	0.078 ± 0.003	0.006 ± 0.001	0.079 ± 0.004	—	—	0.005	0.005	—	0.0691	0.0694	1.072	97.753 ± 0.703	0.326 ± 0.013	2.247 ± 0.703
QML	beta58	0.00	0.020	0.020 ± 0.003	—	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.058 ± 0.003	—	—	0.015	0.015	—	0.0513	0.0534	1.039	97.978 ± 0.667	0.223 ± 0.007	2.022 ± 0.667
UPI	beta41	0.20	0.414	0.214 ± 0.032	1.069 ± 0.160	0.454 ± 0.043	0.674 ± 0.033	0.499 ± 0.049	0.706 ± 0.038	12.474 ± 1.232	3.532 ± 0.191	0.392	0.192	96.22	0.6735	0.7004	1.453	100.000 ± 0.000	3.838 ± 0.205	—
UPI	beta42	0.20	0.351	0.151 ± 0.028	0.753 ± 0.140	0.348 ± 0.034	0.590 ± 0.029	0.369 ± 0.036	0.608 ± 0.033	9.235 ± 0.909	3.039 ± 0.165	0.250	0.050	25.18	0.5939	0.5960	1.426	99.551 ± 0.317	3.294 ± 0.168	—
UPI	beta43	0.20	0.138	-0.062 ± 0.007	-0.312 ± 0.033	0.020 ± 0.002	0.141 ± 0.007	0.024 ± 0.002	0.154 ± 0.008	0.594 ± 0.056	0.771 ± 0.039	0.145	-0.055	-27.41	0.1361	0.1467	1.419	95.281 ± 1.005	0.785 ± 0.035	—
UPI	beta44	0.00	0.091	0.091 ± 0.016	—	0.119 ± 0.013	0.344 ± 0.020	0.127 ± 0.014	0.356 ± 0.022	—	—	0.055	0.055	—	0.2848	0.2901	1.603	100.000 ± 0.000	2.164 ± 0.145	0.000 ± 0.000

Table S277

Simulation 2 - Condition 133: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.031	0.031 ± 0.012	—	0.063 ± 0.007	0.251 ± 0.014	0.064 ± 0.007	0.252 ± 0.015	—	—	0.007	0.007	—	0.1983	0.1984	1.560	100.000 ± 0.000	1.533 ± 0.089	0.000 ± 0.000
UPI	beta46	0.00	-0.103	-0.103 ± 0.014	—	0.084 ± 0.009	0.290 ± 0.016	0.094 ± 0.010	0.307 ± 0.018	—	—	-0.076	-0.076	—	0.2591	0.2701	1.411	99.775 ± 0.224	1.602 ± 0.097	0.225 ± 0.224
UPI	beta47	0.00	-0.072	-0.072 ± 0.011	—	0.053 ± 0.006	0.230 ± 0.012	0.058 ± 0.006	0.241 ± 0.014	—	—	-0.033	-0.033	—	0.2118	0.2143	1.376	99.775 ± 0.224	1.240 ± 0.065	0.225 ± 0.224
UPI	beta51	0.16	0.175	0.015 ± 0.007	0.093 ± 0.044	0.022 ± 0.002	0.149 ± 0.006	0.022 ± 0.002	0.150 ± 0.007	0.877 ± 0.071	0.936 ± 0.044	0.167	0.007	4.30	0.1614	0.1616	1.361	98.876 ± 0.500	0.796 ± 0.030	—
UPI	beta52	0.16	0.240	0.080 ± 0.020	0.501 ± 0.124	0.176 ± 0.017	0.419 ± 0.020	0.182 ± 0.017	0.427 ± 0.022	7.108 ± 0.656	2.666 ± 0.138	0.207	0.047	29.53	0.3694	0.3724	1.693	99.326 ± 0.388	2.783 ± 0.154	—
UPI	beta53	0.16	0.123	-0.037 ± 0.007	-0.234 ± 0.047	0.025 ± 0.002	0.158 ± 0.007	0.026 ± 0.002	0.162 ± 0.008	1.028 ± 0.088	1.014 ± 0.049	0.111	-0.049	-30.78	0.1538	0.1615	1.620	98.652 ± 0.547	1.004 ± 0.055	—
UPI	beta54	0.16	0.218	0.058 ± 0.022	0.365 ± 0.140	0.224 ± 0.021	0.473 ± 0.023	0.227 ± 0.023	0.476 ± 0.026	8.862 ± 0.880	2.977 ± 0.164	0.162	0.002	1.42	0.4257	0.4257	1.586	98.876 ± 0.500	2.942 ± 0.168	—
UPI	beta55	0.00	0.027	0.027 ± 0.014	—	0.088 ± 0.008	0.296 ± 0.014	0.088 ± 0.008	0.297 ± 0.015	—	—	0.018	0.018	—	0.2492	0.2498	1.583	99.326 ± 0.388	1.838 ± 0.099	0.674 ± 0.388
UPI	beta56	0.00	0.014	0.014 ± 0.010	—	0.044 ± 0.004	0.210 ± 0.010	0.044 ± 0.004	0.210 ± 0.011	—	—	0.004	0.004	—	0.1738	0.1738	1.654	99.551 ± 0.317	1.363 ± 0.081	0.449 ± 0.317
UPI	beta57	0.00	-0.040	-0.040 ± 0.009	—	0.033 ± 0.003	0.182 ± 0.009	0.035 ± 0.003	0.187 ± 0.010	—	—	-0.025	-0.025	—	0.1499	0.1520	1.636	100.000 ± 0.000	1.170 ± 0.064	0.000 ± 0.000
UPI	beta58	0.00	-0.031	-0.031 ± 0.008	—	0.031 ± 0.003	0.176 ± 0.009	0.032 ± 0.003	0.178 ± 0.010	—	—	-0.011	-0.011	—	0.1374	0.1378	1.524	100.000 ± 0.000	1.050 ± 0.060	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S278

Simulation 2 - Condition 134: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.281	0.081 ± 0.011	0.405 ± 0.054	0.082 ± 0.005	0.287 ± 0.009	0.089 ± 0.006	0.298 ± 0.011	2.215 ± 0.150	1.488 ± 0.057	0.259	0.059	29.61	0.2606	0.2673	1.230	99.156 ± 0.343	1.382 ± 0.041	—
LSAM	beta42	0.20	0.273	0.073 ± 0.011	0.365 ± 0.055	0.087 ± 0.006	0.295 ± 0.010	0.093 ± 0.006	0.304 ± 0.011	2.313 ± 0.153	1.521 ± 0.057	0.262	0.062	31.02	0.2636	0.2708	1.256	99.297 ± 0.313	1.455 ± 0.043	—
LSAM	beta43	0.20	0.173	-0.027 ± 0.002	-0.136 ± 0.012	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.118 ± 0.007	0.343 ± 0.011	0.169	-0.031	-15.46	0.0666	0.0735	1.178	96.343 ± 0.704	0.291 ± 0.004	—
LSAM	beta44	0.00	0.033	0.033 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.003	0.008 ± 0.001	0.087 ± 0.003	—	—	0.031	0.031	—	0.0784	0.0842	1.169	97.328 ± 0.605	0.371 ± 0.009	2.672 ± 0.605
LSAM	beta45	0.00	0.009	0.009 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	—	—	0.010	0.010	—	0.0666	0.0673	1.125	97.468 ± 0.589	0.315 ± 0.007	2.532 ± 0.589
LSAM	beta46	0.00	-0.044	-0.044 ± 0.004	—	0.011 ± 0.001	0.103 ± 0.004	0.012 ± 0.001	0.112 ± 0.004	—	—	-0.033	-0.033	—	0.0846	0.0909	1.200	98.734 ± 0.419	0.483 ± 0.016	1.266 ± 0.419
LSAM	beta47	0.00	-0.041	-0.041 ± 0.004	—	0.011 ± 0.001	0.104 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	—	—	-0.032	-0.032	—	0.0831	0.0890	1.219	99.437 ± 0.281	0.495 ± 0.017	0.563 ± 0.281
LSAM	beta51	0.16	0.154	-0.006 ± 0.003	-0.035 ± 0.019	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.262 ± 0.015	0.512 ± 0.018	0.152	-0.008	-5.24	0.0809	0.0813	0.952	93.671 ± 0.913	0.305 ± 0.002	—
LSAM	beta52	0.16	0.246	0.086 ± 0.009	0.539 ± 0.054	0.052 ± 0.003	0.229 ± 0.008	0.060 ± 0.004	0.245 ± 0.009	2.336 ± 0.157	1.528 ± 0.057	0.236	0.076	47.74	0.2043	0.2181	1.179	98.594 ± 0.442	1.058 ± 0.033	—
LSAM	beta53	0.16	0.120	-0.040 ± 0.003	-0.251 ± 0.021	0.008 ± 0.000	0.088 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.363 ± 0.022	0.602 ± 0.020	0.121	-0.039	-24.62	0.0852	0.0938	1.139	97.187 ± 0.620	0.392 ± 0.005	—
LSAM	beta54	0.16	0.236	0.076 ± 0.008	0.477 ± 0.051	0.047 ± 0.003	0.218 ± 0.007	0.053 ± 0.003	0.230 ± 0.008	2.074 ± 0.133	1.440 ± 0.052	0.226	0.066	41.53	0.2043	0.2148	1.143	98.312 ± 0.483	0.975 ± 0.031	—
LSAM	beta55	0.00	0.041	0.041 ± 0.005	—	0.015 ± 0.001	0.121 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	—	—	0.040	0.040	—	0.1105	0.1176	1.046	97.609 ± 0.573	0.494 ± 0.012	2.391 ± 0.573
LSAM	beta56	0.00	0.014	0.014 ± 0.003	—	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	—	—	0.009	0.009	—	0.0779	0.0785	1.067	98.594 ± 0.442	0.368 ± 0.006	1.406 ± 0.442
LSAM	beta57	0.00	-0.046	-0.046 ± 0.004	—	0.010 ± 0.001	0.098 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	—	—	-0.037	-0.037	—	0.0772	0.0856	1.144	97.046 ± 0.635	0.441 ± 0.016	2.954 ± 0.635
LSAM	beta58	0.00	-0.035	-0.035 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	—	—	-0.027	-0.027	—	0.0641	0.0697	1.153	98.734 ± 0.419	0.347 ± 0.014	1.266 ± 0.419
QML	beta41	0.20	0.182	-0.018 ± 0.002	-0.092 ± 0.011	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.005	0.306 ± 0.010	0.183	-0.017	-8.69	0.0615	0.0639	0.992	94.515 ± 0.854	0.227 ± 0.001	—
QML	beta42	0.20	0.179	-0.021 ± 0.002	-0.103 ± 0.011	0.003 ± 0.000	0.056 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.005	0.299 ± 0.009	0.184	-0.016	-7.89	0.0618	0.0638	1.058	95.921 ± 0.742	0.232 ± 0.001	—
QML	beta43	0.20	0.203	0.003 ± 0.002	0.014 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.059 ± 0.003	0.243 ± 0.008	0.199	-0.001	-0.27	0.0509	0.0509	1.002	95.921 ± 0.742	0.191 ± 0.001	—
QML	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.003	-0.003	—	0.0660	0.0661	1.052	96.062 ± 0.729	0.264 ± 0.003	3.938 ± 0.729
QML	beta45	0.00	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	0.011	0.011	—	0.0534	0.0546	0.985	95.499 ± 0.778	0.222 ± 0.002	4.501 ± 0.778
QML	beta46	0.00	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.043 ± 0.002	—	—	0.017	0.017	—	0.0359	0.0396	0.975	94.374 ± 0.864	0.147 ± 0.002	5.626 ± 0.864
QML	beta47	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.017	0.017	—	0.0327	0.0369	1.009	95.218 ± 0.800	0.140 ± 0.002	4.782 ± 0.800
QML	beta51	0.16	0.172	0.012 ± 0.003	0.076 ± 0.017	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.214 ± 0.013	0.463 ± 0.016	0.168	0.008	5.23	0.0685	0.0690	1.021	95.077 ± 0.811	0.293 ± 0.002	—
QML	beta52	0.16	0.143	-0.017 ± 0.002	-0.104 ± 0.013	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.007	0.362 ± 0.012	0.148	-0.012	-7.39	0.0589	0.0600	1.098	95.781 ± 0.754	0.239 ± 0.001	—
QML	beta53	0.16	0.151	-0.009 ± 0.002	-0.054 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.123 ± 0.006	0.351 ± 0.011	0.151	-0.009	-5.71	0.0568	0.0575	1.047	95.781 ± 0.754	0.228 ± 0.001	—
QML	beta54	0.16	0.141	-0.019 ± 0.002	-0.117 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.153 ± 0.009	0.391 ± 0.013	0.146	-0.014	-8.60	0.0607	0.0622	1.013	95.077 ± 0.811	0.237 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.002	0.002	—	0.0906	0.0906	1.040	95.921 ± 0.742	0.365 ± 0.006	4.079 ± 0.742
QML	beta56	0.00	0.028	0.028 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.026	0.026	—	0.0605	0.0659	1.043	95.781 ± 0.754	0.260 ± 0.003	4.219 ± 0.754
QML	beta57	0.00	0.014	0.014 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	—	—	0.012	0.012	—	0.0416	0.0432	1.081	96.343 ± 0.704	0.184 ± 0.004	3.657 ± 0.704
QML	beta58	0.00	0.016	0.016 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.013	0.013	—	0.0305	0.0331	1.011	94.233 ± 0.874	0.125 ± 0.002	5.767 ± 0.874
UPI	beta41	0.20	0.404	0.204 ± 0.020	1.019 ± 0.098	0.275 ± 0.023	0.524 ± 0.022	0.316 ± 0.026	0.562 ± 0.025	7.895 ± 0.638	2.810 ± 0.123	0.369	0.169	84.68	0.4791	0.5081	1.207	97.609 ± 0.573	2.480 ± 0.114	—
UPI	beta42	0.20	0.317	0.117 ± 0.015	0.585 ± 0.074	0.157 ± 0.012	0.396 ± 0.015	0.170 ± 0.013	0.413 ± 0.017	4.256 ± 0.314	2.063 ± 0.084	0.293	0.093	46.37	0.3516	0.3637	1.209	96.343 ± 0.704	1.877 ± 0.077	—
UPI	beta43	0.20	0.144	-0.056 ± 0.003	-0.278 ± 0.014	0.005 ± 0.000	0.073 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.210 ± 0.012	0.458 ± 0.014	0.142	-0.058	-29.23	0.0734	0.0938	1.289	90.436 ± 1.103	0.369 ± 0.014	—
UPI	beta44	0.00	0.060	0.060 ± 0.005	—	0.018 ± 0.001	0.133 ± 0.005	0.021 ± 0.002	0.146 ± 0.006	—	—	0.045	0.045	—	0.1136	0.1224	1.455	99.578 ± 0.243	0.758 ± 0.031	0.422 ± 0.243

Table S278

Simulation 2 - Condition 134: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.020	0.020 ± 0.004	—	0.011 ± 0.001	0.106 ± 0.004	0.012 ± 0.001	0.107 ± 0.005	—	—	0.013	0.013	—	0.0891	0.0900	1.446	99.297 ± 0.313	0.599 ± 0.024	0.703 ± 0.313
UPI	beta46	0.00	-0.086	-0.086 ± 0.007	—	0.033 ± 0.002	0.182 ± 0.007	0.040 ± 0.003	0.201 ± 0.008	—	—	-0.071	-0.071	—	0.1514	0.1674	1.232	98.734 ± 0.419	0.877 ± 0.039	1.266 ± 0.419
UPI	beta47	0.00	-0.056	-0.056 ± 0.005	—	0.015 ± 0.001	0.123 ± 0.004	0.018 ± 0.001	0.135 ± 0.005	—	—	-0.048	-0.048	—	0.1065	0.1166	1.258	98.031 ± 0.521	0.605 ± 0.023	1.969 ± 0.521
UPI	beta51	0.16	0.162	0.002 ± 0.003	0.015 ± 0.020	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.283 ± 0.017	0.532 ± 0.019	0.157	-0.003	-2.18	0.0824	0.0825	1.156	96.203 ± 0.717	0.386 ± 0.006	—
UPI	beta52	0.16	0.247	0.087 ± 0.014	0.542 ± 0.087	0.139 ± 0.010	0.372 ± 0.013	0.146 ± 0.010	0.382 ± 0.015	5.698 ± 0.408	2.387 ± 0.096	0.207	0.047	29.45	0.3371	0.3404	1.328	97.750 ± 0.556	1.938 ± 0.074	—
UPI	beta53	0.16	0.110	-0.050 ± 0.004	-0.315 ± 0.022	0.009 ± 0.001	0.096 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	0.456 ± 0.028	0.676 ± 0.023	0.111	-0.049	-30.84	0.0920	0.1044	1.295	95.640 ± 0.766	0.486 ± 0.017	—
UPI	beta54	0.16	0.279	0.119 ± 0.015	0.746 ± 0.096	0.166 ± 0.011	0.408 ± 0.014	0.180 ± 0.013	0.425 ± 0.017	7.041 ± 0.492	2.654 ± 0.104	0.204	0.044	27.50	0.3495	0.3523	1.295	98.172 ± 0.502	2.071 ± 0.087	—
UPI	beta55	0.00	0.038	0.038 ± 0.006	—	0.023 ± 0.002	0.150 ± 0.005	0.024 ± 0.002	0.155 ± 0.006	—	—	0.028	0.028	—	0.1308	0.1337	1.451	99.297 ± 0.313	0.855 ± 0.030	0.703 ± 0.313
UPI	beta56	0.00	0.014	0.014 ± 0.004	—	0.010 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	—	—	0.011	0.011	—	0.0900	0.0906	1.352	99.437 ± 0.281	0.543 ± 0.026	0.563 ± 0.281
UPI	beta57	0.00	-0.042	-0.042 ± 0.005	—	0.019 ± 0.001	0.137 ± 0.005	0.020 ± 0.001	0.143 ± 0.006	—	—	-0.024	-0.024	—	0.1155	0.1179	1.355	99.297 ± 0.313	0.726 ± 0.027	0.703 ± 0.313
UPI	beta58	0.00	-0.045	-0.045 ± 0.005	—	0.017 ± 0.001	0.129 ± 0.005	0.019 ± 0.001	0.136 ± 0.006	—	—	-0.022	-0.022	—	0.1087	0.1110	1.297	99.578 ± 0.243	0.654 ± 0.028	0.422 ± 0.243

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S279

Simulation 2 - Condition 135: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.214	0.014 ± 0.004	0.068 ± 0.022	0.015 ± 0.001	0.123 ± 0.003	0.015 ± 0.001	0.124 ± 0.004	0.385 ± 0.021	0.621 ± 0.020	0.212	0.012	6.18	0.1269	0.1275	1.030	97.253 ± 0.577	0.498 ± 0.005	—
LSAM	beta42	0.20	0.221	0.021 ± 0.005	0.105 ± 0.024	0.018 ± 0.001	0.136 ± 0.003	0.019 ± 0.001	0.137 ± 0.004	0.472 ± 0.023	0.687 ± 0.021	0.220	0.020	10.18	0.1401	0.1416	0.995	95.506 ± 0.732	0.530 ± 0.005	—
LSAM	beta43	0.20	0.195	-0.005 ± 0.002	-0.023 ± 0.011	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.005	0.301 ± 0.010	0.194	-0.006	-2.87	0.0572	0.0575	1.018	95.755 ± 0.712	0.240 ± 0.002	—
LSAM	beta44	0.00	0.011	0.011 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.012	0.012	—	0.0690	0.0701	0.958	94.632 ± 0.796	0.284 ± 0.004	5.368 ± 0.796
LSAM	beta45	0.00	0.007	0.007 ± 0.003	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	0.001	0.001	—	0.0689	0.0689	0.921	95.131 ± 0.760	0.269 ± 0.004	4.869 ± 0.760
LSAM	beta46	0.00	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.007	-0.007	—	0.0487	0.0492	0.925	94.632 ± 0.796	0.187 ± 0.003	5.368 ± 0.796
LSAM	beta47	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.011	-0.011	—	0.0463	0.0476	0.924	94.382 ± 0.814	0.189 ± 0.003	5.618 ± 0.814
LSAM	beta51	0.16	0.161	0.001 ± 0.003	0.009 ± 0.019	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.304 ± 0.016	0.551 ± 0.018	0.161	0.001	0.63	0.0871	0.0871	0.929	93.883 ± 0.847	0.321 ± 0.002	—
LSAM	beta52	0.16	0.178	0.018 ± 0.005	0.110 ± 0.029	0.017 ± 0.001	0.130 ± 0.003	0.017 ± 0.001	0.131 ± 0.004	0.675 ± 0.034	0.822 ± 0.025	0.178	0.018	11.16	0.1329	0.1341	0.989	95.256 ± 0.751	0.506 ± 0.005	—
LSAM	beta53	0.16	0.150	-0.010 ± 0.003	-0.060 ± 0.019	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.306 ± 0.016	0.553 ± 0.017	0.146	-0.014	-9.01	0.0904	0.0916	0.995	94.632 ± 0.796	0.343 ± 0.003	—
LSAM	beta54	0.16	0.173	0.013 ± 0.005	0.082 ± 0.028	0.017 ± 0.001	0.129 ± 0.004	0.017 ± 0.001	0.129 ± 0.004	0.654 ± 0.037	0.809 ± 0.027	0.174	0.014	8.84	0.1271	0.1279	0.992	95.006 ± 0.770	0.501 ± 0.005	—
LSAM	beta55	0.00	0.012	0.012 ± 0.004	—	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	—	—	0.014	0.014	—	0.1020	0.1029	0.957	95.006 ± 0.770	0.406 ± 0.006	4.994 ± 0.770
LSAM	beta56	0.00	0.002	0.002 ± 0.003	—	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	—	—	0.000	0.000	—	0.0845	0.0845	0.934	95.256 ± 0.751	0.346 ± 0.005	4.744 ± 0.751
LSAM	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.015	-0.015	—	0.0545	0.0564	0.931	95.131 ± 0.760	0.214 ± 0.004	4.869 ± 0.760
LSAM	beta58	0.00	-0.007	-0.007 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.002	0.002 ± 0.000	0.049 ± 0.002	—	—	-0.005	-0.005	—	0.0413	0.0417	0.949	94.257 ± 0.822	0.179 ± 0.003	5.743 ± 0.822
QML	beta41	0.20	0.183	-0.017 ± 0.002	-0.086 ± 0.012	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.121 ± 0.006	0.348 ± 0.011	0.183	-0.017	-8.31	0.0696	0.0716	1.074	96.130 ± 0.682	0.284 ± 0.002	—
QML	beta42	0.20	0.185	-0.015 ± 0.003	-0.073 ± 0.013	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.143 ± 0.007	0.378 ± 0.011	0.185	-0.015	-7.48	0.0767	0.0781	1.001	95.006 ± 0.770	0.291 ± 0.002	—
QML	beta43	0.20	0.203	0.003 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.075 ± 0.004	0.273 ± 0.009	0.202	0.002	1.07	0.0525	0.0526	1.031	95.880 ± 0.702	0.221 ± 0.001	—
QML	beta44	0.00	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	—	0.0659	0.0659	0.996	95.506 ± 0.732	0.264 ± 0.004	4.494 ± 0.732
QML	beta45	0.00	0.005	0.005 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0624	0.0624	0.935	94.382 ± 0.814	0.237 ± 0.003	5.618 ± 0.814
QML	beta46	0.00	0.017	0.017 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.043 ± 0.002	—	—	0.012	0.012	—	0.0392	0.0410	0.931	92.634 ± 0.923	0.147 ± 0.003	7.366 ± 0.923
QML	beta47	0.00	0.014	0.014 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.010	0.010	—	0.0347	0.0361	0.927	93.009 ± 0.901	0.140 ± 0.003	6.991 ± 0.901
QML	beta51	0.16	0.163	0.003 ± 0.003	0.022 ± 0.019	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.280 ± 0.015	0.529 ± 0.017	0.163	0.003	1.76	0.0817	0.0817	0.997	96.005 ± 0.692	0.331 ± 0.002	—
QML	beta52	0.16	0.148	-0.012 ± 0.003	-0.078 ± 0.017	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.246 ± 0.014	0.496 ± 0.016	0.147	-0.013	-7.85	0.0790	0.0800	0.991	95.256 ± 0.751	0.305 ± 0.002	—
QML	beta53	0.16	0.155	-0.005 ± 0.003	-0.031 ± 0.016	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.010	0.446 ± 0.014	0.154	-0.006	-3.56	0.0717	0.0719	0.993	94.382 ± 0.814	0.277 ± 0.002	—
QML	beta54	0.16	0.145	-0.015 ± 0.003	-0.092 ± 0.017	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.239 ± 0.012	0.489 ± 0.015	0.152	-0.008	-5.29	0.0774	0.0779	1.005	94.757 ± 0.788	0.303 ± 0.002	—
QML	beta55	0.00	-0.006	-0.006 ± 0.003	—	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.003	—	—	-0.002	-0.002	—	0.0900	0.0900	1.011	95.880 ± 0.702	0.367 ± 0.007	4.120 ± 0.702
QML	beta56	0.00	0.014	0.014 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	—	—	0.008	0.008	—	0.0702	0.0707	0.936	94.132 ± 0.830	0.282 ± 0.004	5.868 ± 0.830
QML	beta57	0.00	0.010	0.010 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	0.006	0.006	—	0.0415	0.0419	0.987	93.633 ± 0.863	0.179 ± 0.005	6.367 ± 0.863
QML	beta58	0.00	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.009	0.009	—	0.0331	0.0343	0.936	94.132 ± 0.830	0.135 ± 0.002	5.868 ± 0.830
UPI	beta41	0.20	0.224	0.024 ± 0.005	0.120 ± 0.025	0.020 ± 0.001	0.140 ± 0.004	0.020 ± 0.001	0.142 ± 0.005	0.502 ± 0.032	0.709 ± 0.025	0.218	0.018	8.88	0.1362	0.1374	1.289	98.752 ± 0.392	0.706 ± 0.026	—
UPI	beta42	0.20	0.234	0.034 ± 0.005	0.168 ± 0.026	0.022 ± 0.001	0.149 ± 0.005	0.023 ± 0.002	0.153 ± 0.006	0.585 ± 0.038	0.765 ± 0.028	0.228	0.028	13.90	0.1504	0.1530	1.298	98.752 ± 0.392	0.760 ± 0.032	—
UPI	beta43	0.20	0.190	-0.010 ± 0.002	-0.051 ± 0.011	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.005	0.307 ± 0.010	0.185	-0.015	-7.38	0.0634	0.0651	1.076	95.131 ± 0.760	0.256 ± 0.004	—
UPI	beta44	0.00	0.014	0.014 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	—	—	0.014	0.014	—	0.0787	0.0798	1.251	97.004 ± 0.602	0.388 ± 0.014	2.996 ± 0.602

Table S279

Simulation 2 - Condition 135: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.007	0.007 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	—	—	0.010	0.010	—	0.0742	0.0748	1.196	97.628 ± 0.538	0.370 ± 0.022	2.372 ± 0.538
UPI	beta46	0.00	-0.014	-0.014 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.011	-0.011	—	0.0522	0.0533	1.210	97.628 ± 0.538	0.283 ± 0.014	2.372 ± 0.538
UPI	beta47	0.00	-0.018	-0.018 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.015	-0.015	—	0.0516	0.0537	1.265	97.878 ± 0.509	0.290 ± 0.016	2.122 ± 0.509
UPI	beta51	0.16	0.160	-0.000 ± 0.003	-0.001 ± 0.019	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.282 ± 0.015	0.531 ± 0.017	0.157	-0.003	-1.83	0.0813	0.0813	0.982	95.256 ± 0.751	0.327 ± 0.002	—
UPI	beta52	0.16	0.178	0.018 ± 0.005	0.115 ± 0.029	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.131 ± 0.004	0.673 ± 0.038	0.820 ± 0.027	0.179	0.019	11.74	0.1354	0.1367	1.072	95.880 ± 0.702	0.546 ± 0.008	—
UPI	beta53	0.16	0.148	-0.012 ± 0.003	-0.077 ± 0.019	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.283 ± 0.015	0.532 ± 0.017	0.143	-0.017	-10.43	0.0849	0.0865	1.043	94.632 ± 0.796	0.344 ± 0.003	—
UPI	beta54	0.16	0.181	0.021 ± 0.005	0.133 ± 0.030	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.139 ± 0.005	0.755 ± 0.042	0.869 ± 0.029	0.175	0.015	9.08	0.1348	0.1355	1.081	96.629 ± 0.638	0.582 ± 0.011	—
UPI	beta55	0.00	0.006	0.006 ± 0.004	—	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	—	—	0.004	0.004	—	0.0976	0.0977	1.114	97.129 ± 0.590	0.470 ± 0.011	2.871 ± 0.590
UPI	beta56	0.00	0.003	0.003 ± 0.003	—	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	—	—	0.004	0.004	—	0.0819	0.0821	1.031	97.129 ± 0.590	0.370 ± 0.006	2.871 ± 0.590
UPI	beta57	0.00	-0.012	-0.012 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.012	-0.012	—	0.0547	0.0560	1.034	96.754 ± 0.626	0.241 ± 0.006	3.246 ± 0.626
UPI	beta58	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.010	-0.010	—	0.0412	0.0423	1.051	96.879 ± 0.614	0.214 ± 0.006	3.121 ± 0.614

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S280

Simulation 2 - Condition 136: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.208	0.008 ± 0.002	0.041 ± 0.012	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.138 ± 0.006	0.371 ± 0.010	0.209	0.009	4.43	0.0748	0.0753	0.994	95.469 ± 0.667	0.288 ± 0.002	—
LSAM	beta42	0.20	0.219	0.019 ± 0.003	0.097 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.166 ± 0.008	0.407 ± 0.012	0.215	0.015	7.71	0.0790	0.0805	0.990	94.439 ± 0.735	0.307 ± 0.002	—
LSAM	beta43	0.20	0.195	-0.005 ± 0.001	-0.026 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.039 ± 0.002	0.196 ± 0.005	0.193	-0.007	-3.29	0.0393	0.0399	0.967	93.409 ± 0.796	0.148 ± 0.001	—
LSAM	beta44	0.00	0.007	0.007 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.006	0.006	—	0.0385	0.0390	0.963	93.924 ± 0.767	0.156 ± 0.001	6.076 ± 0.767
LSAM	beta45	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.000	0.000	—	0.0375	0.0375	0.962	95.469 ± 0.667	0.150 ± 0.001	4.531 ± 0.667
LSAM	beta46	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.007	-0.007	—	0.0242	0.0251	0.963	93.409 ± 0.796	0.098 ± 0.001	6.591 ± 0.796
LSAM	beta47	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.009	-0.009	—	0.0252	0.0267	0.956	92.997 ± 0.819	0.101 ± 0.001	7.003 ± 0.819
LSAM	beta51	0.16	0.157	-0.003 ± 0.002	-0.018 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.315 ± 0.009	0.155	-0.005	-2.83	0.0502	0.0504	1.004	95.057 ± 0.696	0.198 ± 0.001	—
LSAM	beta52	0.16	0.177	0.017 ± 0.002	0.106 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.237 ± 0.011	0.487 ± 0.013	0.179	0.019	11.66	0.0773	0.0795	0.985	95.366 ± 0.675	0.294 ± 0.002	—
LSAM	beta53	0.16	0.151	-0.009 ± 0.002	-0.055 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.154	-0.006	-3.88	0.0513	0.0516	1.026	95.057 ± 0.696	0.208 ± 0.001	—
LSAM	beta54	0.16	0.177	0.017 ± 0.002	0.107 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.237 ± 0.010	0.486 ± 0.013	0.179	0.019	12.07	0.0766	0.0790	0.987	94.439 ± 0.735	0.294 ± 0.001	—
LSAM	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.010	0.010	—	0.0566	0.0576	1.005	94.954 ± 0.702	0.228 ± 0.002	5.046 ± 0.702
LSAM	beta56	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.001	0.001	—	0.0456	0.0457	0.975	94.542 ± 0.729	0.196 ± 0.002	5.458 ± 0.729
LSAM	beta57	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.010	-0.010	—	0.0277	0.0294	0.962	94.027 ± 0.761	0.114 ± 0.001	5.973 ± 0.761
LSAM	beta58	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.008	-0.008	—	0.0244	0.0257	0.964	93.100 ± 0.813	0.098 ± 0.001	6.900 ± 0.813
QML	beta41	0.20	0.182	-0.018 ± 0.001	-0.091 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.240 ± 0.006	0.184	-0.016	-8.25	0.0471	0.0499	1.016	94.336 ± 0.742	0.177 ± 0.002	—
QML	beta42	0.20	0.188	-0.012 ± 0.001	-0.061 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.002	0.238 ± 0.006	0.186	-0.014	-7.16	0.0483	0.0504	1.019	94.645 ± 0.722	0.184 ± 0.004	—
QML	beta43	0.20	0.203	0.003 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.201	0.001	0.40	0.0371	0.0371	0.980	94.233 ± 0.748	0.139 ± 0.001	—
QML	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0388	0.0388	1.002	94.954 ± 0.702	0.162 ± 0.004	5.046 ± 0.702
QML	beta45	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0369	0.0369	0.989	95.057 ± 0.696	0.144 ± 0.001	4.943 ± 0.696
QML	beta46	0.00	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.011	0.011	—	0.0214	0.0239	0.970	93.306 ± 0.802	0.083 ± 0.001	6.694 ± 0.802
QML	beta47	0.00	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.011	0.011	—	0.0211	0.0238	1.026	93.409 ± 0.796	0.087 ± 0.005	6.591 ± 0.796
QML	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.096 ± 0.004	0.309 ± 0.009	0.159	-0.001	-0.36	0.0498	0.0498	1.033	95.675 ± 0.653	0.200 ± 0.001	—
QML	beta52	0.16	0.150	-0.010 ± 0.002	-0.065 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.308 ± 0.009	0.151	-0.009	-5.71	0.0489	0.0497	0.995	94.851 ± 0.709	0.188 ± 0.001	—
QML	beta53	0.16	0.156	-0.004 ± 0.001	-0.026 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.073 ± 0.003	0.270 ± 0.007	0.157	-0.003	-1.85	0.0438	0.0439	1.020	95.469 ± 0.667	0.172 ± 0.001	—
QML	beta54	0.16	0.150	-0.010 ± 0.002	-0.060 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.152	-0.008	-5.20	0.0494	0.0501	0.987	94.439 ± 0.735	0.189 ± 0.001	—
QML	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0548	0.0548	1.018	95.984 ± 0.630	0.220 ± 0.002	4.016 ± 0.630
QML	beta56	0.00	0.014	0.014 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.013	0.013	—	0.0432	0.0450	0.971	93.306 ± 0.802	0.173 ± 0.001	6.694 ± 0.802
QML	beta57	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.007	0.007	—	0.0252	0.0262	0.967	94.954 ± 0.702	0.105 ± 0.001	5.046 ± 0.702
QML	beta58	0.00	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.007	0.007	—	0.0199	0.0212	0.948	93.718 ± 0.779	0.081 ± 0.001	6.282 ± 0.779
UPI	beta41	0.20	0.214	0.014 ± 0.003	0.068 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.161 ± 0.008	0.401 ± 0.012	0.212	0.012	5.92	0.0777	0.0786	1.048	96.910 ± 0.555	0.325 ± 0.003	—
UPI	beta42	0.20	0.232	0.032 ± 0.003	0.160 ± 0.013	0.007 ± 0.000	0.083 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.198 ± 0.010	0.445 ± 0.013	0.226	0.026	12.92	0.0855	0.0893	1.085	95.984 ± 0.630	0.354 ± 0.004	—
UPI	beta43	0.20	0.190	-0.010 ± 0.001	-0.052 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.039 ± 0.002	0.197 ± 0.005	0.189	-0.011	-5.34	0.0371	0.0386	0.977	93.512 ± 0.790	0.145 ± 0.001	—
UPI	beta44	0.00	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.007	0.007	—	0.0402	0.0408	1.006	95.881 ± 0.638	0.174 ± 0.002	4.119 ± 0.638

Table S280

Simulation 2 - Condition 136: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0381	0.0381	1.013	96.292 ± 0.606	0.162 ± 0.002	3.708 ± 0.606
UPI	beta46	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.008	-0.008	—	0.0250	0.0264	0.999	95.572 ± 0.660	0.115 ± 0.002	4.428 ± 0.660
UPI	beta47	0.00	-0.015	-0.015 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.013	-0.013	—	0.0275	0.0305	1.055	94.336 ± 0.742	0.124 ± 0.002	5.664 ± 0.742
UPI	beta51	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.157	-0.003	-1.92	0.0499	0.0500	1.026	95.263 ± 0.682	0.199 ± 0.001	—
UPI	beta52	0.16	0.179	0.019 ± 0.003	0.120 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.270 ± 0.013	0.520 ± 0.015	0.180	0.020	12.67	0.0792	0.0818	0.992	95.469 ± 0.667	0.315 ± 0.002	—
UPI	beta53	0.16	0.148	-0.012 ± 0.002	-0.076 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.149	-0.011	-6.77	0.0509	0.0520	1.035	93.100 ± 0.813	0.205 ± 0.001	—
UPI	beta54	0.16	0.184	0.024 ± 0.003	0.153 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.291 ± 0.014	0.540 ± 0.015	0.185	0.025	15.40	0.0804	0.0841	1.013	95.057 ± 0.696	0.329 ± 0.003	—
UPI	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.011	0.011	—	0.0535	0.0545	1.033	96.189 ± 0.614	0.239 ± 0.002	3.811 ± 0.614
UPI	beta56	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.004	0.004	—	0.0471	0.0474	1.008	95.263 ± 0.682	0.200 ± 0.002	4.737 ± 0.682
UPI	beta57	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.011	-0.011	—	0.0301	0.0320	0.980	94.542 ± 0.729	0.125 ± 0.001	5.458 ± 0.729
UPI	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.010	-0.010	—	0.0261	0.0279	0.979	94.130 ± 0.754	0.111 ± 0.002	5.870 ± 0.754

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S281

Simulation 2 - Condition 137: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.147 ± 0.007	0.384 ± 0.011	0.198	-0.002	-1.20	0.0751	0.0751	1.008	95.335 ± 0.672	0.304 ± 0.002	—
LSAM	beta42	0.20	0.206	0.006 ± 0.003	0.028 ± 0.013	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.166 ± 0.007	0.407 ± 0.011	0.204	0.004	2.15	0.0806	0.0807	1.002	95.842 ± 0.636	0.319 ± 0.002	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.193	-0.007	-3.45	0.0470	0.0475	0.967	94.625 ± 0.718	0.183 ± 0.001	—
LSAM	beta44	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.000	0.000	—	0.0456	0.0456	0.962	94.726 ± 0.712	0.193 ± 0.002	5.274 ± 0.712
LSAM	beta45	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.003	-0.003	—	0.0463	0.0464	0.981	93.712 ± 0.773	0.186 ± 0.002	6.288 ± 0.773
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0291	0.0292	0.918	93.915 ± 0.761	0.112 ± 0.001	6.085 ± 0.761
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.003	-0.003	—	0.0281	0.0283	0.903	92.495 ± 0.839	0.112 ± 0.001	7.505 ± 0.839
LSAM	beta51	0.16	0.165	0.005 ± 0.002	0.031 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.181 ± 0.009	0.426 ± 0.012	0.161	0.001	0.83	0.0675	0.0675	0.936	94.219 ± 0.743	0.249 ± 0.001	—
LSAM	beta52	0.16	0.158	-0.002 ± 0.003	-0.014 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.263 ± 0.013	0.513 ± 0.015	0.157	-0.003	-1.96	0.0810	0.0810	1.004	95.842 ± 0.636	0.323 ± 0.002	—
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.008	0.418 ± 0.012	0.158	-0.002	-1.56	0.0642	0.0642	0.946	93.408 ± 0.790	0.248 ± 0.001	—
LSAM	beta54	0.16	0.166	0.006 ± 0.003	0.037 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.267 ± 0.013	0.516 ± 0.015	0.161	0.001	0.34	0.0797	0.0797	0.991	94.422 ± 0.731	0.320 ± 0.002	—
LSAM	beta55	0.00	0.005	0.005 ± 0.002	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.005	0.005	—	0.0754	0.0756	0.907	93.408 ± 0.790	0.275 ± 0.003	6.592 ± 0.790
LSAM	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.003	-0.003	—	0.0571	0.0572	0.901	93.205 ± 0.801	0.232 ± 0.002	6.795 ± 0.801
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.003	-0.003	—	0.0310	0.0311	0.971	94.422 ± 0.731	0.130 ± 0.001	5.578 ± 0.731
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.004	-0.004	—	0.0277	0.0279	0.871	92.089 ± 0.860	0.109 ± 0.001	7.911 ± 0.860
QML	beta41	0.20	0.187	-0.013 ± 0.002	-0.064 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.004	0.316 ± 0.008	0.188	-0.012	-6.09	0.0627	0.0639	1.024	95.335 ± 0.672	0.248 ± 0.003	—
QML	beta42	0.20	0.193	-0.007 ± 0.002	-0.036 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.004	0.321 ± 0.009	0.193	-0.007	-3.70	0.0644	0.0648	1.015	95.436 ± 0.665	0.254 ± 0.001	—
QML	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.197	-0.003	-1.42	0.0453	0.0454	0.973	94.523 ± 0.725	0.179 ± 0.001	—
QML	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	—	—	-0.004	-0.004	—	0.0446	0.0448	0.979	95.233 ± 0.679	0.191 ± 0.002	4.767 ± 0.679
QML	beta45	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.004	-0.004	—	0.0445	0.0447	0.989	93.712 ± 0.773	0.181 ± 0.002	6.288 ± 0.773
QML	beta46	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.005	0.005	—	0.0271	0.0275	0.914	93.408 ± 0.790	0.104 ± 0.002	6.592 ± 0.790
QML	beta47	0.00	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.004	0.004	—	0.0253	0.0256	0.898	93.306 ± 0.796	0.102 ± 0.001	6.694 ± 0.796
QML	beta51	0.16	0.165	0.005 ± 0.002	0.033 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.177 ± 0.008	0.420 ± 0.012	0.162	0.002	1.20	0.0662	0.0662	0.969	94.625 ± 0.718	0.255 ± 0.001	—
QML	beta52	0.16	0.149	-0.011 ± 0.002	-0.072 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.184 ± 0.009	0.429 ± 0.012	0.149	-0.011	-7.12	0.0683	0.0692	1.006	95.639 ± 0.650	0.267 ± 0.001	—
QML	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.012	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.382 ± 0.011	0.157	-0.003	-2.14	0.0591	0.0592	0.955	93.712 ± 0.773	0.229 ± 0.001	—
QML	beta54	0.16	0.155	-0.005 ± 0.002	-0.031 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.178 ± 0.008	0.422 ± 0.012	0.152	-0.008	-5.05	0.0658	0.0663	1.012	94.929 ± 0.699	0.267 ± 0.002	—
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.000	0.000	—	0.0729	0.0729	0.933	94.118 ± 0.749	0.265 ± 0.003	5.882 ± 0.749
QML	beta56	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	—	0.0537	0.0537	0.930	93.306 ± 0.796	0.218 ± 0.003	6.694 ± 0.796
QML	beta57	0.00	0.005	0.005 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.004	0.004	—	0.0302	0.0304	0.978	94.625 ± 0.718	0.123 ± 0.001	5.375 ± 0.718
QML	beta58	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0256	0.0256	0.874	92.292 ± 0.849	0.100 ± 0.001	7.708 ± 0.849
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.012	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.137 ± 0.006	0.370 ± 0.010	0.197	-0.003	-1.52	0.0730	0.0731	1.035	96.146 ± 0.613	0.301 ± 0.002	—
UPI	beta42	0.20	0.206	0.006 ± 0.003	0.029 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.158 ± 0.007	0.397 ± 0.011	0.205	0.005	2.43	0.0786	0.0787	1.025	95.639 ± 0.650	0.318 ± 0.002	—
UPI	beta43	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.196	-0.004	-2.22	0.0465	0.0467	0.963	94.219 ± 0.743	0.182 ± 0.001	—
UPI	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.002	-0.002	—	0.0484	0.0485	0.971	95.538 ± 0.658	0.200 ± 0.002	4.462 ± 0.658

Table S281

Simulation 2 - Condition 137: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.004	-0.004	—	0.0476	0.0478	0.976	94.828 ± 0.705	0.192 ± 0.002	5.172 ± 0.705
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.002	-0.002	—	0.0281	0.0281	0.924	94.422 ± 0.731	0.113 ± 0.001	5.578 ± 0.731
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.003	-0.003	—	0.0276	0.0277	0.916	94.320 ± 0.737	0.113 ± 0.001	5.680 ± 0.737
UPI	beta51	0.16	0.165	0.005 ± 0.002	0.030 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.176 ± 0.008	0.419 ± 0.012	0.163	0.003	1.68	0.0672	0.0673	0.962	94.219 ± 0.743	0.253 ± 0.001	—
UPI	beta52	0.16	0.157	-0.003 ± 0.003	-0.017 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.252 ± 0.012	0.502 ± 0.014	0.157	-0.003	-1.83	0.0798	0.0799	1.008	95.943 ± 0.628	0.317 ± 0.002	—
UPI	beta53	0.16	0.160	0.000 ± 0.002	0.002 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.411 ± 0.012	0.156	-0.004	-2.57	0.0640	0.0641	0.949	93.205 ± 0.801	0.245 ± 0.002	—
UPI	beta54	0.16	0.165	0.005 ± 0.003	0.034 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.259 ± 0.012	0.509 ± 0.014	0.162	0.002	1.01	0.0775	0.0776	0.998	94.929 ± 0.699	0.318 ± 0.002	—
UPI	beta55	0.00	0.005	0.005 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.004	0.004	—	0.0740	0.0741	0.941	94.320 ± 0.737	0.278 ± 0.003	5.680 ± 0.737
UPI	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.004	-0.004	—	0.0596	0.0598	0.914	93.712 ± 0.773	0.232 ± 0.002	6.288 ± 0.773
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.004	-0.004	—	0.0294	0.0296	0.975	94.320 ± 0.737	0.128 ± 0.001	5.680 ± 0.737
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.004	-0.004	—	0.0272	0.0274	0.882	92.394 ± 0.844	0.109 ± 0.001	7.606 ± 0.844

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S282

Simulation 2 - Condition 138: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.018 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.206	0.006	2.76	0.0486	0.0490	0.985	95.395 ± 0.663	0.186 ± 0.001	—
LSAM	beta42	0.20	0.203	0.003 ± 0.002	0.013 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.203	0.003	1.29	0.0510	0.0511	0.986	95.996 ± 0.620	0.193 ± 0.001	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.199	-0.001	-0.67	0.0294	0.0295	0.991	95.696 ± 0.642	0.116 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0292	0.0292	0.981	94.394 ± 0.728	0.114 ± 0.001	5.606 ± 0.728
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0268	0.0268	0.983	93.594 ± 0.775	0.112 ± 0.001	6.406 ± 0.775
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0165	0.0167	0.929	93.093 ± 0.802	0.064 ± 0.001	6.907 ± 0.802
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0168	0.0170	0.905	92.192 ± 0.849	0.063 ± 0.001	7.808 ± 0.849
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.267 ± 0.007	0.158	-0.002	-1.31	0.0440	0.0441	0.941	93.493 ± 0.780	0.158 ± 0.001	—
LSAM	beta52	0.16	0.164	0.004 ± 0.002	0.023 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.307 ± 0.008	0.163	0.003	2.07	0.0492	0.0493	1.034	95.596 ± 0.649	0.199 ± 0.001	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.157	-0.003	-1.89	0.0427	0.0428	0.958	93.594 ± 0.775	0.156 ± 0.001	—
LSAM	beta54	0.16	0.166	0.006 ± 0.002	0.035 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.320 ± 0.009	0.166	0.006	3.49	0.0514	0.0517	0.994	94.494 ± 0.722	0.198 ± 0.001	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.004	0.004	—	0.0441	0.0442	0.940	93.994 ± 0.752	0.163 ± 0.001	6.006 ± 0.752
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0370	0.0370	0.950	94.094 ± 0.746	0.141 ± 0.001	5.906 ± 0.746
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0190	0.0192	0.943	93.493 ± 0.780	0.076 ± 0.001	6.507 ± 0.780
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.001	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0170	0.0171	0.912	92.693 ± 0.823	0.066 ± 0.001	7.307 ± 0.823
QML	beta41	0.20	0.192	-0.008 ± 0.001	-0.038 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.201 ± 0.005	0.193	-0.007	-3.25	0.0397	0.0402	0.990	95.095 ± 0.683	0.154 ± 0.001	—
QML	beta42	0.20	0.192	-0.008 ± 0.001	-0.041 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.192	-0.008	-3.92	0.0404	0.0411	0.989	95.495 ± 0.656	0.157 ± 0.000	—
QML	beta43	0.20	0.203	0.003 ± 0.001	0.013 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.202	0.002	0.99	0.0287	0.0287	0.994	95.596 ± 0.649	0.113 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.002	-0.002	—	0.0297	0.0297	0.996	94.294 ± 0.734	0.117 ± 0.001	5.706 ± 0.734
QML	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0265	0.0266	0.995	93.794 ± 0.763	0.111 ± 0.001	6.206 ± 0.763
QML	beta46	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.005	0.005	—	0.0160	0.0166	0.937	92.593 ± 0.829	0.061 ± 0.001	7.407 ± 0.829
QML	beta47	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.004	0.004	—	0.0163	0.0168	0.889	92.092 ± 0.854	0.060 ± 0.001	7.908 ± 0.854
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.158	-0.002	-1.06	0.0442	0.0442	0.955	93.894 ± 0.758	0.159 ± 0.001	—
QML	beta52	0.16	0.154	-0.006 ± 0.001	-0.036 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.155	-0.005	-3.34	0.0407	0.0411	1.039	95.596 ± 0.649	0.166 ± 0.000	—
QML	beta53	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.006	0.157	-0.003	-1.65	0.0388	0.0389	0.965	93.894 ± 0.758	0.144 ± 0.000	—
QML	beta54	0.16	0.156	-0.004 ± 0.001	-0.026 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.157	-0.003	-2.12	0.0421	0.0422	1.004	94.595 ± 0.715	0.167 ± 0.000	—
QML	beta55	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.001	0.001	—	0.0431	0.0431	0.946	94.494 ± 0.722	0.161 ± 0.001	5.506 ± 0.722
QML	beta56	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.005	0.005	—	0.0349	0.0352	0.964	94.695 ± 0.709	0.135 ± 0.001	5.305 ± 0.709
QML	beta57	0.00	0.004	0.004 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.003	0.003	—	0.0183	0.0185	0.938	93.393 ± 0.786	0.073 ± 0.001	6.607 ± 0.786
QML	beta58	0.00	0.004	0.004 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.003	0.003	—	0.0156	0.0159	0.905	92.593 ± 0.829	0.062 ± 0.001	7.407 ± 0.829
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.017 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.056 ± 0.003	0.238 ± 0.007	0.205	0.005	2.69	0.0475	0.0478	0.984	94.695 ± 0.709	0.183 ± 0.001	—
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.014 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.204	0.004	1.87	0.0489	0.0490	0.995	95.495 ± 0.656	0.190 ± 0.001	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.198	-0.002	-0.97	0.0290	0.0291	0.994	95.395 ± 0.663	0.115 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0295	0.0295	0.975	94.094 ± 0.746	0.115 ± 0.001	5.906 ± 0.746

Table S282

Simulation 2 - Condition 138: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0273	0.0273	0.983	94.494 ± 0.722	0.113 ± 0.001	5.506 ± 0.722
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0167	0.0168	0.923	93.794 ± 0.763	0.064 ± 0.001	6.206 ± 0.763
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0162	0.0163	0.903	92.392 ± 0.839	0.063 ± 0.001	7.608 ± 0.839
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.070 ± 0.003	0.265 ± 0.007	0.158	-0.002	-1.18	0.0441	0.0442	0.953	93.994 ± 0.752	0.159 ± 0.001	—
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.018 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.162	0.002	1.50	0.0481	0.0482	1.038	95.896 ± 0.628	0.197 ± 0.001	—
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.158	-0.002	-1.50	0.0423	0.0424	0.954	93.493 ± 0.780	0.155 ± 0.001	—
UPI	beta54	0.16	0.165	0.005 ± 0.002	0.032 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.165	0.005	3.26	0.0508	0.0510	0.995	95.195 ± 0.677	0.197 ± 0.001	—
UPI	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.004	0.004	—	0.0420	0.0422	0.947	93.794 ± 0.763	0.164 ± 0.001	6.206 ± 0.763
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0349	0.0350	0.962	94.394 ± 0.728	0.142 ± 0.001	5.606 ± 0.728
UPI	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	-0.003	-0.003	—	0.0188	0.0190	0.945	93.694 ± 0.769	0.075 ± 0.001	6.306 ± 0.769
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.001	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0165	0.0166	0.910	92.492 ± 0.834	0.065 ± 0.001	7.508 ± 0.834

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S283

Simulation 2 - Condition 139: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.246	0.046 ± 0.026	0.232 ± 0.128	0.009 ± 0.002	0.096 ± 0.016	0.011 ± 0.003	0.103 ± 0.021	0.267 ± 0.078	0.516 ± 0.103	0.228	0.028	14.06	0.1262	0.1292	1.450	100.000 ± 0.000	0.544 ± 0.053	—
LSAM	beta42	0.20	0.234	0.034 ± 0.027	0.168 ± 0.135	0.010 ± 0.005	0.101 ± 0.034	0.011 ± 0.006	0.103 ± 0.036	0.264 ± 0.142	0.514 ± 0.182	0.225	0.025	12.42	0.0816	0.0853	1.330	100.000 ± 0.000	0.526 ± 0.069	—
LSAM	beta43	0.20	0.177	-0.023 ± 0.025	-0.117 ± 0.126	0.009 ± 0.003	0.094 ± 0.020	0.009 ± 0.002	0.094 ± 0.020	0.220 ± 0.061	0.469 ± 0.100	0.184	-0.016	-7.88	0.0895	0.0908	0.914	92.857 ± 6.883	0.338 ± 0.019	—
LSAM	beta44	0.00	0.036	0.036 ± 0.030	—	0.012 ± 0.004	0.112 ± 0.027	0.013 ± 0.004	0.114 ± 0.026	—	—	0.047	0.047	—	0.1024	0.1129	1.246	92.857 ± 6.883	0.545 ± 0.072	7.143 ± 6.883
LSAM	beta45	0.00	0.050	0.050 ± 0.034	—	0.016 ± 0.005	0.128 ± 0.026	0.018 ± 0.007	0.133 ± 0.034	—	—	0.026	0.026	—	0.1118	0.1148	0.941	85.714 ± 9.352	0.471 ± 0.042	14.286 ± 9.352
LSAM	beta46	0.00	-0.114	-0.114 ± 0.044	—	0.027 ± 0.007	0.165 ± 0.028	0.038 ± 0.013	0.195 ± 0.040	—	—	-0.076	-0.076	—	0.1276	0.1484	1.141	92.857 ± 6.883	0.736 ± 0.136	7.143 ± 6.883
LSAM	beta47	0.00	-0.065	-0.065 ± 0.028	—	0.011 ± 0.003	0.106 ± 0.020	0.015 ± 0.006	0.121 ± 0.030	—	—	-0.054	-0.054	—	0.0999	0.1134	1.223	100.000 ± 0.000	0.507 ± 0.089	0.000 ± 0.000
LSAM	beta51	0.16	0.081	-0.079 ± 0.043	-0.496 ± 0.266	0.025 ± 0.006	0.159 ± 0.025	0.030 ± 0.010	0.173 ± 0.037	1.168 ± 0.392	1.081 ± 0.234	0.128	-0.032	-20.20	0.1217	0.1259	0.774	92.857 ± 6.883	0.484 ± 0.032	—
LSAM	beta52	0.16	0.216	0.056 ± 0.043	0.352 ± 0.266	0.025 ± 0.009	0.159 ± 0.038	0.027 ± 0.010	0.163 ± 0.042	1.042 ± 0.406	1.021 ± 0.264	0.170	0.010	6.53	0.0861	0.0868	0.886	92.857 ± 6.883	0.552 ± 0.052	—
LSAM	beta53	0.16	0.133	-0.027 ± 0.019	-0.171 ± 0.116	0.005 ± 0.001	0.070 ± 0.011	0.005 ± 0.002	0.072 ± 0.017	0.205 ± 0.073	0.453 ± 0.106	0.148	-0.012	-7.62	0.0838	0.0847	1.692	100.000 ± 0.000	0.462 ± 0.073	—
LSAM	beta54	0.16	0.215	0.055 ± 0.034	0.342 ± 0.214	0.016 ± 0.004	0.128 ± 0.019	0.018 ± 0.005	0.135 ± 0.028	0.713 ± 0.215	0.845 ± 0.173	0.211	0.051	32.10	0.1318	0.1415	0.991	92.857 ± 6.883	0.498 ± 0.055	—
LSAM	beta55	0.00	0.049	0.049 ± 0.038	—	0.020 ± 0.006	0.141 ± 0.027	0.021 ± 0.007	0.145 ± 0.034	—	—	0.045	0.045	—	0.1419	0.1488	1.530	100.000 ± 0.000	0.849 ± 0.133	0.000 ± 0.000
LSAM	beta56	0.00	-0.024	-0.024 ± 0.039	—	0.022 ± 0.006	0.147 ± 0.026	0.021 ± 0.006	0.144 ± 0.032	—	—	0.002	0.002	—	0.1422	0.1423	1.285	100.000 ± 0.000	0.743 ± 0.123	0.000 ± 0.000
LSAM	beta57	0.00	-0.084	-0.084 ± 0.064	—	0.057 ± 0.018	0.238 ± 0.050	0.060 ± 0.021	0.244 ± 0.058	—	—	-0.057	-0.057	—	0.1779	0.1869	0.885	100.000 ± 0.000	0.826 ± 0.211	0.000 ± 0.000
LSAM	beta58	0.00	-0.051	-0.051 ± 0.026	—	0.009 ± 0.004	0.096 ± 0.027	0.011 ± 0.003	0.105 ± 0.021	—	—	-0.076	-0.076	—	0.1078	0.1320	1.441	100.000 ± 0.000	0.541 ± 0.083	0.000 ± 0.000
QML	beta41	0.20	0.216	0.016 ± 0.019	0.078 ± 0.094	0.005 ± 0.002	0.070 ± 0.018	0.005 ± 0.002	0.069 ± 0.021	0.120 ± 0.053	0.347 ± 0.104	0.207	0.007	3.32	0.0780	0.0783	1.191	100.000 ± 0.000	0.327 ± 0.016	—
QML	beta42	0.20	0.205	0.005 ± 0.015	0.027 ± 0.075	0.003 ± 0.001	0.056 ± 0.015	0.003 ± 0.001	0.054 ± 0.015	0.073 ± 0.029	0.270 ± 0.075	0.208	0.008	3.83	0.0753	0.0757	1.440	100.000 ± 0.000	0.315 ± 0.016	—
QML	beta43	0.20	0.200	0.000 ± 0.025	0.001 ± 0.125	0.009 ± 0.003	0.093 ± 0.020	0.008 ± 0.003	0.090 ± 0.022	0.202 ± 0.067	0.449 ± 0.111	0.202	0.002	0.79	0.0748	0.0749	0.821	85.714 ± 9.352	0.300 ± 0.014	—
QML	beta44	0.00	-0.016	-0.016 ± 0.030	—	0.012 ± 0.004	0.112 ± 0.022	0.012 ± 0.004	0.109 ± 0.028	—	—	-0.005	-0.005	—	0.0938	0.0939	1.032	100.000 ± 0.000	0.452 ± 0.045	0.000 ± 0.000
QML	beta45	0.00	0.036	0.036 ± 0.032	—	0.014 ± 0.005	0.120 ± 0.029	0.015 ± 0.007	0.121 ± 0.037	—	—	0.000	0.000	—	0.0957	0.0957	0.830	85.714 ± 9.352	0.391 ± 0.030	14.286 ± 9.352
QML	beta46	0.00	-0.043	-0.043 ± 0.022	—	0.007 ± 0.002	0.083 ± 0.015	0.008 ± 0.003	0.091 ± 0.019	—	—	-0.025	-0.025	—	0.0663	0.0707	1.075	78.571 ± 10.966	0.351 ± 0.050	21.429 ± 10.966
QML	beta47	0.00	-0.018	-0.018 ± 0.018	—	0.004 ± 0.001	0.066 ± 0.010	0.004 ± 0.001	0.066 ± 0.013	—	—	-0.016	-0.016	—	0.0660	0.0680	1.159	92.857 ± 6.883	0.298 ± 0.028	7.143 ± 6.883
QML	beta51	0.16	0.105	-0.055 ± 0.033	-0.345 ± 0.203	0.015 ± 0.004	0.122 ± 0.023	0.017 ± 0.006	0.130 ± 0.030	0.655 ± 0.230	0.809 ± 0.186	0.113	-0.047	-29.07	0.1159	0.1249	1.013	92.857 ± 6.883	0.483 ± 0.030	—
QML	beta52	0.16	0.198	0.038 ± 0.020	0.240 ± 0.127	0.006 ± 0.002	0.076 ± 0.020	0.007 ± 0.003	0.083 ± 0.024	0.267 ± 0.119	0.516 ± 0.150	0.167	0.007	4.55	0.0909	0.0912	1.146	92.857 ± 6.883	0.341 ± 0.022	—
QML	beta53	0.16	0.153	-0.007 ± 0.018	-0.046 ± 0.112	0.005 ± 0.001	0.067 ± 0.011	0.004 ± 0.001	0.065 ± 0.013	0.166 ± 0.042	0.407 ± 0.084	0.146	-0.014	-8.93	0.0839	0.0851	1.357	100.000 ± 0.000	0.358 ± 0.021	—
QML	beta54	0.16	0.179	0.019 ± 0.026	0.117 ± 0.162	0.009 ± 0.002	0.097 ± 0.014	0.009 ± 0.002	0.095 ± 0.020	0.353 ± 0.095	0.594 ± 0.124	0.160	0.000	0.25	0.1006	0.1006	0.836	92.857 ± 6.883	0.317 ± 0.015	—
QML	beta55	0.00	0.021	0.021 ± 0.026	—	0.010 ± 0.003	0.099 ± 0.024	0.009 ± 0.003	0.097 ± 0.022	—	—	0.019	0.019	—	0.1077	0.1094	1.522	92.857 ± 6.883	0.588 ± 0.075	7.143 ± 6.883
QML	beta56	0.00	-0.043	-0.043 ± 0.026	—	0.010 ± 0.003	0.099 ± 0.022	0.011 ± 0.003	0.104 ± 0.023	—	—	-0.016	-0.016	—	0.0816	0.0833	1.354	100.000 ± 0.000	0.523 ± 0.061	0.000 ± 0.000
QML	beta57	0.00	-0.019	-0.019 ± 0.026	—	0.010 ± 0.004	0.098 ± 0.032	0.009 ± 0.004	0.096 ± 0.027	—	—	-0.026	-0.026	—	0.0860	0.0899	0.940	100.000 ± 0.000	0.362 ± 0.067	0.000 ± 0.000
QML	beta58	0.00	-0.016	-0.016 ± 0.012	—	0.002 ± 0.000	0.045 ± 0.007	0.002 ± 0.001	0.046 ± 0.009	—	—	-0.038	-0.038	—	0.0623	0.0731	1.338	100.000 ± 0.000	0.238 ± 0.020	0.000 ± 0.000
UPI	beta41	0.20	0.113	-0.087 ± 0.074	-0.436 ± 0.371	0.077 ± 0.023	0.278 ± 0.054	0.079 ± 0.031	0.282 ± 0.072	1.982 ± 0.770	1.408 ± 0.361	0.265	0.065	32.64	0.7202	0.7232	1.237	92.857 ± 6.883	1.347 ± 0.279	—
UPI	beta42	0.20	0.223	0.023 ± 0.129	0.114 ± 0.645	0.233 ± 0.157	0.483 ± 0.275	0.217 ± 0.163	0.466 ± 0.260	5.418 ± 4.064	2.328 ± 1.302	0.265	0.065	32.57	0.3932	0.3986	0.802	78.571 ± 10.966	1.518 ± 0.431	—
UPI	beta43	0.20	0.176	-0.024 ± 0.049	-0.119 ± 0.247	0.034 ± 0.015	0.185 ± 0.058	0.032 ± 0.016	0.179 ± 0.062	0.805 ± 0.407	0.897 ± 0.311	0.159	-0.041	-20.39	0.1325	0.1386	1.107	100.000 ± 0.000	0.801 ± 0.143	—
UPI	beta44	0.00	0.012	0.012 ± 0.085	—	0.101 ± 0.031	0.318 ± 0.066	0.094 ± 0.030	0.306 ± 0.074	—	—	0.121	0.121	—	0.4240	0.4409	1.886	100.000 ± 0.000	2.347 ± 0.960	0.000 ± 0.000
UPI	beta45	0.00	-0.128	-0.128 ± 0.077	—	0.084 ± 0.038	0.290 ± 0.095	0.094 ± 0.054	0.307 ± 0.115	—	—	0.037	0.037	—	0.3107	0.3129	1.377	100.000 ± 0.000	1.564 ± 0.527	0.000 ± 0.000
UPI	beta46	0.00	-0.081	-0.081 ± 0.117	—	0.192 ± 0.084	0.438 ± 0.135	0.185 ± 0.100	0.430 ± 0.154	—	—	-0.044	-0.044	—	0.6579	0.6594	1.409	100.000 ± 0.000	2.421 ± 0.630	0.000 ± 0.000

Table S283

Simulation 2 - Condition 139: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.122	-0.122 ± 0.197	—	0.541 ± 0.282	0.736 ± 0.277	0.517 ± 0.310	0.719 ± 0.293	—	—	-0.120	-0.120	—	1.0117	1.0188	1.303	100.000 ± 0.000	3.758 ± 1.830	0.000 ± 0.000
UPI	beta51	0.16	0.287	0.127 ± 0.059	0.792 ± 0.369	0.049 ± 0.020	0.221 ± 0.065	0.061 ± 0.031	0.248 ± 0.079	2.400 ± 1.196	1.549 ± 0.497	0.279	0.119	74.55	0.1431	0.1863	1.659	100.000 ± 0.000	1.438 ± 0.525	—
UPI	beta52	0.16	0.127	-0.033 ± 0.043	-0.209 ± 0.267	0.025 ± 0.007	0.160 ± 0.030	0.025 ± 0.007	0.157 ± 0.035	0.968 ± 0.289	0.984 ± 0.217	0.119	-0.041	-25.55	0.2078	0.2118	1.679	100.000 ± 0.000	1.051 ± 0.207	—
UPI	beta53	0.16	0.005	-0.155 ± 0.080	-0.970 ± 0.497	0.089 ± 0.034	0.298 ± 0.079	0.106 ± 0.054	0.326 ± 0.104	4.153 ± 2.112	2.038 ± 0.650	0.140	-0.020	-12.42	0.2076	0.2086	1.399	71.429 ± 12.074	1.632 ± 0.643	—
UPI	beta54	0.16	0.132	-0.028 ± 0.162	-0.175 ± 1.010	0.365 ± 0.141	0.605 ± 0.158	0.340 ± 0.147	0.583 ± 0.172	13.286 ± 5.732	3.645 ± 1.072	0.164	0.004	2.58	0.3229	0.3230	1.344	100.000 ± 0.000	3.186 ± 0.892	—
UPI	beta55	0.00	0.052	0.052 ± 0.088	—	0.109 ± 0.040	0.329 ± 0.084	0.103 ± 0.045	0.322 ± 0.096	—	—	0.054	0.054	—	0.2223	0.2288	2.229	100.000 ± 0.000	2.878 ± 1.086	0.000 ± 0.000
UPI	beta56	0.00	-0.054	-0.054 ± 0.056	—	0.045 ± 0.019	0.211 ± 0.064	0.044 ± 0.023	0.210 ± 0.074	—	—	-0.005	-0.005	—	0.1484	0.1485	2.290	100.000 ± 0.000	1.894 ± 0.569	0.000 ± 0.000
UPI	beta57	0.00	-0.072	-0.072 ± 0.165	—	0.382 ± 0.142	0.618 ± 0.152	0.360 ± 0.134	0.600 ± 0.156	—	—	0.003	0.003	—	0.3775	0.3775	1.655	100.000 ± 0.000	4.011 ± 1.295	0.000 ± 0.000
UPI	beta58	0.00	-0.138	-0.138 ± 0.126	—	0.224 ± 0.071	0.473 ± 0.100	0.227 ± 0.072	0.476 ± 0.107	—	—	-0.014	-0.014	—	0.2703	0.2707	1.610	100.000 ± 0.000	2.984 ± 0.752	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S284

Simulation 2 - Condition 140: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.265	0.065 ± 0.019	0.325 ± 0.093	0.005 ± 0.001	0.069 ± 0.012	0.009 ± 0.003	0.093 ± 0.020	0.217 ± 0.080	0.466 ± 0.099	0.239	0.039	19.52	0.0985	0.1060	1.117	92.857 ± 6.883	0.303 ± 0.014	—
LSAM	beta42	0.20	0.157	-0.043 ± 0.026	-0.215 ± 0.132	0.010 ± 0.002	0.099 ± 0.015	0.011 ± 0.003	0.105 ± 0.019	0.273 ± 0.067	0.523 ± 0.095	0.186	-0.014	-6.83	0.1181	0.1189	0.872	92.857 ± 6.883	0.338 ± 0.017	—
LSAM	beta43	0.20	0.204	0.004 ± 0.021	0.021 ± 0.105	0.006 ± 0.002	0.079 ± 0.017	0.006 ± 0.002	0.076 ± 0.019	0.145 ± 0.050	0.380 ± 0.094	0.180	-0.020	-9.95	0.0969	0.0989	0.693	78.571 ± 10.966	0.214 ± 0.007	—
LSAM	beta44	0.00	0.046	0.046 ± 0.019	—	0.005 ± 0.001	0.072 ± 0.013	0.007 ± 0.003	0.083 ± 0.020	—	—	0.037	0.037	—	0.0696	0.0787	0.977	92.857 ± 6.883	0.276 ± 0.018	7.143 ± 6.883
LSAM	beta45	0.00	0.014	0.014 ± 0.016	—	0.004 ± 0.001	0.062 ± 0.013	0.004 ± 0.001	0.061 ± 0.014	—	—	0.019	0.019	—	0.0872	0.0891	0.938	100.000 ± 0.000	0.227 ± 0.012	0.000 ± 0.000
LSAM	beta46	0.00	-0.124	-0.124 ± 0.020	—	0.006 ± 0.002	0.077 ± 0.020	0.021 ± 0.006	0.144 ± 0.022	—	—	-0.081	-0.081	—	0.0735	0.1097	1.101	78.571 ± 10.966	0.330 ± 0.025	21.429 ± 10.966
LSAM	beta47	0.00	-0.012	-0.012 ± 0.020	—	0.006 ± 0.002	0.076 ± 0.015	0.006 ± 0.002	0.074 ± 0.017	—	—	-0.052	-0.052	—	0.0940	0.1075	1.093	92.857 ± 6.883	0.327 ± 0.031	7.143 ± 6.883
LSAM	beta51	0.16	0.124	-0.036 ± 0.021	-0.224 ± 0.134	0.006 ± 0.002	0.080 ± 0.016	0.007 ± 0.003	0.085 ± 0.020	0.282 ± 0.100	0.531 ± 0.124	0.115	-0.045	-28.06	0.0707	0.0837	0.909	85.714 ± 9.352	0.285 ± 0.008	—
LSAM	beta52	0.16	0.189	0.029 ± 0.019	0.182 ± 0.118	0.005 ± 0.002	0.071 ± 0.014	0.005 ± 0.002	0.074 ± 0.017	0.214 ± 0.072	0.463 ± 0.104	0.199	0.039	24.13	0.0661	0.0765	1.056	92.857 ± 6.883	0.292 ± 0.011	—
LSAM	beta53	0.16	0.156	-0.004 ± 0.010	-0.026 ± 0.063	0.001 ± 0.000	0.037 ± 0.007	0.001 ± 0.000	0.036 ± 0.008	0.052 ± 0.015	0.227 ± 0.050	0.155	-0.005	-2.98	0.0521	0.0523	1.572	100.000 ± 0.000	0.231 ± 0.006	—
LSAM	beta54	0.16	0.166	0.006 ± 0.016	0.037 ± 0.099	0.003 ± 0.001	0.059 ± 0.011	0.003 ± 0.001	0.057 ± 0.013	0.128 ± 0.040	0.357 ± 0.083	0.164	0.004	2.44	0.0676	0.0677	1.093	100.000 ± 0.000	0.253 ± 0.011	—
LSAM	beta55	0.00	0.029	0.029 ± 0.024	—	0.008 ± 0.002	0.089 ± 0.018	0.008 ± 0.003	0.090 ± 0.023	—	—	0.010	0.010	—	0.0587	0.0597	1.134	100.000 ± 0.000	0.395 ± 0.035	0.000 ± 0.000
LSAM	beta56	0.00	-0.018	-0.018 ± 0.022	—	0.007 ± 0.002	0.083 ± 0.018	0.007 ± 0.002	0.082 ± 0.019	—	—	0.002	0.002	—	0.0724	0.0724	1.023	92.857 ± 6.883	0.334 ± 0.017	7.143 ± 6.883
LSAM	beta57	0.00	-0.046	-0.046 ± 0.021	—	0.006 ± 0.002	0.077 ± 0.022	0.008 ± 0.002	0.087 ± 0.016	—	—	-0.065	-0.065	—	0.0709	0.0962	1.017	92.857 ± 6.883	0.307 ± 0.025	7.143 ± 6.883
LSAM	beta58	0.00	-0.051	-0.051 ± 0.018	—	0.004 ± 0.001	0.066 ± 0.009	0.007 ± 0.002	0.082 ± 0.014	—	—	-0.031	-0.031	—	0.0748	0.0810	0.991	85.714 ± 9.352	0.256 ± 0.023	14.286 ± 9.352
QML	beta41	0.20	0.207	0.007 ± 0.011	0.036 ± 0.054	0.002 ± 0.000	0.041 ± 0.008	0.002 ± 0.001	0.040 ± 0.009	0.039 ± 0.013	0.199 ± 0.046	0.201	0.001	0.35	0.0616	0.0616	1.119	100.000 ± 0.000	0.178 ± 0.005	—
QML	beta42	0.20	0.178	-0.022 ± 0.017	-0.112 ± 0.086	0.004 ± 0.001	0.065 ± 0.015	0.004 ± 0.001	0.066 ± 0.014	0.110 ± 0.032	0.331 ± 0.070	0.201	0.001	0.69	0.0694	0.0694	0.771	85.714 ± 9.352	0.195 ± 0.006	—
QML	beta43	0.20	0.213	0.013 ± 0.018	0.067 ± 0.089	0.004 ± 0.001	0.066 ± 0.015	0.004 ± 0.001	0.065 ± 0.015	0.107 ± 0.032	0.326 ± 0.074	0.192	-0.008	-3.97	0.0507	0.0514	0.700	85.714 ± 9.352	0.182 ± 0.004	—
QML	beta44	0.00	0.026	0.026 ± 0.014	—	0.003 ± 0.001	0.052 ± 0.012	0.003 ± 0.001	0.056 ± 0.015	—	—	0.014	0.014	—	0.0401	0.0423	1.188	92.857 ± 6.883	0.241 ± 0.017	7.143 ± 6.883
QML	beta45	0.00	0.009	0.009 ± 0.010	—	0.001 ± 0.001	0.038 ± 0.009	0.001 ± 0.000	0.037 ± 0.009	—	—	0.019	0.019	—	0.0459	0.0496	1.295	100.000 ± 0.000	0.192 ± 0.007	0.000 ± 0.000
QML	beta46	0.00	-0.054	-0.054 ± 0.010	—	0.001 ± 0.000	0.037 ± 0.009	0.004 ± 0.001	0.064 ± 0.009	—	—	-0.043	-0.043	—	0.0401	0.0585	1.062	71.429 ± 12.074	0.153 ± 0.009	28.571 ± 12.074
QML	beta47	0.00	-0.026	-0.026 ± 0.012	—	0.002 ± 0.000	0.046 ± 0.006	0.003 ± 0.001	0.051 ± 0.009	—	—	-0.037	-0.037	—	0.0590	0.0696	1.025	78.571 ± 10.966	0.184 ± 0.015	21.429 ± 10.966
QML	beta51	0.16	0.145	-0.015 ± 0.019	-0.096 ± 0.116	0.005 ± 0.001	0.070 ± 0.014	0.005 ± 0.002	0.069 ± 0.016	0.184 ± 0.062	0.430 ± 0.102	0.117	-0.043	-26.94	0.0883	0.0983	1.032	92.857 ± 6.883	0.281 ± 0.009	—
QML	beta52	0.16	0.168	0.008 ± 0.013	0.053 ± 0.080	0.002 ± 0.001	0.048 ± 0.009	0.002 ± 0.001	0.047 ± 0.011	0.086 ± 0.029	0.294 ± 0.070	0.166	0.006	4.04	0.0515	0.0519	1.098	100.000 ± 0.000	0.206 ± 0.004	—
QML	beta53	0.16	0.166	0.006 ± 0.011	0.038 ± 0.067	0.002 ± 0.000	0.040 ± 0.006	0.002 ± 0.000	0.039 ± 0.009	0.059 ± 0.017	0.244 ± 0.054	0.177	0.017	10.94	0.0644	0.0667	1.360	100.000 ± 0.000	0.213 ± 0.006	—
QML	beta54	0.16	0.145	-0.015 ± 0.012	-0.096 ± 0.073	0.002 ± 0.001	0.044 ± 0.009	0.002 ± 0.001	0.045 ± 0.010	0.079 ± 0.024	0.281 ± 0.061	0.159	-0.001	-0.51	0.0474	0.0474	1.095	100.000 ± 0.000	0.188 ± 0.004	—
QML	beta55	0.00	0.014	0.014 ± 0.016	—	0.003 ± 0.001	0.059 ± 0.010	0.003 ± 0.001	0.058 ± 0.012	—	—	0.031	0.031	—	0.0589	0.0667	1.431	100.000 ± 0.000	0.329 ± 0.013	0.000 ± 0.000
QML	beta56	0.00	-0.020	-0.020 ± 0.021	—	0.006 ± 0.002	0.079 ± 0.014	0.006 ± 0.002	0.079 ± 0.016	—	—	-0.008	-0.008	—	0.0851	0.0855	0.977	92.857 ± 6.883	0.304 ± 0.020	7.143 ± 6.883
QML	beta57	0.00	-0.022	-0.022 ± 0.011	—	0.002 ± 0.000	0.041 ± 0.007	0.002 ± 0.001	0.045 ± 0.008	—	—	-0.037	-0.037	—	0.0489	0.0616	1.121	92.857 ± 6.883	0.179 ± 0.011	7.143 ± 6.883
QML	beta58	0.00	-0.022	-0.022 ± 0.010	—	0.001 ± 0.000	0.038 ± 0.006	0.002 ± 0.000	0.042 ± 0.007	—	—	-0.014	-0.014	—	0.0450	0.0471	0.987	92.857 ± 6.883	0.147 ± 0.009	7.143 ± 6.883
UPI	beta41	0.20	0.238	0.038 ± 0.167	0.189 ± 0.836	0.391 ± 0.132	0.625 ± 0.139	0.365 ± 0.136	0.604 ± 0.157	9.115 ± 3.392	3.019 ± 0.783	0.293	0.093	46.30	0.5669	0.5745	1.058	100.000 ± 0.000	2.594 ± 0.550	—
UPI	beta42	0.20	0.884	0.684 ± 0.316	3.418 ± 1.582	1.401 ± 0.387	1.184 ± 0.222	1.768 ± 0.789	1.330 ± 0.358	44.197 ± 19.718	6.648 ± 1.791	0.312	0.112	56.15	0.5919	0.6025	0.966	78.571 ± 10.966	4.483 ± 1.806	—
UPI	beta43	0.20	-0.021	-0.221 ± 0.092	-1.107 ± 0.460	0.118 ± 0.049	0.344 ± 0.099	0.159 ± 0.076	0.399 ± 0.121	3.971 ± 1.907	1.993 ± 0.603	-0.024	-0.224	-112.00	0.2825	0.3605	1.129	92.857 ± 6.883	1.522 ± 0.484	—
UPI	beta44	0.00	-0.012	-0.012 ± 0.084	—	0.100 ± 0.045	0.316 ± 0.101	0.093 ± 0.044	0.304 ± 0.099	—	—	-0.026	-0.026	—	0.2177	0.2193	1.183	100.000 ± 0.000	1.464 ± 0.436	0.000 ± 0.000

Table S284

Simulation 2 - Condition 140: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	-0.042	-0.042 ± 0.063	—	0.055 ± 0.026	0.235 ± 0.076	0.053 ± 0.028	0.230 ± 0.082	—	—	0.011	0.011	—	0.1449	0.1453	1.241	100.000 ± 0.000	1.142 ± 0.376	0.000 ± 0.000
UPI	beta46	0.00	-0.318	-0.318 ± 0.232	—	0.753 ± 0.208	0.868 ± 0.159	0.801 ± 0.310	0.895 ± 0.224	—	—	-0.181	-0.181	—	1.3504	1.3625	1.158	100.000 ± 0.000	3.940 ± 0.977	0.000 ± 0.000
UPI	beta47	0.00	-0.520	-0.520 ± 0.424	—	2.515 ± 0.983	1.586 ± 0.423	2.606 ± 1.218	1.614 ± 0.498	—	—	-0.182	-0.182	—	1.3003	1.3130	1.328	100.000 ± 0.000	8.258 ± 3.075	0.000 ± 0.000
UPI	beta51	0.16	0.175	0.015 ± 0.055	0.092 ± 0.342	0.042 ± 0.013	0.205 ± 0.043	0.039 ± 0.013	0.198 ± 0.047	1.526 ± 0.492	1.235 ± 0.295	0.176	0.016	10.26	0.2251	0.2257	0.752	85.714 ± 9.352	0.603 ± 0.118	—
UPI	beta52	0.16	0.441	0.281 ± 0.155	1.758 ± 0.967	0.335 ± 0.108	0.579 ± 0.124	0.390 ± 0.194	0.624 ± 0.187	15.233 ± 7.560	3.903 ± 1.169	0.269	0.109	68.14	0.2895	0.3093	0.714	100.000 ± 0.000	1.620 ± 0.550	—
UPI	beta53	0.16	0.118	-0.042 ± 0.030	-0.261 ± 0.189	0.013 ± 0.001	0.113 ± 0.011	0.014 ± 0.003	0.117 ± 0.021	0.532 ± 0.126	0.730 ± 0.132	0.056	-0.104	-65.24	0.1435	0.1775	0.607	64.286 ± 12.806	0.269 ± 0.056	—
UPI	beta54	0.16	0.059	-0.101 ± 0.031	-0.631 ± 0.195	0.014 ± 0.006	0.117 ± 0.033	0.023 ± 0.008	0.151 ± 0.030	0.891 ± 0.296	0.944 ± 0.190	0.020	-0.140	-87.50	0.1670	0.2179	1.109	71.429 ± 12.074	0.507 ± 0.087	—
UPI	beta55	0.00	0.033	0.033 ± 0.069	—	0.066 ± 0.030	0.257 ± 0.083	0.062 ± 0.028	0.250 ± 0.077	—	—	0.009	0.009	—	0.1715	0.1717	0.911	92.857 ± 6.883	0.917 ± 0.262	7.143 ± 6.883
UPI	beta56	0.00	-0.075	-0.075 ± 0.032	—	0.014 ± 0.005	0.119 ± 0.032	0.019 ± 0.009	0.137 ± 0.040	—	—	-0.044	-0.044	—	0.1545	0.1606	1.286	92.857 ± 6.883	0.602 ± 0.127	7.143 ± 6.883
UPI	beta57	0.00	-0.454	-0.454 ± 0.191	—	0.510 ± 0.139	0.714 ± 0.130	0.679 ± 0.304	0.824 ± 0.218	—	—	-0.302	-0.302	—	0.6329	0.7011	1.100	100.000 ± 0.000	3.079 ± 1.031	0.000 ± 0.000
UPI	beta58	0.00	-0.052	-0.052 ± 0.085	—	0.101 ± 0.068	0.318 ± 0.178	0.097 ± 0.076	0.311 ± 0.183	—	—	0.040	0.040	—	0.2119	0.2157	1.403	100.000 ± 0.000	1.750 ± 1.188	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S285

Simulation 2 - Condition 141: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.207	0.007 ± 0.003	0.034 ± 0.013	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.005	0.296 ± 0.011	0.206	0.006	2.85	0.0612	0.0615	1.051	96.667 ± 0.772	0.243 ± 0.001	—
LSAM	beta42	0.20	0.205	0.005 ± 0.003	0.024 ± 0.014	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.003	0.106 ± 0.007	0.326 ± 0.013	0.202	0.002	0.97	0.0644	0.0644	0.993	95.185 ± 0.921	0.253 ± 0.002	—
LSAM	beta43	0.20	0.196	-0.004 ± 0.003	-0.020 ± 0.013	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.006	0.300 ± 0.012	0.195	-0.005	-2.66	0.0577	0.0580	0.969	94.444 ± 0.986	0.227 ± 0.001	—
LSAM	beta44	0.00	0.015	0.015 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	0.012	0.012	—	0.0713	0.0724	0.965	95.185 ± 0.921	0.284 ± 0.003	4.815 ± 0.921
LSAM	beta45	0.00	0.006	0.006 ± 0.003	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	—	—	0.009	0.009	—	0.0674	0.0680	0.922	93.148 ± 1.087	0.246 ± 0.002	6.852 ± 1.087
LSAM	beta46	0.00	-0.026	-0.026 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	-0.027	-0.027	—	0.0820	0.0864	0.963	93.519 ± 1.059	0.337 ± 0.004	6.481 ± 1.059
LSAM	beta47	0.00	-0.024	-0.024 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	-0.029	-0.029	—	0.0824	0.0875	0.962	93.519 ± 1.059	0.338 ± 0.005	6.481 ± 1.059
LSAM	beta51	0.16	0.159	-0.001 ± 0.004	-0.003 ± 0.024	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	0.309 ± 0.020	0.556 ± 0.022	0.157	-0.003	-1.93	0.0885	0.0885	0.917	91.852 ± 1.177	0.320 ± 0.002	—
LSAM	beta52	0.16	0.162	0.002 ± 0.003	0.011 ± 0.017	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.009	0.406 ± 0.015	0.162	0.002	1.29	0.0661	0.0662	1.021	96.667 ± 0.772	0.260 ± 0.002	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.003	-0.013 ± 0.018	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	0.180 ± 0.012	0.424 ± 0.017	0.155	-0.005	-3.05	0.0648	0.0650	0.977	94.630 ± 0.970	0.260 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.003	0.013 ± 0.018	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.003	0.175 ± 0.012	0.418 ± 0.017	0.167	0.007	4.63	0.0631	0.0636	0.977	94.074 ± 1.016	0.256 ± 0.002	—
LSAM	beta55	0.00	0.007	0.007 ± 0.004	—	0.011 ± 0.001	0.104 ± 0.004	0.011 ± 0.001	0.104 ± 0.005	—	—	0.009	0.009	—	0.0995	0.0999	0.931	96.296 ± 0.813	0.378 ± 0.005	3.704 ± 0.813
LSAM	beta56	0.00	0.004	0.004 ± 0.004	—	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	0.006	0.006	—	0.0921	0.0923	0.953	94.815 ± 0.954	0.347 ± 0.004	5.185 ± 0.954
LSAM	beta57	0.00	-0.019	-0.019 ± 0.004	—	0.009 ± 0.001	0.096 ± 0.004	0.010 ± 0.001	0.098 ± 0.004	—	—	-0.024	-0.024	—	0.0895	0.0925	0.943	93.889 ± 1.031	0.356 ± 0.005	6.111 ± 1.031
LSAM	beta58	0.00	-0.020	-0.020 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.021	-0.021	—	0.0831	0.0858	1.049	95.185 ± 0.921	0.327 ± 0.004	4.815 ± 0.921
QML	beta41	0.20	0.204	0.004 ± 0.002	0.021 ± 0.012	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.004	0.271 ± 0.010	0.204	0.004	1.81	0.0575	0.0576	1.045	96.667 ± 0.772	0.222 ± 0.001	—
QML	beta42	0.20	0.202	0.002 ± 0.003	0.011 ± 0.013	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.090 ± 0.006	0.300 ± 0.012	0.201	0.001	0.35	0.0613	0.0613	0.967	94.815 ± 0.954	0.228 ± 0.001	—
QML	beta43	0.20	0.198	-0.002 ± 0.003	-0.012 ± 0.013	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.006	0.293 ± 0.011	0.197	-0.003	-1.45	0.0570	0.0571	0.957	94.444 ± 0.986	0.220 ± 0.001	—
QML	beta44	0.00	0.003	0.003 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	—	—	0.003	0.003	—	0.0654	0.0654	0.983	94.074 ± 1.016	0.280 ± 0.004	5.926 ± 1.016
QML	beta45	0.00	0.000	0.000 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	—	—	0.002	0.002	—	0.0616	0.0617	0.932	94.444 ± 0.986	0.241 ± 0.002	5.556 ± 0.986
QML	beta46	0.00	-0.010	-0.010 ± 0.003	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.013	-0.013	—	0.0548	0.0565	1.006	92.222 ± 1.153	0.230 ± 0.003	7.778 ± 1.153
QML	beta47	0.00	-0.013	-0.013 ± 0.003	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.018	-0.018	—	0.0565	0.0594	0.951	90.926 ± 1.236	0.224 ± 0.003	9.074 ± 1.236
QML	beta51	0.16	0.162	0.002 ± 0.004	0.011 ± 0.024	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.301 ± 0.019	0.549 ± 0.021	0.160	0.000	-0.22	0.0882	0.0882	0.969	94.259 ± 1.001	0.334 ± 0.003	—
QML	beta52	0.16	0.160	-0.000 ± 0.003	-0.000 ± 0.016	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.008	0.383 ± 0.013	0.161	0.001	0.48	0.0641	0.0641	1.055	97.037 ± 0.730	0.253 ± 0.002	—
QML	beta53	0.16	0.160	0.000 ± 0.003	0.002 ± 0.018	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	0.170 ± 0.011	0.412 ± 0.017	0.161	0.001	0.77	0.0613	0.0613	0.984	94.815 ± 0.954	0.255 ± 0.001	—
QML	beta54	0.16	0.160	0.000 ± 0.003	0.002 ± 0.017	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.003	0.155 ± 0.011	0.393 ± 0.016	0.164	0.004	2.20	0.0572	0.0574	0.967	94.074 ± 1.016	0.239 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.004	—	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	—	—	0.003	0.003	—	0.0951	0.0951	0.979	96.111 ± 0.832	0.374 ± 0.005	3.889 ± 0.832
QML	beta56	0.00	0.000	-0.000 ± 0.004	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	—	—	0.000	0.000	—	0.0866	0.0866	0.987	93.519 ± 1.059	0.324 ± 0.003	6.481 ± 1.059
QML	beta57	0.00	-0.009	-0.009 ± 0.003	—	0.004 ± 0.000	0.065 ± 0.003	0.004 ± 0.000	0.066 ± 0.003	—	—	-0.014	-0.014	—	0.0627	0.0643	0.984	92.037 ± 1.165	0.252 ± 0.004	7.963 ± 1.165
QML	beta58	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.014	-0.014	—	0.0538	0.0557	1.035	92.222 ± 1.153	0.213 ± 0.003	7.778 ± 1.153
UPI	beta41	0.20	0.210	0.010 ± 0.003	0.049 ± 0.015	0.005 ± 0.000	0.070 ± 0.003	0.005 ± 0.000	0.070 ± 0.003	0.124 ± 0.010	0.352 ± 0.016	0.210	0.010	5.13	0.0719	0.0726	1.345	98.519 ± 0.520	0.368 ± 0.015	—
UPI	beta42	0.20	0.218	0.018 ± 0.003	0.090 ± 0.016	0.006 ± 0.000	0.074 ± 0.003	0.006 ± 0.000	0.076 ± 0.004	0.146 ± 0.012	0.382 ± 0.018	0.217	0.017	8.27	0.0866	0.0881	1.356	96.111 ± 0.832	0.395 ± 0.015	—
UPI	beta43	0.20	0.188	-0.012 ± 0.003	-0.061 ± 0.014	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.107 ± 0.007	0.327 ± 0.012	0.185	-0.015	-7.64	0.0742	0.0757	1.151	96.667 ± 0.772	0.290 ± 0.008	—
UPI	beta44	0.00	0.021	0.021 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	—	—	0.021	0.021	—	0.0913	0.0938	1.279	97.593 ± 0.660	0.450 ± 0.029	2.407 ± 0.660
UPI	beta45	0.00	0.004	0.004 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.075 ± 0.003	—	—	0.005	0.005	—	0.0811	0.0813	1.084	95.926 ± 0.851	0.320 ± 0.010	4.074 ± 0.851
UPI	beta46	0.00	-0.015	-0.015 ± 0.006	—	0.020 ± 0.002	0.143 ± 0.006	0.021 ± 0.002	0.143 ± 0.007	—	—	-0.014	-0.014	—	0.1248	0.1255	1.256	98.333 ± 0.551	0.702 ± 0.026	1.667 ± 0.551

Table S285

Simulation 2 - Condition 141: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.032	-0.032 ± 0.006	—	0.023 ± 0.002	0.150 ± 0.006	0.024 ± 0.002	0.153 ± 0.007	—	—	-0.037	-0.037	—	0.1437	0.1483	1.461	97.778 ± 0.634	0.860 ± 0.077	2.222 ± 0.634
UPI	beta51	0.16	0.159	-0.001 ± 0.004	-0.004 ± 0.023	0.007 ± 0.000	0.086 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	0.287 ± 0.018	0.535 ± 0.021	0.155	-0.005	-2.88	0.0874	0.0876	1.024	95.370 ± 0.904	0.344 ± 0.005	—
UPI	beta52	0.16	0.164	0.004 ± 0.003	0.024 ± 0.018	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.184 ± 0.012	0.429 ± 0.017	0.166	0.006	3.90	0.0733	0.0736	1.206	98.148 ± 0.580	0.324 ± 0.009	—
UPI	beta53	0.16	0.155	-0.005 ± 0.003	-0.030 ± 0.019	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.188 ± 0.012	0.434 ± 0.017	0.151	-0.009	-5.34	0.0674	0.0679	1.095	95.741 ± 0.869	0.297 ± 0.005	—
UPI	beta54	0.16	0.166	0.006 ± 0.003	0.037 ± 0.020	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	0.224 ± 0.016	0.474 ± 0.020	0.172	0.012	7.79	0.0720	0.0731	1.203	95.741 ± 0.869	0.356 ± 0.015	—
UPI	beta55	0.00	0.009	0.009 ± 0.004	—	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	—	—	0.012	0.012	—	0.1029	0.1036	1.137	97.778 ± 0.634	0.459 ± 0.018	2.222 ± 0.634
UPI	beta56	0.00	0.002	0.002 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	—	—	0.006	0.006	—	0.0982	0.0985	1.100	97.222 ± 0.707	0.406 ± 0.009	2.778 ± 0.707
UPI	beta57	0.00	-0.008	-0.008 ± 0.005	—	0.015 ± 0.001	0.123 ± 0.005	0.015 ± 0.001	0.123 ± 0.006	—	—	-0.018	-0.018	—	0.1166	0.1180	1.205	97.963 ± 0.608	0.582 ± 0.026	2.037 ± 0.608
UPI	beta58	0.00	-0.022	-0.022 ± 0.006	—	0.018 ± 0.002	0.134 ± 0.006	0.019 ± 0.002	0.136 ± 0.006	—	—	-0.024	-0.024	—	0.1080	0.1107	1.406	99.444 ± 0.320	0.741 ± 0.045	0.556 ± 0.320

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S286

Simulation 2 - Condition 142: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.002	0.015 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.007	0.202	0.002	0.79	0.0375	0.0375	0.980	94.462 ± 0.910	0.148 ± 0.000	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.019 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.195 ± 0.007	0.202	0.002	1.15	0.0387	0.0387	1.013	95.411 ± 0.832	0.154 ± 0.001	—
LSAM	beta43	0.20	0.194	-0.006 ± 0.001	-0.032 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.186 ± 0.006	0.194	-0.006	-3.24	0.0373	0.0378	0.976	94.937 ± 0.872	0.140 ± 0.000	—
LSAM	beta44	0.00	0.015	0.015 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.016	0.016	—	0.0414	0.0444	0.993	94.304 ± 0.922	0.167 ± 0.001	5.696 ± 0.922
LSAM	beta45	0.00	0.006	0.006 ± 0.002	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.006	0.006	—	0.0377	0.0382	0.983	94.304 ± 0.922	0.149 ± 0.001	5.696 ± 0.922
LSAM	beta46	0.00	-0.030	-0.030 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.029	-0.029	—	0.0481	0.0562	0.985	89.715 ± 1.208	0.200 ± 0.002	10.285 ± 1.208
LSAM	beta47	0.00	-0.028	-0.028 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.027	-0.027	—	0.0484	0.0556	1.035	90.032 ± 1.192	0.199 ± 0.002	9.968 ± 1.192
LSAM	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.013	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.099 ± 0.006	0.315 ± 0.011	0.158	-0.002	-1.09	0.0487	0.0487	1.005	94.146 ± 0.934	0.199 ± 0.001	—
LSAM	beta52	0.16	0.166	0.006 ± 0.002	0.037 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.004	0.251 ± 0.009	0.165	0.005	3.22	0.0394	0.0397	1.018	95.886 ± 0.790	0.159 ± 0.001	—
LSAM	beta53	0.16	0.155	-0.005 ± 0.002	-0.031 ± 0.010	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.067 ± 0.004	0.259 ± 0.009	0.153	-0.007	-4.38	0.0407	0.0413	0.989	95.095 ± 0.859	0.160 ± 0.001	—
LSAM	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.004	0.244 ± 0.009	0.161	0.001	0.51	0.0366	0.0367	1.024	95.411 ± 0.832	0.157 ± 0.000	—
LSAM	beta55	0.00	0.014	0.014 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.014	0.014	—	0.0588	0.0605	0.959	92.405 ± 1.054	0.222 ± 0.001	7.595 ± 1.054
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.001	-0.001	—	0.0524	0.0524	1.004	95.570 ± 0.819	0.208 ± 0.001	4.430 ± 0.819
LSAM	beta57	0.00	-0.020	-0.020 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.023	-0.023	—	0.0528	0.0577	0.967	92.247 ± 1.064	0.213 ± 0.002	7.753 ± 1.064
LSAM	beta58	0.00	-0.019	-0.019 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.023	-0.023	—	0.0487	0.0537	0.989	93.196 ± 1.002	0.197 ± 0.001	6.804 ± 1.002
QML	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.002	0.176 ± 0.006	0.198	-0.002	-0.77	0.0350	0.0350	0.993	94.937 ± 0.872	0.137 ± 0.001	—
QML	beta42	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.002	0.176 ± 0.006	0.200	0.000	0.18	0.0362	0.0362	1.020	95.886 ± 0.790	0.141 ± 0.001	—
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.020 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.002	0.180 ± 0.006	0.196	-0.004	-1.96	0.0356	0.0358	0.978	94.620 ± 0.897	0.137 ± 0.001	—
QML	beta44	0.00	0.007	0.007 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.010	0.010	—	0.0405	0.0417	1.016	94.304 ± 0.922	0.168 ± 0.002	5.696 ± 0.922
QML	beta45	0.00	0.001	0.001 ± 0.002	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0367	0.0368	0.990	95.411 ± 0.832	0.146 ± 0.001	4.589 ± 0.832
QML	beta46	0.00	-0.018	-0.018 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.018	-0.018	—	0.0329	0.0377	0.995	88.924 ± 1.248	0.137 ± 0.001	11.076 ± 1.248
QML	beta47	0.00	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.020	-0.020	—	0.0340	0.0394	1.062	88.449 ± 1.271	0.139 ± 0.005	11.551 ± 1.271
QML	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.097 ± 0.005	0.311 ± 0.011	0.159	-0.001	-0.38	0.0470	0.0471	1.033	95.253 ± 0.846	0.202 ± 0.001	—
QML	beta52	0.16	0.164	0.004 ± 0.002	0.028 ± 0.010	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.244 ± 0.009	0.164	0.004	2.77	0.0379	0.0382	1.001	95.253 ± 0.846	0.152 ± 0.001	—
QML	beta53	0.16	0.157	-0.003 ± 0.002	-0.018 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.004	0.253 ± 0.009	0.155	-0.005	-3.12	0.0397	0.0400	0.986	94.462 ± 0.910	0.156 ± 0.001	—
QML	beta54	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.009	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.003	0.233 ± 0.009	0.158	-0.002	-1.04	0.0353	0.0353	1.023	95.728 ± 0.804	0.149 ± 0.001	—
QML	beta55	0.00	0.013	0.013 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	0.014	0.014	—	0.0574	0.0589	0.979	93.196 ± 1.002	0.222 ± 0.001	6.804 ± 1.002
QML	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	—	—	-0.006	-0.006	—	0.0494	0.0498	1.021	95.253 ± 0.846	0.199 ± 0.001	4.747 ± 0.846
QML	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.015	-0.015	—	0.0386	0.0414	0.983	90.032 ± 1.192	0.152 ± 0.002	9.968 ± 1.192
QML	beta58	0.00	-0.012	-0.012 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.015	-0.015	—	0.0331	0.0364	1.014	91.772 ± 1.093	0.135 ± 0.003	8.228 ± 1.093
UPI	beta41	0.20	0.207	0.007 ± 0.002	0.034 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.054 ± 0.003	0.233 ± 0.009	0.206	0.006	3.11	0.0506	0.0510	1.261	97.627 ± 0.605	0.228 ± 0.008	—
UPI	beta42	0.20	0.214	0.014 ± 0.002	0.070 ± 0.010	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	0.069 ± 0.005	0.263 ± 0.011	0.212	0.012	5.79	0.0521	0.0534	1.270	97.627 ± 0.605	0.253 ± 0.009	—
UPI	beta43	0.20	0.186	-0.014 ± 0.002	-0.070 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.210 ± 0.007	0.183	-0.017	-8.46	0.0445	0.0476	1.109	94.937 ± 0.872	0.172 ± 0.003	—
UPI	beta44	0.00	0.020	0.020 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.024	0.024	—	0.0497	0.0552	1.134	95.411 ± 0.832	0.212 ± 0.003	4.589 ± 0.832

Table S286

Simulation 2 - Condition 142: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.008	0.008 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.002	—	—	0.006	0.006	—	0.0403	0.0408	1.062	95.253 ± 0.846	0.170 ± 0.002	4.747 ± 0.846
UPI	beta46	0.00	-0.033	-0.033 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	—	—	-0.035	-0.035	—	0.0786	0.0859	1.194	95.570 ± 0.819	0.415 ± 0.016	4.430 ± 0.819
UPI	beta47	0.00	-0.042	-0.042 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	—	—	-0.042	-0.042	—	0.0823	0.0924	1.239	96.994 ± 0.679	0.435 ± 0.015	3.006 ± 0.679
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.012	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.097 ± 0.005	0.312 ± 0.011	0.154	-0.006	-3.47	0.0510	0.0513	1.058	95.570 ± 0.819	0.207 ± 0.001	—
UPI	beta52	0.16	0.169	0.009 ± 0.002	0.056 ± 0.011	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	0.078 ± 0.005	0.280 ± 0.010	0.168	0.008	4.99	0.0449	0.0456	1.183	97.310 ± 0.644	0.203 ± 0.005	—
UPI	beta53	0.16	0.150	-0.010 ± 0.002	-0.063 ± 0.011	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.004	0.274 ± 0.009	0.145	-0.015	-9.22	0.0465	0.0488	1.075	95.411 ± 0.832	0.180 ± 0.002	—
UPI	beta54	0.16	0.168	0.008 ± 0.002	0.049 ± 0.012	0.002 ± 0.000	0.047 ± 0.002	0.002 ± 0.000	0.048 ± 0.002	0.089 ± 0.006	0.299 ± 0.012	0.167	0.007	4.20	0.0489	0.0494	1.238	96.519 ± 0.729	0.229 ± 0.007	—
UPI	beta55	0.00	0.016	0.016 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.016	0.016	—	0.0634	0.0655	1.007	93.671 ± 0.969	0.244 ± 0.003	6.329 ± 0.969
UPI	beta56	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	0.002	0.002	—	0.0548	0.0548	1.059	96.994 ± 0.679	0.223 ± 0.002	3.006 ± 0.679
UPI	beta57	0.00	-0.020	-0.020 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	—	—	-0.021	-0.021	—	0.0729	0.0760	1.117	96.519 ± 0.729	0.335 ± 0.009	3.481 ± 0.729
UPI	beta58	0.00	-0.030	-0.030 ± 0.003	—	0.007 ± 0.001	0.085 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	-0.031	-0.031	—	0.0770	0.0832	1.176	97.152 ± 0.662	0.393 ± 0.013	2.848 ± 0.662

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S287

Simulation 2 - Condition 143: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.002	0.233 ± 0.007	0.199	-0.001	-0.38	0.0475	0.0475	0.992	95.696 ± 0.642	0.181 ± 0.001	—
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.201	0.001	0.28	0.0489	0.0489	0.978	94.595 ± 0.715	0.186 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.201	0.001	0.55	0.0475	0.0475	0.964	93.694 ± 0.769	0.177 ± 0.001	—
LSAM	beta44	0.00	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	—	—	0.007	0.007	—	0.0549	0.0553	0.922	92.793 ± 0.818	0.199 ± 0.001	7.207 ± 0.818
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0417	0.0417	1.009	94.895 ± 0.696	0.181 ± 0.001	5.105 ± 0.696
LSAM	beta46	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.012	-0.012	—	0.0555	0.0567	0.996	93.293 ± 0.791	0.229 ± 0.001	6.707 ± 0.791
LSAM	beta47	0.00	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.011	-0.011	—	0.0572	0.0583	0.983	93.393 ± 0.786	0.227 ± 0.001	6.607 ± 0.786
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.192 ± 0.009	0.438 ± 0.012	0.156	-0.004	-2.44	0.0702	0.0703	0.903	93.393 ± 0.786	0.248 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.327 ± 0.009	0.162	0.002	1.28	0.0531	0.0531	0.976	93.994 ± 0.752	0.200 ± 0.001	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.159	-0.001	-0.33	0.0544	0.0544	0.957	94.094 ± 0.746	0.204 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.018 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.165	0.005	2.91	0.0512	0.0514	0.972	94.294 ± 0.734	0.200 ± 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	—	—	0.001	0.001	—	0.0677	0.0677	0.989	95.095 ± 0.683	0.261 ± 0.002	4.905 ± 0.683
LSAM	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.003	-0.003	—	0.0623	0.0624	0.988	94.294 ± 0.734	0.249 ± 0.002	5.706 ± 0.734
LSAM	beta57	0.00	-0.006	-0.006 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.005	-0.005	—	0.0633	0.0635	0.992	94.294 ± 0.734	0.249 ± 0.001	5.706 ± 0.734
LSAM	beta58	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.004	-0.004	—	0.0636	0.0637	0.949	93.894 ± 0.758	0.232 ± 0.001	6.106 ± 0.758
QML	beta41	0.20	0.200	0.000 ± 0.001	0.001 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.200	0.000	0.00	0.0463	0.0463	0.995	95.596 ± 0.649	0.178 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.200	0.000	0.15	0.0482	0.0482	0.985	95.195 ± 0.677	0.182 ± 0.001	—
QML	beta43	0.20	0.202	0.002 ± 0.001	0.009 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.201	0.001	0.53	0.0472	0.0472	0.967	93.694 ± 0.769	0.176 ± 0.001	—
QML	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	—	—	0.003	0.003	—	0.0544	0.0545	0.941	93.694 ± 0.769	0.201 ± 0.001	6.306 ± 0.769
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.003	-0.003	—	0.0419	0.0420	1.014	95.495 ± 0.656	0.181 ± 0.001	4.505 ± 0.656
QML	beta46	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.009	-0.009	—	0.0459	0.0467	1.015	93.894 ± 0.758	0.188 ± 0.001	6.106 ± 0.758
QML	beta47	0.00	-0.006	-0.006 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.009	-0.009	—	0.0466	0.0474	0.989	93.794 ± 0.763	0.186 ± 0.001	6.206 ± 0.763
QML	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.156	-0.004	-2.45	0.0718	0.0719	0.939	93.994 ± 0.752	0.255 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.162	0.002	1.36	0.0519	0.0519	0.986	94.695 ± 0.709	0.200 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.114 ± 0.005	0.337 ± 0.009	0.159	-0.001	-0.91	0.0532	0.0532	0.965	94.494 ± 0.722	0.204 ± 0.001	—
QML	beta54	0.16	0.163	0.003 ± 0.002	0.016 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.165	0.005	3.05	0.0518	0.0520	0.970	94.695 ± 0.709	0.198 ± 0.001	—
QML	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.002	-0.002	—	0.0651	0.0652	1.009	95.796 ± 0.635	0.263 ± 0.002	4.204 ± 0.635
QML	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.004	-0.004	—	0.0605	0.0606	1.005	95.295 ± 0.670	0.245 ± 0.002	4.705 ± 0.670
QML	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.005	-0.005	—	0.0520	0.0522	1.010	94.094 ± 0.746	0.208 ± 0.002	5.906 ± 0.746
QML	beta58	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.004	-0.004	—	0.0513	0.0514	0.958	93.794 ± 0.763	0.190 ± 0.001	6.206 ± 0.763
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.199	-0.001	-0.57	0.0473	0.0473	0.985	95.295 ± 0.670	0.180 ± 0.001	—
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.240 ± 0.006	0.201	0.001	0.45	0.0478	0.0479	0.979	94.895 ± 0.696	0.185 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.010 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.201	0.001	0.53	0.0470	0.0470	0.968	94.294 ± 0.734	0.179 ± 0.001	—
UPI	beta44	0.00	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	—	—	0.004	0.004	—	0.0573	0.0574	0.934	93.594 ± 0.775	0.205 ± 0.001	6.406 ± 0.775
UPI	beta45	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.002	-0.002	—	0.0456	0.0456	1.003	94.795 ± 0.703	0.185 ± 0.001	5.205 ± 0.703
UPI	beta46	0.00	-0.007	-0.007 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.008	-0.008	—	0.0609	0.0614	1.026	94.895 ± 0.696	0.255 ± 0.002	5.105 ± 0.696

Table S287

Simulation 2 - Condition 143: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.007	-0.007 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.007	-0.007	—	0.0633	0.0637	1.015	94.995 ± 0.690	0.253 ± 0.002	5.005 ± 0.690
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.186 ± 0.009	0.431 ± 0.012	0.156	-0.004	-2.64	0.0705	0.0706	0.932	93.594 ± 0.775	0.252 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.161	0.001	0.81	0.0525	0.0525	0.972	93.694 ± 0.769	0.199 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.159	-0.001	-0.41	0.0549	0.0549	0.959	94.795 ± 0.703	0.205 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.020 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.165	0.005	3.29	0.0525	0.0527	0.968	94.394 ± 0.728	0.199 ± 0.001	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	—	—	0.001	0.001	—	0.0664	0.0664	1.007	95.696 ± 0.642	0.265 ± 0.002	4.304 ± 0.642
UPI	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.003	-0.003	—	0.0609	0.0610	1.013	94.995 ± 0.690	0.253 ± 0.002	5.005 ± 0.690
UPI	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.004	-0.004	—	0.0656	0.0657	1.002	95.596 ± 0.649	0.260 ± 0.002	4.404 ± 0.649
UPI	beta58	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0642	0.0642	0.973	95.395 ± 0.663	0.250 ± 0.002	4.605 ± 0.663

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S288

Simulation 2 - Condition 144: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	0.10	0.0301	0.0301	0.989	95.200 ± 0.676	0.113 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.200	0.000	-0.06	0.0311	0.0311	1.000	95.000 ± 0.689	0.117 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.199	-0.001	-0.37	0.0293	0.0293	1.020	95.200 ± 0.676	0.112 ± 0.000	—
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.002	0.002	—	0.0314	0.0315	1.023	95.000 ± 0.689	0.126 ± 0.000	5.000 ± 0.689
LSAM	beta45	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0286	0.0286	1.016	95.500 ± 0.656	0.114 ± 0.000	4.500 ± 0.656
LSAM	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.008	-0.008	—	0.0375	0.0384	0.978	92.700 ± 0.823	0.143 ± 0.000	7.300 ± 0.823
LSAM	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.006	-0.006	—	0.0366	0.0371	0.974	94.300 ± 0.733	0.143 ± 0.000	5.700 ± 0.733
LSAM	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.160	0.000	-0.10	0.0412	0.0412	0.988	94.700 ± 0.708	0.157 ± 0.001	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.006	0.160	0.000	-0.17	0.0313	0.0313	1.000	95.100 ± 0.683	0.125 ± 0.000	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.005	0.159	-0.001	-0.59	0.0330	0.0330	1.028	95.400 ± 0.662	0.129 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.201 ± 0.005	0.159	-0.001	-0.54	0.0336	0.0337	0.989	95.700 ± 0.641	0.125 ± 0.000	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.004	0.004	—	0.0442	0.0443	0.956	93.800 ± 0.763	0.162 ± 0.001	6.200 ± 0.763
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	—	0.0374	0.0375	0.990	95.300 ± 0.669	0.154 ± 0.001	4.700 ± 0.669
LSAM	beta57	0.00	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.008	-0.008	—	0.0414	0.0421	0.992	94.400 ± 0.727	0.154 ± 0.001	5.600 ± 0.727
LSAM	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0364	0.0366	1.007	94.100 ± 0.745	0.145 ± 0.000	5.900 ± 0.745
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.200	0.000	-0.11	0.0296	0.0296	0.989	95.100 ± 0.683	0.112 ± 0.000	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	-0.08	0.0303	0.0303	0.998	95.300 ± 0.669	0.115 ± 0.000	—
QML	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.200	0.000	-0.13	0.0294	0.0294	1.021	95.600 ± 0.649	0.111 ± 0.000	—
QML	beta44	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	—	0.0313	0.0313	1.027	95.100 ± 0.683	0.127 ± 0.001	4.900 ± 0.683
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0284	0.0284	1.019	95.300 ± 0.669	0.114 ± 0.000	4.700 ± 0.669
QML	beta46	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.007	-0.007	—	0.0311	0.0317	0.991	92.000 ± 0.858	0.118 ± 0.001	8.000 ± 0.858
QML	beta47	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.006	-0.006	—	0.0288	0.0294	0.977	93.500 ± 0.780	0.117 ± 0.001	6.500 ± 0.780
QML	beta51	0.16	0.160	0.000 ± 0.001	0.003 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.160	0.000	-0.12	0.0407	0.0407	1.001	95.200 ± 0.676	0.159 ± 0.001	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.160	0.000	-0.29	0.0302	0.0302	1.004	95.000 ± 0.689	0.124 ± 0.000	—
QML	beta53	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.005	0.159	-0.001	-0.40	0.0332	0.0332	1.025	95.400 ± 0.662	0.128 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.159	-0.001	-0.60	0.0328	0.0328	0.992	95.600 ± 0.649	0.124 ± 0.000	—
QML	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.003	0.003	—	0.0434	0.0435	0.963	93.700 ± 0.768	0.163 ± 0.001	6.300 ± 0.768
QML	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0371	0.0371	0.991	95.500 ± 0.656	0.152 ± 0.001	4.500 ± 0.656
QML	beta57	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.008	-0.008	—	0.0331	0.0340	1.011	94.100 ± 0.745	0.128 ± 0.001	5.900 ± 0.745
QML	beta58	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.004	-0.004	—	0.0305	0.0307	1.019	94.100 ± 0.745	0.119 ± 0.001	5.900 ± 0.745
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.200	0.000	-0.08	0.0303	0.0303	0.987	95.100 ± 0.683	0.113 ± 0.000	—
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.199	-0.001	-0.48	0.0298	0.0298	0.998	94.600 ± 0.715	0.115 ± 0.000	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.200	0.000	-0.09	0.0291	0.0291	1.021	95.300 ± 0.669	0.112 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.000	0.000	—	0.0317	0.0317	1.019	95.000 ± 0.689	0.127 ± 0.000	5.000 ± 0.689

Table S288

Simulation 2 - Condition 144: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0294	0.0294	1.009	95.600 ± 0.649	0.115 ± 0.000	4.400 ± 0.649
UPI	beta46	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.005	-0.005	—	0.0385	0.0388	0.992	94.500 ± 0.721	0.150 ± 0.001	5.500 ± 0.721
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.003	-0.003	—	0.0378	0.0379	0.994	94.800 ± 0.702	0.151 ± 0.001	5.200 ± 0.702
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.252 ± 0.007	0.160	0.000	-0.22	0.0408	0.0408	1.000	95.100 ± 0.683	0.158 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.159	-0.001	-0.36	0.0312	0.0312	1.000	94.900 ± 0.696	0.124 ± 0.000	—
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.200 ± 0.005	0.159	-0.001	-0.34	0.0332	0.0332	1.028	95.400 ± 0.662	0.129 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.201 ± 0.005	0.159	-0.001	-0.55	0.0335	0.0335	0.988	95.300 ± 0.669	0.125 ± 0.000	—
UPI	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.002	0.002	—	0.0445	0.0446	0.952	93.900 ± 0.757	0.163 ± 0.001	6.100 ± 0.757
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	—	0.0371	0.0371	0.997	95.300 ± 0.669	0.155 ± 0.001	4.700 ± 0.669
UPI	beta57	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.005	-0.005	—	0.0414	0.0417	0.993	95.200 ± 0.676	0.157 ± 0.001	4.800 ± 0.676
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0375	0.0376	1.008	95.000 ± 0.689	0.150 ± 0.001	5.000 ± 0.689

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Additional Distributional Conditions: Differences from Baseline

Figure S1

Additional Distributional Conditions Difference: Relative Bias for Simulation 1 (Epsilon)

Figure S2

Additional Distributional Conditions Difference: Relative Bias for Simulation 1 (Zeta)

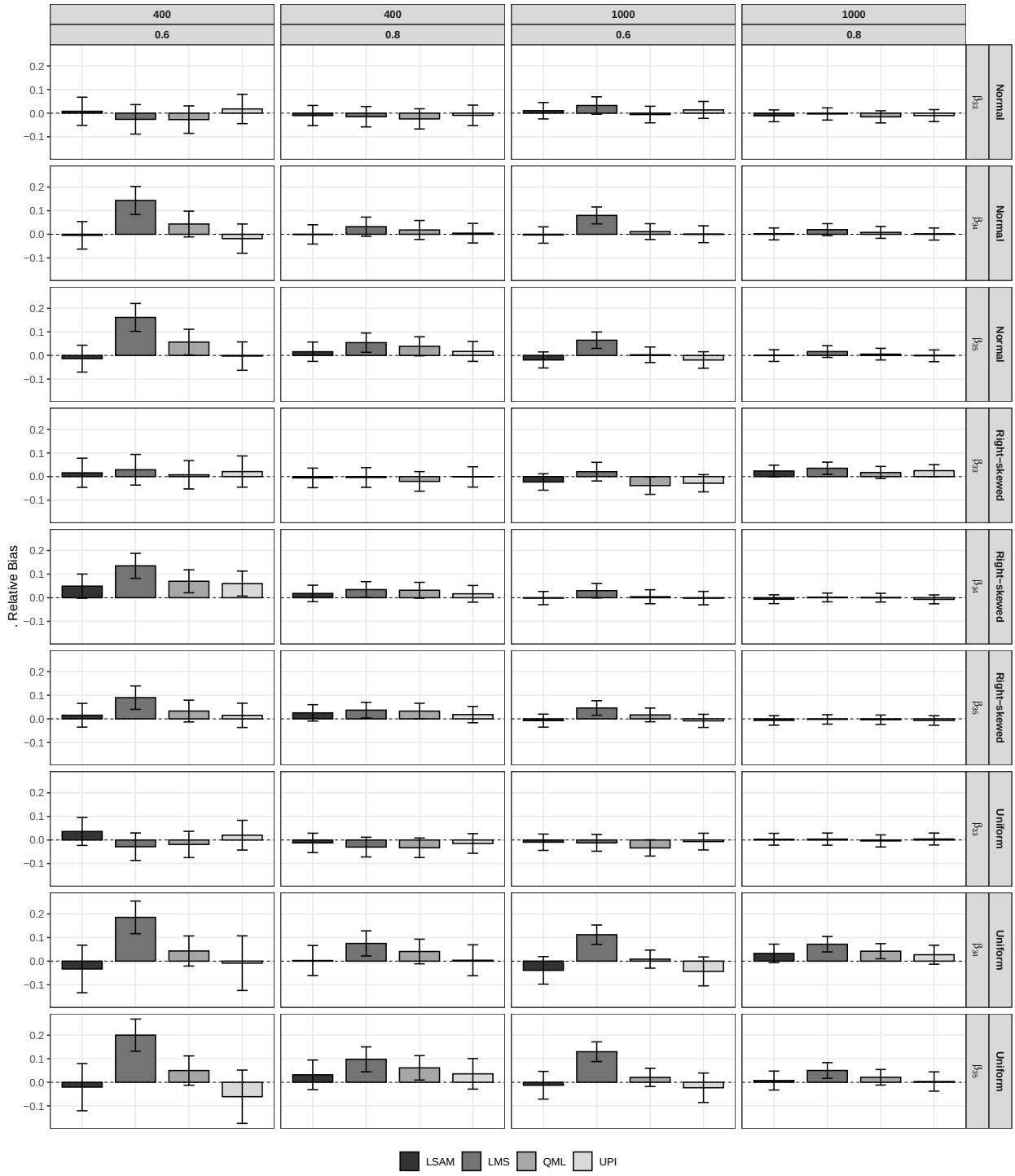


Figure S3

Additional Distributional Conditions Difference: Relative Bias for Simulation 1 (Epsilon & Zeta)

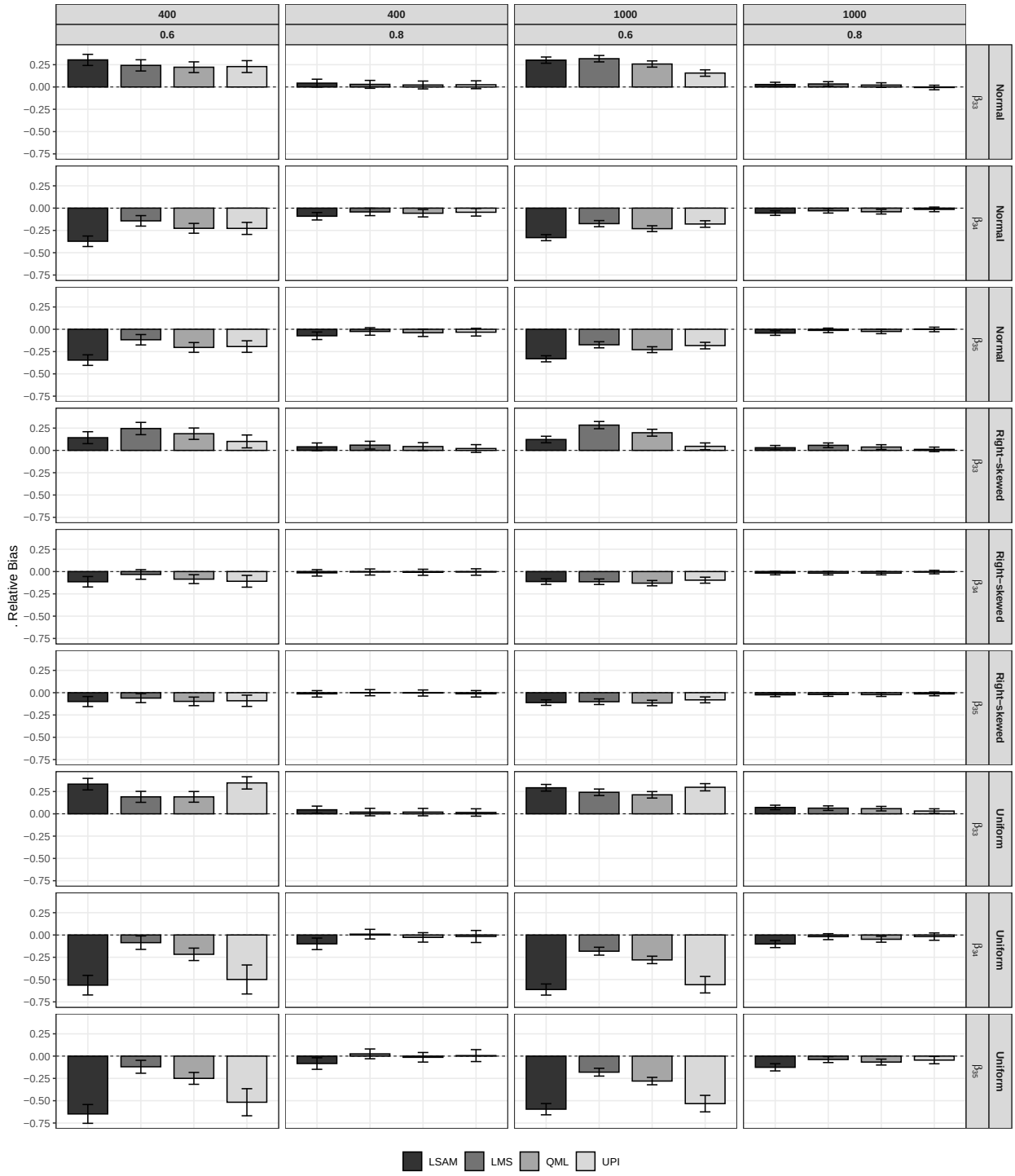


Figure S4

Additional Distributional Conditions Difference: Relative RMSE for Simulation 1 (Epsilon)

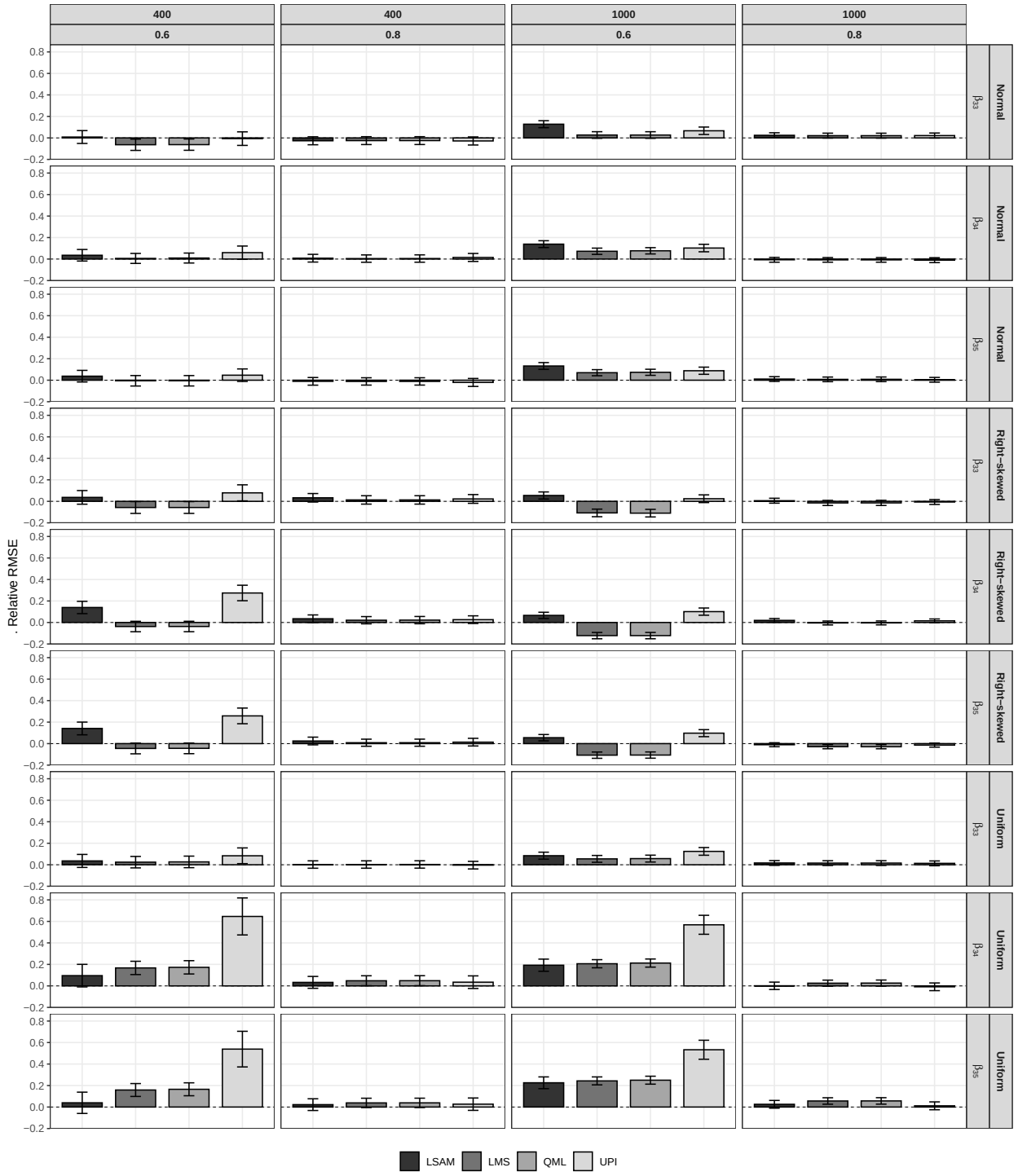


Figure S5

Additional Distributional Conditions Difference: Relative RMSE for Simulation 1 (Zeta)

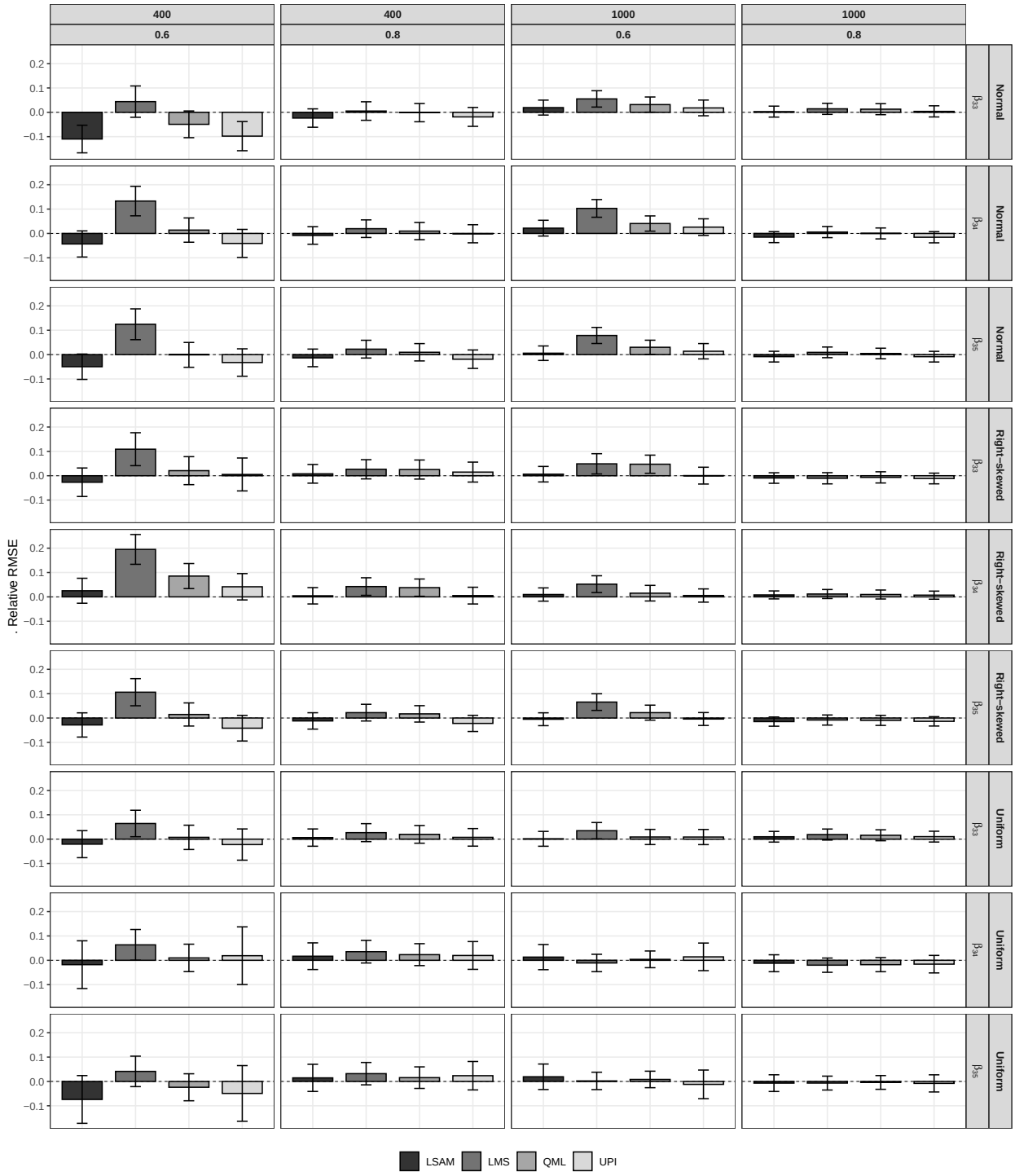


Figure S6

Additional Distributional Conditions Difference: Relative RMSE for Simulation 1 (Epsilon & Zeta)

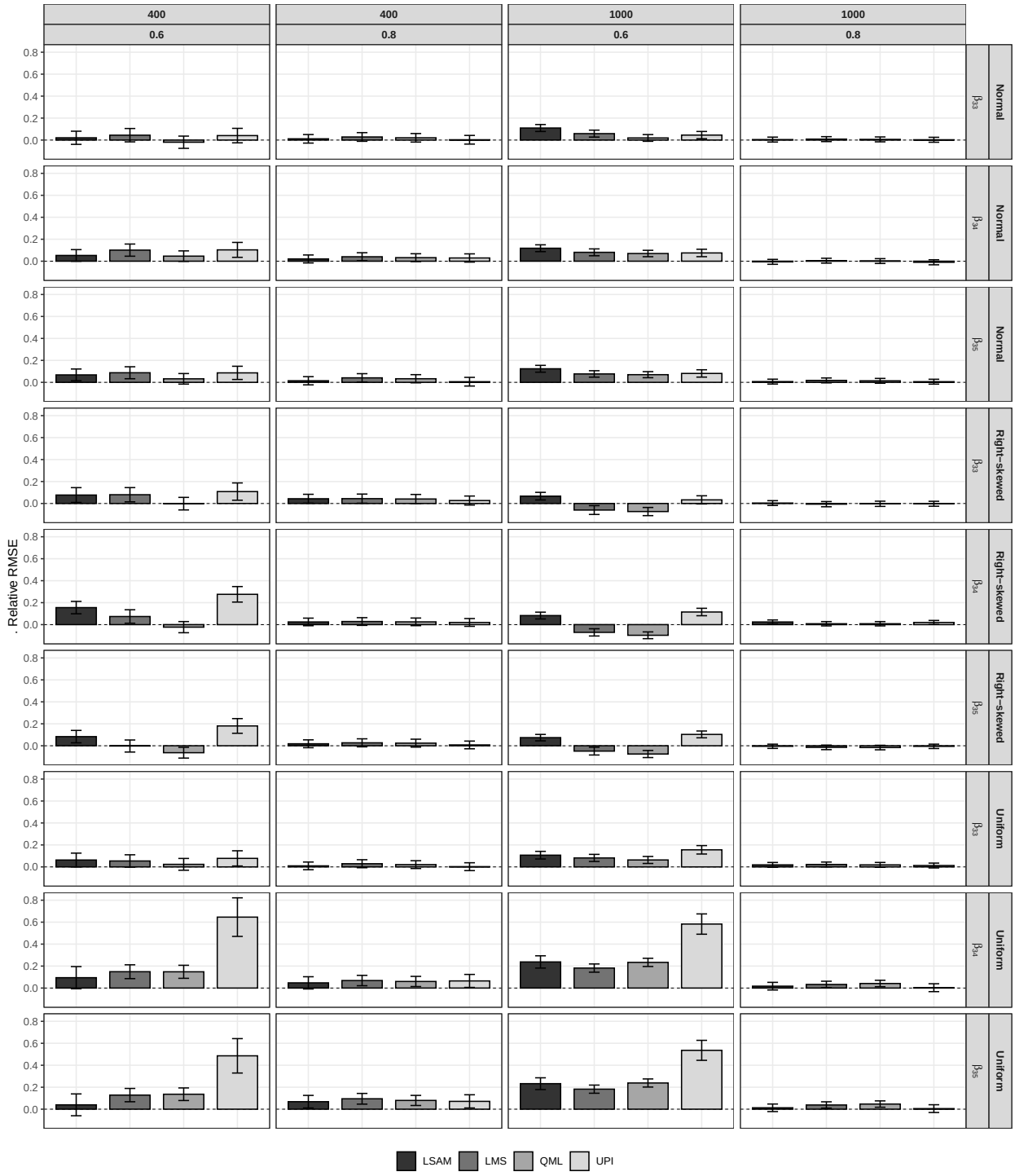


Figure S7

Additional Distributional Conditions Difference: Coverage Rate for Simulation 1 (Epsilon)

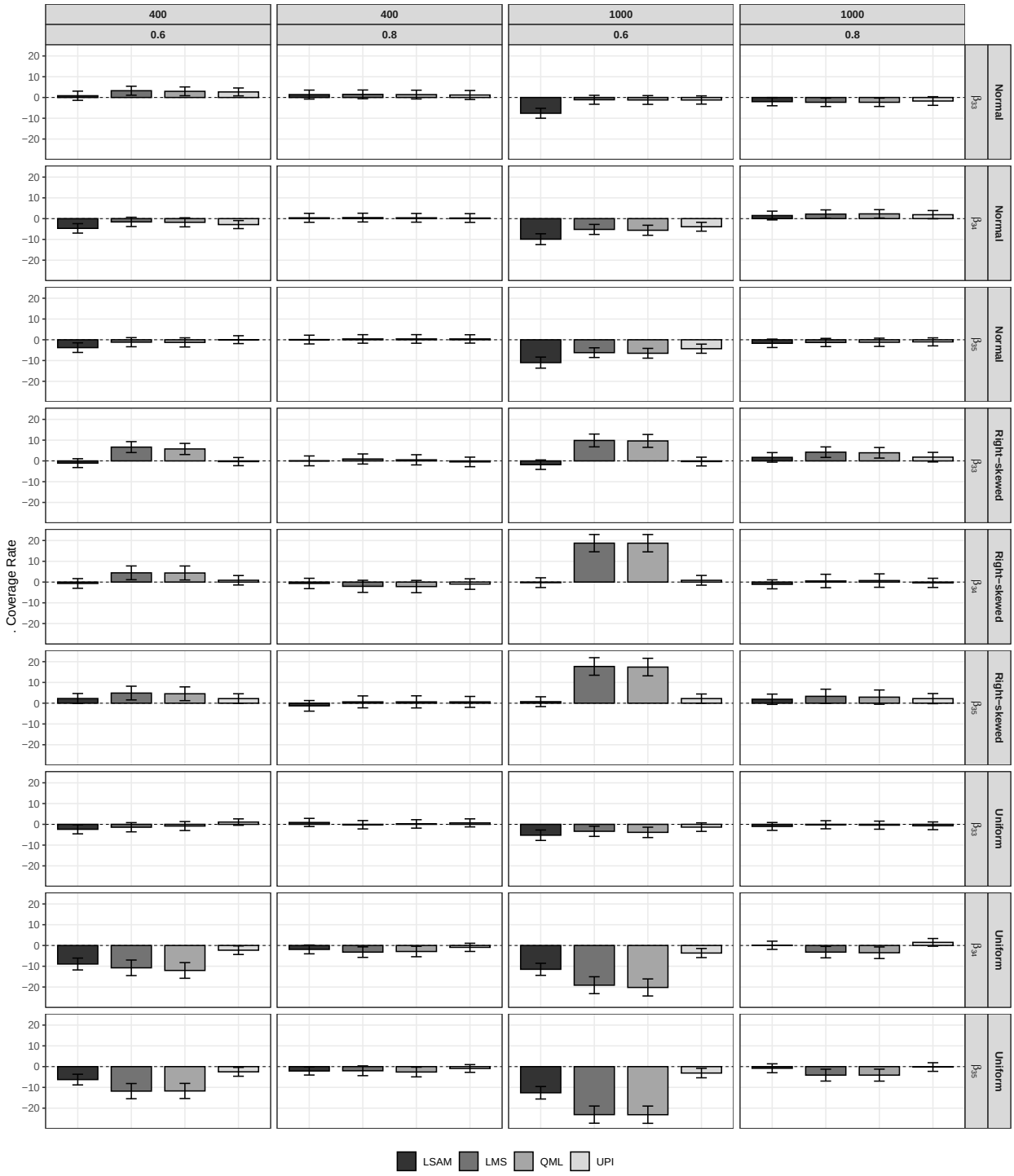


Figure S8

Additional Distributional Conditions Difference: Coverage Rate for Simulation 1 (Zeta)

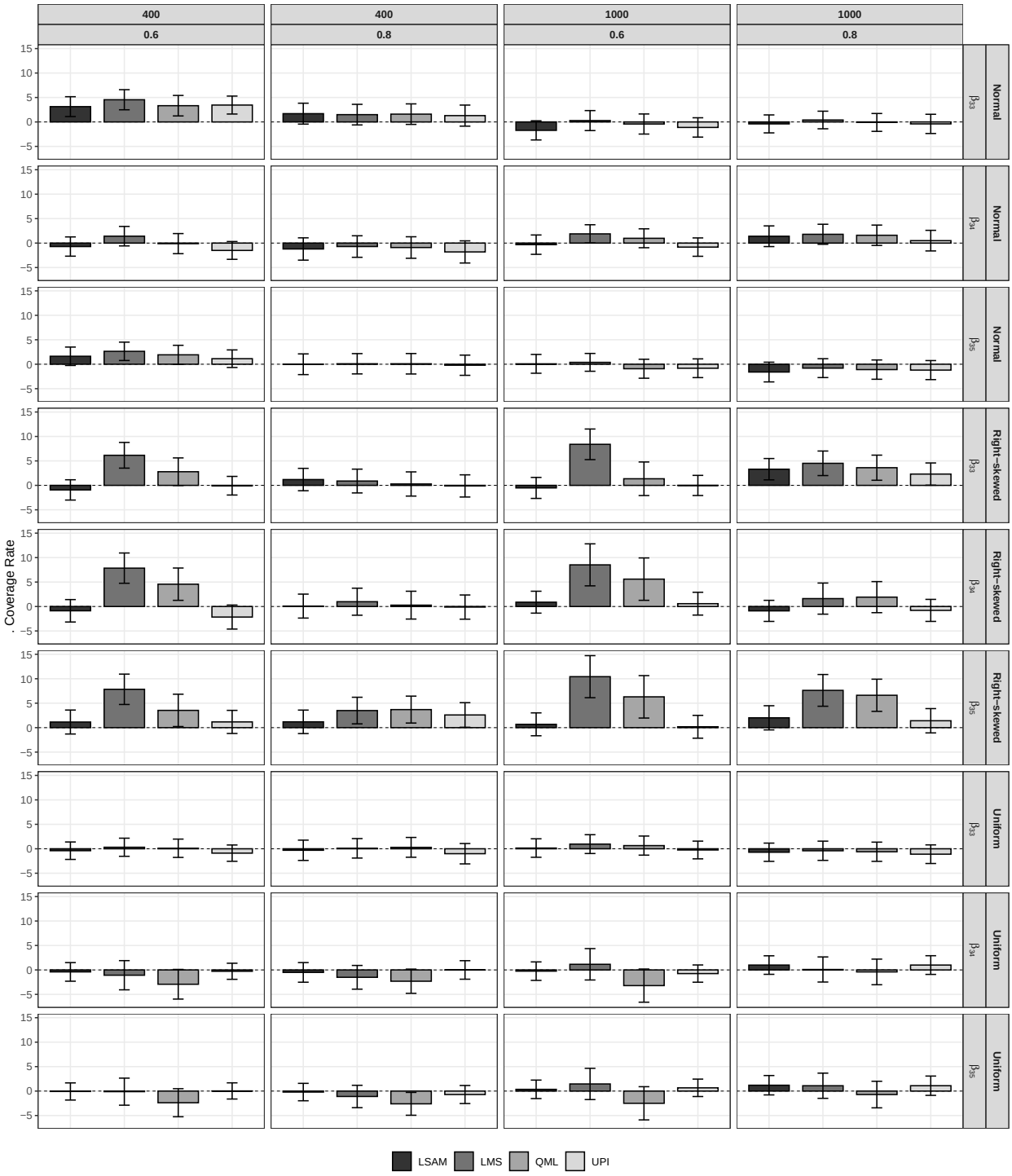


Figure S9

Additional Distributional Conditions Difference: Coverage Rate for Simulation 1 (Epsilon & Zeta)

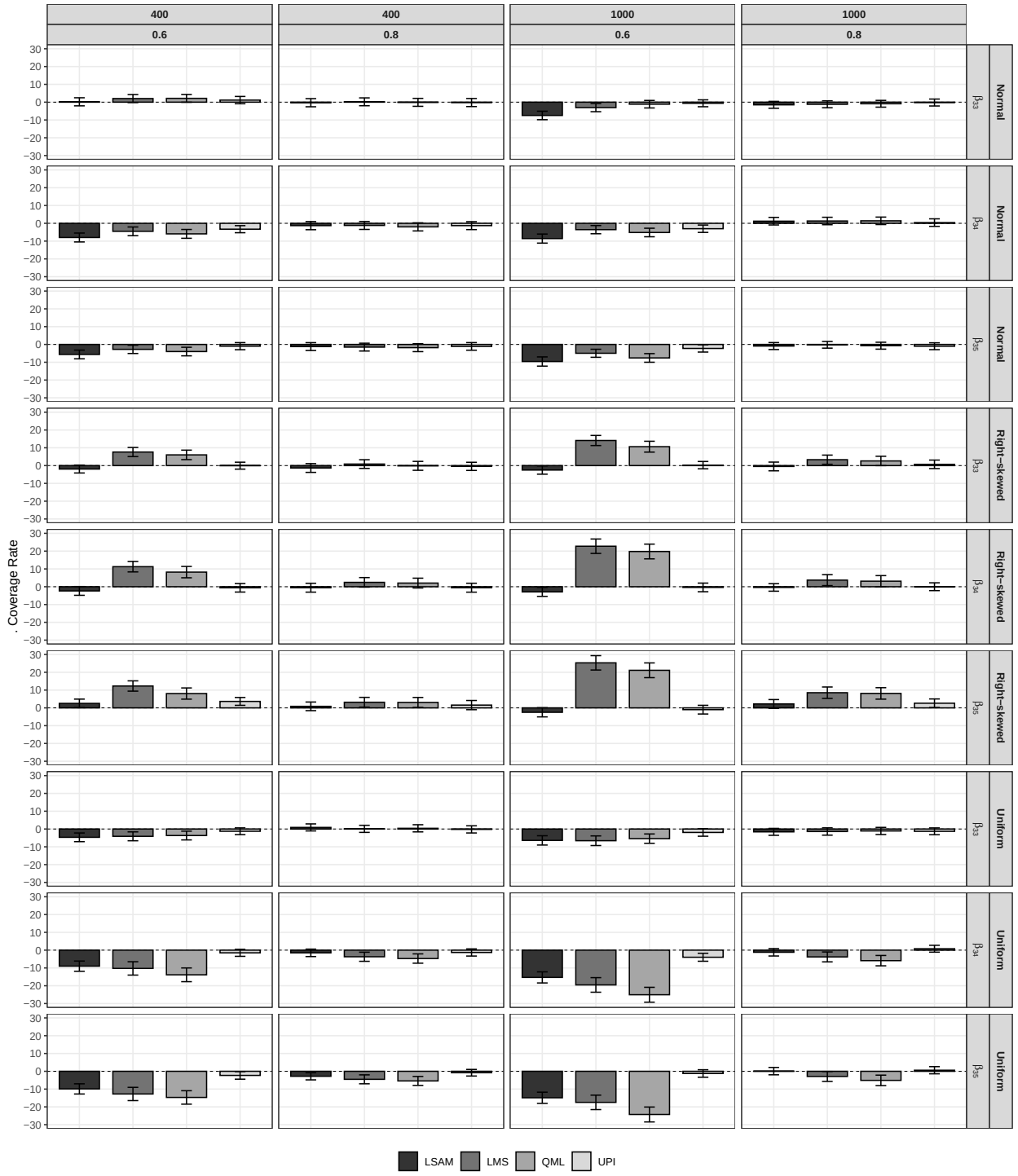


Figure S10

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 1 (Epsilon)

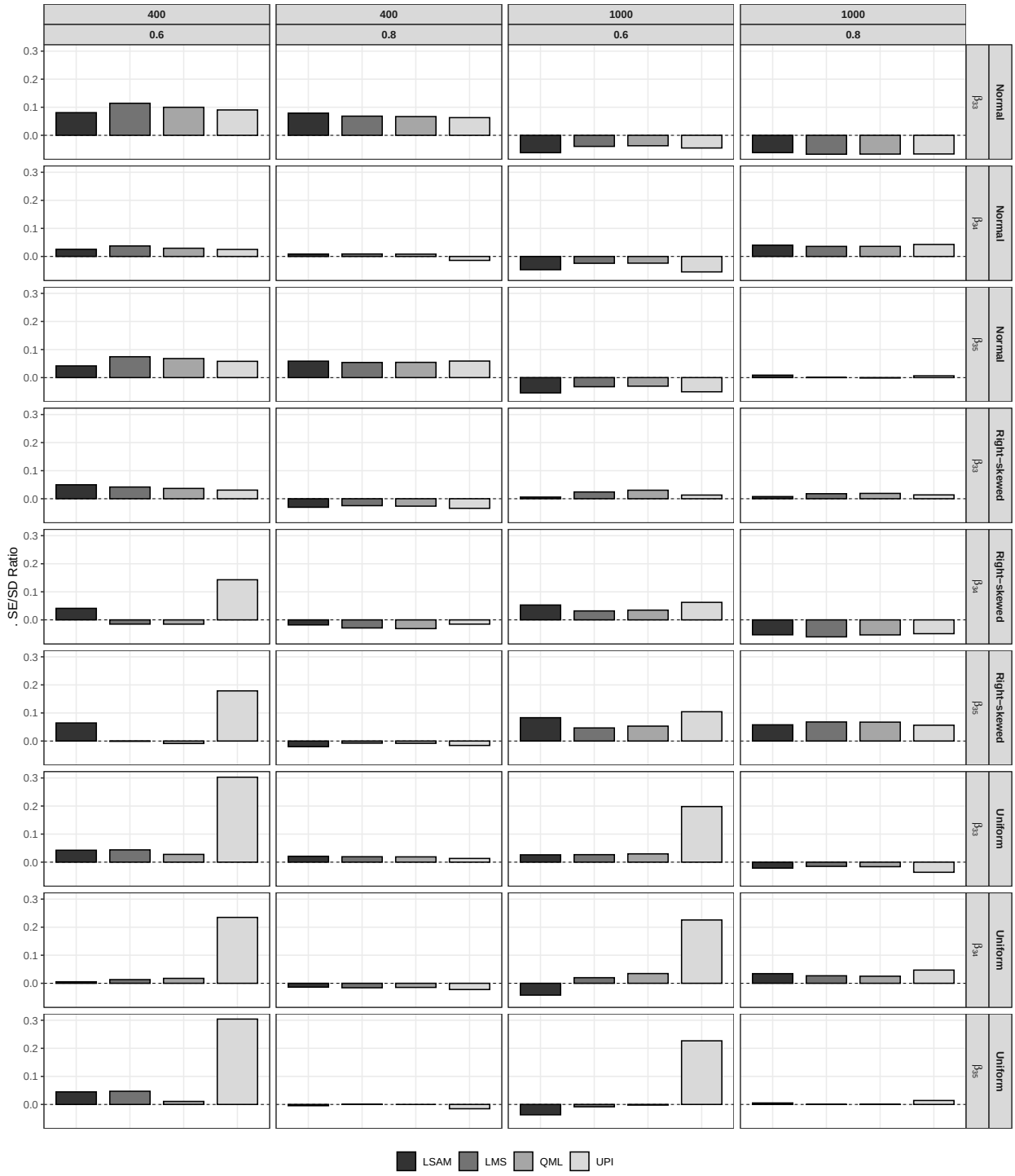


Figure S11

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 1 (Zeta)

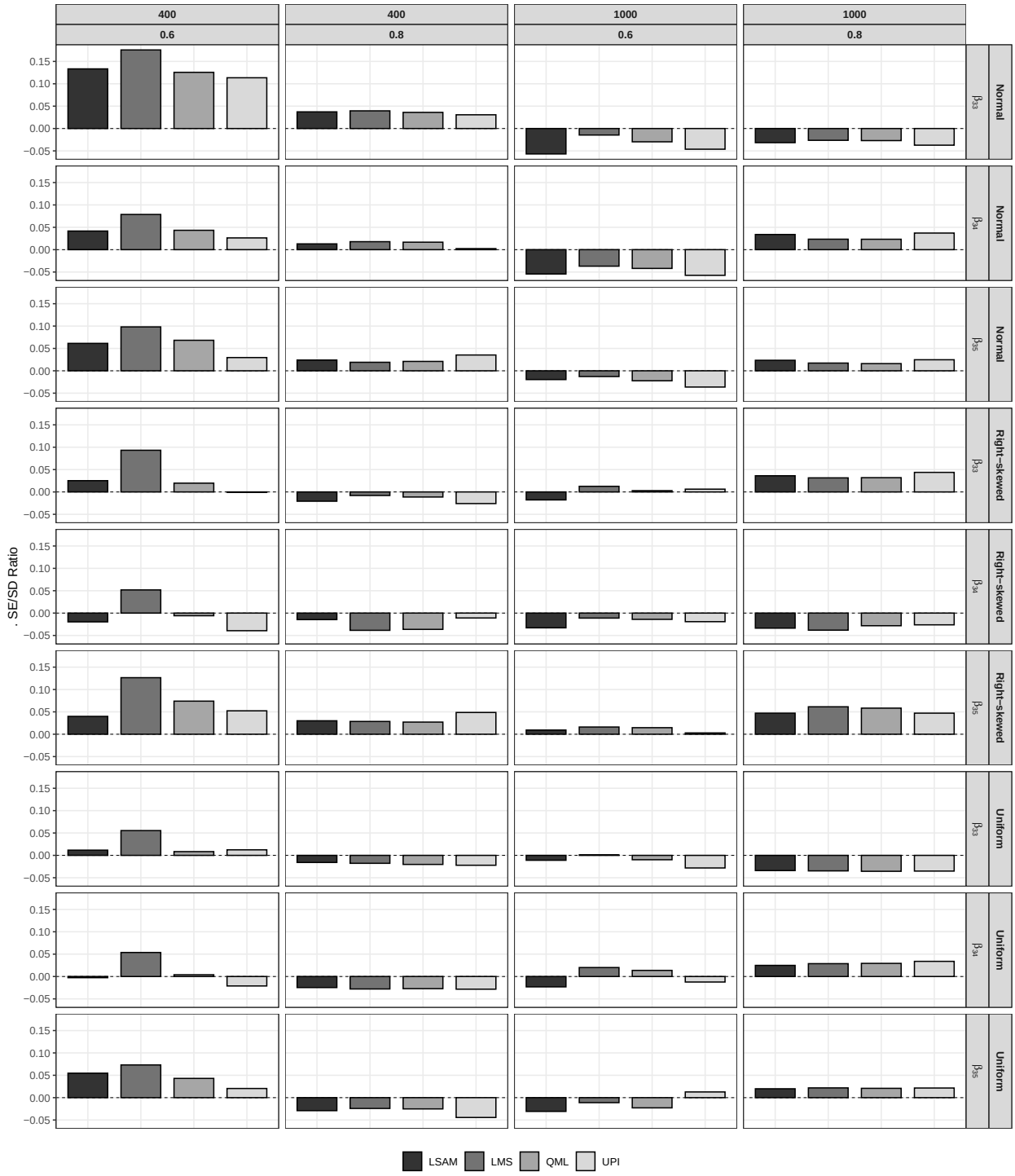


Figure S12

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 1 (Epsilon & Zeta)

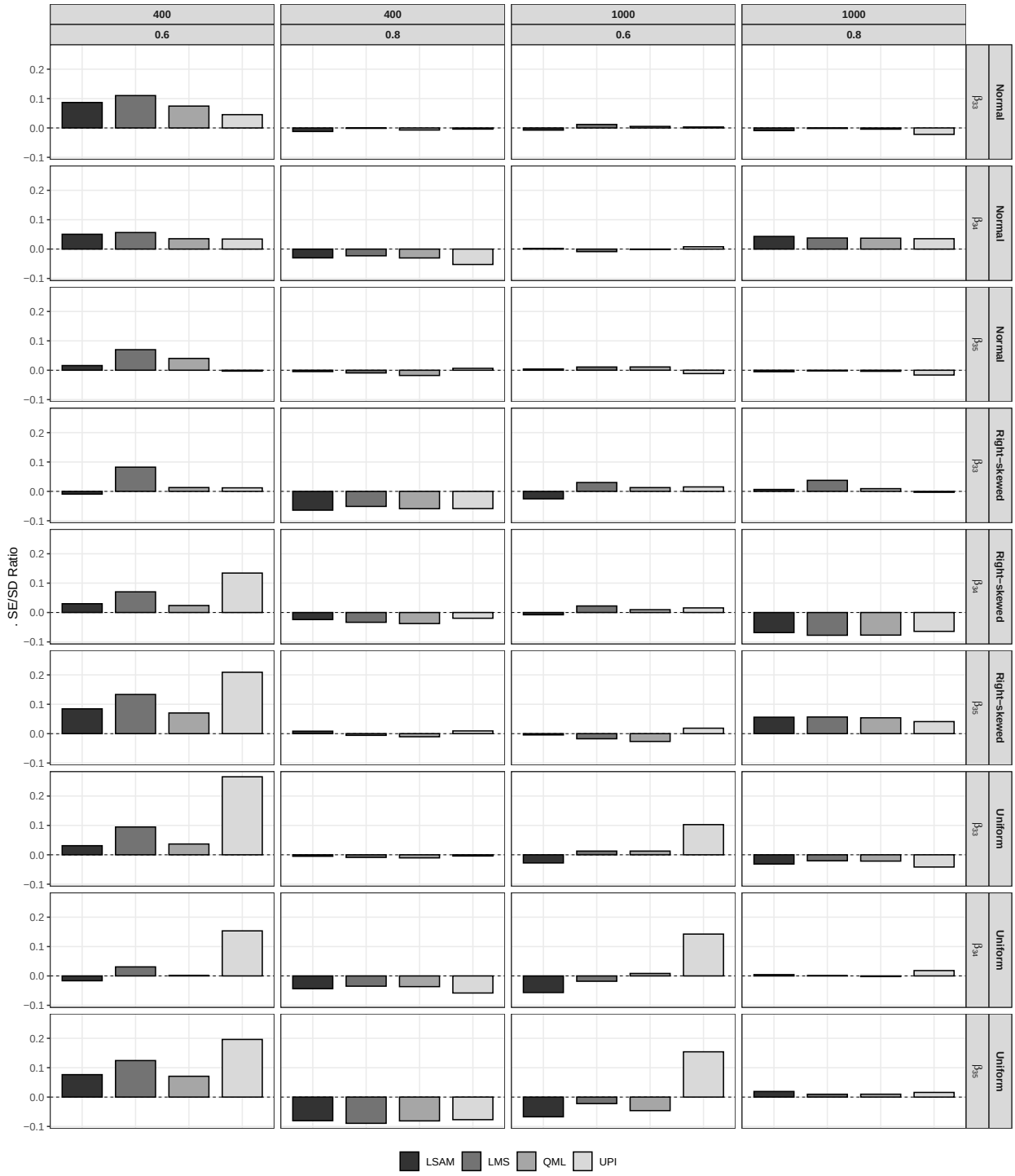


Figure S13

Additional Distributional Conditions Difference: Power for Simulation 1 (Epsilon)

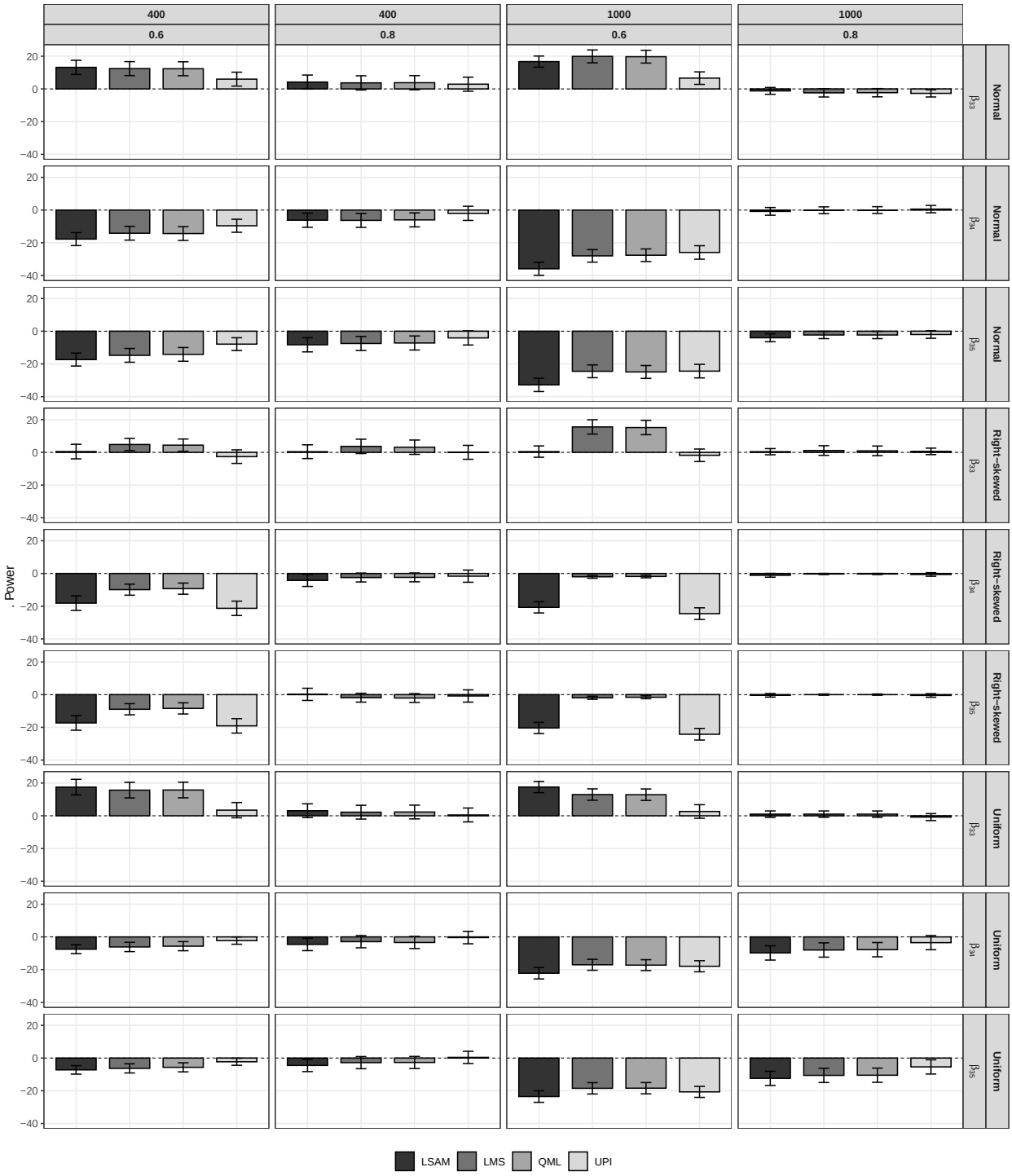


Figure S14

Additional Distributional Conditions Difference: Power for Simulation 1 (Zeta)

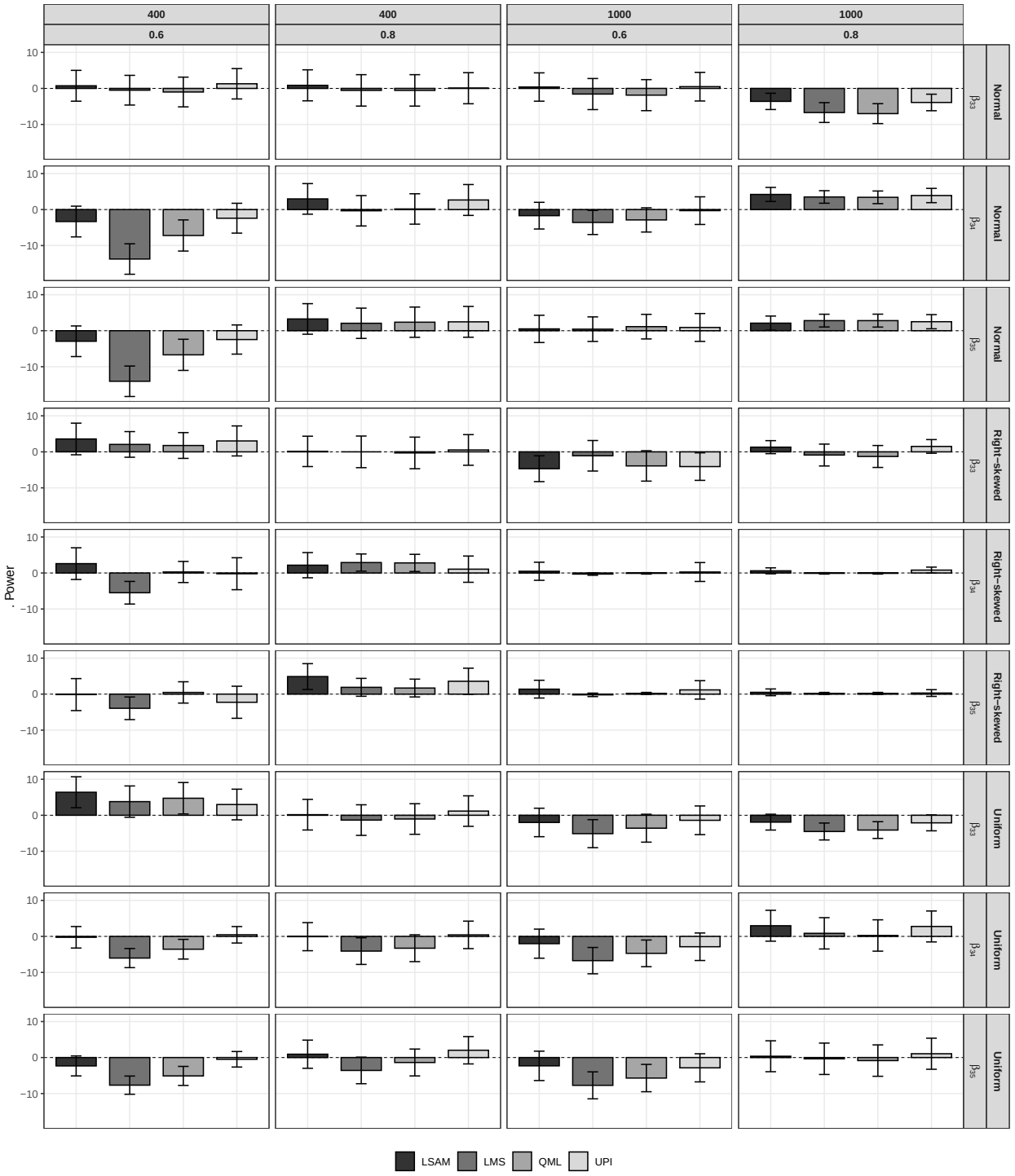


Figure S15

Additional Distributional Conditions Difference: Power for Simulation 1 (Epsilon & Zeta)

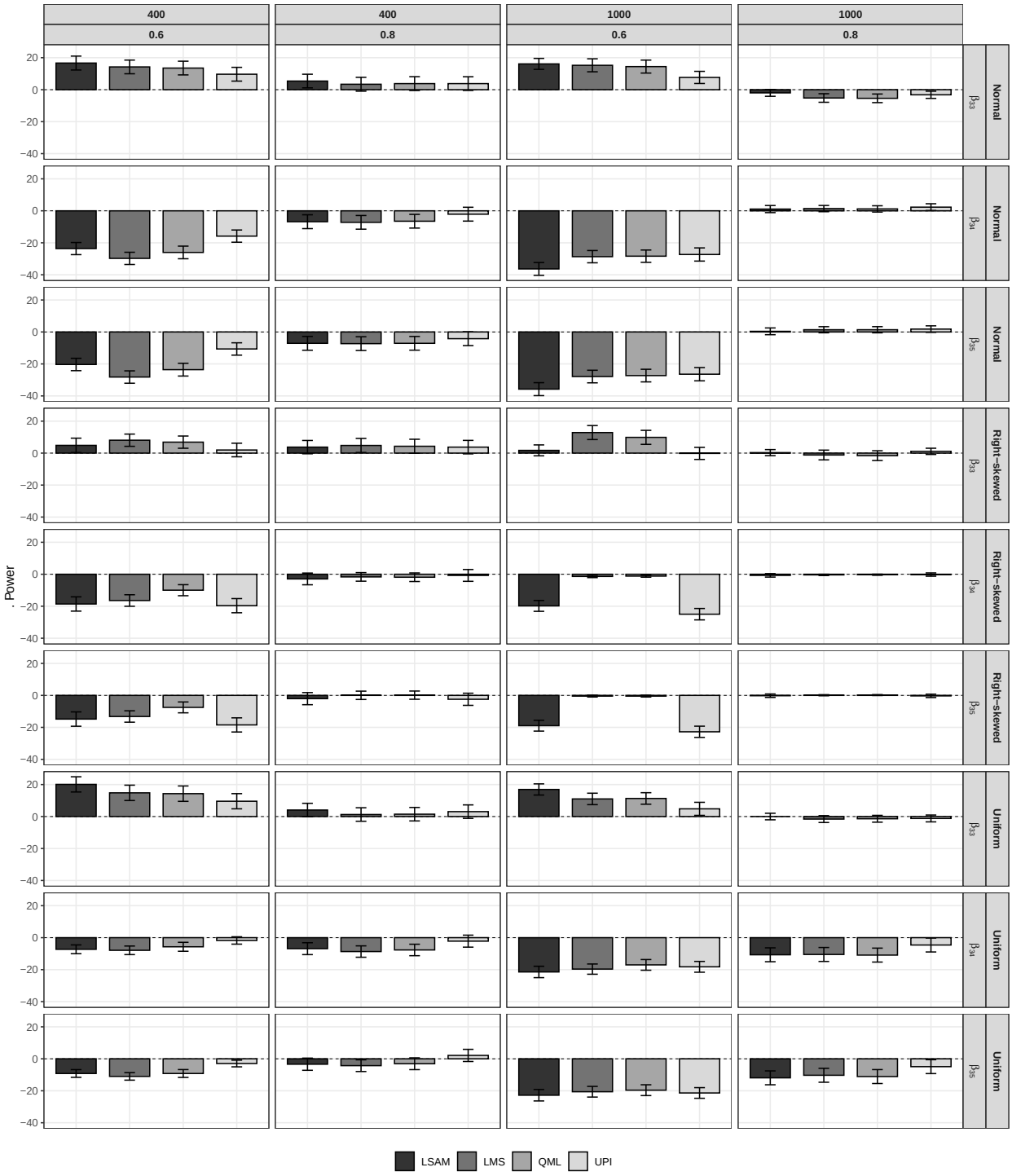


Figure S16

Additional Distributional Conditions Difference: Type I Error for Simulation 1 (Epsilon)

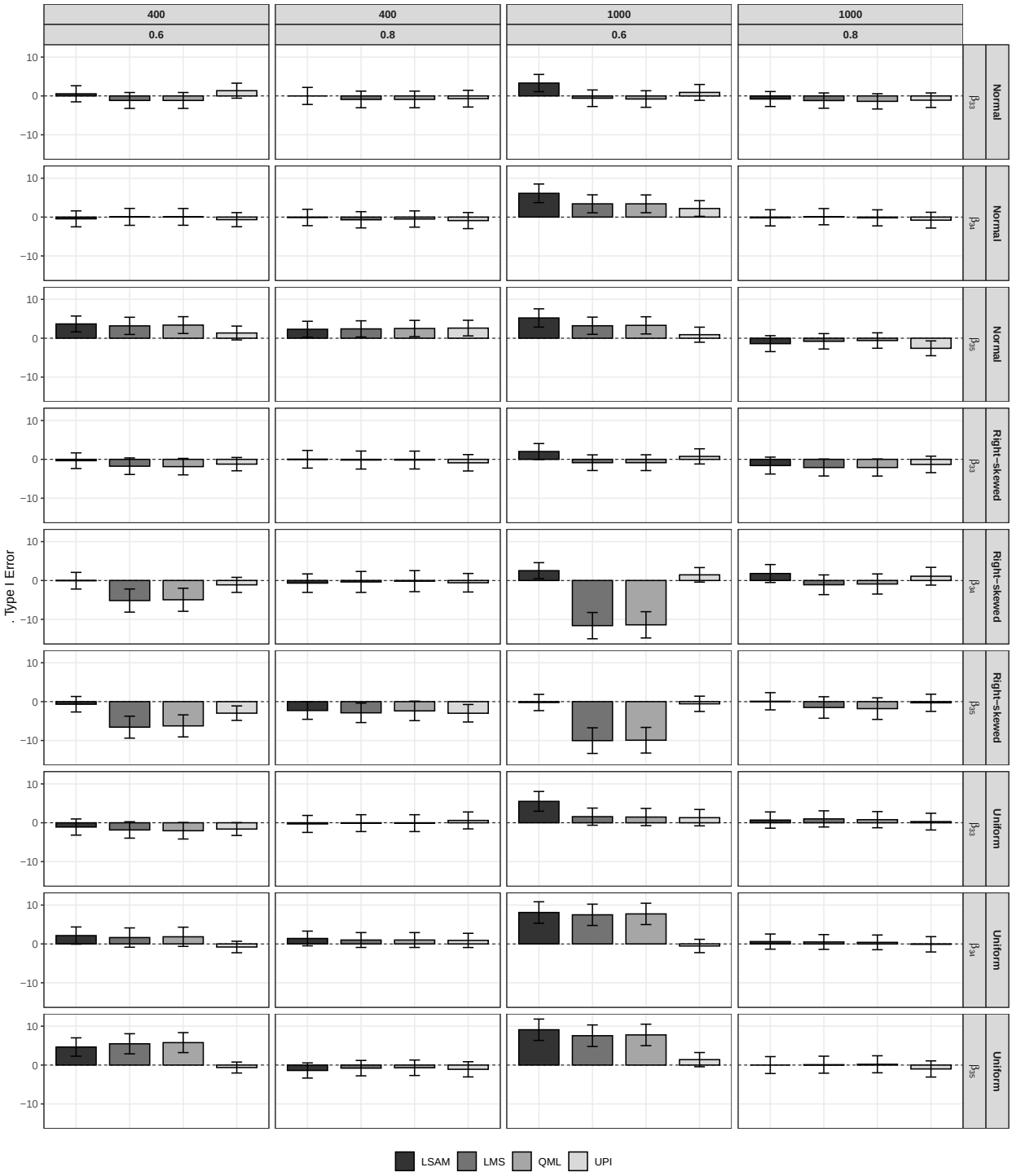


Figure S17

Additional Distributional Conditions Difference: Type I Error for Simulation 1 (Zeta)

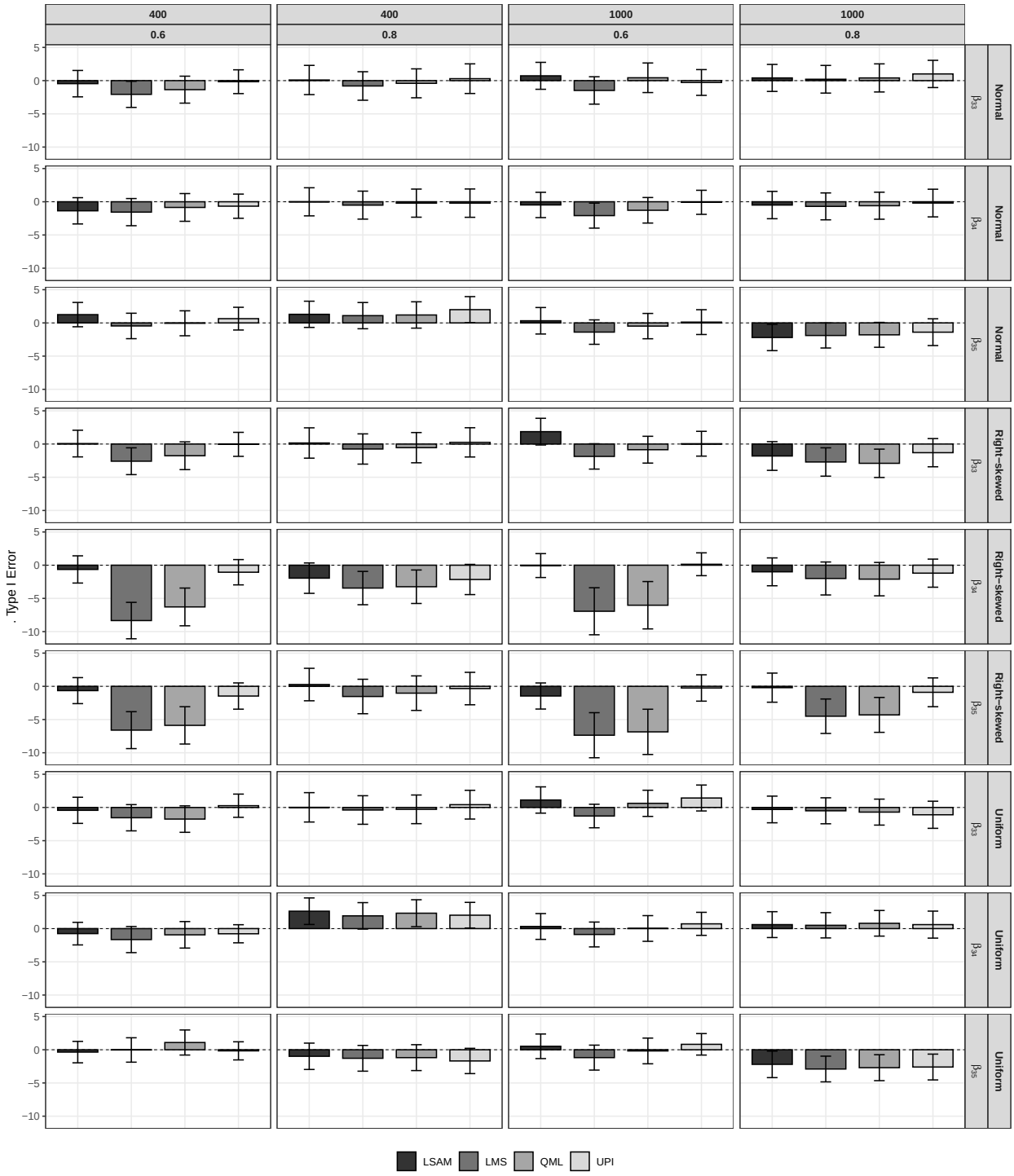


Figure S18

Additional Distributional Conditions Difference: Type I Error for Simulation 1 (Epsilon & Zeta)

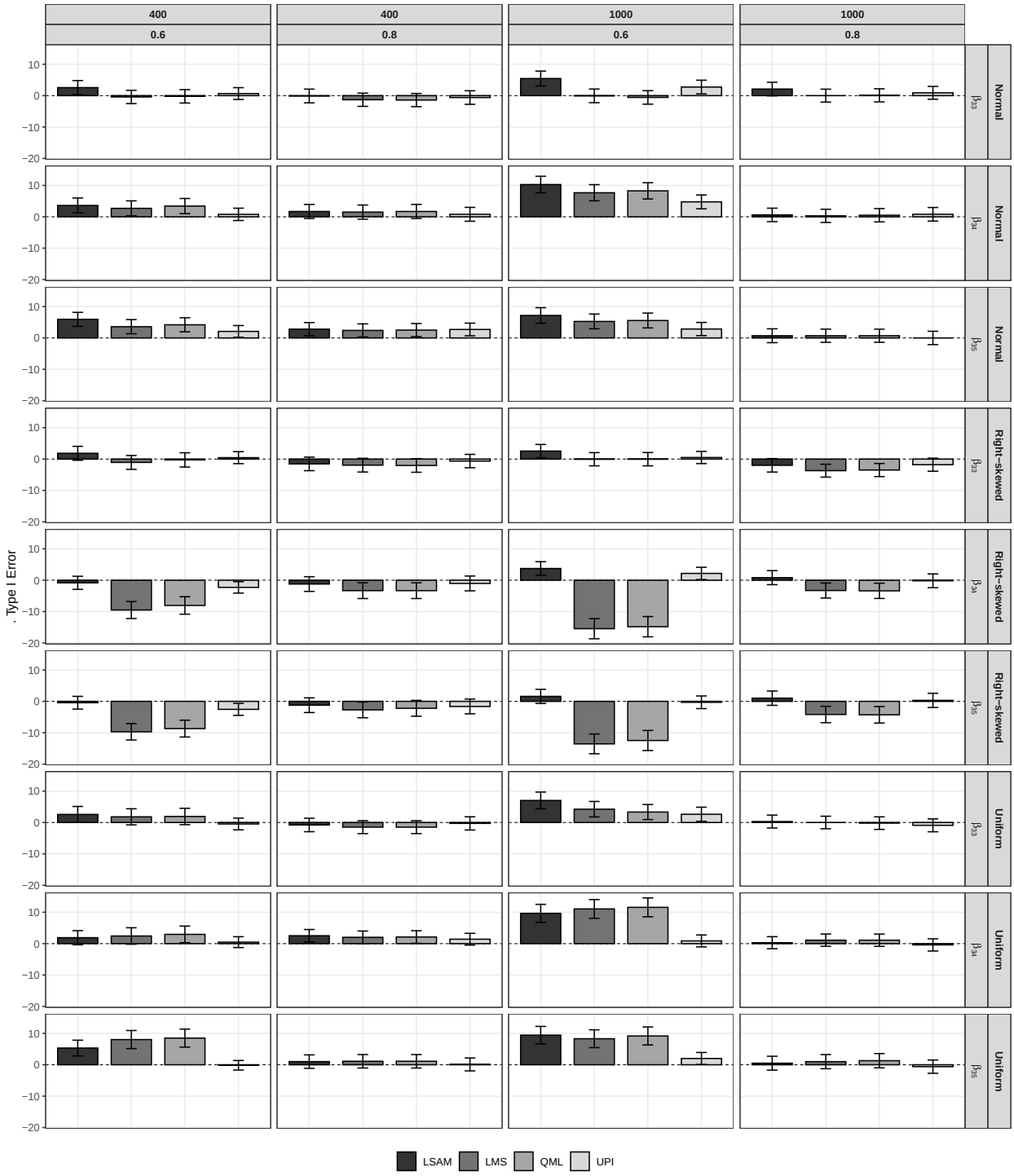


Figure S19

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Normal (Epsilon)

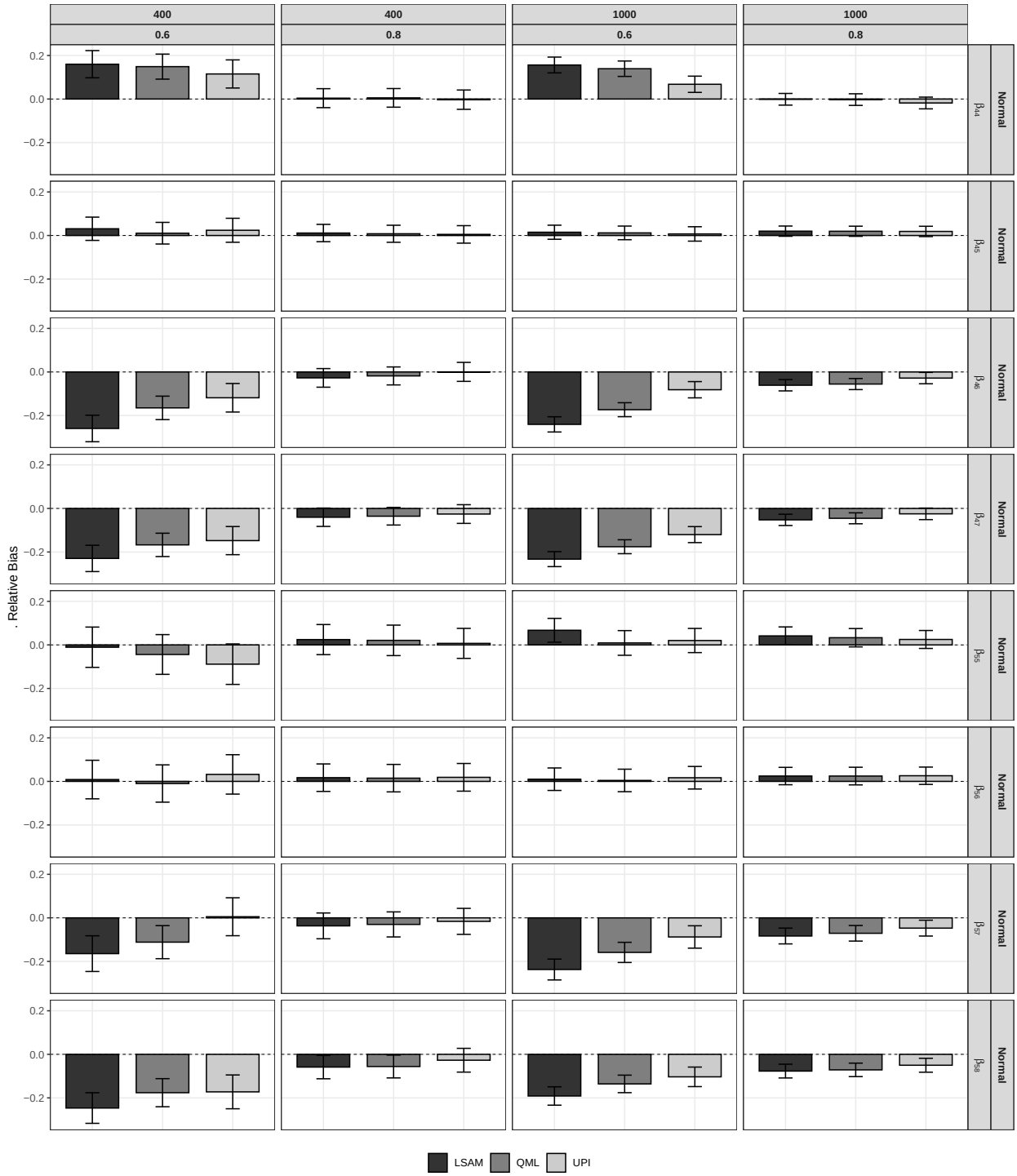


Figure S20

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Normal (Zeta)

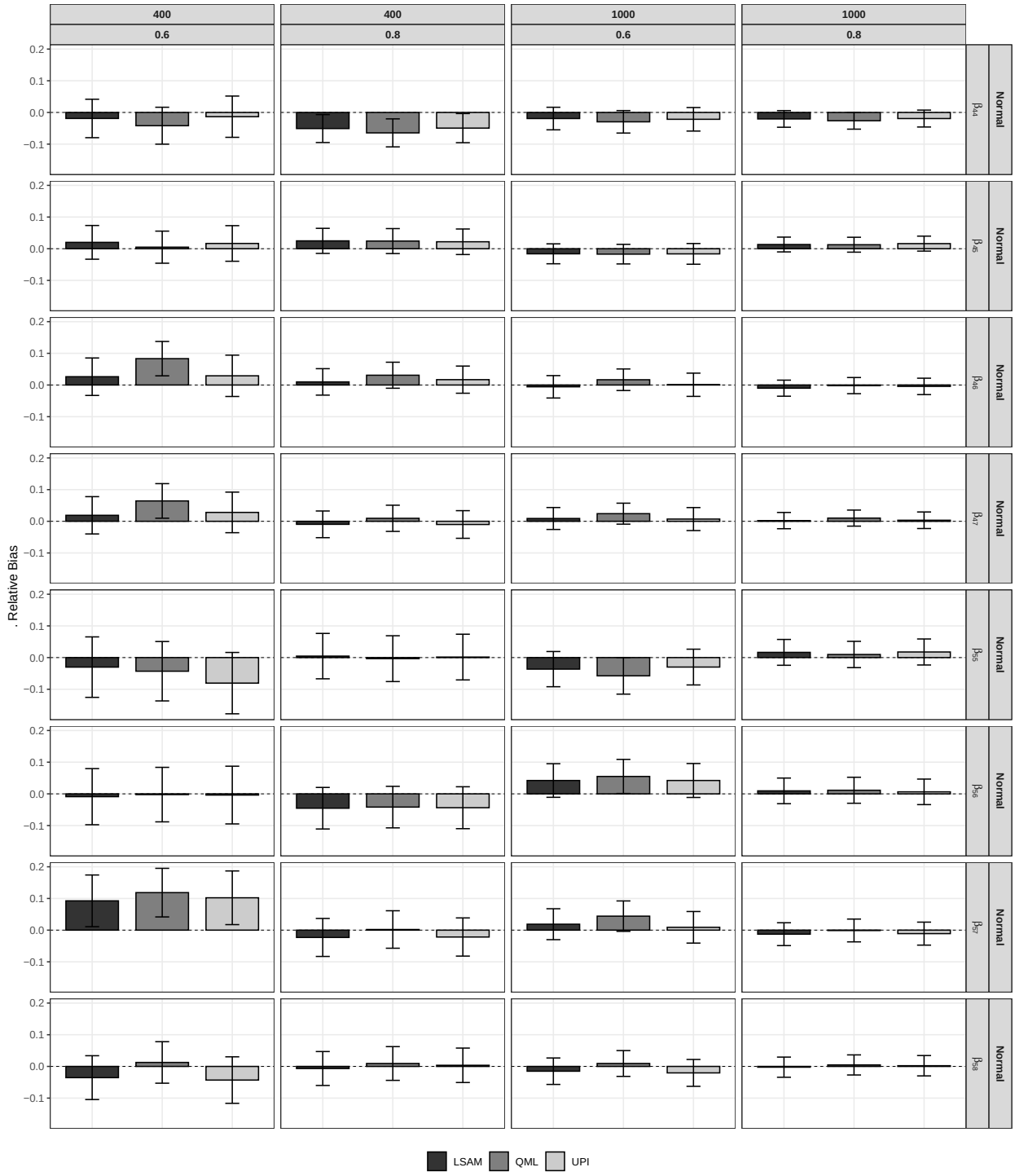


Figure S21

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Normal (Epsilon & Zeta)

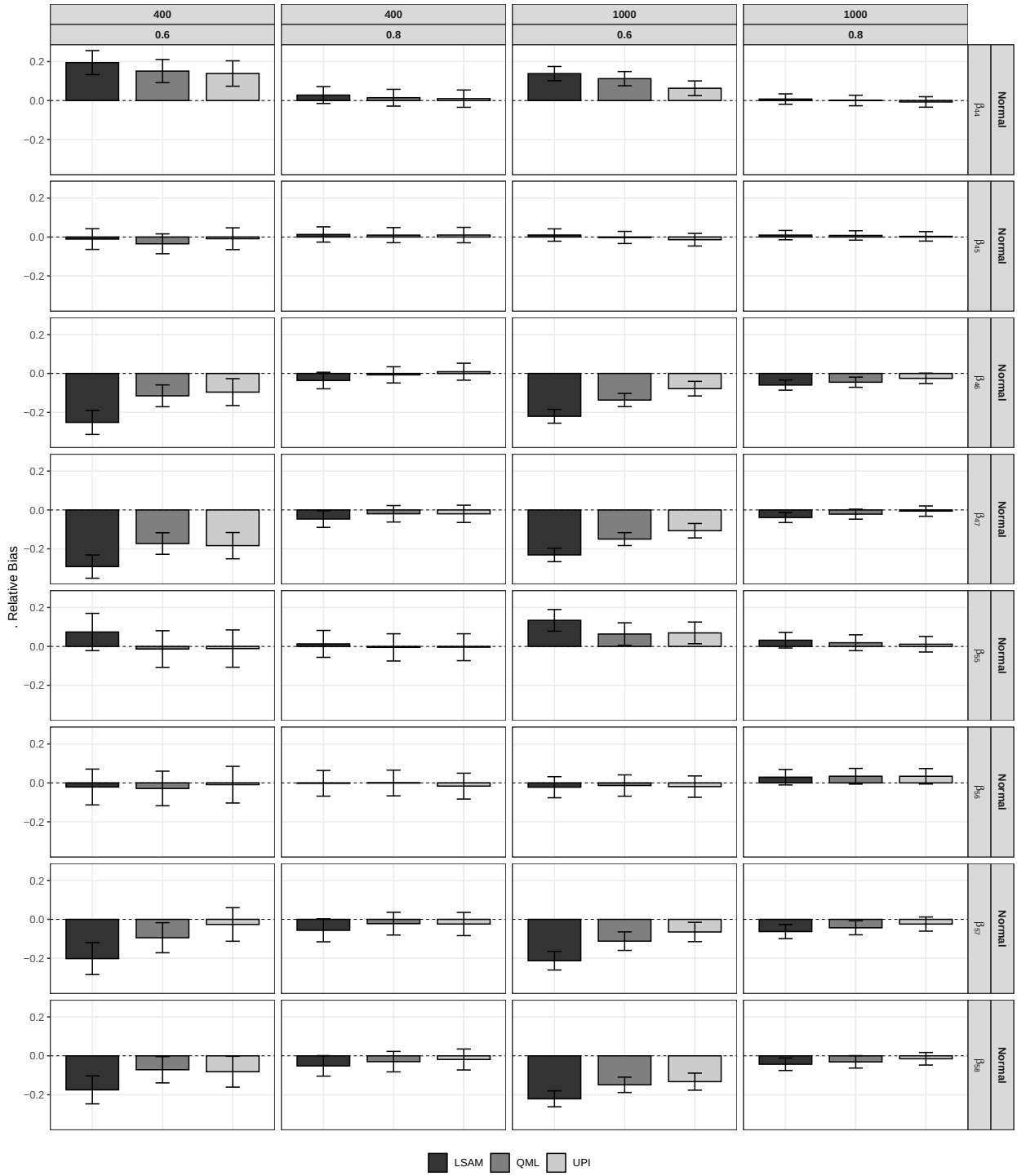


Figure S22

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Normal (Epsilon)

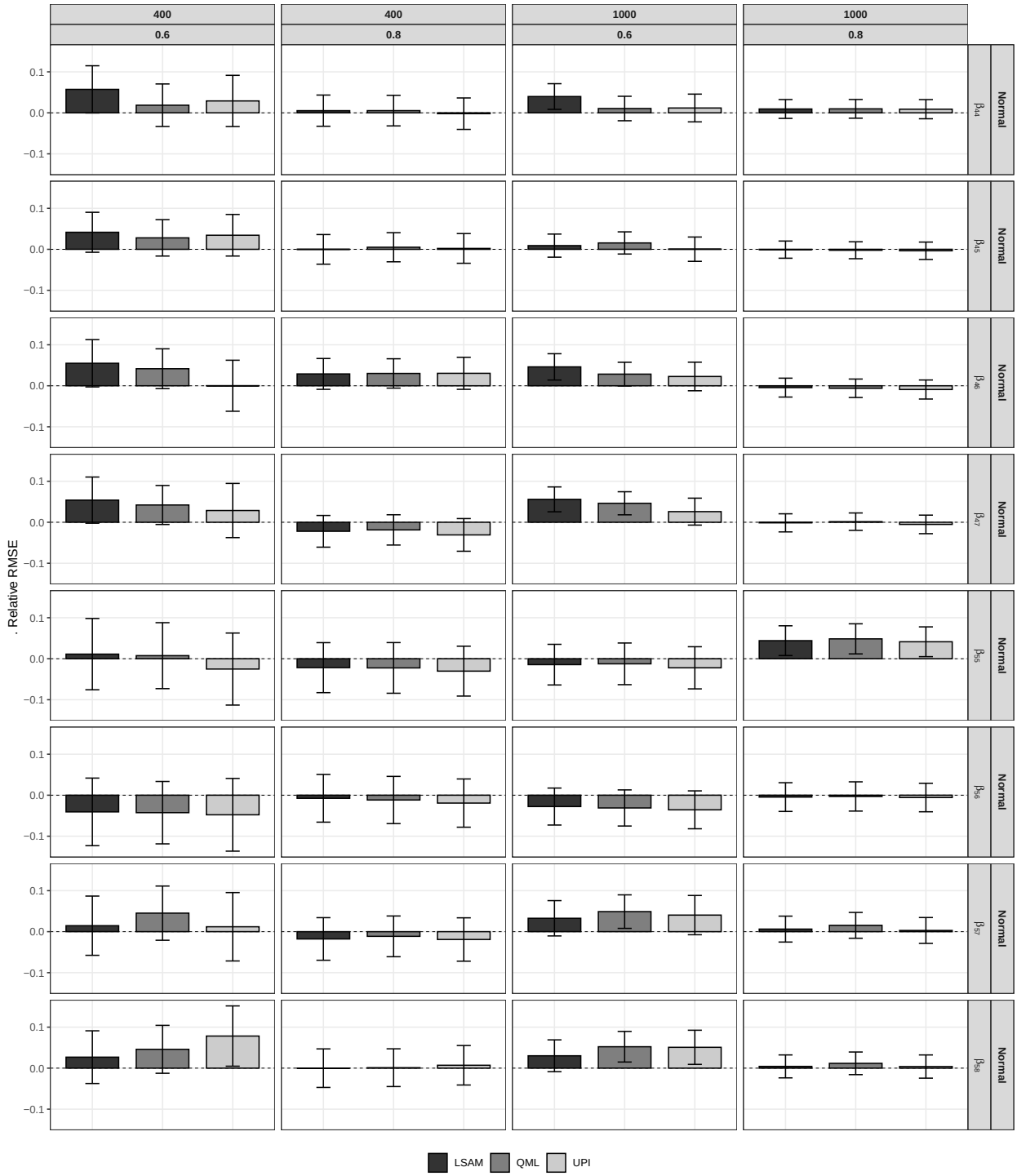


Figure S23

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Normal (Zeta)

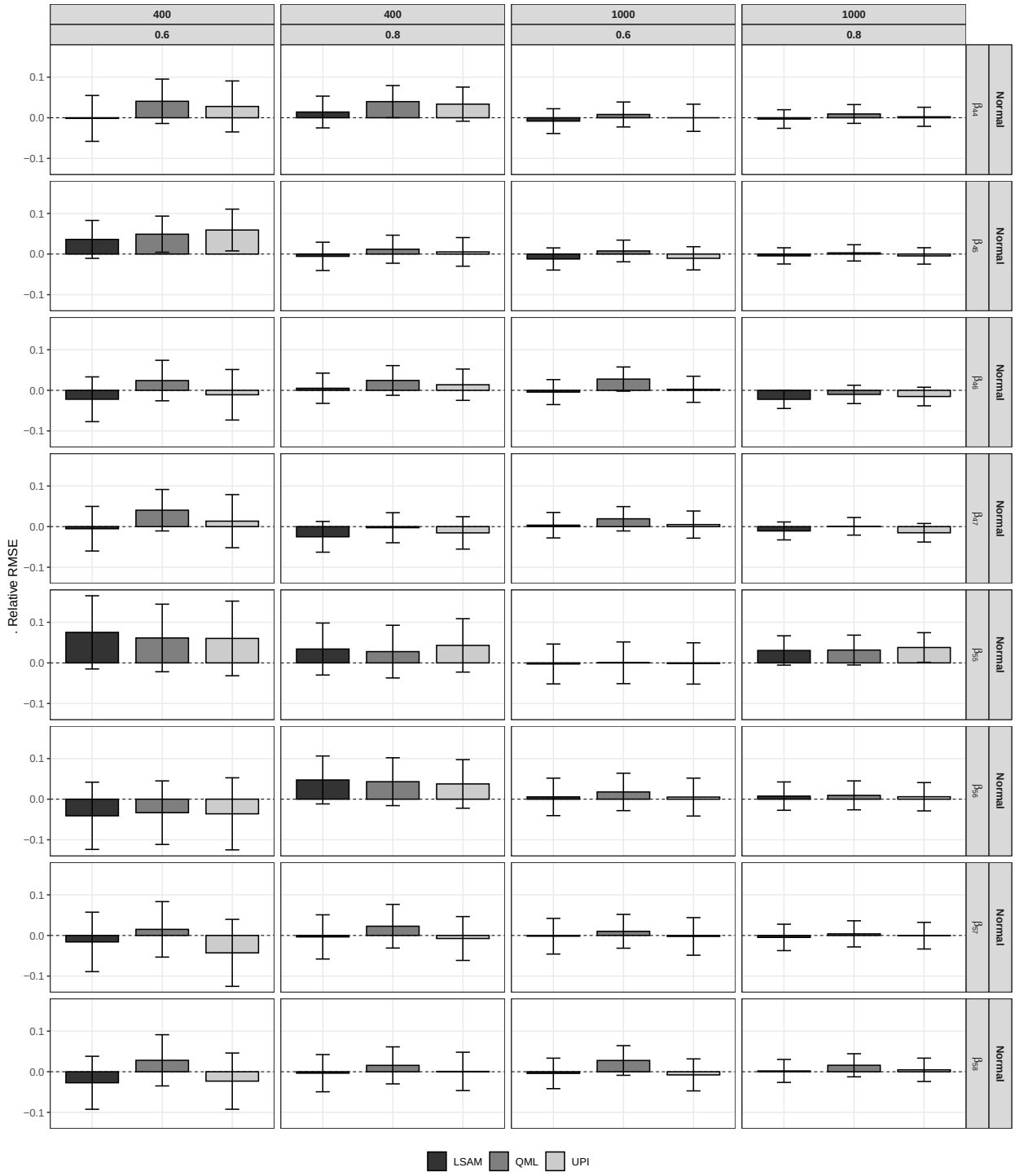


Figure S24

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Normal (Epsilon & Zeta)

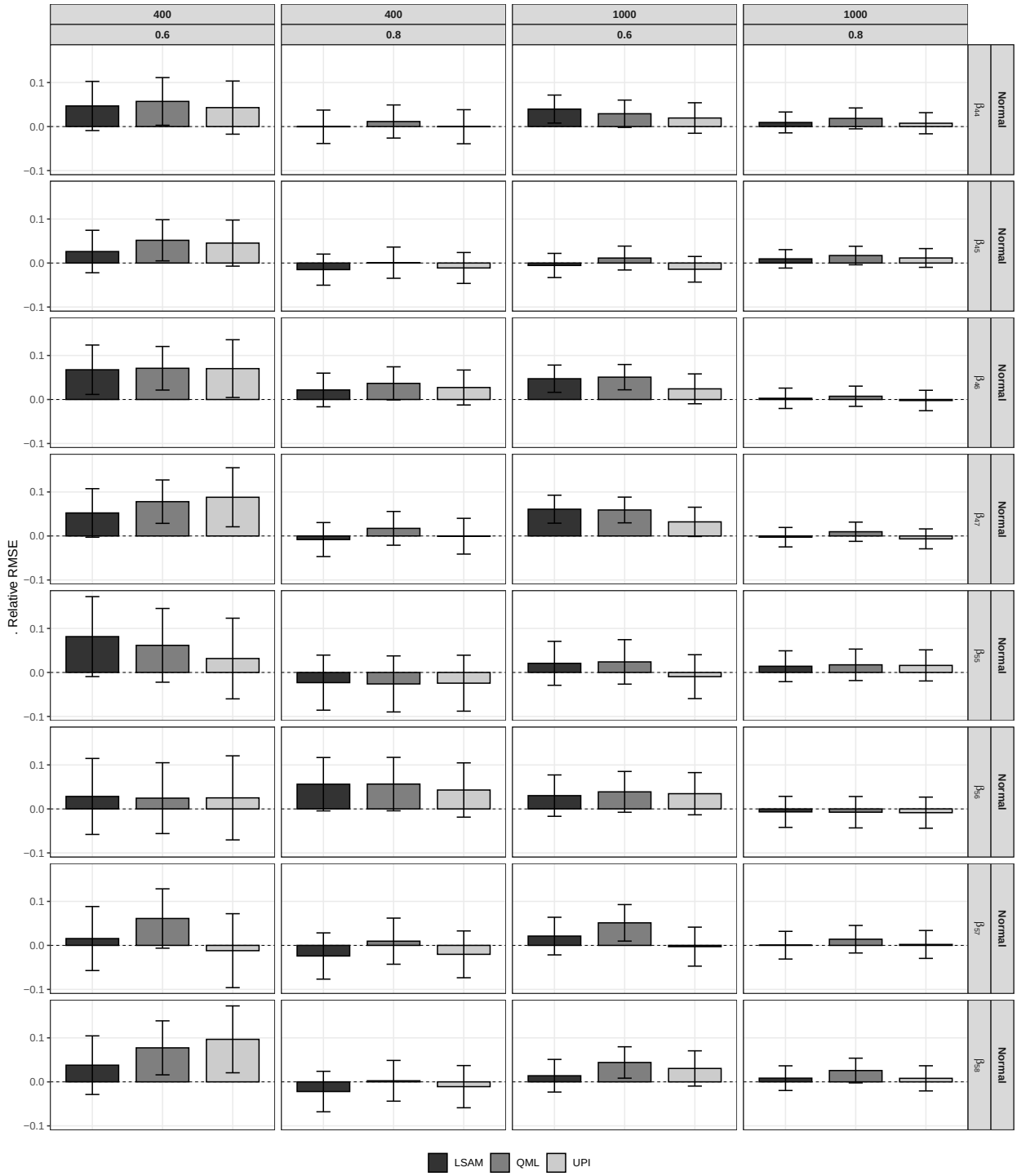


Figure S25

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Normal (Epsilon)

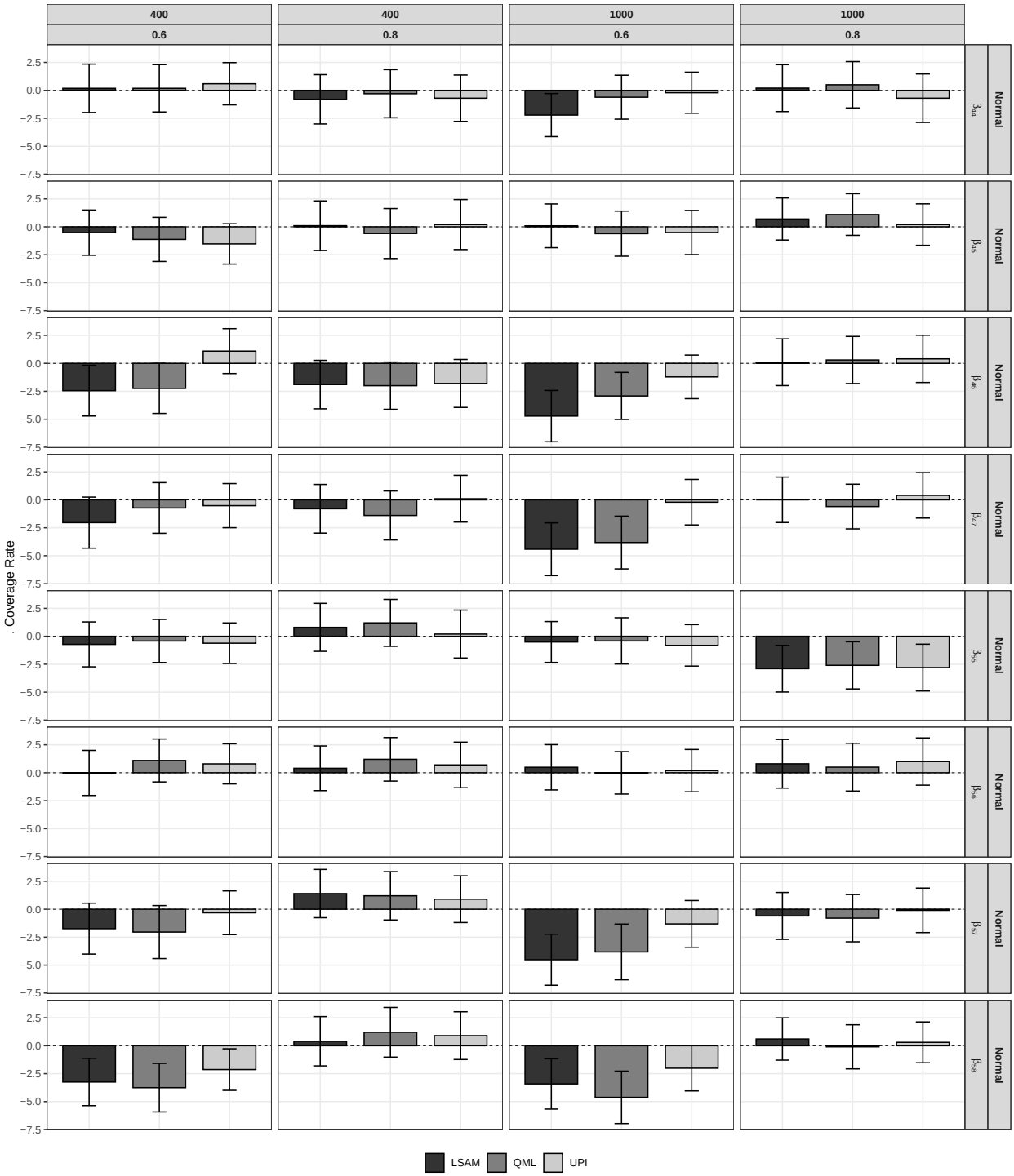


Figure S26

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Normal (Zeta)

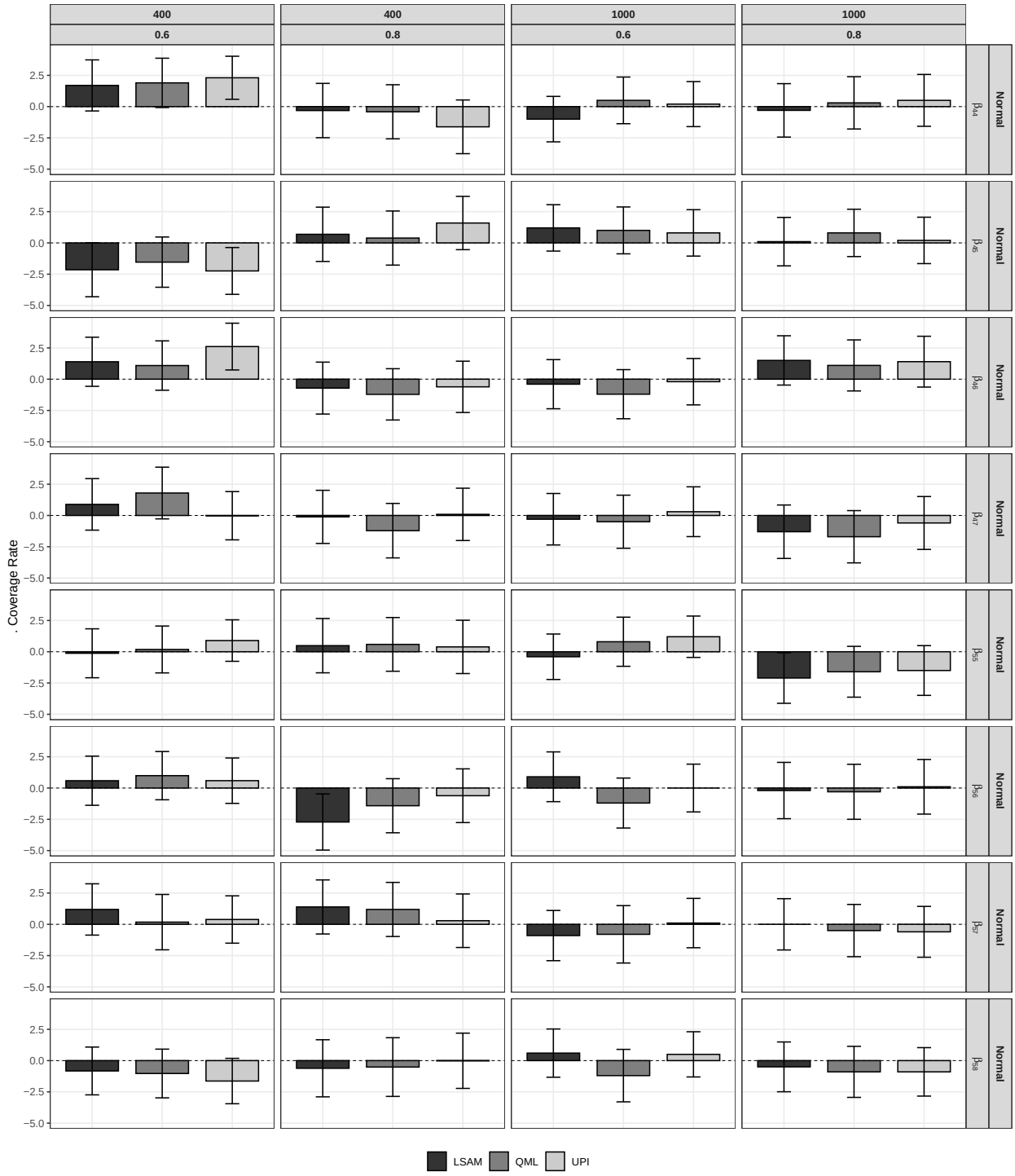


Figure S27

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Normal (Epsilon & Zeta)

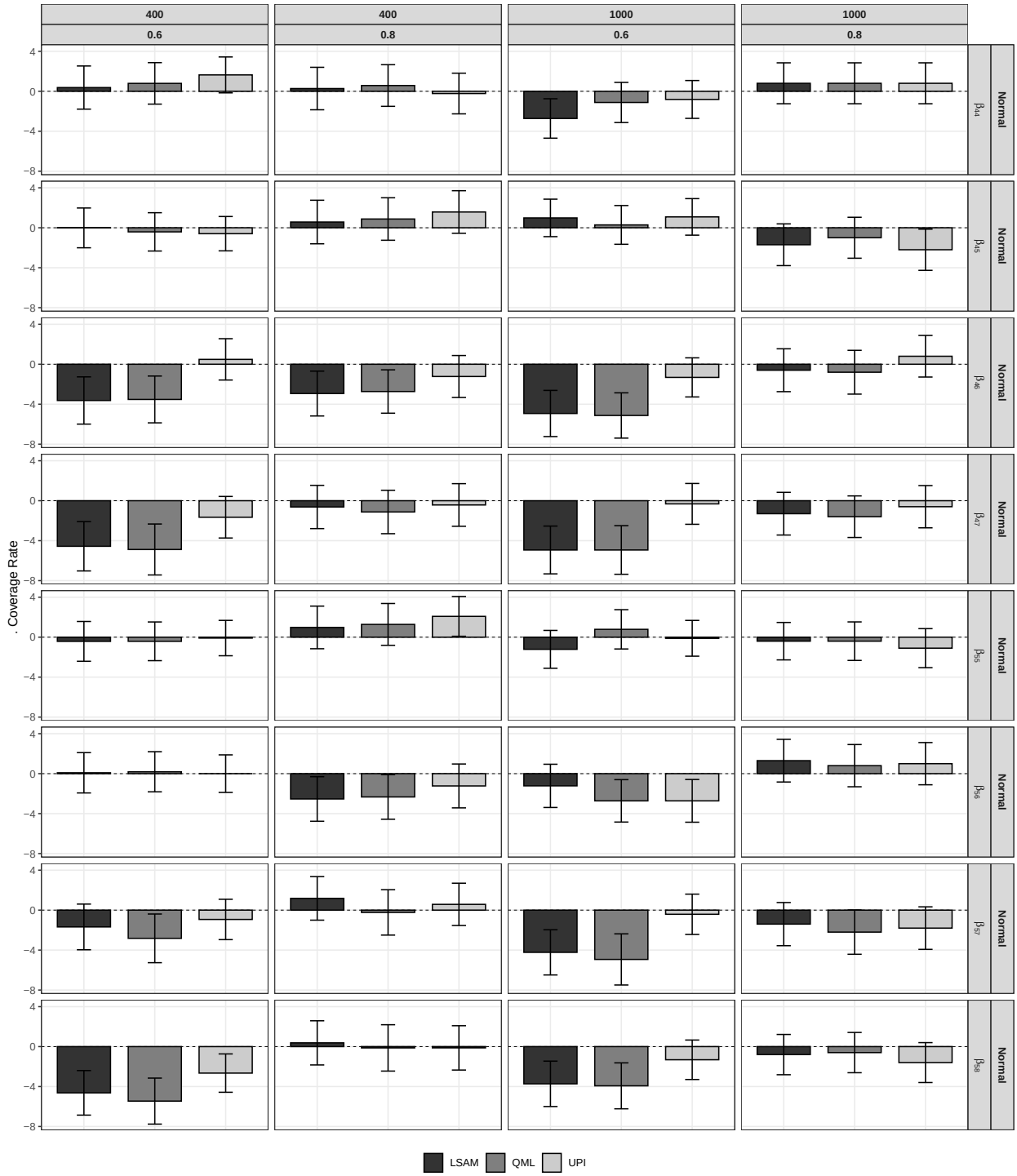


Figure S28

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Normal (Epsilon)

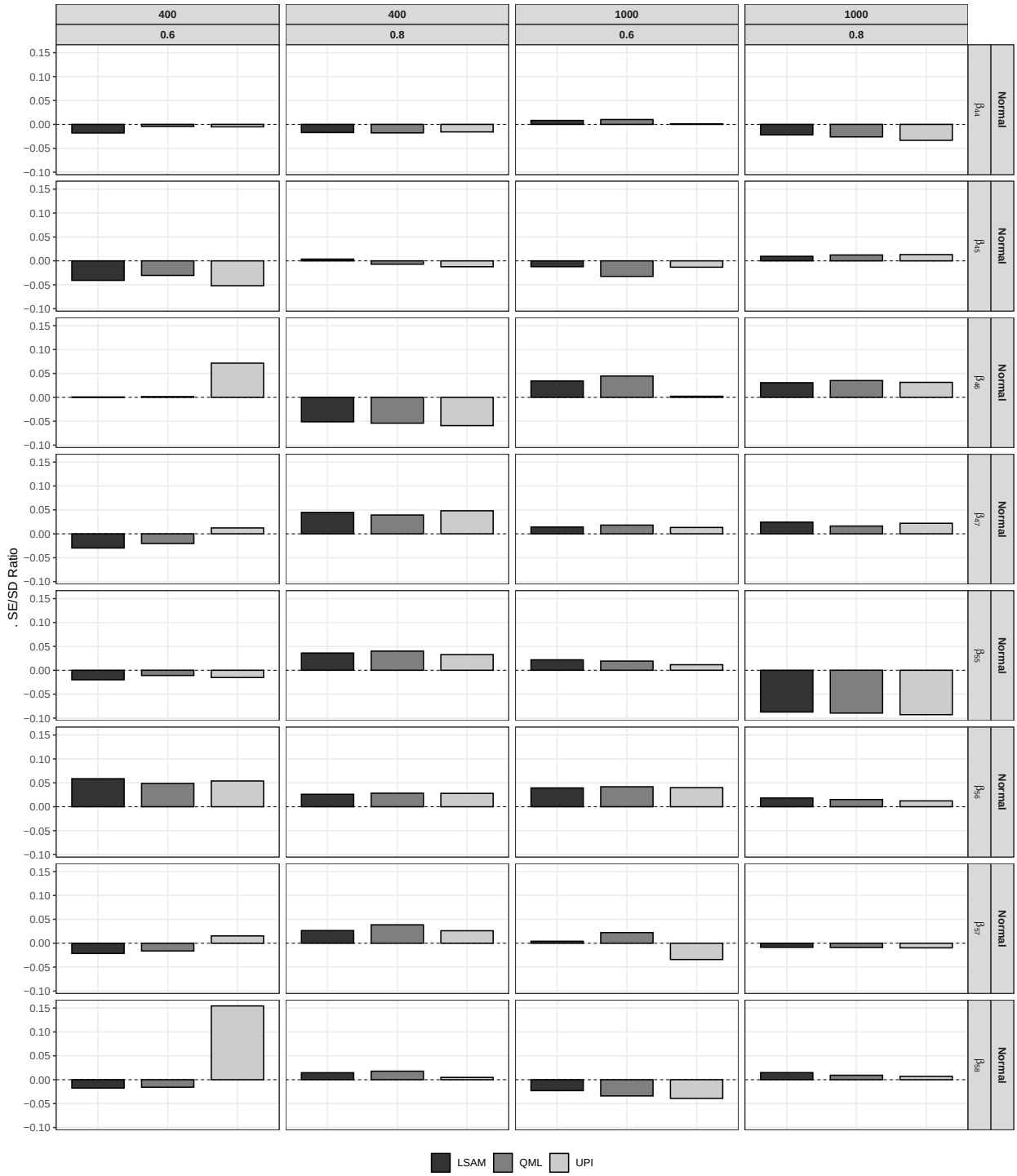


Figure S29

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Normal (Zeta)

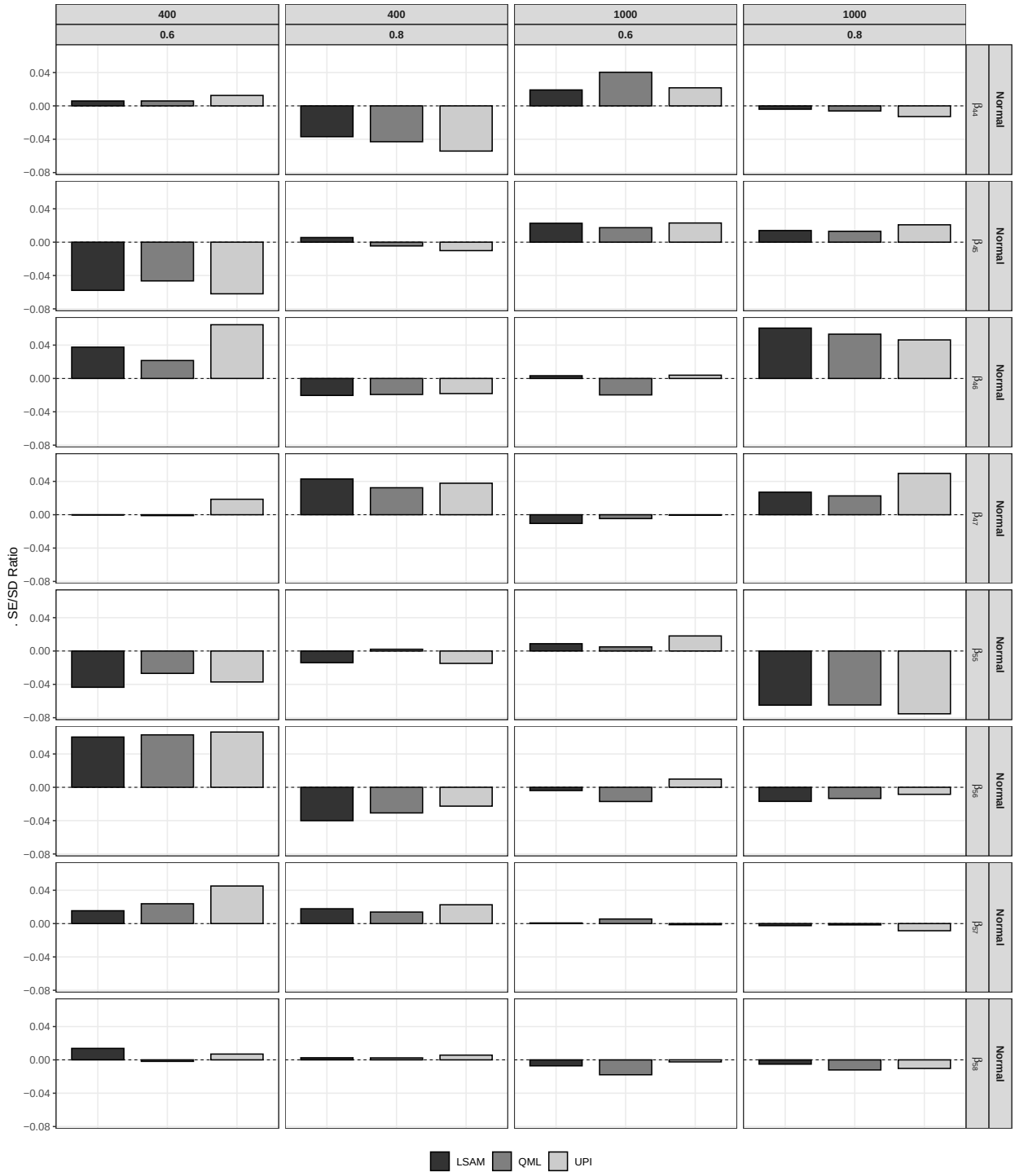


Figure S30

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Normal (Epsilon & Zeta)

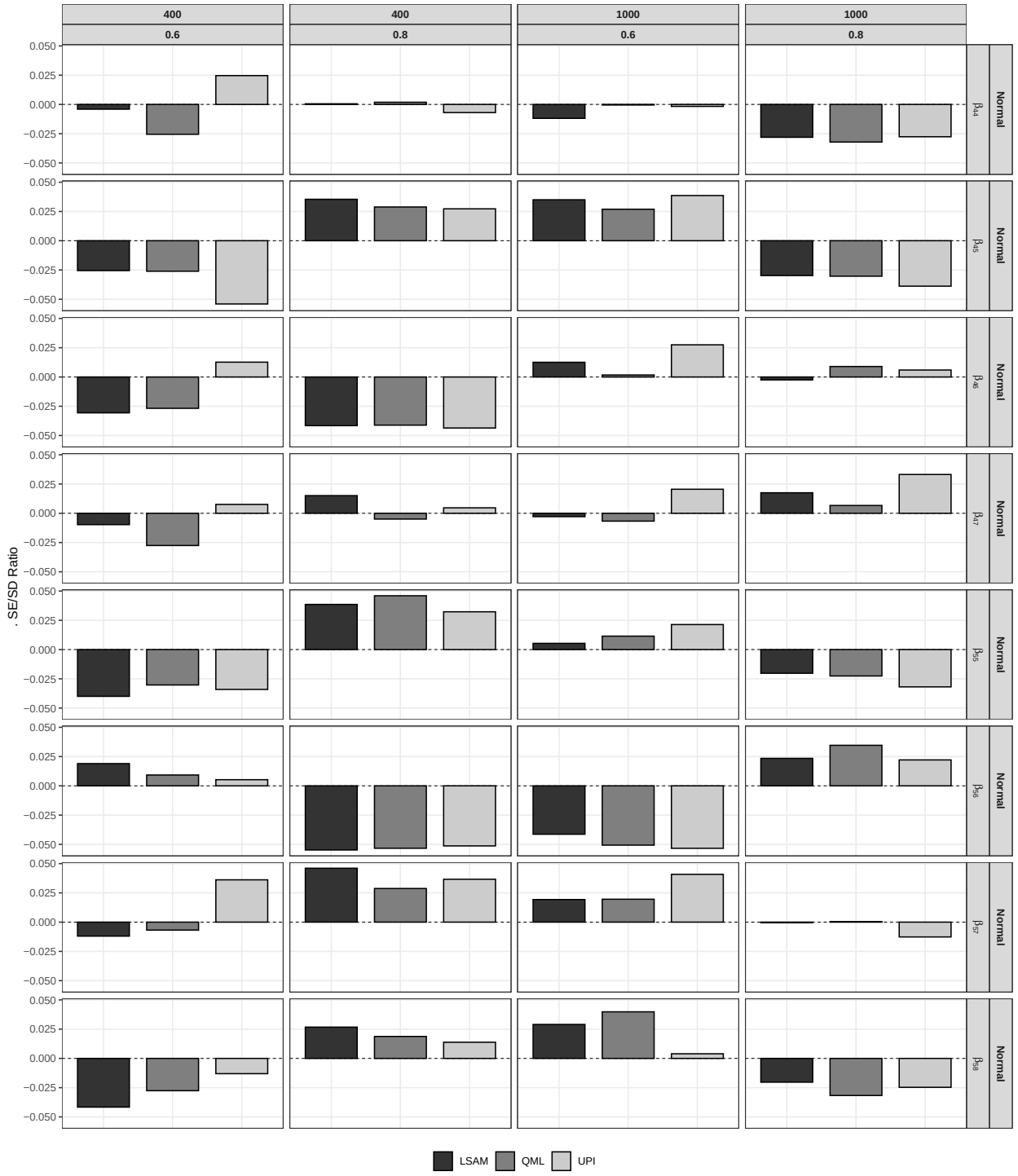


Figure S31

Additional Distributional Conditions Difference: Power for Simulation 2 – Normal (Epsilon)

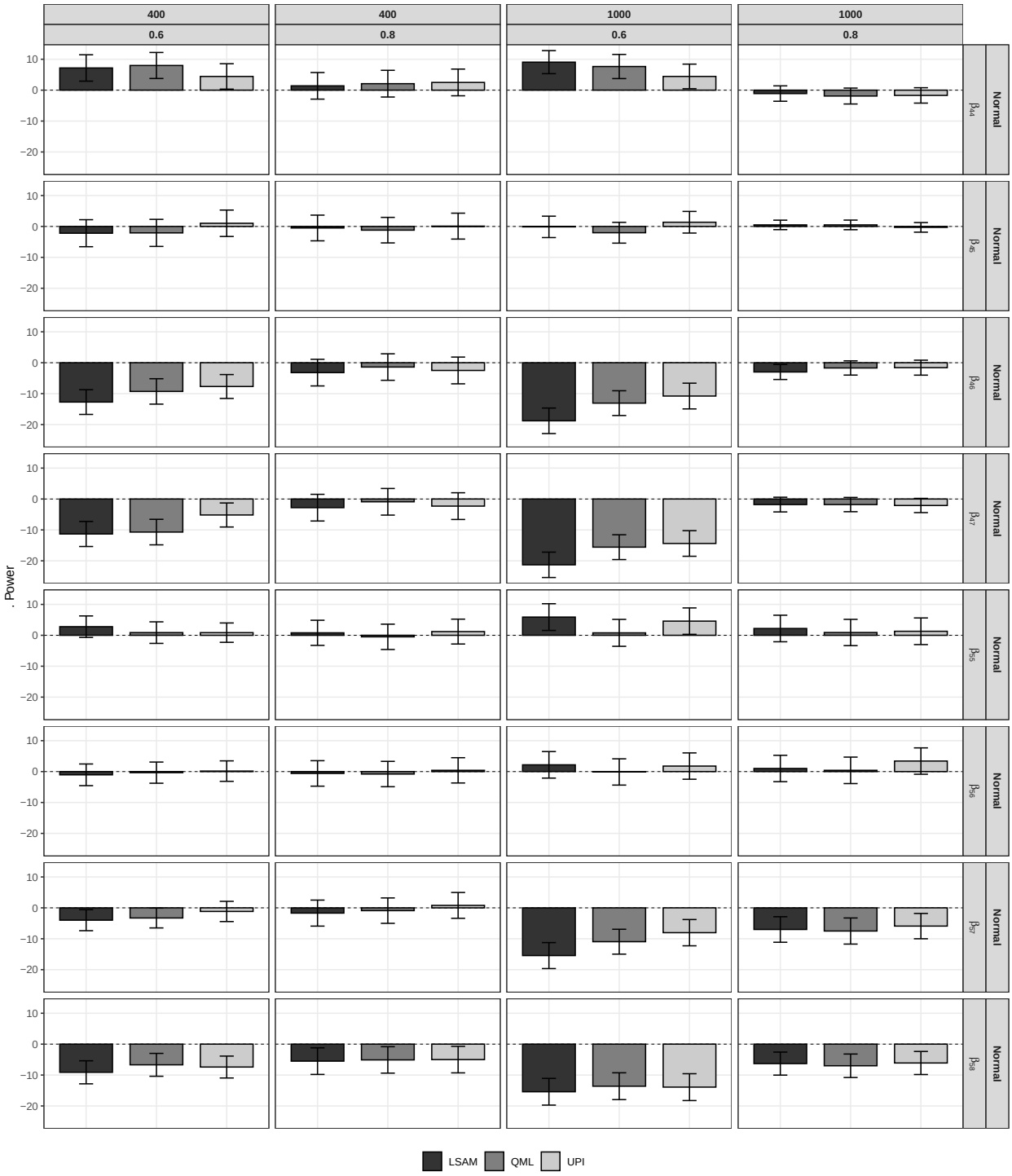


Figure S32

Additional Distributional Conditions Difference: Power for Simulation 2 – Normal (Zeta)

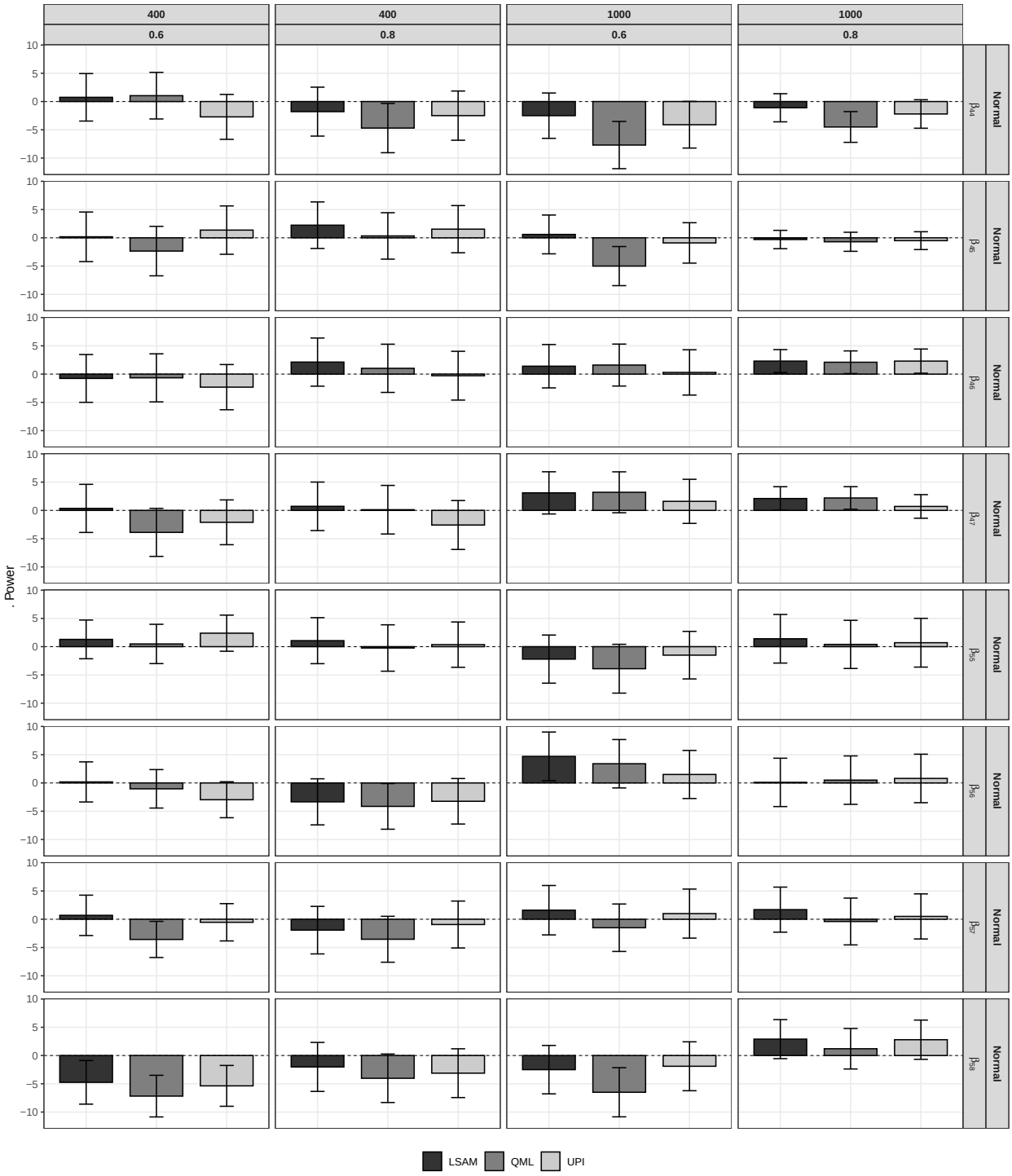


Figure S33

Additional Distributional Conditions Difference: Power for Simulation 2 – Normal (Epsilon & Zeta)

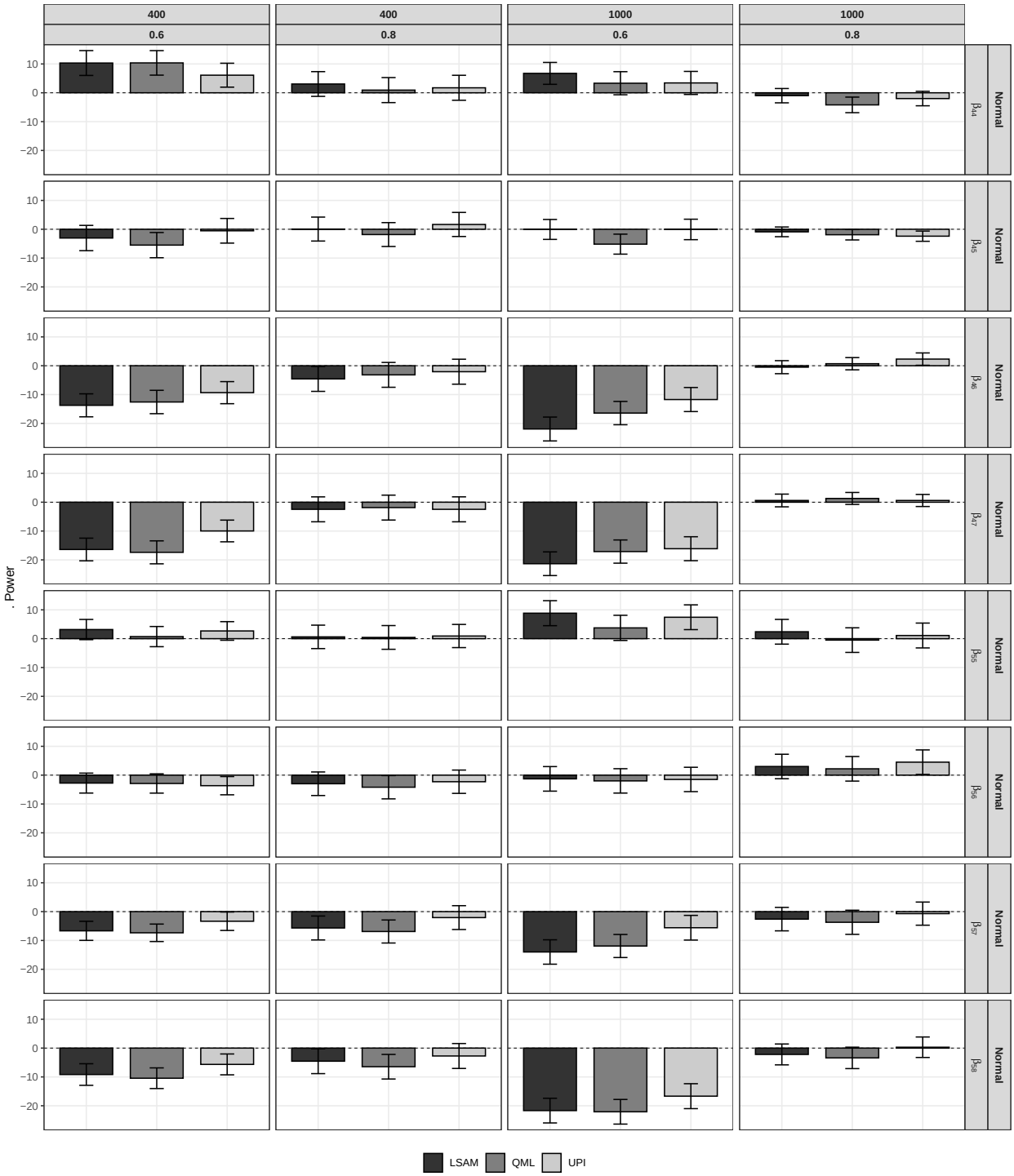


Figure S34

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Normal (Epsilon)

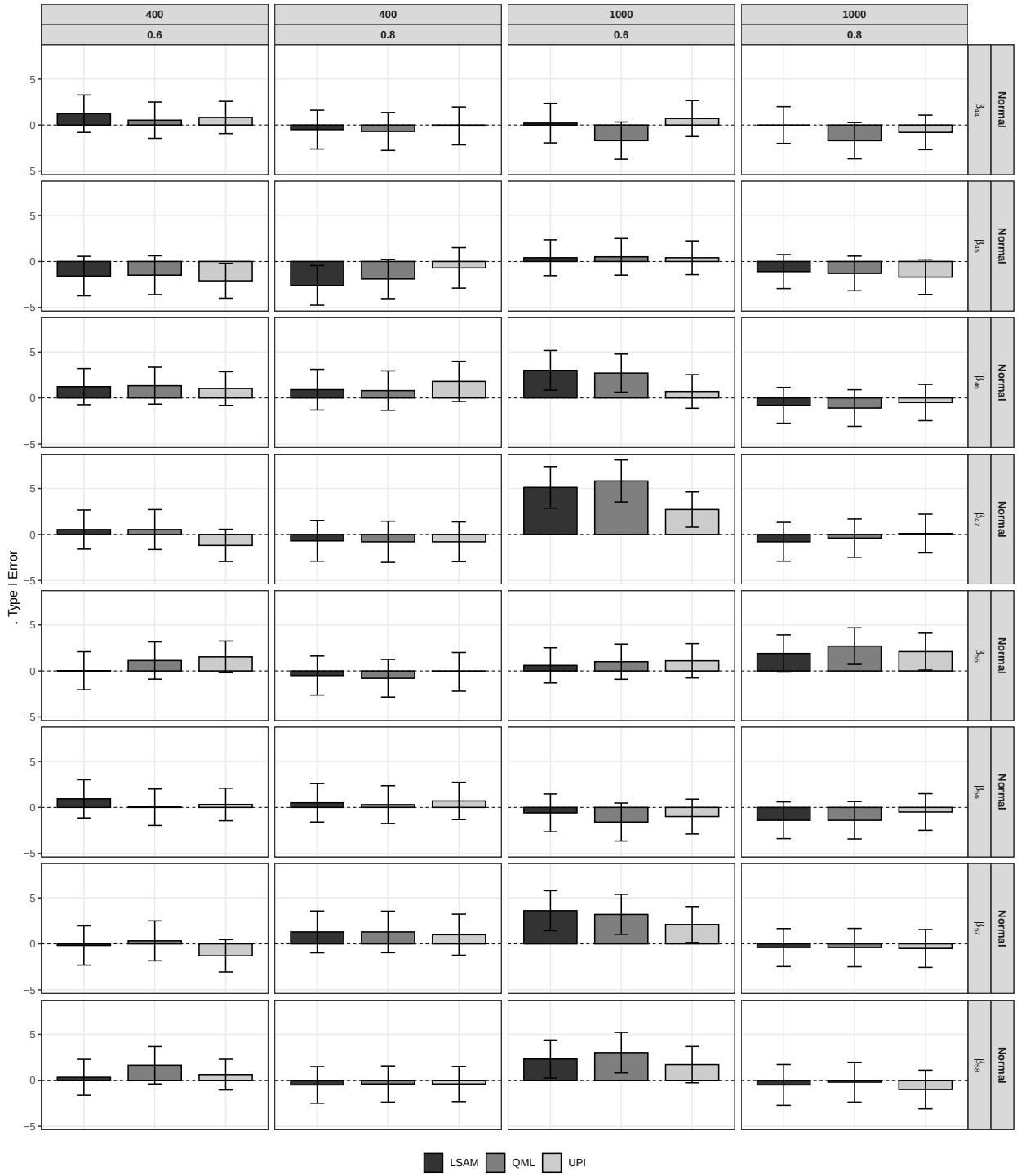


Figure S35

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Normal (Zeta)

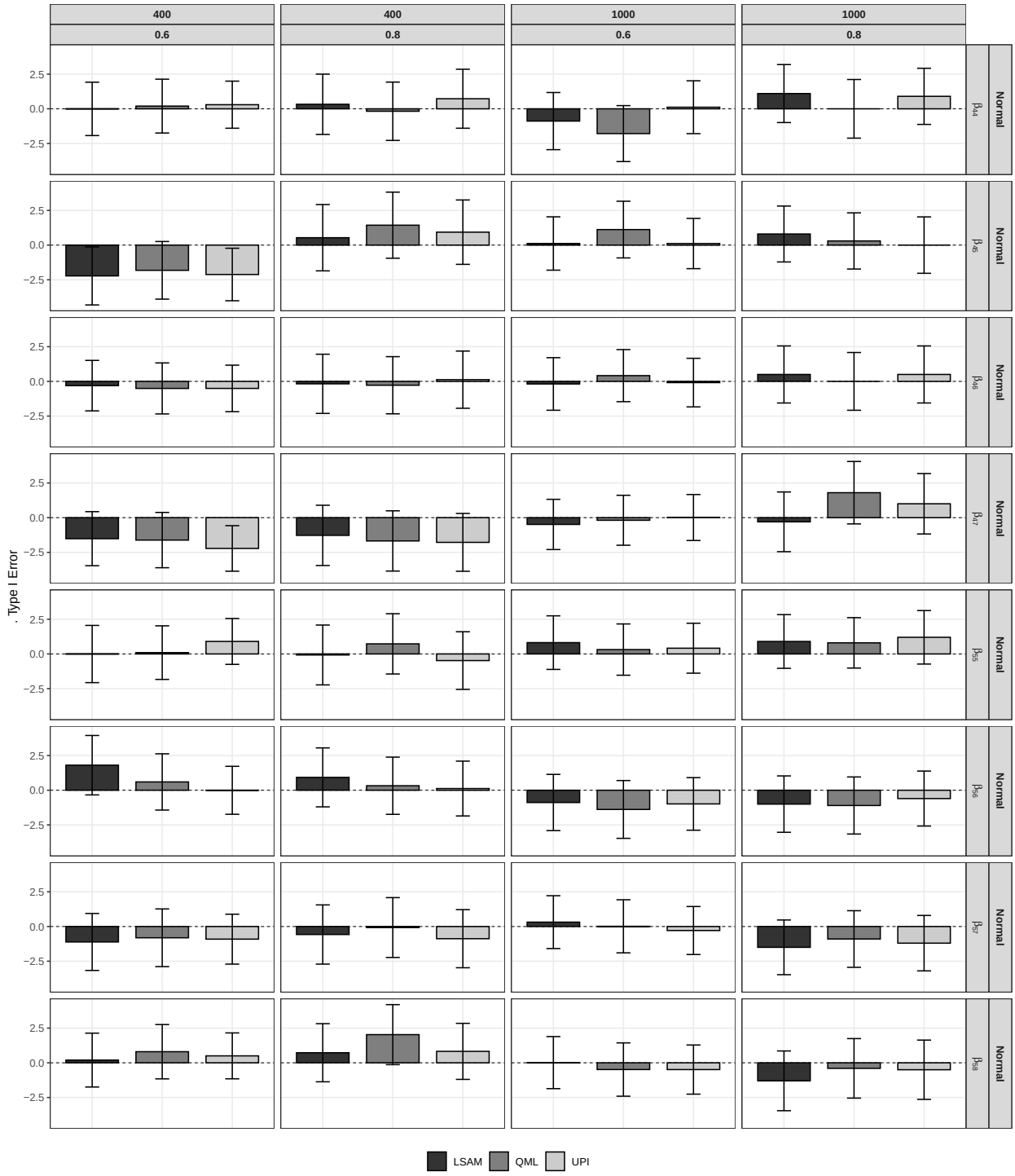


Figure S36

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Normal (Epsilon & Zeta)

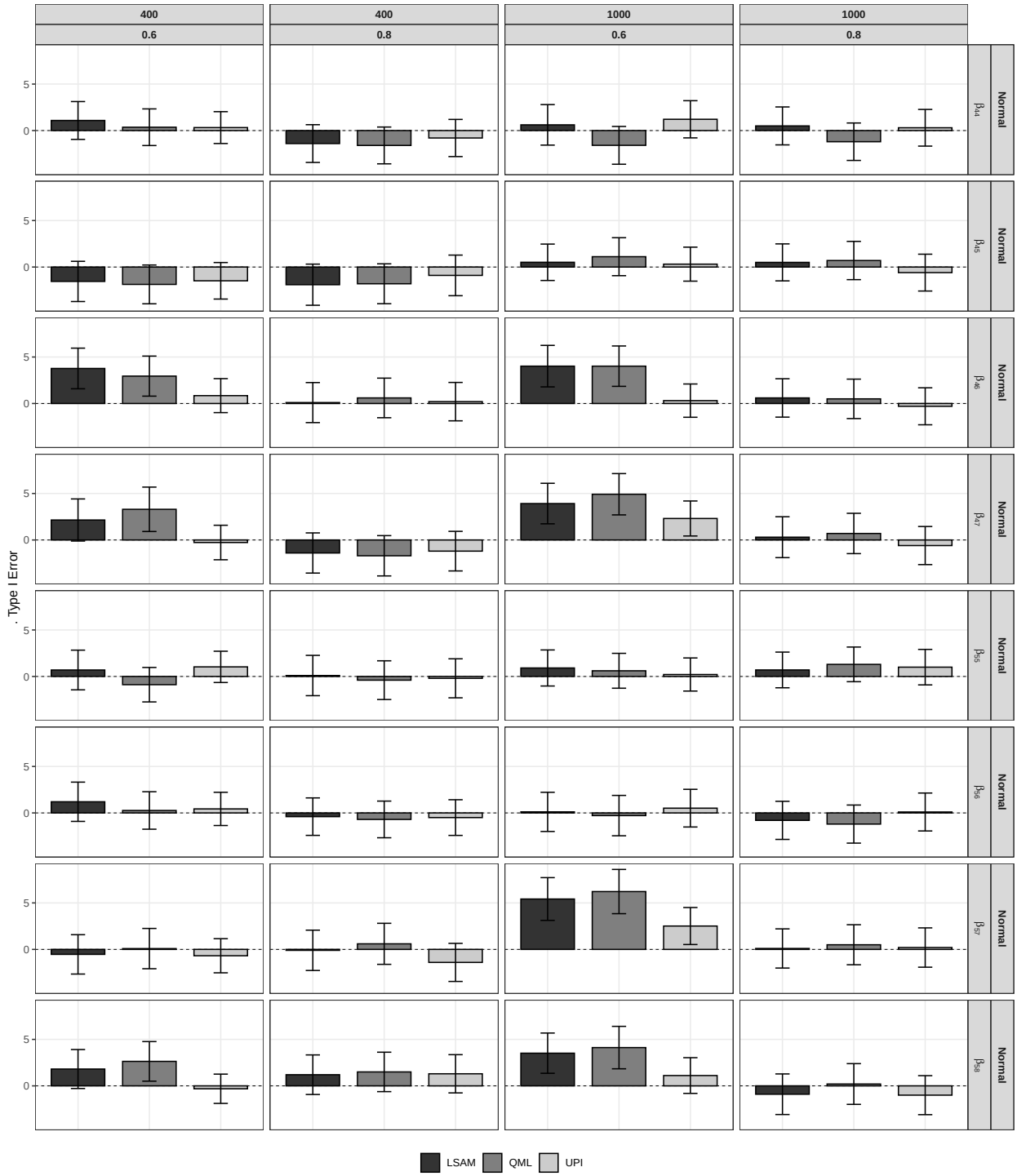


Figure S37

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Right-skewed (Epsilon)

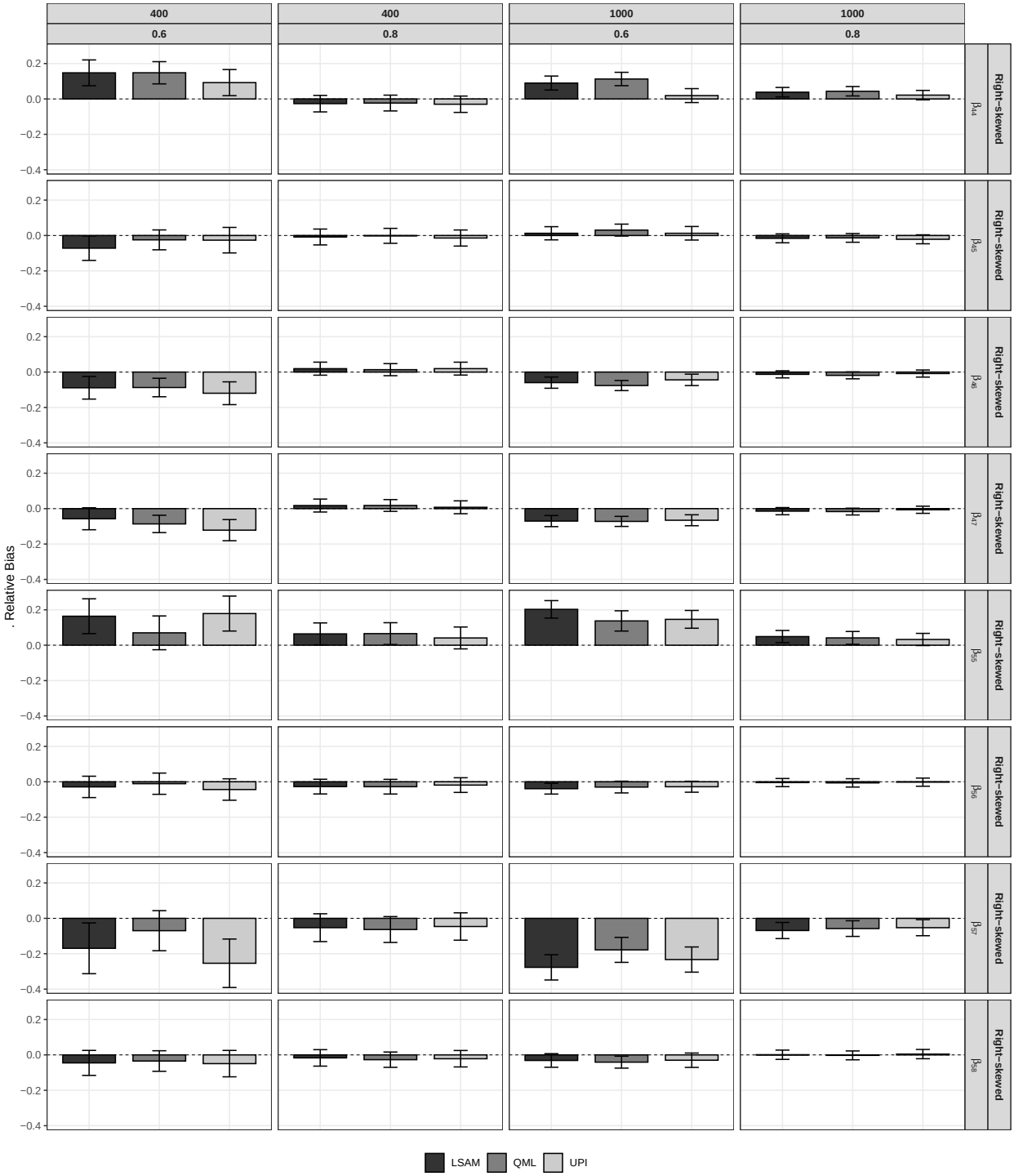


Figure S38

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Right-skewed (Zeta)

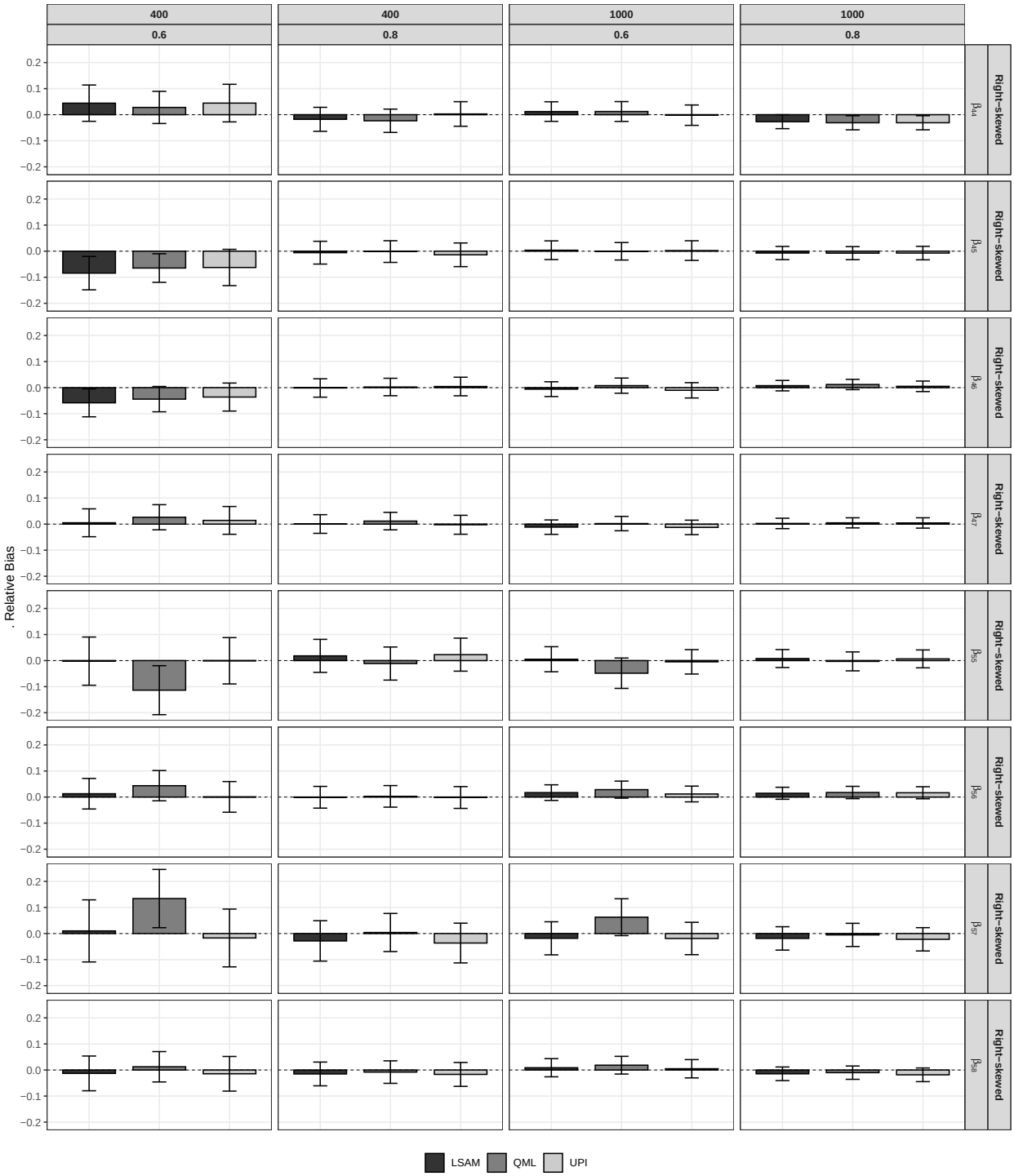


Figure S39

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Right-skewed (Epsilon & Zeta)

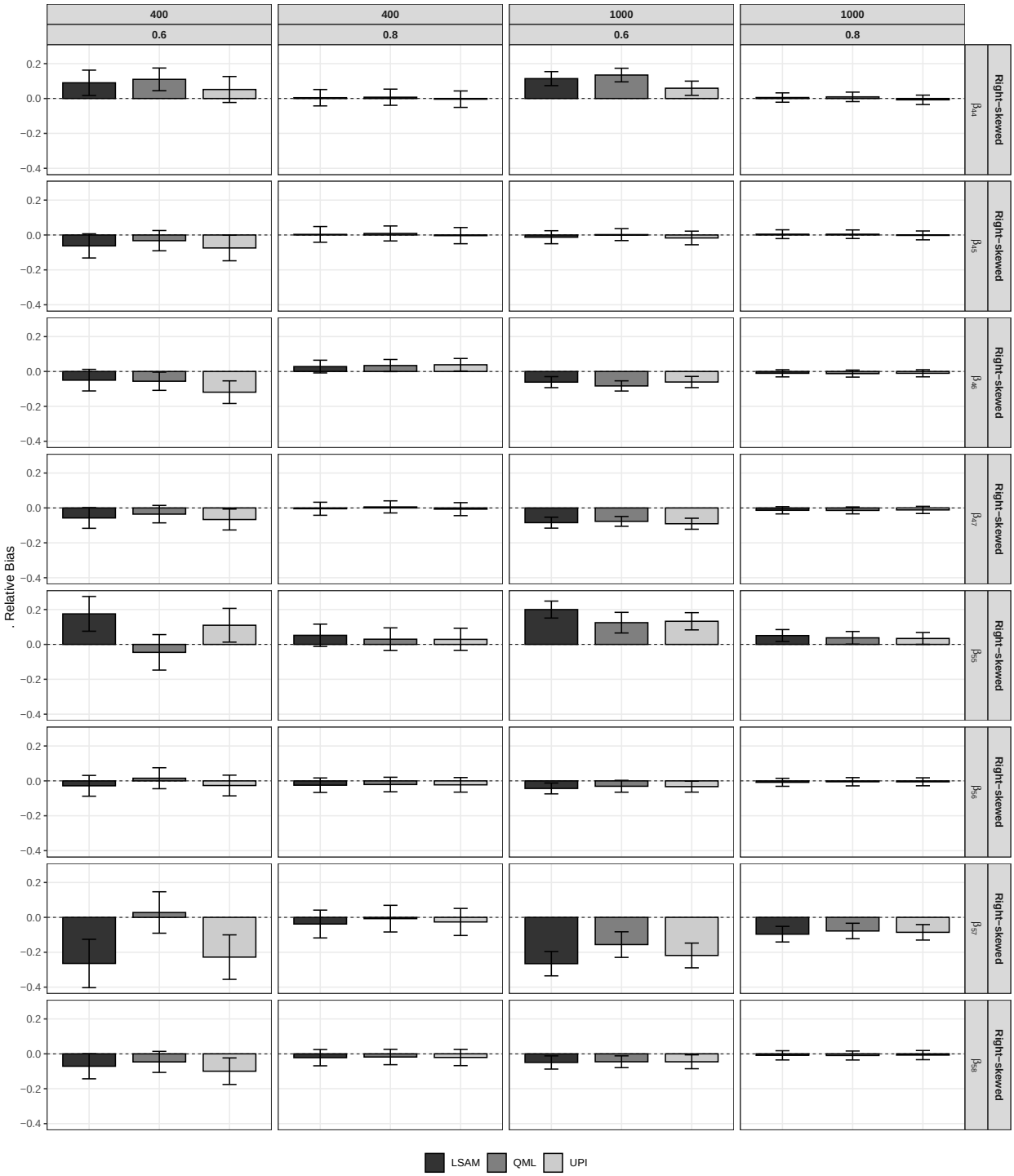


Figure S40

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Right-skewed (Epsilon)

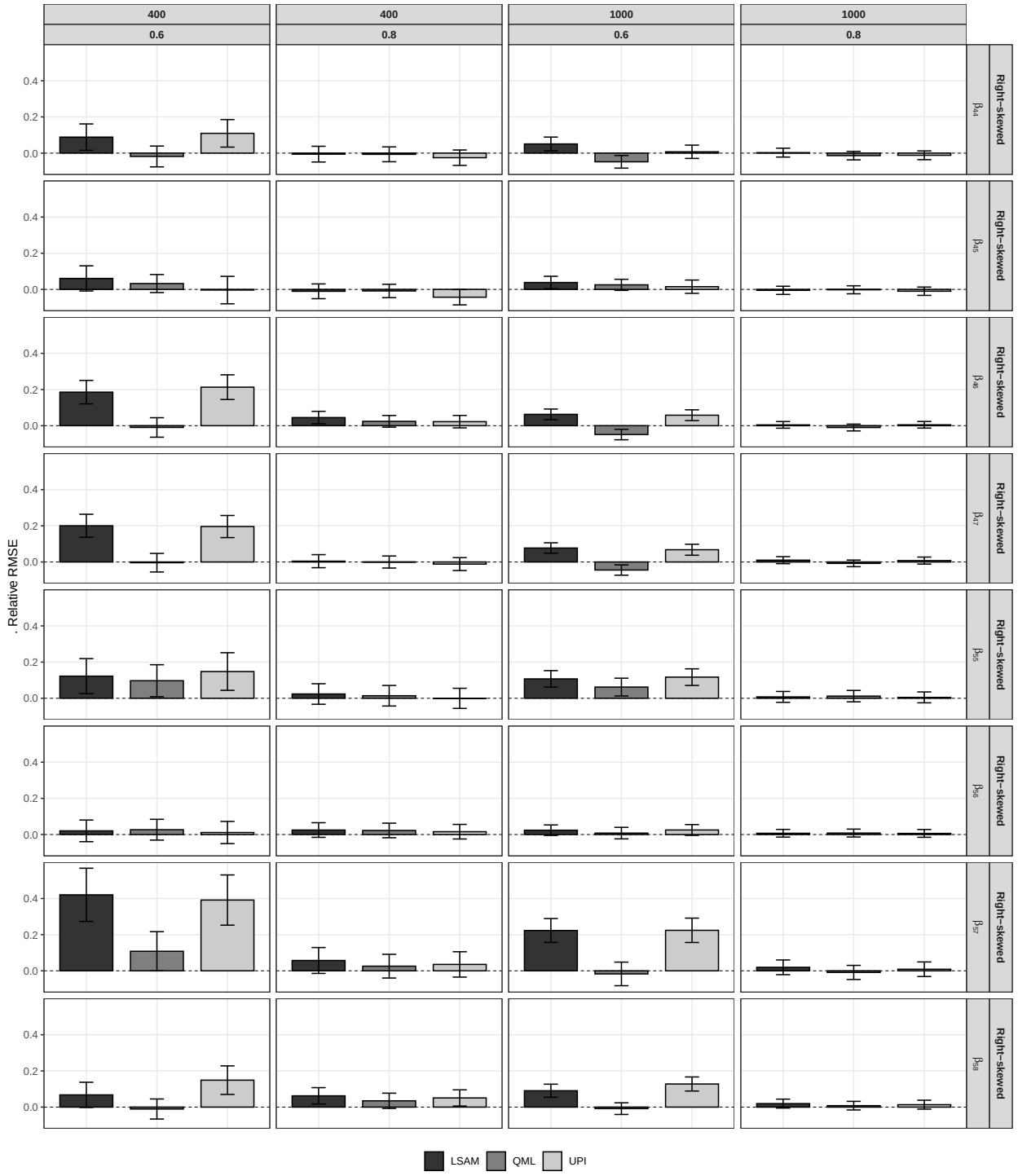


Figure S41

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Right-skewed (Zeta)

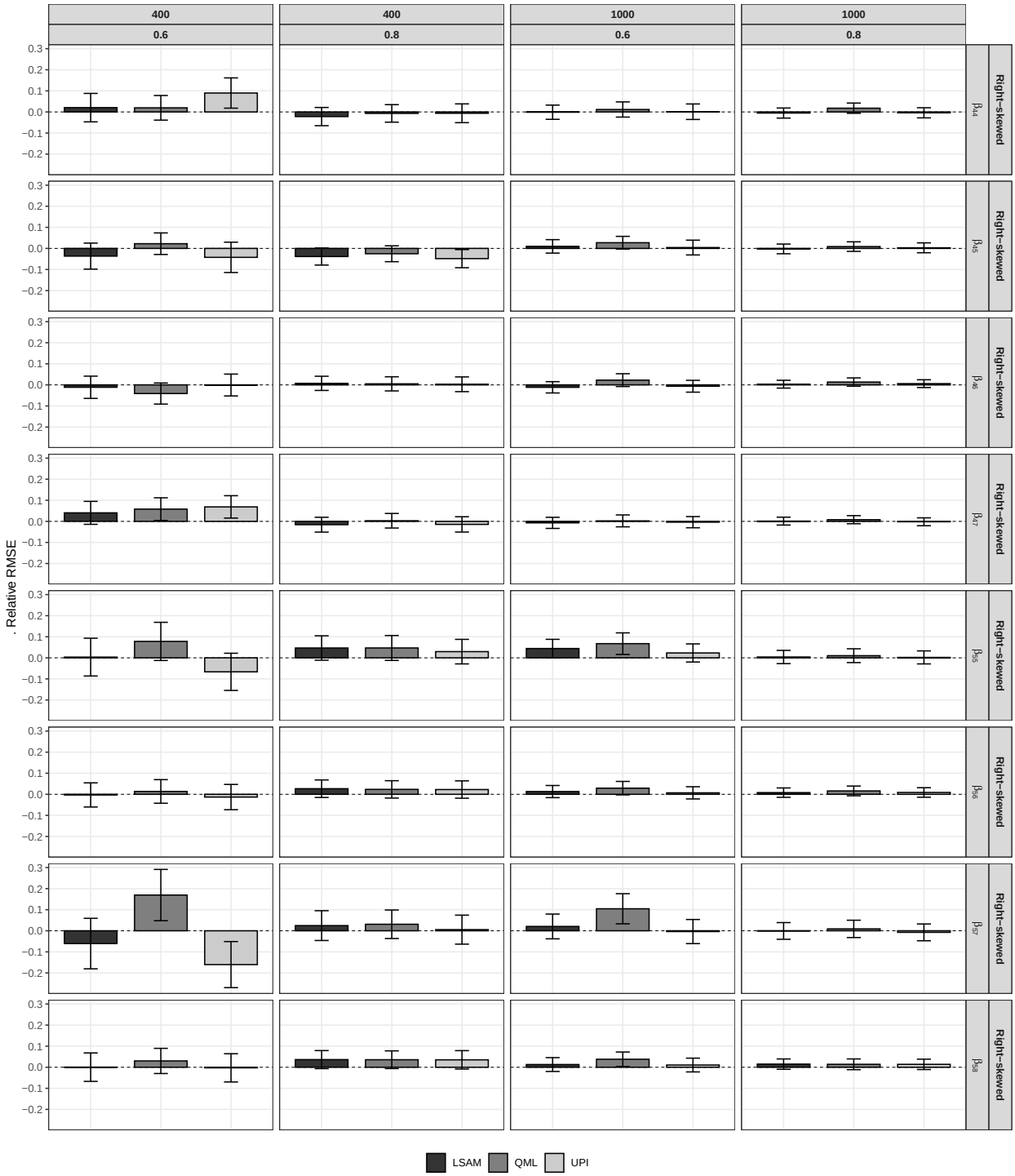


Figure S42

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Right-skewed (Epsilon & Zeta)

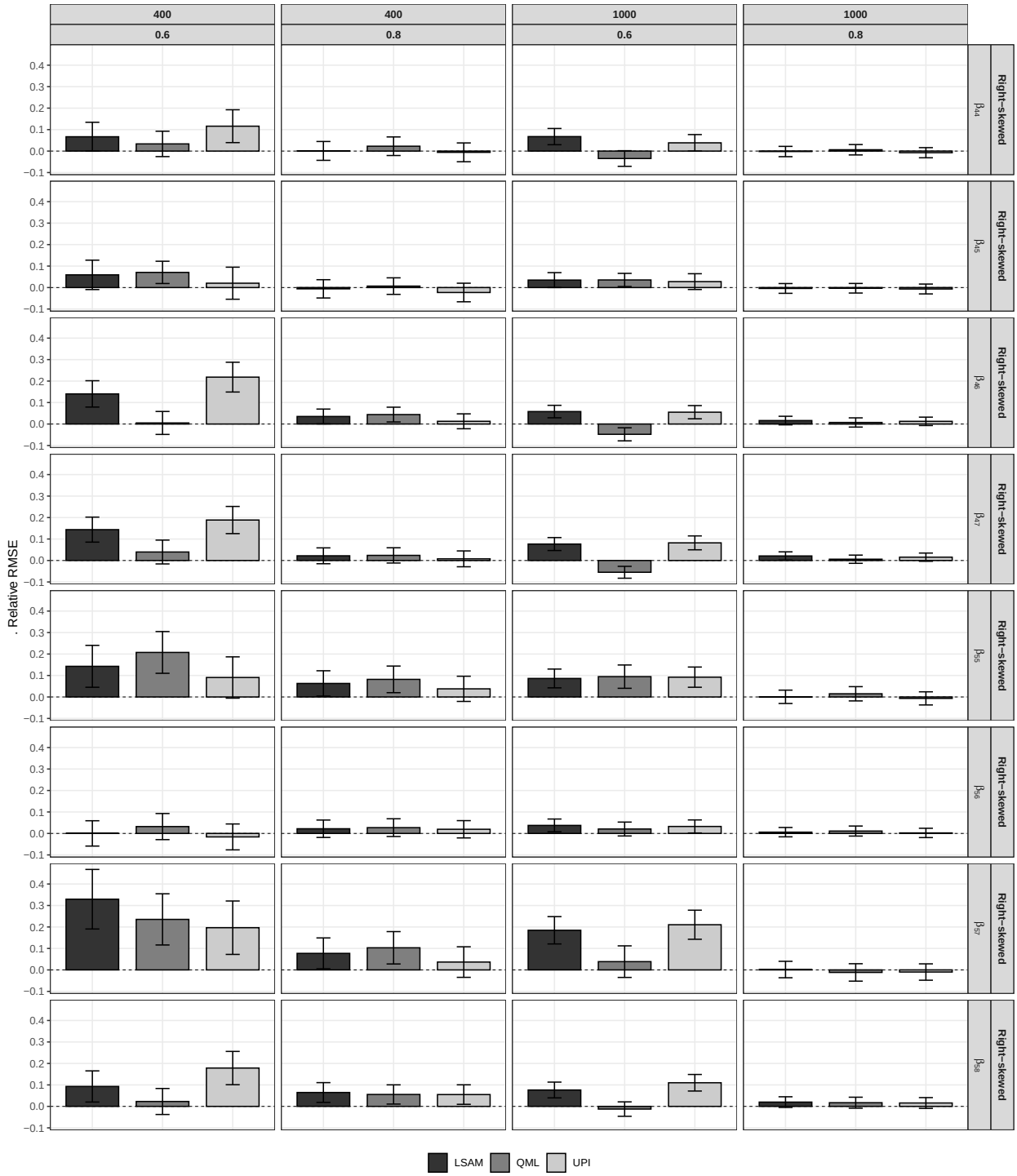


Figure S43

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Right-skewed (Epsilon)

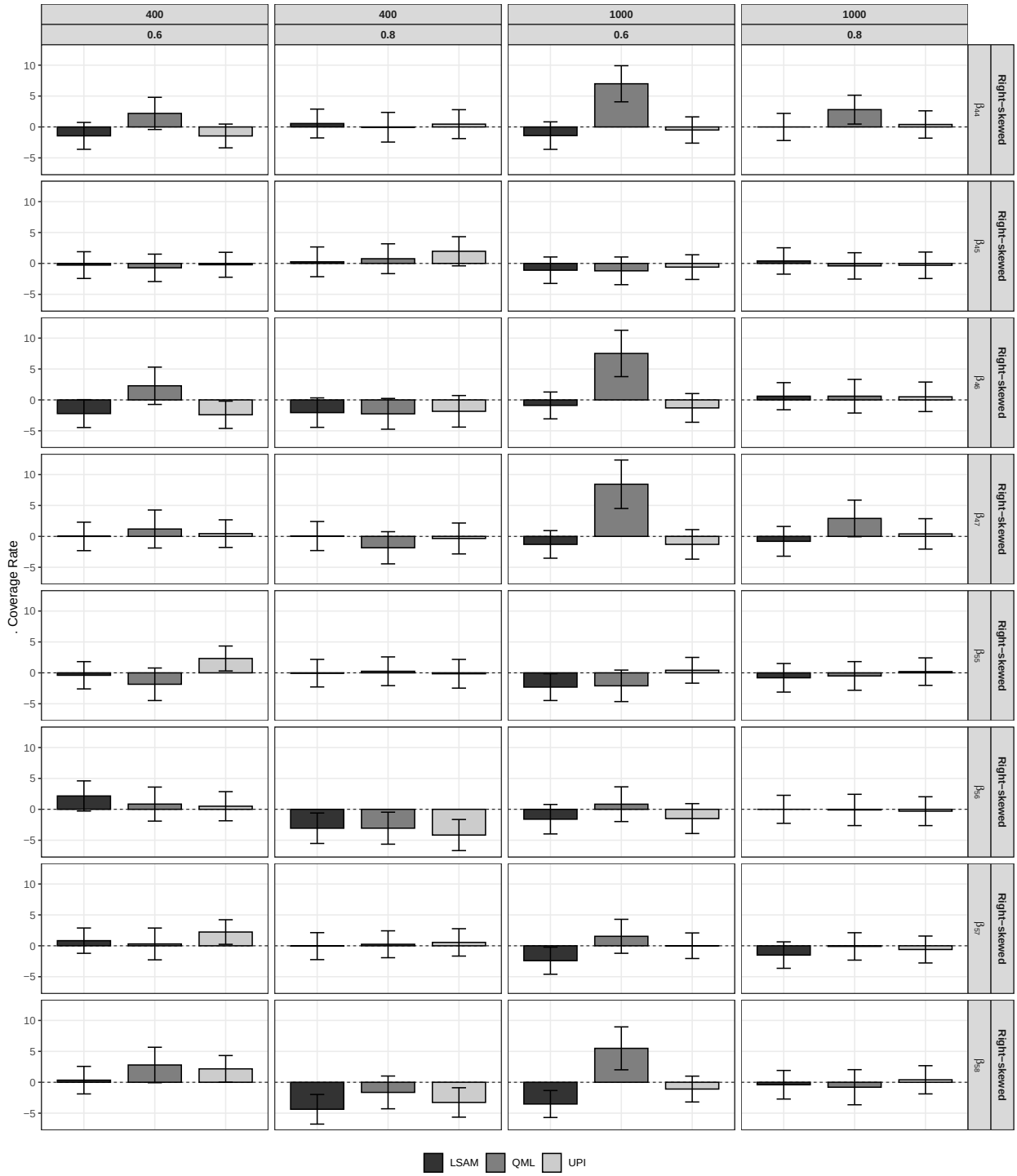


Figure S44

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Right-skewed (Zeta)

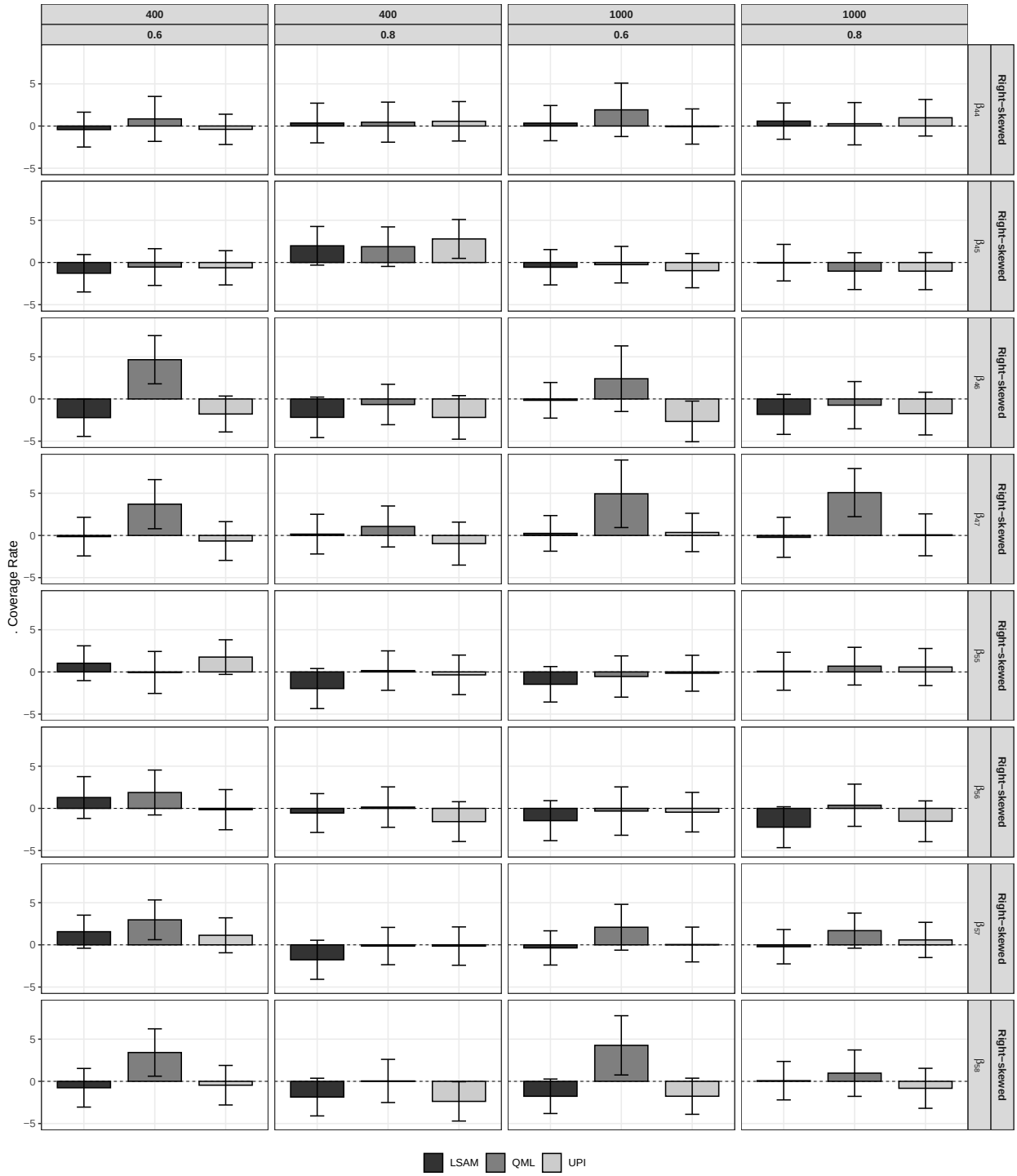


Figure S45

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Right-skewed (Epsilon & Zeta)

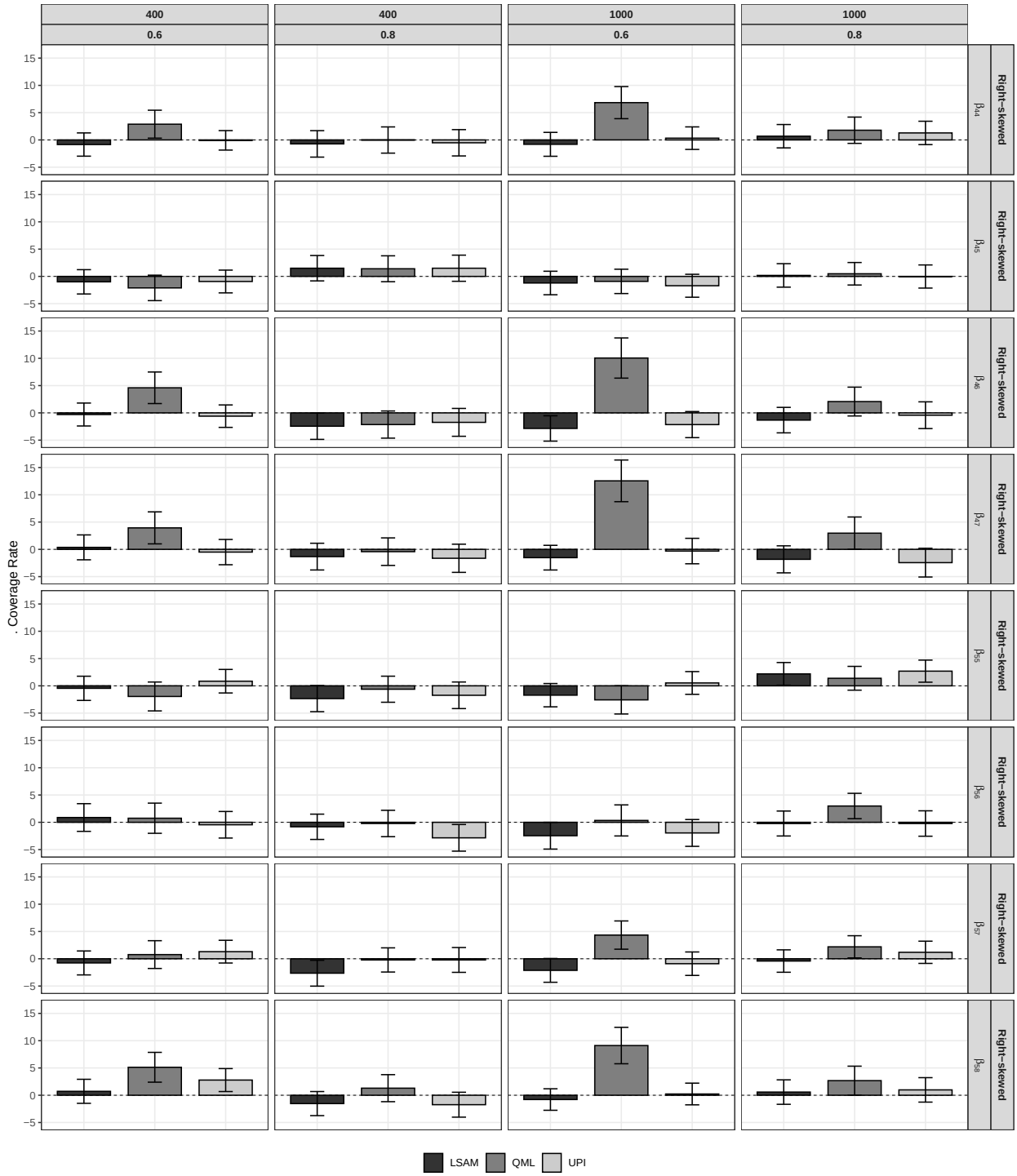


Figure S46

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Right-skewed (Epsilon)

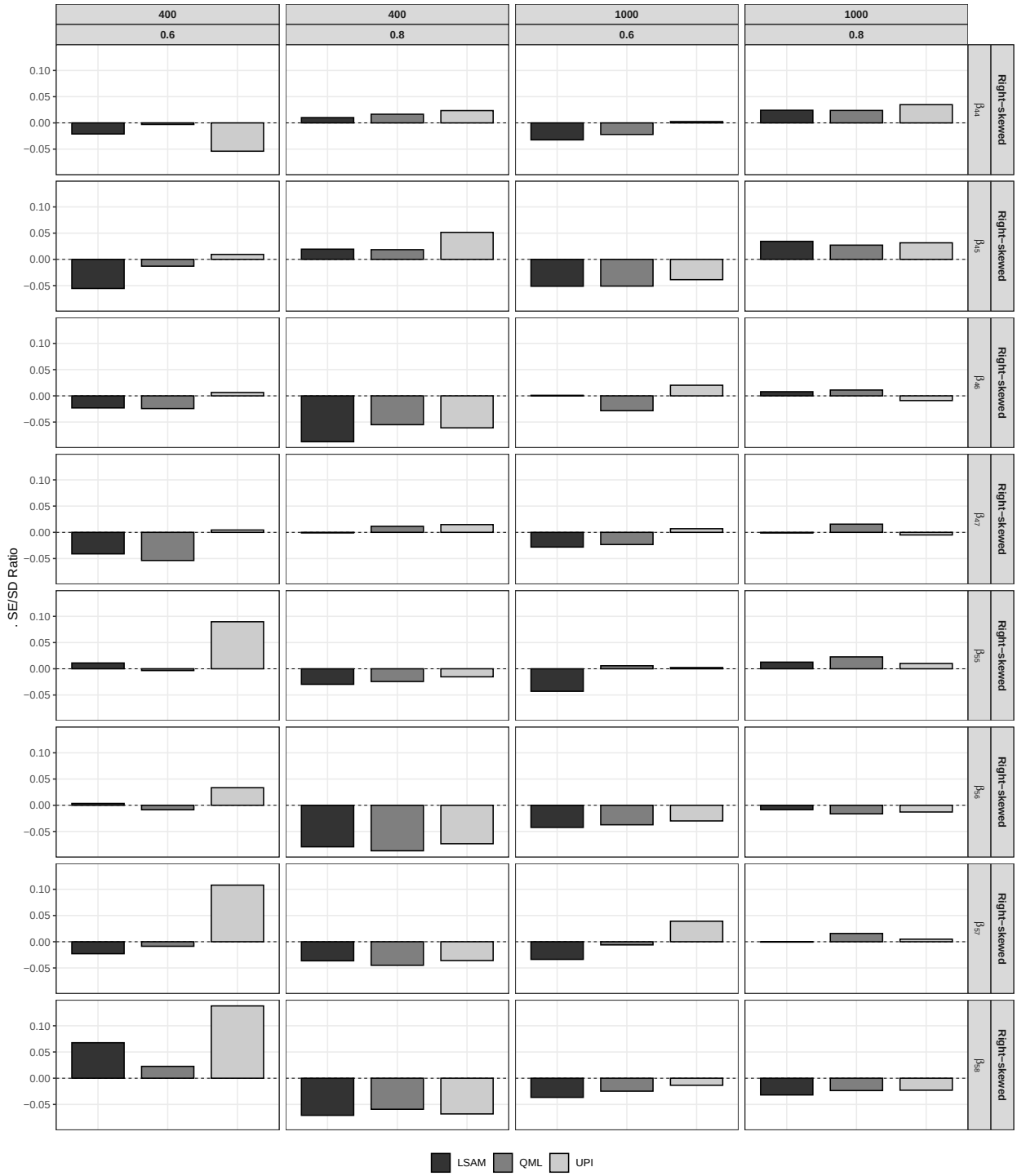


Figure S47

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Right-skewed (Zeta)

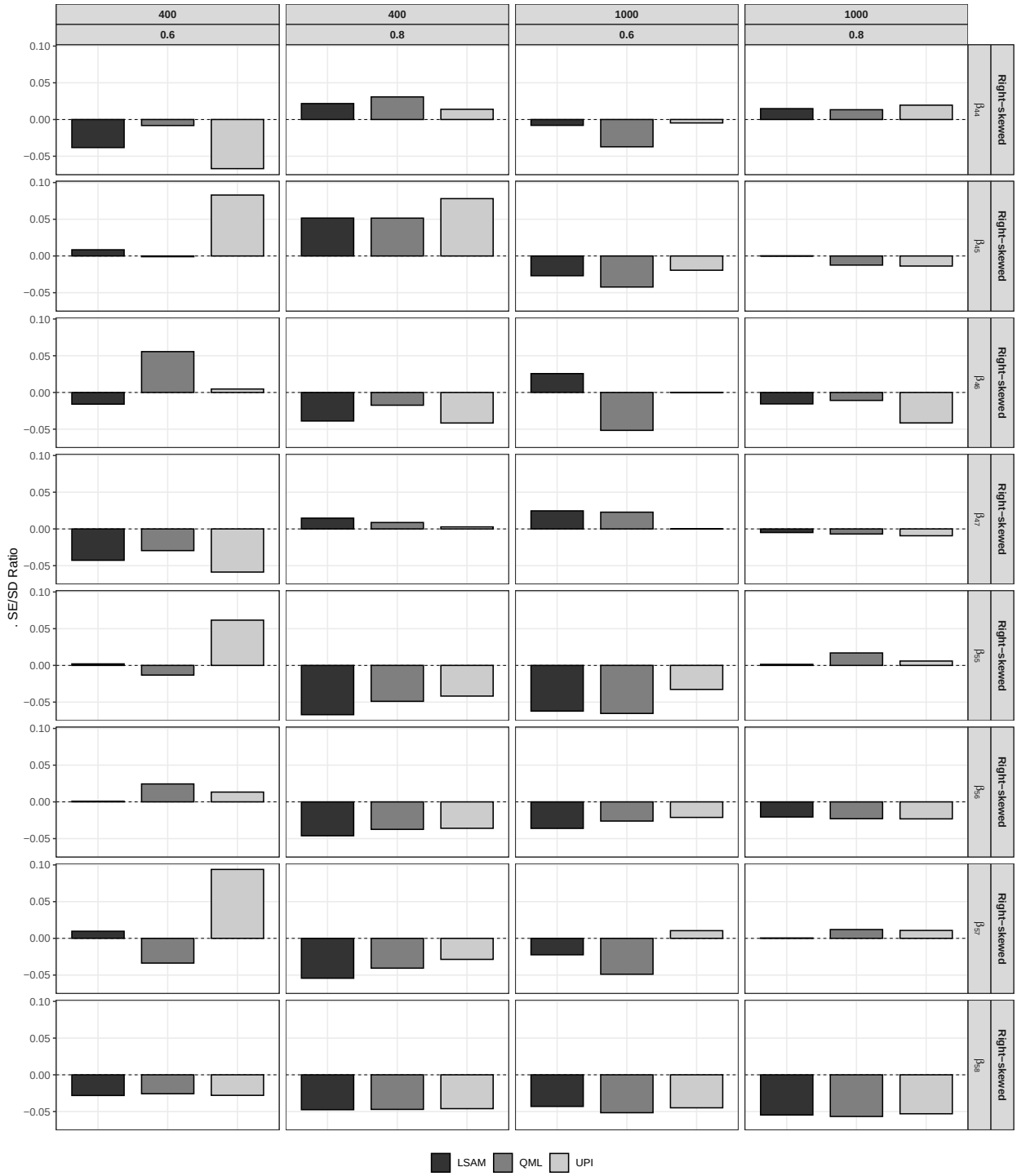


Figure S48

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Right-skewed (Epsilon & Zeta)

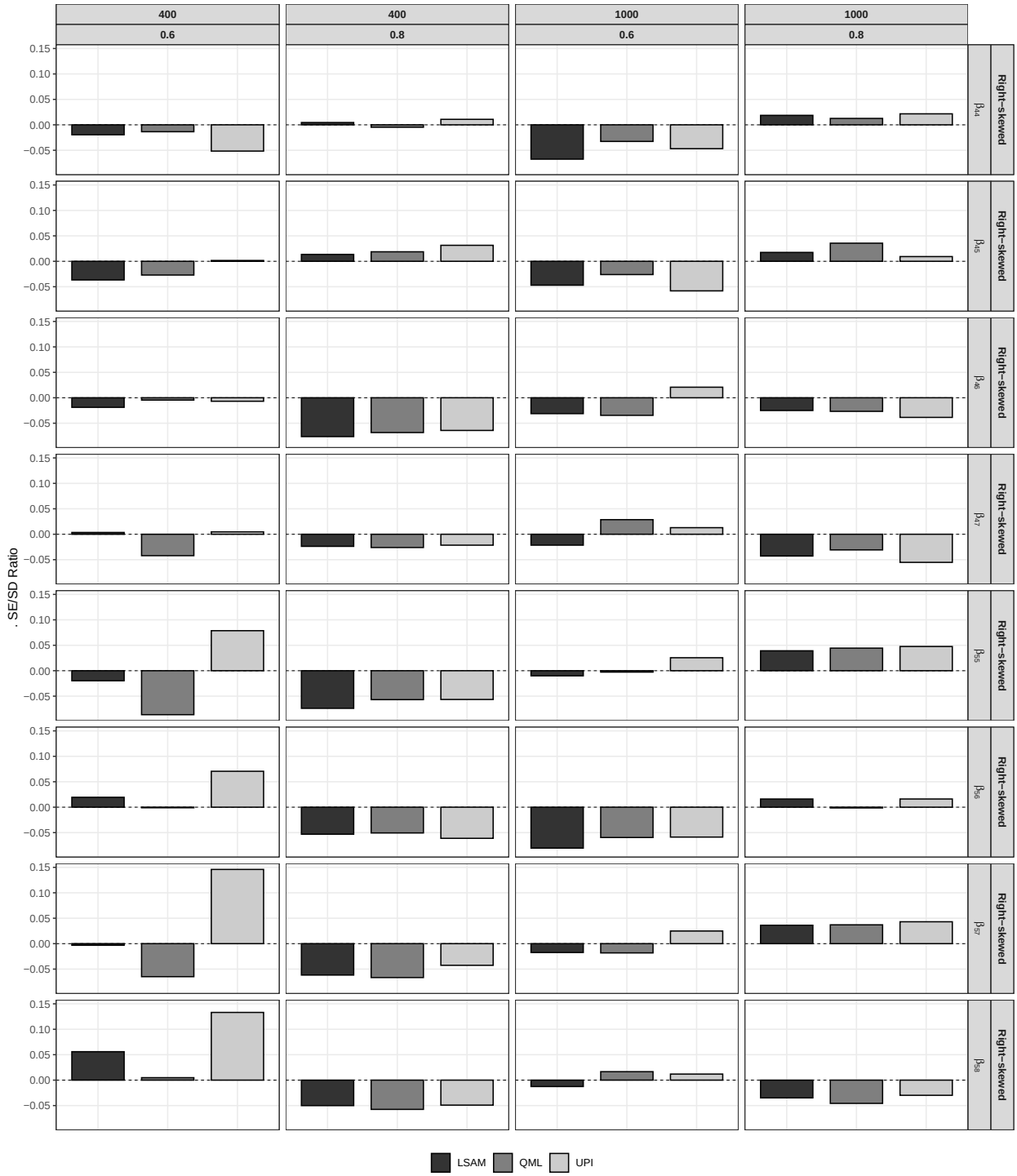


Figure S49

Additional Distributional Conditions Difference: Power for Simulation 2 – Right-skewed (Epsilon)

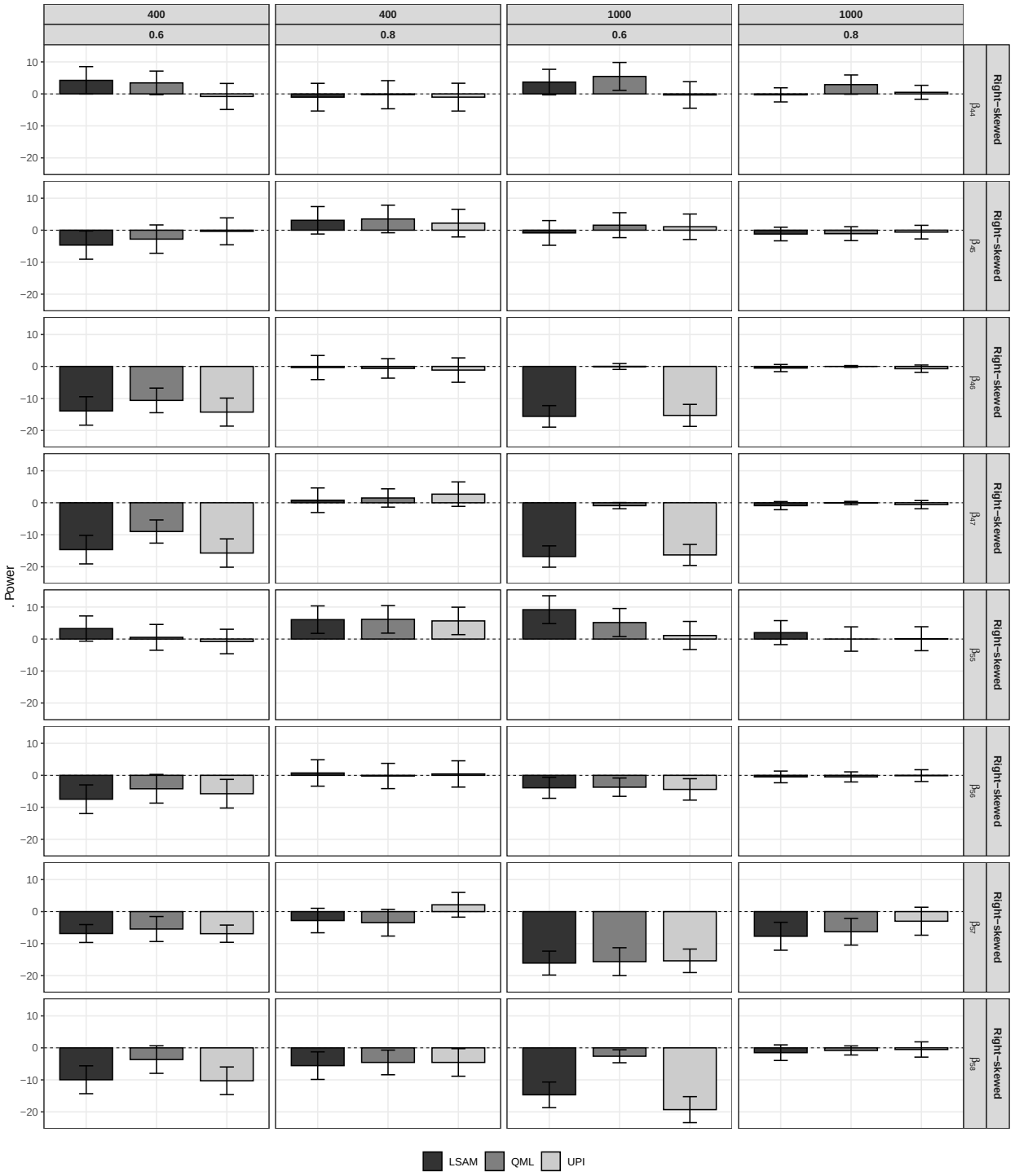


Figure S50

Additional Distributional Conditions Difference: Power for Simulation 2 – Right-skewed (Zeta)

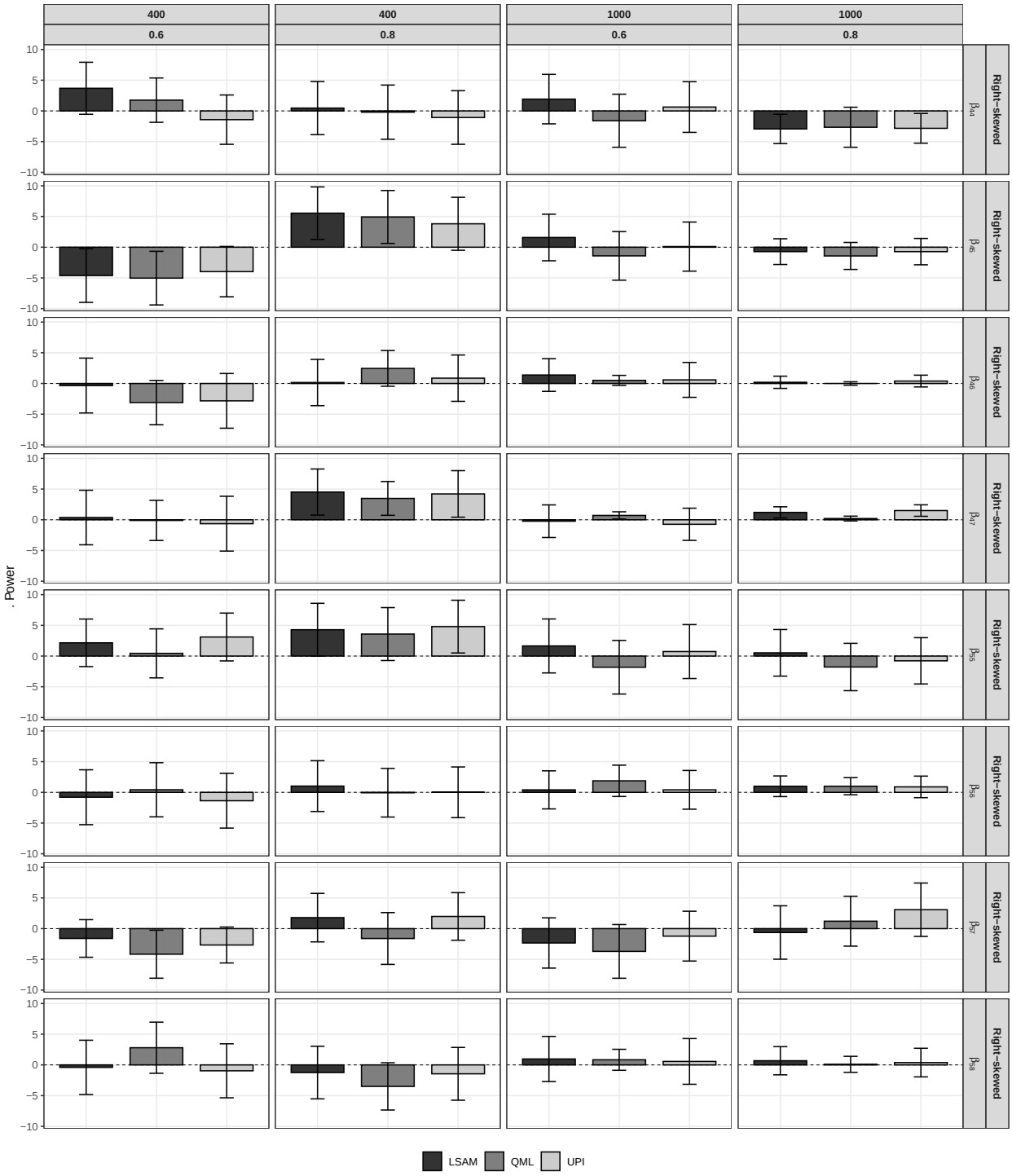


Figure S51

Additional Distributional Conditions Difference: Power for Simulation 2 – Right-skewed (Epsilon & Zeta)

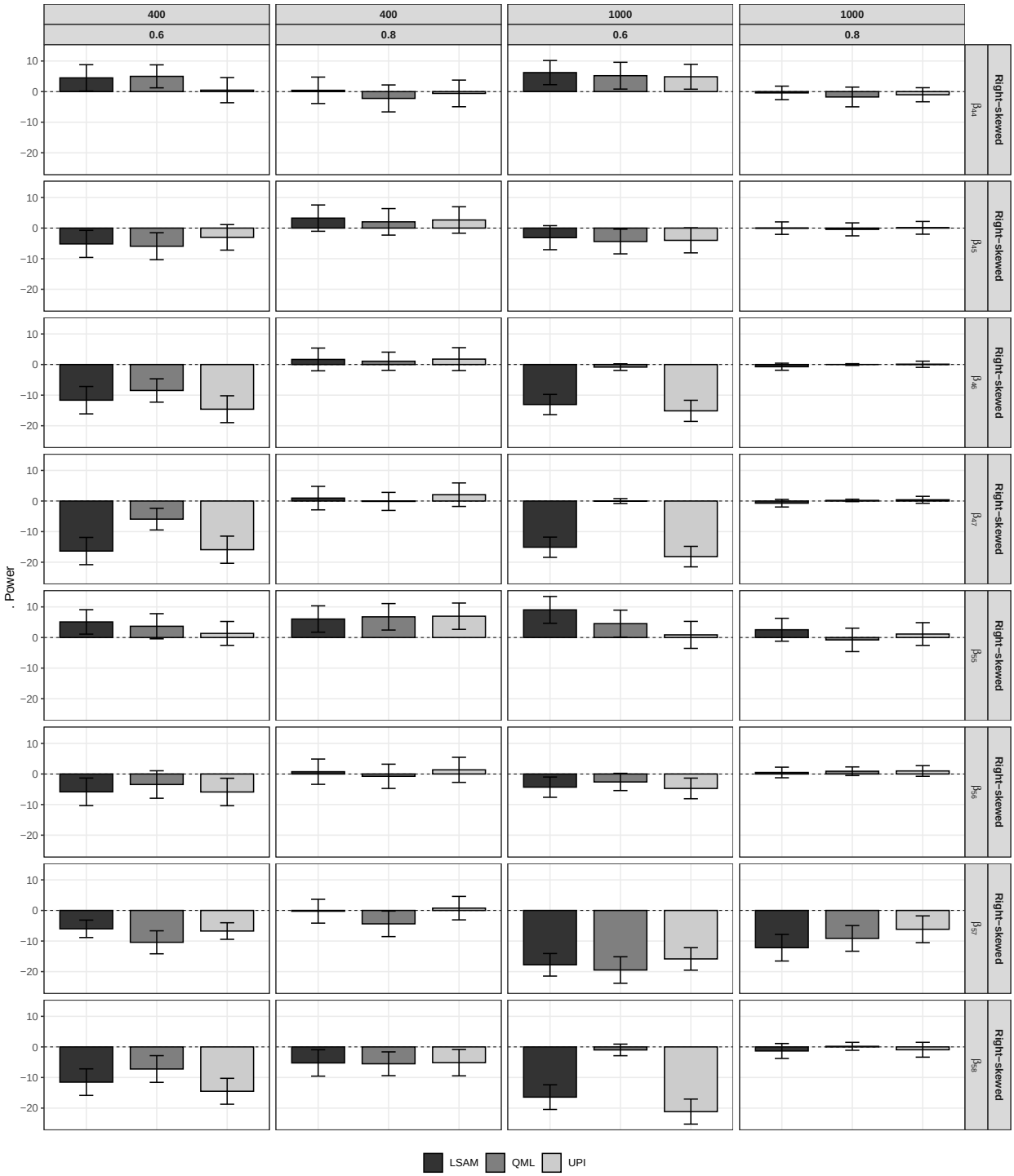


Figure S52

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Right-skewed (Epsilon)

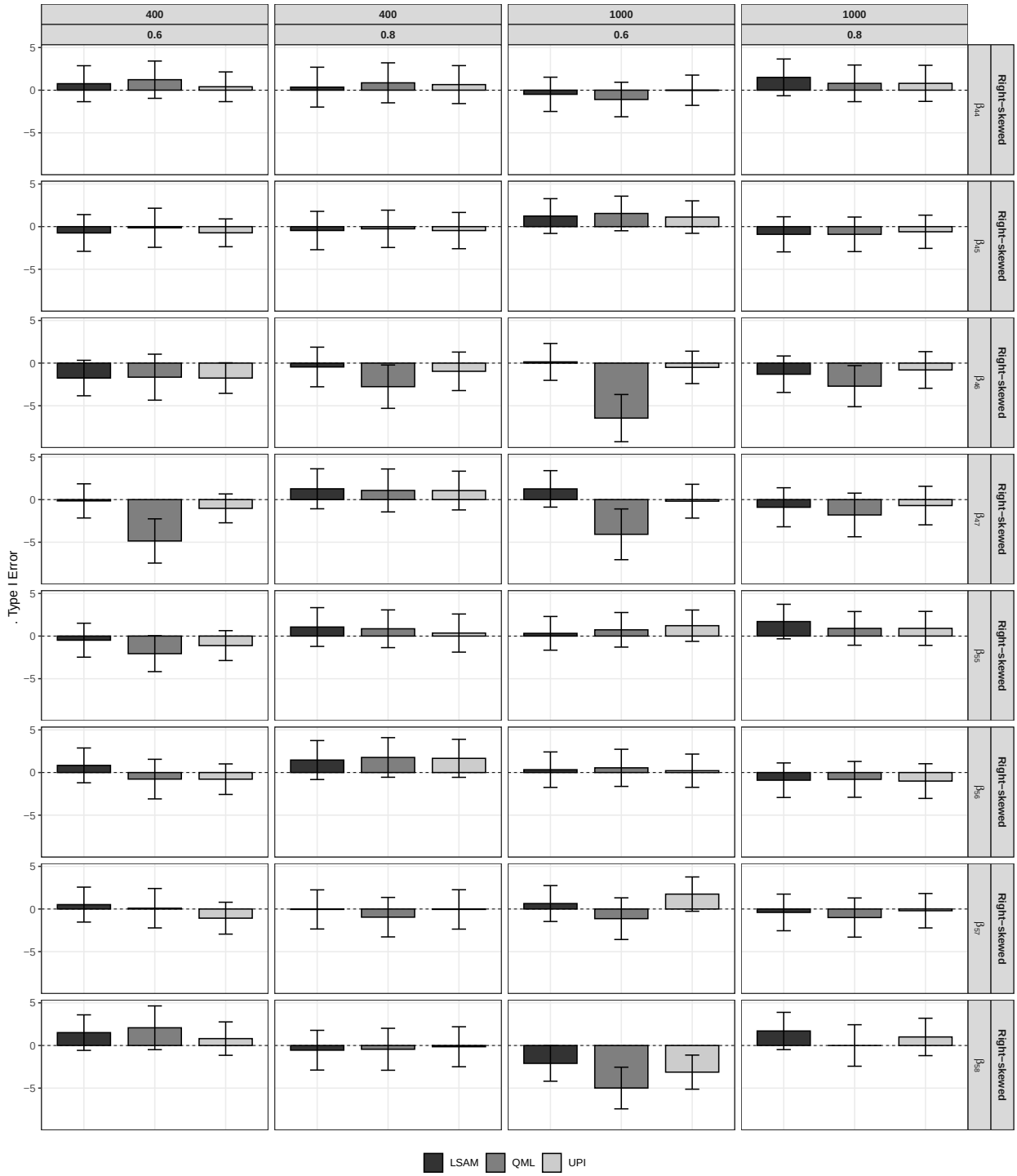


Figure S53

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Right-skewed (Zeta)

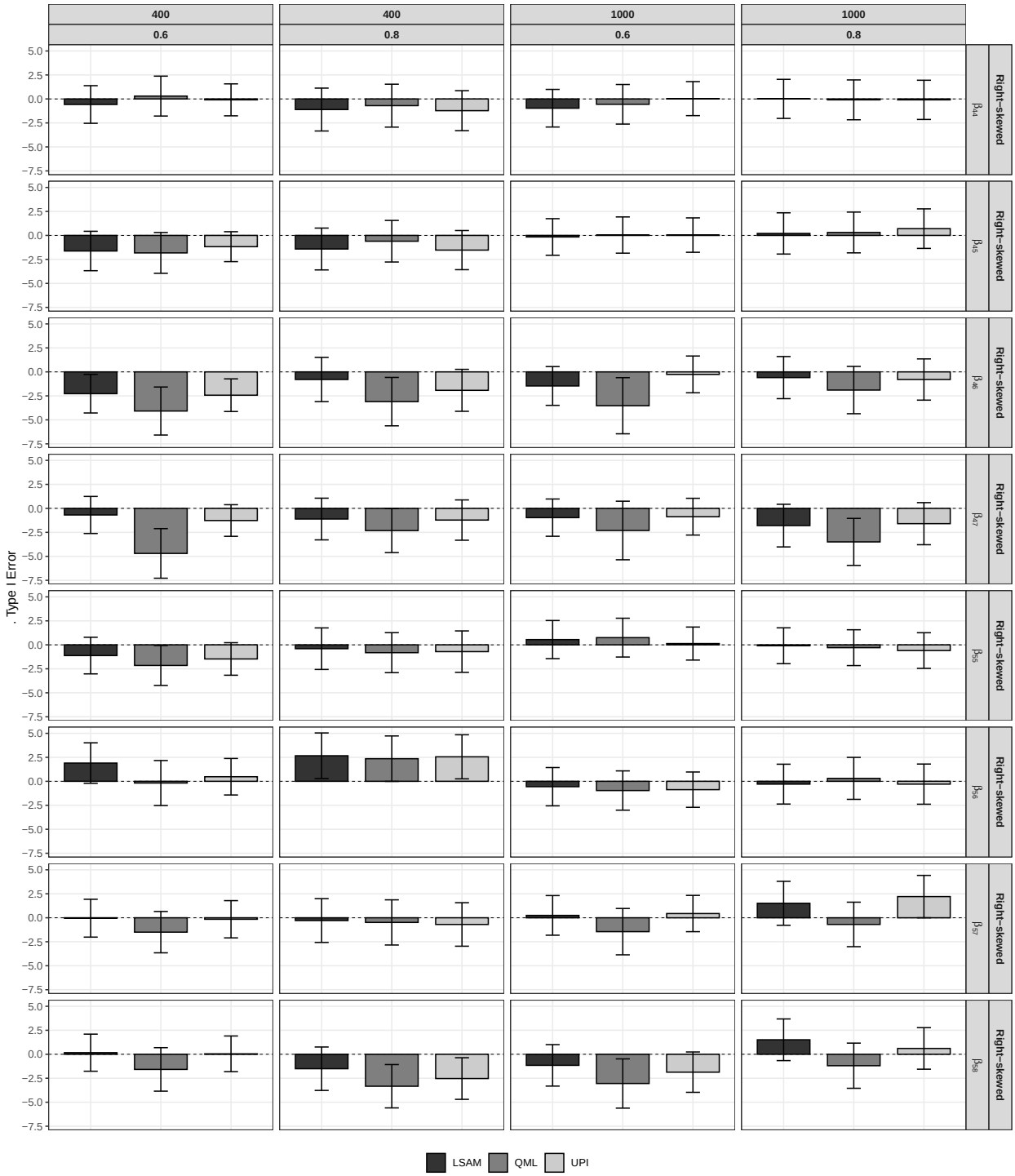


Figure S54

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Right-skewed (Epsilon & Zeta)

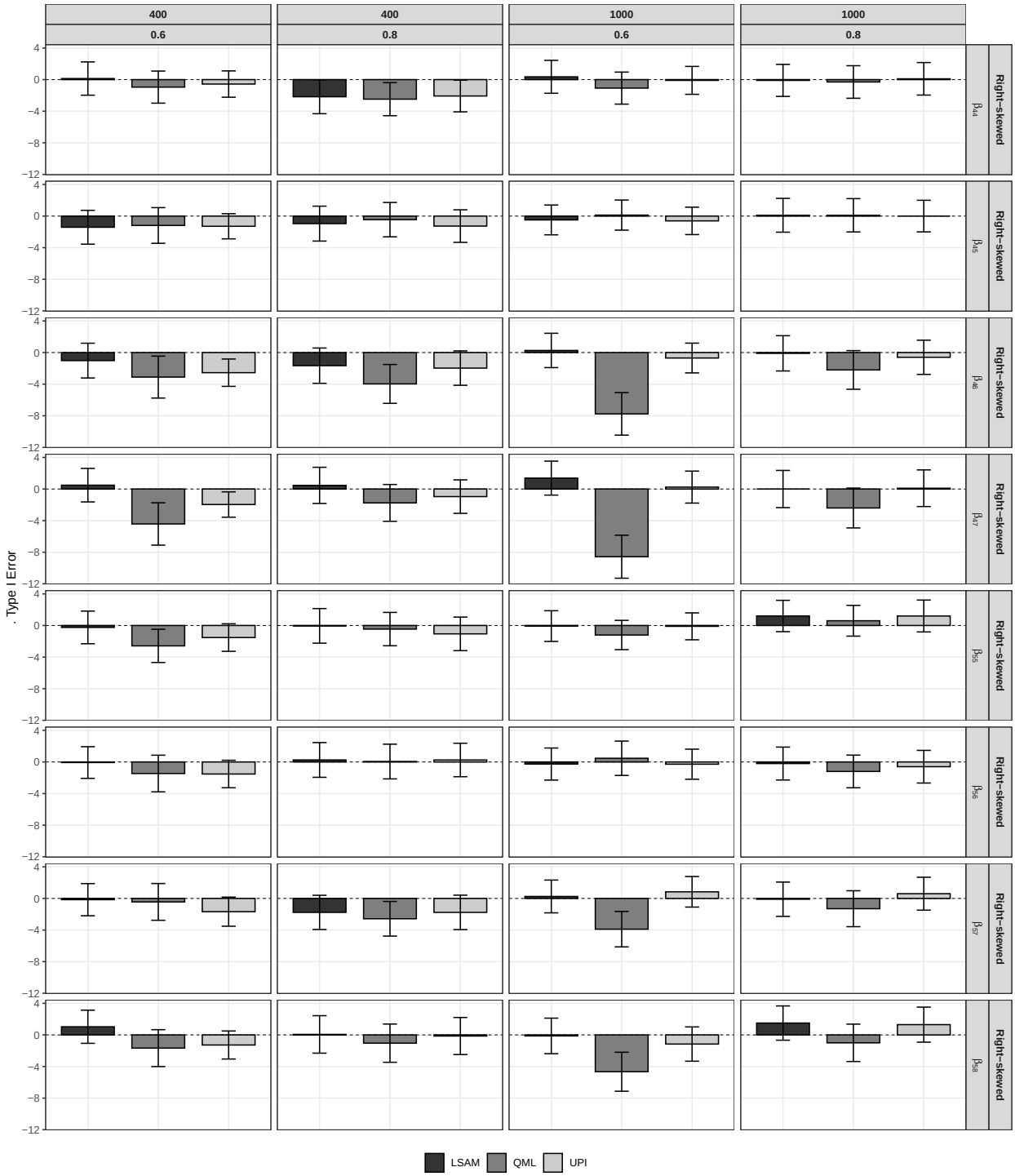


Figure S55

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Uniform (Epsilon)

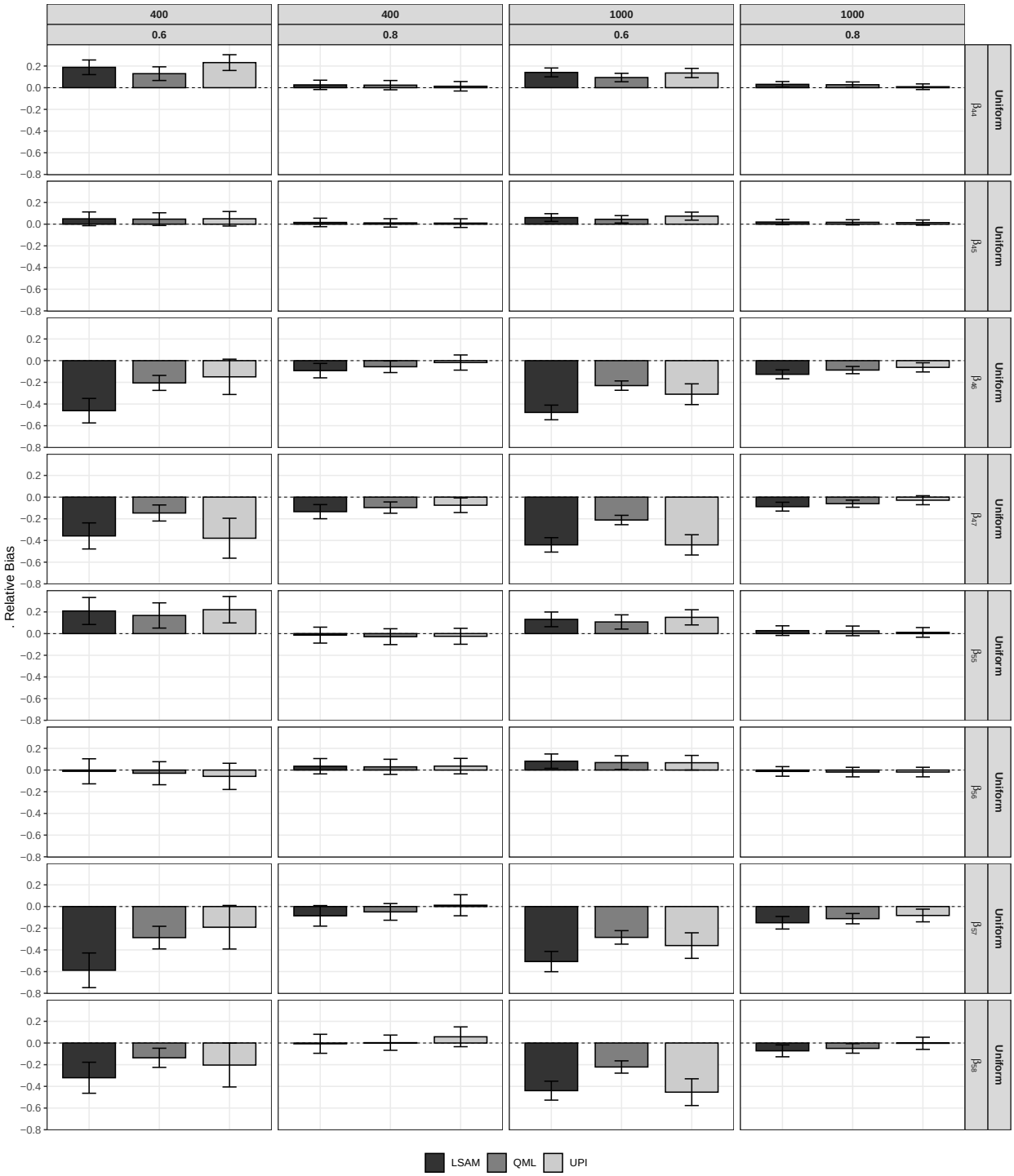


Figure S56

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Uniform (Zeta)

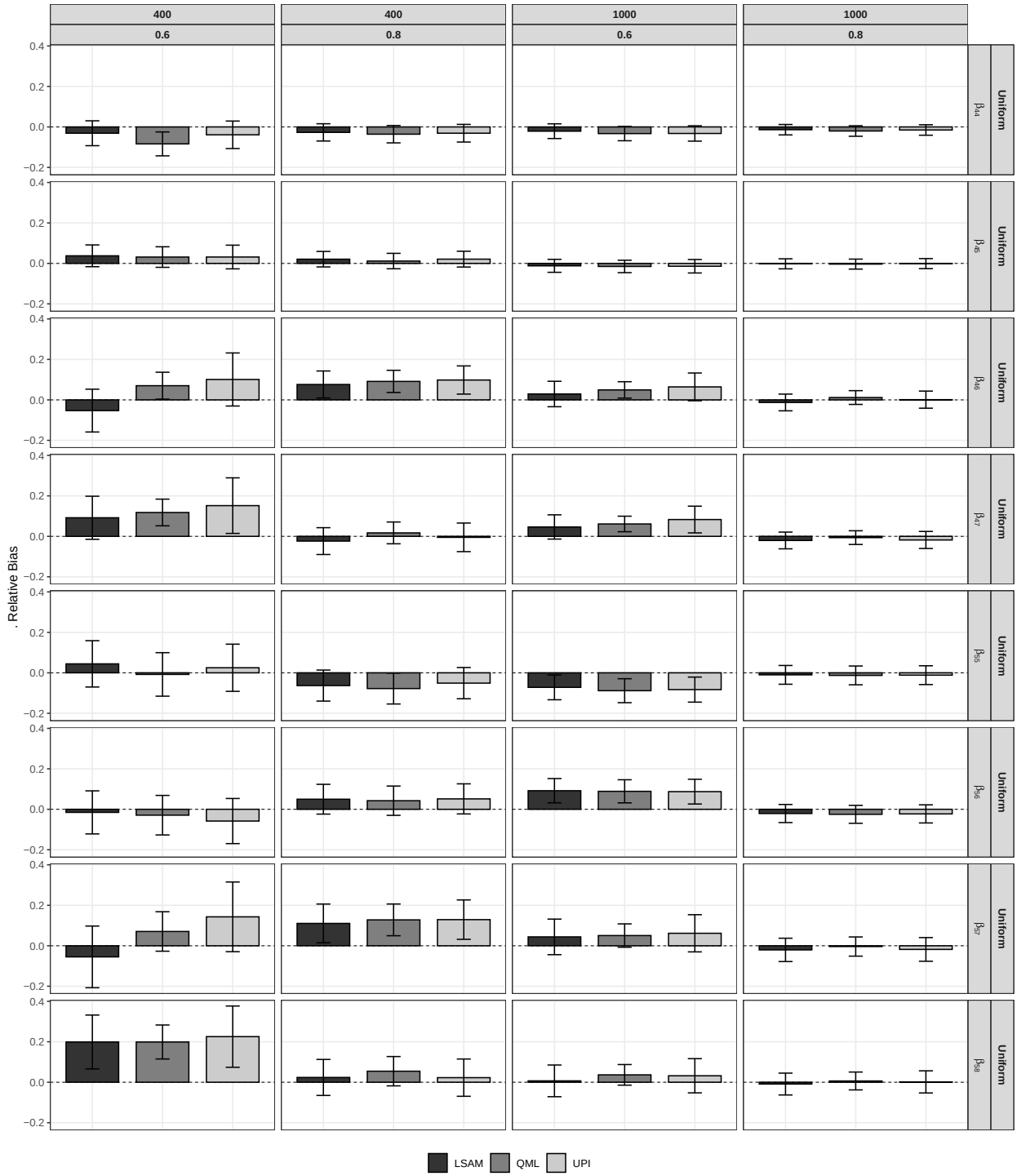


Figure S57

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Uniform (Epsilon & Zeta)

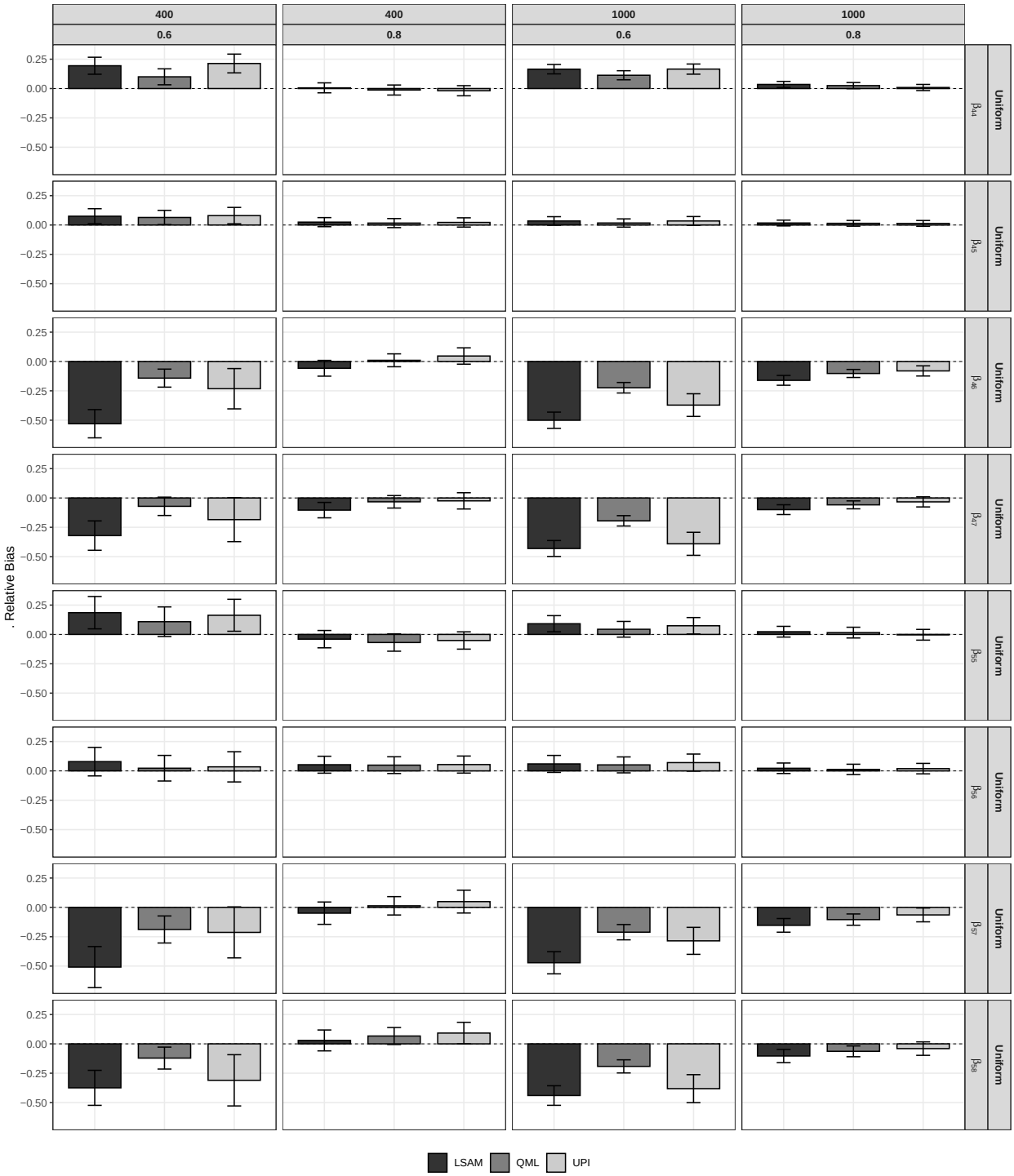


Figure S58

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Uniform (Epsilon)

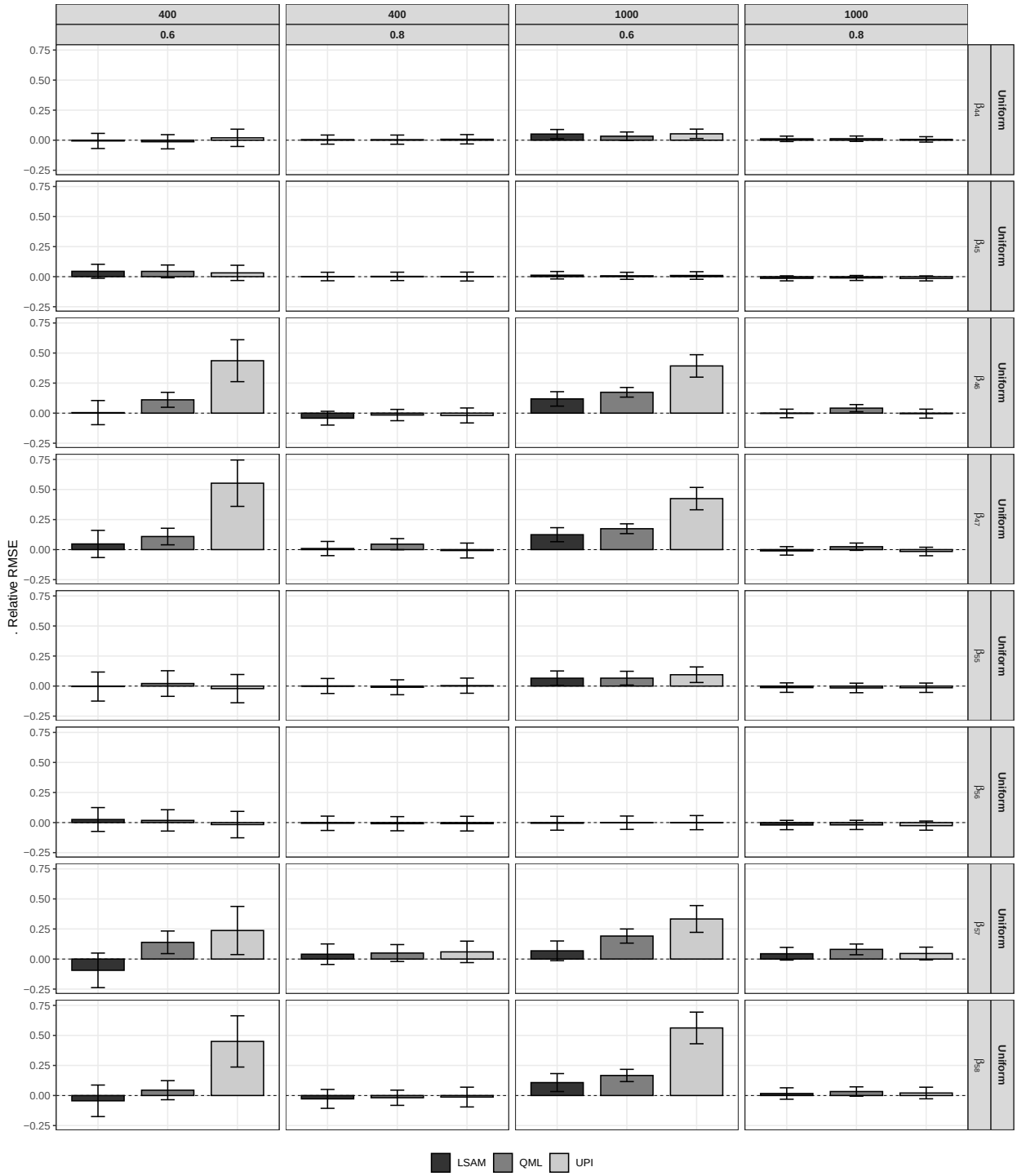


Figure S59

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Uniform (Zeta)

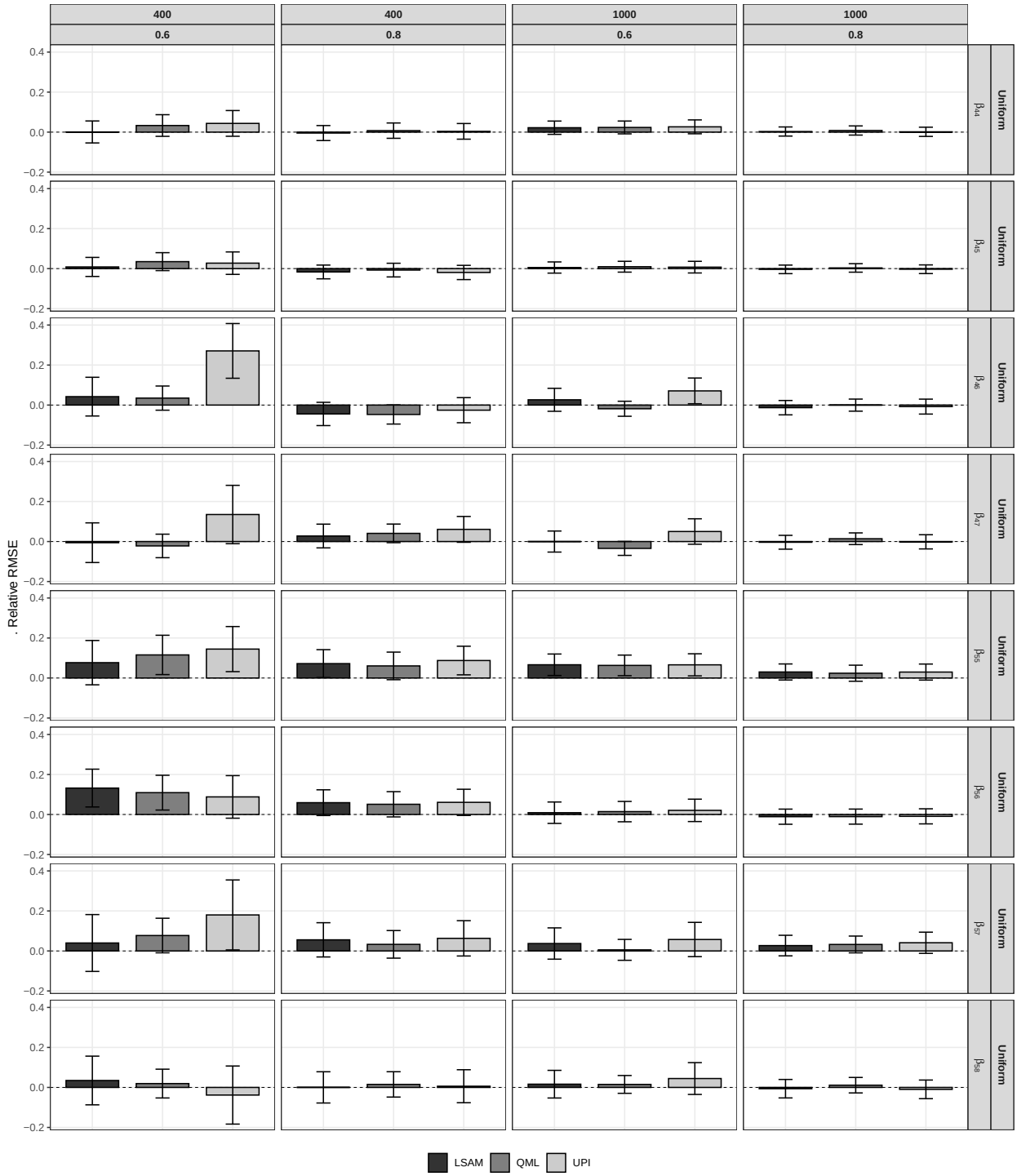


Figure S60

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Uniform (Epsilon & Zeta)

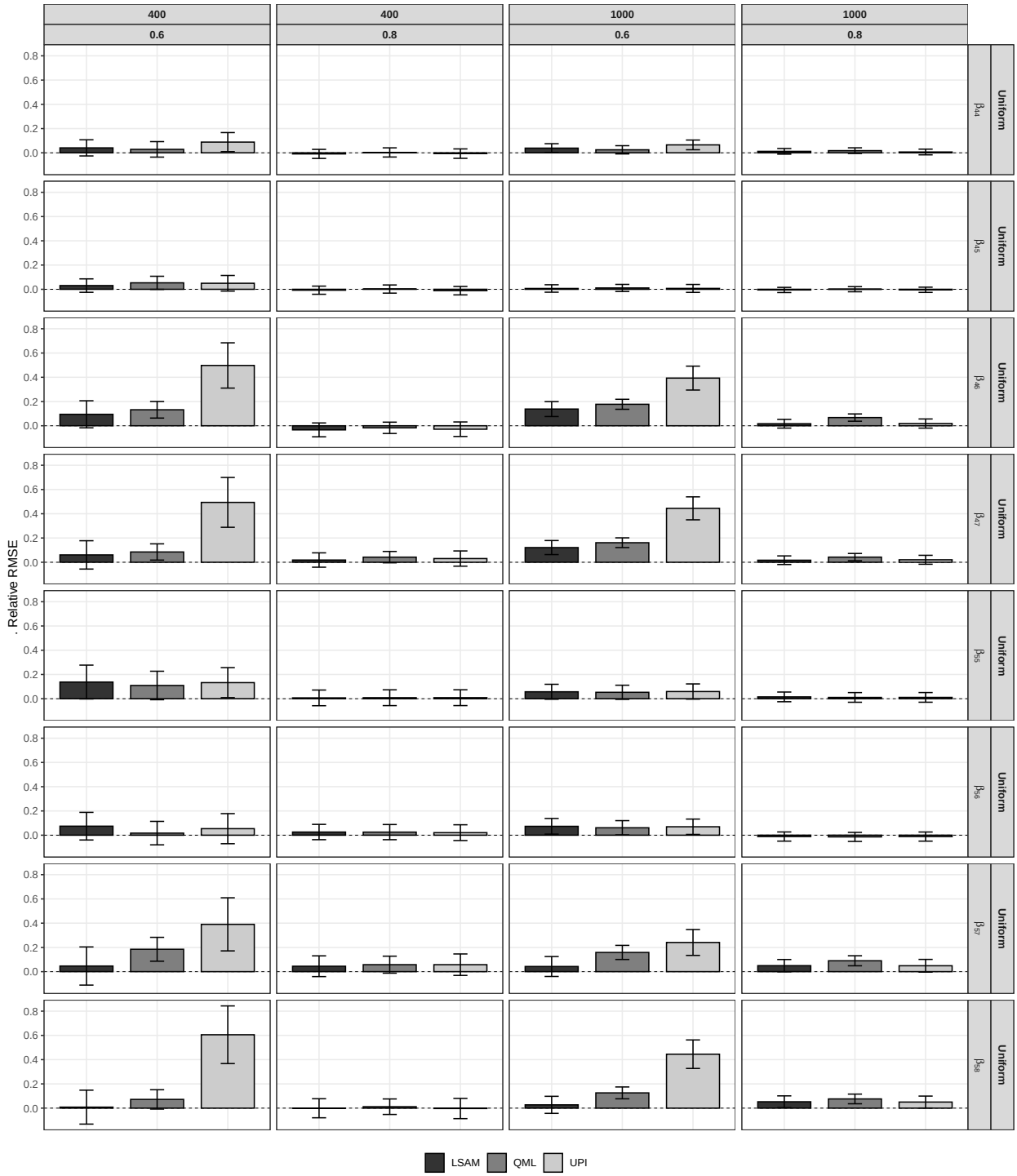


Figure S61

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Uniform (Epsilon)

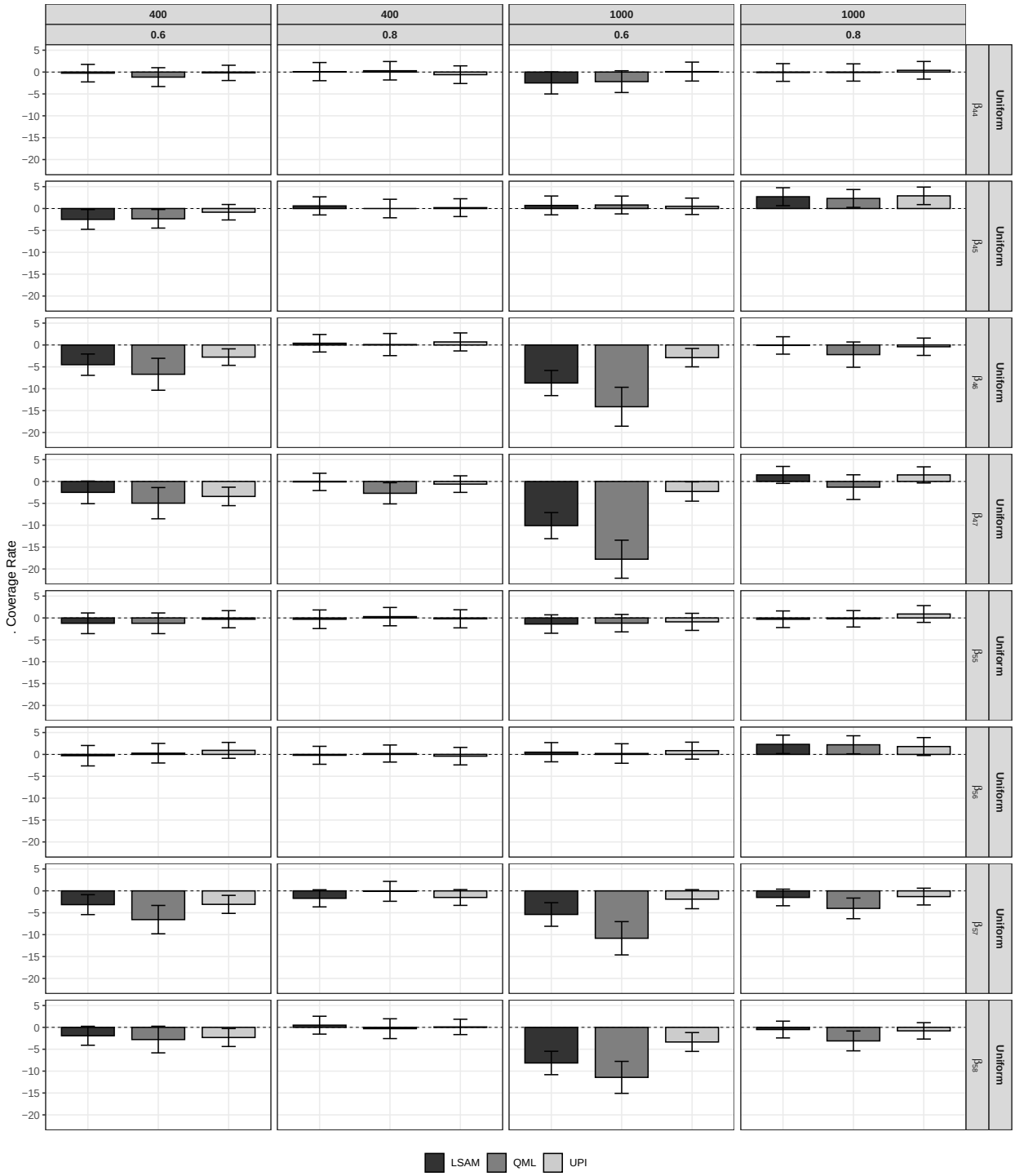


Figure S62

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Uniform (Zeta)

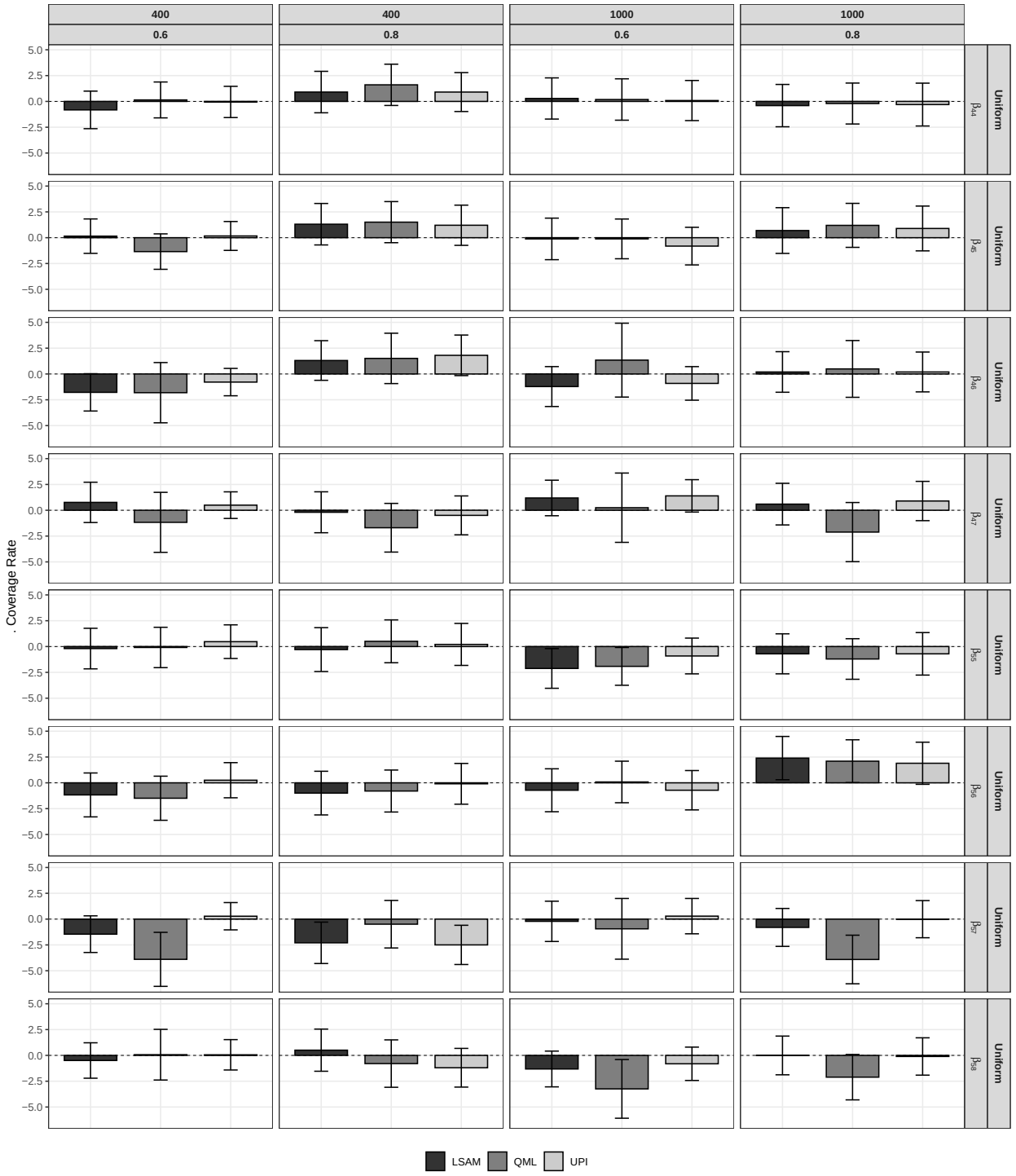


Figure S63

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Uniform (Epsilon & Zeta)

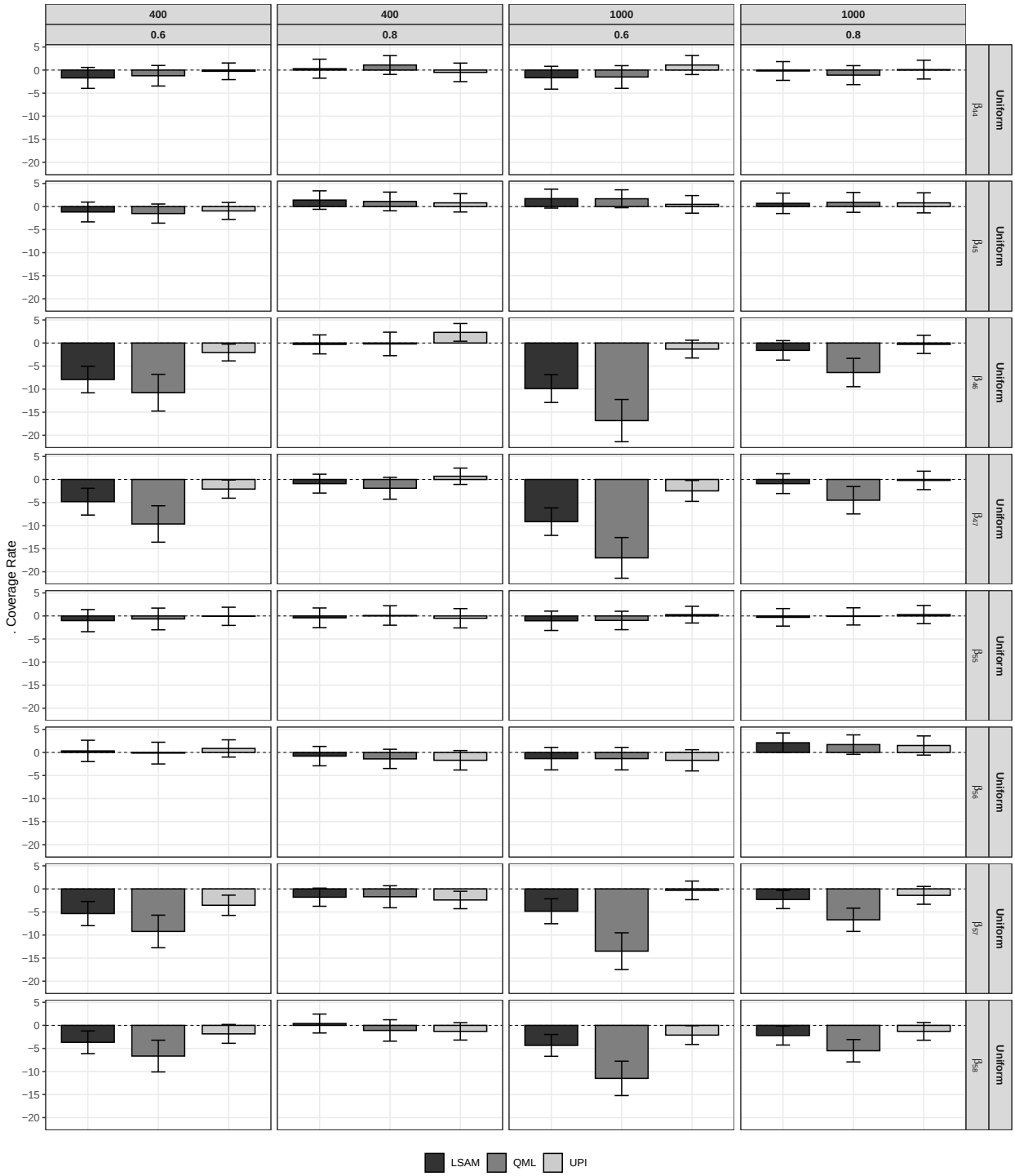


Figure S64

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Uniform (Epsilon)

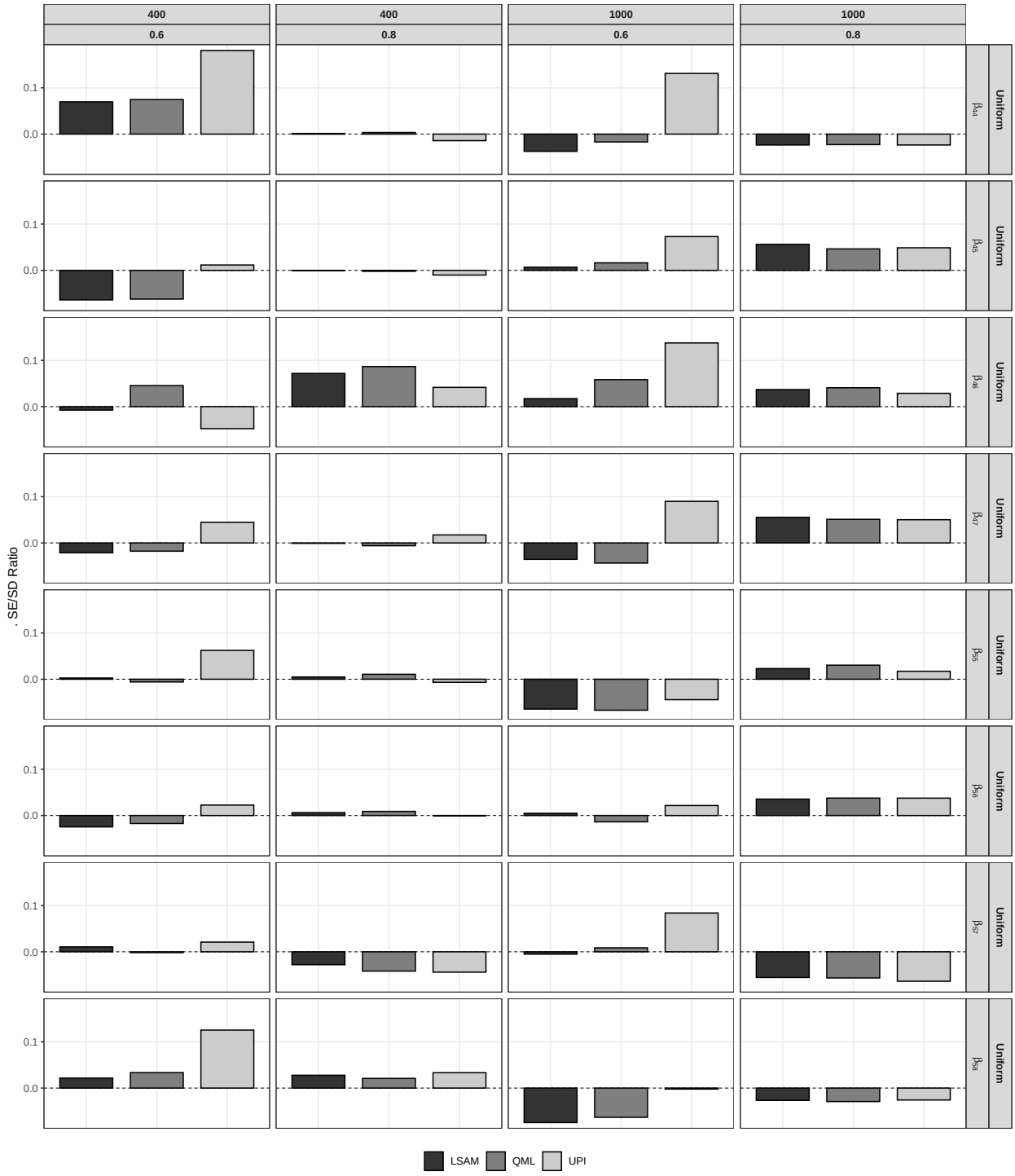


Figure S65

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Uniform (Zeta)

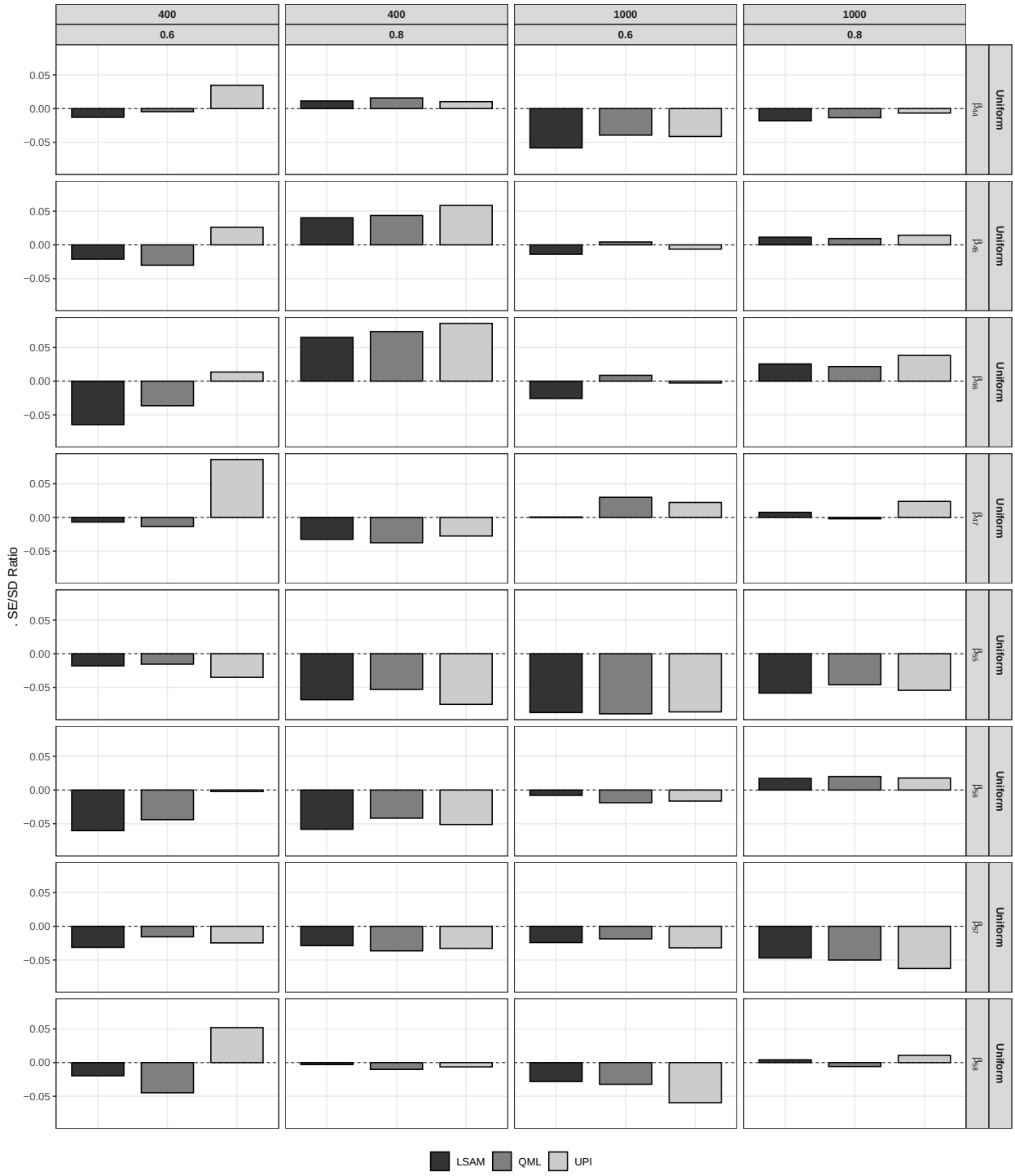


Figure S66

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Uniform (Epsilon & Zeta)

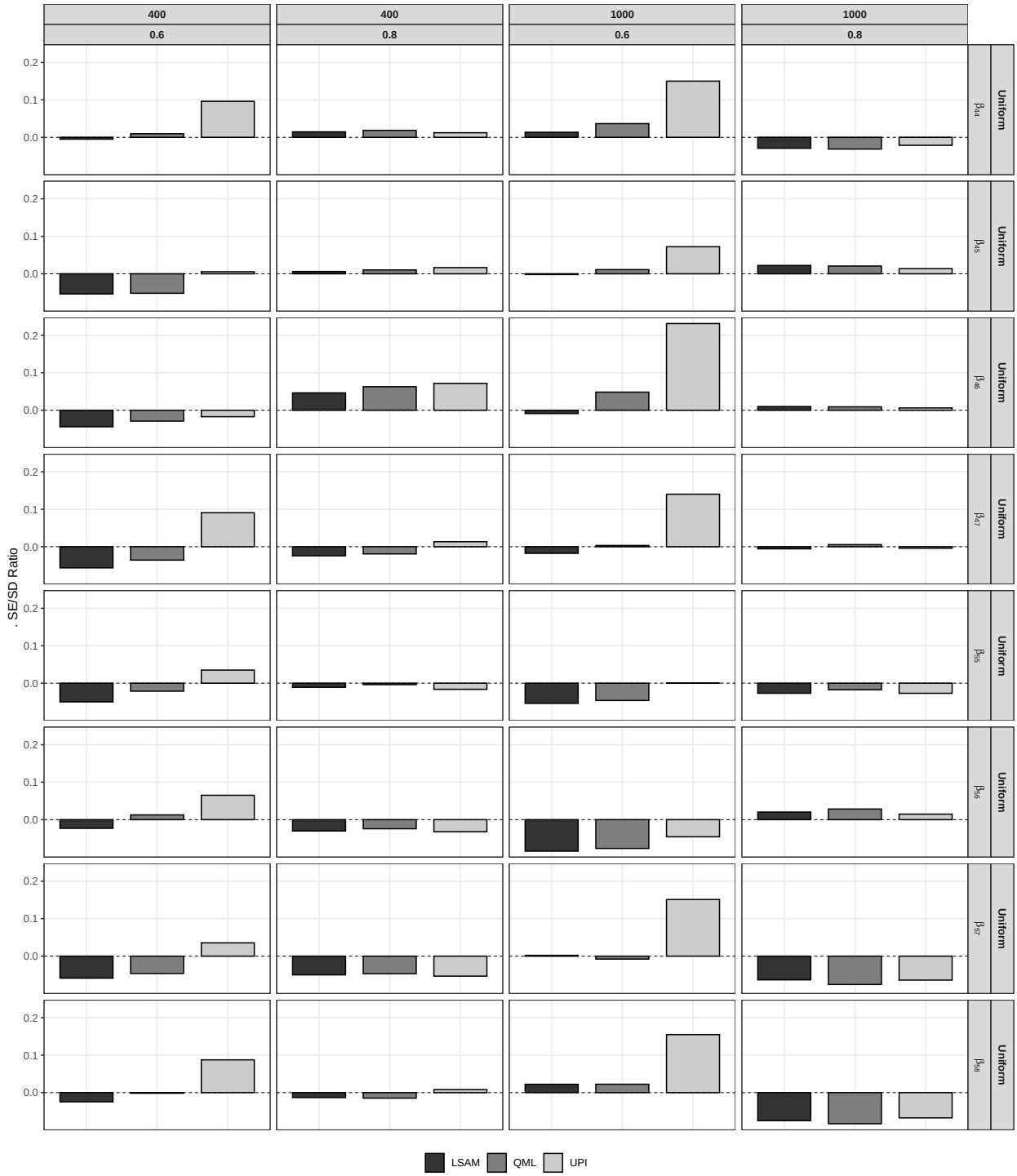


Figure S67

Additional Distributional Conditions Difference: Power for Simulation 2 – Uniform (Epsilon)

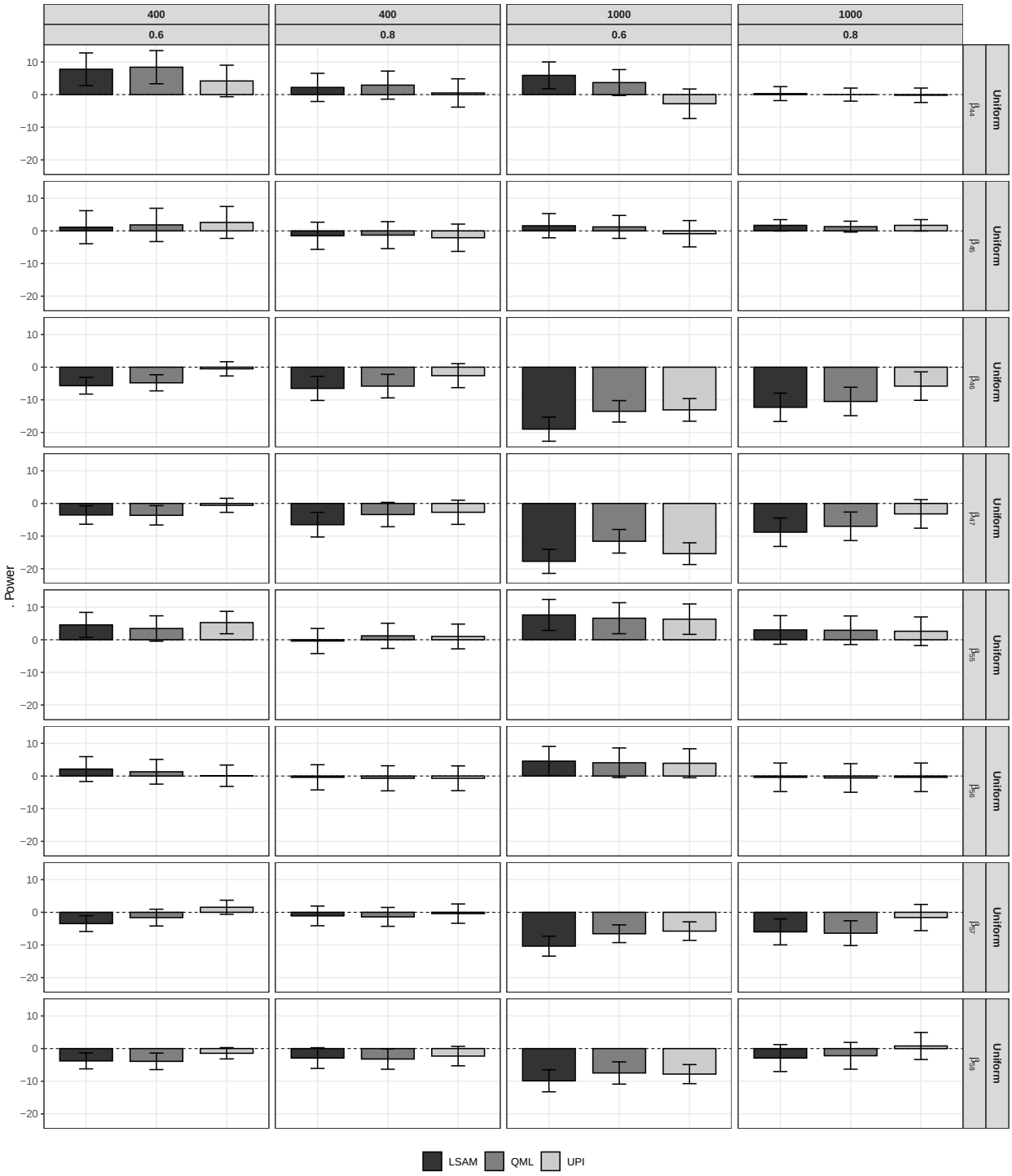


Figure S68

Additional Distributional Conditions Difference: Power for Simulation 2 – Uniform (Zeta)

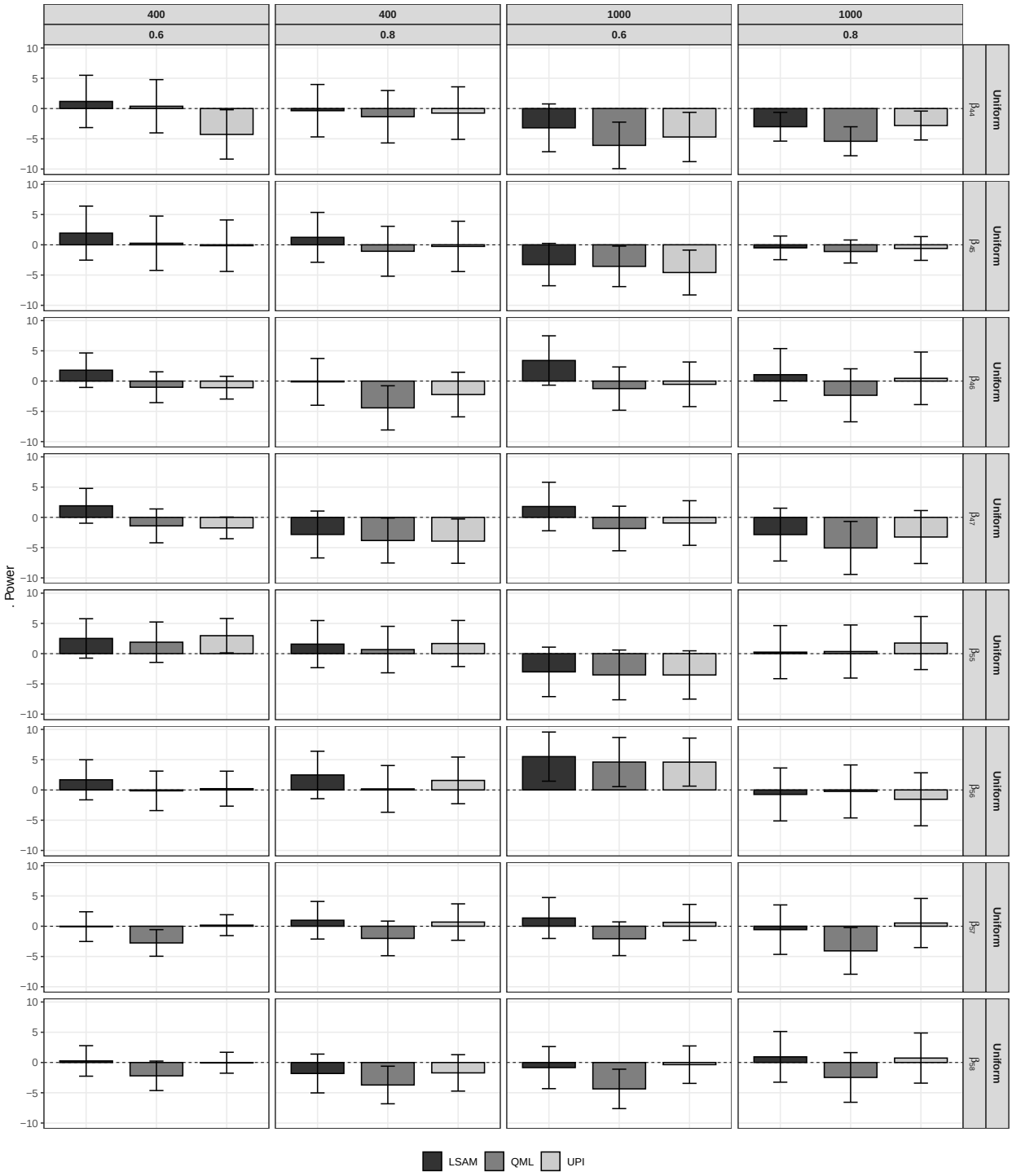


Figure S69

Additional Distributional Conditions Difference: Power for Simulation 2 – Uniform (Epsilon & Zeta)

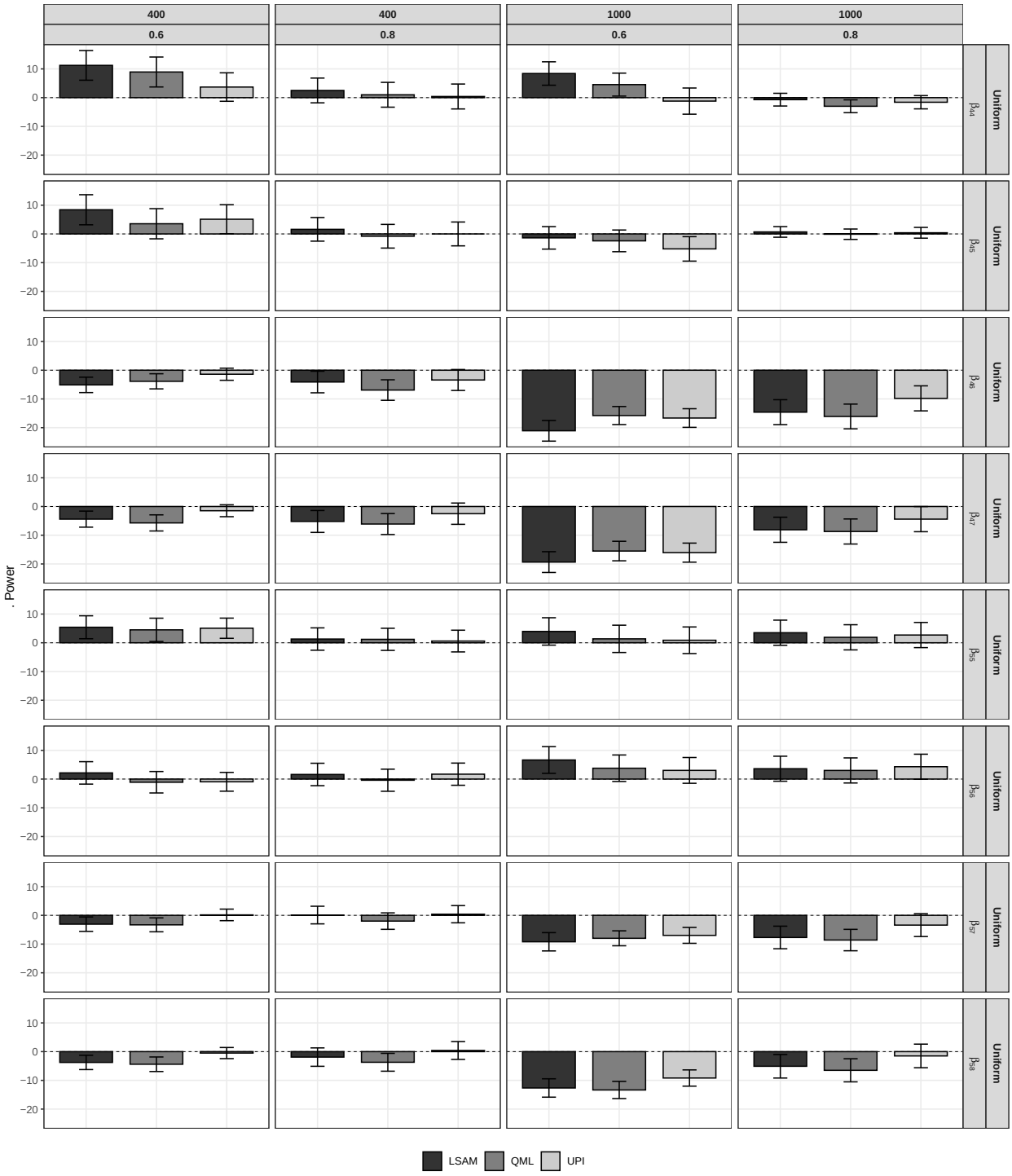


Figure S70

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Uniform (Epsilon)

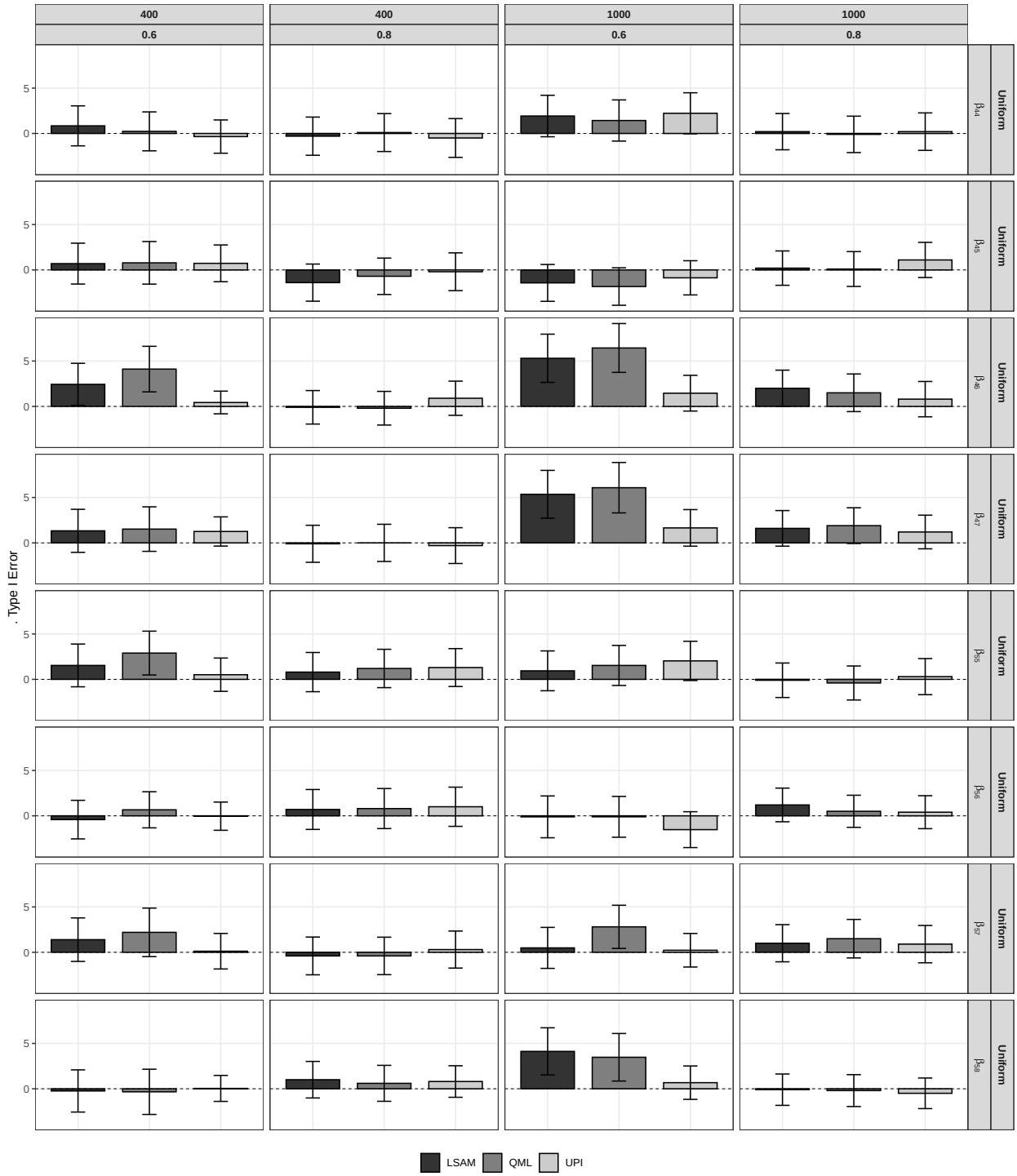


Figure S71

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Uniform (Zeta)

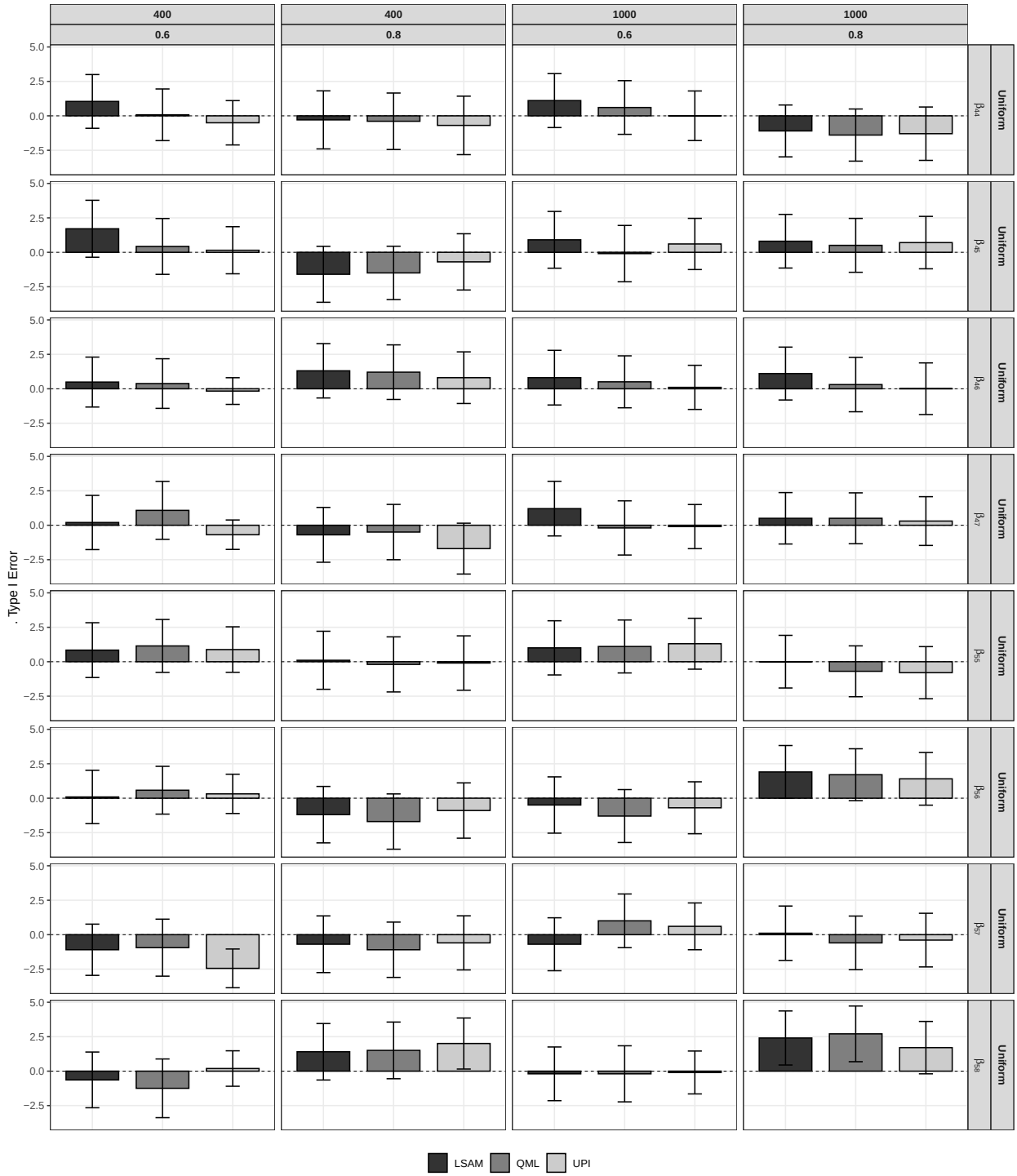
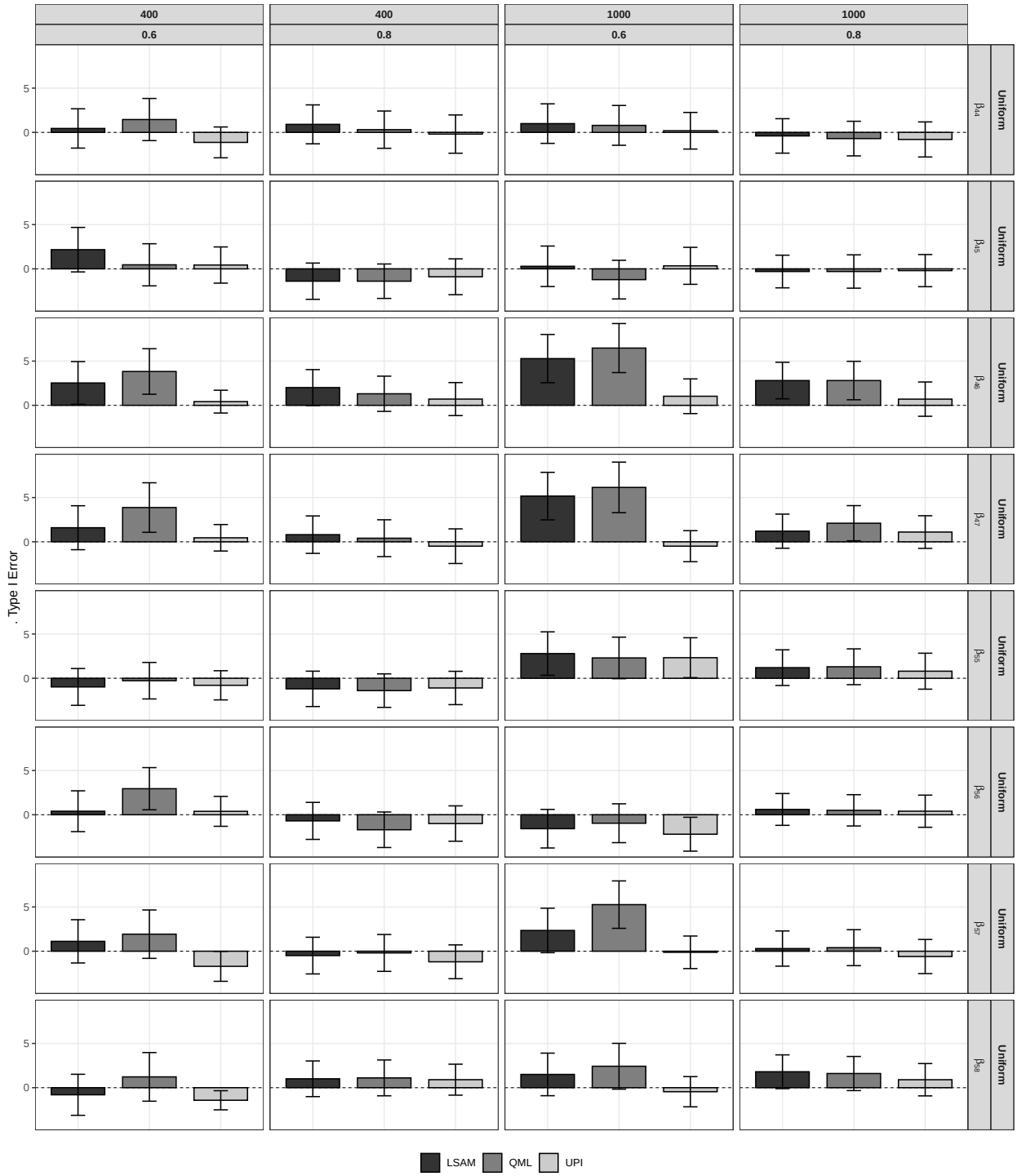


Figure S72

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Uniform (Epsilon & Zeta)